

CONGRUENCES FOR THE NUMBER OF SUBGROUPS OF ORDER p^k IN A FINITE GROUP

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The method used by Serre [1, p. 147] to prove the Sylow theorem gives results more general than that theorem. We spell them out here.

We use the following notation: p denotes a prime number; for every integer n , $v(n)$ is the exponent of the largest power of p dividing n ; finally, G is a finite group of order g .

For completeness, recall the following lemma (Lazard [2, p. 71]).

Lemma 1.

a) If $n = \sum c_i p^i$, with $0 \leq c_i < p$ the base- p expansion of n , then

$$v(n!) = \frac{1}{p-1} \left(n - \sum c_i \right).$$

b)

$$v\binom{r}{s} \geq v(r) - v(s).$$

For a), one has

$$v(p^k!) = \sum_{i \leq k} i(p^{k-i} - p^{k-i-1}) = \sum_{i=0}^{k-1} p^i = \frac{p^k - 1}{p - 1},$$

which proves the assertion in this case. It is clear that, if $0 \leq c < p$, then $v((cp^k)!) = cv(p^k!)$, and that, if $m < p^{v(n)}$, then

$$v((n+m)!) = v(n!) + v(m!).$$

These additivity properties are the same as those of the right-hand side, whence the formula.

Part b) follows easily by returning to the definitions and taking account of the algorithm which computes the digits c_i of a sum from the digits of its summands. Equality holds when r is a power of p .

Proposition 2.

$$\binom{p^k n}{p^k m} \equiv \binom{n}{m} \pmod{p^{v(n)+2}}$$

if $p \neq 2$, or if n is odd. The congruence is always true modulo $p^{v(n)+1}$.

Proof. Let X and Y be two indeterminates. One has

$$(X + Y)^{p^k} = X^{p^k} + Y^{p^k} + pQ,$$

where Q is a polynomial of degree $< p^k$ in each variable. The binomial formula then gives

$$(1) \quad (X + Y)^{p^k n} = (X^{p^k} + Y^{p^k})^n + np(X^{p^k} + Y^{p^k})^{n-1}Q$$

modulo the largest power of p which divides all the numbers $p^i \binom{n}{i}$, $i \geq 2$. These have valuation at least

$$i + v(n) - v(i) \geq v(n) + 2$$

if $p \neq 2$. If $p = 2$, one always has $i > v(i)$ (Cantor), which gives the exponent $v(n) + 1$.

The binomial coefficient $\binom{p^k n}{p^k m}$ is the coefficient of $X^{p^k m} Y^{p^k(n-m)}$. The second term on the right-hand side of (1) contains no term of this type, since none of its exponents is divisible by p^k . Thus the coefficient of this monomial on the right-hand side of (1) is $\binom{n}{m}$. This proves the assertion when $p \neq 2$. If $p = 2$, the preceding already proves the congruence modulo $2^{v(n)+1}$. In addition, $Q \equiv X^{2^{k-1}} Y^{2^{k-1}} \pmod{2}$, so $Q^2 \equiv X^{2^k} Y^{2^k} \pmod{4}$, and the next term in the binomial expansion gives

$$\binom{2^k n}{2^k m} \equiv \binom{n}{m} + 2n(n-1) \binom{n-2}{m-1} \pmod{2^{v(n)+2}},$$

which gives the asserted improvement when n is odd. $\square$

Theorem 3.

Let S be a Sylow p -subgroup of G , let $s = v(g)$, and let d_k be the number of subgroups of G of order p^{s-k} , for $0 \leq k \leq s$. Then:

- (i) $d_k \equiv 1 \pmod{p}$;
- (ii) if S is cyclic or abelian of type $(p, \dots, p)$, then d_k is congruent modulo p^{k+1} to the number of subgroups of S of index p^k ;
- (iii) if S is not cyclic, then

$$d_1 \equiv 1 + p \pmod{p^2}.$$

Let G act by left translations on the set of its subsets with p^k elements. If such a subset P of G has stabilizer H , then P is a union of right cosets of H , and hence H is a group of order p^l with $l \leq k$; one has $l = k$ if and only if P is a translate, on the right or on the left, of a subgroup of order p^k . The number of elements in the orbit of P is divisible by p^{s-l} . Therefore $\binom{g}{p^k}$ is congruent modulo p^{s-k+1} to the number of subsets of G which are translates of a subgroup of order p^k . By Proposition 1, it is also congruent to gp^{-k} . There are $d_{s-k} gp^{-k}$ such subsets. Thus

$$gp^{-k} \equiv d_{s-k} gp^{-k} \pmod{p^{s-k+1}},$$

and the first assertion follows.

We now consider case (ii), first when S is abelian of type $(p, \dots, p)$. Let $G_{i,j}$ be the number of subgroups of order p^i in a group of order p^j of this type; this is the number of points of a Grassmannian. For $i > j$, $G_{i,j} = 0$; for $i \leq j$,

$$G_{i,j} = \frac{(p^j - 1) \cdots (p^j - p^{i-1})}{(p^i - 1) \cdots (p^i - p^{i-1})} = G_i G_{j-i} G_j^{-1},$$

where

$$G_i = [(p^i - 1) \cdots (p - 1)]^{-1} \quad (i \geq 0), \quad G_i = 0 \quad (i < 0).$$

We know that p^i divides the number u_i of subsets of G with p^i elements whose stabilizer has order p^{s-i} . Put $g = p^s h$. Counting in two ways the pairs formed by a subgroup of order p^{s-i} and a subset with p^i elements fixed by it gives, for $0 \leq i, j \leq s$,

$$(2) \quad d_i \binom{p^s h}{p^i} = \sum_j G_{s-i, s-j} u_j,$$

$$(3) \quad p^i \mid u_i,$$

$$(4) \quad d_s = 1.$$

Now work in the localization of $\mathbf{Z}$ at p , or in $\mathbf{Z}_p$. Let e_i be the quotient of the left-hand side of (2) by $\binom{p^s h}{p^i} G_i$. By Proposition 1, and since G_i is prime to p for $i \geq 0$, there are numbers v_i , $0 \leq i \leq s$, such that

$$(5) \quad d_i \equiv G_i e_i \pmod{p^{i+1}},$$

$$(2') \quad e_i = \sum_j G_{i-j} v_j,$$

$$(6) \quad p^i \mid v_i,$$

$$(7) \quad e_s = G_s^{-1}.$$

Analogous identities hold in the group S . To show that they determine d_i modulo p^{i+1} , it is enough to check that the identities

$$(8) \quad e_i = \sum_j G_{i-j} v_j,$$

$$(9) \quad p^i \mid v_i,$$

$$(10) \quad e_s = 0$$

for $0 \leq i, j \leq s$ imply $p^{i+1} \mid e_i$.

From (2'') one obtains

$$e_i - e_{i+1} = \sum_j (G_{i-j} - G_{i+1-j}) v_j.$$

Since trivially $G_k^{-1} \equiv G_{k+1}^{-1} \pmod{p^{k+1}}$, one also has

$$p^{i+1-j} \mid G_{i-j} - G_{i+1-j}, \quad p^{i+1} \mid e_i - e_{i+1}.$$

The conclusion follows from (4'').

The same technique, in a simpler form, applies when S is cyclic.

We turn to case (iii). The subgroups of S of index p are all normal. Their intersection S' is the Frattini subgroup of S . For every subgroup T of S , if $TS' = S$, then $T = S$. In particular, if S is not cyclic, then S/S' is not cyclic either, and the subgroups of S of index p are identified with the hyperplanes of a vector space over $\mathbf{Z}/p$ of dimension > 1 ; their number is congruent to $1 + p \pmod{p^2}$.

The preceding equations leave

$$\begin{cases} d_0 h = u_0, \\ d_1 \binom{p^s h}{p} = (1+p)u_0 + u_1 \pmod{p^2}, \\ \binom{p^s h}{p^s} = u_0 + u_1 \pmod{p^2}. \end{cases}$$

By Proposition 1, $\binom{p^s h}{p^s} \equiv h \pmod{p^2}$. Hence

$$d_1 h \equiv (1+p)d_0 h + (h - d_0 h) \pmod{p^2}, \quad \text{and } d_1 \equiv 1 + p d_0 \pmod{p^2}.$$

The assertion follows because $d_0 \equiv 1 \pmod{p}$.

REFERENCES

- [1] SERRE, J.-P., *Corps locaux. Act. sc. et ind.* Hermann, 1962.
- [2] LAZARD, M., Groupes analytiques p -adiques. *Publ. Math. I.H.E.S.*, no. 26, 1965.

APPENDIX: COHOMOLOGY WITH PROPER SUPPORT AND THE CONSTRUCTION OF THE FUNCTOR $f^!$

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This is a completed version of a letter from P. Deligne to R. Hartshorne (letter of March 3, 1966). The footnotes were added by the copyist.

Verdier showed that, in the topological case, the formalism of Poincaré duality reduces to local questions upstairs (see [1]). In order to transpose his construction to the schematic setting, one needs a theory of cohomology “with proper support” for coherent sheaves.

Unless explicitly stated otherwise, all preschemes considered below are Noetherian, and all sheaves are quasi-coherent.

NO. 1. THE SORITES OF PRO-OBJECTS

Proposition 1. *Let C be a U -category¹ admitting finite colimits. Let h be a functor $C^\circ \rightarrow (\text{Sets})$. The following conditions are equivalent:*

- (i) *h is a filtered colimit, indexed by a small² directed ordered set, of representable functors.*
- (ii) *h is a filtered colimit, indexed by a small filtered category, of representable functors.*
- (iii) *h transforms finite projective limits into finite inductive limits, and there exists a small set of objects such that every element of every $h(X)$ factors through one of them.*

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and (iii) $\Rightarrow$ (ii) is standard (h is the limit of the h_F indexed by the category of pairs (F, α) with $\alpha \in h(F)$ and F ranging through the full subcategory of C stable under finite projective limits and generated by the given small set). It remains to prove that (ii) $\Rightarrow$ (i). If $\mathcal{L}$ and $\mathcal{M}$ are filtered categories, a functor $F: \mathcal{L} \rightarrow \mathcal{M}$ is said to be cofinal if $\varinjlim_{\mathcal{L}} h_{F(L)}$ is the terminal functor. For every $G: \mathcal{M} \rightarrow \mathcal{N}$ one then has $\varinjlim_{\mathcal{M}} G = \varinjlim_{\mathcal{L}} GF$. Thus it suffices to prove the following.

Lemma 1. *For every small filtered category $\mathcal{L}$, there exists a cofinal functor from a small directed ordered set into $\mathcal{L}$.*

The first projection $\mathcal{L} \times \mathbb{N} \rightarrow \mathcal{L}$ is cofinal ($\mathbb{N}$ being ordered in the usual way); this reduces the problem to the case in which the preorder on $\text{Ob } \mathcal{L}$ defined by $\text{Hom}(X, Y) \neq \emptyset$ has no greatest element. Order by inclusion the set E of finite subcategories of $\mathcal{L}$ having a unique final object (“finite” means that the set of arrows is finite). One gets a functor from E to $\mathcal{L}$ by sending such a subcategory to its final object. Under the above hypothesis, E is filtered and this functor is cofinal.

The functors satisfying the equivalent conditions of Proposition 1 are the Ind-objects of C . They form a full subcategory $\text{Ind } C$ of $\underline{\text{Hom}}(C^\circ, (\text{Sets}))$, stable under colimits. The assignment $C \mapsto \text{Ind } C$ is a functor $(\text{cat}) \rightarrow (\text{cat})$. If filtered colimits exist in C , then for every $h \in \text{Ind } C$ the functor $X \mapsto \text{Hom}(h, h_X)$ is corepresentable; hence there is a functor, denoted $\varinjlim$, from $\text{Ind } C$ to C . In particular one always has a functor $\text{Ind } \text{Ind } C \rightarrow \text{Ind } C$, and every functor $\tilde{C} \rightarrow \text{Ind } D$ extends to a functor $\text{Ind } C \rightarrow \text{Ind } D$. In $\text{Ind } C$, filtered colimits are exact and will again be denoted by

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¹ U denotes a fixed universe throughout.

²“small” means “ $\in U$ ”.

“ $\varinjlim$ ”; in particular, identifying C with a full subcategory of $\text{Ind } C$, if $(X_i)_{i \in I}$ is a filtered inductive system in C ,

$$\varinjlim_{i \in I} X_i = \varinjlim_{i \in I} \text{“} \varinjlim \text{” } X_i.$$

For a functor on $\text{Ind } C$ to be representable it is necessary and sufficient that it transform finite projective limits into finite inductive limits, filtered inductive limits into filtered inductive limits, and that its restriction to C satisfy the smallness condition of Proposition 1(iii) (indeed, Proposition 1 then shows that its restriction to C is an Ind -object, to which it is equal everywhere by the condition on limits).

Everything above, except Proposition 1(iii) and the preceding assertion, remains true if one uses throughout limits of sequences (or countable limits, which is the same thing).

By duality one defines the category $\text{pro } C$ of pro-objects (a full subcategory of $\underline{\text{Hom}}(C, (\text{Sets}))^\circ$).

Proposition 2. *Let X be a quasi-compact quasi-separated prescheme (not necessarily Noetherian). The category of quasi-coherent sheaves on X is equivalent to the category of Ind -objects of the category of finitely presented quasi-coherent sheaves on X .*

The functor is “ $\varinjlim$ ” $(\mathcal{F}_i) \mapsto \varinjlim \mathcal{F}_i$. If $\mathcal{F}$ is finitely presented, then for every filtered inductive system $(\mathcal{G}_i)_{i \in I}$,

$$\text{Hom}_X(\mathcal{F}, \varinjlim \mathcal{G}_i) = \varinjlim \text{Hom}_X(\mathcal{F}, \mathcal{G}_i)$$

(by a finite gluing argument commuting with $\varinjlim$, one reduces to the affine case). The preceding functor is therefore fully faithful. Its image is stable under filtered colimits and finite sums. It remains to prove that for every $x \in X$, every sheaf $\mathcal{F}$ on X , and every $s \in \mathcal{F}_x$, there exists a finitely presented sheaf $\mathcal{G}$ and a morphism $\mathcal{G} \rightarrow \mathcal{F}$ whose image contains s :

$\mathcal{F}$ will then be a limit of images of finitely presented sheaves, and each such image a quotient of a finitely presented sheaf by a limit of finite-type subsheaves.

Let $f: \mathcal{C} \rightarrow \mathcal{F}$ be a morphism on some quasi-compact neighborhood U of x , with $\mathcal{C}$ finitely presented and s lying in the image. It suffices to extend $\mathcal{C}$ and f to all of X ; proceeding step by step, it is enough to know how to extend from U to $U \cup V$ when V is affine. This reduces to extending from $U \cap V$ to V :

Lemma 2. *Let $\mathcal{C}$ be a finitely presented sheaf on a quasi-compact open subset U of an affine scheme X , let $\mathcal{F}$ be a sheaf on X , and let f be a morphism from $\mathcal{C}$ to $\mathcal{F}|_U$. Then $\mathcal{C}$ and f can be extended to $\bar{\mathcal{C}}$ and $\bar{f}$ defined on all of X (with $\bar{\mathcal{C}}$ finitely presented).*

Let ι be the inclusion of U into X and let $\mathcal{C}_1$ be the fiber product of $\iota_* \mathcal{C}$ and $\mathcal{F}$ over $\iota_* \iota^* \mathcal{F}$. Then $\mathcal{C}_1$ extends $\mathcal{C}$ and maps to $\mathcal{F}$. If one writes $\mathcal{C}_1$ as a limit of its finite-type subsheaves, then, because U is quasi-compact and $\mathcal{C}$ is finitely presented, one of them already extends $\mathcal{C}$. Writing this subsheaf as a quotient of $\mathcal{O}^n$ by a limit of finite-type subsheaves of $\mathcal{O}^n$, one sees similarly that $\mathcal{C}_1$ can be replaced by a finitely presented sheaf.

Corollary 1. *For a contravariant additive functor $F: (\text{q coh}) \rightarrow \text{Ab}$ to be representable, it is necessary and sufficient that it be left exact and commute with filtered colimits.*

Corollary 2. *Every finitely presented sheaf defined on a quasi-compact open subset U of X extends to a finitely presented sheaf on X .*

Let $\mathcal{A}$ be an abelian category. One will have to work in an essential way inside $\text{pro } D^b \mathcal{A}$, the full subcategory of $\text{pro } D(\mathcal{A})$ consisting of the “ $\varinjlim$ ” of K_i whose cohomology is uniformly bounded with respect to i . The functors $\mathbb{H}^p$ extend to functors $\text{pro } D(\mathcal{A}) \rightarrow \text{pro } \mathcal{A}$.

Proposition 3. *The functors $\mathbb{H}^p$ form a conservative system³ of functors on $\text{pro } D^b \mathcal{A}$.*

³i.e. if u is a morphism such that all the $\mathbb{H}^p(u)$ are invertible, then u itself is invertible.

Let $f: K \rightarrow L$ be a morphism in $\text{pro } D^b \mathcal{A}$ such that the maps $\mathbb{H}^p(K) \rightarrow \mathbb{H}^p(L)$ are isomorphisms (in $\text{pro } \mathcal{A}$). One has to verify that for every M in $D^b(\mathcal{A})$,

$$\text{Hom}(K, M) \leftarrow \text{Hom}(L, M)$$

is an isomorphism. Write $K = \varprojlim K_\alpha$. There is a convergent spectral sequence

$$\text{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \text{Ext}^{p+q}(K_\alpha, M),$$

hence, after passing to the limit, $\varinjlim \text{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \varinjlim \text{Ext}^{p+q}(K_\alpha, M)$, still convergent because q is uniformly bounded. Using the fact that every functor (here Ext) passes to pro-objects, together with the functor $\varinjlim: \text{Ind } \text{Ab} \rightarrow \text{Ab}$, one can rewrite this as

$$\varinjlim \text{Ext}^p(\mathbb{H}^q K, M) \Longrightarrow \varinjlim \text{Ext}^{p+q}(K, M) = \text{Hom}(K[-p-q], M).$$

This spectral sequence remains valid for $M \in D(\mathcal{A})$. The proposition follows immediately from its existence.

NO. 2. FUNDAMENTAL LEMMAS

Recall that the preschemes considered are Noetherian.

Proposition 4 (Duality theorem for an open immersion). *Let U be an open subset of X , let j denote the inclusion, let $\mathcal{J}$ be an ideal defining the complementary closed subset, let $\mathcal{F}$ be a coherent sheaf on U , and let $\mathcal{G}$ be a quasi-coherent sheaf on X . If $\bar{\mathcal{F}}$ is a coherent extension of $\mathcal{F}$, then*

$$(1) \quad \text{Hom}_U(\mathcal{F}, j^* \mathcal{G}) = \text{Hom}_X(\varprojlim \mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}).$$

The map $\varinjlim \text{Hom}_X(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) \rightarrow \text{Hom}_U(\mathcal{F}, \mathcal{G})$ is injective: if the image under f of $\mathcal{J}^n \bar{\mathcal{F}}$ in $\mathcal{G}$ has support contained in the complement of U , then it is annihilated by a power of $\mathcal{J}$, say $\mathcal{J}^k$, and the image of $\mathcal{J}^{n+k} \bar{\mathcal{F}}$ is zero.

Assume now that X is affine, and let $f \in \text{Hom}_U(\mathcal{F}, \mathcal{G})$. For k sufficiently large, every section s of $\mathcal{J}^k \bar{\mathcal{F}}$ over X has an image (over U) extending to a section of $\mathcal{G}$ over X . Replacing $\bar{\mathcal{F}}$ by $\mathcal{J}^k \bar{\mathcal{F}}$, one may suppose given a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & R & \longrightarrow & \mathcal{O}^n & \longrightarrow & \bar{\mathcal{F}} \longrightarrow 0 \\ & & \vdots & & \parallel & & \vdots \\ 0 & \longrightarrow & R_1 & \longrightarrow & \mathcal{O}^n & \longrightarrow & \mathcal{G} \end{array}$$

in which the dotted arrows are defined over U .

For ℓ large enough, by Krull's theorem and the fact that $R/(R \cap R_1)$ is supported outside U , one has $R \cap \mathcal{J}^\ell \cdot \mathcal{O}^n \subset R_1$; this allows one to extend f to $\bar{f}: \mathcal{J}^\ell \bar{\mathcal{F}} \rightarrow \mathcal{G}$.

In the general case, let U_i be a finite affine covering of X and let U_{ijk} be finite affine coverings of the intersections $U_{ij} = U_i \cap U_j$. Since filtered colimits commute with finite products, one gets

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_U(\mathcal{F}, j^* \mathcal{G}) & \longrightarrow & \prod_i \text{Hom}_{U_i \cap U}(\mathcal{F}, j^* \mathcal{G}) & \rightrightarrows & \prod_{ijk} \text{Hom}_{U_{ijk} \cap U}(\mathcal{F}, j^* \mathcal{G}) \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \varinjlim \text{Hom}_X(\mathcal{J}^n \bar{\mathcal{F}}, j^* \mathcal{G}) & \longrightarrow & \prod_i \varinjlim \text{Hom}_{U_i}(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) & \rightrightarrows & \prod_{ijk} \varinjlim \text{Hom}_{U_{ijk}}(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) \end{array}$$

which completes the proof.

Proposition 4 gives an explicit formula for the functor $(\text{coh on } U) \rightarrow \text{pro}(\text{coh on } X)$ that is left adjoint to j^* . We denote this functor by $j_!$ ("extension by zero")⁴. By Krull's theorem it is exact;

⁴This terminology and notation obviously conflict with the generally accepted ones (Godement's book, SGA 1962, ...). The reader may prefer to read $\mathfrak{J}_!$ instead of $j_!$.

this also follows from the fact that, because X is Noetherian, an injective quasi-coherent sheaf on X remains injective and quasi-coherent after restriction to U .

Proposition 5 (Independence of the compactification).

$$\begin{array}{ccc} U & \hookrightarrow & X \\ \parallel & & \downarrow f \\ U & \hookrightarrow & Y \end{array}$$

Let $f: X \rightarrow Y$ be a proper morphism inducing an isomorphism from the open subset U of Y onto $f^{-1}(U)$. Let $\mathcal{J}$ be a sheaf of ideals defining $Y \setminus U$, and let $\mathcal{I} = f^* \mathcal{J}$. If $\mathcal{F}$ is coherent on X , then

- (i) if $k > 0$, the projective system $\mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is essentially zero (i.e. it defines the zero projective object);
- (ii) if $k = 0$, then for n sufficiently large, $f_* \mathcal{I}^{n+1} \mathcal{F} = \mathcal{J} \cdot f_* \mathcal{I}^n \mathcal{F}$.

One has

$$\bigoplus_{n=0}^{\infty} \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \text{ is a finitely generated } \bigoplus_{n=0}^{\infty} \mathcal{J}^n \text{-module (EGA III),}$$

so for n sufficiently large the map $\mathcal{J} \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \rightarrow \mathbb{R}^k f_* \mathcal{I}^{n+1} \mathcal{F}$ is surjective; this gives (ii). If $k > 0$, then $\mathbb{R}^k f_*$ vanishes on U ; hence there exists N such that $\mathcal{J}^N \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is zero for every n . The diagram

$$\begin{array}{ccccc} \mathcal{J}^p \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^{n+p} \mathcal{F} & \longrightarrow & 0 \quad (\text{if } n \geq n_0) \\ \downarrow & & \downarrow & & \\ \mathcal{J}^p \cdot \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & & \end{array}$$

shows that for N sufficiently large and $k > 0$,

$$\mathbb{R}^k f_* \mathcal{I}^{n+N} \mathcal{F} \rightarrow \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \text{ is zero, which is (i).}$$

NO. 3. DEFINITION OF $\mathbb{R}f_!$

A morphism f (respectively a pair of composable morphisms, respectively a triple, and so on) will be called compactifiable if one can respectively find commutative diagrams

$$\begin{array}{ccc} \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & \nearrow & \\ S & & \end{array} & \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & & \downarrow g \\ Y & \hookrightarrow & \overline{Y} \\ h \downarrow & \nearrow & \\ S & & \end{array} & \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & & \downarrow g \\ Y & \hookrightarrow & \overline{Y} \\ h \downarrow & & \downarrow \\ Z & \hookrightarrow & \overline{Z} \\ \downarrow & \nearrow & \\ S & & \end{array} \end{array}$$

in which the horizontal arrows are open immersions and the oblique arrows are proper. Every compactifiable morphism is separated of finite type. Nagata asserts in [2] that his proof shows the converse for Noetherian integral schemes, but the hypotheses he places on the base are not clear.

We wish, for each compactifiable f , to define

$$\mathbb{R}f_! : \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

Recall that $D_{\text{coh}}^b(S)$, the bounded derived category of the category of coherent sheaves on S , is a full subcategory of the derived category of all sheaves (not necessarily quasi-coherent) on S , whose essential image consists of complexes with coherent and bounded cohomology.

If f is proper, one has $\mathbb{R}f_*: D_{\text{coh}}^b(X) \rightarrow D_{\text{coh}}^b(S)$ of finite cohomological dimension, hence an induced functor

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

If f is an open immersion, then

$$f_!: (\text{coh on } X) \rightarrow \text{pro}(\text{coh on } S)$$

induces

$$D^b(\text{coh on } X) \rightarrow D^b \text{pro}(\text{coh on } S),$$

and its image lies in $\text{pro } D_{\text{coh}}^b(S)$; this extends to

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

If the coherent complex K on X is extended to $\bar{K}$ on S , its image is “ $\varprojlim$ ” $\mathcal{J}^n \bar{K}$, where $\mathcal{J}$ defines $S \setminus X$.

For a composite $f = gh$ with g proper and h an open immersion, define $\mathbb{R}f_! = \mathbb{R}g_! \mathbb{R}h_!$. One has to verify:

Independence of the compactification.

Consider a commutative diagram

$$\begin{array}{ccc} & & X' \\ & \nearrow j' & \downarrow g \\ X & & \\ & \searrow j'' & \downarrow \\ & & X'' \\ \downarrow f & \nwarrow \bar{f} & \\ S & & \end{array}$$

where j' and j'' are open immersions and $\bar{f}$ and g are proper maps. One has $j'X = g^{-1}j''X \cap j'X$. One must prove that

$$\mathbb{R}\bar{f}_* \mathbb{R}j_!'' = \mathbb{R}(\bar{f}g)_* \mathbb{R}j_!'$$

or, since $\mathbb{R}(\bar{f}g)_* = \mathbb{R}\bar{f}_* \mathbb{R}g_*$, equivalently that

$$\mathbb{R}j_!'' = \mathbb{R}g_* \mathbb{R}j_!'$$

One may reduce to the case where X is dense in X' , so as to be in the situation of Proposition 5. Let K be a bounded coherent complex on X , extended to $\bar{K}$ on X' . Then $g_* \bar{K}$ extends K on X'' , and there is a morphism

$$(2) \quad g_* \mathcal{J}'^n \bar{K} \rightarrow \mathbb{R}g_* \mathcal{J}''^n \bar{K},$$

where $\mathcal{J}'$ denotes an ideal sheaf defining $X' \setminus X$. Proposition 5(ii) shows that

$$j_!'' K = \varprojlim g_* \mathcal{J}''^n \bar{K}.$$

Passing to the “limit” one gets

$$\varprojlim \mathbb{R}^p g_* \mathcal{J}''^n \bar{K}^q \implies \varprojlim \mathbb{R}^{p+q} g_* \mathcal{J}''^n \bar{K};$$

Proposition 5(i) shows that this spectral sequence degenerates, and it follows from Proposition 3 that the morphism induced from (1) by passage to the “limit” is an isomorphism. This is exactly the desired formula.

It remains to verify

(a) for every diagram, a compatibility

$$\begin{array}{ccc} X & \longrightarrow & X' \\ f \downarrow & & \downarrow h \\ S & \longrightarrow & \bar{X} \end{array} \quad \mathbb{R}\bar{f}_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}g_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}g_* \mathbb{R}h_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}(gh)_* \mathbb{R}j'_! = \mathbb{R}f_* \mathbb{R}j'_!;$$

(b) for every compactifiable pair, an identification

$$\mathbb{R}(fg)_! = \mathbb{R}f_! \mathbb{R}g_!;$$

(c) for every compactifiable triple, a compatibility

$$\begin{array}{ccc} \mathbb{R}(fgh)_! & = & \mathbb{R}(fg)_! \mathbb{R}h_! \\ \parallel & & \parallel \\ \mathbb{R}f_! \mathbb{R}(gh)_! & = & \mathbb{R}f_! \mathbb{R}g_! \mathbb{R}h_! \end{array}$$

One then obtains:

Theorem 1. *For a compactifiable morphism $f: X \rightarrow Y$, for $K \in \text{pro } D_{\text{coh}}^b(X)$ and $L \in \text{pro } D_{\text{coh}}^b(Y)$, let*

$$\text{Hom}_f(K, L) = \text{Hom}_Y(\mathbb{R}f_! K, L).$$

Modulo compactifiability questions, this makes the categories $\text{pro } D_{\text{coh}}^b(X)$ into a cofibration over the category of Noetherian preschemes (with compactifiable morphisms as arrows, and so on).

NO. 4. DEFINITION OF $\mathbb{R}f^!$

It follows from Verdier's general formalism [1] that the functor $\mathbb{R}f_*: D(X) \rightarrow D(S)$ has a right adjoint $D^+(S) \rightarrow D^+(X)$. It deserves the name $\mathbb{R}f^!$ only when f is proper, and its definition is not immediately suited to computation.

One must first make explicit a procedure for computing $\mathbb{R}f_*$ at the level of complexes, of the form $\mathbb{R}f_* K = f_* C^*(K)$, where C^* is a finite acyclic resolution depending functorially and exactly on the object to which it is applied and compatible with filtered colimits. Then, for every injective I on S , the functor $\text{Hom}(f_* C^p \mathcal{F}, I)$ is exact in quasi-coherent sheaves on X and transforms $\varinjlim$ into $\varprojlim$, hence is representable by some $f_p^! I$, which is injective and quasi-coherent. The $f_p^! I$ form a complex, and this defines

$$\mathbb{R}f^!: D^+ S \rightarrow D^+ X,$$

right adjoint to

$$\mathbb{R}f_*: D(X) \rightarrow D(S).$$

Here is how one may define such a resolution.

- (1) If S is separated, choose a finite covering of X by open subsets affine over S (for example affine open subsets) and put

$$\mathbb{R}f_* K = f_* \check{C}(\mathcal{U}, K)$$

(the alternating Čech complex).

- (2) Otherwise, the canonical flasque resolutions have here two defects.

- (a) $f_* C^* \mathcal{F}$ is no longer quasi-coherent. This does not matter as long as, as here, injective quasi-coherent sheaves are injective when regarded simply as sheaves.
- (b) C^* does not commute with filtered colimits. It is easy to correct this defect: write a quasi-coherent sheaf as $\mathcal{F} = \varinjlim \mathcal{F}_i$ with the $\mathcal{F}_i$ finitely presented, and set

$$C'^* \mathcal{F} = \varinjlim C^* \mathcal{F}_i.$$

This is independent of the choices made.

For an open immersion one takes $\mathbb{R}f^!K = f^*K$.

For a composite morphism $f = gh$ with g proper and h an open immersion, one takes $\mathbb{R}f^! = \mathbb{R}g^!\mathbb{R}h^!$. Putting together Proposition 4 and the preceding discussion one obtains:

Theorem 2 (Duality). *Let $f: X \rightarrow S$ be compactifiable, with $f = gh$. The functors*

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S)$$

and

$$\mathbb{R}f^!: D^+(S) \rightarrow D^+(X)$$

are adjoint to one another.

One should still verify the independence of the compactification and a list of compatibilities and identifications analogous to those for $\mathbb{R}f_!$. Some special cases of the formalism follow immediately from the adjunction formula (uniqueness of an actual adjoint, and so on).

In the general case one may rely on the following proposition.

Proposition 6. *Let $f: X \rightarrow S$ be compactifiable and let $L \in D_{\text{qc}}^+(S)$. The stalk at $x \in X$ of $\mathcal{H}^p\mathbb{R}f^!L$ is given by*

$$(\mathcal{H}^p\mathbb{R}f^!L)_x = \varinjlim_{x \in U} \text{Hom}_S^p(\mathbb{R}f_!(U, \mathcal{O}_U), L).$$

One may assume X proper and L given as a complex of injective quasi-coherent sheaves. Let U be an affine open subset of X and let $\mathcal{J}$ define $X \setminus U$. Then

$$\begin{aligned} (\mathcal{H}^p\mathbb{R}f^!L)(U) &= \mathbb{H}^p(\mathbb{R}f^!L(U)) = \mathbb{H}^p \text{Hom}_U(\mathcal{O}, \mathbb{R}f^!L) \\ &= \mathbb{H}^p \varinjlim \text{Hom}_X(\mathcal{J}^n, \mathbb{R}f^!L) = \varinjlim \text{Hom}_S^p(\mathbb{R}f_*\mathcal{J}^n, L) \\ &= \text{Hom}_S^p(\mathbb{R}f_!(U, \mathcal{O}_U), L), \end{aligned}$$

and the proposition follows.

We shall henceforth assume for $\mathbb{R}f^!$ a formalism of variance analogous to that of Theorem 1 for $\mathbb{R}f_!$.

If one wants to make the definition of $\mathbb{R}f^!$ in the proper case more explicit, one may remark that if a quasi-coherent sheaf $\mathcal{F}$ represents a functor F , then for every open subset U and every ideal $\mathcal{J}$ defining $X \setminus U$ one has

$$\mathcal{F}(U) = \varinjlim \text{Hom}(\mathcal{J}^n, \mathcal{F}) = \varinjlim F(\mathcal{J}^n).$$

NO. 5. COMPUTATION OF $\mathbb{R}f^!$

For a finite morphism, $\mathbb{R}f^!$ is what one expects; to see this it suffices to take, in the computation of $\mathbb{R}f_*$, the identity as the resolving functor. For projective space, uniqueness of an adjoint leaves no choice (one uses the fact that the adjunction formula holds for $\mathbb{R}f_! = \mathbb{R}f_*: D(X) \rightarrow D(S)$); we return to this morphism. For an open immersion, $\mathbb{R}f^! = \mathbb{R}f^*$.

For every quasi-projective open subset U of X over S , one can therefore compute the restriction of $\mathbb{R}f^!$ to U . In particular, if one can verify locally that $\mathbb{R}^q f^!K = 0$ for $q \neq p$, then one knows $\mathbb{R}f^!K$, because an object of the derived category having only one nonzero cohomology sheaf is determined by that sheaf. Let us recall an important case in which this condition is satisfied, with $K = \mathcal{O}$.

Proposition 7. *Let $f: X \rightarrow Y$ be compactifiable. Assume that locally (on X and on Y) one can find a diagram*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow g & \swarrow h \\ & S & \end{array}$$

in which g and h are local complete intersections of relative dimensions d' and d'' , and put $d = d' - d''$.

Then $\mathbb{R}f^!\mathcal{O}$ has only one nonzero cohomology sheaf, located in degree $-d$, and that sheaf is invertible.

Indeed, $\mathbb{R}g^!\mathcal{O} = \mathbb{R}f^!\mathbb{R}h^!\mathcal{O}$. One knows that both $\mathbb{R}g^!\mathcal{O}$ and $\mathbb{R}h^!\mathcal{O}$ are of the indicated form, concentrated in degrees $-d'$ and $-d''$, by local computations in projective space. Since everything is local, one may suppose $\mathbb{R}^{-d''}h^!\mathcal{O} \cong \mathcal{O}$, and the assertion follows.

To pass from this to more general complexes one would first need formulas of the shape

$$\mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L) = \mathbb{R}f^!(K) \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L.$$

Only the definition of the morphism presents any difficulty, because its being an isomorphism is a local problem. Degree restrictions are of course needed. I have not checked in detail any of what follows.

First method for defining the morphism.

Assume K and L lie in $D_{\text{coh}}^b(S)$ (and are not merely quasi-coherent), and moreover that

- (a) $\mathbb{R}f^!K$ is bounded in degree (it is automatically coherent; that is a local problem),
- (b) $K \overset{\mathbb{L}}{\otimes} L$ is bounded in degree,
- (c) $\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L$ is bounded in degree.

To define

$$\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L \longrightarrow \mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L),$$

it suffices by duality to define

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L) \longrightarrow K \overset{\mathbb{L}}{\otimes} L.$$

Now by K"unneth,

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L) = \mathbb{R}f_!\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} L,$$

and the desired map is induced by the adjunction morphism $\mathbb{R}f_!\mathbb{R}f^!K \rightarrow K$.

Second method.

Assume f compactifiable, flat, and relatively local complete intersection. Then $\mathbb{R}f^!\mathcal{O} = \omega[d]$ with ω invertible. Let $\mathbb{R}f_!$ denote some complex-level procedure computing $\mathbb{R}f_!$ (that is, such a procedure for a compactification $\bar{f}: \bar{X} \rightarrow S$ of f), assumed to be of the sort described in No. 4 and such that $C^p\mathcal{F}$ vanishes for $p > d$ (which can be achieved by truncation). There is then a morphism

$$\mathbb{R}^d\bar{f}_*\mathcal{F}[-d] \rightarrow \mathbb{R}\bar{f}_*\mathcal{F} \quad \text{for every sheaf } \mathcal{F} \text{ on } \bar{X}$$

(the last cohomology sheaf maps to its complex). Hence for any injective I on S one gets morphisms of complexes

$$\text{Hom}_S(\mathbb{R}\bar{f}_*\mathcal{F}, I) \rightarrow \text{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}[-d], I).$$

Since $\mathbb{R}^d\bar{f}_*[-d]$ is right exact in $\mathcal{F}$, the functor in $\mathcal{F}$ on the right is representable: define $I_!$ by

$$\text{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}, I) = \text{Hom}_{\bar{X}}(\mathcal{F}, I_!) \quad \text{and} \quad \mathbb{R}f_!I = I_![d].$$

The preceding construction defines

$$\mathbb{R}f^!I \rightarrow \mathbb{R}f_!^!I,$$

and for a complex of injectives the functor $\mathbb{R}f_!^!$ is defined termwise.

It remains to evaluate $\mathbb{R}f_!^!I$ on affine open subsets U of X ; this is given by $\varinjlim \text{Hom}(\mathbb{R}^d f_!\mathcal{J}^n, I)$. Since this problem is local upstairs, it is easy to show by projective methods that the dual of " $\varinjlim$ " $\mathbb{R}^d f_!\mathcal{J}^n = \mathbb{R}^d f_!(U, \mathcal{O})$, which is an ind-object, is $f_*(U, \omega)$, represented as the limit of its submodules of sections having poles of order at least n outside U ($n \rightarrow \infty$).

This gives $\mathbb{R}f_!^* I = \omega \otimes f^* I[d]$, and hence the general formula

$$\mathbb{R}f^! K = f^* K \otimes \omega[d].$$

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ORDINARY ABELIAN VARIETIES OVER A FINITE FIELD

PIERRE DELIGNE

We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbf{F}_q$. The result was inspired by Ihara [2], Chapter V; see also [3].

1. CONVENTIONS

Let p be a prime number, let $\mathbf{F}_p = \mathbf{Z}/(p)$, let $\overline{\mathbf{F}}_p$ be an algebraic closure of $\mathbf{F}_p$, and, for every power q of p , let $\mathbf{F}_q$ be the subfield of $\overline{\mathbf{F}}_p$ with q elements. For every algebraic extension k of $\overline{\mathbf{F}}_p$, let $W_0(k)$ be the henselian discrete valuation ring essentially of finite type over $\mathbf{Z}$, absolutely unramified, with residue field k . Let $W(k)$ be the ring of Witt vectors over k , that is, the completion of $W_0(k)$; put $W = W(\overline{\mathbf{F}}_p)$, and choose an embedding

$$\varphi : W \hookrightarrow \mathbf{C}.$$

Let $\mathbf{Z}(1)$ denote the subgroup $2\pi i \mathbf{Z}$ of $\mathbf{C}$. The exponential map defines an isomorphism

$$\mathbf{Z}(1) \otimes \mathbf{Z}_l \simeq \mathbf{Z}_l(1)(\mathbf{C}) = \varprojlim \mu_{l^n}(\mathbf{C}).$$

For an abelian variety A , write A^* for its dual. For every field k , write $\bar{k}$ for an algebraic closure.

2. ORDINARY ABELIAN VARIETIES

Let A be an abelian variety of dimension g over a field k of characteristic p . Recall that A is called *ordinary* if the following equivalent conditions hold:

- (I) A has p^g points over $\bar{k}$ of order dividing p .
- (II) The Hasse–Witt matrix

$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \rightarrow H^1(A, \mathcal{O}_A)$$

is invertible.

- (III) The identity component of the group scheme A_p , the kernel of multiplication by p , is of multiplicative type; equivalently, after extension of scalars it is isomorphic to a power of μ_p .

If $k = \mathbf{F}_q$, if F is the Frobenius endomorphism of A , and if $P_{c_A}(F; x)$ is its characteristic polynomial, these conditions are also equivalent to

- (IV) At least half of the roots of $P_{c_A}(F; x)$ in $\overline{\mathbf{Q}}_p$ are p -adic units. Equivalently, if $n = \dim A$, the reduction modulo p of $P_{c_A}(F; x)$ is not divisible by x^{n+1} .

3. THE CANONICAL LIFT

Let A be an ordinary abelian variety over $\overline{\mathbf{F}}_p$. Let $\tilde{A}$ denote the Serre–Tate canonical lift [4] of A over W . Recall that $\tilde{A}$ depends functorially on A and is characterized by the fact that the p -divisible group $T_p(\tilde{A})$ over W attached to $\tilde{A}$ [5] is the product of the p -divisible groups, uniquely determined by 2.(III), lifting respectively the identity component and the largest étale quotient of $T_p(A)$. Equivalently, $\tilde{A}$ is the unique lift of A to which every endomorphism of A lifts.

Let $T(A)$ be the integral homology of the complex abelian variety A_c obtained from $\tilde{A}$ and φ by extension of scalars from W to $\mathbf{C}$:

$$T(A) = H_1(\tilde{A} \otimes_W \mathbf{C}).$$

One knows that $\tilde{A}$ descends uniquely to $W_0(\overline{\mathbf{F}}_p)$; hence A_c depends only on A and on the restriction of φ to $W_0(\overline{\mathbf{F}}_p)$. The free $\mathbf{Z}$ -module $T(A)$ has rank $2 \dim A$, and it is functorial in A . Moreover, if $l \neq p$ is a prime number, there is a functorial identification

$$(1) \quad T(A) \otimes \mathbf{Z}_l = T_l(A).$$

The canonical lift of the dual abelian variety A^* is dual to $\tilde{A}$; hence $(A_c)^* = \tilde{A}^*$, and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbf{Z}(1)$:

$$(2) \quad T(A) \otimes T(A^*) \longrightarrow \mathbf{Z}(1).$$

It is essential to use $\mathbf{Z}(1)$ rather than $\mathbf{Z}$ in order to obtain a theory invariant under complex conjugation. The pairings (2) are compatible, via (1), with the pairings

$$T_l(A) \otimes T_l(A^*) \longrightarrow \mathbf{Z}_l(1).$$

A morphism $\phi : A \rightarrow A^*$ defines a polarization of A if and only if the lifted morphism

$$\tilde{\phi} : \tilde{A} \rightarrow \tilde{A}^*$$

defines a polarization of A_c .

Put

$$T'_p(A) = \text{Hom}(\mathbf{Q}_p/\mathbf{Z}_p, A(\overline{\mathbf{F}}_p))$$

and

$$T''_p(A) = \text{Hom}_{\mathbf{Z}_p}(T'_p(A^*), \mathbf{Z}(1) \otimes \mathbf{Z}_p).$$

These $\mathbf{Z}_p$ -modules are covariant functors of A .

By the definition of the canonical lift, the p -divisible group $T_p(\tilde{A})$ is the sum of the constant pro-etale group $T'_p(A)$ and the Cartier dual of $T'_p(A^*)$.¹ For a morphism $u : A \rightarrow B$, the induced morphism $u : T_p(A) \rightarrow T_p(B)$ identifies with the sum of

$$u|_{T'_p(A)} : T'_p(A) \rightarrow T'_p(B)$$

and the Cartier transpose of

$${}^t u|_{T'_p(B^*)} : T'_p(B^*) \rightarrow T'_p(A^*).$$

Over $\mathbf{C}$ there is a canonical isomorphism

$$\mathbf{Z}(1)/(p^n) \simeq \mu_{p^n},$$

hence an isomorphism of functors

$$(3) \quad T_p(A) = T(A) \otimes \mathbf{Z}_p = T'_p(A) \oplus T''_p(A).$$

4. FINITE SUBGROUPS AND LATTICES

Recall that if $\varphi : X \rightarrow Y$ is an isogeny of complex abelian varieties, the homotopy exact sequence reduces to a short exact sequence

$$0 \rightarrow H_1(X) \rightarrow H_1(Y) \rightarrow \text{Ker}(\varphi) \rightarrow 0.$$

The quotient abelian varieties of X by a finite subgroup, and the finite subgroups themselves, correspond bijectively to the overlattices of $H_1(X)$ in $H_1(X) \otimes \mathbf{Q}$.

Let A be an ordinary abelian variety over $\overline{\mathbf{F}}_p$. If n is prime to p , the finite subgroup schemes of order n of A , of $\tilde{A}$, and of A_c correspond bijectively, and they also correspond to the overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p = T'_p(A) \otimes \mathbf{Q}_p, \quad V''_p(A) = T''_p(A) \otimes \mathbf{Q}_p.$$

The finite subgroup schemes of A of order p^k are products of an etale subgroup and an infinitesimal subgroup. The etale subgroups of order p^k of A correspond to those subgroups of order p^k of A_c such that the corresponding overlattice R of $T(A)$ is contained in

$$T'_p(A) + V''_p(A).$$

By duality, the infinitesimal subgroups of A correspond to overlattices R of $T(A)$ which are p -isogenous to $T(A)$, i.e. for which $[R : T(A)]$ is a power of p , and which are contained in

$$T''_p(A) + V'_p(A).$$

Altogether, the finite subgroups of A , or the quotient abelian varieties of A , correspond bijectively to the overlattices R of $T(A)$ such that

$$(4) \quad R \otimes \mathbf{Z}_p = (R \otimes \mathbf{Z}_p \cap V'_p) + (R \otimes \mathbf{Z}_p \cap V''_p).$$

¹We write T_p for the p -adic Tate module and T'_p for the etale component of the p -divisible group.

5. RELATIVE FROBENIUS

In particular, $A^{(p)}$, which is the quotient of A by the largest infinitesimal subgroup of A killed by p when A is ordinary, is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p -isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbf{Z}_p = T'_p(A) + V''_p(A).$$

6. DESCENT TO $\mathbf{F}_q$

Let A be an abelian variety over $\mathbf{F}_q$, and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined by the pair $(\bar{A}, F)$ obtained from (A, F) by extension of scalars from $\mathbf{F}_q$ to $\bar{\mathbf{F}}_q$. The endomorphism F of $\bar{A}$ factors through the relative Frobenius

$$\mathrm{Fr}(q) : \bar{A} \rightarrow \bar{A}^{(q)}$$

and an isomorphism

$$F' : \bar{A}^{(q)} \rightarrow \bar{A}.$$

If A is ordinary, let $T(A)$ denote the $\mathbf{Z}$ -module $T(\bar{A})$ endowed with the endomorphism F induced by the Frobenius endomorphism of $\bar{A}$. By 5, by the preceding discussion, and by (3), the lattices $T(A)$ and $F(T(A))$ are p -isogenous and one has

$$(5) \quad F(T'_p(A)) = T'_p(A),$$

$$(6) \quad F(T''_p(A)) = q T''_p(A).$$

7. THE EQUIVALENCE OF CATEGORIES

Theorem 1. *The functor*

$$A \longmapsto (T(A), F)$$

is an equivalence from the category of ordinary abelian varieties over $\mathbf{F}_q$ to the category of finite-rank free $\mathbf{Z}$ -modules T equipped with an endomorphism F satisfying the following conditions:

- (a) *F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$.*
- (b) *At least half of the roots in $\bar{\mathbf{Q}}_p$ of the characteristic polynomial of F are p -adic units. Equivalently, if T has rank d , the reduction modulo p of $Pc_T(F; x)$ is not divisible by $x^{[d/2]+1}$.*
- (c) *There exists an endomorphism V of T such that $FV = q$.*

If condition (a) holds, conditions (b) and (c) are equivalent to

- (d) *The module $T \otimes \mathbf{Z}_p$ admits a decomposition, stable under F , into two $\mathbf{Z}_p$ -submodules T'_p and T''_p of equal dimension, such that $F|_{T'_p}$ is invertible and $F|_{T''_p}$ is divisible by q .*

Proof. (A) We first prove that (a)+(b)+(c) imply (d). If α is a complex eigenvalue of F , then $\bar{\alpha}$ is another eigenvalue, with the same multiplicity, and

$$\alpha \bar{\alpha} = q.$$

Except for those eigenvalues equal to $\pm q^{1/2}$, the eigenvalues of F in $\mathbf{C}$, hence in $\bar{\mathbf{Q}}_p$, arrange themselves in pairs α and q/α . The two roots α and q/α cannot both be p -adic units. It follows from (b) that $\pm q^{1/2}$ is not an eigenvalue of F , that half of the eigenvalues of F in $\bar{\mathbf{Q}}_p$ are p -adic units, say

$$\alpha_1, \dots, \alpha_{d/2},$$

and that the other half are

$$\beta_1 = q/\alpha_1, \dots, \beta_{d/2} = q/\alpha_{d/2}.$$

Let

$$T_p = T \otimes \mathbf{Z}_p, \quad V_p = T \otimes \mathbf{Q}_p.$$

Let V'_p be the kernel of $\prod_i (F - \alpha_i)$ on V_p , and let V''_p be the kernel of

$$\psi = \prod_i (F - \beta_i).$$

Then

$$V_p = V'_p \oplus V''_p.$$

Let T'_p be the projection of T_p onto V'_p , and put $T''_p = T_p \cap V''_p$. Since ψ kills V'_p and preserves T , it sends T_p into $T_p \cap V''_p \subset T''_p$. On the other hand,

$$\det(\psi|V'_p) = \prod_{i,j} (\alpha_i - \beta_j)$$

is a p -adic unit. Hence $\psi(T_p) = T_p$ and $T_p \cap V'_p = \psi T_p$, so

$$T_p = T'_p \oplus T''_p.$$

(B) Full faithfulness. Let A and B be abelian varieties over $\mathbf{F}_q$, and consider

$$\psi : \mathrm{Hom}(A, B) \rightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A), T(B)).$$

By Tate's theorem [7] and (1), the map

$$\psi_l : \mathrm{Hom}(A, B) \otimes \mathbf{Z}_l \rightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A), T(B)) \otimes \mathbf{Z}_l$$

is an isomorphism for $(l, p) = 1$. Thus $\psi \otimes \mathbf{Q}$ is an isomorphism. The group $\mathrm{Hom}(A, B)$ is torsion-free, so ψ is injective. Let $u : A \rightarrow B$ be a morphism such that $T(u)$ is divisible by n . Then the induced morphism

$$u_c : A_c \rightarrow B_c$$

is divisible by n , hence so is

$$\tilde{u} : \tilde{A} \rightarrow \tilde{B}$$

over the generic point of W . The kernel of multiplication by n is flat over W ; since $\tilde{u}$ vanishes on this kernel, both $\tilde{u}$ and u are divisible by n . Therefore ψ is bijective.

(C) Necessity. The fact that $(T(A), F)$ satisfies (a) follows from Weil. Condition (d), which implies (b) and (c), follows from 6.

(D) Isogenies. Let (T_0, F) satisfy (a)–(d), and let T be a lattice in $T_0 \otimes \mathbf{Q}$, stable under F and still satisfying (d). Suppose that (T_0, F) is the image of an abelian variety $\bar{A}$ over $\bar{\mathbf{F}}_q$; we prove that (T, F) comes from an isogenous abelian variety. Replacing T by $\frac{1}{n}T$, which is isomorphic to it, we may assume $T \supset T_0$. Condition (d) implies that T satisfies (4). Thus T defines a subgroup H of $\bar{A}$, defined over $\mathbf{F}_q$, and

$$(T, F) = T(\bar{A}/H).$$

(E) Essential surjectivity. The functor T induces a functor $T_{\mathbf{Q}}$ from the category of ordinary abelian varieties over $\mathbf{F}_q$ up to isogeny to the category of finite-dimensional $\mathbf{Q}$ -vector spaces endowed with an automorphism F satisfying (a) and (b). By (D), it is enough to prove that $T_{\mathbf{Q}}$ is essentially surjective. It is even enough to show that every simple object (V, F) of the image category is reached. By Honda [1], see also [6], there exists an abelian variety $\bar{A}$ over $\bar{\mathbf{F}}_q$ such that the characteristic polynomial of the Frobenius F_A of A is a power of the characteristic polynomial of F . The third characterization in 2 of ordinary abelian varieties shows that $\bar{A}$ is ordinary. Moreover $(T(A) \otimes \mathbf{Q}, F)$ is a sum of copies of (V, F) . By (B), the abelian variety up to isogeny $A \otimes \mathbf{Q}$ is a sum of copies of an abelian variety B such that

$$T(B) \otimes \mathbf{Q} = (V, F).$$

□

8. THE COMPLEX STRUCTURE

Let (T, F) be a pair satisfying the hypotheses of the theorem, and let $2g$ be the rank of T . Let A be the corresponding abelian variety over $\mathbf{F}_q$, and let A_c be the associated complex abelian variety constructed in 3. Then

$$T = H_1(A_c),$$

so $T \otimes \mathbf{R}$ identifies with the Lie algebra of A_c , and as such it is endowed with a complex structure. Following J.-P. Serre, here is how to reconstruct this complex structure in terms of T , F , and the restriction of φ to $W_0(\bar{\mathbf{F}}_p)$.

Proposition 2. *The complex structure just described on $T \otimes \mathbf{R}$ is characterized by the following properties:*

- (I) *The endomorphism F is $\mathbf{C}$ -linear.*
- (II) *Let v be the valuation of the algebraic closure $\bar{\mathbf{Q}}$ of $\mathbf{Q}$ inside $\mathbf{C}$ extending the valuation of $W_0(\bar{\mathbf{F}}_p)$. The valuations of the g eigenvalues of this endomorphism are strictly positive.*

Condition (I) is evident, and condition (II) follows because the action of F on the Lie algebra of $\tilde{A}$ is congruent to zero modulo p . The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) for (T, F) .

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HODGE THEORY I

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The purpose of this note is to give a heuristic dictionary between statements in l -adic cohomology and statements in Hodge theory. This dictionary has, among its sources, Serre's work [3] and Grothendieck's conjectural theory of motives [2]. Up to now it has mainly served to formulate conjectures in Hodge theory, and at times it has suggested proofs of them.

1.

Definition 1. A mixed Hodge structure H consists of:

- (a) a finitely generated $\mathbf{Z}$ -module $H_{\mathbf{Z}}$, the integral lattice;
- (b) a finite increasing filtration W on $H_{\mathbf{Q}} = H_{\mathbf{Z}} \otimes \mathbf{Q}$, the weight filtration;
- (c) a finite decreasing filtration F on $H_{\mathbf{C}} = H_{\mathbf{Z}} \otimes \mathbf{C}$, the Hodge filtration.

These data are subject to the following axiom: on $\mathrm{Gr}_W(H_{\mathbf{C}})$ there is a unique bigrading by subspaces $H^{p,q}$ such that

$$(i) \quad \mathrm{Gr}_W^n(H_{\mathbf{C}}) = \bigoplus_{p+q=n} H^{p,q};$$

- (ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbf{C}})$ the filtration

$$F^p = \bigoplus_{p' \geq p} H^{p',q'};$$

- (iii) $H^{p,q} = \overline{H^{q,p}}$.

A morphism $f : H \rightarrow H'$ is a homomorphism $f : H_{\mathbf{Z}} \rightarrow H'_{\mathbf{Z}}$ such that $f : H_{\mathbf{Q}} \rightarrow H'_{\mathbf{Q}}$ and $f_{\mathbf{C}} : H_{\mathbf{C}} \rightarrow H'_{\mathbf{C}}$ are compatible respectively with the filtrations W and F .

The Hodge numbers of H are the integers

$$(1) \quad h^{p,q} = \dim_{\mathbf{C}} H^{p,q} = h^{q,p}.$$

We say that H is pure of weight n if $h^{p,q} = 0$ for $p + q \neq n$, equivalently if $\mathrm{Gr}_i^W(H) = 0$ for $i \neq n$. We also call such an object a Hodge structure of weight n .

The Tate Hodge structure $\mathbf{Z}(1)$ is the Hodge structure of weight -2 , purely of type $(-1, -1)$, for which $\mathbf{Z}(1)_{\mathbf{C}} = \mathbf{C}$ and

$$\mathbf{Z}(1)_{\mathbf{Z}} = 2\pi i \mathbf{Z} = \mathrm{Ker}(\exp : \mathbf{C} \rightarrow \mathbf{C}^*) \subset \mathbf{C}.$$

Put $\mathbf{Z}(n) = \mathbf{Z}(1)^{\otimes n}$.

Date: Actes, Congres intern. math., 1970, Tome 1, pp. 425–430.

The mixed Hodge structures form an abelian category. If $f : H \rightarrow H'$ is a morphism, then $f_{\mathbf{Q}}$ and $f_{\mathbf{C}}$ are strictly compatible with the filtrations W and F [1, 2.3.5].

2.

Let A be a normal integral ring of finite type over $\mathbf{Z}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K . Let K^{nr} be the largest subextension of $\overline{K}$ unramified at every prime ideal of A . One knows, or one defines,

$$\pi_1(\mathrm{Spec} A, \overline{K}) = \mathrm{Gal}(K^{nr}/K).$$

For every closed point x of $\mathrm{Spec} A$, defined by a maximal ideal m_x of A , the residue field $k_x = A/m_x$ is finite. The point x defines a conjugacy class of Frobenius substitutions $\varphi_x \in \pi_1(\mathrm{Spec} A, \overline{K})$. Put $q_x = \#k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p , let $\overline{K}$ be an algebraic closure of K , let $l \neq p$ be prime, and let H be a finitely generated $\mathbf{Z}_l$ -module, or $\mathbf{Q}_l$ -vector space, endowed with a continuous action ρ of $\mathrm{Gal}(\overline{K}/K)$. We shall always suppose below that there exists A as above, with l invertible in A , such that ρ factors through

$$\pi_1(\mathrm{Spec} A, \overline{K}) = \mathrm{Gal}(K^{nr}/K).$$

We say that H is pure of weight n if, for every closed point x of a nonempty open subset of $\mathrm{Spec} A$, the eigenvalues α of F_x acting on H are algebraic integers all of whose complex conjugates have absolute value $q_x^{n/2}$.

Principle 2. *If the Galois module H comes from algebraic geometry, then on $H \otimes_{\mathbf{Z}_l} \mathbf{Q}_l$ there is a unique increasing filtration W , invariant under Galois, such that $\mathrm{Gr}_n^W(H)$ is pure of weight n .*

One may also expect $\mathrm{Gr}_n^W(H)$ to be semisimple.

When resolution of singularities is available, one can often give a conjectural definition of W , whose correctness follows from the Weil conjectures [5]; see section 6.

Let μ_{l^∞} be the subgroup of $\overline{K}^*$ formed by the roots of unity. The Tate module

$$\mathbf{Z}_l(1) = \mathrm{Hom}(\mathbf{Q}_l/\mathbf{Z}_l, \mu_{l^\infty})$$

is pure of weight -2 . Put $\mathbf{Z}_l(n) = \mathbf{Z}_l(1)^{\otimes n}$.

It is immediate that every morphism $f : H \rightarrow H'$ is strictly compatible with the weight filtration.

Principle 2 agrees with the fact that every extension of $\mathbf{G}_m$ (weight -2) by an abelian variety (weight $-1 > -2$) is trivial.

3. TRANSLATION

The Galois modules that occur in l -adic cohomology have, over $\mathbf{C}$, mixed Hodge structures as their analogues. We have moreover the following dictionary:

module pure of weight n	Hodge structure of weight n
weight filtration	weight filtration
Galois-compatible homomorphism	morphism
Tate module $\mathbf{Z}_l(1)$	Tate Hodge structure $\mathbf{Z}(1)$

4.

Let X be a complex algebraic variety, i.e. a scheme of finite type over $\mathbf{C}$, assumed separated. There is a subfield K of $\mathbf{C}$, of finite type over $\mathbf{Q}$, over which X can be defined; thus X is obtained by extension of scalars from a K -scheme X' . Let $\bar{K}$ be the algebraic closure of K in $\mathbf{C}$. Then $\text{Gal}(\bar{K}/K)$ acts on the l -adic cohomology groups $H^*(X_{\bar{K}}, \mathbf{Z}_l)$, and

$$H^*(X(\mathbf{C}), \mathbf{Z}) \otimes \mathbf{Z}_l = H^*(X, \mathbf{Z}_l) = H^*(X'_{\bar{K}}, \mathbf{Z}_l).$$

By section 3, one should expect the groups $H^n(X(\mathbf{C}), \mathbf{Z})$ to carry natural mixed Hodge structures. This can in fact be proved; see [1, 3.2.5] for the case in which X is smooth. The proof is algebraic, starting from classical Hodge theory [6]. For X projective and smooth, the Weil conjectures imply that $H^n(X_{\bar{K}}, \mathbf{Q}_l)$ is pure of weight n , while classical Hodge theory endows $H^n(X(\mathbf{C}), \mathbf{Z})$ with a Hodge structure of weight n . For every morphism $f : X \rightarrow Y$ and for K large enough, the pullback $f^* : H^*(Y_{\bar{K}}, \mathbf{Z}_l) \rightarrow H^*(X_{\bar{K}}, \mathbf{Z}_l)$ commutes with Galois. Likewise, $f^* : H^*(Y(\mathbf{C}), \mathbf{Z}) \rightarrow H^*(X(\mathbf{C}), \mathbf{Z})$ is a morphism of mixed Hodge structures.

For X smooth, the cohomology class in $H^{2n}(X_{\bar{K}}, \mathbf{Z}_l(n))$ of an algebraic cycle Z of codimension n , defined over K , is Galois invariant, i.e. it defines

$$c(Z) \in \text{Hom}_{\text{Gal}}(\mathbf{Z}_l(-n), H^{2n}(X_{\bar{K}}, \mathbf{Z}_l)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbf{C}), \mathbf{Z})$ is purely of type (n, n) , i.e. it corresponds to

$$c(Z) \in \text{Hom}_{\text{Hodge}}(\mathbf{Z}(-n), H^{2n}(X(\mathbf{C}), \mathbf{Z})).$$

5.

If $f : H \rightarrow H'$ is a Galois-compatible morphism between $\mathbf{Q}_l$ -vector spaces of different weights, then $f = 0$. Likewise, if $f : H \rightarrow H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful observation is the following.

Scholium 3. *Let H and H' be Hodge structures of weights n and n' with $n > n'$. Let $f : H_{\mathbf{Q}} \rightarrow H'_{\mathbf{Q}}$ be a homomorphism such that $f_{\mathbf{C}} : H_{\mathbf{C}} \rightarrow H'_{\mathbf{C}}$ respects F . Then $f = 0$.*

6.

Let $\overline{X}$ be a smooth projective variety over $\mathbf{C}$, let $D = \bigcup D_i$ be a normal crossings divisor in $\overline{X}$, a sum of smooth divisors, and let j be the inclusion in $\overline{X}$ of $U = \overline{X} - D$. For $Q \subset [1, n]$, put $D_Q = \bigcap_{i \in Q} D_i$.

In l -adic cohomology we have canonically

$$(2) \quad \mathcal{R}^q j_* \mathbf{Z}_l = \bigoplus_{|Q|=q} (i_Q)_* \mathbf{Z}_l(-q),$$

and the Leray spectral sequence for j is

$$(3) \quad {}'E_2^{p,q} = \bigoplus_{|Q|=q} H^p(D_Q, \mathbf{Q}_l) \otimes \mathbf{Z}_l(-q) \Rightarrow H^{p+q}(U, \mathbf{Q}_l).$$

By the Weil conjectures [5], $H^p(D_Q, \mathbf{Q}_l)$ is pure of weight p , hence $'E_2^{p,q}$ is pure of weight $p + 2q$. Being a quotient of a subobject of $'E_2^{p,q}$, $'E_r^{p,q}$ is again pure of weight $p + 2q$. By Scholium 3, $d_r = 0$ for $r \geq 3$, since the weights $p + 2q$ and $p + 2q - r + 2$ of $'E_r^{p,q}$ and $'E_r^{p+r, q-r+1}$ are different. Thus $'E_3^{p,q} = 'E_\infty^{p,q}$. Up to a reindexing, the weight filtration of $H^*(U, \mathbf{Q}_l)$ is the abutment of (3):

$$(4) \quad \mathrm{Gr}_n^W(H^k(U, \mathbf{Q}_l)) = 'E_2^{2k-n, n-k}.$$

7.

For the usual topology and integral cohomology, the Leray spectral sequence for j is

$$(5) \quad {}'E_2^{p,q} = \bigoplus_{|Q|=q} H^p(D_Q, \mathbf{Z}) \Rightarrow H^{p+q}(U, \mathbf{Z}).$$

Since each D_Q is a nonsingular projective variety, $'E_2^{p,q}$ carries a Hodge structure of weight p . Put $E_2^{p,q} = {}'E_2^{p,q} \otimes \mathbf{Z}(-q)$, a Hodge structure of weight $p + 2q$. As an abelian group, $E_2^{p,q} = {}'E_2^{p,q}$; it is useful to regard (5) as a spectral sequence with initial term $E_2^{p,q}$. By section 3, one expects $d_2 : E_2^{p,q} \rightarrow E_2^{p+2, q-1}$ to be a morphism of Hodge structures. This is proved by interpreting d_2 as a Gysin morphism. Thus $E_3^{p,q}$ has a Hodge structure of weight $p + 2q$. Again by section 3, one expects that, modulo torsion, the spectral sequence (3) degenerates at E_3 and that the vanishing of d_r for $r \geq 3$ is an application of Scholium 3. This program is carried out in [1, 3.2]. The weight filtration of $H^*(U, \mathbf{Q})$ is defined there as the abutment of (5), up to the reindexing (4).

In fact, in order to put mixed Hodge structures on cohomology groups H^* , the key point so far has always been to find a spectral sequence E abutting to H^* such that the l -adic analogue of $E_2^{p,q}$ is conjecturally pure, of weight $p + 2q$. Then $E_2^{p,q}$ should carry a natural Hodge structure of weight $p + 2q$, and the filtration W is the abutment of E .

8.

Let $\text{Spec}(V)$ be the spectrum of a Henselian discrete valuation ring, with fraction field K and residue field k of finite type over the prime field of characteristic p . Let $\bar{K}$ be an algebraic closure of K , and let H be a finite-dimensional $\mathbf{Q}_l$ -vector space, $l \neq p$, on which $\text{Gal}(\bar{K}/K)$ acts continuously. By Grothendieck [4, Appendix], a finite-index subgroup of the inertia group I acts unipotently. After replacing V by a finite extension, we reduce to the case where all of I acts unipotently, the semistable case. The action then factors through the largest pro- l quotient I_l of I , canonically isomorphic to $\mathbf{Z}_l(1)$.

Principle 4. *In the semistable case, if the Galois module H comes from algebraic geometry, then there is a unique increasing filtration W of H , the weight filtration, such that I acts trivially on $\text{Gr}_n^W(H)$ and such that $\text{Gr}_n^W(H)$, as a Galois module under $\text{Gal}(\bar{k}/k) \subset \text{Gal}(\bar{K}/K)/I$, is pure of weight n .*

Compare this with Principle 2 and with the appendix to [4]. When resolution of singularities is available, one can sometimes give a conjectural definition of W whose validity follows from the Weil conjectures. Using resolution and Weil, it is often easy to show at least that H can be broken up into Galois modules, under $\text{Gal}(\bar{k}/k)$, that are pure.

Assume that H is semistable. For $T \in I$, define $\log T$ by

$$-\sum_{n>0} (\text{Id} - T)^n / n.$$

The map $(T, x) \mapsto \log T(x)$ identifies with a homomorphism

$$(6) \quad M : \mathbf{Z}_l(1) \otimes_{\mathbf{Z}_l} H \rightarrow H.$$

Since $\mathbf{Z}_l(1)$ has weight -2 , necessarily

$$(7) \quad M(\mathbf{Z}_l(1) \otimes W_n(H)) \subset W_{n-2}(H),$$

and M induces

$$(8) \quad \text{Gr}_n(M) : \mathbf{Z}_l(1) \otimes \text{Gr}_n^W(H) \rightarrow \text{Gr}_{n-2}^W(H).$$

Definition 5. If X is a nonsingular projective variety over an algebraically closed field k , define

$$L : \mathbf{Z}_l(-1) \otimes H^*(X, \mathbf{Z}_l) \rightarrow H^*(X, \mathbf{Z}_l)$$

to be cup product with the cohomology class of a hyperplane section.

There is a formal analogy between L and M . Just as M is defined by an action of $\mathbf{Z}_l(1)$, one may regard L as defined by an action of $\mathbf{Z}_l(-1)$. The operator L raises degree by 2, and $\text{Gr}_n M$ in (8) lowers it by 2.

9.

Let D be the unit disk, $D^* = D - \{0\}$, and let

$$\begin{array}{ccc} X & \xrightarrow{\subset} & \mathbf{P}^r(\mathbf{C}) \times D \\ f \downarrow & & \downarrow \text{pr}_2 \\ D & \xlongequal{\quad} & D \end{array}$$

be a family of projective varieties parametrized by D , with f proper and $f|_{D^*}$ smooth.

Keep the notation of section 8. In the analogy between a Henselian trait and a small neighborhood of 0 in the complex line, one has the following dictionary:

D	$\text{Spec}(V)$
D^*	$\text{Spec}(K)$
a universal cover $\tilde{D}^*$ of D^*	$\text{Spec}(\bar{K})$
$\pi_1(D^*)$	inertia group I
$\pi_1(D^*) = \mathbf{Z} \simeq \mathbf{Z}(1)$	$I_l = \mathbf{Z}_l(1)$
X	projective scheme over $\text{Spec}(V)$
$X^* = f^{-1}(D^*)$	X_K
$X_D = X \times_D \tilde{D}^*$	$X_{\bar{K}}$
$\mathcal{H}^i f_* \mathbf{Z} _{D^*}$	$H^i(X_{\bar{K}}, \mathbf{Z}_l)$
$H^i(X_t, \mathbf{Z})$	$H^i(X_{\bar{K}}, \mathbf{Z}_l)$

Note that X_D is homotopy equivalent to each fiber $X_t = f^{-1}(t)$ for $t \in D^*$. Thus $H^i(X_t, \mathbf{Z})$ again corresponds to $H^i(X_{\bar{K}}, \mathbf{Z}_l)$, and the action of I corresponds to the monodromy transformation T .

Here too, a finite-index subgroup of $\pi_1(D^*)$ acts unipotently on $H^i(X_t, \mathbf{Q}) = H^i(X_D, \mathbf{Q})$. Put ourselves in the semistable case, where all of $\pi_1(D^*)$ acts unipotently; this amounts to replacing D by a finite cover. Let T be the action of the canonical generator of $\pi_1(D^*)$.

By sections 3 and 8, one expects $H^i(X_t, \mathbf{Q}) \simeq H^i(X_D, \mathbf{Q})$ to be endowed with an increasing filtration W , the groups $\text{Gr}_n^W H^i(X_t, \mathbf{Q})$ to be endowed with Hodge structures of weight n , and

$$\log T(W_n) \subset W_{n-2},$$

with $\log T$ inducing a morphism of Hodge structures

$$M : \mathbf{Z}(-1) \otimes \text{Gr}_n^W(H^i) \rightarrow \text{Gr}_{n-2}^W(H^i).$$

One would like (6), not only (7) and (8), to have an analogue. In fact, for each tangent vector u to D at 0, one can define a mixed Hodge structure $\mathcal{H}_u$ on $H^i(X_t, \mathbf{Z})$. The filtration W and the Hodge structures on $\text{Gr}_n^W(\mathcal{H}_u)$ are independent of u , and the dependence of $\mathcal{H}_u$ on u is expressed simply in terms of T . In analogy with (6), one finds that, for every u , $\log T$ induces a

homomorphism of mixed Hodge structures

$$M : \mathbf{Z}(1) \otimes H^i(X, \mathbf{Z}) \rightarrow H^i(X, \mathbf{Z}).$$

Finally, the analogy with Definition 5 is not misleading, though here the assumption that $f|_{D^*}$ is proper and smooth is probably essential. One proves that

$$(\log T)^k : \mathrm{Gr}_{n+k}^W(H^n(X_t, \mathbf{Q})) \rightarrow \mathrm{Gr}_{n-k}^W(H^n(X_t, \mathbf{Q}))$$

is an isomorphism for every k ; see [6, IV, 6, corollary to Theorem 5]. This characterizes the filtration W . Up to now, an analogue of the Hodge positivity theorem [6, IV, 7, corollary to Theorem 7] is available only in very special cases. One hopes that the mixed structures $\mathcal{H}$ determine the asymptotic behavior, as $t \rightarrow 0$, of the family of pure structures $H^i(X_t, \mathbf{Z})$ for $t \in D^*$.

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Paper 10 - Working English translation

Working full-body English translation generated from the cleaned source PDF. Mathematical formulae and source line structure are preserved where possible. This is a source-aligned working translation, not yet a polished final LaTeX edition.

HODGE THEORY II

PIERRE DELIGNE

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Introduction

[Polished translation of the opening paragraphs]

This paper develops the formalism of filtrations and then uses it to construct mixed Hodge structures on the cohomology of smooth algebraic varieties that need not be compact. The key point is that, although the cohomology of a nonsingular quasiprojective variety is no longer pure, it can still be organized as a successive extension of pure Hodge structures whose weights lie between n and $2n$.

The proof is essentially algebraic. It combines classical Hodge theory with Hironaka-style resolution of singularities, which allows one to express the cohomology of a smooth quasiprojective variety, via a spectral sequence, in terms of the cohomology of smooth projective varieties.

Section 1 collects some general facts on filtrations and proves two key results: the theorem on mixed Hodge structures, used later through its corollary on strictness, and the lemma of the two filtrations. Section 2 recalls classical Hodge theory and introduces mixed Hodge structures. The heart of the paper is Section 3.2, where the mixed Hodge structure on $H^*(X, \mathbb{C})$ is defined and the fundamental spectral-sequence degenerations are proved. Section 4 gives applications, especially the fixed-part theorem, semisimplicity, and consequences for homomorphisms of abelian schemes.

[Source-aligned literal body continues]

After Hodge, l'espace de cohomologie $H^n(X, \mathbb{C})$ d'une variété kahle-rienne compacte X est muni d'une « Hodge structure » de poids n , i.e. d'une bigraduation naturelle

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}$$

vérifiant $H^{p,q} = H^{q,p}$. On montre ici que la cohomologie complexe d'une variété algébrique nonsingulière, non nécessairement compacte, est munie d'une structure d'espèce à peu près générale, qui fait apparaître $H^n(X, \mathbb{C})$ comme « extension successive » de Hodge structures de poids décroissants, contenus entre $2n$ et n , dont les nombres de Hodge $h_{p,q} = \dim H^{p,q}$ sont nuls pour $p > n$ ou $q > n$.

Le lecteur trouvera exposé dans [14] le yoga qui sous-tend cette construction.

Travail présenté à l'Université d'Orsay.

1

2

PIERRE DELIGNE

La preuve, essentiellement algébrique, repose d'une part sur la théorie de Hodge, d'autre part sur la résolution des singularités à la Hironaka qui permet, via une suite spectrale, « d'exprimer » la cohomologie d'une variété algébrique nonsingulière quasi-projective en termes de la cohomologie de variétés projectives nonsingulières.

Le SS I, outre quelques sorites sur les filtrations réunis pour la commodité du lecteur, contient deux résultats-clés :

a) Le theorem (1.2.10), qui ne will be utilise que via son corollary (2.3.5), donnant the properties fondamentales of << Hodge structures mixtes >>.

b) Le << lemma of deux filtrations >> (1.3.16).

Le SS 2 recalls the Hodge theory and introduces the Hodge structures mixtes. Le coeur of ce travail is the no 3.2, qui defines the Hodge structure mixte of $H^*(X, \mathbb{C})$ and etablit quelques degenerations of spectral sequences.

Le SS 4 gives diverses maps, all deduites of (4.1.1) and of the theorie of the (K/A) -trace for the Hodge structures qui en resulte (4.1.2). Les principales are (4.2.6) and (4.4.15).

1. Filtrations

1.1. Filtered objects. (1.1.1) Let A a categorie abelienne.

On aura a considerer of filtrations of type $\mathbb{Z}$, en general finies, on the objects of A :

Definition 1.1. Une filtration decroissante (resp. croissante) F d'un object A of A is a famille $(F_n(A))_{n \in \mathbb{Z}}$ (resp. $(F_n(A))_{n \in \mathbb{Z}}$) of sous-objets of A , verifiant

$$n, m \in \mathbb{Z} \Rightarrow F_n(A) \supset F_m(A)$$

(resp. $n, m \in \mathbb{Z} \Rightarrow F_n(A) \subset F_m(A)$).

Un object filtre is a object muni d'une filtration. Quand il n'y aura pas danger of confusion, one designera often par a meme lettre of filtrations on of objects differents of A .

If F is a filtration decroissante (resp. croissante) on A , one pose $F(A) = 0$ and $F_{-\infty}(A) = A$ (resp. $F_{-\infty}(A) = 0$ and $F_{\infty}(A) = A$).

Les filtrations decalees d'une filtration decroissante W are definies par

$$W[n] = W_{n-\square} \quad (n, \square \in \mathbb{Z}).$$

(1.1.3) If F is a filtration decroissante (resp. croissante) of A , then the $F_n(A) = F_{-n}(A)$ (resp. the $F_n(A) = F_{-n}(A)$) forment a filtration croissante (resp. decroissante) of A . Ceci permet en principe of ne considerer que of filtrations decroissantes ; sauf mention explicite of the contraire, par << filtration >> one entendra dorenavant << filtration decroissante >>.

(1.1.4) Une filtration F of A is dite finie s'il existe n and m tels que $F_n(A) = A$ and $F_m(A) = 0$.

(1.1.5) Un morphism d'un object filtre (A, F) in a object filtre (B, F) is a morphism f of A in B qui satisfies $f(F_n(A)) \subset F_n(B)$ for $n \in \mathbb{Z}$.

Les objects filtres (resp. filtres of filtration finie) of A forment a categorie additive in laquelle existent the limites inductives and projectives finies (and therefore the noyaux, conoyaux, images and coimages d'un morphism).

HODGE THEORY II

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Un morphism $f : (A, F) \rightarrow (B, F)$ is dit strict, or strictement compatible aux filtrations, if the fleche canonique of $\text{Coim}(f)$ in $\text{Im}(f)$ is a isomorphism d'objets filtres (cf. (1.1.11)).

(1.1.6) Let θ the functor contravariant identique of A in the categorie duale A^0 . If (A, F) is a object filtre of A , the $(A/F_n(A))^0$ s'identifient a of sous-objets of A^0 . La filtration on A^0 duale of F is definie par

$$F_n(A^0) = (A/F_{-n}(A))^0.$$

Le bidual of (A, F) s'identifie a (A, F) . Cette construction identifie the duale of the categorie of objects filtres of A a the categorie of objects filtres of A^0 .

(1.1.7) If (A, F) is a object filtre of A , son gradue associe is l'objet of A^0 defini par

$$\text{Gr}_n(A) = F_n(A)/F_{n+1}(A).$$

La convention (1.1.6) is justifiee par the formule simple

$$\text{Gr}_n(A^0) = \text{Gr}_{-n}(A)^0$$

qui se satisfies on the diagramme autodual

$$(1.1.7.1) \quad \begin{array}{ccc} A/F_{-n}(A) & \xleftarrow{\quad} & A/F_{-n+1}(A) \\ | & & | \\ \text{Gr}_n(A) & & F_n(A) \\ | & \cdot & | \\ F_{n+1}(A) & \xrightarrow{\quad} & 0 \end{array}$$

(1.1.8) Let (A, F) a object filtre and $j : X \rightarrow A$ a sous-objet of A . La filtration induite on X par F is l'unique filtration on X telle que j let strictly compatible aux filtrations ; one a

$$F_n(X) = j^{-1}(F_n(A)) = X \cap F_n(A).$$

Dualement, the filtration quotient on A/X (unique filtration telle que $p : A \rightarrow A/X$ let strictly compatible aux filtrations) is given par

$$F_n(A/X) = p(F_n(A)) = (X + F_n(A))/X = F_n(A)/(X \cap F_n(A)).$$

Lemme 1.2. If X and Y are deux sous-objets of A , with $X \subset Y$, then on $Y/X = \text{Ker}(A/X \rightarrow A/Y)$, the filtration quotient of celle of Y coincide with celle induite par celle of A/X . Dans the diagramme

$$\begin{array}{ccc} A/Y & \xleftarrow{\quad} & A/X \\ | & & | \\ Y/X & & \end{array}$$

$$\begin{array}{ccc} & & | \\ X & \xrightarrow{\quad} & Y \xrightarrow{\quad} A \end{array}$$

the fleches are strictes.

(1.1.10) On appellera the filtration (1.1.9) on Y/X the filtration induite par celle of A . After (1.1.9), sa definition is autoduale.

$$\begin{array}{ccc} & f & g \\ & \downarrow & \downarrow \end{array}$$

En particulier, if $S : A \rightarrow B$

$$\rightarrow B$$

$\rightarrow C$ is a θ -sequence, and if B is filtre, then

$H(S) = \text{Ker}(g)/f = \text{Ker}(\text{Coim}(g) \rightarrow \text{Coim}(g))$ is muni d'une filtration induite canonical.

Le lecteur verifiera que :

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Proposition 1.3. (i) Let $f : (A, F) \rightarrow (B, F)$ a morphism d'objets filtres of filtration finie. For que f let strict, il faut and il suffices que the sequence $0 \rightarrow \text{Gr}(\text{Ker}(f)) \rightarrow \text{Gr}(A) \rightarrow \text{Gr}(B) \rightarrow \text{Gr}(\text{Coker}(f)) \rightarrow 0$

let exacte.

(ii) Soit

$$\begin{array}{ccc} & f & g \\ S : (A, F) \rightarrow (B, F) & \rightarrow & (C, F) \end{array}$$

a θ -sequence of morphisms stricts. On a then

$$H(\text{Gr}(S)) = \text{Gr}(H(S)).$$

En particulier, if S is exacte in A , then $\text{Gr}(S)$ is exacte in $A \mathbb{Z}$.

Dans a categorie of modules, dire qu'un morphism $f : (A, F) \rightarrow (B, F)$ let strict signifie que all $b \in B$, of filtration n ($b \in F_n(B)$) qui is in l'image of A , is already in l'image of $F_n(A)$:

$$f(F_n(A)) = f(A) \cap F_n(B).$$

(1.1.12) If $\otimes : A_1 \times \dots \times A_n \rightarrow B$ is a functor multiadditif exact a droite, and if A_i is a object of filtration finie of A_i ($1 \leq i \leq n$), one defines a filtration on $\otimes A_i$, par

$$F_{\square}(\otimes A_i) = \bigoplus_{k_1 + \dots + k_n = \square} (\otimes F_{k_i}(A_i) \rightarrow \otimes A_i)$$

(sum of sous-objets).

Dualement, if H is exact a gauche, one pose

$$F_{\square}(H(A_i)) = \bigcap_{n} \text{Ker}(H(A_i) \rightarrow H(A_i / F_n(A_i)))$$

($k_i = \square$).

For H exact, the deux definitions are equivalent.

On etend ces definitions a of functors contravariants en certaines variables par (1.1.6). En particulier, for the functor exact a gauche Hom , one pose

$$F_n(\text{Hom}(A, B)) = \{f : A \rightarrow B \mid f(F_{\square}(A)) \subset F_{\square+n}(B)\}.$$

On a therefore

$$\text{Hom}((A, F), (B, F)) = F_0(\text{Hom}(A, B)).$$

Sous the hypotheses precedentes, one dispose of morphisms evidents

$$\otimes \text{Gr}(A_i) \rightarrow \text{Gr}(\otimes A_i)$$

and

$$\text{Gr}(H(A_i)) \rightarrow H(\text{Gr}(A_i)).$$

For H exact, ce are of isomorphisms inverses l'un of l'autre.

Ces constructions are compatibles a the composition of functors, en a sens qu'on laisse au lecteur the soin of ne pas expliciter.

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1.2. Opposed filtrations. (1.2.1) Let A a object of A muni of deux filtrations F and F . Par definition, $\text{GrnF}(A)$ is a quotient d'un sous-objet of A , and, en tant que tel, se trouve canonically plonge in A . On pose

$$\begin{array}{c} \text{GrnF Grm} \\ F \\ (A) = \text{GrnF}(\text{Grm} \\ F \\ (A)). \end{array}$$

C'est a sous-quotient of A , quotient of $F_n F(A)$ and sous-objet of $\text{GrnF}(A)$. On a of formules duales :

$$\begin{array}{c} \text{GrnF Grm} \\ F \\ (A) = \end{array} \quad \begin{array}{c} F_n(A) \cap F(A) \\ m \\ F_{n+1}(A) \cap F(A) + F_n(A) \cap F_{m+1}(A) \end{array}.$$

(1.2.2) On appelle filtration n-opposee a a filtration F a filtration F telle que

$$\text{GrpF GrqF } (A) = 0 \text{ for } p + q \neq n.$$

If F admet a filtration n-opposee, one dit que F is n-opposable.

Proposition 1.4. Let F and F deux filtrations n-opposees of A. Then :

(i) F and F are finies.

$$(ii) \text{ For all } p, F_p(A) \otimes F_{n-p+1}(A) \rightarrow A.$$

$$(iii) \text{ For all } p \text{ and } q \text{ with } p + q = n + 1, F_p(A) \otimes F_q(A) \rightarrow A.$$

(iv) Les gradues $\text{GrpF GrqF } (A) (p + q = n)$ are the facteurs d'une decomposition directe of A.

Demonstration. (i) resulte of (iii) with $q = n - p + 1$.

(ii) and (iii) equivalent a dire que l'application canonical

$$F_p(A) \otimes F_q(A) \rightarrow A$$

(for $p + q = n + 1$) is injective and surjective. L'assertion d'injectivite

is equivalent a dire que $F_p(A) \cap F_q(A) = 0$, i.e. que $F_p F_q(A) = 0$. C'est a case particulier of l'hypothese. L'assertion of surjectivite is equivalent a dire que

$$F_p(A) + F_q(A) = A.$$

For the prouver, one procede par recurrence descendante on p (supposons F and F décroissantes). For p assez grand, $F_p(A) = A$ and l'assertion is claire.

For passer of p a $p - 1$, one utilise l'exactitude of

$$0 \rightarrow \text{Grp-1} F(A) \rightarrow A/F_p(A) \rightarrow A/F_{p-1}(A) \rightarrow 0$$

and l'isomorphisme

$$\text{Grp-1} F(A) = F_{n-p+1}(A)/F_p(A) = F_{p-1}(F_{n-p+1}(A)).$$

L'hypothese permet d'identifier ce dernier a

$$F_{p-1}(A) \cap F_{n-p+1}(A) = F_p(A) \cap F_{n-p+1}(A)$$

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and l'application $A/F_p(A) \rightarrow A/F_{p-1}(A)$ a l'application quotient of A par $F_{p-1}(A) + F_p(A)$. L'assertion resulte then of l'hypothese of recurrence and of l'exactitude of the sequence

$$0 \rightarrow F_{p-1}(A) \cap F_{n-p+1}(A) \rightarrow F_p(A) \cap F_{n-p+1}(A) \rightarrow F_p(A) \rightarrow F_{p-1}(A) \rightarrow 0.$$

(iv) After (ii), the $F_p F_q(A) (p \in \mathbb{Z})$ forment a filtration of A. On

$$a F_p F_q(A)/F_{p+1} F_q(A) = \text{GrpF Grn-p} F_q(A). \quad \#$$

(1.2.4) Let Apq a famille d'objets of A, zero sauf for a nombre fini of couples (p, q) tels que $p + q = n$. If one pose

$$A = \bigoplus_{p+q=n} \text{Apq},$$

then, one defines deux filtrations finies n-opposees of A =

$\bigoplus \text{Apq}$ en posant

$$(1.2.1) F_p(A) = \bigoplus_{p' \leq p} \text{Apq},$$

$$(1.2.2) F_q(A) = \bigoplus_{q' \leq q} \text{Apq}.$$

On a

$$\text{GrpF GrqF } (A) = \text{Apq}.$$

Reciproquement :

Proposition 1.5. (i) Let F and F deux filtrations finies on A . For que F and F let n -opposees, il faut and il suffices que

$$p, q \quad p + q = n + 1 \rightarrow F_p(A) \otimes F_q(A) \rightarrow A.$$

(ii) If F and F are n -opposees, and if one pose

$$A = \bigoplus_{p+q=n} F_p(A) \otimes F_q(A)$$

then $A = \bigoplus_{p+q=n} F_p(A) \otimes F_q(A)$, and F and F are the filtrations (1.2.4) definies par cette decomposition.

Demonstration. Prouvons (i). La condition is necessary after (1.2.3)

(iii). Reciproquement, supposons-la verifiee and prove que $\text{GrpF GrqF}(A) = 0$ for $p + q \neq n$. Par dualite, il suffices of traiter the case or $p + q > n$. Posons $r = p$, $s = n - p + 1$ and $t = q - (n - p + 1) = p + q - n > 0$. On a $r + s = n + 1$, d'ou

$$F_r(A) \otimes F_s(A) \rightarrow A.$$

On en deduit que

$$F_r(A) \otimes F_s(A) = F_r(A) \otimes F_s(A) \otimes F_t(A) = 0,$$

d'ou $\text{GrpF GrqF}(A) = 0$ puisque $F_p(A) \otimes F_q(A) \subset F_r(A) \otimes F_s(A)$.

Prouvons (ii). After (i), applique a F and F decalees, one a

$$F_p(A) \otimes F_q(A) = 0 \quad \text{and} \quad F_p(A) + F_q(A) = A,$$

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d'ou $A = \bigoplus_{p+q=n} F_p(A) \otimes F_q(A)$. On satisfies then que $F_p(A) = \bigoplus_{q' \leq p} F_{q'}(A)$ and $F_q(A) = \bigoplus_{p' \leq q} F_{p'}(A)$.

(1.2.6) If F and F are deux filtrations n -opposees on A , one pose

$$\text{Apq} = \text{GrpF GrqF}(A) \quad (\text{zero for } p + q \neq n).$$

After (1.2.5), one a

$$A = \bigoplus_{p+q=n} \text{Apq}.$$

On appelle cette decomposition the decomposition of A en type (p, q) . If $a \in A$ is of type (p, q) , i.e. $a \in \text{Apq}$, then

$$a \in F_p(A) \otimes F_q(A).$$

Definition 1.6. Trois filtrations finies W , F and F on a object A of A are dites opposees if :

(i) For all n , the filtrations induites par F and F on $\text{GrnW}(A)$ are n -opposees.

(ii) For all p, q, n , one a

$$\text{GrnW GrpF GrqF}(A) = 0 \quad \text{for } p + q \neq n.$$

(1.2.8) Let W , F , F trois filtrations opposees on A . Posons

$$\text{Apqn} = \text{GrnW GrpF GrqF}(A).$$

After (1.2.7) (ii), $\text{Apqn} = 0$ for $p + q \neq n$. On pose

$$\text{Apq} = \text{Apq, } p+q = \text{GrpF GrqF Grp+q W}(A).$$

On a then

$$\text{GrnW}(A) = \bigoplus_{p+q=n} \text{Apq}$$

and the filtrations induites par F and F on $\text{GrnW}(A)$ are given par (1.2.4).

(1.2.9) Let W , F , F trois filtrations opposees on A , and let W' a quatrieme filtration. On dit que W' is compatible au systeme (W, F, F) if elle is compatible a chacune of trois filtrations W , F , F . If W' is compatible, then the filtrations induites par F and F on $\text{Grm } W'(A)$ restent m -opposees, and the trois filtrations W , F , F induites on $\text{Grm } W'(A)$ restent

opposees.

Theoreme 1.7. Let A a categorie abelienne, and designons provisoirement par A_f the categorie of objects of A munis of trois filtrations finies opposees.

(i) A_f is a categorie abelienne.

(ii) Les functors $\ll$ oubli of filtrations $\gg$, GrnW , GrpF , GrFq , GrnW GrpF , GrnW GrqF , GrpF GrFq and GrnW GrpF GrqF of A_f in A (resp. in A_Z) are exacts.

(iii) Tout morphism $f : A \rightarrow B$ in A_f admettant a kernel (resp. co-kernel, image, coimage) in A admet a kernel (resp. conoyau,

image, coimage) in $A f$. Le gradue of ce kernel (resp. ...) is the kernel (resp. ...) of gradues.

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- (iv) Toute sequence $A \rightarrow B \rightarrow C$ in $A f$, exacte in A , is exacte in $A f$.
- (v) Le functor $A \rightarrow (A, W, F, F)$ of $A f$ in the categorie of objects of A munis of deux filtrations opposees admet a adjoint a gauche and a droite.

Esquisse of proof. Il suffices of prouver (iii) for the noyaux. Let $f : A \rightarrow B$ a morphism in $A f$. Let K a kernel of f in A . Muni of filtrations induites par W , F and F , il s'agit of voir que ces trois filtrations are opposees. La condition (1.2.7) (ii) is clearly verifiee (the $Apqn$ are the noyaux of $f : Apqn \rightarrow B pqn$). Verifions (1.2.7) (i). Les filtrations induites par F and F on $GrnW(K)$ are the filtrations induites par W , F and F of K are therefore opposees and K is a kernel of f in $A f$. #

(1.2.11) Le theorem (1.2.10) permet of define the notion of systeme of trois filtrations opposees on a object d'une categorie abelienne A on the- quelle one ne fait no hypothese. Let en effet A' the categorie of objects of A munis d'une filtration finie. Le functor $A' \rightarrow (A, W, F)$ of $A f$ in the categorie of objects of A munis of deux filtrations finies admet a adjoint a gauche L and a droite R ((1.2.10) (v)). Le functor $A' \rightarrow (LA, W, F, F)$ of $A f$ in $A f$ is adjoint a gauche au functor identique, therefore is pleinement fidele. On en deduit que $A f$ s'identifie a the categorie of objects of A' munis d'une filtration W opposable.

(1.2.12) Let (A, W, F, F) a object of $A f$. On pose
 $Apqn = GrnW GrpF GrFq(A)$.

After (1.2.7) (ii), $Apqn = 0$ for $p + q \neq n$. On pose
 $Apq = Apq, p+q$.

On a

$$GrnW(A) = \bigoplus_{p+q=n} Apq, \quad GrpF(A) = \bigoplus_{q,n} Apqn, \quad GrFq(A) = \bigoplus_{p,n} Apqn.$$

(1.2.13) Let W , F and F trois filtrations finies on A . Les conditions suivantes are equivalent :

- (i) Les filtrations W , F and F are opposees.
- (ii) For all n , the filtrations induites par F and F on $GrnW(A)$ are n -opposees, and for all p , the filtrations induites par W and F on $GrpF(A)$ are p -opposees.
- (iii) For all n , the filtrations induites par F and F on $GrnW(A)$ are n -opposees, and for all q , the filtrations induites par W and F on $GrqF(A)$ are q -opposees.

En effet, (ii) signifie que $GrnW GrpF GrqF(A) = 0$ for $p + q \neq n$ and que $GrpF GrnW GrFq(A) = 0$ for $n + q \neq p$.

De meme, (iii) signifie que $GrnW GrpF GrqF(A) = 0$ for $p + q \neq n$ and que $GrFq GrnW GrpF(A) = 0$ for $n + p \neq q$.

On se ramene a supposer que S is connexe non vide, of point generique e , and que X_e is a variety abelienne simple on $k(e)$. If (i) or (ii) is verifiee, $D = \text{End}(X_e) \otimes Q$ is then a corps, and, for $s \in S$, $(R1 f* Q)s$ is a representation irreductible of $\pi_1(S, s)$.

For prouver que (ii) $\Rightarrow$ (i), one peut supposer Y_e simple, and $(R1 g* Q)s$ isotypique of type $(R1 f* Q)s$. On a then a isomorphism of familles continues of Q -Hodge structures (D being of type $(0, 0)$)

$$R1 g* Q \rightarrow R1 f* Q \otimes_D \text{Hom}(R1 f* Q, R1 g* Q).$$

On a $D \otimes C \simeq M_n(Z \otimes C)$. If $e \in D \otimes C$ se transforme par a tel isomorphism en lidempotent e_{11} , one a again a isomorphism of vectoriels bigradues

$$(R1 g* C)s \rightarrow (R1 f* C)s \otimes_{Z \otimes C} e(\text{Hom}(R1 f* C, R1 g* C)).$$

If the condition (ii) is verifiee, and if $e(\text{Hom}(R1 f* C, R1 g* C))$ nest pas purement of type $(0, 0)$, then $(R1 g* C)s$ ne pourrait pas be purement of type $(-1, 0) + (0, -1)$, ce qui is absurde. On en deduit que $\text{Hom}(R1 f* Z, R1 g* Z)$ is purement of type $(0, 0)$, and (i) en resulte par (4.4.3) and (2.1.11.1).

Let Z a extension quadratique totalement imaginaire dun corps of nombres totalement reel $Z \neq 0$. If $Z \otimes R \subset C \otimes \dots \otimes C$, one defines a action par homotheties of S on $Z \otimes R$ en envoyant $s \in S(R) \simeq C^*$ on $(s^2)^{-1}$, $1, \dots, 1$. Le couple form of Z and of cette action of S on $Z \otimes R$ is a Hodge structure polarisable Zp of type $(-1, 1) + (0, 0) + (1, -1)$, qui depend of Z and of the place choisie $r : Z \rightarrow C$.

Let X a schema abelien on S , tel que X_e let simple. If the centre Z of $\text{End}(R1 f* Q)$ nest pas totalement reel, il is extension quadratique totalement imaginaire dun corps of nombres totalement reel. If the seconde hypothese of (ii) is violee, the facteur direct $R1 f* Q \otimes_Z Zp$ of the famille continue of Hodge structures polarisable $R1 f* Q \otimes_Q Zp$ is purement of type $(-1, 0) + (0, -1)$.

Dapres (4.4.3), the choix dun reseau entier in $R_1 f^* Q \otimes_{\mathbb{Z}} \mathbb{Z}_p$ defines a schema abelien $g : Y \rightarrow S$, and $\text{Hom}(R_1 f^* Q, R_1 g^* Q)$ nest pas purement of type $(0, 0)$, ce qui viole (i).

Corollaire 1 (4.4.13). Let S a schema smooth connexe of point generique e and $f : X \rightarrow S$ a schema abelien on S of dimension relative $g \leq 3$. On suppose que on no revetement etale fini of S , X nadmet of sous-schema abelien constant. Then, for all schema abelien $g : Y \rightarrow S$, one a

$$\text{Hom}(X, Y) \xrightarrow{\sim} \text{Hom}(R_1 f^* Z, R_1 g^* Z)$$

and

$$\text{Hom}(Y, X) \xrightarrow{\sim} \text{Hom}(R_1 g^* Z, R_1 f^* Z).$$

Par dualite, one se ramene a ne prouver que the premiere assertion. On se ramene aussi a supposer X simple. Verifions que X satisfait a lune of conditions (c) of (4.4.11).

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Posons $H = \text{Mr}(D)$, D being a corps gauche of rang b_2 on son centre Z , que lon sait be totalement reel or totalement imaginaire. Posant $a = [Z : \mathbb{Q}]$, on a

$$(4.4.13.1) \quad ab_2 \mid c \mid 2g \leq 6.$$

1) If Z is totalement imaginaire, and non quadratique imaginaire, one doit avoir $a = 2g = 4$. For $s \in S$, the commutant of Z agissant on $(R_1 f^* Q)_s$ is then commutatif, of sorte que l'action of $\pi_1(S, s)$ is abelienne, therefore se factorise par a group fini ((4.2.8), (iii), b) or (4.2.9)). Ceci is absurde ((4.1.3.3) et (4.4.3)).

Avec the notations of (4.4.8), one pourrait aussi noter que the T_p are then of rang a , of sorte que the condition (4.4.8), b) is verifiee en chaque place and que the Hodge structure is localement constante.

2) If Z is totalement reel and if the conditions (c1) and (c2) are violees, then $a \geq 2$ and $b \geq 2$, ce qui is absurde.

Verifions que X satisfait aux conditions of (4.4.12). If Z is quadratique imaginaire, and sil existe $r : Z \rightarrow \mathbb{C}$ tel que $R_1 f^* Q \otimes_{\mathbb{Z}, r} \mathbb{C}$ let purement of type of Hodge $(-1, 0)$, then the Hodge structure of $R_1 f^* Q$ is localement constante, ce qui is absurde. Ceci proves (4.4.13).

(4.4.14) Let $f : X \rightarrow S$ a schema abelien polarise of dimension g on a base smooth and connexe S . If $s \in S$, the fundamental group $\pi_1(S, s)$ agit on the fibre $H_1(X_s, \mathbb{Z})$ of $R_1 f^* \mathbb{Z}$ en s . La polarisation of X defines a form alternee on $H_1(X_s, \mathbb{Z})$, a values in $\mathbb{Z}(-1)$, non degeneratese on $\mathbb{Q}$, and l'action of $\pi_1(S, s)$ se fait en respectant cette form. Considerons lhypothese : (4.4.14.1)

Le morphism defini par X of S in lune of composantes irreductibles of the schema of modules of varieties abeliennes polarisees correspondant is a morphism dominant.

Le result suivant ma been suggere par J.-P. Serre.

Corollaire 2 (4.4.15). Let S a schema smooth connexe and $f : X \rightarrow S$ a schema abelien polarise on S qui satisfies (4.4.14.1). Then, for all schema abelien $g : Y \rightarrow S$, one a

$$\text{Hom}(X, Y) \xrightarrow{\sim} \text{Hom}(R_1 f^* Z, R_1 g^* Z).$$

Let $s \in S$. Dapres (4.4.11) and (4.4.12), il suffices of show que the representation of $\pi_1(S, s)$ on $(R_1 f^* Q)_s$ is absolument irreductible, of sorte que $H = \mathbb{Q}$. Cest ce qui resulte of the lemme suivant :

Lemme 3 (4.4.16). If X satisfies (4.4.14.1), then l'image of $\pi_1(S, s)$ in the group of automorphismes symplectiques of $H_1(X_s, \mathbb{Z})$ is d'indice fini.

Let n a entier. Quitte a remplacer S par a revetement etale surjectif, defini par a sous-groupe d'indice fini of $\pi_1(S, s)$, one peut supposer que the systeme local $nX = \text{Ker}(n \cdot \text{id} : X \rightarrow X)$ is trivial on S . Remplacant X

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par a schema abelien isogene, one se ramene ensuite and of more a supposer X polarise of the serie principale. Supposons $n \geq 3$.

Le schema abelien X defines then a morphism dominant x of S in the schema of modules $\ll$ of Jacobi $\gg$ M_n of schemas abeliens polarises of the serie principale munis dun isomorphism symplectique entre nX and $(\mathbb{Z}/n)^{2g}$; of more, X is a image reciproque par x of the schema abelien universel on M_n . On sait que $\pi_1(M_n, x(s))$ senvoie isomorphiquement on the sous-groupe of $\text{Sp}(H_1(X_s, \mathbb{Z}))$ kernel of l'application of reduction mod n .

Il reste therefore a verifier que :

Lemme 4 (4.4.17). Let $p : S \rightarrow M$ a morphism dominant of schemas smooth connexes. Le sous-groupe $p^*(\pi_1(S, s))$ of $\pi_1(M, p(s))$ is then d'indice fini.

Let t a point ferme of the fibre generique of p , and let T ' son adherence

in S . Il existe a ouvert of Zariski dense U of M tel que $T = p^{-1}(U) \cap T'$
 let a revetement etale of U .

On ne restreint pas the generalite en prenant $s \in T$. Dans the diagramme

$$\pi_1(T, s) \xrightarrow{\quad} \pi_1(S, s)$$

ay

c
y

b

$$\pi_1(U, p(s)) \xrightarrow{\quad} \pi_1(M, p(s))$$

the fleche a a a image dindice fini, car T is a revetement fini etale, and the
 fleche b is surjective, car $M \setminus U$ is of codimension reelle ≥ 2 . La fleche c
 a therefore a image dindice fini.

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A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

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Let p be a prime number. For every nonzero integer m , let $v(m)$ denote the exponent of the highest power of p dividing m . We also put $v(0) = \infty$.

For a k -tuple $\mathbf{n} = (n_1, \dots, n_k)$ of integers ≥ 0 , set

$$|\mathbf{n}| = \sum_1^k n_i, \quad a\mathbf{n} = (an_1, \dots, an_k), \quad X^{\mathbf{n}} = \prod_1^k X_i^{n_i}, \quad \mathbf{n}! = \prod_1^k n_i!.$$

Let $C(\mathbf{n})$ be the coefficient of the monomial $X^{\mathbf{n}}$ in

$$\left(\sum_1^k X_i \right)^{|\mathbf{n}|};$$

these are the multinomial coefficients. Thus

$$(1) \quad \left(\sum_1^k X_i \right)^n = \sum_{|\mathbf{n}|=n} C(\mathbf{n}) X^{\mathbf{n}}$$

$$(2) \quad C(\mathbf{n}) = \frac{|\mathbf{n}|!}{\mathbf{n}!}.$$

For $k = 2$, these are the binomial coefficients:

$$(3) \quad C(n_1, n_2) = \binom{n_1 + n_2}{n_1}.$$

Proposition 1. *Let $\mathbf{n}$ be a k -tuple of integers ≥ 0 , and put $n = |\mathbf{n}|$. If p is a prime number ≥ 5 , then*

$$(4) \quad C(p^l \mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+3}}.$$

EXAMPLE. — For $p = 5$, $k = 2$, $l = 1$, and $\mathbf{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3.$$

PROOF. — It is known, or follows from (2), that

$$(5) \quad \left(\sum_1^k X_i \right)^{p^l} = \left(\sum_1^k X_i^{p^l} \right) + pQ,$$

where Q is a polynomial with integer coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$(6) \quad \left(\sum_1^k X_i \right)^{p^l n} = \left(\sum_1^k X_i^{p^l} \right)^n + \sum_1^n p^i \binom{n}{i} R_i,$$

$$(7) \quad R_i = \left(\sum_1^k X_i^{p^l} \right)^{n-i} Q^i.$$

For two polynomials A and B with integer coefficients, write $A \equiv B \pmod{r}$ if the coefficients of $X^{p^l n}$ in A and B are congruent modulo r . With this notation, (4) is rewritten as

$$(8) \quad \left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n \pmod{p^{v(n)+3}}.$$

Lemma 1.

$$(9) \quad v\left(\binom{n}{i}\right) \geq v(n) - v(i).$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation. Thus each $x \in \mathbb{Z}/(n)$ determines the permutation $y \mapsto x + y$ of $\mathbb{Z}/(n)$. If a subset P with i elements in $\mathbb{Z}/(n)$ has stabilizer H , that is, if $H \cdot P = P$, then P is a union of cosets of H , and $\#H$ divides i . The cardinal

$$r_P = n/\#H$$

of the orbit of P under $\mathbb{Z}/(n)$, acting on the power set $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_P.$$

Since the set of subsets of $\mathbb{Z}/(n)$ with i elements is a disjoint union of orbits, it follows that

$$p^{v(n)-v(i)} \mid \binom{n}{i}.$$

Lemma 2. *For $p \neq 2, 3$ and $i \geq 3$, one has*

$$(10) \quad p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}}.$$

Indeed,

$$v\left(p^i \binom{n}{i}\right) = i + v\left(\binom{n}{i}\right) \geq v(n) + (i - v(i))$$

by Lemma 1. If $3 \leq i < p$, then $v(i) = 0$ and $i - v(i) \geq 3$. If $i \geq p$, then

$$i - v(i) \geq i - \log i / \log p.$$

The last function of i is increasing for $i \geq p$, and its value at $i = p$ is $p - 1 \geq 3$.

Lemma 3. *The coefficient of $X^{p^l n}$ in R_1 is zero.*

No monomial $X^{\mathfrak{r}}$ appearing in Q has an exponent $\mathfrak{r}$ divisible by p^l . The polynomial

$$R_1 = \left(\sum_1^k X_i^{p^l} \right)^{n-1} Q$$

inherits this property.

By Lemmas 2 and 3,

$$(11) \quad \left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n + \binom{n}{2} p^2 R_2 \pmod{p^{v(n)+3}}.$$

This congruence already implies, by Lemma 1, that

$$(12) \quad C(p^l n) \equiv C(n) \pmod{p^{v(n)+2}}.$$

The only monomials $X^{\mathfrak{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} X_j^{p^l} \quad (i < j).$$

The coefficient of each of them is

$$(13) \quad \sum_{i \neq 0, p^l} \binom{p^l}{i} \binom{p^l}{p^l - i} = \sum \binom{p^l}{i}^2 - 2 = \binom{2p^l}{p^l} - 2.$$

By (12),

$$\binom{2p^l}{p^l} - 2 \equiv \binom{2p}{p} - 2 \pmod{p^3}.$$

Thus

$$\left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n + \binom{n}{2} \left[\binom{2p}{p} - 2 \right] R_2 \pmod{p^{v(n)+3}},$$

with R_2 a polynomial with integer coefficients.

Since

$$v\left(\binom{n}{2}\right) \geq v(n)$$

by Lemma 1, the proposition follows from the next lemma.

Lemma 4. *For every prime $p \geq 5$,*

$$\binom{2p}{p} \equiv 2 \pmod{p^3}.$$

For $1 \leq i \leq p-1$, one has

$$p^{-1} \binom{p}{i} \equiv (-1)^{i-1} i^{-1} \pmod{p}.$$

Hence

$$(14) \quad p^{-2} \left[\binom{2p}{p} - 2 \right] = \sum_1^{p-1} p^{-2} \binom{p}{i}^2 \equiv \sum_1^{p-1} (1/i)^2 \pmod{p}.$$

Define

$$z \in \mathbb{Z}/(p), \quad z = \sum_{i \in \mathbb{Z}/(p)^*} 1/i^2.$$

For $u \in \mathbb{Z}/(p)^*$, one has

$$z = \sum_{i \in \mathbb{Z}/(p)^*} 1/(u^{-1}i)^2 = u^2 z,$$

and therefore

$$(u^2 - 1)z = 0.$$

Since $p \neq 2, 3$, there exists u such that $u^2 \neq 1$, and hence $z = 0$.

Formula (14) consequently becomes

$$p^{-2} \left[\binom{2p}{p} - 2 \right] \equiv 0 \pmod{p},$$

which completes the proof.

UNIRATIONAL VARIETIES THAT ARE NOT RATIONAL

[after M. Artin and D. Mumford]

by Pierre Deligne

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This report contains a description of the variety constructed by Artin and Mumford, together with the proof that it is unirational but not rational. It also states the theorems proved by Clemens and Griffiths [4], and by Manin and Iskovskih [6], which yield other examples. I attended talks by Artin and Mumford on the same subject, and I have drawn heavily on them.

Throughout, the expression “algebraic variety” means “a separated scheme of finite type over $\mathbf{C}$, reduced and irreducible.” We write $k(X)$ for the field of rational functions on an algebraic variety X .

1. RATIONALITY AND UNIRATIONALITY

Let K be a finitely generated extension of $\mathbf{C}$. Then there exists an algebraic variety X such that K is isomorphic to $k(X)$. One calls X a *model* of K . By [5], K always admits a smooth projective model.

Let K and L be finitely generated extensions of $\mathbf{C}$, with models X and Y . Homomorphisms $a^*: K \rightarrow L$ correspond bijectively to dominant rational maps $a: Y \dashrightarrow X$, by the rule $a^*(f) = f \circ a$. By [5], if X and Y are smooth and projective, every rational map $a: Y \dashrightarrow X$ admits a factorization

$$(1) \quad a = b \circ c_n^{-1} \circ \cdots \circ c_1^{-1}$$

where b and the c_i are everywhere defined morphisms and where Y_{i+1} is obtained from Y_i by blowing up a smooth subvariety Z_i of codimension ≥ 2 .

If $a^*: K \rightarrow L$ is an isomorphism, one can apply the preceding result to (a^{*-1}, Y_{n+1}, X) . Iterating, one obtains a commutative diagram of scheme morphisms

$$(2) \quad \begin{array}{ccccccc} Y'_n & \xrightarrow{y_n} & \cdots & \xrightarrow{y_2} & Y'_1 & \xrightarrow{y_1} & Y \\ \downarrow a_n & & & & \downarrow a_1 & & \downarrow a \\ X & \xleftarrow{x_n} & X'_n & \xleftarrow{x_2} & \cdots & \xleftarrow{x_1} & X'_1 \xleftarrow{\quad} X \end{array}$$

In this diagram, the horizontal arrows x_i, y_i are composites of blow-ups with smooth centers, as in (1); the a_i and b_i are birational.

An invariant of a smooth projective variety X is said to be *birational* if it depends only on $k(X)$. One often verifies birational invariance using (2) (for a typical example, see Proposition 2.1). Invariants of a field K are usually defined as the value of some birational invariant of any smooth projective model of K .

Definition 1.1. Let X be an algebraic variety of dimension n . One says that X is *unirational* if the following equivalent conditions hold:

- (i) $k(X)$ is a subfield of a purely transcendental extension $\mathbf{C}(T_1, \dots, T_r)$ of $\mathbf{C}$;
- (ii) a finite extension of $k(X)$ is isomorphic to $\mathbf{C}(T_1, \dots, T_n)$;
- (iii) if X is smooth and projective, there exists a diagram (1) with $Y = \mathbf{P}^r(\mathbf{C})$;
- (iv) if X is smooth and projective, there exists a diagram (1) with $Y = \mathbf{P}^n(\mathbf{C})$.

One may suppose X smooth and projective, and one has

$$(ii) \Rightarrow (i) \Rightarrow (iii) \Rightarrow (iv) \Rightarrow (ii).$$

Definition 1.2. One says that X is *rational* if, equivalently,

- (i) $k(X)$ is a purely transcendental extension of $\mathbf{C}$;
- (ii) if X is smooth and projective, there exists a diagram (2) with $Y = \mathbf{P}^n(\mathbf{C})$.

The most obvious birational invariants do not distinguish unirational varieties from rational varieties.

Proposition 1.3. *Let X be a smooth projective unirational algebraic variety of dimension $n > 0$.*

- (i) *The plurigenera $P_k = \dim H^0(X, (\Omega_X^n)^{\otimes k})$ ($k > 0$) vanish. More generally, $H^0(X, (\Omega_X^1)^{\otimes k}) = 0$ for $k > 0$ (see [9, Remark, p. 102]).*
- (ii) *X is simply connected [10].*

For $n = 1$ or 2 , property (i) already characterizes rational varieties. Thus a unirational variety of dimension at most 2 is rational, and a subextension of $\mathbf{C}(T_1, \dots, T_k)$ of transcendence degree at most 2 is automatically purely transcendental. More precisely:

- a) If $n = 1$ and $H^0(X, \Omega_X^1) = 0$, then $X \simeq \mathbf{P}^1(\mathbf{C})$. Indeed, by Riemann–Roch, for every $x \in X$, the line bundle $\mathcal{O}(x)$ defines an embedding of X into $\mathbf{P}^1(\mathbf{C})$.
- b) For $n = 2$, the Castelnuovo–Enriques criterion says that X is rational if $P_a = P_2 = 0$, i.e. if $H^0(X, \Omega_X^2) = H^0(X, (\Omega_X^2)^{\otimes 2}) = 0$; see [9].

For $n > 2$, it has been very difficult to define or compute birational invariants capable of distinguishing rational from unirational varieties.

2. A FEW BIRATIONAL INVARIANTS

A) Artin and Mumford.

Proposition 2.1. *The torsion subgroup $H^3(X, \mathbf{Z})_{\text{tors}}$ of $H^3(X, \mathbf{Z})$ is a birational invariant of the smooth projective variety X .*

If X_2 is obtained from a smooth variety X_1 by blowing up a smooth subvariety $Z \subset X_1$ of codimension $r \geq 2$, then (SGA 5, VII)

$$H^n(X_2, \mathbf{Z}) \simeq H^n(X_1, \mathbf{Z}) \oplus \bigoplus_{i=1}^{r-1} H^{n-2i}(Z, \mathbf{Z}).$$

In particular, $H^3(X_2, \mathbf{Z}) \simeq H^3(X_1, \mathbf{Z}) \oplus H^1(Z, \mathbf{Z})$. Since $H^1(Z, \mathbf{Z})$ is torsion-free, one gets

$$H^3(X_1, \mathbf{Z})_{\text{tors}} \xrightarrow{\sim} H^3(X_2, \mathbf{Z})_{\text{tors}}.$$

Assume $k(X) \simeq k(Y)$, and choose a diagram (2). One obtains

$$\begin{array}{ccccccc} H^3(X, \mathbf{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X'_1, \mathbf{Z})_{\text{tors}} & \longleftarrow & H^3(X'_n, \mathbf{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X, \mathbf{Z})_{\text{tors}} \\ a_1^* \uparrow & & \uparrow b_1^* & & \uparrow b_n^* & & \uparrow a_2^* \\ H^3(Y, \mathbf{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(Y'_1, \mathbf{Z})_{\text{tors}} & \longleftarrow & H^3(Y'_n, \mathbf{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(Y, \mathbf{Z})_{\text{tors}} \end{array}$$

Since $a_2^* \circ b_n^*$ is bijective, b_n^* is injective. Since $b_n^* \circ a_n^*$ is bijective and b_n^* is injective, a_n^* is bijective, and the assertion follows.

Construction 2.2. We shall construct a smooth projective unirational variety X of dimension 3 with $H^3(X, \mathbf{Z})_{\text{tors}} \neq 0$ (in fact, $H^3(X, \mathbf{Z}) \simeq \mathbf{Z}/(2)$). Multiplying this variety by $\mathbf{P}^r(\mathbf{C})$ yields an analogous example in every dimension ≥ 3 .

B) Clemens and Griffiths. Let X be a smooth projective variety of dimension 3 such that $H^0(X, \Omega_X^3) = 0$. Then the Hodge decomposition of $H^3(X, \mathbf{C})$ reduces to

$$H^3(X, \mathbf{C}) = H^{2,1} \oplus H^{1,2},$$

and, by Weil [11], the complex torus

$$J(X) = H^3(X, \mathbf{Z}) \backslash H^3(X, \mathbf{C}) / H^{2,1}$$

is an abelian variety: the *intermediate Jacobian* of X . Let

$$\Phi : H^3(X, \mathbf{Z}) \otimes H^3(X, \mathbf{Z}) \rightarrow H^6(X, \mathbf{Z}) \simeq \mathbf{Z}$$

be the cup product. By Poincaré duality, the induced alternating form on $H^3(X, \mathbf{Z})/\text{torsion}$ has discriminant one. If $H^1(X, \mathbf{Z}) = 0$, it follows from Weil [11] that this defines a principal polarization on $J(X)$.

Proposition 2.3 (. 4] *If X is rational, the principally polarized abelian variety $J(X)$ is a product of Jacobians.*

In the sketch of proof below, the sign $\simeq$ denotes an isomorphism of principally polarized abelian varieties.

a) $J(\mathbf{P}^3(\mathbf{C})) = 0$.

b) If Y'' is obtained from a smooth projective variety Y' by blowing up a smooth curve Z with Jacobian $J(Z)$ (resp. a point), one has [4, 3.11]

$$J(Y'') \simeq J(Y') \times J(Z)$$

(resp. $J(Y'') \simeq J(Y')$).

c) If a morphism $b: Y' \rightarrow X$ is birational, then $J(X)$ is a direct factor of $J(Y')$: there exists a principally polarized abelian variety B from the main series such that

$$J(Y') \simeq J(X) \times B.$$

d) Let $(A_i)_{1 \leq i \leq n}$ be nonzero principally polarized abelian varieties. Assume that none of the A_i admits a decomposition $A_i \simeq A'_i \times A''_i$ (for example, assume A_i is a Jacobian). Then any decomposition of $A = \prod A_i$ is of the form

$$A \simeq \left(\prod_{i \in I} A_i \right) \times \left(\prod_{i \notin I} A_i \right) \quad \text{for } I \subset [1, n]$$

([4, 3.23]).

If X is rational, there exists a diagram (2) with $Y = \mathbf{P}^3(\mathbf{C})$ and b birational. By a) and b), $J(Y'_{n+1})$ is a product of Jacobians. By c) and d), so is $J(X)$.

Now take for X a smooth cubic hypersurface in $\mathbf{P}^4(\mathbf{C})$. One has

$$H^1(X, \mathbf{Z}) = H^3(X, \mathbf{Z})_{\text{tors}} = H^0(X, \Omega_X^1) = 0,$$

$$\dim J(X) = \dim H^1(X, \Omega_X^2) = 5,$$

and X is unirational by [7]. Let S be the surface parametrizing the lines contained in X , and let $\alpha: S \rightarrow \text{Alb}(S)$ be the Albanese map. One of the essential results of [4] is the following.

Theorem 2.4 (. 4] *There exists an isomorphism $J(X) \xrightarrow{\sim} \text{Alb}(S)$ carrying the divisor Θ of $J(X)$ to the image of $S \times S$ under the map $(x, y) \mapsto \alpha(x) - \alpha(y)$.*

Clemens and Griffiths then use results from [1] to prove that $(\text{Alb}(S), \theta(S \times S))$ cannot be the polarized Jacobian of a curve. Another proof of the irrationality of X is suggested in the appendix to [4]; it amounts to proving the following.

Theorem 2.5. *The singular locus of the divisor Θ of $J(X)$ has dimension 0, hence codimension > 4 . In fact it is reduced to the image under θ of the diagonal of $S \times S$.*

This is known never to happen for the theta divisor of a Jacobian.

C) Manin and Iskovskih. The group of birational automorphisms of a variety X (equivalently, the group of $\mathbf{C}$ -automorphisms of $k(X)$) is a birational invariant of X . It is this invariant that Manin and Iskovskih use to show that a smooth quartic hypersurface in $\mathbf{P}^4(\mathbf{C})$ is never rational.

Theorem 2.6 ([6]). *Let X and Y be smooth quartic hypersurfaces in $\mathbf{P}^4(\mathbf{C})$. Every birational isomorphism $a: X \dashrightarrow Y$ is automatically biregular.*

Since some quartic hypersurfaces are unirational [7], this theorem provides a new example of unirational but nonrational varieties.

3. THE ARTIN–MUMFORD EXAMPLE

Consider in $\mathbf{P}^2(\mathbf{C})$ a configuration consisting of

- a) two smooth cubic curves C_1 and C_2 , with equations $f_1 = 0$ and $f_2 = 0$, meeting transversally;
- b) a smooth conic Q , with equation $q = 0$, meeting each C_i at three distinct tangency points.

We shall construct

- a) a rank-3 vector bundle V on $\mathbf{P}^2(\mathbf{C})$;
- b) a quadratic form $\Phi: V \rightarrow \mathcal{O}$ ($\mathcal{O}$ an invertible sheaf) with discriminant $f_1 f_2$ (if Δ , a section of $\mathcal{L}^{\otimes 3} \otimes (\Lambda^3 V)^{\otimes (-2)}$, denotes the discriminant, then the subscheme of $\mathbf{P}^2(\mathbf{C})$ cut out by $\Delta = 0$ is $C_1 \cup C_2$).

Let $X \subset \mathbf{P}(V^*)$ be the conic bundle over $\mathbf{P}^2(\mathbf{C})$, degenerating along $C_1 \cup C_2$, defined by the homogeneous equation $\Phi = 0$. The following conditions will hold.

- c) Φ has rank ≥ 2 at every point: for $s \in C_1 \cup C_2$, the fiber $X_s = p^{-1}(s)$ is the union of two concurrent lines; the only singular points of X are the singular points of the fibers X_s for $s \in C_1 \cap C_2$.
- d) Let S be the double cover of $\mathbf{P}^2(\mathbf{C})$ branched along Q , and write $X_S = X \times_{\mathbf{P}^2(\mathbf{C})} S$. Then X_S/S admits a section.
- e) For $s \in (C_1 \cup C_2) - Q$, the two points $\varphi(t)$ with $\varphi(t) = s$ lie on the two irreducible components of X_s .

Locally on $C_1 \cup C_2$ (for the usual topology), the restriction of the bundle X to $C_1 \cup C_2$ is obtained by gluing two $\mathbf{P}^1$ -bundles along a section. However, when one goes around a loop in $C_1 \cup C_2$, these two bundles may be interchanged: there is an unramified double covering $(C_1 \cup C_2)^*$ of $C_1 \cup C_2$ whose two local sheets correspond locally to the two $\mathbf{P}^1$ -bundles whose union is $X|_{C_1 \cup C_2}$. Let $\pi_i: C_i^* \rightarrow C_i$ be the double covering of C_i induced by $(C_1 \cup C_2)^*$.

Lemma 3.1. *C_i^* is irreducible.*

By e), C_i^* is the normalization of the double cover of C_i induced by S . The lemma follows from the fact that the section q of $\mathcal{O}(2)|_{C_i}$ is not the square of a section of $\mathcal{O}(1)|_{C_i}$; indeed, if q were a square, the three intersection points of Q with C_i would be collinear.

The double covering $C_i^* \rightarrow C_i$ corresponds to an order-2 character of the fundamental group of C_i and to a local system A_i , locally isomorphic to $\mathbf{Z}$, defined by the exact sequence of sheaves on C_i

$$(3) \quad 0 \longrightarrow A_i \longrightarrow \pi_{i*} \mathbf{Z} \xrightarrow{\text{Tr}} \mathbf{Z} \longrightarrow 0,$$

where Tr is “the sum of the values on the two sheets of the covering.”

One has

$$(4) \quad H^0(C_i, A_i) = 0, \quad H^1(C_i, A_i) \simeq \mathbf{Z}/2, \quad H^2(C_i, A_i) = \mathbf{Z}/2.$$

Lemma 3.2. *X is unirational.*

The surface S is a quadric in $\mathbf{P}^3(\mathbf{C})$, hence is rational. Since X_S/S admits a section, there is a birational isomorphism $X_S \dashrightarrow S \times \mathbf{P}^1(\mathbf{C})$, and X is an image of the rational variety X_S .

The nine singular points of X are ordinary quadratic points. One resolves each of them as follows (cf. [3]).

a) Blow it up. This resolves X and replaces the singular point by a quadric P , isomorphic to $\mathbf{P}^1(\mathbf{C}) \times \mathbf{P}^1(\mathbf{C})$.

b) Choose a projection $P \rightarrow \mathbf{P}^1(\mathbf{C})$, and contract inside the blow-up $\tilde{X}$ of X the fibers of this projection. This is possible because $\tilde{X}$ is smooth, its fibers are isomorphic to $\mathbf{P}^1(\mathbf{C})$, and the normal bundle to P in $\tilde{X}$ induces on each fiber a line bundle $\mathcal{O}(-1)$ [8].

Let $\bar{X}$ be the resulting variety. There is a morphism from $\bar{X}$ to X , and $\bar{X}$ is obtained from X by replacing each singular point by a projective line.

The fibers of the projection $p: \bar{X} \rightarrow \mathbf{P}^2(\mathbf{C})$ are of the following three kinds:

$s \notin C_1 \cup C_2$: a conic;

$s \in C_1 \cup C_2, s \notin C_1 \cap C_2$: two concurrent lines;

$s \in C_1 \cap C_2$: three lines arranged as indicated.

When one specializes from the generic point ξ of $\mathbf{P}^2(\mathbf{C})$ to the generic point η_i of C_i ($i = 1, 2$), and then to a point $s \in C_1 \cap C_2$, the irreducible components specialize as indicated in the following diagram:

$$(5) \quad (3.4) \quad \begin{array}{ccccc} & \eta_1 & & \xi & & \eta_2 \\ & \diagdown & & \diagup & & \diagdown \\ & & \times & & \times & \\ & \diagup & & \diagdown & & \diagup \\ & \eta_2 & & \eta_1 & & \eta_2 \end{array}$$

(the crosses represent irreducible components; all multiplicities are one). The rather tedious verification of this local result will be omitted. One deduces that $R^0 p_* \mathbf{Z} = \mathbf{Z}$, $R^1 p_* \mathbf{Z} = 0$, and the following exact sequence of sheaves.

Lemma 3.3. *One has*

$$0 \longrightarrow A_1 \oplus A_2 \longrightarrow R^2 p_* \mathbf{Z} \xrightarrow{\text{Tr}} \mathbf{Z} \longrightarrow 0.$$

One can now use the Leray spectral sequence of p to compute $H^*(\bar{X}, \mathbf{Z})$. The essential fact is the following, which implies the irrationality of X .

Lemma 3.4. *One has $H^3(\bar{X}, \mathbf{Z}) = \mathbf{Z}/(2)$.*

Let us compute the E_2 -terms of the Leray spectral sequence of p . Since $R^0 p_* \mathbf{Z} = \mathbf{Z}$, we have $E_2^{p,0} = \mathbf{Z}, 0, \mathbf{Z}, 0, \mathbf{Z}$ for $p = 0$ to 4; and since $R^1 p_* \mathbf{Z} = 0$, we have $E_2^{p,1} = 0$.

To compute $E_2^{p,2} = H^p(R^2p_*\mathbf{Z})$, one uses the long exact sequence attached to the short exact sequence of sheaves in Lemma 3.3. One finds

$$0 \longrightarrow E_2^{0,2} \xrightarrow{\alpha} \mathbf{Z} \longrightarrow E_2^{1,2} \longrightarrow 0,$$

$$0 \longrightarrow \mathbf{Z}/2 \oplus \mathbf{Z}/2 \longrightarrow E_2^{2,2} \longrightarrow 0,$$

$$E_2^{3,2} = 0 \quad \text{and} \quad E_2^{4,2} = \mathbf{Z}.$$

One checks easily that $\alpha(E_2^{0,2}) = 2\mathbf{Z}$ (this is not in fact necessary for the proof that $H^3(\bar{X}, \mathbf{Z})_{\text{tors}} \neq 0$). Thus the $E_2^{p,q}$ are as follows:

q						
2	$\mathbf{Z}$	$\mathbf{Z}/2$	$E_2^{2,2} \supset 0$	0	0	$\mathbf{Z}$
1	0	0	0	0	0	0
0	$\mathbf{Z}$	0	$\mathbf{Z}$	0	0	$\mathbf{Z}$
	0	1	2	3	4	p

The differentials d_r can only vanish, so $E_2^{p,q} = E_\infty^{p,q}$, and the lemma follows.

One sees that it was essential to have two curves $C_i \subset \mathbf{P}^2(\mathbf{C})$ along which X degenerates, because one kills a $\mathbf{Z}/2$ in passing from

$$H^1(\text{Ker}(R^2p_*\mathbf{Z} \rightarrow \mathbf{Z})) = H^1(R^2p_*\mathbf{Z}) = E_2^{1,2}$$

to $H^1(R^2p_*\mathbf{Z}) = E_2^{1,2}$.

It remains to carry out the promised constructions. Artin and Mumford are able to prove *a priori* the existence of a conic bundle of the required type. I shall content myself with giving equations that define one.

1) The image of f_1f_2 in $H^0(Q, \mathcal{O}(6))$ is the square of an element $g \in H^0(Q, \mathcal{O}(3))$, because Q is rational and the zeros of f_1f_2 on Q are double. This g lifts to an element still denoted $g \in H^0(\mathbf{P}^2(\mathbf{C}), \mathcal{O}(3))$ because $H^1(\mathbf{P}^2(\mathbf{C}), \mathcal{O}(1)) = 0$. Since $f_1f_2 - g^2$ vanishes on Q , it is a multiple of q :

$$f_1f_2 = g^2 + qd.$$

2) Take $V = \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O}$, and the quadratic form

$$\Phi: V \rightarrow \mathcal{O}: (x, y, z) \longmapsto qx^2 + 2gxy - dy^2 - z^2.$$

Then $\Delta = g^2 + qd = f_1f_2$. At a point where Φ had rank ≤ 1 , one would have $\Delta = q = d = 0$, hence $g = 0$, and $\Delta = g^2 + qd$ would have a double zero. This is impossible because $C_1 \cap C_2 \cap Q = \emptyset$. Finally, S is identified with the subscheme of X cut out by the equation $y = 0$, and d) and e) follow.

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Paper 20 - Working English translation

Working full-body English translation generated from the cleaned source PDF. Mathematical formulae and source line structure are preserved where possible. This is a source-aligned working translation, not yet a polished final LaTeX edition.

THE WEIL CONJECTURE I

PIERRE DELIGNE

Table of contents

[Polished translation of the opening paragraphs]

In this paper Deligne proves the Weil conjecture on the eigenvalues of Frobenius endomorphisms. A precise statement is given in (1.6). The argument is presented in as geometric and elementary a form as possible, with many reminders of background material. The genuinely new ingredients are the results of sections 3, 6, 7, and 8.

A sequel develops refinements of the intermediate results and applications, among them the hard Lefschetz theorem for repeated cup-product with the cohomology class of a hyperplane section.

The text follows closely six lectures delivered in Cambridge in July 1973. Deligne thanks Nicholas Katz for permission to use his notes.

[Source-aligned literal body continues at Section 1]

1. Grothendieck's theory: cohomological interpretation of L-functions

(1.1) Let X a schema of type fini on Z , $|X|$ l'ensemble of points fermes of X and, for $x \in |X|$, let $N(x)$ the nombre d'elements of the corps residuel $k(x)$ of X en x . La zeta function of Hasse-Weil of X is

$$(1.1) \quad z_X(s) = \prod_{x \in |X|} (1 - N(x)s^{-1})^{-1}$$

(this product converge absolument for $R(s)$ assez grand). For $X = \text{Spec}(Z)$, $z_X(s)$ is the zeta function of Riemann.

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Nous considererons exclusivement the case or X is a schema on a finite field F_q .

For $x \in |X|$, we ecrivons q_x plutot que $N(x)$. Posant $\deg(x) = [k(x) : F_q]$, $q_x = q^{\deg(x)}$. Il y a interet a introduire the variable $t = q^{-s}$. Posons

$$(1.2) \quad Z(X_0; t) = \prod_{x \in |X_0|} (1 - t^{\deg(x)})^{-1};$$

this product infini converge for $|t|$ assez petit, and one a

$$(1.3) \quad z_X(s) = Z(X_0; q^{-s}).$$

(1.2) Dwork (On the rationality of the zeta function of an algebraic variety, Amer. J. Math., 82, 1960, p. 631-648) and Grothendieck ([1] and SGA 5) ont demontre que $Z(X, t)$ is a fonction rationnelle of t .

For Grothendieck, c'est the a corollary of results generaux en cohomologie ℓ -adique (ℓ designe a nombre premier different of the caracteristique p of F_q). Ceux-ci fournissent a interpretation cohomologique of zeros and of poles of $Z(X; t)$, and a functional equation lorsque X is proper and smooth. Les methodes of Dwork are p -adiques. For X a nonsingular hypersurface in l'espace projectif, elles lui fournissaient egalement a interpretation cohomological of zeros and of poles, and l'functional equation. Elles ont inspire the theorie cristalline of Grothendieck and Berthelot, qui for X proper and smooth fournit a interpretation cohomologique p -adique of zeros and poles, and l'equation fonctionnelle. Se basant on of idees of Washnitzer, Lubkin a cree a variante of cette theorie, valable seulement for X proper, smooth and releuable en caracteristique zero (A p -adic proof of Weil's conjectures, Ann. of Math., 87, 1968, p. 105-255).

Nous ferons a usage essentiel of results of Grothendieck, and the let us recall ci-dessous.

(1.3) Let X a algebraic variety on a corps algebraically closed k of caracteristique p , i.e. a schema separe of type fini on k . On n'exclut pas the case $p = 0$. For all nombre premier $\ell \neq p$, Grothendieck a defini of cohomology groups ℓ -adiques $H^i(X, Q_\ell)$. Il a aussi defini of groups of cohomology with proper support $H^i_c(X, Q_\ell)$. For X proper, the deux coincident. Les $H^i(X, Q_\ell)$ are of spaces vectoriels of dimension finie on Q_ℓ , zero for $i > 2 \dim(X)$.

(1.4) Let X_0 a algebraic variety on F_q , F_q a cloture algebrique of F_q and X the algebraic variety on F_q deduite of X_0 par extension of scalaires of F_q a F_q . Dans the langage of Weil or Shimura, one

exprimerait cette situation par : "Let X a algebraic variety definie on F_q ". Let $F : X \rightarrow X$ the Frobenius morphism ; il envoie the point of coordonnees x vers the point of coordonnees xq ; en d'autres termes, for $U \subset X$ a ouvert of Zariski of X_0 , definissant a ouvert U of X , one a $F^{-1}(U) = U$; for $x \in H^0(U, \mathcal{O})$, one a $F * x = xq$. Identifions l'ensemble

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$|X|$ of closed points of X a $X_0(F_q)$ (l'ensemble $\text{Hom}_{F_q}(\text{Spec}(F_q), X_0)$ of points of X_0 a coefficients in F_q), and let $ph \in \text{Gal}(F_q/F_q)$ the Frobenius substitution : $ph(x) = xq$. L'action of F on $|X|$ s'identifie a l'action of ph on $X_0(F_q)$. Des lors :

a) L'ensemble X^F of closed points of X fixes sous F s'identifie a l'ensemble $X_0(F_q) \subset X_0(F_q)$ of points of X definis on F_q . Ceci exprime simplement que, for $x \in F_q$, one a $xq = x$ if and seulement if $x \in F_q$.

b) De meme, l'ensemble X^F of closed points of X fixes sous the nieme itere of F s'identifie a $X_0(F_{q^n})$.

c) L'ensemble $|X_0|$ of closed points of X_0 s'identifie a l'ensemble $|X|/F$ of orbites of F (or of ph) in $|X|$. Le degre $\deg(x)$ of $x \in |X_0|$ is the nombre d'elements of l'orbite correspondante.

d) De b) and c) resulte the formule

$$(1.4) \quad \#X^F = \#X_0(F_{q^n}) = \sum_{x \in |X_0|} \frac{\deg(x)}{\deg(x)|n}$$

(for $x \in |X_0|$ and $\deg(x) \mid n$, x defines $\deg(x)$ points a coordonnees in F_{q^n} , all conjugues on F_q).

(1.5) Le morphism F is fini, en particulier proper. Il induit therefore of maps $F^* : H^i(X, \mathbb{Q}) \rightarrow H^i(X, \mathbb{Q})$. Grothendieck a proves the formule of Lefschetz

$$\#X^F = \sum_i (-1)^i \text{Tr}(F^*, H^i(X, \mathbb{Q}))$$

the membre of droite, a priori a nombre $\mathbb{Q}$ -adique, is entier, and egal au membre of gauche. On notera qu'une telle formule n'est raisonnable que parce que $dF = 0$, meme a l'infini (X n'est pas suppose proper) ; the relation $dF = 0$ implique que the points fixes of F are of multiplicity a.

Une formule analogue vaut for the iterates of F :

$$(1.5) \quad \#X^F = \#X_0(F_{q^n}) = \sum_i (-1)^i \text{Tr}(F^{*n}, H^i(X, \mathbb{Q}))$$

Prenons the derivee logarithmique of (??) :

$$(1.6) \quad \frac{d}{dt} \log Z(X_0, t) = \sum_{x \in |X_0|} \frac{-t \deg(x)}{1 - t \deg(x)} = \sum_{x \in |X_0|} \sum_{n \geq 0} \frac{\deg(x)^n}{\deg(x)^n} t^n \deg(x) = \sum_{n \geq 0} \#X_0(F_{q^n}) t^n$$

For F a endomorphism d'un space vectoriel V , one a a identite of series formelles

$$(1.7) \quad \frac{d}{dt} \log(\det(1 - F t, V) - 1) = \sum_{n \geq 0} \text{Tr}(F^n, V) t^n$$

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(the verifier for $\dim(V) = 1$, and observer que the deux membres are additifs en V in a sequence exacte courte). Substituant (??) in (??) and appliquant (??), one trouve

$$(1.8) \quad \frac{d}{dt} \log Z(X_0, t) = \sum_i (-1)^i \frac{d}{dt} \log \det(1 - F * t, H^i(X, \mathbb{Q})) - 1$$

let

$$Z(X_0, t) = \prod_{i=0}^{\dim X} \det(1 - F * t, H^i(X, \mathbb{Q}))^{(-1)^{i+1}}$$

Le membre of droite is a element of $\mathbb{Q}[[t]]$. La formule affirme que son developpement of Taylor en $t = 0$, a priori a serie formelle in $\mathbb{Q}[[t]]$ of terme constant a, is in $\mathbb{Z}[[t]]$, and is egal au membre of gauche, lui aussi considers as a serie formelle en t. Cette formule is l'interpretation cohomological of Grothendieck of the fonction Z.

Notre result principal is the suivant.

Theoreme 1.1 (1.6). Let X_0 a variety projective nonsingular (=smooth) on $\mathbb{F}_q$. For chaque i, the polynome caracteristique $\det(1 - F^* t, H^i(X, \mathbb{Q}))$ is a coefficients entiers independants of q ($q \neq p$). Les roots complexes a of ce polynome (the conjugues complexes of eigenvalues of F^*) are of value absolue $|a| = q^{i/2}$.

Montrons already que ?? resulte of the result apparemment more faible suivant.

Lemme 1.2 (1.7). For chaque i, and chaque $q \neq p$, the eigenvalues of l'endomorphism F^* of $H^i(X, \mathbb{Q})$ are of nombres algebriques dont all the conjugues complexes a satisfy $|a| = q^{i/2}$.

Preuve of (??) $\Rightarrow$ (??). -- Regardons $Z(X_0, t)$ as a P serie for-

melle of terme constant 1, element of $\mathbb{Z}[[t]]$: $Z(X_0, t) = \sum_{n=0}^{\infty} a_n t^n$.

After (??), l'image of $Z(X_0, t)$ in $\mathbb{Q}[[t]]$ is the developpement of Taylor d'une fraction rationnelle. Ceci signifie que for N and M assez grands ($\geq$ the degres of numerateurs and denominateurs) the determinants of Hankel

$$H_k = \det((a_{i+j+k})_{0 \leq i, j \leq M}) \quad (k > N)$$

are zero. Cette nullite is vraie in $\mathbb{Q}$ if and seulement if elle l'est in $\mathbb{Q}$; $Z(X_0, t)$ is therefore the developpement in serie of Taylor d'un element of $\mathbb{Q}(t)$. En d'autres termes,

$$Z(X_0, t) \in \mathbb{Q}(t) \cap \mathbb{Z}[[t]].$$

Ecrivons $Z(X_0, t) = P/Q$, with $P, Q \in \mathbb{Z}[t]$, premiers entre eux, and of terme constant positif. After a lemma of Fatou, que $Z(X_0, t)$ let in $\mathbb{Z}[[t]]$ and of terme constant a implique que the termes constants of P and Q are 1. Posons

$$P_i(t) = \det(1 - F^* t, H^i(X, \mathbb{Q})).$$

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After l'hypothese (??), the P_i are premiers entre eux. Le membre of droite of (??) is therefore in the form irreductible, and

$$P(t) = \prod_{i \text{ impair}} P_i(t), \quad Q(t) = \prod_{i \text{ pair}} P_i(t).$$

Let K the sous-corps d'une cloture algebrique $\mathbb{Q}$ of $\mathbb{Q}$ engendre on $\mathbb{Q}$ par the roots of $R(t) = P(t)Q(t)$. Les roots of $P_i(t)$ are celles of roots of $R(t)$ ayant the property que all leurs conjugues complexes are of value absolue $q^{-i/2}$. Cet ensemble is stable sous $\text{Gal}(K/\mathbb{Q})$. Le polynome $P_i(t)$ is therefore a coefficients rationnels. After the lemma of Gauss (or parce que the roots of P_i , being roots of $R(t)$, are of inverses d'entiers algebriques), il is meme a coefficients entiers. La description ci-dessus of roots of $P_i(t)$ is independante of q ; the polynome $P_i(t)$ lui-meme is therefore independant of q .

La sequence of cet article is consacree a the proof of (??).

(1.8) The theory of Grothendieck fournit a interpretation cohomologique non seulement of fonctions zeta, mais again of L-functions. Les results are the suivants.

(1.9) Let X a algebraic variety on a corps k . For the definition d'un $\mathbb{Q}$ -sheaf constructible on X , je renvoie a SGA 5 VI. Qu'il suffice of dire que :

a) If F is a $\mathbb{Q}$ -sheaf constructible on X , il existe a partition finie of X en parties localement fermees X_i telles que $F|_{X_i}$ let constant tordu.

b) Supposons X connexe and let x a point geometric of X . For F constant tordu, $\pi_1(X, x)$ agit on the fibre F_x ; the functor fibre en x is a equivalence of categorie

($\mathbb{Q}$ -sheaves constructibles constants tordus on X) - (representations continues of $\pi_1(X, x)$

Une telle representation ne se factorise pas, en general, a travers a quotient fini of $\pi_1(X, x)$.

c) If $k = \mathbb{C}$, the $\mathbb{Q}$ -sheaves constructibles on X s'identifient aux sheaves of $\mathbb{Q}$ -spaces vectoriels F on X an, tels qu'il existe a partition finie of X en parties Zariski-localement fermees X_i , and for chaque i a systeme local of $\mathbb{Z}$ -Modules libres of type fini L_i on X_i , with $F|_{X_i} = L_i \otimes_{\mathbb{Z}} \mathbb{Q}$.

Nous ne considererons que of $\mathbb{Q}$ -sheaves constructibles, and the appellerons simplement $\mathbb{Q}$ -sheaves.

(1.10) Supposons k algebraically closed, and let F a $\mathbb{Q}$ -sheaf on X . Grothendieck a defini of cohomology groups $H^i(X, F)$ and $H_{ci}(X, F)$. Les $H_{ci}(X, F)$ are of spaces vectoriels of dimension finie on $\mathbb{Q}$, zero for $i > 2 \dim(X)$.

For $k = \mathbb{C}$, the $H^i(X, F)$ and $H_{ci}(X, F)$ are the groups of cohomologie usuels (resp. support proper) of X an, a coefficients in F .

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(1.11) Let X_0 a algebraic variety on F_q , X the variety on F_q correspondante, and F_0 a sheaf d'ensembles on X_0 (for the topologie etale). On note F son image reciproque on X . Outre the morphism of Frobenius $F : X \rightarrow X$, one dispose then d'un isomorphism canonical

$\tilde{F} : F * F \rightarrow F$. En voici a description. On regarde F_0 as a space etale on X_0 , i.e. one identifie F_0 a l'espace algebrique $[F_0]$, muni d'un morphism etale $f : [F_0] \rightarrow X_0$, tel que F_0 let the sheaf of sections locales of $[F_0]$. L'espace analogue $[F]$, etale on X , se deduit of $[F_0]$ par extension of scalaires. On dispose therefore d'un diagramme commutatif

$$\begin{array}{ccc} [F] & & / [F] \\ f \downarrow & & \downarrow f \\ X & \square & \square \\ & F & /X \end{array}$$

d'ou a morphism $[F] \rightarrow X \times F$ $[F] = [F * F]$, qui is a isomorphism

car f is etale. Son inverse defines l'isomorphisme $F * F \rightarrow F$ cherche.

Cette construction se generalise aux $\mathbb{Q}$ -sheaves.

(1.12) Let X_0 a algebraic variety on F_q , F_0 a $\mathbb{Q}$ -sheaf on X_0 , (X, F) deduit of (X_0, F_0) par extension of scalaires of F_q a F_q ,

$F : X \rightarrow X$ and $\tilde{F} : F * F \rightarrow F$. Le morphism fini F and $\tilde{F}$ definissent a endomorphism

$$F^* : H_{ci}(X, F) \rightarrow H_{ci}(X, F).$$

For $x \in |X|$, F^* defines a morphism $Fx^* : FF(x) \rightarrow Fx$. For $x \in X \times F$, c'est a endomorphism of Fx . Grothendieck a demontre the formule of Lefschetz

$$\sum_{x \in X \times F} \text{Tr}(Fx^*, Fx) = \sum_i (-1)^i \text{Tr}(F^*, H_{ci}(X, F)).$$

Une formule analogue vaut for the iterates of F : the nieme itere of F^* defines of morphisms $Fx^{*n} : FF^n(x) \rightarrow Fx$; for x fixe sous F^n , Fx^{*n} is a endomorphism, and

$$(1.9) \quad \sum_{x \in X \times F^n} \text{Tr}(Fx^{*n}, Fx) = \sum_i (-1)^i \text{Tr}(F^{*n}, H_{ci}(X, F)).$$

(1.13) Let $x_0 \in |X_0|$, Z l'orbite correspondante of F in $|X|$ and $x \in Z$. L'orbite Z a $\deg(x_0)$ elements (1.4). On note Fx_0 l'endomorphism $* \deg(x)$

of Fx , and one pose

$$\det(1 - Fx_0 t, F_0) = \det(1 - Fx^* \deg(x) t, Fx).$$

A isomorphism pres, (Fx, Fx_0) ne depend pas of the choix of x . Ceci justifie d'omettre x of the notation. On utilisera a notation analogue for d'autres fonctions of (Fx, Fx_0) .

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(1.14) Definissons $Z(X_0, F_0, t) \in \mathbb{Q}[[t]]$ par the product

$$(1.10) \quad Z(X_0, F_0, t) = \prod_{x \in |X_0|} \det(1 - Fx t \deg(x), F_0)^{-1}.$$

For F the sheaf constant $\mathbb{Q}$, one retrouve (??). After (??), the derivee logarithmique of Z is

$$(1.11) \quad \frac{d}{dt} \log Z(X_0, F_0, t) = \sum_{n \geq 1} \frac{1}{n} \sum_{x \in X} \text{Tr}(F^n, Fx) t^{n-1} = \sum_{x \in X} \text{Tr}(F, Fx) t^0.$$

Substituant (??) in (??), one trouve par the meme calcul qu'en (1.5) the generalisation suivante of (??)

$$\prod_{i=1}^Y \det(1 - F^* t, H_{ci}(X, F))^{(-1)^i}.$$

$$(1.12) \quad Z(X_0, F_0, t) =$$

Cette formule is a identite in $\mathbb{Q}[[t]]$.

(1.15) Il is sometimes commode d'utiliser a langage galoisien plutot que geometric. Voici the dictionnaire.

If F_1 and F_2 are deux clotures algebriques of F_0 , (X_1, F_1) on F_0 defines par extension of scalaires (X_1, F_1) on F_0 and (X_2, F_2) on F_0 .
 $1 \sim 2$

Tout F_0 -isomorphism $s : F_1 \rightarrow F_2$ induit a isomorphism

$$H^i(X_1, F_1) \rightarrow H^i(X_2, F_2).$$

En particulier, for $F_0 = F_0$ (note F_0), one trouve que $\text{Gal}(F_0/F_0)$ agit on $H^i(X, F)$ (action par transport of structure). Let ph the substitution of Frobenius. On satisfies que

$$F^* = \text{ph}^{-1}$$

(in $\text{End}(H^i(X, F))$).

Ceci amene a define the Frobenius geometric $F \in \text{Gal}(F_0/F_0)$ as being ph^{-1} . On a

$$(1.13) \quad F^* = F.$$

Let x a point geometric of X_0 , localise en $x_0 \in |X_0|$. Par transport of structure, the group $\text{Gal}(k(x)/k(x_0))$ agit on the fibre $(F_0)_x$ of F_0 en x ; en particulier, the Frobenius geometric relatif $F_x \in \text{Gal}(k(x)/k(x_0))$, agit. For x defini par a point ferme, again note x , of X , one a $(F_0)_x = F_x$, and

$$(1.14) \quad F_x = F_x \cdot \deg(x) \quad (\text{in } \text{End}(F_x)).$$

En notations galoisiennes, (??) s'ecrit

$$\det(1 - F_x t \deg(x), F_0) = \prod_{x \in |X_0|} \det(1 - F t, H^i(X, F))^{i+1} (-1)^i.$$

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2. Grothendieck's theory: Poincare duality

(2.1) For expliquer the relation entre roots of l'unite and orientations, je vais au prealable redire in a langage farfelu deux case classiques.

a) Varieties differentiables. -- Let X a variety differentiable purement of dimension n . Le sheaf d'orientation Z_0 on X is the sheaf localement isomorphe au sheaf constant Z , dont the sections inversibles on a ouvert U of X correspond aux orientations of U . Une orientation of X is a isomorphism of Z_0 on the sheaf constant Z . La classe fondamentale of X is a morphism $\text{Tr} : H^{2n}(X, Z_0) \rightarrow Z$; if X is orientee, il s'identifie a a morphism $\text{Tr} : H^{2n}(X, Z) \rightarrow Z$. La Poincare duality s'exprime a l'aide of the classe fondamentale.

b) Varieties complexes. -- Let C a cloture algebrique of $\mathbb{R}$. Une variete algebrique complexe smooth, or plutot the variety differentiable sous-jacente, is always orientable. For l'orienter, il suffices d'orienter C lui-meme. Ceci revient au choix :

- a) a choisir l'une of deux roots of l'equation $X^2 = -1$; one l'appelle $+i$;
- b) a choisir a isomorphism of $\mathbb{R}/Z$ with $U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$; $+i$ is l'image of $1/4$;
- c) a choisir l'un of deux isomorphisms $x \mapsto \exp(+2\pi i x)$ of $\mathbb{Q}/Z$ on the group of roots of l'unite of C , qui se prolonge par continuite en a isomorphism of $\mathbb{R}/Z$ on $U(1)$.

On note $Z(1)$ a Z -module libre of rang a dont l'ensemble a deux elements of generateurs let en correspondance canonical with l'un of ensembles a deux elements a), b), c). Le more simple is of prendre $Z(1) = \text{Ker}(\exp : C \rightarrow C^*)$. Au generateur $y = +2\pi i$ corresponds l'isomorphisme c) : $x \mapsto \exp(xy)$.

Let $Z(r)$ the puissance tensorielle r ieme of $Z(1)$. If X is a variete algebrique complexe smooth purement of dimension complexe r , the sheaf d'orientation of X is the sheaf constant of value $Z(r)$.

(2.2) For "orienter" the algebraic varieties on k algebraically closed of caracteristique 0 , il faut choisir a isomorphism of $\mathbb{Q}/Z$ on the group of roots of l'unite of k . L'ensemble of ces isomorphisms is a space principal homogene sous Z^* (non more sous Z^*). Lorsqu'on s'interesse seulement a the cohomology ℓ -adique, il suffices of considerer the roots of l'unite d'ordre a puissance of ℓ , and of supposer the caracteristique p of k differente of ℓ . On note $Z/\ell^n(1)$ the group of roots of l'unite of k d'ordre divisant ℓ^n . For n variable, the $Z/\ell^n(1)$

formant a systeme projectif, d'applications of transition the

$$Z^{n+1, n} : Z/\ell^{n+1}(1) \rightarrow Z/\ell^n(1) : x \mapsto x\ell = x\ell^n.$$

On pose $Z_\ell(1) = \lim_{\leftarrow} Z/\ell^n(1)$ and $Q_\ell(1) = Z_\ell(1) \otimes_{Z_\ell} Q_\ell$.

On note $Q_\ell(r)$ the puissance tensorielle r ieme of $Q_\ell(1)$; for $r \in \mathbb{Z}$, r negatif, one pose again $Q_\ell(r) = Q_\ell(-r)$.

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En tant qu'espace vectoriel on Q_ℓ , $Q_\ell(1)$ is isomorphe a Q_ℓ . Toutefois the group of automorphismes of k agit non trivialement on $Q_\ell(1)$: il agit via the caractere $\text{kh} : \text{Gal}(k/k) \rightarrow \mathbb{Z}^*$ a values in $\mathbb{Z}^*$ qui gives son action on the roots of l'unite. En particulier, if $k = \mathbb{F}_q$, the Frobenius substitution $\text{ph} : x \mapsto x^q$ agit par multiplication par q .

Let X a algebraic variety smooth purement of dimension n on k . Le sheaf d'orientation of X en cohomologie ℓ -adique is the Q_ℓ -sheaf constant $Q_\ell(n)$. La classe fondamentale is a morphism

$$\text{Tr} : \text{Hc}2n(X, Q_\ell(n)) \rightarrow Q_\ell,$$

let again

$$\text{Tr} : \text{Hc}2n(X, Q_\ell) \rightarrow Q_\ell(-n).$$

Theoreme 2.1 (2.3). (Poincare duality). For X proper and smooth purement of dimension n , the form bilineaire

$$\text{Tr}(x \cup y) : \text{H}^i(X, Q_\ell) \otimes \text{H}^{2n-i}(X, Q_\ell(n)) \rightarrow Q_\ell(-n)$$

is a dualite parfaite (elle identifie $\text{H}^i(X, Q_\ell)$ au dual of $\text{H}^{2n-i}(X, Q_\ell(n))$).

(2.4) Let X_0 a algebraic variety proper and smooth on $\mathbb{F}_q$, purement of dimension n , and X on $\mathbb{F}_q$ deduit of X_0 par extension of scalaires. Le morphism (??) is compatible a l'action of $\text{Gal}(\mathbb{F}_q/\mathbb{F}_q)$. If the a_j are the eigenvalues of the Frobenius geometric F agissant on $\text{H}^i(X, Q_\ell)$, the eigenvalues of F agissant on $\text{H}^{2n-i}(X, Q_\ell)$ are therefore the $(q^{-n} a_{j-1})$.

(2.5) Supposons for simplifier X connexe. La proof of (??) se transpose as suit in a langage geometric, plutot que galoisien (cf. (1.15)).

- Le cup-produit met $\text{H}^i(X, Q_\ell)$ and $\text{H}^{2n-i}(X, Q_\ell)$ en dualite parfaite a values in $\text{H}^{2n}(X, Q_\ell)$, qui is of dimension a .
- Le cup-produit commute a l'image reciproque F^* par the morphisme of Frobenius $F : X \rightarrow X$.
- Le morphism F is fini and of degre q^n : on $\text{H}^{2n}(X, Q_\ell)$, F^* is the multiplication par q^n .
- Les eigenvalues of F^* ont therefore the property (2.4).

(2.6) On pose $\text{kh}(X) = \sum_i (-1)^i \dim \text{H}^i(X, Q_\ell)$. If n is impair, the form $\text{Tr}(x \cup y)$ on $\text{H}^n(X, Q_\ell)$ is alternee; l'entier $n\text{kh}(X)$ is therefore always pair. On deduit facilement of (??) and of (??), (2.4) que

$$Z(X_0, t) = e \cdot q^{n\text{kh}(X)/2} \cdot \text{tkh}(X) \cdot Z(X_0, q^{-n}t-1)$$

or $e = \pm 1$. If n is impair, notons N the multiplicite of the value proper $q^{n/2}$ of F^* agissant on $\text{H}^n(X, Q_\ell)$ (i.e. the dimension of the sous-espace proper generalise correspondant). On a

$$e = \begin{cases} 1 & \text{if } n \text{ is impair,} \\ (-1)^N & \text{if } n \text{ is pair.} \end{cases}$$

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C'est the the formulation of Grothendieck of l'functional equation of fonctions Z .

(2.7) Nous aurons besoin d'autres forms of the theorem of dualite. Le case of courbes we suffirait in celles-ci. If F is a Q_ℓ -sheaf on a algebraic variety X on k algebraically closed, we noterons $F(r)$ the sheaf $F \otimes_{Q_\ell} Q_\ell(r)$. Ce sheaf is (non canonically) isomorphe a F .

Theoreme 2.2 (2.8). Supposons X smooth purement of dimension n and F constant tordu. Let F^* the dual of F . La form bilineaire

$$\text{Tr}(x \cup y) : \text{H}^i(X, F) \otimes \text{Hc}2n-i(X, F^*(n)) \rightarrow \text{Hc}2n(X, Q_\ell(n)) \rightarrow Q_\ell$$

Tr

is a dualite parfaite.

(2.9) Supposons X connexe and let x a point ferme of X . Le functor $F \mapsto F_x$ is a equivalence of the categorie of Q_ℓ -sheaves constants tordus with celle of representations ℓ -adiques of $\pi_1(X, x)$. Via cette equivalence, $\text{H}^0(X, F)$ s'identifie aux invariants of $\pi_1(X, x)$ agissant in F_x :

$$(2.1) \quad \text{H}^0(X, F) = F_x^{\pi_1(X, x)}.$$

After (??), for X smooth and connexe of dimension n , one a therefore

$$\text{Hc}2n(X, F) = \text{H}^0(X, F^*(n)) = ((F^*(n))^{\pi_1(X, x)}).$$

La dualite echange invariants (more grand sous-espace invariant) and coinvariants (more grand quotient invariant). Cette formule se recrit therefore

$$Hc2n(X, F) = (F_x)_{n1}(X, x) (-n).$$

Nous ne l'utiliserons que for $n = 1$.

Scholie 2.3 (2.10). Let X a courbe smooth and connexe on k algebriquement clos, x a point ferme of X and F a $\mathbb{Q}$ -sheaf constant tordu. On a :

(i) $Hc0(X, F) = 0$ if X is affine.

(ii) $Hc2(X, F) = (F_x)_{n1}(X, x) (-1)$.

L'assertion (i) signifie simplement que F n'a pas of section a support fini.

(2.11) Let X a courbe projective, smooth and connexe on k algebriquement clos, U a ouvert of X , complement d'un ensemble fini S of closed points of X , j l'inclusion $U \rightarrow X$ and F a $\mathbb{Q}$ -sheaf constant tordu on U . Let $j_! F$ the $\mathbb{Q}$ -sheaf constructible image directe of F . Sa fibre en $x \in S$ is of rang inferieur or egal au rang of sa fibre en a point general ; c'est a space d'invariants sous a group of monodromie locale.

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Theoreme 2.4 (2.12). La form bilineaire

$$\text{Tr}(xuy) : H^i(X, j_! F) \otimes H^{2-i}(X, j_* F(1)) \rightarrow H^2(X, j_! F \otimes j_* F(1)) \rightarrow H^2(X, \mathbb{Q}(1)) \xrightarrow{\text{Tr}}$$

is a dualite parfaite.

(2.13) Il we will be commode of disposer of $\mathbb{Q}$ -sheaves $\mathbb{Q}(r)$ on a quelconque schema X or ` is inversible. Le all is of define the $Z/n(1)$. Par definition, $Z/n(1)$ is the sheaf etale of roots (n)-iemes of l'unite.

(2.14) Indications bibliographiques on the SSSS 1 and 2.

A) Tous the results importants en cohomologie etale se demontrent d'abord for of sheaves of torsion. L'extension aux $\mathbb{Q}$ -sheaves se fait par of passages a the limite formels. Dans ce qui suit, for chaque theorem cite, je renverrai non point a a reference or il is demontre, mais a a reference or son analogue for the sheaves of torsion l'est.

B) A l'exception of the formule of Lefschetz and of (2.12), the results of cohomologie etale utilises in cet article are all demontres in SGA 4. For ceux already enonces, the references are : definition of H^i : VII ; definition of Hci : XVII 5.1 ; theorem of finitude : XIV 1, complete en XVII 5.3 ; dimension cohomologique : X ; dualite of Poincare : XVIII.

C) La relation entre the divers Frobenius ((1.4), (1.11), (1.15)) is expliquée en detail in SGA 5, XV, SSSS 1, 2.

D) L'interpretation cohomological of fonctions Z (1.14.3) is clairement exposee in [1] ; toutefois, the formule of Lefschetz (1.12), for X a courbe projective and smooth, y is utilisee, mais non demontree. For the proof, il faut, helas, renvoyer a SGA 5.

E) La form (2.12) of the theorem of Poincare duality resulte of the resultat general SGA 4, XVIII (3.2.4.1) (for $S = \text{Spec}(k)$, $X = X$, $K = j_! F$, $L = \mathbb{Q}$) par a calcul local qui n'est pas difficile. L'enonce figurera explicitement in the version definitive of SGA 5. Dans the case or we l'utiliserons (ramification of F moderee), one pourrait l'obtenir par voie transcendante, en relevant X and F en caracteristique 0.

3. The fundamental estimate

Le result of ce paragraphe a been catalyse par the lecture of Rankin [3].

(3.1) Let U_0 a courbe on F_q , complement in P^1 d'un ensemble fini of closed points, U the courbe on F_q qui s'en deduit, u a point ferme of U , F_0 a $\mathbb{Q}$ -sheaf constant tordu on U_0 and F son image reciproque on U .

Let $b \in \mathbb{Q}$. Nous dirons que F_0 is of weights b if for all $x \in |U_0|$, the eigenvalues of F_x agissant on F_0 (1.13) are of nombres algebriques dont all the conjuges complexes are of value absolue $b/2$.

qx . Par exemple, $\mathbb{Q}(r)$ is of weights $-2r$.

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Theoreme 3.1 (3.2). Faisons the hypotheses suivantes :

(i) F_0 is muni d'une form bilineaire alternee non degeneratese

$$\text{ps} : F_0 \otimes F_0 \rightarrow \mathbb{Q}(-b).$$

(ii) L'image of $n_1(U, u)$ in $GL(F_u)$ is a sous-groupe ouvert of the group symplectique $Sp(F_u, \text{psu})$.

(iii) For all $x \in |U_0|$, the polynome $\det(1 - F_x t, F_0)$ is a coefficients rationnels.

Then, F is of weights b .

On peut supposer, and we supposons, que U is affine and que $F \neq 0$.

Lemme 3.2 (3.3). Let $2k$ a entier pair and notons $F^{\otimes 2k}$ the puissance tensorielle $(2k)$ -ieme of F . For $x \in |U|$, the derivee logarithmique

$$\frac{d}{dt} \log(\det(1 - Fx \deg(x), F^{\otimes 2k})^{-1})$$

is a serie formelle a coefficients rationnels positifs.

L'hypothese (iii) assure que, for all n , $\text{Tr}(F^n, F^{\otimes 2k}) \in \mathbb{Q}$. Le nombre

$$\text{Tr}(F^n, F^{\otimes 2k}) = \text{Tr}(F^n, F^{\otimes 2k})$$

is therefore rationnel positif, and one applique (??).

Lemme 3.3 (3.4). Les facteurs locaux $\det(1 - Fx \deg(x), F^{\otimes 2k})^{-1}$ are of series formelles a coefficients rationnels positifs.

La serie formelle $\log \det(1 - Fx \deg(x), F^{\otimes 2k})^{-1}$ is without terme constant ; after (??), ses coefficients are ≥ 0 ; the coefficients of son exponentielle are therefore egalement positifs.

P

Lemme

3.4 (3.5). Let $f = \sum_{n=0}^{\infty} a_n t^n$ the product of series formelles $f_i = \sum_{n=0}^{\infty} a_{i,n} t^n$, all of terme constant a , and a coefficients reels positifs.

On suppose

que l'ordre of $f_i - 1$ tend vers l'infini with i , and one pose

$f = \prod_i f_i$. Then, the rayon of convergence absolue of f_i is au less egal a celui of f .

P

If $f = \sum_{n=0}^{\infty} a_n t^n$, one a en effet $a_{i,n} \leq a_n$.

Lemme 3.5 (3.6). Sous the hypotheses of (??), if f and the f_i are the developpements en serie of Taylor of fonctions meromorphes, then

$$\inf\{|z| \mid f(z) = \infty\} \leq \inf\{|z| \mid f_i(z) = \infty\}.$$

Ces nombres are en effet the rayons of convergence absolue.

(3.7) For chaque partition P of $[1, 2k]$ en parties a deux elements $\{i_a, j_a\}$ ($i_a < j_a$), one defines

$$\text{ps}_P : F^{\otimes 2k} \rightarrow \bigotimes_{a \in P} (-k_a) : x_1 \otimes \dots \otimes x_{2k} \mapsto \prod_{a \in P} \text{ps}(x_{i_a}, x_{j_a}).$$

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Let x a point ferme of X . L'hypothese (ii) assure que the coinvariants of $\pi_1(U, u)$ in $F^{\otimes 2k}$ are the coinvariants in $F^{\otimes 2k}$ of the group symplectique all entier

$$\text{Sp}(F_u, \text{ps}_u) \subset \text{GL}(F_u).$$

(Le group symplectique agit on $F^{\otimes 2k}$; l'ensemble of coinvariants is l'ensemble of invariants of the dual). After H. Weyl (The classical groups, Princeton University Press, chap. VI, SS 1), for $P \in \mathcal{P}$ convenable, dependant of $\dim(F_u)$, the ps_P for $P \in \mathcal{P}$ definissent a isomorphism

$$(F^{\otimes 2k})^{\text{Sp}} \xrightarrow{\sim} \bigotimes_{P \in \mathcal{P}} (-k_P)^{\otimes N_P}.$$

Let N the nombre d'elements of $\mathcal{P}$. After (??), the formule ci-dessus gives

$$H_{2k}(U, F^{\otimes 2k}) = \bigotimes_{P \in \mathcal{P}} (-k_P)^{\otimes N_P}.$$

Puisque $H_{2k}(U, F^{\otimes 2k}) = 0$, the formule (??) se reduit a

$$Z(U, F^{\otimes 2k}, t) = \det(1 - F * t, H_{2k}(U, F^{\otimes 2k}))^{-1}.$$

Cette fonction Z is therefore the developpement en serie of Taylor d'une fonction rationnelle n'ayant of pole qu'en $t = 1/q^{kb}$. Nous utiliserons seulement the fait que the poles are of value absolue $1/q^{kb}$ in $\mathbb{C}$. Cela pourrait se deduire d'arguments generaux on the groups reductifs. If a is a value proper of $F *$ on $F^{\otimes 2k}$, then a^{2k} is a value proper of $F *$ on $F^{\otimes 2k}$. Notons again a a quelconque conjugue complexe of a .

La puissance inverse $1/a^{2k \deg(x)}$ is a pole of $\det(1 - Fx \deg(x), F^{\otimes 2k})$. After (??) and (??), one a therefore

$$|1/a^{2k}| \leq 1/q^{kb}$$

let

$$|a| \geq q^{b/2}.$$

Par ailleurs, l'existence of ps assure que q^{b-a-1} is egalement value proper, d'ou l'inegalite

$$|q^{b-a-1}| \geq q^{b/2}$$

let

$$q^{b/2} \geq |a|.$$

Ceci acheve the proof.

Corollaire 3.6 (3.8). Let a a value proper of $F *$ agissant on $H_{2k}(U, F)$. Then, a is a nombre algebrique, dont all the conjugues complexes satisfy

$$q^{(b+1)/2} \leq |a| \leq q^{(b+1)/2}.$$

La formule (??) for F_0 se réduit a
 $Z(U_0, F_0, t) = \det(1 - F * t, H^1(U, F))$.
 Le premier membre is a serie formelle a coefficients rationnels, vu
 son developpement en product and l'hypothese (iii). Le second membre

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is therefore a polynome a coefficients rationnels ; $1/a$ en is root. Ceci
 proves already que a is algebrique. For achever the proof, il
 suffices of verifier que the product infini qui defines $Z(U_0, F_0, t)$ converge
 absolument (therefore is non zero) for $|t| < q^{-(b+1)/2}$.
 Let N the rang of F , and posons

$$\det(1 - Fx t, F) = \prod_{i=1}^N (1 - a_{i,x} t).$$

After (??), $|a_{i,x}| = q^x$. La convergence of the product infini Z resulte
 of celle of

$$\prod_{x,n} |a_{i,x}|^n |t|^{n \cdot \deg(x)}.$$

For $|t| = q^{-(b+1)/2 - \epsilon}$ ($\epsilon > 0$), one a

$$\prod_{i,x} |a_{i,x}|^n |t|^{n \cdot \deg(x)} = N \prod_x q^{-\deg(x)(1+\epsilon)}.$$

Sur the droite affine, il y a q^n points a values in F_q^n , therefore au more
 q^n closed points of degree n . On a therefore

$$\prod_n q^n * q^{-n(1+\epsilon)} = \prod_n q^{-n\epsilon} < \infty,$$

ce qui acheve the proof.

Corollaire 3.7 (3.9). Let j_0 l'inclusion of U_0 in $P_1 F_q$, j celle of
 U in P_1 , and a a value proper of F * agissant on $H^1(P_1, j_* F)$.
 Then a is a nombre algebrique, dont all the conjugues complexes
 satisfy

$$q^{(b+1)/2-1} \leq |a| \leq q^{(b+1)/2}.$$

Un segment of the sequence longue of cohomologie definie par the sequence
 exacte courte

$$0 \rightarrow j_! F \rightarrow j_* F \rightarrow j_* F / j_! F \rightarrow 0$$

($j_! =$ prolongement par 0) s'ecrit

$$H^0(U, F) \rightarrow H^1(P_1, j_! F) \rightarrow H^1(P_1, j_* F) \rightarrow 0.$$

La value proper a apparait therefore already in $H^1(U, F)$, and is justiciable
 de (??) :

$$q^{(b+1)/2-1} \leq |a| \leq q^{(b+1)/2}.$$

La Poincare duality (2.12) assure que $q^{b+1-a-1}$ is egalement value
 proper, d'ou l'inegalite

$$|q^{b+1-a-1}| \leq q^{(b+1)/2}$$

and the corollary.

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4. Lefschetz theory: local theory

(4.1) Sur C , the results locaux of Lefschetz are the suivants.

Let $D = \{z \mid |z| < 1\}$ the disque unite, $D^* = D - \{0\}$, and $f : X \rightarrow D$
 a morphism d'espaces analytiques. On suppose que :

- a) X is non singulier, and purement of dimension $n + 1$;
- b) f is proper ;
- c) f is smooth en dehors d'un point x of the fibre speciale $X_0 = f^{-1}(0)$;
- d) en x , f presente a point quadratique non degenerates.

Let $t \in D$ and $X_t = f^{-1}(t)$ "the" fibre generale. Aux given
 precedentes, one associe :

- a) of morphisms of specialisation $sp : H^i(X_0, Z) \rightarrow H^i(X_t, Z)$:
 X_0 is a retracte par deformation of X , and sp is the fleche com-
 $\sim$
 $\rightarrow H^i(X_0, Z) \leftarrow$
 $\sim$
 $\rightarrow H^i(X, Z) \rightarrow H^i(X_t, Z)$;
- b) of transformations of monodromie $T : H^i(X_t, Z) \rightarrow H^i(X_t, Z)$,
 qui decrivent l'effet on the cycles singuliers of X_t of "faire tourner
 t autour of 0". C'est again l'action on $H^i(X_t, Z)$, fibre en t of the
 systeme local $Ri f_* Z|D^*$, of the generateur positif of $\pi_1(D^*, t)$.

The theory of Lefschetz decrit a) and b) en terme of the cycle evanescent
 $d \in H^n(X_t, Z)$. Ce cycle is bien defini au signe pres. For $i \neq n, n + 1$,
 one a

$$H^i(X_0, Z) \xrightarrow{\sim} H^i(X_t, Z) \quad (i \neq n, n + 1).$$

For $i = n, n + 1$, one has a sequence exacte

$$0 \rightarrow H^n(X_0, Z) \rightarrow H^n(X_t, Z) \xrightarrow{x \mapsto x + d} H^n(X_t, Z) \rightarrow H^{n+1}(X_0, Z) \rightarrow H^{n+1}(X_t, Z) \rightarrow 0.$$

For $i \neq n$, the monodromie T is l'identite. For $i = n$, one a

$$T x = x + (x, d)d.$$

Les values of (x, d) , of $T d$, and of (d, d) are the suivantes :

$n \bmod 4$	0	1	2	3
(x, d)	+	-	-	+
$T d$	2	0	-2	0
(d, d)	d	d	-d	-d

La transformation of monodromie T respecte the form d'intersection $\text{Tr}(x \cup y)$ on $H^n(X_t, Z)$. For n impair, c'est a transvection symplectique. For n pair, c'est a symetrie orthogonale.

(4.2) Voici l'analogie of (4.1) en geometrie algebrique abstraite. Le disque D is remplace par the spectre d'un anneau of valuation discrete henselien A a residue field algebraically closed. Let S ce spectre, e son point generique (spectre of the corps of fractions of A), s son point ferme (spectre of the residue field). Le role of t is joue par a point generique geometric e (spectre d'une cloture algebrique of the corps of fractions of A).

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Let $f : X \rightarrow S$ a morphism proper, with X regulier purement of dimension $n + 1$. On suppose f smooth, sauf for a point quadratique ordinaire x in the fibre speciale X_s .

Let ℓ a nombre premier different of the caracteristique residuelle p of S . Notant X_e the fibre generique geometric, one dispose again d'un morphism of specialisation

$$(4.1) \quad \text{sp} : H^i(X_s, \mathbb{Q}_\ell) \xrightarrow{\sim} H^i(X, \mathbb{Q}_\ell) \rightarrow H^i(X_e, \mathbb{Q}_\ell).$$

Le role of T is joue par l'action of the group d'inertie $I = \text{Gal}(e/e)$ agissant on $H^i(X_e, \mathbb{Q}_\ell)$ par transport of structure (cf. (1.15)) :

$$(4.2) \quad I = \text{Gal}(e/e) \rightarrow \text{GL}(H^i(X_e, \mathbb{Q}_\ell)).$$

Les given (??), (??) decrivent entierement the sheaves $R_i f_* \mathbb{Q}_\ell$ on S .

(4.3) Posons $n = 2m$ for n pair, and $n = 2m + 1$ for n impair. (??)

and (??) se decrivent again en terme d'un cycle evanescent

$$(4.3) \quad d \in H^n(X_e, \mathbb{Q}_\ell)(m).$$

Ce cycle is bien defini au signe pres.

For $i \neq n, n + 1$, one a

$$(4.4) \quad H^i(X_s, \mathbb{Q}_\ell) \xrightarrow{\sim} H^i(X_e, \mathbb{Q}_\ell) \quad (i \neq n, n + 1).$$

For $i = n, n + 1$, one a a sequence exacte

$$(4.5) \quad 0 \rightarrow H^n(X_s, \mathbb{Q}_\ell) \rightarrow H^n(X_e, \mathbb{Q}_\ell) \xrightarrow{x \mapsto x + d} \mathbb{Q}_\ell(m-n) \rightarrow H^{n+1}(X_s, \mathbb{Q}_\ell) \rightarrow H^{n+1}(X_e, \mathbb{Q}_\ell) \rightarrow 0.$$

L'action (??) of I (the monodromie locale) is triviale if $i \neq n$. For $i = n$, elle se decrit as suit.

A) n impair. -- On dispose d'un homomorphisme canonical

$$t : I \rightarrow \mathbb{Z}/(1),$$

and l'action of $s \in I$ is

$$sx = x + t(s)(x, d)d.$$

B) n pair. -- Ce case ne we servira pas. Disons seulement que, if $p \neq 2$, il existe a unique caractere d'ordre deux

$$e : I \rightarrow \{\pm 1\},$$

and que l'on a

$$sx = x \text{ si } e(s) = +1, \\ sx = x + (x, d)d \text{ if } e(s) = -1.$$

Les signes $\pm$ in A) and B) are the memes qu'en (4.1).

(4.4) Ces results apportent the informations suivantes on the $R_i f_* \mathbb{Q}_\ell$.

a) Si $d \neq 0$:

$6 \leq n$, the sheaf $R_i f_* \mathbb{Q}_\ell$ is constant.

1) Pour $i =$

2) Let j l'inclusion of e in S . On a $R^n f_* \mathbb{Q}_\ell = j_* j^* R^n f_* \mathbb{Q}_\ell$.

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b) If $d = 0$: (C'est the a case exceptionnel. Puisque $(d, d) = \pm 2$ for n pair, il ne peut se produire que for n impair.)

1) For $i \neq n + 1$, the sheaf $R_i f_* \mathbb{Q}_\ell$ is constant.

2) Let $\mathbb{Q}_\ell(m-n)$ the sheaf $\mathbb{Q}_\ell(m-n)$ on $\{s\}$, prolonge par zero on S . On a a sequence exacte

$0 \rightarrow Q \rightarrow (m-n)s \rightarrow R_{n+1} \rightarrow j_* j^* R_{n+1} \rightarrow 0$,
or $j_* j^* R_{n+1} \rightarrow 0$ is a sheaf constant.

5. Lefschetz theory: global theory

(5.1) Sur C , the results of Lefschetz are the suivants. Let P a space projectif of dimension > 1 , and P^* l'espace projectif dual ; ses points parametrisent the hyperplans of P , and one note H_t l'hyperplan defini par $t \in P^*$. If A is a sous-espace lineaire of codimension 2 of P , the hyperplans contenant A are parametrises par the points d'une droite $D \subset P^*$, the duale of A . Ces hyperplans $(H_t)_{t \in D}$ forment the pinceau d'axe A .

Let $X \subset P$ a variety projective nonsingular connexe and of dimension $n+1$. Let $X \subset X \times D$ l'ensemble of couples (x, t) tels que $x \in H_t$. Les maps premiere and seconde coordonnee forment a diagramme

$$(5.1) \quad \begin{array}{ccc} X & & /X \\ f \downarrow & & \downarrow \\ D & \xrightarrow{\quad} & / \text{Spec}(C) \end{array}$$

La fibre of f en $t \in D$ is the section hyperplane $X_t = X \cap H_t$ of X .

Fixons X , and prenons A assez general. Then :

- A) A is transverse a X , and X se deduit of X par eclatement of $A \cap X$. En particulier, X is non singulier.
- B) Il existe a partie finie S of D and for chaque $s \in S$ a point $x_s \in X_s$, tels que f let smooth en dehors of x_s .
- C) Les x_s are of points critiques non degenerates of f .

For chaque $s \in S$, the theorie of Lefschetz locale (4.1) s'applique therefore a a petit disque D_s autour of s and a $f^{-1}(D_s)$.

(5.2) On pose $U = D - S$. Let $u \in U$, and choisissons of lacets disjoints $(g_s)_{s \in S}$ partant of u , with g_s tournant a fois autour of s :

Ces lacets engendrent the fundamental group $\pi_1(U, u)$. Ce group agit on $H^i(X_u, \mathbb{Z})$, fibre en u of the systeme local $R_i \otimes \mathbb{Z}[U]$. After the theorie locale (4.1), a chaque $s \in S$ corresponds a cycle evanescent $ds \in H^{n+1}(X_u, \mathbb{Z})$; ces cycles dependent of the choix of g_s . For $i \neq n$, l'action of $\pi_1(U, u)$ on $H^i(X_u, \mathbb{Z})$ is triviale. For $i = n$, one a

$$(5.2) \quad g_s x = x + (-1)^n (x, ds) ds.$$

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Let E the sous-espace of $H^n(X_u, \mathbb{Q})$ engendre par the ds (partie evanescente of the cohomology).

Proposition 5.1 (5.3). E is stable sous l'action of the group of monodromie $\pi_1(U, u)$. L'orthogonal $E^\perp$ of E (for the form d'intersection $\text{Tr}(x \cup y)$) is the sous-espace of invariants of the monodromie in $H^n(X_u, \mathbb{Q})$.

Les g_s engendrant the group of monodromie, c'est clair on (??). Theoreme 5.2 (5.4). Les cycles evanescents $\pm ds$ (pris au signe pres) are conjugues sous l'action of $\pi_1(U, u)$.

Let $X \subset P$ the variety duale of X : c'est l'ensemble of $t \in P$ tels que H_t let tangent a X , i.e. tels que X_t let singulier, or que $X \subset H_t$. La variety X is irreductible. Let $Y \subset X \times P$ l'espace of couples (x, t) tels que $x \in H_t$. On dispose d'un diagramme

$$\begin{array}{ccc} Y & & /X \\ g \downarrow & & \downarrow \\ P & \xrightarrow{\quad} & / \text{Spec}(C) \end{array}$$

La fibre of g en $t \in P$ is the section hyperplane $X_t = X \cap H_t$ of X , and g is smooth en dehors of l'image reciproque of X .

On retrouve the situation of (??) en remplaçant P par a droite $D \subset P$ and Y par $g^{-1}(D)$. On a $S = D \cap X$.

After a theorem of Lefschetz, for D assez generale, $\pi_1(D - S, u) \rightarrow \pi_1(P - X, u)$

is surjective. Il suffices therefore of show que the $\pm ds$ are conjugues sous $\pi_1(P - X)$.

For x in the lieu smooth of codimension 1 of X , let ch a chemin of u a x in $P - X$, and g_x the lacet qui suit ch jusqu'au voisinage of X , tourne a fois autour of X , puis revient a u par ch . Les lacets g_x (for ch variable) are conjugues entre eux. Puisque X is irreductible, deux points of the lieu smooth of X peuvent always, in X , be joints par a chemin qui ne quitte pas the lieu smooth. Il en resulte que the classe of conjugaison of g_x ne depend pas of x . En particulier, the g_s are conjugues entre eux. On lit on (??) que this implique the conjugaison des $\pm ds$.

Corollaire 5.3 (5.5). L'action of $\pi_1(U, u)$ on $E/(E \cap E^\perp)$ is absolument irreductible.

Let $F \subset E/(E \cap E)$ a sous-espace stable sous the monodromie. If $F \not\subset (E \cap E)$, il existe $x \in F$ and $s \in S$ tels que $(x, ds) \neq 0$. On a then $gsx - x = \pm(x, ds)ds \in F$,

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and $\pm ds \in F$. After (??), all the $\pm ds$ are then in F , and $F = E$. Ceci proves (??).

(5.6) Ces results se transposent as suit en geometrie algebrique abstraite.

Let P a space projectif of dimension ≥ 1 on k algebriquement clos of caracteristique p and $X \subset P$ a variety projective nonsingular connexe and of dimension $n + 1$.

For A a sous-espace lineaire of P of codimension 2, one defines as en (5.1) D and the pinceau $(H_t)_{t \in D}$, the X_t , X and the diagramme (??). On dit que the $(X_t)_{t \in D}$ forment a pinceau of Lefschetz of sections hyperplanes if the conditions suivantes are verifiees :

A) L'axe A is transverse a X . L'espace X se deduit then of X par eclatement of $A \cap X$, and is smooth.

B) Il existe a partie finie S of D and for chaque $s \in S$ a point $x_s \in X_s$, tels que f let smooth en dehors of x_s .

C) x_s is a point singulier quadratique ordinaire of X_s .

For chaque $s \in S$, the theorie of Lefschetz locale of the SS_4 s'applique au spectre D_s of l'henselise of l'anneau local of D en s , and a $X_s = X \times_D D_s$.

(5.7) Let N the dimension of P , r a entier ≥ 1 , $\square$ and $i(r)$ the plongement of P in l'espace projectif of dimension $N+r-1$, of coordonnees homogenes the monomes of degre r en the coordonnees homogenes of P . Les sections hyperplanes of $i(r)(P)$ are the hypersurfaces of degre r of P .

If $p \neq 0$, il se peut qu'aucun pinceau of sections hyperplanes of X ne let of Lefschetz. Toutefois, if $r \geq 2$ and qu'on remplace the plongement projectif gives $i : X \rightarrow P$ par $i_0 = i(r) * i$, then, in ce nouveau plongement, all pinceau assez general is of Lefschetz. En d'autres termes, if $r \geq 2$, a pinceau assez general of sections hypersurfaces of degre r of X is always of Lefschetz.

(5.8) Dans the sequence of cette discussion, we etudions a pinceau of Lefschetz of sections hyperplanes of X , en excluant the case $p = 2$, n pair. Le case or n is impair we suffirait for the sequence. On pose $U = D - S$. Let $u \in U$ and ℓ a nombre premier $\neq p$. Les results locaux of the SS_4 montrent que $R_n f^* Q_\ell$ is moderelement ramifie en chaque $s \in S$.

Le fundamental group modere of U is a quotient of the complete profini of the fundamental group transcendant analogue (relevement en caracteristique 0 of revetements moderes, and theorem d'existence of Riemann). La situation algebrique is of lors all pareille a the if-tuation transcendant, and the transposition of results of Lefschetz se fait par of arguments standards. Dans the proof of (??), the theorem of Lefschetz on the n_1 devient the theorem of Bertini, and one doit invoquer the lemma d'Abhyankar for controler the ramification of $R_i g^* Q_\ell$ the long of the lieu smooth of codimension a of X .

Les results are the suivants.

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a) If the cycles evanescents are non zero :

$\neq 0$, the sheaf $R_i f^* Q_\ell$ on D is constant.

1) Pour $i =$

2) Let j l'inclusion of U in D . On a $R_n f^* Q_\ell = j^* j_* R_n f^* Q_\ell$.

3) Let $E \subset H^n(X_u, Q_\ell)$ the sous-espace of the cohomology engendre par the cycles evanescents. Ce sous-espace is stable sous $\pi_1(U, u)$, and $E = H^n(X_u, Q_\ell) \cap \pi_1(U, u)$. La representation of $\pi_1(U, u)$ on $E/(E \cap E)$ is absolument irreductible, and l'image of π_1 in $GL(E/(E \cap E))$ is engendree (topologiquement) par the $x \mapsto x \pm (x, ds)ds$ ($s \in S$) (the signe $\pm$ being determine as en (4.1)).

b) If the cycles evanescents are zero : (C'est the a case exceptionnel.

Puisque $(d, d) = \pm 2$ for n pair, il ne peut se produire que for n impair : $n = 2m + 1$. On notera que if a cycle evanescent is zero, ils the are all, car conjugues.)

1) For $i \neq n + 1$, the sheaf $R_i f^* Q_\ell$ is constant.

2) On a a sequence exacte

$$0 \rightarrow M \rightarrow Q_\ell(m - n)_S \rightarrow R_{n+1} f^* Q_\ell \rightarrow F \rightarrow 0,$$

$S \in S$

or F is constant.

3) $E = 0$.

(5.9) Le sous-espace $E \cap E$ of E is the kernel of the restriction a E of the form d'intersection $Tr(x \cup y)$. Cette form induit therefore a form bilineaire non degeneratese

$\rho : E/(E \otimes E) \otimes E/(E \otimes E) \rightarrow Q(-n)$,
 alternee for n impair, and symetrique for n pair. Cette form is
 respectee par the monodromie ; for n impair, the representation of mo-
 nodromie induit therefore

$$r : \pi_1(U, u) \rightarrow \text{Sp}(E/(E \otimes E), \rho).$$

Theoreme 5.4 (5.10). (Kajdan-Margulis). L'image of r is ouverte.

L'image of r is a sous-groupe compact, therefore analytique $\mathbb{A}$ -adique,
 of $\text{Sp}(E/(E \otimes E), \rho)$. Il suffices of show que son algebre of Lie
 L is egale a $\text{sp}(E/(E \otimes E), \rho)$. L'analogue transcendant of cette
 algebre of Lie is l'algebre of Lie of l'adherence of Zariski of the group
 of monodromie.

On deduit of (5.8) que L is engendree par the transformations of
 carre zero

$$N_s : x \mapsto x + (x, ds) ds \quad (s \in S)$$

and que $E/(E \otimes E)$ is a representation absolument irreductible of
 L . Le theorem resulte of the lemma suivant.

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Lemme 5.5 (5.11). Let V a space vectoriel of dimension finie
 on a corps k of caracteristique 0 , ρ a form alternee non degeneratese
 and L a sous-algebre of Lie of $\text{sp}(V, \rho)$. On suppose que :

- (i) V is a representation simple of L .
- (ii) L is engendree par a famille d'endomorphisms of V of the
 form $x \mapsto x + \rho(x, d)d$.

Then, $L = \text{sp}(V, \rho)$.

On peut supposer and one supposera que V , therefore ρ , is non zero. Let
 $W \subset V$ l'ensemble of $d \in V$ tels que $N(d) : x \mapsto x + \rho(x, d)d$ let in
 L .

- a) W is stable par homothetie (car L is a sous-espace vectoriel
 de $\text{gl}(V)$).
- b) If $d \in W$, $\exp(tN(d))$ is a automorphisme of (V, ρ, ρ) , therefore
 transforme W en lui-meme. If $d \neq 0$, $d \in W$, one a therefore $\exp(tN(d)) \cdot d \neq 0$
 $= d + t\rho(d, d)d \in W$; if $\rho(d, d) \neq 0$, the sous-espace vectoriel
 tendu par d and d is in W .
- c) Il en resulte que W is reunion of ses sous-espaces lineaires maxi-
 maux W_α , and que ceux-ci are deux a deux orthogonaux. Chaque
 W_α is of lors stable sous the $N(d)$ ($d \in W$), therefore stable sous
 L . Vu l'hypothese (i), $W \neq 0$ and L contient all the $N(d)$ for
 $d \in V$. On conclut en notant que l'algebre of Lie $\text{sp}(V, \rho)$ is
 engendree par the $N(d)$ ($d \in V$).

Remarque 5.6 (5.12). (inutile for the sequence). Il is maintenant facile
 of prouver (??) for a hypersurface of dimension impaire n in
 $\mathbb{P}^{n+1}$

F_q .

Let X_0 a telle hypersurface, and X l'hypersurface on F_q qui s'en
 deduit par extension of scalaires. On a

$$H^{2i}(X, Q) = Q(-i) \quad (0 \leq i \leq n);$$

$2i$

$H(X, Q(i))$ is engendre par the ieme cup-puissance of x , the classe of
 cohomologie $c_1(0(1))$ d'une section hyperplane. On a therefore

$$Z(X_0, t) = \det(1 - F_t, H(X, Q)) \cdot \prod_{i=0}^n (1 - q^i t)^{-1}$$

and $\det(1 - F_t, H(X, Q))$ is a polynome a coefficients entiers inde-
 n

pendants of $\mathbb{A}$.

Faisons varier X_0 in a pinceau of Lefschetz d'hypersurfaces, qui
 let defini on F_q (cf. (5.7) for $X = \mathbb{P}^{n+1}$; l'existence d'un tel pin-
 ceau n'est pas claire ; if one voulait completer l'argument esquisse ici, il
 faudrait recourir aux arguments qui will be donnees en (7.1)). On satisfies
 que E coincide ici with the H^n all entier, and (??) fournit the conjecture
 of Weil for all the hypersurfaces of the pinceau, en particulier for
 X_0 .

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(5.13) Indications bibliographiques on the SSSS 4 and 5.

A) Les results of Lefschetz (4.1) and (5.1) a (5.5) are contenus in
 son livre [2]. For the theorie locale (4.1), il peut be more commode of
 consulter SGA 7, XIV (3.2).

B) Les results of the SS 4 are demontres in the exposes XIII, XIV and

XV of SGA 7.

C) (5.7) is demontre in SGA 7, XVII.

D) (5.8) is demontre in SGA 7, XVIII. Le theoreme d'irreductibilite y is demontre for E, mais seulement sous l'hypothese que $E \cap E = \{0\}$. La proof in the case general (for $E/(E \cap E)$) is pareille.

6. A rationality theorem

(6.1) Let P_0 a space projectif of dimension > 1 on F_q , $X_0 \subset P_0$ a variety projective nonsingular, $A_0 \subset P_0$ a sous-espace lineaire of codimension deux, $D_0 \subset P_0$ the droite duale, F_q a cloture algebrique of F_q and P , X , A , D on F_q deduits of P_0 , X_0 , A_0 , D_0 par extension of scalaires. Le diagramme (??) of (5.6) provient d'un diagramme analogue on F_q :

$$(6.1) \quad \begin{array}{ccc} X_0 & & / X_0 \\ f_0 & & \\ \square & & \square \\ D_0 & & / \text{Spec}(F_q) \end{array}$$

On suppose que X is connexe of dimension paire $n + 1 = 2m + 2$, and que the pinceau $(X_t)_{t \in D}$ of sections hyperplanes of X defini par D is a pinceau of Lefschetz.

L'ensemble S of $t \in D$ tels que X_t let singulier is defini on F_q , i.e. provient of $S_0 \subset D_0$. On pose $U_0 = D_0 - S_0$ and $U = D - S$.

Let $u \in U$. La partie evanescente of the cohomology $E \subset H^n(X_u, \mathbb{Q})$ is stable sous $\pi_1(U, u)$, therefore defines on U a sous-systeme local E of $R_n f_* \mathbb{Q}$. Ce dernier is defini on F_q : $R_n f_* \mathbb{Q}$ is l'image reciproque of the $\mathbb{Q}$ -sheaf $R_n f_0^* \mathbb{Q}$ on D_0 , and, on U , E is l'image reciproque d'un sous-systeme local $E_0 \subset R_n f_0^* \mathbb{Q}$.

Le cup-produit is a form alternee

$$ps : R_n f_0^* \mathbb{Q} \otimes R_n f_0^* \mathbb{Q} \rightarrow \mathbb{Q}(-n).$$

Notant E_0 l'orthogonal of E_0 relativement a ps , in $R_n f_0^* \mathbb{Q} |_{U_0}$, one voit que ps induit a dualite parfaite

$$ps : E_0 / (E_0 \cap E_0) \otimes E_0 / (E_0 \cap E_0) \rightarrow \mathbb{Q}(-n).$$

Theoreme 6.1 (6.2). For all $x \in |U_0|$, the polynome $\det(1 - F_x^* t, E_0 / (E_0 \cap E_0))$ is a coefficients rationnels.

Corollaire 6.2 (6.3). Let j_0 l'inclusion of U_0 in D_0 , and j celle of U in D . Les eigenvalues of F^* agissant on $H^1(D, j_* E_0 / (E_0 \cap E_0))$ are of nombres algebriques dont all the conjugues complexes a satisfy

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E_0)) are of nombres algebriques dont all the conjugues complexes a satisfy

$$q^{(n+1)/2-1} \leq |a| \leq q^{(n+1)/2}.$$

After (??) and (??), the hypotheses of (??) are en effet verifiees par $(U_0, E_0 / (E_0 \cap E_0), ps)$, for $b = n$, and one applique (??).

Lemme 6.3 (6.4). Let G_0 a $\mathbb{Q}$ -sheaf constant tordu on U_0 , tel que son image reciproque G on U let a sheaf constant. Il existe then in $\mathbb{Q}$ of unites a_i telles que, for all $x \in |U_0|$, one ait

$$\det(1 - F_x^* t, G_0) = \prod_i^{\deg(x)} (1 - a_i t).$$

Ce lemma exprime que G_0 is l'image reciproque d'un sheaf on $\text{Spec}(F_q)$, a savoir son image directe on $\text{Spec}(F_q)$. Ce dernier s'identifie a a representation ℓ -adique G_0 of $\text{Gal}(F_q/F_q)$, and one prend

$$\det(1 - F t, G_0) = \prod_i^Y (1 - a_i t).$$

Le lemma (??) s'applique aux $R f_0^* \mathbb{Q}$ ($i \neq n$), a $R_n f_0^* \mathbb{Q} / E_0$ and a

$E_0 \cap E_0$.

For $x \in |U_0|$, the fibre $X_x = f_0^{-1}(x)$ is a variety on the finite field $k(x)$. If x is a point of U au-dessus of x , X_x se deduit of X_x par extension of scalaires of $k(x)$ a sa cloture algebrique $k(x) = F_q$, and $H^i(X_x, \mathbb{Q})$ is the fibre of $R_i f_* \mathbb{Q}$ en x . La formule (??) for the variety X_x on $k(x)$ s'ecrit therefore

$$Z(X_x, t) = \prod_i^Y \det(1 - F_x^* t, R_i f_0^* \mathbb{Q})^{(-1)^{i+1}}$$

and $Z(X_x, t)$ is the product of

Y

$$Z_1 = \det(1 - F_x * t, R_n f * Q / E_0) * \det(1 - F_x * t, E_0 n E_0) * \det(1 - F_x * t, R_i f_0 * Q) (-1)^{i+1}$$

i+1

$$i_6 = n$$

par

$$Z_m = \det(1 - F_x * t, E_0 / (E_0 n E_0)).$$

Posons $E_0 = E_0 / (E_0 n E_0)$, $E_e = E / (E n E)$ and appliquons (??) aux facteurs of Z_1 . On trouve qu'il existe of unites $\mathbb{A}$ -adiques a_i ($1 \leq i \leq N$) and b_j ($1 \leq j \leq M$) in Q tels que for all $x \in |U_0|$

$$Z(Xx, t) = i \prod_{i=1}^N (1 - a_i t)^{\deg(x)} * \prod_{j=1}^M (1 - b_j t)^{\deg(x)} \frac{\det(1 - F_x t, E_0)^j}{\det(1 - F_x t, E_e)}$$

and qu'en particulier the second membre is in $Q(t)$. If a a_i coincide with a b_j , il is loisible of supprimer simultanement cet a_i of the liste of a and ce b_j of the liste of b. On peut therefore supposer, and we supposerons, que $a_i \neq b_j$ for all i and all j.

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(6.5) Il suffices of prouver que the polynomes $i \prod_{i=1}^N (1 - a_i t)$ and $j \prod_{j=1}^M (1 - b_j t)$ are a coefficients rationnels, i.e. que the famille of a_i (resp. the famille of b_j) is definie on Q . Nous the deduirons of propositions suivantes.
Proposition 6.4 (6.6). Let (g_i) ($1 \leq i \leq P$) and (d_j) ($1 \leq j \leq Q$) deux familles d'unites $\mathbb{A}$ -adiques in Q . On suppose que $g_i \neq d_j$. If K is a ensemble fini assez grand d'entiers $n \geq 1$, and L a partie of densite θ assez grande of $|U_0|$, then, if $x \in |U_0|$ satisfies $k \deg(x)$ (for all $k \in K$) and $x \in L$, the denominateur of

$$\det(1 - F_x * t, E_0)^{-1} * \prod_{i=1}^P (1 - g_i t)^{\deg(x)} \prod_{j=1}^Q (1 - d_j t)^{\deg(x)}$$

(6.2)

ecrit in the form irreductible, is $j \prod_{j=1}^Q (1 - d_j t)^{\deg(x)}$.

La proof will be given en (6.10-13). After (??) ci-dessous, (??) fournit a description intrinseque of the famille of d_j en terme of the famille of fractions rationnelles (??) for $x \in |U_0|$.

Lemme 6.5 (6.7). Let K a ensemble fini d'entiers $n \geq 1$ and (d_j) ($1 \leq j \leq Q$) and (e_j) ($1 \leq j \leq Q$) deux familles d'elements d'un corps. If, for all n assez grand, non divisible par no of $k \in K$, the famille of d_{jn} coincide with celle of e_{jn} (a l'ordre pres), then the famille of d_j coincide with celle of e_j (a l'ordre pres).

On procede par recurrence on Q . L'ensemble of entiers n tels que $dQ = e_{jn}$ is a ideal (n_j) . Prouvons qu'il existe J_0 tel que $dQ = e_{J_0}$. Sinon the n_j seraient distincts of 1, and il existerait of entiers arbitrairement grands n , non divisibles par no of n_j , ni par no of $k \in K$. On aurait $dQ = e_{jn}$ for all j , and this contredirait l'hypothese.

Il existe therefore J_0 tel que $dQ = e_{J_0}$. On conclut en appliquant l'hypothese of recurrence aux familles (d_j) ($j \neq J_0$) and (e_j) ($j \neq J_0$).

Proposition 6.6 (6.8). Let (g_i) ($1 \leq i \leq P$) Q and (d_j) ($1 \leq j \leq Q$) deux familles d'unites $\mathbb{A}$ -adiques in Q , $R(t) = i \prod_{i=1}^P (1 - g_i t)$ and $S(t) =$

$$j \prod_{j=1}^Q (1 - d_j t). \text{ Supposons que for all } x \in |U_0|, \prod_{i=1}^P (1 - g_i t)^{\deg(x)} \prod_{j=1}^Q (1 - d_j t)^{\deg(x)}$$

Then $S(t)$ divide $R(t)$.

Supprimons of familles (g_i) and (d_j) of paires d'elements communs jusqu'a verifier l'hypothese of (??). Appliquons (??). Par hypothese, the fractions rationnelles (??) are of polynomes. Aucun d n'a therefore subsiste, ce qui signifie que $S(t)$ divide $R(t)$.

Cette proposition fournit a caracterisation intrinseque of $R(t)$ en terme of the famille of polynomes Q $i \prod_{i=1}^P (1 - g_i t)$. C'est the ppcm of polynomes $S(t) = j \prod_{j=1}^Q (1 - d_j t)$ verifiant l'hypothese

(6.9) Prouvons (6.5) and therefore (??) (modulo (??)). Faisons in (??) $(g_i) = (a_i)$ and $(d_j) = (b_j)$. On trouve a caracterisation intrinseque of the famille of b_j en terme of the famille of fractions rationnelles $Z(X_x, t)$ ($x \in |U_0|$). Celles-ci being in $Q(t)$, the famille of b_j is definie on Q .

$Q \quad \deg(x)$

Les polynomes $j(1 - b_j t)$ are therefore in $Q[t]$. La proposition (??) fournit a description intrinseque of the famille of a_i en terme of cette famille of polynomes. La famille of a_i is therefore definie on Q .

(6.10) Preliminaires. -- Let $u \in U$ and E_u the fibre of E en u . Le fundamental group arithmetique $\pi_1(U_0, u)$, extension of $Z = \text{Gal}(F_q/F_q)$ (generateur : ph) par the fundamental group geometric $\pi_1(U, u)$, agit on E_u par similitudes symplectiques :

$$r : \pi_1(U_0, u) \rightarrow \text{CSp}(E_u, \text{psu}).$$

Nous notons $\mu(g)$ the multiplicateur d'une similitude symplectique g . Let

$$H \subset Z^* \times \text{CSp}(E_u, \text{psu})$$

the sous-groupe defini par l'equation

$$q^{-n} = \mu(g)$$

(q being a unite ℓ -adique, $q \in Z^*$ is defini for all $n \in \mathbb{Z}$). Le fait que ps let a values in Q_ℓ^* ($-n$) s'exprime en disant que l'application of π_1 in $Z^* \times \text{CSp}$, of coordonnees the projection canonical on Z and r , se factorise par

$$r_1 : \pi_1(U_0, u) \rightarrow H.$$

Lemme 6.7 (6.11). L'image H_1 of r_1 is ouverte in H .

En effet, $\pi_1(U_0, u)$ se projette on Z , and l'image of $\pi_1(U, u) = \text{Ker}(\pi_1(U_0, u) \rightarrow Z)$ in $\text{Sp}(E_u, \text{psu}) = \text{Ker}(H \rightarrow Z^*)$ is ouverte (5.10).

Lemme 6.8 (6.12). For $z \in Q_\ell^*$ a unite ℓ -adique, l'ensemble Z of $(n, g) \in H_1$ tels que $z \neq \text{let value proper of } g$ is a ferme of mesure zero.

Il is clair que Z is ferme. For chaque $n \in Z$, let CSp_n l'ensemble of $g \in \text{CSp}(E_u, \text{psu})$ tels que $\mu(g) = q^{-n}$, and let Z_n l'ensemble of $g \in \text{CSp}_n$ tels que $z \neq \text{let value proper of } g$. Then, CSp_n is a space homogene sous Sp , and one satisfies que Z_n en is a sous-espace algebrique proper, therefore of mesure 0. After (??), $H_1 \cap (\{n\} \times Z_n)$ is therefore of mesure 0 in l'image inverse in H_1 of n , and one applique Fubini a the projection $H_1 \rightarrow Z$.

(6.13) Prouvons (??). For chaque i and j , l'ensemble of entiers n tels que $g_{in} = d_{jn}$ is l'ensemble of multiples d'un entier fixe n_0 (one n'exclut pas $n_0 = 0$). Par hypothese, $n_0 \neq 1$.

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After (??) and the theorem of densite of Cebotarev, l'ensemble of let value proper of F_x^* , agissant on E_{e_0} is

$$\deg(x)$$

$x \in |U_0|$ tels qu'un g_i of densite 0. On prend for K l'ensemble of n_0 and for L l'ensemble of x ci-dessus.

7. End of the proof of (1.7)

Lemme 7.1 (7.1). Let X_0 a variety projective nonsingular absolutely irreducible of dimension paire d on F_q . Let X on F_q deduit of X_0 par extension of scalaires and a a value proper of F^* agissant on $H^d(X, Q_\ell)$. Then, a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy

$$(7.1) \quad q^{d/2-1} \leq |a| \leq q^{d/2+1}.$$

On procede par recurrence on d (always suppose pair). Le case $d = 0$ is trivial, meme without supposer X_0 absolument irreducible ; one suppose desormais $d \geq 2$. On pose $d = n + 1 = 2m + 2$.

If Fr is a extension of degre r of F_q and que $X_0 \otimes Fr$ se deduit of X_0/F_q par extension of scalaires, l'assertion (??) for X_0/F_q equivaut a (??) for $X_0 \otimes Fr/F_r$: of meme que q is replace par q^r , the eigenvalues of F^* are remplacees par leurs puissances r -iemes.

After (5.7), in a plongement projectif convenable $i : X \rightarrow P$, X admet a pinceau of Lefschetz of sections hyperplanes. La remark precedente we permet of supposer i and ce pinceau definis on F_q (quitte a remplacer F_q par a extension finie).

Supposons therefore qu'il existe on F_q a plongement projectif $X_0 \rightarrow P_0$ and a sous-espace of codimension deux A_0 of P_0 qui definisse a pinceau of Lefschetz. On reprend the notations of (6.1) and (6.3). Une nouvelle extension of scalaires we ramene a supposer que :

a) Les points of S are definis on F_q .

- b) Les cycles evanescents en x_s ($s \in S$) are definis on F_q (puisque seul $+d$ is intrinseque, ils pourraient n'etre definis que on a extension quadratique).
- c) Il existe a point rationnel $u_0 \in U_0$. On prend the point correspondant u of U as point base.
- d) $Xu_0 = f_0^{-1}(u_0)$ admet a section hyperplane smooth Y_0 , definie on F_q . On pose $Y = Y_0 \otimes F_q$.

Puisque X se deduit of X par eclatement of the sous-variety smooth of codimension deux $A \cap X$, one a

$$H^i(X, Q) \sim H^i(X, Q) \\ = H^i(X, Q)$$

(en fait, $H^i(X, Q) \sim H^i(X, Q) \otimes H^{i-2}(A \cap X, Q)(-1)$). Il suffices therefore of prouver (??) for the eigenvalues a of F * agissant on $H^d(X, Q)$.

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La spectral sequence of Leray for f s'ecrit

$$E_{p,q}^2 = H^p(D, R^q f_* Q) \Rightarrow H^{p+q}(X, Q).$$

Il suffices of prouver (??) for the eigenvalues a of F * agissant on

$E_{p,q}^\infty$ for $p + q = d = n + 1$.

Ce are :

$$1, n-1$$

A) $E_{1,n-1}^\infty$. After (5.8), $R^{n-1} f_* Q$ is constant. After (??), one a therefore

$$E_{1,n-1}^\infty = H^{n-1}(Xu, Q)(-1),$$

After the theorem of Lefschetz faible (corollary of SGA 4, XIV (3.2), and of the Poincare duality, SGA 4, XVIII), one a

$$H^{n-1}(Xu, Q)(-1) \sim H^{n-1}(Y, Q)(-1),$$

and one applique l'hypothese of recurrence a Y_0 .

$$0, n+1$$

B) $E_{0,n+1}^\infty$.

If the cycles evanescents are non zero, $R^{n+1} f_* Q$ is constant and

$$0, n+1$$

$$E_{0,n+1}^\infty = H^{n+1}(Xu, Q).$$

Le morphism of Gysin

$$H^{n-1}(Y, Q)(-1) \rightarrow H^{n+1}(Xu, Q)$$

is surjectif (cet argument is dual of A)), and one applique l'hypothese of recurrence a Y_0 .

If the cycles evanescents are zero, the sequence exacte of (5.8) b) fournit a sequence exacte

$$M \\ Q(m-n) \rightarrow E_{0,n+1}^\infty \rightarrow H^{n+1}(Xu, Q).$$

$$s \in S$$

La value proper of F agissant on $Q(m-n)$ is $q d/2$, and H^{n+1} se traite as more haut.

$$2, n$$

C) $E_{2,n}^\infty$. If one disposait of the theorem of Lefschetz "difficile", one saurait que $E_{0,n}^\infty$ is zero, and que $R^n f_* Q$ is sum directe of $j^* E$ and d'un sheaf constant. Le H^1 d'un sheaf constant on P^1 is zero, and il suffirait d'appliquer (??).

Faute d'avoir already demontre Lefschetz "difficile", il we va falloir deviser. If the cycles evanescents are zero, $R^n f_* Q$ is constant ((5.8)

$$2, n$$

b)) and $E_{2,n}^\infty = 0$. On peut therefore supposer, and one supposera, the cycles evanescents non zero. Filtrons $R^n f_* Q = \sum j^* E$ (5.8) par the sous-sheaves $j^* E$ and $j^*(E + E)$.

If the cycles evanescents d ne are pas in $E \cap E$, one dispose of sequences exactes :

$$(7.2) \quad 0 \rightarrow j^* E \rightarrow R^n f_* Q \rightarrow \text{sheaf constant} \rightarrow 0,$$

$$(7.3) \quad 0 \rightarrow \text{sheaf constant} \rightarrow j^* E / (E \cap E) \rightarrow j^* E \rightarrow 0.$$

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If, a Dieu ne plaise, the d are in $E \cap E$, one a $E \subset E$ and of sequences exactes :

$$(7.4)$$

$$0 \rightarrow \text{the sheaf constant } j^* E \rightarrow R^n f_* Q \rightarrow \text{a sheaf } F \rightarrow 0,$$

$$(7.5)$$

$$0 \rightarrow F \rightarrow \text{the sheaf constant } j^* j_* F \rightarrow Q(n-m)s \rightarrow 0.$$

Dans the premier case, the sequences exactes longues of cohomologie four-nissent

$$(7.1.20) \quad H^1(D, j^* E) \rightarrow H^1(R^n f_* Q) \rightarrow 0,$$

$$(7.1.30) \quad 0 \rightarrow H^1(D, j^* E) \rightarrow H^1(D, j^* (E/(E \otimes E))),$$

and one applique (??).

Dans the second case, elles fournissent

$$(7.1.40) \quad 0 \rightarrow H^1(D, R^n f_* Q) \rightarrow H^1(D; F),$$

M

$$(7.1.50) \quad Q(n-m) \rightarrow H^1(D, F) \rightarrow 0,$$

s ∈ S

and one remark que F agit on $Q(n-m)$ par multiplication par $q^{d/2}$.

Lemme 7.2 (7.2). Let X_0 a variety projective nonsingular absolute irreducible of dimension d on F_q . Let X on F_q deduit of X_0 par extension of scalaires and a a value proper of F^* agissant on $H^d(X, Q)$. Then, a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy $|a| = q^{d/2}$.

Prouvons already que (??) ⇒ (??). For X_0 projective nonsingular on F_q and i a entier, il faut prouver l'assertion suivante :

$W(X_0, i)$. Let X deduit of X_0 par extension of scalaires of F_q a F_q . If a is a value proper of F^* agissant on $H^i(X, Q)$, then a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy $|a| = q^{i/2}$.

- If F is an extension of degree n of F_q , and que $X_0 \otimes F$ se deduit of X_0/F_q par extension of scalaires, then $W(X_0, i)$ is equivalent a $W(X_0 \otimes F, i)$: etendre the scalaires revient a remplacer a par an and q par q^n .
- If X_0 is purement of dimension n , $W(X_0, i)$ is equivalent a $W(X_0, 2n-i)$: this resulte of the Poincare duality.
- If X_0 is sum of varieties X_{0a} , $W(X_0, i)$ is equivalent a the conjunction of $W(X_{0a}, i)$.
- If X_0 is purement of dimension n , Y_0 a section hyperplane smooth of X_0 , and que $i < n$, then $W(Y_0, i) = W(X_0, i)$: this resulte of the theorem of Lefschetz faible.

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For prouver the assertions $W(X_0, i)$, one se ramene successivement :

- par c), a supposer X_0 purement d'une dimension n ;
- par b), a supposer of more $0 \leq i \leq n$;
- par a) and d), a supposer of more $i = n$;
- par a) and c), a supposer of more X_0 absolutely irreducible.

Ce case is justiciable of (??).

(7.3) Prouvons (??). For all entier k , a_k is value proper of F^* agissant on $H^{kd}(X_k, Q)$ (formule of Kunneth). For k pair, X_k is justiciable of (??), d'ou

$$q^{kd/2-1} \leq |a_k| \leq q^{kd/2+1}$$

and

$$q^{kd/2-k} \leq |a_k| \leq q^{kd/2+k}.$$

Faisant tendre k vers l'infini, one trouve (??).

8. First applications

Theoreme 8.1 (8.1). Let $X_0 \subset P^{n+r}$ F_q a intersection complete non singuliere on F_q , of dimension n and of multidegree $(d_1, \dots, d_r)$. Let b_0 the nieme nombre of Betti of intersections completes nonsingular complexes, of memes dimension and multidegree. Posons $b = b_0$ for n impair, and $b = b_0 - 1$ for n pair. Then

$$|\#X_0(F_q) - \#P^n(F_q)| \leq b \cdot q^{n/2}.$$

Let X/F_q deduit of X_0 , and $Q^i \otimes x_i$ the droite in $H^{2i}(X, Q)$ engendree par the ieme cup-puissance of the cohomology class d'une section hyperplane. Sur cette droite, F^* agit par multiplication par q^i . La cohomologie of X is sum of $Q^i \otimes x_i$ ($0 \leq i \leq n$) and of the partie primitive of $H^n(X, Q)$, of dimension b . After (1.5), il existe therefore b nombres algebriques a_j , the eigenvalues of F^* agissant on cette cohomologie primitive, tels que

$$\#X_0(F_q) = \sum_{i=0}^n q^i + (-1)^n \sum_{j=1}^b a_j.$$

After (??), $|a_j| = q^{n/2}$ and

$$|\#X_0(F_q) - \#P^n(F_q)| = |\sum_{i=0}^n q^i - \sum_{j=1}^b a_j| \leq \sum_{j=1}^b |a_j| \leq b \cdot q^{n/2}.$$

i=0 j=1 j=1

Theoreme 8.2 (8.2). Let N a entier ≥ 1 , $e : (Z/N)^* \rightarrow C^*$ a caractere, k a entier ≥ 2 and f a form modulaire holomorphe on $G_0(N)$, of weights k and of caractere $e : f$ is a

□ fonction
□ holomorphe

on the demi-plan of Poincare, telle que for $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, Z)$ with

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$c \not\equiv 0 \pmod{N}$, one ait

$$f\left(\frac{az+b}{cz+d}\right) = e(d)(cz+d)^k f(z).$$

On suppose f cuspidale and primitive ("new" au sens d'Atkin-Lehner and of Miyake), en particulier

P vecteur propre of operateurs of Hecke T_p $(p \nmid N)$. Posons $f = \sum_{n=1}^{\infty} a_n q^n$, with $f = e^{2\pi i z}$ (and $a_1 = 1$). Then, for p premier ne divisant pas N ,

$$|a_p| \leq 2 * p^{(k-1)/2}.$$

En d'autres termes, the roots of l'equation

$$T^2 - a_p T + e(p) * p^{k-1}$$

are of value absolue $p^{(k-1)/2}$.

Ces roots are en effet of eigenvalues of Frobenius agissant on the H^{k-1} d'une variety projective nonsingular of dimension $k-1$ definie on F_p .

Sous of hypotheses restrictives, ce fait is demontre in mon expose Bourbaki (Formes modulaires and representations p -adiques, expose 355, fevrier 1969, in : Lecture Notes in Mathematics, 179). Le case general n'est pas beaucoup more difficile.

Remarque 8.3 (8.3). J. P. Serre and moi-meme avons recemment of- shows que (??) reste vrai for $k=1$. La proof is all differente.

L'application suivante m'a been suggeree par E. Bombieri.

Theoreme 8.4 (8.4). Let Q a polynome a n variables and of degree d on F_q , Q_d the partie homogene of degree d of Q and $\psi : F_q \rightarrow C^*$ a caractere additif non trivial on F_q . On suppose que :

- (i) d is premier a p ;
 - (ii) l'hypersurface H_0 in P^{n-1} definie par Q_d is smooth.
- Then

$$\sum_{x_1, \dots, x_n \in F_q} \psi(Q(x_1, \dots, x_n)) \leq (d-1)n * q^{n/2}.$$

Quitte a remplacer Q par a multiple scalaire, one peut supposer (and one supposera) que

$$(8.1) \quad \psi(x) = \exp(2\pi i \text{Tr}_{F_q/F_p}(x)/p).$$

Let X_0 the revetement etale of l'espace affine A_0 of dimension n on F_q d'equation $T^p - T = Q$, and let α the projection of X_0 on A_0 :

$$\alpha : X_0 = \text{Spec}(F_q[x_1, \dots, x_n, T]/(T^p - T - Q)) \rightarrow A_0.$$

Le revetement X_0 is galoisien, of group of Galois Z/p ; $i \in Z/p = F_p$ agit par $T \mapsto T + i$.

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Let $x \in A_0(F_q)$ and calculons l'Frobenius endomorphism of the fibre of X_0/A_0 en x . Posons $q = p^f$ and let F_q a cloture algebrique of F_q . For $(x, T) \in X_0(F_q)$ au-dessus of x , one a $F((x, T)) = (x, T^q)$, and

$$T = T + \sum_{i=0}^{f-1} (T^{p^{i+1}} - T^{p^i}) = T + \sum_{i=0}^{f-1} (Q(x))^{p^i} = T + \text{Tr}_{F_q/F_p}(Q(x)).$$

C'est l'action of l'element $\text{Tr}_{F_q/F_p}(Q(x))$ of the group of Galois.

Let E the corps of roots p -iemes of l'unite and l a place finie of E premiere a p . Nous travaillerons en cohomologie l -adique. For $j \in Z/p$, let $L_{j,0}$ the El -systeme local of rang a on A_0 defini par X_0 and $\psi(-jx) : Z/p \rightarrow E^* \subset Cl^* : \text{one dispose of } \alpha : X_0 \rightarrow L_{j,0} \text{ and } t(i * x) = \psi(-ji)L_j(x)$. Notons without 0 the objects deduits of A_0 , X_0 , $L_{j,0}$ par extension of scalaires a F_q . La formule of traces (??) for

Lj,0 s'ecrit

$$(8.2) \quad \sum_{x \in F_n} \sum_{q=0}^n \sum_{i=0}^n \text{ps}(Q(x_1, \dots, x_n)) = (-1)^i \text{Tr}(F^*, \text{Hci}(A, L_j)).$$

$$(8.3) \quad \text{Hci}(X, Q^j) \otimes_{Q^j} \text{El} = \sum_j \text{Hci}(A, L_j).$$

For $j = 0$, L_j is the sheaf constant El ; ce facteur corresponds a l'inclusion, par image reciproque, of the cohomology of A in celle of X .

Lemme 8.5 (8.5). (i) For $j \neq 0$, $\text{Hci}(A, L_j)$ is zero for $i \neq n$; for $i = n$, cet space of cohomologie is of dimension $(d-1)n$.
(ii) For $j \neq 0$, the cup-produit

$$\text{Hcn}(A, L_j) \otimes \text{Hcn}(A, L_{-j}) \rightarrow \text{Hc}_{2n}(A, \text{El}(-n))$$

is a dualite parfaite.

(iii) X_0 is a ouvert d'une variety projective nonsingular Z_0 .

Deduisons (??) of (??). Let $j_0 : X_0 \rightarrow Z_0$ and $j : X \rightarrow Z$ deduit par extension of scalaires a F_q . After (??), (i), and (??) for Z_0 , il suffices of prouver l'injectivite of

$$\text{Hcn}(A, L_j) \rightarrow \text{Hcn}(X, L_j) = \text{Hcn}(X, \text{El}) \rightarrow \text{H}^n(Z, \text{El}).$$

On a $\text{Tr}(a^* u \cup a^* v) = i \text{Tr}(j_0^! a^* u \cup j_0^! a^* v)$, therefore cette injectivite resulte of (ii).

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(8.6) Prouvons (??) (iii). Let P_0 l'espace projectif on F_q complete of A_0 par adjonction d'un hyperplan a l'infini P_0^∞ , $H_0 \subset P_0^\infty$ d'equation $Q_d = 0$ and Y_0 the revetement of P_0 normalise of P_0 in X_0 .

$$(8.4) \quad \frac{A_0}{P_0} \rightarrow \frac{P_0^\infty}{H_0}$$

Etudions Y_0/P_0 pres of l'infini, localement for the topologie etale.

Lemme 8.6 (8.7). Y_0 is smooth en dehors of l'image reciproque of H_0 .

Le diviseur of the fonction rationnelle Q on P_0 is sum of sa partie finie $\sum (Q)_f$ and of $(-d)$ fois l'hyperplan a l'infini. On a :

$$\sum (Q)_f = \sum (Q)_f - d * P_0^\infty, \\ \sum (Q)_f \cap P_0^\infty = H_0.$$

A distance finie, $Y_0 = X_0$ is etale on A_0 , therefore smooth. A l'infini, mais en dehors of l'image reciproque of H_0 , il existe of coordonnees locales $(z_1, \dots, z_n)$ telles que $Q = z_1 - d$ (this utilise que $(d, p) = 1$). Dans ces coordonnees, Y_0 apparait as product d'une courbe and d'un space smooth (correspondant aux coordonnees $z_2, \dots, z_n$). Par normalite, il is smooth.

Lemme 8.7 (8.8). Au voisinage etale d'un point au-dessus of H_0 , Y_0 is smooth on a surface singuliere normale, always the meme.

On peut cette fois trouver of coordonnees locales telles que $Q = z_1 - d * z_2$. En effet, puisque H_0 is smooth, $\sum (Q)_f$ is smooth au voisinage of l'infini and coupe P_0^∞ transversalement. Cette form is independante of the point choisi, and n'utilise que deux coordonnees, d'ou l'assertion.
(8.9) On sait que the procede suivant (dit a Zariski) permet of resoudre the singularites of surfaces : alternativement, one normalise and one eclate the lieu singulier (reduit). Les operateurs en jeu commutent a the localisation etale and au product par a space smooth. Le procede of Zariski permet therefore again of resoudre the singularites d'un space qui, as Y_0 , is, localement for the topologie etale, smooth on a surface. La resolution obtenue of Y_0 is the Z_0 cherche.

If T is a courbe on a surface S , contenant the lieu singulier, and que T_0 is l'image reciproque of T in the resolution of Zariski S_0 of S , one sait que if one eclate of facon iteree in S_0 the lieu singulier (reduit) of $(T_0)_{\text{red}}$, one obtains a surface S_{00} telle que l'image reciproque reduite $(T_{00})_{\text{red}}$ of T in S_{00} let a diviseur a croisements normaux.

A nouveau, the operations en jeu commutent a the localisation etale and au product par a space smooth. Raisonnant as more haut, and observant que $(Y_0, \text{l'infini})$ is localement smooth on a (S, T) , one peut meme trouver Z_0 tel que $Z_0 - X_0$ let a diviseur a croisements normaux.

(8.10) Prouvons (??) (i), (ii). Ce are of assertions geometrics ; this we permet of travailler desormais on F_q . Let $S \subset \mathbb{A}^n$ l'espace affine on F_q qui parametriser les polynomes a n variables of degree $\leq d$, and let S l'ouvert of $S \subset \mathbb{A}^n$ correspondant aux polynomes dont the partie homogene of degree d is of discriminant non zero. On note $Q_S \in H^0(S, \mathcal{O}_S[x_1, \dots, x_n])$ the polynome universel on S , and X_S the revetement galoisien etale of $A_S = A_n \times_S d'equation T^p - T = Q_S$, of group of Galois Z/p . Let $P_S = P_n \times_S$ l'espace projectif on S complete of A_S and Y_S the normalise of P_S in X_S . On dispose, on S , d'un diagramme analogue a (??).

Les expressions of Q en coordonnees locales given en (??) and (??) restent valables in the situation presente, with parametres, of sorte que, localement for the topologie etale on Y_S , Y_S/S is isomorphe au product of S (qui is smooth) with a fibre. La methode canonical of resolution utilisee en (8.9) we fournit a compactification relative Z_S/S of X_S/S , with $Z_S - X_S$ a diviseur a croisements normaux relatifs on S

$$X_S \quad / \quad Z_S$$

$$\begin{array}{ccc} \square & & \square \\ AS & & /S \\ & \searrow & \\ & \{\{\} & \\ & \{\{\} & \\ & \{\{\} & \\ & \}\} & \\ \square \{ & & \\ S & & \end{array}$$

(f proper and smooth, u plongement ouvert, $Z_S - X_S$ diviseur a croisements normaux relatifs).

Let $L_{j,S}$ le $\mathcal{O}_S$ -sheaf on A_S deduit as en (??) of X_S/A_S . On a $a^* \mathcal{E}_l = j_* L_{j,S}$, therefore

$$R_i(f_u)! \mathcal{E}_l = \sum_j^M R_i f! (u^* L_{j,S}).$$

Les properties of Z_S assurent que the $R_i(f_u)! \mathcal{E}_l = R_i f! (u^* \mathcal{E}_l)$ are of sheaves localement constants on S . Des lors, the $R_i f! (u^* L_{j,S})$ egalement are localement constants.

Puisque S is connexe, il suffices of prouver (??) (i), (ii) for a polynome Q particulier. On prendra $Q = \sum x_i^d$. Ce polynome satisfies the condition of non-singularite car $(d, p) = 1$. For ce polynome, the variables se separant in the sum exponentielle of (??). Ceci correspond au fait que L_j is the product tensoriel of images reciproques of sheaves analogues $L_{j,i}$ on the facteurs of dimension 1 of $A = A_n$. Par the formule of Kunneth

$$H^0_c(A, L_j) = \sum_i H^0_c(A_1, L_{j,i}).$$

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Ceci ramene the proof of (??) (i), (ii) au case or $n = 1$ and or Q is x^d .

(8.11) Traitons ce case particulier. Le revetement X of A is irreductible, of sorte que for $i = 0, 2$

$$\begin{array}{c} \sim \\ H^i_c(A, \mathcal{E}_l) \rightarrow H^i_c(X, \mathcal{E}_l). \end{array}$$

For $i \neq 0$ and $j \neq 0$, one a therefore $H^i_c(A, L_j) = 0$.

L'assertion (ii) resulte of (2.8) or (2.12) and of ce que $L_j \otimes L_{-j} = \mathcal{O} = \mathcal{E}_l$. For prouver (i), il reste a verifier que

$$kh_c(A, L_j) = 1 - d.$$

After the formule d'Euler-Poincare (voir l'expose Bourbaki 286 of fevrier 1965, par M. Raynaud), this is equivalent au lemma suivant. Lemme 8.8 (8.12). Le conducteur of Swan of L_j a l'infini is egal a d .

Cet enonce is equivalent au suivant.

Lemme 8.9 (8.13). Let k a finite field of caracteristique p , $y \in k[[x]]$ a element of valuation d premiere a p , L l'extension of $K = k((x))$ engendree par the roots of $T^p - T = y^{-1}$ and kh the caractere suivant of $\text{Gal}(L/K)$ a values in Z/p :

$$kh(s) = sT - T.$$

Then, kh is of conducteur $d + 1$.

Par extension of the residue field, one se ramene a supposer k algebriquement clos plutot que fini, and one applique : J. P. Serre, Sur the corps locaux a residue field algebraically closed, Bull. Soc. Math. France,

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WEIGHTS IN THE COHOMOLOGY OF ALGEBRAIC VARIETIES

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1. Let X be a complex algebraic variety, that is, a separated scheme of finite type over $\mathbf{C}$. We again write X for the usual topological space $X(\mathbf{C})$ underlying it. In this lecture I describe a remarkable filtration of the rational cohomology groups of X , the *weight filtration*, and give a collection of results about its properties. Its definition will be given in §12. For proofs, I refer to the works cited in the bibliography, where the theorems are often proved in more general settings. Working over $\mathbf{Q}$ allows one to have at the same time Hodge theory, actions of Galois groups, and resolution of singularities.

The weight filtration is a finite increasing filtration. I denote it by W . It is also defined in relative situations and in cohomology with proper supports. It depends not only on the topological space X , but also on the way in which that space is realized as an algebraic variety. It is compatible with Künneth isomorphisms: under the isomorphism

$$H^*(X \times Y) = H^*(X) \otimes H^*(Y)$$

one has

$$W_j(H^*(X \times Y)) = \sum_{k+l=j} W_k(H^*(X)) \otimes W_l(H^*(Y)).$$

It is functorial for algebraic morphisms. More precisely, if $f : X \rightarrow Y$ is algebraic, then

$$f^* : H^*(Y) \rightarrow H^*(X)$$

is strictly compatible with the weight filtrations of $H^*(Y)$ and $H^*(X)$.¹ More generally, every natural map is strictly compatible with the weight filtrations.

The weight filtration is a discrete invariant: it is invariant under deformation of the algebraic structure of X . More precisely, one has the following theorem.

THEOREM 1. *Let $f : X \rightarrow S$ be a morphism. For $t \in S$, let $X_t = f^{-1}(t)$. If the sheaf $R^i f_* \mathbf{Q}$ is locally constant, that is, a local system, then there is a weight filtration W of $R^i f_* \mathbf{Q}$ by local subsystems such that the maps*

$$r_t : (R^i f_* \mathbf{Q})_t \rightarrow H^i(X_t)$$

are strictly compatible with the weight filtrations, and in particular such that r_t is an isomorphism and W induces the weight filtration of $H^i(X_t)$.

3. Roughly speaking, the weight filtration expresses how the cohomology of X may be built from the cohomology of nonsingular projective varieties. Here are several examples.

EXAMPLE 2 (3.1). If X is proper, equivalently compact for example projective, and smooth, equivalently nonsingular, then $H^i(X)$ is pure of weight i :

$$\mathrm{Gr}_j^W H^i(X) = 0 \quad (j \neq i).$$

Equivalently, $W_{i-1} H^i(X) = 0$ and $W_i H^i(X) = H^i(X)$.

EXAMPLE 3 (3.2). Let X be smooth, and let X be open in a nonsingular compactification $\overline{X}$ with complement a divisor with normal crossings. If the irreducible components of $\overline{X} - X$ are denoted Y_α , and if $Y_I = \bigcap_{\alpha \in I} Y_\alpha$, then the graded pieces of $H^n(X)$ are controlled by the cohomology of the Y_I . In particular the possible weights of $H^n(X)$ lie between n and $2n$.

¹If a class $x \in H^*(X)$ lies in the image of f^* , then it has filtration $\leq i$ if and only if it is the image of a class of filtration $\leq i$.

EXAMPLE 4 (3.3). If X is proper, possibly singular, then the possible weights of $H^n(X)$ are at most n .

EXAMPLE 5 (3.4). If X is arbitrary, then the possible weights of $H^n(X)$ lie between 0 and $2n$.

4. Let H be a finite-dimensional vector space over $\mathbf{Q}$, provided with a finite increasing filtration W . A *grading* of W is a direct-sum decomposition

$$H = \bigoplus H_i$$

such that

$$W_j H = \bigoplus_{i \leq j} H_i.$$

The filtration W is always gradable, but noncanonically. In the situations considered above, it is often useful to know more: the grading can sometimes be chosen to respect additional algebraic structures.

5. Suppose, for example, that H is the cohomology of a differential graded algebra $\mathcal{M}$, or of a minimal model in rational homotopy theory. One seeks a grading compatible not only with the vector spaces H^i , but also with products and differentials. In favorable cases, one obtains a grading of the whole model whose induced grading on cohomology is the weight grading. Such a grading imposes severe restrictions on the rational homotopy type.

6. Let X be smooth. Then $H^n(X)$ is controlled by the part of the square $[0, n] \times [0, n]$ above the second diagonal, in the following sense. The mixed Hodge structure on $H^n(X)$ has nonzero Hodge numbers only for pairs (p, q) in a region dictated by smoothness. In particular the weights range from n to $2n$.

7. More explicitly, for the cohomology groups considered below, the following regions $\mathcal{C}$ in the square $[0, n] \times [0, n]$ control the possible Hodge numbers and weights.

(7.1) If X is proper, $H^n(X)$ is controlled by

$$\{(p, q) \in [0, n] \times [0, n] \mid p + q \leq n\}.$$

(7.2) If X is smooth, $H^n(X)$ is controlled by

$$\{(p, q) \in [0, n] \times [0, n] \mid p + q \geq n\}.$$

(7.3) If $N = \dim X \leq n$, the square $[0, n] \times [0, n]$ may be replaced by the square $[n - N, N] \times [n - N, N]$.

8. Variants.

(8.1) The group with proper supports $H_c^n(X)$ is controlled by the part of the square $[0, n] \times [0, n]$ below the second diagonal. For $N = \dim X \leq n$, this square may again be replaced by $[n - N, N] \times [n - N, N]$. In particular $H_c^{2N}(X)$ is pure of weight $2N$.

(8.2) The “duality” between the proper and smooth cases follows from Poincaré duality. If X is smooth and purely of dimension N , then

$$H^n(X) = \text{Hom}(H_c^{2N-n}(X), H_c^{2N}(X)) = (H_c^{2N-n}(X)^\vee)(-N).$$

This argument extends the results stated for smooth X to the case where X is a rational homology manifold. For such an X , the image of $H_c^n(X)$ in $H^n(X)$ is pure of weight n .

(8.3) If X is assumed only normal, it remains true that $H^1(X)$ is controlled by

$$\{(0, 1), (1, 0), (1, 1)\}.$$

9. Let X be a connected variety, let $x \in X$, and let $\Gamma^{(n)}$ be the largest torsion-free nilpotent quotient of length n of $\pi_1(X, x)$. It is known that sufficiently small finite-index subgroups H of $\Gamma^{(n)}$ have the following property.

(*) There exists a nilpotent Lie algebra of length n , $(\mathcal{G}, [])$, with $\mathcal{G}$ a free $\mathbf{Z}$ -module, such that the Campbell–Hausdorff formula

$$x \circ y = x + y + [x, y]/2 + [x, [x, y]]/12 + [y, [y, x]]/12 + \cdots$$

turns $\mathcal{G}$ into a group isomorphic to H .

Moreover, the Lie algebra

$$\mathcal{G}^{(n)} = \mathcal{G} \otimes \mathbf{Q}$$

depends only on $\Gamma^{(n)}$, not on H .

THEOREM 6 (Morgan). *If X is normal, then $\mathcal{G}^{(n)}$ admits a grading W , in negative degrees, such that $\mathcal{G}^{(n)}$ is generated by elements of degrees -1 and -2 , subject only to relations of degrees -2 , -3 , and -4 , together with the vanishing of n -fold iterated commutators.*

It is more convenient to work with the projective system $\mathcal{G}^{(\infty)}$ of the $\mathcal{G}^{(n)}$. Defining

$$H^i(\mathcal{G}^{(\infty)}) = \varinjlim H^i(\mathcal{G}^{(n)}),$$

one has

$$H^1(\mathcal{G}^{(\infty)}) = H^1(X), \quad H^2(\mathcal{G}^{(\infty)}) \subset H^2(X).$$

Furthermore, a grading E of $\mathcal{M}$ as in §5 defines compatible gradings of $\mathcal{G}^{(\infty)}$, of its cohomology, and of the cohomology of X . By (8.3), respectively (7.1), $H^1(X)$ has weights 1 and 2, and $H^2(X)$ has weights at most 4. Hence $H^1(\mathcal{G}^{(\infty)})$, the dual of the group $\mathcal{G}^{(\infty)}/[\mathcal{G}^{(\infty)}, \mathcal{G}^{(\infty)}]$ of generators of $\mathcal{G}^{(\infty)}$, has weights 1 and 2, while $H^2(\mathcal{G}^{(\infty)})$, the dual of the group of relations among the generators, has weights at most 4. This proves the theorem.

11. For a proper smooth variety, the weight filtration reduces to very little (3.1). The fact that it can be graded in the sense of §4 implies the vanishing of all Massey products. The fact that it can be graded in the sense of §5 even implies that the rational homotopy type of X can be read off from its cohomology algebra; see [?].

12. Hodge theory. The cohomology group $H^i(X)$ of every complex algebraic variety is endowed with a mixed Hodge structure (W, F) . Here W is a filtration of $H^i(X)$ and F is a filtration of

$$H^i(X) \otimes \mathbf{C} = H^i(X, \mathbf{C}),$$

with W and F satisfying suitable axioms; see [?]. The weight filtration is, by definition, this filtration W .

13. l -adic theory. An algebraic variety X is defined by finitely many polynomial equations. The coefficients of these equations generate a subring R of $\mathbf{Q}$ of finite type, with fraction field K , and X is obtained by extension of scalars to $\mathbf{C}$ from a scheme over K , indeed over R . The statements below become true after replacing R by $R[1/f]$, with $f \in R$ sufficiently divisible.

- (13.1) Let l be a prime number. The group $H^*(X) \otimes \mathbf{Q}_l$ identifies with the l -adic cohomology group $H^*(X, \mathbf{Q}_l)$. The latter is defined in a purely algebraic way, and therefore, by transport of structure, carries an action of $\text{Aut}(\mathbf{C}/K)$.
- (13.2) Let $\overline{K}$ be the algebraic closure of K inside $\mathbf{C}$. The action factors through $\text{Gal}(\overline{K}/K)$. It is unramified over $R[1/l]$, that is, it factors through the Galois group of the largest extension $K_{\text{nr}} \subset \overline{K}$ of K unramified over $R[1/l]$.
- (13.3) Let $\mathfrak{m}$ be a maximal ideal of $R[1/l]$, and put $N(\mathfrak{m}) = \#R/\mathfrak{m}$, so that $R/\mathfrak{m}$ is a finite field. To $\mathfrak{m}$ corresponds a conjugacy class of Frobenius substitutions $\varphi_{\mathfrak{m}} \in \text{Gal}(K_{\text{nr}}/K)$, whose inverses are the geometric Frobenius elements

$$F_{\mathfrak{m}} = \varphi_{\mathfrak{m}}^{-1}.$$

THEOREM 7. *For f sufficiently divisible, the following hold.*

- (i) *For every l and every $\mathfrak{m}$, the eigenvalues α of $F_{\mathfrak{m}}$, considered in a suitable finite extension of $\mathbf{Q}_l$, are algebraic integers. For each α , there is an integer $w(\alpha)$ such that all complex conjugates of α have absolute value*

$$N(\mathfrak{m})^{w(\alpha)/2}.$$

- (ii) *Choose a geometric Frobenius $F_{\mathfrak{m}}$, and let ${}_{\mathfrak{m}}W_j$ be the sum of the generalized eigenspaces corresponding to eigenvalues α of $F_{\mathfrak{m}}$ such that $w(\alpha) = j$. The filtration by the ${}_{\mathfrak{m}}W_j$ is independent of $\mathfrak{m}$ and of the choice of $F_{\mathfrak{m}}$; it is rational and its trace on $H^*(X)$ is independent of l . It is the weight filtration.*

PRINCIPLE OF THE PROOF. One expresses the cohomology of X , by means of a spectral sequence, in terms of the cohomology of proper smooth varieties, as in [?]. The spectral sequence converges to a filtration W of $H^*(X)$, which is by definition the weight filtration in Hodge theory. This spectral sequence has an l -adic analogue, so the l -adic extension W_l of W is stable under a Galois group. The Weil conjecture implies that the eigenvalues of $F_{\mathfrak{m}}$ on $\text{Gr}_j^{W_l}$ are algebraic integers with complex absolute values $N(\mathfrak{m})^{j/2}$, and the theorem follows.

15. In the cases considered in (7.1), (7.2), and (7.3), the region $\mathcal{C}$ described there controls not only the weights but also the Hodge numbers and the divisibility of the eigenvalues of the $F_{\mathfrak{m}}$. More precisely, in (7.1), (7.2), and (7.3), a region $\mathcal{C}$ was assigned to a group H , and:

- (a) the nonzero Hodge numbers $h^{p,q}$ of the Hodge structures $\mathrm{Gr}_j^W(H)$ satisfy $(p, q) \in \mathcal{C}$;
- (b) useful only in case (7.3), if the pairs $(p, q) \in \mathcal{C}$ with $p + q = j$ satisfy $p, q \geq k$, then, again for f sufficiently divisible, the eigenvalues of $F_{\mathfrak{m}}$ with complex absolute value $N(\mathfrak{m})^{j/2}$ are divisible by $N(\mathfrak{m})^k$.

One hopes that this is a special case of a general principle; see [?].

16. The following result is clear from the viewpoint of Hodge theory.

(16.1) For j odd, $\dim \mathrm{Gr}_j^W H^n(X)$ is even.

By contrast, I do not yet know how to prove §§2 and 5 except by the l -adic method. For §5, the method of §14 gives an l -adic grading; a trick of Sullivan then implies the existence of rational gradings of the required type.

The functoriality properties of the weight filtration are evident from the l -adic point of view, because Galois commutes with everything that can be defined. It is nevertheless useful to prove them from the Hodge-theoretic point of view, in order to obtain analogous properties for the filtration F . Morgan has obtained many results in this direction, enough to prove a slightly weaker form of §10 by Hodge theory.

17. As §5 suggests, a weight filtration also exists on the homotopy groups, after tensoring with $\mathbf{Q}$, under hypotheses of simple connectedness.

It also exists on groups of vanishing cycles (cf. [?]), and I hope that these filtrations will help us understand them better.

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COMPLEX VECTOR BUNDLES WITH DISCRETE STRUCTURE GROUP

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Every complex vector bundle whose structure group can be reduced to a discrete group, and whose base is a compact polyhedron, becomes trivial after passage to a finite covering of the base.

Theorem. *Let V be a finite-dimensional complex local system of rank n on a compact polyhedron X , and let $\mathcal{V}$ be the corresponding complex vector bundle. There exists a finite surjective covering $\pi : \tilde{X} \rightarrow X$ such that the inverse image of $\mathcal{V}$ on $\tilde{X}$ is trivial as a vector bundle.*

We may assume that X is connected and has a base point p . Once a basis of V_p has been chosen, the local system V corresponds to a homomorphism

$$\rho : \pi_1(X, p) \rightarrow \mathrm{GL}(n, \mathbf{C}).$$

Since $\pi_1(X, p)$ is finitely generated, there is a subring A of $\mathbf{C}$, finitely generated over $\mathbf{Z}$, such that ρ factors through $\mathrm{GL}(n, A)$: choose a finite set of generators (g_i) for $\pi_1(X, p)$ and let A be generated by the matrix coefficients of the g_i . Let m_1 and m_2 be two maximal ideals of A such that the finite fields A/m_1 and A/m_2 have distinct characteristics. We take for $\tilde{X}$ the covering of X corresponding to the subgroup of $\pi_1(X, p)$ consisting of the elements g such that $\rho(g) \equiv 1$ modulo both m_1 and m_2 . Its index divides the order of $\mathrm{GL}(n, A/m_1) \times \mathrm{GL}(n, A/m_2)$. Showing that this $\tilde{X}$ works is equivalent to proving the following proposition.

Proposition. *Suppose that V is defined by a representation $\rho : \pi_1(X, p) \rightarrow \mathrm{GL}(n, A)$, where A is a subring of $\mathbf{C}$ finitely generated over $\mathbf{Z}$. If there are two maximal ideals m_1 and m_2 of A , of distinct residue characteristics, such that ρ is trivial modulo m_1 and modulo m_2 , then $\mathcal{V}$ is trivial.*

Let d and N be integers with $\dim X \leq d$ and N large enough, say $N \geq d/2$. The Grassmannian $G_{n,N}$ of n -planes in $\mathbf{C}^{n+N}$ is a sufficiently good approximation to $B\mathrm{GL}(n, \mathbf{C})$ that

$$[X, G_{n,N}] = [X, B\mathrm{GL}(n, \mathbf{C})],$$

and $\mathcal{V}$ corresponds to a homotopy class of maps $v : X \rightarrow G_{n,N}$. Let X' be the fibration over X whose fiber over $x \in X$ is the space, $(2N+1)$ -connected and hence $(d+1)$ -connected, of linear injections from $\mathcal{V}_x$ into $\mathbf{C}^{n+N}$. Let

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$v' : X' \rightarrow G_{n,N}$ send an injection to its image. The projection $\pi : X' \rightarrow X$ has a section s , unique up to homotopy, and $v = v's$.

$$\begin{array}{ccc} X' & \xrightarrow{v'} & G_{n,N} \\ \uparrow \pi \quad s & \nearrow v & \\ X & & \end{array}$$

The following conditions are equivalent: (a) $\mathcal{V}$ is trivial; (b) $v \sim 0$, i.e. v is homotopic to a constant map; (c) v' is homotopic to zero on the d -skeleton of X' , i.e. for every $w : K \rightarrow X'$ with $\dim K \leq d$ (or just for K the d -skeleton of a triangulation of X'), one has $v'w \sim 0$; (d) the composite

$$v'' : X' \xrightarrow{v'} G_{n,N} \rightarrow \text{cosq}_d(G_{n,N}),$$

where $\text{cosq}_d(G_{n,N})$ is the d -th stage of the Postnikov decomposition of $G_{n,N}$, having the same homotopy groups π_i for $i \leq d$, is homotopic to zero.

The equivalence of (c) and (d) is general for any map v' . The equivalence of (b) and (c) follows from the fact that, for every complex K of dimension $\leq d$, the map $[K, X'] \rightarrow [K, X]$ is an isomorphism.

Since $G_{n,N}$ is simply connected, it follows from the Hasse principle for maps [4, Theorem 3.1] that $v'' \sim 0$ if and only if, for every l , its l -adic completion

$$v''_l : X'^{\wedge}_l \rightarrow \text{cosq}_d(G_{n,N})^{\wedge}_l = \text{cosq}_d((G_{n,N})^{\wedge}_l)$$

is homotopic to zero. The proposition follows from the following lemma.

Lemma. *Suppose V is defined by $\rho : \pi_1(X, p) \rightarrow \text{GL}(n, A)$, where A is a subring of $\mathbf{C}$ finitely generated over $\mathbf{Z}$. If there is a maximal ideal m of A , of residue characteristic different from the prime number l , such that ρ is trivial modulo m , then $v''_l \sim 0$.*

We may suppose that X is a union of faces of the simplex $\Delta \subset \mathbf{R}^M$ spanned by the basis vectors. For each face σ of Δ , let $\sigma_{\mathbf{R}}$ be the affine subspace spanned by σ , and let $\sigma_{\mathbf{Z}}$ be the affine subvariety of affine space $\mathbf{A}_{\mathbf{Z}}^M$ over $\mathbf{Z}$ whose real points are $\sigma_{\mathbf{R}}$. Let $X_{\mathbf{Z}}$ be the union of the $\sigma_{\mathbf{Z}}$ for $\sigma \subset X$. This is a scheme over $\mathbf{Z}$; for any ring B , denote by X_B the scheme over B obtained from it by extension of scalars. The inclusion of X into $X_{\mathbf{Z}}(\mathbf{C}) \subset \mathbf{C}^M$ is a homotopy equivalence. For $l \neq p$, the mod l cohomology of X , or of $X_{\mathbf{Z}}(\mathbf{C})$, agrees with the mod l etale cohomology of $X_{\overline{\mathbf{F}}_p}$. By Mayer-Vietoris, this follows from the corresponding fact for affine space (SGA 4, XV 2.2).

The representation ρ gives a vector bundle $\mathcal{V}_A$ on X_A . The bundle $\mathcal{V}$ is obtained by extension of scalars to $\mathbf{C}$, and its reduction modulo m is trivial. The definition of X' is purely algebraic, and gives a diagram of schemes over A :

$$X_A \xleftarrow{s_A} X'_A \xrightarrow{v'_A} (G_{n,N})_A.$$

Use a subscript 1 for the analogous objects attached to the unit representation of $\pi_1(X, p)$ in $\mathrm{GL}(n, A)$. The morphisms of schemes v'_A and v'_{1A} become isomorphic after reduction modulo m . If k is an algebraic closure of A/m , and if B is the strict Henselization of A at k , then there is a chain of isomorphisms between morphisms of l -completed homotopy types:

$$v'_l \sim (v'_C)_l \sim (v'_B)_l \sim (v'_k)_l \sim (v'_{1k})_l \sim (v'_{1B})_l \sim (v'_{1C})_l \sim v'_{1l}.$$

Therefore

$$v''_l \sim v''_{1l} \sim 0.$$

In this chain, the two extreme terms are completions of morphisms between ordinary spaces [4, Section 3]; the other terms are completions of morphisms between étale homotopy types [1].

Remarks. (a) If N is large and $2p \geq d$, the map induced by Chern classes

$$\mathrm{cosq}_d(G_{n,N}) \rightarrow \mathrm{cosq}_d\left(\prod_1^n K(\mathbf{Z}, 2i)\right)$$

is an isomorphism after localization at p . One may deduce that the hypothesis of the proposition can be replaced by the following one: ρ is trivial modulo a maximal ideal m of residue characteristic p such that $H_*(X)$ has no p -torsion and $2p \geq \dim X$.

(b) In (a), and in the proposition, one may replace “ ρ is trivial modulo m ” by “modulo m , the image of ρ is contained in the unipotent radical of a Borel subgroup.”

Remarks. (c) The theorem strengthens the differential-geometric theorem [3] asserting that the real Chern classes of a vector bundle with integrable connection are trivial. Its proof recalls Grothendieck’s algebraic proof of that fact [2].

(d) The problem was suggested to one of us by [5]; the theorem extends the range of application of computations made there (Theorems C.2 and D.2).

(e) The analogue of the theorem for real vector bundles with discrete structure group in $\mathrm{SL}(n, \mathbf{R})$ is false: the real Euler class may be nonzero already for $n = 2$ and X a Riemann surface.

Problem. Does the real analogue of the theorem hold for bundles whose structure group can be reduced to $\mathrm{SL}(n, \mathbf{Z})$ [6], or for n odd? Is the real Euler class the only obstruction?

Remarks. (f) The technique of reducing modulo two maximal ideals, and a problem of Atiyah about vector bundles, are described in the introduction to [1]. The algebraic classifying map, namely the diagram above, and the methods of [5] used here, solve Atiyah’s problem.

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THE SUPPORT OF THE CHARACTER OF A SUPERCUSPIDAL REPRESENTATION

PIERRE DELIGNE

The character of a supercuspidal representation of a semisimple group over a local field is supported on the union of the compact open subgroups.

1. REVIEW OF MONOGENIC GROUPS

Let V be a finite-dimensional vector space over a field k , and let $g \in \mathrm{GL}(V)$. If k is perfect, the Zariski closure $\langle g \rangle$ of the group generated by g is the product of a multiplicative group $\langle g \rangle_m$ and a unipotent group $\langle g \rangle_{\mathrm{un}}$. If $g = su = us$ is the Jordan decomposition of g into its semisimple and unipotent parts, then

$$\langle g \rangle_m = \langle s \rangle, \quad \langle g \rangle_{\mathrm{un}} = \langle u \rangle.$$

Without any hypothesis on k , define the subgroup $\langle g \rangle_m$ of $\langle g \rangle$ as the Zariski closure of the union of the subgroups of $\langle g \rangle$ that are kernels of multiplication by n , where n is prime to the characteristic exponent of k . This definition is compatible with extension of scalars, and for k perfect it is equivalent to the preceding one. Over an algebraic closure $\bar{k}$ of k , the character group of $\langle g \rangle_m$ is the subgroup X of $\bar{k}^*$ generated by the eigenvalues of g , and

$$s \in \langle g \rangle_m(\bar{k}) = \mathrm{Hom}(X, \bar{k}^*)$$

corresponds to the identity inclusion of X .

Assume that k is a complete valued field, with uniformizer π . Let v be the valuation of k , with values in $\mathbf{Q}$, normalized by $v(\pi) = 1$, and let $n > 0$ be an integer such that $nv(X) \subset \mathbf{Z}$. The homomorphism $nv(\chi) : X \rightarrow \mathbf{Z}$ transposes to a homomorphism

$$m_g : \mathbf{G}_m \rightarrow \langle g \rangle_m.$$

For a linear algebraic group G over k and an element $g \in G(k)$, the preceding constructions may be repeated after choosing an embedding $G \subset \mathrm{GL}(V)$. The homomorphism $m_g : \mathbf{G}_m \rightarrow \langle g \rangle_m \subset G$ does not depend on the embedding, once n is fixed. For every linear representation W of G , it defines an action of $\mathbf{G}_m$ on W , equivalently a grading of W (SGA 3, I, 4.7.3). The part of degree d is the sum of the generalized eigenspaces of g corresponding to eigenvalues $\alpha \in \bar{k}$ such that $nv(\alpha) = d$.

Assume that G is reductive. Every homomorphism $m : \mathbf{G}_m \rightarrow G$ defines two opposite parabolic subgroups of G , namely the parabolics contracted

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and expanded by m . Let P_g^+ be the parabolic contracted by m_g , or by g . It has the following equivalent characterizations:

- (a) An element $x \in G(\bar{k})$ lies in P_g^+ , respectively in its unipotent radical U_g^+ , if and only if the morphism

$$\lambda \mapsto m_g(\lambda)xm_g(\lambda)^{-1} : \mathbf{G}_m \rightarrow G$$

extends to $\mathbf{G}_a$, respectively if this extension has limit e as $\lambda \rightarrow 0$. Equivalently, $\lim_{n \rightarrow \infty} g^n x g^{-n}$ exists, respectively equals e .

- (b) $\text{Lie } P_g^+$, respectively $\text{Lie } U_g^+$, is the sum in $\text{Lie } G$ of the generalized eigenspaces of $\text{ad } g$ whose eigenvalues have valuation ≥ 0 , respectively > 0 .

Put $P_g^- = P_{g^{-1}}^+$, the parabolic expanded by g and contracted by $m_{g^{-1}} = m_g^{-1}$, and let U_g^- be its unipotent radical. The parabolics P_g^+ and P_g^- are opposite, and put $M_g = P_g^+ \cap P_g^-$.

2. STATEMENT AND PROOF OF THE THEOREM

Assume that k is locally compact and that G is reductive. Let τ be the projection from G to its adjoint group G^{ad} . An admissible representation $(\mathcal{H}, \rho)$ of $G(k)$ is *supercuspidal* if its matrix coefficients are supported on inverse images of compact subsets of $G^{\text{ad}}(k)$.

If K is a compact subgroup of $G(k)$, let dk be the normalized Haar measure on K and let $[K]$ be its image in the algebra of compactly supported distributions on $G(k)$. The operator $\rho(K) = \int_K \rho([K])$ is the projector from $\mathcal{H}$ onto the subspace $\mathcal{H}^K$ of K -invariants. Saying that ρ is supercuspidal is equivalently saying that, for every compact open subgroup K , one has $\rho(K)\rho(g)\rho(K) = 0$ for $\tau(g) \in G^{\text{ad}}$ outside a suitable compact set depending on K . Let G_c^{ad} denote the union of the compact subgroups, or equivalently of the compact open subgroups, of $G^{\text{ad}}(k)$.

Theorem. *If $(\mathcal{H}, \rho)$ is an admissible supercuspidal representation of $G(k)$, then its character is supported on $\tau^{-1}(G_c^{\text{ad}})$.*

Choose local coordinates near e : a lattice $L \subset \text{Lie}(G)$ and an analytic isomorphism α from L to a neighborhood of e , such that

- (1) $\alpha(0) = e$, and the differential of α at 0 is the identity on $\text{Lie}(G)$.

Let $g \in G(k)$. We choose L so that

- (2) $L = (L \cap \text{Lie } U_g^+) \oplus (L \cap \text{Lie } M_g) \oplus (L \cap \text{Lie } U_g^-)$. Write this as $L^+ \oplus L^0 \oplus L^-$.

- (3) $\text{ad } g(L^+) \subset L^+$, $\text{ad } g(L^0) = L^0$, and $\text{ad } g^{-1}(L^-) \subset L^-$.

Since P_g^+ and P_g^- are stable under $\text{ad } g$, condition (3) may also be written as

- (3') $\text{ad } g(L \cap \text{Lie } P_g^+) \subset L$ and $\text{ad } g^{-1}(L \cap \text{Lie } P_g^-) \subset L$.

Denote by an index i the intersection with $G_i = \alpha(\pi^i L)$. Under hypotheses (2) and (3), there is i_0 such that, for $i \geq i_0$,

- (4) G_i is a subgroup of $G(k)$, and $G_i = U_{gi}^+ M_{gi} U_{gi}^-$;
- (5) $\text{Int } g(U_{gi}^+) \subset U_{gi}^+$, $\text{Int } g(M_{gi}) = M_{gi}$, and $\text{Int } g^{-1}(U_{gi}^-) \subset U_{gi}^-$.

This follows from the decomposition

$$U_g^+ \times M_g \times U_g^- \hookrightarrow G, \quad (u^+, m, u^-) \mapsto u^+ m u^-,$$

of a neighborhood of e .

Moreover, after increasing i_0 if necessary, there is a neighborhood V of g such that (2)–(5) remain true after replacing g by any $g' \in V$. For (2) and (3'), one uses the continuous dependence of P_g^+ and $\text{ad } g$ on g ; for (4) and (5), one uses the analytic dependence of m_g , and hence of $P_g^\pm$, on g , so that in the decomposition $x = u^+ m u^-$ the components u^+ , m , and u^- are analytic functions of the pair (x, g) .

Lemma. *Assume (1)–(5), let $i \geq i_0$, and suppose $\tau(g) \notin G_c^{\text{ad}}$. Then $\rho(G_i)\rho(g)\rho(G_i)$ is nilpotent.*

For the big-cell decomposition $U_g^+ \times M_g \times U_g^- \rightarrow G$, Haar measures satisfy $dg = \lambda(m) du^+ dm du^-$, where λ is a character of M_g . On every compact subgroup of M_g , $\lambda = 1$, so

$$[G_i] = [U_{gi}^+] \star [M_{gi}] \star [U_{gi}^-].$$

We prove by induction on N that

$$([G_i]g[G_i])^N = [G_i]g^N[G_i]$$

in the algebra of compactly supported distributions on G , writing g for δ_g . Indeed,

$$\begin{aligned} ([G_i]g^N[G_i])([G_i]g[G_i]) &= [G_i]g^N[G_i]g[G_i] \\ &= [G_i]g^N[U_{gi}^+][M_{gi}][U_{gi}^-]g[G_i] \\ &= [G_i][g^N U_{gi}^+ g^{-N}][g^N M_{gi} g^{-N}]g^N g[g^{-1} U_{gi}^- g][G_i] \\ &= [G_i]g^{N+1}[G_i]. \end{aligned}$$

If $\tau(g) \notin G_c^{\text{ad}}$, then the set of $\tau(g^N)$, $N \geq 1$, is not relatively compact. Since ρ is supercuspidal, there is an N such that

$$(\rho(G_i)\rho(g)\rho(G_i))^N = \rho(G_i)\rho(g^N)\rho(G_i) = 0.$$

Proof of the theorem. Let $g \in G(k)$ be such that $\tau(g) \notin G_c^{\text{ad}}$. Choose L , α , i_0 , and V as above, with $gG_{i_0} \subset V$. Let Θ be the character of ρ . For every $g' \in gG_i$ and every $i \geq i_0$, we have

$$\Theta(g'[G_i]) = \Theta([G_i]g'[G_i]) = \text{Tr}(\rho(G_i)\rho(g')\rho(G_i)) = 0$$

by the lemma. Since the subgroups G_i form a fundamental system of neighborhoods, Θ is zero on gG_i ; hence its support does not contain g . $\square$

Remarks. (1) The same theorem, with the same proof, holds for an analytic group that is a finite étale cover of an open subgroup of G . The hypothesis becomes: the image in G^{ad} of the support of the matrix coefficients is relatively compact. The conclusion becomes: the image in G^{ad} of the support of Θ lies in G_c^{ad} .

(2) Let p be the residue characteristic of k . The proof uses Haar measures $[K]$ in an essential way only for small K , hence in particular for a p -group. Therefore the theorem and its proof also hold for representations with coefficients in any field of characteristic $\neq p$.

Once a Haar measure dg on G is chosen, the character Θ of ρ is the distribution on G whose value on a test function φ is

$$\text{Tr} \left(\int \varphi(g) \rho(g) dg \right).$$

More intrinsically, it is the generalized function, i.e. the linear form on the space of measures $\varphi(g) dg$ with φ locally constant of compact support and dg a Haar measure, given by the same formula.

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GROUP THEORY

Non-residually finite central extensions of arithmetic groups. Note (*) by **Pierre Deligne**,
Member of the Academy.

A variant of the known methods for treating the congruence subgroup problem gives results such as the following. Let $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ be the inverse image of $\mathrm{Sp}(2n, \mathbf{Z})$ in the universal covering of $\mathrm{Sp}(2n, \mathbf{R})$; this is a central extension of $\mathrm{Sp}(2n, \mathbf{Z})$ by

$$\pi_1 \mathrm{Sp}(2n, \mathbf{R}) = \mathbf{Z}.$$

If $n \geq 2$, then $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ is not separated for the topology of subgroups of finite index. More precisely, every subgroup of finite index in $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ contains $2\pi_1 \mathrm{Sp}(2n, \mathbf{R})$.

1. S -arithmetic groups. Let G be an almost simple simply connected isotropic group over a global field F . To simplify the statements, we shall also assume that $G(F)$ is perfect, that is, equal to its commutator subgroup; this is the case except possibly when G is a trialitarian form of D_4 . Let S be a finite set of places of F , containing all the archimedean places, and suppose that

$$(i) \quad \sum_{v \in S} \mathrm{rank}(G/F_v) \geq 2.$$

For each place v of F , put $G_v = G(F_v)$. For each finite set T of places, put similarly

$$G_T = \prod_{v \in T} G_v, \quad G_{T'} = \prod_{v \notin T} G_v$$

where the second product is restricted. Thus, for $\mathbf{A}$ the adèle ring of F ,

$$G(\mathbf{A}) = G_T \times G_{T'}.$$

A subgroup of $G(F)$ is called S -congruence if it is the inverse image of a compact open subgroup of G_S . It is called S -arithmetic if it is commensurable with an S -congruence subgroup, equivalently if it contains, with finite index, a subgroup of finite index in an S -congruence subgroup. The S -congruence subgroups, respectively the S -arithmetic subgroups, form a fundamental system of neighborhoods of the identity for group topologies $\mathcal{T}(c)$ and $\mathcal{T}(a)$ on $G(F)$. For a fixed S -congruence subgroup Γ , one obtains the same fundamental systems by considering only the S -congruence subgroups normal in Γ , respectively only the normal subgroups of finite index in Γ . This ensures the existence of the completions

$$G(F)^\wedge(c), \quad G(F)^\wedge(a)$$

of $G(F)$ for $\mathcal{T}(c)$ and $\mathcal{T}(a)$. The topology $\mathcal{T}(c)$ is the topology induced from G_S . Since $G(F)$ is dense in G_S by strong approximation, one has

$$G(F)^\wedge(c) = G_S.$$

The completion $G(F)^\wedge(a)$ is an extension of G_S by the profinite group

$$C(S, G) = \varprojlim \Gamma / \Delta,$$

where the inverse limit is over pairs $\Delta \subset \Gamma$ with Δ normal of finite index in Γ and Γ S -congruence.

For G of rank at least 2, Raghunathan ⁽¹⁾ proved that

$$(ii) \quad G(F)^\wedge(a) \text{ is a central extension of } G(F)^\wedge(c) = G_S,$$

and he finishes checking that the same conclusion holds under the sole hypothesis (i).

Let $\widetilde{G}_S$ be a topological central extension of G_S by a discrete group A , and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in $\widetilde{G}_S$. A subgroup of $\widetilde{G(F)}$ is called S -congruence if it is the inverse image of an S -congruence subgroup of $G(F)$, and is called S -arithmetic if it is commensurable with an S -congruence subgroup. We again denote the corresponding topologies by $\mathcal{T}(c)$ and $\mathcal{T}(a)$, and the separated completions by

$$\widetilde{G(F)}^\wedge(c), \quad \widetilde{G(F)}^\wedge(a).$$

Again $\widetilde{G(F)}^\wedge(c) = G_S$, while $\widetilde{G(F)}^\wedge(a)$ is an extension of G_S by a profinite group $\widetilde{C}(S, G)$, itself an extension of $C(S, G)$ by

$$\widehat{A} = \varprojlim A/(A \cap \Gamma) \quad (\Gamma \text{ } S\text{-arithmetic}).$$

Proposition 1. *If condition (ii) holds, then $\widetilde{G(F)}^\wedge(a)$ is a central extension of $\widetilde{G(F)}^\wedge(c) = G_S$.*

Recall that, if a group $\widetilde{H}$ is a central extension of a group H , then the commutator map $\widetilde{H} \times \widetilde{H} \rightarrow \widetilde{H}$ factors through a map, still denoted $(,) : H \times H \rightarrow \widetilde{H}$. In particular, since $\widetilde{G(F)}^\wedge(a)$ is a central extension of $G(F)^\wedge(a)$ by $\widehat{A}$, the commutator map

$$\widetilde{C}(S, G) \times \widetilde{G(F)}^\wedge(a) \rightarrow \widetilde{G(F)}^\wedge(a)$$

factors through $C(S, G) \times G(F)^\wedge(a)$. By (ii), it factors through a bimultiplicative map

$$C(S, G) \times G(F)^\wedge(a) \rightarrow \widehat{A}.$$

Since $G(F)$ is perfect and dense in $G(F)^\wedge(a)$, this map is trivial.

More concretely, the proposition says that, for every S -arithmetic subgroup Δ , the following hold.

(1a) For every conjugate Δ^g of Δ , with $g \in \widetilde{G(F)}$, there exists an S -congruence subgroup Γ such that

$$\Gamma \cap \Delta = \Gamma \cap \Delta^g.$$

(1b) $\widetilde{G(F)}$ acts trivially on the finite group

$$C_\Delta = \varprojlim \Gamma/(\Gamma \cap \Delta).$$

These conditions are also equivalent to

(1c) For every $g \in G(F)$, there exists an S -congruence subgroup Γ such that $(g, \Gamma) \subset \Delta$.

One has $\widetilde{C}(S, G) = \varprojlim C_\Delta$.

2. Central extensions of adelic groups. From now on we assume that

(iii) $\widetilde{G}_S$ is perfect.

The most interesting case would be that in which $\widetilde{G}_S$ is the universal central extension of G_S by a discrete group, the “universal covering” of G_S , but we do not want to assume the existence of such a universal extension.

A topological central extension E of $G(\mathbf{A})$ by a discrete group X is determined by the extensions E_S and $E_{S'}$ of G_S and $G_{S'}$ by X obtained by inverse image in E . Since G_S is perfect, one has

$$E \simeq (E_S \times E_{S'}) / \{(x, -x) \mid x \in X\}.$$

If the extension E_S is dominated by $\widetilde{G}_S$, that is, if it is obtained from $\widetilde{G}_S$ by pushout through a map $f : A \rightarrow X$, then f is unique, and giving E is the same as giving $E_{S'}$ together with f ; this is an equivalence of categories. In this case

$$E \simeq \widetilde{G}_S \times E_{S'} / \{(a, -f(a)) \mid a \in A\}.$$

Let $\mathcal{L}$ be the category of topological central extensions of $G(\mathbf{A})$ by a finite group, split above $G(F)$, and such that E_S is dominated by $\widetilde{G}_S$:

$$\begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E & \longrightarrow & G(\mathbf{A}) \longrightarrow 1 \\ & & & & \downarrow & & \\ & & & & G(F) & & \end{array}$$

It would amount to the same to require only that the extension be splittable over $G(F)$, since $G(F)$ is perfect and a splitting, if it exists, is unique. The category $\mathcal{L}$ is cofiltered. For (E, s) in $\mathcal{L}$, the closure in E of

$$(\text{image of } \widetilde{G}_S) \cdot sG(F)$$

maps to $G(\mathbf{A})$; it maps to a closed subset, because X is finite, and it contains $G_S G(F)$, which is dense by strong approximation. Thus the full subcategory L of $\mathcal{L}$ consisting of those (E, s) for which $(\text{image of } \widetilde{G}_S) \cdot sG(F)$ is dense in E is cofinal. Since there is at most one arrow between two objects of L , it is just a filtered ordered set.

Proposition 2. *Under hypotheses (ii) and (iii), there is a natural isomorphism between $\tilde{C}(S, G)$ and the inverse limit, over $\mathcal{L}$ or over L , of the groups X . Via this isomorphism, the natural map from A to $\tilde{C}(S, G)$ is the inverse limit of the maps $-f : A \rightarrow X$, where f is such that E_S is obtained from $\tilde{G}_S$ by pushout through f .*

Let E be a topological central extension of $G(\mathbf{A})$ by a discrete group X . To give a splitting s of E above $G(F)$ is to give an isomorphism between the extension of $G(F)$ by X obtained from E_S , and the opposite of the extension obtained from $E_{S'}$. If E_S is dominated by $\tilde{G}_S$ and f is as above, this is the same as giving a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E_{S'} & \longrightarrow & G_{S'} \longrightarrow 1 \\ & & \uparrow -f & & \uparrow i & & \uparrow \\ 1 & \longrightarrow & A & \longrightarrow & \widetilde{G(F)} & \longrightarrow & G(F) \longrightarrow 1. \end{array}$$

It follows that $\mathcal{L}$ is equivalent to the category $\mathcal{L}'$ of commutative diagrams

$$(1) \quad \begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E_{S'} & \xrightarrow{q} & G_{S'} \longrightarrow 1 \\ & & & & \uparrow i & & \uparrow \\ & & & & \widetilde{G(F)} & \longrightarrow & G(F) \end{array}$$

where $E_{S'}$ is a topological central extension of $G_{S'}$ by a finite group. One reconstructs E as

$$\tilde{G}_S \times E_{S'} / \{(a, t(a)) \mid a \in A\},$$

and the splitting over $G(F)$ by

$$s(g) = (\tilde{g}, \tilde{t}\tilde{g}) \bmod (a, t(a)),$$

for $\tilde{g} \in \widetilde{G(F)} \subset \tilde{G}_S$ mapping to $g \in G(F)$. The subcategory L corresponds to the subcategory L' of diagrams (1) in which t has dense image.

Let $(E_{S'}, t)$ be in L' , and let $\mathcal{T}$ be the topology on $\widetilde{G(F)}$ induced by that of $E_{S'}$. If K is a compact open subgroup of $E_{S'}$ disjoint from X , then the intersections of $\Delta = t^{-1}K$ with the S -congruence subgroups form a fundamental system of neighborhoods for $\mathcal{T}$. For Γ an S -congruence subgroup defined by a compact open subgroup $K_1 \subset qK$, one moreover has

$$(2) \quad X \simeq XK/K \simeq q^{-1}K_1/(K \cap q^{-1}K_1) \simeq \Gamma/(\Delta \cap \Gamma) \simeq C_\Delta.$$

Since X is finite, Δ is S -arithmetic. Conversely, Proposition 1 ensures that a fundamental system of S -arithmetic groups is obtained in this way. The filtered ordered set L' is identified with the set of germs, for the topology of S -congruence subgroups, of S -arithmetic groups, and the isomorphism of Proposition 2 is defined as the inverse limit of the isomorphisms (2).

3. Quasi-split groups. Let μ be the group of roots of unity of F . For each non-complex place v of F , let μ_v be the group of roots of unity in F_v ; for a complex v , put $\mu_v = \{1\}$. Deodhar proved that, for G quasi-split, there exists a universal topological central extension $\tilde{G}_v$ of G_v by a discrete group, the “universal covering”, and that for v not real its kernel $\pi_1(G_v)$ is canonically a quotient of μ_v . I have completed this result by showing that in fact $\pi_1(G_v) = \mu_v$. For v real, one has $\pi_1(G_v) = \mathbf{Z}$ or $\mathbf{Z}/2$. In all cases, μ_v occurs as a quotient of $\pi_1(G_v)$.

For almost all v , the extension $\tilde{G}_v$ splits over the natural maximal compact subgroup of G_v . This makes it possible to define the restricted product of the $\tilde{G}_v$, an extension of $G(\mathbf{A})$ by the direct sum of the $\pi_1(G_v)$. Push it out by

$$R : \bigoplus \pi_1(G_v) \rightarrow \bigoplus \mu_v \hookrightarrow \mu, \quad \sigma((x_v)) = \prod_v x_v^{[\mu_v : \mu]}.$$

One obtains a central extension of $G(\mathbf{A})$ by μ , and it follows from Deodhar ⁽²⁾ and C. Moore ⁽³⁾ that this is the universal topological central extension of $G(\mathbf{A})$ by a discrete group which splits over $G(F)$:

$$(3) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \widetilde{G(\mathbf{A})} & \longrightarrow & G(\mathbf{A}) \longrightarrow 1 \\ & & & & \searrow & \uparrow & \\ & & & & & G(F) & \end{array}$$

Take $\widetilde{G}_S$ to be the universal covering of G_S , and write R_S for the restriction of R to

$$\pi_1 G_S = \prod_{v \in S} \pi_1 G_v.$$

Proposition 2 identifies the completion $\widetilde{G(F)}^\wedge(a)$ of $\widetilde{G(F)}$ with the inverse image $\widetilde{G(\mathbf{A})}_S$ of $G_S = \widetilde{G(F)}^\wedge(c)$ in (3), and gives an exact commutative diagram

$$(4) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \widetilde{G(\mathbf{A})}_S & \longrightarrow & G_S \longrightarrow 1 \\ & & \uparrow -R_S & & \uparrow i & & \uparrow \\ \prod_{v \in S} \pi_1(G_v) & \longrightarrow & \widetilde{G(F)}^\wedge(a) & \longrightarrow & G(F)^\wedge(a) & \longrightarrow & 1. \end{array}$$

In particular:

Theorem 1. *Let G/F be quasi-split, and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in the universal covering of G_S . If*

$$\sum_{v \in S} \text{rank}(G_v) \geq 2,$$

then the intersection of the S -arithmetic subgroups of $\widetilde{G(F)}$ is the kernel of

$$R_S : \prod_{v \in S} \pi_1(G_v) \rightarrow \mu.$$

For $\text{Sp}(2n, \mathbf{Q})$ and $S = \infty$, this is the result announced in the introduction. For $\text{SL}(n, K)$, where K is real quadratic and $S = \infty$, the result obtained generalizes that of J. Millson ⁽⁴⁾.

Problem 1. Does every discrete subgroup of finite covolume in the universal covering of $\text{Sp}(2n, \mathbf{R})$, with $n \geq 2$, contain $2\pi_1 \text{Sp}(2n, \mathbf{R})$? ⁽⁵⁾.

4. The factor system of theta functions. When condition (i) fails, Proposition 1 is no longer available. It is nevertheless still possible to study those S -arithmetic subgroups satisfying a condition of type (1a).

Let S be a nonempty finite set of places, containing all archimedean places, and let $T \supset S$ be such that (G, T) satisfies condition (ii). The interesting case is that in which S consists of a single place at which G has rank 1. Let $\widetilde{G}_S$ again be a topological central extension of G_S by a discrete group, and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in $\widetilde{G}_S$. For an S -arithmetic subgroup Δ of $\widetilde{G(F)}$, consider the condition:

($\star$) there exists a T -arithmetic subgroup U of $\widetilde{G(F)}$ such that, for every conjugate Δ^g of Δ with $g \in U$, there exists an S -congruence subgroup Γ with

$$\Gamma \cap \Delta = \Gamma \cap \Delta^g$$

(U -invariance of the germ of Δ).

Proposition 3. *The completion $\widetilde{G(F)}^\wedge(a^*)$ of $\widetilde{G(F)}$ for the topology of S -arithmetic subgroups satisfying ($\star$) is a central extension of $\widetilde{G(F)}^\wedge(c) = G_S$.*

More concretely, we have to verify that if (U, Δ) satisfies ($\star$), then Δ satisfies (1c). To prove this, we may first shrink U and Δ . In particular, we may suppose $\Delta \subset U$ by replacing Δ with $\Delta \cap U$. Then U acts by conjugation on the finite group

$$C_\Delta^U = \varprojlim (\Gamma \cap U) / (\Gamma \cap \Delta)$$

where Γ runs over the S -congruence subgroups. Replacing U by the kernel of this action, and Δ by $\Delta \cap \Gamma$, we may also suppose that U acts trivially on C_Δ^U .

Since U is T -arithmetic, its completion $U(c) \subset G_T$ is of finite index in a subgroup $G_{T-S} \times L$, where $L \subset G_T$ is compact open. Since G_{T-S} has no open subgroup of finite index, $U(c)$ itself is of the form $G_{T-S} \times L$. Condition $(\star)$ implies that the $\Delta \cap \Gamma$, with Γ S -congruence, form a fundamental system of neighborhoods of the identity for a group topology on U . The corresponding completion $U(\Delta)$ is an extension of $U(c)$ by C_Δ^U , hence a central extension of $G_{T-S} \times L$. Replace L by an open subgroup L_1 over which this extension is trivial, and replace U and Δ by the inverse images of L_1 . Since G_{T-S} is perfect, a central extension of $G_{T-S} \times L$ is determined by its restrictions to the two factors. This reduces us to the case in which the central extension $U(\Delta)$ of $G_{T-S} \times L$ is the inverse image of a central extension $\tilde{G}_{T-S}$ of G_{T-S} by C_Δ^U . By construction, U lifts into $\tilde{G}_{T-S}$ and Δ is the inverse image of a compact open subgroup K of $\tilde{G}_{T-S} \times L$:

$$(5) \quad \begin{array}{ccc} K & \hookrightarrow & \tilde{G}_{T-S} \times L \\ \uparrow & & \uparrow \\ \Delta & \longrightarrow & U \end{array}$$

Let $\widetilde{G(F)}_T$ be the central extension of $G(F)$ obtained as the inverse image of $G(F)$ in

$$\tilde{G}_T = \tilde{G}_S \times \tilde{G}_{T-S},$$

and let $\tilde{U}$ be the lifting (5) of U into $\widetilde{G(F)}_T$. For $g \in G(F)$, Proposition 1 gives $(g, \Gamma_T) \subset \tilde{U}$ for a T -congruence subgroup Γ_T defined by a sufficiently small $L' \subset L$. If the S -congruence subgroup Γ defined by $K' \subset G_{T-S} \times L'$, with inverse image K'' in $\tilde{G}_{T-S} \times L'$, is small enough that $(g, K') \subset K$, then $(g, \Gamma) \subset \Delta$, which proves (1c).

The kernel

$$\tilde{C}^{\sim*}(S, G) = \text{Ker}(\widetilde{G(F)}^{\wedge}(a^*) \rightarrow G(F)^{\wedge}(c))$$

is still described by Proposition 2, with the same proof, and the calculation can be carried out in the quasi-split case, as in no. 3. Here is an application, with $G = \text{SL}(2, \mathbf{Q})$, $S = \{\infty\}$, and $T = \{\infty, p\}$. Let $\widetilde{\text{SL}(2, \mathbf{R})}$ be the double cover of $\text{SL}(2, \mathbf{R})$. The factor systems of theta functions, restricted to arbitrarily small congruence subgroups of $\text{SL}(2, \mathbf{Z})$, may be interpreted as defining a germ τ , for the congruence subgroup topology, of liftings of $\text{SL}(2, \mathbf{Z})$ into $\widetilde{\text{SL}(2, \mathbf{R})}$. This germ is invariant under $\text{GL}(2, \mathbf{Q})$. If one defines a modular form of half-integral weight $k/2$ to be a function on the Poincare upper half-plane which, under the action of a congruence subgroup of $\text{SL}(2, \mathbf{Z})$, transforms like the theta function of a quadratic form in k variables, this amounts to saying that if

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbf{Q})$$

has positive determinant and f is modular of weight $k/2$, then $f(-z^-)$ and

$$(cz + d)^{k/2} f\left(\frac{az + b}{cz + d}\right)$$

are also modular. Conversely, one has:

Proposition 4. *For every prime p , the germ τ is the unique germ of a lifting of $\text{SL}(2, \mathbf{Z})$ into $\widetilde{\text{SL}(2, \mathbf{R})}$ which is invariant under a subgroup of finite index of $\text{SL}(2, \mathbf{Z}[1/p])$.*

The arguments leading to the theorem of no. 3 show that every arithmetic subgroup of $\text{SL}(2, \mathbf{Z})$ whose germ is invariant under a subgroup of finite index of $\text{SL}(2, \mathbf{Z}[1/p])$ is congruence.

Let $\Gamma \subset \text{SL}(2, \mathbf{Z})$ be a congruence subgroup, and let τ_1 and τ_2 be two liftings of Γ into $\widetilde{\text{SL}(2, \mathbf{R})}$. Let Δ be the subgroup of Γ on which $\tau_1 = \tau_2$. If the germs of τ_1 and τ_2 are distinct and invariant under a $\{p\}$ -arithmetic group, then Δ satisfies $(\star)$ and is not a congruence subgroup.

(*) Session of 3 July 1978.

(¹) M. S. RAGHUNATHAN, *Publ. Math. I.H.E.S.* 46 (1976), 107–162.

- (²) V. V. DEODHAR, *On central extensions of rational points of algebraic groups* (to appear).
- (³) C. C. MOORE, *Publ. Math. I.H.E.S.* 35 (1968), 5–70.
- (⁴) J. MILLSON, *Real Vector Bundles with Discrete Structure Groups* (to appear).
- (⁵) I learn from Mostow that such a subgroup Γ at least contains a subgroup of finite index in $\pi_1 \mathrm{Sp}(2n, \mathbf{R})$: one checks that the projection of Γ into $\mathrm{Sp}(2n, \mathbf{R})$ is discrete; otherwise the Lie algebra of its closure would be nontrivially central.

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Values of Abelian L -functions at Negative Integers over Totally Real Fields

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§0 Introduction

The purpose of this paper is to interpolate p -adically the values at negative integers of abelian L -functions for totally real fields, by using the method of Hilbert modular forms. We recall that the p -adic study of L -values by means of p -adic modular forms was initiated by Serre in his Antwerp paper [33] and was suggested by Siegel's use of Eisenstein series to prove the rationality of L -values [38]. Serre used modular forms of one variable in his theory, and constructed p -adic zeta functions, as well as p -adic L -functions for powers of the Teichmüller character, over totally real fields. Soon after, Katz (unpublished) and Queen [27] observed that Serre's theory applied more generally to the construction of p -adic L -functions attached to abelian characters over totally real fields. Here the point was to introduce p -adic modular forms with level, i.e., on $\Gamma_0(N)$. Already in Serre's paper, it was clear that the use of forms of one variable was insufficient for a complete theory. For example, one could not rule out the possibility of a pole at $s = 1$ for the p -adic L -functions attached to certain non-trivial powers of the Teichmüller character. At the close of [33], Serre suggested that one develop a theory of p -adic Hilbert modular forms to overcome these difficulties.

This problem was taken up by the first author of this paper, who described in letters to Serre [5] how the required congruences among L -values would follow from a (conjectural) theory of p -adic Hilbert forms. This theory was in turn seen as a consequence of a construction over $\mathbb{Z}$ of certain Hilbert-Blumenthal moduli schemes, whose fibres in characteristic p were further conjectured to be geometrically irreducible. The geometric irreducibility amounted to the surjectivity of a certain Galois representation, and was already known in the case of one-variable modular forms (i.e., with the totally real field taken to be $\mathbb{Q}$) [12].

The second author entered the picture by proving (in 1974) the conjectured irreducibility from the Galois-representation point of view [29]. Strictly speaking, the schemes being proved irreducible did not yet exist.

In 1976, however, Rapoport constructed in his thesis [28] the schemes in question. The work of Rapoport contains the foundational material necessary to the theory of p -adic Hilbert forms, and in particular the all-important q -expansion principle. Thus, among other things, Rapoport gave algebraic meaning to the q -expansion of a Hilbert modular form, so that one could define the q -expansion of a Hilbert modular form over an arbitrary ring R , at an arbitrary unramified cusp over R . The q -expansion principle then states that each q -expansion of a Hilbert modular form F over a ring R completely determines that form, and furthermore that F is a form over a subring R_0 of R if and only if the q -expansion coefficients of F lie in R_0 . Rapoport's thesis completed the program begun by Serre and Deligne and thus removed the last tangible obstacle to the writing of the present paper.

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Before summarizing the contents of this paper, we wish to mention related work by other authors. First of all, the “moduli” approach to modular forms and its application to the construction of p -adic L -functions has been developed in a series of papers by Katz [15–19]. Especially, in [19], Katz uses the theory of p -adic Hilbert modular forms in a construction of “ $2n$ -variable” p -adic L -functions for CM fields. His paper begins with an excellent summary of the theory, on which our §5 is modeled.

Secondly, Barsky [1] and Cassou-Noguès [2] have given alternate approaches to p -adic L -functions over totally real fields, based on the explicit formulas of Shintani [37]. The results of Barsky/Cassou-Noguès and those of the present paper were compared in [30]. The two sets of results are nearly identical; the only difference between them is that we obtain certain “extra” 2-adic divisibilities in the presence of functions with parity, which have not yet been proven by the method which uses Shintani’s formulas.

As we have already suggested, our method for studying L -values is based on a general theorem about p -adic Hilbert modular forms. It states, roughly speaking, that such a form which at one cusp has a p -adically integral q -expansion will have a p -adically integral q -expansion at every cusp. We apply this theorem to certain linear combinations of Eisenstein series and thus obtain integrality statements for corresponding linear combinations of L -values. These integrality statements include, for example, the “axioms” introduced by Coates in [3] and therefore lead in particular to the construction of p -adic L -functions over totally real fields.

For the convenience of the reader, we now briefly state our main results concerning modular forms and integrality of L -values. We then explain how these relate to p -adic L -functions.

q -Expansions of Hilbert Modular Forms

Let K be a totally real field, which for simplicity we choose different from the rational field. Let $r = [K : \mathbb{Q}]$ be its degree. Let

$$\mathfrak{H} = \{\tau \in K \otimes \mathbb{C} \mid \Im \tau \gg 0\}$$

be the usual Hilbert upper half plane associated to K . Let k be a non-negative integer. We define an action $|_k$ of $\mathrm{SL}(2, K \otimes \mathbb{R})$ on the space of $\mathbb{C}$ -valued functions $F(\tau)$ on $\mathfrak{H}$ by the usual formula

$$(F|_k \begin{pmatrix} a & b \\ c & d \end{pmatrix})(\tau) = \mathcal{N}(c\tau + d)^{-k} F\left(\frac{a\tau + b}{c\tau + d}\right),$$

in which $\mathcal{N} : K \otimes \mathbb{C} \rightarrow \mathbb{C}$ denotes the norm.

We now let N denote a positive integer. Let $\Gamma_0(N)$ denote the subgroup of $\mathrm{SL}(2, K)$ consisting of those matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ which satisfy the conditions :

$$a, d \in 1 + N\mathcal{O}, \quad b \in \mathfrak{D}^{-1}, \quad c \in N\mathfrak{D},$$

where $\mathcal{O}$ and $\mathfrak{D}$ denote respectively the integer ring and the different of K . A (Hilbert) modular form of weight k on $\Gamma_0(N)$ is a holomorphic function $F : \mathfrak{H} \rightarrow \mathbb{C}$ such that

$$F|_k M = F$$

for each $M \in \Gamma_0(N)$.

The condition $K \neq \mathbb{Q}$ implies that F is “holomorphic at infinity,” as is well known. Since F is invariant under translations $\tau \mapsto \tau + b$ ($b \in \mathfrak{D}^{-1}$), we may expand F as a Fourier series

$$c(0) + \sum_{\substack{\mu \in \mathfrak{D}^{-1} \\ \mu \gg 0}} c(\mu) q^\mu,$$

where

$$q^\mu = e^{2\pi i \cdot \text{tr}(\mu \cdot \tau)}.$$

This series is the standard q -expansion of F . To obtain the others, it is convenient to introduce the action on F of the group $\text{SL}(2, \mathbb{A})$, where $\mathbb{A}$ is the ring of finite adeles of K . For this, let $\bar{\Gamma}$ be the closure in $\text{SL}(2, \mathbb{A})$ of the group $\Gamma = \Gamma_0(N)$. Using the strong approximation theorem for $\text{SL}(2)$, we write

$$\text{SL}(2, \mathbb{A}) = \bar{\Gamma} \cdot \text{SL}(2, K).$$

Given $M \in \text{SL}(2, \mathbb{A})$, we write $M = M_1 M_2$, with $M_1 \in \bar{\Gamma}$ and $M_2 \in \text{SL}(2, K)$. We then define $F|_k M$ to be $F|_k M_2$. For each invertible element α of $\mathbb{A}$ (i.e., for each “finite idele”) we let

$$F_\alpha = F|_k \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix}.$$

We may write F_α as a Fourier series

$$c(0, \alpha) + \sum_{\substack{\mu \text{ totally positive} \\ \mu \in (\alpha)^2 \mathfrak{D}^{-1}}} c(\mu, \alpha) q^\mu,$$

where the sum is restricted to totally positive elements of K which lie in the square of the ideal (α) of K “generated” by α^2 . We call this series the q -expansion of F at the cusp determined by α . We are interested in comparing the various q -expansions of a form, or of a collection of forms. For example, we recall the following result of Rapoport [28] :

Theorem 0.1 ((0.1)). *Suppose that the q -expansion coefficients of F at one cusp are rational numbers, i.e., that there is an $\alpha \in K^*$ such that $c(\mu, \alpha) \in \mathbb{Q}$ for all μ . Then the q -expansion coefficients of F are rational for each $\alpha \in K^*$.*

We now consider forms F_k ($k \geq 0$) on $\Gamma_0(N)$ of respective weights $0, 1, 2, \dots$, all but finitely many of which are zero. We assume that each form satisfies the rationality condition of (0.1). For each k and each $\alpha \in K^*$, let $F_{k, \alpha}$ be the q -expansion of F_k at the cusp determined by α . We view these q -expansions as formal power series over $\mathbb{Q}$, in the variables q^μ (with $\mu = 0$ or μ totally positive). Let p be a prime. For $\alpha \in K^*$, the p -th component α_p of α is the image of α in $K \otimes \mathbb{Q}_p$; its norm $\mathcal{N}\alpha_p$ is an element of $\mathbb{Q}_p$. When $\alpha \in K^*$, the sum

$$S(\alpha) = \sum_{k \geq 0} \mathcal{N}\alpha_p^k \cdot F_{k, \alpha}$$

is thus a formal power series with coefficients in $\mathbb{Q}_p$.

Theorem 0.2 ((0.2)). *If $S(\alpha)$ has coefficients in $\mathbb{Z}_p$ for one α , then $S(\alpha)$ has coefficients in $\mathbb{Z}_p$ for each α .*

2. We recall that for x an invertible element of K , we denote by (x) the ideal “generated” by x .

Corollary 0.3 ((0.3)). *Let α be such that $S(\alpha)$ has $\mathbb{Z}_p$ coefficients, except possibly for its constant coefficient. Then for all $\beta \in K^*$, $S(\beta)$ again has non-constant coefficients in $\mathbb{Z}_p$, and the difference between the constant coefficients of $S(\alpha)$ and $S(\beta)$ lies in $\mathbb{Z}_p$.*

These assertions are special cases of (5.13), (5.14) as explained in (5.15).

Kummer Congruences

When the F_k are Eisenstein series, (0.3) yields congruences among L -values. We now state the principal such congruence (cf. (8.2)). Let K and p be as above, and let $\mathfrak{f}$ be a non-zero integral ideal of K . Let $G_{\mathfrak{f}}$ be the strict ray class group of K mod $\mathfrak{f}$. Let G be the ray class group of conductor $\mathfrak{f}p^\infty$, corresponding via class field theory to the largest abelian extension of K which is unramified at all finite places of K prime to $\mathfrak{f}p$. When ε is a complex-valued function on $G_{\mathfrak{f}}$, we define as usual

$$L(s, \varepsilon) = \sum_{\mathfrak{a}} \varepsilon(\mathfrak{a}) \mathcal{N}\mathfrak{a}^{-s} \quad (\operatorname{Re} > 1),$$

taking the sum over prime-to- $\mathfrak{f}$ integral ideals of K . One knows that $L(s, \varepsilon)$ may be continued to a meromorphic function on $\mathbb{C}$, holomorphic except for a possible simple pole at $s = 1$. In particular, for each integer $k \geq 1$, values $L(1 - k, \varepsilon)$ are defined.

According to a fundamental theorem of Siegel [38], the association $\varepsilon \mapsto L(1 - k, \varepsilon)$ is rational in the sense that $L(1 - k, \varepsilon)$ is a rational number whenever ε is $\mathbb{Q}$ -valued. This being the case, we define by linearity values $L(1 - k, \varepsilon) \in V$ for each function ε on $G_{\mathfrak{f}}$ with values in a $\mathbb{Q}$ -vector space V . This construction applies especially when V is a field of characteristic zero, for example $\mathbb{Q}_p$.

We now let $\mathcal{N} : G \rightarrow \mathbb{Z}_p^*$ be that continuous character whose value on the image in G of a prime-to- $\mathfrak{f}p$ ideal is its norm. For $c \in G$, $k \geq 1$, and ε a $\mathbb{Q}_p$ -valued function on $G_{\mathfrak{f}}$, we define

$$\Delta_c(1 - k, \varepsilon) = L(1 - k, \varepsilon) - \mathcal{N}c^k \cdot L(1 - k, \varepsilon_c) \in \mathbb{Q}_p,$$

where ε_c is the function $g \mapsto \varepsilon(cg)$ and the product cg is computed in $G_{\mathfrak{f}}$. Let $\varepsilon_1, \varepsilon_2, \dots$ be a sequence of such functions, only finitely many of which are non-zero. For $\mathfrak{x}$ a prime-to- $\mathfrak{f}$ ideal, set

$$\varphi(\mathfrak{x}) = \sum_{k \geq 1} \varepsilon_k(\mathfrak{x}) \mathcal{N}\mathfrak{x}^{k-1} \in \mathbb{Q}_p.$$

Theorem 0.4 ((0.4)). *Suppose that $\varphi(\mathfrak{x}) \in \mathbb{Z}_p$ for each prime-to- $\mathfrak{f}$ ideal $\mathfrak{x}$. Then for all $c \in G$, we have $\Delta \in \mathbb{Z}_p$, where*

$$\Delta = \sum_{k \geq 1} \Delta_c(1 - k, \varepsilon_k).$$

Our main theorem (8.2) is equivalent to (0.4). We prove also some further assertions concerning the case where $p = 2$ and the ε_k satisfy certain parity conditions. Briefly, if $p = 2$ and if ε_k is an odd (resp. even) function when k is odd (resp. even), then $\Delta \in 2\mathbb{Z}_2$. We may furthermore give necessary and sufficient conditions for the divisibility of Δ by 2^r . (See (8.11), (8.12).)

p -adic L -functions

It is now well known that the integrality theorem (0.4) may be used to construct p -adic L -functions [3, 30, 34]. In the above situation, we take $\mathfrak{f}$ to be divisible by each prime of K lying over p . Whenever $\mathfrak{f}'$ is an ideal divisible by $\mathfrak{f}$ and dividing $(\mathfrak{f}p)^n$ for some $n \geq 1$, the ideals of K prime to $\mathfrak{f}'$ are just those prime to $\mathfrak{f}$. Hence for ε a locally constant function on G with values in a $\mathbb{Q}$ -vector space V , we may unambiguously define values $L(1 - k, \varepsilon) \in V$ by selecting an $\mathfrak{f}'$ such that ε factors through $G \rightarrow G_{\mathfrak{f}'}$, and applying the above procedure, with $\mathfrak{f}$ replaced by $\mathfrak{f}'$. We may thus view the functions

$$\varepsilon \mapsto L(1 - k, \varepsilon) \quad (k \geq 1)$$

as $\mathbb{Q}$ -valued distributions on G . For $k \geq 1$ and $c \in G$, the map

$$\mu_{c,k} : \varepsilon \mapsto \Delta_c(1 - k, \varepsilon)$$

is then a $\mathbb{Q}_p$ -valued distribution on G .

From (0.4), we immediately obtain (cf. [30, 4.1])

Theorem 0.5 ((0.5)). *The distribution $\mu_{c,k}$ is in fact a measure with values in $\mathbb{Z}_p$. We have the formula³*

$$\mu_{c,k} = \mathcal{N}^{k-1} \cdot \mu_{c,1}.$$

Following a normalization of Serre, we put $\Delta_c = \mathcal{N}^{-1} \mu_{c,1}$ for $c \in G$. For $k \geq 1$ and for ε locally constant, we have

$$\int \varepsilon \mathcal{N}^{k-1} d\Delta_c = \Delta_c(1 - k, \varepsilon).$$

As in [34], we regard the various measures Δ_c as elements of the ring $\Lambda = \Lambda_G$ of $\mathbb{Z}_p$ -valued measures on G ; this ring may alternately be described as a sort of completed group ring $\mathbb{Z}_p[[G]]$. Choose c so that $1 - c$ is not a zero-divisor in Λ and let

$$\lambda_c = (1 - c)^{-1} \Delta_c$$

in the total quotient ring of Λ . One sees easily that $(1 - c')\lambda_c = \Delta_{c'}$ for $c' \in G$, so that λ_c is a pseudo-measure in the sense of [34]. With k and ε as above, the integral against λ_c of $\varepsilon \mathcal{N}^{k-1}$ is $L(1 - k, \varepsilon)$. We thus obtain a result cited by Serre in his discussion of L -values [34, 3.5]. As Serre notes, one now obtains the usual p -adic L -functions of K (attached to characters whose conductors divide powers of $\mathfrak{f}$) by integrating suitable characters against λ_c . (See also [30, §4].)

The plan of this paper is as follows. §1 begins by recalling the Kummer congruences in the case $K = \mathbb{Q}$. Although we do not prove these congruences here, they may easily be derived from explicit expressions for values $L(1 - k, \varepsilon)$ in terms of Bernoulli polynomials. Our aim is to motivate what follows by reinterpreting the congruences in terms of measures on $\mathbb{Z}$ and on $\mathbb{Q}$. In §2 and §3, we study the analogues A and I of these two spaces for the case of an arbitrary totally real field $K \neq \mathbb{Q}$. Especially, we prove a functional equation for L -functions which involves a somewhat curious Fourier transform for functions on I .

3. In this formula, the right-hand member represents the product of the function $\mathcal{N}^{k-1}$ and the measure $\mu_{c,1}$. Such products are to be distinguished from the (convolution) product of two measures.

In the next two §§, we prove an irreducibility theorem for the Hilbert-Blumenthal moduli problem and give applications to p -adic modular forms. We obtain (0.2), in particular. Then in §6 and §7 we construct Eisenstein series and theta series for the Hilbert group and compute their q -expansions. From (0.3) we can then obtain information about L -values. In the final § we make explicit this information and use it to prove (0.4) and its 2-adic refinements. For this, we work with measures on the space I .

As suggested above, most of the work presented in this paper was in principle completed in 1976. However, the material concerning theta series grew out of work done in 1977–78. The second author wishes to thank the Sloan Foundation and the NSF for financial support and the I.H.E.S. for its continuing hospitality.

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1 Review of Kummer Congruences over $\mathbb{Q}$

Let ε be a complex valued function defined on the set $\mathbb{Z}/f\mathbb{Z}$ of integers mod f . Viewing ε as a function on $\mathbb{Z}$, periodic mod f , we set

$$L(s, \varepsilon) = \sum_{n \geq 1} \varepsilon(n) n^{-s} \quad (1)$$

for $s \in \mathbb{C}$ of real part $\text{Re} > 1$. As is well known, this L -function may be continued to a meromorphic function of s , analytic except for a possible simple pole at $s = 1$. We are interested in its values at the non-negative integral points $s = 1 - k$ with $k \geq 1$.

Define Bernoulli polynomials $B_k(x) \in \mathbb{Q}[x]$, $k \geq 0$, by the formal expansion

$$\frac{ze^{xz}}{e^z - 1} = \sum_{k \geq 0} \frac{B_k(x)}{k!} z^k.$$

Theorem 1.1 ((1.2)). *For $k \geq 1$, we have*

$$L(1 - k, \varepsilon) = -\frac{f^{k-1}}{k} \sum_{t \in \mathbb{Z}/f\mathbb{Z}} \varepsilon(t) B_k\left(\frac{t}{f}\right). \quad (2)$$

Example 1.2. If $k = 1$, the theorem gives

$$L(0, \varepsilon) = - \sum_{t \in \mathbb{Z}/f\mathbb{Z}} \varepsilon(t) \left(\frac{t}{f}\right).$$

In the special case where $\sum \varepsilon(t) = 0$, the value $L(0, \varepsilon)$ is thus the Cesaro sum of the series $\sum \varepsilon(t)$: we have

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^n \varepsilon(t) = - \sum_{t=1}^f \varepsilon(t) \left(\frac{t}{f}\right) = \frac{1}{f} \sum_{t=1}^f t \varepsilon(t).$$

The general formula for $L(0, \varepsilon)$ may be recovered from this special one after one knows that $\zeta(0) = -1/2$, where ζ is the Riemann zeta function. This latter fact is just Euler's formula $\sum_{n \geq 1} 1 = 0$, as we see by rewriting the sum as $1 + 2 \sum_{n \geq 1} 1 = 1 + 2\zeta(0)$.

Discussions of the analytic continuation of $L(s, \varepsilon)$ and its evaluation at negative integers may be found for example in [11 ; I, pp.72–78], [13, 23, 42].

Since the numbers $B_k(\frac{t}{f})$ are rational, (2) may be used as the definition of values $L(1 - k, \varepsilon)$ for ε now a function mod f with values in an arbitrary $\mathbb{Q}$ -vector space V :

$$L(1 - k, \varepsilon) = -\frac{f^{k-1}}{k} \sum_{t \in \mathbb{Z}/f\mathbb{Z}} \varepsilon(t) B_k\left(\frac{t}{f}\right). \quad (3)$$

Furthermore, we can unambiguously define such values for ε a periodic function $\mathbb{Z} \rightarrow V$ (i.e. a function defined mod f for some f) whose modulus of periodicity f is not specified along with ε . [To verify this it suffices to examine the case $V = \mathbb{Q}$ and thus *a fortiori* it is sufficient to treat the case $V = \mathbb{C}$; then one simply observes that the number f does not intervene in (1).] In other terms, (3) defines for a given $k \geq 1$ a $\mathbb{Q}$ -linear map

$$\varepsilon \mapsto L(1 - k, \varepsilon)$$

from (periodic functions $\mathbb{Z} \rightarrow \mathbb{Q}$) to $\mathbb{Q}$, and by tensoring with a $\mathbb{Q}$ -vector space V we obtain

$$(\text{periodic functions } \mathbb{Z} \rightarrow V) \rightarrow V.$$

Since restriction from $\mathbb{Z}$ to $\mathbb{Z}$ identifies the locally constant functions on $\mathbb{Z}$ with the periodic functions on $\mathbb{Z}$, the map $\varepsilon \mapsto L(1-k, \varepsilon)$ may be viewed as a distribution T_k on $\mathbb{Z}$ with values in $\mathbb{Q}$. Such a distribution may alternately be regarded as a function on the compact open subsets U of $\mathbb{Z}$, here explicitly given by the formula

$$T_k(U) = \sum_{\substack{n \in U \\ n \geq 1}} n^{k-1} \cdot n^s \Big|_{s=1-k} = \sum_{\substack{n \in U \\ n \geq 1}} n^{k-1}. \quad (4)$$

(The summation is extended over the positive integers $n \in U$.) From (4) we may deduce immediately the “invariance”

$$T_k(nU) = n^{k-1} T_k(U) \quad (5)$$

for positive integers n . This invariance implies that T_k has a unique extension to a distribution T_k on $\mathbb{Q} = \mathbb{Z} \otimes \mathbb{Q}$ which satisfies the invariance

$$T_k(nU) = n^{k-1} T_k(U) \quad (6)$$

for n a positive rational number and U a compact-open subset of $\mathbb{Q}$. This distribution is again defined by (4), with the summation now over all positive $n \in \mathbb{Q}$.

Another way to view distributions on $\mathbb{Z}$ is to regard them as collections of functions α_f on the groups $\mathbb{Z}/f\mathbb{Z}$ ($f \geq 1$), linked by a suitable compatibility relation (cf. [26]). The value of $\alpha_f(x)$ on the class of a modulo f is the value of the distribution on the open subset $a + f\mathbb{Z}$ of $\mathbb{Z}$. For T_k , if we take a in the range $1 \leq a \leq f$ then (3) states that α_f maps $(a \bmod f)$ to the rational number $-\frac{f^{k-1}}{k} B_k(\frac{a}{f})$. This formula, or else the invariance (5), implies that the maps α_f associated to T_1 are each obtained by composing the natural maps $\mathbb{Z}/f\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z}$ with a single map $\alpha : \mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}$. In the language of Kubert-Lang ([21]), T_1 is thus an ordinary distribution.

Twists

We now introduce “twists.” If c is a positive integer and ε a periodic function on $\mathbb{Z}$ with values in a $\mathbb{Q}$ -vector space, we write ε_c for the function $x \mapsto \varepsilon(cx)$. If we are regarding ε as a function modulo f , then c will typically be taken prime to f . Set

$$\Delta_c(1-k, \varepsilon) = L(1-k, \varepsilon) - c^k L(1-k, \varepsilon_c). \quad (7)$$

Now let $\varepsilon_1, \dots, \varepsilon_t$ be functions mod f with values in a p -adic field $\mathbb{Q}_p$, and let $k_1, \dots, k_t$ be positive integers. The following result summarizes the “generalized Kummer congruences” as considered by Mazur and others (cf. [20, 24, 25]).

Theorem 1.3 ((1.9)). *Suppose for all $n \geq 1$ that we have*

$$\sum_{i=1}^t \varepsilon_i(n) n^{k_i-1} \in \mathbb{Z}_p.$$

Then for each $c \geq 1$, prime to pf , we have

$$\sum_{i=1}^t \Delta_c(1-k_i, \varepsilon_i) \in \mathbb{Z}_p.$$

This elementary theorem may be proved directly from (3). It is the analogue for $K = \mathbb{Q}$ of (0.4). As mentioned in the Introduction, we now wish to rephrase (1.9) in terms of measures.

First, taking a more “distribution-theoretic” point of view, we eliminate the modulus f from the statement of the theorem. For this, it is desirable to allow twisting by c which are simultaneously prime to all moduli f : these will be elements of the group $\mathbb{Z}^* \subset \hat{\mathbb{Z}}$.

So let ε be a locally constant function on $\hat{\mathbb{Z}}$ with values in a $\hat{\mathbb{Q}}$ -module V ($\hat{\mathbb{Q}} = \mathbb{Q} \otimes \hat{\mathbb{Z}}$ is the ring of finite adeles for $\mathbb{Q}$). Then for $c \in \hat{\mathbb{Z}}^*$ we again define ε_c and elements $\Delta_c(1 - k, \varepsilon) \in V$. We regard $V = \mathbb{Q}_p$ as a $\hat{\mathbb{Q}}$ -module via the projection $\hat{\mathbb{Q}} \rightarrow \mathbb{Q}_p$.

The hypothesis of (1.9) may be rephrased as follows : the locally constant functions $\varepsilon_i : \hat{\mathbb{Z}} \rightarrow \mathbb{Q}_p$ and positive integers k_i ($i = 1, \dots, t$) are such that for all $n \in \mathbb{Z}$ we have $\varphi(n) \in \mathbb{Z}_p$ where

$$\varphi(n) = \sum_{i=1}^t \varepsilon_i(n) n^{k_i-1}.$$

Indeed, φ is a continuous function $\hat{\mathbb{Z}} \rightarrow \mathbb{Q}_p$ and is thus $\mathbb{Z}_p$ -valued if and only if its values on positive integers lie in $\mathbb{Z}_p$. The conclusion may be rephrased as follows : for all $c \in \hat{\mathbb{Z}}^*$, we have

$$\sum_{i=1}^t \Delta_c(1 - k_i, \varepsilon_i) \in \mathbb{Z}_p.$$

In fact, one sees immediately that this statement for a given $c \in \hat{\mathbb{Z}}^*$ and the conclusion of (1.9) for a given $d \geq 1$ are equivalent if $c \equiv d \pmod{fp^n}$, where f is a modulus for all the ε_i and where p^n is a common denominator for the (finitely many) numbers

$$L(1 - k_i, \varepsilon_{ic}) \quad (i = 1, \dots, t; c \bmod f).$$

Having eliminated the f , we may also eliminate the p . Namely, the following is trivially equivalent to (1.9).

Theorem 1.4 ((1.9a)). *Let $\varepsilon_1, \dots, \varepsilon_t$ be locally constant functions on $\hat{\mathbb{Z}}$ with values in $\mathbb{Q}$. Let $k_1, \dots, k_t$ be positive integers. Suppose that*

$$\sum_{i=1}^t \varepsilon_i(x) x^{k_i-1} \in \mathbb{Z}$$

for all $x \in \hat{\mathbb{Z}}$. Then for each $c \in \hat{\mathbb{Z}}^$ we have*

$$\sum_{i=1}^t \Delta_c(1 - k_i, \varepsilon_i) \in \mathbb{Z}.$$

We now rephrase this result in terms of measures on $\hat{\mathbb{Z}}$. Let $c \in \hat{\mathbb{Z}}^*$. For each locally constant function $\varepsilon : \hat{\mathbb{Z}} \rightarrow \mathbb{Q}$, the quantity $\Delta_c(0, \varepsilon)$ lies in $\mathbb{Z}$ by (1.9a). The map $\varepsilon \mapsto \Delta_c(0, \varepsilon)$ is thus a measure μ_c on $\hat{\mathbb{Z}}$ with values in $\mathbb{Z}$, so that a quantity $\int \varepsilon d\mu_c \in \mathbb{Z}$ is defined for all continuous functions $\varepsilon : \hat{\mathbb{Z}} \rightarrow \mathbb{Z}$.

In particular, let x be the identity function ($a \mapsto a$) on $\hat{\mathbb{Z}}$, and let ε be locally constant. Then (1.9a) implies that we have

$$\int \varepsilon x^{k-1} d\mu_c = \Delta_c(1 - k, \varepsilon)$$

for all $k \geq 1$. Indeed, to verify this it suffices to check that the difference between the two sides is congruent to 0 mod N , for each $N \geq 1$. Let N be given, and let $\eta : \hat{\mathbb{Z}} \rightarrow \mathbb{Z}$ be a locally constant function which is congruent to x^{k-1} mod N . Because μ_c is a measure we have

$$\int \varepsilon x^{k-1} d\mu_c \equiv \int \varepsilon \eta d\mu_c \pmod{N \cdot \mathbb{Z}}.$$

Thus it suffices to check that $\Delta_c(0, \varepsilon \eta) \equiv \Delta_c(1-k, \varepsilon) \pmod{N}$. This follows from (1.9a) and the congruence $\varepsilon(x)\eta(x) \equiv \varepsilon(x)x^{k-1}$ for $x \in \hat{\mathbb{Z}}$.

Summarizing, we have obtained from the Kummer congruences the following result.

Theorem 1.5 ((1.9b)). *For each $c \in \hat{\mathbb{Z}}^*$, the map $\varepsilon \mapsto \Delta_c(0, \varepsilon)$ is a measure μ_c on $\hat{\mathbb{Z}}$ with values in $\mathbb{Z}$. For all $k \geq 1$, the product $x^{k-1} \cdot \mu_c$ is the distribution $\varepsilon \mapsto \Delta_c(1-k, \varepsilon)$ (which is consequently a measure as well).*

Conversely, this theorem immediately implies (1.9a). For suppose that we are given ε_i , k_i and c as in the hypothesis of (1.9a). Then we have, assuming (1.9b),

$$\sum_{i=1}^t \Delta_c(1-k_i, \varepsilon_i) = \int \left(\sum_{i=1}^t \varepsilon_i x^{k_i-1} \right) d\mu_c \in \mathbb{Z}.$$

In yet another variant, we consider locally constant functions ε with compact support on $\mathbb{Q}$. For these, we define values $L(1-k, \varepsilon)$ as the integrals of ε against the distributions T_k introduced above. Formally,

$$L(1-k, \varepsilon) = \sum_{\substack{n > 0 \\ n \in \mathbb{Q}}} \varepsilon(n) n^{k-1}. \quad (8)$$

Introducing twisted values $\Delta_c(1-k, \varepsilon)$ as in (7), we obtain from the invariance (6) the following integrality properties :

(1.11) For $c \in \hat{\mathbb{Z}}^*$, the map $\mu_c : \varepsilon \mapsto \Delta_c(0, \varepsilon)$ is a $\mathbb{Z}$ -valued measure on $\mathbb{Q}$.

(1.12) For $k \geq 1$, the distribution $x^{k-1} d\mu_c$ on $\mathbb{Q}$ is the map $\varepsilon \mapsto \Delta_c(1-k, \varepsilon)$.

Note that the map x^{k-1} is not bounded when $k > 1$. Hence $x^{k-1} d\mu_c$ will not be a measure on $\mathbb{Q}$ for $k > 1$.

Extra divisibility at 2

Finally, we discuss an “extra divisibility” at 2 which arises formally from parity considerations. For simplicity, we discuss only functions ε supported on $\mathbb{Z}$.

Let δ be the Dirac measure at 0 on $\hat{\mathbb{Z}}$:

$$\delta : \varepsilon \mapsto \varepsilon(0).$$

Lemma 1.6 ((1.13)). $\mu_{-1} = -\delta$.

Démonstration. We must verify that $L(0, \varepsilon) + L(0, \varepsilon_{-1}) = -\varepsilon(0)$. This follows immediately from the formula

$$L(0, \varepsilon) = - \sum_{t=1}^f \varepsilon(t) \left(\frac{t}{f} - \frac{1}{2} \right),$$

which is a special case of (2). □

Corollary 1.7 ((1.14)). For $k > 1$, we have $x^{k-1} \cdot \mu_{-1} = 0$, i.e.,

$$L(1-k, \varepsilon) = (-1)^k L(1-k, \varepsilon_{-1}).$$

Démonstration. We have $x^{k-1} \cdot (-\delta) = 0$. □

Remark 1.8. (1.13) and (1.14) summarize the presence of trivial zeros of L -functions for complex valued characters ε .

Definition 1.9. A function φ on $\hat{\mathbb{Z}}$ with values in a $\mathbb{Q}$ -vector space V is *odd* (resp. *even*) if it satisfies $\varphi(-x) = -\varphi(x)$ (resp. $\varphi(-x) = \varphi(x)$) for $x \in \hat{\mathbb{Z}}$.

From (1.14) we see that if k is even then $L(1-k, \varepsilon) = 0$ if a locally constant ε is odd, whereas $L(1-k, \varepsilon) = 0$ for even functions if $k > 1$ is odd. For $k = 1$, we have $L(1-k, \varepsilon) = 0$ for even ε which vanish at 0 (1.13).

Theorem 1.10 ((1.15)). Let $\varphi : \hat{\mathbb{Z}} \rightarrow \mathbb{Z}$ be an odd function. Then we have

$$\int \varphi d\mu_c \in 2\mathbb{Z}$$

for all $c \in \hat{\mathbb{Z}}^*$.

Démonstration. Let α be a locally constant function which is congruent to $\varphi \pmod{4}$ and which vanishes at 0. Set

$$\alpha_+(x) = \frac{\alpha(x) + \alpha(-x)}{2}, \quad \alpha_-(x) = \frac{\alpha(x) - \alpha(-x)}{2}.$$

Since α_+ is even and vanishes at 0, we have $\int \alpha_+ d\mu_c \equiv 0 \pmod{2\mathbb{Z}}$, from the above discussion. Moreover, we have $\varphi \equiv \alpha_- \pmod{2}$, since

$$2\varphi(x) = \varphi(x) - \varphi(-x) \equiv \alpha(x) - \alpha(-x) = 2\alpha_-(x) \pmod{4}.$$

Hence we have $\int \varphi d\mu_c \equiv \int \alpha_- d\mu_c \pmod{2\mathbb{Z}}$.

These computations show that we may assume that φ is in fact locally constant, say $\pmod{f}$. We make this assumption.

Since φ is odd, we may find a function $g : \mathbb{Z}/f\mathbb{Z} \rightarrow \mathbb{Z}$, vanishing at 0, such that $\varphi(x) = g(x) - g(-x)$. We find

$$\int \varphi(x) d\mu_c = \Delta_c(0, g) - \Delta_c(0, g_{-1}) = 2\Delta_c(0, g) \in 2\mathbb{Z}. \quad \square$$

Corollary 1.11 ((1.16)). Let $\varepsilon_1, \dots, \varepsilon_t$ and $k_1, \dots, k_t$ be as in (1.9a). Suppose further for each $i = 1, \dots, t$ that ε_i and k_i have the same parity (namely : ε_i is even if k_i is even and odd if k_i is odd). Then for all $c \in \hat{\mathbb{Z}}^*$ we have

$$\sum_{i=1}^t \Delta_c(1-k_i, \varepsilon_i) \in 2\mathbb{Z}.$$

2 The Measure Spaces A and I

Let K be a totally real field, which will be fixed for the remainder of this paper. In order to avoid certain technical difficulties later on, we will assume right away that K is different from the rational field $\mathbb{Q}$. We introduce the following notation and conventions, which will be in force for the remainder of this paper :

- r is the degree of K over $\mathbb{Q}$.
- $\mathcal{N}$ is the norm map $K \rightarrow \mathbb{Q}$ or any norm map derived from it, such as the norm map on ideals of K or a map of the type $K \otimes R \rightarrow R$, where R is a $\mathbb{Q}$ -algebra.
- An ideal of K is tacitly understood to be non-zero and fractional (as opposed to integral), unless otherwise described. A conductor is a (non-zero) integral ideal. The group of ideals is denoted by I_0 and the sub-monoid of integral ideals by A_0 .
- We denote the ring of integers of K by $\mathcal{O}$ and its different by $\mathfrak{D}$. The unit ideal (1) is sometimes abbreviated 1.
- For $\alpha \in K$, the symbol $\alpha \gg 0$ means that α is totally positive. For $\alpha \in K \otimes \mathbb{R}$, we write $\alpha > 0$ if the image of α under each $\mathbb{R}$ -algebra homomorphism $K \otimes \mathbb{R} \rightarrow \mathbb{R}$ is positive. Thus, if we number the embeddings $K \hookrightarrow \mathbb{R}$, so that $K \otimes \mathbb{R}$ may be written $\mathbb{R}^r$, then $\alpha > 0$ means that α is an r -tuple of positive numbers.

We let $\mathbb{A}$ denote the ring of adeles of K and write either $\hat{K}$ or $\mathbb{A}_f$ for the ring of finite adeles of K . If $x \in \hat{K}^*$ is a “finite idele” we write (x) for the ideal “generated” by x , namely

$$(x) = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\text{ord}_{\mathfrak{p}}(x_{\mathfrak{p}})},$$

the product being extended over the prime ideals $\mathfrak{p}$ of K . Also, when $\mathfrak{a}$ is an ideal, we write $\hat{\mathfrak{a}}$ for the completion of $\mathfrak{a}$, i.e., the closure of $\mathfrak{a}$ in $\hat{K}$.

Definition 2.1 ((2.1)). Let $\mathfrak{a}$ and $\mathfrak{b}$ be ideals, and let $\mathfrak{f}$ be a conductor. The ideals $\mathfrak{a}$ and $\mathfrak{b}$ are *equivalent* (or *congruent*) mod $\mathfrak{f}$ if the ideal $\mathfrak{a}\mathfrak{b}^{-1}$ may be written as the principal ideal (α) for some totally positive α in the set $1 + \mathfrak{f}\mathfrak{b}^{-1}$.

We write $\sim_{\mathfrak{f}}$ for the equivalence relation introduced above. If $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent mod $\mathfrak{f}$, we write $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$ and sometimes say that $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent mod $\mathfrak{f}$. The following results follow easily from the definition of $\sim_{\mathfrak{f}}$.

- (2.2) Suppose that $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$ and that $\mathfrak{b}$ is an ideal containing $\mathfrak{f}$. Then $\mathfrak{b}$ contains $\mathfrak{a}$ if and only if $\mathfrak{b}$ contains $\mathfrak{a}$.
- (2.3) **Corollary.** Any ideal equivalent mod $\mathfrak{f}$ to an integral ideal is again integral.
- (2.4) If $\mathfrak{a}$ and $\mathfrak{b}$ are integral ideals which are equivalent mod $\mathfrak{f}$, then the greatest common divisors $(\mathfrak{a}, \mathfrak{f})$ and $(\mathfrak{b}, \mathfrak{f})$ are equal.
- (2.5) Let $\mathfrak{a}$ and $\mathfrak{b}$ be ideals and let $\mathfrak{b}$ be an integral ideal. We have $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$ if and only if $\mathfrak{a}\mathfrak{b} \sim_{\mathfrak{f}\mathfrak{b}} \mathfrak{b}\mathfrak{b}$.
- (2.6) The set $A_{\mathfrak{f}}$ of mod $\mathfrak{f}$ classes of integral ideals is a monoid under a multiplication $A_{\mathfrak{f}} \times A_{\mathfrak{f}} \rightarrow A_{\mathfrak{f}}$ induced by multiplication of ideals. The invertible elements are those classes represented by ideals which are relatively prime to $\mathfrak{f}$. Further, the group $G_{\mathfrak{f}}$ of invertible elements of $A_{\mathfrak{f}}$ is precisely the group of strict ray classes of K modulo $\mathfrak{f}$: when $\mathfrak{a}$ and $\mathfrak{b}$ are prime-to- $\mathfrak{f}$ integral ideals, we have $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$ if and only if $\mathfrak{a}$ and $\mathfrak{b}$ represent the same strict ray class mod $\mathfrak{f}$.

Using the above results, and especially (2.5), (2.6), we obtain the following description of $A_{\mathfrak{f}}$. It is a disjoint union of copies of the groups $G_{\mathfrak{f}/\mathfrak{b}}$, $\mathfrak{b}$ running over the divisors of $\mathfrak{f}$. The image of $G_{\mathfrak{f}/\mathfrak{b}}$ in $A_{\mathfrak{f}}$ consists of those classes whose greatest common divisor with $\mathfrak{f}$ is $\mathfrak{f}\mathfrak{b}^{-1}$. This decomposition generalizes the fact that, for $\mathfrak{f} \mid f$, $\mathbb{Z}/f\mathbb{Z}$ is a disjoint union of copies of $(\mathbb{Z}/d\mathbb{Z})^*$, with d running over the divisors of f .

(2.7) Suppose that $\mathfrak{a}$ and $\mathfrak{b}$ are integral ideals divisible by $\mathfrak{f}$. Then $\mathfrak{a}$ is equivalent to $\mathfrak{b}$ mod $\mathfrak{f}$ if and only if $\mathfrak{a}$ is equivalent to $\mathfrak{b}$ mod (1) .

We have already introduced the notation $A_{\mathfrak{f}}$ for the (finite) set of classes of integral ideals mod $\mathfrak{f}$. Now let $I_{\mathfrak{f}}$ be the set of equivalence classes of all ideals mod $\mathfrak{f}$. For each ideal $\mathfrak{b}$ containing $\mathfrak{f}$, let $I_{\mathfrak{f}}^{\mathfrak{b}}$ be the set of classes mod $\mathfrak{f}$ of ideals contained in $\mathfrak{b}$, cf. (2.2).

(2.8) Each set $I_{\mathfrak{f}}^{\mathfrak{b}}$ is finite. Indeed, if $\mathfrak{a}$ is the inverse of an integral ideal $\mathfrak{a}'$, the map on ideals $\mathfrak{x} \mapsto \mathfrak{a}\mathfrak{x}$ induces a bijection $I_{\mathfrak{f}}^{\mathfrak{b}} \xrightarrow{\sim} A_{\mathfrak{f}/\mathfrak{b}\mathfrak{a}}$.

We now allow $\mathfrak{f}$ to vary. Clearly, if $\mathfrak{f}' \mid \mathfrak{f}$ and if $\mathfrak{a} \sim_{\mathfrak{f}'} \mathfrak{b}$, then $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$. There are thus natural maps $I_{\mathfrak{f}'} \rightarrow I_{\mathfrak{f}}$, inducing maps $A_{\mathfrak{f}'} \rightarrow A_{\mathfrak{f}}$ and $G_{\mathfrak{f}'} \rightarrow G_{\mathfrak{f}}$. We define

$$I = \varprojlim I_{\mathfrak{f}}, \quad A = \varprojlim A_{\mathfrak{f}}, \quad G = \varprojlim G_{\mathfrak{f}}.$$

(We note that G no longer has the meaning given it in the Introduction.) In the prohibited case $K = \mathbb{Q}$, these limits become respectively $\mathbb{Q}$, $\hat{\mathbb{Z}}$, and $\hat{\mathbb{Z}}^*$. Also, we recognize G as the Galois group of the maximal abelian extension K^{ab} of K . In what follows, we will gradually establish parallels between A and $\mathbb{Z}$ and between I and $\mathbb{Q}$. A typical element of I will be denoted $(\mathfrak{a}_{\mathfrak{f}}) = \mathfrak{a}$, indicating that $\mathfrak{a}_{\mathfrak{f}}$ is the $\mathfrak{f}$ -component of $\mathfrak{a}$.

For $\mathfrak{b} \in I_0$, we let

$$\begin{aligned} I^{\mathfrak{b}} &= \{(\mathfrak{a}_{\mathfrak{f}}) \in I \mid \mathfrak{a}_{\mathfrak{f}} \subseteq \mathfrak{b} \text{ for all } \mathfrak{f} \in \mathfrak{D}\} \\ &= \{(\mathfrak{a}_{\mathfrak{f}}) \in I \mid \mathfrak{a}_{\mathfrak{f}} \subseteq \mathfrak{b} \text{ for some } \mathfrak{f} \subseteq \mathfrak{b}\}; \end{aligned}$$

the equivalence of the two definitions resulting from (2.2). It is clear that we have

$$I^{\mathfrak{b}_1\mathfrak{b}_2} \subseteq I^{\mathfrak{b}_1} \cap I^{\mathfrak{b}_2}.$$

Proposition 2.2 ((2.9)). *Let $\mathfrak{b}_1$ and $\mathfrak{b}_2$ be ideals. Suppose that $\mathfrak{f}_1 \subseteq \mathfrak{b}_1$ and $\mathfrak{f}_2 \subseteq \mathfrak{b}_2$ are conductors such that*

$$\mathfrak{f}_1\mathfrak{b}_2, \mathfrak{f}_2\mathfrak{b}_1 \subseteq \mathfrak{f}.$$

Then multiplication of ideals induces a map

$$I_{\mathfrak{f}_1}^{\mathfrak{b}_1} \times I_{\mathfrak{f}_2}^{\mathfrak{b}_2} \rightarrow I_{\mathfrak{f}}^{\mathfrak{b}_1\mathfrak{b}_2 + \mathfrak{f}\mathfrak{b}_1\mathfrak{b}_2}.$$

Using (2.9), we find a multiplication

$$I \times I \rightarrow I.$$

Specifically, to calculate a product $\mathfrak{a} \cdot \mathfrak{b}$ mod $\mathfrak{f}$, suppose that $(\mathfrak{a}_{\mathfrak{f}}) \in I^{\mathfrak{b}_1}$, $(\mathfrak{b}_{\mathfrak{f}}) \in I^{\mathfrak{b}_2}$. If $\mathfrak{f}_1 \subseteq \mathfrak{b}_1$ and $\mathfrak{f}_2 \subseteq \mathfrak{b}_2$ are such that $\mathfrak{f}_1\mathfrak{b}_2, \mathfrak{f}_2\mathfrak{b}_1 \subseteq \mathfrak{f}$, then $\mathfrak{a}_{\mathfrak{f}_1} \cdot \mathfrak{b}_{\mathfrak{f}_2}$ gives $\mathfrak{a}\mathfrak{b}$ mod $\mathfrak{f}$. It is clear that $\mathfrak{b}_1 \cdot I^{\mathfrak{b}_2} = I^{\mathfrak{b}_1\mathfrak{b}_2}$ for ideals $\mathfrak{b}_1$ and $\mathfrak{b}_2$. (We write again $\mathfrak{b}$ for the image of an ideal $\mathfrak{b}$ in I .) Namely, $\mathfrak{b}_1 \subseteq \mathfrak{b}_2$ for all $\mathfrak{f}$ sufficiently small if and only if $\mathfrak{b}_1 \subseteq \mathfrak{b}_2$ for all $\mathfrak{f}$ sufficiently small.

We observe, incidentally, that the natural map $I_0 \rightarrow I$ is injective. In fact, if $\mathfrak{a}$ and $\mathfrak{b}$ are ideals which are equivalent modulo a conductor $\mathfrak{f} \subseteq \mathfrak{a} \cap \mathfrak{b}$, then by (2.2) we have $\mathfrak{a} \subseteq \mathfrak{b}$ and $\mathfrak{b} \subseteq \mathfrak{a}$.

Proposition 2.3 ((2.10)). *Let I^* be the set of invertible elements of I under the multiplication. Then $I^* = I_0 \times G$.*

Démonstration. It is immediate from (2.6) that if we have $\mathfrak{a} \cdot \mathfrak{b} = 1$ with $\mathfrak{a}, \mathfrak{b} \in A$, then both $\mathfrak{a}$ and $\mathfrak{b}$ belong to G . In general, if $\mathfrak{a} \cdot \mathfrak{b} = 1$ with $\mathfrak{a}, \mathfrak{b} \in I$, multiply $\mathfrak{a}$ by an ideal (and $\mathfrak{b}$ by its inverse) to have $\mathfrak{a} \in A$. We need not have $\mathfrak{b} \in A$; let $\mathfrak{b}' \in I_0$ be the denominator of $\mathfrak{b}$. Then $\mathfrak{b}' \in I^{*1}$. Calculating $\mathfrak{a} \cdot \mathfrak{b}' \bmod 1$, we find $1 \sim \mathfrak{a}\mathfrak{b}\mathfrak{b}'$, giving that $\mathfrak{a}\mathfrak{b}'$ is divisible by 1. Hence $\mathfrak{a} \in I^{\mathfrak{b}'}$, so that in the expression

$$(\mathfrak{b}'^{-1}\mathfrak{a}) \cdot (\mathfrak{b}'\mathfrak{b}) = 1,$$

both terms in the product lie in A . Thus $\mathfrak{b}'^{-1}\mathfrak{a} \in G$, so $\mathfrak{a} \in I_0 \times G$.

[It is obvious that the intersection $I_0 \cap G$ is trivial because an element $(\mathfrak{a}_{\mathfrak{f}})$ of G is such that $\mathfrak{a}_{\mathfrak{f}} \in I_{\mathfrak{f}}$, $(\mathfrak{a}_{\mathfrak{f}}, \mathfrak{f}) = 1$ for each $\mathfrak{f}$.] $\square$

Proposition 2.4 ((2.11)). *Let $\mathfrak{f}$ be a positive integer, and suppose that $\mathfrak{a}, \mathfrak{b} \in I_0$ are congruent mod $\sim_{\mathfrak{f}}$ (i.e., mod $\sim_{\mathfrak{f}\mathcal{O}}$). Let $\mathfrak{b}$ be a denominator for $\mathfrak{a}$ (and hence for $\mathfrak{b}$). Then*

$$\mathcal{N}\mathfrak{a} \equiv \mathcal{N}\mathfrak{b} \pmod{(\mathcal{N}\mathfrak{b})^{-1}\mathfrak{f}},$$

i.e.,

$$\mathcal{N}\mathfrak{a} - \mathcal{N}\mathfrak{b} \in (\mathcal{N}\mathfrak{b})^{-1}\mathfrak{f} \cdot \mathbb{Z}.$$

Corollary 2.5 ((2.12)). *The norm map $I_0 \rightarrow \mathbb{Q}_{>0}$ extends to a continuous map $\mathcal{N} : I \rightarrow \mathbb{Q}$, such that $\mathcal{N}(\mathfrak{a}\mathfrak{b}) = \mathcal{N}(\mathfrak{a}) \cdot \mathcal{N}(\mathfrak{b})$. We have $\mathcal{N}(\mathfrak{a}) \geq 1$ for $\mathfrak{a} \in A$.*

Proof of (2.11). We may assume that $\mathfrak{b} = 1$. Indeed, under the hypothesis that $\mathfrak{b}$ is a denominator, the ideals $\mathfrak{a}\mathfrak{b}$ and $\mathfrak{b}\mathfrak{b}$ are integral and congruent mod $\mathfrak{f}$ (in fact, mod $\mathfrak{f}\mathcal{O}$). The assertion for $\mathfrak{b} = 1$ gives $\mathcal{N}(\mathfrak{a}\mathfrak{b}) \equiv \mathcal{N}(\mathfrak{b}\mathfrak{b}) \pmod{\mathfrak{f}}$, as desired.

Now with $\mathfrak{a}, \mathfrak{b} \in A_0$, set $\mathfrak{d} = (\mathfrak{f}, \mathfrak{a}) = (\mathfrak{f}, \mathfrak{b})$. We have $\mathfrak{a}\mathfrak{d}^{-1} \sim \mathfrak{b}\mathfrak{d}^{-1} \bmod \mathfrak{f}\mathfrak{d}^{-1}$; let f' be the largest integer dividing the ideal $\mathfrak{f}\mathfrak{d}^{-1}$. Then $\mathfrak{f}$ divides $f' \cdot \mathcal{N}\mathfrak{b}$, so that the assertion

$$\mathcal{N}(\mathfrak{a}\mathfrak{b}^{-1}) \equiv \mathcal{N}(\mathfrak{b}\mathfrak{b}^{-1}) \pmod{f'}$$

implies the proposition. We are thus reduced to the case where $(\mathfrak{f}, \mathfrak{a}) = (\mathfrak{f}, \mathfrak{b}) = 1$. Then, finally, let α be as in the definition of $\sim_{\mathfrak{f}}$ for $\mathfrak{a}, \mathfrak{b}$. We have $\mathcal{N}\mathfrak{a} = \mathcal{N}\alpha \cdot \mathcal{N}\mathfrak{b}$, with $\alpha \equiv 1 \pmod{\mathfrak{f}}$. We see immediately that $\mathcal{N}\alpha \equiv 1 \pmod{\mathfrak{f}}$, giving $\mathcal{N}\mathfrak{a} \equiv \mathcal{N}\mathfrak{b} \pmod{\mathfrak{f}}$. $\square$

We now wish to compare the space I with the ring $\hat{K}$ of finite adeles of K . For this, we regard $\hat{K}$ as the inverse limit $\varprojlim K/\mathfrak{f}$ and consider an element α of $\hat{K}$ as a sequence $(\alpha_{\mathfrak{f}})$ of totally positive numbers in K , compatible with the transition maps $K/\mathfrak{f}' \rightarrow K/\mathfrak{f}$ for $\mathfrak{f}' \subseteq \mathfrak{f}$. Let $i(\alpha)$ be the sequence of principal ideals $((\alpha_{\mathfrak{f}}))$.

Proposition 2.6 ((2.13)). *$i(\alpha)$ is an element of I which depends only on α , and not on the choice of the $\alpha_{\mathfrak{f}}$.*

Démonstration. This follows immediately from the following lemma, which in turn is an immediate consequence of the definition of $\sim_{\mathfrak{f}}$. $\square$

Lemma 2.7 ((2.14)). *Let $\alpha, \beta \in \hat{K}$ with $\alpha_{\mathfrak{f}}, \beta_{\mathfrak{f}} \gg 0$. Then $(\alpha_{\mathfrak{f}}) \sim (\beta_{\mathfrak{f}}) \bmod \mathfrak{f}$ if and only if there is a totally positive unit $u \in K$ such that $\alpha_{\mathfrak{f}} \equiv u\beta_{\mathfrak{f}} \pmod{\mathfrak{f}}$.*

Now let U^+ be the closure in $\hat{\mathcal{O}}^*$ of the group of totally positive units of K .

Proposition 2.8 ((2.15)). *Let $\alpha, \beta \in \hat{K}$. Then $i(\alpha) = i(\beta)$ if and only if $\alpha = \beta u$ for some $u \in U^+$.*

Démonstration. The lemma shows that if $\alpha = \beta u$ with $u \in U^+$, then $(\alpha_f) \sim (\beta_f) \pmod{f}$ for all f , so $i(\alpha) = i(\beta)$. Conversely, suppose $i(\alpha) = i(\beta)$, and let us prove that $\alpha = \beta u$ for some u . We do have $\alpha_f \equiv u_f \beta_f \pmod{f}$ for each f , where u_f is a totally positive unit. However, the sequence $(u_f \pmod{f})$ may not be compatible because the units u_f may not be unique mod f .

To deal with this problem, we multiply α and β by a positive integer so as to have $\alpha_f, \beta_f \in \mathcal{O}$. (This changes neither the hypothesis nor the conclusion of the proposition.) For each f , let $\mathfrak{d}_f = \gcd(f, \alpha_f^{-1}) = \gcd(f, \beta_f^{-1})$. Then the unit u_f is determined modulo $f\mathfrak{d}_f^{-1} =: \mathfrak{g}_f$. For $f' \subseteq f$, we have $\mathfrak{g}_{f'} \subseteq \mathfrak{g}_f$. Thus we need only to know that U^+ maps into $\varprojlim C_{\mathfrak{g}}$, where $C_{\mathfrak{g}}$ is the group of totally positive units taken mod $\mathfrak{g}$. This is obvious, for example because U^+ is compact and maps onto each $C_{\mathfrak{g}}$. $\square$

Proposition 2.9 ((2.16)). *$\mathcal{N}(i(\alpha)) = \mathcal{N}\alpha$ for all $\alpha \in \hat{K}$.*

Démonstration. Indeed, $\mathcal{N}((\alpha_f)) = \mathcal{N}\alpha_f$ because $\alpha_f \gg 0$. $\square$

Convention. Let $\mathfrak{a} \in I$, $\alpha \in \hat{K}$. We write $(\mathfrak{a} \cdot \alpha)$, or simply $\mathfrak{a} \cdot \alpha$, for the product $\mathfrak{a} \cdot i(\alpha)$ in I .

Note that for $\alpha > 0$ in K we have $(\alpha) = i(\alpha)$ in I . Thus $(\mathfrak{a} \cdot \alpha) \in I$.

Lemma 2.10 ((2.17)). *For $\mathfrak{a} \in I$, the quantity $(\mathfrak{a} \cdot \mathcal{O})$ depends only on the strict ideal class of $\mathfrak{a}_1$: for $\mathfrak{b} \in I$, we have $(\mathfrak{a} \cdot \mathcal{O}) = (\mathfrak{b} \cdot \mathcal{O})$ if and only if $\mathfrak{a}_1$ and $\mathfrak{b}_1$ belong to the same strict ideal class of K .*

Démonstration. For $t > 0$, we have $(t\mathfrak{a} \cdot \mathcal{O}) = (\mathfrak{a} \cdot t\mathcal{O}) = (\mathfrak{a} \cdot \mathcal{O})$. Hence we can assume that $\mathfrak{a}, \mathfrak{b} \in A$, in which case the assertion to be proved is: $(\mathfrak{a} \cdot \mathcal{O}) = (\mathfrak{b} \cdot \mathcal{O})$ if and only if $\mathfrak{a}_1 \sim \mathfrak{b}_1 \pmod{1}$. For each f , let α_f be a totally positive number divisible by f . Modulo f , $(\mathfrak{a} \cdot \mathcal{O})$ is $\mathfrak{a}_f \alpha_f$ and $(\mathfrak{b} \cdot \mathcal{O})$ is $\mathfrak{b}_f \alpha_f$. By (2.7) we see that $\mathfrak{a}_f \alpha_f \sim \mathfrak{b}_f \alpha_f \pmod{f}$ if and only if $\mathfrak{a}_f \alpha_f \sim \mathfrak{b}_f \alpha_f \pmod{1}$, which is true if and only if $\mathfrak{a}_1$ and $\mathfrak{b}_1$ are in the same strict ideal class. $\square$

Proposition 2.11 ((2.18)). *Let $\mathfrak{a} \in I$. Suppose that $(\mathfrak{a} \cdot \mathcal{O}) = (1 \cdot \mathcal{O})$. Then $\mathfrak{a} = (1 \cdot \alpha)$ for some $\alpha \in \hat{K}$.*

Démonstration. As usual, we can multiply $\mathfrak{a}$ by some $t > 0$ to assume that $\mathfrak{a} \in A$. Then by hypothesis we have, for each f , $\mathfrak{a}_f = (\alpha_f)$, where $\alpha_f \in K$ is a totally positive integer, which by (2.14) is well defined, mod f , modulo the action of the group of totally positive units. Let C_f be the set of integers mod f , taken modulo this action. Then the compact $\hat{\mathcal{O}}$ maps onto each C_f and hence onto the limit $\varprojlim C_f$. This gives what is needed. $\square$

Corollary 2.12 ((2.19)). *Let $\mathfrak{a} = (\alpha_f) \in I$, and let $\mathfrak{b} \in I_0$ be an ideal in the same ideal class as $\mathfrak{a}_1$. Then $\mathfrak{a} = (\mathfrak{b} \cdot \alpha)$ for some $\alpha \in \hat{K}$.*

Démonstration. By (2.17), $(\mathfrak{a}\mathfrak{b}^{-1} \cdot \mathcal{O}) = (1 \cdot \mathcal{O})$. Hence $\mathfrak{a}\mathfrak{b}^{-1}$ may be written $(1 \cdot \alpha)$ for some $\alpha \in \hat{K}$. $\square$

Proposition 2.13 ((2.20)). *For each $x \in \hat{K}^*$, we have $(x) \cdot x^{-1} \in G$.*

Démonstration. It suffices to prove that $(x) \cdot x^{-1} \in A$, using the symmetry and the fact that an element of A with an inverse in A must in fact belong to G . This follows from the following result, whose proof we omit. $\square$

Lemma 2.14 ((2.21)). *Let $\mathfrak{a}, \mathfrak{b} \in I_0$, and $\alpha \in \hat{K}$. Then $(\mathfrak{a} \cdot \alpha) \in I^{\mathfrak{b}}$ if and only if α belongs to the completion $\hat{\mathfrak{a}}^{-1}\mathfrak{b}$ of the ideal $\mathfrak{a}^{-1}\mathfrak{b}$.*

As a complement to (2.21), we mention the following fact :

(2.22) Let $\mathfrak{a}$ be an ideal and let $\alpha, \beta \in \hat{K}$. We have $(\mathfrak{a} \cdot \alpha) \sim (\mathfrak{a} \cdot \beta) \bmod \mathfrak{f}$ if and only if there is a totally positive unit u such that we have

$$\alpha \equiv u\beta \pmod{\hat{\mathfrak{a}}^{-1}\mathfrak{f}}.$$

Referring now to (2.20), we let $j(x) = (x) \cdot x^{-1}$ for $x \in \hat{K}^*$. Then j may be viewed as a homomorphism

$$j : \hat{K}^* \rightarrow G$$

which is trivial on the set $\hat{K}_{>0}$ of totally positive elements of $\hat{K}$. On the other hand, for each conductor $\mathfrak{f}$ there is a natural map

$$\psi_{\mathfrak{f}} : \hat{K}^* \rightarrow G_{\mathfrak{f}},$$

trivial on $\hat{K}_{>0}$, which maps each $x \in \hat{K}^*$ such that $x \equiv 1 \pmod{\mathfrak{f}}$ to the class in $G_{\mathfrak{f}}$ of the ideal (x) generated by x , cf. [22, pp. 146–147]. (Here we should recall that $G_{\mathfrak{f}}$ is the ray class group of $K \bmod \mathfrak{f}$, so that each prime-to- $\mathfrak{f}$ ideal of K , integral or not, has a well defined image in $G_{\mathfrak{f}}$. From the point of view of the equivalence $\sim_{\mathfrak{f}}$, only the integral prime-to- $\mathfrak{f}$ ideals map to $G_{\mathfrak{f}}$.) The resulting map

$$\psi : \hat{K}^* \rightarrow G = \varprojlim G_{\mathfrak{f}}$$

is surjective. By abuse of language, we may refer to it as an Artin symbol.

Proposition 2.15 ((2.23)). *We have $\psi = j$.*

Démonstration. Since both ψ and j vanish on $\hat{K}_{>0}$, it suffices to check that $\psi_{\mathfrak{f}}$ and the composition of j and the quotient $G \rightarrow G_{\mathfrak{f}}$ agree on elements of the set $1 + \hat{\mathfrak{f}}$. Now, $\bmod \mathfrak{f}$, $j(x)$ is represented by any ideal of the form $(\alpha)(x)$, where α is a totally positive element of K with $\alpha \equiv x^{-1} \pmod{\mathfrak{f}}$. We must verify that

$$(\alpha)(x) \sim (x),$$

and it suffices to see that

$$\alpha \equiv 1 + (x)^{-1}\mathfrak{f}.$$

This, finally, follows from the defining property of α , together with the fact that $1 - x^{-1} = x^{-1}(x - 1) \in x^{-1}\mathfrak{f}$. $\square$

Frobenius Elements (2.24)

Let $\mathfrak{f}$ be a conductor, and let v be a real place of K . Let $\alpha \in 1 + \mathfrak{f}$ be a number which is negative at v and positive at each real place of K different from v . The class σ_v of (α) in $G_{\mathfrak{f}}$ is independent of α and has order 1 or 2. It is trivial if and only if α may be chosen to be a unit. If we interpret $G_{\mathfrak{f}}$ as the Galois group over K of the ray class field $K_{\mathfrak{f}}$ of K of conductor $\mathfrak{f}$, then σ_v becomes complex conjugation after we choose an embedding $K_{\mathfrak{f}} \hookrightarrow \mathbb{C}$ which induces v on K .

For fixed v and varying $\mathfrak{f}$, the elements σ_v piece together to give a Frobenius element in G , which we again call σ_v . Each σ_v now has order 2, and in fact the subgroup Σ of G generated by the σ_v is an elementary 2-group whose order is 2^r . Each σ_v has norm -1 . If we let Σ_0 be the kernel of the norm map

$$\mathcal{N} : \Sigma \rightarrow \{\pm 1\},$$

then Σ_0 has order 2^{r-1} and is generated by products $\sigma_v \sigma_w$, with v and w running over the real places of K . For $\mathfrak{f}$ a conductor, we let $\Sigma_{\mathfrak{f}}^*$ (resp. $\Sigma_{\mathfrak{f}}$) be the image of Σ^* (resp. Σ) in $G_{\mathfrak{f}}$. We will be especially interested in the case $\mathfrak{f} = (1)$.

In a natural way, the σ_v lead to the notion of functions with parity. Let φ be a function on G with values in an abelian group $\mathcal{W}$. For each real place v , let a_v be one of the two integers 0, 1. We say that φ has parity (a_v) if we have

$$\varphi(\sigma_v g) = (-1)^{a_v} \varphi(g)$$

for all $g \in G$. When the a_v are all equal to 0 (resp. 1) we say that φ is even (resp. odd). The phrase “ φ has parity $(-1)^k$ ” means that φ is odd if k is odd and even if k is even.

Invertible Modules (2.25)

As is well known, the group of ideal classes of K may be interpreted as the group of isomorphism classes of invertible $\mathcal{O}$ -modules. We shall now similarly interpret the constructions of this §. For this, we must introduce the notion of a signed $\mathcal{O}$ -module.

Let $\mathcal{E}$ be an invertible $\mathcal{O}$ -module. For each real place v of K , let $\mathcal{E}_v$ be the tensor product $\mathcal{E} \otimes_{\mathcal{O}} \mathbb{R}$, with $\mathbb{R}$ viewed as an $\mathcal{O}$ -module via v . Then $\mathcal{E}_v$ is a free $\mathbb{R}$ -module of rank 1. A positivity for $\mathcal{E}$ is the choice, for each v , of an element of

$$\text{Isom}(\mathcal{E}_v, \mathbb{R}) \cong \mathbb{R}^*,$$

taken modulo the action of the group of positive real numbers. There are precisely 2^r possible positivities for $\mathcal{E}$.

A pair consisting of an invertible $\mathcal{O}$ -module $\mathcal{E}$, together with a positivity $+$ for $\mathcal{E}$, is a signed $\mathcal{O}$ -module. Two such modules $(\mathcal{E}, +)$, $(\mathcal{E}', +')$ are isomorphic if there is an $\mathcal{O}$ -isomorphism $\mathcal{E} \xrightarrow{\sim} \mathcal{E}'$ compatible with $+$, $+$ '. It is clear that the isomorphism classes of signed $\mathcal{O}$ -modules are the strict ideal classes of K . More precisely, this works as follows. Each ideal $\mathfrak{a}$ of K has a canonical positivity $+\text{can}$ arising from the canonical isomorphism $\mathfrak{a} \otimes_{\mathcal{O}} K = K$. With $\mathfrak{a}$ and $\mathfrak{b}$ ideals, we have $(\mathfrak{a}, +\text{can}) \cong (\mathfrak{b}, +\text{can})$ if and only if $\mathfrak{a}$ and $\mathfrak{b}$ are in the same strict class. Finally, any pair $(\mathcal{E}, +)$ is isomorphic to $(\mathfrak{a}, +\text{can})$ for some ideal $\mathfrak{a}$.

Now let $\mathfrak{f}$ be a conductor. We consider triples $(\mathcal{E}, +, \varphi)$, where φ is an $\mathcal{O}$ -linear map $\mathcal{E} \otimes \hat{\mathcal{O}} \rightarrow \hat{K}/\mathfrak{f}$. Two such triples $(\mathcal{E}, +, \varphi)$, $(\mathcal{E}', +', \varphi')$ are said to be isomorphic if there is

an isomorphism $(\mathcal{E}, +) \xrightarrow{\sim} (\mathcal{E}', +')$ so that the diagram

$$\begin{array}{ccc} \mathcal{E} \otimes \hat{\mathcal{O}} & \longrightarrow & \hat{K}/\mathfrak{f} \\ \downarrow & \nearrow \varphi' & \\ \mathcal{E}' \otimes \hat{\mathcal{O}} & & \end{array}$$

is commutative. For $\mathfrak{a} \in I_0$, we let φ_{can} be the map $\mathfrak{a} \otimes \hat{\mathcal{O}} \rightarrow \hat{K}/\mathfrak{f}$. A computation shows that $(\mathfrak{a}, +_{\text{can}}, \varphi_{\text{can}})$ and $(\mathfrak{b}, +_{\text{can}}, \varphi_{\text{can}})$ are isomorphic if and only if $\mathfrak{a} \sim_{\mathfrak{f}} \mathfrak{b}$.

Furthermore, let $(\mathcal{E}, +, \varphi)$ be a given triple, and let $\mathfrak{a}$ be an ideal such that $(\mathcal{E}, +)$ and $(\mathfrak{a}, +_{\text{can}})$ are isomorphic. Choosing an isomorphism we find a map $\mathfrak{a} \rightarrow \hat{K}/\mathfrak{f}$, necessarily given by a multiplication $\gamma : \mathfrak{a} \rightarrow \gamma\mathfrak{a}$ (with $\gamma \in K$), followed by the reduction $\hat{K} \rightarrow \hat{K}/\mathfrak{f}$. We may choose γ to be totally positive. Then the isomorphism $\mathfrak{a}\gamma^{-1} \xrightarrow{\sim} \mathfrak{b} \cdot \mathfrak{f}^{-1}$ shows that we have $(\mathcal{E}, +, \varphi) \cong (\mathfrak{a}\gamma, +_{\text{can}}, \varphi_{\text{can}})$. Hence $I_{\mathfrak{f}}$ is just the set of isomorphism classes of triples $(\mathcal{E}, +, \varphi)$.

To describe I , we consider triples $(\mathcal{E}, +, \varphi)$, where φ is now an $\mathcal{O}$ -linear map $\mathcal{E} \otimes \hat{\mathcal{O}} \rightarrow \hat{K}$. Such triples give rise to elements of I as follows. Choosing an isomorphism $(\mathcal{E}, +) \cong (\mathfrak{a}, +_{\text{can}})$, we find an $\mathcal{O}$ -linear $\mathfrak{a} \rightarrow \hat{K}$, which is necessarily of the form “multiplication by γ ” for some $\gamma \in K$. The product $(\mathfrak{a} \cdot \gamma)$ in I is easily seen to depend only on $(\mathcal{E}, +, \varphi)$. Note that the image mod $\mathfrak{f}$ of this product is represented by $(\mathcal{E}, +)$ together with the map “ $\varphi \bmod \mathfrak{f}$ ” :

$$\mathcal{E} \rightarrow \mathcal{E} \otimes \hat{\mathcal{O}} \xrightarrow{\varphi} \hat{K} \rightarrow \hat{K}/\mathfrak{f} = K/\mathfrak{f}.$$

In either situation, the action of a real Frobenius σ_v is as follows : we have $\sigma_v \cdot (\mathcal{E}, +, \varphi) = (\mathcal{E}, +', \varphi)$, where $+'$ is deduced from $+$ by changing $+$ at v .

3 A Functional Equation

In this § we define the L -series attached to a (complex valued) Schwartz function on the measure space I . We then prove a functional equation for such L -series, involving a Fourier transform on the space I . Defining this Fourier transform is the first order of business.

We shall view the ring $\mathbb{A}$ of adeles of K as the product $\mathbb{A}_f \times K_{\infty}$, where $\mathbb{A}_f = \hat{K}$ is the ring of finite adeles of K and $K_{\infty} = K \otimes \mathbb{R}$ is the product of the real completions of K . For computing, we number these real places of K , so that $K_{\infty} = \mathbb{R}^r$. Also, for $x \in \mathbb{A}$, we write x_f (resp. x_{∞}) for the image of x in $\mathbb{A}_f$ (resp. K_{∞}).

Let $|\cdot| : K_{\infty} \rightarrow \mathbb{R}$ be the standard absolute value, namely

$$(x_1, \dots, x_r) \mapsto \prod_{i=1}^r |x_i|.$$

We write instead $\|\cdot\|$ for the standard absolute value on $\mathbb{A}_f$, defined for example by the equation

$$d(\gamma x) = \|\gamma\| \cdot dx,$$

where dx is a Haar measure on $\mathbb{A}_f$. For $y \in K^*$, we have $\|y\| = \mathcal{N}(y)^{-1}$, where (y) is the ideal of K generated by y .

Let $\psi : \mathbb{A} \rightarrow \mathbb{C}^*$ be the standard additive character, trivial on K , whose restriction to a completion K_v of K is given as follows :

- For v real, $\psi : x \mapsto e^{2\pi i x}$;
- For v p -adic, $\psi : x \mapsto e^{-2\pi i \operatorname{tr}(x)}$,

where $\operatorname{tr} : K_v \rightarrow \mathbb{Q}_p$ is the trace, and the exponential $e^{2\pi i t}$ for $t \in \mathbb{R}$ is defined in the usual way. The restriction of ψ to $\mathbb{A}_f$ (resp. K_∞) is denoted ψ_f (resp. ψ_∞), or simply ψ , according to the context. For ε a Schwartz function on $\mathbb{A}_f$ (resp. K_∞ , resp. $\mathbb{A}$), we define the Fourier transform $\hat{\varepsilon}$ of ε by the formula

$$\hat{\varepsilon}(y) = \int \varepsilon(x) \psi(xy) dx. \quad (9)$$

Here dx denotes Haar measure, normalized so that the formula $\hat{\varepsilon}(x) = \varepsilon(-x)$ holds, cf. §3.3 and §4.1 of [39].

We now turn our attention to Schwartz functions on $\hat{K}$ and on I . We recall the following “variance” formula for $\hat{K}$:

$$\text{If } \eta(x) = \varepsilon(\gamma x) \text{ with } \gamma \in \hat{K}^*, \text{ then } \hat{\eta}(y) = \hat{\varepsilon}(\gamma^{-1}y) \|\gamma\|^{-1}. \quad (10)$$

Now let ε be a locally constant function with compact support on I . Specifically, assume that ε is supported on $I^{\mathfrak{b}}$ and defined modulo $\sim_{\mathfrak{f}}$. If $\mathfrak{a} \in I_0$, let $\varepsilon^{\mathfrak{a}} : \hat{K} \rightarrow \mathbb{C}$ be the function $x \mapsto \varepsilon(\mathfrak{a} \cdot x)$. By (2.21) and (2.22), $\varepsilon^{\mathfrak{a}}$ is supported on $\hat{\mathfrak{a}}^{-1}\mathfrak{b}$ and defined modulo $\hat{\mathfrak{a}}^{-1}\mathfrak{f}$. Let $\hat{\varepsilon}^{\mathfrak{a}}$ denote the Fourier transform of $\varepsilon^{\mathfrak{a}}$. For $v \in K$ totally positive, one has the formula

$$\hat{\varepsilon}^{\mathfrak{a}v}(x) = \hat{\varepsilon}^{\mathfrak{a}}(v^{-1}x) \cdot \|v\|^{-1}, \quad (11)$$

as follows easily from (10).

Definition. We define the Fourier transform $T\varepsilon : I \rightarrow \mathbb{C}$ of ε by the formula

$$(T\varepsilon)(\mathfrak{a}^{-1} \cdot x) = \mathcal{N}\mathfrak{a} \cdot \hat{\varepsilon}^{\mathfrak{a}}(x), \quad (12)$$

for $\mathfrak{a} \in I_0$, $x \in \hat{K}$. [We observe that each element of I is a product $\mathfrak{a}^{-1} \cdot x$, by (2.19)] To check that $T\varepsilon$ is well defined, suppose that $\mathfrak{a}^{-1} \cdot \alpha = \mathfrak{b}^{-1} \cdot \beta$; we then wish to verify that

$$\mathcal{N}\mathfrak{a} \cdot \hat{\varepsilon}^{\mathfrak{a}}(\alpha) = \mathcal{N}\mathfrak{b} \cdot \hat{\varepsilon}^{\mathfrak{b}}(\beta).$$

The equation $\mathfrak{a}^{-1} \cdot \alpha = \mathfrak{b}^{-1} \cdot \beta$ gives $(\mathfrak{a}^{-1} \cdot \alpha) = (\mathfrak{b}^{-1} \cdot \beta)$, so that $\alpha = v\beta$ for some $v \gg 0$ by (2.17).

We find that $(\mathfrak{a}^{-1} \cdot \alpha) = (\mathfrak{b}^{-1} \cdot v^{-1}\beta)$, so that $\alpha = v^{-1}\beta t$ for some $t \in U^+$ (2.15). By (11), we have

$$\hat{\varepsilon}^{\mathfrak{a}}(\alpha) = \hat{\varepsilon}^{\mathfrak{a}}(v^{-1}\beta t) = \hat{\varepsilon}^{\mathfrak{a}v}(\beta t) \cdot \mathcal{N}v^{-1} = \hat{\varepsilon}^{\mathfrak{b}}(\beta t) \cdot \mathcal{N}v^{-1}.$$

The required formula then follows from (10), since $\varepsilon^{\mathfrak{a}}$ is invariant under U^+ , and $\|t\| = 1$.

We now give some formal properties of T that follow directly from the definition.

Proposition 3.1 ((3.5)). $T(T\varepsilon)(x) = \varepsilon(x \cdot (-1))$.

Démonstration. Let $\eta = T\varepsilon$. For $(\mathfrak{a}^{-1} \cdot \alpha) \in I$, we have

$$\begin{aligned} (T\eta)(\mathfrak{a}^{-1} \cdot \alpha) &= \mathcal{N}\mathfrak{a} \cdot \hat{\eta}^{\mathfrak{a}}(\alpha) \\ &= \mathcal{N}\mathfrak{a} \int \eta(\mathfrak{a} \cdot x) \psi(x\alpha) dx \\ &= \mathcal{N}\mathfrak{a} \int \hat{\varepsilon}^{\mathfrak{a}}(x) \psi(x\alpha) dx \\ &= \varepsilon^{\mathfrak{a}}(-\alpha) = \varepsilon(\mathfrak{a} \cdot (-\alpha)) \\ &= \varepsilon((\mathfrak{a}^{-1} \cdot \alpha) \cdot (-1)). \end{aligned} \quad \square$$

Proposition 3.2 ((3.6)). *If $\eta(x) = \varepsilon(\mathfrak{b}x)$ for some $\mathfrak{b} \in I_0$, then*

$$(T\eta)(\mathfrak{a}) = (T\varepsilon)(\mathfrak{a}\mathfrak{b}^{-1}) \cdot \mathcal{N}\mathfrak{b}.$$

Démonstration. We have

$$\begin{aligned} \hat{\eta}^{\mathfrak{a}}(\alpha) &= \int \eta(\mathfrak{a} \cdot x) \psi(x\alpha) dx \\ &= \int \varepsilon(\mathfrak{b} \cdot (\mathfrak{a} \cdot x)) \psi(x\alpha) dx \\ &= (T\varepsilon)((\mathfrak{a}\mathfrak{b})^{-1} \cdot \alpha) \cdot \mathcal{N}(\mathfrak{a}\mathfrak{b})^{-1} \cdot \mathcal{N}\mathfrak{a} \\ &= (T\varepsilon)(\mathfrak{a}^{-1} \cdot \alpha \cdot \mathfrak{b}^{-1}) \cdot \mathcal{N}\mathfrak{b}^{-1}. \end{aligned} \quad \square$$

Corollary 3.3 ((3.7)). *Suppose that $\eta(x) = \varepsilon(cx)$ with $c \in G$. Then $(T\eta)(y) = (T\varepsilon)(c^{-1}y)$.*

Démonstration. As noted in §2, each $c \in G$ may be written $(\gamma)\gamma^{-1}$ for some $\gamma \in \hat{K}^*$. Since $\|\gamma^{-1}\| \cdot \mathcal{N}(\gamma) = 1$, the result follows from (3.6), (3.3). $\square$

Using the above properties, we will now verify that $T\varepsilon$ is locally constant and compactly supported. For the application that we ultimately have in mind, the ideal $\mathfrak{b}$ governing the support of ε is (1) : that is, ε is supported on A . Let us assume that this is the case, for simplicity. [If $\mathfrak{b} \neq 1 = (1)$, one may use (3.6) to reduce to the case $\mathfrak{b} = (1)$.] We suppose (as above) that ε is defined mod $\sim_{\mathfrak{f}}$.

Let η be defined by the formula

$$\eta(\mathfrak{a}) = (T\varepsilon)(\mathfrak{a}\mathfrak{D}^{-1}).$$

Theorem 3.4 ((3.8)). *The function η is supported on $I^{\mathfrak{D}^{-1}\mathfrak{f}}$ and defined mod $\sim_1$.*

Démonstration. We first verify the periodicity. If $x, y \in I$ are congruent mod 1, we may write $x = (\mathfrak{a}^{-1} \cdot \alpha)$, $y = (\mathfrak{a}^{-1} \cdot \beta)$ for some $\mathfrak{a} \in I_0$ and elements $\alpha, \beta \in \hat{\mathcal{O}}$ with $\alpha - \beta \in \hat{\mathcal{O}}$. Thus the periodicity property asserted for $T\eta$ reduces to the statement that $\hat{\varepsilon}^{\mathfrak{a}}$ is periodic mod $\hat{\mathcal{O}}$. As already noted, $\varepsilon^{\mathfrak{a}}$ is supported on $\mathfrak{a}^{-1}\mathfrak{D}^{-1}$. Looking at the formula defining $\hat{\varepsilon}^{\mathfrak{a}}$, we see that the required periodicity follows from the fact that the additive character ψ is trivial on $\hat{\mathfrak{D}}^{-1}$.

For the question of support, it is now enough to check that η vanishes on each ideal $\mathfrak{a}$ with $\mathfrak{a}\mathfrak{D} \not\subseteq \mathfrak{f}^{-1}\mathfrak{D}^{-1}$. (This remark follows from the periodicity and the fact that for $x \sim y$ mod 1 we have $x \in I^{\mathfrak{D}^{-1}\mathfrak{f}} \iff y \in I^{\mathfrak{D}^{-1}\mathfrak{f}}$.) So we must check that $(T\varepsilon)(\mathfrak{a}) = 0$ provided that $\mathfrak{a} \not\subseteq \mathfrak{f}^{-1}\mathfrak{D}^{-1}$. The vanishing of $(T\varepsilon)(\mathfrak{a})$ amounts to the vanishing of the integral

$$\int \varepsilon(\mathfrak{a}^{-1} \cdot x) \psi(x) dx.$$

But the hypothesis on $\mathfrak{a}$ insures the existence of an $\alpha \in \hat{\mathcal{O}}\mathfrak{f}$ with $\psi(\alpha) \neq 1$. The integral thus vanishes because of the formula

$$\varepsilon(\mathfrak{a}^{-1} \cdot (x + \alpha)) = \varepsilon(\mathfrak{a}^{-1} \cdot (x)),$$

cf. (2.22). $\square$

Corollary 3.5 ((3.9)). *The function $T\varepsilon$ is supported on $I^{\mathfrak{D}^{-1}\mathfrak{f}\mathfrak{D}^{-1}}$. It is defined modulo $\mathfrak{D}^{-1}$ in the sense that $(T\varepsilon)(x) = (T\varepsilon)(y)$ whenever $\mathfrak{D}x \sim \mathfrak{D}y$ mod 1.*

Remark 3.6 ((3.10)). The introduction of η is a consequence of our failure to have defined $\sim_{\mathfrak{f}}$ for $\mathfrak{f}$ a fractional ideal. For later use we now define two elements x, y of I to be congruent mod $\mathfrak{D}^{-1}$ if $\mathfrak{D}x$ and $\mathfrak{D}y$ are congruent mod 1.

We now begin our study of L -functions. Let $\varepsilon : I \rightarrow \mathbb{C}$ be locally constant and compactly supported. We set

$$L(s, \varepsilon) = \sum_{\mathfrak{a} \in I_0} \varepsilon(\mathfrak{a}) \mathcal{N}\mathfrak{a}^{-s}$$

for s with large real part, where the sum runs over all (fractional) ideals of K . It is easy to rewrite $L(s, \varepsilon)$ as a linear combination of functions of the form $\mathcal{N}\mathfrak{b}^{-s} \zeta_f(s, c)$, where $\mathfrak{b}$ is some ideal of K and $\zeta_f(s, c)$ is the partial zeta function of a class c mod $\mathfrak{f}$ in the sense of Siegel. Using results of Hecke, we may thus admit *a priori* that $L(s, \varepsilon)$ may be continued to a meromorphic function on $\mathbb{C}$, with at worst a simple pole at $s = 1$ and with no other poles. We are interested in obtaining a functional equation linking $L(s, \varepsilon)$ and $L(s, T\varepsilon)$.

For this, we shall now assume that ε has parity (a_v) for some collection of integers $a_v = 0$ or 1 , cf. (2.24). By (3.7), $T\varepsilon$ again has parity (a_v) . Let $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(\frac{s}{2})$, and set

$$\gamma(s) = \prod_v \Gamma_{\mathbb{R}}(s + a_v), \quad (13)$$

$$\Lambda(s, \varepsilon) = \gamma(s) L(s, \varepsilon), \quad (14)$$

with the product taken over the real places of K .

Theorem 3.7 ((3.12)). *We have*

$$\Lambda(s, \varepsilon) = i^a \cdot 2^{ra/2} \cdot \Lambda(1 - s, T\varepsilon),$$

where $a = \sum a_v$.

In fact, the case of primary interest for us is that where all a_v are either 0 (ε even) or 1 (ε odd) and s is an integer $k \geq 1$ with the same parity as ε . In this case we will obtain the following

Corollary 3.8 ((3.13)). *If ε has parity $(-1)^k$ and $k \geq 1$, then*

$$L(k, \varepsilon) = 2^r \cdot c_k \cdot L(1 - k, T\varepsilon),$$

where

$$c_k = \frac{(2\pi i)^{rk}}{(k-1)!^r}.$$

Before beginning the proof of (3.12), (3.13), we remark that these equations are essentially those discussed by Siegel [38].

Proof of (3.12). We will in fact establish an identity between the partial L -function obtained by summing $\varepsilon(\mathfrak{a}) \mathcal{N}\mathfrak{a}^{-s}$ over a given wide ideal class of K and that made by summing $(T\varepsilon)(\mathfrak{a}) \mathcal{N}\mathfrak{a}^{s-1}$ over the inverse of the class. Let $\mathfrak{g}$ be an ideal. We will prove that

$$\sum_{\beta \in K^*/U} \varepsilon(\beta \mathfrak{g}) |\beta|^{-s} = h(s) \sum_{\beta \in K^*/U} (T\varepsilon)(\beta \mathfrak{g}^{-1}) |\beta|^{s-1}, \quad (15)$$

where the notation is as follows :

- U = unit group of K
- $h(s) = i^a 2^{ra/2} \pi^{r(s-1)/2} \Gamma_{\mathbb{R}}(1-s+a)^r \Gamma_{\mathbb{R}}(s+a)^{-1}$
- $|\beta|$ is the archimedean absolute value of β , i.e., $|\beta| = |\mathcal{N}\beta|$.

Let ρ be the function on $(K \otimes \mathbb{R})^*$ given by $\rho(\beta) = \prod_v (\text{sgn}_v \beta)^{a_v}$. We have, by the parity hypothesis,

$$\varepsilon(\beta \mathfrak{g}) = \varepsilon(\mathfrak{g} \cdot \beta) \rho(\beta) = \varepsilon_{\mathfrak{g}}(\beta) \rho(\beta), \quad (16)$$

$$(T\varepsilon)(\beta \mathfrak{g}^{-1}) = (T\varepsilon)(\mathfrak{g}^{-1} \cdot \beta) \rho(\beta) = (T\varepsilon)_{\mathfrak{g}^{-1}}(\beta) \rho(\beta). \quad (17)$$

Thus the formula to be verified reads :

$$\sum_{\beta \in K^*/U^+} \varepsilon_{\mathfrak{g}}(\beta) |\beta|^{-s} \rho(\beta) = h(s) \sum_{\beta \in K^*/U^+} (T\varepsilon)_{\mathfrak{g}^{-1}}(\beta) |\beta|^{s-1} \rho(\beta), \quad (18)$$

where U^+ is the group of totally positive units. (We prefer to sum over this latter unit group because $\varepsilon_{\mathfrak{g}}$, $\beta \mathfrak{g}$ and ρ (as well as $|\cdot|$) are invariant by it, whereas only the products $\varepsilon_{\mathfrak{g}} \cdot \rho$, $\hat{\varepsilon}_{\mathfrak{g}} \rho$ need be invariant by U . In (15), we have the right to replace U by any subgroup of finite index in it.)

Let χ be a quasicharacter $K_{\infty}^* \rightarrow \mathbb{C}^*$, and let φ_{∞} be a function such that both it and its Fourier transform are rapidly decreasing at infinity (cf. [39], §2.4). For $\text{Re } s > 0$, the integral

$$\int_{K_{\infty}^*} \chi(x) |x|^s \varphi_{\infty}(x) d^*x \quad (19)$$

converges to an analytic function of s which has a meromorphic continuation to all of $\mathbb{C}$. Furthermore, there is a non-zero meromorphic function $\alpha(\chi, s)$ of s , independent of φ_{∞} , such that

$$\alpha(\chi, s) \int_{K_{\infty}^*} \chi^{-1}(x) |x|^{1-s} \hat{\varphi}_{\infty}(x) d^*x = \int_{K_{\infty}^*} \chi(x) |x|^s \varphi_{\infty}(x) d^*x. \quad (20)$$

This formula, which is just a semi-local variant of the functional equation of [39, loc. cit], expresses the fact that the Fourier transform of the distribution $\chi|\cdot|^{-s} = \chi_s$ on K_{∞} is proportional to $\chi^{-1}|\cdot|^{1-s}$.

Let us take χ equal to the “sign” character ρ , viewed as a function on K_{∞}^* . If we write the collection of numbers (a_v) as a tuple $(a_1, \dots, a_r)$, using the chosen numbering of the real places, then

$$\rho(x) = \prod_{i=1}^r \text{sgn}(x_i)^{a_i}.$$

Following Tate, we compute $\alpha(\rho, s)$ by choosing $\varphi_{\infty}(x) = \prod_v (x_v^{a_v} e^{-\pi x_v^2})$. With this choice for φ_{∞} , each integral in (20) may be rewritten as a product of r different elementary integrals of the type computed on p. 317 of [39]. We find that

$$\alpha(\rho, s) = h(s),$$

with h as in (15). Thus (15) is a special case of □

Theorem 3.9 ((3.19)). *Let $\chi : K_{\infty}^* \rightarrow \mathbb{C}^*$ be a quasicharacter invariant under a subgroup U_0 of finite index in U . Let ε be a locally constant compactly supported function on $\hat{K}$ which is also invariant under U_0 . Then we have the equality of meromorphic functions*

$$\sum_{\beta \in K^*/U_0} \varepsilon(\beta) \chi(\beta) |\beta|^{-s} = \alpha(\chi^{-1}, 1-s) \sum_{\beta \in K^*/U_0} \hat{\varepsilon}(\beta) \chi(\beta) |\beta|^{s-1}.$$

Remarks. 1. More generally, let K be a global field, and let S be a finite set of primes of K which contains the archimedean primes. Let χ be a quasicharacter of $\prod_{v \in S} K_v^*$, invariant by a subgroup U_0 of finite index in the group of S -units of K . The Fourier transform $\hat{\chi}_S$ of χ is a multiple $\alpha(\chi)|\cdot|^{-1}\chi^{-1}$ of $|\cdot|^{-1}\chi^{-1}$. Let ε be a U_0 -invariant locally constant function with compact support on $\prod_{v \in S} K_v^*$, and let $\hat{\varepsilon}$ be its Fourier transform. The proof given below yields the identity

$$\sum_{x \in K^*/U_0} \chi(x)\varepsilon(x) = \sum_{x \in K^*/U_0} \hat{\chi}_S(x)\hat{\varepsilon}(x), \quad (21)$$

where both sides are defined by analytic continuation in the family $\chi|\cdot|^s$.

Suppose that we (formally) allow S to be the set of all places of K , and take $U_0 = K^*$. Then (21) yields the identity $\alpha(\chi) = 1$ for χ a grossencharacter. This identity is Tate's global functional equation, as rewritten by Weil [40].

2. Our method of proof is similar to that given in Sato-Shintani [31]. A common generalization should be possible.

Proof of (3.19). For simplicity, we will write U instead of U_0 for the unit group appearing in (3.19). Furthermore, let us continue to write $\chi_s = \chi \cdot |\cdot|^s$.

Step I. Let φ_∞ be a function on K_∞ as in (19), chosen so that

$$\int_{K_\infty^*} \chi_s(x)\varphi_\infty(x) d^*x$$

is not identically zero. Let φ be the function $\varepsilon \otimes \varphi_\infty$ on $\mathbb{A}$:

$$\varphi(x) = \varphi_\infty(x_\infty)\varepsilon(x_f).$$

For each $x \in \mathbb{A}_f^*$ we consider the function on $\mathbb{A}$

$$a \mapsto \varphi(xa) \quad (a \in \mathbb{A}),$$

whose Fourier transform is

$$a \mapsto |x|^{-1}\hat{\varphi}(x^{-1}a).$$

We apply the Poisson summation formula to this function and obtain

$$\sum_{\beta \in K} \varphi(x\beta) = \sum_{\beta \in K} |x|^{-1}\hat{\varphi}(x^{-1}\beta), \quad (22)$$

cf. [39, p. 333].

Step II. We have an equality of meromorphic functions

$$\int_{K_\infty^*/U} \chi_s(x) \sum_{\beta \in K^*} \varphi(x\beta) d^*x = \int_{K_\infty^*/U} \chi_s(x^{-1})|x| \sum_{\beta \in K^*} \hat{\varphi}(x\beta) d^*x. \quad (23)$$

[Note that the integrands are U -invariant. E.g., if $\gamma \in U$ and $x' = \gamma x$, then $\varphi(x'\beta)$ is by definition $\varphi(x\gamma^{-1}\beta)$. Since ε is invariant under U , we may rewrite this as $\varphi(x\beta)$.]

Let $K_{\infty,+}^*$ (resp. $K_{\infty,-}^*$) be the subset of K_∞^* consisting of elements with $|x| > 1$ (resp. $|x| < 1$). Over $K_{\infty,+}^*$, both integrals converge to analytic functions of s because of the rapid decrease of $\varphi, \hat{\varphi}$ at ∞ . Over $K_{\infty,-}^*$, the left-hand integral is well behaved for

$\operatorname{Re} < 0$ and the right hand integral for $\operatorname{Re} > 0$. Let L and R , respectively, denote the integrands of these integrals.

Suppose that we multiply (22) by $\chi_s(x)$, integrate over $K_{\infty,-}^*/U$, change variables $x \mapsto x^{-1}$ on the right-hand side, and isolate the terms corresponding to $\beta = 0$. We obtain

$$\int_{K_{\infty,-}^*/U} L = \int_{K_{\infty,-}^*/U} R - \varphi(0) \int_{K_{\infty,-}^*/U} \chi_s(x) d^*x - \hat{\varphi}(0) \int_{K_{\infty,-}^*/U} \chi_s(x^{-1})|x| d^*x,$$

for $\operatorname{Re} > 0$. Similarly, for $\operatorname{Re} < 0$ we have

$$\int_{K_{\infty,-}^*/U} R = \int_{K_{\infty,-}^*/U} L + \varphi(0) \int_{K_{\infty,-}^*/U} \chi_s(x) d^*x - \hat{\varphi}(0) \int_{K_{\infty,-}^*/U} \chi_s(x^{-1})|x| d^*x.$$

Now the point is that each integral multiplying $\varphi(0)$ and $\hat{\varphi}(0)$ is meromorphic on $\mathbb{C}$, so that $\int_{K_{\infty}^*/U} L$, $\int_{K_{\infty}^*/U} R$ are continuable to meromorphic functions on $\mathbb{C}$. Furthermore, when we take the difference of the two equations just written, the $\varphi(0)$ and the $\hat{\varphi}(0)$ terms cancel, giving the desired equality (23).

For example, let us show that $\int_{K_{\infty}^*/U} \chi_s(x) d^*x$ and $\int_{K_{\infty}^*/U} \chi_s(x) d^*x$ are meromorphic in s , and that each is the negative of the other. We let $\pi : K_{\infty}^*/U \rightarrow (0, \infty)$ be the surjection $x \mapsto |x|$. By Dirichlet's theorem, the fibres of π are compact. Hence for each $t \in (0, \infty)$, the integral

$$\int_{\pi^{-1}(t)} \chi_s(x) d^*x$$

is defined for all s . Because π and χ_s are homomorphisms, the integral vanishes if χ (and hence χ_s) is non-trivial on the kernel of π . In that case, by Fubini's theorem, the two integrals to be calculated are identically zero. So we can assume that χ is trivial on $\operatorname{Ker} \pi$, giving that $\chi(x) = |x|^{s_0}$ for some $s_0 \in \mathbb{C}$. We then find :

$$\int_{K_{\infty,+}^*/U} \chi_s(x) d^*x = \frac{V}{s - s_0}$$

and

$$\int_{K_{\infty,-}^*/U} \chi_s(x) d^*x = -\frac{V}{s - s_0},$$

where V is the volume of $\pi^{-1}(1)$. Hence our claim, and thus (23), is verified.

Step III. We apply Fubini's theorem to the left hand side of (23), obtaining

$$\sum_{\beta \in K^*/U} \int_{K_{\infty}^*} \chi_s(x) \varphi(x\beta) d^*x.$$

We replace x by $\beta^{-1}x$, i.e., fix β , in the integral. It becomes

$$\varepsilon(\beta) \chi_s(\beta)^{-1} \int_{K_{\infty}^*} \chi_s(x) \varphi_{\infty}(x) d^*x,$$

so that the left-hand side of (23) may be rewritten

$$\left[\sum_{\beta \in K^*/U} \varepsilon(\beta) \chi_s(\beta)^{-1} \right] \cdot \int_{K_{\infty}^*} \chi_s(x) \varphi_{\infty}(x) d^*x.$$

This decomposition gives a meromorphic continuation to the first factor, i.e., the sum : the product of the two factors is meromorphic by the above, whereas the integral is meromorphic (and non-zero by the choice of φ_∞). We similarly write the right-hand side of (23) as

$$\left[\sum_{\beta \in K^*/U} \hat{\varepsilon}(\beta) \chi_s(\beta) |\beta|^{-1} \right] \cdot \int_{K_\infty^*} \chi(x^{-1}) |x|^{1-s} \varphi_\infty(x) d^*x.$$

This gives, because of (20), the equality

$$\sum_{\beta \in K^*/U} \varepsilon(\beta) \chi^{-1}(\beta) |\beta|^{-s} = \alpha(\chi^{-1}, 1-s) \sum_{\beta \in K^*/U} \hat{\varepsilon}(\beta) \chi(\beta) |\beta|^{s-1},$$

which is just (3.19) with χ^{-1} and $-s$ instead of χ and s . □

Proof of (3.13). The factor $\alpha(\chi, s)$ is in this case

$$\prod_v \begin{cases} \frac{\Gamma_{\mathbb{R}}(1-s)}{\Gamma_{\mathbb{R}}(s)} & \text{if } k \text{ is even,} \\ \frac{\Gamma_{\mathbb{R}}(2-s)}{\Gamma_{\mathbb{R}}(1+s)} & \text{if } k \text{ is odd.} \end{cases}$$

However, we have

$$\frac{\Gamma_{\mathbb{R}}(s)}{\Gamma_{\mathbb{R}}(1-s)} = 2^{1-s} \cos \frac{\pi s}{2} \Gamma(s), \quad \frac{\Gamma_{\mathbb{R}}(1+s)}{\Gamma_{\mathbb{R}}(2-s)} = 2^{1-s} \sin \frac{\pi s}{2} \Gamma(s),$$

as remarked in [39, p. 317]. The assertion now follows on setting $s = k$ in each case. □

Trivial Zeros

Proposition 3.10 ((3.23)). *The function $\Lambda(s, \varepsilon)$ of (14) is holomorphic on $\mathbb{C}$ except in the case where ε is even, i.e., where all a_v vanish. In this case, $\Lambda(s, \varepsilon)$ is holomorphic for all $s \neq 0, 1$ and has at worst simple poles at these points.*

As is well known, by considering the poles of Γ one deduces the existence of “trivial zeros” for $L(s, \varepsilon)$:

Corollary 3.11 ((3.24)). *We have $L(1-k, \varepsilon) = 0$ for all even integers $k \geq 1$ except when ε is an even function. Similarly, if ε is not an odd function, we have $L(1-k, \varepsilon) = 0$ for all odd integers $k \geq 1$.*

[The assertion concerning $k = 1$ would be false in the excluded case $K = \mathbb{Q}$. Note that in (3.24) the function ε is still assumed to have some parity, as in (3.12)]

Proof of (3.23). We review the proof of (3.19), making the normalization $\chi = \rho$. The proof of the equality (23) shows that the functions in (23) are everywhere holomorphic, except for the case where χ is trivial, in which case there may be simple poles only at the points $s = 0, 1$. We now make in Step III of the proof of (3.19) the choice

$$\varphi_\infty = \prod_v |x_v|^{a_v} e^{-\pi x_v^2}.$$

Then we may explicitly calculate the integrals appearing in Step III, obtaining in particular the equation

$$\int_{K_\infty^*} \chi_s(x) \varphi_\infty(x) d^*x = \prod_v \Gamma_{\mathbb{R}}(a_v - s).$$

Except for the change of sign in s , the product is the Γ -factor $\gamma(s)$ used to define $\Lambda(s, \varepsilon)$. We may conclude that the product of $\gamma(s)$ by the L -series in the statement of (3.19) has the property asserted of $\Lambda(s, \varepsilon)$: it is holomorphic if the a_v are not all zero, and otherwise has possible simple poles at 0 and 1 and no other poles. Expressing $\Lambda(s, \varepsilon)$ in terms of such series, we find that $\Lambda(s, \varepsilon)$ has this property as well. $\square$

4 An Irreducibility Theorem

This § concerns the “Hilbert-Blumenthal” moduli problem of classifying abelian varieties with real multiplication. We wish to complement the work of Rapoport [28] with an irreducibility theorem in characteristic p .

We recall that a Hilbert-Blumenthal abelian variety (or : HBAV) relative to the integer ring $\mathcal{O}$ of K over a base S is an abelian scheme X/S , furnished with a homomorphism $m : \mathcal{O} \hookrightarrow \text{End } X$ making $\text{Lie}(X/S)$ into a locally free $\mathcal{O} \otimes \mathcal{O}_S$ -module of rank 1. It follows in particular from this definition that the relative dimension of X over S is the degree r of K over $\mathbb{Q}$. Conversely, if S is of characteristic 0 and X/S is an abelian scheme of relative dimension r furnished with an $m : \mathcal{O} \hookrightarrow \text{End } X$, then the condition on $\text{Lie}(X/S)$ is automatically satisfied. Also, if S is of characteristic $p > 0$, then the condition is satisfied at least whenever X is ordinary.

Let X be a HBAV over an algebraically closed field k . If N is prime to the characteristic of k , the group scheme X_N of N -division points of X is étale, and so may be identified with the group of its points : a free $\mathcal{O}/N\mathcal{O}$ -module of rank 2. Also, we attach to X its polarization module $\mathcal{P}(X)$; this is the invertible $\mathcal{O}$ -module consisting of the symmetric $\mathcal{O}$ -linear homomorphisms $X \rightarrow X^*$, where X^* is the dual of X . This module has a natural positivity, in which we declare positive those homomorphisms defined by a polarization [28, §1].

In the special case $k = \mathbb{C}$, we may interpret X_N and $\mathcal{P}(X)$ in terms of the homology group $T(X) = H_1(X(\mathbb{C}), \mathbb{Z})$. For X_N , we have

$$X_N = \frac{1}{N} T(X) / T(X) \xrightarrow{\sim} T(X) / NT(X). \quad (24)$$

For $\mathcal{P}(X)$, we regard $T(X)$ and $T(X^*)$ as being paired over $\mathbb{Z}$ into $\mathbb{Z}(1) = 2\pi i\mathbb{Z}$. We have

$$T(X^*) = \text{Hom}_{\mathcal{O}}(T(X), \mathfrak{D}^{-1}(1)),$$

which leads to

$$\mathcal{P}(X) = \text{Hom}_{\mathcal{O}}(\wedge_{\mathcal{O}}^2 T(X), \mathfrak{D}^{-1}(1)). \quad (25)$$

(The exterior power is taken over $\mathcal{O}$.) The notion of positivity on $\mathcal{P}(X)$ results from the fact that the real vector space $T(X) \otimes_{\mathbb{Z}} \mathbb{R}$ has a natural complex structure [28, 1.26].

If we combine (24) and (25), we obtain an isomorphism

$$\wedge_{\mathcal{O}}^2 X_N \xrightarrow{\sim} \text{Hom}_{\mathcal{O}}(\mathcal{P}(X), \mathfrak{D}^{-1}(1)) \otimes \mathbb{Z}/N\mathbb{Z}. \quad (26)$$

Over an arbitrary base S , the groups $(X_{\bar{s}})_N$ (with N invertible on S) and $\mathcal{P}(X_{\bar{s}})$ fit together to give local systems X_N , $\mathcal{P}(X)$ (for the étale topology). From the e_m -pairings of Weil, we may construct an analogue of (26) :

$$\wedge_{\mathcal{O}}^2 X_N \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}}(\mathcal{P}(X), \mathfrak{D}^{-1}(1)) \otimes \mu_N. \quad (27)$$

Let $N \geq 3$ be an integer, and let $\mathcal{G}$ be an invertible $\mathcal{O}$ -module of rank 1, given with a positivity. For each base S , we consider triples (X, λ, α) over S consisting of : a HBAV X/S , a positive isomorphism $\lambda : \mathcal{P}(X) \xrightarrow{\sim} \mathcal{G}$, and a level- N structure $\alpha : (\mathcal{O}/N\mathcal{O})^2 \xrightarrow{\sim} X_N$. Let $F[\mathcal{G}, N]$ be the functor

$$S \mapsto \{\text{isomorphism classes of triples } (X, \lambda, \alpha) \text{ over } S\}.$$

According to [28], $F[\mathcal{G}, N]$ is represented by an algebraic space $\mathcal{M} = \mathcal{M}_N[\mathcal{G}]$, smooth over $\mathbb{Z}[\frac{1}{N}]$. One can find a compactification $\bar{\mathcal{M}}$ of $\mathcal{M}$, proper and smooth over $\mathbb{Z}[\frac{1}{N}]$, so that the complement of $\mathcal{M}$ in $\bar{\mathcal{M}}$ is a relative divisor with normal crossings. (We note in passing that the level- N structure is imposed in order to eliminate automorphisms.) It follows that $\mathcal{M}$ has the “same” geometric connected components in characteristic $p \nmid N$ as over $\mathbb{C}$. These components are parameterized by the set of “invertible” elements of the free $\mathcal{O}/N\mathcal{O}$ -module

$$H = (\mathrm{Hom}_{\mathcal{O}}(\mathcal{G}, \mathfrak{D}^{-1}) \otimes \mathbb{Z}/N\mathbb{Z})(1).$$

Explicitly, given a triple (X, λ, α) , we obtain an element of H according to the following recipe. From α and (27) we obtain an isomorphism

$$\mathcal{O}/N\mathcal{O} = \wedge_{\mathcal{O}}^2(\mathcal{O}/N\mathcal{O})^2 \xrightarrow{\sim} \wedge_{\mathcal{O}}^2 X_N \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}}(\mathcal{P}(X), \mathfrak{D}^{-1}) \otimes \mu_N = H',$$

and using λ we have also an isomorphism $H \xrightarrow{\sim} H'$. Composing the two, we obtain an isomorphism $\mathcal{O}/N\mathcal{O} \xrightarrow{\sim} H$, and hence a canonically given “basis vector” of H .

Let $p \nmid N$ be a prime, and let $p^n > 1$ be a power of p . Let $\mathcal{M}^0 = \mathcal{M}_N^0[\mathcal{G}, p]$ be the open subset of the reduction of $\mathcal{M}$ modulo p which corresponds to ordinary abelian varieties. (We recall that X is ordinary if X_p is an extension of an étale group scheme by a group scheme of multiplicative type.) We shall see below that $\mathcal{M}^0$ meets each connected component of $\mathcal{M} \bmod p$, so that it is dense in $\mathcal{M} \bmod p$.

If X/S is an ordinary Hilbert abelian variety in characteristic p , then X_{p^n} is locally (over S for the étale topology) an extension of an étale group scheme, $\mathcal{O}$ -isomorphic to $\mathcal{O}/p^n\mathcal{O}$, by a group scheme of multiplicative type, isomorphic to $\mathcal{O} \otimes \mu_{p^n}$:

$$(X_{p^n})^{\mathrm{\acute{e}t}} \sim \mathcal{O}/p^n\mathcal{O}, \quad (X_{p^n})^0 \sim \mathcal{O} \otimes \mu_{p^n}.$$

Let $J = J[\mathcal{G}, N, p^n]$ be the covering of $\mathcal{M}^0$ defined by :

$$J = \mathrm{Isom}_{\mathcal{M}^0}(\mathcal{O}/p^n\mathcal{O}, (X_{p^n})^{\mathrm{\acute{e}t}}).$$

$X/\mathcal{M}^0$ being the universal Hilbert-Blumenthal abelian variety. It is a principal homogeneous space (“torsor”) over $\mathcal{M}^0$, with structural group $(\mathcal{O}/p^n\mathcal{O})^*$. It may alternately be obtained as the reduction (mod p) of the space J' associated to the moduli problem of classifying quadruples $(X, \lambda, \alpha, \beta)$, with X, λ, α as above, and β an $(\mathcal{O}$ -linear) immersion

$$\mathcal{O} \otimes \mu_{p^n} \hookrightarrow X.$$

The space J' is étale and quasi-finite over $\mathcal{M}$, but not finite.

Theorem 4.1 ((4.5)). *The covering $J[\mathcal{G}, N, p^n]$ of $\mathcal{M}_N^0[\mathcal{G}, p]$ is geometrically irreducible. I.e., if $\bar{\mathbb{F}}$ is an algebraic closure of $\mathbb{F}_p$, then J defines an irreducible covering of each connected component of $\mathcal{M}^0 \otimes_{\mathbb{F}_p} \bar{\mathbb{F}}$.*

Démonstration. Let q be large enough so that all the geometric connected components of $\mathcal{M}^0$ are defined over $\mathbb{F}_q$. (By the above discussion, it suffices in fact to have $\mathbb{F}_q$ containing the N -th roots of 1.) Let $\mathcal{M}_i^0 \subseteq \mathcal{M}^0 \otimes_{\mathbb{F}_p} \bar{\mathbb{F}}_q$ be one of the components, and let $m \in \mathcal{M}_i^0(\mathbb{F}_q)$ be a base point of $\mathcal{M}_i^0$. The covering of $\mathcal{M}_i^0$ induced by J is determined (up to unique isomorphism) by its fibre J_m and the monodromy action of $\pi_1(\mathcal{M}_i^0, m)$ on J_m . This action is given by a character

$$\rho : \pi_1(\mathcal{M}_i^0, m) \rightarrow (\mathcal{O}/p^n \mathcal{O})^*.$$

Irreducibility of the covering, after extension from $\mathbb{F}_q$ to $\bar{\mathbb{F}}_q$, amounts to the surjectivity of $\rho|_{\pi_1(\mathcal{M}_i^0 \otimes_{\mathbb{F}_q} \bar{\mathbb{F}}_q)}$. Geometric irreducibility amounts to surjectivity for all $a \geq 1$: any geometric connected component of J is defined over some $\mathbb{F}_{q^a}$.

Each closed point x of $\mathcal{M}^0 \otimes \mathbb{F}_{q^a}$ defines a Frobenius element F_x in $\pi_1(\mathcal{M}_i^0 \otimes \mathbb{F}_{q^a})$. (Although F is in fact not well-defined, its image under ρ is well defined.) We will prove the required surjectivity by using these Frobenius elements and their powers. (The Cebotarev density theorem tells us that we will be able to do this if the theorem is true.) The powers come in because of the following “base change” formula : Let b be a multiple of a , and let $x \in \mathcal{M}_i^0(\mathbb{F}_{q^b})$. Then x is a closed point of $\mathcal{M}^0 \otimes \mathbb{F}_{q^a}$ but defines as well a closed point x' of $\mathcal{M}^0 \otimes \mathbb{F}_{q^b}$. Let $\mathbb{F}_{q^a}$ be the residue field of x' , so that $a \mid b$. Then we have $F_x^{b/a} = F_{x'}$ in $\pi_1(\mathcal{M}_i^0 \otimes \mathbb{F}_{q^b})$.

Suppose that $x \in \mathcal{M}_i^0(\mathbb{F}_{q^b})$ corresponds to a triple (X, λ, α) over $\mathbb{F}_{q^b}$. Then $\rho(F_x) \in (\mathcal{O}/p^n \mathcal{O})^*$ is the number giving the action of Frobenius on the étale group scheme $\text{Hom}(\mathcal{O} \otimes \mu_{p^n}, (X_{\bar{x}})^0)$. Equivalently, we write the Frobenius endomorphism of the formal group $\hat{X}$ of X as a product $q^b u$, where $u \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$. Then $\rho(F_x)$ is the image of u in $(\mathcal{O}/p^n \mathcal{O})^*$, cf. [15, (42.1)]. (Depending on conventions chosen, one may prefer to take u^{r-1} instead of u at this point ; this problem is irrelevant for our purposes.)

Notice that the recipe just given for $\rho(F_x)$ does not involve either the level N -structure α or the isomorphism λ . On the other hand, the action of $\text{GL}(2, \mathcal{O}/N\mathcal{O})$ on $\mathcal{M}^0$ deduced from

$$g : (X, \lambda, \alpha) \mapsto (X, \lambda, \alpha \circ g^{-1})$$

permutes the geometric connected components of $\mathcal{M}^0$. This enables us to ignore the distinction between $\mathcal{M}^0(\mathbb{F}_{q^a})$ and $\mathcal{M}^0(\mathbb{F}_{q^b})$ in finding a triple (X, λ, α) such that the associated u maps mod p^n to a desired element $\bar{u}$ of $(\mathcal{O}/p^n \mathcal{O})^*$; we are required only to find an ordinary $X/\mathbb{F}_{q^b}$, for some multiple b of a , with the following properties :

- (i) X admits a level- N structure α over $\mathbb{F}_{q^b}$;
- (ii) The number $u \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$ computed for X maps to the given $\bar{u}$ in $(\mathcal{O}/p^n \mathcal{O})^*$;
- (iii) The polarization module $\mathcal{P}(X)$ is isomorphic to the given $\mathcal{G}$ (as an invertible $\mathcal{O}$ -module with positivity).

Carrying out this construction will prove the surjectivity, and thus the theorem.

We eliminate consideration of requirement (iii) by the following trick. Let M be a positive integer, prime to pN , such that the ideal $M\mathcal{O}$ is divisible by some ideal of $\mathcal{O}$ in each strict ideal class of K . Assume that $X/\mathbb{F}_{q^b}$ satisfies (ii) and admits a level NM structure over $\mathbb{F}_{q^b}$, but does not necessarily satisfy (iii). Choose a level NM structure for X , and view it as a level M structure plus a level N structure. If $\mathfrak{c}$ is an ideal with $M\mathcal{O} \subseteq \mathfrak{c}\mathcal{O}$, we may divide X by the image in X of the group $(\mathcal{O}/\mathfrak{c}\mathcal{O}) \bmod M$ under

the immersion $(\mathcal{O}/M\mathcal{O})^2 \hookrightarrow X_M$. We obtain a variety $X_{\mathcal{G}}$, which satisfies (i) and (ii). It is easily seen (and at least guessed from (25)) that the polarization module $\mathcal{P}(X')$ is isomorphic to $\mathcal{P}(X) \otimes \mathfrak{c}^{-1}$. Since we may choose $\mathfrak{c}$ in any strict ideal class of K , we may arrange $\mathcal{P}(X')$ to be isomorphic to the given $\mathcal{G}$. We may thus forget about condition (iii), after replacing N by a suitable multiple.

To construct varieties X satisfying (i) and (ii), we use the description given in [4] of the category of ordinary abelian varieties over a finite field. The equivalence of categories furnished by [4] identifies ordinary Hilbert-Blumenthal abelian varieties with pairs $(\mathcal{F}, F)$, where $\mathcal{F}$ is a locally free $\mathcal{O}$ -module of rank 2, and F is a “Frobenius” endomorphism. One requires that F satisfy :

- (a) $\det F = q^b$, and $(\mathrm{tr} F)^2 - 4 \det F < 0$;
- (b) the completion $\mathcal{F} \otimes \mathbb{Z}_p$ of $\mathcal{F}$ decomposes as the sum $\mathcal{F}' \oplus \mathcal{F}''$ of two F -stable free $(\mathcal{O} \otimes \mathbb{Z}_p)$ -modules of rank 1, with F acting as $q^{1/b}u$ on $\mathcal{F}'$ and as u^{-1} on $\mathcal{F}''$ for some $u \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$.

[The number u appearing in (b) is the number u associated above to the abelian variety corresponding to $(\mathcal{F}, F)$.] Varieties which admit level- N structures correspond to pairs $(\mathcal{F}, F)$ which satisfy in addition :

- (c) We have $F \equiv 1 \pmod{N}$.

We will construct pairs $(\mathcal{F}, F)$ by considering irreducible polynomials $X^2 + cX + d$ ($c, d \in \mathcal{O}$) and taking $\mathcal{F}$ to be the integer ring of $K[X]/(X^2 + cX + d)$ and F to be multiplication by X . To satisfy conditions (a) and (b), we need :

- (1) $d = q^b$;
- (2) $c^2 - 4q^b \ll 0$;
- (3) $c \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$.

For (c), we require that $\frac{X-1}{N}$ be integral :

- (4) $N \mid (2 + c)$, $N^2 \mid (1 + q^b + c)$.

Then u^{-1} , being the unit root of $X^2 + cX + d$, will be congruent to $-c$, mod q^b .

To complete the proof of the theorem, we pick $c \in \mathcal{O}$ so that $c \equiv -2 \pmod{N^2}$ and $c \equiv -\bar{u}^{-1} \pmod{p^n}$. Next we choose b so that $c^2 - 4q^b \ll 0$, $q^b \equiv 1 \pmod{N}$, and $p^n \mid q^b$. Conditions (1) through (4) are satisfied, and we have $u^{-1} \equiv -c \equiv -\bar{u}^{-1} \pmod{p^n}$ as required. $\square$

We now define a $\Gamma_0(N)$ structure on a Hilbert-Blumenthal abelian variety to be an $\mathcal{O}$ -linear immersion $\mathcal{O} \otimes \frac{1}{N}\mathbb{Z}/\mathbb{Z} \hookrightarrow X$. (Here N is a positive integer, not necessarily prime to p .) Let $\mathcal{G}$ be a locally free rank-one $\mathcal{O}$ -module with positivity, and let $F[\mathcal{G}, N]$ be the functor

$$S \mapsto \{\text{isomorphism classes of triples } (X, \lambda, \alpha) \mid \begin{array}{l} X \text{ is a Hilbert-Blumenthal} \\ \text{abelian variety over } S; \lambda \text{ is a positive isomorphism } \mathcal{P}(X) \cong \mathcal{G}; \alpha \\ \text{is a } \Gamma_0(N)\text{-structure} \end{array}\}.$$

For N sufficiently large, such triples have no automorphisms, and $F[\mathcal{G}, N]$ is represented by an algebraic space $\mathcal{M}[\mathcal{G}, N]$, smooth over $\mathbb{Z}$.

Corollary 4.2 ((4.6)). *For each prime p , the reduction mod p of $\mathcal{M}[\mathcal{G}, N]$ is geometrically irreducible.*

Démonstration. Let us write $N = p^n N'$, where $p \nmid N'$. Since $\mathcal{M}[\mathcal{G}, N]$ is surjective mod p when N is a multiple of N' , we may prove the corollary with N replaced by any multiple of N' . Hence we may assume that N' is so large that the moduli space $\mathcal{M}[\mathcal{G}, N']$ exists.

Over the algebraic closure $\bar{\mathbb{F}}$ of $\mathbb{F}_p$, let us fix an embedding $\mu_N \hookrightarrow (\mathbb{Z}/N'\mathbb{Z})^2$. After tensoring with $\mathcal{O}$, we get a map $g : \mathcal{O} \otimes \mu_N \hookrightarrow (\mathcal{O}/N'\mathcal{O})^2$. Via g , a level N' structure on a Hilbert-Blumenthal abelian variety gives rise to a $\Gamma_0(N)$ -structure on the variety, so that we have a map $\mathcal{M}_{N'}[\mathcal{G}] \rightarrow \mathcal{M}[\mathcal{G}, N']$ over $\bar{\mathbb{F}}$. If Γ is the subgroup of $\mathrm{GL}(2, \mathcal{O}/N'\mathcal{O})$ fixing g , we find an isomorphism $\mathcal{M}_{N'}[\mathcal{G}]/\Gamma \xrightarrow{\sim} \mathcal{M}[\mathcal{G}, N']$ over $\bar{\mathbb{F}}$. Now Γ acts transitively on the geometric connected components of $\mathcal{M}_{N'}[\mathcal{G}]$ because $\det \Gamma = (\mathcal{O}/N'\mathcal{O})^*$; this gives the irreducibility of $\mathcal{M}[\mathcal{G}, N']$ over $\bar{\mathbb{F}}$, proving the corollary in case $N = N'$. If $p^n > 1$ we remark that similarly there is an isomorphism

$$\mathcal{M}_{N'}[\mathcal{G}, p^n]/\Gamma \xrightarrow{\sim} \mathcal{M}[\mathcal{G}, N] \quad (\text{over } \bar{\mathbb{F}}).$$

The theorem implies that Γ permutes the geometric connected components of $\mathcal{M}_{N'}[\mathcal{G}, p^n]$ (they are the same as those of $\mathcal{M}_{N'}[\mathcal{G}]$), so we again get the statement of the corollary. $\square$

Completion of §4. An Irreducibility Theorem (continued)

[Theorem (4.5) was stated on page 255. Its proof was in progress. We resume at the point where varieties satisfying conditions (i), (ii), (iii) are being constructed.]

To construct varieties X satisfying (i) and (ii), we use the description given in [4] of the category of ordinary abelian varieties over a finite field. The equivalence of categories furnished by [4] identifies ordinary Hilbert–Blumenthal abelian varieties with pairs $(\mathcal{E}, F)$, where $\mathcal{E}$ is a locally free $\mathcal{O}$ -module of rank 2, and F is a “Frobenius” endomorphism. One requires that F satisfy :

- (a) $\det F = q^b$, and $(\operatorname{tr} F)^2 - 4 \det F < 0$,
- (b) the completion $\mathcal{E} \otimes \mathbb{Z}_p$ of $\mathcal{E}$ decomposes as the sum $\mathcal{E}' \oplus \mathcal{E}''$ of two F -stable free $(\mathcal{O} \otimes \mathbb{Z}_p)$ -modules of rank 1, with F acting as $q^{1/b}u$ on $\mathcal{E}'$ and as u^{-1} on $\mathcal{E}''$ for some $u \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$.

[The number u appearing in (b) is the number u associated above to the abelian variety corresponding to $(\mathcal{E}, F)$.] Varieties which admit level- N structures correspond to pairs $(\mathcal{E}, F)$ which satisfy in addition :

- (c) We have $F \equiv 1 \pmod{N}$.

We will construct pairs $(\mathcal{E}, F)$ by considering irreducible polynomials $X^2 + cX + d$ ($c, d \in \mathcal{O}$) and taking $\mathcal{E}$ to be the integer ring of $K[X]/(X^2 + cX + d)$ and F to be multiplication by X . To satisfy conditions (a) and (b), we need :

- (1) $d = q^b$;
- (2) $c^2 - 4q^b \ll 0$;
- (3) $c \in (\mathcal{O} \otimes \mathbb{Z}_p)^*$.

For (c), we require that $\frac{X-1}{N}$ be integral :

- (4) $N \mid (2 + c)$, $N^2 \mid (1 + q^b + c)$.

Then u , being the unit root of $X^2 + cX + d$, will be congruent to $-c$, mod q^b .

To complete the proof of the theorem, we pick $c \in \mathcal{O}$ so that $c \equiv -2 \pmod{N^2}$ and $c \equiv -\tilde{u} \pmod{p^n}$. Next we choose b so that $c^2 - 4q^b \ll 0$, $q^b \equiv 1 \pmod{N}$, and $p^n \mid q^b$. Conditions (1) through (4) are satisfied, and we have $u^{-1} \equiv -c^{-1} \equiv -\tilde{u}^{-1} \pmod{p^n}$ as required. □

We now define a $\Gamma_0(N)$ structure on a Hilbert–Blumenthal abelian variety to be an $\mathcal{O}$ -linear immersion $\mathcal{O} \otimes \mu_N \hookrightarrow X$. (Here N is a positive integer, not necessarily prime to p .) Let $\mathcal{L}$ be a locally free rank-one $\mathcal{O}$ -module with positivity, and let $\mathcal{F}[\mathcal{L}, N]$ be the functor

$S \longmapsto \{\text{isomorphism classes of triples } (X, \lambda, \alpha) \mid X \text{ is a Hilbert-Blumenthalabelian variety over } S; \lambda \text{ is}$

For N sufficiently large, such triples have no automorphisms, and $\mathcal{F}[\mathcal{L}, N]$ is represented by an algebraic space $\mathcal{M}[\mathcal{L}, N]$, smooth over $\mathbb{Z}$.

Corollaire 0.1 ((4.6)). *For each prime p , the reduction mod p of $\mathcal{M}[\mathcal{L}, N]$ is geometrically irreducible.*

Démonstration. Let us write $N = p^n N'$, where $p \nmid N'$. Since $\mathcal{M}[\mathcal{L}, N]$ is surjective mod p when N is a multiple of N' , we may prove the corollary with N replaced by any multiple of N' . Hence we may assume that N' is so large that the moduli space $\mathcal{M}[\mathcal{L}, N']$ exists.

Over the algebraic closure $\overline{\mathbb{F}}$ of $\mathbb{F}_p$, let us fix an embedding $\mu_{N'} \hookrightarrow (\mathbb{Z}/N'\mathbb{Z})^2$. After tensoring with $\mathcal{O}$, we get a map $g : \mathcal{O} \otimes \mu_{N'} \hookrightarrow (\mathcal{O}/N'\mathcal{O})^2$. Via g , a level N' structure on a Hilbert–Blumenthal abelian variety gives rise to a $\Gamma_0(N')$ -structure on the variety, so that we have a map $\mathcal{M}_{N'}[\mathcal{L}] \rightarrow \mathcal{M}[\mathcal{L}, N']$ over $\overline{\mathbb{F}}$. If Γ is the subgroup of $\mathrm{GL}(2, \mathcal{O}/N'\mathcal{O})$ fixing g , we find an isomorphism $\mathcal{M}_{N'}[\mathcal{L}]/\Gamma \xrightarrow{\sim} \mathcal{M}[\mathcal{L}, N']$ over $\overline{\mathbb{F}}$. Now Γ acts transitively on the geometric connected components of $\mathcal{M}_{N'}[\mathcal{L}]$ because $\det \Gamma = (\mathcal{O}/N'\mathcal{O})^*$; this gives the irreducibility of $\mathcal{M}[\mathcal{L}, N']$ over $\overline{\mathbb{F}}$, proving the corollary in case $N = N'$. If $p^n > 1$ we remark that similarly there is an isomorphism

$$\mathcal{M}_{N'}[\mathcal{L}, p^n]/\Gamma \xrightarrow{\sim} \mathcal{M}[\mathcal{L}, N] \quad (\text{over } \overline{\mathbb{F}}).$$

The theorem implies that Γ permutes the geometric connected components of $\mathcal{M}_{N'}[\mathcal{L}, p^n]$ (they are the same as those of $\mathcal{M}_{N'}[\mathcal{L}, N']$), so we again get the statement of the corollary. $\square$

1 Hilbert Modular Forms

In this §, we review the connections among complex, algebraic, and p -adic Hilbert modular forms (for K), emphasizing the q -expansions of these forms. Our irreducibility theorem (4.6) implies a q -expansion principle for such forms, which states (roughly speaking) that a form is determined by any one of its q -expansions.

In our exposition, we will follow very closely Chap. I of Katz’s paper [19]. In order to make our discussion consistent with that of [19], we shall slightly change the definition of a $\Gamma_0(N)$ -structure on a HBAV X/S . Namely, we shall now understand such a structure to mean an embedding

$$i : (N^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1})(1) \hookrightarrow X$$

rather than an embedding

$$\alpha : ((\mathcal{O}/N\mathcal{O})^2 \hookrightarrow X$$

as above. This change has little importance for what follows, in light of the correspondence $X \mapsto X \otimes \mathfrak{D}^{-1}$ between HBAV’s provided with a $\Gamma_0(N)$ structure in the previous sense and those provided with a $\Gamma_0(N)$ -structure in the new sense.

In order to define modular forms, we choose (and fix) an ideal $\mathfrak{c}$ of K , which will serve as polarization module. We view $\mathfrak{c}$ as a signed $\mathcal{O}$ -module (2.25), endowing it with its canonical positivity $+\mathrm{can}$. This choice being made, modular forms are functions of quadruples (X, λ, ω, i) , where :

- X is a HBAV over a ring R ,
- $\lambda : \mathcal{P}(X) \xrightarrow{\sim} \mathfrak{c}$ is a positive isomorphism,
- ω is a basis of the $\mathcal{O} \otimes R$ -module $\underline{\omega}_{X/R}$, apriori a locally free $\mathcal{O} \otimes R$ -module of rank 1.
- $i : N^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1} \hookrightarrow X$ is a $\Gamma_0(N)$ -structure (over R).

Note that ω amounts to an $\mathcal{O}$ -linear isomorphism

$$\mathrm{Lie}(X/R) \xrightarrow{\sim} \mathfrak{D}^{-1} \otimes R. \tag{1}$$

For k a positive integer, we let $M_k(\Gamma_0(N), R)$ be the R -module of (Hilbert) modular forms of weight k on $\Gamma_0(N)$ over R , for the polarization module $\mathfrak{c}$. Recall that an element of $M_k(\Gamma_0(N), R)$ is a function F defined on quadruples (X, λ, ω, i) over R -algebras R' such that :

- For (X, λ, ω, i) over R' , the value $F((X, \lambda, \omega, i))$ is an element of R' which depends only on the R' -isomorphism class of (X, λ, ω, i) .
- For (X, λ, ω, i) over R' and $\alpha \in (\mathcal{O} \otimes R')^*$, we have

$$F((X, \lambda, \alpha\omega, i)) = \mathcal{N}\alpha^{-k} \cdot F((X, \lambda, \omega, i)).$$

- For (X, λ, ω, i) over R' and $f : R' \rightarrow R''$ a homomorphism of R -algebras, we have

$$F((X, \lambda, \omega, i)_{/R''}) = f(F((X, \lambda, \omega, i))).$$

That is, the function F commutes with any extension of scalars $R' \rightarrow R''$.

An (unramified) cusp on $\Gamma_0(N)$ over R is determined by the following data :

- Two ideals $\mathfrak{a}, \mathfrak{b}$ such that $\mathfrak{a}\mathfrak{b}^{-1} = \mathfrak{c}$.
- An $\mathcal{O}$ -linear isomorphism $\varepsilon : N^{-1}\mathcal{O}/\mathcal{O} \xrightarrow{\sim} N^{-1}\mathfrak{a}^{-1}/\mathfrak{a}^{-1}$.
- An $\mathcal{O} \otimes R$ -linear isomorphism $j : \mathfrak{a}^{-1} \otimes R \xrightarrow{\sim} \mathcal{O} \otimes R$.

Roughly speaking, these three data give rise, respectively to :

- A HBAV analogue $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$ of the Tate curve [15].
- A $\Gamma_0(N)$ -structure i on $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$.
- A basis vector $\omega = \omega(i)$ for $\underline{\omega}_{\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}}$,

all over a suitable power series ring. Further, $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$ is provided with a canonical λ . By evaluating a modular form $F \in M_k(\Gamma_0(N), R)$ on the resulting quadruple, we obtain a formal power series

$$F(\mathfrak{a}, \mathfrak{b}, \varepsilon, j) = \sum_{\substack{\mu \in \mathfrak{a}\mathfrak{b} \\ \mu \gg 0 \text{ or } \mu=0}} c_\mu q^\mu$$

whose coefficients $c_\mu = c_\mu(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ are elements of R . This is the q -expansion of F at the cusp (determined by) $\mathfrak{a}, \mathfrak{b}, \varepsilon, j$.

Now suppose that $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ is a cusp, and suppose that α is an invertible element of $\mathcal{O} \otimes R$. Let $\alpha \cdot j$ be the composite

$$\mathfrak{a}^{-1} \otimes R \xrightarrow{j} \mathcal{O} \otimes R \xrightarrow{\text{mult. by } \alpha} \mathcal{O} \otimes R.$$

Then, since F is of weight k , we have the formula

$$F(\mathfrak{a}, \mathfrak{b}, \varepsilon, \alpha \cdot j) = \mathcal{N}\alpha^{-k} F(\mathfrak{a}, \mathfrak{b}, \varepsilon, j). \quad (2)$$

Exemple 1.1 ((5.3)). If R is a $\mathbb{Q}$ -algebra, the equality

$$\mathfrak{D}^{-1} \otimes \mathbb{Q} = \mathcal{O} \otimes \mathbb{Q}$$

provides us with a canonical isomorphism j_{can} between $\mathfrak{a}^{-1} \otimes R$ and $\mathcal{O} \otimes R$. Suppose, further, that we are given a finite idele $\alpha \in \mathbb{A}_K^*$. We may then take $\mathfrak{a} = (\alpha)$ to be the ideal generated by α , $\mathfrak{b} = \mathfrak{c}\mathfrak{a}^{-1}$, and ε to be the isomorphism

$$N^{-1}\mathcal{O}/\mathcal{O} \xrightarrow{\sim} N^{-1}\mathfrak{a}^{-1}/\mathfrak{a}^{-1} = N^{-1}\alpha^{-1}\mathcal{O}/\alpha^{-1}\mathcal{O}.$$

We shall refer to the cusp $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j_{\text{can}})$ simply as the cusp determined by α .

Our irreducibility theorem (4.6) implies the following “ q -expansion principles” cf. [6, VII, §3], [28, §6]) :

Théorème (5.4). *Let $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ be a cusp on $\Gamma_0(N)$ over R . Suppose that $F(\mathfrak{a}, \mathfrak{b}, \varepsilon, j) = 0$, where $F \in M_k(\Gamma_0(N), R)$. Then $F = 0$.*

Théorème (5.5). *Further, let R_0 be a subring of R , and suppose that $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ is in fact a cusp on $\Gamma_0(N)$ over R_0 . Suppose that the q -expansion coefficients c_μ of $F(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ all lie in R_0 . Then F is an element of $M_k(\Gamma_0(N), R_0)$.*

[In connection with the conclusion of (5.5), we remark that there is a natural map $M_k(\Gamma_0(N), R_0) \rightarrow M_k(\Gamma_0(N), R)$ because every R -algebra R' is in particular an R_0 -algebra. The injectivity of this map may be deduced from (5.4).]

Complex Hilbert Modular Forms

Over the complex field $\mathbb{C}$, we may express quadruples (X, λ, ω, i) in terms of lattices $\mathcal{E}$ in the $\mathbb{C}$ -vector space $K \otimes_{\mathbb{Q}} \mathbb{C}$. (We tacitly suppose all such lattices to be stable under $\mathcal{O}$.) Specifically, the data (X, λ, ω, i) amount to the giving of :

- A lattice $\mathcal{E} \subset K \otimes \mathbb{C}$.
- An ($\mathcal{O}$ -linear) isomorphism $\lambda : \bigwedge_{\mathcal{O}} \mathcal{E} \xrightarrow{\sim} 2\pi i \cdot \mathfrak{D}^{-1} \mathfrak{c}^{-1}$.
- An $\mathcal{O}$ -linear embedding $i : N^{-1} \mathfrak{D}^{-1} / \mathfrak{D}^{-1} \hookrightarrow N^{-1} \mathcal{E} / \mathcal{E}$.

We may view λ as an $\mathcal{O}$ -linear alternating form $\lambda(x, y)$ on $\mathcal{E} \times \mathcal{E}$, giving rise by linearity to a $K \otimes \mathbb{R}$ -linear alternating form on $K \otimes \mathbb{C}$. For $\alpha \in K \otimes \mathbb{C}$, we have

$$\lambda(\alpha x, y) = (\alpha \otimes 1) \cdot \lambda(x, y),$$

where $\alpha \otimes 1$ is the conjugate of α in $K \otimes \mathbb{C}$.

From this point of view, an element of $M_k(\Gamma_0(N), \mathbb{C})$ is simply a holomorphic function F on the space of triples $(\mathcal{E}, \lambda, i)$ which satisfies :

$$F(\alpha \mathcal{E}, (\alpha \otimes \alpha)^{-1} \lambda, i) = \mathcal{N} \alpha^{-k} F(\mathcal{E}, \lambda, i) \quad (3)$$

for $\alpha \in (K \otimes \mathbb{C})^*$. Given such a function, we may obtain the q -expansion of the corresponding modular form at a cusp of the form $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j_{\text{can}})$ (cf. (5.3)) by evaluating the function on a suitable triple $(\mathcal{E}, \lambda, i)$.

This triple is constructed as follows. First, we introduce the lattice $\mathcal{E} = 2\pi i(\mathfrak{c}^{-1} \mathfrak{a}^{-1} + \mathfrak{b} \tau)$, where τ is a “variable” element of the “upper half plane”

$$\mathfrak{H} = \{\tau \in K \otimes \mathbb{C} \mid \Im \tau \gg 0\}.$$

Second, we define an alternating form $\lambda = \lambda_{\text{can}}$ by the formula

$$\lambda((2\pi i)(\alpha + \beta \tau), (2\pi i)(\gamma + \delta \tau)) = \alpha \delta - \beta \gamma.$$

Finally, we let i be the embedding

$$N^{-1} \mathfrak{D}^{-1} / \mathfrak{D}^{-1} \xrightarrow{\varepsilon} N^{-1} \mathfrak{a}^{-1} / \mathfrak{a}^{-1} \hookrightarrow N^{-1} \mathcal{E} / \mathcal{E},$$

the first map being the isomorphism deduced from ε and the second being induced by multiplication by $2\pi i$.

If we evaluate an F as above on $(\mathcal{E}, \lambda, i)$, we obtain a holomorphic function $F(\tau)$ which is invariant under translations by elements of $\mathfrak{a}^{-1}\mathfrak{b}^{-1} \cap \mathfrak{b}^{-1}$. It may consequently be expanded as a Fourier series

$$F(\tau) = F_{\mathfrak{a}, \mathfrak{b}, \varepsilon}(\tau) = \sum_{\substack{\mu \in \mathfrak{ab} \\ \mu \gg 0 \text{ or } \mu=0}} c_{\mu} e^{2\pi i \cdot \text{tr}(\mu \cdot \tau)}$$

the summation being restricted to those elements μ of $\mathfrak{ab}$ which are either 0 or totally positive. (Here tr is the trace map $K \otimes \mathbb{C} \rightarrow \mathbb{C}$.) Then the numbers c_{μ} are precisely the coefficients $c_{\mu}(\mathfrak{a}, \mathfrak{b}, \varepsilon, j_{\text{can}})$ of the q -expansion of the modular form determined by F , at the cusp (over $\mathbb{C}$) determined by $\mathfrak{a}, \mathfrak{b}$, and ε . It is usual to make the formal substitution

$$q^{\mu} = e^{2\pi i \cdot \text{tr}(\mu \cdot \tau)}.$$

this underscores the equality between the Fourier expansion of $F(\tau)$ and the q -expansion of the corresponding modular form.

The Modular Forms of § ??

According to (5.7), the weight- k modular forms defined in the Introduction are precisely the standard q -expansions $F_1(\tau)$ of forms $F \in M_k(\Gamma_0(N), \mathbb{C})$, with the polarization module $\mathfrak{c}$ chosen to be $\mathcal{O}$. The “other” q -expansions of these forms were defined by a formula which is now justified by (5.8). Therefore (0.1) is a special case of the q -expansion principle (5.4), (5.5).

p -adic Modular Forms

Let p be a prime. A $\Gamma_0(Np^{\infty})$ -structure on a HBAV X is a compatible system of $\Gamma_0(Np^n)$ -structures, for $n \geq 0$. Let R be a p -adic ring : $R = \varprojlim R/p^n R$. A p -adic Hilbert modular form on $\Gamma_0(N)$ over R is a function F of triples (X, λ, i) , where :

- X is a HBAV over a p -adic R -algebra R' ,
- $\lambda : \mathcal{P}(X) \xrightarrow{\sim} \mathfrak{c}$ is a positive isomorphism,
- i is a $\Gamma_0(Np^{\infty})$ -structure on X .

It is required that $F(X, \lambda, i)$ depend only on the R' -isomorphism class of $(X, \lambda, i)/R'$ and that F commute with extensions of scalars $R' \rightarrow R''$.

We let $V(\Gamma_0(N), R)$ be the R -module of p -adic Hilbert modular forms over R .

The key point concerning p -adic modular forms is that there is a canonical way to associate a p -adic modular form to each (Hilbert) modular form, of any weight, over R . We recall that whenever X/R has a $\Gamma_0(p^{\infty})$ -structure i , we may deduce from i an isomorphism of formal groups

$$\mathfrak{D}^{-1} \otimes \hat{\mathbb{G}}_m \xrightarrow{\sim} \hat{X}$$

and hence an isomorphism of their Lie algebras

$$\mathfrak{D}^{-1} \otimes R \xrightarrow{\sim} \text{Lie}(X/R).$$

There is thus a canonically chosen basis $\omega(i)$ of $\underline{\omega}_{X/R}$.

The map on test objects

$$(X, \lambda, i) \longmapsto (X, \lambda, \omega(i), i)$$

induces a map

$$M_k(\Gamma_0(N), R) \longrightarrow V(\Gamma_0(N), R)$$

for each k . We denote this map by η_k , or simply η .

To make q -expansions for p -adic modular forms, let $\mathfrak{a}$ and $\mathfrak{b}$ be ideals of K with $\mathfrak{a}\mathfrak{b}^{-1} = \mathfrak{c}$ and let $\varepsilon = (\varepsilon_n)$ be a compatible system of isomorphisms

$$\varepsilon_n : p^{-n}N^{-1}\mathcal{O}/\mathcal{O} \xrightarrow{\sim} p^{-n}N^{-1}\mathfrak{a}^{-1}/\mathfrak{a}^{-1}.$$

These data provide us with a “Tate variety” $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$, as before, together with a $\Gamma_0(Np^\infty)$ -structure on $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$, which we denote by $i(\varepsilon)$. Given a p -adic modular form F , its q -expansion $F(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ is obtained by evaluating F on the triple consisting of $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$, the canonical isomorphism

$$\lambda : \mathcal{P}(\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}) \xrightarrow{\sim} \mathfrak{c},$$

and the level structure $i(\varepsilon)$. (We refer to $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ as a “ p -adic cusp”.)

It is easy to describe the differential $\omega(i(\varepsilon))$ on $\mathcal{X}^{\mathfrak{a}, \mathfrak{b}}$ in terms of an isomorphism

$$j = j(\varepsilon) : \mathfrak{a}^{-1} \otimes R \xrightarrow{\sim} \mathcal{O} \otimes R.$$

Namely, the ε_n give isomorphisms

$$p^{-n}\mathcal{O}/\mathcal{O} \xrightarrow{\sim} p^{-n}\mathfrak{a}^{-1}/\mathfrak{a}^{-1}$$

for each n , and hence an isomorphism

$$\mathcal{O} \otimes \mathbb{Z}_p \xrightarrow{\sim} \mathfrak{a}^{-1} \otimes \mathbb{Z}_p.$$

Tensoring with R gives an isomorphism between $\mathfrak{a}^{-1} \otimes R$ and $\mathcal{O} \otimes R$, which is just the inverse of the desired map j . This gives the compatibility :

Théorème (5.10). *If $F \in M_k(\Gamma_0(N), R)$ is a modular form of weight k , and $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ is a p -adic cusp, then*

$$(\eta_k F)(\mathfrak{a}, \mathfrak{b}, \varepsilon) = F(\mathfrak{a}, \mathfrak{b}, \varepsilon_0, j(\varepsilon)).$$

Exemple 1.5 ((5.11)). Let $\alpha \in \mathbb{A}_K^*$. A p -adic cusp $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ is defined by α as follows. We take $\mathfrak{a} = (\alpha)$ and $\mathfrak{b} = \alpha\mathfrak{c}^{-1}$, as in (5.3), and for each $n \geq 0$ we define ε_n to be the composite

$$p^{-n}N^{-1}\mathcal{O}/\mathcal{O} \xrightarrow{\alpha} N^{-1}p^{-n}\mathfrak{a}^{-1}/\mathfrak{a}^{-1} = p^{-n}N^{-1}\alpha^{-1}\mathcal{O}/\alpha^{-1}\mathcal{O},$$

cf. (5.3). Let α_p be the image of α in $(K \otimes \mathbb{Q}_p)^*$. Multiplication by α_p induces an isomorphism

$$\mathfrak{a}^{-1} \otimes \mathbb{Z}_p \xrightarrow{\sim} \mathcal{O} \otimes \mathbb{Z}_p. \tag{4}$$

Tensoring with R , we obtain an isomorphism between $\mathfrak{a}^{-1} \otimes R$ and $\mathcal{O} \otimes R$, which is just the map $j(\alpha)$ discussed above.

We now consider the situation where R is a flat $\mathbb{Z}_p$ -module. We define $V(\Gamma_0(N), E)$, where $E = R \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$, to be the tensor product

$$V(\Gamma_0(N), R) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p.$$

Then for each k we have a map

$$\eta_k : M_k(\Gamma_0(N), E) = M_k(\Gamma_0(N), R) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \longrightarrow V(\Gamma_0(N), E),$$

obtained by tensoring the previous η_k with $\mathbb{Q}_p$. Given data $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ as above, we may define for each $F \in V(\Gamma_0(N), E)$ its q -expansion $F(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ at $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$. Then we have the following q -expansion principle.

Théorème (5.13). *Let $F \in V(\Gamma_0(N), E)$, and let $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ be a p -adic cusp. Then the q -expansion $F(\mathfrak{a}, \mathfrak{b}, \varepsilon)$, a priori a power series with coefficients in E , in fact has coefficients in R if and only if F is an element of $V(\Gamma_0(N), R)$.*

Corollaire 1.7 ((5.14)). *Suppose that $F \in V(\Gamma_0(N), E)$ has at one cusp $(\mathfrak{a}, \mathfrak{b}, \varepsilon)$ a q -expansion whose non-constant coefficients are all elements of R . Then the difference between the constant terms of the q -expansions of F at any two cusps is an element of R .*

Démonstration. Let $t \in E$ be the constant term of $F(\mathfrak{a}, \mathfrak{b}, \varepsilon)$. We may view t as an element of $M_0(\Gamma_0(N), E)$ and hence in particular as an element of $V(\Gamma_0(N), E)$. The q -expansion of t at each cusp is the constant power series t . By the hypothesis on F , and because of (5.13), $F - t$ belongs to $V(\Gamma_0(N), R)$. The q -expansion of this form at each cusp has coefficients in R , and so in particular its constant term belongs to R . This proves the required “integrality.” $\square$

Théorème (5.15). *Let $F \in M_k(\Gamma_0(N), E)$, and for $\alpha \in \mathbb{A}_K^*$ let F_α denote the q -expansion of F at the cusp determined by α (5.3). Then the q -expansion of $\eta_k F$ at the p -adic cusp determined by α (5.11) is given by*

$$\mathcal{N}\alpha_p^{-k} \cdot F_\alpha,$$

where $\alpha_p \in (K \otimes \mathbb{Q}_p)^*$ is the component of α at p .

To see this, we write the p -adic q -expansion as

$$F(\mathfrak{a}, \mathfrak{b}, \varepsilon_0, j),$$

where j is the isomorphism $\mathfrak{a}^{-1} \otimes E \xrightarrow{\sim} \mathcal{O} \otimes E$ induced by multiplication by α_p , viewed as a map $\mathfrak{a}^{-1} \otimes \mathbb{Z}_p \xrightarrow{\sim} \mathcal{O} \otimes \mathbb{Z}_p$. (The ideals $\mathfrak{a}$ and $\mathfrak{b}$, and the map ε_0 , are defined in (5.11).) The q -expansion F_α is

$$F(\mathfrak{a}, \mathfrak{b}, \varepsilon_0, j_{\text{can}}),$$

where j_{can} is the isomorphism $\mathfrak{a}^{-1} \otimes E \xrightarrow{\sim} \mathcal{O} \otimes E$ derived from the equality $\mathfrak{a}^{-1} \otimes \mathbb{Q} = \mathcal{O} \otimes \mathbb{Q}$. We thus have $j = \alpha_p \cdot j_{\text{can}}$, so that the desired formula follows from (5.2).

Now for each $k \geq 1$, let F_k be an element of $M_k(\Gamma_0(N), E)$. Assume that $F_k = 0$ for k sufficiently large. Then, by the above discussion, the p -adic modular form

$$\sum_{k \geq 0} \eta_k F_k$$

has q -expansion

$$\sum_{k \geq 0} \mathcal{N}\alpha_p^{-k} \cdot F_{k, \alpha}$$

at the p -adic cusp determined by α . (Here we have written $F_{k, \alpha}$ for the q -expansion $(F_k)_\alpha$ of F_k at the cusp determined by α .) By (5.13), we see that this power series has coefficients in R for one α if and only if it has coefficients in R for each α . Similarly, by (5.14), if the non-constant coefficients of one of these series all lie in R , then the difference between the constant coefficients of any two of these series lies in R .

We thus see, in particular, how (0.2) and (0.3) are special cases of (5.13) and (5.14), respectively.

Variant : Forms on $\Gamma_0(\mathfrak{f})$

We suppose that $\mathfrak{f}$ is a conductor, i.e., a non-zero integral ideal of K . A $\Gamma_0(\mathfrak{f})$ -structure on a HBAV X is an embedding

$$(\mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1})(1) \hookrightarrow X.$$

A modular form of weight k on $\Gamma_0(\mathfrak{f})$ over R is a function of tuples (X, λ, ω, i) as before, with i now a $\Gamma_0(\mathfrak{f})$ -structure on X over R . We denote the space of such forms by $M_k(\Gamma_0(\mathfrak{f}), R)$.

Let N be a positive integer divisible by $\mathfrak{f}$. Then an element of $M_k(\Gamma_0(\mathfrak{f}), R)$ is simply an element of $M_k(\Gamma_0(N), R)$ which enjoys the following extra invariance : its value on a tuple (X, λ, ω, i) , where i is a $\Gamma_0(N)$ -structure on X , depends only on X, λ, ω , and the composite

$$(\mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1})(1) \hookrightarrow (N^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1})(1) \xrightarrow{i} X.$$

In particular, a q -expansion $F(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ is defined for each form F on $\Gamma_0(\mathfrak{f})$ and each cusp $(\mathfrak{a}, \mathfrak{b}, \varepsilon, j)$ on $\Gamma_0(N)$. When R is a $\mathbb{Q}$ -algebra, we may thus speak of the q -expansion of F at the cusp determined by an $\alpha \in \mathbb{A}_K^*$.

We may similarly define p -adic modular forms on $\Gamma_0(\mathfrak{f})$.

2 Eisenstein Series

Let $\mathfrak{B}$ be an ideal of K . In this § we shall construct Eisenstein series for the Hilbert group, following Hecke and Siegel. The polarization module $\mathfrak{c}$ of §1 will be the ideal $\mathfrak{B}\mathfrak{D}^{-1}$.

For ε a Schwartz function on I , we define the *modification* of ε (with respect to $\mathfrak{B}$) to be the unique Schwartz function $\tilde{\varepsilon}$ on I which is supported on $I^{\mathfrak{B}}$ and whose value on an ideal $\mathfrak{B}\mathfrak{a}\mathfrak{c}$ is given by $\varepsilon(\mathfrak{a}^{-1} \cdot \mathfrak{c})$. Since this quantity depends only on the strict ideal class of $\mathfrak{a}$, $\tilde{\varepsilon}$ is defined mod $\mathfrak{f}$ for every conductor $\mathfrak{f} \subseteq \mathfrak{B}$.

For $c \in G$, we define ε_c to be $(\tilde{\varepsilon})_c$, i.e., the twist by c of the modification of ε .

Théorème 2.1 ((6.1)). *Let $k \geq 1$ be an integer, and let $\varepsilon : I \rightarrow \mathbb{C}$ be a function with parity $(-1)^k$ which is supported on A and defined modulo the conductor $\mathfrak{f}$. Then there exists a modular form*

$$G_{k,\varepsilon} \in M_k(\Gamma_0(\mathfrak{f}), \mathbb{C})$$

whose q -expansion at the cusp determined by each $\alpha \in K^*$ is given by the formulas

$$\text{If } k > 1 : \quad \mathcal{N}\mathfrak{a}^{-k} \left\{ 2^{-r}L(1-k, \varepsilon_c) + \sum_{\substack{\mu \gg 0 \\ \mu \in \mathfrak{a}\mathfrak{b}}} \left(\sum_{\substack{\omega \in \mathfrak{f}^{-1}\mathfrak{a}\mathfrak{c} \\ \omega \gg 0}} \varepsilon(\omega \cdot \mathfrak{c}^{-1}) \mathcal{N}(\omega \mu^{-1})^{k-1} \right) q^\mu \right\}. \quad (5)$$

$$\text{If } k = 1 : \quad \mathcal{N}\mathfrak{a}^{-1} \left\{ 2^{-r}L(0, \varepsilon_c + \tilde{\varepsilon}_c) + \sum_{\substack{\mu \gg 0 \\ \mu \in \mathfrak{a}\mathfrak{b}}} \left(\sum_{\substack{\omega \in \mathfrak{f}^{-1}\mathfrak{a}\mathfrak{c} \\ \omega \gg 0}} \varepsilon(\omega \cdot \mathfrak{c}^{-1}) \right) q^\mu \right\}. \quad (6)$$

Here :

— $\mathfrak{a} = (\alpha)$ is the ideal of K “generated” by α ,

— $\mathbf{c} = (\mathbf{a} \cdot \alpha \mathbf{c}^{-1})$ is the element of G defined by α , cf. (2.23).

Démonstration. We will construct $G_{k,\varepsilon}$ as a function of $(\mathcal{O})$ -lattices $\mathcal{E}$ in $K \otimes \mathbb{C}$. We are required from this point of view to produce a complex number $G_{k,\varepsilon}(\mathcal{E})$ each time that we are given a lattice $\mathcal{E}$ together with the supplementary data of a polarization and a $\Gamma_0(\mathfrak{f})$ structure on $\mathcal{E}$. Given such data, we first ignore the polarization completely (!). We then view the $\Gamma_0(\mathfrak{f})$ structure as the giving of an overlattice $\mathcal{E}' \supseteq \mathcal{E}$ together with an isomorphism

$$\mathcal{E}'/\mathcal{E} = \mathfrak{D}^{-1}\mathfrak{f}^{-1}/\mathfrak{D}^{-1}.$$

We consider the various $\mathcal{O}$ -submodules $\mathcal{M}$ of $\mathcal{E}'$ which are invertible, i.e., of rank 1. Given an $\mathcal{M}$, we choose arbitrarily a positivity $+$ for $\mathcal{M}$ (2.25). We will then evaluate the Fourier transform $T\varepsilon$ of ε on an element of I which is defined mod $\mathfrak{D}^{-1}$ in the sense of (3.9). This element is obtained as in (2.25) from the triple consisting of $\mathcal{M}$, the positivity $+$, and the map given as the composite

$$\mathcal{M} \hookrightarrow \mathcal{E}' \twoheadrightarrow \mathcal{E}'/\mathcal{E} = \mathfrak{D}^{-1}\mathfrak{f}^{-1}/\mathfrak{D}^{-1} \hookrightarrow K/\mathfrak{f}\mathfrak{D}^{-1}.$$

We write $T\varepsilon(\mathcal{M} \rightarrow K/\mathfrak{f}\mathfrak{D}^{-1})$ for the value of $T\varepsilon$ on this element.

We further associate a volume to $(\mathcal{M}, +)$. Namely, let $\mathfrak{a}$ be an ideal which defines the same strict ideal class as $(\mathcal{M}, +)$. Choose a positive isomorphism $(\mathfrak{a}, +_{\text{can}}) \simeq (\mathcal{M}, +)$. This isomorphism induces an inclusion $K \hookrightarrow K \otimes \mathbb{C}$, and if $m \in K \otimes \mathbb{C}$ is the image of $1 \in K$, then $\mathcal{M} = \mathfrak{a} \cdot m$. We define

$$\text{vol}(\mathcal{M}, +) = \mathcal{N}\mathfrak{a} \cdot \mathcal{N}m \in \mathbb{C},$$

the product of the norms of the ideal $\mathfrak{a}$ and the number $m \in K \otimes \mathbb{C}$. The volume is well defined because replacing $(\mathcal{M}, +) : (\mathfrak{a}, +_{\text{can}})$ by another isomorphism would only replace $\mathfrak{a}$ by $\alpha\mathfrak{a}$ and m by γm , where γ is totally positive, and in particular of positive norm.

Similarly, if we modify $+$ at one real place v , then $\text{vol}(\mathcal{M}, +)$ is replaced by its negative, so that $\text{vol}(\mathcal{M}, +)^k$ is multiplied by $(-1)^k$. However, because ε has parity $(-1)^k$, so that $T\varepsilon$ again has this parity, changing $+$ at v multiplies $T\varepsilon(\mathcal{M}, +, \mathcal{M} \rightarrow K/\mathfrak{f}\mathfrak{D}^{-1})$ by this same factor. Hence the quotient

$$\frac{T\varepsilon(\mathcal{M}, +, \mathcal{M} \rightarrow K/\mathfrak{f}\mathfrak{D}^{-1})}{\text{vol}(\mathcal{M}, +)^k}$$

is independent of the choice of $+$, so that a quantity

$$\frac{T\varepsilon(\mathcal{M}, \mathcal{M} \rightarrow K/\mathfrak{D}^{-1})}{(\text{vol } \mathcal{M})^k}$$

is well defined. We define the value of $G_{k,\varepsilon}$ on $\mathcal{E}$ to be $[(-1)^r(k-1)!]^r \cdot S$, where S is the sum

$$S = \sum'_{\mathcal{M} \subset \mathcal{E}'} \frac{T\varepsilon(\mathcal{M}, \mathcal{M} \rightarrow K/\mathfrak{f}\mathfrak{D}^{-1})}{(\text{vol } \mathcal{M})^k}, \quad (7)$$

computed in the sense of analytic continuation. [The sum is absolutely convergent for $k > 2$; the continuation is necessary only for $k = 1, 2$.]

To verify that this sum defines a modular form whose q -expansion is that given in (6.2), we must compute the value of $G_{k,\varepsilon}$ at the cusp defined by $\mathbf{c}, \alpha$. That is, we take

$$\mathcal{E} = 2\pi i(\mathbf{c}^{-1}\mathbf{a}^{-1} + \mathbf{b}\tau), \quad \mathcal{E}' = 2\pi i(\mathbf{c}^{-1}\mathbf{a}^{-1}\mathfrak{f}^{-1} + \mathbf{b}\tau)$$

together with the isomorphism

$$\mathcal{E}'/\mathcal{E} = \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{a}^{-1}\mathfrak{D}^{-1} = \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1} \xrightarrow{\sim} \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1}$$

induced by multiplication by $\alpha : \mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{a}^{-1}\mathfrak{D}^{-1} \xrightarrow{\sim} \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1}$. The map $\mathcal{M} \rightarrow \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1}$ is then just the composition of the three maps :

- division by $2\pi i$ and projection : $\mathcal{M} \subset 2\pi i(\mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1} + \mathfrak{a}\mathfrak{B}\tau) \rightarrow \mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1}$;
- multiplication : $\mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1} \xrightarrow{\cdot\alpha} \mathfrak{f}^{-1}\mathfrak{D}^{-1} \xrightarrow{\cdot\alpha} \mathfrak{f}^{-1}\mathfrak{D}^{-1}$;
- the canonical map : $\mathfrak{f}^{-1}\mathfrak{D}^{-1} \twoheadrightarrow \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1}$.

Hence, for $\mathcal{M} \subset \mathcal{E}'$, $T\varepsilon(\mathcal{M}, +, \mathcal{M} \rightarrow \mathfrak{f}^{-1}\mathfrak{D}^{-1}/\mathfrak{D}^{-1})$ is the value of $T\varepsilon$ on the element of I given by the triple

$$(\mathcal{M}, +, \mathcal{M} \hookrightarrow \mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1} \xrightarrow{\cdot\alpha} K).$$

The sum (6.3) may now be written

$$S = (2\pi i)^{-kr} \sum_{\substack{\mathcal{M} \subset K + \mathfrak{a}\mathfrak{B}\tau \\ (\mathcal{M} \text{ invertible})}} \frac{T\varepsilon(\mathcal{M}, +, \mathcal{M} \hookrightarrow K)}{(\text{vol } \mathcal{M}, +)^k},$$

where the “pr” is simply projection $\mathcal{M} \rightarrow K$ onto the first factor. Indeed, let $\mathcal{M} \subset K + \mathfrak{a}\mathfrak{B}\tau$ be an invertible $\mathcal{O}$ -module. If $\text{pr}(\mathcal{M})$ is the ideal $\mathfrak{b}$, then $(\mathcal{M}, +, \mathcal{M} \xrightarrow{\text{pr}} K)$ is simply $(\mathfrak{b} \cdot \mathfrak{a})$ for a suitable choice of $+$. Since $(\mathfrak{b} \cdot \mathfrak{a}) \in I^{\mathfrak{f}^{-1}\mathfrak{D}^{-1}}$ if and only if $\mathfrak{b}\mathfrak{a} \subset \mathfrak{f}^{-1}\mathfrak{D}^{-1}$, $\mathcal{M}$ gives a non-zero contribution to the sum only if it is contained in $\mathfrak{a}^{-1}\mathfrak{f}^{-1}\mathfrak{D}^{-1} + \mathfrak{a}\mathfrak{B}\tau = (2\pi i)^{-1}\mathcal{E}'$.

Let $\eta(x) = (T\varepsilon)(x \cdot \alpha)$, $\mathfrak{b} = \mathfrak{f}\mathfrak{B}\mathfrak{a}$. Then we have

$$\frac{T\varepsilon(\mathcal{M}, +, \mathcal{M} \xrightarrow{\text{pr}} K)}{(\text{vol } \mathcal{M}, +)^k} = (2\pi i)^{kr} \mathcal{N}\mathfrak{a}^{-k} \cdot \frac{\eta(\mathcal{M}, +, \mathcal{M} \xrightarrow{\text{pr}} K)}{(\text{vol } \mathcal{M}, +)^k}$$

calculated (if necessary) by analytic continuation.

Let $\mathcal{M} \subset K + \mathfrak{a}\mathfrak{B}\tau$, and suppose that $\mathfrak{a}$ is an ideal isomorphic to $\mathcal{M}$ as an invertible $\mathcal{O}$ -module. Then $\mathcal{M} = \mathfrak{a} \cdot (\beta + \lambda\tau)$ for some $\beta, \lambda \in K$, and the pair (β, λ) is well defined up to multiplication by a unit of K . Fix one choice (β, λ) and endow $\mathcal{M}$ with the positivity induced by $+_{\text{can}}$ on $\mathfrak{a}$ via $(\beta + \lambda\tau) : \mathfrak{a} \xrightarrow{\sim} \mathcal{M}$. For this positivity we have $\text{vol } \mathcal{M} = (\mathcal{N}\mathfrak{a})(\mathcal{N}(\beta + \lambda\tau))$, and the element of I given by $(\mathcal{M}, \mathcal{M} \rightarrow K)$ is $(\mathfrak{a} \cdot \beta)$. Hence S may be written

$$S = \mathcal{N}\mathfrak{a}^{-k} \sum_{\{\mathfrak{a}\}} \sum_{\substack{\beta \in K/\mathcal{O}^* \\ \lambda \in \mathfrak{a}^{-1}\mathfrak{b} \\ \beta + \lambda\tau \in \mathcal{E}'}} \frac{\eta(\mathfrak{a} \cdot \beta)}{|\mathcal{N}(\beta + \lambda\tau)|^k},$$

where the outer sum is over a set $\{\mathfrak{a}\}$ of representatives for the set of wide ideal classes of K , and where $\eta(\mathfrak{a} \cdot \beta) = \eta((\mathfrak{a}^{-1}\beta))$ as in §3.

Thus S is naturally the sum of two terms, the first being

$$(2\pi i)^{-kr} L(k, \eta) = (2\pi i)^{-kr} L(k, (T\varepsilon)(\mathfrak{B} \cdot \mathfrak{a})).$$

Changing variables, we rewrite this as

$$\begin{aligned} & (2\pi i)^{-kr} L(k, T\varepsilon(\mathfrak{c}\mathfrak{a}^{-1} \cdot \mathfrak{a})) \cdot \mathcal{N}(\mathfrak{c}\mathfrak{a}^{-1})^{-k} \quad (\mathfrak{c} = \mathfrak{a} \cdot \alpha\mathfrak{c}^{-1}) \\ &= \frac{[(-1)^r(k-1)!]^r}{(2\pi i)^{kr}} L(1-k, T(T\varepsilon(\mathfrak{c}\mathfrak{a}^{-1} \cdot \mathfrak{a}))) \quad (\text{by (3.13)}) \\ &= \frac{[(-1)^r(k-1)!]^r}{(2\pi i)^{kr}} L(1-k, \varepsilon(\mathfrak{a} \cdot \mathfrak{a}^{-1})) \quad (\text{by the invariance (3.7)}) \\ &= \frac{[(-1)^r(k-1)!]^r}{(2\pi i)^{kr}} \cdot 2^{-r} L(1-k, \varepsilon_c) \quad (\text{by the parity of } \varepsilon). \end{aligned}$$

Since $G_{k,\varepsilon} = [(-1)^r(k-1)!]^r \cdot S$, this first contribution to S gives precisely the term $2^{-r}\mathcal{N}\mathfrak{a}^{-k}L(1-k, \varepsilon_c)$ in (6.2).

For the second term, we must find the value at $s = 0$ of the series

$$\sum'_{0 \neq \lambda \in \mathfrak{ba}^{-1}} \sum_{\beta \in K} \frac{\eta(\mathfrak{a}\beta)}{|\mathcal{N}(\beta + \lambda\tau)|^{k+s}}.$$

(We write x^{k+s} for $x^k|x|^s$.) As Katz has recently recalled [19], Hecke's technique for treating such expressions is to apply the Poisson summation formula to β , with $\Re(s) > 0$, and then to show that one obtains the value of the resulting double series by setting $s = 0$ in each term.

More precisely, for $\Re(s) > 0$, our series may be written

$$\sum'_{0 \neq \lambda \in \mathfrak{ba}^{-1}} \sum_{\beta \in K \pmod{U}} \frac{\eta(\mathfrak{a}\beta)}{\mathcal{N}\lambda^{k+s}} \int_{K \otimes \mathbb{R}} e^{2\pi i \operatorname{tr}(t \cdot y)} \frac{dt}{|\mathcal{N}(\lambda^{-1}(t + \lambda\tau))|^{k+s}}.$$

[The summation modulo U continues to be well defined, as one can verify explicitly from the formula

$$\eta(\zeta x) = (\operatorname{sgn} \zeta)^{kr} \eta(x) \quad (8)$$

for all units ζ of K . (Here we write $\operatorname{sgn} \zeta$ for the sign of $\mathcal{N}\zeta$.) The formula (6.4) is an immediate consequence of the parity of ε .]

We consider the term in the inner sum with $y = 0$:

$$\sum'_{0 \neq \lambda \in \mathfrak{ba}^{-1}} \frac{\eta(\mathfrak{a} \cdot 0)}{\mathcal{N}\lambda^{k+s}} \int_{K \otimes \mathbb{R}} \frac{dt}{|\mathcal{N}(\lambda^{-1}t)|^{k+s}}. \quad (9)$$

By (6.4), we have $\eta(\mathfrak{a} \cdot 0) = 0$ if there is a unit ζ with $(\operatorname{sgn} \zeta)^{kr} = -1$. We assume from now on, therefore, that this is not the case.

In the integral, we write y for the totally positive number $\Im\tau \in (K \otimes \mathbb{R})^*$. Then the integral becomes $\mathcal{N}\lambda^{k+s} \cdot \varphi_k(s) \cdot \mathcal{N}y^{-1}$, where

$$\varphi_k(s) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} \frac{dt}{|(i+t)|^{k+s}}.$$

(We have, in fact $|y| = \mathcal{N}y$ because y is totally positive.) Then (6.5) may be rewritten

$$\varphi_k(s) \cdot \mathcal{N}y^{-k-s} \sum_{0 \neq \lambda \in \mathfrak{ba}^{-1} \pmod{U}} (\operatorname{sgn} \lambda)^{kr} \cdot \mathcal{N}\lambda^{-k-s}. \quad (10)$$

By writing the sum in terms of partial zeta functions (for example), we see that it represents a meromorphic function of s with a possible simple pole at $s = 2 - k$. Thus if $k > 2$, (6.5) represents a function of s which is holomorphic at least in the half-plane $\Re(s) > -1$. Further, its value at $s = 0$ is then zero, because we easily compute $\varphi_k(0) = 0$.

Now Hecke [9, p. 393] yields the expressions

$$\varphi_2(s) = -\frac{\pi^{r/2}\Gamma(s/2+1)}{\Gamma(s/2+2)}, \quad \varphi_1(s) = \frac{\pi^{r/2}\Gamma(-s/2+1)}{s}.$$

For $k = 2$, we thus find again that (6.5) is holomorphic for $\Re(s) > -\frac{1}{2}$ and has the value 0 at $s = 0$. (Because $r > 1$, $\varphi_2(s)$ has at least a double zero at $s = 2 - k = 0$.)

For $k = 1$, the possible pole at $s = 1$ of the series

$$\sum'_{0 \neq \lambda \in \mathfrak{ba}^{-1} \pmod{U}} \frac{(\operatorname{sgn} \lambda)^r}{\mathcal{N}\lambda^s} \quad (11)$$

does not appear. Indeed, (6.6) is, up to the factor $\mathcal{N}(\mathfrak{ba}^{-1})^s$, the L -series attached to an odd function on the group of strict ideal classes of K . Hence (6.5) is again holomorphic for $\Re(s) > -\frac{1}{2}$, and its value at $s = 0$ is

$$\frac{(2\pi i)^r}{2} \cdot \mathcal{N}\mathfrak{a} \cdot \sum_{\substack{\mu \in \mathfrak{ba}^{-1} \\ \mu \gg 0}} \eta(\mathfrak{a} \cdot 0) \cdot q^{\mu \cdot \lambda} \Big|_{s=0}.$$

Thus the term we are considering becomes

$$\frac{(2\pi i)^r}{2} \mathcal{N}\mathfrak{a} \cdot \sum_{\substack{\mu \in \mathfrak{ba}^{-1} \\ \mu \gg 0}} \eta(\mathfrak{a} \cdot 0) \cdot q^\mu. \quad (12)$$

For $\mu \in K$ we have

$$\operatorname{tr}(\mu \cdot i \cdot y) = (\operatorname{sgn} \mu) \cdot \mathcal{N}\mu \cdot \varphi_1(0) = (\operatorname{sgn} \mu) \cdot \mathcal{N}\mu \cdot (2\pi i)^r / 2.$$

Hence (6.7) may be rewritten as the L -value which is formally

$$(2\pi i)^r \mathcal{N}\mathfrak{a} \cdot \eta(\mathfrak{a} \cdot 0).$$

Now $\eta(x) = T\varepsilon(x \cdot \alpha)$. Hence

$$T^{-1}\eta(\omega) = \varepsilon(\omega \cdot \alpha^{-1}) \cdot (-1)^r \mathcal{N}\mathfrak{a}^{-1}.$$

Thus (6.7) becomes

$$\mathcal{N}\mathfrak{a} \cdot 2^{-r} \cdot \eta(\mathfrak{a} \cdot 0) = \mathcal{N}\mathfrak{a} \cdot 2^{-r} \cdot \varepsilon(\omega \cdot \alpha^{-1}).$$

Since $\mathfrak{c} = (\mathfrak{a} \cdot \alpha \mathfrak{c}^{-1})$, this becomes

$$2^{-r} \mathcal{N}\mathfrak{a} \cdot L(0, \tilde{\varepsilon}_c) = 2^{-r} \mathcal{N}\mathfrak{a} \cdot L(0, \tilde{\varepsilon}_c).$$

If we now take into account the factor $[(-1)^r(k-1)!]^r$ in $G_{k,\varepsilon}$, we finally arrive at the conclusion that the term we are calculating contributes to the q -expansion (6.2) a term

$$(-1)^r 2^{-r} \mathcal{N}\mathfrak{a} L(0, \tilde{\varepsilon}_c)$$

when $k = 1$. Under the assumption made above that all units of K have norm 1, we can ignore the $(-1)^r$, and we obtain precisely the perturbing term in (6.2) for $k = 1$. If there is a unit of K with norm -1 , then as we have already noted, (6.5) vanishes. On the other hand, $\tilde{\varepsilon} = 0$ for odd ε if there is a unit of K of negative norm. Hence the perturbing term of (6.4) vanishes as well.

We must finally consider the behavior for $s \rightarrow 0$ of the sum

$$\sum'_{0 \neq \lambda \in \mathfrak{ba}^{-1}} \sum_{\substack{\beta \in K \\ \beta \neq 0}} \frac{\eta(\mathfrak{a}\beta)}{\mathcal{N}\lambda^{k+s}} \int_{K \otimes \mathbb{R}} \frac{e^{-2\pi i \operatorname{tr}(t \cdot y)} dt}{|\mathcal{N}(\lambda^{-1}(\beta + t))|^{k+s}}. \quad (13)$$

As Katz explains, it suffices to continue each integral as a holomorphic function of s , set $s = 0$, and evaluate the resulting double sum. Referring to Katz's calculation ([19], 3.2.32), we find that

$$\int_{K \otimes \mathbb{R}} e^{-2\pi i \operatorname{tr}(ty)} dt = \begin{cases} \frac{[(k-1)!]^r}{(2\pi i)^{kr}} e^{-2\pi i \operatorname{tr}(\mu \cdot y)} \mathcal{N}\mu^{k-1} & \text{if } \mu \gg 0, \\ 0 & \text{otherwise.} \end{cases}$$

The inner sum of (6.8) may thus be written

$$\frac{[(k-1)!]^r}{(2\pi i)^{kr}} \sum_{\substack{\mu \in \mathfrak{ba}^{-1} \\ \mu \gg 0}} \eta(\mathfrak{a}\mu) \mathcal{N}\mu^{k-1} q^{\mu \cdot \lambda}.$$

because on the substitution $\beta \mapsto -\beta$ the functions $\varepsilon^{\mathfrak{a}}(\beta)$ and $\mathcal{N}\beta^k$ both take on factors $(-1)^{kr}$. Thus (6.8) gives to S the contribution

$$\frac{[(k-1)!]^r}{(2\pi i)^{kr}} \sum_{\substack{\mu \in \mathfrak{ba}^{-1} \\ \mu \gg 0}} \sum_{0 \neq \lambda \in \mathfrak{ba}^{-1}} \eta(\mathfrak{a}\mu) \mathcal{N}\mu^{k-1} q^{\mu \cdot \lambda}. \quad (14)$$

We recognise the bracketed term as $(T\eta)(\beta\mathfrak{a}^{-1})$. Setting $\rho = \beta/\lambda$, we change the order of summation in (6.9) and rewrite it as

$$\mathcal{N}\mathfrak{a}^{-k} \sum_{\substack{\mu \in K \\ \mu \gg 0}} \sum_{\substack{\omega \in \mathfrak{f}^{-1}\mathfrak{a}\mathfrak{c} \\ \omega \gg 0}} \varepsilon(\omega \cdot \mathfrak{c}^{-1}) \mathcal{N}(\omega\mu^{-1})^{k-1} q^{\mu}.$$

We have $(T\eta)(\mu\mathfrak{c}) = \varepsilon(\mu \cdot \alpha^{-1}) \cdot \mathcal{N}\mathfrak{a} \cdot (-1)^{kr}$. Multiplying by the factor $(-1)^{kr} [(k-1)!]^r$ which appears in $G_{k,\varepsilon}$, we have then

$$\begin{aligned} & \mathcal{N}\mathfrak{a}^{-k} \sum_{\substack{\mu \gg 0 \\ \mu \in \mathfrak{ab}}} \left(\sum_{\substack{\omega \in \mathfrak{f}^{-1}\mathfrak{a}\mathfrak{c} \\ \omega \gg 0}} \varepsilon(\omega \cdot \mathfrak{c}^{-1}) \mathcal{N}(\omega\mu^{-1})^{k-1} \right) q^{\mu} \\ &= \mathcal{N}\mathfrak{a}^{-k} \sum_{\substack{\mu \gg 0 \\ \mu \in \mathfrak{ab}}} \left(\sum_{\substack{\omega \in \mathfrak{f}^{-1}\mathfrak{a}\mathfrak{c} \\ \omega \gg 0}} \varepsilon(\omega \cdot \mathfrak{c}^{-1}) \mathcal{N}(\omega\mu^{-1})^{k-1} \right) q^{\mu}. \end{aligned}$$

The proof of (6.1) is complete. $\square$

3 Theta Series

In this §, we study certain series of weight one, with the polarization module $\mathfrak{c}$ of §1 taken to be the ring of integers of K .

Let u be a totally positive element of K , and let $L = K(\sqrt{-u})$ be the corresponding totally imaginary quadratic extension of K . Let

$$\omega : G \rightarrow \{-1, +1\}$$

be the character of $G = \operatorname{Gal}(K^{\text{ab}}/K)$ corresponding to L/K . By composing ω with the “Artin” map j of (2.23) we obtain a character

$$K^* \rightarrow \{\pm 1\},$$

again denoted by ω .

Théorème 3.1 ((7.1)). *For suitable $N \geq 1$, there exists a weight-one modular form F over $\mathbb{C}$ on $\Gamma_\omega(N)$ whose q -expansion at the cusp determined by each $\alpha \in K^*$ is the series*

$$\omega(\alpha) \mathcal{N}\mathfrak{a} \sum_{x,y \in \mathfrak{a}} q^{x^2+uy^2},$$

where $\mathfrak{a} = (\alpha)$ is the ideal generated by α .

Remarque 3.2. It is well known that the series $\sum_{x,y \in \mathbb{Z}} q^{x^2+uy^2}$ is the standard q -expansion of a weight-one modular form, cf. e.g. [7]. Our interest is in calculating the “other” q -expansions of this form. We do this by analyzing the action (by “slashing”) of $\mathrm{SL}(2, K)$ on this form, in terms of the Weil representation.

Proof of (7.1). Let $\psi : \mathbb{A} \rightarrow \mathbb{C}^*$ be the additive character of §3. We recall that the choice of ψ as an additive character on $\mathbb{A}$, trivial on K , determines a certain representation, the Weil representation, of $\mathrm{SL}(2, \mathbb{A})$ on the Schwartz space $\mathcal{S}(L_{\mathbb{A}})$ of the adelization $L_{\mathbb{A}} = L \otimes_K \mathbb{A}_K$ of L . (See [14, 35, 41].) We denote this representation by r . This representation is derived from local Weil representations r_v of the various groups $\mathrm{SL}(2, K_v)$ on the corresponding Schwartz spaces $\mathcal{S}(L_v)$. This local representations may in turn be described by certain explicit formulas giving the actions of $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ and of matrices of the form $\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$ of [35].

For $M \in \mathcal{S}(L_{\mathbb{A}})$ one defines

$$\Theta(M) = \sum_{x \in L} M(x).$$

This function is left invariant under $\mathrm{SL}(2, K)$: for $s \in \mathrm{SL}(2, K)$ we have

$$\Theta(r(s)M) = \Theta(M).$$

For fixed M , the function $s \mapsto \Theta(r(s)M)$ is then a continuous function on $\mathrm{SL}(2, \mathbb{A})$ which is invariant on the left by $\mathrm{SL}(2, K)$ [35, §2].

We shall make a particular choice for the infinite component of M (cf. [8, 2.35]). Let Nm denote the norm from L to K . Let $L_\infty = L \otimes_{\mathbb{Q}} \mathbb{R}$. For $x \in L_\infty$, set

$$M_\infty(x) = \exp\{-2\pi \cdot \mathrm{tr}_{K \otimes \mathbb{R}/\mathbb{R}}(\mathrm{Nm}(x))\}.$$

Then M_∞ is the product of functions M_v on the completions $L_v = L \otimes_K K_v$ of L at the infinite primes of K (we use $v : K \hookrightarrow \mathbb{R}$ to make the tensor product), each given by the same formula

$$M_v(x) = e^{-2\pi|x|^2}.$$

If we view L_v as the complex field $\mathbb{C}$, the function M_v is simply

$$z \mapsto e^{-\pi|z|^2}.$$

For $\theta \in \mathbb{R}$, let $\rho(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R})$. If v is an infinite place of K , we write $\rho_v(\theta)$ for $\rho(\theta)$ if we think of $\mathrm{SL}(2, \mathbb{R})$ as $\mathrm{SL}(2, K_v)$. One has the formula

$$r_v(\rho_v(\theta))M_v = e^{i\theta}M_v \tag{15}$$

cf. [36, Lemma 1.2].

Théorème (7.3). *On the other hand, let $a, b \in \mathbb{R}$, $b > 0$. The explicit formulas which define r_v show that*

$$r_v \begin{pmatrix} \sqrt{b} & a/\sqrt{b} \\ 0 & 1/\sqrt{b} \end{pmatrix}^{-1} M_v$$

is the function

$$x \mapsto b^{1/2} e^{2\pi i(a+bi) \text{Nm}(x)}.$$

For τ in the upper halfplane

$$\mathfrak{H} = \{\tau \in K \otimes \mathbb{C} \mid \Im \tau \gg 0\},$$

and for $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, K \otimes \mathbb{R})$, we write $g \cdot \tau$ for the quotient $\frac{a\tau+b}{c\tau+d}$.

Writing $\mathcal{N}$ for the norm map $K \otimes \mathbb{C} \rightarrow \mathbb{C}$, we let $j(g, \tau)$ be the “factor of automorphy”

$$\mathcal{N}(c\tau + d).$$

One verifies easily the formula

$$j(gh, \tau) = j(g, h\tau) \cdot j(h, \tau) \quad (16)$$

for $g, h \in \text{SL}(2, K \otimes \mathbb{R})$.

We now consider the finite part $L_f = L \otimes_K \mathbb{A}_f$ of $L_{\mathbb{A}}$, and its Schwartz space $\mathcal{S}(L_f)$. A “finite” Weil representation gives an action of $\text{SL}(2, \mathbb{A}_f)$ on $\mathcal{S}(L_f)$, which we again denote r . The “multiplication” map

$$\text{SL}(2, \mathbb{A}_f) \times \mathcal{S}(L_f) \rightarrow \mathcal{S}(L_f)$$

is continuous, so that the stabilizer in $\text{SL}(2, \mathbb{A}_f)$ of each $\varepsilon \in \mathcal{S}(L_f)$ is open. For $\varepsilon \in \mathcal{S}(L_f)$, we let $M_\varepsilon = \varepsilon \otimes M_\infty$ be the product of ε with the function M_∞ defined above. Let

$$\varphi_\varepsilon : \text{SL}(2, \mathbb{A}) \rightarrow \mathbb{C}$$

be the corresponding function $s \mapsto \Theta(r(s) \cdot M_\varepsilon)$ on $\text{SL}(2, \mathbb{A})$.

For $g \in \text{SL}(2, K \otimes \mathbb{R})$, we write simply $\varphi_\varepsilon(g)$ for the value of φ_ε on the image $1 \times g$ of g in $\text{SL}(2, \mathbb{A})$.

For each ε , we define a function F_ε on $\mathfrak{H}$ by the series

$$F_\varepsilon(\tau) = \sum_{x \in L} \varepsilon(x_f) M_\infty(x_\infty) e^{2\pi i \text{tr}(\text{Nm}(x) \cdot \tau)}.$$

Proposition 3.4 ((7.5)). *We have $F_\varepsilon(\tau) = \varphi_\varepsilon(g) j(g, i)^{-1}$ for all $g \in \text{SL}(2, K \otimes \mathbb{R})$ such that $g \cdot i = \tau$.*

Démonstration. If g is an element such that $g \cdot i = \tau$, then any other element is obtained by multiplying g on the right by rotations $\rho_v(\theta) \in \text{SL}(2, K_v) \hookrightarrow \text{SL}(2, K \otimes \mathbb{R})$. Since $j(\rho_v(\theta), i) = e^{i\theta}$, we see by (7.2) and by (7.4) that the assertion is true for one g if and only if it is true for each g . We write $\tau = a + bi$, with $a, b \in K \otimes \mathbb{R}$. Since $\tau \in \mathfrak{H}$, b is totally positive. Thus b has a unique totally positive square root $\sqrt{b}$. It is clear that the matrix $g = \begin{pmatrix} \sqrt{b} & a/\sqrt{b} \\ 0 & 1/\sqrt{b} \end{pmatrix}$ is such that $g \cdot i = \tau$. For this choice of g however, the formula $F_\varepsilon(\tau) = \varphi_\varepsilon(g) j(g, i)^{-1}$ is an immediate consequence of the definition of $\varphi_\varepsilon(g)$ and (7.3). $\square$

We now recall the “slashing” operator (with the weight k taken to be 1)

$$(F|h)(\tau) = F(h \cdot \tau) j(h, \tau)^{-1}, \quad h \in \mathrm{SL}(2, K \otimes \mathbb{R}).$$

Proposition 3.5 ((7.6)). *For $h \in \mathrm{SL}(2, K \otimes \mathbb{R})$, we have*

$$(F_\varepsilon|h)(\tau) = \varphi_\varepsilon(hg) j(g, i)^{-1}$$

for each g such that $g \cdot i = \tau$.

Démonstration. This formula follows immediately from (7.4) and (7.5). $\square$

For $A \in \mathrm{SL}(2, \mathbb{A})$, we let A_f (resp. A_∞) denote the image of A in $\mathrm{SL}(2, \mathbb{A}_f)$ (resp. $\mathrm{SL}(2, K \otimes \mathbb{R})$). Regarding $\mathrm{SL}(2, \mathbb{A}_f)$ and $\mathrm{SL}(2, K \otimes \mathbb{R})$ as subgroups of $\mathrm{SL}(2, \mathbb{A})$, we have $A = A_f \cdot A_\infty$.

Théorème 3.6 ((7.7)). *For all $A \in \mathrm{SL}(2, K)$, we have*

$$F_\varepsilon|A = F_{r(A_f)^{-1}\varepsilon}.$$

Démonstration. Let $\tau \in \mathfrak{H}$, and choose $g \in \mathrm{SL}(2, K \otimes \mathbb{R})$ so that $g \cdot i = \tau$. Since $|A$ represents the action of a matrix in $\mathrm{SL}(2, K \otimes \mathbb{R})$, it is $(F_\varepsilon|A_\infty)(\tau)$ which we must compute. By (7.6), this number is

$$\varphi_\varepsilon(A_\infty^{-1}g) j(g, i)^{-1} = \varphi_\varepsilon(A_f^{-1}g) j(g, i)^{-1},$$

with the equality resulting from the left-invariance of φ_ε under $\mathrm{SL}(2, K)$. Since $g \in \mathrm{SL}(2, K \otimes \mathbb{R})$, A_f and g commute in $\mathrm{SL}(2, \mathbb{A})$. Commuting the two, and recalling the definition of φ_ε , we find that

$$\varphi_\varepsilon(A_f^{-1}g) = \varphi_{r(A_f)^{-1}\varepsilon}(g).$$

The asserted equality now follows from (7.5). $\square$

Since the stabilizer of ε in $\mathrm{SL}(2, \mathbb{A}_f)$ is open, (7.7) implies that F_ε is invariant under some congruence subgroup of $\mathrm{SL}(2, \mathcal{O})$. Hence, by strong approximation, $F_\varepsilon|A$ may be defined for all $A \in \mathrm{SL}(2, \mathbb{A}_f)$. By continuity, we see that the definition thus obtained is the obvious one : for all $A \in \mathrm{SL}(2, \mathbb{A}_f)$, we have

$$F_\varepsilon|A = F_{r(A^{-1})\varepsilon}. \quad (17)$$

We now make a specific choice of ε . Let $\mathcal{R} \subset L$ be the lattice $\mathcal{O} + \mathcal{O}\sqrt{-u}$, and let ε be the characteristic function of the subset $\mathcal{R} \otimes \hat{\mathbb{Z}} = \mathcal{O} + \mathcal{O}\sqrt{-u}$ of L_f . Then

$$F_\varepsilon = \sum_{x,y \in \mathcal{O}} q^{x^2+uy^2}.$$

To show that the holomorphic function $F = F_\varepsilon$ is (the q -expansion of) a modular form of weight 1 on $\Gamma_\omega(N)$ over $\mathbb{C}$, we must show that F is invariant under the group of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, d \in 1 + N\mathcal{O}$, $b \in \mathfrak{D}^{-1}$, $c \in N\mathfrak{D}$. Any such matrix is the product of a transvection $\begin{pmatrix} 1 & \varepsilon \\ 0 & 1 \end{pmatrix}$ ($\varepsilon \in \mathfrak{D}^{-1}$) and a matrix which satisfies $b \in N\mathfrak{D}^{-1}$ in addition to

the other requirements. Since F is clearly invariant under the transvections, it is enough to show that F is invariant under matrices with

$$a, d \in 1 + N\mathcal{O}, \quad b \in N\mathfrak{D}^{-1}, \quad c \in N\mathfrak{D}.$$

What is required, for (7.1), is merely that F be invariant under $\Gamma(N)$ for some N . But this is simply a restatement of the fact that F_ε is invariant under some congruence subgroup of $\mathrm{SL}(2, \mathcal{O})$, which we have already established.

To conclude the proof of (7.1), we note by (5.8) that the q -expansion of F at the cusp corresponding to α is $F|A$, where $A = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \in \mathrm{SL}(2, \mathbb{A}_f)$. By (7.8), we may rewrite this $F_{r(A)^{-1}\varepsilon}$. Now A^{-1} is such that $r(A^{-1})$ is given by a particularly simple formula [35, (2.1)]. We find that $r(A)^{-1}\varepsilon$ is the function on L_f

$$x \mapsto \omega(\alpha)^{-1} \|\alpha\|^{-1} \varepsilon(\alpha^{-1}x).$$

Now $\omega(\alpha)^{-1} = \omega(\alpha)$, and $\|\alpha\|^{-1} = \mathcal{N}\mathfrak{a}$. Finally, $\varepsilon(\alpha^{-1}x)$ is the characteristic function of $\mathfrak{a} + \mathfrak{a}\sqrt{-u}$. Hence

$$F_{r(A)^{-1}\varepsilon} = \omega(\alpha)\mathcal{N}\mathfrak{a} \sum_{x,y \in \mathfrak{a}} q^{x^2+uy^2},$$

and

$$F|A = \omega(\alpha)\mathcal{N}\mathfrak{a} \sum_{x,y \in \mathfrak{a}} q^{x^2+uy^2},$$

as desired. □

4 Congruences for L -values

For $\varepsilon : I \rightarrow \mathbb{C}$ a Schwartz function, we have defined $L(s, \varepsilon)$ as the sum

$$\sum_{\mathfrak{a} \in I_0} \varepsilon(\mathfrak{a})\mathcal{N}\mathfrak{a}^{-s}$$

in §3. By writing $L(s, \varepsilon)$ in terms of partial zeta functions, we see that Siegel's rationality theorem [37, 38] is equivalent to :

Théorème 4.1 ((8.1)). *If ε takes values in $\mathbb{Q}$, then for each integer $k \geq 1$ we have*

$$L(1 - k, \varepsilon) \in \mathbb{Q}.$$

In the following discussion, we will admit (8.1) as known. The reader will observe that it is in fact a consequence of (0.1), in view of the calculations with Eisenstein series below.

As in §0, Theorem (8.1) allows us to define, by linearity, values

$$L(1 - k, \varepsilon) \in V$$

for each ε with values in a $\mathbb{Q}$ -vector space V . We may view each association

$$\varepsilon \mapsto L(1 - k, \varepsilon) \quad (k \geq 1)$$

as a $\mathbb{Q}$ -valued distribution T_k on I . Each T_k has an obvious homogeneity property with respect to substitutions

$$\varepsilon(x) \mapsto \varepsilon(\mathfrak{a}x)$$

with $\mathfrak{a}$ an ideal of K . In particular, T_1 is invariant under such substitutions.

If V is now a $\mathbb{Q}$ -module and $c \in G$, we define

$$\Lambda_c(1 - k, \varepsilon) = L(1 - k, \varepsilon) - \mathcal{N}c^k L(1 - k, \varepsilon_c),$$

where ε_c is the function $\mathfrak{a} \mapsto \varepsilon(c\mathfrak{a})$, cf. §0. For each c and k , the map

$$\mu_{c,k} : \varepsilon \mapsto \Lambda_c(1 - k, \varepsilon)$$

is a distribution on I with values in $\mathbb{Q}$. Let us denote $\mu_{c,1}$ simply by μ_c .

Théorème 4.2 ((8.2) Main Theorem). *For each $c \in G$, μ_c is a measure on I with values in $\mathbb{Z}$. If ε is a Schwartz function on I and k a positive integer, we have*

$$\int \varepsilon \cdot \mathcal{N}^{k-1} d\mu_c = \Lambda_c(1 - k, \varepsilon).$$

[Note that in the above equation, $\mathcal{N}$ is the norm function $I \rightarrow \mathbb{Q}$.]

The principal purpose of this § is to derive (8.2). Along the way, we will prove certain further assertions concerning the integrals against μ_c of odd functions. These are summarized in (8.11), (8.12). Also, we wish to point out how (0.4) is a consequence of (8.2).

Proposition 4.3 ((8.3)). *Let ε be a Schwartz function on I with values in a $\mathbb{Q}$ -vector space V . Suppose that $k \geq 1$ and that c is an element of the subgroup $\mathfrak{Z}$ of G , cf. (2.24). Then we have*

$$L(1 - k, \varepsilon_c) = \mathcal{N}c^{-k} L(1 - k, \varepsilon).$$

Démonstration. We easily reduce to the case where ε is a complex-valued function which has parity (a_v) for some set of integers $a_v = 0, 1$. For such ε , the proposition amounts to the vanishing of $L(1 - k, \varepsilon)$ whenever we have $(-1)^{a_v} \neq (-1)^{kr}$ for some v . This is the vanishing given by (3.24). $\square$

Théorème 4.4 ((8.4) The “Eisenstein congruences”). *Let $\varepsilon_1, \dots, \varepsilon_t$ be Schwartz functions on I with values in $\mathbb{Q}$. Let $\mathfrak{B}$ be an ideal of K . Suppose that only finitely many of the ε_k are non-zero and that ε_k has parity $(-1)^k$ for each k . Set*

$$\varphi : \sum_{k \geq 1} \varepsilon_k \mathcal{N}^{k-1} : I \rightarrow \mathbb{Q}.$$

Suppose for each totally positive element μ of K that the finite sum

$$\sum_{\substack{\mathfrak{a} \in I_0 \\ \mathfrak{a} \subset \mathcal{O}}} \varphi(\mathfrak{a}\mu)$$

lies in $\mathbb{Z}$. Then for each $c \in G$, we have

$$\Lambda_c(0, \tilde{\varepsilon}_1) + \sum_{k \geq 1} \Lambda_c(1 - k, \varepsilon_k) \in \mathbb{Z}, \tag{18}$$

where $\tilde{\varepsilon}_1$ is the modification of ε_1 with respect to $\mathfrak{B}$ (§6).

Démonstration. For a given collection of functions (ε_k) , the hypothesis and the conclusion of the theorem depend only on the strict ideal class of $\mathfrak{B}$. Also, if $\mathfrak{b}$ is an ideal of K , the theorem for (ε_k) and for $\mathfrak{B}$ is equivalent to the theorem for the ideal $\mathfrak{B}\mathfrak{b}$ and the collection of functions

$$\varepsilon \mapsto \varepsilon_k(\mathfrak{a}\mathfrak{b})\mathcal{N}\mathfrak{b}^{k-1}.$$

It thus suffices to treat the case where the ε_k are supported on A . At the same time, we may as well assume that the ideal $\mathfrak{B}$ is equal to the ideal $\mathcal{O}$. (This latter assumption will prove to be convenient because the modular forms in §6 were made with polarization module $\mathfrak{c}$ implicitly taken to be $\mathcal{O}$.) Furthermore, we observe that it suffices to treat the case where the ε_k are $\mathbb{Q}$ -valued. Indeed, it is clear that the hypothesis to (8.4) does not change if we replace the ε_k by functions which are congruent to them modulo $\mathbb{Z}$. As for the conclusion of (8.4), we may specify in advance a positive integer D such that the conclusion is unchanged if we replace the ε_k by functions congruent to them mod $D\mathbb{Z}$. (The point is that the L -values in question are a priori given as linear combinations of certain rational numbers — values of partial zeta functions — which may be specified as soon as one knows a common modulus of definition for the ε_k .)

So let functions ε_k be given as in (8.4), under the further assumption that they are $\mathbb{Q}$ -valued and supported on A . Assume that the ideal $\mathfrak{B}$ is equal to $\mathcal{O}$. For each $k \geq 1$, let F_k be the Eisenstein series G_{k,ε_k} of (6.1). From (8.1) and equation (6.2), we see that the F_k each satisfy the rationality condition of (0.1). For each $\alpha \in K^*$, let

$$S(\alpha) = \sum_{k \geq 1} \mathcal{N}\alpha_p^{-k} \cdot F_{k,\alpha};$$

this is a priori a formal series with coefficients in $\mathbb{Q}$.

Now the non-constant coefficients of $S(1)$ are rational integers and thus in particular are in $\mathbb{Z}$. Hence, by (0.3), applied for each prime p , the difference between the constant coefficients of $S(1)$ and $S(\alpha)$ lies again in $\mathbb{Z}$, for $\alpha \in K^*$.

Given α , set $c = j(\alpha) = (\alpha) \cdot \alpha^{-1}$. Looking at (6.2), and using the equation

$$\mathcal{N}c = \mathcal{N}(\alpha) \cdot \mathcal{N}\alpha^{-1},$$

we see that the conclusion of (8.4) holds for c . But j is surjective; hence the conclusion holds for each $c \in G$. $\square$

[In fact, it is clear that the conclusion of (8.4) depends a priori only on the image of c in $G_{\mathfrak{f}}$, where $\mathfrak{f}$ is a suitable conductor. Hence all that is needed is that $G_{\mathfrak{f}}$ is a quotient of K^* .]

Although (8.4) and (8.2) are quite similar, there are two obstacles which prevent us from deducing (8.2) as an immediate consequence of (8.4). First of all, there is a parity assumption on the ε_k in (8.4). Secondly, (8.4) contains a perturbing term, involving the modification of ε_1 . These difficulties may account for the complexity of the calculations that follow.

We shall suppose below without comment that ε is a Schwartz function on I with values in $\mathbb{Z}$.

Proposition 4.5 ((8.6)). *If K has some unit of norm -1 (e.g., if r is odd), and if ε is an odd function, then $\Lambda_c(0, \varepsilon) \in 2\mathbb{Z}$.*

Démonstration. By (8.4), it suffices to show that $\tilde{\varepsilon} = 0$. However, the unit hypothesis shows that there is an element $\mathfrak{a}$ of $\mathfrak{Z}_1$, not in $\mathfrak{Z}_1^+$, whose image in G_1 is trivial. It follows that the restriction to the set of elements of I of the form $(\omega \cdot 0)$ of any odd function on I is identically zero. Hence, by definition, $\tilde{\varepsilon} = 0$ in this case. $\square$

We now consider the case where no unit of K has norm -1 . In G_1 , the Frobenius group $\mathfrak{Z}_1$ contains as a subgroup of index 2 the group $\mathfrak{Z}_1^+$ generated by products of pairs of Frobenius elements. We let $\mathfrak{X} = G_1/\mathfrak{Z}_1^+$ and let σ be the real Frobenius element of $\mathfrak{X}$. The order of σ is 2. (We may view $\mathfrak{X}$ as the Galois group over K of the largest abelian extension of K which is unramified at the finite places and either totally real or else a “CM field.” The fact that K has no unit of norm -1 means that this extension is in fact a CM field. In this optic, σ is the canonical “complex conjugation” of the field.)

Lemme 4.6 ((8.7)). *Let ε be a function $\mathfrak{X} \rightarrow \mathbb{Z}$ which is odd [$\varepsilon(\sigma x) = -\varepsilon(x)$] and which satisfies*

$$\sum_{x \in \mathfrak{X}} \varepsilon(x) \equiv 0 \pmod{2}.$$

Then ε is the sum of functions of the form $h(x) - h(\sigma x)$, where $h : \mathfrak{X} \rightarrow \mathbb{Z}$ is odd and where $\sigma \in \mathfrak{X}$.

Démonstration. For each $\sigma \in \mathfrak{X}$, define an odd function $h_\sigma : \mathfrak{X} \rightarrow \mathbb{Z}$ by :

$$h_\sigma(x) = \begin{cases} 1 & \text{if } x = \sigma, \\ -1 & \text{if } x = \sigma\sigma, \\ 0 & \text{otherwise.} \end{cases}$$

By subtracting from ε multiples of the functions $h_\sigma(x) - h_\sigma(\sigma x)$, we may assume that ε is supported on $\{1, \sigma\}$. The hypothesis on ε then means that ε takes even values. The equation

$$\varepsilon(x) = \varepsilon(x)/2 - \varepsilon(\sigma x)/2$$

exhibits ε as a function of the form $h(x) - h(\sigma x)$. $\square$

Suppose now that $\varphi : I \rightarrow \mathbb{Z}$ is a function which is odd (or, more generally, which has parity (a_v) for some collection of integers $a_v = 0$ or 1). We will define an invariant $\delta(\varphi)$ in $\mathbb{Z}/2\mathbb{Z}$. The restriction of φ to the set of elements of the form $(x \cdot 0)$ in I may be viewed as a function

$$r : G_1 \rightarrow \mathbb{Z}.$$

Composing r with the natural map $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$, we obtain a function r' on G_1 which is even, i.e., $\mathfrak{Z}_1$ -invariant. We set

$$\delta(\varphi) = \sum_{g \in G_1/\mathfrak{Z}_1} r'(g),$$

the summation running over elements g of G_1 , taken modulo $\mathfrak{Z}_1$.

Proposition 4.7 ((8.8)). *Suppose that ε is an odd function and that $\delta(\varepsilon) = 0$. Then*

$$\Lambda_c(0, \varepsilon) \in 2\mathbb{Z}.$$

Démonstration. Because of (8.6), we may assume that no unit of K has norm -1 . The function $r : G_1 \rightarrow \mathbb{Z}$ constructed as above for ε is $\mathfrak{Z}_1$ -invariant and consequently may be viewed as a function on $\mathfrak{X}$. By (8.7), we may find odd locally constant functions $H : A \rightarrow \mathbb{Z}$ and ideals $\mathfrak{a}$ of K such that the difference ε' between ε and the sum of the functions

$$H(\mathfrak{a}) - H(\sigma\mathfrak{a})$$

vanishes on $\mathfrak{X}$. By (8.4), we get then

$$\Lambda_c(0, \varepsilon') + \Lambda_c(0, \varepsilon') \in \mathbb{Z}.$$

On the other hand, we have $\Lambda_c(0, H(\mathfrak{a}\mathfrak{c})) = \Lambda_c(0, H(\sigma\mathfrak{a}\mathfrak{c}))$ for each H , so that

$$\Lambda_c(0, \varepsilon) = \Lambda_c(0, \varepsilon').$$

This proves what is wanted. □

Corollaire 4.8 ((8.9)). *For each odd ε , we have $\Lambda_c(0, \varepsilon) \in 2^{r-1}\mathbb{Z}$.*

We now introduce the notion of an *exceptional field*. We say that K is exceptional if the following two conditions are satisfied :

- i) All units of K have norm $+1$,
- ii) There are units of K of all signatures compatible with (i).

When K is exceptional, we have $\mathfrak{Z}_1 = \{1\}$, and $\mathfrak{Z}_1^*$ consists of two elements. A field is exceptional if its strict Hilbert class field is a CM field.

Proposition 4.9 ((8.10)). *Suppose that K is not exceptional. Then for each odd ε we have*

$$\Lambda_c(0, \varepsilon) \in 2^r\mathbb{Z}.$$

Démonstration. In view of (8.6), we may again suppose that K satisfies (i). The hypothesis that K is not exceptional then means that there is a principal ideal (α) generated by an integer α of positive norm, whose strict ideal class is non-trivial. Let η be the locally constant compactly supported function on I whose value on an ideal $\mathfrak{a}$ is given by

$$\eta(\mathfrak{a}) = \begin{cases} 0 & \text{if } \mathfrak{a} \neq (\alpha), \\ \text{sgn } \alpha & \text{if } \mathfrak{a} = (\alpha). \end{cases}$$

Here

$$\text{sgn } \alpha = \begin{cases} 0 & \text{if } \alpha \text{ is non-principal,} \\ +1 & \text{if } \alpha = (\beta) \text{ with } \mathcal{N}\beta > 0, \\ -1 & \text{if } \alpha = (\beta) \text{ with } \mathcal{N}\beta < 0. \end{cases}$$

Note that sgn is well defined precisely because there is no unit of negative norm in K .

Let us check that η satisfies the hypothesis to (8.4) with $\mathfrak{B} = \mathcal{O}$, i.e.,

$$\sum_{\substack{\mathfrak{a} \in I_0 \\ \mathfrak{a} \subset \mathcal{O}}} \eta(\mathfrak{a}) \mathcal{N}\mathfrak{a}^{-k} \in \mathbb{Z},$$

for all $\mu \gg 0$. We observe that $\mathfrak{a} \mapsto \alpha\mathfrak{a}^{-1}\mu$ gives an involution on the set of $\mathfrak{a}$ in the second sum, and that $\eta(\mathfrak{a}) = \eta(\alpha\mathfrak{a}^{-1}\mu)$. Hence η does indeed satisfy the hypothesis if there are

no fixed points of the involution on which η takes non-zero values. But if $\mu = \alpha\mu^{-1}\lambda$ and if $\mathfrak{a}$ is principal, then $(\alpha\mu^{-1}\lambda)$ is the square of a principal ideal and hence in the trivial strict ideal class of K . This is contrary to the choice of α and the fact that $\mu \gg 0$. Hence $\Lambda_c(0, \eta) + \Lambda_c(0, \tilde{\eta}) \in \mathbb{Z}$, by (8.4).

However $\tilde{\eta}$ is obviously the “same” function as η , except that its support is $\mathcal{O}$ rather than (α) . More precisely, we have $\tilde{\eta}(\mathfrak{a}) = \eta((\alpha)\mathfrak{a})$. It follows that $\Lambda_c(0, \eta) = \Lambda_c(0, \tilde{\eta})$. Thus

$$\Lambda_c(0, 2\tilde{\eta}) \in \mathbb{Z}.$$

The function $\tilde{\eta} : 2\tilde{\eta}$ is odd and $\mathbb{Z}$ -valued (it is just the function sgn with support on A). It satisfies $\delta(\tilde{\eta}) = 1$ because, on the $(\omega \cdot 0)$, $\tilde{\eta}$ vanishes on all elements except those with wide class 1; in the sum defining $\delta(\tilde{\eta})$ there is only one term. This gives (8.10). $\square$

Proof of (8.2). For $c \in G$, we must first show that μ_c is a measure. This means showing that $\Lambda_c(0, \varepsilon) \in \mathbb{Z}$ for all ε . Given ε , we may define an odd function ε' by the formula

$$\varepsilon'(\mathfrak{a}) = 2^{-r} \sum_{\sigma \in \mathfrak{Z}_1} \varepsilon(\sigma\mathfrak{a}) \in 2^{-r}\mathbb{Z},$$

in which the summation runs over the group $\mathfrak{Z}_1$ (whose cardinality is 2^r). It is apparent from (8.3) that $\Lambda_c(0, \varepsilon) = \Lambda_c(0, \varepsilon')$. By (8.8) and (8.10), it suffices to show that $\delta(2^r \varepsilon') = 0$ whenever K is exceptional.

Under this hypothesis on K , the restriction of ε to the $(x \cdot 0)$ is already invariant under $\mathfrak{Z}_1^+$, since the image of this group in G_1 is trivial. Hence the restriction of ε' to the $(x \cdot 0)$ takes values in $2^{r-1}\mathbb{Z}$. Since $r > 1$, the restriction of r' to the $(x \cdot 0)$ thus takes even values. In particular, $\delta(2^r \varepsilon') = 0$, as required.

Thus μ_c is indeed a measure with values in $\mathbb{Z}$.

It remains now to show that

$$\int \varepsilon \cdot \mathcal{N}^{k-1} d\mu_c = \Lambda_c(1 - k, \varepsilon)$$

for $k > 1$, and all Schwartz functions ε . We define ε' as above, and we similarly define ε'' by omitting the factor $\mathcal{N}\mathfrak{a}^{-1}$ in the sum defining ε' . Let ε^* be either ε' or ε'' according as k is odd or even, so that ε^* has the same parity as k . By (8.3), we have

$$\Lambda_c(1 - k, \varepsilon) = \Lambda_c(1 - k, \varepsilon^*),$$

and

$$\int \varepsilon \cdot \mathcal{N}^{k-1} d\mu_c = \int \varepsilon^* \cdot \mathcal{N}^{k-1} d\mu_c.$$

Hence there is no loss of generality in supposing that ε has parity $(-1)^k$, which we now do. Making a change of variables, we shall suppose also that ε is supported on A .

Let m be a positive integer. Choose a locally constant function

$$\eta : A \rightarrow \mathbb{Z}$$

which is odd and congruent modulo m to the norm function $\mathcal{N}$ on A . By (8.4), we obtain from

$$\sum_{k \geq 1} \varepsilon_k \eta^{k-1} \equiv \sum_{k \geq 1} \varepsilon_k \mathcal{N}^{k-1} \pmod{m\mathbb{Z}}$$

the congruence

$$\Lambda_c(1 - k, \varepsilon) \equiv \Lambda_c(0, \varepsilon \eta^{k-1}) + \Lambda_c(0, \tilde{\varepsilon} \eta^{k-1}) \pmod{m\mathbb{Z}}.$$

Now $\mathcal{N}$ vanishes on the $(x \cdot 0)$, so that η takes values on these elements which are divisible by m . Hence $\tilde{\varepsilon} \eta^{k-1} \equiv 0 \pmod{m}$, so that its integral against μ_c is again divisible by m . Thus we have simply

$$\Lambda_c(1 - k, \varepsilon) \equiv \Lambda_c(0, \varepsilon \eta^{k-1}) \pmod{m}.$$

On the other hand, because μ_c is a measure, the right hand member of this congruence is congruent mod m to the integral against μ_c of $\varepsilon \mathcal{N}^{k-1}$. Hence the two numbers we wish to prove equal are congruent mod m . But m was an arbitrary positive integer, so that the proof of (8.2) is complete. $\square$

We now summarize what we can say about the integral of an odd function. Let $\varphi : I \rightarrow \mathbb{Z}$ be odd, continuous, and compactly supported.

Théorème 4.10 ((8.11)). *We have $\int \varphi d\mu_c \in 2^{r-1}\mathbb{Z}$. Moreover, this integral lies in $2^r\mathbb{Z}$ unless K is exceptional and $\delta(\varphi) = 1$.*

Démonstration. Choose an odd Schwartz function $\varepsilon : I \rightarrow \mathbb{Z}$ which is congruent to $\varphi \pmod{2^r}$. Then $\delta(\varphi) = \delta(\varepsilon)$ and the integrals against μ_c of ε and φ are congruent mod 2^r . Hence our assertions follow from (8.8), (8.9), and (8.10). $\square$

Suppose now that K is exceptional. For $c \in G$ and φ with $\delta(\varphi) = 1$, the quantity

$$2^{1-r} \int \varphi d\mu_c \pmod{2}$$

is an element $\varepsilon(c)$ of $\mathbb{Z}/2\mathbb{Z}$ which, by (8.11), is independent of φ . To compute it, we can take φ to be locally constant. The equation

$$\Lambda_{cc'}(0, \varphi) = \Lambda_c(0, \varphi) + \Lambda_{c'}(0, \mathcal{N}c \cdot \varphi_c)$$

then shows that ε is a homomorphism $G \rightarrow \mathbb{Z}/2\mathbb{Z}$. Its kernel corresponds, then, to some extension of K of degree 1 or 2 which, by (8.3) is easily seen to be totally real. After looking at a small number of numerical examples, Lenstra predicted that this extension would prove to be the field M obtained by adjoining to K the square roots of all positive units. (That this field is a quadratic extension follows from the fact that K is exceptional.)

His prediction was correct.

Théorème 4.11 ((8.12)). *If K is exceptional and $\delta(\varphi) = 1$, then we have*

$$\int \varphi d\mu_c \in 2^r\mathbb{Z}$$

if and only if the image of c in $\text{Gal}(M/K)$ is trivial.

Démonstration. Let ε be the “sign” function with support on A , as in the proof of (8.10). We have already noted in that proof that ε is an odd function with $\delta(\varepsilon) = 1$. Also, with $\mathfrak{B} = \mathcal{O}$, we have $\tilde{\varepsilon} = \varepsilon$.

Let E be the Eisenstein series $G_{1,\varepsilon}$, made with $\mathfrak{B} = \mathcal{O}$. Next, choose u to be a totally positive unit which is not a square in K , so that

$$M = K(\sqrt{u}).$$

Let F be the theta series of (7.1), made with this choice of u . The two forms F and E satisfy the rationality property of (0.1), as we see by looking at the q -expansions of F and E at the standard cusp. (For the constant term of F , we need also (8.1).) An elementary argument shows that the non-constant terms of the standard q -expansions of E and $2F$ are congruent modulo 2. In other words, the form $E/2 - F/4$ has a standard q -expansion whose non-constant terms all lie in $\mathbb{Z}$. Arguing now as in the proof of (8.4), we find for all $c \in G$ a congruence

$$\frac{A_c(0, \varepsilon)}{2} + \frac{A_c(0, \tilde{\varepsilon})}{2} \equiv \frac{1 - \mathcal{N}c \cdot \omega(c)}{2} \pmod{2},$$

with ω as in (7.1). On the other hand, we have already remarked that ε and $\tilde{\varepsilon}$ are equal. Hence the integral in (8.12) lies in $2^r\mathbb{Z}$ if and only if

$$\mathcal{N}c \cdot \omega(c) \equiv 1 \pmod{4}.$$

Now, mod 4, $\mathcal{N}$ is the quadratic character of G corresponding to the extension

$$K(\sqrt{-1})/K,$$

whereas ω corresponds to the quadratic extension

$$K(\sqrt{-u})/K.$$

Hence the product of these two quadratic characters corresponds to the extension M/K , as desired. $\square$

Proof of (0.4)

Let (ε_k) be a sequence of functions as in (0.4). We may view these as functions on I , defined mod $\mathfrak{f}$ and supported on A . From this point of view, the ε_k furthermore vanish on integral ideals which are not prime to $\mathfrak{f}$. Hence if we form the sum $\sum_k \varepsilon_k \mathcal{N}^{k-1}$ as a function $I \rightarrow \mathbb{Q}_p$, the hypothesis to (0.4) implies that this function is in fact $\mathbb{Z}_p$ -valued. (Note that $\mathcal{N}$ is a priori $\mathbb{Z}$ -valued, while ε_k is $\mathbb{Q}_p$ -valued, so we may consider the product $\varepsilon_k \mathcal{N}^{k-1}$ as a number in $\mathbb{Q}_p$.) For each $c \in G$, the integral against μ_c of this sum is again an element of $\mathbb{Z}_p$. Using the second part of (8.2), we get then

$$\sum_{k \geq 1} A_c(1 - k, \varepsilon_k) \in \mathbb{Z}_p,$$

as needed for (0.4).

We must observe that the group called G in §0 is not the full group $\text{Gal}(K^{\text{ab}}/K)$ as in the succeeding §§, but rather a quotient of it. The explanation is that for ε a function mod $\mathfrak{f}$ with values in $\mathbb{Q}_p$, the quantities $A_c(1 - k, \varepsilon) \in \mathbb{Q}_p$ depend only on the image of c in the quotient group used in §0.

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ABSOLUTE HODGE CYCLES AND THE PERIODS OF INTEGRALS OF ABELIAN VARIETIES

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ABSTRACT. We explain here how a suitable theory of “motives” recovers certain results of Shimura on the periods of abelian varieties of CM type [5]. It seemed instructive to begin with an informal description of the properties of motives and of their realizations; this is done in Section 1. Section 2 interprets certain results of Shimura by means of motives. Section 3 defines absolute Hodge cycles, and the corresponding theory of motives. Section 4 gives an upper bound for the transcendence degree of the field generated by the periods of an arbitrary abelian variety.

1. THE LANGUAGE OF MOTIVES

Let k be a field of characteristic 0. There are several definitions of “motives over k ” — it is not known whether they are equivalent — and each has its own virtues. In Section 3 we shall specify the one that is useful for us.

The motives over k form an additive category equipped with a tensor product. Every smooth projective variety X over k defines a motive $h(X)$ over k , the *motivic cohomology* of X , and every motive over k is obtained from a motive of this form (by cutting into pieces and Tate twisting). Finally, various cohomology theories for algebraic varieties correspond to realization functors on motives.

Every abelian variety A over k defines a motive $T(A)$ over k , its motivic H_1 , and the functor $A \mapsto T(A)$ is fully faithful. The advantage of embedding the category of abelian varieties into the category of motives is that one can then take tensor products. The resulting objects are, however, no longer geometric: there are no “functions” on a motive.

The *rank* of a motive is the dimension of any one of its realizations, which we now describe.

If $k = \mathbb{C}$, the *Betti realization* of a motive M is a free abelian group of finite rank $H_B(M)$ endowed with a *Hodge structure*, i.e. a decomposition

$$H_B(M) \otimes_{\mathbb{Z}} \mathbb{C} = \bigoplus_{p+q=n} H^{p,q} \quad (\overline{H}^{p,q} = H^{q,p}).$$

Let us illustrate this with the motive $T(A)$, namely the H_1 of an abelian variety A . One defines $H_B(T(A)) = H_1(A, \mathbb{Z})$, where $A = A(\mathbb{C})$ is regarded as a topological space.

Let $\text{Lie}(A)$ be the Lie algebra of the analytic group A , and let $J : H_1(A, \mathbb{Z}) \rightarrow \text{Lie}(A)$ be the homomorphism “integration along a 1-cycle.” There is an exact sequence of complex analytic groups

$$0 \longrightarrow H_1(A, \mathbb{Z}) \xrightarrow{J} \text{Lie}(A) \xrightarrow{\exp} A \longrightarrow 0.$$

The Hodge structure on $H_1(A, \mathbb{C})$ is such that $H_1(A, \mathbb{C}) = H^{0,-1} \oplus H^{-1,0}$. One obtains $H^{0,-1}$ as the kernel of the map $H_1(A, \mathbb{C}) \rightarrow \text{Lie}(A)$ obtained by extending $J : H_1(A, \mathbb{Z}) \rightarrow \text{Lie}(A)$ $\mathbb{C}$ -linearly.

Now let k be any field of characteristic 0 and fix an algebraic closure $\bar{k}$ of k . For each prime number ℓ , the ℓ -*adic realization* of a motive M is a free $\mathbb{Z}_\ell$ -module of finite type $H_\ell(M)$ (equipped moreover with an action of the Galois group of $\bar{k}$ over k). When $k = \mathbb{C}$, one has an isomorphism

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$H_\ell(M) \cong H_B(M) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. If A is an abelian variety over k , and if $\bar{k}$ is an algebraic closure of k , the ℓ -adic realization of $T(A)$ is denoted $T_\ell(A)$; it is the $\mathbb{Z}_\ell$ -module

$$T_\ell(A) = \varprojlim_n A_{\ell^n}(\bar{k}) \quad (\text{where } A_{\ell^n}(\bar{k}) \text{ is the group of points of order } \ell^n \text{ in } A(\bar{k})).$$

If $k = \mathbb{C}$, the isomorphism $T_\ell(A) \cong H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is easy to describe. For every n , the projection of $\gamma \otimes x \in H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ to $A_{\ell^n}(\mathbb{C})$ is computed as follows: if $m \in H_B(T(A))$ is such that $x - m \in \ell^n(H_B(T(A)) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)$, the exponential of the element $m/\ell^n \in H_1(A, \mathbb{Q})$ is the desired point of order ℓ^n .

The *de Rham realization* of M is a finite-dimensional vector space over k , denoted $H_{DR}(M)$, endowed with a decreasing filtration $F^\bullet$ (the *Hodge filtration*). When $k = \mathbb{C}$ there is an isomorphism $H_B(M) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(M)$. Under this isomorphism the Hodge filtration is such that $F^p = \bigoplus_{p' \geq p} H^{p', q}$. In the case of the motive $T(A)$, recall from [7] the existence of a “universal extension” E of A by a vector group, fitting into an exact sequence of algebraic groups

$$0 \longrightarrow H^1(A, \mathcal{O}_A)^* \longrightarrow E \longrightarrow A \longrightarrow 0.$$

One has $H_{DR}(T(A)) = \text{Lie}(E)$, and the Hodge filtration is given by

$$0 = F^1 \subset F^0 = H^1(A, \mathcal{O}_A)^* \subset F^{-1} = \text{Lie}(E).$$

If $k = \mathbb{C}$, one has $H_1(E, \mathbb{Z}) \xrightarrow{\sim} H_1(A, \mathbb{Z})$, hence a morphism

$$H_1(A, \mathbb{C}) \xleftarrow{\sim} H_1(E, \mathbb{C}) \xrightarrow{J} \text{Lie}(E) = H_{DR}(T(A)).$$

This is the desired isomorphism.

Examples. 1) Here $k = \mathbb{Q}$. We define the *Tate motive* $\mathbb{Z}(1)$; it is the H_1 of the algebraic group $\mathbb{G}_m$. To find its various realizations, one substitutes $\mathbb{G}_m$ for A in the definition of $T(A)$. One obtains:

- $H_B(\mathbb{Z}(1)) = 2\pi i \cdot \mathbb{Z}$, the kernel of $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$;
- $H_B(\mathbb{Z}(1)) \otimes_{\mathbb{Z}} \mathbb{C} = H^{-1, -1}$;
- $H_\ell(\mathbb{Z}(1)) = \mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_m) = \varprojlim_n \mu_{\ell^n}(\bar{\mathbb{Q}})$;
- $H_{DR}(\mathbb{Z}(1)) = \mathbb{Q}$.

The isomorphism $H_B(\mathbb{Z}(1)) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(\mathbb{Z}(1)) \otimes_{\mathbb{Q}} \mathbb{C}$ sends $(2\pi i) \otimes 1$ to $2\pi i$. Recall the isomorphisms $H_1(\mathbb{G}_m) \cong H_1(\mathbb{P}^1 \setminus \{0, \infty\}) \cong H_2(\mathbb{P}^1)$. Moreover, for every smooth proper connected algebraic curve over $\mathbb{Q}$, $H_2(X)$ is isomorphic to $H_2(\mathbb{P}^1)$ (hence to $\mathbb{Z}(1)$). For every integer d , one defines a motive $\mathbb{Z}(d)$ so that $\mathbb{Z}(d) = \mathbb{Z}(d-1) \otimes_{\mathbb{Z}} \mathbb{Z}(1)$. One has $\mathbb{Z}(m+n) \cong \mathbb{Z}(m) \otimes_{\mathbb{Z}} \mathbb{Z}(n)$ for $m, n \in \mathbb{Z}$.

2) Let E be an elliptic curve over $\mathbb{Q}$. The motive $T(E)$ has rank 2. Hence the motive $\wedge^2 T(E)$ has rank 1, and one has $\wedge^2 T(E) \cong \wedge^2 H_1(E) \cong H_2(E) \cong \mathbb{Z}(1)$.

Let us make explicit the isomorphism $\wedge^2 T(E) \cong \mathbb{Z}(1)$ in the different realizations.

On $H_1(E(\mathbb{C}), \mathbb{R}) \cong \text{Lie}(E)$, the complex structure defines an orientation. If (e_1, e_2) is an oriented $\mathbb{Z}$ -basis of $H_B(T(E)) = H_1(E(\mathbb{C}), \mathbb{Z})$, then $e_1 \wedge e_2$ is a basis of $H_B(\wedge^2 T(E))$; it is sent to $2\pi i \in H_B(\mathbb{Z}(1))$.

The isomorphism $\wedge^2(T_\ell(E)) \cong \mathbb{Z}_\ell(1)$ is given by the alternating Weil pairing [6]. Finally, one has $\wedge^2 H_{DR}(T(E)) \cong \text{Hom}(H_{DR}^2(E), \mathbb{Q})$. Let ω and η be the usual differentials of the first and second kind. $H_{DR}^2(E)$ is generated by the cup-product $\omega \wedge \eta$ of their cohomology classes. Under the isomorphism $H_B(\wedge^2 T(E)) \otimes_{\mathbb{Z}} \mathbb{C} \rightarrow H_{DR}(\wedge^2 T(E)) \otimes_{\mathbb{Q}} \mathbb{C}$, the element $e_1 \wedge e_2$ maps to the homomorphism that sends $\omega \wedge \eta \in H_{DR}^2(E)$ to

$$\begin{vmatrix} \omega_1 & \omega_2 \\ \eta_1 & \eta_2 \end{vmatrix} \in \mathbb{C}.$$

Compatibility between the Betti and de Rham realizations of the isomorphism $\Lambda^2(T(E)) \cong \mathbb{Z}(1)$ is equivalent to the *Legendre relation*

$$\omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i.$$

3) Let A be an abelian variety over $\mathbb{Q}$. To the cohomology class of a hyperplane section of A in a projective embedding there corresponds a morphism of motives $\Lambda^2 T(A) \rightarrow \mathbb{Z}(1)$. In Betti cohomology this corresponds to an alternating form $2\pi i \cdot E$ on $H_B(T(A))$, with values in $2\pi i \cdot \mathbb{Z}$. In de Rham cohomology one likewise has an alternating form E_{DR} on $H_{DR}(T(A))$ with values in $\mathbb{Q}$. It is classical that F^0 is a totally isotropic subspace. It follows that one can find differentials of the first kind $\omega_1, \dots, \omega_g$ on A and differentials of the second kind $\eta_1, \dots, \eta_g$ on A such that

$$E_{DR}(x, y) = \sum_{i=1}^g ((\omega_i \cdot x)(\eta_i \cdot y) - (\omega_i \cdot y)(\eta_i \cdot x)).$$

Compatibility between E and E_{DR} is expressed as follows. Let $(e_1, \dots, e_{2g})$ be a basis of $H_B(T(A))$, let Ω be the $g \times 2g$ matrix with entries $\Omega_{ij} = \int_{e_j} \omega_i$, and let N be the $g \times 2g$ matrix with entries $N_{ij} = \int_{e_j} \eta_i$. Then

$${}^t N \cdot \Omega - \Omega \cdot N = 2\pi i \cdot E.$$

Finally, in the ℓ -adic realization one finds an alternating form on $T_\ell(A)$ with values in $\mathbb{Z}_\ell(1)$: it is the Weil pairing attached to the divisor of the hyperplane section [6].

2. MOTIVES ATTACHED TO CM ABELIAN VARIETIES

Let k be an algebraically closed subfield of $\mathbb{C}$ — for example $\bar{\mathbb{Q}}$ — and let $\mathcal{A}$ be the full subcategory of the category of motives over k generated by the $T(A)$ of abelian varieties, under tensor product, duals, and direct summands. The functor “Betti realization of the motive obtained from M/k after extension of scalars to $\mathbb{C}$ ” sends $\mathcal{A}$ to the category of Hodge structures. It follows formally from Theorem 1 of Section 3 that this functor is fully faithful.

Let E be a CM field, let S be the set of its complex embeddings, and let $- : S \rightarrow S$ denote complex conjugation. To each function $\varphi : S \rightarrow \mathbb{Z}$ satisfying

$$(*) \quad \varphi(\sigma) + \varphi(\bar{\sigma}) \text{ is independent of } \sigma \in S,$$

attach a Hodge structure H_φ endowed with a module structure over the ring of integers $\mathcal{O}_E$ of E : $H_\varphi = \mathcal{O}_E$, with its evident module structure, and in the decomposition $H_\varphi \otimes_{\mathbb{Z}} \mathbb{C} = E \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^S$, the factor $\mathbb{C}$ indexed by $\sigma \in S$ is assigned Hodge type $(\varphi(\sigma), \varphi(\bar{\sigma}))$.

One checks that the H_φ belong to the image of $\mathcal{A}$, indeed already to the image of the subcategory generated by CM abelian varieties. This allows one to define the motive M_φ over $\bar{\mathbb{Q}}$ as the motive whose Betti realization is H_φ .

The de Rham realization $H_{DR}(M_\varphi)$ is a vector space over $\bar{\mathbb{Q}}$ endowed with an action of E . For $\sigma \in S$, let $H_{DR}(M_\varphi)_\sigma$ be the $\bar{\mathbb{Q}}$ -subspace on which $x \in E$ acts as $\sigma(x) \in \bar{\mathbb{Q}}$. Each of these spaces is one-dimensional, generated by an element ω_σ , and $H_{DR}(M_\varphi)$ is their direct sum. Via the isomorphism $H_B(M_\varphi) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{DR}(M_\varphi) \otimes_{\bar{\mathbb{Q}}} \mathbb{C}$, an element e of $E = H_B(M_\varphi) \otimes_{\mathbb{Z}} \mathbb{Q}$ corresponds to a family (e_σ) with $e_\sigma \in H_{DR}(M_\varphi)_\sigma$. One checks immediately that the complex number $p(\sigma, \varphi)$ defined by

$$p(\sigma, \varphi) e_\sigma = \omega_\sigma$$

is well defined up to multiplication by an element of $\bar{\mathbb{Q}}^*$, independently of both e and the choice of ω_σ . This justifies the notation.

The relation between these periods and Shimura’s invariants $p_E(\sigma, \varphi)$ (see [5]) is

$$p_E(\sigma, \varphi) = \frac{p(\sigma, \varphi)}{(2\pi i)^{\varphi(\sigma)}} \pmod{\bar{\mathbb{Q}}^*},$$

which is simply a reformulation of the definitions.

The identities between Shimura's periods correspond to algebraic properties of the motives M_φ . Before summarizing them in a table, let us introduce some notation for an extension F/E of CM fields. Write S_F (resp. S_E) for the set of complex embeddings of F (resp. E). For $\sigma \in S_F$, write $\sigma|_E \in S_E$ for the restriction of σ to E .

For $\varphi : S_E \rightarrow \mathbb{Z}$ satisfying (*), let $\text{Inf}(\varphi)$ be the function on S_F defined by $\text{Inf}(\varphi)(\sigma) = \varphi(\sigma|_E)$. This function again satisfies (*). For $\varphi : S_F \rightarrow \mathbb{Z}$ satisfying (*), let $\text{Res}(\varphi)$ be the function on S_E defined by $\text{Res}(\varphi)(\sigma) = \sum_{\tau|_E=\sigma} \varphi(\tau)$; it too satisfies (*).

Identities among periods	Relations among motives
1. $p(\sigma, \varphi' + \varphi'') = p(\sigma, \varphi') p(\sigma, \varphi'')$	1. $M_{\varphi' + \varphi''} \cong M_{\varphi'} \otimes_E M_{\varphi''}$
2. For a CM extension F/E , $\sigma \in S_F$, and $\varphi : S_E \rightarrow \mathbb{Z}$ satisfying (*); $p(\sigma, \text{Inf}(\varphi)) = p(\sigma _E, \varphi)$.	2. $M_{\text{Inf}(\varphi)} \cong M_\varphi \otimes_E F$
3. For $\sigma \in S_E$ and $\varphi : S_F \rightarrow \mathbb{Z}$ satisfying (*); $p(\sigma, \text{Res} \varphi) = \prod_{\tau _E=\sigma} p(\tau, \varphi)$.	3. M_φ is an F -motive of rank 1, hence an E -motive of rank $d = [F : E]$, and $M_{\text{Res}(\varphi)} \cong \bigwedge_E^d M_\varphi$.
4. $p(\sigma, \underline{1}) = 2\pi i$	4. $M_{\underline{1}} = \mathbb{Z}(-1)$, the dual of the Tate motive.

These identities among periods recover Theorems 1–4 of Shimura's article [5]. Note that in [5] Shimura relates the periods $p_E(\sigma, \varphi)$ to the values at a special point of Hilbert modular forms rational over $\bar{\mathbb{Q}}$.

3. ABSOLUTE HODGE CYCLES AND MOTIVES

Let X be a smooth projective variety over $\mathbb{C}$. A *Hodge cycle* of codimension d on X is an element of $H^{2d}(X, \mathbb{Q})$ of type (d, d) , or equivalently an element of

$$H^{2d}(X, \mathbb{Q})(d) \stackrel{\text{def}}{=} H^{2d}(X, \mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Z}(d)$$

of type $(0, 0)$. Note that for $x \in H^{2d}(X, \mathbb{Q})(d)$ it is equivalent to say that x is of type $(0, 0)$ or that x lies in F^0 . A cohomology class coming from an algebraic subvariety of X is a Hodge cycle. Hodge conjectured that every Hodge cycle should be a linear combination of cohomology classes of subvarieties. Since that conjecture seems inaccessible, one introduces a notion that “arithmetizes” the notion of Hodge cycle.

The cycles we define will possess all the visible attributes of algebraic cycles, namely realizations in ℓ -adic and de Rham cohomology. An *absolute Hodge cycle* of codimension d on a smooth projective variety X over a field k of characteristic 0 is a collection of cohomology classes in the groups $H_{\text{et}}^{2d}(X_{\bar{k}}, \mathbb{Q}_\ell)(d)$ (resp. $H_{DR}^{2d}(X)(d)$) satisfying the following two conditions:

- the element of $H_{DR}^{2d}(X)(d)$ lies in F^0 ;
- for every embedding $k \hookrightarrow \mathbb{C}$, the various cohomology classes correspond (via the isomorphisms of Section 1) to a rational class in the Betti cohomology of $X \otimes_k \mathbb{C}$.

Thus one has inclusions

$$(\text{algebraic cycles}) \subset (\text{absolute Hodge cycles}) \subset (\text{Hodge cycles}).$$

Using this definition, let us now specify the category of motives used in Sections 1 and 2. Let $\mathcal{V}(k)$ first denote the category of smooth projective varieties over k . One constructs an *additive* category $\mathcal{M}(k)$ in which the groups $\text{Hom}(M, N)$ are $\mathbb{Q}$ -vector spaces, and which is equipped with the following data:

- a) a tensor product $\otimes$, associative, commutative, and distributive with respect to direct sums of objects;
- b) a contravariant functor H^* from $\mathcal{V}(k)$ to $\mathcal{M}(k)$, bijective on objects, compatible with sums, and taking products to tensor products.

For connected X and Y one has in $\mathcal{M}(k)$

$$\mathrm{Hom}(H^*(X), H^*(Y)) = Z_{h.a.}^{\dim(Y)}(X \times Y),$$

where $Z_{h.a.}^d(Z)$ denotes the $\mathbb{Q}$ -vector space of absolute Hodge cycles of codimension d on Z . The category of motives $\mathcal{M}(k)$ is then constructed exactly as in [2], with algebraic cycles replaced by absolute Hodge cycles. Thanks to this new definition, the Künneth components of the diagonal of $X \times X$ are absolute Hodge cycles, and hence one has in $\mathcal{M}(k)$ a decomposition

$$H^*(X) = \bigoplus_i H^i(X).$$

The following result was presented at the I.H.É.S. in 1979.

Theorem 1. *Every Hodge cycle on an abelian variety is absolute Hodge.*

We also note the following.

Corollary 1 (Borovoi). *Let A be an abelian variety over a field k of characteristic 0, and let ξ be a Hodge cycle on A . Then the ℓ -adic realizations ξ_ℓ of ξ are fixed by an open subgroup of $\mathrm{Gal}(\bar{k}/k)$.*

4. PERIODS OF INTEGRALS OF ABELIAN VARIETIES

Let A be a complex abelian variety, and let

$$H_1(A, \mathbb{C}) = H^{0,-1} \oplus H^{-1,0}$$

be the Hodge decomposition of $H_B(A)$ (cf. Section 1). Regard $\mathrm{GL}(H_1(A))$ as an algebraic group over $\mathbb{Q}$. There is then a morphism of complex algebraic groups

$$\mu : \mathbb{G}_m \rightarrow \mathrm{GL}(H_1(A, \mathbb{C}))$$

such that $\mu(z)$ acts by the homothety of ratio z on $H^{-1,0}$ and by the identity on $H^{0,-1}$. The *Hodge group* G of A is the Zariski closure of the subgroup $\mu(\mathbb{C}^*)$ of $\mathrm{GL}(H_1(A))$. In other words, G is an algebraic subgroup over $\mathbb{Q}$ of $\mathrm{GL}(H_1(A))$. A $\mathbb{Q}$ -rational regular function on a Zariski-open subset of G vanishes on G if and only if it vanishes on $\mu(\mathbb{C}^*)$. One checks easily that G contains the scalar homotheties of $H_1(A)$. A second description of G , very useful in practice, is obtained by considering all rational tensors of type $(-N/2, -N/2)$ in the tensor powers $\bigotimes^N H_1(A)$. Then G is the subgroup of $\mathrm{GL}(H_1(A))$ consisting of those linear transformations g for which there exists a scalar μ such that

$$g \cdot t = \mu^m t$$

for every tensor t of type (m, m) . Finally, one shows that G is the group fixing all absolute Hodge cycles of type $(0, 0)$ in the mixed tensor spaces built from $H_1(A)$.

Now assume that A is defined over a subfield k of $\mathbb{C}$. Put $g = \dim(A)$.

Theorem 2. *Let $\omega_1, \dots, \omega_g$ be differentials of the first kind on A , and let $\eta_1, \dots, \eta_g$ be differentials of the second kind, such that*

$$(\omega_1, \dots, \omega_g, \eta_1, \dots, \eta_g)$$

is a basis of $H_{DR}^1(A)$. Let K be the subfield of $\mathbb{C}$ generated by k and by the periods

$$\int_\gamma \omega_i, \quad \int_\gamma \eta_i \quad (1 \leq i \leq g, \gamma \in H_1(A, \mathbb{Z})).$$

Then the transcendence degree of K over k is bounded above by the dimension of G .

Proof. K is the field of definition of the isomorphism

$$H_{1,B}(A) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{1,DR}(A) \otimes_k \mathbb{C}$$

given by integration along cycles. This isomorphism preserves the cohomology classes of absolute Hodge cycles. Let P be the algebraic variety of isomorphisms with this property. By the definition of G , together with Theorem 1, P is a principal homogeneous space under G , defined over k , so $\dim(P) = \dim(G)$. One concludes by the following lemma.

Lemma 1. *Let z be a point of $\mathbb{A}^N(\mathbb{C})$. The transcendence degree of $k(z)$ over k is equal to the dimension of the Zariski closure of $\{z\}$ over k .*

This is tautological. □

Remark 1. From Example 3 of Section 1 it follows that $2\pi i \in K$. One can moreover show that the transcendence degree of K over $k(2\pi i)$ is at most $\dim(G) - 1$.

Corollary 2. *With the notation of Theorem 2, assume moreover that A is simple of CM type. Then the transcendence degree of K over k is at most $g + 1$.*

Proof. It suffices to show that $\dim(G) \leq g + 1$. Now $\mu(\mathbb{C}^*)$ commutes with the action of the field E on $H_1(A, \mathbb{Q})$. It follows that G also commutes with this action, and therefore G is an algebraic torus. Moreover, G preserves, up to scalar, the alternating form associated with a hyperplane section (see Example 3 of Section 1). Hence the claim. □

Note that in this last situation one also has the lower bound

$$\dim(G) \geq 2 + \log_2(g),$$

due to Ribet [3].

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THE FUNDAMENTAL GROUP OF THE COMPLEMENT OF A PLANE CURVE WITH ONLY ORDINARY DOUBLE POINTS IS ABELIAN

PIERRE DELIGNE

INTRODUCTION

Theorem 1. Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a plane curve whose only singularities are ordinary double points with distinct tangents. Then the fundamental group of $\mathbb{P}^2(\mathbb{C}) \setminus C$ is abelian.

This theorem was stated by Zariski in 1929 [8]; see also [9, Chapter VIII]. His proof used Severi's theorem asserting that the space of irreducible plane curves of fixed degree with a prescribed number of double points is irreducible (*Vorlesungen über algebraische Geometrie*, 1921, Appendix F). It is still unknown whether Severi's assertion is true.

The group π_1 appearing in the theorem is the ordinary topological fundamental group. If one replaces $\mathbb{C}$ by an arbitrary algebraically closed field k of characteristic 0, the same statement holds for the algebraic fundamental group classifying finite étale coverings: by the Lefschetz principle one may reduce to $k = \mathbb{C}$, and then the algebraic fundamental group is the profinite completion of the ordinary one. In arbitrary characteristic one has the following variant.

Theorem 2 (Fulton [4]). Let k be an algebraically closed field, and let $C \subset \mathbb{P}_k^2$ be a plane curve whose only singularities are ordinary double points with distinct tangents. Then the tame fundamental group of $\mathbb{P}_k^2 \setminus C$ is abelian.

In other words, coverings in the sense of §0.2 of $\mathbb{P}_k^2$, ramified only along C and only tamely ramified there, are abelian. Recall that tameness can be tested after base change to the localizations of $\mathbb{P}_k^2$ at the generic points of the irreducible components of C ; these localizations are traits (that is, spectra of discrete valuation rings).

Assume now that k has characteristic $p > 0$. If C is as in Theorem 2, there always exists in the relative projective plane over $\text{Spec}(W(k))$ a relative normal-crossings divisor $\tilde{C}$ whose special fiber is C (see, for example, Mumford, appendix to Chapter VIII of [9]). This allows one to deduce Theorem 2 from Theorem 1 by specialization.

Once one knows that $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C)$ is abelian, homological arguments immediately yield its structure: if C is the union of irreducible curves of degrees $d_1, \dots, d_r$, then

$$\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C) \cong \mathbb{Z}^r / \mathbb{Z} \cdot (d_1, \dots, d_r).$$

In his proof of Theorem 2, Fulton uses on the one hand methods introduced by Abhyankar [1] to handle the special case where the irreducible components of C are smooth—methods that Serre explained very clearly in [7]—and on the other hand the following connectedness theorem. An analysis of Fulton's proof then allows the present writer to give a variant that yields Theorem 1.

Abhyankar and Fulton both exploit instances of the principle that, in algebraic geometry, anything sufficiently mobile is connected. Let $C \subset \mathbb{P}^2$ be as in Theorem 2, let D be an irreducible component of C , and let

$$f: X \rightarrow \mathbb{P}^2$$

be a covering ramified only along C and only tamely ramified there. Since $D \subset \mathbb{P}^2$ is very mobile, $f^{-1}(D)$ is connected. More precisely, D is ample in $\mathbb{P}^2$, hence $f^{-1}(D)$ is ample in X because f

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is finite, and one applies Bertini. A local study of f shows moreover that if D is smooth then $(f^{-1}(D))_{\text{red}}$ is smooth, hence $f^{-1}(D)$ is irreducible.

In the general case let $g: \tilde{D} \rightarrow D$ be the normalization of D . The same local study shows that the reduced fiber product

$$(X \times_{\mathbb{P}^2} \tilde{D})_{\text{red}}$$

is smooth, hence is the normalization of $f^{-1}(D)$. If one can prove that it is connected, it follows that $f^{-1}(D)$ is irreducible, and one may then repeat the end of Abhyankar's argument as presented in [7].

Recall the definition of the fiber product: if

$$h: X \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2$$

is the product of $f: X \rightarrow \mathbb{P}^2$ and $g: \tilde{D} \rightarrow D \subset \mathbb{P}^2$, then $X \times_{\mathbb{P}^2} \tilde{D}$ is the inverse image $h^{-1}(\Delta)$ of the diagonal. Since $\mathbb{P}^2$ has many automorphisms, the diagonal $\Delta \subset \mathbb{P}^2 \times \mathbb{P}^2$ is very mobile, and therefore $h^{-1}(\Delta)$ is connected. This finishes the argument. More precisely, one applies to h the special case $r = 2$, $m = 2$ of the following theorem.

Theorem 3 (Fulton–Hansen [5]). Let

$$P = (\mathbb{P}^m)^r$$

be the product of r copies of projective m -space, let

$$\delta: \mathbb{P}^m \rightarrow P$$

be the diagonal map, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be a complete irreducible variety, and let

$$h: Z \rightarrow P.$$

Write $n = \dim h(Z)$ and $c = (r - 1)m$, the codimension of Δ in P . If $n - c \geq 1$, then $h^{-1}(\Delta)$ is connected.

This theorem has many other applications; for them I refer to [4] and to the paper of Gaffney and Lazarsfeld [5].

To end the introduction, let us explain on an example the method used to transform Fulton's proof of Theorem 2 into a proof of Theorem 1. The example is the case where C is smooth (hence irreducible).

Let U be a tubular neighborhood of C in $\mathbb{P}^2(\mathbb{C})$, put $U^* = U \setminus C$, and choose $u \in U^*$. For any covering

$$f: X \rightarrow \mathbb{P}^2(\mathbb{C})$$

ramified only along C , one has

$$(f^{-1}(C) \text{ connected}) \iff (f^{-1}(U) \text{ connected}) \iff (f^{-1}(U^*) \text{ connected}).$$

Thus the key result—that $f^{-1}(C)$ is connected for every such covering X —is equivalent to the following: for every finite quotient H of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$, the composite map

$$\pi_1(U^*, u) \xrightarrow{j_*} \pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u) \rightarrow H$$

is surjective. Morse theory proves more: the map

$$j_*: \pi_1(U^*, u) \rightarrow \pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$$

is itself surjective.

Once a Riemannian metric on $\mathbb{P}^2(\mathbb{C})$ has been chosen, one may, for sufficiently small $\varepsilon > 0$, identify U with the image under the exponential map of the radius- ε subbundle of the normal bundle of C in $\mathbb{P}^2(\mathbb{C})$. For sufficiently small ε , the exponential map identifies U^* with an oriented punctured-disc bundle over C , and hence there is an exact sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, u) \rightarrow \pi_1(C, \bar{u}) \rightarrow 1$$

(where $\bar{u}$ denotes the image of u in C). The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop γ going once positively around C . Since j_* is surjective, γ defines a central element of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$. The quotient of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$ by the normal subgroup generated by γ is

$$\pi_1(\mathbb{P}^2(\mathbb{C}), u) = \{e\},$$

so the theorem follows in this special case.

0. TERMINOLOGY

0.1. “Connected” means connected and nonempty. Likewise for “irreducible.”

0.2. A *covering* of a normal connected scheme S is a morphism $f: T \rightarrow S$ with T normal and connected, whose restriction to some dense open subset of T is étale.

0.3. Throughout the rest of the paper we work over $\mathbb{C}$. We simply write $\mathbb{P}^n$ for $\mathbb{P}^n(\mathbb{C})$. All schemes are assumed separated.

1. LEFSCHETZ– AND BERTINI-TYPE THEOREMS

The special case of Assertion 1.1 below in which $Z \subset \mathbb{P}^n$ is the complement of a hypersurface was proved by Hamm and Lê [6] by Morse theory. Our goal in this section is to deduce from it the variant, Theorem 1.6, of Theorem 3.

Assertion 1.1. Let $Z \subset \mathbb{P}^n$ be a nonempty Zariski-open subset. If H is a general hyperplane, the natural morphism

$$\pi_i(Z \cap H) \rightarrow \pi_i(Z)$$

is bijective for $i < n - 1$ and surjective for $i = n - 1$.

Assertion 1.1 is stated in [2], but Cheniot’s proof covers only the case where the complement of Z has codimension at least 2. Under that assumption, Z and $Z \cap H$ are simply connected, so it is harmless to work, as he does, with homology rather than homotopy groups. The case $i = 1$ may also be deduced from [6] or [2 bis].

Recall that the adjective “general,” when used of an element H of an irreducible algebraic variety S (here the dual projective space $\mathbb{P}^{n\vee}$), is a quantifier: it means that there exists a nonempty Zariski-open subset $S' \subset S$ such that the desired conclusion holds for every $H \in S'$.

1.2. Let

$$Y \subset \mathbb{P}^n \times \mathbb{P}^{n\vee}$$

be the set of pairs (z, H) such that $z \in Z$ and $z \in H$. Applying the isotopy theorem to the second projection

$$\text{pr}_2: Y \rightarrow \mathbb{P}^{n\vee}$$

gives a nonempty Zariski-open subset $U \subset \mathbb{P}^{n\vee}$ over which Y is a topologically locally trivial fibration. It is enough to take for U the set of hyperplanes transverse to the strata of an algebraic Whitney stratification of the complement of Z in $\mathbb{P}^n$.

Locally on U , after choosing a trivialization of this fibration, the inclusion of the fiber $Z \cap H = \text{pr}_2^{-1}(H)$ into Z appears as a map depending continuously on H from a fixed space into Z . Hence the conclusion of Assertion 1.1 either holds for every $H \in U$ or for none of them; in other words, Assertion 1.1 says precisely that it holds for all $H \in U$.

Suppose now that Z depends on parameters: let S be an irreducible scheme, let Z be an open subset of $S \times \mathbb{P}^n$, write

$$Z(s) = \text{pr}_2(\text{pr}_1^{-1}(\{s\})),$$

and assume each $Z(s)$ is nonempty. Let

$$Y \subset S \times \mathbb{P}^n \times \mathbb{P}^{n\vee}$$

be the set of triples (s, z, H) with $z \in Z(s)$ and $z \in H$. Applying the isotopy theorem to the projection of Y to $S \times \mathbb{P}^{n\vee}$ (respectively of Z to S) yields a nonempty Zariski-open subset $U \subset$

$S \times \mathbb{P}^{n^\vee}$ (respectively $V \subset S$) over which Y (respectively Z) is a topologically locally trivial fibration. As above, if $s \in V$ and $(s, H) \in U$, then the conclusion of Assertion 1.1 holds for $Z(s)$ and H . This parameterized variant will be used implicitly in the proof of Lemma 1.4.

I conjecture more generally the following.

Conjecture 1.3. Let Z be smooth and connected of dimension n , let

$$f: Z \rightarrow \mathbb{P}^N$$

be a morphism with finite fibers, let c be an integer, and let $L \subset \mathbb{P}^N$ be a linear subspace of codimension c . Set $n^* = n - c$. Choose a Riemannian metric on $\mathbb{P}^N$, and let

$$V(\varepsilon) = \{p \in \mathbb{P}^N \mid d(p, L) < \varepsilon\}.$$

Then, for sufficiently small ε , the natural morphism

$$\pi_i(f^{-1}(V(\varepsilon))) \rightarrow \pi_i(Z)$$

is bijective for $i < n^*$ and surjective for $i = n^*$.

For general L , the inclusion of $f^{-1}(L)$ in $f^{-1}(V(\varepsilon))$ is, for ε sufficiently small, a homotopy equivalence; thus Conjecture 1.3 is indeed a generalization of Assertion 1.1.

If one drops the assumption that f is quasi-finite and if the set of points of $\mathbb{P}^N$ over which the fiber of f has dimension k has dimension $\varphi(k)$, then one should redefine

$$n^* = 2 \dim Z - \sup(2k + \varphi(k) + \inf(\varphi(k), c - 1)) - 1.$$

The conjecture should moreover follow from a more precise statement to the effect that Z is obtained from $f^{-1}(V(\varepsilon))$ by attaching handles that kill spheres of dimension at least n^* .

We shall content ourselves with weaker results than Conjecture 1.3, deduced from the special case of Assertion 1.1 where Z is the complement of a hypersurface.

Lemma 1.4. Let Z be normal and connected, let

$$f: Z \rightarrow \mathbb{P}^N$$

be a morphism, let $n = \dim f(Z)$, and let c be an integer. Assume $n - c \geq 1$. If L is a general linear subspace of codimension c in $\mathbb{P}^N$, then $f^{-1}(L)$ is connected and, for every $z \in f^{-1}(L)$, the natural morphism

$$\pi_1(f^{-1}(L), z) \rightarrow \pi_1(Z, z)$$

is surjective.

One may replace Z by any nonempty Zariski-open subset Z' : for general L , the scheme $f^{-1}(L)$ is again normal, and Z' (respectively $Z' \cap f^{-1}(L)$) is obtained from Z (respectively $f^{-1}(L)$) by removing an algebraic subset with empty interior. In view of normality, this does not change π_0 and can only enlarge π_1 , as is seen from the commutative diagram

$$\begin{array}{ccc} \pi_1(Z' \cap f^{-1}(L), z) & \longrightarrow & \pi_1(f^{-1}(L), z) \\ \downarrow & & \downarrow \\ \pi_1(Z', z) & \longrightarrow & \pi_1(Z, z). \end{array}$$

In particular, we may and do assume that Z is smooth.

Shrinking Z , we may also assume that $f(Z)$ is locally closed and that Z is a topologically locally trivial fibration over $f(Z)$.

If $c = 1$ and f is dominant, then $f(Z)$ is a Zariski-open subset of $\mathbb{P}^N$ —after shrinking Z we may assume it is the complement of a hypersurface—and the lemma follows from Assertion 1.1 together with the long exact homotopy sequences for $Z \rightarrow f(Z)$ and $f^{-1}(L) \rightarrow L \cap f(Z)$.

Continue to assume $c = 1$. If f is not dominant, let A be a linear subspace of dimension $N - n - 1$, and let B be the projective space of linear subspaces of dimension $N - n$ containing A . Define

$$q: \mathbb{P}^N \setminus A \rightarrow B, \quad x \mapsto \text{the linear subspace containing } A \text{ and } x.$$

If A is disjoint from $f(Z)$, the restriction of q to $f(Z)$ is finite, hence dominant because $\dim f(Z) = \dim B$, and therefore $qf: Z \rightarrow B$ is dominant. Applying the already proved case of the lemma to qf and $c = 1$, we obtain the conclusion for a general hyperplane L among those containing A . Since A itself is only required to lie in a suitable dense open subset of a Grassmannian, the conclusion also holds for a general L in the usual sense (cf. 1.2).

The proof is completed by induction on c : a general linear subspace of codimension $c - 1$ inside a general hyperplane is a general linear subspace of codimension c (again cf. 1.2).

Lemma 1.5. Let Z , f , n , and c be as in Lemma 1.4. Every linear subspace L of codimension c in $\mathbb{P}^N$ admits a fundamental system of open neighborhoods V such that $f^{-1}(V)$ is connected and, for every $z \in f^{-1}(V)$, the natural morphism

$$\pi_1(f^{-1}(V), z) \rightarrow \pi_1(Z, z)$$

is surjective.

Let Gr be the Grassmannian of linear subspaces of codimension c in $\mathbb{P}^N$, let

$$Y \subset Z \times \text{Gr}$$

be the set of pairs (z, L) such that $f(z) \in L$, and let $G \subset \text{Gr}$ be a nonempty Zariski-open subset over which Y is a topologically locally trivial fibration. By 1.2, the conclusion of Lemma 1.4 holds for all $L \in G$.

For every open subset $U \subset \text{Gr}$, write U^* for the open subset of $\mathbb{P}^N$ that is the union of the $L \in U$. We claim that if U is connected, then Lemma 1.5 holds for $V = U^*$.

Indeed, if U is connected then $f^{-1}(U^*)$ is connected. The intersection $U' = U \cap G$ is connected, since it is obtained from U by deleting a triangulable subset of real codimension at least 2. By Lemma 1.4, the union $f^{-1}(U'^*)$ of the sets $f^{-1}(L)$ for $L \in U'$ is therefore connected. We now show that it is dense in $f^{-1}(U^*)$.

The image $f(Z)$ is not contained in the complement F of G^* , since $n - c \geq 0$ implies that the members of G meet $f(Z)$. If $z \in Z$, the image of any neighborhood of z cannot be contained in F , otherwise the image of all of Z would be, by analytic continuation. Thus if $z \in f^{-1}(U^*)$, there exist points z' arbitrarily close to z such that $f(z') \in G^*$. Among the $L \in \text{Gr}$ containing z' , the intersection with G is dense, because it is nonempty. If $z \in f^{-1}(L)$ with $L \in U$, this allows one to find z' as above, moreover lying in some $f^{-1}(L')$ with $L' \in G$ arbitrarily close to L . In particular, one may choose $L' \in U'$, so $z' \in f^{-1}(U'^*)$. This proves the density of $f^{-1}(U'^*)$ in $f^{-1}(U^*)$ and hence the connectedness of $f^{-1}(U^*)$.

Finally, if $L \in U'$, then $f^{-1}(L) \subset f^{-1}(U^*) \subset Z$. By Lemma 1.4, $\pi_1(f^{-1}(L))$ maps onto $\pi_1(Z)$, and therefore so does $\pi_1(f^{-1}(U^*))$.

We now derive a variant of Theorem 3.

Theorem 1.6. Let

$$P = (\mathbb{P}^m)^r$$

be the product of r copies of projective m -space, let

$$\delta: \mathbb{P}^m \rightarrow P$$

be the diagonal map, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be normal and connected, and let

$$h: Z \rightarrow P.$$

Write $n = \dim h(Z)$ and $c = (r - 1)m$, the codimension of Δ in P . If $n - c \geq 1$, then Δ admits a fundamental system of open neighborhoods V such that $h^{-1}(V)$ is connected. Whenever $h^{-1}(V)$

is connected and $z \in h^{-1}(V)$, the natural morphism

$$\pi_1(h^{-1}(V), z) \rightarrow \pi_1(V, z)$$

is surjective.

As in Lemma 1.4, we may shrink Z , and in particular we may assume it smooth.

Choose a family of hyperplanes $(H_\alpha)_{0 \leq \alpha \leq m}$ in $\mathbb{P}^m$ such that:

- (a) their intersection is empty, so in suitable projective coordinates they are given by $X_\alpha = 0$;
- (b) $h(Z)$ is contained in none of the inverse images $\text{pr}_i^{-1}(H_\alpha)$. After shrinking Z we may and do assume that $h(Z)$ is disjoint from every $\text{pr}_i^{-1}(H_\alpha)$;
- (c) $h(Z) \cap \Delta$ is not contained in the union of the $\delta(H_\alpha)$. Since Δ belongs to a connected family of cycles filling all of P , one has $h(Z) \cap \Delta \neq \emptyset$, and one can choose the H_α so that (c) holds.

Set

$$A_\alpha = \mathbb{P}^m \setminus H_\alpha, \quad B_\alpha = A_\alpha^r \subset P, \quad \Delta_\alpha = \delta(A_\alpha),$$

the diagonal in B_α . The spaces A_α and B_α are affine spaces and Δ_α is an affine subspace of B_α . The projective completion of A_α is $\mathbb{P}^m$; denote by $\overline{B}_\alpha$ the completion of B_α , and by $\overline{\Delta}_\alpha \subset \overline{B}_\alpha$ the closure of Δ_α .

Fix α , and for simplicity write $H, A, B, \Delta_A, \overline{B}$, and $\overline{\Delta}_A$ for $H_\alpha, A_\alpha, B_\alpha, \Delta_\alpha, \overline{B}_\alpha$, and $\overline{\Delta}_\alpha$.

Lemma 1.7. For every neighborhood U of Δ in P , there exists a neighborhood V of $\overline{\Delta}_A$ in $\overline{B}$ such that

$$V \cap B \subset U \cap B.$$

Choose on A a Hermitian vector-space structure compatible with its affine structure. A fundamental system of neighborhoods of Δ (respectively of $\overline{\Delta}_A$) has, upon intersection with B , the form

$$U \cap B = \left\{ (x_i) \left| \begin{array}{l} \text{if } |x_i| \leq R \text{ for some } i, \text{ then } |x_j - x_k| < \varepsilon \forall j, k, \\ \text{if all } |x_i| \geq R, \text{ then the angle between the lines } \mathbb{C}x_j, \mathbb{C}x_k \\ \text{is } < \varepsilon \forall j, k \end{array} \right. \right\},$$

$$V \cap B = \left\{ (x_i) \left| \begin{array}{l} \text{if } |x_i| \leq R \text{ for some } i, \text{ then } |x_j - x_k| < \varepsilon \forall j, k, \\ \text{if all } |x_i| \geq R, \text{ then } |x_j - x_k| \leq \varepsilon \inf_i |x_i| \forall j, k \end{array} \right. \right\}.$$

The lemma follows immediately.

In the case $r = m = 1$, the two compactifications $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ of $\mathbb{A}^2$ are dominated by a third one obtained by blowing up two boundary points; the original paper gives a schematic figure illustrating this and indicating the neighborhoods U and V .

We now return to the proof of Theorem 1.6. Apply Lemma 1.5 to

$$h: Z \rightarrow \overline{B}$$

and to the affine diagonal $\Delta_A \subset \overline{B}$. We find that $\overline{\Delta}_A$ has arbitrarily small open neighborhoods V_α such that

$$h^{-1}(V_\alpha) = h^{-1}(V_\alpha \cap B_\alpha)$$

is connected and has fundamental group mapping onto that of Z . By Lemma 1.7, given any neighborhood U of Δ in P , one can choose each V_α so that

$$V_\alpha \cap B_\alpha \subset U.$$

Then the union V of the opens $V_\alpha \cap B_\alpha$ is a neighborhood, arbitrarily small, of Δ in P . Hypothesis (b) on the hyperplanes H_α ensures that the intersection of the sets $V_\alpha \cap B_\alpha$ meets $h(Z)$. Hence

$$h^{-1}(V),$$

being a union of connected open sets $h^{-1}(V_\alpha)$ with nonempty mutual intersection, is connected. Since the π_1 of one of the $h^{-1}(V_\alpha)$ maps onto $\pi_1(Z)$, the same is true a fortiori for $\pi_1(h^{-1}(V))$, which completes the proof.

2. PROOF OF THEOREM 1

2.1. Let $C \subset \mathbb{P}^2$ be a reduced plane curve, let D be an irreducible component of C , let $C' = (C - D)$ be the union of the other components, let $\text{Sing}(C)$ be the set of singular points of C , and let

$$D^* = D - \text{Sing}(C).$$

Choose a base point $O \in \mathbb{P}^2 - C$, and let γ be a loop in $\mathbb{P}^2 - C$ of the following form: γ goes from O to a point p' near a point $p \in D^*$ by a path ch , winds once positively around D^* , and returns to O by ch^{-1} . Since D^* is connected, all loops of this type are mutually conjugate. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the normal subgroup they generate is

$$\pi_1(\mathbb{P}^2 - C').$$

Proposition 2.2. Assume that the singular points of C lying on D are ordinary double points with distinct tangents. Then γ is central in $\pi_1(\mathbb{P}^2 - C, O)$.

Let j be the inclusion of $\mathbb{P}^2 - C$ into $\mathbb{P}^2$, let $g: \tilde{D} \rightarrow D$ be the normalization of D , and apply Theorem 1.6 to

$$h = j \times g: (\mathbb{P}^2 - C) \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2.$$

To describe conveniently the inverse image of certain neighborhoods of the diagonal, we choose a Riemannian structure on $\mathbb{P}^2$ such that, in a neighborhood of every point $P \in D \cap \text{Sing}(C)$, the surface $\mathbb{P}^2$ is Euclidean and the two branches of C through P are orthogonal real 2-planes.

If $(u, v) \in (\mathbb{P}^2 - C) \times \tilde{D}$ lies in the inverse image of a sufficiently small neighborhood of the diagonal—that is, if u and v are sufficiently close—then u lies near D , and one can drop from u a short perpendicular to D . When u lies near $\text{Sing}(D)$ there are even two such perpendiculars, one for each branch. We choose the one whose foot $\bar{u}$ is close to v on $\tilde{D}$; then $\bar{u} \in D^*$.

Conversely, let $\varepsilon > 0$ be sufficiently small, let

$$q: U^* \rightarrow D^*$$

be the punctured-disc subbundle of the normal bundle to D^* in $\mathbb{P}^2$ consisting of nonzero normal vectors of length $< \varepsilon$, and let E be the set of pairs (x, v) with $x \in U^*$, $v \in \tilde{D}$, and

$$d(q(x), v) < \varepsilon$$

(the distance being computed on $\tilde{D}$). The projection

$$\text{pr}_1: E \rightarrow U^*$$

is a homotopy equivalence, with section given by the subspace of pairs $(x, q(x))$. The map

$$\text{exp}: (x, v) \mapsto (\text{exp}(x), v)$$

from E to $(\mathbb{P}^2 - C) \times \tilde{D}$ identifies E with the inverse image of a neighborhood of the diagonal, and as $\varepsilon \rightarrow 0$ one obtains the inverse images of a fundamental system of neighborhoods. Applying Theorem 1.6, we find that for any $O \in U^*$ the maps exp and q induce a surjection

$$(\text{exp}, q)_*: \pi_1(U^*, O) \rightarrow \pi_1(\mathbb{P}^2 - C, \text{exp}(O)) \times \pi_1(\tilde{D}, q(O)).$$

There is an exact homotopy sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, O) \rightarrow \pi_1(D^*, q(O)) \rightarrow 1.$$

The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop turning once positively around D^* . Since $(\text{exp}, q)_*$, and hence exp_* , is surjective, this loop determines a central element of $\pi_1(\mathbb{P}^2 - C, \text{exp}(O))$; this is precisely γ .

2.3. Proof of Theorem 1. For every irreducible component D of C , let $\gamma(D)$ be the corresponding central element of $\pi_1(\mathbb{P}^2 - C, O)$. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the central subgroup generated by the $\gamma(D)$ is

$$\pi_1(\mathbb{P}^2, O) = \{e\}.$$

Therefore $\pi_1(\mathbb{P}^2 - C, O)$ is abelian. □

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Postscript (1 October 1980). I have just received a beautiful survey setting out the methods studied here, their variants, and their many applications:

W. Fulton and R. Lazarsfeld, *Connectivity and its applications in algebraic geometry*.

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Duality for Representations of a Reductive Group over a Finite Field

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1

The purpose of this paper is to construct a duality operation for representations of a reductive group over a finite field. Its effect, very roughly speaking, is to interchange irreducible representations of small degree with ones of large degree (for example, the unit and Steinberg representation). At the level of characters, this operation has been also considered by Alvis [1], Curtis [2], and Kawanaka [4].

We shall now fix some notation. G will denote a connected reductive group defined over a finite field $\mathbb{F}_q$, (W, S) its Weyl group, and $\tilde{S}$ the set of orbits of the Frobenius map on S . The subsets of $\tilde{S}$ parametrize the classes of parabolic subgroup of G which are defined over $\mathbb{F}_q$; let $\mathcal{P}_I$ be the class corresponding to $I \subset \tilde{S}$. (Thus $\mathcal{P}_\emptyset$ is the class of Borel subgroups and $\mathcal{P}_{\tilde{S}} = \{G\}$.) We denote by G the group of $\mathbb{F}_q$ -rational points of G and by $\mathcal{P}_I$ the set of parabolic subgroups in $\mathcal{P}_I$ which are defined over $\mathbb{F}_q$. Similarly, if $P \in \mathcal{P}_I$, we denote by P its group of $\mathbb{F}_q$ -rational points and by U_P the group of $\mathbb{F}_q$ -rational points of its unipotent radical. Let K be an algebraically closed field of characteristic zero. All G -modules will be over K .

Let E be a G -module. For each $I \subset \tilde{S}$ we define

$$E_{(I)} = \bigoplus_{P \in \mathcal{P}_I} E^{U_P},$$

where E^{U_P} denotes the set of U_P -invariant vectors of E . We regard $E_{(I)}$ as a G -module in a natural way. If $I \subset I'$, there is a canonical linear map $\varphi_I^{I'} : E_{(I)} \rightarrow E_{(I')}$. It is defined as follows: if $e \in E^{U_P}$ ($P \in \mathcal{P}_I$), e can be also regarded as an element of $E^{U_{P'}}$, where $P' \in \mathcal{P}_{I'}$ is uniquely defined by the condition $P \subset P'$. (Note that $U_{P'} \supset U_P$, hence $E^{U_P} \subset E^{U_{P'}}$.) The map $\varphi_I^{I'}$ takes $e \in E^{U_P}$ to e regarded as an element of $E^{U_{P'}}$. It is clear that if $I \subset I' \subset I''$, then

$$\varphi_{I'}^{I''} \varphi_I^{I'} = \varphi_I^{I''}. \quad (1)$$

Now let $\tilde{E}_{(I)} = E_{(I)} \otimes \Lambda^{|I|}(K^I)$. (We regard $\Lambda^{|I|}(K^I)$ as a G -module with trivial action.) If $I \subset I'$ and $|I'| = |I| + 1$, we have a natural map $\varepsilon_I^{I'}: \Lambda^{|I|}(K^I) \rightarrow \Lambda^{|I'|}(K^{I'})$ given by $\omega \mapsto \omega \wedge f$ where f is the unique element in $I' - I$. We define $\psi_I^{I'}: \tilde{E}_{(I)} \rightarrow \tilde{E}_{(I')}$ by $\psi_I^{I'} = \varphi_I^{I'} \otimes \varepsilon_I^{I'}$.

We now consider the sequence of maps of G -modules:

$$0 \rightarrow \tilde{E}_{(\emptyset)} \rightarrow \bigoplus_{|I|=1} \tilde{E}_{(I)} \rightarrow \bigoplus_{|I|=2} \tilde{E}_{(I)} \rightarrow \cdots \rightarrow \tilde{E}_{(\tilde{S})} \rightarrow 0 \quad (2)$$

where the maps d have components $\psi_I^{I'}$ ($I \subset I'$, $|I'| = |I| + 1$). Using (1), we see that (2) is a complex.

2

We have the following:

Theorem 1. *Assume that E is irreducible. Let i_0 be the smallest integer ≥ 0 such that $\bigoplus_{|I|=i_0} \tilde{E}_{(I)} \neq 0$. Then the sequence*

$$\bigoplus_{|I|=i_0} \tilde{E}_{(I)} \xrightarrow{d_{i_0}} \bigoplus_{|I|=i_0+1} \tilde{E}_{(I)} \rightarrow \cdots \rightarrow \tilde{E}_{(\tilde{S})} \rightarrow 0 \quad (3)$$

is exact.

In particular, the homology of the complex (2) is concentrated in a single degree.

3

We shall prove that the conclusion of Theorem 2 holds in the case where instead of assuming E irreducible we shall assume that:

$$E \text{ admits a direct sum decomposition } E = \bigoplus_{P \in \mathcal{P}_{I_0}} E_P \text{ (for some } I_0 \subset \tilde{S}) \quad (4)$$

such that

- (a) $gE_P = E_{gPg^{-1}}$ ($\forall g \in G, P \in \mathcal{P}_{I_0}$);
- (b) for each $P \in \mathcal{P}_{I_0}$, the P -module E_P factors through P/U_P and is irreducible, cuspidal as a P/U_P -module.

This would imply the theorem since any irreducible G -module is a direct summand of a G -module E as in (4).

In this and the following section, we assume that E is as in (4).

Given $J \subset \tilde{S}$, we define J to be the set of G -orbits σ on $\mathcal{P}_J \times \mathcal{P}_{I_0}$ which have the following property. There exists $P \in \mathcal{P}_{I_0}$, $Q \in \mathcal{P}_J$, $(Q, P) \in \sigma$ such that Q contains a Levi subgroup of P .

If $Q \in \mathcal{P}_J$, we have

$$E^{U_Q} = \left(\bigoplus_{P \in \mathcal{P}_{I_0}} E_P \right)^{U_Q} = \bigoplus_{\substack{P \in \mathcal{P}_{I_0} \\ (Q \cap P)U_P = P}} E_P^{U_Q} \oplus \bigoplus_{\substack{P \in \mathcal{P}_{I_0} \\ (Q \cap P)U_P \subsetneq P}} E_P^{U_Q}.$$

The second summand is zero (since E_P is cuspidal for P/U_P , so that $E_P^{U_P^g \cap P} = 0$). Thus

$$E_{(J)} = \bigoplus_{Q \in \mathcal{P}_J} E^{U_Q} = \bigoplus_{Q \in \mathcal{P}_J} \bigoplus_{\substack{P \in \mathcal{P}_{I_0} \\ (Q \cap P)U_P = P}} E_P^{U_Q} = \bigoplus_{\sigma \in J} E^\sigma \quad (5)$$

where

$$E^\sigma = \bigoplus_{\substack{(Q,P) \in \mathcal{P}_J \times \mathcal{P}_{I_0} \\ (Q,P) \in \sigma \\ (Q \cap P)U_P = P}} E_P^{U_Q} \quad (\sigma \in J).$$

Now assume that $J \subset J'$ and $\sigma \in J$; we define $\varphi_J^{J'}(\sigma) = \sigma' = \{(Q', P) \in \mathcal{P}_{J'} \times \mathcal{P}_{I_0} \mid \exists Q \in \mathcal{P}_J, Q \subset Q', (Q, P) \in \sigma\}$. Then $\sigma' \in J'$ and we say that σ' is the face of type J' of σ . In this case, we define a linear map $\psi_J^{J'}: E^\sigma \rightarrow E^{\sigma'}$ as follows. If $e = (e_{Q,P}) \in E^\sigma$, $e_{Q,P} \in E_P \forall (Q, P) \in \sigma$, we set $\psi_J^{J'}(e) = (e'_{Q',P}) \in E^{\sigma'}$ where, for any $(Q', P) \in \sigma'$, we have

$$e'_{Q',P} = \sum_{\substack{Q \in \mathcal{P}_J \\ Q \subset Q' \\ (Q,P) \in \sigma}} e_{Q,P} \in E_P.$$

Let us admit the following

Lemma 1. *For any σ, σ' as above, the map $\psi_J^{J'}: E^\sigma \rightarrow E^{\sigma'}$ is an isomorphism.*

Note also that given $J \subset J' \subset J''$ and $\sigma \in J, \sigma' \in J'$ as above we can define $\sigma'' \in J''$ to be the face of type J'' of σ (or of σ') and we have

$$\psi_{J'}^{J''} \psi_J^{J'} = \psi_J^{J''}.$$

Using (5), the complex (2) can be rewritten as

$$\cdots \rightarrow \bigoplus_{|J|=|I_0|+1} \bigoplus_{\sigma \in J} E^\sigma \otimes \Lambda^{|J|}(K^J) \rightarrow \bigoplus_{|J|=|I_0|} \bigoplus_{\sigma \in J} E^\sigma \otimes \Lambda^{|J|}(K^J) \rightarrow \cdots \quad (6)$$

where the maps d have components $\psi_J^{J'} \otimes \varepsilon_J^{J'}$ ($J \subset J', |J'| = |J| + 1, \sigma \in J, \sigma' \in J'$ the face of type J' of σ). Let ρ be the unique element in $\tilde{s}$. We have $E^\rho = E$. Using Lemma 1, we may identify E^σ with E for any $\sigma \in J$, via the isomorphism ψ_J^ρ ; the complex (6) becomes the tensor product of E with the complex

$$0 \rightarrow \cdots \rightarrow \bigoplus_{|J|=|I_0|+1} K^J \otimes \Lambda^{|J|}(K^J) \rightarrow \bigoplus_{|J|=|I_0|} K^J \otimes \Lambda^{|J|}(K^J) \rightarrow \cdots \quad (7)$$

whose differential has components $\psi_J^{J'} \otimes \varepsilon_J^{J'}$ ($J \subset J', |J'| = |J| + 1$). This complex has the following simple interpretation. Let T be an $\mathbb{F}_q$ -split torus of maximal possible dimension in the adjoint group G_{ad} . Let Y be the lattice of one-parameter subgroups of T . The root hyperplanes in the real vector space $Y \otimes \mathbb{R}$ give rise to a partition of $Y \otimes \mathbb{R}$ into simplicial cones. These are in a natural 1–1 correspondence with the parabolic subgroups in G which are defined over $\mathbb{F}_q$ and whose image in G_{ad} contain T . Let us fix $P_0 \in \mathcal{P}_{I_0}$. It corresponds to a cone in $Y \otimes \mathbb{R}$, which spans a linear subspace $L \subset Y \otimes \mathbb{R}$ of codimension $|I_0|$. The cones contained in L are in 1–1 correspondence with those parabolic subgroups $Q \subset G$ (defined over $\mathbb{F}_q$) which contain a Levi subgroup of P_0 and whose image in G_{ad} contain T , hence they are in 1–1 correspondence with $\bigcup_{J \supset I_0} J$. Thus $\bigcup_{J \supset I_0} J$ can be regarded as the set of simplices in a triangulation of a unit sphere with centre 0 in L ; and (7) is the (reduced) chain complex of this triangulation. It follows that its homology is concentrated in a single degree (corresponding to $|J| = |I_0|$) where it is $\pm K$.

4

It remains to prove Lemma 1. We will give the proof in six steps. Parts of the argument are similar in spirit to arguments in [3].

Step 1. Given $\sigma \in_J$ and $(Q, P) \in \sigma$, we consider the parabolic subgroup $R = (Q \cap P)U_Q \subset Q$. Then $R \in \mathcal{P}_{\tilde{J}} \in_{\tilde{J}}$ for some $\tilde{J} \subset J$ and σ is the face of type J of some element of $\tilde{J}$. Also R, P have a common Levi subgroup and hence are associated. We first show that

$$\varphi: \bigoplus_{\substack{R \in \mathcal{P}_{\tilde{J}} \\ (R, P) \in \tilde{J} \\ R \subset Q}} E_P^{U_R} \longrightarrow E_P^{U_Q}$$

is an isomorphism.

Now the pairs (R, P) such that $(R, P) \in \tilde{J}$ are in 1–1 correspondence with the pairs (Q, P) such that $(Q, P) \in \sigma$. Hence φ is injective. If we fix $Q \in \mathcal{P}_J$, there is a natural 1–1 correspondence between the set $\{R \in \mathcal{P}_{\tilde{J}} \mid R \subset Q\}$ and the set of orbits of U_Q on the set $\{P \in \mathcal{P}_{I_0} \mid (Q, P) \in \sigma\}$. Hence the space E^σ is a direct sum over all pairs $R \subset Q$ ($R \in \mathcal{P}_{\tilde{J}}, Q \in \mathcal{P}_J$) of pieces of form $(\bigoplus_{P \in B} E_P)^{U_Q}$, where P runs through a fixed U_Q -orbit. But this piece has the same dimension as E_P . (Note that for P in that U_Q -orbit we have $P \cap U_Q \subset U_P$ and U_P acts trivially on E_P .) Thus $\dim E^\sigma = \dim E_P \cdot \#(\mathcal{P}_J) = \dim E_{\rho} \cdot \#(\mathcal{P}_{I_0}) = \dim E$. Similarly, $\dim E^{\sigma'} = \dim E$. It follows that φ is an isomorphism.

Step 2. Assume that $|I_0| = |\tilde{S}| - 1$. Let $\mathcal{P}_J$ be the class of parabolic subgroups opposed to $\mathcal{P}_{I_0}$, and let $\sigma = \{(Q, P) \in \mathcal{P}_J \times \mathcal{P}_{I_0} \mid Q \text{ opposed to } P\}$. We will show that $\psi_J^{\tilde{S}}: E^\sigma \rightarrow E^\rho = E$ is an isomorphism. It is certainly non-zero. Thus, if E is irreducible, $\psi_J^{\tilde{S}}$ is indeed an isomorphism. If E is reducible, it has exactly two composition factors, and J is necessarily equal to I_0 . For each $P \in \mathcal{P}_{I_0}$, there exists an isomorphism

$$\eta_P: E_P \xrightarrow{\sim} \bigoplus_{(P', P) \in \sigma} E_{P'}^{U_P}$$

of P/U_P -modules unique up to a scalar. We may assume that $\eta_{gPg^{-1}} = g\eta_P g^{-1}$ ($\forall P \in \mathcal{P}_{I_0}, \forall g \in G$). These maps together define an isomorphism of G -modules $\eta = \bigoplus_{P \in \mathcal{P}_{I_0}} \eta_P$:

$$E = \bigoplus_{P \in \mathcal{P}_{I_0}} E_P \xrightarrow{\sim} \left(\bigoplus_{(P', P) \in \sigma} E_{P'} \right)^{U_P}.$$

The composition $\psi_J^{\tilde{S}} \circ \eta$ is a semisimple G -endomorphism of E . It clearly has trace zero (its diagonal blocks are zero). Let λ', λ'' be its eigenvalues on the two irreducible pieces E', E'' of E . Then $\lambda' \dim E' + \lambda'' \dim E'' = 0$. Since $\dim E' \neq 0, \dim E'' \neq 0$, λ' and λ'' are either both zero or both non-zero. They cannot be both zero since $\psi_J^{\tilde{S}} \circ \eta \neq 0$. Hence they are both non-zero. It follows that $\psi_J^{\tilde{S}} \circ \eta$ and hence $\psi_J^{\tilde{S}}$ are isomorphisms.

Step 3. Assume that $|I_0| < |\tilde{S}| - 1$. Let $J \subset \tilde{S}$ be such that $|J| = |I_0| + 1$, and let $\tau \in_J$. There are precisely two elements $\sigma' \in_{J'}$, $\sigma'' \in_{J''}$ such that $J', J'' \subset J, |J'| = |J''| = |I_0|$, and such that τ is the J -face of σ' and the J -face of σ'' . We shall prove that $\psi_{I_0}^{J'}$ and $\psi_{I_0}^{J''}$ are isomorphisms. One of these maps, say $\psi_{I_0}^{J'}$, is of the type considered in Step 1 ($\sigma = \tau$); hence it is known to be an isomorphism. We now turn to $\psi_{I_0}^{J''}$. Define

$$E' = \bigoplus_{R \in \mathcal{P}_J} E'_R$$

where, by definition,

$$E'_R = \left(\bigoplus_{\substack{P \in \mathcal{P}_{I_0} \\ (R \cap P) U_P = P}} E_P \right)^{U_R}.$$

Then E' is a G -module in a natural way. It satisfies a property similar to property (4) for E . Let α be the set of pairs $(R'', R') \in \mathcal{P}_{J''} \times \mathcal{P}_{J'}$ such that R'', R' are both contained in the same $Q \in \mathcal{P}_J$ and are opposed in Q . We define E'^α and $\psi_{J''}^{J'}: E'^\alpha \rightarrow E'$ in the same way as E^σ and $\psi_{J'}^{J'}: E^\sigma \rightarrow E^{\sigma'}$. We have a natural commutative diagram

$$\begin{array}{ccc} E'^\alpha & \xrightarrow{\psi_{J''}^{J'}} & E' (= E^{\sigma'}) \\ \downarrow \cong & & \uparrow \psi_{I_0}^{J'} \\ E^{\sigma''} & \xrightarrow{\psi_{I_0}^{J''}} & E^\sigma (= E^\tau) \end{array}$$

(Note that, given $(R'', P) \in \sigma''$ there is a unique $R' \in \mathcal{P}_{J'}$ such that $(R'', R') \in \alpha$, $(R', P) \in \sigma'$; conversely, if $(R'', R') \in \alpha$, $(R', P) \in \sigma'$ then $(R'', P) \in \sigma''$. This gives rise to the isomorphism $E'^\alpha \xrightarrow{\sim} E^{\sigma''}$.) Now $\psi_{J''}^{J'}$ is known to be an isomorphism; moreover, by Step 2, $\psi_{J''}^{J'}$ is an isomorphism. It follows that $\psi_{I_0}^{J''}$ is an isomorphism.

Step 4. Let $\sigma \in_J$, $|J| = |I_0|$. (Thus σ corresponds to an open cone in L .) We show that $\psi_{I_0}^J: E^\sigma \rightarrow E$ is an isomorphism. We can find a sequence $\sigma = \sigma_1, \sigma_2, \dots, \sigma_n = \rho$ of elements corresponding to open cones in L such that (when regarded as cones in L) two consecutive ones have a common face of codimension 1; moreover, we take n as small as possible. Let τ be the element corresponding to the common face of codimension 1 of $\sigma = \sigma_1$ and σ_2 . By induction on n , we may assume that $\psi_{I_0}^{J_2}: E^{\sigma_2} \rightarrow E$ is an isomorphism. From Step 3, we know that $\psi_{I_0}^J$ and $\psi_{J_2}^{J_2}$ are isomorphisms. From $\psi_{I_0}^{J_2} = \psi_{J_2}^{J_2} \circ \psi_{I_0}^J$, it follows that $\psi_{I_0}^J$ is an isomorphism. From $\psi_{I_0}^{\tilde{S}} = \psi_{J_2}^{\tilde{S}} \circ \psi_{I_0}^J$ it follows that $\psi_{I_0}^{\tilde{S}}$ is an isomorphism.

Step 5. Let $J \subset \tilde{S}$ and $\sigma \in_J$. We want to show that $\psi_{I_0}^J$ is an isomorphism. We can find $I \subset J$ and $\delta \in_I$ such that $|I| = |I_0|$, such that σ is the face of type J of δ , and such that $\psi_{I_0}^I$ is of the type considered in Step 4 (hence an isomorphism). We have $\psi_I^J \circ \psi_{I_0}^I = \psi_{I_0}^J$. By Step 4, $\psi_{I_0}^I$ is an isomorphism. It follows that $\psi_{I_0}^J$ is an isomorphism.

Step 6. Let $J \subset J'$ and $\sigma \in_J$; let $\sigma' \in_{J'}$ be the face of type J' of σ . We want to show that $\psi_{J'}^{J'}$ is an isomorphism. We have $\psi_{I_0}^{J'} \circ \psi_{J'}^J = \psi_{I_0}^J$. By Step 5, $\psi_{I_0}^J$ and $\psi_{I_0}^{J'}$ are isomorphisms. It follows that $\psi_{J'}^{J'}$ is an isomorphism.

This completes the proof of the Lemma 1 and hence that of the theorem.

5

We have

Corollary 1 (see Alvis [1]). *In the setting of the theorem, let $E^\#$ denote the kernel of d_{i_0} in (3). Then*

- (a) $E^\#$ is an irreducible G -module,
- (b) $(E^\#)^\#$ is isomorphic to E as a G -module,
- (c) $(-1)^{i_0} E^\# = \sum_{I \subset \tilde{S}} (-1)^{|I|} E_{(I)}$ in the Grothendieck group of virtual G -modules.

Statement (c) is obvious.

Let E' be a G -module of the type considered in (4) which has a direct sum decomposition $E' = E_1 \oplus \cdots \oplus E_n$ with E_i irreducible G -modules and $E_1 \cong E$. The proof of Theorem 2 shows that $(E')^\#$ can be defined in the same way as $E^\#$, and that it has the following properties:

$$(E')^\# \cong E_1^\# \oplus \cdots \oplus E_n^\#, \quad E'^\# \subset E'. \quad (8)$$

For each $1 \leq i, j \leq n$, we have from (c)

$$\langle E_i^\#, E_j^\# \rangle = \langle (E_i^\#)^\#, E_j \rangle = \sum_{I \subset \tilde{S}} (-1)^{|I|+i_0} \langle E_{i(I)}, E_{j(I)} \rangle \quad (9)$$

(where $P_I \in \mathcal{P}_I$). It follows that

$$\langle E_i^\#, E' \rangle = \sum_j \langle E_i^\#, E_j^\# \rangle = \sum_j \langle E_i, E_j^\# \rangle = \langle E_i, (E')^\# \rangle = \langle E_i, E' \rangle$$

and, in particular, $E_i^\# \neq 0$. If $E_i^\#$ is not irreducible for some i , then $E_1^\# \oplus \cdots \oplus E_n^\#$ would have more irreducible components than $E_1 \oplus \cdots \oplus E_n$, contradicting (8). Thus, $E_i^\#$ is irreducible for each i . Hence, there exists a permutation π of $\{1, 2, \dots, n\}$ such that $E_i^\# = E_{\pi(i)}$ for all i . From (9), we have

$$\langle E_{\pi^{-1}(i)}, E_i \rangle = \langle (E_i^\#)^\#, E_i \rangle = \langle E_i^{\#\#}, E_i^\# \rangle = \langle E_{\pi(i)}, E_{\pi(i)} \rangle = 1.$$

It follows that $E_{\pi^{-1}(i)}$ is isomorphic to E_i . The corollary is proved.

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Duality for Representations of a Reductive Group over a Finite Field, II

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This paper is a continuation of [6] whose notations and references we shall preserve.

6

Let l be a prime number, $(l, q) = 1$, and let $\bar{\mathbb{Q}}_l$ be an algebraic closure of $\mathbb{Q}_l$. Let $\mathcal{R}(G)$ be the Grothendieck group of virtual G -modules over $\bar{\mathbb{Q}}_l$. We denote $D_G: \mathcal{R}(G) \rightarrow \mathcal{R}(G)$ the homomorphism such that

$$D_G(E) = \sum_i (-1)^i E_i,$$

for any finite-dimensional G -module E over $\bar{\mathbb{Q}}_l$.

Let T be a maximal torus of G , defined over $\mathbb{F}_q$ and let T be the group of its $\mathbb{F}_q$ -rational points. For any one dimensional T -module θ (over $\bar{\mathbb{Q}}_l$) we have defined in [5, 1.20] a virtual representation $R_T^G(\theta)$. Let $\sigma(G)$ be the $\mathbb{F}_q$ -rank of G .

We shall prove the following:

Theorem 0.1.

$$D_G(R_T^G(\theta)) = (-1)^{\sigma(G) - \sigma(T)} R_T^G(\theta).$$

Let us recall the definition of $R_T^G(\theta)$ or, more generally, that of “twisted induction” from Levi subgroups (cf. [5, 7]).

Let P be a parabolic subgroup of G , with unipotent radical U and let L be a Levi subgroup of P . We assume that L is defined over $\mathbb{F}_q$, but we make no assumption on P . Let L be the group of $\mathbb{F}_q$ -rational points of L . Let $F: G \rightarrow G$ be the Frobenius map corresponding to the $\mathbb{F}_q$ -rational structure of G . The variety

$$X_{L,P} = \{g \in G \mid g^{-1}F(g) \in P\}$$

has a $G \times L$ action

$$(g_0, l): g \mapsto g_0 g l^{-1}.$$

Hence we have a $G \times L$ -action $((g_0, l)^*)^{-1}$ of $G \times L$ on the cohomology groups $H_c^i(X_{L,P})$. (Here $H_c^i(\)$ denotes étale cohomology with compact support with coefficients in $\bar{\mathbb{Q}}_l$.) Let π be a finite-dimensional L -module over $\bar{\mathbb{Q}}_l$. Then $(H_c^i(X_{L,P}) \otimes \pi)_L$, with L acting diagonally, is in a natural way a G -module. By definition,

$$R_{L \subset P}^G(\pi) = \sum_i (-1)^i (H_c^i(X_{L,P}) \otimes \pi)_L \in \mathcal{R}(G).$$

In particular, if B is a Borel subgroup containing T , with unipotent radical V , the virtual representation $R_{T \subset B}^G(\theta)$ is defined. It is independent of B [5, 4.3] and is, by definition, $R_T^G(\theta)$.

Let $\langle \ , \ \rangle_G: \mathcal{R}(G) \times \mathcal{R}(G) \rightarrow \mathbb{Z}$ be the bilinear form defined by $\langle E, E' \rangle_G = \dim \text{Hom}_G(E, E')$ for any two finite-dimensional G -modules E, E' over $\bar{\mathbb{Q}}_l$.

The following result is a generalization of the orthogonality formula [5, 6.8].

Theorem 0.2.

$$\langle R_T^G(\theta), R_{L \subset P}^G(\pi) \rangle_G = \sum_{\substack{n \in G \\ n^{-1}Tn \subset L}} \langle R_{n^{-1}Tn \cap L}^L({}^n\theta), \pi \rangle_L,$$

where, for n in the sum, ${}^n\theta$ denotes the representation of $n^{-1}Tn$ defined by ${}^n\theta(n^{-1}tn) = \theta(t)$, $t \in T$.

7

We shall now give the proof of Theorem 6 assuming Theorem 7. If $G = T$, the theorem is obvious. Assume now that $G \neq T$ and that the theorem is already proved for groups of dimension $< \dim G$. We shall distinguish two cases.

Case 1. T is contained in a proper parabolic subgroup P of G , defined over $\mathbb{F}_q$. Let L be a Levi subgroup of P , containing T and defined over $\mathbb{F}_q$. Let L, P denote the groups of $\mathbb{F}_q$ -rational points of L, P . According to [5, 8.2] we have $R_T^G(\theta) = \text{Ind}(R_T^L(\theta))$, where $\text{Ind}: \mathcal{R}(L) \rightarrow \mathcal{R}(G)$ is the homomorphism defined by lifting a representation of L to P and then inducing it from P to G . According to [1], we have $\text{Ind} \circ D_L = D_G \circ \text{Ind}$. Since the theorem is assumed to be true for L , we have

$$\begin{aligned} D_G(R_T^G(\theta)) &= D_G \text{Ind}(R_T^L(\theta)) = \text{Ind}(D_L(R_T^L(\theta))) \\ &= \text{Ind}((-1)^{\sigma(L) - \sigma(T)} R_T^L(\theta)) = (-1)^{\sigma(G) - \sigma(T)} R_T^G(\theta), \end{aligned}$$

since $\sigma(G) = \sigma(L)$.

Case 2. For any proper parabolic subgroup P defined over $\mathbb{F}_q$, we have $T \not\subset P$.

Given such P , let L be a Levi subgroup of P defined over $\mathbb{F}_q$ and let P, L, Ind be defined as in Case 1. Our hypothesis shows that the set $\{n \in G \mid n^{-1}Tn \subset L\}$ is empty. By Theorem 7, we then have $\langle R_{L \subset P}^G(\pi), R_T^G(\theta) \rangle_G = 0$, for any L -module π of finite dimension. An argument entirely similar to that in the proof of [5, 8.2] shows that $R_{L \subset P}^G(\pi) = \text{Ind } \pi$. Thus we have

$$\langle \text{Ind } \pi, R_T^G(\theta) \rangle_G = 0$$

for any L -module π of finite dimension or, equivalently,

$$(R_T^G(\theta))^U = 0,$$

where $M \mapsto M^U$ is the homomorphism $\mathcal{R}(G) \rightarrow \mathcal{R}(L)$ such that, whenever M is a G -module, M^U is the space of vectors of M , invariant under U ($=$ the unipotent radical of P). From the definition of D_G , we see then that $D_G(R_T^G(\theta)) = (-1)^{\sigma(G)} R_T^G(\theta)$. It remains to use the fact that, for our T , we have $\sigma(G) - \sigma(T) \equiv 0 \pmod{2}$.

8

9

Let X be a scheme of finite type over an algebraic closure of $\mathbb{F}_q$ and let $f: X \rightarrow X$ be a morphism. We set

$$\mathcal{P}(f, X) = \sum_i (-1)^i \text{Tr}(f^*, H_c^i(X)).$$

In the proof of Theorem 7, we shall need the following:

Lemma 0.1. *Assume that f is an automorphism (of finite order) of X , and let T be a torus acting on X such that f commutes with each element of T . Then $\mathcal{P}(f, X) = \mathcal{P}(f, X^T)$, where X^T is the fixed point set of T on X .*

Proof. Write $f = f_u \cdot f_s = f_s \cdot f_u$, where f_s (resp. f_u) is a power of f and has order prime to p (resp. power of p). According to [5, 3.2], we have $\mathcal{P}(f, X) = \mathcal{P}(f_u, X^{f_s})$, $\mathcal{P}(f, X^T) = \mathcal{P}(f_u, (X^T)^{f_s}) = \mathcal{P}(f_u, (X^{f_s})^T)$. Thus, we are reduced to the case where f has order a power of p . We can find an element $t \in T$ such that $X^t = X^T$. Assume that we know that

$$\mathcal{P}(f, X) = \mathcal{P}(ft, X). \quad (1)$$

Using again [5, 3.2], we have $\mathcal{P}(ft, X) = \mathcal{P}(f, X^t) = \mathcal{P}(f, X^T)$; hence, by (1), we have $\mathcal{P}(f, X) = \mathcal{P}(f, X^T)$, as desired. It remains to prove (1).

We choose $\mathbb{F}_q$ -rational structures on T and X with Frobenius maps $F: T \rightarrow T$, $F: X \rightarrow X$, such that $F(t'x) = F(t')F(x)$, for all $t' \in T$, $x \in X$, $F(f) = f$ and $F(tx) = tF(x)$ for all $x \in X$. Using the trace formula for Frobenius maps, we see that (1) would be a consequence of

$$|X^{F^n f}| = |X^{F^n ft}| \quad \text{for all integers } n \geq 1. \quad (2)$$

Given $n \geq 1$, we can find $t_0 \in T$ such that $t_0^{-1} F^n(t_0) = t$. Then the map $x \mapsto t_0^{-1} x$ is a bijection $X^{F^n f} \xrightarrow{\sim} X^{F^n ft}$. This completes the proof of the lemma. $\square$

10

We shall now prove Theorem 7. As in the proof of [7, Theorem 8] we see that the left-hand side of (7.1) is equal to

$$|T|^{-1} |L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j(S)) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi), \quad (3)$$

where

$$S = \{(x, x', y) \in V \times U \times G \mid xF(y) = yx'\}$$

and $T \times L$ acts on S by

$$(t, l): (x, x', y) \mapsto (txt^{-1}, lx'l^{-1}, tyl^{-1}).$$

Let $C = \{n \in G \mid n^{-1}Tn \subset L\}$ and let $C_1, C_2, \dots, C_m$ be the orbits of L acting on C by right translation. For each i , $1 \leq i \leq m$, let $G_i = F^{-1}(V)C_iF^{-1}(P) = F^{-1}(V)C_iF^{-1}(U)$. Then G_i ($1 \leq i \leq m$) are the double cosets of G with respect to $F^{-1}(B)$ and $F^{-1}(P)$.

We set $S_i = \{(x, x', y) \in S \mid y \in G_i\}$. The S_i form a partition of S , stable under the action of $T \times L$. Corresponding to this partition, we have a decomposition of expression (3) into the sum of m terms, the terms being obtained from (3) by replacing S by S_i ($1 \leq i \leq m$). Let (3)(i) be the term corresponding to i . Let

$$\tilde{S}_i = \{(x, x', v, n, u) \in V \times U \times F^{-1}(V) \times C_i \times F^{-1}(U) \mid xF(vnu) = vlux'\}.$$

The map $\tilde{S}_i \rightarrow S_i$ defined by $(x, x', v, n, u) \mapsto (x, x', vnu)$ is a locally trivial fibration all of whose fibres are affine spaces of fixed dimension. Moreover, this map is $T \times L$ -equivariant, for the action of $T \times L$ on $\tilde{S}_i$ given by

$$(t, l): (x, x', v, n, u) \mapsto (txt^{-1}, lx'l^{-1}, tvnt^{-1}, lnl^{-1}, lul^{-1}). \quad (4)$$

It follows that expression (3)(i) is equal to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j(\tilde{S}_i)) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi). \quad (5)$$

We now make the change of variable $xF(v) \mapsto x$, $x'F(u)^{-1} \mapsto x'$. Then $\tilde{S}_i$ becomes

$$\{(x, x', v, n, u) \in V \times U \times F^{-1}(V) \times C_i \times F^{-1}(U) \mid xF(n) = vnx'\}.$$

The action of $T \times L$ is given by the same formula (4) as in the old coordinates. Next, we note that the same formula (4) defines an action of the group

$$H_i = \{(t, l) \in Z \times Z' \mid Z^{-1}F(l) = F(n_i)^{-1}t^{-1}F(t)F(n_i)t\}$$

on $\tilde{S}_i$, where Z (resp. Z') is the identity component of the centre of $n_iLn_i^{-1}$ (resp. of L) and n_i is any element of C_i . (Clearly, H_i does not depend on the choice of n_i in C_i .)

We shall now determine the fixed point set $(\tilde{S}_i)^{H_i^0}$ of the torus H_i^0 (= identity component of H_i) on $\tilde{S}_i$.

By Lang's theorem, the homomorphism $H_i \rightarrow Z$, $((t, l) \mapsto t)$ is onto, hence its restriction $H_i^0 \rightarrow Z$ must be also onto since Z is connected. Similarly, the homomorphism $H_i^0 \rightarrow Z'$, $((t, l) \mapsto l)$ is onto. Therefore, if $(x, x', v, n, u) \in (\tilde{S}_i)^{H_i^0}$, we must have

$$\begin{aligned} txt^{-1} &= x, & tvnt^{-1} &= v & \text{for all } t \in Z, \\ lx'l^{-1} &= x', & lul^{-1} &= u & \text{for all } l \in Z'. \end{aligned}$$

Since L is the centralizer of Z' and $n_iLn_i^{-1}$ is the centralizer of Z , it follows that $x, v \in V \cap n_iLn_i^{-1}$ and $x', u \in U \cap L = \{1\}$. We then have $xF(n) = vn$, hence $F(n) = x^{-1}vn \in n_iLn_i^{-1}n$. Since $n \in C_i$, we have $F(n) \in n_iL$, hence $F(C_i) = C_i$. In particular, $(\tilde{S}_i)^{H_i^0}$ is empty, unless $F(C_i) = C_i$. Assume now that $F(C_i) = C_i$. Then $n_i \in C_i$ may be chosen so

that $n_i \in G^F$. The equation defining H_i may now be written: $F(ln_i^{-1}t^{-1}n_i) = ln_i^{-1}t^{-1}n_i$. It follows that

$$H_i^0 = \{(t, l) \in Z \times Z' \mid l = n_i^{-1}tn_i\}$$

and

$$(\tilde{S}_i)^{H_i^0} = \{(x, 1, v, n_i, 1) \mid x, v \in V \cap n_iLn_i^{-1}, y \in L, y^{-1}F(y) = n_i^{-1}v^{-1}xn_i\}.$$

Since the action of $T \times L$ on $\tilde{S}_i$ commutes with the action of the torus H_i^0 , expression (5) is equal, by Lemma 9, to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j((\tilde{S}_i)^{H_i^0})) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi).$$

In the case where $F(C_i) = C_i$, the map $(\tilde{S}_i)^{H_i^0} \rightarrow \{y \in L \mid y^{-1}F(y) \in n_i^{-1}Vn_i\}$ given by $(x, 1, v, n_iy^{-1}, 1) \mapsto y$ is a locally trivial fibration all of whose fibres are affine spaces of the same dimension. The action of (t, l) on $(\tilde{S}_i)^{H_i^0}$ corresponds to the action $y \mapsto lyn_i^{-1}tn_i$. Hence expression (3)(i) is equal to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0 \\ F(C_i)=C_i}} (-1)^j \text{Tr}((t, l), H_c^j(\{y \in L \mid y^{-1}F(y) \in n_i^{-1}Vn_i\})) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi)$$

which is clearly equal to the right hand side of (7.1). This completes the proof of Theorem 6.

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DIVISION ALGEBRAS AND THE HAUSDORFF-BANACH-TARSKI PARADOX

by Pierre DELIGNE and Dennis SULLIVAN

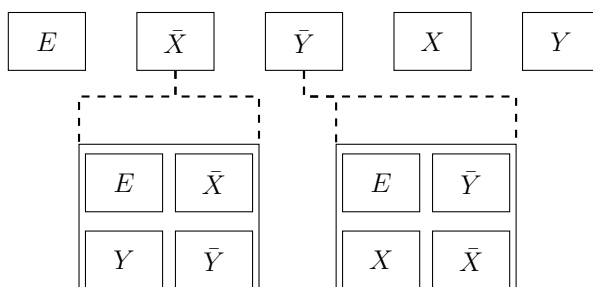
In this note we observe that a question raised by Dekker (1956) about rotations inspired by the Hausdorff-Banach-Tarski paradox can be answered using algebraic number theory. For motivation, we recall a form of the paradox.

Partition the free group in two generators F into the five sets $E, X, \bar{X}, Y, \bar{Y}$ consisting respectively of the identity and of the elements which, when written in reduced form, begin with x, x^{-1}, y or y^{-1} . If F acts freely on a sphere S , by rigid rotations, and if, using the axiom of choice, we choose a transversal T , i.e. a set with exactly one point in each orbit, the five subsets $T, XT, \bar{X}T, YT, \bar{Y}T$ form a partition of S . For economy of notation, we will write again $E, X, \bar{X}, Y, \bar{Y}$ for $T, XT, \bar{X}T, YT, \bar{Y}T$. The rotation by x moves $\bar{X}$ onto

$$S - X = E \cup \bar{X} \cup Y \cup \bar{Y}.$$

Similarly, y moves $\bar{Y}$ onto

$$S - Y = E \cup X \cup \bar{X} \cup \bar{Y}.$$



Thus we can reassemble from these 11 (actually 5) pieces 2 congruent spheres plus one congruent copy of the set E . This is a form of the Hausdorff-Banach-Tarski paradox which comes quickly from a free action of a free group on two generators (see Appendix A for the more precise form). Thus we have the question of Dekker (communicated by Jan Mycielski): do such actions really exist? The sphere must be odd-dimensional for topological reasons: a fixed point free map $f : S^d \rightarrow S^d$ must have vanishing Lefschetz number

$$L(f) = \sum (-1)^i \text{Trace}(f; H_i(S^d)) = 1 + (-1)^d \deg(f).$$

For a free action of a free group, the square of a non-trivial element will act by a map of degree $+1$ and this forces d odd. The dimension must be > 1 . These are the only conditions.

Theorem 1. *For $n \geq 2$, there is a free non-abelian group of rigid rotations acting freely on the odd dimensional sphere S^{2n-1} .*

Remark. The corresponding orthogonal matrices can be chosen to have algebraic entries, and the group of matrices corresponds to a subgroup of the non-zero elements in a division algebra over a number field.

Remark. The theorem was proved by Dekker for n even [D].

Remark. Let $u : O(p) \times O(q) \rightarrow O(p+q)$ be the natural embedding:

$$u(A, B) = \text{matrix} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

If $A \in O(p)$ and $B \in O(q)$ don't have any nonzero fixed vector, neither has $u(A, B)$. If $\sigma_1 : F \rightarrow O(p)$ and $\sigma_2 : F \rightarrow O(q)$ define free actions on S^{p-1} and S^{q-1} , $u(\sigma_1, \sigma_2)$ hence defines a free action of F on S^{p+q-1} . Using this remark, one could reduce the theorem to the two particular cases $n = 2$ and $n = 3$.

Proof. Let $k \subset \mathbf{R}$ be a real algebraic number field, and $k' \subset \mathbf{C}$ be a quadratic extension of k . We assume that $k' \not\subset \mathbf{R}$, i.e. that $k' \otimes_k \mathbf{R} = \mathbf{C}$. Let D be a division algebra of dimension n^2 over its center k' , equipped with an anti-involution $*$ inducing on k' the complex conjugation. The $\mathbf{R}$ -algebra $D \otimes_k \mathbf{R}$ is a simple algebra over its center $k' \otimes_k \mathbf{R} = \mathbf{C}$, hence isomorphic to $M(n, \mathbf{C})$. We assume that, for a suitable isomorphism between $D \otimes_k \mathbf{R}$ and $M(n, \mathbf{C})$, $*$ becomes transpose conjugate.

In term of an isomorphism as above, the elements u of D satisfying $uu^* = 1$ become unitary matrices. They operate on the unit sphere in $\mathbf{C}^n$. Furthermore, if $u \neq 1$, $u - 1$ is invertible in D so that the corresponding matrix does not have 1 as an eigenvalue. It hence acts without fixed point on the sphere.

The group $\Gamma := \{u \in D \mid uu^* = 1\}$ is the group of k -rational points of a k -form of the real algebraic group $U(n)$. For $n > 1$, the perfect subgroup $SU(n)$ of $U(n)$ is not trivial, and $U(n)$ is not solvable. The group Γ is dense in $U(n)$: skew adjoint elements of D are dense in the skew adjoint matrices in $M(n, \mathbf{C})$, and the Cayley transform $t \mapsto \frac{t-1}{t+1}$ is an homeomorphism from the space of skew-adjoint matrices in $M(n, \mathbf{C})$ to an open dense subset of $U(n)$, carrying skew adjoint elements of D into Γ . From this density, it results that, if $n > 1$, the linear group Γ is not solvable. By [Tits], it contains a non abelian free subgroup.

It remains to construct pairs $(D, *)$. A division algebra D with center k' admits an anti-involution $*$ inducing on k' the non trivial element $\text{Gal}(k'/k)$, if and only if its class $\text{cl}(D)$ in the Brauer group $\text{Br}(k')$ of k' has a trivial image by the norm map $N_{k'/k} : \text{Br}(k') \rightarrow \text{Br}(k)$ —see Appendix B. Class field theory provides an explicit computation of $\text{Br}(k)$, and of $N_{k'/k}$, and tells which elements of $\text{Br}(k')$ come from division algebras. From the explicit description it provides, existence of such D follows. A direct construction is given in Appendix C. When we choose an isomorphism of $D \otimes_k \mathbf{R}$ with $M(n, \mathbf{C})$, the involution $*$ becomes adjunction with respect to some hermitian form ϕ on $\mathbf{C}^n$, not necessarily positive definite: $\phi(ax, y) = \phi(x, a^*y)$. If h is self adjoint in D , $\text{int}(h^{-1}) \circ *$ is adjunction, with respect to the form $\phi_h(x, y) = \phi(hx, y)$. For suitable h , ϕ_h is positive definite and $(D, \text{int}(h^{-1}) \circ *)$ is of the type sought.

Appendix A

Consider $\phi : S' \cup S'' \rightarrow S - E$ as in the introduction, with S' and S'' two copies of the sphere S , and $\psi : S \rightarrow S'$ the obvious bijection. Consider as in the Schröder-Bernstein theorem the set S_e of points p in S with an even number of ancestors, namely for which there exists an integer $n \geq 0$ with $p \in \text{Im}(\phi \circ \psi)^n$ and $p \notin \text{Im}(\psi \circ (\phi \circ \psi)^n)$. Consider also the set S_o of those p in S for which there exists $n \geq 0$ with $p \in \text{Im}(\psi \circ (\phi \circ \psi)^n)$ and $p \notin \text{Im}(\phi \circ \psi)^{n+1}$, and finally the set S_∞ of those p such that $p \in \text{Im}(\phi \circ \psi)^n$ for any $n \geq 0$. Consider similarly

$$S' \cup S'' = (S' \cup S'')_e \cup (S' \cup S'')_o \cup (S' \cup S'')_\infty.$$

Then ψ induces a bijection from $S_e \cup S_\infty$ onto $(S' \cup S'')_o \cup (S' \cup S'')_\infty$ and ϕ^{-1} from S_o onto $(S' \cup S'')_e$. Combining these two we have a bijection $\chi : S \rightarrow S' \cup S''$ and a partition of S into finitely many pieces, the restriction of χ to each of these being a rotation.

Appendix B

Let K be a separable quadratic extension of a field k . We denote $x \mapsto \bar{x}$ the non trivial element $\text{Gal}(K/k)$. Let D be a simple algebra with dimension n^2 over its center K . We will check the criterion of the text, for the existence of an involution of the second kind on D , i.e. of an anti-involution $*$ of D , inducing $x \mapsto \bar{x}$ on K . The criterion is that $N_{K/k} \text{cl}(D) = 0$ in $\text{Br}(k)$.

Let us localize, for the étale topology, over $\text{Spec}(k)$. This means making large enough étale extensions of scalars $\otimes_k k'$, and keeping track of the functoriality in k' . The field K becomes the separable quadratic extension $K' = K \otimes_k k'$ of k' . The algebra D becomes $D' = D \otimes_k k'$, and is of the form $D' = \text{End}_{K'}(V')$, for V' a free K' -module. The module V' is not determined uniquely by D' , only up to homotheties (the corresponding projective space is uniquely determined).

For any K -module M , let $\bar{M}$ be the module deduced from M by the extension of scalars $\bar{\cdot} : K \rightarrow K$, i.e. the module, unique up to unique isomorphism, provided with an anti-linear isomorphism $x \mapsto \bar{x} : M \xrightarrow{\sim} \bar{M}$. Similarly for K' -modules. If $D' = \text{End}(V')$, then $\bar{D}' = \text{End}(\bar{V}')$, and

$$(D \otimes_K \bar{D})' = \text{End}(V' \otimes \bar{V}').$$

Let W' be the fixed subspace of the anti-linear automorphism of $V' \otimes \bar{V}'$ defined by $v \otimes \bar{w} \mapsto w \otimes \bar{v}$. It is the space of Hermitian forms on the dual of V' . One has $W' \otimes_{k'} K' = V' \otimes \bar{V}'$. If $D_1 \subset D \otimes_K \bar{D}$ is the fixed subspace of the anti-linear automorphism of $D \otimes_K \bar{D}$ defined by $x \otimes \bar{y} \mapsto y \otimes \bar{x}$, then D'_1 is the k' -form of the K' -algebra $(D \otimes_K \bar{D})' = \text{End}(V' \otimes \bar{V}')$ deduced from the k' -form W' of the K' -module $V' \otimes \bar{V}'$: $D'_1 = \text{End}_{k'}(W')$.

Involutions of the second kind on D' correspond one to one to non degenerate Hermitian forms on V' , taken up to a factor (in k'^*). Those, in turn, by the “dual form” construction, correspond to “non degenerate” elements of W' . Again, one has to take them up to a factor. The projective space $\mathbf{P}(W')$ over k' is determined up to unique isomorphism by D' . It is hence (this is the point of localisation) defined over k : $\mathbf{P}(W') = P \otimes_k k'$, functorially in k' . The k -points of P (rather, the non degenerate points) parametrize the involutions of the second kind on D .

The functorial isomorphism $D'_1 = \text{End}_{k'}(W')$ shows that P is the form of projective space (Severi-Brauer variety) attached to D_1 . It has a rational point, and is then the ordinary projective space, if and only if D_1 is a matrix algebra. This shows that D has

involutions of the second kind if and only if the class of D_1 in $\text{Br}(k)$ is trivial. This class is the required norm $N_{K/k}(\text{cl}(D))$. In the localization spirit, this can be deduced from the fact that the homothety by $\lambda \in K'^*$ of V' induces on W' the homothety by $N_{K'/k'}(\lambda) \in k'^*$.

Appendix C

For $n \geq 3$, examples can be obtained as follows: take $k' = \mathbf{Q}[\zeta]$, with $\zeta = \exp(2\pi i/n)$, and $k = k' \cap \mathbf{R}$. Fix $a, b \in k^*$ and let D be the k' -algebra generated by X, Y , subject to

$$X^n = a, \quad Y^n = b$$

$$XY = \zeta YX.$$

It admits the anti-involution $*$, inducing complex conjugation on k' , defined by $\zeta^* = \zeta^{-1}$, $X^* = X, Y^* = Y$. The algebra D is of the type we require, provided it is a division algebra. This happens already with $a, b \in \mathbf{Z}$: take for a a prime congruent to 1 mod n , and for b an integer whose residue mod a has in the cyclic group of order n $(\mathbf{Z}/(a))^*/(\mathbf{Z}/(a))^{*n}$ an image of exact order n . For instance $n = 3, a = 7, b = 2$. For $n = 2$, one proceeds similarly with $k' = \mathbf{Q}[i], \zeta = -1, a$ congruent to 1 mod 4 and b not a square mod a . For instance, $a = 5$, and $b = 2$. In each case, the assumption on a ensures that k' embed in the a -adic completion $\mathbf{Q}_a$ of $\mathbf{Q}$, and the fact that D is a division algebra can be seen locally at a : $D \otimes_{k'} \mathbf{Q}_a$ is a division algebra with center $\mathbf{Q}_a$.

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TRACE PALEY-WIENER THEOREM FOR REDUCTIVE p -ADIC GROUPS

By

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Statement of the theorem

1.1

Let G be a reductive p -adic group. A smooth representation (π, E) of the group G on a complex vector space E is called a G -module. Usually we shorten the notation and write π or E .

Let $\mathcal{M}(G)$ be the category of G -modules, $\text{Irr } G$ the set of equivalence classes of irreducible G -modules, and $R(G)$ the Grothendieck group of G -modules of finite length; $R(G)$ is a free abelian group with basis $\text{Irr } G$.

We fix a minimal parabolic subgroup $P_0 \subset G$ and its Levi decomposition $P_0 = M_0 \cdot U_0$. By a standard Levi subgroup we mean a subgroup $M \supset M_0$ which is a Levi component of the parabolic subgroup $P = M \cdot P_0$ (notation $M < G$). For any standard Levi subgroup $M < G$ the functors $i_{G,M} : \mathcal{M}(M) \rightarrow \mathcal{M}(G)$ and $r_{M,G} : \mathcal{M}(G) \rightarrow \mathcal{M}(M)$ define morphisms $i_{G,M} : R(M) \rightarrow R(G)$, $r_{M,G} : R(G) \rightarrow R(M)$ (see [2] or [1, 2.5]).

Let $\mathcal{X}(G) \subset \{\psi : G \rightarrow \mathbb{C}^*\}$ be the group of unramified characters of G . It acts naturally on $\text{Irr } G$ and $R(G)$ by $\psi : \pi \mapsto \psi\pi$. This group has a natural structure of complex algebraic group (isomorphic to $(\mathbb{C}^*)^d$).

1.2

Let $\mathcal{H}(G)$ be the Hecke algebra of G (algebra of locally constant complex valued measures on G with compact support). Each measure $h \in \mathcal{H}(G)$ defines a linear form $f_h : R(G) \rightarrow \mathbb{C}$ by $f_h(\pi) = \text{tr } \pi(h)$.

It is easy to see that the form $f = f_h$ satisfies the following conditions:

- (i) For any standard Levi subgroup $M < G$ and $\sigma \in \text{Irr } M$, the function $\psi \mapsto f(i_{G,M}(\psi\sigma))$ is a regular function on the complex algebraic variety $\mathcal{X}(M)$.
- (ii) There exists an open compact subgroup $K \subset G$ which dominates f , i.e., f is nonzero only on the G -modules E , which have a nontrivial space E^K of K -invariant vectors.

We want to prove that the conditions (i)–(ii) characterize the trace forms $\{f_h\}$. Namely, let $R^*(G) = \text{Hom}_{\mathbb{Z}}(R(G), \mathbb{C}) = \text{Map}(\text{Irr } G, \mathbb{C})$ be the space of all linear forms on $R(G)$. We call a form $f : R(G) \rightarrow \mathbb{C}$ *good* if it satisfies conditions (i)–(ii)¹ and *trace form* if $f = f_h$ for some $h \in \mathcal{H}(G)$. We denote the subspaces of trace and good forms by $F_{\text{tr}} = F_{\text{tr}}(G) \subset F_{\text{good}} = F_{\text{good}}(G)$.

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Theorem 1 (Trace Paley-Wiener Theorem). $F_{\text{tr}} = F_{\text{good}}$.

Remark 1. This theorem describes the image of the natural morphism $\text{tr} : \mathcal{H}(G) \rightarrow R^*(G)$. One can show that $\text{Ker tr} = [\mathcal{H}(G), \mathcal{H}(G)]$. See Theorem B in [8].

1.3

The paper has the following structure. In Section 2 we recall the basic properties of the central algebra of a p -adic group $\mathcal{Z}(G)$. In Section 3 we formulate three technical results, which are proven in Section 5. In Section 4 we prove the theorem, using these results. Section 6 contains some remarks on the proof.

The central algebra $\mathcal{Z}(G)$

In this section we recall some definitions and results about the central algebra $\mathcal{Z}(G)$ of a p -adic group G . This algebra is analogous to the center of the universal enveloping algebra for a real group. We follow [1] but we use a slightly different terminology in order to emphasize this analogy.

2.1 Infinitesimal Characters

We call a *cuspidal pair* a pair (M, ρ) where $M < G$ is a standard Levi subgroup, $\rho \in \text{Irr } M$ is a cuspidal irreducible M -module. We denote by $\mathcal{O}(G)$ the set of all cuspidal pairs up to conjugation by G . A point $\theta \in \mathcal{O}(G)$ is called an *infinitesimal character* of G .

For any cuspidal pair (M, ρ) the image of the map $\psi \mapsto (M, \psi\rho)$, $\psi : \mathcal{X}(M) \rightarrow \mathcal{O}(G)$, is called a *connected component* of $\mathcal{O}(G)$. This component has the natural structure of a complex affine algebraic variety as a quotient of $\mathcal{X}(M)$ by a finite subgroup. We consider $\mathcal{O}(G)$ as a complex “algebraic variety”, consisting of an infinite number of connected components $\mathcal{O}$ and we will write $\mathcal{O}(G) = \bigcup \mathcal{O}$.

For each $\omega \in \text{Irr } G$ there exists a cuspidal pair (M, ρ) , such that π is a sub-quotient of $i_{G,M}(\rho)$. This pair is uniquely defined up to conjugation by G and hence defines a point $\theta \in \mathcal{O}(G)$, which is called the *infinitesimal character* of ω (notation: $\theta = \text{infch}(\omega)$). The map $\text{infch} : \text{Irr } G \rightarrow \mathcal{O}(G)$ is onto and finite-to-one.

Define an algebraic action of $\mathcal{X}(G)$ on $\mathcal{O}(G)$ by $\psi : (M, \rho) \mapsto (M, \psi|_M \cdot \rho)$. Then infch is a $\mathcal{X}(G)$ -equivariant map. We can decompose $R(G) = \bigoplus_{\theta \in \mathcal{O}(G)} R(\theta)$, where $R(\theta)$ is the subgroup, generated by irreducible G -modules with infinitesimal character θ . More generally, for each subset $S \subset \mathcal{O}(G)$ we put

$$R(S) = \bigoplus_{\theta \in S} R(\theta).$$

2.2 The Central Algebra $\mathcal{Z}(G)$

Definition 1. The central algebra $\mathcal{Z}(G)$ is the algebra of regular functions on the complex algebraic variety $\mathcal{O}(G)$.

The decomposition $\mathcal{O}(G) = \bigcup \mathcal{O}$ shows that $\mathcal{Z}(G) = \prod_{\mathcal{O}} \mathcal{Z}(\mathcal{O})$, where $\mathcal{Z}(\mathcal{O})$ is the algebra of regular functions on the component $\mathcal{O}$. We denote by $\mathcal{Z}^f(G)$ the ideal $\bigoplus_{\mathcal{O}} \mathcal{Z}(\mathcal{O}) \subset \mathcal{Z}(G)$, consisting of functions supported on a finite number of components.

We can (and will) interpret infinitesimal characters $\theta \in \mathcal{O}(G)$ as algebra homomorphisms $\theta : \mathcal{Z}(G) \rightarrow \mathbb{C}$, nontrivial on $\mathcal{Z}^f(G)$.

Theorem 2 (see [1].) *On each G -module E there exists a natural action of $\mathcal{Z}(G)$ such that*

- (i) $z : E \rightarrow E$ is a morphism of G -modules for each $z \in \mathcal{Z}(G)$.
- (ii) Each G -module morphism $\alpha : E \rightarrow E'$ is a $\mathcal{Z}(G)$ -morphism.
- (iii) On each irreducible G -module E the action of $z \in \mathcal{Z}(G)$ is given by $z = \inf \text{ch}(E)(z) \cdot 1_E$.

Remark 2. (1) The system of actions of $\mathcal{Z}(G)$ on G -modules is uniquely determined by properties (i)–(iii).

- (2) Consider $\mathcal{H}(G)$ as a G -module with respect to the left action of G . Then the corresponding action of $\mathcal{Z}(G)$ on $\mathcal{H}(G)$ identifies $\mathcal{Z}(G)$ with the algebra of all endomorphisms of $\mathcal{H}(G)$, invariant with respect to left and right actions of G (see [1]).

For each G -module π , $h \in \mathcal{H}(G)$, $z \in \mathcal{Z}(G)$ we have $\pi(zh) = z \circ \pi(h)$. Hence the action of $\mathcal{Z}(G)$ on π can be described in terms of its action on $\mathcal{H}(G)$.

2.3 Decomposition Theorem

Let E be a G -module. Then for each connected component $\mathcal{O} \subset \mathcal{O}(G)$ its characteristic function $1_{\mathcal{O}} \in \mathcal{Z}(G)$ acts on E as a projector on a G -submodule $E_{\mathcal{O}}$, and $E = \bigoplus_{\mathcal{O}} E_{\mathcal{O}}$. Moreover, for each open compact subgroup $K \subset G$ there exists an open subset $\mathcal{O}_K(G) \subset \mathcal{O}(G)$, which is a union of a finite number of components, such that for $\mathcal{O} \notin \mathcal{O}_K(G)$, $E_{\mathcal{O}} = 0$ for all G -modules E (see [1]).

This theorem implies that the Hecke algebra $\mathcal{H}(G)$ can be decomposed as a direct sum of two-sided ideals $\mathcal{H}(G)_{\mathcal{O}}$ and for each open compact subgroup $K \subset G$ the corresponding decomposition $\mathcal{H}_K(G) = \bigoplus_{\mathcal{O}} \mathcal{H}(G)_{\mathcal{O}}$ has only a finite number of nonzero terms (here $\mathcal{H}_K(G)$ is the algebra of K -bi-invariant measures).

2.4 Harish-Chandra Homomorphism

Let $M < G$ be a standard Levi subgroup. Define the morphism $i_{G,M}^* : \mathcal{O}(M) \rightarrow \mathcal{O}(G)$ by $(L, \rho) \mapsto (L, \rho)$. This is a finite morphism of algebraic varieties (it is not an inclusion, since cuspidal pairs conjugate under G may be non-conjugate under M). We call the corresponding morphism $i_{G,M}^* : \mathcal{Z}(G) \rightarrow \mathcal{Z}(M)$ the *Harish-Chandra homomorphism*. As for real groups $\mathcal{Z}(M)$ is a finitely generated $\mathcal{Z}(G)$ -module.

It is easy to check that the map $i_{G,M}^* : \mathcal{O}(M) \rightarrow \mathcal{O}(G)$ is compatible with the functors $i_{G,M}$ and $r_{M,G}$, i.e., for each subset $S_M \subset \mathcal{O}(M)$, $i_{G,M} R(S_M) \subset R(i_{G,M}(S_M))$ and for each subset $S \subset \mathcal{O}(G)$, $r_{M,G} R(S) \subset R(i_{G,M}(S))$. In fact, a more precise statement is true.

Proposition 1. *Let $z \in \mathcal{Z}(G)$, $z_M = i_{G,M}^*(z) \in \mathcal{Z}(M)$. Then for each M -module π , $i_{G,M}(z_M) = z$ on $i_{G,M}(\pi)$ and for each G -module π , $r_{M,G}(z\pi) = z_M$ on $r_{M,G}(\pi)$.*

The proof can be found in [1].

2.5 Finiteness Theorem

Theorem 3 (see [1]). *Let $K \subset G$ be an open compact subgroup. Then for any finitely generated G -module E the space E^K of K -invariant vectors is a finitely generated $\mathcal{Z}^f(G)$ -module. In particular, the algebra $\mathcal{H}_K(G)$ is a finitely generated $\mathcal{Z}^f(G)$ -module.*

This is an analogue of Harish-Chandra's finiteness theorem for real groups (see [5]).

Preparation for induction: discrete G -modules and discrete forms

3.1 Discrete G -modules

Let $R^I(G) \subset R(G)$ be the subgroup of “strictly induced” G -modules, defined by $R^I(G) = \sum_{M \not\leq G} i_{G,M}(R(M))$ with $M \not\leq G$. An irreducible G -module ω is called *discrete* if $\omega \notin R^I(G)$. An infinitesimal character θ is called *discrete* if it is the infinitesimal character of a discrete G -module (or, equivalently, if $R(\theta) \not\subset R^I(G)$). We denote the set of all discrete infinitesimal characters by $\mathcal{O}_{\text{disc}}(G)$.

Proposition 2 (see 5.1–5.2). *For each connected component $\mathcal{O} \subset \mathcal{O}(G)$ the set $\mathcal{O}_{\text{disc}} = \mathcal{O} \cap \mathcal{O}_{\text{disc}}(G)$ consists of a finite number of $\mathcal{X}(G)$ -orbits.*

Let $M < G$, $\sigma \in \text{Irr } M$. The family $\{i_{G,M}(\psi\sigma)\}$ parametrized by $\mathcal{X}(M)$ is called a *standard family* of G -modules.

Corollary 1. *For each connected component $\mathcal{O} \subset \mathcal{O}(G)$ the group $R(\mathcal{O})$ is generated by a finite number of standard families $\{i_{G,M}(\psi\sigma)\}$, corresponding to discrete M -modules σ .*

Indeed, it is clear that $R(\mathcal{O})$ is generated by standard families, corresponding to discrete M -modules σ with infinitesimal characters in $i_{G,M}(\mathcal{O}) \subset \mathcal{O}(M)$. Since $i_{G,M}(\mathcal{O})$ consists of a finite number of components and the map $\text{inf ch} : \text{Irr } M \rightarrow \mathcal{O}(M)$ is finite-to-one, the proposition shows that there are only a finite number of such families.

3.2

Let $i_{G,M}^* : R^*(G) \rightarrow R^*(M)$ and $r_{M,G}^* : R^*(M) \rightarrow R^*(G)$ be morphisms adjoint to morphisms $i_{G,M}$ and $r_{M,G}$.

Proposition 3 (see 5.3). *(i) $i_{G,M}^*(F_{\text{good}}(G)) \subset F_{\text{good}}(M)$;*

(ii) $r_{M,G}^(F_{\text{tr}}(M)) \subset F_{\text{tr}}(G)$.*

3.3

A linear form $f \in R^*(G)$ is called *discrete* if $f(R^I(G)) = 0$, i.e., $i_{G,M}^*(f) = 0$ for all $M \not\leq G$.

Lemma 1 (Combinatorial Lemma, see 5.4–5.5). *There are constants $c_M \in \mathbb{Q}$ for each $M \not\leq G$ such that for each linear form $f \in R^*(G)$ the form $f_d = f - \sum_M c_M r_{M,G}^* \circ i_{G,M}^*(f)$, $M \not\leq G$, is discrete.*

Proof of Theorem 1.2

4.1

The central algebra $\mathcal{Z}(G)$ naturally acts on $R^*(G)$ by $(zf)(\pi) = \text{infch}(\omega)(z) \cdot f(\pi)$, $z \in \mathcal{Z}(G)$, $f \in R^*(G)$, $\omega \in \text{Irr } G$. Using Remark 2 in 2.2 we see that $\text{tr} : \mathcal{H}(G) \rightarrow R^*(G)$ is a morphism of $\mathcal{Z}(G)$ -modules, so its image F_{tr} is a $\mathcal{Z}(G)$ -submodule. It is also easy to check that the subspace $F_{\text{good}} \subset R^*(G)$ is a $\mathcal{Z}(G)$ -submodule (one should only notice that for each standard family $\{i_{G,M}(\psi\sigma)\}$ the corresponding map $\psi : \mathcal{X}(M) \rightarrow \mathcal{O}(G)$ is a morphism of algebraic varieties).

The group $\mathcal{X}(G)$ naturally acts on $R^*(G)$ by $\psi(f)(\pi) = f(\psi\pi)$. Clearly the subspace F_{good} is $\mathcal{X}(G)$ -invariant. In order to prove that F_{tr} is $\mathcal{X}(G)$ -invariant it is sufficient to observe that the morphism $\text{tr} : \mathcal{H}(G) \rightarrow R^*(G)$ is $\mathcal{X}(G)$ -equivariant, where $\mathcal{X}(G)$ acts on $\mathcal{H}(G)$ by $\psi : h \mapsto \psi h$.

4.2

Consider any subset $S \subset R(G)$. We call the restrictions to S of good and trace linear forms good and trace functions on S . We denote the corresponding spaces of functions by $F_{\text{good}}[S]$, $F_{\text{tr}}[S]$.

Lemma 2. *For a finite set S the restriction to S of any linear form $f : R(G) \rightarrow \mathbb{C}$ is a trace function. In particular, $F_{\text{tr}}[S] = F_{\text{good}}[S]$.*

This is just another way of saying that characters of irreducible G -modules are linearly independent functionals on $\mathcal{H}(G)$.

Proposition 4. *Let $S \subset \text{Irr } G$ be a union of a finite number of $\mathcal{X}(G)$ -orbits. Then $F_{\text{tr}}[S] = F_{\text{good}}[S]$.*

Proof. The set S has the natural structure of algebraic variety, as a union of a finite number of $\mathcal{X}(G)$ -orbits. By definition of good forms $F_{\text{tr}}[S] \subset F_{\text{good}}[S] \subset \mathcal{R}(S)$, where $\mathcal{R}(S)$ is the algebra of regular functions on S .

Choose a cocompact lattice Λ in the center $Z(G)$ of the group G and put $Y = \mathcal{X}(\Lambda) = \text{Hom}(\Lambda, \mathbb{C}^*)$. Then the restriction map $r : \mathcal{X}(G) \rightarrow Y$ is a finite epimorphism of algebraic groups.

Define a map $c' : \text{Irr } G \rightarrow Y$ by $c' : \pi \mapsto \pi|_{\Lambda}$. Its restriction to S , $c : S \rightarrow Y$ is a finite $\mathcal{X}(G)$ -equivariant submersive morphism of algebraic varieties. In particular, $\mathcal{R}(S)$ is a finitely generated $\mathcal{R}(Y)$ -module. We claim that $F_{\text{tr}}[S]$, $F_{\text{good}}[S]$ are $\mathcal{R}(Y)$ -submodules of $\mathcal{R}(S)$. For $F_{\text{good}}[S]$ it is clear. For $F_{\text{tr}}[S]$ one can either directly describe the action of $\mathcal{R}(Y)$, which is after all just the group algebra of Λ , on $F_{\text{tr}}[S]$, or use the homomorphism $\mathcal{R}(Y) \rightarrow \mathcal{Z}(G)$, corresponding to $c'' : \mathcal{O}(G) \rightarrow Y$.

Let $y \in Y$, $\mathfrak{M}_y \subset \mathcal{R}(Y)$ the corresponding maximal ideal. For each $\mathcal{R}(Y)$ -module we put $\mathcal{F}_y = \mathcal{F}/\mathfrak{M}_y\mathcal{F}$ – the fiber of $\mathcal{F}$ at y . Since $c : S \rightarrow Y$ is finite and submersive, the set $S_y = c^{-1}(y)$ is finite and the fiber $\mathcal{R}(S)_y$ coincides with $\mathcal{R}(S_y)$. By the lemma above $F_{\text{tr}}[S_y] = F_{\text{good}}[S_y]$, which gives

$$F_{\text{good}}[S] \subset F_{\text{tr}}[S] + \mathfrak{M}_y\mathcal{R}(S).$$

Put $\mathcal{F} = \mathcal{R}(S)/F_{\text{tr}}[S]$, $\mathcal{F}' = F_{\text{good}}[S]/F_{\text{tr}}[S] \subset \mathcal{F}$. Then the inclusion above can be written as

$$\mathcal{F}' \subset \mathfrak{M}_y\mathcal{F} \quad \text{for each } y \in Y. \quad (*)$$

Since $\mathcal{F}$ is a finitely generated $\mathcal{R}(Y)$ -module it is locally free at almost every point $y \in Y$. Since it is $\mathcal{X}(G)$ -equivariant, it is locally free everywhere. Hence (*) implies that $\mathcal{F}' = 0$, i.e., $F_{\text{tr}}[S] = F_{\text{good}}[S]$. $\square$

4.3 Proof of Theorem 1.2

Let $f \in F_{\text{good}}(G)$. By definition f is dominated by an open compact subgroup $K \subset G$ (see condition (ii) in 1.2). Using 2.3 we can write $f = \sum 1_{\mathcal{O}} \cdot f$, where $1_{\mathcal{O}} \cdot f = 0$ for all but a finite number of components $\mathcal{O}$. It is sufficient to prove that $1_{\mathcal{O}} \cdot f \in F_{\text{tr}}(G)$ for each component $\mathcal{O}$, so we can fix $\mathcal{O}$ and assume that $1_{\mathcal{O}} \cdot f = f$.

By Proposition 3.1 the group $R(\mathcal{O}_{\text{disc}})$ is generated by a subset $S \subset \text{Irr } G$, which is a union of a finite number of $\mathcal{X}(G)$ -orbits. Using Proposition 4.2, we can find a trace form f_h such that $f = f_h$ on $R(\mathcal{O}_{\text{disc}})$. Replacing f by $f - 1_{\mathcal{O}} \cdot f_h$ we can assume that f is zero on the subgroup $R_{\text{disc}} = R(\mathcal{O}_{\text{disc}}(G))$.

Consider the form $f_d = f - \sum_M c_M r_{M,G}^* \circ i_{G,M}^* f$, defined in 3.3. This form is discrete, i.e., it equals 0 on $R^I(G)$. It also equals 0 on R_{disc} , since $f(R_{\text{disc}}) = 0$ and the operators $i_{G,M}^*$, $r_{M,G}$ preserve R_{disc} because they preserve $R(\mathcal{O})$ for each $\mathcal{O} \in \mathcal{O}(G)$. Since $R(G) = R_{\text{disc}} + R^I(G)$, $f_d = 0$, i.e.,

$$f = \sum_M c_M r_{M,G}^* \circ i_{G,M}^* f, \quad M \not\leq G.$$

Let $M \not\leq G$. By induction in $\dim M$ we can assume that $F_{\text{tr}}(M) = F_{\text{good}}(M)$. Then $i_{G,M}^* f \in F_{\text{good}}(M) = F_{\text{tr}}(M)$ and hence $r_{M,G}^* \circ i_{G,M}^* f \in r_{M,G}^* F_{\text{tr}}(M) \subset F_{\text{tr}}(G)$ (see 3.2(i), (ii)). This shows that $f \in F_{\text{tr}}(G)$.

Proofs of Propositions 3.1, 3.2 and 3.3

5.1

Fix a connected component $\mathcal{O} \subset \mathcal{O}(G)$. We want to prove that the subset $\mathcal{O}_{\text{disc}} \subset \mathcal{O}$ of discrete infinitesimal characters consists of a finite number of $\mathcal{X}(G)$ -orbits.

Lemma 3. $\mathcal{O}_{\text{disc}}$ is a constructible subset of $\mathcal{O}$, i.e., a union of a finite number of subsets, locally closed in Zariski topology.

This lemma is “morally obvious”, since the whole situation is algebraic. But, as usual in such cases, the proof is a little tiresome.

Proof. Step 1. Let $\mathcal{A}$ be a commutative algebra over $\mathbb{C}$, E a G - $\mathcal{A}$ -module, i.e., a G -module together with a homomorphism $\mathcal{A} \rightarrow \text{End}_G(E)$. Suppose E is finitely generated as a G - $\mathcal{A}$ -module and for each open compact subgroup $K \subset G$ the $\mathcal{A}$ -module E^K is projective and finitely generated. In this situation we call E an $\mathcal{A}$ -family of G -modules. For any homomorphism of algebras $\mathcal{A} \rightarrow \mathcal{A}'$ we denote by $E_{\mathcal{A}'}$ the induced $\mathcal{A}'$ -family of G -modules $E_{\mathcal{A}'} = \mathcal{A}' \otimes_{\mathcal{A}} E$.

Usually we consider $\mathcal{A}$ to be the algebra of regular functions on an algebraic variety S , and then call E an S -family of G -modules. For each morphism $\phi : S' \rightarrow S$ we denote by $E_{S'}$ an induced S' -family of G -modules. For instance, for any point $s \in S$ the corresponding G -module E_s is the specialization of the family E at s .

For an S -family of G -modules E we define a function $\nu_E : S \rightarrow R(G)$ by $\nu_E(s) = E_s$. We call functions of this type *regular*. A regular function $\nu : S \rightarrow R(G)$ is called *irreducible* if $\nu(S) \subset \text{Irr } G$. Two irreducible functions ν, ν' are called *disjoint* if for each $s \in S$, $\nu(s) \neq \nu'(s)$.

Step 2. Let $\nu : S \rightarrow R(G)$ be a regular function. Then there exists a dominant étale morphism $\phi : V \rightarrow S$, irreducible regular functions $\{\lambda_j : V \rightarrow R(G)\}$ and $n_j \in \mathbb{Z}_+$ such that $\phi^*(\nu) = \sum n_j \lambda_j$.

Let E be an S -family of G -modules, representing ν . We can find an open compact subgroup $K \subset G$ such that E is generated by E^K as a G -module. If we choose K to be good, then any nonzero subquotient E' of E has a nonzero space E'^K (see [1]). This implies that we can everywhere replace G -modules E' by $\mathcal{H}_K$ -modules E'^K , where $\mathcal{H}_K = \mathcal{H}_K(G)$, i.e., essentially we study families of finite-dimensional $\mathcal{H}_K$ -modules.

We can assume that S is irreducible. Let $\mathcal{A} = \mathcal{R}(S)$, L be the field of fractions of $\mathcal{A}$. There are two possibilities.

(i) $\mathcal{H}_K \subset \text{End}_L(E_L^K)$ generates the whole space as L -module. Then, after replacing S by an open subset, we can assume that $\mathcal{H}_K$ generates $\text{End}_{\mathcal{A}}(E^K)$ as a $\mathcal{A}$ -module, i.e., E_s is irreducible for each $s \in S$.

(ii) $\mathcal{H}_K$ does not generate $\text{End}_L(E_L^K)$ as an L -module. Then for some finite extension L' of L the representation of $\mathcal{H}_K$ in E_L^K is reducible. Replacing S by some V , étale over S , we may assume that $L' = L$.

Let $E_0^K \subset E_L^K$ be a nontrivial $\mathcal{H}_K \otimes L$ submodule. Replacing S by an open subset we can assume that $E_0^K = (E_0)_L$, where $E_0 \subset E^K$ is an $\mathcal{H}_K \otimes \mathcal{A}$ submodule, such that E_0 and E^K/E_0 are projective $\mathcal{A}$ -modules. Induction on $\dim_{\mathcal{A}} E^K$ finishes the proof.

Step 3 (Decomposition in irreducible functions). Let $\nu_1, \dots, \nu_r : S \rightarrow R(G)$ be regular functions. Then there exist a dominant étale morphism $\phi : V \rightarrow S$, irreducible regular functions $\lambda_j : V \rightarrow R(G)$ and integers $n_{ij} \in \mathbb{Z}_+$ such that $\phi^*(\nu_i) = \sum_j n_{ij} \lambda_j$. Moreover, we can choose λ_j to be disjoint.

Indeed, repeating Step 2 we can find λ_j . If λ, λ' are two irreducible functions, then the set of points $v \in V$ such that $\lambda(v) \sim \lambda'(v)$ is closed in Zariski topology. Hence, replacing V by an open subset, we can assume that each two functions λ_i, λ_j either coincide, or are disjoint.

Step 4. Let $\nu_1, \dots, \nu_r, \mu_1, \dots, \mu_m : S \rightarrow R(G)$ be regular functions. Consider the subset $S_0 \subset S$ consisting of all points s , such that $\{\nu_1(s), \dots, \nu_r(s)\}$ lie in a subgroup of $R(G)$, generated by $\{\mu_1(s), \dots, \mu_m(s)\}$. Then there exists a dominant étale morphism $\phi : V \rightarrow S$ such that $\phi^{-1}(S_0)$ is either empty or the whole V .

It follows immediately from Step 3.

Step 5 Proof of the Lemma. In order to prove that the subset $\mathcal{O}_{\text{disc}} \subset \mathcal{O}$ is constructible it is sufficient to check the following condition.

(*) Let $S \subset \mathcal{O}$ be a locally closed subvariety, $S_0 = S \setminus (S \cap \mathcal{O}_{\text{disc}})$. Then there exists a dominant étale morphism $\phi : V \rightarrow S$ such that $\phi^{-1}(S_0)$ is either empty or the whole V .

Let (N, ρ) be a cuspidal pair, corresponding to $\mathcal{O}$, $\psi_{\mathcal{O}} : \mathcal{X}(N) \rightarrow \mathcal{O}$ the corresponding morphism. For each standard Levi subgroup $M < G$, which contains N , consider the regular function $\nu_M : \mathcal{X}(N) \rightarrow R(M)$ given by $\psi \mapsto i_{M,N}(\psi\rho)$.

Choose a subvariety $V_1 \subset \mathcal{X}(N)$ such that $\psi_{\mathcal{O}}(V_1) \subset S$ and $\psi_{\mathcal{O}}|_{V_1} : V_1 \rightarrow S$ is dominant étale, and denote by ν_M the restrictions of functions ν_M to V_1 . Using Step 1 we can replace V_1 by V_2 , étale over V_1 , such that on V_2 for each M we have a decomposition $\nu_M = \sum n_i \lambda_{M,i}$ with irreducible $\lambda_{M,i}$.

Now consider on V_2 two systems of regular functions: the first is $\{\lambda_{M,i}\}$ and the second

is $\{i_{G,M}(\lambda_{M,j}) \mid M \not\prec G\}$. It is clear that the set $(V_2)_0$ of points v for which the second system generates the first one is exactly the pre-image $\phi_2^{-1}(S_0)$, where $\phi_2 : V_2 \rightarrow S$ is the natural étale morphism. So Step 4 proves $(*)$ and the lemma. $\square$

5.2 Proof of Proposition 3.1

For each G -module π denote by π^+ its Hermitian contragradient G -module. This defines an involutive map $+ : R(G) \rightarrow R(G)$, which maps $\text{Irr } G$ into itself. This involution is compatible with morphisms $i_{G,M}$. For each subgroup $M < G$ the involution $+$ acts on $\mathcal{X}(M)$ and on the set of cuspidal pairs (M, ρ) . This defines an action of $+$ on $\mathcal{O}(G)$, which is compatible with the map infch . It is easy to see that the action of $+$ on the algebraic varieties $\mathcal{X}(M)$, $\mathcal{O}(G)$ is antiholomorphic (more precisely, antialgebraic).

By Langlands theory (see [3]) the group $R(G)$ is generated by standard families $\{i_{G,M}(\psi\sigma)\}$ in which σ is a tempered M -module. Hence each point $\theta \in \mathcal{O}_{\text{disc}}$ can be written as $\theta = \text{infch}(\psi\pi)$, with $\psi \in \mathcal{X}(G)$, and π an irreducible tempered G -module. Since π is tempered, it is unitary, i.e., $\pi^+ = \pi$. This implies

$$\theta^+ \in \mathcal{X}(G) \cdot \theta. \quad (+)$$

Suppose for a moment that $\mathcal{X}(G) = \{1\}$, i.e., G has compact center. Then the subset $\mathcal{O}_{\text{disc}} \subset \mathcal{O}$ is constructible by 5.1 and is pointwise fixed by the antialgebraic involution $+$. This implies that this subset is finite.

For arbitrary $\mathcal{X}(G)$ the same proof, applied to the subset $\tilde{\mathcal{O}}_{\text{disc}} = \mathcal{O}_{\text{disc}}/\mathcal{X}(G)$ of the algebraic quotient variety $\tilde{\mathcal{O}} = \mathcal{O}/\mathcal{X}(G)$, shows that $\tilde{\mathcal{O}}_{\text{disc}}$ is finite, i.e., $\mathcal{O}_{\text{disc}}$ consists of a finite number of $\mathcal{X}(G)$ -orbits.

5.3 Proof of Proposition 3.2

Fix $M < G$. The inclusion $i_{G,M}^*(F_{\text{good}}(G)) \subset F_{\text{good}}(M)$ is straightforward. We prove the inclusion $r_{M,G}^*(F_{\text{tr}}(M)) \subset F_{\text{tr}}(G)$ using a modification of the method used in [4].

Let $P = MU$ be the standard parabolic subgroup, corresponding to M , $P^- = MU^-$ the opposite parabolic. Choose a small good subgroup $K \subset G$, i.e., an open compact subgroup such that $K = (K \cap U^-) \cdot (K \cap M) \cdot (K \cap U)$. Put $\Gamma = K \cap M$ and consider Hecke algebras $\mathcal{H}_K(G)$ and $\mathcal{H}_\Gamma(M)$. We want to prove that for each $h \in \mathcal{H}_\Gamma(M)$ the form $r_{M,G}^*(f_h) \in R^*(G)$ is a trace form. Let J denote the subspace of $h \in \mathcal{H}_\Gamma(M)$ for which it is true. We should prove that $J = \mathcal{H}_\Gamma(M)$.

Step 1. Fix a central element $a \in Z(M)$ such that $\text{Ad } a|_U$ is a strictly contracting operator. Then for each $h \in \mathcal{H}_\Gamma(M)$, $a^n \cdot h \in J$ for large n .

Proof. Let e_Γ, e_K be units in $\mathcal{H}_\Gamma(M)$ and $\mathcal{H}_K(G)$. For each $m \in M$ put $h(m) = e_\Gamma \cdot \delta_m \cdot e_\Gamma$, $h'(m) = \text{mod}_U^{1/2}(m) \cdot e_\Gamma \cdot \delta_m \cdot e_K$, where δ_m is the δ -distribution at m and $\delta_U = \text{mod}_U \in \mathcal{H}(M)$ (this factor appears since we use the normalized functor $r_{M,G}$, see [1]).

We can assume that a given measure h is of the form $h(m)$ for some $m \in M$. Replacing a by its power and m by $a^r m$ we can assume that a, m lie in

$$M^+ = \{m \in M \mid m(K \cap U)m^{-1} \subset K \cap U \text{ and } m^{-1}(K \cap U^-)m \subset K \cap U^-\}.$$

Then (see [4])

$$h(a^n m) = h(a)^n \cdot h(m) \quad \text{and} \quad h'(a^n m) = h'(a)^n \cdot h'(m) = h'(m) \cdot h'(a)^n.$$

Let π be an admissible G -module, $\sigma = r_{M,G}(\pi)$. Then, arguing as in [4], we see that for large n

$$\mathrm{tr} \pi(h'(a^n m)) = \mathrm{tr} \sigma(h(a^n m)).$$

Since both functions of n are Z -finite, for $n \gg 1$, equality holds already for $n = 1$ (see [4]).

This proves that $r_{M,G}^*(f_{h(m)}) = f_{h'(m)}$ is a trace form, i.e., $h(am) = a \cdot h(m) \in J$. $\square$

By the finiteness theorem 2.5, $\mathcal{H}_\Gamma(M)$ is finitely generated as $\mathcal{Z}(M)$ -module and hence is finitely generated as $\mathcal{Z}(G)$ -module. Since $J \subset \mathcal{H}_\Gamma(M)$ is a $\mathcal{Z}(G)$ -submodule, we can deduce from step 1 that for some $n \in \mathbb{Z}$, $a^n \mathcal{H}_\Gamma(M) \subset J$. But $a^n \mathcal{H}_\Gamma(M) = \mathcal{H}_\Gamma(M)$, so $J = \mathcal{H}_\Gamma(M)$.

Step 2.

5.4 Relations Between Morphisms i and r

Let $W_G = \mathrm{Norm}(M_0, G)/M_0$ be the Weyl group of G . For $M < G$ we consider W_M as a subgroup of W_G .

We call two standard Levi subgroups $M, N < G$ *associate* if $N = wMw^{-1}$ for some $w \in W_G$. Each such element w defines an isomorphism $w : R(M) \rightarrow R(N)$, which depends only on the class $w \cdot W_M = W_N \cdot w$ of w . If $w' : R(L) \rightarrow R(M)$ is defined, we have $ww' = w \circ w' : R(L) \rightarrow R(N)$.

Lemma 4. *The system of groups $R(M)$ for $M < G$, isomorphisms w and morphisms $i_{G,M}$, $r_{M,G}$ for $M < N < G$ satisfies the following relations.*

(i) For $L < M < N < G$, $i_{G,M} \circ i_{M,L} = i_{G,L}$, $r_{L,M} \circ r_{M,G} = r_{L,G}$.

(ii) Let $M, N < G$, $W_G^{M,N}$ be the set of representatives of $W_N \backslash W_G / W_M$ of minimal length. Then

$$r_{N,G} \circ i_{G,M} = \sum_w i_{N,N_w} \circ w \circ r_{M_w,M}$$

with $w \in W_G^{M,N}$, $M_w = w^{-1}Nw \cap M$, $N_w = wM_w w^{-1} = N \cap wMw^{-1}$.

(iii) If $N = wMw^{-1}$, then

$$i_{G,N} \circ w(\sigma) = i_{G,M}(\sigma) \quad \text{for } \sigma \in R(M).$$

Proof. (i) is standard, (ii) is the reformulation of [2, 2.11]. The statement (iii) is well known, but we were unable to find a reference. So we give a proof of it.

It is sufficient to check (iii) on generators of $R(M)$. Hence by Langlands theory (see [3]) we can assume that $\sigma = i_{M,L}(\xi\omega)$, where $L < M$, $\xi \in \mathcal{X}(L)$, ω is a tempered irreducible L -module. Since $i_{G,M}(\sigma) = i_{G,L}(\xi\omega)$ and $i_{G,N} \circ w(\sigma) = i_{G,wLw^{-1}}(w(\xi\omega))$, we can just assume that $L = M$, $\sigma = \xi\omega$.

Since characters of both parts in (iii) are regular functions in ξ it is sufficient to check (iii) for a generic character ξ .

Using the formula

$$\mathrm{Hom}_G(i_{G,M}(\xi\omega), i_{G,N}(w(\xi\omega))) = \mathrm{Hom}_N(r_{N,G} \circ i_{G,M}(\xi\omega), w(\xi\omega))$$

and using (ii) we see that this Hom space is 1-dimensional, since for generic ξ all subquotients of $r_{N,G} \circ i_{G,M}(\xi\omega)$ except $w(\xi\omega)$ have central characters different from the central character of $w(\xi\omega)$. Hence it is sufficient to check that the G -module $i_{G,M}(\xi\omega)$ (and similarly $i_{G,N}(w(\xi\omega))$) is irreducible for generic ξ .

The same proof as above shows that for generic ξ , $\text{End}_G(i_{G,M}(\xi\omega)) = \mathbb{C}$. If ξ is unitary, the G -module $i_{G,M}(\xi\omega)$ is unitary and, hence, irreducible. Since irreducibility is an open condition on ξ , $i_{G,M}(\xi\omega)$ is irreducible for generic ξ . $\square$

Corollary 2. *For each $M < G$ denote by T_M the operator $T_M = i_{G,M} \circ r_{M,G} : R(G) \rightarrow R(G)$. Then*

$$(i) \quad T_N \circ i_{G,M} = \sum_w i_{G,M_w} \circ r_{M_w,M}, \quad w \in W_G^{M,N}, \quad M_w = M \cap w^{-1}Nw.$$

$$(ii) \quad T_N \circ T_M = \sum_w T_{M_w}, \quad w \in W_G^{M,N}, \quad M_w = M \cap w^{-1}Nw.$$

Proof. (i) $T_N \circ i_{G,M} = i_{G,N} \circ r_{N,G} \circ i_{G,M} = \sum_w i_{G,N} \circ i_{N,N_w} \circ w \circ r_{M_w,M} = \sum_w i_{G,M_w} \circ r_{M_w,M}$.

(ii) $T_N \circ T_M = T_N \circ i_{G,M} \circ r_{M,G} = \sum_w i_{G,M_w} \circ r_{M_w,M} \circ r_{M,G} = \sum_w i_{G,M_w} \circ r_{M_w,G} = \sum_w T_{M_w}$. $\square$

5.5 Proof of the Combinatorial Lemma 3.3

For each $M < G$ put $d(M) = \dim \mathcal{X}(M)$ – the “depth” of M in G . Define a decreasing filtration $\{R^i \subset R(G)\}$ by $R^i = \sum_{d(M) \geq i} i_{G,M}(R(M))$. By definition $R^i = R(G)$ for $i \leq d(G)$, $R^{d(G)+1} = R^I(G)$ – the subgroup of strictly induced G -modules and $R^i = 0$ for $i > d(M_0)$.

Corollary 5.4(i) shows that each operator T_N preserves this filtration and gives an explicit description of the action of T_N on quotients. For instance, if we put $P(N) = \#(\text{Norm}(N, W_G)/W_N)$, then for $d = d(N)$ the action of T_N on R^d/R^{d+1} is given by

$$T_N i_{G,M}(\sigma) = P(N) \cdot i_{G,M}(\sigma) \quad \text{if } M \sim N,$$

$$T_N i_{G,M}(\sigma) = 0 \quad \text{if } M \not\sim N, \quad d(M) = d.$$

This implies that the operator $A_d = \prod_{d(N)=d} (T_N - P(N))$ preserves the filtration $\{R^i\}$ and annihilates R^d/R^{d+1} .

Put $A = A_{d(M_0)} \circ A_{d(M_0)-1} \circ \cdots \circ A_{d(G)}$. Then $A(R^I(G)) = 0$ and by Corollary 5.4(ii) A takes the form $A = P(1 - \sum_M c_M T_M)$, $M \not\sim G$, with $c_M \in \mathbb{Q}$, $P \in \mathbb{Z}$, $P \neq 0$.

Then the adjoint operator $A^* : R^*(G) \rightarrow R^*(G)$ maps $R^*(G)$ into the orthogonal complement to $R^I(G)$, i.e., into the subspace of discrete forms, and is of the form $A^* = P(1 - \sum_M c_M r_{M,G}^* \circ i_{G,M}^*)$, $M \not\sim G$ with $c_M \in \mathbb{Q}$, $P \in \mathbb{Z}$, $P \neq 0$. But this is exactly the statement of Lemma 3.3.

Miscellaneous remarks

6.1

Let $M < G$, $\sigma \in \text{Irr } M$. Then the natural morphism $\psi' : \mathcal{X}(M) \rightarrow \mathcal{O}(G)$, $\psi \mapsto \text{infch } i_{G,M}(\psi\sigma)$ is a finite morphism of algebraic varieties. Indeed, it is a composition of the morphism $\psi'' : \mathcal{X}(M) \rightarrow \mathcal{O}(M)$, $\psi \mapsto \text{infch}(\psi\sigma)$ which is finite by 4.2, and a finite morphism $i_{G,M} : \mathcal{O}(M) \rightarrow \mathcal{O}(G)$.

Fix a connected component $\mathcal{O} \subset \mathcal{O}(G)$ and consider a finite set of standard families, given by $\{(M_i, \sigma_i)\}$, generating $R(\mathcal{O})$. Put $X = \bigcup_i \mathcal{X}(M_i)$ and consider the natural map $\nu : X \rightarrow R(\mathcal{O})$. Using ν we realize $R^*(\mathcal{O})$ as some space of functions on X . Clearly, the space $F_{\text{good}}(\mathcal{O})$ lies in the subspace $\mathcal{R}(X)$ of regular functions on X . Since $\mathcal{R}(X)$ is a finitely generated $\mathcal{Z}(\mathcal{O})$ -module, $F_{\text{good}}(\mathcal{O})$ is also a finitely generated $\mathcal{Z}(\mathcal{O})$ -module.

In an earlier version of the paper we used this realization in order to prove the theorem. Let us sketch the proof.

First, using 3.2 and 3.3, we reduced the proof to the statement that any discrete form $f \in F_{\text{good}}(\mathcal{O})$ is a trace form. Since $\mathcal{R}(X)$ is a finitely generated $\mathcal{Z}(\mathcal{O})$ -module, it is sufficient to check that for each point $\theta \in \mathcal{O}$ the function $f \in \mathcal{R}(X)$ can be approximated by trace functions modulo any power of the maximal ideal $\mathfrak{M}_\theta \subset \mathcal{Z}(\mathcal{O})$.

Assume $\mathcal{X}(G) = \{1\}$ (in the general case one should argue as in 4.2). By 4.1 f is equal to a trace function f' on a finite subset $\nu^{-1}(\theta) \subset X$. Now we can apply the following simple statement from linear algebra.

(*) Let $A : V \rightarrow V$ be an endomorphism of a finite-dimensional vector space. Then for each $n \in \mathbb{Z}_+$ there exists a polynomial $P(t) \in \mathbb{C}[t]$ such that

$$\text{tr } P(A + X) = \text{tr } A + O(\|X\|^n) \quad \text{for } X \in \text{End } V.$$

This statement allows us to replace f' by a trace form f'' , which coincides with f' and f on $\nu^{-1}(\theta)$ and is a constant modulo a high power of $\mathfrak{M}_\theta$. Since f is discrete, it is constant on each component of X , and hence f'' approximates f modulo a high power of $\mathfrak{M}_\theta$.

6.2

Discrete G -modules and discrete forms, which appeared in our proof, apparently play a very important role in harmonic analysis on G . Some interesting results can be found in [6].

6.3

It would be interesting to study more thoroughly the combinatorial structures, described in 5.4–5.5. They have some relations with Hopf algebras (see [7]). Here we want only to note that the operators T_M in 5.4 do not coincide (and even do not commute) for associate Levi subgroups.

Example 1. Let $G = GL(3)$, $M = GL(2) \times GL(1)$, $N = GL(1) \times GL(2)$. Then $T_M T_N(\pi) \neq T_N T_M(\pi)$ if $\pi \in R(G)$ is the trivial G -module.

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Monodromy of Hypergeometric Functions and Non-Lattice Integral Monodromy

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Introduction

The hypergeometric series

$$F(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a, n)(b, n)}{(c, n) n!} x^n, \quad |x| < 1 \quad (0.1)$$

where $(a, n) = \prod_{i=0}^{n-1} (a+i)$, defined for c not an integer ≤ 0 , was first introduced by Euler in 1778 as a solution of the hypergeometric differential equation

$$x(1-x)y'' + (c - (a+b+1)x)y' - aby = 0. \quad (0.2)$$

$F(a, b; c; x)$ represents the unique solution of (??) which is holomorphic at $x = 0$ and takes the value 1 at $x = 0$. If neither b nor $c-b$ is an integer < 0 , Euler knew the integral representation

$$F(a, b; c; x) = \frac{\Gamma(b)\Gamma(c-b)}{\Gamma(c)} \int_0^1 z^{b-1} (1-z)^{c-b-1} (1-zx)^{-a} dz. \quad (0.3)$$

Replacing z by u^{-1} , (??) also has the form

$$\int_1^{\infty} u^{-a} (u-1)^{c-b-1} (u-x)^{-b} du. \quad (3')$$

If we integrate instead from g to h with g and h in $\{0, 1, \infty, x\}$ we get other solutions of (??), a fact discovered independently by Hermite [?], Pochhammer [?], and Schöffli [?]. Even when the integral diverges, it yields solutions of (??) when taken as its Hadamard “finite part”, provided that the integrand does not have a pole of integral order at g or h .

The hypergeometric equation (??) is the unique second order linear differential equation with regular singularities, singular only at $0, 1, \infty$ with exponents $(0, 1-c)$, $(0, c-a-b)$, (a, b) respectively. “Exponents (α, α') at a singularity p ” means that suitable linearly independent linear combinations of the branches of a solution have the form $(x-p)^\alpha f_1(x)$, $(x-p)^{\alpha'} f_2(x)$ with f holomorphic around p ; if $\alpha - \alpha' \in \mathbb{Z}$ an additional logarithmic term is allowed. In 1857, Riemann proved that the solutions of (??) are the only multivalued functions with exactly two linearly independent branches, branching only at $0, 1, \infty$ with exponents as above (cf. [?]). Riemann’s proof assumed that none of the exponent

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differences at a singularity was an integer and proceeded by first computing global monodromy. The above characterization of the hypergeometric equation is the basis of Fuchs' proof of Riemann's theorem (cf. [?]).

In his seminal paper [?], which seeks to determine the values a, b, c for which the hypergeometric function is an algebraic function of x , Schwarz considered the map $x \mapsto w_2(x)/w_1(x)$ where w_1 and w_2 form a base of W , the two dimensional linear space of all solutions of (??). Let $\tilde{Q}$ denote the universal covering space of $Q := \mathbb{P}^1(\mathbb{C}) - \{0, 1, \infty\}$. The map

$$w : x \mapsto w_2(x)/w_1(x) \quad (0.4)$$

is a multivalued map from Q to the projective space $\mathbb{P}(W^*)$ of lines in the dual space W^* of W , i.e. w may be construed as a single-valued map

$$\tilde{w} : \tilde{Q} \rightarrow \mathbb{P}(W^*). \quad (0.5)$$

The fundamental group $\pi_1(Q)$ acts on $\mathbb{P}(W^*)$ ("monodromy action") and the map is $\pi_1(Q)$ -equivariant. Let Γ denote the image of $\pi_1(Q)$ via the monodromy action. For Schwarz's original problem, the question reduces to "when is Γ finite?" Schwarz also solved the problem : when can (??) be inverted to provide a univalued map from an open domain to Q ? When this happens Γ has a fundamental domain for its action on either (i) $\mathbb{P}(W^*)$, (ii) $\mathbb{P}(W^*)$ minus a point, or (iii) a disc in $\mathbb{P}(W^*)$, and is consequently discrete in $\text{PGL}_2(\mathbb{C})$.

It is case (iii) of this latter question which Picard generalized in [?] to a two variable analogue of the hypergeometric function. In the more general d variable case, this function is best defined by its integral representation

$$F(x_2, \dots, x_{d+1}) = \int_1^\infty u^{-\mu_0} (u-1)^{-\mu_1} \prod_{i=2}^{d+1} (u-x_i)^{-\mu_i} du. \quad (0.6)$$

Let b_∞ be the order of the pole of the integrand at ∞ . When the sum is extended over all the μ 's, one has $\sum b_\mu = 2$. In this introduction, we assume that none of the μ_i is an integer.

The function G obtained from F by holding fixed all variables but one was investigated by Pochhammer [?]. He formed the $(d+1)$ -order linear differential equation satisfied by G and characterized its solutions as the multivalued functions with exactly $d+1$ linearly independent branches and ramification of a prescribed type. At each finite ramification point, d branches of G are holomorphic, and at ∞ the same holds true after multiplying G by a suitable power of the variable.

The function F defined by (??) has a power series expansion

$$\sum_{\substack{n_i \geq 0 \\ 2 \leq i \leq d+1}} \frac{\prod_{i=2}^{d+1} (\mu_i, n_i) \cdot (1 - \sum \mu_i, \sum n_i)}{(1 - \mu_0, \sum n_i) \cdot (\sum n_i)!} \prod_{i=2}^{d+1} x_i^{n_i}. \quad (0.7)$$

For $d = 2$, this series has been investigated by Appell [?] who showed that it satisfies a system of linear partial differential equations expressing each second derivative of F in terms of its first derivatives. In [?] Picard characterized the solutions of this system as the multivalued functions of two variables x, y with exactly three linearly independent branches and with ramification of a prescribed type along the seven lines x or $y = 0, 1, \infty$, $x = y$. The function F is the only solution holomorphic at $(0, 0)$.

For d arbitrary, the series (??) has been investigated by Lauricella [?], and Terada [?] obtained results parallel to those of Picard [?].

The existence of differential or partial differential equations of Fuchsian type as above is to be expected if one considers (??) as a period integral. The equations are satisfied not only by F but also by any integral of the same integrand taken from g to h where g and h are in $\{\infty, 0, 1, x_2, \dots, x_{d+1}\}$. There are $d + 1$ linearly independent such integrals, and, following Schwarz, it is natural to take them as the projective coordinates of a point in projective d -space $\mathbb{P}^d$. This yields a map

$$\tilde{Q} \rightarrow \mathbb{P}^d \quad (4'')$$

where $\tilde{Q}$ is the universal covering of the space $Q \subset (\mathbb{P}^1)^d$ defined as

$$Q = \{(x_i) \mid x_i \neq 0, 1, \infty \text{ and } x_i \neq x_j \text{ for } i \neq j\}.$$

The action of $\pi_1(Q)$ on $\mathbb{P}^d$ is called the monodromy action. The map (4'') is $\pi_1(Q)$ -equivariant. Let Γ denote the image of $\pi_1(Q)$ in $\mathrm{PGL}(d + 1, \mathbb{C})$. In [?] and [?] Picard gives a criterion for the multivalued map $w : \tilde{Q} \rightarrow \mathbb{P}^d$ ($d = 2$) defined by (4'') to invert as a univalued map from a ball in $\mathbb{P}^d$ to a partial compactification of Q . When this happens, Γ is discrete in a $\mathrm{PU}(1, d)$ subgroup of $\mathrm{PGL}(d + 1, \mathbb{C})$.

As pointed out in [?] the proof of discreteness of Γ that Picard sketches in [?] leads to an obstacle and is inadequate. Our first objective in this paper is to give a correct proof for $d = 2$ and also for general d . In order to carry this out in modern concepts, we need only deal with rational μ_i and multivalued integrands on $\mathbb{P}^1$ which are single valued on a finite ramified covering of $\mathbb{P}^1$. However, in order to provide a framework for dealing with arbitrary parameters μ_i , we have introduced local systems on $\mathbb{P}^1 - \{\infty, 0, 1, x_2, \dots, x_{d+1}\}$. This necessitates an *ab ovo* development of holomorphic cohomology in order to validate the corresponding Hodge decomposition. In the end, this viewpoint does have the advantage of directness.

The two main theorems of this paper are Theorems (11.4) and (12.11). The first says in effect : If $0 < \mu_i < 1$ for all i ($0 \leq i \leq d + 1$ or $i = \infty$) and $(1 - \mu_i - \mu_j)^{-1}$ is an integer for all $i \neq j$ such that $\mu_i + \mu_j < 1$, then the monodromy group of the hypergeometric function (??) is a lattice in the projective unitary group $\mathrm{PU}(1, d)$, i.e. is discrete and of finite covolume.

Theorem (12.11) combined with (12.9) says in effect : For each non-arithmetic lattice arising in Theorem (11.4), there is an algebraic family defined over $\mathbb{Q}$ of algebraic curves X whose monodromy group is a subgroup of $\mathrm{Aut} H^1(X_0, \mathbb{Z})$ which is not of finite covolume in its Zariski-closure.

The example 15 of §(14.3) corresponds for instance to the family of curves, depending on the parameters x, y , with equation

$$U_1^{19} = U_0^6(U - 1)^5(U - x)^4(U - y)^4$$

(a cyclic covering of order 19 of $\mathbb{P}^1$).

The integrality condition that $(1 - \mu_i - \mu_j)^{-1}$ be an integer ensures that the key map of this paper is

'etale in codimension one. A model situation in which integrality is used in such a way is the following. Let D be the unit disc, $D^* := D - \{0\}$, and $\zeta = r/s$ a rational number > 0 , written as a reduced fraction. Let $\tilde{D}^*$ be the finite covering of D^* on which the multivalued function $z \mapsto z^\zeta$ is defined. We complete it to the ramified covering $\tilde{D} = \tilde{D}^* \cup \{0\}$ of D .

The multivalued function $z^{1/s}$ on D^* “is” a uniformizing parameter for $\tilde{D}$ at 0. The map $z \mapsto z^\zeta$ from $\tilde{D}$ to D is
 ’etale at 0 (i.e.

’etale in codimension 1) if and only if ζ^{-1} is an integer. Indeed $z^\zeta = (z^{1/s})^r$. For a description of how this enters the proof, we refer to the comments after (3.11).

The lists in Section 14 provide examples of non-arithmetic lattices in $\mathrm{PU}(1, d)$ for $d = 2, 3$ —both cocompact and non-cocompact for $d = 2$. For $d > 5$, Theorem (11.4) yields no lattice at all, by (14.2).

The case $d = 1$ is treated in (12.5.5) and (12.6.3). The lattices arising from Theorem (11.4) in this case are the triangle groups (p, q, r) generated by rotations through double the angles with centers the vertex of a geodesic triangle in the Poincaré disc, with angles $\pi/p, \pi/q, \pi/r$ when p, q, r are positive integers (or ∞) satisfying

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \leq 1.$$

These lattices are described in Fricke-Klein [?] and are infinite in number.

However, the number of arithmetic lattices arising is finite. An explicit list can be deduced from results of Takeuchi on arithmetic triangle groups [?].

In the final Section 15, we make some comments about the list of 102 solutions of R. Levavasseur in [?] to Picard’s integrality conditions for discreteness in the case $d = 2$.

In the paper “Generalized Picard lattices arising from half-integral conditions”, printed after this one, the second named author is able to relax the integrality condition on $(1 - \mu_i - \mu_j)^{-1}$ and obtains lattices for $d < 9$ [?].

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Notations

P	a complex projective line
N	an integer ≥ 0
S	a set of N points of P
$\alpha = (\alpha_s)_{s \in S}$	a family of complex numbers indexed by S
L	rank one local system
$H^*(P - S, L)$	cohomology with coefficients in L
$H_c^*(P - S, L)$	cohomology with compact support
M	space of injective maps $m : S \rightarrow P$
Q	quotient $\mathrm{PGL}(2)$
$\backslash M$	
$B(\alpha)_M$	flat projective space bundle
$\tilde{w}$	holomorphic section
Γ	monodromy group
Q_{st}	stable compactification
Q_{sst}	semi-stable compactification
$B^+(\alpha)$	complex ball bundle
INT	integrality condition

1 Cohomology of a rank one local system on a punctured projective line

1.1

Let us start with the following data :

- P : a complex projective line ;
- N : an integer ≥ 0 ;
- S : a set of N points of P ;
- $\alpha = (\alpha_s)_{s \in S}$: a family of complex numbers indexed by S , satisfying the condition $\prod_{s \in S} \alpha_s = 1$.

We will be mainly interested in the case when $N > 3$ and none of the α_s is 1.

If a base point $o \in P - S$ is given, the functor “fibre at o ” is an equivalence of the category of complex local systems (= flat vector bundles) on $P - S$ with that of complex vector spaces provided with an action of $\pi_1(P - S, o)$. In particular, rank one local systems correspond to homomorphisms $\mathbb{H}_1(P - S) = \pi_1(P - S, o)^{\mathrm{ab}} \rightarrow \mathbb{C}^*$. The group $\mathbb{H}_1$ is generated by small positive loops γ_s around each $s \in S$, with the relation $\prod \gamma_s = 1$ as the only relation. Up to isomorphism, there is hence a unique one dimensional complex local system L on $P - S$ such that the monodromy of L around each $s \in S$ is multiplication by α_s , or, as we will say, of monodromy α . By definition, if v_0 is in the fibre of L at a point x near $s \in S$, and if we let x turn counter clockwise once around s , and push v_0 horizontally along the path of x , then when coming back, v_0 becomes $\alpha_s \cdot v_0$. In other words, in term of a local coordinate z centered at s , and of a multivalued section $e(z)$ of L defined in a neighborhood of s , one has

$$e(\exp(2\pi i u) \cdot z)_{u=1} = \alpha_s \cdot e(\exp(2\pi i u) \cdot z)_{u=0}.$$

The complex local system L has automorphisms : $\text{Aut}(L) = \mathbb{C}^*$, with $\nu \in \mathbb{C}^*$ corresponding to multiplication by the scalar ν . Because of them, even though the isomorphism class of L is uniquely determined by the α_s , L is not determined up to unique isomorphism.

Let us fix one L . In (2.2)–(2.20), we will review some of the descriptions of the cohomology of $P - S$ with coefficients in L .

1.2 Combinatorial description

Let us fix a triangulation of $P - S$. One can then identify $H^*(P - S, L)$ with the cohomology of the complex of L -valued cochains of : cochains c for which the value of c on an oriented simplex σ is a horizontal section of L on σ (thus $c(\sigma) \in H^0(\sigma, L)$). To make sense of the formula $(dc)(\tau) = c(\partial\tau)$, one uses that a horizontal section of L on a face of τ extends uniquely as a horizontal section on τ . This complex is the dual of the complex of chains of with coefficients in the dual local system $L^\vee$: chains C which are finite sums $\sum c_\sigma \cdot \sigma$ with $c_\sigma \in H^0(\sigma, L^\vee)$ and σ an oriented simplex of . The sign rule $\langle dc, C \rangle = -\langle c, \partial C \rangle$ is however unusual for duality of complexes. We write $H_*(P - S, L^\vee)$ for the corresponding homology.

The cohomology with compact support $H_c^*(P - S, L)$ is the cohomology of the complex of compactly supported L -valued cochains of . The dual complex is the complex of locally finite chains with coefficient in $L^\vee$: chains C which are possibly infinite “sums” $\sum c_\sigma \cdot \sigma$. We write $H_*^{\text{lf}}(P - S, L^\vee)$ for the corresponding homology.

Let X be obtained from P by deleting a small open disc around each $s \in S$. “Small” means “small enough”. What matters here is that the closure of the discs be disjoint. A homotopy argument shows that ($\dot{X}$ denoting $X - \partial X$)

$$H^*(P - S, L) \xrightarrow{\sim} H^*(X, L) \xrightarrow{\sim} H^*(\dot{X}, L)$$

and that $H^*(X \bmod \partial X, L) \xrightarrow{\sim} H_c^*(\dot{X}, L) \xrightarrow{\sim} H_c^*(P - S, L)$.

To compute cohomology, one can hence use a triangulation of X , instead of one of $P - S$. As X is compact, the triangulation is finite, and this combinatorial description makes clear that the Euler-Poincaré characteristic

$$\chi(P - S, L) := \sum (-1)^i \dim H^i(P - S, L)$$

(resp. $\chi_c(P - S, L) := \sum (-1)^i \dim H_c^i(P - S, L)$) is independent of L . Each H^i (resp. H_c^i) is indeed expressed as the i -th cohomology group of a finite dimensional chain complex K , and $\dim K^i$ is independent of L . In the case of cohomology with compact support, the additivity of χ_c (deduced from the long exact sequence $\cdots \rightarrow H_c^i(P - S, \mathbb{C}) \rightarrow H^i(P, \mathbb{C}) \rightarrow H^i(S, \mathbb{C}) \rightarrow \cdots$) gives

$$\chi_c(P - S, L) = \chi_c(P - S, \mathbb{C}) = \chi_c(P) - \chi_c(S) = 2 - N. \quad (1.1)$$

For an algebraic variety Y , one always has $\chi_c(Y) = \chi(Y)$. In the case at hand ($Y = P - S$), this can be deduced from Poincaré duality. One gets

$$\chi(P - S, L) = 2 - N. \quad (1.2)$$

1.3 De Rham (C^∞) description

$H^*(P - S, L)$ is the cohomology of the de Rham complex of L -valued C^∞ differential forms on $P - S$, and $H_c^*(P - S, L)$ that of the subcomplex of compactly supported forms.

If the triangulation of (2.2) is smooth, integration $\omega \mapsto c$, $c(\sigma) = \int_{\sigma} \omega$ is defined. It is a map of complexes from the de Rham complex to the cochain complex (both with or without support condition), inducing an isomorphism on cohomology.

The Poincar

'e duality pairing obtained by integration on $P - S : \alpha, \beta \mapsto \int_{P-S} \alpha \wedge \beta$ induces a perfect pairing

$$H^i(P - S, L) \otimes H_c^{2-i}(P - S, L^\vee) \rightarrow \mathbb{C},$$

the composite of cup product with value in $H_c^2(P - S, \mathbb{C})$, and of the trace, or integration, map $H_c^2(P - S, \mathbb{C}) \rightarrow \mathbb{C}$.

If at least one of the α_s is not 1, there can be no global horizontal section : $H^0(P - S, L) = 0$, and a fortiori none with compact support : $H_c^0(P - S, L) = 0$ (for this, $i = 0$ suffices). The same applies to the dual $L^\vee$ of L , with monodromy α^{-1} , and Poincar 'e duality gives vanishing for H^2 and H_c^2 . The Euler-Poincar 'e characteristic being known ((??) and (??)), we get

Proposition 1.1. *If $\alpha_s \neq 1$ for at least one s , then $H^i(P - S, L)$ and $H_c^i(P - S, L)$ vanish for $i \neq 1$, and*

$$\dim H^1(P - S, L) = \dim H_c^1(P - S, L) = N - 2.$$

1.4 Currents

Instead of using the C^∞ de Rham complex, consisting of forms written in local coordinates as $\sum_i f_I dx_{i_1} \wedge \cdots \wedge dx_{i_k} \cdot e$, with e an horizontal section of L and f a C^∞ function, one can as well use the complex of currents, where f is allowed to be a generalized function (distribution). This complex can be used to express Poincar 'e duality as a cap-product isomorphism between homology and cohomology : for C a $L^\vee$ -valued chain, there is a unique current $\delta(C)$ such that $\int_C \omega = \int \delta(C) \wedge \omega$; one has $d\delta(C) = (-1)^k \delta(\partial C)$, and the map $C \mapsto \delta(C)$ provides isomorphisms

$$H_k(P - S, L^\vee) \xrightarrow{\sim} H_c^{2-k}(P - S, L) \quad \text{and} \quad H_k^{\text{lf}}(P - S, L^\vee) \xrightarrow{\sim} H^{2-k}(P - S, L).$$

It is often more convenient to use currents than the chains of a fixed triangulation.

If β is a rectifiable proper map from an open, semi-open or closed interval I to $P - S$, and for $e \in H^0(I, \beta^* L^\vee)$, we let $(e \cdot \beta)$ be the $L^\vee$ -valued current for which

$$\int (e \cdot \beta) \wedge \omega = \int_I \langle e, \beta^* \omega \rangle.$$

If $\beta : [0, 1] \rightarrow P$ maps 0 and 1 to S , and $]0, 1[$ to $P - S$, then for $e \in H^0(]0, 1[, \beta^* L^\vee)$, $e \cdot \beta$ is a cycle and as such defines an homology class in $H_1^{\text{lf}}(P - S, L^\vee)$. We will use such cycles to construct convenient bases of $H_1^{\text{lf}}(P - S, L^\vee)$.

1.5

Assume the following is given :

- a) a partition of S into two subsets S_1 and S_2 ;
- b) trees T_1 and T_2 (a tree is a contractible CW complex of dimension ≤ 1), and an embedding $\beta : T_1 \amalg T_2 \hookrightarrow P$, mapping the set of vertices of T_1 (resp. T_2) onto S_1 (resp. S_2);

c) for each (open) edge a of T_1 or T_2 , an orientation of a , and $t(a) \in H^0(a, \beta^* L^\vee)$.
For each edge a , $t(a) \cdot \beta|_a$ is then a locally finite cycle on $P - S$, with coefficient in L .

Proposition 1.2. *If $\prod_{s \in S_i} \alpha_s \neq 1$, the elements $t(a) \cdot \beta|_a$, for a an edge of T_1 or T_2 , form a basis of $H_1^{\text{lf}}(P - S, L^\vee)$.*

Démonstration. Let T'_i be the disjoint union of the (open) edges of T_i . We have $(P - S) - \beta(T'_1 \amalg T'_2) = P - \beta(T_1 \amalg T_2)$, hence a long exact sequence

$$\cdots \rightarrow H_k^{\text{lf}}(T'_1 \amalg T'_2, \beta^* L^\vee) \rightarrow H_k^{\text{lf}}(P - S, L) \rightarrow H_k(P - \beta(T_1 \amalg T_2), L^\vee) \rightarrow \cdots$$

The group $H_1^{\text{lf}}(T'_1 \amalg T'_2, \beta^* L^\vee)$ has as basis the $t(a) \cdot a$. It remains to be proved that $H_1^{\text{lf}}(P - \beta(T_1 \amalg T_2), L) = 0$.

The space $P - \beta(T_1) - \beta(T_2)$ is the complement of two disjoint trees in S^2 . It is essentially an annulus. The monodromy of “turning around T_1 ” is the product of the α_s ($s \in S_1$). In $P - S$, a loop around T_1 can indeed be homotoped to a product of loops around each $s \in S_1$.

The group $H_1^{\text{lf}}(P - \beta(T_1 \amalg T_2), L^\vee)$ is dual to $H_c^1(P - \beta(T_1 \amalg T_2), L)$, which is the Poincaré dual of $H^{2-1}(P - \beta(T_1 \amalg T_2), L^\vee)$. One can homotope $P - \beta(T_1 \amalg T_2)$ to S^1 , and the group becomes $H^{2-1}(S^1, L^\vee)$, for $L^\vee$ a non trivial local system on S^1 . It remains to use the

Lemme 1.3. *If L is a non trivial rank one local system on S^1 , then $H^*(S^1, L) = 0$.*

The H^0 is 0 by non triviality, and $\chi(S^1, L) = \chi(S^1) = 0$. □

Remarque 1.4. As $T_1 \amalg T_2$ has $N-2$ edges, (2.5.2) reproves that $\dim H_c^1(P - S, L) = N-2$ when the α_s are not all 1.

1.6 Sheaf cohomology

We will always identify a local system L with its sheaf of locally constant sections. Let j denote the inclusion of $P - S$ in P . Then, $H_c^*(P - S, L)$ is the cohomology of P with coefficients in $j_! L$, the extension of L by 0. It is the hypercohomology on P of any complex of sheaves K with ${}^0(K) = j_! L$ and ${}^i(K) = 0$ for $i \neq 0$. On the other hand, if L^* is any resolution of L whose components are acyclic for j_* ($R^q j_* L^k = 0$ for $q > 0$), then $H^*(P - S, L)$ is the hypercohomology on P of $j_* L^*$. To prove it, one reduces to the case where L^* (and hence $j_* L^*$) is soft, one has $\Gamma(P - S, L^*) = \Gamma(P, j_* L^*)$, and

$$H^*(P - S, L) := H^* \Gamma(P - S, L^*) = H^* \Gamma(P, j_* L^*) =: H^*(P, j_* L^*).$$

Proposition 1.5. *If all α_s are different from 1, then*

$$H_c^*(P - S, L) \xrightarrow{\sim} H^*(P - S, L).$$

Démonstration. Let L^* be a soft resolution of L , for instance the L -valued C^∞ de Rham complex of sheaves. It suffices to check that $j_! L^*$ is a resolution of $j_! L$, and this results from the local fact that for D any small disc around $s \in S$, one has $H^* \Gamma(D - s, L^*) = H^*(D - s, L) = 0$. To check this vanishing, one can replace $D - s$ by a homotopic S^1 and apply (2.5.2). □

From the point of view of homology, this implies that a locally finite 1-cycle with coefficient in $L^\vee$ is homologous to a compactly supported one. Here is a drawing of how it can be done :

(figure : a locally finite cycle being pushed to a compactly supported one)

This is reminiscent of the arguments used in Whittaker and Watson (12.22) (Hankel's expression of $\Gamma(z)$ as a contour integral) or (12.43) [?].

1.7 The de Rham (holomorphic) description

The holomorphic L -valued de Rham complex $\Omega^*(L) : \mathcal{O}(L) \rightarrow \Omega^1(L)$ is a resolution of L on $P - S$. One can hence interpret $H^*(P - S, L)$ as the hypercohomology on $P - S$ of $\Omega^*(L)$. As $S \neq \emptyset$, $P - S$ is Stein, hence $H^q(P - S, \Omega^p(L)) = 0$ for $q > 0$, and this gives

$$H^*(P - S, L) = H^*\Gamma(P - S, \Omega^*(L)). \quad (1.3)$$

Similarly, $R^q j_* \Omega^p(L) = 0$ for $q > 0$ hence

$$H^*(P - S, L) = H^*(P, j_* \Omega^*(L))$$

(hypercohomology of P , with coefficients in the complex $j_* \Omega^*(L)$; see (2.6)).

After the preliminaries (2.8), (2.9), we will show how to replace $j_* \Omega^*(L)$ by smaller complexes of sheaves.

1.8

We will often describe a section of $\mathcal{O}(L)$ on a connected open set U as the product of a multivalued function with a multivalued section of L . This has the following meaning : U should be provided with a base point 0, and a multivalued section of a sheaf ($\mathcal{O}$, or L , ...) is a section of the pull back of on the universal covering $(\tilde{U}, 0)$ of $(U, 0)$. Products are taken on $\tilde{U}$. A multivalued section, which is the pull-back of a section on U , is then identified with that section. The role of the base point can be played by a contractible subset of U . A section of L at 0 extends uniquely to a multivalued section and will be denoted by the same symbol. A multivalued section of $\mathcal{O}$ is determined by its germ at 0, and will be denoted by the same symbol. As multivalued sections of $\mathcal{O}$, we will mainly use products $\prod g^{\mu_i}$. When g is not real and negative, the principal determination of g^μ is $\exp(2\pi i \mu \log g)$, with $|\operatorname{Im} \log g| < \pi$. As a rule, g^μ will denote the multivalued function whose germ at the base contractible set is the principal determination when g^μ is not real and negative on it. When the determination used is irrelevant, we will not take the trouble of specifying which one we mean.

1.9

Fix $s \in S$ and let z be a local coordinate, which we take to be an isomorphism, carrying s to 0, of a small neighborhood D of s with a disc in $\mathbb{C}$ centered at 0. If $\mu \in \mathbb{C}$ is such that $\alpha_s = \exp(2\pi i \mu)$, the monodromy of $z^{-\mu}$ around s is the inverse of that of a horizontal section of L . Any section u of $\mathcal{O}(L)$ (resp. $\Omega^1(L)$) on D^* can hence be written $u = z^{-\mu} \cdot e \cdot f$ (resp. $u = z^{-\mu} \cdot e \cdot f dz$) with e a non zero multivalued section of L and f a

holomorphic function on D^* . We define u to be meromorphic at s iff f is, and we define its valuation by

$$v_s(u) = v_0(f).$$

These definitions are independent of the choices of local coordinate and of μ .

1.10 The de Rham (meromorphic at S) description

Let us write $j_*^m \Omega^*(L)$ for the subcomplex of $j_* \Omega^*(L)$ of meromorphic forms. A local computation around each $s \in S$ shows that the inclusion of $j_*^m \Omega^*(L)$ in $j_* \Omega^*(L)$ induces an isomorphism on the cohomology sheaves. This implies that

$$H^*(P, j_*^m \Omega^*(L)) \xrightarrow{\sim} H^*(P, j_* \Omega^*(L)) = H^*(P - S, L). \quad (1.4)$$

It follows from (2.11) below that $j_*^m \Omega^p(L)$ is an inductive limit of line bundles whose degrees tends to ∞ . From this one concludes first that $H^q(P, j_*^m \Omega^p(L)) = 0$ for $q > 0$, and then that $H^*(P, j_*^m \Omega^*(L))$ is simply $H^* \Gamma(P, j_*^m \Omega^*(L))$, the cohomology of the complex of L -valued forms holomorphic on $P - S$ and meromorphic at each $s \in S$:

$$H^*(P - S, L) = H^* \Gamma(P, j_*^m \Omega^*(L)). \quad (1.5)$$

1.11

Let $(\mu_s)_{s \in S}$ be a family of complex numbers, such that $\exp(2\pi i \mu_s) = \alpha_s$. We define the line bundle $\mathcal{O}(\sum \mu_s \cdot s)(L)$ as the subsheaf of $j_*^m \mathcal{O}(L)$ whose local holomorphic sections are the local sections u of $j_*^m \mathcal{O}(L)$ such that for $s \in S$, the integer $v_s(u) + \mu_s$ is > 0 (in short : $v_s(u) \geq -\mu_s$). With the local coordinate notation (2.9), $z^{-\mu_s} \cdot e$ determines an invertible section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ near $s \in S$. The degree of a meromorphic section u of a line bundle at a point x is the order of the zero (or minus the order of the pole) of u at x ; for z a local coordinate centered at x , it is the supremum of the integers n such that $z^{-n}u$ is a holomorphic section of at x . The degree of is the sum of the degrees at all points of any non zero meromorphic section of . If u is a meromorphic section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ and $x \notin S$, one has $\deg_x(u) = v_x(u)$. If one defines $\mu_x = 0$ for $x \in P - S$, the same holds at any $x \in P$.

Proposition 1.6. *The line bundle $\mathcal{O}(\sum \mu_s \cdot s)(L)$ is of degree $\sum \mu_s$.*

Démonstration. We may assume $P = \mathbb{P}^1$ and $\infty \notin S$. Let e be a multivalued horizontal section of L . The product

$$u = \prod_{s \in S} (z - s)^{-\mu_s} \cdot e$$

is an invertible section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ on the affine line. At ∞ , u is meromorphic, of valuation $\sum \mu_s$. The degree of $\mathcal{O}(\sum \mu_s \cdot s)(L)$, equal to the sum at all $x \in P$ of the degree of any non zero meromorphic section, is hence $\sum \mu_s$. $\square$

We define $\Omega^1(\sum \mu_s \cdot s)(L) := \mathcal{O}(\sum \mu_s \cdot s)(L) \otimes \Omega^1$. It results from (2.11.1) that

$$\deg \Omega^1(\sum \mu_s \cdot s)(L) = -2 + \sum \mu_s. \quad (1.6)$$

In particular, if $\sum \mu_s = 2$, the line bundle $\Omega^1(\sum \mu_s \cdot s)(L)$ is of degree 0, hence isomorphic to $\mathcal{O}$, the trivial line bundle.

Corollaire 1.7. *If $\sum \mu_s = 2$, there is up to a constant factor one and only one non zero form $\omega \in \Gamma(P, j_*^m \Omega^1(L))$ whose valuation at $s \in S$ is $\geq -\mu_s$. One has $v_s(\omega) = -\mu_s$, and ω is invertible on $P - S$.*

For $P = \mathbb{P}^1$, $\infty \notin S$, and e a multivalued horizontal section of L , one has (up to a factor) $\omega = \prod_{s \in S} (z - s)^{-\mu_s} \cdot e \, dz$. If $\infty \in S$, one has $\omega = \prod_{s \in S - \{\infty\}} (z - s)^{-\mu_s} \cdot e \, dz$.

Proposition 1.8. *Assume that $\sum \mu_s = 2$, $N \geq 3$, and that none of the α_s is 1 (i.e. that none of the μ_s is an integer). Then the cohomology class of the form ω of (2.12) is not zero.*

Démonstration. One has $\sum v_s(\omega) = \sum -\mu_s = -2 > 1 - N$. The proposition hence results from the

Proposition 1.9. *Assume that none of the α_s is 1. If a non zero form ω is such that $\sum v_s(\omega) \geq 1 - N$, its cohomology class is not zero.*

Démonstration. By (2.10.2), it suffices to verify that the equation $\omega = du$ has no solution $u \in \Gamma(P, j_*^m \mathcal{O}(L))$.

For any local section u of $j_*^m \mathcal{O}(L)$ near s , one has $v_s(du) > v_s(u) - 1$, with equality if $v_s(u)$ is not 0, or if $v_s(du)$ is not a positive (meaning > 0) integer. The integer case being excluded, a solution u would be a section of $\mathcal{O}(-\sum (v_s(\omega) + 1)s)(L)$, a line bundle of degree $-\sum (v_s(\omega) + 1) \leq -1$, a contradiction. $\square$

$\square$

1.12

Here is how (2.13), (2.14) has to be modified for integral μ 's. As the result will not be needed, except for some historical comments in §15, we will be sketchy.

(2.15.1) When some α_s are 1, (2.6.1) is not true, one has to distinguish between H_c^1 and H^1 , and even to introduce some intermediate groups : for $S' \subset S$, the family $\Phi(S')$ of subsets of $P - S$ closed in $P - S'$ is a family of supports, and $H_{\Phi(S')}^*(P - S, L)$ is defined. For j' the inclusion of $P - S$ into $P - S'$, it is $H^*(P - S', j'_! L)$. For $T \subset S$, define $T(1) := \{s \in T \mid \alpha_s = 1\}$. The proof of (2.6.1) shows that

$$H_{\Phi(S')}^*(P - S, L) \xrightarrow{\sim} H_{|S'|}^*(P - S, L).$$

These groups are also the hypercohomology groups, on P , of the subcomplex of $j_*^m \Omega^*(L)$ consisting of the u in $j_*^m \mathcal{O}(L)$ and of the ω in $j_*^m \Omega^1(L)$ such that $v_s(\omega) > 0$ and $v_s(u) > 0$ for $s \in (S - S')(1)$. If the μ_s are such that $\alpha_s = \exp(2\pi i \mu_s)$, and that $S'(1)$ is the set of s for which μ_s is integral > 0 , the groups $H_{\Phi(S')}^*(P - S, L)$ are also the hypercohomology groups of the sub-complex $\mathcal{O}(\sum (\mu_s - 1)s)(L) \rightarrow \Omega^1(\sum \mu_s \cdot s)(L)$ of $j_*^m \Omega^*(L)$.

(2.15.2) If $\omega \in \Gamma(P, j_*^m \Omega^1(L))$, and if all s with $\alpha_s = 1$ and $v_s(\omega) < 0$ are in S' , then ω defines a cohomology class in $H_{\Phi(S')}^*(P - S, L)$. If in addition S' contains none of the s for which simultaneously $\alpha_s = 1$ and $v_s(\omega) \geq 0$, and if further $\sum v_s(\omega) > 1 - N$ (which amounts to ω having at most $N - 3$ zeros, counted with multiplicities, on $P - S$), the proof of (2.14) still shows that the class of ω in $H_{\Phi(S')}^*(P - S, L)$ is not zero. This applies to the form ω of (2.12), with $S' = S$, provided $N > 3$ and $\mu_s > 0$ whenever $\mu_s \in \mathbb{Z}$.

Remarque 1.10. Let us assume that none of the α_s is 1. If ω is an holomorphic section of $\Omega^1(L)$ on $P - S$, by (2.6.1), the class $[\omega]$ of ω in $H^1(P - S, L)$ is the image of a unique class, again denoted $[\omega]$, in $H_c^1(P - S, L)$. For any locally finite cycle or current C with coefficient in $L^\vee$, $\langle [C], [\omega] \rangle$ is defined : $H_1^{\text{lf}}(P - S, L^\vee)$ is indeed paired with $H_c^1(P - S, L)$. We want to prove that

$$\langle [C], [\omega] \rangle = \int_C \omega.$$

This is not completely obvious, especially since the integral may be divergent. The problem is rather to understand which value the cohomological formalism attaches to the integral.

1st answer : One replaces C by a finite homologous cycle C' , as pictured in (2.6), and $\langle [C], [\omega] \rangle = \int_{C'} \omega$.

2nd answer : If D_s is a small open disk around $s \in S$, one has $H^1(D_s - s, L) = 0$ (cf. (2.6)). On $D_s^* := D_s - s$, ω is hence of the form du_s , for a unique section u_s of $\mathcal{O}(L)$ on D_s^* . Let φ_s be a compactly supported C^∞ function on D_s , equal to 1 in a neighborhood of s . The C^∞ 1-form $\omega - \sum d(\varphi_s u_s)$ is cohomologous to ω and is compactly supported. One has hence

$$\langle [C], [\omega] \rangle = \int_C \omega - \sum \int_C d(\varphi_s u_s).$$

It is sometimes more convenient to take for φ_s the (discontinuous) characteristic function of D_s . The current $\omega - \sum d(\varphi_s u_s)$ is again compactly supported and cohomologous to ω .

These formulae can also be used to define the integral of ω on a locally finite chain or current, which is a cycle only near each $s \in S$. Given for instance a path $\beta : [0, 1] \rightarrow P$, with $\beta([0, 1])$ in $P - S$ and $\beta(1) = s \in S$, and given $e \in H^0([0, 1[, \beta^* L^\vee)$, the formula for the “finite part” $\int_{[0, 1]} e \cdot \beta^* \omega - d(\varphi_s u_s)$ can be rewritten

$$\text{P.V.} \int_{[0, 1]} \langle e, \beta^* \omega \rangle - \langle e(1 - \varepsilon), \beta^* u_s(1 - \varepsilon) \rangle \quad (\varepsilon \rightarrow 0).$$

For a meromorphic ω , this agrees with Hadamard’s “finite part” of a divergent integral.

Exemple 1.11. We take $P = \mathbb{P}^1$, $S = \{0, 1, \infty\}$ and we assume that none of the μ_s is an integer, and that $\sum \mu_s = 2$. For $\lambda_s = 1 - \mu_s$, this means $\sum \lambda_s = 1$. We normalize L by fixing a section e of it on $]0, 1[$, and we let $e^\vee$ be the dual base of $L^\vee$ on $]0, 1[$. Take

$$\omega = z^{\lambda_0 - 1} (1 - z)^{\lambda_1 - 1} e dz;$$

the “principal” determination of $z^{\lambda_0 - 1}$ and of $(1 - z)^{\lambda_1 - 1}$ on $]0, 1[$ is used (cf. (2.8)). By (2.5.1), the homology $H_1^{\text{lf}}(P - S, L^\vee)$ is one-dimensional, generated by the current $e^\vee \cdot]0, 1[$. The content of (2.13) here is that

$$\text{P.V.} \int_0^1 z^{\lambda_0 - 1} (1 - z)^{\lambda_1 - 1} dz \neq 0. \quad (1.7)$$

The generalization (2.15) of (2.13) can be used to extend this to the case when λ_0 and λ_1 , if integers, are > 0 , and $\lambda_\infty = 1 - \lambda_0 - \lambda_1$, if an integer, is < 0 .

Of course, (2.17.1) is easy to deduce from the formula for the B-function in terms of Γ -functions.

1.13

We now assume that each α_s is of absolute value 1, and not equal to 1. There is then a horizontal positive Hermitian structure $(,)$ on L . Let us choose one. One can view $(,)$ as a perfect pairing between L and the complex conjugate local system $\bar{L}$. As such, it induces a perfect pairing

$$\psi_0 : H_c^1(P - S, L) \otimes \overline{H_c^1(P - S, L)} \rightarrow H_c^2(P - S, \mathbb{C}) = \mathbb{C}$$

((2.3), (2.6.1)). The vector space $\overline{H_c^1(P - S, L)}$ is the complex conjugate of $H_c^1(P - S, L)$, and $\psi_0(u, \bar{v})$ defines a non degenerate skew-Hermitian form ψ on $H_c^1(P - S, L)$. Anticommutativity of the cup product shows that

$$\psi(v, u) = -\overline{\psi(u, v)}, \quad \text{i.e.} \quad \psi(u, v) = -\overline{\psi(v, u)}. \quad (1.8)$$

Writing $(u, v) := \frac{1}{2\pi i} \psi(u, v) = \frac{1}{2\pi i} \psi_0(u, \bar{v})$, we have a Hermitian form

$$(u, v) = \overline{(v, u)}. \quad (1.9)$$

A section ω of $j_*^m \Omega^1(L)$ is said to be of the *first kind* if $v_s(\omega) > -1$ for each $s \in S$. Those are the forms for which the integral $\int_{P-S} \omega \wedge \bar{\omega}$ is convergent. The integrand is defined as follows : if, locally, $\omega_i = e_i \cdot f_i$ ($i = 1, 2$, f_i a 1-form, e_i a section of L), then $\omega_1 \wedge \bar{\omega}_2$ is the $(1, 1)$ -form $(e_1, e_2) f_1 \wedge \bar{f}_2$.

We define $H^{1,0}(P - S, L)$ as the vector space of forms of the first kind in $\Gamma(P, j_*^m \Omega^1(L))$, and $H^{0,1}(P - S, L)$ as the complex conjugate of $H^{1,0}(P - S, L)$. It can be viewed as the space of antiholomorphic L -valued forms, whose complex conjugate is of the first kind. Such forms also define cohomology classes.

Proposition 1.12. *If ω_1 and ω_2 are each either in $H^{1,0}(P - S, L)$ or in $\overline{H^{0,1}(P - S, L)}$, one has, for the corresponding cohomology classes $[\omega_1]$ and $[\omega_2]$,*

$$\psi([\omega_1], [\omega_2]) = \int_{P-S} \omega_1 \wedge \omega_2.$$

Démonstration. Let us assume, for definiteness, that ω_1 is in $H^{1,0}(P - S, L)$. Choose D_s , $(u_s)_{s \in S}$, as in (2.16), and let φ_s be the characteristic function of D_s . It follows from (2.16) that the class of ω_1 in $H_c^1(P - S, L)$ is also the class of the compactly supported current $\omega_1 - \sum d(\varphi_s u_s)$. Hence (2.3)

$$\psi([\omega_1], [\omega_2]) = \psi_0([\omega_1], [\bar{\omega}_2]) = \int_{P-S} (\omega_1 - \sum d(\varphi_s u_s)) \wedge \omega_2.$$

When we let the D_s shrink, the first term converges to $\int_{P-S} \omega_1 \wedge \omega_2$. We want to show that all others tend to 0. When the radius r of D_s tends to zero, the size of u_s on ∂D_s is $O(r^{v_s(u_s)+\varepsilon})$. One has $v_s(u_s) = v_s(\omega_1) + 1$. The forms ω_1 and ω_2 being of the first kind, the size is $O(r^\beta)$, with $\beta > -1$. On the other hand, the length of ∂D_s is $O(r)$; the boundary term is hence $O(r^{\beta+1})$ and tends to zero. $\square$

Proposition 1.13. *Under the assumption (2.18) that $|\alpha_s| = 1$, $\alpha_s \neq 1$ for $s \in S$, the natural map*

$$H^{1,0}(P - S, L) \oplus \overline{H^{0,1}(P - S, L)} \rightarrow H^1(P - S, L) = H_c^1(P - S, L)$$

is an isomorphism. The Hermitian form $(,)$ is positive definite on $H^{1,0}$, negative definite on $H^{0,1}$ and the decomposition is orthogonal.

Démonstration. If ω is in $H^{1,0}(P - S, L)$ (resp. $\overline{H^{0,1}(P - S, L)}$), the integrand $\frac{1}{2\pi i} \omega \wedge \bar{\omega}$ is > 0 (resp. < 0), and vanishes only for $\omega = 0$. If ω_1 and ω_2 are one in $H^{1,0}$ and the other in $H^{0,1}$, the integrand $\frac{1}{2\pi i} \omega_1 \wedge \bar{\omega}_2$ vanishes. This implies that $H^{1,0} \oplus H^{0,1}$ injects into $H^1(P - S, L)$. It remains only to prove the surjectivity of $H^{1,0} \oplus H^{0,1} \rightarrow H^1$.

We will check surjectivity by counting dimensions. Let μ_s be the number between 0 and 1 such that $\alpha_s = \exp(2\pi i \mu_s)$. From the definitions it follows that $H^{1,0}(P - S, L) = \Gamma(P, \Omega^1(\sum \mu_s \cdot s)(L))$. By (2.8), $\deg \Omega^1(\sum \mu_s \cdot s)(L) = -2 + \sum \mu_s$. It is an integer > -2 and hence

$$\dim H^{1,0}(P - S, L) = \deg \Omega^1(\sum \mu_s \cdot s)(L) + 1 = -1 + \sum \mu_s. \quad (1.10)$$

Let us apply this to $\bar{L}$. One has to replace α_s by $\bar{\alpha}_s = \alpha_s^{-1}$ and μ_s by $1 - \mu_s$. One gets

$$\dim H^{0,1}(P - S, L) = -1 + \sum (1 - \mu_s). \quad (1.11)$$

The sum of the dimensions is $-2 + N = N - 2$, as required (2.3.1). $\square$

Corollaire 1.14. *Notation being as above ($\alpha_s = \exp(2\pi i \mu_s)$ with $0 < \mu_s < 1$), the signature of the Hermitian form $(\cdot, \cdot)$ on $H^1(P - S, L)$ is $(\sum \mu_s - 1, \sum (1 - \mu_s) - 1)$.*

Remarque 1.15. (2.20) is a special case of Hodge theory for the cohomology of a curve with values in a polarized variation of Hodge structures : a local system, provided with an horizontal positive Hermitian structure, can be viewed as a polarized variation of type $(0, 0)$. The artificial counting argument above enabled us to shortcut the general theory, for which the reader may consult S. Zucker [?].

1.14

In our applications, the α_s will be roots of unity. When this is the case, (2.20) can also be deduced from the Hodge theory of suitable coverings of P . Let X be an irreducible abelian covering of P , with covering group G , ramified only at S . If, for definiteness, we take $P = \mathbb{P}^1$ and $\infty \notin S$, this means that the function field $\mathbb{C}(X)$ of X is a subextension of the extension $\mathbb{C}(\mathbb{P}^1)((z - s)_{s \in S - \{\infty\}})^{1/d}$ of $\mathbb{C}(z) = \mathbb{C}(\mathbb{P}^1)$, for suitable d .

Let π denote the projection of X onto P . The Galois group $G = \text{Aut}(X/P)$ acts (by transport of structure) on $\pi_* \mathbb{C}$ (by abuse of notation, we also denote by $\mathbb{C}$ the constant sheaf with fiber $\mathbb{C}$), and at a point $z \in P$ the representation of G on $(\pi_* \mathbb{C})_z = H^0(\pi^{-1}(z), \mathbb{C})$ is a regular representation of G . For each character χ of G , let L_χ be the subsheaf of $\pi_* \mathbb{C}$ on which G acts by χ . One has

$$\pi_* \mathbb{C} = \bigoplus_{\chi} L_{\chi}$$

and, outside of S , L_χ is a rank one local system.

Let $g_s \in G$ be the natural generator of the inertial (= decomposition) group at s : if $x(t)$ ($0 < t < 1$) is a path in X , such that $\pi x(t)$ in P stays near s in $P - S$, and turn once around s , then $x(1) = g_s x(0)$. Let d_s be the order of g_s . If t is a local coordinate centered at s , near s , X is a sum of copies of the Riemann surface of t^{1/d_s} . If $\chi(g_s) = 1$, L_χ is a local system at s . If $\chi(g_s) \neq 1$, the fibre of L_χ at s is zero, and the monodromy of L_χ around s is the d_s -th root of unity $\chi(g_s)$.

If we take X to be the largest covering such that G is killed by d , G is the abelian group generated by the g_s , with the only relations $g_s^{d_s} = 1$ and $\prod g_s = 1$: $G = (\mathbb{Z}/d)^S / (\mathbb{Z}/d)$.

If $p \in \mathbb{P}^1$ and $\infty \in S$, X corresponds to the extension $\mathbb{C}(\mathbb{P}^1)((z-s)_{s \in S-\{\infty\}})^{1/d}$ of $\mathbb{C}(\mathbb{P}^1)$. This shows that any system of roots of unity (α_s) , with $\prod \alpha_s = 1$, is of the form $(\chi(g_s))_{s \in S}$ for suitable X and χ .

Let $H^*(X, \mathbb{C})_\chi$ be the subspace of $H^*(X, \mathbb{C})$ on which G acts (by transport of structure) as χ . The isomorphism $H^*(X; \mathbb{C}) = H^*(P, \pi_* \mathbb{C})$ is compatible with the action of G , hence induces isomorphisms

$$H^*(X, \mathbb{C})_\chi = H^*(P, L_\chi). \quad (1.12)$$

On $P-S$, the sheaf $\mathcal{O}(L_\chi)$ is the direct factor $(\pi_* \mathcal{O})_\chi$ of $\pi_* \mathcal{O}$ consisting of the holomorphic functions on X such that $f(g^{-1}x) = \chi(g)f(x)$. Similarly, $\Omega^1(L_\chi) = (\pi_* \Omega^1)_\chi$. Let j' be the inclusion of $X - \pi^{-1}(S)$ in X . On P , one has again

$$j_*^m(L_\chi) = (\pi_* j'^m \mathcal{O})_\chi \quad \text{and} \quad (1.13)$$

$$j_*^m \Omega^1(L_\chi) = (\pi_* j'^m \Omega^1)_\chi. \quad (1.14)$$

Let us assume $\chi(g_s) \neq 1$ for $s \in S$, so that $H^*(P, L_\chi) = H_c^*(P-S, L_\chi)$. Differential forms of the first kind on $P-S$ correspond then, by (2.23.3), to the differential forms of the first kind on X , on which G acts by χ :

$$H^{1,0}(P-S, L_\chi) = \Gamma(X, \Omega_X^1)_\chi = H^{1,0}(X)_\chi$$

and (2.20) is the trace on $H^1(P-S, L_\chi) = H^1(X, \mathbb{C})_\chi$ of the Hodge decomposition of $H^1(X, \mathbb{C})$.

2 Reformulation of Picard's Theorem

2.1

We now let the punctures s move, while the monodromy remains the same. The starting data will be :

- P : a complex projective line ;
- N : an integer, ≥ 3 ;
- S : a set with N elements, for instance $[1, N] \subset \mathbb{N}$;
- $\alpha = (\alpha_s)_{s \in S}$: a family of complex numbers indexed by S , satisfying $\prod \alpha_s = 1$, and such that none of the α_s is 1.

We will be mainly interested in the case $|\alpha_s| = 1$.

For any space X mapping to $\mathbb{P}^S$, we will write P_X for the pull back on X of the universal punctured line

$$P_0 = \{(p, m) \in \mathbb{P} \times \mathbb{P}^S \mid p \notin m(S)\},$$

and π for the projection $P_X \rightarrow X$. Suppose a group H acts freely on X , with quotient Y , and that a lifting of the action to P_X is given. We will then write P_Y for P_X/H and π for the projection $P_Y \rightarrow Y$. The H -equivariant fiber space P_X over X is then the pull back of P_Y over Y .

Let $M \subset \mathbb{P}^S$ be the space of injective maps $m : S \rightarrow \mathbb{P}$. Then,

$$P_M = \{(p, m) \in \mathbb{P} \times M \mid p \notin m(S)\}$$

is a topologically locally trivial fiber space above M . On each fiber $P_m = P - m(S)$, there is a rank one local system with monodromy α , i.e. with monodromy α_s around $m(s)$, for $s \in S$. However, this local system not being unique up to unique isomorphism, one cannot conclude that on P_M there is a unique rank one local system L such that

$$\text{For } m \text{ in } M, \text{ the local system } L_m \text{ induced by } L \text{ on } P_m \text{ has monodromy } \alpha. \quad (2.1)$$

If L satisfies (3.1.1), all local systems satisfying (3.1.1) are of the form $L \otimes \pi^*T$, for T a rank one local system M . We will see in (3.12), (3.13) that there is an L satisfying (3.1.1). If N were 2, the analogous existence assertion would be false for $\alpha \neq \pm 1$.

2.2

Suppose L satisfies (3.1.1). The projection π being topologically locally trivial, the $H^1(P_m, L_m)$ ($= H_c^1(P_m, L_m)$ by (2.6.1)) organize themselves into a local system $R^1\pi_!L$ ($= R^1\pi_*L$) on M .

We will be interested mainly in the corresponding flat projective space bundle $B(\alpha)_M := \mathbb{P}R^1\pi_!L$, the fiber space with fiber at $m \in M$ the projective space $\mathbb{P}H^1(P_m, L_m) := (H^1(P_m, L_m) - \{0\})/\mathbb{C}^*$, and with flat structure, that deduced from $R^1\pi_!L$. For any vector space V and non-zero element $v \in V$, we will denote by $\mathbb{P}V$ the projective space of one-dimensional subspaces of V and by $\mathbb{P}_v$ the image of v in $\mathbb{P}V$.

If $L' = L \otimes \pi^*T$ is another local system satisfying (3.1.1),

$$R^1\pi_!L' = (R^1\pi_!L) \otimes T,$$

and $\mathbb{P}R^1\pi_!L'$ is canonically isomorphic to $\mathbb{P}R^1\pi_!L$, i.e. the flat projective bundle $B(\alpha)_M$ depends only on α . Another explanation of the same fact : the automorphisms of a local system L_m on P_m with monodromy α act trivially on $\mathbb{P}H^1(P_m, L_m)$. Hence, although L_m is unique up to only a non unique isomorphism, $\mathbb{P}H^1(P_m, L_m)$ is defined up to unique isomorphism.

2.3

Locally on M , the existence of L satisfying (3.1.1) poses no problem : if U is a contractible neighborhood of $m \in M$, a local system L_m on P_m with monodromy α extends uniquely to a local system L_U on $\pi^{-1}(U)$ and this extension has, fiber by fiber, monodromy α . The flat projective space bundle $\mathbb{P}R^1\pi_!L_U$ on U is independent (up to unique isomorphism) of the choice of L_U ; hence for variable U , they glue into a flat projective space bundle on the whole of M . This enables us to define $B(\alpha)_M$ without having to assume the existence of a global L .

For $o \in M$, the flat structure of $B(\alpha)_M$ defines an action of $\pi_1(M, o)$ by projective transformations on the fibre $B(\alpha)_o$. A choice of L enables one to lift this projective representation of $\pi_1(M, o)$ on $\mathbb{P}H^1(P_o, L_o)$ to a linear representation on $H^1(P_o, L_o)$.

2.4

Fix a system of complex numbers $(\mu_s)_{s \in S}$ such that $\alpha_s = \exp(2\pi i \mu_s)$, and that $\sum \mu_s = 2$. For each $m \in M$, there is then up to a factor a unique non-zero section ω of $\Omega^1(L_m)$ on P_m , meromorphic on P , and such that $v_s(\omega) > -\mu_s$ ($s \in S$) (2.12). Its cohomology class

is not trivial (2.13). This construction hence defines a section $w_\alpha(m)$ of the projection $B(\alpha)_M \rightarrow M$.

The flat structure of the projective space bundle $B(\alpha)$ induces a structure of holomorphic projective space bundle. With respect to it, we have :

Lemme 2.1. *The section w_α is holomorphic.*

Démonstration. The question is local on M . Fix $m \in M$, and choose on the fibre P_m a system of cycles C_i with coefficient in $L^\vee$ giving rise to a basis of $H_1(P_m, L^\vee)$. They have support in some compact K and, for U a contractible neighborhood of m , small enough so that $K \times U$ is isomorphic to the pull back of $L|_K$ by a unique isomorphism which is the identity above m . This enables us, for $m' \in U$, to consider C_i as a cycle with coefficients in $L^\vee$ on $P_{m'}$; and each linear form $\langle -, C_i \rangle$ on the $H_1(P_{m'}, L_{m'}^\vee)$ ($m' \in U$) is a horizontal linear form on $R^1\pi_!L$. This provides a horizontal system of projective coordinates on $\mathbb{P}R^1\pi_!L$.

Fix a coordinate $z : \mathbb{P} \rightarrow \mathbb{P}^1$, such that, for m' in a possibly smaller neighborhood U of m , $z^{-1}(\infty) \subset P_{m'}$, and let us trivialize L along $z^{-1}(\infty)$ by a section e . We can take (cf. (2.12))

$$\omega = \prod_{s \in S} (z - z(m'(s)))^{-\mu_s} \cdot dz \cdot e,$$

and the projective coordinates of w are the

$$\int_{C_i} \prod_{s \in S} (z - z(m'(s)))^{-\mu_s} \cdot dz \cdot e.$$

This is clearly holomorphic in m' . □

Remarque 2.2. A more general method to get horizontal linear forms on $R^1\pi_!L$ is to start with a C^∞ diffeomorphism $\varphi_{m,m'} : (P_m, L) \xrightarrow{\sim} (P_{m'}, L_{m'})$, and with $L^\vee$ -valued cycle or current C on P_m , and to take $\int_{\varphi_{m,m'}(C)}$. For instance, if C is a path from $m(s)$ to $m(t)$, together with a section e of $L^\vee$ on it, one deforms it with m' so that $C(m')$ remains a path from $m'(s)$ to $m'(t)$.

2.5

The group G of automorphisms of $\mathbb{P}$ is isomorphic to $\mathrm{PGL}(2)$. Its action on $\mathbb{P}$ induces an action on $M \subset \mathbb{P}^S$, on the space P_M , and on the flat projective bundle $B(\alpha)_M$ on M . The section w_α is preserved. The action on M is free. Let $Q = M/G$. The fiber bundle P_M , the flat bundle $B(\alpha)_M$ and w_α being equivariant, they descend to a fiber bundle P_Q , a flat projective space bundle $B(\alpha)_Q$ and a section w_α of $B(\alpha)_Q \rightarrow Q$.

For $\mathbb{P} = \mathbb{P}^1$, and for a, b, c distinct in S , let $M_0 \subset M$ be the space of m such that $m(a) = 0$, $m(b) = 1$, $m(c) = \infty$. One has $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$. The quotient map hence induces an isomorphism $M_0 \xrightarrow{\sim} Q$, and via this isomorphism, $B(\alpha)_Q$ and w_α can be identified with the restriction of $B(\alpha)_M$ and w_α to M_0 .

2.6

Fix a base point $o \in Q$. The section w_α can then be viewed as a multivalued map from Q to $B(\alpha)_o$. More precisely, let $\tilde{Q}$ be the covering of Q corresponding to the kernel

of the monodromy action ρ of $\pi_1(Q, o)$ on $B(\alpha)_o$, i.e. let $\tilde{Q}$ be the quotient by (ρ) of the universal covering of (Q, o) . The pull back of $B(\alpha)_Q$ on $\tilde{Q}$ is $\tilde{Q} \times B(\alpha)_o$, and w_α becomes a $\pi_1(Q, o)$ -equivariant map $\tilde{w}$ from $\tilde{Q}$ to $B(\alpha)_o$.

Proposition 2.3. $\tilde{w} : \tilde{Q} \rightarrow B(\alpha)_o$ is
'etale.

Démonstration. We may take $\mathbb{P} = \mathbb{P}^1$ and replace Q by M_0 , as in (3.7). The problem is local. Take $m \in M_0$, let e be a trivialization of L —near m —on $\mathbb{R}^+$ near ∞ . We may take

$$\omega = \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz \cdot e$$

(cf. (2.8)). Choose cycles C_i as in (3.5, 3.6). Differentiating in m , one has, with $d_{M_0}\omega = \sum \frac{\partial \omega}{\partial m(s)} dm(s)$,

$$d_{M_0}\omega = \sum_s \mu_s \frac{dm(s)}{z - m(s)} \cdot \omega.$$

The spaces $\tilde{Q}$ and $B(\alpha)_o$ have the same dimension $N - 3$. The map $\tilde{w}$ is hence 'etale at m if and only if $d(\tilde{w})$ is injective, i.e. if and only if for no tangent vector $v \neq 0$ at m is the family $\langle \partial_v \omega, C_i \rangle$ proportional to the family of integrals $\langle \omega, C_i \rangle$, i.e. if and only if for no v the cohomology class of $\partial_v \omega$ is proportional to that of ω . This means that the cohomology class of any non-trivial linear combination

$$\sum b_s \frac{\omega}{z - m(s)} \quad (b_s \in \mathbb{C}, \sum b_s = 0)$$

should be non-trivial. If $b_s \neq 0$, $\frac{\omega}{z - m(s)}$ is of valuation exactly $v_s(\omega) - 1$ at s , and hence $\neq 0$. If $b_s = 0$, the valuation is $\geq v_s$. This shows that $\sum b_s \frac{\omega}{z - m(s)} \neq 0$ and that

$$\sum_s v_s \left(\sum_t b_t \frac{\omega}{z - m(t)} \right) = \sum_{s: b_s \neq 0} (v_s(\omega) - 1) + \sum_{s: b_s = 0} v_s(\omega) \geq \sum v_s(\omega) - (N - 2) > 1 - N,$$

so that the non vanishing of the class of $\tilde{\omega}$ results from (2.14). $\square$

2.7

We will now assume that $|\alpha_s| = 1$ and that the numbers μ_s defined by $\alpha_s = \exp(2\pi i \mu_s)$, $0 < \mu_s < 1$, satisfy $\sum \mu_s = 2$.

Locally on M , L admits a horizontal positive definite Hermitian form $(,)$, unique up to a positive real factor (2.18, 3.3). It induces on $R^1\pi_1 L$ a horizontal Hermitian form $(,) = \frac{-1}{2\pi i} \psi$ (2.18), of signature $(1, N - 3)$ (2.21). The vectors on which $(,)$ is positive define a horizontal family of complex balls $B^+(\alpha)_M \subset B(\alpha)_M$, and the section w_α is in $B^+(\alpha)_M$ (2.20) (for the meaning of the term “complex ball”, see §5).

The family $B^+(\alpha)_M$ is equivariant, hence descends to $B^+(\alpha)_Q \subset B(\alpha)_Q$. The 'etale map of (3.9) is a map

$$\tilde{w} : \tilde{Q} \rightarrow B^+(\alpha)_o \tag{2.2}$$

equivariant with respect to the action of $\pi_1(Q, o)$ on $B^+(\alpha)_o$. This action is a morphism

$$\rho : \pi_1(Q, o) \rightarrow \text{PU} \left(H^1(P_o, L_o), \frac{-1}{2\pi i} \psi \right). \tag{2.3}$$

Notation being as above, our main result is the

Théorème 2.4. *Assume that*

(INT) *The numbers μ_s defined by $\alpha_s = \exp(2\pi i \mu_s)$, $0 < \mu_s < 1$, satisfy $\sum \mu_s = 2$ and, for all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer.*

Then, the image Γ of (3.10.2) is a lattice in the projective unitary group

$$\mathrm{PU} \left(H^1(P_o, L_o), \frac{-1}{2\pi i} \psi \right) = \mathrm{PU}(1, N - 3).$$

The theorem will be proved in section 11, where it is restated as theorem (11.4).

The strategy of the proof is presented in the next paragraphs.

In §4, we define a partial compactification Q_{st} of Q . We also define a compactification Q_{sst} , with $Q \subset Q_{\mathrm{st}} \subset Q_{\mathrm{sst}}$ and $Q_{\mathrm{sst}} - Q_{\mathrm{st}}$ finite. In §8, we recall a construction of R. H. Fox to define the completion $\tilde{Q}_{\mathrm{st}}$ (resp. $\tilde{Q}_{\mathrm{sst}}$) of $\tilde{Q}$ over Q_{st} (resp. Q_{sst}). When the condition (INT) is satisfied, one can show that each point y of Q_{st} admits open neighborhoods U such that the inverse image of U in $\tilde{Q}$ is a disjoint sum of finite coverings of $U \cap Q$. The completion $\tilde{Q}_{\mathrm{st}}$ is then a normal analytic space; it is the normal ramified covering of Q_{st} extending $\tilde{Q}$.

The results of §6 allow us, in §8, to extend $\tilde{w}$ to a map from $\tilde{Q}_{\mathrm{st}}$ to $B^+(\alpha)_o$. We will again write $\tilde{w}$ for the extended map. In §9, we show that the condition INT is tantamount to requiring this extended map to be

'etale in codimension one. Since the projection $\tilde{Q}_{\mathrm{st}} \rightarrow Q_{\mathrm{st}}$ is locally (on Q_{st}) finite to one, and Q_{st} is a normal analytic space, it follows from §6 that the extended map $\tilde{w} : \tilde{Q}_{\mathrm{st}} \rightarrow B^+(\alpha)_o$ is locally finite to one. By the purity of the branch locus theorem, if $\tilde{w}$ is

'etale in codimension one, it is then

'etale everywhere. Actually, we give two proofs of this fact in §10, the second proof not requiring the theorem on purity of branch locus. Additional work, relying on the compactness of Q_{sst} , shows that $\tilde{w} : \tilde{Q}_{\mathrm{st}} \rightarrow B^+(\alpha)_o$ is a topological covering map. The ball being simply connected, it is an isomorphism. In other words, the inverse of the multivalued map w_α is a single valued map $w_\alpha^{-1} : B^+(\alpha)_o \rightarrow \tilde{Q}_{\mathrm{st}}$.

The homeomorphism $\tilde{w}$ transforms the fibers of the projection $\tilde{Q}_{\mathrm{st}} \rightarrow Q_{\mathrm{st}}$ into the orbits of Γ . Those orbits are hence discrete, and Γ is a discrete subgroup of the Lie group $\mathrm{PU}(1, N - 3)$ of isometries of $B^+(\alpha)_o$. When $Q_{\mathrm{st}} = Q_{\mathrm{sst}}$, the quotient $B^+(\alpha)_o / \Gamma \cong Q_{\mathrm{sst}}$ is compact, and Γ is cocompact. In the general case, a local analysis near the cusps $p \in Q_{\mathrm{sst}} - Q_{\mathrm{st}}$ shows that it has finite covolume.

For $N = 3$, the group $\mathrm{PU}(1, 0)$ is trivial and the statement (3.11) uninteresting. For $N \geq 4$, it is not at once clear that $(1 - \mu_i - \mu_j)^{-1}$ must be an integer for all $i \neq j$ under condition INT (cf. (14.2.1)). It is easy to see however that condition INT implies that all μ_i are rational. We show this, even though it is not used in our proof.

Lemme 2.5. *Assume that $\mathrm{card} S > 4$. If positive real numbers $(\nu_s)_{s \in S}$ sum up to 2 and if for each of the pairs of elements $s \neq t$ of S for which $\nu_s + \nu_t \leq 1$, $\nu_s + \nu_t$ is rational, then all ν_s are rational.*

We may and shall assume that S is the set of integer from 1 to N , $N \geq 4$, and that $\nu_1 \leq \dots \leq \nu_N$.

Case 1. — $\nu_1 + \nu_2 > 1$. For any distinct $i, j < N$ one can find $k < N$ distinct from i and j . One has $\nu_k + \nu_N \geq \nu_1 + \nu_2 > 1$, hence

$$\nu_i + \nu_j = 2 - (\nu_k + \nu_N) < 1$$

is rational. If three numbers a, b, c are such that $a + b$, $b + c$ and $c + a$ are rational, they are rational. Applying this to the ν_i ($i < N$), we find that they are rational. So is ν_N , because $\sum \nu_i = 2$.

Case 2. — $\nu_1 + \nu_2 \leq 1$. In this case, for any $i \neq 1$, $\nu_1 + \nu_i \leq \nu_1 + \nu_2 \leq 1$ is rational. Summing, one concludes that $(N - 2)\nu_1 + \sum \nu_i = (N - 2)\nu_1 + 2$ is rational : ν_1 is rational, and so are the ν_i ($i \neq 1$).

2.8

Next, we explain how local systems L with the prescribed monodromy in the fiber direction can be constructed globally above M on P_M . Fix three distinct elements $a, b, c \in S$. For each m in M , there is then a unique isomorphism $\mathbb{P} \xrightarrow{\sim} \mathbb{P}^1$ mapping $m(a), m(b), m(c)$ respectively to $0, 1, \infty$. Let $z : P_M \rightarrow \mathbb{P}^1$ be the resulting map. When no ambiguity results, we will write $z(x)$ for $z(x, m)$. Let $A : M \rightarrow \mathbb{R}_+$ be a large enough continuous function. What is needed is that $[A(m), \infty[$ be disjoint from $z(m(S))$. For any $U \subset M$, let I_U denote the locus of $(x, m) \in \pi^{-1}(U)$ such that $z(x, m) \in [A(m), \infty[$. If $U \subset M$ is contractible, then $I_U \subset P_M$ is too. For $U \subset M$ contractible, if L is a local system on $\pi^{-1}(U)$ with the prescribed monodromy α (cf. (3.3)), then L_U admits an horizontal section $e_U \neq 0$ on I_U . The pair (L_U, e_U) is unique up to unique isomorphism, hence there is no problem in glueing. One gets a global local system L on P_M , provided with an horizontal section e on I_M .

Let $M_0 = \{m \in M \mid m(a) = 0, m(b) = 1, m(c) = \infty\}$ as in (3.7). The product decomposition $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$ induces an isomorphism $\mathrm{PGL}(2) \times P_{M_0} \xrightarrow{\sim} P_M$; and if one takes $A(m)$ constant on the $\mathrm{PGL}(2)$ orbits, so that I_M is the pullback of I_{M_0} , the system (L, e) is the pull back of its restriction to P_{M_0} .

2.9

We give two other descriptions of the same L . Fix numbers μ_s such that $\alpha_s = \exp(2\pi i \mu_s)$. Let L_μ be the subsheaf of $\mathcal{O}$ (cf. §2) consisting of those functions which are constant multiples of $\prod_{s \in S} (z - z(m(s)))^{\mu_s}$. This function is multivalued but the ratio of two determinations is a constant, so that the definition of L_μ makes sense. If μ' is another choice of μ_s with $\alpha_s = \exp(2\pi i \mu'_s)$, the multiplication by the univalued function $\prod_{s \in S} (z - z(m(s)))^{\mu'_s - \mu_s}$ is an isomorphism of L_μ with $L_{\mu'}$.

The principal determination of the multivalued function $\prod_{s \in S} (z - z(m(s)))^{\mu_s}$ on I_M is defined as $\exp(\sum \mu_s \log(z - z(m(s))))$ with $|\arg \log(z - z(m(s)))| < \pi$ for z real and large enough. This provides a section of L_μ on I_M . One easily checks that L_μ has the monodromy α on each fibre.

An identical but more algebraic description of L_μ is : the local system of horizontal sections of $\mathcal{O}$, provided with the connection ∇_μ for which

$$f^{-1} \nabla_\mu f = \frac{df}{f} - \sum_s \mu_s \frac{d(z - z(m(s)))}{z - z(m(s))}.$$

2.10

We close section 3 with some remarks about liftings of the projective representation

$$\rho : \pi_1(Q, o) \rightarrow \mathrm{PGL}(H^1(P_o, L_o))$$

to a linear representation.

As in (3.7) and (3.13), fix three elements a, b, c in S and denote by M_0 the subset of M with $m(a) = 0$, $m(b) = 1$ and $m(c) = \infty$. Let $M(c)$ denote the subset of $m \in M$ with $m(c) = \infty$, and let B denote the stabilizer of ∞ in $\mathrm{PGL}(2)$. One has $M_0 \xrightarrow{\sim} Q$ and $B \times M_0 \xrightarrow{\sim} M(c)$, $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$.

As is well known,

$$\pi_1(\mathrm{PGL}(2)) = \pi_1((3)) = \pi_1((2)/\{\pm 1\}) = \mathbb{Z}/2$$

and $\pi_1(B) = \pi_1(\mathbb{C}^*) = \mathbb{Z}$, so that

$$\pi_1(M) \cong \pi_1(Q, o) \oplus \mathbb{Z}/2, \quad (2.4)$$

$$\pi_1(M(c)) \cong \pi_1(Q, o) \oplus \mathbb{Z}. \quad (2.5)$$

Both those isomorphisms depend on the choices of a, b, c .

For X any of Q , $M(c)$ or M , each choice of a local system L on P_X with fiberwise monodromy α leads to a lifting

$$\rho' : \pi_1(X, o) \rightarrow \mathrm{GL}(H^1(P_o, L_o))$$

of ρ . If L carries a unitary structure, ρ' has values in $(H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$. For instance, the L of (3.13) carries a unique unitary structure for which e is of length one, and its restriction to $M_0 \cong Q$ provides a lifting $\rho' : \pi_1(Q, o) \rightarrow (H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$.

The local system L itself provides a lifting $\rho' : \pi_1(M, o) \rightarrow (H^1(P_o, L_o), \psi)$ which is trivial on the second factor of the decomposition (3.15.1).

Let $L(c)$ denote the local system on $P_{M(c)}$ which is given by the subsheaf of $\mathcal{O}$ consisting of those functions which are constant multiples of

$$\prod_{s \in S - \{c\}} (z - m(s))^{-\mu_s}.$$

Here, z is a fixed coordinate on $\mathbb{P}$ composed with $\mathbb{P} \times M(c) \rightarrow \mathbb{P}$ and not as in (3.13) and (3.13). The local system $L(c)$ provides a lifting $\rho' : \pi_1(M(c)) \rightarrow (H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$ of ρ .

Inasmuch as $M(c)$ is the space of distinct $(N - 1)$ -uples of points in the plane ($N = \mathrm{card} S$), $\pi_1(M(c))$ is by definition the colored braid group on $(N - 1)$ strands. For any $s, t \in S - \{c\}$, let $\gamma_{s,t}$ in $\pi_1(M(c))$ be a path in which t comes near s , makes a positive turn around s and comes back to its original position. The colored braid group is generated by such elements. The lifting ρ' provided by $L(c)$ has the virtue that each $\rho'(\gamma_{s,t})$ is a pseudo-reflection (i.e. $\rho'(\gamma_{s,t}) - I$ is of rank one)—cf. (9.1), (12.3).

3 The compactified quotient space Q_{st}

3.1

As in section 3, we fix a complex projective line $\mathbb{P}$, an integer $N \geq 3$ and a finite set S with N elements. We further fix a family $\mu = (\mu_s)_{s \in S}$ of real numbers with $0 < \mu_s < 1$ and $\sum \mu_s = 2$. The complex numbers $\alpha_s := \exp(2\pi i \mu_s)$ satisfy $|\alpha_s| = 1$, $\alpha_s \neq 1$ and $\prod \alpha_s = 1$.

As in section 3, $M \subset \mathbb{P}^S$ is the space of injective maps $\gamma : S \rightarrow \mathbb{P}$. We denote by $\mathrm{PGL}(2)$ the group of automorphisms of $\mathbb{P}$.

These conventions will hold throughout this section as well as in sections 6 to 12. In this section, from (4.2) on, $\mathbb{P}$ will be the standard projective line $\mathbb{P} := \mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ and M the space of S -uples of distinct points of the Riemann sphere. This is no loss of generality.

3.2

The group $\mathrm{PGL}(2)$ operates diagonally on $\mathbb{P}^S$. We shall define a compactification of the quotient space $Q := \mathrm{PGL}(2) \backslash M$. The definition is taken from D. Mumford's theory of quotient varieties of reductive groups (cf. (4.6)). The account here is self-contained.

A point $y \in \mathbb{P}^S$ is called μ -stable (resp. μ -semi-stable) if and only if for all $z \in \mathbb{P}$,

$$\sum_{y(s)=z} \mu_s < 1 \quad (\text{resp. } \leq 1). \quad (3.1)$$

The set of all μ -stable points (resp. μ -semi-stable points) is denoted M_{st} (resp. M_{sst}) and we set $M_{0\mathrm{usp}} = M_{\mathrm{sst}} - M_{\mathrm{st}}$.

For each partition $\{S_1, S_2\}$ of S with $\sum_{s \in S_i} \mu_s = 1$ ($i = 1, 2$), the points y in $\mathbb{P}^S$ for which $y(S_1) \cap y(S_2) = \emptyset$ and y constant on S_1 or S_2 are in $M_{0\mathrm{usp}}$. All points in $M_{0\mathrm{usp}}$ are obtained in this way, each from a unique partition.

On M_{sst} we define a relation $\sim$ via

$y \sim y'$ if and only if either

- a) $y, y' \in M_{\mathrm{st}}$ and $y' \in \mathrm{PGL}(2)y$; or
- b) $y, y' \in M_{0\mathrm{usp}}$ and the partitions of S corresponding to y and y' coincide.

It is clear that $\sim$ is an equivalence relation. Set

$$Q_{\mathrm{st}} = M_{\mathrm{st}} / \mathrm{PGL}(2), \quad Q_{\mathrm{sst}} = M_{\mathrm{sst}} / \sim, \quad Q_{0\mathrm{usp}} = M_{0\mathrm{usp}} / \sim$$

each with its quotient topology. The elements of $Q_{0\mathrm{usp}}$ are uniquely determined by their partitions. Thus $Q_{0\mathrm{usp}}$ is a finite set.

Exemple 3.1. The cross-ratio $c(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_3 - z_2} \cdot \frac{z_1 - z_4}{z_4 - z_2}$ of four points in $\mathbb{P} := \mathbb{P}^1$ is defined and is a continuous map into $\mathbb{P}^1$ as long as no three of z_1, z_2, z_3, z_4 are equal. When two of the z_i are equal, the cross-ratio takes one of the values $0, 1, \infty$, the value depending only on the partition of $\{1, 2, 3, 4\}$ consisting of the subset $\{i, j\}$ with $z_i = z_j$ and its complement.

Suppose $S = \{1, 2, 3, 4\}$ and $\mu_s = \frac{1}{2}$ for all $s \in S$. One can show that the cross-ratio map $M_{\mathrm{sst}} \rightarrow \mathbb{P}^1$ descends to Q_{sst} and yields a homeomorphism onto the projective line.

3.3

Let M' be the space of $y \in \mathbb{P}^S$ such that $y(S)$ has at least three distinct points, and let $Q' = \mathrm{PGL}(2) \backslash M'$ be the corresponding quotient orbit space. Fix a, b, c in S , let U be the set of all $y \in M'$ taking distinct values on a, b, c , and let U_0 be the subset of all $y \in M$ for which $y(a) = 0, y(b) = 1, y(c) = \infty$. Then U is in M , stable under the action of $\mathrm{PGL}(2)$, and $\mathrm{PGL}(2) \times U_0 \xrightarrow{\sim} U$. Hence

$$U_0 \xrightarrow{\sim} \mathrm{PGL}(2) \backslash U \hookrightarrow Q'.$$

As one varies a, b, c , the images of U_0 in Q' cover Q' . Consequently Q' is a manifold. However, this manifold is not Hausdorff if $\mathrm{card} S > 3$. For $\mathrm{card} S = 4$ (cf. (4.2)) one gets the projective line with $0, 1$ and ∞ doubled. One infers from (4.1.1) that $M_{\mathrm{st}} \subset M'$. The quotient Q_{st} being open in Q' is a manifold. The following Lemma implies that it is Hausdorff.

Lemme 3.2. Q_{sst} is Hausdorff.

Démonstration. Let F denote the set of injective maps of $\{1, 2, 3, 4\}$ into S . For any $f \in F$, let M_f denote the set of all $y \in M_{\text{sst}}$ such that for all $z \in \mathbb{P}$, $\text{card}(yf)^{-1}(z) < 2$ and additionally, if $y \in M_{0\text{usp}}$, $\text{card } f^{-1}(S_i) = 2$ for $i = 1, 2$ where $\{S_1, S_2\}$ is the partition of S corresponding to y . The additional condition guarantees that $M_f \cap M_{f'}$ as well as $M_f \cap M_{f'}$ is $\sim$ -saturated.

For any $f \in F$, define $c_f : M_f \rightarrow \mathbb{P}$ via the cross-ratio :

$$c_f(y) = c(yf(1), yf(2), yf(3), yf(4));$$

the function c_f is constant on $\sim$ -equivalence classes of the $\sim$ -saturated set M_f . Inasmuch as c_f is continuous on M_f , it descends to a continuous function on $M_f / \sim$. To prove the lemma, it suffices to prove :

For any $y \not\sim y' \bmod \sim$, there is an $f \in F$ such that y and y' are in M_f , and $c_f(y) \neq c_f(y')$. (3.2)

One is free to replace y and y' by y_1 and y'_1 , with $y \sim y_1$, $y' \sim y'_1$.

To each y in M_{sst} we attach a partition of S as follows : for $y \in M_{0\text{usp}}$, attach the partition defined in (4.1); for $y \in M_{\text{st}}$, attach the partition $T = \{C_1, \dots, C_k\}$ such that $y(s) = y(t)$ if and only if s and t lie in the same coset of T . If $y \sim y_1$, the same partition is attached to y and y_1 . In the proof of (4.3.1), we may and shall assume that if y or y' is in $M_{0\text{usp}}$, then it is constant on the cosets of its partition.

We first treat the case that y and y' are in $M_{0\text{usp}}$. Let $\{S_1, S_2\}$ and $\{S'_1, S'_2\}$ be their corresponding partitions of S . Since $y \not\sim y' \bmod \sim$, $S_i \cap S'_j$ is not empty for $i, j \in \{1, 2\}$; otherwise, if say $S_1 \cap S'_2 = \emptyset$, we would have $S_1 \subset S'_1$, $S_1 \neq S'_1$ and

$$\sum_{s \in S'_1 - S_1} \mu_s = 0,$$

contradicting $0 < \mu_s$ for each s . Choose f so that $\text{card } f^{-1}(S_i \cap S'_j) = 1$ for each choice of $i, j \in \{1, 2\}$. Then for suitable choice of f we have

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, a, b, b, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', b', a', b', \end{aligned}$$

with $a \neq b$, $a' \neq b'$. Hence $c_f(y) = \infty$, $c_f(y') = 0$.

Suppose next that $y \in M_{\text{st}}$ and $y' \in M_{0\text{usp}}$. Let $T = \{C_i\}$ be the partition of S corresponding to y' by (4.2). Each coset in the partition of S corresponding to y must meet at least two distinct cosets of T . We can thus choose an $f \in F$ so that

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, a, b, b, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', b', c, d, \end{aligned}$$

with $a \neq b$, $a' \neq b'$, and $c \neq d$. It follows at once that y and y' are in M_f , $c_f(y) = 1$, $c_f(y') \neq 1$.

It remains only to consider the case where y and y' are in M_{st} ; let T and T' be the corresponding partitions of S . If $T = T'$, then for any $f \in F$, $y \in M_f$ if and only if $y' \in M_f$. Replacing y' by an element in its $\text{PGL}(2)$ -orbit, we can assume that there are

three distinct cosets C_1, C_2, C_3 of T with $y(s) = y'(s)$ for $s \in C_1 \cup C_2 \cup C_3$. If $c_f(y) = c_f(y')$ for all f with $y \in M_f$ then, clearly $y = y'$. One is thus reduced to the case where $T \neq T'$. Then there is a $C_1 \in T$ and $C'_1 \in T'$ such that $C_1 \cap C'_1$ is not empty and $C_1 \neq C'_1$. Either $C_1 - C'_1$ or $C'_1 - C_1$ is not empty—say $C'_1 - C_1 \neq \emptyset$. Choose $C'_2 \in T'$ so that $C'_2 \cap C_1 \neq \emptyset$. Choose $C'_3 \in T'$ with $C'_3 \neq C'_2$. Next choose $C_2 \in T$ so that $C'_2 \cap C_2 \neq \emptyset$. Then $C'_3 \cup C_2 \neq S$; otherwise, since y' and y are in M_{st} ,

$$2 < \sum_{s \in C'_1} \mu_s + \sum_{s \in C_2} \mu_s.$$

Choose $f : \{1, 2, 3, 4\} \rightarrow S$ with

$$f(1) \in C_1 \cap C'_1, \quad f(2) \in C'_2 \cap C_1, \quad f(3) \in C'_2 \cap C_2, \quad f(4) \in C'_3 \cup C_2.$$

Any such f is injective, and

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, b, c, d \quad \text{with } a \neq b, c \neq d, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', a', b', c' \quad \text{with } a' \neq b', a' \neq c'. \end{aligned}$$

Hence $c_f(y') = 1$, $c_f(y) \neq 1$. This completes the proof. $\square$

3.4

Lemme 3.3. *Fix a Riemannian metric on $\mathbb{P}$. For $y \in \mathbb{P}^S$, define*

$$d(y) = \inf_{\sum_{s \in T} \mu_s > 1} \text{diam } y(T)$$

and

$$a(y) = \sup_{g \in \text{PGL}(2)} d(gy).$$

Then there is $a > 0$ such that $(y) \geq a$ for all $y \in M_{\text{st}}$.

Démonstration. The validity of the lemma is independent of the chosen metric. We will use the Fubini metric (for which $\mathbb{P}$ is isometric to the sphere of radius 1) and prove the lemma with $a = \pi/4$.

If $y \in M_{0\text{usp}}$ corresponds to the partition $\{S_1, S_2\}$ of S , with y constant on S_1 , let us take g to fix $y(S_1)$ and the antipodal point u of $\mathbb{P}$ and to carry the complement of the ε -neighborhood of $y(S_1)$ into the ε -neighborhood of u . Taking $\varepsilon \rightarrow 0$, one gets $d(y) = \pi$.

For any ε there is a compact $K \subset \text{PGL}(2)$ such that for $g \notin K$, g maps the complement of the ε -neighborhood of some point u into the ε -neighborhood of some point v . If $y \in M_{\text{st}}$ and if ε is small enough, then, for all u , $\{s \in S \mid (y(s), u) > \varepsilon\}$ is in $\mathcal{C}$. This implies $d(gy) \rightarrow 0$ for $g \rightarrow \infty$ in $\text{PGL}(2)$. The sup defining (y) is hence attained. Fix $y \in M_{\text{st}}$ with $(y) = d(y)$. We have to prove $d(y) > a = \pi/4$.

Suppose to the contrary that $d(y) < a$. Fix $T_1 \in \mathcal{C}$ with $\text{diam } y(T_1) < a$ and $t \in T_1$. For any T_1, T_2 in $\mathcal{C}$, $T_1 \cap T_2 \neq \emptyset$. If S' is the union of all $T \in \mathcal{C}$ with $\text{diam } y(T) < a$, $y(S')$ is hence contained in a half-sphere of radius $r < 2a = \pi/2$ with center $y(t)$. In particular, $y(S')$ is contained in an open halfsphere. Identify $\mathbb{P}$ with the Riemann sphere $\mathbb{C} \cup \{\infty\}$, with the unit disc the halfsphere, in such a way that $y(S')$ lies in $D = \{z \mid |z| < b\}$, $b < 1$. Take for g the multiplication by $1 + \varepsilon$ ($\varepsilon > 0$). For ε small enough (relative to b), distances in D are increased by g . For any $T \in \mathcal{C}$ with $\text{diam } y(T) < a$, one hence has $\text{diam } gy(T) > \text{diam } y(T)$. For $T \in \mathcal{C}$ with $\text{diam } y(T) > a$, $\text{diam } y(T)$ can decrease, but not much for ε small, so that $d(gy) > d(y)$. This contradicts the choice of y . $\square$

Lemme 3.4. Q_{sst} is compact.

Démonstration. Fix $a > 0$ as in (4.4.1), and define $M' \subset \mathbb{P}^S$ to be the set of y such that $\text{diam } y(T) > a$ for each $T \subset S$ with $\sum_{s \in T} \mu_s > 1$. The subset M' of $\mathbb{P}^S$ is closed, hence compact. It is contained in M_{sst} and, by (4.4.1), it maps onto Q_{sst} . Compactness follows. $\square$

3.5

By contrast with Q_{st} which is a manifold, the points of $Q_{0\text{usp}}$ may be singular.

Let π denote the natural projection of M_{sst} onto Q_{sst} . Let $\{S_1, S'_1\}$ be the partition corresponding to a point $y_0 \in M_{0\text{usp}}$. We shall describe a neighborhood of $\pi(y_0)$ in Q_{sst} . We can assume that $y_0(s) = 0$ for all $s \in S_1$ and $y_0(s) = \infty$ for all $s \in S'_1$. Fix elements $a \in S_1$ and $b \in S'_1$, and define V as the set of all $y \in M_{\text{sst}}$ such that $y(a) = 0$, $y(b) = \infty$ and

$$\sup_{s \in S_1} |y(s)| < \inf_{s \in S'_1} |y(s)|.$$

Then, $\pi(V)$ is an open neighborhood of $\pi(y_0)$ in Q_{sst} and is the quotient of V by the equivalence relation on V induced by $\sim$. The punctured neighborhood $\pi(V) - \pi(y_0)$ is the quotient of $V - \pi^{-1}\pi(y_0)$ by the stabilizer in $\text{PGL}(2)$ of the points 0 and ∞ :

$$\pi(V) - \pi(y_0) = (V - \pi^{-1}\pi(y_0))/\mathbb{C}^*.$$

Set $S_2 := S_1 - \{a\}$, $S'_2 := S'_1 - \{b\}$, $m = \text{card}(S_2)$, $m' = \text{card}(S'_2)$ and consider the composite map

$$\begin{aligned} \Phi : V &\rightarrow \mathbb{C}^{S_2} \times \mathbb{C}^{S'_2} = \mathbb{C}^m \times \mathbb{C}^{m'} \rightarrow \mathbb{C}^{mm'} \\ y &\mapsto ((y(s))_{s \in S_2}, (y(s)^{-1})_{s \in S'_2}) \mapsto (y(s) \cdot y(t)^{-1})_{s \in S_2, t \in S'_2}. \end{aligned}$$

The map Φ descends to an embedding of $\pi(V)$ into $\mathbb{C}^{mm'}$, whose image is the intersection of the cone over the Veronese embedding of $\mathbb{P}^{m-1} \times \mathbb{P}^{m'-1}$ in $\mathbb{P}^{mm'-1}$ with the polydisc (of radius < 1 , $|x_{s,t}| < 1$).

If $m = 1$ or $m' = 1$, the Veronese embedding is an isomorphism with $\mathbb{P}^{mm'-1}$. As a consequence

If $\text{card}(S_1) = 2$ or $\text{card}(S'_1) = 2$, the cusp point with partition $\{S_1, S'_1\}$ is non singular. (3.3)

One can define an algebraic structure on Q_{sst} by using as a chart at a point $\text{PGL}(2)y$ of Q_{st} the orbit space

$$\text{PGL}(2) \backslash \{y' \in M_{\text{st}} \mid y'(i) \neq y'(j) \text{ if } y(i) \neq y(j)\}$$

and at a cusp point with partition $\{S_1, S'_1\}$ the Zariski-open subset of the Veronese cone of $\mathbb{P}^{m-1} \times \mathbb{P}^{m'-1}$ in $\mathbb{C}^{mm'}$ given by $x_{s,t} = y(s)/y(t) \neq 1$, for all $s \in S_1$, $t \in S'_1$.

Exemple 3.5. 1.— $S = \{1, 2, 3, 4, 5\}$, $\mu_s = \frac{2}{5}$ for all $s \in S$. Here there are no cusp points : $Q_{\text{sst}} = Q_{\text{st}} = \text{PGL}(2) \backslash M_{\text{st}}$. Each pair s, t in S defines the diagonal line $L_{s,t} : y(s) = y(t)$ in M_{st} and under the projection to Q_{st} , one gets 10 lines. The map of M_{st} to $(\mathbb{P}^1)^2$ given by

$$(y_1, y_2, y_3, y_4, y_5) \mapsto (\infty, 0, 1, c(y_1, y_2, y_3, y_4)^{-1}, c(y_1, y_2, y_3, y_5)^{-1})$$

(where c denotes cross-ratio) descends to a map p of Q_{sst} which is biregular except at $L_{0,\infty} = p^{-1}(0, 0)$, $L_{0,\infty} = p^{-1}(1, 1)$, $L_{\infty,\infty} = p^{-1}(\infty, \infty)$. Here, Q_{sst} is a blowup of $\mathbb{P}^1 \times \mathbb{P}^1$.

Each of the curves $p^{-1}(0, 0)$, $p^{-1}(1, 1)$, $p^{-1}(\infty, \infty)$ are exceptional, that is, each has self-intersection -1 . Inasmuch as the set of lines $\{L_{ij}, i \neq j, i, j \in S\}$ are permuted transitively by permutation of coordinates in M_{st} , their images in Q_{st} are also permuted transitively and thus each of the 10 lines is exceptional. The 10 lines of Q_{st} consist of $p^{-1}(0, 0)$, $p^{-1}(1, 1)$, $p^{-1}(\infty, \infty)$, and the 7 lifts of the seven lines on $\mathbb{P}^1 \times \mathbb{P}^1$:

$$x = 0, x = 1, x = \infty, \quad y = 0, y = 1, y = \infty, \quad x = y$$

where $x = c(y_1, y_2, y_3, y_4)^{-1}$, $y = c(y_1, y_2, y_3, y_5)^{-1}$.

2.— $S = \{1, 2, 3, 4, 5\}$, $\mu = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3})$. Here there are four points in Q_{0usp} corresponding to the partitions $\{S_i, S_i^c\}$ where $S_i = \{i, 5\}$, $1 \leq i \leq 4$. Q_{sst} is a non-singular manifold by (4.5.1). This Q_{sst} can be obtained from the Q_{sst} of Example 1 by blowing down the four lines $L_{i,5}$ ($1 \leq i \leq 4$). It can be obtained from $\mathbb{P}^1 \times \mathbb{P}^1$ by blowing up the point (∞, ∞) and then blowing down the lines $x = \infty$ and $y = \infty$. The resulting space is $\mathbb{P}^2$.

3.6

We conclude this section by relating our Q_{sst} to Mumford's when all the μ_s are rational.

Let $\mathcal{O}(n)$ be the line bundle on $\mathbb{P}$ of degree n , and let Ω be the canonical line bundle on $\mathbb{P}$; i.e. $\Omega \cong \mathcal{O}(-2)$. For any line bundle we write $^{\otimes n}$ for the dual line bundle and $^{\otimes n} = (^{\otimes(-1)})^{\otimes(-n)}$ for $n < 0$.

Let D be the common denominator for $\{\mu_s, s \in S\}$. On $\mathbb{P}^S$ define the line bundle

$$\bigotimes_{s \in S}^* (\Omega^{\otimes \mu_s \cdot D} \otimes \mathcal{O}(2)^{\otimes D}).$$

It admits a $\text{PGL}(2)$ -action coming from the action on the tangent bundle $\Omega^{\otimes(-1)} \cong \mathcal{O}(2)$. The stable and semi-stable points of $\mathbb{P}^S$ defined in (4.1) are the same as those defined in Mumford's "Geometric Invariant Theory" for the action of $\text{PGL}(2)$ on the linear space of holomorphic sections of this line bundle; Q_{sst} is the underlying topological space of Mumford's "quotient variety" for $\text{PGL}(2) \backslash \mathbb{P}^S$ [?].

4 The complex ball

4.1

Let V be a complex vector space, and $(,)$ be a Hermitian form on V , of signature $(1, \dim V - 1)$. The complex ball B^+ in $\mathbb{P}(V) := \{1\text{-dimensional subspaces of } V\}$ corresponding to $(,)$ is the set of lines in V which are spanned by a vector v with $(v, v) > 0$. The form $(,)$ is determined, up to a positive real factor, by B^+ . The closure $\overline{B^+}$ of B^+ in $\mathbb{P}(V)$ is the set of lines spanned by a non zero vector v such that $(v, v) \geq 0$.

4.2

The hyperbolic angle $\theta \in \mathbb{R}_+$ between lines $\ell_1, \ell_2 \in B^+$, spanned by vectors v_1, v_2 , is defined by

$$|(v_1, v_2)| = \cosh(\theta)(v_1, v_1)^{1/2}(v_2, v_2)^{1/2}. \quad (4.1)$$

It remains unchanged when $(,)$ is replaced by a positive multiple. The angle $\theta(\ell_1, \ell_2)$ is a Riemannian distance on B^+ . It is invariant by the action of the unitary group (V) on B^+ and renders B^+ a Hermitian symmetric space. By continuity we extend the function $\theta(\ell_1, -)$ to a function $\theta(\ell_1, -)$ from $\overline{B^+}$ to $\mathbb{R}_+ \cup \{\infty\} : \theta(\ell_1, \ell) = \infty$ for $\ell \in \partial B^+$.

4.3

Fix $\ell \in \partial B^+$. For $v \neq 0$ in ℓ , we define a “distance from ℓ ” function on B^+ by

$$d_\ell(\ell_1) = |(v, v_1)|/(v_1, v_1)^{1/2} \quad \text{for } \ell_1 \in B^+, \text{ any } v_1 \neq 0 \text{ in } \ell_1; \quad (4.2)$$

$$d_\ell(\ell) = 0, \quad d_\ell(\ell') = \infty \text{ for } \ell' \in \partial B^+, \ell' \neq \ell. \quad (4.3)$$

The function d_ℓ is continuous and ≥ 0 on $\overline{B^+} - \{\ell\}$. One has $d_{\lambda\ell}(\ell_1) = |\lambda| d_\ell(\ell_1)$, and the family of functions d_ℓ ($v \in \ell - \{0\}$) is stable by the stabilizer of ℓ in (V) .

4.4

On $\overline{B^+}$, we will use the coarsest topology for which the functions d_ℓ and $\theta(\ell_1, -)$ are continuous. A fundamental system of neighborhoods of $\ell \in B^+$ (resp. of $\ell \in \partial B^+$ —on which we choose $v \neq 0$) consists of the V_ε ($\varepsilon > 0$) defined by $V_\varepsilon = \{\ell_1 \in B^+ \mid \theta(\ell, \ell_1) < \varepsilon\}$ (resp. $V_\varepsilon = \{\ell_1 \mid \ell_1 = \ell \text{ or } \ell_1 \in B^+ \text{ and } d_\ell(\ell_1) < \varepsilon\}$).

This topology is finer than the one induced by the topology of $\mathbb{P}(V)$. It induces the usual topology on B^+ , and the discrete one on ∂B^+ . It is respected by (V) .

Proposition 4.1. *Fix $\ell \in \partial B^+$ and let A be a discrete subgroup of the projective unitary group $\text{PU}(V)$ stabilizing ℓ and respecting a “distance to ℓ ” function d_ℓ . If ℓ has a neighborhood V (in the topology 5.4) such that V/A is compact, the volume of $(V \cap B^+)/A$ is finite.*

Démonstration. The subgroup of $\text{PU}(V)$ respecting ℓ and d_ℓ is the isomorphic image of the stabilizer H of v in (V) . It is an extension of the unitary group $(v^\perp/v)$ by the unipotent subgroup N of H consisting of the $n \in (V)$ respecting the filtration $V \supset v^\perp \supset \mathbb{C}v \supset 0$ and acting trivially on the successive quotients. The extension is split : H and N act transitively on each horocycle $d_\ell(\cdot) = C$ and the stabilizer of ℓ_1 is a lifting of $(v_1^\perp/v_1)$.

Fix an isotropic vector v' such that $(v, v') = 1$, and let $v_1 = v \cdot t^{-2} - v'$. The line ℓ_1 spanned by v_1 is in B^+ on the horocycle $d_{\ell_1} = t$. Let V' be spanned by v and v' , and let V'' be the orthogonal complement ($\supset v^\perp/v$). Any $h \in H$ fixing ℓ_1 fixes each point of V' ; hence the stabilizer of ℓ_1 in H is $(V'') \subset (V)$. Let $A \subset (V') \subset (V)$ be the group of unitary transformations $a(\lambda) : v \mapsto \lambda^{-1}v, v' \mapsto \lambda v'$ ($\lambda \in \mathbb{R}_+$) and fixing each point of V'' . It normalizes H : one has $d_\ell \circ a(\lambda) = d_\ell \cdot |\lambda|^{-1} = \lambda d_\ell$. Let $A_R = \{a(\lambda) \mid \lambda < R\}$. The map $g \mapsto g\ell_1$ maps isomorphically $A_R H/(V'') = A_R N$ onto $\{\ell_1 \in B^+ \mid d_\ell(\ell_1) < R\}$. The horocycles are closed in $\overline{B^+}$ (for the topology 5.4). The compactness assumption implies that for R small enough, $\{\ell_1 \mid d_\ell(\ell_1) = R\}/A$ is compact. One has

$$A \backslash a(R)H/(V'') \xrightarrow{\sim} \{\ell_1 \mid d_\ell(\ell_1) = R\}/A$$

and A is hence cocompact in H . It remains to evaluate the volume of

$$A \backslash A_R H/(V'') \xrightarrow{\sim} \{\ell_1 \in B^+ \mid d_\ell(\ell_1) < R\}/A$$

for a volume element on $A_R H / (V'')$ which is A - H -invariant, hence a multiple of the image of $da \cdot dh$ on $A_R H$.

$$\int_{A \setminus A_R H} da' dh = \int_{\lambda \leq R} \int_{A \setminus a(\lambda) H} d(a(\lambda)^{-1} h a(\lambda)) d\lambda$$

and

$$\int_{A \setminus a(\lambda) H} dh = \int_{A \setminus H} d(a(\lambda)^{-1} h a(\lambda)) = |\det(a(\lambda)^{-1}, H)| \int_{A \setminus H} dh.$$

This leads us to evaluate $|\det(a(\lambda)^{-1}, H)|$. One checks that the character $\det(a(\lambda)^{-1}, H)$ of $\lambda \in \mathbb{R}_+^*$ tends to zero for $\lambda \rightarrow 0$, and the integral is hence finite.

The computation above is of course exactly the one used to prove that Siegel domains have finite volume ([?], Lemma 1.9). $\square$

5 Stable points

5.1

Notation and assumptions (4.0) are in force in this section as well as the assumption $N \geq 4$. Our aim is to investigate the asymptotic behaviour of the section w_α of $B(\alpha)_M$, for $y \in M$ converging to a stable point $m \in M_{\text{st}}$.

Fix $m \in M_{\text{st}}$. To make a local study of w_α near m , it will be convenient to identify $\mathbb{P}$ with $\mathbb{P}^1$, in such a way that $\infty \notin m(S)$. Let U be an open neighborhood of m in M_{st} , such that $\infty \notin y(S)$ for $y \in U$. On the inverse image P_U of U in P_M (3.1), there is a rank one local system, trivialized at ∞ by a section e over $\{\infty\} \times U$, and with monodromy α on P_U for $y \in U \cap M$. A model of it is : the local system of constant multiples of $\prod_{s \in S} (z - y(s))^{\mu_s}$; the μ_s summing up to an integer, there is no ramification at ∞ . A trivialisation for large positive real z is provided by the principal determination of $\prod_{s \in S} (z - y(s))^{\mu_s}$; it extends by continuity to a trivialisation e at ∞ . We normalize the Hermitian form $(,)$ on L so that $(e, e) = 1$.

For a general $y \in U$, the monodromy of L_U on P_y around $p \in y(S)$ is $\prod_{y(s)=p} \alpha_s = \exp(2\pi i \sum_{y(s)=p} \mu_s)$. The stability of y ensures that $0 < \sum_{y(s)=p} \mu_s \leq 1$, so that this local monodromy is never trivial.

Fiber by fiber, we take as section ω of $\Omega^1(\sum \mu_s \cdot y(s))(L)$

$$\omega_y = \prod_{s \in S} (z - y(s))^{-\mu_s} \cdot e dz.$$

On $U \cap M$, it depends holomorphically on y . The assumption of stability of y amounts to saying that this form is of the first kind.

5.2

The sheaf $R^1 \pi_! L_U$, for π the projection $P_U \rightarrow U$, is no longer a local system if $m \notin M$. Its fiber at m is $H_c^1(P_m, L_m)$ ($\xrightarrow{\sim} H^1(P_m, L_m)$ by (2.6.1)). For each $h \in H_c^1(P_m, L_m)$, there is hence a unique germ $h^\sim$ of a section of $R^1 \pi_! L$ inducing h on P_m . The vector space $H_c^1(P_m, L_m)$ being finite dimensional, this provides us on U suitably small, with a map

$$(\text{constant sheaf } H_c^1(P_m, L_m)) \hookrightarrow R^1 \pi_! L_U. \quad (5.1)$$

This map is compatible with the non degenerate Hermitian form $(\cdot, \cdot)$, and hence injective. Compatibility can be checked on the description given below. In particular, each $h \in H_c^1(P_m, L_m)$ defines a horizontal section $h^\sim$ of $R^1\pi_!L$ on $U \cap M$.

We describe $h^\sim$, for h the class of a L_m -valued compactly supported C^∞ form η , whose support is contained in a connected compact K containing ∞ , and for U connected and small enough for K to be disjoint from $y(S)$ for $y \in U$. On $K \times U \subset P_U$, L is uniquely isomorphic to ${}^*_1(L_m)$, by an isomorphism which is the identity above m . It hence make sense, in each fibre P_y , to take the same form η ; it defines $h^\sim$.

5.3

If we apply this to ω_m , we get an horizontal section $(\omega_m)^\sim$ of $R^1\pi_!L_U$ on $U \cap M$. As $(\omega_m, \omega_m) > 0$ (2.20), the same positivity holds for $(\omega_m)^\sim$, and, projecting from $R^1\pi_!L_U$ to $B(\alpha)$, we get a section $(\omega_m)^\sim$ of $B^+(\alpha)_M$. We will compare it with the holomorphic section w_α , using fiber by fiber the metric (5.2).

Proposition 5.1. $\theta(w_\alpha, (\omega_m)^\sim)$ tends to 0 for $y \in M$ tending to m .

Démonstration. This will result at once from the two statements

$$(\omega_y, \omega_y) \rightarrow (\omega_m, \omega_m) \quad \text{for } y \rightarrow m; \quad (5.2)$$

$$(\omega_y, (\omega_m)^\sim) \rightarrow (\omega_m, \omega_m) \quad \text{for } y \rightarrow m. \quad (5.3)$$

If it made sense, we would like to say that $\omega_y \rightarrow \omega_m$ in the L^2 -norm, for $y \rightarrow m$, but ω_y and ω_m do not live in the same space, having different ramification. Let K_R be the complement in $\mathbb{P}^1$ of discs of radius R around each point of $m(S)$. One assumes R small enough for these discs to be disjoint; on a connected neighborhood U of y , small enough for $y(S)$ to stay disjoint from K_R , one can compare $\omega_y|_{K_R}$ and $\omega_m|_{K_R}$: on $K_R \times U$, $L \xrightarrow{\sim} {}^*_1 L_m$. There, one has a C^∞ isotopy of ω_y to ω_m . To prove (6.4.1), it hence suffices to obtain a bound r uniform in y , close to m , for the L^2 -norm of the restriction of ω_y to a disk D_R of radius R around $p \in m(s)$, for any $s \in S$, this bound being such that $r \rightarrow 0$ for $R \rightarrow 0$.

We may assume R and U small enough so that $y(s) \subset D_{2R}$ if $m(s) \neq p$. This ensures

$$\prod_{m(s) \neq p} |z - y(s)|^{-\mu_s} < \text{constant} \cdot \prod_{m(s) \neq p} |z - y(s)|^{-\mu_s}$$

and the required estimate follows from the following lemma.

Lemme 5.2. Let D be a disc of radius R , $(a_i)_{i \in I}$ be a family of points of D and $(\mu_i)_{i \in I}$ a family of real numbers, such that $0 < \mu_i < 1$ and $\sum \mu_i < 1$. Then, writing $z = x + iy$,

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy < \int_{|z| < R} |z|^{-2\sum \mu_i} dx dy = \frac{2\pi R^{2(1-\sum \mu_i)}}{2(1-\sum \mu_i)}.$$

Démonstration. We may and shall assume D centered at 0. If the real numbers p_i are such that $1 > \frac{1}{p_i} > \mu_i$ and $\sum \frac{1}{p_i} = 1$, Hölder's inequality gives

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy \leq \prod_i \left[\int_D |z - a_i|^{-2\mu_i p_i} dx dy \right]^{1/p_i}.$$

One has that the integral of $|z - a_i|^{-2\mu_i p_i} dx dy$ is bigger when taken on the disc $D_R(a_i)$ of radius R and center a_i than when taken on $D_R(0)$: results from the fact that $|z - a_i|^{-2\mu_i p_i}$ is bigger on $D_R(a_i) - D_R(0)$ than on $D_R(0) - D_R(a_i)$: one has $|z - a_i| < R$ on $D_R(a_i) - D_R(0)$, and $|z - a_i| \geq R$ on $D_R(0) - D_R(a_i)$.

Using that $\sum 1/p_i = 1$, we have

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy \leq \prod_i [2\pi \dots]^{1/p_i} = 2\pi \cdot R^{2(1-\sum \mu_i)} \cdot C^{-1}$$

with $C = \prod_i (2 - 2\mu_i p_i)^{1/p_i}$. Taking the p_i such that the $\mu_i p_i$ are all equal, i.e. $p_i = \mu_i^{-1} \sum \mu_i$, one gets the result. $\square$

Proof of (6.4.2).— For each $p \in m(S)$, let D_p be a disc of radius R around p , u_p the solution on $D_p^* = D_p - \{p\}$ of $\omega_m = du$ and φ_p the characteristic function of D_p . The current with compact support $\omega_m - \sum d(\varphi_p u_p)$ can be used to compute $(\omega_m)^\sim$. One gets

$$(\omega_y, (\omega_m)^\sim) = \int_{P \cup D_p} \omega_y \wedge \overline{\omega_m} - \sum_p \int \omega_y \wedge \overline{d(\varphi_p u_p)}.$$

For $y \rightarrow m$, and $R \rightarrow 0$ in such a way that the disc of radius $2R$ around p contains all $y(s)$ for $m(s) = p$, the first integral converges to (ω_m, ω_m) , while the others each are $O(R^{2\beta})$, $\beta = 1 - \sum_{m(s)=p} \mu_s$. This concludes the proof. $\square$

5.4

In the neighborhood of a stable point $y \in Q_{\text{st}}$, the projection of $\tilde{Q}_{\text{st}}$ to Q_{st} has a section. Taking a pull-back by such a section, we get the following :

Corollaire 5.3. *For $y \in Q_{\text{st}}$, there exists a neighborhood V of y in Q_{st} and a horizontal section b_V of $B(\alpha)$ on $Q \cap V$ such that $\theta(b_V(y'), w_\alpha(y')) \rightarrow 0$ for $y' \rightarrow y$ ($y' \in Q$).*

For $y \in Q$, the corollary just tells once more that the section w is continuous at y .

5.5

A partition T of S is *stable* if for each $C \in T$, $\mu(T)_C := \sum_{s \in C} \mu_s < 1$.

Define $\alpha(T)_C = \exp(2\pi i \mu(T)_C) = \prod_{s \in C} \alpha_s$. For T stable, let $M_T \subset M_{\text{st}}$ be the space of maps y from S to $\mathbb{P}$, such that $y(s) = y(t)$ if and only if s and t are in the same coset of T . It is the inverse image in M_{st} of a subspace Q_T of Q_{st} . The M_T (resp. Q_T) for T stable form a partition of M_{st} (resp. Q_{st}) and each M_T (resp. Q_T) can be identified with the analogue $M(T)$ of M (resp. $Q(T)$ of Q) for T , $\mu(T)$, and its closure with the analogue $M_{\text{st}}(T)$ (resp. $Q_{\text{st}}(T)$) of M_{st} (resp. Q_{st}).

Fix m, U, L as in (6.1), with $m \in M_T$. On $M_T \cap U$, the $H_c^1(P_y, L_y)$ form a local system. By (6.2.1), this local system, which is trivial as U is small enough, extends as a trivial sublocal system of $R^1 \pi_! L_U$. The ω_y ($y \in M_T \cap U$) define a non-zero holomorphic section of this local system on $M_T \cap U$. The corresponding ray can be identified with the section $w_{\alpha(T)}$ of $B(\alpha(T))$ on $M(T)$.

On Q_{st} , this gives us in a neighborhood V of $y \in Q_T$:

- a) A horizontal sub-ball-bundle of $B(\alpha)$, which extends as a constant horizontal bundle on V . Its restriction to Q_T can be identified with $B(\alpha(T))$ on $Q(T)$.

b) Via this identification, the horizontal section b_V of (6.8) is the horizontal section through $w_{\alpha(T)}(y)$.

Proposition 5.4. *If μ satisfies condition INT of (3.11) : $0 < \mu_s < 1$, $\sum \mu_s = 2$ and for all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer, then any subset $C \subset S$ for which $\sum_{s \in C} \mu_s < 1$ has at most three elements, and for any such subset with at least two elements, $(1 - \sum_{s \in C} \mu_s)^{-1}$ is an integer.*

The condition INT implies that $\mu_s + \mu_t$ is always $> 1/2$ for $s \neq t$. Averaging over the pairs in $C \subset S$, we get that for $\text{card}(C) \geq 2$, the mean of the μ_s ($s \in C$) is $> 1/4$. From $\sum_{s \in C} \mu_s < 1$ there results $\text{card}(C) < 4$.

The case $\text{card}(C) = 2$ is trivial by INT. Suppose $C = \{a, b, c\}$. The relations

$$1 - \mu_a - \mu_b = \frac{1}{n_{ab}}, \quad 1 - \mu_b - \mu_c = \frac{1}{n_{bc}}, \quad 1 - \mu_c - \mu_a = \frac{1}{n_{ca}}$$

sum up to $3 - 2(\mu_a + \mu_b + \mu_c) = \frac{1}{n_{ab}} + \frac{1}{n_{bc}} + \frac{1}{n_{ca}}$, hence $\frac{1}{n_{ab}} + \frac{1}{n_{bc}} + \frac{1}{n_{ca}} > 1$.

We now use the elementary fact that a triple of integers whose reciprocals have their sum strictly greater than 1 must be one of

$$(2, 2, n), \quad (2, 3, 3), \quad (2, 3, 4), \quad (2, 3, 5).$$

The excesses of the sums over 1 are respectively $1/n$, $1/6$, $1/12$, $1/30$.

It follows at once that $(1 - \mu_a - \mu_b - \mu_c)^{-1}$ is an integer.

Corollaire 5.5. *If μ satisfies INT and T is a stable partition of S (cf. (6.9)), the family $\mu(T)$ defined by*

$$\mu(T)_C = \sum_{s \in C} \mu_s \quad (C \in T)$$

again satisfies INT.

6 Semi-stable points

6.1

The notation and assumptions (4.0) are in force in this section ; we assume also $N \geq 4$. Let $(S(1), S(2))$ be a partition of S , such that $\sum_{s \in S(i)} \mu_s := 1$. We want to investigate the asymptotic behavior on Q of w_α , for y converging to a point in $Q_{0\text{usp}}$ of type $(S(1), S(2))$.

Fix $a \in S(1)$ and $b \in S(2)$, identify $\mathbb{P}$ with $\mathbb{P}^1$, and let W be the space of $y \in \mathbb{P}^S$ such that $y(a) = 0$, $y(b) = \infty$, and

$$\sup\{|y(s)| \mid s \in S(1)\} < \inf\{|y(s)| \mid s \in S(2)\}$$

(as in section 4 where we wrote V instead of W). We put $A(y) := \sup\{|y(s)| \mid s \in S(1)\}$, $B(y) = \inf\{|y(s)| \mid s \in S(2)\}$, $C(y) = A(y)/B(y)$. The quotient map $\pi : W \rightarrow Q_{\text{sst}}$ sends a point y with $C(y) = 0$ to the semistable point J of type $(S(1), S(2))$ and identifies $(W - \pi^{-1}(J))/\mathbb{C}^*$ with a punctured neighborhood of J . One has convergence of $\pi(y)$ to J if and only if $C(y)$ converges to 0.

On P_W , there is, up to a unique isomorphism, a unique rank one local system L , trivialized on the annulus $A(y) < |z| < B(y)$ by a section e , and which for $y \in M$, has

fiberwise monodromy α (hence trivial monodromy around the annulus, as $\prod_{s \in S(i)} \alpha_s = 1$). A model of it is the local system of constant multiples of

$$\prod_{s \in S(1)} (z - y(s))^{\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{\mu_s}.$$

We normalize the Hermitian form $(\cdot, \cdot)$ on L by $(e, e) = 1$.

For any y , fix R such that $A(y) < R < B(y)$, and let γ be the L -valued homology class represented by the cycle $|z| = R$ in P_y , positively oriented and provided with e , the trivializing section of L . It can also be viewed as a class in $H_1^{\text{lf}}(P_y, L_y)$. For $y = j$, defined by $j(S(1)) = 0$, $j(S(2)) = \infty$, it is a generator of this H_1^{lf} . The γ provide a horizontal section of $R^1\pi_!L$. Since $(\omega_j, \omega_j) = 0$, the line (ω_j) spanned by ω_j is in the boundary of the ball $B^+(\alpha)$ and the functions d_{ω_j} of (5.3) provide a horizontal family of “distance to $\mathbb{P}\omega_j$ ” functions. We will use them to compare $\mathbb{P}\omega_j$ with the holomorphic section w_α of $B^+(\alpha)$.

Proposition 6.1. *For y variable in W , if $C(y)$ tends to 0, the $d_{\omega_j}(w_\alpha(y))$ tend to 0.*

Démonstration. Fiber by fiber w_α is the line spanned by the cohomology class of

$$\omega_y = \prod_{s \in S(1)} (z - y(s))^{-\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{-\mu_s} \cdot dz \cdot e.$$

The form ω_y , as a function of y , is invariant by the action of $\mathbb{C}^*$ on (W, L) . It hence suffices to treat the case where $A(y) \rightarrow 0$, $B(y) \rightarrow \infty$. In that case, the numerator of $d_{\omega_j}(w_\alpha)$ clearly tends to 1 : it is the absolute value of

$$\frac{1}{2\pi i} \int_{|z|=R} \prod_{s \in S(1)} (z - y(s))^{-\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{-\mu_s} \cdot \frac{dz}{z}$$

which tends to 1.

We have to prove that $(\omega_y, \omega_y) \rightarrow \infty$. Indeed,

$$(\omega_y, \omega_y) = \int_{\mathbb{C}} \prod_{s \in S(1)} |z - y(s)|^{-2\mu_s} \cdot \prod_{s \in S(2)} \left|1 - \frac{z}{y(s)}\right|^{-2\mu_s} |dz \wedge d\bar{z}|$$

and this integral is bounded below by

$$\int_{A(y) < |z| < B(y)} |z|^{-2} |dz \wedge d\bar{z}| = 2\pi \cdot 2|\log C(y)|.$$

□

On Q , the proposition (7.2) has the following corollary :

Corollaire 6.2. *For each semi-stable point y of Q_{usp} , there is on a neighborhood V of y a horizontal family of functions d on the balls $B(\alpha)_{y'}$ ($y' \in V \cap Q$)—each a “distance to a point of $\partial B(\alpha)_y$ ” function—such that $d(w_\alpha(y')) \rightarrow 0$ for $y' \rightarrow y$.*

7 Extending $\tilde{w}$ by continuity

Let X, Y be complex algebraic varieties which are separated as algebraic varieties, i.e. which are Hausdorff in their complex topology.

7.1

One knows that any quasi-finite map $f : X \rightarrow Y$ —i.e. a map whose fibers are finite—admits a factorization $X \xrightarrow{j} \mathcal{X} \xrightarrow{\bar{f}} Y$, with j an open embedding, and $\bar{f}$ a finite map (= quasi-finite and proper). If X and Y are connected and normal, of the same dimension, there is an unique such factorization for which $\mathcal{X}$ is normal. One calls this $\mathcal{X}$ the normalization of Y in X .

For more general spaces, and for some maps with totally disconnected but possibly infinite fibers, a simple topological generalization of this construction has been given by R. H. Fox [?].

The spaces X and Y are taken to be Hausdorff and locally connected. A continuous map $f : X \rightarrow Y$ is called a *spread* if, for any x in X , one obtains a fundamental system of neighborhoods of x by taking, for each neighborhood V of $f(x)$, the connected component of $f^{-1}(V)$ containing x . A spread f is called *complete* if, for any $y \in Y$,

$$f^{-1}(y) \cong \varprojlim \pi_0(f^{-1}(V)),$$

the projective limit being taken on the neighborhoods V of y .

Fox proves that each spread $f : X \rightarrow Y$ can be embedded in a complete spread $\bar{f} : \bar{X} \rightarrow Y$ with the following universal property : any commutative diagram of solid arrows

$$\begin{array}{ccc} X & \longrightarrow & X' \\ f \downarrow & \circlearrowleft & \downarrow f' \\ Y & \xlongequal{\quad} & Y \end{array}$$

with f' a complete spread, can be completed as shown. For any $y \in Y$, one has

$$\bar{f}^{-1}(y) \cong \varprojlim \pi_0(f^{-1}(V)),$$

the projective limit being taken on the neighborhoods V of y . It is clear from the universal property that $\bar{X}$ is unique up to unique isomorphism, and functorial. We will call it the *completion of X over Y* .

It contains X as a dense subspace, and the topology of X is induced by that of $\bar{X}$. The space $\bar{X}$ is locally connected.

7.2

We define a space X to be connected if it is non-empty and has no proper open and closed subset.

A map $f : X \rightarrow Y$ is called a *covering map* if each $y \in Y$ has a neighborhood V for which there exists a discrete set D (possibly empty), and a commutative diagram

$$\begin{array}{ccc} D \times V & \longrightarrow & f^{-1}(V) \\ & \searrow & \downarrow \\ & & V \end{array}$$

If Y is connected and X non-empty, such a map is onto.

A map $f : X \rightarrow Y$ is called a *local homeomorphism* at $x \in X$ if x has a neighborhood U which maps homeomorphically onto a neighborhood of $f(x)$. The map f is called a *local homeomorphism* if this holds at all points $x \in X$.

Assume that X is a connected covering space of a connected locally simply connected open subset U of Y . The composed map $X \rightarrow U \hookrightarrow Y$ is then a spread. Assume further that each y in Y has a fundamental system y of open neighborhoods such that

$$\text{for } V \text{ in } y, V \cap U \text{ is connected;} \quad (7.1)$$

$$\text{for } V' \subset V'' \text{ in } y, \pi_1(V' \cap U) \rightarrow \pi_1(V'' \cap U). \quad (7.2)$$

As f is a local homeomorphism, X is open in $\bar{X}$. As X is a covering of U , one has further $\bar{X} = f^{-1}(U)$.

Fix a base point $o \in U$. One knows that the functor $(g : \tilde{U} \rightarrow U) \mapsto g^{-1}(o)$ is an equivalence of the category of coverings of U with that of discrete sets on which $\pi_1(U, o)$ acts. In particular, the connected components of $\tilde{U}$ correspond to the orbits of $\pi_1(U, o)$ on $g^{-1}(o)$. For W open in U , connected and containing o , the same applies to W and the restriction functor $(g : \tilde{U} \rightarrow U) \mapsto (g^{-1}(W) \rightarrow W)$ from coverings of U to coverings of W corresponds to the functor : restriction of the action of $\pi_1(U, o)$ to $\pi_1(W, o)$.

In particular, if $\tilde{U}$ corresponds to $\pi_1(U, o)/K$, $\pi_0(g^{-1}W) = \pi_1(W, o) \backslash \pi_1(U, o)/K$. This applies to X . If a lifting $\tilde{o}$ of o is chosen in X , the map $\pi_1(U, o) \rightarrow f^{-1}(o) : \sigma \mapsto \sigma \cdot \tilde{o}$ identifies $f^{-1}(o)$ with a suitable homogeneous space $\pi_1(U, o)/K$ of $\pi_1(U, o)$.

Fix $y \in Y$, and a neighborhood $V \in y$. Let us assume at first that $o \in V$. The decomposition group D_V is then defined as the image of $\pi_1(V \cap U, o)$ in $\pi_1(U, o)$ and one has

$$\pi_0(f^{-1}(V \cap U)) = D_V \backslash \pi_1(U, o)/K.$$

For $V' \subset V$ in y , (8.2.1) and (8.2.2) imply that $\pi_0(f^{-1}(V' \cap U)) \rightarrow \pi_0(f^{-1}(V \cap U))$ hence

$$f^{-1}(y) = D_V \backslash \pi_1(U, o)/K. \quad (7.3)$$

For a $\sigma \in \pi_1(U, o)/K$, $\sigma \cdot \tilde{o}$ is a lifting of o in the connected component $f^{-1}(V)_\sigma$ of $f^{-1}(V)$ corresponding to the double coset $D_V \sigma K$. If we use it as a base point, the fibre at o of the covering $f^{-1}(V)_\sigma$ of $V \cap U$ is identified with $D_V \sigma K / K \cong D_V / D_V \cap \sigma K \sigma^{-1}$. To the component $f^{-1}(V)_\sigma$ there corresponds a point $\tilde{x}$ in $f^{-1}(y)$. The $f^{-1}(V)_\sigma \cap f^{-1}(V')$, for V' a neighborhood of y , are the trace on X of a fundamental system of neighborhood of $\tilde{x}$ in $\bar{X}$.

For o not necessarily in V , one chooses o' in $V \cap U$ and a path p from o to o' in U . This path lifts as a path from $\tilde{o}$ to some lifting $\tilde{o}'$ of o' . It defines an isomorphism of $\pi_1(U, o)$ with $\pi_1(U, o')$ via which the above constructions, which make sense for o' , can be pushed back into $\pi_1(U, o)$. For instance, the decomposition group $D_V \subset \pi_1(U, o)$ is defined as the image of the decomposition group $D_V \subset \pi_1(U, o')$. It depends on the path p . One still has (8.2.3).

(8.2.4) A covering map $f : X \rightarrow U$ is normal if the $\pi_1(U, o)$ homogeneous space $f^{-1}(o)$ is isomorphic to a homogeneous space $\pi_1(U, o)/K$, with K a normal subgroup. If $f : X \rightarrow U$ is a normal covering map, the group $\pi_1(U, o)$ acts on X by deck-transformations, and this action extends to $\bar{X}$ by functoriality. We leave to the reader to check that, under the conditions (8.2.1), (8.2.2), one has

1. $\pi_1(U, o) \backslash \bar{X} \xrightarrow{\sim} Y$;
2. for each $y \in Y$, the stabilizer in $\pi_1(U, o)$ of a point $\tilde{x} \in \bar{f}^{-1}(y)$ is the conjugate of the decomposition group D_V determined by a path in U corresponding to $\tilde{x}$.

Remarque 7.1. Let K denote the kernel of the action of $\pi_1(U, o)$ on X . Then $\pi_1(U, o)/K$ acts properly discontinuously on X but not necessarily on $\bar{X}$.

7.3

Let w be a continuous function from X to a topological space B . We assume B regular, i.e. that any neighborhood of any point contains a closed neighborhood. The function w then extends as a continuous function on $\bar{X}$ if and only if, for any $\tilde{y} \in \bar{Y}$, $w(x)$ has a limit for $x \rightarrow \tilde{y}$ in X (convergence of w on the filter of traces on X of neighborhoods of $\tilde{y}$ in $\bar{X}$).

In the situation (8.2) (with $o \in V$ for simplicity), one can treat w as a multivalued function on U . If $\tilde{x}$ in $f^{-1}(y)$ corresponds to the connected component $f^{-1}(V)_\sigma$ of $f^{-1}(V)$, the restriction of w to $f^{-1}(V)_\sigma$ is a multivalued function w_σ on $V \cap U$. That $w(x)$ has a limit for $x \rightarrow \tilde{x}$ means that there is b in B , and for each neighborhood W of b a neighborhood V' of y , such that all determinations of $w_\sigma(u)$, for $u \in V' \cap U$, are in W .

Exemple 7.2. If we take for Y the unit disk $D \subset \mathbb{C}$, for U the punctured unit disk D^* and for X the universal covering of D^* , the completion of X over D is deduced from X by adding one point above 0 : $\bar{X} = X \cup \{0\}$. For α , real > 0 , the function z^α has a limit for $z \rightarrow 0$ in X , but it has no limit for non real α , with $(\alpha) > 0$. Similarly, $z^\alpha \log z$ has no limit.

7.4

With the assumptions (8.2.1) and (8.2.2) on (Y, U) , let π be the flat fiber space on U defined by an action θ of $\pi_1(U, o)$ on a regular topological space B , and let (X, o) be the smallest covering of (U, o) on which π becomes trivial : one has $f^{-1}(o) = \pi_1(U, o)/(\theta)$. A continuous section w of π defines by pull-back to X an equivariant continuous map from X to B (and conversely). We write w for this map. When does it extend to $\bar{X}$? Unravelling (8.3), we find that w has a limit for $x \rightarrow \tilde{x}$ above y if and only if there is a horizontal section b of π over the trace on U of a neighborhood $V \in \mathcal{V}_y$ of y , toward which w converges for $y' \rightarrow y$ ($y' \in U$), in the following sense. For $y_1 \in V \cap U$ and W_1 a neighborhood of $b(y_1)$ in the fibre $\pi^{-1}(y_1)$ of π at y_1 , there is a neighborhood V' of y such that for any $y_2 \in V' \cap U$, and for W_2 a neighborhood of $b(y_2)$ in the fiber of π at y_2 , obtained by horizontal transport of W_1 along any path in $V \cap U$ from y_1 to y_2 , one has $w(y_2) \in W_2$. This holds if b has a system of horizontal neighborhoods W which are fiber arbitrarily small, such that $w(y_1)$ is in W for y_1 close enough to y . Of course, b defines the limit of w .

In the rest of section 8, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are in force.

7.5

As in (3.8), fix a base point $o \in Q$, and let $p : (\tilde{Q}, o) \rightarrow (Q, o)$ be the smallest covering of (Q, o) on which the flat projective space bundle $B(\alpha)_Q$ becomes trivial. For $\tilde{Q} \subset Y \subset Q_{\text{sst}}$ and Y open in Q_{sst} , the composed map $\tilde{Q} \rightarrow Y$ is a spread. We will write $\tilde{Y}$ for the completion of $\tilde{Q}$ over Y . The assumptions (8.2.1) and (8.2.2) are satisfied by (Y, Q) .

It follows from (8.2.4) that

$$\pi_1(Q, o) \text{ acts on } \tilde{Q}_{\text{sst}}; \quad (7.4)$$

$$\pi_1(Q, o) \backslash \tilde{Q}_{\text{sst}} = Q_{\text{sst}}. \quad (7.5)$$

Proposition 7.3. *The map $\tilde{w}$, of (3.8) extends as a continuous map from $\tilde{Q}_{\text{sst}}$ to the closed ball $\overline{B^+(\alpha)_o}$, provided with the topology (5.4). The inverse image of $\partial B^+(\alpha)_o$ consists of the stable points.*

Démonstration. We are in the situation envisioned in (8.5), if we take for the flat fiber space with fiber $\overline{B^+(\alpha)}$, completing $B^+(\alpha)$. That the extendability criterion of (8.6) applies is (6.8) for y stable and (7.3) for y semi-stable. Further, the limit occurring in criterion (8.5) lies in $\partial B^+(\alpha)_o$ if and only if y is in Q_{usp} . $\square$

The extended map $\tilde{Q}_{\text{sst}} \rightarrow \overline{B^+(\alpha)_o}$ is also denoted $\tilde{w}$. It is $\pi_1(Q, o)$ -equivariant.

7.6

For T a stable partition of S (cf. (6.9)), the system of the $Q_{T'}$, for T' finer than T , is topologically constant along Q_T , and the inverse image $\tilde{Q}_T$ of Q_T in $\tilde{Q}_{\text{sst}}$ is hence a covering of Q_T ($\cong Q_*(T)$). By (6.9), on each connected component $\tilde{Q}_T^c$ of $\tilde{Q}_T$, the flat ball bundle $B(\alpha(T))^+$ trivializes, and can be identified with a flat sub-ball bundle of the constant bundle $B(\alpha)^+$, in such a way that

$$\tilde{w}|_{\tilde{Q}_T^c} = w_{\alpha(T)}. \quad (7.6)$$

From (3.9), we hence get :

Proposition 7.4. *For any component $\tilde{Q}_T^c$ of $\tilde{Q}_T$, $\tilde{w}|_{\tilde{Q}_T^c}$ is an étale map from $\tilde{Q}_T^c$ to a sub-ball of $B(\alpha)^+$.*

8 Codimension 1 'etateness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

Let T be a stable partition of S (cf. (6.9)), with $\text{card } T = \text{card } S - 1$; one coset of T has two elements and all others only one. Our aim in this section is to describe the flat bundle $B(\alpha)_Q$ over Q (cf. (3.7)) and the holomorphic section w_α in the neighborhood of a point in $Q_T \subset Q_{\text{st}}$ (cf. (6.9)).

8.1

The subspace Q_T of Q_{sst} , defined in (6.9), is a locally closed purely one codimensional complex submanifold of the complex manifold Q_{sst} and $Q \cap Q_T$ is open in Q_{st} . The monodromy around Q_T is the following conjugacy class in $\pi_1(Q)$.

Let $\varphi : D \rightarrow Q \cup Q_T$ be an embedding of the unit disc with $\varphi(D)$ transversal to Q_T and $\varphi^{-1}(Q_T) = \{0\}$; φ induces an embedding of $D^* := D - \{0\}$ in Q . The fundamental group $\pi_1(D^*)$ is canonically isomorphic to $\mathbb{Z}$, generated by the loop $[0, 1] \rightarrow D^* : t \mapsto z_0 \cdot \exp(2\pi it)$. The decomposition group at $\varphi(0)$ is the image of $\pi_1(D^*)$ in $\pi_1(Q)$ and the monodromy around Q_T is defined as the image of the positive generator of $\pi_1(D^*)$. Both are well defined up to conjugacy and independent of φ . Let us fix a base point o in Q . The fundamental group $\pi_1(Q, o)$ acts on $B(\alpha)_o$. We will compute the action of the monodromy around Q_T .

Let $\{s_1, s_2\}$ be the two element coset of T and choose additional elements $s_3, s_4 \in S$. We assume that $\mathbb{P}$ is the standard projective line $\mathbb{P}^1$ and we choose in it two distinct points $b, c \neq 0$. Let M_0 be the space of injective maps $y : S \rightarrow \mathbb{P}$ with $y(s_1) = 0$, $y(s_2) = b$, $y(s_4) = c$. Let M_{0T} be the space of maps $S \rightarrow \mathbb{P}$ with $y(s) = y(t)$ if and only if s and t are in the same coset of T , and $y(s_1) = y(s_2) = 0$, $y(s_3) = b$, $y(s_4) = c$. The quotient map $M_{\text{st}} \rightarrow Q_{\text{st}}$ induces isomorphisms $M_0 \xrightarrow{\sim} Q$, $M_{0T} \xrightarrow{\sim} Q_T$ and $M_0 \cup M_{0T} \xrightarrow{\sim} Q \cup Q_T$. We will work on $M_0 \cup M_{0T}$, rather than on $Q \cup Q_T$.

The monodromy around Q_T is represented by the following loop in M_0 : it starts from a point y_0 with, for some r , $y_0(S_1)$ in the disc D_r of radius r around 0 and the $y_0(s_i)$ ($i \neq 1, 2$) outside of this disc ; it is given by the map from $[0, 1]$ to M_0 , $t \mapsto y_t$, with $y_t(s_i)$ constant for $i \neq 2$, and $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$.

Above this loop, one can construct on P_{y_t} a local system L with fiberwise monodromy α , as follows : we fix a base point d on $\mathbb{P}$, outside of $y_0(S)$ and of D_r , and take for L the local system trivialized on the constant section d and with the required monodromy. The action of monodromy on $B(\alpha)_{y_0} = \mathbb{P}H^1(P_{y_0}, L)$ is deduced from the action of monodromy on $H^1(P_{y_0}, L)$.

Proposition 8.1. *In a suitable basis of $\mathbb{P}H^1(P_{y_0}, L)$, the above action of the monodromy around Q_T is given by a diagonal matrix $\text{diag}(\alpha_1\alpha_2, 1, \dots, 1)$.*

Démonstration. Set $S_1 = \{s_1, s_2\}$ and $S_2 = S - S_1$. Let T_1, T_2 be trees as in (2.5) with the vertices of T_i in S_i ($i = 1, 2$) and let $\beta : T_1 \amalg T_2 \hookrightarrow \mathbb{P}$ be an embedding with $\beta|_S = y_0$. We may and shall assume that $\beta|_{T_1}$ is the straight line segment from $y_0(s_1) = 0$ to $y_0(s_2)$ and that $\beta(T_2)$ is outside of the disc D_r . For each (open) edge a of T_1 or T_2 , we choose one of its orientations, and a non zero section $t(a)$ of β^*L above it. As in (2.4), (2.5), each edge then defines a closed L -valued current $t(a) \cdot \beta|_a$, and the cohomology classes of those currents form a basis of $H^1(P_{y_0}, L)$. It is the basis we will use.

To transport our basis elements around the loop y_t ($0 \leq t \leq 1$) horizontally, it suffices to deform β with $\beta_t|_S = y_t$, dragging along the $t(a)$. We will keep $\beta_t|_{T_2}$ fixed and take for $\beta_t|_{T_1}$ the family of straight line segments from $y_t(s_1) = 0$ to $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$. Outside of $D_r \subset \mathbb{P}$, the fibers P_{y_t} and L remain constant, and the $t(a) \cdot \beta_t|_a$ for a an edge of T_2 are, as currents, independent of t . It follows that the monodromy is trivial on the $t(a) \cdot \beta_t|_a$ for a an edge of T_2 .

Let a be the unique edge of T_1 . When $\beta_t(a)$ ($0 \leq t \leq 1$) has made a complete turn around 0, $t(a)$ is multiplied by $\alpha_1\alpha_2$. Indeed, if $p_t(x) : [0, 1] \rightarrow \mathbb{P}$ is a path from the base point d on $\mathbb{P}$ to a point d_t on $\beta_t(a)$, and if this path deforms with t so as to always avoid $\beta_t(S)$, one can take $t(a)$ to be obtained at d_t by transporting along p_t the trivialising section e of L at d . As the following picture shows, p_1 can be chosen to differ from p_0 by a loop around $y_0(s_1)$ and $y_0(s_2)$. Around this loop, L has monodromy $\alpha_1\alpha_2$, and the claim follows.

$$\begin{array}{ccccc}
& y_0(s_1) & y_0(s_2) & d & \\
t = 0 : & \bullet & \bullet & \bullet & \\
&) & | & / & \\
& & y_t(s_2) & & \\
& & \bullet & & \\
t \text{ small} : & y_0(s_1) & | & d & \\
& \bullet & | & \bullet & \\
& y_0(s_1) & y_0(s_2) & d & \\
t = 1 : & \bullet & \bullet & \bullet & \\
&) & | & / &
\end{array}$$

□

Remarque 8.2. In the computation (9.2), we have made use of the assumption $\mu_1 + \mu_2 < 1$ only in the weaker form : each $\mu_i < 1$. In the case $\mu_1 + \mu_2 > 1$, the monodromy when s_2 turns around s_1 would be given by the same formula, but the basis e in which it holds would have $(e_1, e_1) > 0$ instead of $(e_1, e_1) < 0$: it would be a pseudo-reflection with respect to a hyperplane not meeting the ball $B^+(\alpha)$.

More precisely, one checks that if the section $t(a)$ of L , for a the only edge of T_1 , is chosen of length one : $(t(a), t(a)) = 1$, then

$$(e_1, e_1) = 2\pi i \cdot \frac{2}{1 - \alpha_1 \alpha_2} (1 + \mu_1 + \mu_2 - 1)$$

We will not need this formula.

Remarque 8.3. For later computations, it will be more convenient to work with the basis of the dual space $H_1^{\text{lf}}(P - y_0(S), L^\vee)$ to $H^1(P - y_0(S), L)$ given by a choice of non zero sections $t^\vee(a)$ of $\beta^* L^\vee$ above the edges of T_1 or T_2 . The same arguments as in (9.2), show that on such a basis of the dual space, monodromy around Q_T acts by $\text{diag}((\alpha_1 \alpha_2)^{-1}, 1, \dots, 1)$.

8.2

Fix $y \in Q_T$. We will describe the section w_α of $B(\alpha)_Q$ near y . A neighborhood of y in $Q \cup Q_T$ and local coordinates are chosen as follows. First, one replaces $Q \cup Q_T$ by $M_0 \cup M_{0T}$, as in (9.1). It is convenient to choose b and c , in the definition of M_0 , so that $\infty \notin y(S)$ and that the discs of radius 1 around the points in $y(S)$ are disjoint. We do so. The chosen neighborhood U is then identified with the space of $m \in M_0 \cup M_{0T}$ such that $|m(s) - y(s)| < 1$. The chosen coordinates are the $m(s) - y(s)$ ($s \neq s_1, s_2, s_4$). If $S' := S - \{s_1, s_2, s_3, s_4\}$, this system of coordinates identifies U with $D \times D^{S'}$, and $U \cap M_0$ with $D^* \times D^{S'}$.

We choose L as in (9.1), using a base point d on $\mathbb{P}$ at distance > 1 from $y(S)$ —for instance d a large positive real number. In $U \cap M_0$, we choose a base point m_0 —for instance with $m_0(s_2) = -1$ and $m_0(s) = y(s)$ for $s \neq s_2$. We view w_α as a multivalued map from $U \cap M_0$ to $B(\alpha)_{m_0} = \mathbb{P}H^1(\mathbb{P} - m_0(S), L)$, and on $H^1(\mathbb{P} - m_0(S), L)$ we use the coordinates provided by a basis (9.4) of the dual space. The multivalued map w_α then lifts to the multivalued map with values in $\mathbb{C}^{N-3}$ having as coordinates (up to multiplicative

constants)

$$\int_{C_{s_1}} t^\vee(z) \cdot \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

as first coordinate and as the other coordinates the integrals

$$\int \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

taken along a path remaining outside the disc $|z| < 1$ from one $m(s)$ to another ($s \neq s_1, s_2$). When $m(s_2)$ turns once around $m(s_1)$, the first coordinate gets multiplied by $(\alpha_1 \alpha_2)^{-1} = \exp(2\pi i(1 - \mu_1 - \mu_2))$ while the others regain their value (9.4). This allows us to write the first coordinate as $(m(s_2) - m(s_1))^{1-\mu_1-\mu_2} I(m)$, with $I(m)$ an ordinary function on $U \cap M_0$. When m tends to $m_1 \in M_{0T}$, $I(m)$ tends to the limit

$$\int_{C_{s_1}} \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot z^{-\mu_1-\mu_2} dz,$$

which is non zero (2.17.1), hence $I(m)$ extends as a holomorphic function on U , invertible on $U \cap M_{0T}$. The other coordinates extend as holomorphic functions on U , having as restriction to $U \cap M_{0T}$ integrals

$$\int z^{-\mu_1-\mu_2} \cdot \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} dz.$$

By (3.9) applied to $\mu(T)$, these functions on $U \cap M_{0T}$ are the projective coordinates of an 'etale map from $U \cap M_{0T}$ to the projective $[(N-2)-3]$ -space (cf. (8.9)).

8.3

In suitable local coordinates, the multivalued map w_α from $U \cap M_0$ to $B(\alpha)_{m_0}$ has the form

$$(z_0, z) \mapsto (z_0^{1-\mu_1-\mu_2} I(z_0, z), J(z_0, z));$$

here, (z_0, z) are local coordinates on U coming from the product decomposition above, $U \cong D \times D^n$, $U \cap M_0 \cong D^* \times D^n$; the local coordinates on the image of U in $B(\alpha)_{m_0}$ are selected on an open set V with $V \cong D \times D^n$ and $\{0\} \times D^n$ lying in a hyperplane; and finally, $I : D \times D^n \rightarrow \mathbb{C}$ and $J : D \times D^n \rightarrow D^n$ are holomorphic maps, with $I(0, z) \neq 0$ and $J(0, z) : D^n \rightarrow D^n$

'etale.

At a point of $\{0\} \times D^n \subset D \times D^n$, the holomorphic functions $z_0 \cdot I(z_0, z)^{1/(1-\mu_1-\mu_2)}$ and $J(z_0, z)_i$ ($1 \leq i \leq n$) form a system of coordinates. In those coordinates, the multivalued map w_α is

$$(z_0, z_1, \dots, z_n) \mapsto (z_0^{1-\mu_1-\mu_2}, z_1, \dots, z_n).$$

Suppose $1 - \mu_1 - \mu_2$ is rational, set $1 - \mu_1 - \mu_2 = k/\ell$, in reduced terms. If $\tilde{D}$ is another copy of the disc, with coordinate u , mapping to D by $u \mapsto u^\ell$, the pull back to $\tilde{D} \times D^n$ of the multivalued map above is the holomorphic map

$$(u, z_1, \dots, z_n) \mapsto (u^k, z_1, \dots, z_n).$$

This map is

'etale in a neighborhood of $\{0\} \times D^n \subset \tilde{D} \times D^n$ if and only if $k = 1$, i.e. if and only if $(1 - \mu_1 - \mu_2)^{-1}$ is an integer.

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7.1

As in (3.8), fix a base point $o \in Q_*$ and let $p : (\tilde{Q}_*, o) \rightarrow (Q_*, o)$ be the smallest covering of (Q_*, o) on which the flat projective space bundle $B(\alpha)_q$ becomes trivial. For $Q_* \subset Y \subset Q_{*st}$ and Y open in Q_{*st} , the composed map $\tilde{Q}_* \rightarrow Y$ is a spread. We will write $\tilde{Y}$ for the completion of $\tilde{Q}_*$ over Y . The assumptions (8.2.1) and (8.2.2) are satisfied by (Y, Q_*) .

It follows from (8.2.4) that

$$\pi_1(Q_*, o) \text{ acts on } \tilde{Q}_{*st}, \quad (1)$$

$$\pi_1(Q_*, o) \backslash \tilde{Q}_{*st} = Q_{*st}. \quad (2)$$

Proposition 7.1. *The map $\tilde{w}$ of (3.8) extends as a continuous map from $\tilde{Q}_{*st}$ to the closed ball $B(\alpha)_0^+$, provided with the topology (5.4). The inverse image of $B(\alpha)_0^+$ consists of the stable points.*

Démonstration. We are in the situation envisioned in (8.5), if we take for $\mathcal{E}$ the flat fiber space with fiber $B(\alpha)^+$, completing $B(\alpha)^+$. That the extendability criterion of (8.6) applies is (6.8) for y stable and (7.3) for y semi-stable. Further, the limit occurring in criterion (8.5) lies in $\partial B(\alpha)^+$ if and only if y is in Q_{cusp} . $\square$

The extended map $\tilde{Q}_{*st} \rightarrow B(\alpha)_0^+$ is also denoted $\tilde{w}$. It is $\pi_1(Q_*, o)$ -equivariant.

7.2

For T a stable partition of S (cf. (6.9)), the system of the $Q_{T'}$, for T' finer than T , is topologically constant along Q_T , and the inverse image $\tilde{Q}_T$ of Q_T in $\tilde{Q}_{*st}$ is hence a covering of Q_T ($\cong Q_*(T)$). By (6.9), on each connected component $\tilde{Q}_T^0$ of $\tilde{Q}_T$, the flat ball bundle $B(\alpha(T))^+$ trivializes, and can be identified with a flat sub-ball bundle of the constant bundle $B(\alpha)^+$, in such a way that

$$\tilde{w} | \tilde{Q}_T^0 = I. \quad (3)$$

From (3.9), we hence get :

Proposition 7.2. *For any component $\tilde{Q}_T^0$ of $\tilde{Q}_T$, $\tilde{w} | \tilde{Q}_T^0$ is an étale map from $\tilde{Q}_T^0$ to a sub-ball of $B(\alpha)^+$.*

8 Codimension 1 étaleness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

Let T be a stable partition of S (cf. (6.9)), with $\text{card } T = \text{card } S - 1$; one coset of T has two elements and all others only one. Our aim in this section is to describe the flat bundle $B(\alpha)_q$ over $\tilde{Q}_*$ (cf. (3.7)) and the holomorphic section $\tilde{w}_q$ in the neighborhood of a point in $\tilde{Q}_T \subset \tilde{Q}_{*st}$ (cf. (6.9)).

8.1

The subspace Q_T of Q_{*st} , defined in (6.9), is a locally closed purely one codimensional complex submanifold of the complex manifold Q_{*st} and $Q_* \cup Q_T$ is open in Q_{*st} . The monodromy around Q_T is the following conjugacy class in $\pi_1(Q_*)$.

Let $\varphi : D \rightarrow Q_* \cup Q_T$ be an embedding of the unit disc with $\varphi(D)$ transversal to Q_T and $\varphi^{-1}(Q_T) = \{o\}$; φ induces an embedding of $D^* := D - \{o\}$ in Q_* . The fundamental group $\pi_1(D^*)$ is canonically isomorphic to $\mathbb{Z}$, generated by the loop $[0, 1] \rightarrow D^* : t \mapsto z_0 \cdot \exp(2\pi it)$. The decomposition group at $\varphi(o)$ is the image of $\pi_1(D^*)$ in $\pi_1(Q_*)$ and the monodromy around Q_T is defined as the image of the positive generator of $\pi_1(D^*)$. Both are well defined up to conjugacy and independent of φ .

Let us fix a base point o in Q_* . The fundamental group $\pi_1(Q_*, o)$ acts on $B(\alpha)_o$. We will compute the action of the monodromy around Q_T .

Let $\{s_1, s_2\}$ be the two element coset of T and choose additional elements $s_3, s_4 \in S$. We assume that P is the standard projective line $\mathbb{P}^1$ and we choose in it two distinct points $b, c \neq o$. Let M_0 be the space of injective maps $y : S \rightarrow P$ with $y(s_1) = o$, $y(s_3) = b$, $y(s_4) = c$. Let M_{0T} be the space of maps $S \rightarrow P$ with $y(s) = y(t)$ if and only if s and t are in the same coset of T , and $y(s_1) = y(s_2) = o$, $y(s_3) = b$, $y(s_4) = c$.

The quotient map $M_0^{st} \rightarrow Q_{*st}$ induces isomorphisms $M_0 \xrightarrow{\sim} Q_*$, $M_{0T} \xrightarrow{\sim} Q_T$ and $M_0 \cup M_{0T} \xrightarrow{\sim} Q_* \cup Q_T$. We will work on $M_0 \cup M_{0T}$, rather than on $Q_* \cup Q_T$.

The monodromy around Q_T is represented by the following loop in M_0 : it starts from a point y_0 with, for some r , $y_0(S_1)$ in the disc D_r of radius r around o and the $y_0(s_i)$ ($i \neq 1, 2$) outside of this disc; it is given by the map from $[0, 1]$ to M_0 , $t \mapsto y_t$, with $y_t(s_i)$ constant for $i \neq 2$, and $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$.

Above this loop, one can construct on $\mathbb{P}^1$ a local system L with fiberwise monodromy, as follows: we fix a base point d on P , outside of $y_0(S)$ and of D_r , and take for L the local system trivialized on the constant section d and with the required monodromy. The action of monodromy on $B(\alpha)_{y_0} = \mathbb{P}H^1(\mathbb{P}_{y_0}, L)$ is deduced from the action of monodromy on $H^1(\mathbb{P}_{y_0}, L)$.

Proposition 8.1. *In a suitable basis of $H^1(\mathbb{P}_{y_0}, L)$, the above action of the monodromy around Q_T is given by a diagonal matrix $\text{diag}(\alpha_1\alpha_2, 1, \dots, 1)$.*

Démonstration. Set $S_1 = \{s_1, s_2\}$ and $S_2 = S - S_1$. Let T_1, T_2 be trees as in (2.5) with the vertices of T_i in S_i ($i = 1, 2$) and let $\beta : T_1 \cup T_2 \hookrightarrow P$ be an embedding with $\beta|_S = y_0$. We may and shall assume that $\beta|_{T_1}$ is the straight line segment from $y_0(s_1) = o$ to $y_0(s_2)$ and that $\beta(T_2)$ is outside of the disc D_r . For each (open) edge a of T_1 or T_2 , we choose one of its orientations, and a non zero section $\ell(a)$ of β^*L above it. As in (2.4), (2.5), each edge then defines a closed L -valued current $\ell(a) \cdot \beta_*[a]$, and the cohomology classes of those currents form a basis of $H^1(\mathbb{P}_{y_0}, L)$. It is the basis we will use.

To transport our basis elements around the loop y_t ($0 < t < 1$) horizontally, it suffices to deform β with $\beta_t|_S = y_t$, dragging along the $\ell(a)$. We will keep $\beta_t|_{T_2}$ fixed and take for $\beta_t|_{T_1}$ the family of straight line segments from $y_t(s_1) = o$ to $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$. Outside of $D_r \subset P$, the fibers $\mathbb{P}_{y_t}$ and L remain constant, and the $\ell(a) \cdot \beta_{t*}[a]$ for a an edge of T_2 are, as currents, independent of t . It follows that the monodromy is trivial on the $\ell(a) \cdot \beta_*[a]$ for a an edge of T_2 .

Let a be the unique edge of T_1 . When $\beta_t(a)$ ($0 \leq t \leq 1$) has made a complete turn around o , $\ell(a)$ is multiplied by $\alpha_1\alpha_2$. Indeed, if $p_t(x) : [0, 1] \rightarrow P$ is a path from the base point d on P to a point d_t on $\beta_t(a)$, and if this path deforms with t so as to always avoid $\beta_t(S)$, one can take $\ell(a)$ to be obtained at d_t by transporting along p_t the trivialising

section e of L at d . As the following picture shows, p_1 can be chosen to differ from p_0 by a loop around $y_0(S_1)$ and $y_0(S_2)$. Around this loop, L has monodromy $\alpha_1\alpha_2$, and the claim follows.

$$\begin{array}{rcccl}
& y_0(s_1) & y_0(s_2) & d & \\
t=0 : &) & I / \dots\dots & & \\
& & y_0(s_2) & & \\
& & y_0(s_1) & & \\
t \text{ small} : & & d & & \\
& & -\circ-\circ-\circ-\circ-\circ- & & \\
& & d & & \\
t=1 : & \circ-\circ-\circ-\circ-\circ- & & &
\end{array}$$

□

Remarque 8.2. In the computation (9.2), we have made use of the assumption $\alpha_1 + \alpha_2 < 1$ only in the weaker form $\alpha_i \neq 1$. In the case $\alpha_1 + \alpha_2 > 1$, the monodromy when s_2 turns around s_1 would be given by the same formula, but the basis e in which it holds would have $(e_1, e_1) > 0$ instead of $(e_1, e_1) < 0$: it would be a pseudo-reflection with respect to a hyperplane not meeting the ball $B(\alpha)^+$.

More precisely, one checks that if the section $\ell(a)$ of L , for a the only edge of T_1 , is chosen of length one : $(\ell(a), \ell(a)) = 1$, then

$$(e_1, e_1) = 2i \frac{\sin \pi(\alpha_1 + \alpha_2) \cdot \sin \pi(1 - \alpha_1 + 1 - \alpha_2)}{\sin \pi(2 - \alpha_1 - \alpha_2)}.$$

We will not need this formula.

Remarque 8.3. For later computations, it will be more convenient to work with the basis of the dual space $H_1(\mathbb{P} - y_0(S), L^\vee)$ to $H^1(\mathbb{P} - y_0(S), L)$ given by a choice of non zero sections $\ell'(a)$ of $\beta^* L^\vee$ above the edges of T_1 or T_2 . The same arguments as in (9.2), show that on such a basis of the dual space, monodromy around Q_T acts by $\text{diag}((\alpha_1\alpha_2)^{-1}, 1, \dots, 1)$.

8.2

Fix $y \in \tilde{Q}_T$. We will describe the section $\tilde{w}_q$ of $B(\alpha)_q$ near y . A neighborhood of y in $\tilde{Q}_* \cup \tilde{Q}_T$ and local coordinates are chosen as follows. First, one replaces $\tilde{Q}_* \cup \tilde{Q}_T$ by $\tilde{M}_0 \cup \tilde{M}_{0T}$, as in (9.1). It is convenient to choose b and c , in the definition of M_0 , so that $\infty \notin y(S)$ and that the discs of radius 1 around the points in $y(S)$ are disjoint. We do so. The chosen neighborhood U is then identified with the space of $m \in M_0 \cup M_{0T}$ such that $|m(s) - y(s)| < 1$. The chosen coordinates are the $m(s) - y(s)$ ($s = s_1, s_2, s_3, s_4$). If $S' := S - \{s_1, s_2, s_3, s_4\}$, this system of coordinates identifies U with $D \times D^{S'}$, and $U \cap M_0$ with $D' \times D^{S'}$.

We choose L as in (9.1), using a base point d on P at distance > 1 from $y(S)$ — for instance d a large positive real number. In $U \cap M_0$, we choose a base point m_0 — for instance with $m_0(s_2) = -1$ and $m_0(s) = y(s)$ for $s \neq s_2$. We view $\tilde{w}_q$ as a multivalued map from $U \cap M_0$ to $B(\alpha)_{m_0} = \mathbb{P}H^1(\mathbb{P} - m_0(S), L)$, and on $H^1(\mathbb{P} - m_0(S), L)$ we use the coordinates provided by a basis (9.4) of the dual space. The multivalued map $\tilde{w}_q$ then lifts to the multivalued map with values in $\mathbb{C}^{S-3}$ having as coordinates (up to multiplicative constants)

$$\int_{C(s_1, s_2)} \ell'(z) \cdot \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

as first coordinate and as the other coordinates the integrals

$$\int_{[s]} \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

taken along a path remaining outside of the disc $|z| < 1$ from one $m(s)$ to another ($s \neq s_1, s_2$). When $m(s_2)$ turns once around $m(s_1)$, the first coordinate gets multiplied by $(\alpha_1 \alpha_2)^{-1} = \exp(2\pi i(1 - \mu_1 - \mu_2))$ while the others regain their value (9.4). This allows us to write the first coordinate as $(m(s_2) - m(s_1))^{1-\mu_1-\mu_2} \cdot I(m)$, with $I(m)$ an ordinary function on $U \cap M_0$. When m tends to $m_1 \in M_{0T}$, $I(m)$ tends to the limit

$$\int_{C(s_1, s_2)} \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot z^{-\mu_1-\mu_2} (z-1)^{-\mu_2} dz,$$

which is non zero (2.17.1), hence $I(m)$ extends as a holomorphic function on U , invertible on $U \cap M_{0T}$. The other coordinates extend as holomorphic functions on U , having as restriction to $U \cap M_{0T}$ integrals

$$\int_{[s]} z^{-\mu_1-\mu_2} \cdot \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot dz.$$

By (3.9) applied to μ_T , these functions on $U \cap M_{0T}$ are the projective coordinates of an étale map from $U \cap M_{0T}$ to the projective $[(N-2)-3]$ -space (cf. (8.9)).

8.3

In suitable local coordinates, the multivalued map $\tilde{w}_q$ from $U \cap \widetilde{M}_0$ to $B(\alpha)_q$ has the form

$$(z_0, z) \mapsto (z_0^{1-\mu_1-\mu_2} \cdot I(z_0, z), J(z_0, z));$$

here, (z_0, z) are local coordinates on U coming from the product decomposition above, $U \cong D \times D^n$, $U \cap \widetilde{M}_0 \cong D^* \times D^n$; the local coordinates on the image of U in $B(\alpha)_q$ are selected on an open set V with $V \cong D \times D^n$ and $\{0\} \times D^n$ lying in a hyperplane; and finally, $I : D \times D^n \rightarrow \mathbb{C}$ and $J : D \times D^n \rightarrow D^n$ are holomorphic maps, with $I(0, z) \neq 0$ and $J(0, z) : D^n \rightarrow D^n$ étale.

At a point of $\{0\} \times D^n \subset D \times D^n$, the holomorphic functions $z_0 \cdot I(z_0, z)^{1/(1-\mu_1-\mu_2)}$ and $J(z_0, z)_i$ ($1 \leq i \leq n$) form a system of coordinates. In those coordinates, the multivalued map $\tilde{w}_q$ is

$$(z_0, z_1, \dots, z_n) \mapsto (z_0^{1-\mu_1-\mu_2}, z_1, \dots, z_n).$$

Suppose $1 - \mu_1 - \mu_2$ is rational, set $1 - \mu_1 - \mu_2 = t/k$, in reduced terms. If $\tilde{D}$ is another copy of the disc, with coordinate u , mapping to D by $u \mapsto u^k$, the pull back to $\tilde{D} \times D^n$ of the multivalued map above is the holomorphic map

$$(u, z_1, \dots, z_n) \mapsto (u^t, z_1, \dots, z_n).$$

This map is étale in a neighborhood of $\{0\} \times D^n \subset \tilde{D} \times D^n$ if and only if $t = 1$, i.e. if and only if $(1 - \mu_1 - \mu_2)^{-1}$ is an integer.

8.4

Let Y be a complex analytic manifold, $\mathcal{F}$ a family of disjoint closed complex analytic submanifolds purely of complex codimension 1, $Z = \bigcup\{H; H \in \mathcal{F}\}$, and $\kappa : \mathcal{F} \rightarrow \mathbb{N}$ a positive integer valued function.

A continuous map $\phi : R \rightarrow Y$ is called a *branched cover with branch locus Z of order κ* if and only if :

1. ϕ induces a covering map from $\phi^{-1}(Y - Z)$ to $Y - Z$.
2. For any $H \in \mathcal{F}$, if we put $k := \kappa(H)$, any $y \in H$ admits a neighborhood V in Y such that the restriction of ϕ to any connected component of $\phi^{-1}(V)$ is topologically equivalent to the map

$$D^n \rightarrow D^n : (z_1, \dots, z_n) \mapsto (z_1^k, z_2, \dots, z_n).$$

A branched cover $\phi : R \rightarrow Y$ has a unique complex analytic manifold structure such that ϕ is holomorphic.

Proposition 8.4. *Let $\phi : R \rightarrow Y$ be a branched cover with branch locus Z . It follows from the definitions that ϕ is the completion of $\phi^{-1}(Y - Z) \rightarrow Y - Z$ over Y . Moreover, given any commutative diagram*

$$\begin{array}{ccc} \tilde{Q}_* & \xrightarrow{\quad} & R \\ & \searrow & \swarrow \phi \\ & Y - Z & \end{array}$$

with p and ϕ spreads, then there is a diagram

$$\begin{array}{ccc} \tilde{Q}_{*st} & \xrightarrow{\quad} & \tilde{R} \\ & \searrow & \swarrow \\ & Y & \end{array}$$

with $\tilde{p}$ and $\tilde{\phi}$ the completions of the spreads above, which is also commutative by the universal property of completions (8.1.1).

8.5

Let Q be an open set in a locally connected space Y . Let $p : \tilde{Q} \rightarrow Q$ be a spread and denote by $\bar{p}$ also the completion $\tilde{Y} \rightarrow Y$ of p . Then for any open set U in Y , the restriction of $\bar{p}$ to a connected component $\bar{p}^{-1}(U)_C$ of $\bar{p}^{-1}(U)$ is the completion of the spread $\bar{p}^{-1}(U)_C \cap \bar{p}^{-1}(U \cap Q) \rightarrow U \cap Q$.

8.6

We now assume that the μ_s are rational (cf. (2.12)). Let $\mathcal{T}_1$ denote the family of stable partitions T of S such that $\text{card } T = \text{card } S - 1$. For T in $\mathcal{T}_1$, and $\{s, t\}$ the coset of T with two elements, let k_T be the denominator of $1 - \mu_s - \mu_t$. Set

$$Q_1 = Q_* \cup \bigcup_{T \in \mathcal{T}_1} Q_T,$$

$\tilde{Q}_1$ = completion of $\tilde{Q}_*$ over Q_1 ,

$$\kappa(Q_T) = k_T \text{ for each } T \in \mathcal{T}_1.$$

Proposition 8.5. *The map $\tilde{Q}_1 \rightarrow Q_1$ is a branched cover with branch locus $\bigcup Q_T$ of order κ .*

Démonstration. By (9.8), the problem is local around each $y \in Q_T$. The claim then results from (9.2) and the fact that the order in the projective unitary group of $\text{diag}(\alpha_s \alpha_t, 1, \dots, 1)$ is the denominator of $1 - \mu_s - \mu_t$. $\square$

Proposition 8.6. *The map $\tilde{w} : \tilde{Q}_1 \rightarrow B^+(\alpha)_0$ is holomorphic.*

Démonstration. For $\tilde{w}$ is holomorphic on $\tilde{Q}_*$ and is continuous on $\tilde{Q}_1$ by Proposition (8.7). Since $\tilde{Q}_1 - \tilde{Q}_*$ has $\mathbb{C}$ -codimension 1, the assertion follows from the theorem on removable singularities. $\square$

Proposition 8.7. *Assume that*

$$\text{INT : } 0 < \mu_s < 1, \quad \sum \mu_s = 2 \text{ and for all } s, t \in S \text{ distinct and with } \mu_s + \mu_t < 1, \\ (1 - \mu_s - \mu_t)^{-1} \text{ is an integer.}$$

Then, $\tilde{w} : \tilde{Q}_1 \rightarrow B^+(\alpha)_0$ is étale.

Démonstration. This follows from the local description (9.6). $\square$

9 Proof of discreteness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

9.1

We will give two proofs that when condition INT is satisfied, the image Γ of $\pi_1(Q_*, o)$ in $\text{PU}(1, N - 2)$ is discrete.

9.2

First proof. We follow the strategy explained in (2.13). The key point in proving that the projection $\tilde{Q}_{*st} \rightarrow Q_{*st}$ is locally (on $\tilde{Q}_{*st}$) finite-to-one is the following lemma.

Lemme 9.1. *Let D_1, D_2 and D_3 be three distinct lines in $\mathbb{C}^2$ passing through o , and n_1, n_2, n_3 integers > 1 such that $\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} \geq 1$. Fix a base point $b \in \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$, and consider the coverings of $(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3), b)$, with ramification index along D_i dividing n_i ($i = 1, 2, 3$) : the monodromy permutation of the sheets, when turning around D_i , is of order dividing n_i . Then the universal such covering is a finite covering.*

Démonstration. There exists a spherical triangle Δ with angles $\frac{\pi}{n_1}, \frac{\pi}{n_2}, \frac{\pi}{n_3}$. Let W be the Coxeter group generated by the reflections along its edges. The $w\Delta$ ($w \in W$) form a tessellation of the sphere, and hence

$$|W| = \text{area}(S^2)/\text{area}(\Delta) = 2 \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}.$$

Let W^+ be the subgroup of index two in W consisting of the orientation preserving elements. If we identify the sphere with the Riemann sphere metrized by its Fubini metric, W^+ becomes a finite subgroup of $\text{PU}(2)$, of order

$$|W^+| = \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}.$$

The quotient $\mathbb{P}^1/W^+$ is of genus 0 and hence is a projective line. Fix an isomorphism $\mathbb{P}^1/W^+ \cong \mathbb{P}^1$ and let $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1/W^+ \cong \mathbb{P}^1$ be the quotient map. The pull-back by f of the line bundle $\mathcal{O}(-1)$ on $\mathbb{P}^1$ is of degree the negative of $d = \deg(f) = |W^+|$. Fix an isomorphism $\mathcal{O}(-1) \cong f^*\mathcal{O}(-1)$. The group W^+ acts naturally on $(\mathbb{P}^1, f^*\mathcal{O}(-1))$. Let H be the group of all automorphisms (h, r) of $(\mathbb{P}^1, f^*\mathcal{O}(-1))$ with $(h, r) \mapsto h \in W^+$. It is a central extension of W^+ by μ_d , the group of roots of unity of order dividing d . As a space, the line bundle $\mathcal{O}(-1)$, minus the 0 section, is $\mathbb{C}^2 - \{0\}$, the bundle map being the natural projection $\mathbb{C}^2 - \{0\} \rightarrow \mathbb{P}^1$. The action of H on $\mathbb{C}^2 - \{0\}$ is induced by a linear action on $\mathbb{C}^2$: the contragredient of its action on $H^0(\mathbb{P}^1, \mathcal{O}(1))$. The quotient map of the space $(\mathbb{P}^1, f^*\mathcal{O}(-1))$ to its μ_d orbits may be identified with a d -th power map of $(\mathbb{P}^1, \mathcal{O}(-1))$ to $(\mathbb{P}^1, \mathcal{O}(-1))$, and the quotient map $g : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ to its H orbits is the composite of the d -th power map to the total space of $f^*\mathcal{O}(-1)$ minus its zero section, followed by the quotient map to W^+ orbits. Thus the quotient maps by H and W^+ give rise to a commutative diagram

$$\begin{array}{ccccc} \mathbb{C}^2 & \longrightarrow & \mathbb{C}^2 - \{0\} & \longrightarrow & \mathbb{P}^1 \\ g \downarrow & & \downarrow & & \downarrow \\ \mathbb{C}^2 & \longrightarrow & \mathbb{C}^2 - \{0\} & \longrightarrow & \mathbb{P}^1 \end{array}$$

The stability groups in W^+ of the vertices P_1, P_2, P_3 of the original spherical triangle are cyclic of order n_1, n_2, n_3 , and the points in the orbits of P_1, P_2, P_3 are the only points with a non trivial stabilizer (this results from the fact that the stabilizer in W of any point of Δ is generated by the reflections along the edges passing through the point ([4] V (3.3) prop. 1), or simply from the fact that Δ is a fundamental domain for W ([20])).

Let $d_i = f(P_i)$ and $D_i \subset \mathbb{C}^2$ be the corresponding line. One easily checks that g ramifies only along D_1, D_2, D_3 and that at each point above a point of $D_i - \{o\}$, the ramification index is exactly n_i . If X is a covering of $\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$ as in (10.3), its pull-back by g can hence be extended to unbranched covering of $\mathbb{C}^2 - \{o\}$. As $\mathbb{C}^2 - \{o\}$ is simply connected, each connected component of this covering of $\mathbb{C}^2 - \{o\}$ is trivial. This implies that the universal covering of type (10.3) is $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)) \rightarrow \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$. It is finite. $\square$

Remarque 9.2.

- (i) The proof showed that the universal covering (10.3) is of order $2 \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}$.

It is a Galois covering, with group a central extension of W^+ by a cyclic group of order $|W^+|^{-1}$.

- (ii) If $\alpha_1 + \alpha_2 + \alpha_3 = 1 - \frac{1}{n_1}$, $\alpha_2 + \alpha_3 = 1 - \frac{1}{n_2}$ and $\alpha_3 + \alpha_1 = 1 - \frac{1}{n_3}$, one has $\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} > 1$ amounts to $\alpha_1 + \alpha_2 + \alpha_3 < 1$, and when it holds $(1 - \alpha_1 - \alpha_2 - \alpha_3)^{-1} \cdot |W^+|$ is an integer, as observed in (6.10).
- (iii) The fundamental group of $\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$ is generated by three elements $\gamma_1, \gamma_2, \gamma_3$ (conjugate to a small positive loop around D_i) with relations those expressing that $\gamma_1 \gamma_2 \gamma_3$ (conjugate to a small loop around o , on a general line D through o) is central. To get H , one imposes the additional relations $\gamma_i^{n_i} = 1$.
- (iv) In the commutative diagram

$$\begin{array}{ccc} g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)) & \hookrightarrow & \mathbb{C}^2 \\ \downarrow & & \downarrow g \\ \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3) & \hookrightarrow & \mathbb{C}^2 \end{array}$$

$\mathbb{C}^2$, at the upper right corner, is the completion of $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3))$ over $\mathbb{C}^2$ — at points of $\mathbb{C}^2 - \{o\}$ by (9.7.1) and at o by direct verification. Such a fact is generally true for a finite morphism g of normal analytic spaces. The completion of $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3))$ over $\mathbb{C}^2$ thus acquires a structure of a normal (in fact non-singular) analytic space — indeed of a non-singular analytic manifold; this structure is in fact the unique one for which the projection to $\mathbb{C}^2$ is analytic.

Remarque 9.3. Let $J = J' \cap J'' \cap J'''$ be a finite index set. For $j \in J'$, let $n_1(j), n_2(j), n_3(j)$ satisfy $\frac{1}{n_1(j)} + \frac{1}{n_2(j)} + \frac{1}{n_3(j)} > 1$; for $j \in J''$, let $n(j)$ be any integer > 1 . We define

$$X(j) = \begin{cases} \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3) \subset \mathbb{C}^2 & \text{for } j \in J', \\ \mathbb{C} - \{o\} \subset \mathbb{C} & \text{for } j \in J'', \\ \mathbb{C} & \text{for } j \in J'''. \end{cases}$$

The product X of the $X(j)$ is the complement of a divisor in $Y := (\mathbb{C}^2)^{J'} \times \mathbb{C}^{J'' \cup J'''}$. The covering $\tilde{X}$ of X , with ramification index along $\text{pr}_j^{-1}(D_i)$ dividing $n_i(j)$, ($j \in J'$), with ramification index along $\text{pr}_j^{-1}(\{o\})$ dividing $n(j)$ ($j \in J''$), and universal with respect to those properties, is the product of the corresponding coverings of each $X(j)$. Hence $\tilde{X}$ is a finite covering of X . The completion $\bar{Y}$ of $\tilde{X}$ over Y is similarly a product.

By (10.4) (iv), $\bar{Y}$ has a unique structure of a normal (in fact non-singular) analytic space for which the map to Y is analytic. If X_1 is a covering space of X and a quotient of $\tilde{X}$, then $X_1 = \tilde{X}/G$ with G a finite group, and the completion $\bar{Y}_1$ of X_1 over Y is $\bar{Y}/G$. It inherits the structure of a normal analytic space from $\bar{Y}$.

Similar remarks apply with X replaced by the trace on X of a ball around o in $(\mathbb{C}^2)^{J'} \times \mathbb{C}^{J'' \cup J'''}$.

Lemme 9.4. *If condition (INT) is satisfied, $\tilde{Q}_{*st}$ admits a structure of normal analytic space, such that each $y_0 \in Q_{*st}$ has an open neighborhood U whose inverse image in $\tilde{Q}_{*st}$ is a disjoint sum of finite ramified coverings of U .*

Démonstration. It suffices to show that each $y_0 \in Q_{*st}$ has an open neighborhood U whose inverse image in $\tilde{Q}_{*st}$ is a disjoint sum of finite (topological) coverings of $U \cap Q_*$ which, in suitable local coordinates at y_0 , are of the type considered in (10.5). This we proceed to show, using the control over ramification in codimension 1 provided by § 9.

Fix $y_0 \in Q_{*st}$, and let T be the corresponding stable partition of S . Fix a, b, c in distinct cosets. One can identify a neighborhood of y_0 in Q_{*st} with a subspace of P^S whose elements take prescribed values on a, b, c . We arrange these values so that $\infty \notin y_0(S)$. Near y_0 , we then have the following system of local coordinates, depending on the choice of a representative $d(C)$ in each coset C :

- a) the $y(d(C)) - y_0(d(C))$, for C not the coset of a, b or c ;
- b) for each coset C , the $y(e) - y(d(C))$ with $e \in C - \{d(C)\}$.

In terms of these coordinates, the condition for y near y_0 to be in Q_* is :

- a) for C a coset with three elements, $(y(e) - y(d(C)))_{e \in C - \{d(C)\}} \in \mathbb{C}^2$ is not on any of the lines $z_1 = 0$, $z_2 = 0$, $z_1 = z_2$;
- b) for C a coset with two elements, $y(e) - y(d(C)) \neq 0$ ($e \in C$, $e \neq d(C)$).

We recall that C has at most three elements (6.10.1), and that if C has three elements x, y, z , the sum of the reciprocal of the integers $(1 - \mu_x - \mu_y)^{-1}$, $(1 - \mu_y - \mu_z)^{-1}$, $(1 - \mu_z - \mu_x)^{-1}$ is > 1 (6.10).

For U a suitable open neighborhood of y in Q_{*st} , we can hence apply (10.5) to the covering of $U \cap Q_*$ induced by $\tilde{Q}_*$. We find that this covering breaks up into a disjoint sum of finite covering of the type required. $\square$

Lemme 9.5. *If condition INT is satisfied, the map $\tilde{w} : \tilde{Q}_{*st} \rightarrow B^+(\alpha)_0$ is étale.*

Démonstration. The map $\tilde{w}$ is holomorphic on $\tilde{Q}_*$ by (3.5) and continuous on $\tilde{Q}_{*st}$ by (8.7). It is hence holomorphic on $\tilde{Q}_{*st}$. By (6.9), it has an injective differential on the strata of a suitable analytic stratification. The fibers of $\tilde{w}$ hence have no component of dimension > 0 and $\tilde{w}$ is locally finite-to-one.

We know from § 9 that $\tilde{w}$ is étale outside of a closed analytic subset of Q_{*st} of complex codimension > 2 . By the purity of the branch locus theorem (the fact that the branch locus is always purely of codimension one), it follows that $\tilde{w}$ is étale everywhere. $\square$

9.3

Let us now assume, in addition to INT, that $Q_{*st} = Q_{*sst}$, i.e. that for no $S(1) \subset S$ is $\sum_{s \in S(1)} \mu_s = 1$. Choose a metric on Q_{*st} . We provide $\tilde{Q}_{*st}$ with the metric for which $d(x, y)$ is the infimum of the lengths of paths from x to y , the length being measured by its projection into Q_{*st} . What we need is a metric invariant by the action of $\pi_1(Q_*, o)$; any such metric will do. The space $\tilde{Q}_{*st}$ is locally compact, and the projection to Q_{*st} is open. Since Q_{*st} is assumed to be compact, there is a compact $K \subset \tilde{Q}_{*st}$ mapping onto Q_{*st} . Since $\tilde{w}$ is étale, there are numbers $r, R > 0$ such that for $k \in K$, the restriction of $\tilde{w}$ to the ball $B_r(k)$ of radius r around k in $\tilde{Q}_{*st}$ is an embedding, and

$$\tilde{w}(B_r(k)) \supset B_R(\tilde{w}(k)). \quad (4)$$

The $\pi_1(Q_*, o)$ translates of K cover $\tilde{Q}_{*st}$. The map $\tilde{w}$ being equivariant, and the action on the ball $B^+(\alpha)_0$ being via isometries, (10.8.1) holds for any $k \in \tilde{Q}_{*st}$. The map $\tilde{w}$ is hence a covering map. The ball being simply connected, $\tilde{w}$ is an isomorphism. This concludes the proof of (3.11) in the cocompact case ($Q_{*st} = Q_{*sst}$), following the strategy outlined there.

9.4

Our second proof will bypass lemma (10.6). The purity of the branch locus theorem can be deduced from the fact that, if Y is a complex submanifold of a complex manifold X of complex codimension ≥ 2 , and B a small ball around $y \in Y$, then $B - Y$ is simply connected. It is the latter fact which we will use directly to prove (10.7).

The reader who merely wants to get the idea of how we complete the proof when $Q_{*st} \neq Q_{*sst}$ can skip to Proposition (10.18.1).

9.5

Our second proof of (10.1.1) comes after proving that $\tilde{w}$ is étale on $\tilde{Q}_{*sst}$, that in turn is proved in stages.

Set $N = \text{card } S$. For i between 0 and $N - 3$, set

$$Q_i = \bigcup Q_T$$

where T runs through all stable partitions of cardinality $\geq \text{card } S - i$. Then $Q_0 = Q_*$, $Q_{N-3} = Q_{*st}$. Set $Q_{N-2} = Q_{*sst}$. Let $\tilde{Q}_i$ denote the completion of $\tilde{Q}_*$ over Q_i ($i = 1, \dots, N - 2$). We shall prove inductively that $\tilde{w}$ is étale on $\tilde{Q}_i$.

Proposition (10.17) below will be used repeatedly to show at each stage that as one adds on a submanifold, the extended map $\tilde{w}$ remains étale. We lead up to it via some general topological remarks, especially Proposition (10.15.1) and Corollary (10.15.4).

Proposition 9.6. *Let $\varphi : X \rightarrow Y$ be a continuous map with Y locally connected and X Hausdorff. Assume that each $y \in Y$ admits a neighborhood V such that each $x \in \varphi^{-1}(V)$ has an open neighborhood U with $\varphi(U) \supset V$ such that φ induces a homeomorphism from U to $\varphi(U)$. Then φ is a covering map.*

We first prove the

Lemme 9.7. *Let $\varphi : X \rightarrow Y$ be a continuous map with Y connected and X Hausdorff. Let X_1 and X_2 be open in X and such that φ induces homeomorphisms from X_i to Y , $i = 1, 2$. Then $X_1 = X_2$ or $X_1 \cap X_2 = \emptyset$.*

Démonstration. Let $s_i : Y \rightarrow X$ be the inverse of $\varphi \mid X_i$. The set of $y \in Y$ with $s_1(y) = s_2(y)$ is closed. It is also open, being the image of $X_1 \cap X_2$ by $\varphi : X_i \rightarrow Y$. It is hence the whole of Y or empty; i.e. $X_1 = X_2$ or $X_1 \cap X_2 = \emptyset$. $\square$

Proof of (10.11). For $V \subset Y$, let $\bar{\varphi}$ be the map induced by φ from $\varphi^{-1}(V)$ to V . We have to show that each $y \in Y$ has a neighborhood V such that $(\varphi^{-1}(V), \bar{\varphi}) \cong (V \times D, \text{pr}_1)$ with D discrete. Take V as in (10.11), open and connected. Replacing the U of (10.11) by $U \cap \varphi^{-1}(V)$, we find that each $x \in \varphi^{-1}(V)$ is contained in an open set U such that φ induces a homeomorphism from U to V . By (10.11.1) applied to $\varphi^{-1}(V) \rightarrow V$, they form a partition of $\varphi^{-1}(V)$ and the claim follows. $\square$

Proposition 9.8. *Let $\varphi : (X, o) \rightarrow (Y, p)$ be a continuous map of metric spaces, with $Y - \{p\}$ locally connected. We write d for the metric on both X and Y . Assume that there exists a neighborhood U of o such that for each neighborhood V of o , one has*

- a) *there exists $r > 0$ such that for all $x \in U - V$, the open ball $B_r(x)$ maps homeomorphically onto an open set of Y ;*

b) for all r there is $\varepsilon > 0$ such that for each $x \in U - V$, $\varphi B_r(x) \supset B_\varepsilon \varphi(x)$.

Then, there exist open neighborhoods X' of o and Y' of p such that $\varphi : X' - \varphi^{-1}(p) \rightarrow Y' - \{p\}$ is a covering map.

Roughly speaking, a), b) mean that on $U - \{o\}$, φ is a local homeomorphism with some uniformity. The uniformity is allowed to get worse for $x \approx o$. It is not assumed that o is isolated in $\varphi^{-1}(p)$. Under additional assumptions, this will be a consequence; see (10.13).

Proof of (10.12). Choose U as in (10.12). Select a neighborhood V of o and $\varepsilon(1) > 0$ such that $B_{\varepsilon(1)}(V) := \{x \in X \mid \exists v \in V d(x, v) < \varepsilon(1)\}$ is contained in U . If $B_r(o) \subset U$, one can take $r = r$ and $V = B_r(o)$. By a), b) there is ε and η such that

for all $x \in U - V$, φ induces a homeomorphism of $B_\varepsilon(x)$ with an open set of Y containing $B_\eta(\varphi(x))$.
(*)

Set $Y' = B_\eta(p)$ and take X' to be the set of all x with $\varphi(x) \in Y'$ such that either

(i) $x \in V$; or

(ii) there is $x_1 \in U - V$ with $d(x, x_1) < \varepsilon$, and $\varphi(x_1) \in Y'$.

In case (ii), $x \in B_\varepsilon(x_1) \cap \varphi^{-1}(Y') \subset X'$ and φ induces a homeomorphism of $B_\varepsilon(x_1) \cap \varphi^{-1}(Y')$ with Y' . This results from (*).

Fix $y \in Y' - p$ and let W be a neighborhood of y , in $Y' - \{p\}$, disjoint from a neighborhood of p . Then $\varphi^{-1}(W)$ is disjoint from a neighborhood of o and by a), b) for r small enough there is $\delta(2)$ such that

for all $x \in U$ with $\varphi(x) \in W$, φ induces a homeomorphism from $B_{\delta(2)}(x)$ to an open subset of Y containing $B_{\delta(2)}(\varphi(x))$.
(**)

We may and shall assume that $\delta(2) < \varepsilon$ and that $B_{\delta(2)}(y) \subset W$. Proposition (10.12) now follows from (10.11) applied to $X' - \varphi^{-1}(p) \rightarrow Y' - \{p\}$ and from the

Lemme 9.9. *Each x in $X' \cap \varphi^{-1}(B_{\delta(2)}(y))$ has an open neighborhood B' in X' which maps homeomorphically onto an open subset of Y containing $B_{\delta(2)}(y)$.*

As $x \in X'$, one of the conditions (i) (ii) holds. In the first case ($x \in V$), one has $B_{\delta(2)}(x) \subset U$, and we claim that $B_{\delta(2)}(x) \cap \varphi^{-1}(Y') \subset X'$. For if $x' \in B_{\delta(2)}(x)$ and $\varphi(x') \in Y'$, then either $x' \in X'$ by virtue of (i) or $x' \in U - V$ and $x' \in X'$ by virtue of (ii) with $x_1 = x'$. In the first case, $B' = B_{\delta(2)}(x) \cap \varphi^{-1}(Y')$ and one uses (*). In the second case, take $B' = B_\varepsilon(x_1) \cap \varphi^{-1}(Y')$ and use (*). $\square$

The assumptions of (10.12) remain valid if we replace X and Y by open neighborhoods of o and p . The X' and Y' of (10.12) can hence be chosen arbitrarily small.

From this, we will deduce the

Corollaire 9.10. *The point o has a fundamental system of open neighborhoods X' such that for Y' a suitable open neighborhood of p , $X' - \varphi^{-1}(p)$ is open and closed in $\varphi^{-1}(Y' - \{p\})$ and is a covering of $Y' - \{p\}$.*

First take X_0, Y_0 as in (10.12), and W a neighborhood of o such that $\bar{W} \subset X_0$. If X', Y' are as in (10.12), with $X' \subset W$, $Y' \subset Y_0$ and if $X'_1 := X_0 \cap \varphi^{-1}(Y')$, then both $X'_1 - \varphi^{-1}(p)$ and $X' - \varphi^{-1}(p) (\subset X'_1 - \varphi^{-1}(p))$ are coverings of $Y' - \{p\}$; hence $X' - \varphi^{-1}(p)$ is open and closed in $X'_1 - \varphi^{-1}(p)$. As $(X' - \varphi^{-1}(p)) \subset \bar{W} \subset X_0$, $X' - \varphi^{-1}(p)$ is also open and closed in $\varphi^{-1}(Y' - \{p\})$.

Corollaire 9.11. *If o is not isolated in X , and if p has a fundamental system of neighborhoods V such that $V - \{p\}$ is connected (in particular, non empty), then φ is open at o .*

If o is not isolated, X' cannot be contained in $\varphi^{-1}(p)$; for applying (10.12) a) to x in X' near o would contradict $\varphi(X') \subset \{p\}$. It follows that $\varphi(X' - \varphi^{-1}(p)) = Y' - \{p\}$ and hence that $\varphi = Y'$. Openness follows.

Proposition 9.12. *In addition to the hypothesis of (10.12), assume that $Y - \{p\}$ is locally simply connected and that*

- c) *p has a fundamental system of open neighborhoods $\mathcal{V}$, such that each $V - \{p\}$ (V in $\mathcal{V}$) is connected and that for $V \subset V'$, both in $\mathcal{V}$, $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(V' - \{p\})$.*
- d) *o has a fundamental system of open neighborhoods N with $N - \{o\}$ connected (non-empty).*

Then, one can find X' and Y' as in (10.12) such that $\varphi' : X' \rightarrow Y'$ has the following additional properties : Y' is in $\mathcal{V}$, $\varphi'^{-1}(\{p\}) = \{o\}$, $X' - \{o\}$ is connected, and X' is the completion of $X' - \{o\}$ above Y' .

Proof of (10.13). Shrinking X and Y , we may and do assume that Y is in $\mathcal{V}$, that $\varphi : X - \varphi^{-1}(p) \rightarrow Y - \{p\}$ is a covering map, and that φ is a local homeomorphism at each point $x \neq o$. Let N be an open neighborhood of o such that $N - \{o\}$ is connected.

Lemme 9.13. *$N - \varphi^{-1}(p)$ is connected.*

Let $A \subset N - \varphi^{-1}(p)$ be open and closed. Each $x \in \varphi^{-1}(p) - \{o\}$ has a neighborhood $W \subset N$ with $W - \{x\} \cong \varphi(W) - \{p\}$ connected. For such a W , $W - \{x\} = W - \varphi^{-1}(p)$ lies either in A or in its complement. It follows that the closure of A in $N - \{o\}$ is again open and closed and, by hypothesis, is empty or the whole of $N - \{o\}$. If it is empty (resp. the whole of $N - \{o\}$), A is empty (resp. the whole of $N - \varphi^{-1}(p)$).

Since $Y - \{p\}$ is locally connected, so is the covering space $X - \varphi^{-1}(p)$. Each connected component of $X - \varphi^{-1}(p)$ is open and closed in $X - \varphi^{-1}(p)$ and is a covering of $Y - \{p\}$. Let N_1 be the component of N containing $N - \varphi^{-1}(p)$.

Lemme 9.14. *$N_1 \cap \varphi^{-1}(p) = \{o\}$.*

Any $x \neq o$ in $\varphi^{-1}(p)$ has an open neighborhood W' such that φ induces a homeomorphism of W' with an open neighborhood of p in Y . Take V in $\mathcal{V}$ with $V \subset \varphi(W')$ and define $W := W' \cap \varphi^{-1}(V)$. One has $\varphi : W \xrightarrow{\sim} V$.

The assumption $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(Y - \{p\})$ implies that for any covering map $\psi : Z \rightarrow Y - \{p\}$, one has $\psi_*(\psi^{-1}(V - \{p\})) \xrightarrow{\sim} \pi_0(Z)$. In particular, as N_1 is connected, so is $N_1 \cap \varphi^{-1}(V)$. Both $N_1 \cap \varphi^{-1}(V)$ and $W - \{x\}$ are connected coverings of $V - \{p\}$, contained in the covering $\varphi^{-1}(V - \{p\})$. They are not equal : x is the only point of $\varphi^{-1}(p)$ in W , while $o \in N_1$. They are hence disjoint and $x \notin N_1$.

Lemma (10.13.2) is proved.

We take $Y' = Y$ and $X' := N_1$. It follows from (10.13.2) that X' is open and closed in X and that $\varphi^{-1}(p) = \{o\}$. By construction, $X' - \varphi^{-1}(p) = X' - \{o\}$ is connected. For each V in $\mathcal{V}$, $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(Y' - \{p\})$ implies as above that $\varphi^{-1}(V - \{p\})$ is connected. It then follows from (10.12.2) that the $\varphi^{-1}(V) \cap X'$ ($V \in \mathcal{V}$) form a fundamental system of neighborhoods of o in X' . Consequently X' is the completion of $X' - \{o\}$ over Y' . $\square$

Corollaire 9.15. *If, in addition to the assumptions of (10.12) and (10.13), the $V - \{p\}$ for V in $\mathcal{V}$ are simply connected, then φ is a local homeomorphism at o .*

Indeed, with the notation of (10.13), $Y' - \{p\}$ is simply connected; hence $\varphi : X' - \{o\} \rightarrow Y' - \{p\}$ is a homeomorphism as well as $\varphi' : X' \rightarrow Y'$.

9.6

We shall require a simpler variant of (10.12) in our extension of the map $\tilde{w}$ to cusp points. In that situation, we shall be dealing with a continuous map $\varphi : (X, o) \rightarrow (Y, p)$. Metrics are given only on $X - \{o\}$ and $Y - \{p\}$, and we assume

- a) The $\varphi^{-1}(V)$, for V a neighborhood of p in Y , form a fundamental system of neighborhoods of o . In particular, $\varphi^{-1}(p) = \{o\}$.
- b) There is a neighborhood U of o such that for any neighborhood $V \subset U$, the conditions a), b) of (10.12) are satisfied.

Fix a neighborhood V_1 of p such that $\varphi^{-1}(V_1) \subset U$. For any neighborhood V_2 of p and $x \in V_1 - V_2$, the map $\varphi_1 : \varphi^{-1}(V_1 - V_2) \rightarrow V_1 - V_2$ induces a homeomorphism of $B_r(x)$ with an open set containing $B_\varepsilon(\varphi(x))$ for suitable ε and δ . The map φ_1 is hence a covering map. As this holds for any V_2 , $\varphi : \varphi^{-1}(V_1 - \{p\}) \rightarrow V_1 - \{p\}$ is also a covering map.

If in addition

- c) p has a fundamental system of neighborhoods V such that $V - \{p\}$ is connected and simply connected,
 - d) o has a fundamental system of neighborhoods N such that $N - \{o\}$ is connected,
- then an easy argument shows that φ is a local homeomorphism at o .

9.7

Actually, we require (10.13.3) for the more general case where o and p are replaced by strata. The analogue of (10.12) is :

Proposition 9.16. *Let $\varphi : (X, X_1, o) \rightarrow (Y, Y_1, p)$ be a continuous map of metric spaces, with $Y - Y_1$ locally connected. Assume that there exists a neighborhood U of o such that for each neighborhood V of X_1 , the conditions a) b) of (10.12) hold. Assume further that X_1 is locally compact and that the map $\varphi_1 : X_1 \rightarrow Y_1$ induced by φ is a local homeomorphism at o .*

Then there are open neighborhoods X' of o and Y' of p such that $\varphi : X' - \varphi^{-1}(Y_1) \rightarrow Y' - Y_1$ is a covering map.

Démonstration. Fix U as above, open and small enough so that for some open set R disjoint from U , $K := X_1 - R$ is compact and so that φ is injective on K . Note that if we take $R = X - \bar{U}$, then $K = X_1 \cap \bar{U}$.

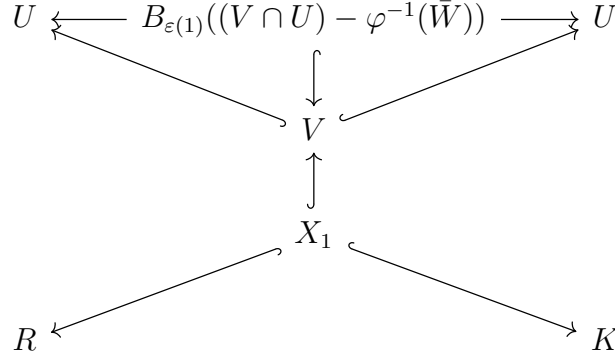
The subset $\varphi(K - U)$ of $Y_1 \subset Y$ is compact and does not contain p . Let W be an open neighborhood of $\varphi(K - U)$ whose closure does not contain p . Then, $\varphi^{-1}(W)$ is an open neighborhood of $K - U$, and $\varphi^{-1}(\bar{W})$ does not contain o .

Select a neighborhood V of X_1 , and $\varepsilon(1) > 0$ such that

$$B_{\varepsilon(1)}((V \cap U) - \varphi^{-1}(\bar{W})) \subset U;$$

this can be done as follows. Since $K - \varphi^{-1}(W)$ is compact and in U , we choose $\varepsilon(1) > 0$ so that $B_{\varepsilon(1)}(K - \varphi^{-1}(W)) \subset U$. One takes $V = B_{\varepsilon(1)}(K - \varphi^{-1}(W)) \cup \varphi^{-1}(W) \cup R$. Then

$(V \cap U) - \varphi^{-1}(W) \subset B_{\varepsilon(1)}(K - \varphi^{-1}(W))$ and $B_{\varepsilon(1)}((V \cap U) - \varphi^{-1}(\bar{W})) \subset U$ by choice of $\varepsilon(1)$.



By a), b), there are r and ε such that

for all $x \in U - V$, φ induces a homeomorphism of $B_\varepsilon(x)$ with an open set of Y containing $B_{2\eta}(\varphi(x))$. (*)

We may and do choose η small enough so that $B_\eta(p)$ is disjoint from $\bar{W}$. Set $Y' = B_\eta(p)$ and take X' to be the set of all x with $\varphi(x) \in Y'$ (hence $x \notin \varphi^{-1}(\bar{W})$) such that either $x \in U \cap V$, or there is $x_1 \in U - V$ with $d(x, x_1) < \varepsilon$ and $\varphi(x_1) \in Y'$.

Any $y \in Y - Y_1$ has a neighborhood W_y disjoint from a neighborhood of Y_1 ; hence, as in (10.12), for r small enough there is $\delta(2)$ such that

for all $x \in U$ with $\varphi(x) \in W_y$, φ induces a homeomorphism from $B_{\delta(2)}(x)$ to an open set of Y containing $B_{\delta(2)}(\varphi(x))$. (**)

One can assume that $\delta(2) < \varepsilon$ and that $B_{\delta(2)}(y) \subset W_y$ and one completes the proof just as in (10.12), using that $\varphi(x) \in Y'$ implies $x \notin \varphi^{-1}(\bar{W})$. □

9.8

As in (10.12.2), X' can be taken as small as we please. If o is not an interior point of X_1 , and if p has a fundamental system of neighborhoods V with $V - Y_1$ connected, φ is open at o . The proof is as in (10.12.3).

Proposition 9.17. *Assume that $Y - Y_1$ is locally simply connected and that*

- c) *p has an open neighborhood V_0 with $V_0 - Y_1$ connected, and each $p' \in Y_1$ close enough to p has a fundamental system of neighborhoods $V \subset V_0$ with $V - Y_1$ connected and $\pi_1(V - Y_1) \xrightarrow{\sim} \pi_1(V_0 - Y_1)$.*
- d) *Each $o' \in X_1$ close enough to o has a fundamental system of neighborhoods N with $N - X_1$ connected.*

Then one can find X' and Y' as in (10.15.1) such that $\varphi' : X' \rightarrow Y'$ has the following additional properties : c) holds with $V_0 = Y'$ for all $p' \in Y_1 \cap V_0$, $X' - X_1$ is connected, $X'_1 := X_1 \cap X'$ is the inverse image of $Y'_1 := Y_1 \cap Y'$, φ' induces a homeomorphism from X'_1 to Y'_1 , and X' is the completion of $X' - X'_1$ above Y' .

Démonstration. Shrinking X and Y , we may and do assume that c) holds for $V_0 = Y$ and for all $p' \in Y_1 \cap V_0$, that $\varphi_1 : X_1 \rightarrow Y_1$ is a homeomorphism, that d) holds for all $o' \in X_1$, that $\varphi : X - \varphi^{-1}(Y_1) \rightarrow Y - Y_1$ is a covering map, and that φ is a local homeomorphism at each point $x \notin X_1$. Let N be an open neighborhood of o with $N - X_1$ connected. As in

(10.13.1), one checks that $N - \varphi^{-1}(Y_1)$ is connected. Let N_1 be the connected component of $X - \varphi^{-1}(Y_1)$ containing $N - \varphi^{-1}(Y_1)$. It is a covering of $Y - Y_1$.

If we shrink Y again without changing the $\pi_1(Y - Y_1)$, and replace X , N , N_1 by their traces on the pull-back of the new Y , we get $N - X_1 \subset N_1$ with N_1 connected and $\bar{N} \supset X_1$.

As in (10.13.2), one sees that $N_1 \cap \varphi^{-1}(Y_1) = X_1$. One takes $Y' = Y$, $X' = \bar{N}_1$ and $\varphi' : X' \rightarrow Y'$ induced by φ . The set X' is open and closed in the (shrunk) X and $\varphi'^{-1}(Y'_1) = X'_1$. All the listed properties of φ' are clear from the construction except for X' being the completion of $X' - X'_1$ over Y' . For $p' \in Y'_1$, $p' = \varphi'(o')$, and V an open neighborhood of p' with $\pi_1(V - Y'_1) \xrightarrow{\sim} \pi_1(Y - Y'_1)$, one has $\pi_0(\varphi'^{-1}(V - Y'_1)) \cong \pi_0(X' - X'_1)$, i.e. $\varphi'^{-1}(V - Y'_1)$ is connected. The $\varphi'^{-1}(V)$ for such V form a fundamental system of neighborhoods of o' and the assertion about completion follows. $\square$

In particular,

Corollaire 9.18. *In addition to the hypothesis of (10.15.1) assume that $Y - Y_1$ is locally simply connected and that*

- c') Each $p' \in Y_1$ close to p has a fundamental system of neighborhood V with $V - Y_1$ connected and simply connected.*
- d) Each $o' \in X_1$ close to o has a fundamental system of neighborhood V with $V - X_1$ connected.*

Then φ is a local homeomorphism at o .

9.9

We will apply (10.15.4) iteratively to some stratified spaces. In the next proposition, “stratified space” means a Hausdorff topological space U , provided with a partition $\mathcal{S}$ into locally closed subsets, the strata, where

- a) each stratum is a manifold;
- b) the closure of any stratum S is the union of strata;

$(U, \mathcal{S})$ is conical in the following sense and locally constant along each stratum : each point x of any stratum S has a neighborhood V isomorphic (with the induced partition) to the product of $V \cap S$ with the cone over $(\Sigma, \mathcal{S}(\Sigma))$, where Σ is a topological space, provided with a partition $\mathcal{S}(\Sigma)$. For simplicity, we assume in addition the existence of a metric d for which $d(x, y)$ is the greatest lower bound of lengths of paths from x to y . The case we will need is $U = Q_{*st}$, $\mathcal{S}$ = the partition of Q_{*st} by the Q_T for T varying over the stable partitions.

Proposition 9.19. *Consider a diagram*

$$\begin{array}{ccccc} \tilde{U}_0 & \hookrightarrow & \tilde{U}_1 & \hookrightarrow & \tilde{U} \\ & \searrow p & \downarrow & \swarrow \varphi & \\ & & U & & \end{array}$$

where

- a) U is locally compact stratified space ; U_0 is a connected open and dense stratum, and $U_1 - U_0$ is a union of strata ; each point has a fundamental system of neighborhoods whose traces on U_0 are connected.

- b) $\tilde{U}_0$ is a covering space of U_0 , and $\tilde{U}_1$ (resp. $\tilde{U}$) is the completion of $\tilde{U}_0$ over U_1 (resp. U).
- c) The covering $\tilde{U}_0$ of U_0 is Galois, with group Γ .
- d) Y is a manifold, provided with a metric and an action of Γ by isometries. The map φ is equivariant.
- e) $\varphi|_{\tilde{U}_1}$ is a local homeomorphism.
- f) For any stratum S , $\tilde{S} := p^{-1}(S)$ is a covering space of S (because U_0 is a stratum). We assume that for $S \subset U - U_1$, each point of $\tilde{S}$ has a neighborhood W in $\tilde{U}$ such that $\varphi|_W$ is a tame embedding, with $\varphi(W)$ a subvariety of codimension > 3 in Y .

Then, φ is a local homeomorphism and the (local) decomposition groups of $\tilde{U}/U$ are finite, i.e. $\tilde{U}$ is locally finite to one.

Démonstration. Let d be a Γ -invariant metric on $\tilde{U}$; for instance take $d(x, y) =$ the greatest lower bound of the lengths, computed in U , of paths in $\tilde{U}$ from x to y .

Let S be an open stratum of $U - U_1$. We first prove that φ is a local homeomorphism at any point $o \in \tilde{p}^{-1}(S)$. For this, replacing U by a suitable open neighborhood V of $p(o)$, $\tilde{U}$ by the connected component of $p^{-1}(V)$ containing o and Γ by the stabilizer of o , we may and shall assume that $U - U_1 \subset S$, that p induces an isomorphism $\tilde{S} \xrightarrow{\sim} S$ and that φ embeds $\tilde{S}$ in Y . Replacing Y by a Γ -stable open neighborhood of $\varphi(o)$, and $\tilde{U}_1$ by its inverse image, one may further assume that $Y_1 := \varphi(\tilde{S}')$ is a tame closed submanifold of Y . Its codimension is ≥ 3 . We will apply (10.15.4) with $X = \tilde{U}$, $X_1 = \tilde{S}$ and o, Y, Y_1, φ as above. Local connectedness and simple connectedness of $Y - Y_1$ results from Y being a manifold. Condition c') of (10.15.4) results from Y_1 being a tame submanifold of codimension ≥ 3 . Condition d) of (10.15.4) results from assumption a). By e) $\varphi|_{(X - X_1)}$ is a local homeomorphism, and it remains to check the uniformity conditions of (10.12). As neighborhood of o , we choose the pull back $p^{-1}(K)$ of a compact neighborhood K of $p(o) \in U$. By definition of the topology of $\tilde{U}$, any neighborhood of $X_1 = \tilde{S}$ contains $p^{-1}(V)$, for V an open neighborhood of S . The required uniformities will follow from the compactness of $K - V$. The function from $X - X_1$ to $\mathbb{R}_+^*$: $x \mapsto$ the greatest lower bound of the ε such that φ is an embedding on $B_\varepsilon(x)$, is > 0 , lower semi-continuous, and Γ -invariant; hence it is of the form $\bar{r} \circ p$ with $\bar{r}$ continuous. On $K - V$, $\bar{r}$ stays away from 0, and this gives (10.12 a)). If r is such that for any $x \in p^{-1}(K - V)$, $\varphi|_{B_{2r}(x)}$ is an embedding, the function from $p^{-1}(K - V)$ to $\mathbb{R}_+^*$: $x \mapsto$ the greatest lower bound of the δ such that $\varphi(B_r(x)) \supset B_\delta(\varphi(x))$ is > 0 , lower semi-continuous and Γ -invariant, hence of the form $\bar{r} \circ p$ and staying away from 0. This gives (10.12 b)).

We now prove that $\varphi : \tilde{U} \rightarrow Y$ is a local homeomorphism. We have just shown that φ is a local homeomorphism on $\tilde{p}^{-1}(U'_1)$, for U'_1 the union of U_1 and of the open strata of $U - U_1$. The assumptions of the Proposition hence hold with U_1 (resp. $\tilde{U}_1$) replaced by U'_1 (resp. $p^{-1}(U'_1)$), and one concludes its proof by induction on $\dim(U - U_1)$.

It remains to show that for any $x \in \tilde{U}$, the stabilizer $\Delta \subset \Gamma$ of x is finite. Fix a neighborhood V of x such that $\varphi|_V$ is an open embedding, fix a Δ -stable compact neighborhood K of $\varphi(x)$ in $\varphi(V)$, and define $V_1 = V \cap \varphi^{-1}(K)$. It is a Δ -invariant compact neighborhood of x . For $y \in V_1 \cap \tilde{U}_0$, $\Gamma \cdot y$ is closed and discrete. Therefore Δ_y , which is in $\Gamma \cdot y \cap V_1$, is discrete and compact, and hence finite. Thus Δ is finite. $\square$

We will use (10.16.1) to give an alternate proof of lemma (10.7), by passing the purity of the branch locus theorem (which was derived in the special case $Q_{*st} = Q_{*sst}$).

Lemme 9.20. *If condition (INT) is satisfied, the map $\tilde{w} : \tilde{Q}_{*st} \rightarrow B^+(\alpha)_0$ is a local homeomorphism.*

Démonstration. We apply (10.16.1) with $U = Q_{*st}$ stratified by the Q_T for T varying over the stable partitions, $U_0 = Q_*$, $U_1 =$ the union of Q_* and of the (complex) codimension one strata, $\tilde{U}_0 = \tilde{Q}_*$; hence $\tilde{U} = \tilde{Q}_{*st}$ and $\varphi = \tilde{w}$. Conditions a) to d) are clear, e) is (9.12) and f) follows from (6.9) and (3.9). $\square$

9.10

As shown in (8.7), the map $\tilde{w}$ extends to a continuous map from $\tilde{Q}_{*sst}$ to $B^+ := B^+(\alpha)_0$, where B^+ has the topology (5.4). Fix $j \in Q_{*sst} - Q_{*st}$ and $o \in \tilde{Q}_{*sst}$ above j . Define $p := \tilde{w}_{*sst}(o)$.

Let V be an open connected neighborhood of j , and $\tilde{V}$ be the connected component containing o of its inverse image in $\tilde{Q}_{*sst}$. For V small enough, one has : o is the only point of $\tilde{V}$ above j ; the stabilizer $\Delta \subset \Gamma$ of o acts on $\tilde{V}$, and $\tilde{V}/\Delta \xrightarrow{\sim} V$; if δ is a distance to p function (5.3) on B^+ , δ descends to a continuous function f on $\tilde{V}$, vanishing only at j . As $\tilde{V}$ is locally compact, it follows, for η small enough, that the open sets $\{v \mid f(v) < r\}$ form a fundamental system of neighborhood of j . As a consequence, the $\tilde{w}^{-1}(A)$, for A a neighborhood of p , form a fundamental system of neighborhood of o .

We now apply (10.14) to prove

Lemme 9.21. *If condition (INT) is satisfied, the map $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ is a local homeomorphism.*

Démonstration. It remains to prove it at a point o as above. We apply (10.14) to $\tilde{w} : (\tilde{V}, o) \rightarrow (B^+, p)$. Condition a) has been proved. Condition c) and d) are clear. The map is a local homeomorphism outside of o . The required uniformity (10.14 b)) follows as before from the local compactness of $\tilde{V}$ and Δ -equivariance. $\square$

Théorème 9.22. *Assume condition (INT). Then the map $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ restricts to a homeomorphism of $\tilde{Q}_{*st}$ onto B^+ and maps $\tilde{Q}_{*sst}$ homeomorphically onto an open subset of B^+ , in the (5.4) topology.*

Démonstration. By Proposition (10.18.1), $\tilde{w}$ is an étale map of $\tilde{Q}_{*sst}$ into B^+ . Let φ denote the restriction of $\tilde{w}$ to $\tilde{Q}_{*sst}$. It suffices to prove that φ is a covering map of $\tilde{Q}_{*sst}$ onto B^+ .

For any $x \in \tilde{Q}_{*sst}$, set

$$f(x) = \sup\{r \mid \varphi \text{ maps a neighborhood of } x \text{ homeomorphically onto } B_r(\varphi(x))\}.$$

Then as in the proof of Lemma (10.16.1), f is continuous, $f(x) > 0$ for $x \in \tilde{Q}_{*st}$ and $f(\gamma \cdot x) = f(x)$ for all $\gamma \in \pi_1(Q_*, o)$; that is, f descends to a positive-valued continuous function on Q_{*st} .

Since $\tilde{w}$ is étale at each point of $\tilde{Q}_{*sst}$ and $Q_{*sst} - Q_{*st}$ consists only of finitely many points, one can deduce from the (5.4) topology of B^+ the existence of an open neighborhood W of Q_{*sst} and an $\varepsilon_1 > 0$ such that $f(x) \geq \varepsilon_1$ for all $x \in p^{-1}(W)$. By § 4, Q_{*sst} is compact. From the compactness of $Q_{*sst} - W$ one deduces that f has a non-zero lower bound on $Q_{*sst} - p^{-1}(W)$. Consequently, f has a non-zero lower bound on Q_{*st} . From this it follows that $B_\eta(\varphi(Q_{*st})) = B^+$ for suitable η and therefore, $\varphi(\tilde{Q}_{*st}) = B^+$.

By (10.11), φ is a covering map. Inasmuch as B^+ is simply connected, φ is a homeomorphism.

From the fact that each point $y \in Q_{*sst}$ has a base of neighborhoods $\{V\}$ with $V - \{y\}$ connected, and the fact that $\tilde{w}$ is a homeomorphism on $\tilde{Q}_{*st}$, it follows that $\tilde{w}$ is injective on $\tilde{Q}_{*sst}$ and hence a homeomorphism onto its image, in the topology of (5.4). $\square$

9.11

Let S be a finite set with at least 3 elements, $N = \text{card } S$, $\mu = (\mu_s)_{s \in S}$ with $0 < \mu_s < 1$ and $\sum \mu_s = 2$. Set $\alpha_s = \exp 2\pi\sqrt{-1}\mu_s$ for each $s \in S$, P the projective line over $\mathbb{C}$, $G = \text{Aut } P$, M the subset of injective elements in P^S , $Q_* = M/G$, $o \in Q_*$ and $\theta : \pi_1(Q_*, o) \rightarrow \text{Aut } B^+(\alpha)_o = \text{PU}(1, N-2)$ the homomorphism of $\pi_1(Q_*, o)$ into the isometry group of the ball $B^+(\alpha)_o$, defined in (3.10.2). Set $\Gamma = \theta(\pi_1(Q_*, o))$, $B^+ = B^+(\alpha)_o$.

Théorème 9.23. *If μ satisfies condition (INT), then Γ is a discrete subgroup of $\text{Aut } B^+$ and discontinuous on B^+ .*

Démonstration. By Theorem (10.17.1), the action of Γ on B^+ is equivalent to the action of Γ on $\tilde{Q}_{*st}$. The latter action is discontinuous on $\tilde{Q}_*$ since Γ is the covering group of the covering map $\tilde{Q}_* \rightarrow Q_*$. It follows at once that Γ is discontinuous on a dense open subset of B^+ . Since Γ acts on B^+ via isometries, it follows that Γ is discrete in $\text{Aut } B^+$, the isometry group of B^+ . Since $\text{Aut } B^+$ operates transitively on B^+ with a compact isotropy group, Γ operates discontinuously on B^+ . $\square$

Corollaire 9.24. *For any $y \in \tilde{Q}_{*sst}$, the stabilizer Γ_y of y in Γ is a finite group.*

Démonstration. Set $p = \tilde{w}_*(y)$. Then $\Gamma_y = \Gamma_p = \Gamma \cap (\text{Aut } B^+)_p$. Since Γ is discrete in $\text{Aut } B^+$, and $(\text{Aut } B^+)_p$ is compact, Γ_p is finite. $\square$

Corollaire 9.25. *$\tilde{Q}_*$ is the normal covering of Q_* universal with respect to the property $[R]$: for each stable partition T of S with $\text{card } T = \text{card } S - 1$, if k_T is defined as in (9.9), the ramification index along Q_T divides k_T .*

Démonstration. Let $\tilde{Q}^*$ denote the universal covering with the property above. Then the map $\tilde{Q}^* \rightarrow \tilde{Q}_*$ extends to a covering map of $\tilde{Q}_1$. By Theorem (10.18.2), $\tilde{Q}_1$ is homeomorphic to the complement in the ball B^+ of a closed subset of complex codimension 2 and is simply connected. Consequently, $\tilde{Q}^* \xrightarrow{\sim} \tilde{Q}_*$. $\square$

Corollaire 9.26. *Above a suitably small neighborhood U of a point $x \in \tilde{Q}_{*sst}$, each connected component of $\tilde{Q}_* \cap p^{-1}(U)$ is the universal covering of $Q_* \cap U$ with respect to the ramification property $[R]$.*

The proof is the same as the preceding one, with the ball replaced by the trace of a neighborhood of a point in B^+ (in the topology of (5.4)).

Corollaire 9.27. *The stabilizer in Γ of a point x in the ball is generated by $\mathbb{C}$ -reflections.*

Démonstration. Choose U as in Corollary (10.19.3). Then $U \cap \tilde{Q}_1$ is simply connected. Hence $\pi_1(Q_* \cap U)$ is generated by loops around the ramification locus. These generators yield $\mathbb{C}$ -reflections by (9.2), generating Γ_x if $x \in B^+$. $\square$

10 Finite measure

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued. We assume moreover that the μ_s are rational.

10.1

Let $j \in Q_{*sst} - Q_{*st}$ be a semi-stable point of type $(S(1), S(2))$ where $\sum_{s \in S(1)} \mu_s = 1$. Let $a \in S(1)$ and $b \in S(2)$. Let W be the neighborhood of j in Q_{*sst} lying below the set of all $y \in P^S$ such that $y(a) = o$, $y(b) = \infty$ and such that

$$\sup\{|y(s)|; s \in S(1)\} \leq \inf\{|y(s)|; s \in S(2)\}.$$

Let $p : \tilde{Q}_{*sst} \rightarrow Q_{*sst}$ and let $o \in p^{-1}(j)$. By (8.2.4), $\pi_1(Q_*, o)$ operates on $\tilde{Q}_{*sst}$, and

$$\pi_1(Q_*, o)_o = D_j = \text{image of } \pi_1(W \cap Q_*) \text{ in } \pi_1(Q_*, o);$$

here, $\pi_1(Q_*, o)_o$ denotes the stabilizer of o and D_j the decomposition group of j .

By (4.5), $W - \{j\}$ is a simply connected manifold if $\text{card } S \geq 5$; hence $\pi_1(W \cap Q_*)$ is generated by circuits around the codimension 1 subspaces Q_T for all stable partitions T finer than $\{S(1), S(2)\}$ with $\text{card } T = \text{card } S - 1$ if $N \geq 5$. By (9.2) the monodromy of such generators are complex reflections which have finite order if μ_s is rational for all $s \in S$. Thus

If $\text{card } S \geq 5$, the quotient $\pi_1(Q_*, o)_o / \text{Ker } \theta$ is generated by elements of finite order; here θ denotes the
(5)

10.2

Assume that condition (INT) is satisfied. Set $V = H^1(P_o, L_o)$, let $\text{PU}(V)$ denote the projective unitary group on V with respect to the hermitian form of (3.10.2). Let $B^+ = B(\alpha)_o^+$, i.e. the set of all $v \in V$ with $(v, v) > 0$ modulo $\mathbb{C}^*$. Let $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ be the map of Proposition (8.7). Set

$$t = \tilde{w}(o), \quad \Gamma_t = \text{stabilizer of } t \text{ in } \Gamma.$$

By Theorem (10.18.1), $\tilde{w}$ is a homeomorphism onto an open subset of B^+ . Consequently $\Gamma_t = \theta(\pi_1(Q_*, o)_o)$, the stabilizer of t in Γ . By (11.1.1) and (10.19.3), we get

Γ_t is generated by $\mathbb{C}$ -reflections of finite order if $\dim B^+ > 1$ and by a unipotent element if $\dim B^+ =$
(6)

10.3

Let $v \in t$, let $\text{PU}(V)_t$ (resp. $\text{PU}(V)_\delta$) denote the stabilizer in $\text{PU}(V)$ of t (resp. δ , (5.3)), and let N denote the unipotent radical of $\text{PU}(V)_t$. As pointed out in the proof of Proposition (5.5),

$$\text{PU}(V)_\delta / N \cong \text{U}(V'')$$

which is compact, and $\mathrm{PU}(V)_t/N = \mathbb{R} \ltimes \mathrm{U}(V'')$. Hence $\mathrm{PU}(V)_t/N$ contains all compact subgroups of $\mathrm{PU}(V)_\delta/N$. Therefore, as is well-known, $\mathrm{PU}(V)_t$ contains all compact subgroups of $\mathrm{PU}(V)_\delta$ (because $\mathrm{PU}(V)_\delta$ contains a maximal reductive subgroup and all maximal reductive subgroups of $\mathrm{PU}(V)_\delta$ are conjugate by an element of N).

From (11.2.1), one infers

$$\Gamma_t \subset \mathrm{PU}(V)_\delta. \quad (7)$$

Remarque 10.1. A more direct proof of (11.3.1) results from (7.3) — more precisely from the fact that on W , we have a horizontal family of functions δ . It follows at once that Γ_p preserves a “distance to t ” function (5.3).

Théorème 10.2. *Let S be a finite set, $0 < \mu_s < 1$, $\alpha_s = \exp 2\pi\sqrt{-1}\mu_s$, $\sum \mu_s = 2$ (all $s \in S$). Let $\theta : \pi_1(Q_*, o) \rightarrow \mathrm{PU}(V)$ be the monodromy action defined in (3.10.2). Assume*

(INT) *For all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer.*

Then Γ , the image of θ , is a lattice in $\mathrm{PU}(V)$.

Démonstration. By Theorem (10.19), Γ is discrete in $\mathrm{PU}(V)$, and by Theorem (10.18.2)

$$\Gamma \backslash \tilde{w}(\tilde{Q}_{*\mathrm{sst}}) = \Gamma \backslash B^+$$

where $\tilde{w}(\tilde{Q}_{*\mathrm{sst}})$ is an open subset of B^+ with respect to the topology of (5.4). Hence $\Gamma \backslash \tilde{w}(\tilde{Q}_{*\mathrm{sst}})$ is the union of a compact quotient of a subset of B^+ and a finite set of neighborhoods W_δ (in the topology of (5.4)) of points $t_\delta \in \partial B^+$, where $t_\delta = \tilde{w}(p^{-1}(j))$, $p : \tilde{Q}_{*\mathrm{sst}} \rightarrow Q_{*\mathrm{sst}}$, and j varies over $Q_{*\mathrm{sst}}$. By § 4, $Q_{*\mathrm{sst}}$ is compact. Any compact subset of B^+ has finite measure. By (11.3.1) and Proposition (5.5), the image in $\Gamma \backslash B^+$ of $W_\delta - t_\delta$ has finite measure. It follows that $\Gamma \backslash B^+$ has finite measure. Thus Γ is a lattice in $\mathrm{PU}(V)$. $\square$

Lemme 10.3. *The subgroup Γ_μ is Zariski-dense in $\mathrm{PU}(V)$.*

The proof will be by induction on $\mathrm{card} S$. By (3.10.2), we can lift $\Gamma_\mu \subset \mathrm{PU}(V)$ to a group $\Gamma'_\mu \subset \mathrm{U}(V)$ which is generated by pseudo-reflections (cf. (9.2), (12.3.2)); recall the definition : u is a pseudo-reflection if and only if $u - 1$ has rank 1. Hence Γ'_μ is irreducible on the linear span L of the one dimensional subspaces of V corresponding to its pseudo-reflections.

It is easy to verify that $L = V$. Thus Γ'_μ stabilizes no proper vector subspace of V .

We start the induction at $N = 4$. Here Γ'_μ operates on the complex 1-ball as a triangle group, rotating through twice the angle at each vertex of a geodesic triangle. Since $0 < \mu_s < 1$ and $\sum \mu_s = 2$, the sum of the angles of the geodesic triangle is less than π and the orbit of each point in B^+ under Γ'_μ is infinite. Let $\bar{\Gamma}'_\mu$, $\bar{\Gamma}_\mu$ denote the Zariski closure of Γ'_μ , Γ_μ in $\mathrm{U}(V)$ and $\mathrm{PU}(V)$ respectively. The group $\bar{\Gamma}'_\mu$ can have no connected normal solvable subgroup non-trivial modulo the center of $\mathrm{U}(V)$, otherwise, some Γ'_μ orbits in B^+ would be finite. Consulting the short well known list of closed complex analytic subgroups in $\mathrm{PU}(1, 1) \cong \mathrm{PGL}(2, \mathbb{R})$, one sees that $\bar{\Gamma}_\mu = \mathrm{PU}(V)$ if $\mathrm{card} S = 4$. Suppose now $\mathrm{card} S > 4$.

By (6.21) and two induction hypothesis, $\bar{\Gamma}_\mu \supset \mathrm{PU}(V_T)$ where T is a stable partition of S with $\mathrm{card} T = \mathrm{card} S - 1$ and V_T is the corresponding subspace of V . Since Γ'_μ is irreducible on V , we infer $\bar{\Gamma}_\mu = \mathrm{PU}(V)$. $\square$

11 Arithmeticity and Integral Monodromy

11.1

Assume that for each $s \in S$, μ_s is a rational number and let d denote the least common denominator of $\{\mu_s \mid s \in S\}$. Set $F = \mathbb{Q}(\sqrt[d]{1})$, $\mathcal{O}_F$ the ring of integers in F , and $V = H^1(P_o, L_o)$ where $P_o = P - o(S)$ as in (3.1). Then the vector space V can be defined over the ring $\mathcal{O}_F$. To see this, let $L(F)$ (resp. $L(\mathcal{O})$) denote a local subsystem of L on P_o (suitably chosen using a base point on P_o) with fiber the subfield F (resp. subring $\mathcal{O}$) of $\mathbb{C}$. The pairing of L with its complex conjugate induces an $\mathcal{O}_F$ -valued pairing on $L(\mathcal{O})$. Then one can define cohomology combinatorially as in (2.2) with coefficients in $L(\mathcal{O})$; one can also define the skew-hermitian cup product

$$\psi_0 : H^1(P_o, L(\mathcal{O})_o) \times H^{2-1}(P_o, \bar{L}(\mathcal{O})_o) \rightarrow H^2(P_o, \mathcal{O}) \cong \mathcal{O}_F$$

combinatorially; set

$$V(\mathcal{O}) = H^1(P_o, L(\mathcal{O})_o), \quad V(F) = H^1(P_o, L(F)_o).$$

Define

$$\psi_0 : V(F) \times V(F) \rightarrow F \tag{8}$$

where $j = b - \bar{b}$ for some $b \in \mathcal{O}_F - \mathcal{O}_F \cap \mathbb{R}$. This ψ_0 is a hermitian form on V defined over $\mathcal{O}$ and may be identified with the form defined in (2.18), up to a real scalar factor.

We lift the map $\theta : \pi_1(Q_*, o) \rightarrow \text{PU}(V)$ of (3.10.2) via a local system L on $M(c)$ to $\theta' : \pi_1(M(c), o) \rightarrow \text{U}(V)$ by (3.14). The image of θ' is in $\text{U}(V)(\mathcal{O})$. Set

$$\Gamma_\mu := \Im \theta, \quad \Gamma'_\mu := \Im \theta' \tag{9}$$

where $\mu : \{\exp 2\pi i \mu_s \mid s \in S\}$. Then

$$\Gamma'_\mu \text{ stabilizes } V(\mathcal{O}). \tag{10}$$

11.2

We collect here some remarks and definitions pertaining to arithmeticity of lattices.

Let Γ be a Zariski-dense subgroup of $G(k)$, for G an adjoint connected semi-simple algebraic group G over k , a field of characteristic 0. Set $E = \mathbb{Q}[\text{Tr Ad } \Gamma]$, the field spanned over $\mathbb{Q}$ by $\{\text{Tr Ad } g \mid g \in \Gamma\}$.

Proposition 11.1.

- (i) *The group G has a faithful matrix representation ρ such that $\rho(\Gamma) \subset \text{GL}_n(E)$ (and hence G has E as a field of definition).*
- (ii) *E remains unchanged when Γ is replaced by a commensurable subgroup of G .*

Démonstration. Let T denote the function $g \mapsto \text{Tr Ad } g$ on G , and let W denote the $\mathbb{C}$ -linear span of the left G -translates of T (we define $(x \cdot f)(y) = f(yx)$ for any function $f : G \rightarrow \mathbb{C}$, x and y in G). Inasmuch as T is a sum of matrix coefficients, W is finite dimensional. Since Γ is Zariski dense in G , there is a finite set of elements $x_1, \dots, x_n$ in Γ such that $\beta = \{x_1 \cdot T, \dots, x_n \cdot T\}$ is a base of W . Let $\rho_W : G \rightarrow \text{Aut } W$ be the representation given by $x \mapsto x \cdot f$ for $x \in G$, $f \in W$. It is well known that the representation ρ_W is

faithful ([10], Theorem (2.1), p. 123). Let $\rho(x)$ denote the matrix of $\rho_W(x)$ with respect to the base β . Then $\rho(x)$ has all its entries in the field E for all $x \in \Gamma$.

To prove (ii) it clearly suffices to consider the case that Γ_0 is a subgroup of Γ of finite index. Replacing Γ_0 by $\bigcap_x x\Gamma x^{-1}$, we can suppose moreover that Γ_0 is normal in Γ . Then Γ_0 is also Zariski-dense in G . Set $E_0 = \mathbb{Q}[\text{Tr Ad } \Gamma_0]$ and select the base $\beta = \{x_1 \cdot T, \dots, x_n \cdot T\}$ with $x_i \in \Gamma_0$ ($i = 1, \dots, n$). Let $\sigma : E \rightarrow \mathbb{C}$ be a monomorphism which is the identity on E_0 . To prove (ii), it suffices to prove that σ leaves each element of E fixed. For any $y \in \Gamma$ and $x \in \Gamma_0$, we have $xyx^{-1} \in \Gamma_0$. Hence, by (12.2.1),

$$\sigma\rho(yxy^{-1}) = \sigma\rho(y) \cdot \sigma\rho(x) \cdot \sigma\rho(y)^{-1} = \rho(y) \cdot \sigma\rho(x) \cdot \rho(y)^{-1}.$$

that is $\rho(y)\sigma\rho(x)\rho(y)^{-1} = \sigma\rho(y)\sigma\rho(x)\sigma\rho(y)^{-1}$. Hence for all $y \in \Gamma_0$, $\sigma\rho(y)^{-1}\rho(y)$ centralizes $\sigma\rho(\Gamma_0)$ and therefore $\sigma\rho(G)$ since Γ_0 is Zariski-dense in G . The center of $\sigma\rho(G)$ is 1 and consequently $\sigma\rho = \rho(y)$ for all $y \in G$. Therefore σ fixes each element of E . $\square$

11.3

Take ρ as in (12.2.1). The Zariski closure of $\rho(E)$ in $\text{GL}_n(E)$ is an algebraic group over E . Because taking Zariski closure commutes with field extensions it is a form of G over E . This provides the group G with an E -structure, called the *natural E -structure* of G . If ρ' is another faithful matrix representation of G with $\rho'(\Gamma) \subset \text{GL}_n(E)$, it leads to the same E -structure. To check this, one compares ρ to ρ' via $\rho \oplus \rho'$.

If F is a subfield of k and G_F an F -structure on G for which $\Gamma \subset G_F(F)$, the field of traces E is contained in F and, by the above construction applied to G_F over F , the given F -structure on G is deduced from the natural E -structure.

11.4

Let A be a semi-simple algebraic linear group defined over the field $\mathbb{Q}$ of rational numbers, let V be a finite dimensional vector space defined over $\mathbb{Q}$, and $\rho : A \rightarrow \text{GL}(V)$ a faithful rational representation defined over $\mathbb{Q}$. By a theorem of Borel-Harish-Chandra [3], the subgroup $\Gamma = \rho^{-1}(\text{GL}(V_{\mathbb{Z}}))$ is a lattice in $A(\mathbb{R})$ for any lattice $V_{\mathbb{Z}}$ in $V_{\mathbb{Q}}$; that is, Γ is a discrete subgroup of $A(\mathbb{R})$ and $A(\mathbb{R})/\Gamma$ has finite measure.

11.5

Let G be an adjoint connected semi-simple real Lie group. It is the topological connected component $G(\mathbb{R})^0$ of $G(\mathbb{R})$, for G an adjoint connected semi-simple algebraic group over $\mathbb{R}$. By definition, a subgroup Γ of G is *arithmetic in G* if and only if there exists an algebraic group A over $\mathbb{Q}$, a compact group K and an analytic isomorphism θ of $A(\mathbb{R})^0$ onto $G \times K$ such that $\theta(A(\mathbb{Z}) \cap A(\mathbb{R})^0)$ has its projection into G commensurable to Γ . The group A is necessarily reductive. It can be assumed connected adjoint semi-simple; namely, replace A by $A' := A/Z(A)^0$. The group A is then a product of $\mathbb{Q}$ -simple groups A_i (cf. [23], p. 46 (3.1.2)). If G is a simple non-compact Lie group, it follows from the definition that all A_i but one, A_1 , are such that $A_i(\mathbb{R})$ is compact. Replacing A by A_1 , one can then assume that A is $\mathbb{Q}$ -simple, i.e. of the form (cf. [23], p. 46 (3.1.2))

$$A = \text{Rest}_{F/\mathbb{Q}} B$$

for B an absolutely simple group over a finite extension F of $\mathbb{Q}$. One has $F \otimes \mathbb{R} = \prod F_v$ (product of the completions of F at the infinite places v) and $A(\mathbb{R}) = B(F \otimes \mathbb{R}) = \prod B(F_v)$. The existence of θ as in (12.2.4) amounts to saying that for all v but one, v_1 , $B(F_v)$ is compact (hence if $v \neq v_1$ the place v is a real place) and that $B(F_{v_1})^0 \cong G$, the isomorphism carrying $B(\mathcal{O}_F)$ to a subgroup of G commensurable to Γ . If G is an absolutely simple Lie group, the place v_1 is a real place, F is totally real, and B is an F -form of G , F being identified with a subfield of $\mathbb{R}$ via v_1 .

SURVEY OF DRINFEL'D MODULES

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Introduction

In Deligne [1971], two-dimensional ℓ -adic representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ are attached to “new” holomorphic modular forms of weight at least two on the Poincaré upper half plane, or equivalently, to certain automorphic representations of the adèle group $GL_2(\mathbb{A}_{\mathbb{Q}})$. The correspondence preserves L -functions.

This theory depends on the properties of moduli varieties of elliptic curves with given level structure. These varieties have a canonical structure over $\mathbb{Q}$ and the ℓ -adic representations are realized in the ℓ -adic H^1 of certain sheaves.

Drinfel’d [1973] transports the theory to the function field case by introducing the concept of elliptic module, which we call a Drinfel’d module, to replace elliptic curves.

Fixed notations throughout the article

C – absolutely irreducible projective and smooth curve over $\mathbb{F}_q$.

∞ – a closed point of C .

F – the function field $\mathbb{F}_q(C)$ of C over $\mathbb{F}_q$.

A – the ring $H^0(C - \infty, \mathcal{O}_C)$ of functions regular on $C \setminus \infty$.

F_{∞} – the completion of F at ∞ with valuation ring $\mathcal{O}_{\infty}$.

C_{∞} – the completion of the algebraic closure of F_{∞} .

$\mathbb{F}_q$ – the finite field of q elements.

Relative to the function field F over $\mathbb{F}_q$, we will define Drinfel’d modules of rank r in the first chapter. Briefly, these are $A = H^0(C - \infty, \mathcal{O}_C)$ module structures on the additive group in characteristic p given by polynomials in Frobenius whose degree is a certain multiple of the rank r . The term elliptic module, which is Drinfel’d’s original term, is used for Drinfel’d modules of rank 2 for these are objects which correspond closely to elliptic curves.

In fact, we have the following dictionary:

elliptic curve	elliptic module (rank 2 Drinfel’d module)
$\mathbb{Q}$	$\mathbb{F}_q(C)$
F	F
infinite place	fixed place ∞
$\mathbb{Z}$	$A = H^0(C \setminus \infty, \mathcal{O}_C)$
scheme	scheme over A
n division point	I division point for I an ideal of A
n level structure	I level structure
moduli space	moduli space
lattice in $\mathbb{C}$	discrete A -module

Most of the above dictionary is explained in chapter 1. Elliptic curves over the complex numbers $\mathbb{C}$ can be interpreted as classes of certain lattices in $\mathbb{C}$. In chapter 2 we describe Drinfel’d modules over C_{∞} in terms of discrete A -modules in C_{∞} defined over A . Drinfel’d modules defined over $\overline{\mathbb{F}}_q$, the algebraic closure of $\mathbb{F}_q$, can be also described by lattices in

C_∞ , and the lattices with Galois invariance properties correspond to Drinfel'd modules over intermediate fields between $\mathbb{F}_q$ and $\overline{\mathbb{F}}_q$.

For elliptic curves, indeed for more generally polarized abelian varieties, and for Drinfel'd modules there are moduli problems for families of these objects and moduli spaces. From the point of view of Shimura varieties, basic information about such moduli problems is collected in triples (F, G, h') consisting of:

- a reductive group G over a global field F ,
- a conjugacy class of maps $h' : \mathbb{G}_m \rightarrow G$.

The above dictionary extends further with the following examples of triples (F, G, h') for the corresponding moduli problems over F :

$$\begin{array}{ll} \left(\mathbb{Q}, GSp_{2g}, \begin{pmatrix} * & * \\ 0 & 1_g \end{pmatrix} \right) & \text{polarized abelian varieties (rank } g) \\ \left(\mathbb{Q}, GL_2, \begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix} \right) & \text{elliptic curves} \\ \left(F, GL_r, \begin{pmatrix} * & * \\ 0 & I_{r-1} \end{pmatrix} \right) & \text{Drinfel'd modules (rank } r) \\ \left(F, GL_2, \begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix} \right) & \text{elliptic module} \end{array}$$

A critical step in the theory over $\mathbb{Q}$ was the calculation of the cohomology of the complex analytic variety of $\mathbb{C}$ -valued points on the moduli scheme and the comparison with the ℓ -adic cohomology. In the function field case Drinfel'd puts a rigid analytic structure on the C_∞ -valued points of the moduli scheme, calculates a rigid analytic H^1 for certain simple sheaves, and compares this H^1 with the ℓ -adic H^1 . This rigid analytic space is the quotient of a C_∞ -analogue of the Poincaré upper half plane by a discrete group Γ , and the cohomology is in the middle of a short exact sequence with subgroup $H^1(\Gamma)$ and quotient the space of coclosed 1-cochains on the tree of $PGL(2, C_\infty)$ invariant under Γ . This in turn is interpreted in terms of representations. Finally, the various representations in H^1 are sorted out with a congruence formula which is analogous to the Eichler–Shimura congruence formula. These considerations are carried out in chapters 3 and 4.

In chapter 5 the global results on automorphic forms for $GL(2, \mathbb{A}_F)$ are applied to a local result conjectured by Langlands. For a local field K of equal characteristic p , there is a natural bijection between irreducible admissible representations of $GL(2, K)$ and two-dimensional representations of the Weil group $W(\overline{K}/K)$. The global theory of automorphic forms given by elliptic modules applies only to representations $\pi = \otimes' \pi_v$ where π_∞ is the special representation. In the form of private letters Drinfel'd has a new theory of “shtuka” which handles all automorphic forms on $GL(1)$ and $GL(2)$ over a function field. Even so it seems that elliptic modules are still worth consideration because the theory for $r = 1$ and $r = 2$ may extend easier for general r than the corresponding theory of “shtuka” which depends on the Selberg trace formula. Another method to relate automorphic forms to ℓ -adic representations is in Drinfel'd [1983]. Recently, elliptic modules were used to describe modular forms in the function field case, see D. Goss [1977].

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1 Algebraic Theory of Drinfel'd Modules

In terms of the notations in the introduction, a Drinfel'd module is an A -module structure on the additive group $\mathbb{G}_a$ in characteristic p . There is a theory of division points and isogenies of Drinfel'd modules which parallels closely the corresponding theory for abelian varieties. Drinfel'd modules have a characteristic v , which is a valuation of $F = \mathbb{F}_q(C)$ over $\mathbb{F}_q$, and they exhibit a singular and sometimes supersingular behavior at v when $v \neq \infty$. Again, we are reminded of elliptic curves in characteristic p .

Drinfel'd formulated and solved a moduli problem for Drinfel'd modules with prescribed level structure. The construction takes place entirely with affine schemes making the existence of the moduli elementary compared with the existence question of the moduli scheme for elliptic curves with level structure. We introduce Drinfel'd's modification of the notion of level structure which allows for a clearer analysis of level structures at singular and supersingular points for the cases of both Drinfel'd modules and elliptic curves. See additional remarks 1.

1.1 Endomorphisms of the Additive Group

The additive group (functor) $\mathbb{G}_a$ over a ring R is represented by the polynomial ring $R[X]$ in one variable with structure morphism $\alpha : R[X] \rightarrow R[X] \otimes_R R[X]$ given by $\alpha(X) = X \otimes 1 + 1 \otimes X$.

A morphism $\phi : \mathbb{G}_a \rightarrow \mathbb{G}_a$ of the underlying schemes over R is given by a polynomial $\phi(X) \in R[X]$ where $\phi^*(a(x)) = a(\phi(x))$ for $a(x) \in R[X]$, and this morphism preserves the group scheme structure if and only if $\phi(X)$ is additive, i.e. $\phi(X+Y) = \phi(X) + \phi(Y)$. Composition $\phi \circ \psi$ of morphisms ϕ and ψ is represented by substitution $\phi(\psi(x))$, and the sum of two morphisms $\mathbb{G}_a \rightarrow \mathbb{G}_a$ is represented by the sum of two polynomials.

For example, $\phi(x) = x^{p^i}$ is additive where p is a prime. An additive polynomial has the form

$$\phi(x) = \sum_{0 \leq i \leq n} a_i x^{p^i} = a_0 x + a_1 x^p + a_2 x^{p^2} + \cdots + a_n x^{p^n}.$$

If $\psi(X)$ is a second additive polynomial, then $\phi \circ \psi(x) = \phi(\psi(x)) = \sum_{i,j} a_i \psi_j^{p^i} x^{p^{i+j}}$ is also additive.

Définition 1.1. Let k be a field of characteristic $p > 0$. The *twisted polynomial ring* $k\{\tau\}$ is $k \otimes_{\mathbb{Z}} \mathbb{Z}[\tau]$ with the twisted tensor product algebra structure satisfying the commutation rule

$$(a \otimes \tau^i) \cdot (b \otimes \tau^j) = ab^{p^i} \otimes \tau^{i+j}.$$

We denote $a \otimes \tau^i$ simply by $a\tau^i$ in $k\{\tau\}$, and then the commutation rule becomes $\tau a = a^p \tau$.

Let k denote a field of characteristic $p > 0$ in the remainder of this section.

Proposition 1.1. *The function θ which assigns to an additive polynomial $\phi(X) = \sum_{0 \leq i \leq n} a_i x^{p^i}$ the element $\theta(\phi(X)) = \sum_{0 \leq i \leq n} a_i \tau^i \in k\{\tau\}$ is an isomorphism $\theta : \text{End}_k(\mathbb{G}_a) \xrightarrow{\sim} k\{\tau\}$ of rings.*

The proof follows from the observation that the relation $(X^{p^j})^{p^i} = X^{p^{i+j}}$ becomes $\tau^j a \tau^i = a^{p^j} \tau^{i+j}$ under θ . Note that the multiplicative structure on $\text{End}_k(\mathbb{G}_a)$ is substitution of additive polynomials and on $k\{\tau\}$ the twisted polynomial multiplication.

We have two degree functions $\deg : \text{End}_k(\mathbb{G}_a) \rightarrow \mathbb{Z}$ and $d : k\{\tau\} \rightarrow \mathbb{Z}$ defined by the relations

$$\deg(a_0x + a_1x^p + \cdots + a_nx^{p^n}) = p^n$$

and

$$d(a_0 + a_1\tau + \cdots + a_n\tau^n) = n$$

where $a_n \neq 0$. The following relations hold for $\phi, \psi \in \text{End}_k(\mathbb{G}_a)$ and $a, b \in k\{\tau\}$:

$$\begin{aligned} \deg(\phi \circ \psi) &= \deg(\phi) \deg(\psi), & \deg(\phi + \psi) &\leq \max(\deg(\phi), \deg(\psi)), \\ d(ab) &= d(a) + d(b), & d(a + b) &\leq \max(d(a), d(b)). \end{aligned}$$

We have height functions $\text{ht} : \text{End}_k(\mathbb{G}_a) \rightarrow \mathbb{Z}$ and $\text{ht} : k\{\tau\} \rightarrow \mathbb{Z}$ defined by the relations

$$\text{ht}(a_hx^{p^h} + \cdots + a_nx^{p^n}) = h$$

and

$$\text{ht}(a_h\tau^h + \cdots + a_n\tau^n) = h$$

where $a_h \neq 0$. Clearly, $\text{ht}(\phi) = \text{ht}(\theta(\phi))$. The following relations hold:

$$\text{ht}(\phi \circ \psi) = \text{ht}(\phi) + \text{ht}(\psi) \quad \text{and} \quad \text{ht}(ab) = \text{ht}(a) + \text{ht}(b)$$

for $\phi, \psi \in \text{End}_k(\mathbb{G}_a)$ and $a, b \in k\{\tau\}$.

We have substitutions $\partial_0 : \text{End}_k(\mathbb{G}_a) \rightarrow k$ and $\partial : k\{\tau\} \rightarrow k$ defined by the relations

$$\partial_0 \left(\sum_i a_i x^{p^i} \right) = a_0 \quad \text{and} \quad \partial \left(\sum_i a_i \tau^i \right) = a_0$$

where ∂_0 is the derivative at the origin and ∂ the value at the origin. Clearly $\partial(\theta(\phi)) = \partial_0(\phi)$ for $\phi \in \text{End}_k(\mathbb{G}_a)$.

Finally, the following properties of additive polynomials seem to have been known for some time.

Proposition 1.2. *Let $H \subset \mathbb{G}_a(k)$ be a finite subgroup, and form the polynomial $P_H(X) = \prod_{h \in H} (X - h)$. Then $P_H(X)$ is an additive polynomial, so $P_H \in \text{End}_k(\mathbb{G}_a)$ with $\deg(P_H) = \text{Card}(H)$, and we are able to recover $H = \ker(P_H)(k)$, the set of all $x \in k$ with $P_H(x) = 0$.*

Proof. To show that P_H is additive, consider $Q_Y(X) = P_H(X + Y) - P_H(Y)$ in $k(Y)[X]$. Since H is a subgroup, $Q_Y(X) = 0$ on H and further $\deg(Q_Y) = \deg(P_H)$. Thus $Q_Y(X)$ and $P_H(X) = P_H(X + Y) - P_H(Y)$ are monic polynomials of the same degree equal to $\text{Card}(H)$ and each equal to zero on $H \subset k(Y)$. It follows that $P_H(X) = Q_Y(X) = P_H(X + Y) - P_H(Y)$, i.e. P_H is additive, and the other statements hold which proves the proposition. $\square$

Remarque 1.1. As a kind of converse of (2.1) observe that for an additive polynomial $f : \mathbb{G}_a \rightarrow \mathbb{G}_a$ over k , the set H of $x \in k$ with $f(x) = 0$ is a subgroup of $\mathbb{G}_a(k)$ with $\text{Card}(H)$ dividing $\deg(f)$. If k is algebraically closed, then $\deg(f) = p^{\text{ht}(f)} \cdot \text{Card}(H)$, and that the group morphism $f : \mathbb{G}_a(k) \rightarrow \mathbb{G}_a(k)$ is surjective for $f \neq 0$. Since $\text{ht}(f) = 0$ if and only if $\partial_0(f) \neq 0$, we see that $\partial_0(f) \neq 0$ implies that $\deg(f) = \text{Card}(\ker(f))$ over an algebraically closed k .

1.2 Definition of Drinfel'd Module over a Field

We return to the basic notations of the introduction, in particular $A = H^0(C \setminus \infty, \mathcal{O}_C)$, and let k denote a field of characteristic p . Before the definition we make some remarks which point out the natural limitations of the definition.

Remarque 1.2. Since $\text{End}_k(\mathbb{G}_a)$ is isomorphic to $k\{\tau\}$ is an integral domain, and since A is a Dedekind domain, any ring morphism $A \rightarrow \text{End}_k(\mathbb{G}_a)$ is either injective or has image contained in the constants $k \subset k\{\tau\}$. The second case being relatively trivial means that we will be interested in morphisms which are injective (or faithful).

To the rational point ∞ , we have associated an absolute value $|a|_\infty$ on the function field F with $\mathbb{F}_{q_\infty}$ as residue class field with q_∞ a power of q and the absolute value normalized such that $|a|_\infty = \text{Card}(A/a)$ for $a \in A \subset F$.

Remarque 1.3. For an injective $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$ the composite $A \xrightarrow{\phi} \text{End}_k(\mathbb{G}_a) \xrightarrow{\deg} \mathbb{Z}$, denoted $\|a\| = \deg(\phi(a))$, satisfies $\|ab\| = \|a\| + \|b\|$ and $\|a + b\| \leq \max(\|a\|, \|b\|)$ from properties of $\deg$. Since $\|a\| \neq 1$ for $a \neq 0$ and for some a , $\|a\| > 1$, the relation $\|a/b\| = \|a\| - \|b\|$ for $a, b \in F$ defines an extension of $\|\cdot\|$ to F as an absolute value. Since $\|a\| > 1$ for all $a \in A$, $a \neq 0$, the absolute value $\|x\|$ on F must be equivalent to $|\cdot|_\infty$, i.e. $\|x\| = |x|_\infty^t$ for some $t > 0$ and all $x \in F$. In fact, $t > 0$ is a natural number, but we defer the proof of this until the next section, see (1.1), and introduce the following basic concept which Drinfel'd called an elliptic module.

Définition 1.2. For a natural number $r > 0$, a *Drinfel'd module over a field k of rank r* for the pointed curve (C, ∞) over $\mathbb{F}_q$ is a morphism of rings $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$ such that G is isomorphic to $\mathbb{G}_a$ and

$$\|a\| = \deg(\phi(a)) = |a|_\infty^r = (\text{Card}(A/a))^r$$

for all nonzero $a \in A$.

Let $\mathcal{k}$ denote the category of commutative k -algebras R , and let Mod_A denote the category of A -modules. To a Drinfel'd module ϕ , we associate a functor $E : \mathcal{k} \rightarrow \text{Mod}_A$ which assigns to a k -algebra R its underlying additive group $\mathbb{G}_a(R)$ together with the A -module structure defined by requiring $\phi(a)(R)$ to be scalar multiplication by a on $\mathbb{G}_a(R)$. Observe that the functor E determines ϕ , and frequently we refer to E as the Drinfel'd module instead of ϕ . We will denote $\phi(a)$ by just ϕ_a occasionally.

Définition 1.3. Let $\phi : A \rightarrow \text{End}_k(G)$ and $\phi' : A \rightarrow \text{End}_k(G')$ be two Drinfel'd modules. A *morphism* $u : \phi \rightarrow \phi'$ is a morphism $u : G \rightarrow G'$ such that $u\phi_a = \phi'_a u$ for all $a \in A$. A nonzero morphism is called an *isogeny*.

Equivalently, u is a morphism of functors $E \rightarrow E'$ associated to ϕ and ϕ' . Thus Drinfel'd modules over k form a category with the composition of morphisms being evident.

Définition 1.4. Let $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$ be a Drinfel'd module and form $\partial_0 \phi : A \rightarrow k$ where $\partial_0 : \text{End}_k(\mathbb{G}_a) \rightarrow k$ is the value of the derivative at the origin. The Drinfel'd module ϕ has *characteristic* ∞ provided $\partial_0 \phi : A \rightarrow k$ is injective and has *characteristic* v , a valuation of F over $\mathbb{F}_q$ different from ∞ , provided $0 \neq |\cdot|_v \in \ker(\partial_0 \phi)$. Since $\ker(\partial_0 \phi)$ is either zero or a maximal ideal of A , every Drinfel'd module has characteristic ∞ or some “finite” $v \neq \infty$.

Remarque 1.4. If $u : \phi \rightarrow \phi'$ is a nonzero morphism (isogeny) between two Drinfel'd modules, then ϕ and ϕ' have the same rank and the same characteristic.

Remarque 1.5. The definition (1.2) of a Drinfel'd module can be formulated in terms of a $(A, \text{End}_k(\mathbb{G}_a))$ -bimodule N which is free of rank 1 over $\text{End}_k(\mathbb{G}_a)$ and satisfying the condition $\|a\| = (\text{Card}(A/a))^r$ for $a \in A$, $a \neq 0$. The choice of a basis element for N identifies N with $\text{End}_k(\mathbb{G}_a)$.

1.3 Division Points

As with abelian varieties, division points of a Drinfel'd module play a basic role in structure and moduli problems. We continue with the notation $E :_k \rightarrow \text{Mod}_A$ equal to the functor associated to a ring morphism $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$.

Définition 1.5. For $a \in A$ (usually $a \neq 0$) the subfunctor $E_a \subset E$ of a -division points is $\ker(\phi_a)$. For an ideal $I \subset A$ the subfunctor $E_I \subset E$ of I -division points is $\bigcap_{a \in I} E_a$. If $a_1, \dots, a_m$ generate an ideal I in A , then $E_I = E_{a_1} \cap \dots \cap E_{a_m}$. So in particular for a principal ideal (b) we have $E_{(b)} = E_b$. Since $E_a = E$ if and only if $a = 0$, the same holds for ideal: $E_I = E$ if and only if $I = 0$, and essentially we consider E_I only for nonzero ideals I . More explicitly, for any k -algebra R , the A -submodule $E_I(R)$ consists of all $x \in E(R)$ such that $ax = 0$ for any $a \in I$. This shows that we can view E_I as a functor defined $E_I :_k \rightarrow \text{Mod}_{A/I}$.

Again as with abelian varieties, $E_I(\bar{k})$ will be a free A/I -module for I prime to the characteristic of E . For the case characteristic ∞ this is no restriction on I and for characteristic v corresponding to the maximal ideal $\mathfrak{p}_v$ in A it means $I \not\subset \mathfrak{p}_v$.

In order to prove this assertion we will use the following lemma on torsion modules over a discrete valuation ring V with local uniformizing parameter π . For a V -module L let $\ell(L)$ denote its length and $\pi_L : L \rightarrow L$ the action of the scalar π on L .

Lemme 1.1. *Let L be a V/π^m -module.*

- (a) *We have $2\ell(\ker(\pi_L)) \geq \ell(L)$.*
- (b) *For nonzero L , the equality holds in (a) if and only if L is a free V/π^m -module, and in this case $\ker(\pi_L)$ is a free V/π^m -module.*

Proof. We can decompose L as the sum of modules N isomorphic to V/π^i with $0 < i \leq 2m$. For $N = V/\pi^i$ ($0 < i \leq 2m$) observe that

$$\ker(\pi^m) = \begin{cases} V/\pi^i & \text{of length } i \text{ for } m \geq i, \\ \pi^{i-m}(V/\pi^i) & \text{of length } m \text{ for } m < i. \end{cases}$$

In both cases $2\ell(\ker(\pi^m)) = 2 \cdot \min(i, m) \geq i = \ell(N)$ so that (a) holds. Also N is free, i.e. $i = 2m$ if and only if $2\ell(\ker(\pi^m)) = \ell(N)$, and $\pi^{m-1} \ker(\pi^m)$ is isomorphic to V/π^m as V/π^m -modules which proves (b). $\square$

In the following basic structure theorem for $E_I(\bar{k})$ over the algebraic closure $\bar{k}$ of k , we also settle the question left at the end of (1.3). Note that the above definitions apply to any morphism $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$ of rings.

Théorème 1.1. *If $\phi : A \rightarrow \text{End}_k(\mathbb{G}_a)$ is a monomorphism of rings, then ϕ is a Drinfel'd module of rank r for an integer r . Moreover, for an ideal I relatively prime to the characteristic of ϕ , the A/I -module $E_I(\bar{k})$ is free of rank r .*

Proof. By (1.3) there exists a real number $t > 0$ with $\deg \phi_a = \text{Card}(A/a)^t$ for all $a \in A$, $a \neq 0$. If $\partial_0 \phi_a = 0$, then by (1.1) $\deg \phi_a = \text{Card } E_a(\bar{k})$ and also

$$\deg(\phi_{a^2}) = \deg(\phi_a)^2 = \text{Card } E_{a^2}(\bar{k}) = \text{Card } E_a(\bar{k})^2.$$

For each irreducible element π of A prime to the characteristic of ϕ , which is a local uniformizing parameter of $A_{\mathfrak{p}} = V$, we apply lemma (1.1) to the V/π^m -module $E_{\pi^m}(\bar{k})$ to prove that $E_{\pi^m}(\bar{k})$ is a free A/π^m -module of rank r where

$$\text{Card } E_{\pi^m}(\bar{k}) = \deg(\phi_{\pi^m}) = \text{Card}(A/\pi^m)^r.$$

From this we deduce that r is an integer and so ϕ is a Drinfel'd module of rank r .

Next, for any nonzero $a \in A$ the primary components of $E_a(\bar{k})$ are of the form $E_{\pi^n}(\bar{k})$, so free A/π^n -module of rank r , from which we deduce that $E_a(\bar{k})$ is a free A/a -module of rank r . Finally, if I is an ideal of A prime to the characteristic of ϕ , then there exists another ideal J in A with $A = I + J$ and $IJ = (e)$, a principal ideal prime to the characteristic. Then $A/e = A/I \oplus A/J$, and $E_e = E_I \oplus E_J$ as functors. Since $E_e(\bar{k})$ is free of rank r over A/e , it follows that $E_I(\bar{k})$ is free of rank r over A/I , and this proves the theorem. $\square$

Remarque 1.6. It remains only to consider for a Drinfel'd module E of rank r and characteristic $v \neq \infty$, the $A/\mathfrak{p}_v^n$ -modules $E_{\mathfrak{p}_v^n}(\bar{k})$ where $V(\mathfrak{p}_v) = v$. In this case $\partial_0 \phi_{\pi^i} = 0$ for all i , and by the discussion in (1.1), the endomorphism ϕ_{π^n} has a height h where $\deg(\phi_{\pi^n}) = p^{nh} \cdot \text{Card } E^{\text{ét}}(\bar{k})$, or more generally

$$p^{nh} = \frac{\deg(\phi_{\pi^n})}{\text{Card } E^{\text{ét}}(\bar{k})} = \frac{(\deg \phi_{\pi})^n}{\text{Card } E_{\mathfrak{p}_v^n}(\bar{k})} = \frac{(p^h \cdot \text{Card } E_{\mathfrak{p}_v}(\bar{k}))^n}{\text{Card } E_{\mathfrak{p}_v^n}(\bar{k})}.$$

Hence again by applying lemma (1.1), we see that $E_{\mathfrak{p}_v^n}(\bar{k})$ is a free module of rank $r - h < r$ over $A/\mathfrak{p}_v^n$. As finite group schemes over k , we have a splitting when k is separable

$$E_{\mathfrak{p}_v^n} = E_{\mathfrak{p}_v^n}^{\text{ét}} \times E_{\mathfrak{p}_v^n}^0$$

where $E_{\mathfrak{p}_v^n}^{\text{ét}}$ is étale of rank $r - h$ and $E_{\mathfrak{p}_v^n}^0$ is infinitesimal of rank h .

Définition 1.6. The *height* of a Drinfel'd module E of characteristic $v \neq \infty$ is the height of ϕ_{π} where π is an irreducible with $V(\pi) = v$. Since (π) is uniquely determined by v , the height is well defined.

Remarque 1.7. Let E be a Drinfel'd module of rank r and characteristic v . Let h be the height when $v = \infty$. For an irreducible π of A , let A_{π} be A localized at π . Then $\varinjlim_n E_{\pi^n}(\bar{k})$ is a divisible A_{π} -module which is isomorphic to

$$\begin{cases} (F_{\pi}/A_{\pi})^r & \text{for } \pi \text{ prime to the characteristic,} \\ (F_{\pi}/A_{\pi})^{r-h} & \text{for } v(\pi) = v. \end{cases}$$

1.4 Isogenies

Recall that for a Drinfel'd A -module (ϕ, E) over k and any field extension k' of k , the group $E(k')$ has a given A -module structure. For two Drinfel'd A -modules over k and $u \in k\{\tau\}$ with $u \neq 0$, the additive $u : \phi \rightarrow \phi'$ is an isogeny if and only if the additive $u : E(\bar{k}) \rightarrow E'(\bar{k})$ is A -linear. This follows from the fact that two polynomials are equal if and only if their associated polynomial maps $E \rightarrow E$ are equal. The kernel of $u : E(k') \rightarrow E'(k')$ is the A -module $\ker(u)(k')$ where $\ker(u)$ is the scheme kernel of $u : \phi \rightarrow \phi'$.

Separable Isogenies

An isogeny $u : \phi \rightarrow \phi'$ is *separable* when $u(X) = b_0x + b_1x^p + \cdots + b_sx^{p^s}$ where $b_0 \neq 0$. Then u is determined by $\ker(u)(\bar{k})$, a finite A -submodule of $E(\bar{k})$, up to constant factor. For a finite A -submodule $H \subset E(\bar{k})$, we form $P_H(x) = \prod_{h \in H} (x - h)$, and there is a unique Drinfel'd A -module ϕ' such that $P_H\phi_a = \phi'_a P_H$ for all $a \in A$. If $a \cdot \ker(u)(\bar{k}) = 0$ for $a \neq 0$, then we can form the isogeny v with kernel $u(\ker(\phi_a)(\bar{k}))$ and $vu = \phi_a$. Note that such a nonzero $a \in A$ with $a \cdot \ker(u)(\bar{k}) = 0$ always exists for $u \neq 0$ since it is a finite A -module. When u is separable, we can always choose v to be separable too. By the same construction, if $u : \phi \rightarrow \phi'$ and $w : \phi \rightarrow \phi''$ are separable isogenies such that $w(\ker(u)(\bar{k})) = 0$, then there is a separable isogeny $v : \phi' \rightarrow \phi''$ with $w = vu$ where $\ker(v)(\bar{k}) = u(\ker(w)(\bar{k}))$.

Purely Inseparable Isogenies

These are of the form $\tau^i \in k\{\tau\}$ or $X \mapsto X^{p^i}$. Since $\phi_a\tau^i = \tau^i\phi_a$ in $k\{\tau\}$ is equivalent to all coefficients of ϕ_a being in $\mathbb{F}_{p^i}$, it follows that a purely inseparable isogeny exists only when the characteristic v of ϕ is unequal to ∞ , and in this case p^i is a power of q_v where q_v is the cardinality of the residue class field of v . The finite group scheme $\ker(u) = \text{Spec}(k[t]/(t^{p^i}))$. This case corresponds to a purely local kernel.

Remarque 1.8. Let H be a finite subgroup scheme of $\mathbb{G}_a$, and let ϕ be a Drinfel'd module structure on $\mathbb{G}_a$. Then H is the kernel of some isogeny $u : \phi \rightarrow \phi'$ if and only if H is stable under the action of A and

$$\ker(u) = \begin{cases} 0 & \text{if the characteristic} = \infty, \\ \text{Spec}(k[t]/(t^{q_v^h})) & \text{if the characteristic} = v \text{ and } q = q_v = \text{Card}(\mathbb{F}_q(v)). \end{cases}$$

For any isogeny $u : \phi \rightarrow \phi'$ there exists an isogeny $v : \phi' \rightarrow \phi$ and $a \in A$ with $\phi_a = vu$.

Let $\delta(u)$ denote the degree of the additive polynomial $u \in \text{End}_k(\mathbb{G}_a)$. The function $\delta : \text{End}_k(\mathbb{G}_a) \rightarrow \mathbb{Z}$ also prolongs under extension of scalars to $\text{End}_k(\mathbb{G}_a) \rightarrow \text{End}_{\bar{k}}(\mathbb{G}_a)$.

Remarque 1.9. We study the A -algebra $\text{End}_k(E)$ with δ using the following properties of $\delta : \text{End}_k(E) \otimes_A F_\infty \rightarrow \mathbb{Q}$:

- (1) $\delta(u) \geq 0$ and $\delta(u) = 0$ if and only if $u = 0$.
- (2) $\delta(au) = |a|_\infty \delta(u)$ for $a \in F$.
- (3) $\delta(u + v) \leq \max(\delta(u), \delta(v))$.

$$(4) \delta(uv) = \delta(v)\delta(u).$$

Moreover, $\text{End}(E) \subset \text{End}_k(E) \otimes_A F_\infty$ will be seen to be a discrete A -module in this normed vector space over F_∞ , see (??)(2). The main step will be to show that $\text{End}(E)$ is a finitely generated A -module. For this we use the following two lemmas.

Lemme 1.2. *Let $A^n \subset X \subset F_\infty^n$ where X is a discrete A -module. Then X is finitely generated.*

Proof. For all i observe that $\mathfrak{p}_\infty^i + A \subset F_\infty$ has finite index, e.g. $k[1/t] + t^i k[[t]] \subset k((t))$ has index $\text{Card}(k)^i$. There exists an i with $X \cap (\mathfrak{p}_\infty^i)^n = 0$ since X is discrete, and thus X embeds in $(F_\infty/\mathfrak{p}_\infty^i)^n$ and X/A^n embeds in $(F_\infty/\mathfrak{p}_\infty^i)^n$ which is finite. Since $(X : A^n)$ is finite and A^n is finitely generated, it follows that X is a finitely generated A -module. $\square$

Proposition 1.3. *For a finite dimensional subspace V over F_∞ of $\text{End}(E) \otimes_A F$, it follows that $V \cap \text{End}(E)$ is a finitely generated A -module which is projective.*

Proof. For $V_\infty = F_\infty \cdot V$, it follows that $X = \text{End}(E) \cap V = \text{End}(E) \cap V_\infty$, and we can assume that S generates V and so V_∞ or replace V by a smaller subspace. Let $x_1, \dots, x_n$ be a basis of V_∞ with $x_i \in X$. Then with this basis $A^n \subset X \subset F_\infty^n = V_\infty$, and we are reduced to the previous lemma since $\text{End}(E)$ is a discrete subspace of $\text{End}(E) \otimes_A F_\infty$. To see that $X = V \cap \text{End}(E)$ is projective, we have only to remark that it is flat since it is torsion free over a Dedekind domain and finitely generated flat modules are projective over a Noetherian ring. $\square$

Corollaire 1.1. *The A -module $\text{End}(E)$ is projective.*

Proof. Let $W = \prod_i W_i$ where $W = \text{End}(E) \otimes_A F$ and $\|w_i\|_{F_\infty \rightarrow i}$ is finite. Then $X_i = \text{End}(E) \cap W_i$ is projective by (1.3) and the restrictions of the projections $W \rightarrow W_i$ to $f : \text{End}(E) \rightarrow \prod_i X_i$ define a morphism onto a projective module with $\ker(f) = 0$. Hence $\text{End}(E)$ is projective. $\square$

For estimates on the rank we use the following lemma.

Lemme 1.3. *Let $a \in A$ be prime to the characteristic of E . Then $\text{End}(E) \otimes_A A/a \rightarrow \text{End}(E_a)$ is injective.*

Proof. If $v \in \text{End}(E)$ and $v(\ker(\phi_a)) = 0$, then as in (1.4), it follows that $v = v'\phi_a$, and $v \neq 0$ implies $v \in \text{End}(E)a$. $\square$

Now we summarize all the basic results in the following theorem and remarks.

Théorème 1.2. *Let E be a Drinfel'd module over an algebraically closed field k of rank r . Then*

- (1) $\text{End}(E)$ is a projective A -module of rank $\leq r^2$, and
- (2) $\text{End}_k(E) \otimes_A F_\infty$ is a field in which $\text{End}(E)$ embeds as a discrete A -module of this normed space over F_∞ .

Proof. (1) The fact that $\text{End}(E)$ is projective is contained in (1.1) and the injectivity of $\text{End}(E) \otimes_A A/a \rightarrow \text{End}(E_a)$ coming from (1.3) bounds the rank by r^2 since E_a is an A/a -module of rank r for a prime to the characteristic.

(2) The existence of δ on $\text{End}_k(E) \otimes_A F$ proves that it is a field, and since F_∞/F is a separable extension $\text{End}_k(E) \otimes_A F_\infty$ is also a field. The subspace where $\delta = 0$ on $\text{End}_k(E) \otimes_A F_\infty$ is zero since $\dim_F \text{End}(E) \otimes_A F_\infty \geq (\text{End}_k(E))$. $\square$

Remarque 1.10. With the notations of the previous theorem, the ring $\text{End}_k(E)$ is commutative for E of characteristic ∞ and further its rank $\leq r$.

Remarque 1.11. For a place v of F we denote by $D_v(E) = \varinjlim_n E_{v^n}(\bar{k})$ and then the Tate module would be by definition $T_v(E) = \text{Hom}(F_v/A_v, D_v(E))$. Then $\text{End}(E) \otimes_A A_v \subset \text{End } T_v(E) = \text{End } D_v(E)$, and the cokernel is without torsion as an A_v -module.

1.5 Drinfel'd Modules over a Base Scheme

Recall that locally free sheaf of rank 1 and invertible sheaf are the same notions, for which we use also the term line bundle.

Définition 1.7. Let S be a scheme in characteristic p . A *Drinfel'd module over S of rank r* is an invertible sheaf $\mathcal{L}$ and a morphism of rings $\phi : A \rightarrow \text{End}(\mathcal{L})$ such that locally over open sets where $\mathcal{L}$ is trivial

$$\phi_a(x) = \sum_{i=0}^m a_i x^{p^i}$$

where $p^m = |a|_\infty^r$ and a_m is a unit. We say that ϕ_a is *strictly of degree p^m* .

In order to analyze the condition that a_m is a unit, we use the following lemma.

Lemme 1.4. Let R be a ring with $p = 0$ and $\phi, \psi, f \in R\{\tau\}$ where $f\phi = \psi f$, $\phi = \sum_{i=0}^M a_i \tau^i$ with a_M a unit, $\psi = \sum_{j=0}^N b_j \tau^j$ with b_N a unit, and $f = \sum_{k=0}^L c_k \tau^k$. Then the leading coefficient $c_L^{p^M}$ is either a unit or zero, and so f is either zero or strictly of degree $\leq M \cdot L$.

Proof. Comparing the coefficients of τ^{M+N} in the relation $\psi f = f\phi$, we deduce $c_L^{p^M} b_N = c_L a_M^{p^L}$ or $c_L^{p^M} (b_N c_L^{1-p^M} - a_M^{p^L}) = 0$. Since b_N is a unit, this can be written $c_L^{p^M} (c_L^{1-p^M} - a_N/b_N^{p^L}) = 0$.

Thus either $c_L = 0$ or c_L a unit with $c_L^{p^M-1} = a_M^{p^L}/b_N$. $\square$

The next lemma is a version of the Weierstrass preparation theorem.

Lemme 1.5. Let R be a ring with $p = 0$. If $\phi = \sum_{i=0}^N a_i \tau^i$ where a_N is a unit and $a_0, \dots, a_{N-1}$ are nilpotent, then after a change of coordinates with $u(\tau) = 1 + \sum_{i=1}^{N-1} c_i \tau^i$ such that $u\phi u^{-1}(\tau) = \sum_{i=0}^N a'_i \tau^i = \psi(\tau)$ with a'_N a unit. The coefficients $c_N, c_{N-1}, \dots, c_1$ are chosen by decreasing induction.

Remarque 1.12. An alternative form of the definition of a Drinfel'd module over a scheme S in characteristic p is to give E/S a group scheme locally isomorphic to $\mathbb{G}_a$ and a morphism of rings $\phi : A \rightarrow \text{End}_S(E)$ with degree $|a|_\infty^r$ locally over S . Using lemma (1.5), we can put an $\mathcal{O}_S$ -linear structure on E such that the action of some ϕ_a for $|a|_\infty > 1$ is locally given by a polynomial expression with highest coefficient a unit and of degree $|a|_\infty^r$. Using lemma (1.4) and the relation $\phi_a \phi_b = \phi_b \phi_a = \phi_{ab}$, it follows that all ϕ_b are given locally by polynomials of strict degree $|b|_\infty^r$. Questions of $\mathcal{O}_S$ -linearity need only be checked over closed subschemes $\text{Spec}(R) \subset S$ where R is a local Artin ring.

Analogue of Characteristic

The function $a \mapsto \partial_0 \phi_a$ defines a morphism $A \rightarrow \mathcal{O}_S$ of rings and hence a morphism of schemes $S \rightarrow \text{Spec}(A)$. For a closed point $s \in S$, the Drinfel'd module over the field $\mathbb{F}(s)$ has characteristic $s(s) \in \text{Spec}(A)$.

1.6 Level Structure and the Moduli Space

Let I be an ideal in A , let $V(I)$ denote the set of prime ideals of A containing I , and let E/S be a Drinfel'd module of rank r over S with characteristic morphism $\theta : S \rightarrow \text{Spec}(A)$. Then form the finite flat group scheme $E_I/S = \bigcap_{a \in I} \ker(E \xrightarrow{\phi_a} E)$ of rank equal to $\text{Card}(A/I)^r$. The scheme E_I is étale outside the characteristics which divide I , i.e. over $\theta^{-1}(\text{Spec}(A) \setminus V(I)) \subset S$.

In the most elementary sense an I -level structure on a Drinfel'd module E should be an isomorphism $\alpha : (I^{-1}/A)^r \xrightarrow{\sim} E_I$ over S , and in fact, this definition works very well away from characteristics dividing I . In order to deal smoothly with characteristics dividing I , we are led to the following definition of Drinfel'd which has also an analogue for elliptic curves with level structures.

Définition 1.8. An I -level structure on a Drinfel'd module E/S of rank r is an A -linear morphism $\alpha : (I^{-1}/A)^r \rightarrow E_I$ such that for all i in I^{-1}/A the corresponding sections $\alpha(i)$ of E_I have the property that as divisors on E

$$\sum_{i \in (I^{-1}/A)^r} (\alpha(i)) = E_I \times_S \cdots \times_S E_I.$$

Locally E is isomorphic to $\mathbb{G}_a$, and in this case E_I is defined as the kernel of a polynomial map $P : \mathbb{G}_a \rightarrow \mathbb{G}_a$ where $P_I(x) = \prod_{i \in I^{-1}/A} (x - \alpha(i))$.

Let $\mathcal{F}_I$ denote the contravariant functor from schemes to sets which assigns to a scheme S the set of isomorphism classes of Drinfel'd modules over S of rank r with I -level structure.

Théorème 1.3. Let I be an ideal in A with $\text{Card } V(I) > 1$. The functor $\mathcal{F}_I$ is representable by an affine scheme of finite type over A .

Proof. For $x \in V(I)$ it suffices to show that the functor restricted to the category of schemes over $\text{Spec}(A) \setminus \{x\}$ is representable. Then for a scheme S over $\text{Spec}(A) \setminus \{x\}$, and a Drinfel'd module E over S with an I -level structure, the choice of nonzero elements for the I -level structure gives a trivialization of E . Over these local pieces E is given by coordinates of ϕ_a for each $a \in A$ and elements $\alpha(i)$ for $i \in (I^{-1}/A)^r$ subject to the relations:

- (a) $\phi_a \phi_b = \phi_b \phi_a$,
- (b) The leading coefficient of ϕ_a is invertible,
- (c) $\phi_a(\alpha(i)) = 0$,
- (d) $P(X) = \prod_{i \in I^{-1}/A} (X - \alpha(i))$.

All of these relations are affine in nature and can be represented by an affine scheme. $\square$

The book of Katz–Mazur [1985] carries the construction of moduli spaces of elliptic curves using Drinfel'd's definition of level structure.

2 Analytic Theory of the Affine Modules of Drinfel'd

The group of points of an elliptic curve over the complex numbers is of the form $\mathbb{C}/L$ where L is a lattice over $\mathbb{Z}$ (so of rank 2) in $\mathbb{C}$. There is a similar assertion for Drinfel'd modules of rank r . The group of C_∞ -valued points is of the form C_∞/L where L is an A -lattice (discrete A -module) of rank r in C_∞^r .

The description of Drinfel'd modules in terms of lattices gives a calculation of the moduli space of Drinfel'd modules over C_∞ in terms of a quotient $GL(r, A) \backslash \text{Hom}_{F_\infty}(F_\infty^r, C_\infty)$. This is similar to the quotient $GL(2, \mathbb{Z}) \backslash \text{Hom}_{\mathbb{R}}(\mathbb{C}, \mathbb{C})$ which classifies elliptic curves plus a differential form. In the last section an adélic description of the points of this moduli space is given.

2.1 Exponential Function Associated to a Lattice

Let X be a subset of a complete nonarchimedean valued field K such that $0 \in X$ and $B(a, r) \cap X$ is finite for any ball $B(a, r) \subset K$ around a of radius $r > 0$. Then the Euler product

$$e_X(t) = t \prod_{Y \in X \setminus \{0\}} \left(1 - \frac{t}{Y}\right)$$

defines an entire function $e_X : K \rightarrow K$ with zeros on X , because the hypothesis implies $\lim_{Y \in X \setminus \{0\}} |t/Y| \rightarrow 0$ from which we deduce that $e_X(t)$ converges uniformly on any ball.

By changing f and t by scalar factors, we can assume that the power series expansion of an entire function has the form $f(t) = \sum_{n \geq 0} a_n t^n$ where $|a_i| \leq 1$, $|a_i| < 1$ for $i < r$, and $|a_r| = 1$; when f is nonconstant, also $r > 0$. By solving congruences modulo powers of the maximal ideal of F , we can factor

$$f(t) = (t^r + c_1 t^{r-1} + \cdots + c_r)(b_0 + b_1 t + \cdots).$$

If, in addition, K is algebraically closed, then it follows that every nonconstant entire function has at least one zero. Thus the Euler product $e_X(t)$ is the unique entire function, up to a constant factor, with one simple zero at each point of X and no other zeros.

Définition 2.1. A subgroup L of the additive group of a complete valued field K is called a *lattice* provided the intersection $L \cap B(a, r)$ is finite for any ball $B(a, r)$.

When K is of characteristic p , then $p \cdot 1 = 0$ and the torsion group L is a limit of finite subgroups H , i.e. $L = \varinjlim_{H \text{ finite} \subset L} H$. If the above Euler expansion

$$e_L(t) = t \prod_{Y \in L \setminus \{0\}} \left(1 - \frac{t}{Y}\right)$$

is additive, i.e. $e_L(x+y) = e_L(x) + e_L(y)$ for any $x, y \in K$, then L is clearly a lattice and $e_L(t)$ is its exponential function.

Proposition 2.1. *Let L be a lattice in a complete valued field K of characteristic p . Then $e_L(x+y) = e_L(x) + e_L(y)$ for any $x, y \in K$.*

Proof. Since $L = \varinjlim_{H \text{ finite} \subset L} H$, it follows that uniformly on any ball

$$e_L(t) = \varinjlim_{H \text{ finite} \subset L} e_H(t).$$

Each $e_H(t)$ is additive since it equals $c \prod_{h \in H} (t - h)$ which is additive by (2.1). Thus $e_L(x+y) = e_L(x) + e_L(y)$ since it is a limit of functions $e_H(t)$ satisfying this property. $\square$

Remarque 2.1. If L is a lattice in a complete valued field K , and if $\lambda \in K^*$, then λL is a lattice also and the exponential function $e_L(t)$ of L determines the exponential function of λL by $e_{\lambda L}(t) = \lambda e_L(\lambda^{-1}t)$. The lattice λL is called the *dilation* of L by $\lambda \in K^*$. Two lattices L and L' are in the same dilation class provided $L' = \lambda L$ for $\lambda \in K^*$. We always normalize the exponential function so that $e_L(t) = t + (\text{higher order terms})$.

2.2 Characterization of Drinfel'd Modules over C_∞

Lattices $\Gamma \subset C_\infty^r$, such that $a\Gamma \subset \Gamma$ for $a \in A \subset C_\infty$ with the induced scalar action by $a \in A$ are also A -modules, which we call A -lattices. A dilation $\lambda\Gamma$ of an A -lattice Γ is again an A -lattice for $\lambda \in C_\infty^*$, and we can speak of dilation classes of A -lattices.

Given an A -lattice Γ in C_∞ , we form the ring morphism $\phi_\Gamma : A \rightarrow \text{End}_{C_\infty}(\mathbb{G}_a)$ where $\phi_{\Gamma,a}$ for $a \in A$ is defined by the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma & \longrightarrow & C_\infty & \xrightarrow{e_\Gamma} & C_\infty \longrightarrow 0 \\ & & \downarrow a & & \downarrow a & & \downarrow \phi_{\Gamma,a} \\ 0 & \longrightarrow & \Gamma & \longrightarrow & C_\infty & \xrightarrow{e_\Gamma} & C_\infty \longrightarrow 0 \end{array}$$

For $\lambda\Gamma$, the relation $\phi_{\lambda\Gamma}(e_{\lambda\Gamma}(t)) = e_{\lambda\Gamma}(at)$ becomes in view of (2.1) simply

$$\begin{aligned} \phi_{\lambda\Gamma}(\lambda e_\Gamma(\lambda^{-1}t)) &= \lambda e_\Gamma(a\lambda^{-1}t) \\ \text{or } \lambda^{-1}\phi_{\lambda\Gamma}(\lambda e_\Gamma(t)) &= e_\Gamma(at). \end{aligned}$$

This implies that Γ and $\lambda\Gamma$ give rise to ϕ_a and $\lambda^{-1}\phi_a\lambda = \phi_a$ for $\lambda \in C_\infty^*$.

Théorème 2.1. *The function, which assigns to each A -lattice Γ , the ring morphism $\phi_\Gamma : A \rightarrow \text{End}_{C_\infty}(\mathbb{G}_a)$ defined above, is a bijection from the set of A -lattice dilation classes in C_∞ determined by Γ which are projective of rank r onto the set of isomorphism classes of affine A -modules ϕ 's over C_∞ of rank r with $\partial\phi_a = a$.*

Proof. First we calculate $\ker(\phi_\Gamma)$. This is isomorphic to $\ker(\phi_\Gamma e_\Gamma)/\Gamma = \ker(e_\Gamma \cdot a)/\Gamma$, and $\ker(e_\Gamma \cdot a)/\Gamma$ is isomorphic to $a^{-1}\ker(e_\Gamma)/\Gamma = a^{-1}\Gamma/\Gamma$. Hence $\ker(\phi_\Gamma)$ is isomorphic to $(A/a)^r$ since Γ is projective of rank r , and moreover $\text{Card}(\ker \phi_a) = \text{Card}(A/a)^r$. Thus $\Gamma \mapsto \phi_\Gamma$ is a well defined function from dilation classes of A -lattices projective of rank r to isomorphism classes of affine A -modules over C_∞ of rank r . The property $\partial\phi_a = a$ follows from $e_\Gamma(t) = t + (\text{higher terms})$, and $\phi_{\lambda\Gamma} = \phi_\Gamma$ from the calculation preceding the statement of the theorem.

To show $\Gamma \mapsto \phi_\Gamma$ is a bijection, we consider an affine module $\phi : A \rightarrow \text{End}_{C_\infty}(\mathbb{G}_a)$ of rank r and construct an additive entire function $e : C_\infty \rightarrow C_\infty$ such that $\phi_a(e(x)) = e(ax)$ and $e(t) = t + (\text{higher terms})$. Then the corresponding A -lattice Γ is the set of $x \in C_\infty$ with $e(x) = 0$.

The first step is to show that there is a unique formal solution $e(T) \in k[[T]]$ such that $\phi_a(e(x)) = e(ax)$ for a given $a \in A$ with $\phi_a(T) = a_0 + a_1\tau + \dots$ and $a = a_0 \neq 0, 1$. The condition $\phi_a(e(x)) = e(ax)$ is equivalent to $\phi_a(e(a^{-1}x)) = e(x) = \sum_{n \geq 0} e_n x^{p^n}$. Equating coefficients of x , we derive from the relation

$$\phi_a(e(a^{-1}x)) = \sum_i a_i \left(\sum_j e_j^{p^i} a^{-p^{i+j}} x^{p^{i+j}} \right) = \sum_n e_n x^{p^n}$$

the formula

$$e_n = \frac{\sum_{i+j=n, i \geq 1} a_i e_j^{p^i}}{a^{p^n} - a},$$

and hence the inductive definition for the coefficients e_n starting with $e_0 = 1$:

$$e_n = \frac{\sum_{i=1}^n a_i e_{n-i}^{p^i}}{a^{p^n} - a}.$$

This shows both the existence and uniqueness of $e(T)$.

The second step is to show that for this $e(T)$, the relation $\phi_b(e(x)) = e(bx)$ holds for any $b \in A$. Consider

$$(\phi_b e b^{-1})(x) = \phi_b(\phi_a e a^{-1})b^{-1}(x) = (\phi_a(\phi_b e b^{-1})a^{-1})(x).$$

From the expression it follows that e and $\phi_b e b^{-1}$ both satisfy the relation $\phi_a e a^{-1} = e$, and from the uniqueness assertion in step one, it follows that $\phi_b e = e b$ for any $b \in A$.

Next we have to show that $e(x) = \sum_{n \geq 0} e_n x^{p^n}$ is an entire function. Assume in the inductive definition of $e(T)$ that $|a| > 1$ and write the above recurrence formula for e_n as

$$e_n(a^{p^n} - a) = \sum_{i=1}^n a_i e_{n-i}^{p^i} \quad \text{for } n \geq 1.$$

Then

$$|a| |e_n|^{p^n} \leq \max_{1 \leq i \leq n} \{|a_i|^{p^{n-i}} r_{n-i}\}$$

where $r_j = |e_j|^{p^j}$. For $a \neq 1$, such that $1/|a| < \delta < 1$ and any $n \geq$ fixed n_0 , each term $|a_i|^{p^{n-i}} < 1 + \delta$, and it follows that

$$r_n \leq \delta \cdot \max_{1 \leq i \leq n} r_{n-i}.$$

Since

$$r_n \leq \frac{\delta}{1 - \delta} \max_{1 \leq i \leq n} r_{n-i},$$

we see that $r_n = |e_n|^{p^n} \rightarrow 0$, which proves that $e(x)$ is an entire function.

Finally, we calculate ϕ_a for this $e(x)$ where Γ equals the subset of $x \in C_\infty$ with $e(x) = 0$. We have

$$\phi_a(x) = a \cdot x \prod_{\gamma \in a^{-1}\Gamma/\Gamma \setminus \{0\}} \left(1 - \frac{x}{e_\Gamma(\gamma)}\right)$$

since $\phi_a(x) = 0$ on $a^{-1}\Gamma/\Gamma$. Thus the degree of ϕ_a equals $\text{Card}(a^{-1}\Gamma/\Gamma) = \text{Card}(A/a)^r = (\text{Card}(A/a))^r$. This proves the theorem. $\square$

For our future considerations I -level structures will play a basic role where I is a nonzero ideal in A . This leads to the following definition.

Définition 2.2. Let Y be a projective A -module of rank r . For a nonzero ideal $I \subset A$ an I -level structure is an isomorphism $\alpha : Y/IY \rightarrow (A/I)^r$ of A -modules (or free A/I -modules). An isomorphism $Y/IY \xrightarrow{\sim} (A/I)^r$ is equivalent to an isomorphism $I^{-1}Y/Y \xrightarrow{\sim} (I^{-1}/A)^r$ since $v = I^{-1} \otimes_A u$.

In the case $Y = \Gamma$ an A -lattice of rank r and the affine A -module $E = (\phi_\Gamma : A \rightarrow \text{End}_{C_\infty}(\mathbb{G}_a))$ corresponding to Γ by (2.1), we calculated $\ker(\phi_\Gamma) = a^{-1}\Gamma/\Gamma$ and thus the subfunctor $E_I = \bigcap_{a \in I} \ker(\phi_a) \subset E$ has the form

$$E_I(\bar{k}) = \bigcap_{a \in I} a^{-1}\Gamma/\Gamma = I^{-1}\Gamma/\Gamma.$$

Thus we immediately deduce the following assertion.

Proposition 2.2. *Using $E_I(\bar{k}) = I^{-1}\Gamma/\Gamma$, we have that an I -level structure on the affine A -module $E = \phi_\Gamma$ is the same as an I -level structure on the projective A -module Γ .*

2.3 Discrete Modules in a Vector Space over a Local Field

The following preliminaries are needed in the next section in order to parametrize A -lattices in C_∞ as homogeneous spaces.

Définition 2.3. Let A be a discrete subring of a local field K with field of fractions $F \subset K$. We assume that K/A is a compact abelian group.

Exemple 2.1. (1) $A = \mathbb{Z} \subset F = \mathbb{Q} \subset K = \mathbb{R}$.

(2) $A = \mathbb{Z} + \mathbb{Z}\omega \subset F = \mathbb{Q}(\sqrt{-d}) \subset K = \mathbb{C}$ for $d > 0$, $d \in \mathbb{Z}$ square free.

(3) $A = \mathbb{F}_q[t] \subset F = \mathbb{F}_q(t) = \mathbb{F}_q(t^{-1}) \subset K = \mathbb{F}_q((t^{-1}))$.

(4) $A = \mathbb{F}_q[C \setminus \infty] \subset F = \mathbb{F}_q(C - \infty) = \mathbb{F}_q(C) \subset K = F_\infty$ where C is an affine curve and F_∞ is the completion of F at ∞ .

Of course (3) is a special case of (4), and (4) is the case of interest in this part.

For any finite dimensional vector space V over K , the topology is well defined and given by a norm. A subgroup H in V is discrete provided there exists a neighbourhood N' of 0 in V with $N' \cap H = 0$. If $N + N \subset N'$, then $N + x \cap N + y = \emptyset$ for $x, y \in H$ if and only if $x \neq y$.

For an A -module H , the rank of H , denoted ${}_A H$, is $\dim_K(K \otimes_A H)$.

Proposition 2.3. *Let H be a discrete A -module contained in a finite dimensional K -vector space V . Then ${}_A(H) \leq \dim_K(V)$.*

Proof. Let $x_1, \dots, x_n$ denote a set of elements H forming a basis of the vector space $H \cdot K = K \cdot H \subset V$. Then $L = Ax_1 + \dots + Ax_n \subset H \subset V$. Since H is discrete in V , there exists a neighbourhood N' of $0 \in V$ with $H \cap N' = 0$ and a neighbourhood N with $N + N \subset N'$. Now $N + L$ is a neighbourhood of 0 in V/L intersecting H/L only at 0. Thus H/L is a discrete subgroup of V/L , and of the compact abelian group

$$V/L = Kx_1 \oplus \dots \oplus Kx_n / Ax_1 \oplus \dots \oplus Ax_n.$$

Thus H/L is finite and $n = \dim_K(K \otimes_A L) \leq \dim_K(K \otimes_A H) = \dim_K(V)$ (which in turn implies ${}_A(H) = \dim_F(F \otimes_A H) \leq \dim_K(V)$). This proves the proposition. $\square$

Proposition 2.4. *Let H be a projective A -submodule contained in a finite dimensional K -vector space V . Then H is discrete in V if and only if $\delta : K \otimes_A H \rightarrow V$ is injective.*

Proof. Assume H is discrete in V . Then $H \subset \text{im}(\delta) \subset V$ is discrete and so ${}_A(H) \leq \dim_K(\text{im}(\delta))$ by (2.3). On the other hand, for any H , we have ${}_A(H) = \dim_K(K \otimes_A H) \geq \dim_K(\text{im}(\delta))$. Hence $\dim_K(K \otimes_A H) = \dim_K(\text{im}(\delta))$ and so δ is injective.

Conversely, assume that δ is injective. There exists H' with $H \oplus H'$ free, and a free module $A^n \subset K^n$ is a discrete A -submodule. Hence H is discrete in $K \otimes_A H$, and since δ is injective, H is discrete in V . This proves the proposition. $\square$

Remarque 2.2. Let C be the completion of the algebraic closure of K . Then for every projective A -module P with ${}_A(P) \leq [C : K]$ there exists an A -monomorphism $P \rightarrow C$ such that P is discrete in C , i.e. by (2.4), the induced $K \otimes_K P \rightarrow C$ is injective.

In example (1), $2 = [C : K]$ and free modules P of rank ≤ 2 over $\mathbb{Z}$ can be embedded as discrete subgroups of $C = \mathbb{C}$. In examples (3) and (4), $[C_\infty : F_\infty] = \infty$ and all projective modules embed as A -lattices of C_∞ .

2.4 Moduli Spaces as Homogeneous Spaces (Local Theory)

Now we return to the notations of the article where $F = \mathbb{F}_q(C)$, $A = \mathbb{F}_q[C - \infty]$, F_∞ completion of F at ∞ , and C_∞ the completion of the algebraic closure $\overline{F}_\infty$ of F_∞ . Let K denote F_∞ a local field as referred to in §2.3 where $A \subset K$ is discrete and K/A compact.

We denote points in projective space $\mathbb{P}_{r-1}(C_\infty)$ by their homogeneous coordinates $y_1 : \dots : y_r$.

We begin by parametrizing F -vector subspaces or equivalently free A -lattices in C_∞ .

Proposition 2.5. *The function*

$$f : C_\infty \setminus (\text{Hom}_{F_\infty}(F_\infty^r, C_\infty) - \{0\}) \longrightarrow \mathbb{P}_{r-1}(C_\infty)$$

given by $f(u) = u(e_1) : \dots : u(e_r)$ where $e_i = (0, \dots, 1, \dots, 0)$ is a bijection which restricts to a bijection

$$C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty^r, C_\infty) \xrightarrow{\sim} \mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes}).$$

Proof. A nonzero element u of $\text{Hom}_{F_\infty}(F_\infty^r, C_\infty)$ is determined by its values $(u(e_1), \dots, u(e_r))$ on the canonical basis elements $e_i \in F_\infty^r$. A dilation of u yields a dilation of the r -tuple $(u(e_1), \dots, u(e_r)) = (x_1, \dots, x_r)$. Next, $(a_1, \dots, a_r) \in \ker(u) \subset F_\infty^r$ if and only if $\sum_{i=1}^r a_i x_i = 0$, i.e. for $(a_1, \dots, a_r) \neq 0$, $(x_1, \dots, x_r)$ is on the F_∞ -rational hyperplane with equation $a_1 x_1 + \dots + a_r x_r = 0$. This proves the proposition. $\square$

For an A -module Y projective of rank r , we have embeddings and isomorphism

$$Y \hookrightarrow F_\infty \otimes_A Y \cong F_\infty^r \quad \text{and} \quad GL_A(Y) \hookrightarrow GL_{F_\infty}(F_\infty \otimes_A Y) \cong GL(r, F_\infty)$$

coming from the tensor product. By taking the image of Y or $F_\infty \otimes_A Y$, and using (2.4), the following horizontal arrows are well defined functions.

Proposition 2.6. *Let Y be a projective module of rank r over A . The following diagram is commutative and the horizontal arrows are bijections:*

$$\begin{array}{ccc}
\left\{ \begin{array}{l} A\text{-sublattices of} \\ C_\infty \text{ isomorphic to } Y \end{array} \right\} & \xrightarrow{\sim} & \text{Hom}_{F_\infty}(F_\infty \otimes_A Y, C_\infty)/GL_A(Y) \\
\downarrow & & \downarrow \\
\left\{ \begin{array}{l} A\text{-sublattices of} \\ C_\infty \text{ isomorphic to } Y \\ \text{modulo dilations in } C_\infty^* \end{array} \right\} & \xrightarrow{\sim} & C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty \otimes_A Y, C_\infty)/GL_A(Y) \\
\uparrow & & \uparrow \\
\left\{ \begin{array}{l} r\text{-dimensional } F\text{-vector} \\ \text{subspaces of } C_\infty \end{array} \right\} & \xrightarrow{\sim} & \text{Hom}_{F_\infty}(F_\infty^r, C_\infty)/GL(r, F_\infty) \\
\uparrow & & \uparrow \\
\left\{ \begin{array}{l} r\text{-dimensional } F\text{-vector} \\ \text{subspaces of } C_\infty \\ \text{modulo dilations in } C_\infty^* \end{array} \right\} & \xrightarrow{\sim} & C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty^r, C_\infty)/GL(r, F_\infty)
\end{array}$$

Now putting (2.5) and (2.6) together along with (2.1), we obtain the following assertion. Let P_A^r denote the set of isomorphism classes of projective A -modules of rank r .

Proposition 2.7. *Using the previous notations, we have the following bijections:*

$$\Omega^r(C_\infty) = \mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})$$

descends to

$$\begin{aligned}
& \bigsqcup_{Y \in P_A^r} (\mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes}))/GL_A(Y) \\
& \sim \\
& \left\{ \begin{array}{l} A\text{-sublattices of } C_\infty \text{ of rank } r \\ \text{modulo dilations in } C_\infty^* \end{array} \right\} \\
& \sim \\
& \bigsqcup_{Y \in P_A^r} C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty \otimes_A Y, C_\infty)/GL_A(Y) \\
& \sim \\
& \left\{ \begin{array}{l} \text{Isomorphism classes of affine } A\text{-modules} \\ \phi \text{ over } C_\infty \text{ of rank } r \text{ with } \partial\phi_a = a \end{array} \right\}
\end{aligned}$$

and taking disjoint unions over $Y \in P_A^r$.

Remarque 2.3. In the case $r = 1$, $\mathbb{P}_0(C_\infty) - (F_\infty\text{-rational hyperplanes})$ is a point, and (2.7) is a statement about bijections between sets with one point. We will be interested particularly in the case $r = 2$ in latter parts. If $a_1x_1 + a_2x_2 = 0$ where $a_1, a_2 \in F_\infty$, then $x_1 : x_2 = -a_2 : a_1 \in \mathbb{P}^1(F_\infty)$. Thus we have $\Omega^2(C_\infty) = \mathbb{P}_1(C_\infty) - (F_\infty\text{-rational lines}) = \mathbb{P}_1(C_\infty) - \mathbb{P}_1(F_\infty)$.

The next step is to modify (2.7) to include the case of A -lattices with I -level structure. For a projective A -module Y of rank r the projection $Y \rightarrow Y/IY$ induces a group morphism $GL_A(Y) \rightarrow GL_A(Y/IY)$ which has kernel $GL_A(Y, I)$.

In the following two statements we have a restatement of (2.6) and a modification taking into account I -level structures. Fix an I -level structure $\alpha_0 : Y/IY \xrightarrow{\sim} (A/I)^r$.

- (1) Each monomorphism $u : F_\infty \otimes_A Y \rightarrow C_\infty$ determines a lattice $u(Y) \subset C_\infty$ and an I -level structure $u_*((u|_Y)^{-1} \bmod I) \circ \alpha$ on $u(Y)$ where $u|_Y : Y \rightarrow u(Y) \subset C_\infty$. Moreover, all A -sublattices of C_∞ , isomorphic to Y together with an I -level structure, come by this construction.
- (2) Two monomorphisms $u, u' : F_\infty \otimes_A Y \rightarrow C_\infty$ determine the same lattice in C_∞ if and only if there exists $h \in GL_A(Y)$ with $u' = uh$. They determine the same lattice in C_∞ with I -level structure if and only if there exists $h \in GL_A(Y, I)$ with $u' = uh$.

This leads to the following modification of the previous proposition.

Théorème 2.2. *Let I be a nonzero ideal in A , and use the previous notations. We have the following bijections:*

$$\begin{aligned}
& \bigsqcup_{Y \in P_A^r} (\mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})) / GL_A(Y, I) \\
& \sim \\
& \left\{ \begin{array}{c} \alpha : I^{-1}Y/Y \rightarrow (I^{-1}/A)^r \\ \text{modulo Aut of } Y \end{array} \right\} \\
& \sim \\
& \left\{ \begin{array}{c} A\text{-sublattices of } C_\infty \text{ of rank } r \text{ with an } I\text{-level} \\ \text{structure modulo dilations in } C_\infty^* \end{array} \right\} \\
& \sim \\
& \bigsqcup_{Y \in P_A^r} C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty \otimes_A Y, C_\infty) / GL_A(Y, I) \\
& \sim \\
& \left\{ \begin{array}{c} \alpha : I^{-1}Y \rightarrow (I^{-1}/A)^r \\ \text{modulo Aut of } Y \end{array} \right\} \\
& \sim \\
& \left\{ \begin{array}{c} \text{Isomorphism classes of affine } A\text{-modules} \\ \phi_a \text{ over } C_\infty \text{ with } I\text{-level structure and of rank } r \\ \text{with } \partial\phi_a = a \end{array} \right\}
\end{aligned}$$

or more briefly the bijection

$$\bigsqcup_{(\alpha, Y) \in P_A^r(I)} \Omega^r(C_\infty) / GL_A(Y, I) \xrightarrow{\sim} \mathcal{M}_I^r(C_\infty).$$

Here $\Omega^r(C_\infty) = \mathbb{P}_{r-1} - \mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})$, also $\mathcal{M}_I^r$ is the moduli functor for affine A -modules rank r with I -level structure, and $P_A^r(I)$ is the set of ordered pairs (α, Y) with $Y \in P_A^r$ and $\alpha : I^{-1}Y/Y \rightarrow (I^{-1}/A)^r$ is an isomorphism.

2.5 Moduli Spaces as Homogeneous Spaces (Adélic Theory)

The basic bijection describing $\mathcal{M}_I^r(C_\infty)$ in (2.2) involves only the local field at ∞ . In this section we will describe $\mathcal{M}_I^r(C_\infty)$ using all the primes of F together with $\mathcal{M}_I^r(C_\infty)$. This adélic description is closely related to the adélic description of vector bundles on a curve which we recall now.

Remarks on Vector Bundles

Over any ringed space X , $(\mathcal{O}_X)$ a vector bundle is a locally free sheaf on X locally isomorphic to $\mathcal{O}_X^n$. If X is a nonempty open set of the curve C over $\mathbb{F}_q$, for example $X = C - \{\infty\}$, then a vector bundle of rank n can be described as a family $(V, L_v)_{v \in X}$ where V is an n -dimensional vector space over $F = \mathbb{F}_q(C)$ and each L_v is a free module of rank n contained in V over the local ring $\mathcal{O}_{(v)}$ at v . We assume that there exists a basis $x_1, \dots, x_n$ of V over F and a finite set $S \subset X$ with $L_v = \mathcal{O}_{(v)}x_1 + \dots + \mathcal{O}_{(v)}x_n$ for all $v \in X - S$. The basis $x_1, \dots, x_n$ is called a trivialization of V on $X - S$.

If $y_1, \dots, y_n$ is any other basis of V , then $y_j = \sum a_{ij}x_i$ and $x_i = \sum b_{ji}y_j$ where $(a_{ij}, b_{ji}) \in GL(n, \mathcal{O}(v))$ for all but a finite number of v . Hence every basis of V is a trivialization of V over some $X - T$ where $T \subset X$ is finite.

To an $\mathcal{O}_X$ -sheaf E , locally isomorphic to $\mathcal{O}_X^n$, we assign $V = E_\eta$ where η is the generic point of C . The $\mathcal{O}(v)$ -submodule $L_v \subset V$ is defined by trivializing $E|_{(X-S)}$ on an open set $X - S$ around v with $x_1, \dots, x_n \in V$ and requiring for all $w \in X - S$ with S finite that $L_w = \mathcal{O}(w)x_1 + \dots + \mathcal{O}(w)x_n$.

These lattices which came up in the above description of a vector bundle can be identified with certain homogeneous spaces. Let F_v denote the completion of F at $v \in X$ with valuation ring $\mathcal{O}_v$. Then using matrix transforms between bases of a lattice in a vector space F_v^n or F^n , we have the following commutative diagram of bijections:

$$\begin{array}{ccc} GL(n, F)/GL(n, \mathcal{O}(v)) & \xrightarrow{\sim} & \{\mathcal{O}(v)\text{-lattices in } F^n\} \\ \downarrow & & \downarrow \\ GL(n, F_v)/GL(n, \mathcal{O}_v) & \xrightarrow{\sim} & \{\mathcal{O}_v\text{-lattices in } F_v^n\} \end{array}$$

Homogeneous Space Description of Trivialized Vector Bundles

A vector bundle E with a trivialization is described by (F^n, L_v) for $v \in X$ where $L_v = \mathcal{O}(v)^n \subset F^n$ for all $v \in X - S$ where S is finite. Thus the vector bundle trivialized on $X - S$ is determined by a finite set of lattices $L_v \subset F^n$ for $v \in S$. This leads to the following bijections:

$$\begin{array}{ccc} \left\{ \begin{array}{l} \text{n-dim. vector bundles on } X \\ \text{trivialized over } X - S \\ \text{all up to isomorphism} \end{array} \right\} & \xrightarrow{\sim} & \prod_{v \in S} GL(n, F_v)/GL(n, \mathcal{O}_v) \\ & & \downarrow \\ & & \prod_{v \in S} GL(n, F_v) \times \prod_{v \in X-S} GL(n, \mathcal{O}_v) / \prod_{v \in X} GL(n, \mathcal{O}_v) \end{array}$$

where the vertical arrow is a homeomorphism of locally compact spaces.

At points of S we can consider a Λ -level structure on the vector bundle E relative to a positive divisor Λ supported on S . This is just an isomorphism of the sheaf $E|_\Lambda(-\Lambda) =$

$E \otimes_{\mathcal{O}_X} (\mathcal{O}/\mathcal{O}(-\Lambda)) \rightarrow (\mathcal{O}/\mathcal{O}(-\Lambda))^n$. In terms of the description of E with the data (F^n, L_v) this is an isomorphism $L_v/\mathfrak{p}_v^{m_v} L_v \xrightarrow{\sim} (\mathcal{O}_v/\mathfrak{p}_v^{m_v})^n$ where $\mathfrak{p}_v^{m_v}$ is the ideal of the uniformizing parameter of $\mathcal{O}(v)$ and hence $\mathcal{O}_v$. If we denote by

$$GL(n, \mathcal{O}_v, \Lambda_v) = \ker(GL(n, \mathcal{O}_v) \rightarrow GL(n, \mathcal{O}_v/\mathfrak{p}_v^{m_v}))$$

and $K(n, \Lambda) = \prod_{v \in X} GL(n, \mathcal{O}_v, \Lambda_v)$, then the previous diagram becomes modified as follows for $(\Lambda) \subset S$:

$$\left\{ \begin{array}{l} \text{n-dim. vector bundles} \\ \text{on } X \text{ with } \Lambda\text{-level} \\ \text{structure, trivialized} \\ \text{over } X - S, \text{ up to iso.} \end{array} \right\} \xrightarrow{\sim} \prod_{v \in S} GL(n, F_v)/GL(n, \mathcal{O}_v, \Lambda_v) \downarrow$$

$$S(X; \vec{\Lambda}) = \prod_{v \in S} GL(n, F_v) \times \prod_{v \in X-S} GL(n, \mathcal{O}_v)/K(n, \Lambda)$$

Vector Bundles on C

The product in the numerator of $S(\vec{X})$ or $S(\vec{X}; \Lambda)$ is one of the terms in the inductive limit which defines the adèle group $GL(n, \mathbb{A}_F)$ where

$$GL(n, \mathbb{A}_F) = \varinjlim_S \prod_{v \in S} GL(n, F_v) \times \prod_{v \in C-S} GL(n, \mathcal{O}_v).$$

Hence considering vector bundles with Λ -level structure and trivialization over some open set $C - S$ of C where $(\Lambda) \subset S$, we obtain the bijection

$$\left\{ \begin{array}{l} \text{vector bundles of dimension } n \\ \text{with a } \Lambda\text{-level structure and} \\ \text{trivialization off the } (\Lambda) \\ \text{all up to isomorphism} \end{array} \right\} \xrightarrow{\sim} GL(n, \mathbb{A}_F)/K(n, \Lambda)$$

Since the various trivializations are all related by the action of $GL(n, F)$ on basis, we have the following quotient of the previous bijection:

$$\left\{ \begin{array}{l} \text{isomorphism classes of} \\ n\text{-dimensional vector bundles} \\ \text{on } C \text{ with } \Lambda\text{-level structure} \end{array} \right\} \xleftarrow{\sim} GL(n, F) \backslash GL(n, \mathbb{A}_F)/K(n, \Lambda)$$

Vector Bundles on $C - \{\infty\}$

The product in the numerator of $S(\vec{X})$ or $S(\vec{X}; \Lambda)$ for $X = C - \{\infty\}$ is one of the terms in the limit which defines the finite adèle group

$$GL(n, \mathbb{A}_F^f) \quad \text{where} \quad \mathbb{A}_F^f = \hat{A} \otimes_A F$$

and $\hat{A} = \varprojlim_J A/J$, the limit being taken over ideals $J \subset A$.

A divisor Λ can be described as an ideal $I \subset A$, and the group $K(n, \Lambda)$ is given by

$$K(n, I) = \ker(GL(n, \hat{A}) \rightarrow GL(n, \hat{A}/I\hat{A}))$$

which we also denote by $GL(n, \hat{A}, I)$ to avoid confusion with the previous case discussed above.

As before we obtain in the injective limit the bijection

$$\left\{ \begin{array}{l} \text{vector bundles of dimension } n \\ \text{on } C - \{\infty\} \text{ with a } \Lambda\text{-level structure} \\ \text{and trivialization off the } (\Lambda) \\ \text{all up to isomorphism} \end{array} \right\} \xrightarrow{\sim} GL(n, \mathbb{A}_F^f)/K(n, I)$$

On the affine $C - \{\infty\}$ with coordinate ring $A = H^0(C - \{\infty\}, \mathcal{O})$ a vector bundle of dimension n is just a projective module Y of rank n over A .

Now we combine the previous bijection with

$$C_\infty \setminus \text{Hom}_{F_\infty}(F_\infty^r, C_\infty) = \Omega^r(C_\infty) = \mathbb{P}_{r-1}(C_\infty) - \bigcup (F_\infty\text{-rational hyperplanes}).$$

We have a bijection

$$\left\{ \begin{array}{l} \text{vector bundles } Y \text{ on } C - \infty \\ \text{of rank } r \text{ with a} \\ \text{trivialization and a } \Lambda\text{-level} \\ \text{structure all up} \\ \text{to isomorphism} \end{array} \right\} \xleftarrow{\sim} (GL(r, \mathbb{A}_F^f)/GL(r, \hat{A}, I)) \times \Omega^r(C_\infty)$$

The trivialization of Y is a basis of $Y \otimes_A F$ over F which defines an embedding $Y \rightarrow F_\infty^r$ as a discrete A -submodule. When this embedding is composed with a monomorphism $F_\infty^r \rightarrow C_\infty$ yields an A -sublattice of C_∞ of rank r with an I -level structure modulo dilations in C_∞^* . Hence by (2.2) we have a map

$$(GL(r, \mathbb{A}_F^f)/GL(r, \hat{A}, I)) \times \Omega^r(C_\infty) \rightarrow \mathcal{M}_I^r(C_\infty)$$

where the fibres correspond to the various bases of $Y \otimes_A F$ over F . Hence factoring out by the action of $GL(r, F)$ yields the adélic description of $\mathcal{M}_I^r(C_\infty)$.

Théorème 2.3. *The above map induces a bijection*

$$GL(r, F) \backslash GL(r, \mathbb{A}_F^f) \times \Omega^r(C_\infty) / GL(r, \hat{A}, I) \xrightarrow{\sim} \mathcal{M}_I^r(C_\infty).$$

Remarque 2.4. The quotient space descriptions in (2.3) and the local version generalize to any open compact subgroup $H \subset GL(r, \mathbb{A}_F^f)$ to give the C_∞ -valued points $\mathcal{M}_H^r(C_\infty)$ of a moduli scheme $\mathcal{M}_H^r$:

$$\mathcal{M}_H^r(C_\infty) = GL(r, F) \backslash GL(r, \mathbb{A}_F^f) \times \Omega^r(C_\infty) / H$$

adélic version

$$= \bigsqcup_{x \in GL(r, \hat{A})} \Omega^r(C_\infty) / (xHx^{-1} \cap GL(r, F))$$

local version.

The moduli scheme $\mathcal{M}_H^r$ arises from E -level structures on Drinfel'd modules.

3 The Rigid Analytic Moduli Space

Now our aim is to describe the rigid analytic structure on the C_∞ -valued points of the moduli space. These points were parametrized by a quotient of $\Omega^r(C_\infty)$ by a discrete group in (2.2). The rigid analytic structure on the moduli space is a quotient of the rigid analytic structure on $\Omega^r(C_\infty)$.

In order to define the rigid analytic structure on $\Omega^r(C_\infty)$, we make use of a natural mapping

$$\lambda : \Omega^r(C_\infty) \longrightarrow |\mathcal{I}(F_\infty^r)|_{\mathbb{Q}}$$

onto the rational points $|\mathcal{I}(F_\infty^r)|_{\mathbb{Q}}$ of the geometric realization $|\mathcal{I}(F_\infty^r)|$ of the building $\mathcal{I}(F_\infty^r)$ of the group $PGL(r)$ over the local field F_∞ . The admissible open sets of the rigid analytic structure are inverse images by λ^{-1} of certain open neighborhoods of skeletons in $|\mathcal{I}(F_\infty^r)|$.

In this discussion we indicate how both $\Omega^r(C_\infty)$ and the building $\mathcal{I}(F_\infty^r)$ are p -adic analogues of real symmetric spaces.

3.1 Norms on Vector Spaces over a Local Field

Définition 3.1. Let K be a local field with valuation ring R , maximal ideal $\mathfrak{p} = R\pi$, and $q = \text{Card}(k)$ where $k = R/\mathfrak{p}$. We normalize $|a|$ on K as $|a| = q^{-\text{ord}(a)}$ so that $|\pi| = q^{-1}$ and $|\pi^n| = q^{-n}$. We consider finite dimensional vector spaces over K , and thus the closed unit ball and unit sphere are compact.

If V is of dimension m over K , then a *lattice* is any R -submodule M free of rank m .

Définition 3.2. A *norm* on a vector space V over K is a function $\alpha : V \rightarrow \mathbb{R}$ satisfying:

- (a) $\alpha(x) \geq 0$ and $\alpha(x) = 0$ if and only if $x = 0$.
- (b) $\alpha(ax) = |a|\alpha(x)$ for $a \in K$ and $x \in V$.
- (c) $\alpha(x + y) \leq \sup(\alpha(x), \alpha(y))$ for $x, y \in V$.

A norm is called *integral* provided $\alpha(V) = |K| = \{0\} \cup q^{\mathbb{Z}}$ and *rational* provided $\alpha(V) \subset \mathbb{Q}$.

If $\alpha(x) < \alpha(y)$, then axiom (c) can be strengthened to the equality $\alpha(x + y) = \alpha(y) = \sup(\alpha(x), \alpha(y))$.

If α is a norm and $t > 0$, then the dilation $t\alpha$ is a norm, and we denote by $\mathcal{N}(V)$ the set of dilation classes of norms on V .

Exemple 3.1. For a basis $x_0, \dots, x_m$ of V and real numbers $r_0 > 0, \dots, r_m > 0$ the function

$$\alpha(a_0x_0 + \dots + a_mx_m) = \sup(r_0|a_0|, \dots, r_m|a_m|)$$

is a norm on V . In fact, every norm can be described by this formula. This is proved by induction on m by considering a nonzero linear form $\lambda : V \rightarrow F$ and choosing $x_0 \neq 0$ in V such that $x \mapsto |\lambda(x)|/\alpha(x)$ defined on the compact projective space $\mathbb{P}(V)$ takes its maximum at x_0 . Then $V \cap Fx_0 \oplus \ker(\lambda)$ for which on $\ker(\lambda)$ the inductive hypothesis applies. See Goldman and Iwahori for the details.

Exemple 3.2. The norm α_M associated to a lattice M in V is given by the formula

$$\alpha_M(x) = \inf\{1/|a| : ax \in M\}.$$

V/π^{2m} -module $E_{\pi^{2m}}(\bar{k})$ to prove that $E_{\pi^m}(\bar{k})$ is a free A/π^m -module of rank d where

$$\text{Card}(E_{\pi^m}(\bar{k})) = \deg(\phi_{\pi^m}) = \text{Card}(A/\pi^m)^r.$$

From this we deduce that r is an integer and so ϕ is a Drinfel'd module of rank r .

Next, for any nonzero $a \in A$ the primary components of $E_a(\bar{k})$ are of the form $E_{\pi^m}(\bar{k})$, so free A/π^m -module of rank r , from which we deduce that $E_a(\bar{k})$ is a free A/a -module of rank r . Finally, if I is an ideal of A prime to the characteristic of ϕ , then there exists another ideal J in A with $A = I + J$ and $IJ = (e)$, a principal ideal prime to the characteristic. Then $A/a = A/I \oplus A/J$, and $E_a = E_I \oplus E_J$ as functors. Since $E_a(\bar{k})$ is free of rank r over A/a , it follows that $E_I(\bar{k})$ is free of rank r over A/I , and this proves the theorem.

Remarque 0.1. It remains only to consider for a Drinfel'd module E of rank r and characteristic $v \neq \infty$, the A/π^n -modules $E_{\pi^n}(\bar{k})$ where $v(\pi) = 1$.

In this case $\partial_0 \phi_{\pi^i} = 0$ for all i , and by the discussion in (1.4), the endomorphism ϕ_π has a height h where $\deg(\phi_\pi) = p^h \cdot \text{Card } E_\pi(\bar{k})$, or more generally

$$\deg(\phi_{\pi^n}) = (\deg(\phi_\pi))^n = (p^h \cdot \text{Card } E_\pi(\bar{k}))^n = p^{nh} \cdot \text{Card } E_{\pi^n}(\bar{k}).$$

Hence again by applying lemma (3.2), we see that $E_{\pi^n}(\bar{k})$ is a free module of rank $r - h < r$ over A/π^n . As finite group schemes over k , we have a splitting when k is separable

$$E_{\pi^i} = E_{\pi^i}^0 \times E_{\pi^i}^{\text{ét}}$$

where $E_{\pi^i}^{\text{ét}}$ is étale of rank $r - h$ and $E_{\pi^i}^0$ is infinitesimal of rank h .

Définition 0.1. The height of a Drinfel'd module E of characteristic $v \neq \infty$ is the height of ϕ_π where π is an irreducible with $v(\pi) = 1$.

Since (π) is uniquely determined by v , the height is well defined.

Remarque 0.2. Let E be a Drinfel'd module of rank r and characteristic v . Let h be the height when $v = \infty$. For an irreducible π of A , let A_π be A localized at π . Then $\varinjlim_n E_{\pi^n}(\bar{k})$ is a divisible A_π -module which is isomorphic to

$$\begin{cases} (F_\pi/A_\pi)^r & \text{for } \pi \text{ prime to the characteristic,} \\ (F_\pi/A_\pi)^{r-h} & \text{for } v(\pi) = 1. \end{cases}$$

1 Isogenies

Recall that for a Drinfel'd A -module (ϕ, E) over k and any field extension k^* of k , the group $E(k^*)$ has a given A -module structure. For two Drinfel'd A -modules over k and $u \in k\{\tau\}$ with $u \neq 0$, the additive $u: \phi \rightarrow \phi'$ is an isogeny if and only if the additive $u: E(\bar{k}) \rightarrow E'(\bar{k})$ is A -linear. This follows from the fact that two polynomials are equal if and only if their associated polynomial maps $\bar{k} \rightarrow \bar{k}$ are equal. The kernel of $u: E(k^*) \rightarrow E'(k^*)$ is the A -module $\ker(u)(k^*)$ where $\ker(u)$ is the scheme kernel of $u: \phi \rightarrow \phi'$.

4.1 Separable isogenies

An isogeny $u: \phi \rightarrow \phi'$ is separable when $u(X) = b_0X + b_1X^p + \cdots + b_sX^{p^s}$ where $b_0 \neq 0$. Then u is determined by $\ker(u)(\bar{k})$, a finite A -submodule of $E(\bar{k})$, up to constant factor. For a finite A -submodule $H \subset E(\bar{k})$, we form $P_H(X) = \prod_{h \in H}(X - h)$, and there is a unique Drinfel'd A -module ϕ' such that $\phi'_a P_H = P_H \phi_a$ for all $a \in A$.

If $a \cdot \ker(u)(\bar{k}) = 0$ for $a \neq 0$, then we can form the isogeny v with kernel $u(\ker(\phi_a)(\bar{k}))$ and $vu = \phi_a$. Note that such a nonzero $a \in A$ with $a \cdot \ker(u)(\bar{k}) = 0$ always exists for $u \neq 0$ since it is a finite A -module. When u is separable, we can always choose v to be separable too. By the same construction, if $u: \phi \rightarrow \phi'$ and $w: \phi \rightarrow \phi''$ are separable isogenies such that $w(\ker(u)(\bar{k})) = 0$, then there is a separable isogeny $v: \phi' \rightarrow \phi''$ with $w = vu$ where $\ker(v)(\bar{k}) = u(\ker(w)(\bar{k}))$.

4.2 Purely inseparable isogenies

These are of the form $\tau^i \in k\{\tau\}$ or $X \mapsto X^{p^i}$. Since $\phi_a \tau^i = \tau^i \phi_a$ in $k\{\tau\}$ is equivalent to all coefficients of ϕ_a being in $\mathbb{F}_{p^i}$, it follows that a purely inseparable isogeny exists only when the characteristic v of ϕ is unequal to ∞ , and in this case p^i is a power of q_v where q_v is the cardinality of the residue class field of v . The finite group scheme $\ker(\tau^i) = \text{Spec}(k[t]/(t^{p^i}))$. This case corresponds to a purely local kernel.

4.3 Remarks

Let H be a finite subgroup scheme of $\mathbb{G}_a$, and let ϕ be a Drinfel'd module structure on $\mathbb{G}_a$. Then H is the kernel of some isogeny $u: \phi \rightarrow \phi'$ if and only if H is stable under the action of A and

$$\ker(u) = \begin{cases} 0 & \text{if the characteristic} = \infty \\ \text{Spec}(k[t]/(t^{q_v^h})) & \text{if the characteristic} = v \text{ and } q_v^h = \text{Card}(F_v). \end{cases}$$

For any isogeny $u: \phi \rightarrow \phi'$ there exists an isogeny $v: \phi' \rightarrow \phi$ and $a \in A$ with $\phi_a = vu$.

Let $\delta(u)$ denote the degree of the additive polynomial $u \in \text{End}_k(\mathbb{G}_a)$. The function $\delta: \text{End}_k(\mathbb{G}_a) \rightarrow \mathbb{Z}$ also prolongs under extension of scalars to

$$\begin{array}{ccc} \text{End}_k(\mathbb{G}_a) & \longrightarrow & \text{End}_{\bar{k}}(\mathbb{G}_a) \\ \downarrow & & \downarrow \cong \\ \text{End}_k(E) \otimes_A F & \longrightarrow & \text{End}_{\bar{k}}(E) \otimes_A F \end{array}$$

4.4 Remarks

We study the A -algebra $\text{End}_{\text{kalg}}(E)$ with δ using the following properties of $\delta: \text{End}_{\text{kalg}}(E) \otimes_{\mathbb{F}_q} F_\infty \rightarrow \mathbb{Q}$:

- (1) $\delta(u) \geq 0$ and $\delta(u) = 0$ if and only if $u = 0$.
- (2) $\delta(au) = |a|\delta(u)$ for $a \in F$.
- (3) $\delta(u + v) \leq \max(\delta(u), \delta(v))$.
- (4) $\delta(uv) = \delta(v)\delta(u)$.

Moreover, $\text{End}(E) \subset \text{End}_{\text{kalg}}(E) \otimes_A F_\infty$ will be seen to be a discrete A -module in this normed vector space over F_∞ , see (4.9)(2). The main step will be to show that $\text{End}(E)$ is a finitely generated A -module. For this we use the following two lemmas.

Lemme 1.1. *Let $A^n \subset X \subset F_\infty^n$ where X is a discrete A -module. Then X is finitely generated.*

Démonstration. For all i observe that $X + \pi^{-i}A^n$ has finite index, e.g. $k[[t]]^n + t^{-i}k[[t]]^n \subset k((t))^n$ has index $\text{Card}(k)^{ni}$. There exists an i with $X \cap \pi^{-i}A^n = 0$ since X is discrete, and thus X/A^n embeds in $(F_\infty/\pi^{-i}A)^n$ which is finite. Since $(X : A^n)$ is finite and A^n is finitely generated, it follows that X is a finitely generated A -module. $\square$

Proposition 1.1. *For a finite dimensional subspace V over F of $\text{End}(E) \otimes_A F$, it follows that $V \cap \text{End}(E)$ is a finitely generated A -module which is projective.*

Démonstration. For $V_\infty = F_\infty \cdot V$, it follows that $X = \text{End}(E) \cap V = \text{End}(E) \cap V_\infty$, and we can assume that V_∞ generates V and so V_∞ or replace V by a smaller subspace.

Let $x_1, \dots, x_n$ be a basis of V_∞ with $x_i \in X$. Then with this basis $A^n \subset X \subset F_\infty^n = V_\infty$, and we are reduced to the previous lemma since $\text{End}(E)$ is a discrete subspace of $\text{End}(E) \otimes_A F_\infty$. To see that $X = V \cap \text{End}(E)$ is projective, we have only to remark that it is flat since it is torsion free over a Dedekind domain and finitely generated flat modules are projective over a Noetherian ring. $\square$

Corollaire 1.1. *The A -module $\text{End}(E)$ is projective.*

Démonstration. Let $W = \text{End}(E) \otimes_A F$ and $W = \bigoplus_i W_i$ where $[W_i : F]$ is finite. Then $X_i = \text{End}(E) \cap W_i$ is projective by (4.6) and the restrictions of the projections $W \rightarrow W_i$ to $X_i = \text{End}(E) \cap W_i$ define a morphism $f: \text{End}(E) \rightarrow \bigoplus_i X_i$ onto a projective module with $\ker(f) = 0$. Hence $\text{End}(E)$ is projective. $\square$

For estimates on the rank we use the following lemma.

Lemme 1.2. *Let $a \in A$ be prime to the characteristic of E . Then $\text{End}(E) \otimes_A A/a \rightarrow \text{End}(E_a)$ is injective.*

Démonstration. If $v \in \text{End}(E)$ and $v(\ker(\phi_a)) = 0$, then as in (4.1), it follows that $v = v'\phi_a$, and $v|_{E_a} = 0$ implies $v \in \text{End}(E)a$. $\square$

Now we summarize all the basic results in the following theorem and remarks.

Théorème 1.1. *Let E be a Drinfel'd module over an algebraically closed field $\bar{k}$ of rank r . Then*

- (1) $\text{End}(E)$ is a projective A -module of rank $\leq r^2$, and
- (2) $\text{End}_{\text{kalg}}(E) \otimes_A F_\infty$ is a field in which $\text{End}(E)$ embeds as a discrete A -module of this normed space over F_∞ .

Démonstration. (1) The fact that $\text{End}(E)$ is projective is contained in (4.7) and the injectivity of $\text{End}(E) \otimes_A A/a \rightarrow \text{End}(E_a)$ coming from (4.8) bounds the rank by r^2 since E_a is an A/a -module of rank r for a prime to the characteristic.

(2) The existence of δ on $\text{End}_{\text{kalg}}(E) \otimes_A F$ proves that it is a field, and since F_∞/F is a separable extension $\text{End}_{\text{kalg}}(E) \otimes_A F_\infty$ is also a field. $\square$

The subspace where $\delta = 0$ on $\text{End}_{\text{kalg}}(E) \otimes_A F_\infty$ is zero since $\dim_F \text{End}(E) \otimes_A F = \text{rank}(\text{End}_{\text{kalg}}(E))$.

Remarque 1.1. With the notations of the previous theorem, the ring $\text{End}_{\text{kalg}}(E)$ is commutative for E of characteristic ∞ and further its rank $\leq r$.

Remarque 1.2. For a place v of F we denote by $D_v(E) = \bigcup_n E_{v^n}(\bar{k})$ and then the Tate module would be by definition

$$T_v(E) = \text{Hom}(F_v/A_v, D_v(E)).$$

Then $\text{End}(E) \otimes_A A_v \subset \text{End } T_v(E) = \text{End } D_v(E)$, and the cokernel is without torsion as an A_v -module.

2 Drinfel'd modules over a base scheme

Recall that locally free sheaf of rank 1 and invertible sheaf are the same notions, for which we use also the term line bundle.

Définition 2.1. Let S be a scheme in characteristic p . A Drinfel'd module over S of rank r is an invertible sheaf L and a morphism of rings $\phi: A \rightarrow \text{End}(L)$ such that locally over open sets where L is trivial, $\phi_a(x) = \sum_{i=0}^m a_i x^{p^i}$ where $p^m = |a|_\infty^r$ and a_m is a unit. We say that ϕ_a is strictly of degree p^m .

In order to analyze the condition that a_m is a unit, we use the following lemma.

Lemme 2.1. Let R be a ring with $p = 0$ and $\phi, \psi, f \in R\{\tau\}$ where $\phi = \sum a_i \tau^i$ with a_N a unit, $\psi = \sum b_i \tau^i$ with b_N a unit, and $f\phi = \psi f$, $\phi = f\psi$. If $f = \sum_{i=0}^M c_i \tau^i$. Then the leading coefficient $c_M^{p^N}$ is either a unit or zero, and so f is either zero or strictly of degree $\leq M$.

Démonstration. Comparing the coefficients of τ^{M+N} in the relation $\psi f = f\phi$, we deduce

$$c_M^{p^N} = c_M a_N^{p^M} \quad \text{or} \quad c_M^{p^N} (c_M^{p^N-1} - a_N b_N^{-1})^{p^M} = 0.$$

Since b_N is a unit, this can be written

$$c_M (c_M^{p^N-1} - a_N b_N^{-1})^{p^M} = 0.$$

Thus either $c_M = 0$ or c_M is a unit with $c_M^{p^N-1} = a_N/b_N$. □

The next lemma is a version of the Weierstrass preparation theorem.

Lemme 2.2. Let R be a ring with $p = 0$. If $\phi = \sum_{i=0}^N a_i \tau^i$ where a_N is a unit and $a_0, \dots, a_{N-1}$ are nilpotent, then after a change of coordinates with $u(T) = 1 + \sum_{i=1}^\infty c_i T^i$ such that

$$c\phi c^{-1}(u(T)) = \sum_{i=0}^N b_i \tau^i \cdot (1 - w\tau)$$

with b_N a unit.

The coefficients $c_N, c_{N-1}, \dots, c_1, c_0$ are chosen by decreasing induction.

Remarque 2.1. An alternative form of the definition of a Drinfel'd module over a scheme S in characteristic p is to give E/S a group scheme locally isomorphic to $\mathbb{G}_a$ and a morphism of rings $\phi: A \rightarrow \text{End}_S(E)$ with $\deg(\phi_a) = |a|_\infty^r$ locally over S .

Using lemma (5.3), we can put an $\mathcal{O}_S$ -linear structure on E such that the action of some ϕ_a for $|a|_\infty > 1$ is locally given by a polynomial expression with highest coefficient a unit and of degree $|a|_\infty^r$. Using lemma (5.2) and the relation $\phi_a\phi_b = \phi_b\phi_a$, it follows that all ϕ_b are given locally by polynomials of strict degree $|b|_\infty^r$. Questions of $\mathcal{O}_S$ -linearity need only be checked over closed subschemes $\text{Spec}(R) \subset S$ where R is a local Artin ring.

5.5 Analogue of characteristic

The function $a \mapsto \partial_0\phi_a$ defines a morphism $A \rightarrow \mathcal{O}_S$ of rings and hence a morphism of schemes $S \rightarrow \text{Spec}(A)$. For a closed point $s \in S$, the Drinfel'd module L_s over the field $F(s)$ has characteristic $s(s) \in \text{Spec}(A)$.

3 Level structure and the moduli space

Let I be an ideal in A , let $V(I)$ denote the set of prime ideals of A containing I , and let E/S be a Drinfel'd module of rank r over S with characteristic morphism $\theta: S \rightarrow \text{Spec}(A)$. Then form the finite flat group scheme $E_I/S = \bigcap_{a \in I} \ker_a(E \rightarrow E)$ of rank equal to $\text{Card}(A/I)^r$. The scheme E_I is étale outside the characteristics which divide I , i.e. over $\theta^{-1}(\text{Spec}(A) - V(I)) \subset S$.

In the most elementary sense an I -level structure on a Drinfel'd module E should be an isomorphism $\alpha: (I^{-1}/A)^r \rightarrow E_I$ over S , and in fact, this definition works very well away from characteristics dividing I . In order to deal smoothly with characteristics dividing I , we are led to the following definition of Drinfel'd which has also an analogue for elliptic curves with level structures.

Définition 3.1. An I -level structure on a Drinfel'd module E/S of rank r is an A -linear morphism

$$\alpha: (I^{-1}/A)^r \longrightarrow E_I$$

such that for all i in I^{-1}/A the corresponding sections $\alpha(i)$ of E_I have the property that as divisors on E

$$\sum_{i \in (I^{-1}/A)^r} (\alpha(i)) = (E_I).$$

Locally E is isomorphic to $\mathbb{G}_a$, and in this case E_I is defined as the kernel of a polynomial map $P: \mathbb{G}_a \rightarrow \mathbb{G}_a$ where

$$P(X) = \prod_{i \in (I^{-1}/A)^r} (X - \alpha(i)).$$

Let F_I^r denote the contravariant functor from schemes to sets which assigns to a scheme S the set of isomorphism classes of Drinfel'd modules over S of rank r with I -level structure.

Théorème 3.1. *Let I be an ideal in A with $\text{Card } V(I) \geq 1$. The functor F_I^r is representable by an affine scheme of finite type over A .*

Démonstration. For $x \in V(I)$ it suffices to show that the functor restricted to the category of schemes over $\text{Spec}(A) - \{x\}$ is representable. Then for a scheme S over $\text{Spec}(A) - \{x\}$, and a Drinfel'd module E over S with an I -level structure, the choice of nonzero elements for the I -level structure gives a trivialization of E . Over these local pieces E is given by coordinates of ϕ_a for each $a \in A$ and elements $\alpha(i)$ for $i \in (I^{-1}/A)^r$ subject to the relations :

- (a) $\phi_a \phi_b = \phi_b \phi_a$.
- (b) The leading coefficient of ϕ_a is invertible.
- (c) $\phi_a(\alpha(i)) = 0$.
- (d) $P(X) = \prod_{i \in (I^{-1}/A)^r} (X - \alpha(i))$.

All of these relations are affine in nature and can be represented by an affine scheme. $\square$

The book of Katz–Mazur [1985] carries the construction of moduli spaces of elliptic curves using Drinfel'd's definition of level structure.

Analytic theory of the affine modules of Drinfel'd

The group of points of an elliptic curve over the complex numbers is of the form $\mathbb{C}/L$ where L is a lattice over $\mathbb{Z}$ (so of rank 2) in $\mathbb{C}$. There is a similar assertion for Drinfel'd modules of rank r . The group of $\mathbb{C}_\infty$ -valued points is of the form $\mathbb{C}_\infty/L$ where L is an A -lattice (discrete A -module) of rank r in $\mathbb{C}_\infty$.

The description of Drinfel'd modules in terms of lattices gives a calculation of the moduli space of Drinfel'd modules over $\mathbb{C}_\infty$ in terms of a quotient $\text{GL}(r, A) \backslash \text{Hom}_{F_\infty}(F_\infty^r, \mathbb{C}_\infty)$. This is similar to the quotient $\text{GL}(2, \mathbb{Z}) \backslash \text{Hom}_{\mathbb{R}}(\mathbb{R}^2, \mathbb{C})$ which classifies elliptic curves plus a differential form. In the last section an adélic description of the points of this moduli space is given.

4 Exponential function associated to a lattice

Let X be a subset of a complete nonarchimedean value field K such that $0 \in X$ and $B(a, r) \cap X$ is finite for any ball $B(a, r) \subset K$ around a of radius $r > 0$. Then the Euler product

$$e_X(t) = t \prod_{Y \in X - \{0\}} \left(1 - \frac{t}{Y}\right)$$

defines an entire function $e_X: K \rightarrow K$ with zeros on X , because the hypothesis implies $\lim_{Y \in X - \{0\}} t/Y = 0$ from which we deduce that $e_X(t)$ converges uniformly on any ball.

By changing t by scalar factors, we can assume that the power series expansion of an entire function has the form $f(t) = \sum_{n \geq 0} a_n t^n$ where $|a_i| \leq 1$, $|a_i| < 1$ for $i < r$, and $|a_r| = 1$; when f is nonconstant, also $r > 0$. By solving congruences modulo powers of the maximal ideal of F , we can factor

$$f(t) = (t^r + c_1 t^{r-1} + \cdots + c_r)(b_0 + b_1 t + \cdots).$$

If, in addition, K is algebraically closed, then it follows that every nonconstant entire function has at least one zero. Thus the Euler product $e_X(t)$ is the unique entire function, up to a constant factor, with one simple zero at each point of X and no other zeros.

Définition 4.1. A subgroup L of the additive group of a complete value field K is called a lattice provided the intersection $L \cap B(a, r)$ is finite for any ball $B(a, r)$.

When K is of characteristic p , then $pL = 0$ and the torsion group L is a limit of finite subgroups H , i.e.

$$L = \varinjlim_{H \text{ finite} \subset L} H.$$

If the above Euler expansion $e_X(t) = t \prod_{Y \in X - \{0\}} (1 - t/Y)$ is additive, i.e. $e_X(x+y) = e_X(x) + e_X(y)$ for any $x, y \in K$, then X is clearly a lattice and $e_X(t)$ is its exponential function.

Proposition 4.1. *Let L be a lattice in a complete value field K of characteristic p . Then $e_L(x+y) = e_L(x) + e_L(y)$ for any $x, y \in K$.*

Démonstration. Since $L = \varinjlim_{H \text{ finite} \subset L} H$, it follows that uniformly on any ball

$$e_L(t) = \varinjlim_{H \text{ finite} \subset L} e_H(t).$$

Each $e_H(t)$ is additive since it equals $c \cdot \prod_{h \in H} (t - h)$ which is additive by 1(1.3). Thus $e_L(x+y) = e_L(x) + e_L(y)$ since it is a limit of functions $e_H(t)$ satisfying this property. $\square$

Remarque 4.1. If L is a lattice in a complete value field K , and if $\lambda \in K^*$, then λL is a lattice also and the exponential function $e_L(t)$ of L determines the exponential function of λL by $e_{\lambda L}(t) = \lambda e_L(\lambda^{-1}t)$. The lattice λL is called the dilation of L by $\lambda \in K^*$.

Two lattices L and L' are in the same dilation class provided $L' = \lambda L$ for $\lambda \in K^*$. We always normalize the exponential function so that

$$e_L(t) = t + (\text{higher order terms}).$$

5 Characterization of Drinfel'd modules over $\mathbb{C}_\infty$

Lattices $\Gamma \subset \mathbb{C}_\infty$ such that $a\Gamma \subset \Gamma$ for $a \in A \subset \mathbb{C}_\infty$ with the induced scalar action by $a \in A$ are also A -modules, which we call A -lattices. A dilation $\lambda\Gamma$ of an A -lattice Γ is again an A -lattice for $\lambda \in \mathbb{C}_\infty^*$, and we can speak of dilation classes of A -lattices.

Given an A -lattice Γ in $\mathbb{C}_\infty$, we form the ring morphism $\phi_\Gamma: A \rightarrow \text{End}_{\mathbb{C}_\infty}(\mathbb{G}_a)$ where $\phi_{\Gamma,a}$ for $a \in A$ is defined by the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma & \longrightarrow & \mathbb{C}_\infty & \xrightarrow{e_\Gamma} & \mathbb{C}_\infty \longrightarrow 0 \\ & & \downarrow a & & \downarrow a & & \downarrow \phi_{\Gamma,a} \\ 0 & \longrightarrow & \Gamma & \longrightarrow & \mathbb{C}_\infty & \xrightarrow{e_\Gamma} & \mathbb{C}_\infty \longrightarrow 0 \end{array}$$

For $\lambda\Gamma$, the relation $\phi_{\lambda\Gamma}(e_{\lambda\Gamma}(t)) = e_{\lambda\Gamma}(at)$ becomes in view of (1.3) simply $\phi_{\lambda\Gamma}(\lambda e_\Gamma(\lambda^{-1}t)) = \lambda e_\Gamma(a\lambda^{-1}t)$, i.e. $\lambda^{-1}\phi_{\lambda\Gamma}(\lambda e_\Gamma(t)) = e_\Gamma(at)$. This implies that $\phi_{\lambda\Gamma} = \phi_\Gamma$ for $\lambda \in \mathbb{C}_\infty^*$.

Théorème 5.1. *The function, which assigns to each A -lattice Γ , the ring morphism $\phi_\Gamma: A \rightarrow \text{End}_{\mathbb{C}_\infty}(\mathbb{G}_a)$ defined above, is a bijection from the set of A -lattice dilation classes in $\mathbb{C}_\infty$ determined by Γ which are projective of rank r onto the set of isomorphism classes of affine A -modules ϕ_Γ over $\mathbb{C}_\infty$ of rank r with $\partial\phi_{\Gamma,a} = a$.*

Démonstration. First we calculate $\ker(\phi_{\Gamma,a})$. This is isomorphic to $\ker(\phi_{\Gamma,a}e_{\Gamma})/\Gamma = \ker(e_{\Gamma} \cdot a)/\Gamma$, and $\ker(e_{\Gamma} \cdot a)/\Gamma$ is isomorphic to $a^{-1}\ker(e_{\Gamma})/\Gamma = a^{-1}\Gamma/\Gamma$. Hence $\ker(\phi_{\Gamma,a})$ is isomorphic to $(A/a)^r$ since Γ is projective of rank r , and moreover $\text{Card}(\ker \phi_{\Gamma,a}) = \text{Card}(A/a)^r$. Thus $\Gamma \mapsto \phi_{\Gamma}$ is a well defined function from dilation classes of A -lattices projective of rank r to isomorphism classes of affine A -modules over $\mathbb{C}_{\infty}$ of rank r . The property $\partial\phi_{\Gamma,a} = a$ follows from $e_{\Gamma}(t) = t + (\text{higher terms})$, and $|\phi_{\Gamma,a}| = |a|_{\infty}^r$ from the calculation preceding the statement of the theorem.

To show $\Gamma \mapsto \phi_{\Gamma}$ is a bijection, we consider an affine module $\phi: A \rightarrow \text{End}_{\mathbb{C}_{\infty}}(\mathbb{G}_a)$ of rank r and construct an additive entire function $e: \mathbb{C}_{\infty} \rightarrow \mathbb{C}_{\infty}$ such that $\phi_a(e(x)) = e(ax)$ and $e(t) = t + (\text{higher terms})$. Then the corresponding A -lattice Γ is the set of $x \in \mathbb{C}_{\infty}$ with $e(x) = 0$.

The first step is to show that there is a unique formal solution $e(T) \in k[[T]]$ such that $\phi_a(e(x)) = e(ax)$ for a given $a \in A$ with $\phi_a(T) = a_0T + a_1T^p + \dots + a_mT^{p^m}$ and $a \cdot \partial_0 \neq 0, 1$. The condition $\phi_a(e(x)) = e(ax)$ is equivalent to $\phi_a(e(a^{-1}x)) = e(x) = \sum e_i x^{p^i}$. Equating coefficients of x^n , we derive from the relation

$$e_n(a^{p^n} - a) = \sum_{i+j=n, i \geq 1} a_i e_j^{p^i}$$

the formula for the coefficients e_n starting with $e_0 = 1$:

$$e_n(1 - a^{p^n-1}) = \sum_{i=1}^n a_i e_{n-i}^{p^i}.$$

This shows both the existence and uniqueness of $e(t)$.

The second step is to show that for this $e(t)$, the relation $\phi_b(e(x)) = e(bx)$ holds for any $b \in A$. Consider

$$\phi_b(e(b^{-1}ax)) = \phi_b\phi_a e_a^{-1}(x) = (\phi_a(\phi_b e_b^{-1})a^{-1})(x).$$

From the expression it follows that e and $\phi_b e_b^{-1}$ both satisfy the relation $\phi_a e_a^{-1} = e$, and from the uniqueness assertion in step one, it follows that $\phi_b e = eb$ for any $b \in A$.

Next we have to show that $e(x) = \sum_{n \geq 0} e_n x^{p^n}$ is an entire function. Assume in the inductive definition of $e(t)$ that $|a| > 1$ and write the above recurrence formula for e_n as

$$e_n(a^{p^n} - a) = \sum_{i=1}^s a_i e_{n-i}^{p^i} \quad \text{for } n \geq s.$$

Then $|a||e_n|^{p^n} \leq \max_{1 \leq i \leq s} \{|a_i|^{p^n} r_{n-i}\}$ where $r_j = |e_j|^{p^j}$. For a such that $1/|a| < \theta < 1$ and any $n \geq$ fixed n_0 , each term $|a_i|^{p^n} < 1 + \theta$, and it follows that

$$r_n < \theta \cdot \max_{1 \leq i \leq s} r_{n-i}.$$

Since $r_n < \theta^{n-n_0} \cdot \max\{r_0, \dots, r_{n_0-1}\}/(1 - \theta)$, we see that $r_n = |e_n|^{p^n} \rightarrow 0$ which proves that $e(x)$ is an entire function.

Finally, we calculate ϕ_a for this $e(x)$ where Γ equals the subset of $x \in \mathbb{C}_{\infty}$ with $e(x) = 0$. We have

$$\phi_a(x) = ax \prod_{\gamma \in a^{-1}\Gamma/\Gamma - \{0\}} \left(1 - \frac{x}{e(\gamma)}\right)$$

since $\phi_a(x) = 0$ on $a^{-1}\Gamma/\Gamma$. Thus the degree of ϕ_a equals $\text{Card}(a^{-1}\Gamma/\Gamma) = \text{Card}(\Gamma/a\Gamma) = (\text{Card}(A/a))^r$. This proves the theorem. $\square$

For our future considerations I -level structures will play a basic role where I is a nonzero ideal in A . This leads to the following definition.

Définition 5.1. Let Y be a projective A -module of rank r . For a nonzero ideal $I \subset A$ an I -level structure is an isomorphism $\alpha: Y/IY \rightarrow (A/I)^r$ of A -modules (or free A/I -modules).

An isomorphism $Y/IY \rightarrow (A/I)^r$ is equivalent to an isomorphism $I^{-1}Y/Y \rightarrow (I^{-1}/A)^r$ since $Y = I^{-1} \otimes_A Y$. In the case $Y = \Gamma$ an A -lattice of rank r and the affine A -module $E = (\phi_\Gamma: A \rightarrow \text{End}_{\mathbb{C}_\infty}(\mathbb{G}_a))$ corresponding to Γ by (2.1), we calculated $\ker(\phi_{\Gamma,a}) = a^{-1}\Gamma/\Gamma$ and thus the subfunctor $E_I = \bigcap_{a \in I} \ker(\phi_{\Gamma,a}) \subset E$ has the form

$$E_I(\mathbb{C}_\infty) = \bigcap_{a \in I} a^{-1}\Gamma/\Gamma = I^{-1}\Gamma/\Gamma.$$

Thus we immediately deduce the following assertion.

Proposition 5.1. Using $E_I(\mathbb{C}_\infty) = I^{-1}\Gamma/\Gamma$, we have that an I -level structure on the affine A -module $E = \phi_\Gamma$ is the same as an I -level structure on the projective A -module Γ .

6 Discrete modules in a vector space over a local field

The following preliminaries are needed in the next section in order to parametrize A -lattices in $\mathbb{C}_\infty$ as homogeneous spaces.

Notations 6.1. Let A be a discrete subring of a local field K with field of fractions $F \subset K$. We assume that K/A is a compact abelian group.

Exemples 6.1. (1) $A = \mathbb{Z} \subset F = \mathbb{Q} \subset K = \mathbb{R}$.

(2) $A = \mathbb{Z} + \mathbb{Z}\omega \subset F = \mathbb{Q}(\omega) \subset K = \mathbb{C}$ for $d > 0$, $d \in \mathbb{Z}$ square free.

(3) $A = \mathbb{F}_q[t] \subset F = \mathbb{F}_q(t) = \mathbb{F}_q(t^{-1}) \subset K = \mathbb{F}_q((t^{-1}))$.

(4) $A = \mathbb{F}_q[C - \infty] \subset F = \mathbb{F}_q(C - \infty) = \mathbb{F}_q(C) \subset K = F_\infty$ where C is an affine curve and F_∞ is the completion of F at ∞ .

Of course (3) is a special case of (4), and (4) is the case of interest in this part.

For any finite dimensional vector space V over K , the topology is well defined and given by a norm. A subgroup H in V is discrete provided there exists a neighbourhood N' of 0 in V with $N' \cap H = \{0\}$. If $N + N \subset N'$, then $N + x \cap N + y = \emptyset$ for $x, y \in H$ if and only if $x \neq y$.

For an A -module H , the rank of H , denoted $\text{rank}_A H$, is $\dim_F(F \otimes_A H)$.

Proposition 6.1. Let H be a discrete A -module contained in a finite dimensional K -vector space V . Then

$$\text{rank}_A(H) \leq \dim_K(V).$$

Démonstration. Let $x_1, \dots, x_n$ denote a set of elements in H forming a basis of the vector space $H \cdot K \subset V$. Then

$$L = Ax_1 \oplus \dots \oplus Ax_n \subset H \subset V.$$

Since H is discrete in V , there exists a neighbourhood N' of $0 \in V$ with $H \cap N' = \{0\}$ and a neighbourhood N with $N + N \subset N'$. Now $N + L$ is a neighbourhood of 0 in V/L

intersecting H/L only at 0. Thus H/L is a discrete subgroup of V/L , and of the compact abelian group

$$V/L = Kx_1 \oplus \cdots \oplus Kx_n / Ax_1 \oplus \cdots \oplus Ax_n.$$

Thus H/L is finite and $n = \dim_K(F \otimes_A L) = \dim_F(F \otimes_A H) = \dim_K(V)$ which in turn implies $\text{rank}_A(H) = \dim_F(F \otimes_A H) \leq \dim_K(V)$. This proves the proposition. $\square$

Proposition 6.2. *Let H be a projective A -submodule contained in a finite dimensional K -vector space V . Then H is discrete in V if and only if $\delta: K \otimes_A H \rightarrow V$ is injective.*

Démonstration. Assume H is discrete in V . Then $H \subset \text{im}(\delta) \subset V$ is discrete and so $\text{rank}_A(H) \leq \dim_K(\text{im}(\delta))$ by (3.3). On the other hand, for any H , we have

$$\text{rank}_A(H) = \dim_K(K \otimes_A H) \geq \dim_K(\text{im}(\delta)).$$

Hence $\dim_K(K \otimes_A H) = \dim_K(\text{im}(\delta))$ and so δ is injective.

Conversely, assume that δ is injective. There exists H' with $H \oplus H'$ free, and a free module $A^n \subset K^n$ is a discrete A -submodule. Hence H is discrete in $K \otimes_A H$, and since δ is injective, H is discrete in V . This proves the proposition. $\square$

Remarque 6.1. Let $\mathbb{C}$ be the completion of the algebraic closure of K . Then for every projective A -module P with $\text{rank}_A(P) \leq [\mathbb{C} : K]$ there exists an A -monomorphism $P \rightarrow \mathbb{C}$.

such that P is discrete in $\mathbb{C}$, i.e. by (3.4), the induced $K \otimes_k P \rightarrow \mathbb{C}$ is injective.

In example (1), $2 = [\mathbb{C} : K]$ and free modules P of rank ≤ 2 over $\mathbb{Z}$ can be embedded as discrete subgroups of $\mathbb{C} = \mathbb{C}$. In examples (3) and (4), $[\mathbb{C}_\infty : F_\infty] = \infty$ and all projective modules embed as A -lattices of $\mathbb{C}_\infty$.

7 Moduli spaces as homogeneous spaces (local theory)

Now we return to the notations of the article where $F = \mathbb{F}_q(C)$, $A = \mathbb{F}_q[C - \infty]$, F_∞ completion of F at ∞ , and $\mathbb{C}_\infty$ the completion of the algebraic closure $\bar{F}_\infty$ of F_∞ . Let K denote F_∞ a local field as referred to in §3 where $A \subset K$ is discrete and K/A compact. We denote points in projective space $\mathbb{P}_{r-1}(\mathbb{C}_\infty)$ by their homogeneous coordinates $y_1 : \cdots : y_r$.

We begin by parametrizing F -vector subspaces or equivalently free A -lattices in $\mathbb{C}_\infty$.

Proposition 7.1. *The function $f: \mathbb{C}_\infty^r \setminus (\text{Hom}_{F_\infty}(F_\infty^r, \mathbb{C}_\infty) - \{0\}) \rightarrow \mathbb{P}_{r-1}(\mathbb{C}_\infty)$ given by $f(u) = u(e_1) : \cdots : u(e_r)$ where $e_i = (0, \dots, 1, \dots, 0)$ is a bijection which restricts to a bijection*

$$\mathbb{C}_\infty^r \setminus \text{Mon}_F(F \otimes_A Y, \mathbb{C}_\infty) \longrightarrow \mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes}).$$

Démonstration. A nonzero element u at $\text{Hom}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty)$ is determined by its values $(u(e_1), \dots, u(e_r))$ on the canonical basis elements $e_i \in F_\infty^r$. A dilation of u yields a dilation of the r -tuple $(u(e_1), \dots, u(e_r)) = (x_1, \dots, x_r)$. Next, $(a_1, \dots, a_r) \in \ker(u) \subset F_\infty^r$ if and only if $\sum_{i=1}^r a_i x_i = 0$, i.e. for $(a_1, \dots, a_r) \neq 0$, $(x_1, \dots, x_r)$ is on the F_∞ -rational hyperplane with equation

$$a_1 x_1 + \cdots + a_r x_r = 0.$$

This proves the proposition. $\square$

For an A -module Y projective of rank r , we have embeddings and isomorphisms $Y \rightarrow F_\infty \otimes_A Y \cong F_\infty^r$ and $\mathrm{GL}_A(Y) \rightarrow \mathrm{GL}_{F_\infty}(F_\infty \otimes_A Y) \cong \mathrm{GL}(r, F_\infty)$ coming from the tensor product. By taking the image of Y in $F_\infty \otimes_A Y$, and using (3.4), the following horizontal arrows are well defined functions.

Proposition 7.2. *Let Y be a projective module of rank r over A . The following diagram is commutative and the horizontal arrows are bijections :*

$$\begin{array}{ccc}
\mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y) & \longrightarrow & \left\{ \begin{array}{l} A\text{-sublattices of } \mathbb{C}_\infty \\ \text{isomorphic to } Y \end{array} \right\} \\
\downarrow & & \downarrow \\
\mathbb{C}_\infty^* \setminus \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y) & \longrightarrow & \left\{ \begin{array}{l} A\text{-sublattices of } \mathbb{C}_\infty \\ \text{isomorphic to } Y \\ \text{modulo dilations in } \mathbb{C}_\infty^* \end{array} \right\} \\
\\
\mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}(r, F) & \longrightarrow & \left\{ \begin{array}{l} r\text{-dimensional} \\ F\text{-vector subspaces} \\ \text{of } \mathbb{C}_\infty \end{array} \right\} \\
\downarrow & & \downarrow \\
\mathbb{C}_\infty^* \setminus \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}(r, F) & \longrightarrow & \left\{ \begin{array}{l} r\text{-dimensional } F\text{-vector} \\ \text{subspaces of } \mathbb{C}_\infty^* \\ \text{modulo dilations in } \mathbb{C}_\infty^* \end{array} \right\}
\end{array}$$

Now putting (4.1) and (4.2) together along with (2.1), we obtain the following assertion. Let $\mathcal{P}_A^r$ denote the set of isomorphism classes of projective A -modules of rank r .

Proposition 7.3. *Using the previous notations, we have the following bijections*

$$\begin{array}{ccc}
(\mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})) / \mathrm{GL}_A(Y) & & \\
\uparrow & & \\
\mathbb{C}_\infty^* \setminus \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y) & \longrightarrow & \left\{ \begin{array}{l} A\text{-sublattices of } \mathbb{C}_\infty \\ \text{isomorphic to } Y \\ \text{modulo dilations in } \mathbb{C}_\infty^* \end{array} \right\},
\end{array}$$

and taking disjoint unions

$$\begin{array}{ccc}
\bigcup_{Y \in \mathcal{P}_A^r} (\mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})) / \mathrm{GL}_A(Y) & & \\
\uparrow & & \\
\bigcup_{Y \in \mathcal{P}_A^r} \mathbb{C}_\infty^* \setminus \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y) & \longrightarrow & \left\{ \begin{array}{l} A\text{-sublattices of } \mathbb{C}_\infty \text{ of rank } r \\ \text{modulo dilations in } \mathbb{C}_\infty^* \end{array} \right\} \\
\downarrow & & \\
\left\{ \begin{array}{l} \text{isomorphism classes of affine} \\ A\text{-modules } \phi_a \text{ over } \mathbb{C}_\infty \text{ of} \\ \text{rank } r \text{ with } \partial \phi_a = a \end{array} \right\}.
\end{array}$$

Remarque 7.1. In the case $r = 1$, $\mathbb{P}_0(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})$ is a point, and (4.3) is a statement about bijections between sets with one point. We will be interested particularly in the case $r = 2$ in latter parts. If $a_1 x_1 + a_2 x_2 = 0$ where $a_1, a_2 \in F_\infty$, then $x_1 : x_2 = a_2 : a_1 \in \mathbb{P}^1(F_\infty)$. Thus we have

$$\mathbb{P}_1(\mathbb{C}_\infty) - (F_\infty\text{-rational lines}) = \mathbb{P}_1(\mathbb{C}_\infty) - \mathbb{P}_1(F_\infty).$$

The next step is to modify (4.3) to include the case of A -lattices with I -level structure. For a projective A -module Y of rank r the projection $Y \rightarrow Y/IY$ includes a group morphism $\mathrm{GL}_A(Y) \rightarrow \mathrm{GL}_A(Y/IY)$ which has kernel $\mathrm{GL}_A(Y, I)$. In the following two statements we have a restatement of (4.2) and a modification taking into account I -level structures. Fix an I -level structure $\alpha_0: I^{-1}Y/Y \rightarrow (I^{-1}/A)^r$.

- (1) Each monomorphism $u: F_\infty \otimes_A Y \rightarrow \mathbb{C}_\infty$ determines a lattice $u(Y) \subset \mathbb{C}_\infty$ and an I -level structure $\alpha = \alpha_0((u|_Y)^{-1} \bmod I)$ on $u(Y)$ where $u|_Y: Y \rightarrow u(Y) \subset \mathbb{C}_\infty$. Moreover, all A -sublattices of $\mathbb{C}_\infty$, isomorphic to Y together with an I -level structure, come by this construction.
- (2) Two monomorphisms $u, u': F_\infty \otimes_A Y \rightarrow \mathbb{C}_\infty$ determine the same lattice in $\mathbb{C}_\infty$ if and only if there exists $h \in \mathrm{GL}_A(Y)$ with $u' = uh$. They determine the same lattice in $\mathbb{C}_\infty$ with I -level structure if and only if there exists $h \in \mathrm{GL}_A(Y, I)$ with $u' = uh$.

This leads to the following modification of the previous proposition.

Théorème 7.1. *Let I be a nonzero ideal in A , and use the previous notations. We have the following bijections :*

$$\bigsqcup_{Y \in \mathcal{P}_A^r} (\mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})) / \mathrm{GL}_A(Y, I)$$

with I -level structure modulo Aut of Y , and

$$\bigsqcup_{Y \in \mathcal{P}_A^r} \mathbb{C}_\infty^* \setminus \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y, I) \longrightarrow \left\{ \begin{array}{l} \text{A-sublattices of } \mathbb{C}_\infty \\ \text{of rank } r \text{ with an } I\text{-level} \\ \text{structure modulo dilations} \\ \text{in } \mathbb{C}_\infty^* \end{array} \right\}$$

with $\alpha: I^{-1}Y/Y \rightarrow (I^{-1}/A)^r$ modulo Aut of Y , and

$$\left\{ \begin{array}{l} \text{isomorphism classes of affine } A\text{-modules } \phi_a \\ \text{over } \mathbb{C}_\infty \text{ with } I\text{-level structure and of} \\ \text{rank } r \text{ with } \partial\phi_a = a \end{array} \right\}$$

or more briefly the bijection

$$\bigsqcup_{(\alpha, Y) \in \mathcal{P}_A^r(I)} \Omega^r(\mathbb{C}_\infty) / \mathrm{GL}_A(Y, I) \cong \mathcal{M}_I^r(\mathbb{C}_\infty).$$

Here $\Omega^r(\mathbb{C}_\infty) = \mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup (F_\infty\text{-rational hyperplanes})$, also $\mathcal{M}_I^r$ is the moduli functor for affine A -modules rank r with I -level structure, and $\mathcal{P}_A^r(I)$ is the set of ordered pairs (α, Y) with $Y \in \mathcal{P}_A^r$ and $\alpha: I^{-1}Y/Y \rightarrow (I^{-1}/A)^r$ is an isomorphism.

8 Moduli spaces as homogeneous spaces (adélic theory)

The basic bijection describing $\mathcal{M}_I^r(\mathbb{C}_\infty)$ in (4.5) involves only the local field at ∞ . In this section we will describe $\mathcal{M}_I^r(\mathbb{C}_\infty)$ using all the primes of F together with $\mathcal{M}_I^r(\mathbb{C}_\infty)$. This adélic description is closely related to the adélic description of vector bundles on a curve which we recall now.

5.1 Remarks on vector bundles

Over any ring space X , a vector bundle E of rank n is a (locally free) sheaf on X locally isomorphic to $\mathcal{O}_X^n$. If X is a nonempty open set of the curve C over $\mathbb{F}_q$, for example $X = C - \{\infty\}$, then a vector bundle of rank n can be described as a family $(V, L_v)_{v \in X}$ where V is an n -dimensional vector space over $F = \mathbb{F}_q(C)$ and each L_v is a free module of rank n contained in V over the local ring $\mathcal{O}_{(v)}$ at v . We assume that there exists a basis $x_1, \dots, x_n$ of V over F and a finite set $S \subset X$ with

$$L_v = \mathcal{O}_{(v)}x_1 + \dots + \mathcal{O}_{(v)}x_n$$

for all $v \in X - S$. The basis $x_1, \dots, x_n$ is called a trivialization of V on $X - S$. If $y_1, \dots, y_n$ is any other basis of V , then

$$y_j = \sum_i a_{i,j} x_i \quad \text{and} \quad x_i = \sum_j b_{j,i} y_j$$

where $(a_{i,j}), (b_{i,j}) \in \text{GL}(n, \mathcal{O}_{(v)})$ for all but a finite number of v . Hence every basis of V is a trivialization of V over some $X - T$ where $T \subset X$ is finite.

To an $\mathcal{O}_X$ -sheaf E , locally isomorphic to $\mathcal{O}_X^n$, we assign $V = E_\eta$ where η is the general point of C . The $\mathcal{O}_{(v)}$ -submodule $L_v \subset V$ is defined by trivializing $E|(X - S)$ on an open set $X - S$ around v with $x_1, \dots, x_n \in V$ and requiring for all $w \in X - S$ with S finite that

$$L_w = \mathcal{O}_{(w)}x_1 + \dots + \mathcal{O}_{(w)}x_n.$$

These lattices which came up in the above description of a vector bundle can be identified with certain homogeneous spaces. Let F_v denote the completion of F at $v \in X$ with valuation ring $\mathcal{O}_v$. Then using matrix transform between basis of a lattice in a vector space F^n or F_v^n , we have the following commutative diagram of bijections :

$$\begin{array}{ccc} \text{GL}(n, F)/\text{GL}(n, \mathcal{O}_{(v)}) & \xrightarrow{\cong} & \{\mathcal{O}_{(v)}\text{-lattices in } F^n\} \\ \downarrow & & \downarrow \\ \text{GL}(n, F_v)/\text{GL}(n, \mathcal{O}_v) & \xrightarrow{\cong} & \{\mathcal{O}_v\text{-lattices in } F_v^n\} \end{array}$$

5.2 Homogeneous space description of trivialized vector bundles

A vector bundle E with a trivialization is described by (F^n, L_v) for $v \in X$ where $L_v = \mathcal{O}_{(v)}^n \subset F^n$ for all $v \in X - S$ where S is finite. Thus the vector bundle trivialized on $X - S$ is determined by a finite set of lattices $L_v \subset F^n$ for $v \in S$. This leads to the following bijections :

$$\begin{array}{ccc} \prod_{v \in S} \text{GL}(n, F_v)/\text{GL}(n, \mathcal{O}_v) & \xrightarrow{\cong} & \left\{ \begin{array}{l} \text{trivialized over } X - S \\ n\text{-dimensional vector bundles on } X \\ \text{all up to isomorphism} \end{array} \right\} \\ \cong \uparrow & & \\ \prod_{v \in S} \text{GL}(n, F_v) \times \prod_{v \in X - S} \text{GL}(n, \mathcal{O}_v)/\text{GL}(n, \mathcal{O}_v) & & \end{array}$$

where the vertical arrow is a homeomorphism of locally compact spaces.

At points of S we can consider a Λ -level structure on the vector bundle E relative to a positive divisor Λ supported on S . This is just an isomorphism of the sheaf $E|_\Lambda(-\Lambda) =$

$E \otimes_{\mathcal{O}_C} (\mathcal{O}_C/\mathcal{O}_C(-\Lambda)) \rightarrow (\mathcal{O}_C/\mathcal{O}_C(-\Lambda))^n$. In terms of the description of E with the data (F^n, L_v) this is an isomorphism

$$L_v/\mathfrak{m}_v^{\lambda_v} L_v \xrightarrow{\cong} (\mathcal{O}_v/\mathfrak{m}_v^{\lambda_v})^n$$

for each $v \in S$ where $\mathfrak{m}_v$ is the maximal ideal of $\mathcal{O}_v$ and hence $\mathcal{O}_v$. If we denote by

$$\mathrm{GL}(n, \mathcal{O}_v, \Lambda_v) = \ker(\mathrm{GL}(n, \mathcal{O}_v) \rightarrow \mathrm{GL}(n, \mathcal{O}_v/\mathfrak{m}_v^{\lambda_v}))$$

and $K(n, \Lambda) = \prod_{v \in X} \mathrm{GL}(n, \mathcal{O}_v, \Lambda_v)$, then the previous diagram becomes modified as follows for $\mathrm{supp}(\Lambda) \subset S$:

$$\begin{array}{ccc} \prod_{v \in S} \mathrm{GL}(n, F_v) / \mathrm{GL}(n, \mathcal{O}_v, \Lambda_v) & \longrightarrow & \left\{ \begin{array}{l} \text{trivialized over } X - S \\ n\text{-dimensional vector bundles} \\ \text{on } X \text{ with } \Lambda\text{-level structure} \end{array} \right\} \\ \uparrow & & \\ \prod_{v \in S} \mathrm{GL}(n, F_v) \times \prod_{v \in X-S} \mathrm{GL}(n, \mathcal{O}_v) / K(n, \Lambda) & & \end{array}$$

Now we consider two special cases for X namely $X = C$ and $X = C - \{\infty\}$ and remove the condition that the vector bundle is trivialized.

5.3 Vector bundles on C

The product in the numerator of $\mathrm{Vect}_S(X)$ or $\mathrm{Vect}_S(X; \Lambda)$ is one of the terms in the inductive limit which defines the adèle group $\mathrm{GL}(n, \mathbb{A}_F)$ where

$$\mathrm{GL}(n, \mathbb{A}_F) = \varinjlim_{S||} \prod_{v \in S} \mathrm{GL}(n, F_v) \times \prod_{v \in C-S} \mathrm{GL}(n, \mathcal{O}_v).$$

Hence considering vector bundles with Λ -level structure and trivialization over some open set $C - S$ of C where $\mathrm{supp}(\Lambda) \subset S$, we obtain the bijection

$$\mathrm{GL}(n, \mathbb{A}_F) / K(n, \Lambda) \xrightarrow{\cong} \left\{ \begin{array}{l} \text{vector bundles of dimension } n \\ \text{with a } \Lambda\text{-level structure and} \\ \text{trivialization off the } \mathrm{supp}(\Lambda) \\ \text{all up to isomorphism} \end{array} \right\}$$

Since the various trivializations are all related by the action of $\mathrm{GL}(n, F)$ on basis, we have the following quotient of the previous bijection

$$\mathrm{GL}(n, F) \backslash \mathrm{GL}(n, \mathbb{A}_F) / K(n, \Lambda) \xrightarrow{\cong} \left\{ \begin{array}{l} \text{isomorphism classes of} \\ n\text{-dimensional vector bundles} \\ \text{on } C \text{ with } \Lambda\text{-level structure} \end{array} \right\}$$

5.4 Vector bundles on $C - \{\infty\}$

The product in the numerator of $\mathrm{Vect}_S(X)$ or $\mathrm{Vect}_S(X; \Lambda)$ for $X = C - \{\infty\}$ is one of the terms in the inductive limit which defines the finite adèle group $\mathrm{GL}(n, \mathbb{A}_F^f)$ where $\mathbb{A}_F^f = \widehat{\prod}_{v \neq \infty} F_v$ and $\widehat{A} = \varprojlim_J A/J$, the limit being taken over ideals $J \subset A$. A divisor Λ can be described as an ideal $I \subset A$, and the group $K(n, \Lambda)$ is given by

$$K(n, \Lambda) = \ker(\mathrm{GL}(n, \widehat{A}) \rightarrow \mathrm{GL}(n, \widehat{A}/I))$$

which we also denote by $\mathrm{GL}(n, \widehat{A}, I)$ to avoid confusion with the previous case discussed in (5.3). As before we obtain in the injective limit the bijection

$$\mathrm{GL}(n, F) \backslash \mathrm{GL}(n, \mathbb{A}_F^f) / \mathrm{GL}(n, \widehat{A}, I) \xrightarrow{\cong} \left\{ \begin{array}{l} \text{vector bundles of dimension } n \text{ on} \\ C - \{\infty\} \text{ with a } \Lambda\text{-level structure} \\ \text{and trivialization off the } \mathrm{supp}(\Lambda) \\ \text{all up to isomorphism} \end{array} \right\}$$

On the affine $C - \{\infty\}$ with coordinate ring $A = H^0(C - \{\infty\}, \mathcal{O}_C)$ a vector bundle of dimension n is just a projective module Y of rank n over A .

Now we combine the previous bijection with

$$\mathbb{C}_\infty^* \backslash \mathrm{Mon}_{F_\infty}(F_\infty \otimes_A Y, \mathbb{C}_\infty) / \mathrm{GL}_A(Y) \cong \Omega^r(\mathbb{C}_\infty) = \mathbb{P}_{r-1}(\mathbb{C}_\infty) - \bigcup \{F_\infty\text{-rational hyperplanes}\}.$$

We have a bijection

$$\left\{ \begin{array}{l} \text{vector bundles } Y \text{ on } C - \infty \text{ of} \\ \text{rank } r \text{ with a trivialization} \\ \text{and a } \Lambda\text{-level structure all up} \\ \text{to isomorphism} \end{array} \right\} \times \Omega^r(\mathbb{C}_\infty) \xrightarrow{\cong} \left\{ \begin{array}{l} \text{isomorphism classes of } A\text{-lattices} \\ \text{in } \mathbb{C}_\infty \text{ of rank } r \text{ with} \\ I\text{-level structure modulo dilations} \end{array} \right\}$$

The trivialization of Y is a basis of $Y \otimes_A F$ over F which defines an embedding $Y \rightarrow F_\infty^r$ as a discrete A -submodule. When this embedding is composed with a monomorphism $F_\infty^r \rightarrow \mathbb{C}_\infty$ yields an A -sublattice of $\mathbb{C}_\infty$ of rank r with an I -level structure modulo dilations in $\mathbb{C}_\infty^*$. Hence by (4.5) we have a map

$$(\mathrm{GL}(r, \mathbb{A}_F^f) / \mathrm{GL}(r, \widehat{A}, I)) \times \Omega^r(\mathbb{C}_\infty) \rightarrow \mathcal{M}_I^r(\mathbb{C}_\infty)$$

where the fibres correspond to the various bases of $Y \otimes_A F$ over F . Hence factoring out by the action of $\mathrm{GL}(r, F)$ yields the adélic description of $\mathcal{M}_I^r(\mathbb{C}_\infty)$.

Théorème 8.1. *The above map induces a bijection*

$$\mathrm{GL}(r, F) \backslash \mathrm{GL}(r, \mathbb{A}_F^f) \times \Omega^r(\mathbb{C}_\infty) / \mathrm{GL}(r, \widehat{A}, I) \xrightarrow{\cong} \mathcal{M}_I^r(\mathbb{C}_\infty).$$

Remarque 8.1. The quotient space descriptions in (5.5) and (5.6) generalize to any open compact subgroup $H \subset \mathrm{GL}(r, \widehat{A})$ to give the $\mathbb{C}_\infty$ -valued points $\mathcal{M}_H^r(\mathbb{C}_\infty)$ of a moduli scheme $\mathcal{M}_H^r$:

$$\begin{aligned} \mathcal{M}_H^r(\mathbb{C}_\infty) &= \mathrm{GL}(r, F) \backslash \mathrm{GL}(r, \mathbb{A}_F^f) \times \Omega^r(\mathbb{C}_\infty) / H \quad \text{adélic version} \\ &= \bigsqcup_{Y \in \mathcal{P}_A^r} \Omega^r(\mathbb{C}_\infty) / (x_Y^{-1} H x_Y \cap \mathrm{GL}(r, F)) \quad \text{local version.} \end{aligned}$$

The moduli scheme $\mathcal{M}_H^r$ arises from E -level structures on Drinfel'd modules.

The rigid analytic moduli space

Now our aim is to describe the rigid analytic structure on the $\mathbb{C}_\infty$ -valued points of the moduli space. These points were parametrized by a quotient of $\Omega^r(\mathbb{C}_\infty)$ by a discrete group in (2.4.5). The rigid analytic structure on the moduli space is a quotient of the rigid analytic structure on $\Omega^r(\mathbb{C}_\infty)$.

In order to define the rigid analytic structure on $\Omega^r(\mathbb{C}_\infty)$, we make use of a natural mapping

$$\lambda: \Omega^r(\mathbb{C}_\infty) \longrightarrow I(F_\infty^r)_\mathbb{Q}$$

onto the rational points $I(F_\infty^r)_\mathbb{Q}$ of the geometric realization $I(F_\infty^r)_\mathbb{R}$ of the building $I(F_\infty^r)$ of the group $\mathrm{PGL}(r)$ over the local field F_∞ . The admissible open sets of the rigid analytic structure are inverse images by λ^{-1} of certain open neighborhoods of skeletons in $I(F_\infty^r)_\mathbb{R}$.

In this discussion we indicate how both $\Omega^r(\mathbb{C}_\infty)$ and the building $I(F_\infty^r)$ are p -adic analogues of real symmetric spaces.

9 Norms on vector spaces over a local field

Notations 9.1. Let K be a local field with valuation ring R , maximal ideal $\mathfrak{p} = R\pi$, and $q = \mathrm{Card}(k)$ where $k = R/\mathfrak{m}$. We normalize $|a|$ on K as $|a| = q^{-\mathrm{ord}(a)}$ so that $|\pi| = q^{-1}$ and $|\pi^i| = q^{-i}$. We consider finite dimensional vector spaces over K , and thus the closed unit ball and unit sphere are compact. If V is of dimension m over K , then a lattice is any R -submodule M free of rank m .

Définition 9.1. A norm on a vector space V over K is a function $\alpha: V \rightarrow \mathbb{R}$ satisfying :

- (a) $\alpha(x) \geq 0$ and $\alpha(x) = 0$ if and only if $x = 0$.
- (b) $\alpha(ax) = |a|\alpha(x)$ for $a \in K$ and $x \in V$.
- (c) $\alpha(x + y) \leq \sup(\alpha(x), \alpha(y))$ for $x, y \in V$.

A norm is called integral provided $\alpha(V) = |K| = \{0\} \cup q^\mathbb{Z}$ and rational provided $\alpha(V) \subset \mathbb{Q}$.

If $\alpha(x) < \alpha(y)$, then axiom (c) can be strengthened to the equality

$$\alpha(x + y) = \alpha(y) = \sup(\alpha(x), \alpha(y)).$$

If α is a norm and $t > 0$, then the dilation $t\alpha$ is a norm, and we denote by $N(V)$ the set of dilation classes of norms on V .

Exemple 9.1 ((I)). For a basis $x_0, \dots, x_m$ of V and real numbers $r_0 > 0, \dots, r_m > 0$ the function

$$\alpha(a_0x_0 + \dots + a_mx_m) = \sup(r_0|a_0|, \dots, r_m|a_m|)$$

is a norm on V . In fact, every norm can be described by this formula. This is proved by induction on m by considering a nonzero linear form $\lambda: V \rightarrow F$ and choosing $x_0 \neq 0$ in V such that $x \mapsto |\lambda(x)|/\alpha(x)$ defined on the compact projective space $\mathbb{P}(V)$ takes its maximum at x_0 . Then $V \cap Fx_0 \oplus \ker(\lambda)$ for which on $\ker(\lambda)$ the inductive hypothesis applies. See Goldman and Iwahori for the details.

Exemple 9.2 ((II)). The norm α_M associated to a lattice M in V is given by the formula $\alpha_M(x) = \inf\{1/|a| : ax \in M\}$.

Observe that M is the unit α_M -ball in V of all $x \in V$ with $\alpha_M(x) \leq 1$, and for $x \neq 0$ we have $\alpha_M(x) = q^m$ where $\mathfrak{p}^m = \{a \in F : ax \in M\}$. For two lattices L and M the inclusion $L \subset M$ holds if and only if $\alpha_L(x) \geq \alpha_M(x)$ for all $x \in V$.

For nonzero $c \in F$ we have $\alpha_{cM}(x) = (1/|c|)\alpha_M(x)$ for all $x \in V$, and in particular, $\alpha_{\pi M}(x) = q\alpha_M(x)$. Finally, if $x_0, \dots, x_m$ is a basis of a lattice M over R , then

$$\alpha_M(a_0x_0 + \dots + a_mx_m) = \sup\{|a_0|, \dots, |a_m|\}.$$

Thus this example is a special case of (1.4).

Going back to $\alpha(a_0x_0 + \cdots + a_mx_m) = \sup(r_0|a_0|, \dots, r_m|a_m|)$, we see that replacing x_i by $\pi^{s_i}x_i$ for $s(i)$ replaces r_i by $q^{s_i}r_i$. In particular, x_i replaced by $\pi^{-1}x_i$ leads to r_i replaced by qr_i . Hence, by rescaling the basis vectors, we can always require the constants r_i to be in an interval of the form $[t, qt)$ or $(t, qt]$ for some given $t > 0$. So when $\alpha(V) = \{0\} \cup q^{\mathbb{Z}}$, we can choose the x_i with each $r_i = 1$, and this gives the next proposition.

Proposition 9.1. *The following are equivalent for $t > 0$ and a norm α on a vector space V :*

- (1) $\alpha = t \cdot \alpha_H$ for some lattice H ,
- (2) $\alpha(V) = |F| \cdot t$, and
- (3) $\alpha(a_0x_0 + \cdots + a_mx_m) = t \sup(|a_0|, \dots, |a_m|)$ for some basis $x_0, \dots, x_m$ of V .

Now we wish to study to what extent the representation

$$\alpha(a_0x_0 + \cdots + a_mx_m) = \sup(r_0|a_0|, \dots, r_m|a_m|)$$

for a norm α is unique by renormalizing and reordering so that

$$q \geq r_0 \geq r_1 \geq \cdots \geq r_{m-1} \geq r_m \geq 1.$$

Then the basis elements are unique up to multiplication by π^{s_i} and cyclic permutation. The requirement that $r_m \geq 1$ removes this ambiguity, and further after a dilation of α we can assume that $r_0 = q$. The set of values $\alpha(V) = \{0\} \cup q^{\mathbb{Z}}r_0 \cup \cdots \cup q^{\mathbb{Z}}r_m$, and thus the numbers r_i are uniquely determined by α . They make up the set $\alpha(V) \cap (1, q]$ when we require $r_m \geq 1$.

With these notations, consider the lattices L_i for $i = m, m-1, \dots, 0$ where

$$L_i = Rx_0 + \cdots + Rx_i + \mathfrak{p}x_{i+1} + \cdots + \mathfrak{p}x_m \quad \text{for } r_i > r_{i+1},$$

and $L_m = Rx_0 + \cdots + Rx_m$.

Lemme 9.1. *The lattice L_i is the open ball $B(0, qr_i)$.*

Démonstration. For $x = \sum a_jx_j \in V$ clearly $x \in L_i$ if and only if $|a_j| \leq 1$ if $j \leq i$ and $|a_j| < q$ if $j > i$, or equivalently, $r_j|a_j| \leq r_j$ for $j \leq i$ and $r_j|a_j| \leq qr_j$ for $j > i$.

Now assume that $r_i > r_{i+1}$, and we see that both inequalities are equivalent to $r_i|a_i| \leq r_iq$. Hence, $x \in L_i$ if and only if $\alpha(x) < qr_i$. If $r_i = r_{i+1}$, then $L_i = L_{i+1}$, and so the result holds by induction from m to 0. $\square$

Finally, we obtain the following structure theorem.

Théorème 9.1. *Let α be a norm on a vector space V with $\alpha(V) \cap (1, q] = S$, and for $r \in S$, let $L(r) = B(0, qr)$ an open ball. Then $\dim V \geq \text{Card}(S)$ and $\alpha = \sup_{r \in S}(r\alpha_{L(r)})$.*

Démonstration. As in (1.3), we can choose a basis $x_0, \dots, x_m$ of V with

$$\alpha(a_0x_0 + \cdots + a_mx_m) = \sup(r_0|a_0|, \dots, r_m|a_m|)$$

and $q \geq r_0 \geq r_1 \geq \cdots \geq r_m \geq 1$. The r_i exhaust the set S with possible repetitions and thus $\dim V \geq \text{Card}(S)$.

By (1.6) the lattice $L_i = L(r_i)$ for $r_i > r_{i+1}$ is given by

$$L_i = Rx_0 + \cdots + Rx_i + \mathfrak{p}x_{i+1} + \cdots + \mathfrak{p}x_m.$$

Then α_{L_i} is given by

$$\alpha_{L_i}(a_0x_0 + \cdots + a_mx_m) = \sup(|a_0|, \dots, |a_i|, q^{-1}|a_{i+1}|, \dots, q^{-1}|a_m|),$$

and since $r_i q^{-1}|a_i| \leq r_{i+1}|a_{i+1}|$, it follows that

$$\alpha(x) = \sup(r_0\alpha_{L_0}(x), \dots, r_m\alpha_{L_m}(x)) = \sup_{r \in S}(r\alpha_{L(r)}(x)).$$

This proves the theorem. □

10 The building for $\mathrm{PGL}(V)$ over a local field

We continue with the notations (1.1) in this section. The dilation (or homothety) class $\{L\}$ of a lattice L is the set of all lattices λL where $\lambda \in F^*$ in the vector space V . Observe that $\{L\}$ is the set of all $\pi^i L$ for i in the integers.

Let $\mathrm{PGL}(V)$ denote $\mathrm{GL}(V)/(\text{scalars})$ for a vector space V .

Définition 10.1. Let V be a vector space over the local field K . The building $I(V)$ for the group $\mathrm{PGL}(V)$ over K is the simplicial complex whose vertices are dilation classes $\{L\}$ of lattices L in V , and whose simplexes $\{v_0, \dots, v_n\}$ are sets of vertices where after reordering $v_i = \{L_i\}$ with $L_0 \supset L_1 \supset \cdots \supset L_n \supset \pi L_0$.

Observe that the ordering of the vertices $v_0, \dots, v_n$ such that representatives $L_i \in v_i$ can be chosen with $L_0 \supset L_1 \supset \cdots \supset L_n \supset \pi L_0$ is unique up to the action of the cyclic group on $n + 1$ elements inside the symmetric group.

The simplicial complex $I(V)$ has dimension equal to $\dim V - 1$, and each simplex is contained in a top dimensional simplex (one whose dimension equals $\dim I(V)$). For $\dim V = 2$, the building $I(V)$ is one dimensional, a graph. In fact, $I(V)$ is simply connected and hence a tree. Each vertex $v = \{L\}$ is contained in $(m + 1)$ 1-simplexes $\{v, v'\}$ corresponding to the $m + 1$ lattices L' with $L \supset L' \supset \pi L$.

This building $I(V)$ is a special case of the buildings (immeubles) which have been associated to general semisimple groups over a local field by Bruhat and Tits. These simplicial complexes are contractible, and the vertices are in natural bijective correspondence with the cosets of $\mathrm{PGL}(V)/(\text{maximal compact subgroup } K)$. Here K is the image of $\mathrm{GL}(L)$ in $\mathrm{PGL}(V)$ for a lattice L of V . This description together with other features suggest that $I(V)$ is an analogue of the symmetric space for real Lie groups.

Now recall some generalities on geometric realizations as applied to $I(V)$. The geometric realization $I(V)_{\mathbb{R}}$ is the subset of $(t_v) \in \prod_{v \in I(V)} [0, 1]$ such that $\{v : t_v \neq 0\}$ is a simplex of $I(V)$ and $\sum_v t_v = 1$. For each simplex $\sigma = \{v_0, \dots, v_n\}$ of $I(V)$ its geometric realization $|\sigma| \subset I(V)_{\mathbb{R}}$ is the subset of $(t_v) \in I(V)_{\mathbb{R}}$ with $t_v = 0$ for $v \notin \sigma$. As a subset of $\prod_{v \in I(V)} [0, 1]$, $|\sigma|$ is compact, we give $I(V)_{\mathbb{R}}$ the inductive (weak) topology where M is closed in $I(V)_{\mathbb{R}}$ if and only if $M \cap |\sigma|$ is closed in $|\sigma|$ for each simplex σ of $I(V)$. We will make use of the dense subset $I(V)_{\mathbb{Q}} \subset I(V)_{\mathbb{R}}$ consisting of (t_v) with each $t_v \in \mathbb{Q}$. The set $I(V)_{\mathbb{Z}} \subset I(V)_{\mathbb{Q}}$ of (t_v) with each $t_v \in \mathbb{Z}$ can be identified with the set of vertices of $I(V)$.

The function which links the considerations of this section with those of the previous section is $\theta: I(V)_{\mathbb{R}} \rightarrow N(V)$ from the geometric realization of the building to the dilation classes of norms on V as follows : Let $\sigma = \{v_0, \dots, v_n\}$ be a simplex and $t = (t_v) \in |\sigma|$ be a point. We can choose an ordering $v_0, \dots, v_n$ such that $t_n > 0$ and lattices $L_i \in v_i$ with

$$L_0 \supset L_1 \supset \cdots \supset L_n \supset \pi L_0. \quad \text{Then } \beta(t) = \alpha$$

where $\alpha = \sup(q^{-t_0}\alpha_{L_0}, \dots, q^{-t_n}\alpha_{L_n})$.

Théorème 10.1. *The function $\theta: I(V)_{\mathbb{R}} \rightarrow N(V)$ is a well-defined bijection which carries the vertices $I(V)_{\mathbb{Z}}$ onto the set of classes containing integral norms and $I(V)_{\mathbb{Q}}$ onto the set of classes containing rational norms.*

Démonstration. The function θ is a bijection by (1.7), the structure theorem for norm. In that theorem we proved that each norm α has a unique representation $\alpha = \sup(r_0\alpha_{L_0}, \dots, r_m\alpha_{L_m})$ where $q = r_0 \geq r_1 \geq \dots \geq r_m \geq 1$ up to dilation. Let $t_0 = 1$, $t_i = \log_q(r_i/r_{i+1})$. Then $(t_0, \dots, t_m)$ determines the unique point in $|I(\{L_0, \dots, L_m\})|$ which maps to α under θ . The remaining statements are clear from the formulas relating the t_i 's and r_i 's. This proves the theorem. $\square$

11 Metric on the building

We continue with the notations of (1.1) in this section.

Définition 11.1. Let V be a vector space over the local field K . For two norms α, β on V , we define the distance $\rho(\alpha, \beta)$ between α and β by the following equivalent formulae :

$$\begin{aligned} \rho(\alpha, \beta) &= \log_q \left(\sup_{x \in V, x \neq 0} \alpha(x)/\beta(x) \right) + \log_q \left(\sup_{x \in V, x \neq 0} \beta(x)/\alpha(x) \right) \\ &= \log_q \sup_{x \in V, x \neq 0} \alpha(x)\beta(x)^{-1} = \log_q \inf_{x \in V, x \neq 0} \alpha(x)/\beta(x). \end{aligned}$$

Remarque 11.1. From the first formula for $\rho(\alpha, \beta)$ we see that $\rho(\alpha, \beta) = \rho(\beta, \alpha)$, and from the second form it follows that $\rho(\alpha, \beta) \geq 0$, and $\rho(\alpha, \beta) = 0$ if and only if $\beta = t\alpha$ for some $t > 0$. Moreover, for $t_1 > 0$ and $t_2 > 0$ we have $\rho(t_1\alpha, t_2\beta) = \rho(\alpha, \beta)$, and thus $\rho(\{\alpha\}, \{\beta\})$ is well defined on dilation classes of norms on V . Finally, it is easy to check the triangle inequality

$$\rho(\alpha, \gamma) \leq \rho(\alpha, \beta) + \rho(\beta, \gamma)$$

using the relations of the form

$$\sup_{x \in V, x \neq 0} \alpha(x)/\beta(x) = \sup_{x \in V, x \neq 0} \alpha(x)/\beta(x) \cdot \sup_{x \in V, x \neq 0} \beta(x)/\gamma(x).$$

Hence ρ is a metric on the space $N(V)$ of dilation classes of norms on V .

Note from the definition of ρ that if $t_1\alpha \leq \beta \leq t_2\alpha$ on V , then $\rho(\alpha, \beta) \leq \log_q(t_1/t_2)$.

Exemple 11.1. Let M be a lattice in V . Then the integral norm α_M is defined as in (1.4), and

$$M = \{x \in V : \alpha_M(x) \leq 1\} \quad \text{and} \quad \mathfrak{p}M = \{x \in V : \alpha_M(x) < 1\}.$$

Thus $M - \mathfrak{p}M$ is the unit α_M -sphere of all x with $\alpha_M(x) = 1$ in V . Moreover, since every $x \in V$, $x \neq 0$, is proportional to some x' with $\alpha_M(x') = 1$, we deduce that

$$\rho(\alpha_M, \alpha_N) = \log_q \left(\sup_{x \in M - \mathfrak{p}M} \alpha_N(x) \right) - \log_q \left(\inf_{x \in M - \mathfrak{p}M} \alpha_N(x) \right).$$

For two lattices M and N in V , there exists a natural number r such that $M \supset N \supset \pi^r M$ and $N \supset M \supset \pi^r N$ so that $\rho(\alpha_M, \alpha_N) \leq 2r$. For a more precise calculation, we reorganize the hypothesis in the next proposition.

Proposition 11.1. *Let $M \supset N \supset \pi^r M$ be lattices in V where r is minimal with this property. Then $r = \rho(\alpha_M, \alpha_N)$.*

Démonstration. From $\alpha_M \leq \alpha_N \leq q^r \alpha_M$ we deduce that $\rho(\alpha_M, \alpha_N) \leq r$. Since $N \not\supset \pi^{r-1}M$, we have $x \in (M - \mathfrak{p}M) \cap N$ and therefore $\alpha_N(x) \leq 1$. Thus $\log_q(\inf_{x \in M - \mathfrak{p}M} \alpha_N(x)) \leq 0$. Since $\pi^{r+1}N \subset M$ and $\pi^{r+1}N \not\subset \mathfrak{p}M$, there exists $x \in M - \mathfrak{p}M$ with $x \in \pi^{r+1}N$ and so $\alpha_N(x) \geq q^r$. Thus $\log_q(\sup_{x \in M - \mathfrak{p}M} \alpha_N(x)) \geq r$. By the example (3.3) it follows that $\rho(\alpha_N, \alpha_M) \geq r$ and hence $\rho(\alpha_M, \alpha_N) = r$. $\square$

Corollaire 11.1. *A set of lattices $M_0, \dots, M_r$, or equivalently, a set of integral norm classes $\alpha_{M_0}, \dots, \alpha_{M_r}$ (for example $\alpha_i = \alpha_{M_i}$), determine a simplex in the building $I(V)_{\mathbb{Z}}$ or $N(V)$ if and only if $\rho(\alpha_i, \alpha_j) \leq 1$ for $i \neq j$.*

Démonstration. Arrange, after reordering and dilation, the lattices as $M_0 \supset M_1 \supset \dots \supset M_{r-1} \supset M_r \supset \pi^{-1}M_0$. Then $1 = \rho(\alpha_{M_i}, \alpha_{M_{i+1}}) \leq 1$ by (3.4), so $\rho(\alpha_i, \alpha_j) \leq 1$, and the classes $\{\alpha_i\}$ form a simplex.

Conversely, $\rho(\alpha_i, \alpha_j) \leq 1$ for $i \neq j$ when the classes determine a simplex again by (3.4). $\square$

12 The mapping from the p -adic symmetric space to the building

Now we return to the basic situation of the function field $F = \mathbb{F}_q(C)$ of the smooth curve $C/\mathbb{F}_q$, the local field F_∞ at ∞ on the curve, and $\mathbb{C}_\infty$ the completion of the algebraic closure of $\bar{F}_\infty$.

The simple critical observation is the following : For $z = (z_1, \dots, z_r) \in \mathbb{C}_\infty^r$ the function on F_∞^r

$$\alpha_z : (a_1, \dots, a_r) \longmapsto |z_1 a_1 + \dots + z_r a_r|$$

is a norm on the F_∞ -vector space F_∞^r provided $|z_1 a_1 + \dots + z_r a_r| = 0$ implies that $a = (a_1, \dots, a_r) = 0$. This is the case exactly for

$$z \in \mathbb{C}_\infty^r - \{\text{all } F_\infty\text{-rational hyperplanes}\}.$$

Further, for $c \in \mathbb{C}_\infty^*$ and $z \in \mathbb{C}_\infty^r$ the relation $\alpha_{cz} = |c| \alpha_z$, which is dilation of norms, holds and this leads to the following definition.

Définition 12.1. The building map λ defined on the p -adic symmetric space $\Omega^r(\mathbb{C}_\infty)$ to the building $N(F_\infty^r) = I(F_\infty^r)_{\mathbb{R}}$ of the group $\text{PGL}(r, F_\infty)$ is given by

$$\lambda(z) = \text{dilation class of } \alpha_z \quad \text{for } z = (z_i) \in \Omega^r.$$

For a representative r -tuple $(z_1, \dots, z_r) = z$ of $z \in \Omega^r$, we represent $\lambda(z)$ as the norm α_z , i.e.

$$\lambda(z)(a_1, \dots, a_r) = |z_1 a_1 + \dots + z_r a_r|.$$

Note, since $|\mathbb{C}_\infty^*| = q^{\mathbb{Q}}$, in fact $\lambda : \Omega^r(\mathbb{C}_\infty) \rightarrow I(F_\infty^r)_{\mathbb{Q}}$.

Proposition 12.1. *The building map $\lambda : \Omega^r(\mathbb{C}_\infty) \rightarrow I(F_\infty^r)_{\mathbb{Q}}$ is $\text{GL}(r, F_\infty)$ -equivariant for right actions. In particular, it is also $\text{GL}(r, F)$ -equivariant.*

Démonstration. For $s \in \text{GL}(r, F_\infty)$ we view the matrix as acting on the left and F_∞^r the action on the right. The norm $\lambda(z)$ acted on by s on the right is $\lambda(z)s : a \mapsto \lambda(z)(sa)$ for $a \in F_\infty^r$, it is $|\langle z | sa \rangle| = |\langle z^t s | a \rangle| = \lambda(z^t s)(a)$. Thus $\lambda(z)s = \lambda(z^t s)$ which proves the proposition. $\square$

For a subgroup $\Gamma \subset \mathrm{GL}(r, F_\infty)$ we have a quotient building mapping

$$\lambda_\Gamma: \Omega^r(\mathbb{C}_\infty)/\Gamma \longrightarrow I(F_\infty^r)_\mathbb{Q}/\Gamma \subset I(F_\infty^r)_\mathbb{R}/\Gamma.$$

In the case Γ is a certain subgroup of $\mathrm{GL}(r, A)$ this is a mapping of the corresponding moduli space associated with Γ to a quotient of the building by the discrete group Γ .

4.3 Remarks

The building map λ is useful for several purposes.

First, the sets λ^{-1} (ball around a vertex or simplex) can be used to describe the rigid analytic structure on the p -adic symmetric space. In the special case $r = 2$ so that $I(F_\infty^2)$ is a tree $\mathcal{T}$, we will describe a topological model for $\Omega^2(\mathbb{C}_\infty) = \mathbb{C}_\infty - F_\infty$ and the map $\lambda: \mathbb{C}_\infty - F_\infty \rightarrow \mathcal{T}$. This is used to calculate the cohomology of $\Omega^2(\mathbb{C}_\infty)$ and the map $\lambda_\Gamma: \Omega^2(\mathbb{C}_\infty)/\Gamma \rightarrow \mathcal{T}/\Gamma$.

There are some coverings of $\mathcal{T}/\Gamma$ which induce back to admissible covers of $\Omega^2(\mathbb{C}_\infty)/\Gamma$, and these give rise to a Leray spectral sequence of the map $\lambda_\Gamma: \Omega^2(\mathbb{C}_\infty)/\Gamma \rightarrow \mathcal{T}/\Gamma$. In this case the spectral sequence which could be thought of as the defining the rigid analytic structure, is just the cohomology $H^*(\mathcal{T}/\Gamma)$ of the group Γ .

In order to illustrate further what is involved in defining the rigid analytic structure, we must calculate $\rho(\lambda(z), \alpha_\Lambda)$ where Λ is the standard lattice $\mathcal{O}_\infty^r \subset F_\infty^r$. Since any other lattice is $\mathrm{GL}(r, F_\infty)$ -equivalent to Λ and since both λ and ρ are $\mathrm{GL}(r, F_\infty)$ -invariant, this calculation leads to $\rho(\lambda(z), \alpha_L)$ for any vertex $\{L\}$.

By (3.3) we have

$$\rho(\lambda(z), \alpha_\Lambda) = \log_q \left(\sup_{(a_1, \dots, a_r) \in S(\Lambda)} |a_1 z_1 + \dots + a_r z_r| \right) - \log_q \left(\inf_{(a_1, \dots, a_r) \in S(\Lambda)} |a_1 z_1 + \dots + a_r z_r| \right)$$

where $S(\Lambda) = \Lambda - \mathfrak{p}\Lambda$, the set of all $(a_1, \dots, a_r) \in F_\infty^r$ with all $|a_i| \leq 1$ and at least one $|a_i| = 1$.

Thus we see that $\rho(\lambda(z), \alpha_\Lambda) < s$ if and only if for all $(a_1, \dots, a_r) \in F_\infty^r$ with all $|a_i| \leq 1$ and at least one $|a_i| = 1$, we have

$$q^{-s} < |a_1 z_1 + \dots + a_r z_r| < q^s.$$

In the case $S(\Lambda) = \Lambda - \mathfrak{p}\Lambda$, the set of all $(a_1, \dots, a_r) \in F_\infty^r$ with all ratios $|z_i|/|z_j| \notin q^\mathbb{Z}$ for $i \neq j$, we have

$$\rho(\lambda(z), \alpha_\Lambda) = \log_q(|z_1|/|z_r|)$$

by an easy straightforward calculation.

When some of the ratios $|z_i|/|z_j| \in q^\mathbb{Z}$, the calculation of $\rho(\lambda(z), \alpha_\Lambda)$ is more complicated, and in fact, it is at the basis for the structure of $\Omega^r(\mathbb{C}_\infty)$ as a rigid analytic space.

13 Filtration of the 1-dimensional p -adic symmetric space

In this section we study the case $r = 2$ in detail where

$$\Omega^2(\mathbb{C}_\infty) = \Omega^1(\mathbb{C}_\infty) = \mathbb{C}_\infty - F_\infty.$$

We use the notation $u \in \Omega$ for $u \in \mathbb{C}_\infty$ and $\mathcal{T} = I(F_\infty^2)$ is the building which in this case is a tree (a contractible 1-dimensional simplicial complex).

The building map $\lambda: \Omega \rightarrow \mathcal{T}$ is given by

$$\lambda(u)(a, b) = |a + bu|$$

and we wish to study

$$\rho(\lambda(u), \alpha_\Lambda) = \log_q \left(\sup_{(a,b) \in S(\Lambda)} |a + bu| \right) - \log_q \left(\inf_{(a,b) \in S(\Lambda)} |a + bu| \right)$$

relative to elements of u in terms of congruence properties of F_∞ . This is formulated by using the following notion.

Définition 13.1. For $u \in \mathbb{C}_\infty$, we define the irrational absolute value

$$|u|_{\text{irr}} = \inf_{a \in F_\infty} |u - a|$$

to be the distance from u to $F_\infty \subset \mathbb{C}_\infty$.

Observe that this is just the distance from u to $F_\infty \subset \mathbb{C}_\infty$. The following properties are easily deduced from the definition.

Proposition 13.1. *The irrational norm satisfies the following :*

- (1) For $u \in \mathbb{C}_\infty$, $|u|_{\text{irr}} = 0$ if and only if $u \in F_\infty$.
- (2) For $u \in \mathbb{C}_\infty$ and $c \in F_\infty$, we have $|cu|_{\text{irr}} = |c| \cdot |u|_{\text{irr}}$.
- (3) For $|u| \leq q$ and $u \in c\mathcal{O}_\infty$ it follows that $|u|_{\text{irr}} = |u|$.
- (4) For $|u| = 1$ with residue class $\bar{u} \in \bar{\mathbb{F}}_q$ the irrational norm $|u|_{\text{irr}} = |u| = 1$ if and only if $\bar{u} \notin \mathbb{F}_q = \mathbb{F}_q$.

From the previous section we see for $|u| \leq 1$ that

$$\rho(\lambda(u), \alpha_\Lambda) = \log_q(\max(|u|, 1/|u|))$$

which for $|u| < 1$ becomes simply $\rho(\lambda(u), \alpha_\Lambda) = -\log_q |u|$.

Proposition 13.2. *The distance from $\lambda(u)$ to the standard vertex α_Λ in the tree $\mathcal{T}$ is given by*

$$\rho(\lambda(u), \alpha_\Lambda) = \begin{cases} -\log_q |u|_{\text{irr}} & \text{for } |u| \leq 1, \\ -\log_q |1/u|_{\text{irr}} & \text{for } |u| \geq 1. \end{cases}$$

Démonstration. For the case $|u| < 1$, we have $\sup_{(a,b) \in S(\Lambda)} |a + bu| = 1$ since $|a + bu| \leq \max(|a|, |b| \cdot |u|) \leq 1$ and $|1 + 0 \cdot u| = 1$. Further,

$$\inf_{(a,b) \in S(\Lambda)} |a + bu| = \inf_{|a| \leq 1, |b|=1} |a + bu| = \inf_{|c| < 1} |c + u| = |u|_{\text{irr}}$$

since $|a + bu| \leq \max(|a|, |b| \cdot |u|) \leq 1$ for $|u| \leq 1$. Hence, we have

$$\rho(\lambda(u), \alpha_\Lambda) = \log_q(1/|u|_{\text{irr}}) = -\log_q |u|_{\text{irr}}.$$

This proves the proposition.

For the case $|u| > 1$, we observe that $\rho(\lambda(u), \alpha_\Lambda) = \rho(\lambda(1/u), \alpha_\Lambda) = -\log_q |1/u|_{\text{irr}}$ by the first part. $\square$

Now we are in a position to describe the inverse image in Ω of closed balls around the standard lattice vertex α_Λ . First, we choose some notation. Let $r: \mathbb{P}_1(\mathcal{O}_\infty) \rightarrow \mathbb{P}_1(\bar{\mathbb{F}}_q)$ be the residue class reduction and $s: \mathbb{P}_1(\bar{\mathbb{F}}_q) \rightarrow \mathbb{P}_1(\mathcal{O}_\infty)$ a cross section such that $r \circ s = \text{id}$ on $\mathbb{P}_1(\bar{\mathbb{F}}_q)$. We choose s so that $s(0) = 0$, $s(\infty) = \infty$.

- (1) Clearly $u \in \lambda^{-1}(\alpha_\Lambda)$ from (4.2) if and only if $|u|_{\text{irr}} = 1$. This gives the following relations for $\lambda^{-1}(\alpha_\Lambda) \subset \Omega$:

$$\begin{aligned} \lambda^{-1}(\alpha_\Lambda) &= \{u \in \mathbb{C}_\infty : |u|_{\text{irr}} = 1\} \\ &= \mathbb{P}_1(\mathbb{C}_\infty) - \bigcup_{\xi \in \mathbb{P}_1(\bar{\mathbb{F}}_q)} B(s(\xi), 1) \\ &= \Omega^2(\mathbb{C}_\infty) - \bigcup_{\eta \in \mathbb{P}_1(\bar{\mathbb{F}}_q)} B(s(\eta), 1) \end{aligned}$$

where $B(v, \epsilon) = \{u \in \mathbb{C}_\infty : |u - v| < \epsilon\}$ is the open ball of radius ϵ around v in $\mathbb{C}_\infty$ for $v \neq \infty$ and $B(\infty, \epsilon) = \{0\} \cup \{u \in \mathbb{C}_\infty : |u| > 1/\epsilon\} \subset \mathbb{P}_1(\mathbb{C}_\infty)$.

For the closed ball $B^*(\star, c)$ with $0 \leq c < 1$ observe that $\lambda^{-1}(B^*(\star, c)) = \Omega^2(\mathbb{C}_\infty) - \bigcup_{\eta \in \mathbb{P}_1(\bar{\mathbb{F}}_q)} B(s(\eta), q^{-c})$. In this case $q^{-c} < 1$.

- (2) In this case $\lambda^{-1}(B^*(\star, c))$ is $\Omega^2(\mathbb{C}_\infty)$ minus $(q+1)$ balls, but this time of slightly smaller radius. Now we see the relation between these balls and the edges of the tree coming out from the vertex v_Λ . All the points of $B(s(\eta), 1) - B(s(\eta), q^{-c})$ project to the edge corresponding to $\eta \in \mathbb{P}_1(\bar{\mathbb{F}}_q)$.
- (3) As the radius c increases in the tree, we come to new vertices each with q new edges coming out as c approaches 1 from $0 < c < 1$. The inverse image of the closed ball $B^*(v_\Lambda, c)$ as c increases is $\Omega^2(\mathbb{C}_\infty)$ minus $(q+1)$ balls of radius q^{-c} centered at the points $s(\eta)$ for $\eta \in \mathbb{P}_1(\bar{\mathbb{F}}_q)$.

As c increases past integer values, the inverse image changes by removing more balls. This describes the rigid analytic structure on $\Omega^2(\mathbb{C}_\infty)$ in terms of the tree $\mathcal{T}$.

HODGE FILTRATIONS AND POLE-ORDER FILTRATIONS FOR SINGULAR HYPERSURFACES

PIERRE DELIGNE AND ALEXANDRU DIMCA

1. STATEMENT OF THE RESULTS

Let V be a hypersurface of degree d in projective space $\mathbf{P}^n$, and let μ be the integer part of n/d . If the base field is a finite field $\mathbf{F}_q$, a theorem of J. Ax [1], refining an earlier theorem of E. Warning, is equivalent to

$$(1.1) \quad \#V(\mathbf{F}_q) \equiv \sum_0^n q^i \pmod{q^\mu}.$$

It would amount to the same thing to take at the right-hand side the sum from 0 to $n-1$. If U is the complement $\mathbf{P}^n - V$ of V , this congruence can be rewritten more simply as

$$(1.2) \quad \#U(\mathbf{F}_q) \equiv 0 \pmod{q^\mu}.$$

In this article we prove a Hodge-theoretic analogue of (1.1) and (1.2). The analogy is based on Grothendieck's conjecture (extended to mixed motives) that if the de Rham realization of a motive M satisfies $M_{\text{DR}} = F^\mu M_{\text{DR}}$, then M is the tensor product of the Tate motive $\mathbf{Q}(-\mu)$ with an effective motive; consequently, after reduction modulo p to $\mathbf{F}_q$, the Frobenius eigenvalues should be divisible by q^μ . This conjecture, together with Dwork's work, lies behind Katz's conjecture (cf. [11, 2.9] and B. Mazur [13]).

Assume now that the base field is $\mathbf{C}$. In rational or complex cohomology, the restriction morphism

$$H^*(\mathbf{P}^n) \rightarrow H^*(V)$$

is injective, except in degree $i = 2n$. Indeed, its composite with the Gysin morphism $H^*(V) \rightarrow H^*(\mathbf{P}^n)$ is multiplication by the cohomology class of V , namely $d\eta$ where $\eta = c_1\mathcal{O}(1)$, and $H^*(\mathbf{P}^n) = \mathbf{Z}[\eta]/(\eta^{n+1})$.

The restriction morphism sits in a long exact sequence of mixed Hodge structures:

$$(1.3) \quad \cdots \rightarrow H_c^i(U) \rightarrow H^i(\mathbf{P}^n) \rightarrow H^i(V) \xrightarrow{\partial} H_c^{i+1}(U) \rightarrow \cdots$$

Our analogue of (1.1) is the following.

Theorem 1. *The cokernel $\text{coker}(H^*(\mathbf{P}^n) \rightarrow H^*(V))$ has Hodge filtration $\geq \mu$.*

A mixed Hodge structure H is said to have Hodge filtration $\geq \mu$ if $H = F^\mu H$. Exact sequence (1.3) shows that the theorem is equivalent to the following analogue of (1.2).

Variant 2 (1). *The compactly supported cohomology $H_c^*(U)$ has Hodge filtration $\geq \mu$.*

The variety U is smooth of pure dimension n . By Poincaré duality, $H_c^i(U)$ and $H^{2n-i}(U)$ are dual to one another, with values in the Hodge structure $\mathbf{Q}(-n)$ of pure Hodge type (n, n) . The theorem is therefore also equivalent to

Variant 3 (2). *The cohomology of U satisfies*

$$(1.4) \quad F^{n-\mu+1}H^*(U) = 0.$$

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If V is the subvariety of $\mathbf{P}^n$ defined by equations $f_i = 0$ of degrees d_i ($1 \leq i \leq r$), J. Ax deduces from his theorem, applied to the hypersurface defined by the product of the f_i , that

$$(1.5) \quad \#V(\mathbf{F}_q) \equiv \sum_0^n q^i \pmod{q^\mu}$$

for

$$\mu = \left\lfloor \frac{n}{\sum d_i} \right\rfloor.$$

For the complement U of V , this is equivalent to

$$(1.6) \quad \#U(\mathbf{F}_q) \equiv 0 \pmod{q^\mu}.$$

Likewise, from Variant 1 of Theorem 1 and the spectral sequence defined by the open covering of U by the open sets $f_i \neq 0$ in $\mathbf{P}^n$, one deduces that over $\mathbf{C}$ the compactly supported cohomology of U has Hodge filtration $\geq [n/\sum d_i]$.

In the case of base field $\mathbf{F}_q$, N. Katz [11] proved that (1.5) and (1.6) remain true if $[n/\sum d_i]$ is replaced by the smallest integer $\geq (n+1-\sum d_i)/\sup(d_i)$. If d_1 is the largest of the d_i , this integer is

$$\left\lceil \frac{n - \sum_2^r d_i}{d_1} \right\rceil.$$

We do not know how to prove the Hodge-theoretic analogue of this refinement — except when V is smooth, in which case the result (SGA 7 XI 2.5) predates [11]. Added during revision: H. Esnault has just proved it when V has codimension r . Her approach is, roughly speaking, dual to ours, and has the advantage of requiring much less precise control over the desingularization procedure (cf. 2.8).

Let X be smooth over $\mathbf{C}$, let D be a divisor, let i denote the inclusion of D in X , and let $j : U \subset X$ be the inclusion of the open complement. By A. Grothendieck [8], one has

$$(1.7) \quad H^*(U, \Omega_U^*) \xrightarrow{\sim} H^*(U).$$

On the left, one uses the Zariski topology on U and the hypercohomology of the algebraic de Rham complex. On the right, cohomology means the complex cohomology of the usual topological space $U(\mathbf{C})$. The arrow is defined by the continuous map $\varepsilon : U(\mathbf{C}) \rightarrow U$ and the natural morphism from $\varepsilon^* \Omega_U^*$ to the resolution $\Omega_{U(\mathbf{C})}^*$ of the constant sheaf $\mathbf{C}$.

Since j is affine, one has $R^i j_* \Omega_U^p = 0$ for $i > 0$, hence $j_* \Omega_U^* \xrightarrow{\sim} Rj_* \Omega_U^*$ and therefore

$$(1.8) \quad H^*(X, j_* \Omega_U^*) \xrightarrow{\sim} H^*(U, \Omega_U^*).$$

The arrow is restriction to U . Altogether, one obtains

$$(1.9) \quad H^*(X, j_* \Omega_U^*) \xrightarrow{\sim} H^*(U).$$

For every subcomplex P of $j_* \Omega_U^*$ one thus has a map

$$H^*(X, P) \rightarrow H^*(U)$$

defined either via (1.9) or via (1.7) followed by restriction to U .

Each sheaf $j_* \Omega_U^p$ is the increasing union of the $\Omega_X^p(aD)$ ($a \geq 0$). The *pole-order filtration* P (or P_D , if the choice of D is ambiguous) on $j_* \Omega_U^*$ is the decreasing filtration defined by

$$P^s j_* \Omega_U^p = \begin{cases} 0, & p < s, \\ \Omega_X^p((p-s+1)D), & p \geq s. \end{cases}$$

It is decreasing, $P^s = 0$ for $s > \dim(X)$, and $j_* \Omega_U^*$ is the union of the P^s .

It is known that if D is smooth, the inclusion of filtered complexes

$$(1.10) \quad (\Omega_X^*(\log D), F) \hookrightarrow (j_* \Omega_U^*, P)$$

is a filtered quasi-isomorphism ([4, II, 3.13], inspired by [9]; the reader will also find there a more complicated analogue when D is a normal-crossings divisor).

Recall that for any normal-crossings divisor D , $\Omega_X^*(\log D) \subset j_*\Omega_X^*$ is the subsheaf of differential forms α such that both α and $d\alpha$ have at most simple poles along D . Another description, when D is smooth, is

$$\Omega_X^*(\log D) = \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_D^*)(D).$$

The Hodge filtration F is the stupid filtration by the truncations $\sigma_{\geq p}$.

Theorem 4. *Let X be smooth proper and let $D \subset X$ be a divisor. With the notation above, for every s , the image of $H^*(X, P^s(j_*\Omega_U^*))$ in $H^*(U, \mathbf{C})$ contains $F^s H^*(U, \mathbf{C})$.*

If D is smooth, it follows from the definitions and from the filtered quasi-isomorphism (1.10) that

$$H^*(X, P^s j_*\Omega_U^*) \xrightarrow{\sim} F^s H^*(U, \mathbf{C}).$$

Let us deduce Variant 2 of Theorem 1 from Theorem 2. It suffices to show that, with the notation of Theorem 1,

$$R\Gamma(\mathbf{P}^n, P^{n-\mu+1} j_*\Omega_U^*) = 0.$$

A fortiori, it suffices to show that for each p one has

$$R\Gamma(\mathbf{P}^n, P^{n-\mu+1} j_*\Omega_U^p) = 0.$$

The sheaves whose cohomology must be taken are either zero or of the form $\Omega^p(a)$ with $0 \leq p \leq n$, $a > 0$, and $a = (p - (n - \mu + 1) + 1)d$. It is known that $R\Gamma \Omega^p(a) = 0$ if $0 < a \leq p$ (R. Bott [3, Prop. 14.4, p. 246]; for an algebraic proof, valid in all characteristics, see SGA 7 XI 1.1). It remains to verify that

$$(1.11) \quad (p - (n - \mu + 1) + 1)d \leq p.$$

As p increases, the left-hand side grows faster than the right-hand side, so it is enough to consider the case $p = n$. Then (1.11) reduces to the inequality $\mu d \leq n$, which is precisely the definition of μ .

2. PROOF OF THEOREM 2

Proposition 5 (2.1). *Let X be smooth, $Y \subset X$ a smooth closed subscheme, i the inclusion of Y into X , $f : \tilde{X} \rightarrow X$ the blow-up of X along Y , and $\tilde{Y} = f^{-1}(Y) \subset \tilde{X}$ the exceptional divisor. Then on $\tilde{X}$ one has*

$$f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = \text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p)$$

and

$$R^i f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = 0 \quad \text{for } i > 0.$$

Proof. Using again i for the inclusion of $\tilde{Y}$ into $\tilde{X}$, one has

$$\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = \text{Ker}(\Omega_{\tilde{X}}^p \rightarrow i_*\Omega_{\tilde{Y}}^p),$$

hence a morphism

$$\text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p) \rightarrow f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}),$$

and it is enough to show that this induces an isomorphism from $\text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p)$, placed in degree 0, to $Rf_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y})$.

The proposition is local for the étale topology. It is therefore enough to treat the case where X is affine space $\mathbf{A}^{n_1+n_2} = \mathbf{A}^{n_1} \times \mathbf{A}^{n_2}$ and $Y = \{0\} \times \mathbf{A}^{n_2}$. Künneth formulas then reduce matters to the case $n_2 = 0$: the pair (X, Y) is obtained from $(\mathbf{A}^{n_1}, 0)$ by product with $\mathbf{A}^{n_2}$, and the direct sum over p of the sheaves under consideration is obtained from the corresponding sum for $(\mathbf{A}^{n_1}, 0)$ by external tensor product with $\bigoplus \Omega_{\mathbf{A}^{n_2}}^p$. We may therefore assume, and do assume, that Y is a point.

When Y is a point, we give two proofs of 2.1. The first has the advantage of working in every characteristic. The second explains why 2.1 is “obvious” from the point of view of Hodge theory.

First proof. Assume Y is a point. For $p > 0$, a local calculation shows that the morphism $f^*\Omega_X^p \rightarrow \Omega_{\tilde{X}}^p$ induces an isomorphism

$$f^*\Omega_X^p \xrightarrow{\sim} \text{Ker}(\Omega_{\tilde{X}}^p \rightarrow i_*\Omega_Y^p).$$

Since Ω_X^p is locally free and $\Omega_Y^p = 0$, the desired statement reduces to

$$\mathcal{O}_X \xrightarrow{\sim} Rf_*\mathcal{O}_{\tilde{X}}.$$

For $p = 0$, one uses the long exact sequence deduced from

$$0 \rightarrow \text{Ker}(\mathcal{O}_{\tilde{X}} \rightarrow i_*\mathcal{O}_{\tilde{Y}}) \rightarrow \mathcal{O}_{\tilde{X}} \rightarrow i_*\mathcal{O}_{\tilde{Y}} \rightarrow 0,$$

together with the same formula $\mathcal{O}_X \xrightarrow{\sim} Rf_*\mathcal{O}_{\tilde{X}}$ and the fact that for the projective space $\tilde{Y}$, the cohomology $H^*(\tilde{Y}, \mathcal{O}_{\tilde{Y}})$ is reduced to $\mathbf{C}$ in degree 0.

Second proof. The relation with Hodge theory is given by the following lemma.

Lemma 6 (2.2). *Let S be smooth projective, let $i : T \subset S$ be a smooth closed subscheme, and put $U := S - T$. The subcomplex $\text{Ker}(\Omega_S^* \rightarrow i_*\Omega_T^*)$ of the de Rham complex has hypercohomology equal to the compactly supported cohomology $H_c^*(U)$; the spectral sequence defined by the stupid filtration $\sigma_{\geq p}$ degenerates at E_1 and abuts to the Hodge filtration on $H_c^*(U)$.*

Apply this lemma to (X, Y) and $(\tilde{X}, \tilde{Y})$, with X smooth projective and Y a point. Because $\tilde{X} - \tilde{Y} \xrightarrow{\sim} X - Y$, one has $H_c^*(X - Y) \xrightarrow{\sim} H_c^*(\tilde{X} - \tilde{Y})$ and, passing to Gr_F , one obtains

$$(2.2.1) \quad R\Gamma(X, \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_Y^*)) \xrightarrow{\sim} R\Gamma(\tilde{X}, \text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*)).$$

The right-hand side is also

$$R\Gamma(X, Rf_*\text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*)).$$

Let C be the cone of the morphism

$$(2.2.2) \quad \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_Y^*) \rightarrow Rf_*\text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*).$$

Applying $R\Gamma$ to the distinguished triangle defining it, one deduces from (2.2.1) that $R\Gamma C = 0$. On the other hand, C — or rather its cohomology sheaves — is supported at the point Y . From $R\Gamma C = 0$ one concludes that $C = 0$: morphism (2.2.2) is an isomorphism. This completes the second proof of 2.1, modulo 2.2.

Lemma 2.2 comes from the fact that the simple complex associated with the double complex $\Omega_S^* \rightarrow i_*\Omega_T^*$, together with the weight filtration

$$0 \subset W_{-1} = s(0 \rightarrow i_*\Omega_T^*) \subset W_0 = s(\Omega_S^* \rightarrow i_*\Omega_T^*),$$

and the Hodge filtration by the subcomplexes

$$s(\sigma_{\geq p}\Omega_S^* \rightarrow \sigma_{\geq p}i_*\Omega_T^*)$$

is a cohomological mixed Hodge complex. This implicitly includes the rational structure given by the filtered complex $\mathbf{Q} \rightarrow i_*\mathbf{Q}$ with $W^{-1} : 0 \rightarrow i_*\mathbf{Q}$.

The morphism

$$(\text{Ker}(\Omega_S^* \rightarrow i_*\Omega_T^*), \sigma_{\geq p}) \rightarrow (s(\Omega_S^* \rightarrow i_*\Omega_T^*), F)$$

is a filtered quasi-isomorphism. This simply expresses the fact that $\Omega_S^* \rightarrow i_*\Omega_T^*$ is an epimorphism. This suffices to establish 2.2 for a functorial mixed Hodge structure on $H_c^*(U)$. This statement, slightly less precise than 2.2, is enough for the application to 2.1. $\square$

2.3. Let X be smooth, let $V \subset X$ be a divisor, and let $Y \subset X$ be a smooth closed subscheme contained in the singular locus of V . Let $f : \tilde{X} \rightarrow X$ be the blow-up of X along Y , let $\tilde{Y} = f^{-1}(Y)$ be the exceptional divisor, and let $\tilde{V} = f^{-1}(V)$. Because Y is contained in the singular locus of V , one has

$$(2.3.1) \quad 2\tilde{Y} \leq \tilde{V}.$$

Let $V_1 = \tilde{V} - \tilde{Y}$. By (2.3.1) one has

$$(V_1)_{\text{red}} = (f^{-1}V)_{\text{red}}.$$

Let $U = X - V = \tilde{X} - \tilde{V}$, let j denote its inclusion into X , and let $\tilde{j}$ denote its inclusion into $\tilde{X}$. We write P for both pole-order filtrations, that on $j_*\Omega_U^*$ relative to V and that on $\tilde{j}_*\Omega_U^*$ relative to V_1 .

Corollary 7 (2.4). *With the hypotheses and notation of 2.3, there exists in the derived category a morphism*

$$(2.4.1) \quad Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow P^sj_*\Omega_U^*$$

inducing the identity on U .

Proof. The complex $P^s\tilde{j}_*\Omega_U^*$ has components

$$\Omega_{\tilde{X}}^p((p-s+1)V_1) \quad (p \geq s),$$

and, by (2.3.1), these are subsheaves of

$$\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y})((p-s+1)\tilde{V}) = \Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) \otimes f^*\mathcal{O}((p-s+1)V).$$

By 2.1 the latter sheaves are acyclic: $R^if_* = 0$ for $i > 0$. Hence the derived direct image of the complex formed by them is obtained simply by applying f_* componentwise; this yields the complex of sheaves $\text{Ker}(\Omega_X^p \rightarrow \Omega_Y^p)((p-s+1)V)$.

Applying Rf_* to the inclusion

$$P^s\tilde{j}_*\Omega_U^* \rightarrow \sigma_{\geq s}(\Omega_{\tilde{X}}^*(\log \tilde{Y})(-\tilde{Y})((s-s+1)\tilde{V})),$$

one obtains a morphism

$$Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow \sigma_{\geq s} \text{Ker}(\Omega_X^* \rightarrow \Omega_Y^p)((s-s+1)V).$$

Composing with the inclusion of $\text{Ker}(\Omega_X^* \rightarrow \Omega_Y^*)$ into Ω_X^* gives the promised morphism

$$Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow P^sj_*\Omega_U^*.$$

□

Corollary 8 (2.5). *With the hypotheses and notation of 2.3, the image of $R^*\Gamma(\tilde{X}, P^s\tilde{j}_*\Omega_U^*)$ in $H^*(U)$ contains the image of $R^*\Gamma(X, P^sj_*\Omega_U^*)$.*

Proof. For any complex of sheaves K on $\tilde{X}$, one has $R\Gamma(X, Rf_*K) = R\Gamma(\tilde{X}, K)$. Applying $R\Gamma(X, \cdot)$ to 2.4 therefore yields a commutative diagram

$$\begin{array}{ccc} R^*\Gamma(\tilde{X}, P^s\tilde{j}_*\Omega_U^*) & \longrightarrow & R^*\Gamma(X, P^sj_*\Omega_U^*) \\ & \searrow & \swarrow \\ & H^*(U) = R^*\Gamma(U, \Omega_U^*) & \end{array}$$

□

2.6. By Hironaka [10], starting from (X, V) one can construct a sequence of blow-ups (X_i, V_i) ($0 \leq i \leq r$), with $(X_0, V_0) = (X, V)$, such that (X_{i+1}, V_{i+1}) is obtained from (X_i, V_i) in the same way as $(\tilde{X}, V_1)$ is obtained from (X, V) in 2.3, and in such a way that $(V_r)_{\text{red}}$ is a normal-crossings divisor in X_r . Let j_i be the inclusion of $U = X - V$ into X_i , and let P denote the pole-order filtration on $j_{i*}\Omega_U^*$ relative to the divisor V_i . Iterating 2.5, one finds that the image of $R^*\Gamma(X, P^s j_*\Omega_U^*)$ in $H^*(U)$ contains the image of $R^*\Gamma(X_r, P^s j_{r*}\Omega_U^*)$. In X_r , the open set U is the complement of the normal-crossings divisor $(V_r)_{\text{red}}$. By definition, $F^s H^*(U)$ is therefore the hypercohomology on X_r of the complex $\sigma_{\geq s}\Omega_{X_r}^*(\log(V_r)_{\text{red}})$.

One has

$$\sigma_{\geq s}\Omega_{X_r}^*(\log(V_r)_{\text{red}}) \subset P^s j_{r*}\Omega_U^*,$$

so $F^s H^*(U)$ is contained in the image of the hypercohomology of $P^s j_{r*}\Omega_U^*$ on X_r , and Theorem 2 follows.

Remark 9 (2.7). Just as 2.5 was deduced from 2.4 on X , one can refine Theorem 2 by a result on X : for X_r, V_r as in 2.6, there exists a morphism

$$Rf_* F^s \Omega_{X_r}^*(\log(V_r)_{\text{red}}) \rightarrow P^s j_* \Omega_U^*$$

that is the identity on U . More precisely, in the filtered derived category of X there is a morphism

$$Rf_*(\Omega_{X_r}^*(\log(V_r)_{\text{red}}), F) \rightarrow (j_* \Omega_U^*, P)$$

inducing the identity on U . The proof is the same, using 2.4 instead of 2.5.

A variant of results of Ph. Du Bois [7] also says that $Rf_*(\Omega_{X_r}^*(\log(V_r)_{\text{red}}), F)$ is independent of the choice of (X_r, V_r) dominating (X, V) , with X_r proper over X , $X_r - V_r \xrightarrow{\sim} X - V$, and such that $V_r = f^{-1}(Y)$ is a normal-crossings divisor in X_r .

Remark 10 (2.8). Our proof used the fact that in Hironaka's embedded resolution theorem one may choose the successive blow-ups with centers contained in the singular locus of $f^{-1}(V)$. This fact is not stated explicitly in [10], but it follows from the proof.

3. APPLICATIONS AND VARIANTS

3.1. Let X be smooth proper of pure dimension n , let D be a divisor, put $\mathcal{L} := \mathcal{O}(D)$, and set $U := X - D$. The arguments in Section 1 also show that if $H^*(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)}) = 0$ for $0 \leq i < \mu$, then $F^{n-\mu+1} H^*(U) = 0$. By Serre duality, the hypothesis may be reformulated as $H^*(X, \Omega^i \otimes \mathcal{L}^{\otimes-(\mu-i)}) = 0$ for $0 \leq i < \mu$. By Poincaré duality, the conclusion is equivalent to saying that $H_c^*(U)$ has Hodge filtration $\geq \mu$.

If one is interested in a single $H^m(U)$, it suffices to consider the groups $H^q(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)})$ with $q + n - i = m$. In particular, if U is affine, so that $H^m(U) = 0$ for $m > n$, it suffices to consider the groups $H^q(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)})$ with $q + n - i \leq n$, i.e. $0 \leq q \leq i < \mu$. For $\mu = 1$ this gives:

Proposition 11 (3.2). *With the above notation, if U is affine and if $\Omega^n(D)$ has no nonzero global section, then $F^n H^*(U) = 0$.*

Here is a more direct proof of 3.2. Consider again $f : \tilde{X} \rightarrow X$ proper with $f^{-1}(U) \xrightarrow{\sim} U$, where $\tilde{X}$ is smooth and $\tilde{D} := f^{-1}(D)$ is a normal-crossings divisor. By definition,

$$F^n H^i(U) = H^{i-n}(\tilde{X}, \Omega_{\tilde{X}}^n(\log \tilde{D})).$$

This group vanishes for $i < n$, and also for $i > n$ because U is affine. For $i = n$, it is the space of n -forms on U with at most a simple pole along $\tilde{D}$. Since f is an isomorphism above a subset of X of codimension at least 2, such a form α has at most a simple pole along D at the generic points of D , hence everywhere. It must therefore be zero, which proves 3.2.

3.3. To use 3.1, one needs vanishing theorems for the cohomology of the twisted sheaves Ω_X^p . In the case of compact Hermitian symmetric spaces, the results of Bott [3] and Kostant [12] reduce the required vanishing to a combinatorial problem (cf. D. M. Snow [14,15]).

The irreducible compact Hermitian symmetric spaces are necessarily of the form $X := G/P$ with G simple and P a proper maximal parabolic. Their Picard group is $\mathbf{Z}$; we denote by $\mathcal{O}(1)$ its ample generator. Let $n = \dim X$. The invertible sheaf Ω_X^n is antiample, of the form $\mathcal{O}(-k)$; the integer k is computed in A. Borel and F. Hirzebruch [2, §16]. Their table shows that k is the dual Coxeter number of G , and in particular depends only on the type of G :

$$\begin{array}{cccccc} A_l & B_l & C_l & D_l & E_6 & E_7 \\ l+1 & 2l-1 & l+1 & 2l-2 & 12 & 18. \end{array}$$

3.4. The case where X is the Grassmannian $G(r, s)$ of r -dimensional subspaces in a vector space of dimension $r+s$, with $r, s > 0$, is the case where $G = \mathrm{SL}(r+s)$, hence of type A_l with $l = r+s-1$, and where P is the parabolic stabilizing an r -dimensional subspace. Its dimension is $n = rs$.

For every subset $R \subset [1, r+s] \subset \mathbf{Z}$ of cardinality r , let $\bar{R}$ denote its complement and define

$$l(R) := \#\{(i, j) \mid i \in R, j \in \bar{R}, j < i\}.$$

The symmetric group on $[1, r+s]$ is the Weyl group of type A_l , and $l(R)$ is the length of the permutation σ of $[1, r+s]$ that is increasing on $[1, r]$ and on $[r+1, r+s]$, with $\sigma([1, r]) = R$ and $\sigma([r+1, r+s]) = \bar{R}$.

Proceeding as in D. M. Snow [14], one deduces from Bott [3] and Kostant [12] that the following condition is necessary and sufficient for the vanishing of $H^*(X, \Omega^p(-a))$.

3.4.1. For every subset R of cardinality r in $[1, r+s]$, with complement $\bar{R}$ and with $l(R) = p$, one has

$$(R+a) \cap \bar{R} \neq \emptyset.$$

Proposition 12 (3.5). *With the above notation:*

- (i) $H^*(G(r, s), \Omega^p(-a)) = 0$ for $0 < a < l - p$.
- (ii) *The same conclusion holds for $0 < a \leq l - p$, except in the following cases: $H^*(G(3, 3), \Omega^3(-2))$, $H^*(G(2, r), \Omega^{r/2}(1 + r/2))$ for r even, and similarly for $G(r, 2) \cong G(2, r)$.*

Proof. We use criterion 3.4.1. Let $R, \bar{R}$ be as in 3.4 with $l(R) = p$. Put $x = \inf(\bar{R}) - 1$ and $y = \sup(R)$. One has $[1, x] \subset R$, $x+1 \in \bar{R}$, hence $(R+a) \cap \bar{R} \neq \emptyset$ if $0 < a \leq x$; likewise $[y+1, r+s] \subset \bar{R}$, $y \in R$, hence $(R+a) \cap \bar{R} \neq \emptyset$ if $0 < a \leq r+s-y$.

Assume therefore that $a > x$ and $a > r+s-y$. Then

$$\inf((R+a) \cup \bar{R}) = \inf \bar{R} = x+1,$$

$$\sup((R+a) \cup \bar{R}) = \sup(R+a) = y+a,$$

and $(R+a) \cup \bar{R} \subset [x+1, y+a]$ has at most $y-x+a$ elements.

The pairs (i, j) with $j = x+1$, $i \in R - [1, x]$, together with the pairs (i, j) with $i = y$, $j \in \bar{R} - [y+1, r+s]$, satisfy $i \in R$, $j \in \bar{R}$, and $j < i$. The pair (y, x) is counted twice. Hence

$$l(R) \geq (r-x) + (s-(r+s-y)) - 1 = y-x-1.$$

Equality holds if and only if R has the form

$$(3.5.1) \quad R = [1, x] \cup [x+2, r] \cup \{y\}, \quad R \neq [1, r].$$

If condition (i) holds, then

$$0 < a < (r+s-1) - (y-x-1) = (r+s) - (y-x)$$

and therefore

$$\#((R+a) \cup \bar{R}) < (y-x) + [(r+s) - (y-x)] = r+s,$$

so $(R + a) \cap \bar{R} \neq \emptyset$, as required.

To prove (ii), if one of the inequalities $a \leq l - p$ or $l(R) \geq y - x - 1$ is strict, the preceding argument still applies. Thus we may assume, and do assume, that $a = (r + s) - (y - x)$ and $l(R) = y - x - 1$. In particular, R has the form (3.5.1). If $x = 0$, then $a \leq r + s - y$. If $y = r + s$, then $a \leq x$. These cases have already been treated, so we may assume that $x \neq 0$ and $y \neq r + s$. Then $1 \leq x \leq r - 1$ and $r + 1 \leq y \leq r + s - 1$.

Assume $(R + a) \cap \bar{R} = \emptyset$ and show that this leads precisely to the exceptional cases in (ii). The translation $z \mapsto z + a$ sends y to $(r + s) + x$, and if $i \in [1, r + s]$, then either $i + a \in [1, r + s]$ or else $i + a - (r + s) \in [1, x] \subset R$. Let qR and $q\bar{R}$ denote the images of R and $\bar{R}$ in $\mathbf{Z}/(r + s)$. Since $(R + a) \cap \bar{R} = \emptyset$, one has $qR + a \subset q\bar{R}$, hence $qR + a = q\bar{R}$ and $q\bar{R} + a = qR$.

Call the connected components of R (resp. of qR) the largest intervals (resp. images of intervals) contained in R (resp. in qR). The connected components of qR are the images of those of R . By (3.5.1), R has two or three connected components. Translation by a permutes the connected components of qR , sending the one containing y to the next one, namely the one containing x . The complement $q\bar{R}$ has the same number of connected components, permuted in the same way.

If R has three connected components, then the component containing y is reduced to y and every component has one element. The same holds for the components of $\bar{R}$: x is a component by itself and every component of $\bar{R}$ has one element. Hence $r = s = 3$, $R = \{1, 3, 5\}$, and $a = 2$. This is the first exceptional case. One has $l(R) = 3$.

Assume now that R has two connected components. There are two cases: either $x = r - 1$ or $y = r + 1$. If $x = r - 1$, the component containing y is reduced to y and every component has one element. Thus $R = \{1, y\}$ and translation by a permutes 1 and y in $\mathbf{Z}/(r + s)$: one gets $r = 2$, s even,

$$R = \left\{1, 2 + \frac{s}{2}\right\}.$$

If $y = r + 1$, the component of x in $\bar{R}$ is reduced to x ; likewise one gets r even, $s = 2$, and

$$R = \left[1, \frac{r}{2}\right] \cup \left[\frac{r}{2} + 2, r + 1\right].$$

This is the second exceptional case. □

By 3.1 this proposition has the following corollary.

Corollary 13 (3.6). *Let $D \subset G(r, s)$ be the divisor defined by the vanishing of a section of $\mathcal{O}(d)$, and let U be the complementary open subset. Put $\mu = [(r + s - 1)/d]$. Except in the exceptional cases below, $H_c^*(U)$ has Hodge filtration $\geq \mu$. The exceptions are $G(3, 3)$ and $G(2, r) \cong G(r, 2)$ with $d = 1$, where the Hodge filtration is only $\geq \mu - 1$.*

Remark 14 (3.7). In the special case $X = \mathbf{P}^n$, Theorem 2 shows that a cohomology class in $H^*(U, \mathbf{C})$ can be represented by a holomorphic differential form on U with controlled pole order along D . Dimca uses this fact in [5, 6] to relate the Betti numbers and Alexander polynomial of a hypersurface with isolated singularities to the defect of associated linear systems.

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HEISENBERG GROUP AND REALITY

P. DELIGNE

1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ) , with group law

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$T: (x, y, \lambda) \mapsto (-\bar{x}, -\bar{y}, \bar{\lambda}) \quad (1.2)$$

is an automorphism of H .

The coordinate system $x, y, \lambda: H \xrightarrow{\sim} \mathbb{C}^2 \times \mathbb{C}^*$ displays the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section one has $T(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, as its lift by this section, a one-parameter subgroup of H , which we again denote by D . The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$, where

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H , and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which is identified set-theoretically with $\mathbb{R}^2 \times \mathrm{U}^1$.

It is known, by the Stone-von Neumann theorem [2], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V in which $\lambda \in \mathrm{U}^1$ acts by multiplication by λ . In the Schrödinger model of V , one has $V = L^2(\mathbb{R})$ and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (X, 0, 1): f(z) &\mapsto f(z + X), \\ (0, y, 1): f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of V one obtains a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V , compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [3], D. Shale [1]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathcal{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V , are the L^2 functions on $\mathbb{R}$ which extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\mathrm{Im} z| < A_1$, one has for every $A_2 \in \mathbb{R}$

$$|f(z)| = O(\exp(-A_2|z|)).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol} one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V . If one did, the projective action of $\text{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\text{SL}(2, \mathbb{C})$ on V^{hol} ; this action would lift to a genuine action of $\text{SL}(2, \mathbb{C})$, since $\text{SL}(2, \mathbb{C})$ is simply connected; and the projective action of $\text{SL}(2, \mathbb{R})$ would lift to a genuine action. This is not the case: only the double cover of $\text{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

2

Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines in $\mathbb{C}^2$. For $D \in S$, lifted into H as in no. 1, let W_D be the space of holomorphic functions on H satisfying

$$\varphi(\lambda d \cdot h) = \lambda \varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by

$$g: \varphi(h) \mapsto \varphi(hg).$$

For $E \not\supset D$, restriction to E is a bijection

$$r_E: W_D \xrightarrow{\sim} \mathcal{O}(E), \tag{2.1}$$

where $\mathcal{O}(E)$ is the space of holomorphic functions on E .

If R is a holomorphic representation of H , on which each $\lambda \in \mathbb{C}^*$ acts by λ , and if ω is a D -invariant linear form on R , then

$$\omega \mapsto \omega(hw) \tag{2.2}$$

is a morphism from R to W_D .

3. Example

Let V be, as in nos. 1 and 2, the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathcal{S} \subset V$ be the space of its C^∞ vectors, and let $\mathcal{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\text{Im } t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t: f \mapsto \int f(x) \exp(-\pi i x^2/t) dx$$

is defined on V , and even on $\mathcal{S}$. The form ω_t is a holomorphic vector in the dual of V , fixed by D_t . In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t , invariance is expressed by

$$\int [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2), ω_t defines

$$Q_t: V^{\text{hol}} \rightarrow W_{D_t},$$

which extends to V and even to $\mathcal{S}'$.

Let E be the x -axis. The map $r_E \circ Q_t: V \rightarrow \mathcal{O}(E)$ sends $f \in V$ to

$$f(x, t) = r_E \circ Q_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\partial_t f = \frac{i}{4\pi} \partial_x^2 f, \quad \text{i.e.} \quad \left(\partial_t - \frac{i}{4\pi} \partial_x^2 \right) f(x, t) = 0.$$

For real t , the form ω_t remains defined on $\mathcal{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $\int f(x) dx$. The homomorphism Q_t remains defined on V^{hol} . For $t = 0$, $r_E \circ Q_0$ is the identity in the Schrödinger model.

Assume $\text{Im}(t) > 0$. Let $\bar{D}_t$ be the complex conjugate line of D_t . In the affine line $S - \{\bar{D}_t\}$, the “half-plane” $\{D_u \mid \text{Im } u > 0\}$ is a disk centered at D_t . The map $r_{\bar{D}_t} \circ Q_t: V \rightarrow W_{D_t} \rightarrow \mathcal{O}(D_t)$ gives the holomorphic model of V : the action of H is given by

$$\begin{aligned} d \in D_t: f(x) &\mapsto f(x + d), \\ e \in D_t: f(x) &\mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned} \quad (3.1)$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d) \overline{f(d')} \exp(2\pi i \langle d, d' \rangle)$$

is invariant under $H_{\mathbb{R}}$ and identifies V , via $r_{\bar{D}_t} \circ Q_t$, with the space of functions f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i \langle d, \bar{d} \rangle)$, the commutator of d and $\bar{d}$, is

$$\exp(2\pi i \langle d, \bar{d} \rangle) = \exp(2\pi i (t - \bar{t}) |z|^2).$$

4

Fix E . The isomorphisms r_E in (2.1) allow one to regard the W_D with $D \not\supset E$ as forming a family of representations of H acting on the fixed space $\mathcal{O}(E)$, the action of H depending holomorphically on D . We spell out the dependence on E of r_E . Choose coordinates as in no. 1, with E the x -axis. Let E' be generated by $(1, u)$ and let D be generated by $(-t, 1)$. We take x as coordinate on both E and E' . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi i u x x')$. Thus $r_{E'} \circ r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$ with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'} \left(\frac{x}{1 + tu} \right) = \exp \left(\frac{\pi t i}{1 + tu} x^2 \right) f_E(x). \quad (4.1)$$

This formula is holomorphic in t : the W_D form a holomorphic vector bundle $\mathcal{V}$ of infinite dimension on S , on which H acts.

To understand (4.1), one may note that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \not\supset D$ and of the coordinate z on E :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| < O(\exp(A|z|^2)). \quad (4.2)$$

Let $W_D^? \subset W_D$ be defined by this condition. The $W_D^?$ form a holomorphic subbundle $\mathcal{V}^?$ in holomorphic representations of H inside $\mathcal{V}$.

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f :

- (i)_T There is a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$ and the initial condition $g(z, 0) = f(z)$;
- (ii)_T $|f(z)| < O(\exp(|z|^2/4T))$;
- (iii)_T $\iint |f(z)\overline{f(\bar{z})}| \exp(-|z|^2/2T) |dz \wedge d\bar{z}| < \infty$.

Then, if $T < T_1$, each of (i)_T, (ii)_{T_1}, (iii)_{T_1} implies each of (i)_{T_1}, (ii)_T, (iii)_T.

Proof. (i)_T $\Rightarrow$ (ii)_T. The heat equation remains valid for the derivatives of g , and therefore

$$\partial_t^n (z^{2n} g) = \partial_z^{2n+1} g = \partial_t^n (z \partial_z^{2n} g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g / n!| < O(1/T_2^n),$$

hence

$$|\partial_z^{2n} g| = O(n! / T_2^n),$$

and an analogous estimate holds for $\partial_z^{2n+1} g$. The Taylor series of f then gives

$$|f(z)| < \sum_n \binom{2n}{n} \frac{|z|^{2n}}{(2n)!} \frac{n!}{T_2^n}.$$

One has

$$\frac{1}{2n+1} \binom{2n}{n} \leq \frac{2^{2n}}{(n!)^2},$$

so

$$|f(z)| < O\left(\sum_n \frac{1}{n!} \left(\frac{|z|^2}{4T}\right)^n\right),$$

and this sum is $\exp(|z|^2/4T)$.

(ii)_T $\Rightarrow$ (i)_T. It suffices to take g given by the heat kernel:

$$g(z, t) = \frac{1}{\sqrt{4\pi t}} \int_{i\mathbb{R}} f(z + x) \exp(-x^2/4t) dx.$$

Changing x to $-ix$ changes the orientation of the cycle $i\mathbb{R}$ and the sign of the integrand, and hence does not change the integral. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: (ii)_{T_1} $\Rightarrow$ (iii)_T, and (iii)_{T_1} $\Rightarrow$ (ii)_T is verified by writing $f(z)$ as the mean of its values on a disk centered at z and of radius $1/|z|$, and by estimating that mean by Cauchy–Schwarz.

6

In homage to the Stone–von Neumann theorem, the bundle $\mathscr{W}$ on S is, projectively, endowed with a holomorphic connection for which the action of H is horizontal.

Let us make precise what “projectively” means.

On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L together with an isomorphism $L^{\otimes 2} \xrightarrow{\sim} \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic, but the isomorphism is not unique: there is a ± 1 ambiguity. This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathscr{V}$, but on

$$\mathscr{V}(-\tfrac{1}{2}) := \mathscr{V} \otimes \mathcal{O}(-\tfrac{1}{2}) \quad (6.1)$$

that there is a connection

$$\nabla: \mathscr{V}(-\tfrac{1}{2}) \rightarrow \mathscr{V}(-\tfrac{1}{2}) \otimes \Omega^1.$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and ∇ is a connection on $\mathscr{V} \otimes \mathcal{O}(-\frac{1}{2})'$, its transport to $\mathscr{V} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism of square roots

$$\alpha: \mathcal{O}(-\tfrac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\tfrac{1}{2})''$$

does not depend on the choice of α , which can be changed locally only by a constant ± 1 . Thus the notion of a “connection on $\mathscr{V}(-\frac{1}{2})$ ” is well defined, independently of the global existence or choice of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D . For s a local section of $\mathscr{V}(-\frac{1}{2})$, we must define $\nabla_t s$ in the fiber of $\mathscr{V}(-\frac{1}{2})$ at D . We proceed as follows. After making an auxiliary choice u , we define $\nabla_t^u s$ for s a local section of $\mathscr{V}$, or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u , with dependence of the form

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) + a \cdot s(D),$$

where a takes opposite values for $\mathscr{V}$ and for $\mathcal{O}(-\frac{1}{2})$. For $s = s_1 \otimes s_2$, with s_1 a local section of $\mathscr{V}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u .

Let $D \in S$ be a line in $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t: D \rightarrow \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D) \otimes D^*$. To $u' \otimes v' \in (\mathbb{C}^2/D) \otimes D^*$ one attaches $t(d) = \langle u, d \rangle v'$, where u is any representative of u' . Our auxiliary datum will be a lift of t , i.e. a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it as $t = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{z} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let ∂_z be the corresponding right-invariant vector field. One has

$$[\partial_x, \partial_y] = -\langle x, y \rangle'$$

with, on the right, the bracket in the Lie algebra of H . For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one obtains

$$[\partial_x, \partial_y] = -2\pi i \langle x, y \rangle (\partial_0, \partial_0, 1). \quad (6.2)$$

Let $\mathscr{W}_\bullet$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta(0, 0, 1) = 1$. In $\mathscr{W}_\bullet$, one deduces from (6.2) that, for $d \in D$,

$$[\partial_u^2, \partial_d] = 2 \cdot 2\pi i \langle u, d \rangle \partial_u = \langle u, d \rangle \partial_u = \langle \bar{t}(d) \rangle. \quad (6.3)$$

Compute to first order around D . For $\varepsilon^2 = 0$ and $\varphi_0(g) \in W_D$, so that $\partial_d \varphi_0 = 0$, one has

$$\partial_{d+\varepsilon t(d)}[\varphi_0(g) + \varepsilon \partial_u^2 \varphi_0(g)/(4\pi i)] = \varepsilon([\partial_d, \partial_u^2] \varphi_0(g) + \partial_d \partial_u^2 \varphi_0(g)) = 0,$$

since $\partial_d(\partial_u^2 \varphi_0/(4\pi i)) = -[-\partial_u^2/4\pi i, \partial_d] \varphi_0$; apply (6.3). To first order around D , the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi_0(g) + \varepsilon(\partial_u^2/4\pi i) \varphi_0(g)$$

is therefore a section of the bundle $\mathscr{V}$. If $\varphi(g, D')$ is a local section of $\mathscr{V}$, this gives the derivative ∇_t^u in W_D :

$$\nabla_t^u = \partial_t \varphi((g, D)) + (\partial_u^2/4\pi i) \varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on u . If u is replaced by $u + x$ with $x \in D$, then in $\mathscr{W}_\bullet$,

$$\partial_{u+x}^2 = (\partial_u + \partial_x)^2 = \partial_u^2 + 2\partial_u \partial_x + [\partial_x, \partial_u] = \partial_u^2 + 2\partial_u \partial_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For $\varphi \in W_D$, one has $\partial_x \varphi = 0$, and (6.6) simplifies to

$$\partial_{u+x}^2 \varphi = \partial_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

hence

$$\partial_{u+x}^2 \varphi/(4\pi i) = \partial_u^2 \varphi/(4\pi i) + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of u also allows one to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d \rangle \partial_u$ to obtain a homogeneous function of degree 1 on D . When u is changed to $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \partial_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t(s^{-1/2}) = -\frac{1}{2} \nabla_t(s) \cdot s^{-3/2},$$

and, for dependence on u , where now s is a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2} \langle u, x \rangle s. \quad (6.9)$$

The dependences on u in (6.8) and (6.9) cancel, giving a derivation ∇_t for $\mathscr{V}(-\frac{1}{2})$.

Since the connection ∇ is defined in terms of right-invariant vector fields on H , it commutes with the action of H .

7

We now write the connection of no. 6 in coordinates. Choose coordinates as in no. 1 and let E be the x -axis. On S , $\mathcal{O}(1)$ has a section s with a simple zero at E . It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1} . By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{V}$. We compute it.

Let D_t be the line generated by $(-t, 1)$. Via r_E in (2.1), a local section of $\mathcal{V}$ is identified with a function $f(z, t)$ holomorphic in z and t . We compute its derivative with respect to t . In no. 6, one may take for t the map

$$D_t \rightarrow \mathbb{C}^2, \quad (-tz, z) \mapsto (-1, 0)z;$$

this is $d \mapsto -\langle (1, 0), d \rangle (1, 0)$.

The corresponding derivation of $d^* \in \mathcal{O}(1)$ annihilates s , so one simply has

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the local horizontal sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through any $s \in W_D^?(-\frac{1}{2})$ there passes a local horizontal section of $\mathcal{V}(-\frac{1}{2})$. It is automatically a section of $\mathcal{V}^?(-\frac{1}{2})$. It exists in a neighborhood of D whose size increases as s grows more slowly. Notice that in infinite dimension a connection need not define a local trivialization, even if through every point there passes a local horizontal section.

The connection ∇ on $\mathcal{V}(-\frac{1}{2})$ defines one on the corresponding projective-space bundle

$$\mathbb{P}(\mathcal{V}) = \mathbb{P}(\mathcal{V}(-\frac{1}{2})).$$

Its local horizontal sections are the images of nonzero horizontal sections of $\mathcal{V}(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{V})$ over S has no global horizontal section.

The argument of no. 1 would already lead one to expect this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{V})$.

First proof. Divide S into two hemispheres S_1 and S_2 and choose $\mathcal{O}(-\frac{1}{2})$ on each S_i . On $S_1 \cap S_2$ one has, locally, an isomorphism

$$\alpha: \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2,$$

defined up to sign; but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$, with monodromy -1 . If there were global horizontal projective sections of $\mathbb{P}(\mathcal{V})$, there would be horizontal sections f_1 and f_2 of $\mathcal{V}(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would agree up to sign. On $S_1 \cap S_2$, this would allow us to normalize

$$\alpha: \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$$

by requiring $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x -axis and D the y -axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D . By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{V})$ passing through f , it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| < O(\exp(\varepsilon|z|)). \quad (8.1)$$

If E' is another line, then by (4.1), the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| < O(\exp(\varepsilon|z|^2)), \quad (8.2)$$

with a suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$. Thus, in every direction except those for which az^2 is purely imaginary, g has quadratic exponential decrease. In all directions it has, uniformly, at most quadratic exponential growth. By Phragmén–Lindelöf, g has quadratic exponential decrease in every direction; hence it is zero, and $f = 0$.

9

In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. For the commutator of d and $\bar{d}$ one has

$$(d, \bar{d}) = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^* :$$

the commutator is real positive, and

$$(d, \bar{d}) > 1 \iff \text{Im}(t) > 0.$$

The lines D for which $(d, \bar{d}) > 1$ for a generator d of D therefore form a disk in S , said to be attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ on which $\langle \cdot, \cdot \rangle$ is real form a homogeneous space under $\text{SL}(2, \mathbb{C})$, with $\text{SL}(2, \mathbb{R})$ the stabilizer of $\mathbb{R}^2$. Each defines a real structure H_L on H , hence a complex conjugation σ_L with fixed points $L \cdot \text{U}^1$.

The disks in S likewise form a homogeneous space under $\text{SL}(2, \mathbb{C})$, the stabilizer of the disk associated with $H_{\mathbb{R}}$ being $\text{SL}(2, \mathbb{R})$. The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S .

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

10

The computations of no. 3 admit the following interpretation. Let X be an open disk in S , corresponding to a real structure L , hence to real forms H_L of H and $\mathrm{SL}(L)$ of $\mathrm{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$ has an H_L -invariant pre-Hilbert structure, and its completion

$$\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})^\wedge$$

is the irreducible unitary representation of H_L . One has

$$\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0}) \subset \Gamma(X, \mathcal{V}|_{V=0}) \subset \Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})^\wedge.$$

The double cover $\mathrm{SL}(L)'$ of $\mathrm{SL}(L)$ acts on $\mathcal{V}(-\frac{1}{2})|_X$, compatibly with its action on $\mathcal{O}(-\frac{1}{2})$, and therefore acts on $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$. This is the action which normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)'$ on H_L .

The subrepresentation $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$ may be interpreted as the space of real-analytic vectors for the action of $\mathrm{SL}(L)'$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of those $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

11

Everything above generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle \cdot, \cdot \rangle$ that is real on $\mathbb{R}^{2n}$; replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle));$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \bar{d} \rangle| > 1$ for $d \neq 0$ in D . The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ is to be replaced by $\det(D)^{-1/2}$.

12. Remark

Alongside representations of H in which $U^1 \subset \mathbb{C}^*$ acts by multiplication by λ , consider similarly representations in which the action is by λ^{-1} . One defines, as in no. 2, a holomorphic bundle $\mathcal{V}'$ of such representations over S ; on $\mathcal{V}'(-\frac{1}{2})$ there is again a connection, given by a heat equation.

On a connected and simply connected open set U of S , consider $\mathcal{W} = \mathcal{V}(-\frac{1}{2})$, its connection, and the action of H . In $\Gamma(U, \mathcal{W})|_{V=0}$, for each $x \in U$ one has a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(\xi)$ for $f \in \mathcal{V}$. For each $x \in U$, there exists a vector $v(x)$ fixed by D_x ; in the Schrödinger model, when x corresponds to $E =$ the x -axis, this is the constant 1. It is unique up to a factor.

Fix the real structure $H_{\mathbb{R}}$, hence a real structure $S_{\mathbb{R}} \subset S$, a circle, and a decomposition of $S - S_{\mathbb{R}}$ into two disks U_1 and U_2 . Then:

- (a) The complex conjugate of $\Gamma(U_1, \mathcal{W})|_{V=0}$ is $\Gamma(U_2, \mathcal{W}')|_{V=0}$.
- (b) The Hermitian form on $\Gamma(U_1, \mathcal{W})|_{V=0}$, invariant under $H_{\mathbb{R}}$, can be reinterpreted as a bilinear pairing between $\Gamma(U_1, \mathcal{W})|_{V=0}$ and $\Gamma(U_2, \mathcal{W}')|_{V=0}$ that is invariant under H .

Question. Let C be a Jordan curve in S , separating $S - C$ into two open sets U_1 and U_2 . Is there a unique separating H -invariant bilinear form pairing $\Gamma(U_1, \mathscr{W})|_{V=0}$ and $\Gamma(U_2, \mathscr{W}')|_{V=0}$, induced by a perfect duality between $\Gamma(U_1, \mathscr{W})|_{V=0}$ and $\Gamma(U_2, \mathscr{W}')|_{V=0}$?

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Real Mixed Hodge Structures

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Real mixed Hodge structures (§2) form a rigid abelian $\otimes$ -category, and the functor “underlying real vector space” is a fiber functor. The functor Gr^W is another. We propose to compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposed filtrations

1.1

This section is a variation on Section 1.2 of P. Deligne, *Théorème de Hodge II*, Publ. Math. IHES **40**, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ equipped with three opposed filtrations W , F , $\bar{F}$. Here “filtration” means “finite filtration” (loc. cit. 1.1.4); the filtration W is assumed increasing, and the filtrations F , $\bar{F}$ decreasing. The opposition condition is written

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q \mathrm{Gr}_n^W A = 0 \quad \text{for } n \neq p + q$$

(loc. cit. 1.2.7, 1.2.13).

For every bigraded object $B = \bigoplus B^{p,q}$, we shall write W , F_W , and $\bar{F}_W$ for the three opposed filtrations

$$\begin{aligned} W_n &= \bigoplus_{p+q \leq n} B^{p,q}, \\ F_W^i &= \bigoplus_{p \geq i} B^{p,q}, \\ \bar{F}_W^i &= \bigoplus_{q \geq i} B^{p,q}. \end{aligned}$$

With this notation, to say that the three filtrations W , F , and $\bar{F}$ of A are opposed is equivalent (loc. cit. 1.2.6) to saying that each $\mathrm{Gr}_n^W A$ admits a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations of $\mathrm{Gr}_n^W A$ induced by F and $\bar{F}$ are given by

$$\begin{aligned} F^i \mathrm{Gr}_n^W A &= \bigoplus_{\substack{p+q=n \\ p \geq i}} A_W^{p,q}, \\ \bar{F}^i \mathrm{Gr}_n^W A &= \bigoplus_{\substack{p+q=n \\ q \geq i}} A_W^{p,q}. \end{aligned}$$

In other words, F and $\bar{F}$ induce on $\mathrm{Gr}_n^W A$ the filtrations F_W and $\bar{F}_W$.

Set

$$A_F^{p,q} := (W_{p+q} \cap F^p) \cap \left((W_{p+q} \cap \bar{F}^q) + \sum_{i>0} (W_{p+q-i} \cap \bar{F}^{q+1-i}) \right)$$

and, interchanging the roles of F and $\bar{F}$ and of p and q ,

$$A_{\bar{F}}^{p,q} := (W_{p+q} \cap \bar{F}^q) \cap \left((W_{p+q} \cap F^p) + \sum_{i>0} (W_{p+q-i} \cap F^{p+1-i}) \right).$$

For $p+q=n$, the projection of W_n onto $\mathrm{Gr}_n^W A$ induces an isomorphism from $A_F^{p,q}$ to $A_W^{p,q}$: loc. cit. 1.2.8, restricted to the special case 1.2.11, where $A_F^{p,q}$ is denoted $A_0^{p,q}$; note that in the formula defining $A_F^{p,q}$, one may replace the sum over $i>0$ by the same sum over $i \geq 2$. It follows (loc. cit.) that the $A_F^{p,q}$, respectively the $A_{\bar{F}}^{p,q}$, form a bigrading of A , and that the sum a_F , respectively $a_{\bar{F}}$, of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

is a bifiltered isomorphism from $(A; W, F)$ to $(\mathrm{Gr}^W A; W, F_W)$, respectively from $(A; W, \bar{F})$ to $(\mathrm{Gr}^W A; W, \bar{F}_W)$.

Lemma 1.1. *The automorphism $d = a_{\bar{F}} a_F^{-1}$ of $\mathrm{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r \leq p \\ s < q}} A_W^{r,s}. \quad (1.1.1)$$

Passing to the associated graded, the isomorphisms a_F and $a_{\bar{F}}$ induce the identity of $\mathrm{Gr}^W A$. Hence $\mathrm{Gr}^W(d) = 1$.

For each subset I of $\mathbb{Z}^2$, and for B a bigraded object, set $B_I = \bigoplus_{(p,q) \in I} B^{p,q}$. Set

$$\begin{aligned} I(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } r < p)\}, \\ J(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } s < q)\}, \\ K(p,q) &= \{(r,s) \mid r+s \leq p+q \text{ and } r \geq p\}. \end{aligned}$$

$$\begin{array}{llll} I(p,q) : & & \times & \circ \\ & & \times \times & \circ \circ \\ J(p,q) : & & \times \times \circ & (p,q) \quad \circ \circ \times (p,q) \\ K(p,q) : \text{ circled} & \circ & \circ & \times \circ \\ & \circ \circ & \circ & \times \times \circ \end{array}$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects of A , the first subobject is $A_F^{I(p,q)}$ and the second is $A_{\bar{F}}^{J(p,q)}$. This follows from the fact that a_F , respectively $a_{\bar{F}}$, is a bifiltered isomorphism. Thus $A_F^{p,q} \subset A_F^{I(p,q)}$, whence

$$dA_F^{p,q} = a_{\bar{F}} a_F^{-1} A_F^{p,q} \subset A_{\bar{F}}^{I(p,q)},$$

and $dA_F^{p,q} \subset A_{\bar{F}}^{J(p,q)}$. Since $\mathrm{Gr}^W(d) = 1$, one has equality $dA_F^{I(p,q)} = A_{\bar{F}}^{I(p,q)}$. Interchanging F and $\bar{F}$, one similarly obtains $d^{-1}A_{\bar{F}}^{J(p,q)} = A_F^{J(p,q)}$, hence $dA_{\bar{F}}^{J(p,q)} = A_F^{J(p,q)}$, and

$$dA_W^{p,q} = dA_W^{I(p,q)} \cap dA_W^{J(p,q)} = A_W^{I(p,q) \cap J(p,q)} = A_W^{K(p,q)} = \bigoplus_{\substack{r \leq p \\ s < q}} A_W^{r,s}.$$

We conclude by using $\mathrm{Gr}^W(d) = 1$.

Proposition 1.2. *The functor $A \mapsto \mathrm{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ equipped with three opposed filtrations W , F , $\bar{F}$, to the category of objects of $\mathcal{A}$ equipped with a finite bigrading and an automorphism d satisfying (1.1.1).*

The isomorphism $a_F : A \xrightarrow{\sim} \mathrm{Gr}^W A$ respects the filtration W , carries F to F_W , and carries $\bar{F}$ to $d^{-1}(F_W)$: one has $a_{\bar{F}} = d^{-1}a_F$. Thus the functor Gr^W has a left inverse, namely the functor

$$s : (B = \bigoplus B^{p,q}, d) \longmapsto (B, W, F_W, d^{-1}(F_W)),$$

equipped with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \mathrm{Gr}^W A$.

Conversely, start from a bigraded object B equipped with an automorphism d , and define W , F , and $\bar{F}$ as above. These filtrations are opposed as soon as $\mathrm{Gr}^W(d) = 1$. For these filtrations,

$$B_F^{p,q} = B^{K(p,q)} \cap d^{-1}(B^{I(p,q)}).$$

If d satisfies (1.1.1), then $dB^{p,q} \subset B^{I(p,q)}$ and $d^{-1}B^{p,q} \subset B^{J(p,q)}$. Since $(B^{p,q})$ and $(B_F^{p,q})$ are bigradings of B , one has equality $B_F^{p,q} = B^{p,q}$. The same argument, applied to d^{-1} and to the bigrading $\bar{F}^{p,q}$, shows that $d^{-1}B^{\bar{F}^{p,q}} = B^{p,q}$. It follows that the evident isomorphism from B to $\mathrm{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \mathrm{Gr}^W \circ s$.

Remark 1.3. If 2 is invertible, d has a unique square root $d^{1/2}$ which still satisfies (1.1.1). Set

$$a := d^{1/2}a_F = d^{-1/2}a_{\bar{F}}.$$

Then $a_{\bar{F}} = d^{1/2}a$ and $a_F = d^{-1/2}a$, so that a carries F , respectively $\bar{F}$, to the filtration $d^{1/2}(F_W)$, respectively $d^{-1/2}(\bar{F}_W)$, of $\mathrm{Gr}^W A$. The isomorphism a is the same for A equipped with $W, F, \bar{F}$ and for A equipped with $W, \bar{F}, F$.

1.2

These constructions are compatible with tensor product. If $\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{A}$ is an exact biadditive functor, we extend it to filtered objects by setting

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a(A_1) \otimes F^b(A_2) \subset A_1 \otimes A_2.$$

The associated graded functor is compatible with $\otimes$, and for bifiltered objects the isomorphism

$$\mathrm{Gr}^{W_1 \otimes W_2}(A_1 \otimes A_2) \xrightarrow{\sim} \mathrm{Gr}^{W_1}(A_1) \otimes \mathrm{Gr}^{W_2}(A_2)$$

carries the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2 .

For vector spaces over a field, this can be deduced from the fact that a bifiltration always underlies a bigrading. It follows that if W_i , F_i , and $\bar{F}_i$ are three opposed filtrations on A_i for $i = 1, 2$, then the filtrations $W_1 \otimes W_2$, $F_1 \otimes F_2$, and $\bar{F}_1 \otimes \bar{F}_2$ on $A_1 \otimes A_2$ are again opposed.

The formation of the bigradings $A_F^{p,q}$ and $A_{\bar{F}}^{p,q}$ of A , and $A_W^{p,q}$ of $\mathrm{Gr}^W(A)$, as well as the isomorphisms $a_F, a_{\bar{F}} : A \xrightarrow{\sim} \mathrm{Gr}^W(A)$ and the automorphism d of $\mathrm{Gr}^W(A)$, is compatible with $\otimes$.

1.3

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k , and let $\mathcal{C}$ be the category of vector spaces equipped with three opposed filtrations W , F , and $\bar{F}$. This is a rigid abelian $\otimes$ -category with $\text{End}(1) = k$. It has the fiber functor ω_0 , the underlying vector space functor, and hence is neutral Tannakian. We also have the fiber functor $\text{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\bar{F}}$ are isomorphisms of fiber functors $\omega_0 \xrightarrow{\sim} \omega_W$. Put $G = \text{Aut}(\omega_W)$. The functorial automorphism d defines a k -point of the pro-algebraic group G .

The functorial bigrading of $\text{Gr}^W(V)$ by the $V_W^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow G.$$

It is a section of the projection

$$\text{pr} : G \rightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion in $\mathcal{C}$ of the category of bigraded vector spaces, via

$$B \longmapsto (B, W, F_W, \bar{F}_W).$$

I wish to normalize α so that $\alpha(\lambda, \mu)v = \lambda^p \mu^{-q}v$ for $v \in V_W^{p,q}$.

By Proposition 1.2, G is also the algebraic group of automorphisms of the underlying-vector-space fiber functor on the category of bigraded vector spaces equipped with an automorphism d satisfying (1.1.1). We are going to “compute” it when k has characteristic 0.

Let $\mathcal{L}$ be the Lie algebra over k freely generated by elements $D^{i,j}$ for $i, j < 0$, and equip it with the unique bigrading for which $D^{i,j}$ has bidegree (i, j) . The $W_n(\mathcal{L})$ are ideals of $\mathcal{L}$, and the quotients $\mathcal{L}/W_n(\mathcal{L})$ are nilpotent Lie algebras. We shall write U for the pro-algebraic group which is the projective limit of the unipotent algebraic groups $\exp(\mathcal{L}/W_n(\mathcal{L}))$ with Lie algebras $\mathcal{L}/W_n(\mathcal{L})$. Let $\mathbb{G}_m \times \mathbb{G}_m$ act on $\mathcal{L}$ so that (λ, μ) acts on an element D of degree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects the filtration W and therefore gives an action of $\mathbb{G}_m \times \mathbb{G}_m$ on U .

Construction 1.4. The group G is the semidirect product $(\mathbb{G}_m \times \mathbb{G}_m) \ltimes U$.

To make the semidirect product act on a finite-dimensional vector space V is the same as to make $\mathbb{G}_m \times \mathbb{G}_m$ and U act, subject to a compatibility condition relating the two actions. The action of $\mathbb{G}_m \times \mathbb{G}_m$ corresponds to a bigrading, namely the one for which $(\lambda, \mu)v^{p,q} = \lambda^p \mu^{-q}v^{p,q}$. The action of U corresponds to an action of $\mathcal{L}$, zero on some W_n and nilpotent, i.e. to the data of endomorphisms $D^{i,j}$, for $i, j < 0$, such that every composite of $D^{i,j}$ of sufficiently large absolute weight is zero. Compatibility says that $D^{i,j}$ has bidegree (i, j) , and this property implies the required vanishing just stated. The data of the $D^{i,j}$ is equivalent to the data of $D := \sum D^{i,j}$. Altogether, the action of the semidirect product on V is identified with the data of a bigrading and of an endomorphism D of V satisfying

$$DV^{p,q} \subset \bigoplus_{i < 0, j < 0} V^{p+i, q+j}.$$

Putting $d = \exp(D)$, this is again identified with the data of a bigrading and an automorphism d satisfying (1.1.1). This construction is compatible with tensor products, giving the desired isomorphism

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes U \xrightarrow{\sim} G \xrightarrow{\sim} \text{Aut}(\omega_W).$$

This isomorphism is of course compatible with α and pr above.

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space equipped with an increasing filtration W , whose complexification $V_{\mathbb{C}}$ is equipped with a decreasing filtration F . One requires that on $V_{\mathbb{C}}$ the filtration $W_{\mathbb{C}}$ obtained from W by extension of scalars, often denoted simply by W , together with F and its complex conjugate $\bar{F}$, form a system of three opposed filtrations.

It is equivalent to give a real vector space or to give a complex vector space equipped with a real structure, i.e. with complex conjugation, an antilinear involution. Under this dictionary, to give a real mixed Hodge structure is equivalently to give a complex vector space equipped with three opposed filtrations W , F , and $\bar{F}$, and with a complex conjugation which respects W and interchanges F and $\bar{F}$. By Proposition 1.2, the functor Gr^W induces an equivalence from the category of these objects to the category of bigraded complex vector spaces $V = \bigoplus V^{p,q}$ equipped with an automorphism d satisfying (1.1.1) and a complex conjugation c which interchanges $V^{p,q}$ and $V^{q,p}$, and for which the complex conjugate $\bar{d}$ of d is d^{-1} :

$$\bar{d} = cdc^{-1} = d^{-1}.$$

The category $\mathcal{H}$ of real mixed Hodge structures is a rigid abelian $\otimes$ -category, with $\mathrm{End}(1) = \mathbb{R}$. It has the fiber functor ω_0 , the underlying vector space functor, and hence is neutral Tannakian. We also have the fiber functor $\mathrm{Gr}^W =: \omega_W$. With the notation of Remark 1.3, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \mathrm{Gr}^W(V_{\mathbb{C}})$$

respects the real structures of both sides because it depends symmetrically on F and $\bar{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, take $k = \mathbb{C}$, and let G_2 be the real form of G defined by the following complex conjugation:

- a) on $\mathbb{G}_m \times \mathbb{G}_m$, it is $(\lambda, \mu) \mapsto (\bar{\mu}, \bar{\lambda})$;
- b) on U , it is the one compatible with the complex conjugation of $\mathcal{L}$ given by

$$D^{i,j} \mapsto -D^{j,i}.$$

As in 1.1, to make G_2 act on a real vector space V is equivalent to giving:

- a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ of $V_{\mathbb{C}}$, such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another, and
- b) an automorphism d of $V_{\mathbb{C}}$, such that $\bar{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{\substack{i < p \\ j < q}} V^{i,j}.$$

Hence:

Proposition 2.1. *The fiber functor ω_W induces an equivalence from $\mathcal{H}$ to the category of representations of G_2 : one has $G_2 = \text{Aut}(\omega_W)$.*

The inverse functor Φ is obtained as follows. From a representation of G_2 on V , one obtains a bigrading of $V_{\mathbb{C}}$ and an automorphism d . To this data one attaches the real vector space V equipped with the filtration W whose complexification is the filtration W associated with the bigrading of $V_{\mathbb{C}}$, and with the filtration

$$F := d^{-1/2}(F_W)$$

of its complexification. The isomorphism $\Phi \circ \text{Gr}^W \xrightarrow{\sim} \text{Id}$ is the functorial isomorphism a .

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The present article reproduces, with minimal changes, the appendix bearing the same title to the preprint of E. Cattani and A. Kaplan, “On the $\text{SL}(2)$ -orbits in Hodge theory” (IHES 1982). A different presentation of part of the results is given on pp. 470–473 of the article by E. Cattani, A. Kaplan, and W. Schmid, “Degeneration of Hodge structures” (Annals of Math. **123** (1986), 457–535).

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What are motives for?

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The first of the “standard conjectures” (Grothendieck [19], Kleiman [20]), the conjecture of Lefschetz type, asks that certain cohomology classes be “direct” images of algebraic cycles. In any case, if motives are the “direct factors” of the cohomology of algebraic varieties X , defined by projectors, namely algebraic cycles on $X \times X$, their definition is reasonable only if one has enough algebraic cycles. On this problem—the construction of interesting algebraic cycles—progress has been meager.

Grothendieck attempted to make a catalogue of projective constructions of cycles (apparent contours), cf. [19, p. 197]. For those whose cohomology class has been computed, this class is expressed in terms of Chern classes of evident vector bundles. Although I know of no counterexample, it seems unlikely to me that the ring of cycle classes on $X \times X$ generated by the diagonal, by divisors, and by the inverse images of the Chern classes of X under the pr_i ($i = 1, 2$), always contains the cycles required by the first standard conjecture, for instance the Kunneth components of the diagonal.

Rather than try to construct cycles, one may try to construct vector bundles and then take their Chern classes. This is in fact how K. Kodaira and D. C. Spencer (1953) prove the Hodge conjecture for divisors (a Lefschetz theorem): the group $\text{Pic}(X) = H^1(X, \mathcal{O}^*)$ is accessible over $\mathbb{C}$ by the exponential exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \rightarrow 0.$$

Similarly, on an abelian variety it is easier to write the functional equation of theta functions (a cocycle for a line bundle) than to define a theta function itself, which would then give the divisor $\theta = 0$.

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Unfortunately, in higher rank we do not know how to construct vector bundles whose Chern classes are interesting any more than we know how to construct interesting cycles.

If $\ell \in \mathbb{C}$, the Hodge conjecture would provide the desired cycles. In the direction of that conjecture, two difficulties have been discovered. The first is that Atiyah and Hirzebruch showed that it can be true only rationally (cf. Atiyah–Hirzebruch [1]).

The second is that, from the viewpoint of Hodge theory, the class of a codimension d cycle Z on a smooth projective X naturally lives not in $H^{d,d} := H^{d,d} \cap H^{2d}$, but in an extension E^d of this group by the intermediate Jacobian

$$J^d(X) := H^{2d-1}(X, \mathbb{C})/F^d + H^{2d-1}(X, \mathbb{Z}),$$

where F^d denotes the d -th term of the Hodge filtration of $H^{2d}(X, \mathbb{C})$.

In the category of mixed Hodge structures, these are respectively $\text{Hom}(\mathbb{Z}(d), H^{2d}(X))$ and $\text{Ext}^1(\mathbb{Z}(d), H^{2d-1}(X))$. The Hodge conjecture asks that the group of cycle classes map to a finite-index subgroup of the quotient $H^{d,d}$. By contrast, we have no idea what the image in E^d should be. In addition to the aesthetic discomfort caused by this ignorance, it generally makes

P. A. Griffiths's method for proving the Hodge conjecture by induction on the dimension of the variety X is inapplicable. This method is inspired by the one Lefschetz used for surfaces, Lefschetz [24]. The idea is that, if $(H_t)_{t \in \mathbf{P}^1}$ is a pencil of hyperplane sections, constructing the cycle Z on X amounts to constructing the cycles $Z_t = Z \cap H_t$. In good cases: (a) the cohomology class $c \in H^{2d}(X)$ that one wants to be the class of a cycle Z has zero restriction to the H_t ; (b) if Z exists, its cohomology class c determines the class of the Z_t in the intermediate Jacobians $J^d(H_t)$; and (c) the construction of Z amounts to the construction of cycles Z_t whose classes define a given section of the family of the $J^d(H_t)$. One can then conclude if every element of $J^d(H_t)$ is the class of a codimension d cycle homologous to zero. By this method, Zucker [32] proves the Hodge conjecture for cubic hypersurfaces in $\mathbf{P}^5$. Note, however, that if every element of $J^d(H_t)$ is the class of a cycle homologous to zero, then $H^{2d-1}(H_t)$ is of Hodge type $\{(d, d-1), (d-1, d)\}$ (see §1.6); the converse is a consequence of the Hodge conjecture applied to the product of H_t with a suitable abelian variety. Moreover, this surjectivity implies the existence of a curve C_t and an algebraic cycle W_t on $H_t \times C_t$ sending $H^1(C_t)$ onto $H^{2d-1}(H_t)$. One can then expect $H^{2d}(X)$ to be controlled by the H^2 of the surface fibered over $\mathbf{P}^1$ with fibers the C_t , and for an H^2 the Hodge conjecture is available in any case.

The purpose of these notes is to show that, despite this absence of progress on the problem of constructing cycles, the philosophy of motives is a powerful tool.

I thank S. Bloch for his comments on a first version of these notes.

1 Motives

Depending on what one can and wants to do, one has several definitions of motives—or none. One should distinguish pure motives, typically supplied by the cohomology of nonsingular projective varieties, from mixed motives, where open or singular varieties are allowed. The notion of a motive over S (a family of motives parameterized by S) raises still other problems.

For k a field, in all cases one would like the category of motives over k to be a $\mathbb{Q}$ -linear abelian category $\mathcal{M}(k)$ whose Hom groups are finite-dimensional. In the pure case, one would like it to be semisimple and graded by weight. In the mixed case, every motive M should have a finite increasing filtration W with $\mathrm{Gr}_n^W(M)$ pure of weight n . Other essential structures are:

- (a) Every algebraic variety X over k should have motivic cohomology groups $H_{\mathrm{mot}}^i(X)$, objects of $\mathcal{M}(k)$. In a pure case one restricts to nonsingular projective varieties X , and $H_{\mathrm{mot}}^i(X)$ is a pure motive of weight i .
- (b) For each of the usual cohomology theories H , one should have a “realization” functor real_H and isomorphisms $H^*(X) \xrightarrow{\sim} \mathrm{real}_H H_{\mathrm{mot}}^*(X)$.
- (c) A tensor product $\otimes$, with which the realization functors are compatible.

1.1 Tannakian structure

Structure (c) makes it possible to apply the theory of Tannakian categories, invented by Grothendieck to study the formalism of motives.

References: Saavedra [28], Deligne–Milne [12], Deligne [15].

For k of characteristic 0, this theory says that the category of motives over k , with its tensor product, should be equivalent to the category of linear representations of an affine group scheme G over $\mathbb{Q}$. Note that two groups G_1 and G_2 that are inner forms of one another have the same representation categories. The motivic Galois group G is therefore not uniquely determined by $\mathcal{M}(k)$. The theory says that the choice of G and of an equivalence $\omega : \mathrm{Rep}(G) \xrightarrow{\sim} \mathcal{M}(k)$ is equivalent to the choice of an exact tensor-compatible functor from $\mathcal{M}(k)$ to $\mathbb{Q}$ -vector spaces: a fiber functor ω . To this fiber functor there corresponds $G := \underline{\mathrm{Aut}}^{\otimes}(\omega)$, the group scheme of

tensor automorphisms of ω . One may view ω , or rather (ω, G) , as a “cohomology theory with values in $\mathbb{Q}$ -vector spaces.”

For k equipped with an embedding into $\mathbb{C}$, one possible cohomology theory is: “singular cohomology of the topological space of complex points.” For $k = \mathbb{Q}$, another possible choice is: “de Rham cohomology.”

In characteristic p , an example of Serre shows that there cannot be a reasonable cohomology theory with values in $\mathbb{Q}$ -vector spaces: if E is a supersingular elliptic curve, the algebra $\text{End}(E) \otimes \mathbb{Q}$ is a quaternion algebra, hence has no two-dimensional linear representation over $\mathbb{Q}$. From the existence of a cohomology theory with values in vector spaces over an extension of $\mathbb{Q}$, it follows nevertheless that $\mathcal{M}(k)$ must be the category of representations of a suitable gerbe.

Tannakian theory is a linear analogue of a set-theoretic theory, that of the profinite π_1 , also due to Grothendieck (SGA1). The analogy is this: if $\mathcal{T}$ is the category of finite étale covers of a connected scheme S (respectively, a Tannakian category over K), and ω is a fiber functor with values in finite sets (respectively, in finite-dimensional K -vector spaces), then $\mathcal{T}$ is naturally equivalent to the category of finite G -sets (respectively, of representations of G), for G the profinite group $\text{Aut}(\omega)$ (respectively, the affine group scheme $\underline{\text{Aut}}^\otimes(\omega)$). This is the origin of the terminology “motivic Galois group,” since the theory of π_1 becomes Galois theory in the case of spectra of fields.

1.2 Size of the motivic Galois group

In general, the motivic Galois group G is enormous. Except when k is algebraic over a finite field, one expects, for example, that it has $\text{PGL}(2)$ as a quotient. It does not “know” that motives M may form a family depending algebraically on a parameter t .

Although G is enormous, its existence makes it easier to handle “motivic” objects.

Example 1. The category $\text{Ind } \mathcal{M}(k)$ of ind-objects of the category $\mathcal{M}(k)$ of motives over k inherits the tensor product of $\mathcal{M}(k)$. This allows one to define a commutative Hopf algebra in $\text{Ind } \mathcal{M}(k)$ as an object H endowed with a product and coproduct, $H \otimes H \rightarrow H$ and $H \rightarrow H \otimes H$, and also with unit, counit, and antipode, $\mathbf{1} \rightarrow H$, $H \rightarrow \mathbf{1}$, $H \rightarrow H$, satisfying the appropriate axioms. The category of motivic affine group schemes is defined to be the opposite of the category of commutative Hopf algebras in $\text{Ind } \mathcal{M}(k)$. If ω is a fiber functor with values in $\mathbb{Q}$ -vector spaces V , with motivic Galois group $G := \underline{\text{Aut}}^\otimes(\omega)$, then ω induces an equivalence of $\text{Ind } \mathcal{M}(k)$ with the category of linear representations—not necessarily finite-dimensional—of G , and $H \mapsto \text{Spec}(\omega(H))$ is an equivalence from the category of motivic affine group schemes to the category of affine group schemes over $\mathbb{Q}$ endowed with an action of G .

Here are three examples of such objects.

1.3.1. The motivic version of the unipotent completion of $\pi_1(X, x)$, cf. Deligne [14], especially §§5, 7, and 10–13.

1.3.2. The motivic Galois group: it has a motivic version G_{mot} such that, for every fiber functor ω , the corresponding motivic Galois group is obtained from G_{mot} by applying ω . This is a construction valid in every Tannakian category, cf. Deligne [15, §8].

In the analogy with the theory of the profinite π_1 , its analogue is the following: as x varies, the $\pi_1(X, x)$ form a local system on X , or more precisely a group object in the category of pro-objects of the category of étale covers of X . The π_1 at x is obtained by taking the fiber at x .

1.3.3. Let $\mathcal{A}$ be a full subcategory of the category of mixed motives over k , stable under $\otimes$, duals, and subquotients, and let $\mathcal{A}_{\text{ss}}$ be the full subcategory of direct sums of pure motives in $\mathcal{A}$. There is an exact tensor-compatible functor $M \mapsto \text{Gr}^W(M)$ from $\mathcal{A}$ to $\mathcal{A}_{\text{ss}}$. There exists in $\text{Ind } \mathcal{A}_{\text{ss}}$ a motivic pro-unipotent group scheme U , acting functorially on $\text{Gr}^W(M)$ for M in $\mathcal{A}$, such that $M \mapsto \text{Gr}^W(M)$ is an equivalence of $\mathcal{A}$ with the category of representations, in $\mathcal{A}_{\text{ss}}$, of U . This is a consequence of Deligne [15, 8.17].

One may define and handle motivic schemes, torsors, and so on in the same way.

1.3 The abelianized motivic Galois group

The abelianization G^{ab} of the motivic Galois group has a more reasonable size. It is independent of the chosen realization functor, and even of the existence of such a functor. Conjecturally, if $k_1 \subset k_2$ are two algebraically closed fields, G^{ab} is the same for k_1 and k_2 , and the corresponding subcategory of motives is generated by the H_{mot}^1 of abelian varieties of CM type.

In characteristic 0, if motives are taken in the sense of absolute Hodge cycles and one restricts to the subcategory generated by the H^1 of abelian varieties, one can compute G^{ab} . For $k = \overline{\mathbb{Q}}$, one can even compute the quotient of G by the derived group of $\text{Ker}(G \rightarrow \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}))$. See Deligne [13] or Schappacher's lecture at this conference.

For $k = \overline{\mathbb{F}}_p$, every motive should be a direct sum of pure motives and the motivic Galois group G should contain an element F , Frobenius, whose powers are dense in G . In particular, G should be commutative. The conjectural calculation of $\mathcal{M}(\overline{\mathbb{F}}_p)$ was made by Grothendieck (unpublished). See Langlands–Rapoport [23] and Milne's lecture at this conference.

1.4 Abelian varieties and 1-motives

One source for 1.1(a), (b) is the following example. A smooth projective curve X defines its Jacobian $J(X) = \text{Pic}^0(X)$. Abelian varieties up to isogeny over k form a semisimple abelian category with $\text{Hom}(A, B)$ finite-dimensional over $\mathbb{Q}$, and for each usual cohomology theory H , the dual $H_1(X)$ of $H^1(X)$ is obtained from $J(X)$ by applying a “realization” functor. Note that $H_1(X)$ is also the Tate twist $H^1(X)(1)$ of $H^1(X)$. For ℓ -adic cohomology, $\text{real}_{\ell}(A)$ is the Tate module $T_{\ell}(A) = \varprojlim A_{\ell^n}(\bar{k}) \otimes \mathbb{Q}_{\ell}$. For k of characteristic 0 and de Rham theory, $\text{real}_{\text{DR}}(A)$ is the Lie algebra of the universal additive extension of A .

Thus one wants to be able to identify abelian varieties with particular motives of weight -1 . For some applications, such as the study of Shimura varieties, it would already be satisfactory to have a category of motives containing the abelian varieties and stable under $\otimes$. In characteristic 0, the theory of absolute Hodge cycles provides such a theory (Deligne–Brylinski [11], Deligne–Milne [12], or Panchishkin's lecture at this conference). If one does not restrict to pure motives, another key example is that of smooth curves which are not necessarily complete. More generally, one may consider a connected smooth projective curve Y , disjoint subsets S and T of Y , and the cohomology $H^1(Y - S, \text{rel}T)$. A 1-motive K is a complex of group schemes concentrated in degrees -1 and 0 , with, over the algebraic closure, K^{-1} a free finitely generated $\mathbb{Z}$ -module and K^0 an extension of an abelian variety by a torus. Assume for simplicity that k is algebraically closed. For each usual cohomology theory, the H^1 under consideration, or rather its Tate twist $H^1(Y - S, \text{rel}T)(1)$, is then obtained from the following 1-motive by applying a realization functor.

Let $J_{S,T}(Y)$ be the generalized Jacobian classifying degree-zero invertible sheaves on Y trivialized on T . It is an extension of the abelian variety $\text{Pic}^0(Y)$ by the torus whose character group is $\text{Ker}(\mathbb{Z}^T \rightarrow \mathbb{Z})$. Each $s \in S$ defines the invertible sheaf $\mathcal{O}(s)$, trivialized on T and of degree 1, whence

$$\text{Ker}(\mathbb{Z}^S \rightarrow \mathbb{Z}) \rightarrow J_{S,T}(Y). \quad (1.5.1)$$

This is the promised 1-motive. See Deligne [9, §10].

Warning 2. For $S, T \neq \emptyset$, the 1-motive (1.5.1) also determines the category of invertible sheaves on $Y - S$, trivialized on T : a point $x \in J_{S,T}(Y)$ defines such an invertible sheaf $\mathcal{L}_x$, and an isomorphism from $\mathcal{L}_x$ to $\mathcal{L}_y$ which is the identity on T is identified with $k \in \text{Ker}(\mathbb{Z}^S \rightarrow \mathbb{Z})$, with $y - x = \delta k$.

Again, one wants to be able to identify 1-motives with particular mixed motives. For some applications, such as the study at infinity of Shimura varieties, it would already be satisfactory

to have a category of motives containing the 1-motives and stable under $\otimes$. In characteristic 0, this is known (Brylinski [7]).

1.5 Moduli of motives and Shimura varieties

Abelian varieties have moduli spaces, and these moduli spaces make it possible to interpret algebraically certain quotients of Hermitian symmetric domains by arithmetic groups. The case of motives is more complicated. If S is a scheme over $\mathbb{C}$ and M is a family of pure motives parameterized by S , then the Hodge realization of M gives a variation of Hodge structures $\mathcal{M}_{\mathbb{C}}$ on $S(\mathbb{C})$, polarized if M is polarized. Hence there is a map φ_0 from $S(\mathbb{C})$ to the space $\mathcal{H}$ classifying polarized Hodge structures with the same Hodge numbers as M .

The variation $\mathcal{M}_{\mathbb{C}}$ gives in particular a local system of complex vector spaces $\mathcal{M}_{\mathbb{C}}$ endowed with a Hodge filtration F varying continuously. The holomorphy of F and the transversality condition discovered by Griffiths [17] (see also Griffiths [18]) may be expressed as follows. Let T be the tangent bundle of S , viewed as a C^∞ variety. The complex structure of S gives a complex structure on T . Let $\varphi : T \otimes_{\mathbb{R}} \mathbb{C} \rightarrow T$ be the $\mathbb{C}$ -linear extension of the identity of T , and let $T'' \subset T \otimes_{\mathbb{R}} \mathbb{C}$ be the kernel of φ . Define the Hodge filtration of $T \otimes \mathbb{C}$ by

$$F^{-1} = T \otimes \mathbb{C}, \quad F^0 = T'', \quad F^1 = 0.$$

For ξ and η smooth sections of $T \otimes \mathbb{C}$ and of $\mathcal{M}_{\mathbb{C}}$, the flat structure of $\mathcal{M}_{\mathbb{C}}$ allows one to define $\nabla_{\xi}\eta$. The condition is that $\xi, \eta \mapsto \nabla_{\xi}\eta$ be compatible with the Hodge filtrations:

ξ in T'' , η in F^i , $\nabla_{\xi}\eta$ in F^i : holomorphy;

ξ in T , η in F^i , $\nabla_{\xi}\eta$ in F^{i-1} : transversality.

On the classifying space $\mathcal{H}$ there is a complex structure and a holomorphic distribution $\tau \subset T$ such that holomorphy and transversality translate into holomorphy of the classifying map $\varphi : S(\mathbb{C}) \rightarrow \mathcal{H}$ and tangency of $\tau(S)$ to τ .

The distribution τ is generally nonintegrable. The set of points of $\mathcal{H}$ corresponding to a direct factor of some $H^i(X)$, with X nonsingular projective, is a countable union of subvarieties tangent to τ . If τ is not the entire tangent bundle, we have no description of this set, even conjectural.

An analogous argument, also due to Griffiths, explains why the points of the intermediate Jacobian $J^d(X)$ cannot all be classes of algebraic cycles, except when $H^{2d-1}(X)$ is of type $\{(d-1, d), (d, d-1)\}$. Let $f : X \rightarrow S$ be a family of nonsingular projective varieties, and put $X_s = f^{-1}(s)$. The intermediate Jacobians $J^d(X_s) = H^{2d-1}(X_s, \mathbb{C})/F^d + H^{2d-1}(X_s, \mathbb{Z})$ form a holomorphic bundle J^d of complex tori, and the transversality axiom ensures that the Gauss–Manin connection, expressing the fact that the $H^{2d-1}(X_s, \mathbb{C})$ form a local system, descends to the quotient to define a differential operator D on the sheaf of sections of J^d , with values in the sheaf of 1-forms valued in the holomorphic bundle $H^{2d-1}(X_s, \mathcal{O})/F^{d+1}$. An algebraic family $(Z_s)_{s \in S}$ of algebraic cycles homologous to zero on the X_s defines a holomorphic section z of J^d . Griffiths’s result is that $Dz = 0$. The section z defines a holomorphic family of mixed Hodge structures, an extension of $\mathbb{Z}$ of type $(0, 0)$ by $H^{2d-1}(X)(d)$, and $Dz = 0$ is equivalent to this family satisfying the transversality axiom.

1.6

When the distribution τ is the entire tangent bundle, $\mathcal{H}$ is an arithmetic quotient of a Hermitian symmetric domain. This description of the complex points of Shimura varieties as moduli of Hodge structures suggests that these varieties are moduli spaces of motives.

Let $\mathbb{S}$ be the real algebraic group $\mathbb{C}^*$: $\mathbb{S} = R_{\mathbb{C}/\mathbb{R}}\mathbb{G}_m$. Giving an action of $\mathbb{S}$ on the real vector space V amounts to giving a decomposition of $V \otimes \mathbb{C}$ as a direct sum of $V^{p,q}$ with $V^{q,p} = \overline{V^{p,q}}$,

where $z \in \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ acts on $V^{p,q}$ by multiplication by $z^{-p}\bar{z}^{-q}$. Define $w : \mathbb{G}_m \rightarrow \mathbb{S}$, respectively $\mu : \mathbb{G}_m \rightarrow \mathbb{S}_{\mathbb{C}}$, defined over $\mathbb{C}$, by the condition that, for every V , $w(\lambda)$, respectively $\mu(\lambda)$, acts on $V^{p,q}$ by multiplication by λ^{p+q} , respectively λ^{-p} .

The data defining a Shimura variety $\text{Sh}_K(G, X)$ are a reductive group G over $\mathbb{Q}$, a $G(\mathbb{R})$ -conjugacy class of morphisms $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$, with $w_h := h \circ w$ central (hence independent of h), and an open compact subgroup K of $G(\mathbb{A}^f)$. Put ourselves in the case where w_X is defined over $\mathbb{Q}$ and $\text{Int } h(i)$ is a Cartan involution of the quotient $G/w_X(\mathbb{G}_m)$.

The reflex field $E(G, X)$ is the field of definition of the conjugacy class of μ_h , for arbitrary $h \in X$. It is a subfield of $\mathbb{C}$. The Shimura variety is defined over $E(G, X)$, with complex points $K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q})$. Its construction in the general case is due to Borovoi [6].

A point of the Shimura variety over a field F containing $E(G, X)$ should correspond to

- (a) an exact tensor functor χ from the category $\text{Rep}(G)$ of representations of G to the category of pure motives over F ;
- (b) an integral structure. In terms of the finite adelic realization, the restricted product of the ℓ -adic realizations, it can be described as an isomorphism of tensor functors

$$\chi(V)_{\mathbb{A}^f} \xrightarrow{\sim} V \otimes \mathbb{A}^f,$$

given up to composition with an element of K .

The following condition must be satisfied. For simplicity, assume here that F can be embedded in $\mathbb{C}$.

- (c) Let σ be an embedding of F into $\mathbb{C}$, extending the identity embedding of $E(G, X)$ into $\mathbb{C}$. There must exist $h \in X$ such that the following tensor functors from $\text{Rep}(G)$ to Hodge structures are isomorphic:
 - (1) $(V, \rho) \mapsto V$ endowed with the Hodge structure defined by $\rho \circ h$;
 - (2) $(V, \rho) \mapsto$ the Hodge realization of $\chi(V)$, after extension of the base field to $\mathbb{C}$ by σ .

That this condition (c) be independent of the chosen complex embedding σ —assumed to extend the inclusion of $E(G, X)$ into $\mathbb{C}$ —is not evident. Sometimes, however, it follows from (b) and from the more algebraic conditions (d), (e), consequences of (c), that follow.

- (d) The de Rham realization defines a fiber functor on $\text{Rep}(G)$: $V \mapsto \chi(V)_{\text{DR}}$, corresponding to a G -torsor P over F . The Hodge filtration of the $\chi(V)_{\text{DR}}$ is exact and compatible with the tensor product, hence comes from a parabolic Q of P and from $\mu_{\text{DR}} : \mathbb{G}_m \rightarrow$ the center of $Q/\text{Ker}(Q \rightarrow P)$, lifting to a conjugacy class of morphisms from $\mathbb{G}_m$ to Q . Since F contains $E(G, X)$, it makes sense to ask that the corresponding conjugacy class of maps from $\mathbb{G}_m$ to G coincide with that of the $h \circ \mu$ ($h \in X$). Condition (c) implies this. This explains the appearance of the reflex field.
- (e) For (V, ρ) a representation of weight 0—that is, $\rho \circ w_h$ is trivial—and $V \otimes V \rightarrow \mathbb{Q}$ an invariant symmetric bilinear form such that, on $V_{\mathbb{R}}$, $B(v, h(i) \cdot w)$ is symmetric positive definite, one asks that $\chi(V) \otimes \chi(V) \rightarrow \mathbf{1}$ be positive for the expected polarization of the category of motives (loc. cit. V, 2.4, p. 276).

1.7

This motivic interpretation of Shimura varieties was a guide in developing the axioms that characterize them, and also in determining their conjugates (Borovoi [6]).

Let p be a prime number such that $G_{\mathbb{Q}_p}$ extends to a reductive group over $\mathbb{Z}_p$, with K the product of $G(\mathbb{Z}_p)$ with a subgroup of the restricted product of the $G(\mathbb{Q}_{\ell})$, $\ell \neq p$. The reflex field $E(G, X)$ is then unramified at p .

One expects the Shimura variety to have a natural reduction modulo p , and one may try to paraphrase the preceding conjectural motivic description in order to conjecture the structure of its set of points over a finite field of characteristic p . There are difficulties: how should one interpret (c) and the p -part of (b), which must be treated jointly with (d)? See, in this direction, Langlands–Rapoport [23].

2 Cohomology theories

2.1 Transfer principle

Principle. A geometric construction possible in one of the usual cohomology theories should have a motivic meaning and therefore have an analogue in the other usual theories.

This principle was crucial in developing the theory of mixed Hodge structures. Grothendieck saw that every $H^i(X)$ should be devissaged in terms of subquotients of the cohomology of nonsingular projective varieties. The mixed motive $H_{\text{mot}}^i(X)$ should therefore have a class in the Grothendieck group of pure motives. In characteristic 0, the Hodge number $h^{p,q}$ of the Hodge realization of a pure motive defines a homomorphism from this Grothendieck group to $\mathbb{Z}$. Thus Hodge numbers $h^{p,q}(H_{\text{mot}}^i(X))$ should have a meaning.

To go further, one had to be convinced that every motive has an increasing weight filtration W with $\text{Gr}_i^W(M)$ pure of weight i —that is, a direct factor of $H_{\text{mot}}^i(X)$ for X nonsingular projective. It was from the H^1 of curves, i.e. from 1-motives, that I became convinced of this; the first test that the definition of mixed Hodge structures had to pass was that they recover as a special case the 1-motives over $\mathbb{C}$ (Deligne [9, §10]).

The same transfer principle made it possible to conjecture the asymptotic behavior of a variation of Hodge structures on a punctured disk, or on a product of punctured disks.

Warning 3. Let (S, η, s) be a trait, with S the spectrum of a complete discrete valuation ring. According to Raynaud, an abelian variety over the generic point η , with semistable reduction, admits a rigid analytic description as the cokernel of a map defining a 1-motive. In parallel, in Hodge theory, if $\mathcal{F}$ is a polarized variation of Hodge structures of type $\{(1,0), (0,1)\}$ on a punctured disk D^* , the monodromy gives on $\mathcal{F}_0$ a weight filtration W which, together with the original Hodge filtration, makes $\mathcal{F}$ a variation of mixed Hodge structures in a punctured neighborhood of 0.

One should not expect analogous behavior for motives when the transversality condition is nontrivial in Hodge theory. In general, an object describing the asymptotic behavior should exist only on the punctured Zariski tangent space (assuming semistable reduction), and it should not suffice to reconstruct the original object. For example, for $\mathcal{F}$ a polarized variation of Hodge structures on $D^* = D - \{0\}$, W. Schmid’s asymptotic nilpotent orbit is a variation of mixed Hodge structures on the punctured tangent space at 0, and, when $\mathcal{F}$ extends over D , it is the constant variation with value the fiber at 0 of the extension.

Morihiro Saito’s theory (Saito [29]), which provides the six operations and vanishing cycles in mixed Hodge theory, is partly inspired by ℓ -adic theory and the motivic viewpoint. Conversely, it suggests that if one wants to consider motives over a base S , then rather than seeking motives M whose ℓ -adic realizations are ℓ -adic sheaves on S , it may be preferable to have motives whose realizations are perverse sheaves. In any case, only in such a framework can one hope for a weight filtration.

It is again the philosophy of motives that led Grothendieck to conjecture the existence of the “mysterious” functor relating the p -adic étale cohomology of a variety defined over $\mathbb{Q}_p$, assumed to have good reduction, to its de Rham cohomology: since it existed for H^1 , i.e. for the motives that are abelian varieties, it had to exist always, compatibly with the tensor product.

2.2 The motivic π_1

Let X_0 be an algebraic variety over k and let X be obtained from X_0 by extension of scalars to an algebraic closure $\bar{k}$ of k . For $x \in X_0(k)$, the profinite fundamental group $\pi_1(X, x)$ is endowed with an action of $\text{Gal}(\bar{k}/k)$. Assume for simplicity that X is normal. The functor “fiber at x ” is then an equivalence of categories from lisse $\mathbb{Q}_\ell$ -sheaves on X_0 to ℓ -adic representations of $\text{Gal}(\bar{k}/k)$ endowed with an equivariant action of $\pi_1(X, x)$.

I know of no sense in which $\pi_1(X, x)$ is motivic. The situation changes for its pro- ℓ completion $\pi_1(X, x)^{(\ell)}$. Its abelianization is $H_1(X, \mathbb{Z}_\ell)$, and its H_2 is contained in that of X . There should exist a motivic group scheme $\pi_1(X, x)^{\text{mot}}$ over k such that

- (a) for each usual cohomology theory H , the corresponding realization of $\pi_1(X, x)^{\text{mot}}$ over $\mathbb{Q}$ has as its linear representations the “local systems” on X , in the sense of H , which are unipotent;
- (b) the functor “fiber at x ” is an equivalence of categories:

$$\begin{array}{c} \text{smooth motives over } X_0, \text{ iterated extensions of inverse images of motives over } k \\ \xrightarrow{\sim} \\ \text{motives over } k \text{ with an action of } \pi_1(X, x)^{\text{mot}}. \end{array}$$

In (a), the local systems considered are not assumed to be motivic. According to (b), those that are motivic nevertheless form a faithful system of representations.

This philosophy inspired the Hodge theory of π_1 : see Steenbrink–Zucker [30].

More generally, suppose that a set A of pure motives over X_0 is given. The preceding case corresponds to $A = \mathbb{Z}$, or, what is the same, to A reduced to the unit motive. There should exist over k a motivic group scheme $\pi_1(X, x)_A$, the motivic π_1 relative to A , such that

- (a') over $\bar{k}$, its relative realization for the theory H has as its representations the “local systems” on X which are iterated extensions of subquotients of tensor products of realizations of objects of A and of their duals;
- (b') the functor “fiber at x ” is an equivalence from smooth motives over X , which are iterated extensions of motivic subquotients of tensor products of objects of A , of their duals, and of inverse images of motives over k , to motives over k endowed with an action of $\pi_1(X, x)_A$.

In (b'), the subquotients are to be taken over X and are not assumed to be motivic.

In Hodge theory, this suggests the following question. Let X be smooth and connected over $\mathbb{C}$, and let $x \in X$. Let A be a set of polarizable variations of Hodge structure on X . Let $\mathcal{V}_A$ be the Tannakian category of $\mathbb{Q}$ -local systems on X generated, under $\otimes$, duals, subquotients, and extensions, by the local systems underlying the objects of A . Let $\mathcal{M}_A$ be the Tannakian category of admissible mixed variations of Hodge structure on X , in the sense of Steenbrink–Zucker [30], generated, under $\otimes$, duals, subquotients, and extensions, by A and by the constant variations whose associated graded for the weight filtration is polarizable.

On $\mathcal{M}_A$ one has the fiber functor “fiber at x ”, denoted ω_x . Let $G = \text{Aut}^\otimes(\omega_x)$. It is the Zariski closure of $\pi_1(X, x)$ in the product of the $\text{GL}(H_x)$, for H an object of $\mathcal{M}_A$. Let $\mathcal{O}(G)$ be the Hopf algebra whose spectrum is G .

For M in $\mathcal{M}_A$, the underlying local system $M_{\mathbb{Q}}$ belongs to $\mathcal{V}_A$, so that G acts on its fiber $(M_{\mathbb{Q}})_x$ at x : this fiber is an $\mathcal{O}(G)$ -comodule,

$$(M_{\mathbb{Q}})_x \rightarrow (M_{\mathbb{Q}})_x \otimes \mathcal{O}(G). \quad (2.2.1)$$

The Hopf algebra $\mathcal{O}(G)$ should carry an ind-mixed Hodge structure. The group G should be an affine group scheme in the Tannakian category of mixed Hodge structures, so that: (a)

the maps (2.2.1) are morphisms of mixed Hodge structures; (b) the functor $M \mapsto M_x$ is an equivalence of $\mathcal{M}_A$ with the category of mixed Hodge structures H whose associated graded is polarizable, endowed with a comodule structure $H \rightarrow H \otimes \mathcal{O}(G)$ which is a morphism of mixed Hodge structures.

3 Absolute cohomology

3.1

In each of the usual cohomology theories, the functors $H^i(X)$ arise from a functor $R\Gamma(X)$ with values in a triangulated category $\mathcal{D}$. This category is endowed with a t -structure, namely the datum of subcategories $\mathcal{D}^{\leq 0}$ and $\mathcal{D}^{\geq 0}$ satisfying appropriate axioms (Beilinson–Bernstein–Deligne [3, 1.3]), whose heart, the abelian subcategory $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$, is the category in which H^0 takes its values, and

$$H^i(X) = H^i R\Gamma(X)$$

in the sense of this t -structure.

One may hope that these functors $R\Gamma$ are the “realizations” of a functor $R\Gamma_{\text{mot}}$ with values in a triangulated category $\mathcal{DM}(k)$, with t -structure whose heart is the category $\mathcal{M}(k)$ of mixed motives over k , such that the $R\Gamma$ in the various theories are obtained by applying realization functors.

A triangulated category $\mathcal{D}$ with a t -structure is not always the derived category of its heart $\mathcal{E}$. Example: let X be a finite connected CW-complex and let $\mathcal{D} \subset D^b(X, \mathbb{Q})$ be the subcategory of the derived category formed by complexes whose cohomology sheaves are locally constant. Its heart is the abelian category of locally constant sheaves of $\mathbb{Q}$ -vector spaces on X . There is a natural functor $D^b(\mathcal{E}) \rightarrow \mathcal{D}$, but this functor is an equivalence only if X is a $K(\pi, 1)$.

In this section, we shall assume the existence of $\mathcal{DM}(k)$ and of “realization” functors. We shall also assume the existence in $\mathcal{DM}(k)$ of a “Tate twist” which is an auto-equivalence of categories and which corresponds, under the realization functors, to the usual Tate twist. Finally, we shall assume that the category $\mathcal{M}(k)$ is Tannakian and that ℓ -adic cohomology is a fiber functor. This implies that if $\varphi : M_1 \rightarrow M_2$ is a morphism of motives whose ℓ -adic realization is an isomorphism, then φ is an isomorphism.

For M_1, M_2 in $\mathcal{M}(k)$, put

$$\text{Ext}^i(M_1, M_2) = \text{Hom}_{\mathcal{DM}(k)}(M_1, M_2[i]).$$

These are the usual Hom and Ext for $i = 0, 1$, but for $i > 1$ they may differ from Yoneda Ext groups, which are the $\text{Hom}(M_1, M_2[i])$ in $D^b(\mathcal{M}(k))$.

For each of the usual cohomology theories, these motivic Ext groups should map to the Ext^i groups between the corresponding realizations of M_1 and M_2 .

3.2

For an algebraic variety X over k , call the following the **absolute cohomology groups** of X :

$$H_{\text{abs}}^i(X, \mathbb{Q}(j)) := \text{Hom}_{\mathcal{DM}(k)}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[i]),$$

where $\mathbf{1}$ is the unit motive. These groups are often called **motivic cohomology groups**. They are vector spaces over $\mathbb{Q}$. One should take care not to confuse them with the $H_{\text{mot}}^i(X)$, which are motives over k . The two are related by a spectral sequence

$$E_2^{p,q} = \text{Ext}^p(\mathbf{1}, H_{\text{mot}}^q(X)(j)) \Rightarrow H_{\text{abs}}^{p+q}(X, \mathbb{Q}(j)). \quad (3.2.1)$$

For a motive M , we shall also write $H_{\text{abs}}^i(M)$ for $\text{Ext}^i(\mathbf{1}, M) = \text{Hom}_{\mathcal{DM}(k)}(\mathbf{1}, M[i])$. With this notation, the spectral sequence (3.2.1) can be rewritten as

$$E_2^{p,q} = H_{\text{abs}}^p(H_{\text{mot}}^q(X)(j)) \Rightarrow H_{\text{abs}}^{p+q}(X, \mathbb{Q}(j)). \quad (3.2.2)$$

The ℓ -adic analogue is the following. A variety X over k , with algebraic closure $\bar{k}$, defines ℓ -adic cohomology groups $H^i(X \otimes_k \bar{k}, \mathbb{Q}_\ell)$, which one hopes are realizations of motives $H_{\text{mot}}^i(X)$. The variety X also has absolute etale cohomology $H^i(X, \mathbb{Q}_\ell)$, which is the abutment of a spectral sequence

$$H^p(\text{Gal}(\bar{k}/k), H^q(X \otimes_k \bar{k}, \mathbb{Q}_\ell)) \Rightarrow H^{p+q}(X, \mathbb{Q}_\ell).$$

For X smooth and projective over k , one expects $R\Gamma_{\text{mot}}(X)$ to be noncanonically isomorphic, in $\mathcal{DM}(k)$, to the direct sum of the $H_{\text{mot}}^i(X)[-i]$. This would follow from the following hypothesis:

For $\mathcal{L}$ an invertible sheaf on X , one has $c_1(\mathcal{L}) : R\Gamma_{\text{mot}}(X) \rightarrow R\Gamma_{\text{mot}}(X)(1)[2]$, whose ℓ -adic realization gives cup product with the Chern class $c_1(\mathcal{L})$ of $\mathcal{L}$: $H^i(X \otimes_k \bar{k}, \mathbb{Q}_\ell) \rightarrow H^{i+2}(X \otimes_k \bar{k}, \mathbb{Q}_\ell(1))$.

Indeed, take $\mathcal{L}$ ample. Since the ℓ -adic realization functor is a fiber functor, it follows from (*) and from the hard Lefschetz theorem that, for X purely of dimension N , the maps

$$c_1(\mathcal{L})^i : H_{\text{mot}}^{N-i}(X) \rightarrow H_{\text{mot}}^{N+i}(X)(i)$$

are isomorphisms. One can then apply Deligne [8].

Analyzing the proof, one can deduce, for each ample invertible sheaf $\mathcal{L}$, a canonical decomposition, indeed several canonical decompositions,

$$\eta_{\mathcal{L}} : R\Gamma_{\text{mot}}(X) \xrightarrow{\sim} \bigoplus H_{\text{mot}}^i(X)[-i].$$

See Deligne, “Decompositions in the derived category,” in this volume.

Warning 4. Since these arguments depend on the hard Lefschetz theorem, they are essentially rational. If one had an integral variant of $\mathcal{DM}(k)$, one should expect that, in that framework, $R\Gamma(X)$ is not always the direct sum of the $H^i(X)[-i]$.

3.3 Motivic classes of cycles

Let X be smooth over k . The various “classes” of an algebraic cycle Z of codimension d should come from a motivic class

$$cl(Z) \in \text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(d)[2d]) = H_{\text{abs}}^{2d}(X, \mathbb{Q}(d)). \quad (3.3.1)$$

The class (3.3.1) should depend only on the linear-equivalence class of Z .

If X is smooth and projective and one has chosen a decomposition

$$R\Gamma_{\text{mot}}(X) \xrightarrow{\sim} \bigoplus H_{\text{mot}}^i(X)[-i] \quad (*)$$

(cf. 3.2), the class (3.3.1) is equivalent to a sequence of classes

$$cl_i(Z) \in \text{Ext}^{2d-i}(\mathbf{1}, H_{\text{mot}}^{2d-i}(X)(d)) = H_{\text{abs}}^{2d-i}(H_{\text{mot}}^{2d-i}(X)(d)). \quad (3.3.2)$$

The first of these classes that is nonzero is independent of the chosen decomposition.

Example 5. In Hodge theory, Z first defines an integral cohomology class of type (d, d) , that is, a morphism of Hodge structures

$$\mathbb{Z} \rightarrow H^{2d}(X)(d).$$

If this class is zero, Z has a class in the intermediate Jacobian

$$J^d(X) = H^{2d-1}(X, \mathbb{C})/F^d + H^{2d-1}(X, \mathbb{Z}),$$

a group which is precisely $\text{Ext}^1(\mathbb{Z}, H^{2d-1}(X)(d))$ in the category of mixed Hodge structures. The story stops there, because the abelian category of rational mixed Hodge structures has cohomological dimension 1.

Example 6. Let X be smooth and projective, purely of dimension N , over an algebraically closed field k , and let $\mathcal{O}(1)$ be an ample invertible sheaf. We show how one can attach to a divisor D a class $cl(D)$ in $\text{Pic}^0(X) \otimes \mathbb{Q}$, a group that the theory of motives interprets as $H_{\text{abs}}^2(X)(1)$.

By a theorem of A. Weil [31, Theorem 7], the map from linear-equivalence classes of divisors to linear-equivalence classes of 0-cycles,

$$E \mapsto E \cdot c_1(\mathcal{O}(1))^{N-1}$$

induces an isogeny

$$\text{Pic}^0(X) \rightarrow J(X),$$

where $J(X)$ is the Albanese variety, or Jacobian, of X . We therefore have a bijection between $\text{Pic}^0(X) \otimes \mathbb{Q}$ and $J(X) \otimes \mathbb{Q}$. We take

$$cl_1(D) = \frac{\deg(c_1(D) \cdot c_1(\mathcal{O}(1))^{N-1})}{\deg(c_1(\mathcal{O}(1))^N)} \cdot c_1(\mathcal{O}(1))^N - c_1(D) \cdot c_1(\mathcal{O}(1))^{N-1}.$$

Example 7. Now work not over the spectrum of a field but over a base S , smooth over an algebraically closed field k . Let $f : X \rightarrow S$ be projective and smooth over S , and let Z be a codimension d cycle. Work in ℓ -adic cohomology, with $\mathbb{Z}$ invertible on S . The cycle Z has a class in

$$H^{2d}(X, \mathbb{Z}_\ell(d)) = \text{Hom}_{D(S)}(\mathbb{Z}_\ell, Rf_*\mathbb{Z}_\ell[2d](d)).$$

If one chooses a decomposition

$$Rf_*\mathbb{Q}_\ell \cong \bigoplus R^i f_*\mathbb{Q}_\ell[-i], \quad (3.6.1)$$

this class gives rise, as in 3.3, to a sequence of classes

$$cl_i(Z) \in \text{Ext}^i(\mathbb{Z}_\ell, R^{2d-i} f_*\mathbb{Q}_\ell(d)) = H^i(S, R^{2d-i} f_*\mathbb{Q}_\ell(d)). \quad (3.6.2)$$

When X is of the form $X_0 \times_{\text{Spec}(k)} S$, and the decomposition (3.6.1) comes from the unique analogous decomposition of the cohomology of X_0 , these classes identify with the Kunneth components of the cohomology class of Z .

3.4

Assume that X is smooth over a field. Decompose the K -theory group $K_n(X) \otimes \mathbb{Q}$ by Adams operations:

$$K_n(X) \otimes \mathbb{Q} = \bigoplus K_n(X)^{(j)},$$

where on $K_n(X)^{(j)}$ the Adams operation ψ^k acts by multiplication by k^j . Recall that $K_0(X)^{(j)}$ is the Chow group of codimension j cycles modulo rational equivalence, tensored with $\mathbb{Q}$.

The example of the usual cohomology theories leads one to hope for Chern classes

$$ch^j : K_n(X)^{(j)} \rightarrow H_{\text{abs}}^{2j-n}(X, \mathbb{Q}(j))$$

factoring through $K_n(X)^{(j)}$.

An optimistic conjecture is that the “Chern class” morphisms are isomorphisms

$$ch^j : K_n(X)^{(j)} \xrightarrow{\sim} H_{\text{abs}}^{2j-n}(X, \mathbb{Q}(j)) := \text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[2j-n]). \quad (3.7.1)$$

This conjecture underlies the terminology “absolute motivic cohomology group” for the $K_n(X)^{(j)}$, and the notation

$$H_{\mathcal{M}}^{2j-n}(X, \mathbb{Q}(j)) := K_n(X)_{\mathbb{Q}}^{(j)}.$$

Bloch [4] proposed an interpretation of the $K_n(X)^{(j)}$ as “higher Chow groups” $\text{Ch}^j(X; n)$. His argument has a gap, an insufficient moving lemma, which he promises to repair. This interpretation makes the definition of ch^j more natural, but the interpretation in terms of K -theory remains necessary for explicit calculation in the case of number fields. Higher Chow groups have the advantage of having a natural integral structure.

3.5

According to Beilinson, conjecture (3.7.1) can be deduced from hypotheses whose essential parts are: (a) the functor $X \mapsto K_*(X)$ factors through $R\Gamma_{\text{mot}}$; (b) the category $\mathcal{DM}(k)$ is the derived category of the category of motives. The conjecture has striking consequences for both sides of (3.7.1). They vanish for trivial reasons in different regions of the (i, j) -plane:

- (a) For M a motive, $\text{Ext}^i(\mathbf{1}, M) = 0$ for $i < 0$, and $\text{Ext}^0(\mathbf{1}, M) = \text{Hom}(\mathbf{1}, M)$. Hence $\text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[N]) = 0$ for $N < 0$, and also for $N = 0$ and $j \neq 0$. In K -theory this gives the Beilinson–Soule conjecture that $K_n(X)^{(j)} = 0$ for $2j < n$, and also for $2j \leq n$ if $j \neq 0$.
- (b) The Chow group $\text{Ch}^j(X)$ of algebraic cycles of codimension j is zero for $j > \dim X$, and similarly $\text{Ch}^j(X; n)$ is zero for $j > \dim(X) + n$. Moreover, $\text{Ch}^j(X; n) = 0$ for $n < 0$. Applying (3.7.1), one would deduce the following vanishing statements.

Vanishings 3.8.1.

- (i) For M a pure motive of weight w , $\text{Hom}^i(\mathbf{1}, M) = 0$ for $i > -w$.
- (ii) For M effective of weight $\leq w$ and $b > 0$, $\text{Hom}^i(\mathbf{1}, M(w+b)) = 0$ for $i > w+b$.

For (i), assume M is a direct factor of $H_{\text{mot}}^a(X)(c)$, with $c \geq 0$, X a smooth projective variety. Then $\text{Ext}^i(\mathbf{1}, M)$ is a direct factor of $\text{Ch}^c(X; 2c-i-a)$, and, if it is nonzero, one has $2c-i-a \geq 0$, i.e. $i \leq 2c-a = -w$.

For (ii), one also uses the weak Lefschetz theorem (ℓ -adic, hence motivic if ℓ -adic cohomology is a fiber functor) to suppose that $\dim(X) \leq a$. Assume $c \geq a$; one must prove the vanishing of $\text{Ext}^i(\mathbf{1}, H_{\text{mot}}^a(X)(c))$ for $i > (c-a)+a = c$. This inequality is equivalent to $c > a + (2c-i-a) \geq \dim(X) + (2c-i-a)$.

That (3.7.1) is an isomorphism would also imply weak Lefschetz theorems for Chow groups, and the conjectures on Chow groups presented at this conference by J. P. Murre.

3.6

For M the unit motive, (3.8.1)(ii) asks that

$$H_{\text{abs}}^n(\text{Spec}(k), \mathbb{Q}(j)) = 0 \quad \text{for } n > j. \quad (3.9.1)$$

It is not clear to me whether one should expect an analogue over $\mathbb{Z}$ of (3.9.1). If the $H_{\text{abs}}^n(\text{Spec}(k), \mathbb{Z}(j))$ are defined as higher Chow groups, then (3.9.1) is trivially true.

A more concrete question: suppose that one has managed to define, in an “independent of ℓ ” way, Galois cohomology classes in the $H^n(\text{Gal}(\bar{k}/k), \mathbb{Z}_{\ell}(j))$. If $n > j$, (3.9.1) asks that these classes be torsion. Should one even hope that they are zero?

Example 8. Let G be an absolutely simple and simply connected algebraic group over a field k . Over the algebraic closure $\bar{k}$ of k , the relations among the cohomology of G , of BG , and of BT , for T a maximal torus, show in each of the usual cohomology theories that $H^i(G) = 0$ for $i = 1, 2$ and that $H^3(G)(2)$ has a natural generator α . Work in ℓ -adic cohomology. Over $\bar{k}$, the absolute ℓ -adic cohomology of G is the direct sum of that of the point and a reduced cohomology $\tilde{H}$, related to that of BG , through a Hochschild–Serre exact sequence. Thus $H^i(G, \mathbb{Z}_\ell(2)) = 0$ for $i \leq 2$, and $H^3(G, \mathbb{Z}_\ell(2))$ is isomorphic to $\mathbb{Z}_\ell$, with a natural generator α . This generator is primitive, hence defines a homomorphism

$$\alpha : G \rightarrow H^3(\mathrm{Spec}(k), \mathbb{Z}_\ell(2)).$$

One can verify that, as predicted by 3.9, this map is zero: by Merkurjev–Suslin [27], the H^3 is torsion-free, and a trace argument shows that the image is torsion: G splits over a finite extension k' of k , and $G(k')$ is equal to its derived group, at least if $|k'| \geq 4$.

The same circle of ideas makes it possible to construct a canonical central extension of $G(k)$ by $H^2(\mathrm{Spec}(k), \mathbb{Z}/n(2))$, which is equal to $K_2(k)/nK_2(k)$ by Merkurjev–Suslin [27]. Recall that for G split, a canonical central extension of $G(k)$ by $K_2(k)$ was constructed by Matsumoto [25, 5.11].

3.7

For a motive M over $\mathbb{Q}$, $H_{\mathrm{abs}}^i(M)$ appears in Beilinson’s conjecture on the behavior at $s = 0$ of the L -function of M : Beilinson [2] and the lectures of Soule, Nekovar, and Scholl at this conference. I refer to those lectures for a description of the conjecture and of what is known about it.

3.8

If F is a number field, (3.7.1) predicts that in the category of motives over F one has

$$\mathrm{Ext}^1(\mathbf{1}, \mathbb{Q}(j)) = K_{2j-1}(F) \otimes \mathbb{Q}, \quad (3.12.1)$$

with vanishing of the Ext^n for $n \geq 2$. The assertion (3.12.1) imposes severe restrictions on the category of motives that are iterated extensions of Tate motives, and on the systems of realizations attached to an object of this category. See Deligne [14], where one example considered in detail is that of the Lie algebra of the unipotent completion of the fundamental group of the projective line minus three points.

Here is a program for proving consequences of (3.12.1) for such systems of realizations.

- (A) Define a Tannakian category $\mathcal{T}$, whose simple objects are the tensor powers $\mathbb{Q}(n)$, $n \in \mathbb{Z}$, of a rank-one object $\mathbb{Q}(1)$, for which (3.12.1) is true “by definition.”

The idea is that Bloch’s higher Chow groups $\mathrm{Ch}^n(\mathrm{Spec}(F), i)$ are given by a simplicial complex graded by the codimension n of cycles, and that the product is almost given by a graded simplicial commutative algebra structure. “Almost,” because the intersection of two cycles is not always defined, only “generically.” Rationally, it should be possible to replace this almost algebra by a genuine commutative bigraded differential algebra, with d of degree $(1, 0)$, and

$$A^{m,n} = 0 \quad \text{if } m < 0,$$

$A^{*,0}$ reduced to $\mathbb{Q}$ in degree 0, and

$$H^i(A^{*,n}) \quad \text{reduced to } K_{2n-i}(F) \otimes \mathbb{Q} \quad \text{in degree } i.$$

One can then define $\mathcal{T}$ as a suitable category of bigraded differential modules and morphisms up to homotopy.

This part of the program was essentially carried out by May [26]. For a cubical variant, see Bloch's lecture in this volume.

- (B) Define, for each of the usual cohomology theories, a fiber functor “realization” on the Tannakian category $\mathcal{T}$ of (A).
- (C) Show that the mixed Tate motives one is interested in, for example $\mathrm{Lie} \pi_1(\mathbf{P}^1 - \{0, 1, \infty\})$ —or rather the systems of their realizations—come from an object of $\mathcal{T}$.

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Decompositions in the Derived Category

Pierre Deligne

Translator's note. This is a direct English working translation from the project TeX transcription. Mathematical notation and numbering have been kept close to the source; the prose has been normalized into contemporary English.

Let $\mathcal{D}$ be a triangulated category equipped with a t -structure [1, §1.3.1], and let $\mathcal{C}$ be its heart (loc. cit.). We use the notation $\mathcal{D}^{\leq a}$, $\mathcal{D}^{\geq b}$, $\tau_{[a,b]}$, and H^i of [1, §1.3]. Let $K \mapsto K(1)$ be an autoequivalence of $\mathcal{D}$, carrying distinguished triangles to distinguished triangles and respecting the t -structure. We write $K \mapsto K(i)$ for the i -th iterate, $i \in \mathbb{Z}$, of this autoequivalence. We will call the functors $K \mapsto K(i)$ “twists.”

We assume that the objects of $\mathcal{D}$ have bounded cohomology: every object belongs to some $\mathcal{D}^{\leq a}$ and to some $\mathcal{D}^{\geq b}$. If this hypothesis is not satisfied, then the objects considered below should be assumed to have bounded cohomology.

Let K be an object of $\mathcal{D}$, equipped with a morphism

$$\eta: K \rightarrow K(1)[2]. \quad (1)$$

We will again write η for the morphisms deduced from η by twisting (i), shifting, and truncating $\tau_{[a,b]}$, and we write η^i for a composite of i such morphisms. For example, we again write η for the morphisms in the abelian category $\mathcal{C}$

$$\eta: H^i K \rightarrow H^{i+2} K(1) \quad (2)$$

deduced from (1).

Assume that η satisfies the following condition.

(L.V.) For every $i \geq 0$, the i -th iterate of η ,

$$\eta^i: H^{-i}(K) \rightarrow H^i(K)(i),$$

is an isomorphism.

The arguments of [2], recalled in paragraph 1, then show that K is isomorphic to the sum of the $H^i(K)$, each placed in its cohomological degree:

$$K \simeq \bigoplus H^i(K)[-i]. \quad (3)$$

In [2], I announced the existence of a canonical isomorphism (3). The purpose of this note is to define one, and in fact several.

1 Review of the proof of [2]

For $i \geq 0$, the primitive part P^{-i} of $H^{-i}(K)$ is the kernel of

$$\eta^{i+1}: H^{-i}(K) \rightarrow H^{i+2}(K)(i+1).$$

It follows from (L.V.) that the iterated morphisms (2) induce a direct-sum decomposition

$$\sum_{j \geq 0} \eta^j: \bigoplus_{j \geq 0} P^{-i-2j}(-j) \xrightarrow{\sim} H^{-i}(K). \quad (4)$$

For $H^i(K)$, this gives an isomorphism

$$\sum_{j \geq 0} \eta^{i+j} : \bigoplus_{j \geq 0} P^{-i-2j}(-i-j) \xrightarrow{\sim} H^i(K). \quad (5)$$

These isomorphisms assemble into

$$\sum \eta^j : \bigoplus_{i,j} P^{-i}(-j) \xrightarrow{\sim} H^*(K), \quad (6)$$

where the sum is over i, j with $0 \leq j \leq i$. In (6), $P^{-i}(-j)$ is a direct summand of $H^{2j-i}(K)$.

Recall that a cohomological functor F from $\mathcal{D}$ to an abelian category $\mathcal{M}$ is an additive functor such that, for every distinguished triangle $K \rightarrow L \rightarrow M \rightarrow K[1]$, the sequence $F(K) \rightarrow F(L) \rightarrow F(M)$ is exact [6, § 3.1]. Put $F^i(K) = F(K[i])$. By rotating the distinguished triangle $K \rightarrow L \rightarrow M \rightarrow K[1]$, one sees that a cohomological functor F gives a long exact sequence

$$\cdots \rightarrow F^i(K) \rightarrow F^i(L) \rightarrow F^i(M) \rightarrow F^{i+1}(K) \rightarrow F^{i+1}(L) \rightarrow \cdots$$

For K in $\mathcal{D}$, the truncations $\tau_{[a,b]}K$ of K give distinguished triangles

$$\rightarrow \tau_{[a,b]}K \rightarrow \tau_{[a,c]}K \rightarrow \tau_{[b+1,c]}K \rightarrow .$$

The system of corresponding long exact sequences is a “spectral object” in the sense of J. L. Verdier and supplies a spectral sequence

$$E_2^{pq} = F^p H^q K \Rightarrow F^{p+q} K.$$

See the appendix. Let ${}^i E_2^{pq}$ be the analogous spectral sequence for $K(i)$. A morphism (1) gives morphisms of spectral sequences

$${}^i E_2^{pq} \rightarrow {}^{i+1} E_2^{p,q+2}.$$

We show that (L.V.) forces these spectral sequences to degenerate. Proceeding by induction on $r \geq 2$, assume that the d_s vanish for $2 \leq s < r$, and prove that $d_r = 0$. For $q \leq 0$, let ${}^i P^{pq}$ be the direct summand $F^p P^q(i)$ of ${}^i E_2^{pq} = {}^i E_r^{pq}$.

By (4) and (5), it is enough to verify that d_r vanishes on the ${}^i P^{pq}$. By definition, ${}^i P^{pq}$ is annihilated by η^{q+1} . The morphism d_r sends ${}^i E_r^{pq}$ to ${}^i E_r^{p+r, q-r+1}$, on which η^{q+1} is injective. The vanishing of d_r follows.

Applying the degeneration of (6) to a cohomological functor $\text{Hom}(X, K)$, with X in $\mathcal{C}$, one finds that every morphism $X \rightarrow H^i K$ lifts to a degree- i morphism from X to K . In particular, for $X = H^i K$, the identity map of $H^i K$ lifts to a degree- i morphism from $H^i K$ to K . The sum of these morphisms

$$\bigoplus H^i K[-i] \xrightarrow{\sim} K$$

is an isomorphism on cohomology, hence an isomorphism.

2 An isomorphism (3)

Let K, η satisfy (L.V.).

Lemma 1. *Let F be a cohomological functor from $\mathcal{D}$ to the category of abelian groups. Let $i \geq 0$ and let $x \in F(\tau_{\geq -i} K)$ be such that $\eta^{i+1}(x) \in F^{2(i+1)}(\tau_{\geq i+2} K(i+1))$ is zero. Then x has a unique lift y in $F(\tau_{\geq -i-1} K)$ such that $\eta^{i+1}(y) \in F^{2(i+1)}(\tau_{\geq i+1} K(i+1))$ is zero.*

Proof. We saw in paragraph 1 that K is isomorphic to the sum of the $H^i(K)[-i]$. It follows that the sequences

$$0 \rightarrow F^a H^a K \rightarrow F\tau_{\geq a} K \rightarrow F\tau_{\geq a+1} K \rightarrow 0$$

are exact, and similarly for the $K(i)$. The morphism η^{i+1} induces a morphism of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^{i+1} H^{-i-1} K & \longrightarrow & F\tau_{\geq -i-1} K & \longrightarrow & F\tau_{\geq -i} K \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F^{i+1} H^{i+1} K(i+1) & \longrightarrow & F^{2(i+1)} \tau_{\geq i+1} K(i+1) & \longrightarrow & F^{2(i+1)} \tau_{\geq i+2} K(i+1) \longrightarrow 0. \end{array}$$

It remains only to apply the snake lemma. $\square$

The element y of Lemma 1 satisfies the hypothesis of Lemma 1 for $i+1$, and one concludes by induction.

Lemma 2. *Under the hypotheses of Lemma 1, there exists a unique lift $y \in F(K)$ such that, for each $s > i$, $\eta^s y$ has zero image in $F^{2s}(\tau_{\geq s} K(s))$.*

2.3

Apply Lemma 2 to the cohomological functor $F(L) := \text{Hom}(P^{-i}[i], L)$ and to the element $x: P^{-i}[i] \rightarrow \tau_{\geq -i} K$ defined by the inclusion of P^{-i} into $H^{-i} K$. The hypothesis $\eta^{i+1} x = 0: P^{-i}[-i-2] \rightarrow \tau_{\geq i+2} K(i+1)$ is satisfied because $\eta^{i+1}: P^{-i} \rightarrow H^{i+2} K(i+1)$ is zero. We obtain the following.

Proposition 3. *Let $i \geq 0$. There exists a unique morphism f_i from $P^{-i}[i]$ to K such that*

- (i) $H^{-i}(f_i)$ is the inclusion of P^{-i} into $H^{-i} K$;
- (ii) for each $s > i$, the morphism induced by $\eta^s f_i$, from $P^{-i}[i]$ to $(\tau_{\geq s} K)(s)[2s]$, is zero.

2.5

From the morphisms f_i of Proposition 3 and from (6), one deduces morphisms $g_i: H^i(K)[-i] \rightarrow K$ by the rule

$$\text{on the direct summand } P^{-i}(-j) \text{ of } H^{2j-i} \quad (0 \leq j \leq i), \quad g_i \text{ is } \eta^j f_{i-2j}. \quad (7)$$

The sum of these morphisms is an isomorphism (3).

2.6

Let

$$\varphi: \bigoplus H^i(K)[-i] \xrightarrow{\sim} K$$

be an isomorphism inducing the identity on cohomology. Put $Q^{i,j} = P^{-i}(-j)[i-2j]$, so that $P^{-i}(-j)$ lies in cohomological degree $2j-i$. Combining φ with the isomorphism (6), one obtains

$$\varphi_1: \bigoplus_{0 \leq j \leq i} Q^{i,j} \xrightarrow{\sim} K. \quad (8)$$

We denote by $\eta_{i,j}^{k,\ell}$ the matrix entries of $\eta: K \rightarrow K(1)[2]$ relative to this decomposition:

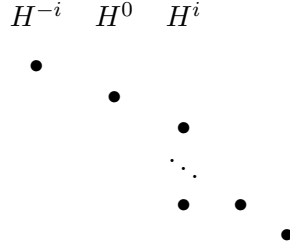
$$\eta_{i,j}^{k,\ell}: Q^{i,j} \rightarrow Q^{k,\ell}(1)[2].$$

These matrix entries are zero for $2j-i < 2\ell-k-2$, since, for A, B in $\mathcal{C}$, one has $\text{Hom}(A[-n], B[-m]) = 0$ for $n < m$. For $2j-i = 2\ell-k-2$, this is the morphism induced by $\eta: H^{2j-i}(K) \rightarrow H^{2j-i+2}(K)(1)$: the identity of $P^{-i}(-j)$ if $k=i$ and $\ell=j+1$, and 0 otherwise.

Proposition 4. For an isomorphism φ as in 2.6 to be the one defined in 2.5, it is necessary and sufficient that

- (i) for $j < i$, the only nonzero $\eta_{i,j}^{k,\ell}$ is the evident identity morphism $\eta_{i,j}^{i,j+1}$;
- (ii) $\eta_{i,i}^{k,\ell}$ is nonzero only for $\ell \leq i$.

The following picture may help one follow the argument. Each $\bullet$ represents a direct summand $Q^{i,j}$. In each row, i is constant. Each column corresponds to one H^n , with $Q^{i,j}$ contributing to H^{2j-i} . Condition (i) determines η , except on the right side of the triangle. Condition (ii) says that $\eta_{i,i}^{k,\ell}$ is nonzero only for k, ℓ in the region to the left of the dotted line.



Notation. We lighten notation by calling an a -morphism from K to L a morphism from K to $L(a)[2a]$. For example, $\eta_{i,j}^{k,\ell}$ is a 1-morphism from $Q^{i,j}$ to $Q^{k,\ell}$.

Proof. Identify K with the sum of the $H^i(K)[-i]$ by means of φ . By definition, the isomorphism φ of 2.5 is characterized by (i) and by the condition

- (*) for every $i \geq 0$ and $s > i$, the s -morphism η^s sends $Q^{i,0}$ into the sum of the $H^n(K)[-n]$ for $n < s$.

Assume (i), so that η is evident on $Q^{i,j}$ for $j < i$. Then condition (*) is equivalent to

- (**) for $t > 0$, the t -morphism η^t sends $Q^{i,i}$ into the sum of the $H^n(K)[-n]$ for $n < i + t$.

*Proof that (**) implies (ii).* We prove by induction on $k - \ell$ that $\eta_{i,i}^{k,\ell} = 0$ if $\ell > i$. Assume that the assertion has been proved for $k - \ell < a$, and prove it for $k - \ell = a$. The restriction of η^{a+1} to $Q^{i,i}$ is the sum

$$\sum \eta^a \circ \eta_{i,i}^{k,\ell}.$$

The terms with $\ell > i$ are controlled by the induction hypothesis and by (i): they are zero if $k - \ell < a$; for $k - \ell \geq a$, they take values in the distinct $Q^{k,\ell+a}$, and the vanishing of $\eta^a \eta_{i,i}^{k,\ell}$ implies the vanishing of $\eta_{i,i}^{k,\ell}$. Note that $Q^{k,\ell+a} \subset H^n[-n]$ for $n = 2\ell + 2a - k$. For $k - \ell = a$, one therefore lands in H^k , and the hypothesis $\ell = k - a > i$ gives $k > i + a$, a value of k forbidden by (**). It remains to show that these terms cannot cancel with those for $\ell \leq i$. It is enough to show that, if $\ell \leq i$, then on $Q^{k,\ell}$ the a -morphism η^a has values in the sum of the $H^n[-n]$ for $n \leq i + a$.

First case: $a \leq k - \ell$. One lands in $H^{-k+2\ell+2a}$, and

$$-k + 2\ell + 2a = (-k + \ell + a) + \ell + a \leq 0 + i + a = i + a.$$

Second case: $a > k - \ell$. Write $\eta^a = \eta^{a-(k-\ell)}\eta^{k-\ell}$, and consider $\eta^{a-(k-\ell)}$ on $Q^{k,k}$. Hypothesis (*) ensures that one lands in the H^n with

$$n < k + (a - (k - \ell)) = a + \ell \leq a + i.$$

Proof that (ii) implies (i). We prove by induction on $a \geq 1$ that the a -morphism η^a restricted to $Q^{i,i}$ has values in the $Q^{k,\ell}$ with $\ell \leq i + a - 1$. For $a = 1$ this is hypothesis (ii). Assume the

claim for a , and consider $\eta^{a+1} = \eta \circ \eta^a$. It is enough to see that, if $\ell \leq i + a - 1$, the restriction of the 1-morphism η to $Q^{k,\ell}$ has values in the sum of the $Q^{m,n}$ with $n \leq i + a$. If $\ell < k$, this follows from (i). If $\ell = k$, it follows from (ii).

Finally $Q^{k,\ell}$ lies in $H^n[-n]$ for $n = 2\ell - k$, and if $\ell \leq i + a - 1$ then

$$n = 2\ell - k = \ell - (k - \ell) \leq \ell \leq i + a - 1.$$

□

3 Another isomorphism (3)

3.1

The opposite category $\mathcal{D}^0$ of $\mathcal{D}$ is again a triangulated category. Its translation functor is $K^0 \mapsto (K[-1])^0$. We equip it with the autoequivalence (1) for which $K^0(1) = K(-1)^0$.

The t -structure of $\mathcal{D}$ supplies one on $\mathcal{D}^0$, and $\eta: K \rightarrow K(1)[2]$ transposes to

$${}^t\eta: K^0 \rightarrow K^0(1)[2].$$

For η to satisfy (L.V.) it is necessary and sufficient that ${}^t\eta$ satisfy (L.V.).

The construction (2.5) is not self-dual, as is seen from its characterization in Proposition 4. The dual construction, obtained from Proposition 3 applied to ${}^t\eta$, gives morphisms

$$f'_i: K \rightarrow P^{-i}(-i)[-i]$$

such that $H^i(f'_i)$ is the projection of $H^i(K)$ onto $P^{-i}(-i)$, and such that, for each $s > i$, the morphism induced by $f'_i\eta^s$, from K to $P^{-i}(-i)[-i](s)[2s]$, is zero on $\tau_{\leq -s}K$. Proceeding as in 2.5, one deduces morphisms $g'_i: K \rightarrow H^i(K)[-i]$ whose sum is an isomorphism from K to the sum of the $H^i(K)[-i]$. This isomorphism is in general not the inverse of the one constructed in 2.5.

In this paragraph we assume that we are in characteristic zero, that is, that the category $\mathcal{D}$ is $\mathbf{Q}$ -linear. Let K, η satisfy (L.V.). Our aim is to construct a self-dual isomorphism (3).

3.2

The grading of H^*K defines an endomorphism

$$h: H^*K \rightarrow H^*K : \text{multiplication by } n \text{ on } H^n K.$$

Passing to cohomology, η gives a morphism of degree 2

$$e: H^*K \rightarrow H^*K(1).$$

Because we are in characteristic zero, one can complete h and e by a morphism of degree -2

$$f: H^*K \rightarrow H^*K(-1)$$

such that h, e, f satisfy the relations between the standard generators of $\mathfrak{sl}(2)$:

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

To give meaning to these formulae, a morphism and those obtained from it by twisting (n) are denoted in the same way.

In characteristic zero, the existence of f is equivalent to (L.V.). In the decomposition (6), each sum over j , $\bigoplus_j P^{-i}(j)$, is stable under h, e, f , and from the factor in H^{i+1} to the factor in H^{i-1} , f is the inverse of e , multiplied by

$$\frac{(i+1)^2}{4} - \frac{n^2}{4}.$$

Let F be one of the bifunctors $\text{Ext}^a(A, B) = \text{Hom}_{\mathcal{D}}(A, B[a])$ on $\mathcal{C}$. Up to canonical isomorphism, the functor $F(A(i), B(j))$ depends only on $n := j - i$. We denote it by $F_n(A, B)$. From h, e, f one again obtains an action of $\mathfrak{sl}(2)$ on $F_*(H^*K, H^*K)$. More precisely, there are morphisms

$$\begin{aligned} h &: F_j(H^*K, H^*K) \rightarrow F_j(H^*K, H^*K), \\ e &: F_j(H^*K, H^*K) \rightarrow F_{j+1}(H^*K, H^*K), \\ f &: F_j(H^*K, H^*K) \rightarrow F_{j-1}(H^*K, H^*K) \end{aligned}$$

satisfying the $\mathfrak{sl}(2)$ relations. The endomorphism h is multiplication by n in degree n , and e , of degree 2, is the bracket with $\eta: H^*K \rightarrow H^*K(1)$.

Let F_j^d be the degree- d part of $F_j(H^*K, H^*K)$: the sum of the $F_j(H^pK, H^qK)$ for $q - p = d$. If $H^iK = 0$ for $|i| > N$, then $F_j^d = 0$ for $|d| > 2N$. On the sum of $\mathbf{Q}$ -vector spaces $\bigoplus_d F_{j+d/2}^d$, where the sum is over d of fixed parity, with j integral for d even and half-integral for d odd, (h, e, f) defines an action of $\mathfrak{sl}(2)$, and e therefore satisfies (L.V.). Reading the corresponding decomposition (6), one obtains the following.

Lemma 5. *With the preceding notation, for $d \leq 1$:*

- (i) $e: F_{j-1}^{d-2} \rightarrow F_j^d$ is injective;
- (ii) F_j^d is the direct sum of the image of e and the kernel of $e^{1-d}: F_j^d \rightarrow F_{j+1-d}^{2-d}$.

3.4

Let φ be an isomorphism

$$\bigoplus H^i(K)[-i] \xrightarrow{\sim} K \quad (9)$$

inducing the identity on cohomology. As in Proposition 4, and using its notation, regard η as a 1-map from K to K , and let $\eta^{(d)}$ be its homogeneous component of degree d in the decomposition (9). With the notation of 3.2, $\eta^{(d)}$ belongs to F_1^d , for F the functor Ext^{2-d} . For $d > 2$, it is therefore zero; for $d = 2$, $\eta^{(2)} \in \text{Hom}(H^*K, H^*K(1))$ is the endomorphism e of 3.2.

Proposition 6. *Assume that $\mathcal{D}$ is $\mathbf{Q}$ -linear and that (K, η) satisfies (L.V.). Then there exists a unique isomorphism (9) for which*

$$\text{for } d \leq 1, \quad e^{1-d}(\eta^{(d)}) = 0. \quad (10)$$

For $d = 1$, condition (10) means that $\eta^{(1)} = 0$.

Proof. By induction on $n \geq -1$, we show that the conditions (10) for $-n < d \leq 1$ determine φ uniquely modulo composition with an endomorphism $1 + w$ of the sum of the $H^i(K)[-i]$, with w of degree $\leq -n - 2$. Note that, with the notation of 3.2, the homogeneous component $\psi^{(d)}$ of degree d of w belongs to F_0^d for F the functor Ext^{-d} .

For $n = -1$, the hypothesis is empty and the conclusion says that φ , assumed to be the identity on cohomology, is defined modulo composition with $1 + \psi$, where ψ has degree ≤ -1 .

We prove the assertion for $n \geq 0$, assuming it known for $n - 1$. Changing φ to $\varphi \circ (1 + \psi)$ conjugates the 1-morphism η – more precisely $\varphi^{-1}\eta\varphi$ – of the sum of the $H^i K[-i]$ to $(1 + \psi)^{-1}\eta(1 + \psi)$. For ψ of degree $\leq -n - 1$, since η has degree ≤ 2 , one has

$$\begin{aligned} (1 + \psi)^{-1}\eta(1 + \psi) &\equiv \eta + [\eta, \psi] \\ &\equiv \eta + e(\psi^{(-n-1)}) \end{aligned}$$

modulo degree $\leq -n$. By Lemma 5, there is a unique choice of $\psi^{(-n-1)}$ giving a new η satisfying (10) for $d > -n$, as required. $\square$

Proposition 7. *Let φ be the isomorphism (9) of Proposition 6. Then, on $P^{-i}[i]$, φ coincides with the morphism f_i of Proposition 3.*

Proof. Identify the sum of the $H^i[-i]$ with K by φ . By the characterization in Proposition 3 of f_i , it is enough to verify that, for $s > i$, the restriction to $P^{-i}[i]$ of the s -morphism η^s has degree $< s + i$. We have a decomposition into homogeneous components

$$\eta = \eta^{(2)} + \sum_{a \leq 0} \eta^{(a)}.$$

Put $e = \eta^{(2)}$. By Proposition 6,

$$(\text{ad } e)^{-a+1}(\eta^{(a)}) = 0.$$

Expand η^s and reorder the factors by repeated application of the rule

$$ex = xe + \text{ad } e(x).$$

One finds that η^s is a sum of terms of the following types:

- (a) e^s ;
- (b) $(\text{ad } e)^{r_k}(\eta^{(a_k)}) \cdots (\text{ad } e)^{r_1}(\eta^{(a_1)})e^{r_0}$ with $\sum_0^k r_k < s$.

If $s > i$, e^s vanishes on $P^{-i}[i]$. Similarly, the terms (b) with $r_0 > i$ vanish on $P^{-i}[i]$. The factor $(\text{ad } e)^{r_k}(\eta^{(a_k)})$ has degree $2r_k + a_k$ and is zero if $r_k > -a_k$. If it is nonzero, its degree is

$$2r_k + a_k = r_k + (r_k + a_k) \leq r_k.$$

Each term of type (b) nonzero on $P^{-i}[i]$ is therefore of degree at most

$$2r_0 + \sum_1^k r_k = r_0 + \sum_0^k r_i < i + s,$$

as required. $\square$

4 ℓ -adic cohomology

4.1

Let k be a field and let ℓ be a prime number different from the characteristic. Take for $\mathcal{D}$ the ℓ -adic derived category with its natural t -structure [5]. The heart $\mathcal{C}$ of $\mathcal{D}$ is the category of ℓ -adic representations (V, ρ) of $\text{Gal}(\bar{k}/k)$: V is finite-dimensional over $\mathbf{Q}$, and $\rho: \text{Gal}(\bar{k}/k) \rightarrow \text{GL}(V)$ is continuous. The $\text{Ext}^a(V, W)$ are the groups $H^a(\text{Gal}(\bar{k}/k), \text{Hom}(V, W))$ computed in terms of continuous cochains.

Let X be projective and smooth over k , purely of dimension N . Let $a: X \rightarrow \text{Spec}(k)$ be the structural morphism. Then $Ra_*\mathbf{Q}_\ell$ is an object of $\mathcal{D}$. Fix an ample invertible sheaf $\mathcal{O}(1)$,

and let $\eta: Ra_*\mathbf{Q}_\ell \rightarrow Ra_*\mathbf{Q}_\ell(1)[2]$ be cup product with $c_1\mathcal{O}(1)$. It is known that $(Ra_*\mathbf{Q}_\ell[N], \eta)$ satisfies (L.V.) (P. Deligne, [4, Theorem 4.1.1]).

The decomposition 2.5 of $Ra_*\mathbf{Q}_\ell[N]$ gives, by translation, a decomposition of $Ra_*\mathbf{Q}_\ell$. Changing notation, we write P^i for the direct summand of $H^i Ra_*\mathbf{Q}_\ell$ given by the kernel of

$$\eta^{N-i+1}: H^i Ra_*\mathbf{Q}_\ell \rightarrow H^{2N-i+2} Ra_*\mathbf{Q}_\ell.$$

In the notation of §1 for $Ra_*\mathbf{Q}_\ell[N]$, this is P^{i-N} . The decomposition (8) deduced from 2.5 translates into a decomposition

$$\bigoplus_{0 \leq j \leq N-i} P^i(-j)[-i-2j] \xrightarrow{\sim} Ra_*\mathbf{Q}_\ell. \quad (11)$$

According to Proposition 4, the only interesting components of the 1-morphism η in this decomposition are the $\eta_{i,N-i}^{k,\ell}$ for $\ell \leq N-i$. One has

$$\eta_{i,N-i}^{k,\ell} \in \text{Ext}^{(2N-i)-(k+2\ell)+2}(P^i(-(N-i)), P^k(-\ell)(1)).$$

One would like to regard these classes as motivic: they have been defined by a procedure uniform in ℓ . On the other hand, optimistic conjectures on the relation between algebraic cycles and the motivic derived category imply [3, §§3.7, 3.8] that, for M an effective motive of weight w and $i \geq 0$, one has

$$\text{Ext}^n(1, \check{M}(i)) = 0 \quad \text{for } n > w + i.$$

The special case which matters to us is the vanishing of this Ext when n is equal to the negative of the weight of $\check{M}(i)$, with M effective and $i > 0$.

In the case at hand, by (L.V.) and Poincaré duality, $\check{P}^k$ is isomorphic to $\check{P}^k(-k)$ and

$$\eta_{i,N-i}^{k,\ell} \in \text{Ext}^n(1, V),$$

where n is the negative of the weight of V , and

$$V = \check{P}^i \otimes \check{P}^k(N-i-k-\ell+1).$$

If the ℓ -adic classes $\eta_{i,N-i}^{k,\ell}$ are motivic, the conjecture therefore implies a positive answer to the following question.

Question. Does one have $\eta_{i,N-i}^{k,\ell} = 0$ for $N-i-k-\ell \geq 0$?

4.2

In the decomposition (11), the cup product

$$Ra_*\mathbf{Q}_\ell \otimes Ra_*\mathbf{Q}_\ell \rightarrow Ra_*\mathbf{Q}_\ell$$

has components

$$c_{i,j;k,\ell}^{m,n} \in \text{Ext}^A(P^i(-j) \otimes P^k(-\ell), P^m(-n)),$$

with A the negative of the weight of V and

$$V = \check{P}^i \otimes \check{P}^k \otimes \check{P}^m(j+k-m-n).$$

Thus one wants vanishing for $j+k-m-n > 0$.

4.3

Let Z be an algebraic cycle of codimension d . It has a class

$$cl(Z): \mathbf{Q}_\ell \rightarrow Ra_* \mathbf{Q}_\ell(d)[2d],$$

with components

$$cl(Z)^{i,j} \in \text{Ext}^A(1, V),$$

where A is the negative of the weight of V and

$$V = \check{P}^i(-j+d) = \check{P}^i(d-i-j).$$

One therefore expects the vanishing of the components $cl(Z)^{i,j}$ for $i+j < d$.

Appendix. Spectral objects, after J. L. Verdier

I learned the formalism presented in this appendix from J. L. Verdier, who said that he was inspired by the treatment of spectral sequences in Cartan–Eilenberg (vol. 1, pp. 318–319).

A.1

Let $\mathcal{D}$ be a triangulated category. A *spectral object* in $\mathcal{D}$ consists of

- (1) a family X_{pq} of objects of $\mathcal{D}$, indexed by pairs of integers $p \leq q$;
- (2) for $p' \leq p$ and $q' \leq q$, a morphism $X_{pq} \rightarrow X_{p'q'}$;
- (3) for $p \leq q \leq r$, a degree-one morphism, called a boundary, $X_{qr} \rightarrow X_{pq}$.

One requires that

- (a) the morphisms (2) define a contravariant functor from the ordered set of pairs (p, q) with $p \leq q$ to $\mathcal{D}$;
- (b) for $p \leq q \leq r$, $p' \leq q' \leq r'$, and $p' \leq p$, $q' \leq q$, $r' \leq r$, the diagram of the morphisms (2) and (3) is commutative;
- (c) for $p \leq q \leq r$, the triangle

$$X_{qr} \rightarrow X_{pr} \rightarrow X_{pq} \xrightarrow{+1} X_{qr}$$

formed by the morphisms (2) and (3) is distinguished.

Example 8. For $\mathcal{D}$ the derived category of an abelian category $\mathcal{A}$ and (K, F) a filtered complex, the objects $X_{pq} = F^p(K)/F^q(K)$ form a spectral object.

Remark 9. A spectral object X is said to have amplitude in $[a, b]$ if the morphisms (2), $X_{pq} \rightarrow X_{p'q'}$, are isomorphisms for $p' \leq p \leq a$ and $q' = q$, and also for $b < q' \leq q$ and $p' = p$. This is equivalent to requiring $X_{p,p+1} = 0$ for $p \notin [a, b]$. A spectral object of amplitude $[0, n]$ with $n = 1$, respectively 2, is identified with a distinguished triangle, respectively an octahedral diagram; see [1, §§ 1.1.6, 1.1.7, 1.1.14].

A.2

Let $\mathcal{A}$ be an abelian category equipped with a “translation functor” $X \mapsto X[1]$, an autoequivalence. Write $X \mapsto X[n]$ for the n -th iterate of the translation functor, and call a morphism of degree n from X to Y a morphism from X to $Y[n]$. A sequence $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$, with α of degree n and β of degree m , is said to be exact if the sequence $X \rightarrow Y[n] \rightarrow Z[n+m]$ is exact.

A *spectral object* of $\mathcal{A}$ consists of data $(1')$, $(2')$, $(3')$ of objects and morphisms of $\mathcal{A}$, as in A.1, satisfying conditions (a') and (b') with the same formulation as A.1(a),(b), and the condition

(c') for $p \leq q \leq r$, the sequence

$$\cdots \rightarrow X_{qr} \rightarrow X_{pr} \rightarrow X_{pq} \xrightarrow{\partial} X_{qr} \rightarrow X_{pr} \rightarrow \cdots$$

formed by the morphisms $(2')$ and $(3')$ is exact.

Let X be a spectral object of $\mathcal{A}$. From (b') , the morphism $(2')$: $X_{p,r} \rightarrow X_{p,r}$ is the identity. From (c') , for $p = q$, one has $X_{pp} = 0$. If $p \leq q'$, the morphism $X_{pq} \rightarrow X_{p'q'}$ is zero, since it factors through X_{pp} . For $p \leq q \leq r \leq s$, the composite of boundary morphisms

$$X_{pq} \rightarrow X_{qr} \rightarrow X_{rs}$$

is zero: the first morphism factors through the boundary morphism to X_{qs} , followed by $X_{qs} \rightarrow X_{qr}$, and the composite $X_{qs} \rightarrow X_{qr} \rightarrow X_{rs}$ is zero by (c') .

If $\mathcal{A}$ is the category of graded objects of an abelian category $\mathcal{B}$, with the translation functor defined by $(X[1])^n = X^{n+1}$, then the data $(1')$ are objects X_{pq}^n of $\mathcal{B}$, the data $(2')$ are morphisms $X_{pq}^n \rightarrow X_{p'q'}^n$, the data $(3')$ are morphisms $X_{pq}^n \rightarrow X_{qr}^{n+1}$, and (c') becomes a long exact sequence

$$\cdots \rightarrow X_{pr}^n \rightarrow X_{pq}^n \xrightarrow{\partial} X_{qr}^{n+1} \rightarrow X_{pr}^{n+1} \rightarrow \cdots$$

Example 10. With the preceding notation, if F is a cohomological functor [6, §3.1] from $\mathcal{D}$ to $\mathcal{B}$, and if F^* is the functor from $\mathcal{D}$ to $\mathcal{A}$ defined by $F^n(X) = F(X[n])$, then F^* carries spectral objects of $\mathcal{D}$ to spectral objects of $\mathcal{A}$.

Remark 11. As in A.1, if X is a spectral object of $\mathcal{A}$, then for the morphisms $(2')$ to be isomorphisms for $p' \leq p \leq a$ and $q' = q$, and also for $b < q' \leq q$ and $p' = p$, it is necessary and sufficient that $X_{p,p+1} = 0$ for $p \notin [a, b]$. One then says that X has amplitude in $[a, b]$. In this case, $X_{-\infty, q}$, respectively $X_{p, \infty}$, denotes any $X_{p, q}$ with $p \leq a$, respectively $b < q$.

A.3

Let X be a spectral object of $\mathcal{A}$. For $p \leq q \leq r \leq s$, put

$$E(pqrs) := \text{Im}(X_{qs} \rightarrow X_{pr}). \quad (12)$$

The morphism $X_{qs} \rightarrow X_{pr}$ factors through X_{qr} , and this factorization gives a cross of exact sequences

$$\begin{array}{ccc} & X_{qr} & \\ X_{qs} \nearrow & & \searrow X_{pr} \\ & X_{qs}/X_{rs} & \end{array}$$

It follows that one also has

$$E(pqrs) = H(X_{qs} \rightarrow X_{qr} \rightarrow X_{pr}). \quad (13)$$

For $k \leq \ell \leq m \leq n \leq p \leq q$, the commutative diagram

$$\begin{array}{ccc} X_{\ell n} & \longrightarrow & X_{nq} \\ \downarrow & & \\ X_{km} & \longrightarrow & X_{mp} \end{array}$$

gives, after passing to images by the vertical morphisms (2'), a degree-one morphism

$$\partial: E(k\ell mn) \rightarrow E(mnpq).$$

For $k \leq \ell \leq m \leq n \leq p \leq q \leq r \leq s$, the composite

$$E(k\ell mn) \xrightarrow{\partial} E(mnpq) \xrightarrow{\partial} E(pqrs) \quad (14)$$

is zero, because the rows in the diagram

$$\begin{array}{ccccc} X_{\ell n} & \longrightarrow & X_{nq} & \longrightarrow & X_{qs} \\ \downarrow & & \downarrow & & \downarrow \\ X_{km} & \longrightarrow & X_{mp} & \longrightarrow & X_{pr} \end{array}$$

have zero composite (see A.2).

Construction. The cohomology of the sequence (14) is $E(\ell npr)$.

The cohomology of (14) is also the cohomology of

$$X_{\ell n} \xrightarrow{\partial} \text{Im}(X_{nq} \rightarrow X_{mp}) \xrightarrow{\partial} X_{pr}.$$

Consider the commutative diagram

$$\begin{array}{ccccc} & & X_{mn} & & \\ & \swarrow & & \searrow & \\ X_{\ell n} & & & & X_{nq} \\ & \searrow & & \swarrow & \\ & & X_{pq} & & X_{mp} \\ & & \searrow & \nearrow & \\ & & X_{pr} & & \end{array}$$

The desired cohomology is still that of

$$X_{\ell n} \rightarrow H(X_{mn} \rightarrow X_{np} \rightarrow X_{pq}) \rightarrow X_{pr},$$

which is equal to

$$H(X_{\ell n} \rightarrow X_{np} \rightarrow X_{pr}),$$

and this gives the required construction by (13).

A.4

For $r \geq 1$, put

$$E_r^p = E(p - r + 1, p, p + 1, p + r).$$

The morphism ∂ of A.3, for $(p - r + 1, p, p + 1, p + r, p + r + 1, p + 2r)$, is a degree-one map $d_r: E_r^p \rightarrow E_r^{p+r}$. One has $d_r^2 = 0$, and A.3 gives

$$E_{r+1}^p = H(E_r^{p-r} \rightarrow E_r^p \rightarrow E_r^{p+r});$$

the E_r^p form a spectral sequence.

If X has finite amplitude, say in an interval $[a, b]$ (A.2), this spectral sequence converges to $X_{-\infty, \infty}$; E_r^p is independent of r for sufficiently large r ($r > b - a$), and identifies with the associated graded of $X_{-\infty, \infty}$ filtered by the images of the $X_{p, \infty}$.

If this filtration is denoted by F , the exact sequence

$$X_{p, \infty} \rightarrow X_{-\infty, \infty} \rightarrow X_{-\infty, p}$$

indeed shows that

$$X_{p, \infty} \twoheadrightarrow F^p \quad \text{and} \quad X_{-\infty, \infty}/F^p \hookrightarrow X_{-\infty, p},$$

so that

$$F^p/F^{p+1} \sim \text{Im}(X_{p, \infty} \rightarrow X_{-\infty, p+1}) = E_{\infty}^p.$$

The spectral sequence constructed in this way starts from E_1 . In the case where $\mathcal{A}$ is the category of graded objects of $\mathcal{B}$, if one wants a spectral sequence starting from E_2 , it is enough to reindex:

$${}_2E_2^{pq} := ({}_1E_1^{-q})^{p+q}.$$

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On the Locus of Hodge Classes

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1 Introduction

Let S be a complex algebraic variety and X a family of nonsingular projective varieties parametrized by S : the X_s are the fibers of $f : \mathcal{X} \rightarrow S$, with X_s projective and f smooth over S . Let moreover $h \in H^{2p}(X_s, \mathbb{Q})$ be a Hodge class, i.e. of type (p, p) in the Hodge decomposition of $H^{2p}(X_s, \mathbb{C})$. On the local system $R^{2p}f_*(\mathbb{Q})$, the Hodge filtration induces a filtration by holomorphic subbundles. Their projective bundle has a flat structure and the local system of the $H^{2p}(X_t, \mathbb{Z})$ modulo torsion is, on a suitable connected neighborhood U of s , trivial on U , so that for $t \in U$ they can all be identified with $H^{2p}(X_s, \mathbb{Z})$. The Hodge filtration F_t of $H^{2p}(X_t, \mathbb{C})$, $t \in U$, can be viewed as a variable filtration on the fixed complex vector space $H^{2p}(X_s, \mathbb{C})$. It varies holomorphically with t .

It follows that the locus $T \subset U$ where h remains of type (p, p) , i.e. in F^p , is a complex analytic subspace of U .

It follows from the (rational) Hodge conjecture that the germ of T at s is algebraic, meaning that its irreducible components are irreducible components of germs at s of algebraic subvarieties of S . Sketch of proof: let T_0 be an irreducible component of T containing s and assume that for all t in T_0 some non zero multiple of h is the class of an algebraic cycle in X_t . A Baire category argument shows that for suitable M, N , the set of $t \in T_0$ for which h is the class of an algebraic cycle of degree $\leq M$ on X_t with $Z^p - Z^{p-}$ effective, $Z^\pm$ effective cycles of degree $\leq M$ on the X_t , is a union of countably many closed algebraic subvarieties of T_0 . On the other hand, because the Chow varieties of effective cycles of degree $\leq M$ on the X_t form an algebraic variety over S , or simply that they form a limited family.

In Corollary 1.2 below, we prove unconditionally that the germ of T at s , as above, is indeed algebraic. Our main result, Theorem 1.1, is slightly more precise and gives as corollary a positive answer to a question of A. Weil [?]: "... whether imposing a certain Hodge class upon a generic member of [such family] amounts to an algebraic condition upon the parameters."

The Hodge conjecture would also imply that if $f : \mathcal{X} \rightarrow S$ can be defined over an algebraically closed subfield of $\mathbb{C}$, then so can the germ of T at s . About this, we are not able to say anything.

The proof will be in the setting of variations of Hodge structures, of which the local system of the $H^{2p}(X_t, \mathbb{Z})$ modulo torsion is an example. For the definition and basic properties of variations of Hodge structures on a complex variety S , we refer to [?] or [?].

Our notations are spelled out in 2.1. Note that in the Hodge decomposition, we do not assume $p, q \geq 0$. If H is a Hodge structure of even weight $2p$, shifting the Hodge filtration, i.e., replacing H by the Tate twist $H(p)$, one obtains a Hodge structure of weight 0. Classes of type (p, p) become of type $(0, 0)$. This allows us to restrict our attention to variations of weight 0 and classes of type $(0, 0)$. For simplicity, we will assume the parameter space S to be nonsingular.

Let S be a nonsingular complex algebraic variety and $\mathbb{V}$ a variation of Hodge structures of weight 0 on S with polarization form Q . If C is the Weil operator: multiplication by i^{p-q} on $\mathbb{V}^{p,q}$, the hermitian form $h(u, v) = Q(Cu, \bar{v})$ is positive definite and makes the Hodge decomposition orthogonal. For u a real element of type $(0, 0)$, $h(u, u) = Q(u, u)$. Fix an integer K and let $S^{(K)}$

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be the space of pairs (s, u) with $s \in S$, $u \in \mathbb{V}_{s\mathbb{Z}}$ integral of type $(0, 0)$, and $Q(u, u) \leq K$. It projects to S and arguments as above show that, locally on S , $S^{(K)}$ is a finite disjoint sum of closed analytic subspaces. Our main result is:

Theorem 1.1. *$S^{(K)}$ is an algebraic variety, finite over S .*

Corollary 1.2. *Fix $s \in S$ and $u \in \mathbb{V}_{s\mathbb{Z}}$ integral of type $(0, 0)$. The germ of analytic subvariety of S where u remains of type $(0, 0)$, is algebraic.*

Proof. The required algebraic subvarieties of S are images in S of irreducible components of $S^{(K)}$, for $K = Q(u, u)$. $\square$

Corollary 1.3. *Let u be a section of the local system $\mathbb{V}_{\mathbb{Z}}$ on a universal covering of S . The set of points in S where some determination of u is of type $(0, 0)$, is an algebraic subvariety of S .*

Proof. Such set is a union of images of connected components of $S^{(K)}$, for $K = Q(u, u)$. $\square$

Corollary 1.4. *Let $\mathbb{V}$ be a polarizable variation of Hodge structures on S , fix $s \in S$ and let $U_{\mathbb{Q}} \subset \mathbb{V}_{s\mathbb{Q}}$ be a rational subspace. The locus where some flat translate of $U_{\mathbb{Q}}$ is a Hodge substructure is an algebraic subvariety of S .*

Proof. Let T be the locus in question, assumed nonempty, and suppose first that $U_{\mathbb{Q}}$ is of dimension one. Then $\mathbb{V}$ is of even weight $2p$ so that, replacing it by $\mathbb{V}(p)$, we may and shall assume it to be of weight 0. Let e be a generator of $U_{\mathbb{Q}} \cap \mathbb{V}_{s\mathbb{Z}}$. Then $U_{\mathbb{Q}}$ is a Hodge substructure if and only if e is of type $(0, 0)$, and one applies 1.3.

Consider now the general case: $U_{\mathbb{Q}}$ of dimension n . A Hodge structure on a rational vector space $H_{\mathbb{Q}}$ gives rise to an action of the real algebraic group $\mathbb{C}^*$ on $H_{\mathbb{R}} = H_{\mathbb{Q}} \otimes \mathbb{R}$, with $z \in \mathbb{C}^*$ acting as multiplication by $z^p \bar{z}^{-q}$ on $H^{p,q}$. A subspace $U_{\mathbb{Q}} \subset H_{\mathbb{Q}}$ is a sub-Hodge structure—i.e., $U_{\mathbb{C}} = U_{\mathbb{Q}} \otimes \mathbb{C}$ is the sum of its intersections with the $H^{p,q}$ —if and only if $U_{\mathbb{R}}$ is stable under $\mathbb{C}^*$. This amounts to $\bigwedge^n U_{\mathbb{R}} \cap \bigwedge^n H_{\mathbb{R}}$ being stable under $\mathbb{C}^*$, i.e. to $\bigwedge^n U_{\mathbb{Q}}$ being a Hodge substructure of $\bigwedge^n \mathbb{V}_{s\mathbb{Q}}$, and reduces us to the one-dimensional case, proving 1.4. $\square$

Let $D^* = D - \{0\}$ be the open punctured disk. Theorem 1.1 will be derived from the following local result.

Theorem 1.5. *Let $\mathbb{V}$ be a polarized variation of Hodge structures of weight 0 on $S = (D^*)^r \times D^m$, with unipotent monodromy. Define $S^{(K)}$ as in 1.1. Then, there is a neighborhood U of 0 in D^{r+m} such that, above U , $S^{(K)}$ is a finite disjoint sum of traces on $S \cap U$ of closed analytic subspaces of U .*

Proof of 1.1 $\Rightarrow$ 1.5: To prove 1.1 one is free to replace S of 1.1 by a finite étale covering $S' \rightarrow S$. We may and shall assume that the monodromy mod k of $\mathbb{V}$ is trivial, for some $k \geq 3$.

Let $\bar{S}$ be a smooth compactification of S , with $\bar{S} - S$ a divisor with normal crossings. The assumption on the monodromy ensures that the local monodromy of $\mathbb{V}$ at infinity is unipotent: in a neighborhood of any point in $\bar{S} - S$, one is in the situation considered in 1.5. One concludes that $S^{(K)}$ can be extended to a space $\bar{S}^{(K)}$ over $\bar{S}$, which locally over $\bar{S}$ is a finite disjoint sum of closed analytic subspaces. By GAGA, $\bar{S}^{(K)}$, finite over $\bar{S}$, is algebraic, and 1.1 follows. $\square$

The rest of the paper is devoted to proving 1.5. To simplify notations, we will assume $m = 0$. The general case is recovered by considering the partition of S , according to which of $z_{r+1}, \dots, z_{r+m}$ vanish.

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2 Notations. Schema of Proof.

2.1

Let $\mathbb{V}$ be a polarized variation of Hodge structures of weight w on a complex manifold S , with polarization form Q . We write $\mathbb{V}_{\mathbb{Z}}$ for the underlying local system of free $\mathbb{Z}$ -modules, $\mathcal{V}_{\mathcal{O}_S}$ for the corresponding holomorphic vector bundle, identified with its sheaf of sections $\mathcal{O}_S \otimes \mathbb{V}_{\mathbb{Z}}$, and $\mathbb{V}$ for the underlying complex vector bundle, identified with its sheaf of C^∞ sections. The Hodge decomposition is a decomposition of this complex vector bundle

$$\mathbb{V} = \bigoplus_{p+q=w} \mathbb{V}^{p,q}$$

and the corresponding Hodge filtration

$$F^p = \bigoplus_{a \geq p} \mathbb{V}^{a,b}$$

is holomorphic. We still write F^p for the corresponding filtration of $\mathcal{V}_{\mathcal{O}_S}$. Griffiths transversality axiom then reads

$$\nabla F^p \subset \Omega^1 \otimes F^{p-1}. \quad (1)$$

The Weil operator C is the endomorphism of $\mathbb{V}$ acting on $\mathbb{V}^{p,q}$ as multiplication by i^{p-q} . The polarization form Q is best viewed as a morphism of Hodge structures $\mathbb{V} \otimes \mathbb{V} \rightarrow \mathbb{Z}(-w)$. It is an integral bilinear form on $\mathbb{V}_{\mathbb{Z}}$, $(-1)^w$ -symmetric, and the form on $\mathbb{V}$

$$h(u, v) = Q(Cu, \bar{v})$$

is hermitian symmetric, positive definite and makes the Hodge decomposition orthogonal. This Hodge metric is, generally, not flat.

2.2

We will need Schmid's theory of nilpotent orbits giving the asymptotic behavior of variations of Hodge structures [?]. We begin with a coordinate-free description. The coordinate-bound translation given in 2.3 suffices for our needs.

Let S be the complement in a smooth variety $\bar{S}$ of smooth divisors E_i meeting transversally. Assume that the monodromy of $\mathbb{V}$ around the E_i is unipotent. Let $\mathcal{V}_{\mathcal{O}_{\bar{S}}}$ be the canonical extension [?] of the holomorphic vector bundle $\mathcal{V}_{\mathcal{O}_S}$ to $\bar{S}$. It is characterized by the property that, in any local basis of $\mathcal{V}_{\mathcal{O}_S}$, the connection matrix (an endomorphism-valued 1-form) has logarithmic poles with nilpotent residues along the E_i . The first result is that

The Hodge filtration of $\mathcal{V}_{\mathcal{O}_S}$ extends to a filtration of $\mathcal{V}_{\mathcal{O}_{\bar{S}}}$ by locally direct factors. (2)

Let E be the intersection of the E_i , L_i the restriction to E of the normal line bundle of E_i , and L_i^* the complement in L_i of the zero section. The product L of the L_i is the normal bundle of E and the product L^* of the L_i^* is obtained by removing from the normal bundle L , the normal bundles of E in the E_i .

The nilpotent orbit $\mathbb{V}^{un}$ (or: deformation to the normal cone) approximating $\mathbb{V}$ around E lives on L^* . It is obtained as follows:

(a) As a filtered holomorphic vector bundle, it is the pullback of the restriction of $(\mathcal{V}_{\mathcal{O}_{\bar{S}}}, F)$ to E .

(b) The connection ∇^{un} on $\mathbb{V}^{un}$ is described as follows. Locally, let $z_i = 0$ be an equation for E_i . It defines a trivialization of L_i and we write again z_i for the corresponding function on L . In

a local basis of $\mathcal{V}_{\mathcal{O}_{\bar{S}}}$, the connection ∇ is $d + \Gamma$, with $\Gamma = \Gamma_0 + \sum_i \nu_i \frac{dz_i}{z_i}$, Γ_0 and ν_i holomorphic. The local basis of $\mathcal{V}_{\mathcal{O}_{\bar{S}}}$ gives one on E , which pulls back to $\mathbb{V}^{un}$; in this basis,

$$\nabla^{un} = d + (\Gamma_0|E) + \sum_i (\nu_i|E) \frac{dz_i}{z_i}.$$

(c) It remains to define the integral lattice. Locally, let φ be an isomorphism from a neighborhood of the zero section of L (identified with E) to a neighborhood of E in $\bar{S}$. Assume it is the identity on E and that, on E , $d\varphi$ induces the identity on the normal bundle of E . Assume further that φ maps L^* into S . Then, the pullback by φ of the local system $\mathbb{V}$ of horizontal sections of $\mathbb{V}$ is canonically isomorphic to $\mathbb{V}_{\mathbb{Z}}$. One defines $(\mathbb{V}^{un})_{\mathbb{Z}}$ to be the pullback of $\mathbb{V}_{\mathbb{Z}}$.

The results are that

On the trace on L^* of a neighborhood U of E , with U depending only on the rank of $\mathbb{V}$, $\mathbb{V}^{un}$ is a variation of (3)

For φ as in (c), defined on U , $\mathbb{V}$ and $\varphi^*\mathbb{V}^{un}$ are close: on any compact $K \subset U$, the distance at $z \in K \cap L^*$ bet (4)

with C_0 depending only on the rank, and C_1 only on the rank, on K and on the chosen distance d .

2.3

We now translate in coordinates. Let $D \subset \mathbb{C}$ be the open unit disk and $D^* = D - \{0\}$. Take $S = (D^*)^r \subset D^r$. Assume that the monodromy of $\mathbb{V}$ is unipotent. Let M_i be the monodromy around $z_i = 0$ and $N_i = \log M_i$.

Using the coordinate z_i , one can view $\mathbb{V}^{un}$ as living on $(\mathbb{C}^*)^r \supset S$. The underlying local system is the unique local system extending $\mathbb{V}_{\mathbb{Z}}$ on S . We continue to denote it by $\mathbb{V}_{\mathbb{Z}}$. Let $V_{\mathbb{Z}}$ be its fiber at 1.

The Poincaré upper half-plane $\mathcal{H}$ is the universal covering of D^* , with covering map $z \mapsto s = e^{2\pi iz}$. Similarly, $\mathcal{H}^r$ is the universal covering of $S = (D^*)^r$ and $(\mathbb{C}^r, 0)$ that of $(\mathbb{C}^{*r}, 1)$. For $z \in \mathbb{C}^r$, we write s for the corresponding point in $\mathbb{C}^{*r}$.

When pulled back to $\mathcal{H}^r$, the variation can be described as a variable Hodge filtration $\Phi(z)$ on the fixed vector space $V_{\mathbb{C}} = V_{\mathbb{Z}} \otimes \mathbb{C}$, with

$$\Phi(z + e_j) = \exp(N_j)\Phi(z)$$

for e_j the j th coordinate vector in $\mathbb{C}^r$. We also view Φ as a holomorphic map $\Phi : \mathcal{H}^r \rightarrow \check{D}(V_{\mathbb{Z}}, Q)$ with values in the appropriate period mapping domain.

Statement (2) translates as

$$\Phi(z) = \exp\left(\sum_j z_j N_j\right)\Psi(s) \quad (5)$$

($s = e^{2\pi iz}$) for some holomorphic map Ψ from D^r to the flag manifold of $V_{\mathbb{C}}$. The nilpotent orbit $\mathbb{V}^{un}$ is given by

$$\Phi_{un}(z) = \exp\left(\sum_j z_j N_j\right)\Psi(0).$$

Statements (3) and (4) translate as the existence of constants C_0, C_2, C_3 , depending only on the rank of $\mathbb{V}$, with the following properties. Let x_i and y_i be the real and imaginary parts of z_i . Then, as soon as $\inf(y_i) \geq C_2$,

$$\Phi_{un}(z) \text{ defines a Hodge structure on } V. \quad (6)$$

The distance between $\Phi(z)$ and $\Phi_{un}(z)$, measured as in (4), is $\leq C_3 e^{-2\pi \inf_j(y_j)} \leq \inf_i(y_i)^{-C_0}$. (7)

Fix $I \subset \{1, \dots, r\}$, with complement J . Let E be the intersection in D^r of the divisors $z_i = 0$, for $i \in I$. It is identified with D^J . Let q_I be the projection $(D^*)^r \rightarrow (D^*)^J$. The nilpotent orbit along E is then given by $\mathbb{V}^{un,I}$

$$\Phi_{un,I}(z) = \exp\left(\sum_j z_j N_j\right) \Psi(q_I(s)).$$

Freezing the z_j ($j \in J$) and applying (6), (7), one sees that $\Phi_{un,I}(z)$ gives a Hodge structure on V as soon as $\inf_{i \in I}(y_i) \geq C_2$, and that the proximity of $\Phi_{un,I}(z)$ to $\Phi(z)$ is controlled by $\inf_{i \in I}(y_i)$ as in (7).

2.4

Any nilpotent endomorphism N of V , $N^{k+1} = 0$, has an associated filtration $W(N)$. This is an increasing filtration

$$0 = W_{-k-1}(N) \subset \dots \subset W_k(N) = V$$

characterized by the properties that

$$NW_\ell \subset W_{\ell-2}$$

and that N^ℓ induces an isomorphism

$$N^\ell : \mathrm{Gr}_\ell^W \xrightarrow{\sim} \mathrm{Gr}_{-\ell}^W.$$

With the notations of 2.3, it is shown in [?] that all elements N in the cone

$$C = \left\{ \sum_j \lambda_j N_j, \lambda_j > 0 \right\}$$

define the same filtration $W(C)$. As $C \times X$ contains endomorphisms defined over $\mathbb{Q}$, the filtration $W := W(C)$ is defined over $\mathbb{Q}$.

Let us call “limiting Hodge filtration” any filtration of the form $\Phi_{un}(z)$ ($z \in \mathbb{C}^r$). It is a consequence of the $\mathrm{SL}(2)$ -orbit theorem [?] that, for any limiting Hodge filtration F , F and $W[-w]$ define a mixed Hodge structure. Here w is the weight of $\mathbb{V}$ and $W[-w]_\ell = W_{\ell-w}$.

The mixed Hodge structure $(W[-w], F)$ is polarized, in the sense of [?], by the polarization form of $\mathbb{V}$ and any N in C .

Theorem 1.5 will be deduced from the following result, where we use the notations of 2.3, 2.4.

Theorem 2.1. *Assume $\mathbb{V}$ of weight 0. Given K , there is a constant A_1 (depending on K and $\mathbb{V}$) such that*

(i) *There are only finitely many $v \in V_{\mathbb{Z}}$ such that: $Q(v, v) \leq K$ and $v \in \Phi^0(z)$ at some point z with $0 \leq x_i \leq 1$ and $\inf(y_i) \geq A_1$.*

(ii) *Any such v is in $W_0 \cap F^0$ for some limiting Hodge filtration F , as well as in F^0 .*

Remark 2.2. If v is, at z , in Φ^0 , it defines a morphism of Hodge structures from the unit Hodge structure $\mathbb{Z}(0)$ to $\mathbb{V}$ at z . By (ii), if $Q(v, v) \leq K$ and $\inf(y_i) \geq A_1$, it will also define a morphism of mixed Hodge structures from $\mathbb{Z}(0)$ to $\mathbb{V}^{un}$ at some point z' . In 2.13, we will deduce from 2.3 that z' can be chosen such that

$$|z' - z| \leq A_2 e^{-2\pi \inf(y_i)}$$

with A_2 depending only on $\mathbb{V}$.

2.5

As a preparation to deducing 1.5 from 2.1, we now fix $v \in V_{\mathbb{Z}}$ which, for some z_0 , is in $W_0 \cap \Phi_{un}^0(z_0)$, and investigate the locus where $v \in \Phi^0(z)$. We make the change of variables $z \mapsto z - z_0$, $s \mapsto s/s_0$, to simplify notations by assuming $z_0 = 0$. This is at the cost of having $\Phi(z)$ defined, in the new coordinates, only for $y_i \geq -\Im(z_{0i})$.

Let F be the filtration $\Phi_{un}(0)$. Let $V^{p,q}$ be the bigrading of $V_{\mathbb{C}}$ associated to the mixed Hodge structure (W, F) (cf. [?, 2.13]). It splits the filtrations W and F : W_{ℓ} (resp. F^p) is the sum of the $V^{a,b}$ for $a + b \leq \ell$ (resp. $a \geq p$). The assumptions on v imply that $v \in V^{0,0}$.

The Lie algebra $\mathfrak{g} = \mathfrak{gl}(V)$ of $\mathrm{GL}(V)$ inherits from V a mixed Hodge structure whose associated bigrading is

$$\mathfrak{g}^{p,q} = \{X \in \mathfrak{gl}(V) \mid X(V^{r,s}) \subset V^{p+r, q+s}\}.$$

We will mainly use the corresponding p -grading, for which V^p (resp. $\mathfrak{g}^p$) is the sum of the $V^{p,b}$ (resp. $\mathfrak{g}^{p,b}$). The isotropy subalgebra of $\mathfrak{g}$ at F is $F^0(\mathfrak{g}) = \bigoplus_{p \geq 0} \mathfrak{g}^p$. It admits as supplement the nilpotent subalgebra $\mathfrak{b} = \bigoplus_{p < 0} \mathfrak{g}^p$. The map $X \mapsto \exp(X) \cdot \bar{F}$ identifies a neighborhood of 0 in $\mathfrak{b}$ with a neighborhood of F in the flag space. As $\Psi(0) = F$, we can, for y_i large enough, rewrite (5) as

$$\Phi(z) = \exp\left(\sum_j z_j N_j\right) \exp(\Gamma(s)) \cdot F \quad (8)$$

($s = e^{2\pi iz}$) with Γ holomorphic at $s = 0$, $\mathfrak{b}$ -valued and such that $\Gamma(0) = 0$.

The N_j are morphisms of mixed Hodge structures $V \rightarrow V(-1)$, hence they lie in $\mathfrak{g}^{-1,-1} \subset \mathfrak{g}^{-1} \subset \mathfrak{b}$. As $v \in V^0$, the equation

$$v \in \Phi^0(z) = \exp\left(\sum_j z_j N_j\right) \exp(\Gamma(s)) \cdot F^0 \quad (9)$$

or, equivalently,

$$(\exp\left(\sum_j z_j N_j\right) \exp(\Gamma(s)))^{-1}(v) \in F^0 = \bigoplus_{p \geq 0} V^p,$$

holds if and only if v is fixed by $\exp(\sum_j z_j N_j) \exp(\Gamma(s))$. Let $\Gamma_p(s)$ be the component of $\Gamma(s)$ in $\mathfrak{g}^p$. Taking the component in V^{-1} of the equation $\exp(\sum_j z_j N_j) \exp(\Gamma(s))(v) = v$, we obtain

$$\left(\sum_j z_j N_j + \Gamma_{-1}(s)\right)(v) = 0. \quad (10)$$

We now use the transversality axiom to prove:

Lemma 2.3. *If at one point of the analytic space Σ where (10) holds one has $v \in \Phi^0(z)$, the same is true on the whole connected component of Σ containing that point.*

Proof. Equation (10) is equivalent to

$$(\exp\left(\sum_j z_j N_j\right) \exp(\Gamma(s)))^{-1}(v) \in F^0 \oplus \bigoplus_{p \leq -2} V^p,$$

i.e., to

$$v \in \exp\left(\sum_j z_j N_j\right) \exp(\Gamma(s)) \left(F^0 \oplus \bigoplus_{p \leq -2} V^p\right),$$

so that 2.3 is a particular case of

Lemma 2.4. *Let $\mathbb{X}$ be a variation of Hodge structures on a complex manifold M . Let U be a supplement to F^{p+1} . For v a horizontal section of $\mathbb{X}$, let Σ be the locus where $v \in F^p \oplus U$. Then, if at one point of Σ one has $v \in F^p$, then the same holds in the whole connected component of Σ containing that point.*

Proof. By assumption, $\mathbb{X} = F^{p+1} \oplus U$. The connection ∇ of $\mathbb{X}$ induces a connection ∇_U on U , not necessarily integrable:

$$\nabla_U(\xi) = \text{projection of } \nabla(\xi).$$

Let $\hat{v}$ be the projection of v to U . Because $\nabla F^p \subset \Omega^1 \otimes F^{p-1}$ on Σ , $\hat{v}$ is a horizontal section of U . The lemma follows. $\square$

This completes the proof of Lemma 2.3. $\square$

2.6

By 2.3, the locus where $v \in \Phi^0(z)$ is a union of connected components of the locus Σ where (10) holds. Each $N_j v$ is in $V_{\mathbb{Q}}$. Writing (10) in a basis of $V_{\mathbb{Q}}$, we obtain a system of equations

$$\sum_j \nu(\alpha)_j z_j + \gamma(\alpha)(s) = 0 \quad (\alpha = 1, \dots, \dim V_{\mathbb{Q}}) \quad (11)$$

with $\nu(\alpha)_j$ rational, $\gamma(\alpha)(s)$, $s = e^{2\pi iz}$, holomorphic at 0 and $\gamma(\alpha)(0) = 0$.

Lemma 2.5. *Near 0 in D^r , there is a closed analytic subspace Δ such that, if Σ_0 is a connected component of Σ (equation (10)), its image in $(D^*)^r$ is a connected component Δ_0 of $\Delta \cap (D^*)^r$. Further, the image of $\pi_1(\Delta_0)$ in $\pi_1((D^*)^r)$ fixes v .*

Proof. If we clear denominators and exponentiate, equations (11) give

$$\prod_i a_{\alpha}(s) \cdot s_i^{n(\alpha,i)} = 1$$

with $a_{\alpha}(s)$ holomorphic at 0, $a_{\alpha}(0) = 1$ and $n(\alpha, i) \in \mathbb{Z}$. That $a_{\alpha}(0) = 1$ springs from the normalization $z_0 = 0$. As equations for Δ , we take

$$a_{\alpha}(s) \cdot \prod_i s_i^{n(\alpha,i)} - \prod_i s_i^{-\min(0, n(\alpha,i))} = 0.$$

An element m of $\pi_1((D^*)^r) = \mathbb{Z}^r$ is in the image of $\pi_1(\Delta_0)$ if and only if Σ_0 contains, together with any point z , the point $z + m$. Subtracting the corresponding equations (11), we obtain

$$\sum_j \nu_{\alpha j} m_j = 0$$

i.e.,

$$\sum_i m_i N_i v = 0.$$

Since m acts on V as $\exp(\sum m_i N_i)$, the Lemma follows. $\square$

2.7

Proof of 2.1 $\Rightarrow$ 1.5: As explained after 1.5, it suffices in 1.5 to consider the case $m = 0$: $S = (D^*)^r$. By 2.1, 2.3, 2.5, there are, in a neighborhood of $0 \in D^r$, finitely many closed analytic subspaces Y such that all connected components of $S^{(K)}$ are obtained by taking a connected component of $Y \cap (D^*)^r$ and a section of $\mathbb{V}_{\mathbb{Z}}$ on it. $\square$

2.8

Proof of 2.2: Expressed in $(D^*)^r$, 2.2 claims that, for s close enough to 0, if $v \in (\mathbb{V}_{\mathbb{Z}})_s$ is of type $(0,0)$ and satisfies $Q(v, v) \leq K$, then at a nearby point s' :

$$|s - s'| \leq A_3 |s| \quad (12)$$

the same v is of type $(0,0)$ for $\mathbb{V}^{un}$. “Same” means: horizontal translate by a path remaining in the neighborhood (12) of s .

As the strip $0 \leq x_i \leq 1$ covers $(D^*)^r$, working on $\mathcal{H}^r$ we may assume that v is as in 2.1. By 2.1 (i), we may treat those v one at a time. We make the same change of variable $z \mapsto z - z_0$ as in 2.7. The equation (10) gives, for a fixed norm on V ,

$$|\sum_j z_j N_j v| \leq A_4 e^{-2\pi \inf(y_j)}.$$

This implies that

$$\sum_j z'_j N_j v = 0, \quad \text{with} \quad |z'_j - z_j| \leq A_5 e^{-2\pi \inf(y_j)} :$$

take $z' \mapsto \sum z'_j N_j v$ to be the image of $\sum z_j N_j v$ by a fixed linear section of $\mathbb{C}^r \rightarrow V$; $v \mapsto \sum z_j N_j v$. The condition $\sum z'_j N_j v = 0$ is equivalent to $v \in \Phi_{un}^0(z')$.

Remark 2.6. The following example shows that the constant A_1 in 2.1 does depend on $\mathbb{V}$. Consider the case of a variation on D^* that extends to a variation on D . The nilpotent orbit is then the constant variation with value $\mathbb{V}_0$. Fix a variation $\mathbb{W}$ on a bigger disk and $v \in \mathbb{W}_{\mathbb{Z}}$ of type $(0,0)$ at 0 and nowhere else. Translating $\mathbb{W}$ by any small ε , we obtain on D $v \in V_{\mathbb{Z}}$ of bounded norm and type $(0,0)$ at ε , but not at 0, and hence nowhere for $\mathbb{V}^{un}$.

Theorem 2.1 is really a statement about what happens when there is a sequence of points $s_\alpha \in (D^*)^r$ tending to 0 in D^r and of v_α integral of type $(0,0)$ and of bounded norm.

2.9

Heuristics for 2.1. Our method of proof forces us to prove a result more general than 2.1, where the assumption $v \in \Phi^0(z)$ is replaced by the assumption that v is close to $\Phi^0(z)$.

In any hermitian space, given α , a quantity Y , a nonzero vector v and a subspace F , we will write

$$v \sim_Y F$$

if the sine of the angle between v and F is bounded by $\exp(-\alpha Y)$, i.e., if $v + w \in F$ with

$$|w| \leq \exp(-\alpha Y) |v|.$$

For $z = (z_1, \dots, z_r)$, with imaginary part y we will write $\sim_z$ for $\sim_{\sup(y_i)}$. We will denote by $\|v\|_{\Phi(z)}$ (or, simply, by $\|v\|$ if no ambiguity is possible) the Hodge norm of $v \in V$ at the point $\Phi(z)$.

Theorem 2.7. Assume $\mathbb{V}$ of weight 0. Given K and $\alpha > 0$, there is a constant A_1 (depending on K , α , and $\mathbb{V}$) such that

- (i) There are only finitely many $v \in V_{\mathbb{Z}}$ such that, at some point z with $0 \leq x_i \leq 1$ and $\inf(y_i) \geq A_1$, $\|v\|_{\Phi(z)}^2 \leq K$ and, relative to the Hodge metric at $\Phi(z)$,

$$v \sim_z \Phi^0(z) \quad (13)$$

- (ii) Any such v is in W_0 .

- (iii) If a fixed v satisfies (i) at a sequence of points z with $0 \leq x_i \leq 1$ and $\inf(y_i) \rightarrow \infty$, then v is in F^0 for some limiting Hodge filtration F .

2.10

Remarks. (i) The proof could be strengthened to show that if $\mathbb{V}$ depends continuously on a parameter τ varying in a compact space, the constant A_1 can be taken independent of τ .

(ii) In (13), we use the Hodge metric at $\Phi(z)$. We could as well have used a fixed metric. Indeed, the ratio between a fixed metric and the Hodge metric is bounded by a power of $\sup(y_i)$ (see (3.8 (i))) so that for any $\alpha' < \alpha$ and for $\inf(y_i)$ large enough, $v \sim_z \Phi^0(z)$ for α and one metric implies $v \sim_z \Phi^0(z)$ for α' and the other metric.

(iii) The condition $v \sim_z \Phi^0(z)$ implies that the ratio $Q(v, v)/\|v\|_{\Phi(z)}^2$ is close to one. Instead of the condition $\|v\|_{\Phi(z)}^2 \leq K$, we could as well have required $Q(v, v) \leq K$.

2.11

To prove 2.7, one would like to be able to replace Φ by Φ_{un} . For $\inf(y_i)$ large, $\Phi(z)$ and $\Phi_{un}(z)$ are close—roughly at a distance $\exp(-2\pi \inf(y_i))$. The case of a variation extending across D^r shows that one cannot hope for anything better. On the other hand, $\exp(\sum z_i N_i)$ is of size $\sup(y_i)^k$, k bounded by the rank of $\mathbb{V}$. If the y_i are of wildly different magnitudes, the product

$$\exp(-2\pi \inf(y_i)) \cdot \prod_i \sup(y_i)^{k_i}$$

need not be small. This leads to difficulties which may be circumvented as follows.

Fix $I \subset [1, r]$, with complement J . Assume that z is such that the y_i ($i \in I$) are of comparable size, and much bigger than the y_j ($j \in J$). Let W^1 be the filtration attached, as in (2.4), to the elements of the cone

$$C_I = \left\{ \sum_{i \in I} \lambda_i N_i, \lambda_i > 0 \right\}.$$

If we freeze the variables z_j ($j \in J$), and consider the asymptotic nilpotent orbit of the resulting variations on $(D^*)^I$, we obtain $\Phi_{un,I}(z)$ which is close to $\Phi(z)$, with a proximity controlled by $\inf_{i \in I}(y_i)$. These nilpotent orbits, for the variables $z = (z_1, \dots, z_r)$, fit into a period map $\Phi_{un,I}$. For z as above, a small angle between v and $\Phi^0(z)$, as in (13), implies a similarly small angle v and $\Phi_{un,I}^0(z)$, as $\inf_{i \in I}(y_i) \sim \sup_{i \in [1, r]}(y_i)$ by assumption.

The first step will be to show that v is in W_0^1 . This cannot be viewed as a consequence of the variant 2.17(i) of 2.7(ii) with parameters, for variations on $(D^*)^I$, as the required set $(D^*)^J$ of parameters is non compact. We have to rely on the $\mathrm{SL}(2)^r$ -orbit theorem [?], which controls $\Phi(z)$ in the whole of regions of the form $y_1 \geq ay_2, y_2 \geq ay_3, \dots, y_{r-1} \geq ay_r$.

Next, one shows that $(\sum_{i \in I} y_i N_i)(v)$ is small, and this allows to find z^* , with the same y_j ($j \in J$) and with y_i^* comparable to $\sup_{j \in J}(y_j)$ such that v is close to $\Phi_{un,I}$ at z^* , with a proximity controlled by $\inf_{i \in I}(y_i) \sim \sup_{i \in [1, r]}(y_i)$. Iterating this process, one eventually finds z^{**} , with all y_i^{**} comparable to $\inf(y_i)$, such that v is close to $\Phi_{un}^0(z^{**})$ (in the sense of (13), possibly for a new α). The next step gives $v \in W_0$ and v close to $\Phi^0(z^{***})$, with z^{***} bounded and a proximity controlled by $\inf(y_i)$. From this 2.7 follows.

2.12

To ease the handling of quantifiers and estimates, but at the cost of effectivity, we will prove 2.7 by contradiction. If (i) fails, we can find sequences $u(n), z(n)$ with the $(u(n), z(n))$ as in 2.7 (i), the $u(n)$ all distinct and $\inf_i(y(n)_i) \rightarrow \infty$. If (ii) fails, we can find sequences as above with each $u(n)$ not in W_0 . If (i) and (ii) hold, but (iii) fails we can find similar sequences with $u(n)$ constant and $u(n)$ not in F^0 for any limiting Hodge filtration F . In each case, a subsequence of the offending sequence is again offending. To prove 2.7 by contradiction, it hence suffices to show that given a sequence $(u(n), z(n))$ with $u(n) \in V_{\mathbb{Z}}$, $\|u(n)\|_{\Phi(z(n))}^2 \leq K$ and $u(n) \sim_z \Phi^0(z(n))$ in the Hodge norm at $z(n)$, $0 \leq x(n)_i \leq 1$, $\inf(y(n)_i) \rightarrow \infty$, it has a subsequence for which $u(n)$ is constant, in W_0 , and in F^0 for some limiting Hodge filtration F .

3 Preliminaries

In 3.2, we comment on what it means and what it takes for a mixed Hodge structure to be close to another. We then recall results of the $SL(2)^r$ -theory of [?] in a form suitable for our purposes.

3.1

Let V be a complex vector space. Fix subspaces A, B of V . If subspaces A', B' are close to A, B in their respective grassmannians, then

$$\dim A' \cap B' \leq \dim A \cap B \quad (14)$$

(upper semicontinuity of the map $(A, B) \mapsto \dim A \cap B$). Further, on the space of pairs of subspaces A, B with $A \cap B$ of fixed dimension, the map $(A, B) \mapsto A \cap B$ is continuous.

Indeed, let A', B' be close to A, B . Some $g \in GL(V)$, close to the identity maps A' to A : we may and shall assume that $A' = A$. Fix a supplement C_1 of $A \cap B$ in A and a supplement of C_2 of $A + B$ in V : for $C = C_1 \oplus C_2$, we have $V = B \oplus C$ and $A = (A \cap B) \oplus (A \cap C)$. Being close to B , B' is the graph of a map $b' : B \rightarrow C$ with $B' \mapsto b'$ continuous. An element $u + b'(u)$ of B' is in A if and only if both $u \in B$ and $b'(u) \in C$ are. This requires $u \in A \cap B$ and the assertion follows. If $\dim A \cap B' = \dim A \cap B$, then $A \cap B'$ is the graph of the restriction of b' to $A \cap B$, continuous in B' .

Let now A, B be finite filtrations of V . We take them decreasing. We consider filtrations A', B' with $\dim A^p \cap B^q = \dim A'^p \cap B'^q$ for all p, q , and show that for such filtrations, if A', B' are close to A, B in the respective flag manifolds, then (A', B') is the image of (A, B) by some $g \in GL(V)$ close to the identity. Arguing as before, we may and shall assume $A = A'$.

Fix a bigrading C splitting the bifiltration (A, B) : A^p (resp. B^q) is the sum of the C^{ij} for $i \geq p$ (resp. $j \geq q$). Fix a basis e_α^{pq} of the C^{pq} . As we assumed that $\dim A^p \cap B^q = \dim A'^p \cap B'^q$, $A'^p \cap B'^q$ is close to $A^p \cap B^q$ and we can find $e'_\alpha{}^{pq} \in A'^p \cap B'^q$ close to e_α^{pq} . The endomorphism $g : e_\alpha^{pq} \mapsto e'_\alpha{}^{pq}$ is close to the identity, hence in $GL(V)$, and maps $A^p \cap B^q$ into, hence onto $A'^p \cap B'^q$. It carries (A, B) to (A', B') .

We now apply this to mixed Hodge structures.

Proposition 3.1. *Let (W, F) be a mixed Hodge structure on a real vector space V . If a filtration F' of $V_\mathbb{C}$ is close to F and such that (W, F') is also a mixed Hodge structure, then there is an automorphism g of $V_\mathbb{C}$, close to the identity, respecting W and carrying F to F' .*

Proof. By (14), we may assume that for all w, p , one has

$$\dim W_w \cap F'^p \leq \dim W_w \cap F^p \quad (15)$$

and it suffices to prove equality.

We proceed by induction on w . If equality in (15) holds for $w - 1$, from

$$\dim \text{Gr}_w^W(F^p) = \dim W_w \cap F^p - \dim W_{w-1} \cap F^{p-1},$$

we obtain

$$\dim \text{Gr}_w^W(F'^p) \leq \dim \text{Gr}_w^W(F^p) \quad (16)$$

and, because of the inductive hypothesis, equality holds if and only if it holds in (15). Taking the sum of (16) for indices (w, p) and $(w, w - p - 1)$, we obtain

$$\dim \text{Gr}_w^W(F'^p) + \dim \text{Gr}_w^W(F'^{w-p-1}) \leq \dim \text{Gr}_w^W(F^p) + \dim \text{Gr}_w^W(F^{w-p-1}).$$

Both sides equal $\dim \text{Gr}_w^W(V)$, implying equality in (16). $\square$

3.2

Let $\mathbb{V}$ be a polarized variation of Hodge structures on $(D^*)^r$, with unipotent monodromy, corresponding to a period mapping Φ on $\mathcal{H}^r$, with values in the filtrations of $V_{\mathbb{C}}$.

Fix $\theta_1, \dots, \theta_d$ in $\mathbb{R}^r$, with $0 \leq \theta_{ij} \leq \dots \leq \theta_{ij}$ and all $\theta_{id} > 0$. We want to control $\Phi(z)$ when z tends to infinity in the following way: the real part x is bounded, the imaginary part can be written as

$$y = \tau_1 \theta_1 + \dots + \tau_d \theta_d + b \quad (17)$$

with b bounded, and where, for

$$t_i = \tau_i / \tau_{i+1} \quad (1 \leq i < d), \quad t_d = \tau_d, \quad (18)$$

one has $t_i \rightarrow \infty$.

For this, we approximate Φ by the nilpotent orbit Φ_{un} and apply [?, (4.20)] to the nilpotent orbit

$$\Phi^*(u_1, \dots, u_d) = \Phi_{un}(\sum u_j \theta_j).$$

Define $T_j = \theta_j N = \sum_i \theta_{ij} N_i$. Adding 1 to u_j transforms Φ^* by $\exp(T_j)$. The T_j are not, in general, rational. In the context of [?], where real variations are considered, this does not matter.

3.3

The $\mathrm{SL}(2, \mathbb{R})^d$ -theory of [?] approximates period mappings, in suitable sectors, by simpler ones which we begin by describing.

On D^* , the family of elliptic curves $\mathbb{C}^*/q^{\mathbb{Z}}$ gives rise, by taking H^1 , to a variation of Hodge structures of rank 2 and type $\{(0, 1), (1, 0)\}$. In a suitable basis of $V_{\mathbb{Z}}$, the corresponding period mapping

$$\Phi : \mathcal{H} \rightarrow \{\text{lines in } \mathbb{R}^2 \otimes \mathbb{C}\}$$

assigns to $z \in \mathcal{H}$ the line spanned by $\begin{pmatrix} z \\ 1 \end{pmatrix}$, with

$$\Phi(z + 1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \Phi(z).$$

On $(D^*)^d$, one can then consider real variations direct sum of variations of the following kind:

$$\mathrm{pr}_j^* \mathrm{Sym}^{n_j}(\mathbb{V}_1) \otimes H$$

for H a fixed Hodge structure. These are the simpler variations announced. Here is an alternative description of them.

The Hodge structure $\Phi(i)$ on $\mathbb{R}^2$ induces a Hodge structure of weight 0 on $\mathfrak{sl}(2, \mathbb{R}) \subset \mathrm{End}(\mathbb{R}^2)$. Let ρ_j be the representation of $\mathfrak{sl}(2, \mathbb{R})^d$ on $\mathbb{R}^d$ via its j -th factor. If A is a real Hodge structure, whose underlying real vector space is a representation of $\mathfrak{sl}(2, \mathbb{R})^d$, and if the representation map

$$\rho : \mathfrak{sl}(2, \mathbb{R})^d \rightarrow \mathrm{End}(A)$$

is a morphism of Hodge structures, where each $\mathfrak{sl}(2, \mathbb{R})$ factor is given the Hodge structure induced by $\Phi(i)$, then, (A, ρ) is isomorphic to a sum of tensor products

$$\mathrm{Sym}^{n_j}(\rho_j) \otimes H$$

where $\mathbb{R}^2$ —the representation space of ρ_j —is given the Hodge structure $\Phi(i)$ and where H is a Hodge structure with trivial action. Indeed, the isomorphism of representations of $\mathfrak{sl}(2, \mathbb{R})^d$:

$$\mathrm{Sym}^{n_j}(\rho_j) \otimes_{\mathfrak{sl}(2, \mathbb{R})^d} \mathrm{Hom}_{\mathfrak{sl}(2, \mathbb{R})^d}(\mathrm{Sym}^{n_j}(\rho_j), A) \xrightarrow{\sim} A$$

is compatible with mixed Hodge structures.

In view of our later applications, we write $\hat{T}_j$ for the image under the representation ρ of the element $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ in the j -th factor of $\mathfrak{sl}(2, \mathbb{R})^d$. For F the Hodge filtration of A , the variations we are considering correspond to period maps of the form

$$\Phi(z) = \exp\left(\sum_j z_j \hat{T}_j\right) F. \quad (19)$$

Denoting by Y_j the image of the element $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ in the j -th factor of $\mathfrak{sl}(2, \mathbb{R})^d$, one has for z purely imaginary

$$\Phi(iy) = \exp\left(-\sum_j \log(y_j) Y_j / 2\right) F_{\sharp} \quad (20)$$

where

$$F_{\sharp} := \Phi(i) := \Phi(i, \dots, i). \quad (21)$$

3.4

Given a nilpotent orbit Φ on $\mathcal{H}^d$ and an ordering of the variables, the $\mathrm{SL}(2, \mathbb{R})^d$ -orbit theorem provides a period mapping Φ_{SL} of the type 3.4 which approximates it. Let T_j be the monodromies for Φ :

$$\Phi(z + e_j) = T_j \Phi(z)$$

and define $C(j)$ to be the cone

$$\left\{ \sum_{i=1}^j \lambda_i T_i, \lambda_i > 0 \right\}$$

and $W^j := W(C(j))$ as in (2.4). Define $t_j = y_j / y_{j+1}$ for $j < d$, and $t_d = y_d$. One has [?]:

(3.5.1) For x bounded and $t_i \rightarrow \infty$, the invariant distance between $\Phi(z)$ and $\Phi_{\mathrm{SL}}(z)$ tends to zero.

(3.5.2) The Y_j define a $\mathbb{Z}^d$ grading of V , with Y_j acting as multiplication by ℓ_j on V_{ℓ} , $\ell = (\ell_1, \dots, \ell_d)$. One has

$$W_w^j = \bigoplus_{\ell_1 + \dots + \ell_j \leq w} V_{\ell}.$$

(3.5.3) The construction is compatible with tensor products. In particular, $\mathrm{SL}(2, \mathbb{R})^d$ respects the polarization form.

One should beware that Φ and Φ_{SL} don't have the same transformation law for $z_i \mapsto z_i + 1$: Φ is transformed by $\exp(T_i)$, while Φ_{SL} is transformed by $\exp(\hat{T}_i)$. The two are related as follows: $\hat{T}_i$ is the degree zero component, for the Y_j , $j < i$, of T_i . In particular

$$T_1 = \hat{T}_1. \quad (22)$$

In addition, (3.5.2) tells that the monodromy weight filtration W^j is also the monodromy weight filtration for $\sum_j \lambda_j \hat{T}_j$ when all $\lambda_i > 0$.

3.5

We now apply this to a period mapping Φ on $\mathcal{H}^r$, to approximate $\Phi(z)$ where z is as in 3.3. With the notations of 3.3, the distance between the following pairs of Hodge structures tends to zero, when $t_i \rightarrow \infty$:

- (a) $\Phi(z)$ and $\Phi_{un}(z)$: by (7),

- (b) $\Phi_{un}(z)$ and $\Phi^*(i\tau)$: one has $\Phi^*(i\tau) = \Phi_{un}(\sum_j i\tau_j \theta_j)$; the hyperbolic distance between z and $\sum_j i\tau_j \theta_j$ is in $O(1/\inf(y_i))$ and one uses the distance decreasing property of period maps,
- (c) $\Phi^*(i\tau)$ and its $\mathrm{SL}(2, \mathbb{R})^d$ -orbit approximation Φ_{SL} : by (3.5.1).

One concludes that the hyperbolic distance between $\Phi(z)$ and $\Phi_{\mathrm{SL}}(i\tau)$ tends to zero.

With the notations of 3.5, we set, as in (21), $F_{\sharp} = \Phi_{\mathrm{SL}}(i)$ and define

$$e(\tau) = \exp\left(\sum_j \log(\tau_j) Y_j / 2\right). \quad (23)$$

It acts by multiplication by $\tau_j^{\ell_j/2}$ on V_{ℓ} (notation of (3.5.2)). It respects the polarization form, hence induces an isometry of the period mapping domain. Because of (20),

$$e(\tau) \Phi_{\mathrm{SL}}^*(i\tau) = \Phi_{\mathrm{SL}}^*(i) = F_{\sharp}$$

and, consequently,

$$e(\tau) \Phi(z) \rightarrow F_{\sharp} \quad (24)$$

for z, τ as in 3.3, $t_i \rightarrow \infty$.

3.6

Let $I(j) \subset [1, r]$ be the set of i for which $\theta_{ij} \neq 0$. It is increasing with j . Let $C(j)$ be the cone

$$C(j) := \left\{ \sum_{i \in I(j)} \lambda_i N_i, \lambda_i > 0 \right\}$$

and W^j be the corresponding filtration $W(C(j))$. Those filtrations W^j coincide with those of 3.5, for $\Phi^*(u)$ and the monodromies $T_j = \theta_j N$.

Some of the above results can be expressed just in terms of the W^j .

(3.7.1) There is a $\mathbb{Z}^d$ -grading A , splitting all the W^j : W_w^j is the sum of A_{ℓ} with $\ell_1 + \dots + \ell_j \leq w$. A may be chosen rational and compatible with the polarization form.

Indeed, (3.5.2) gives one such grading. The statement (3.7.1) is equivalent to the statement that the W^j generate a distributive lattice of subspaces. As the filtrations W_w^j are rational, there exists a rational $\mathbb{Z}^d$ -grading A as in (3.7.1). The grading (3.5.2) is in addition compatible with the polarization form: V_{ℓ} orthogonal to V_m for $\ell + m = 0$.

If A, B are two $\mathbb{Z}^d$ -gradings as in (3.7.1), A_{ℓ} and B_{ℓ} are both canonically isomorphic to an iterated graded of V by the $\mathrm{Gr}_{w_j}^{W^j}$, $w_j = \ell_1 + \dots + \ell_j$. Order irrelevant, because the W_w^j generate a distributive lattice. If g is the direct sum of the resulting isomorphisms $A_{\ell} \xrightarrow{\sim} B_{\ell}$, one has

$$(g-1)(A_{\ell}) \subset \bigoplus_m A_m \text{ with } m \neq \ell, m_1 + \dots + m_j \leq \ell_1 + \dots + \ell_j. \quad (25)$$

If we transport the grading A by the polarization form, viewed as an isomorphism from V to V^* , and dualize, we get another grading A' , equal to A if and only if A is compatible with the polarization. We have $A' = gA$ where g obeys (25). The grading $g^{1/2}A$ is then compatible with the polarization. This shows the existence of rational gradings of the W^j , as in (3.7.1), compatible with the polarization form.

Fix any grading A as in (3.7.1) and define $e_A(\tau)$ to be the multiplication by $\tau_j^{\ell_j/2}$ on A_{ℓ} . If $A_{\ell} = gV_{\ell}$, with g as above, $e_A(\tau) = ge(\tau)g^{-1}$.

Proposition 3.2. *Notations being as above,*

(i) $e_A(\tau)\Phi(z)$ tends to $gF_\sharp$.

(ii) $e_A(\tau)[\text{Hodge metric at } \Phi(z)]$ tends to the transform by g of the Hodge metric at $F_\sharp$.

Proof. For A the decomposition by the V_ℓ , g is the identity, (i) is (24) and (ii) follows as $e(\tau)$ respects the polarization form. In general,

$$e_A(\tau) = ge(\tau)g^{-1} = g(e(\tau)g^{-1}e(\tau)^{-1})e(\tau)$$

and, because of (25), $e(\tau)g^{-1}e(\tau)^{-1}$ tends to the identity, on each A_ℓ . The proposition follows. $\square$

3.7

Remark. (i) Fix a metric $\|\cdot\|_a$. By 3.2(ii), the Hodge metric at $\Phi(z)$ is comparable to the orthogonal direct sum of those metrics, multiplied by $\tau^{\ell/2}$:

$$\|v\|_{\Phi(z)}^2 \sim \sum_\ell \tau^\ell \|v_\ell\|_a^2 \quad (26)$$

i.e., the ratios of both members of (26) are bounded.

(ii) Suppose $v \in V$ is in W_w^1 . Let v_1 be its image in $\text{Gr}_w^{W^1}$. The Hodge filtration $\Phi_{un}(z)$ induces a Hodge filtration on $\text{Gr}_w^{W^1}$. We want to compare the Hodge norm of v and v_1 at z . Decompose V as in (3.7.1), the A_ℓ with $\ell_1 = w$ then project to a similar decomposition B of $\text{Gr}_w^{W^1}$. Numbering: $B_\ell = 0$ if $\ell_1 \neq w$, $B_{0,\ell_2,\dots,\ell_d} = B_{w,\ell_2,\dots,\ell_d}$. Applying (26) both to V and $\text{Gr}_w^{W^1}$, we obtain

$$\|v_1\|_k^2 \leq c\tau_1^{-\ell_1/2} \|v\|_k^2$$

for an appropriate constant c .

Proposition 3.3. *Let A be a real Hodge structure of weight 0 and a representation of $\mathfrak{sl}(2, \mathbb{R})$. One assumes that*

$$\rho : \mathfrak{sl}(2, \mathbb{R}) \rightarrow \text{End}(A)$$

is a morphism of Hodge structures, with $\mathfrak{sl}(2, \mathbb{R})$ being given the Hodge structure of 3.4. Let W be the monodromy weight filtration for $\rho \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then

$$A_{\mathbb{R}} \cap W_0 \cap F^0 \subset A_{\mathbb{R}}^{\mathfrak{sl}(2, \mathbb{R})}.$$

Proof. A can be decomposed as a direct sum

$$\bigoplus_n \text{Sym}^n(\mathbb{R}^2) \otimes H_n, \quad (27)$$

where $\mathbb{R}^2$ is the standard representation of $\mathfrak{sl}(2, \mathbb{R})$, with the Hodge structure of 3.4, and where H_n is a Hodge structure of weight $-n$. For e, f the standard basis of $\mathbb{R}^2$, $\text{Sym}^n(\mathbb{R}^2)$ is the space of homogeneous polynomials of degree n in e, f , $P(e, f)$ is in W_0 if it is divisible by f^m , $m = [(n+1)/2]$ and in F^p if divisible by $(e + if)^p$.

We may and shall assume A reduced to one of the summands (27). Take x real in $W_0 \cap F^0$. Choose a basis h_α of $H_{\mathbb{C}}$, compatible with the Hodge decomposition. Write $x = \sum x_\alpha h_\alpha$. If h_α is of type (p, q) , x_α must be of type (p, q) , i.e. a multiple of $(e + if)^p(e - if)^q$. It can be in W_0 only for $x_\alpha = 0$. $\square$

Corollary 3.4. *Let A be a real Hodge structure of weight 0 and a representation of $\mathfrak{sl}(2, \mathbb{R})^d$. One assumes that*

$$\rho : \mathfrak{sl}(2, \mathbb{R})^d \rightarrow \text{End}(A)$$

is a morphism of Hodge structures, with $\mathfrak{sl}(2, \mathbb{R})$ being given the Hodge structure of 3.4. Let W^j be the monodromy weight filtration for $\rho \circ \Delta_j$, where Δ_j is the diagonal embedding of $\mathfrak{sl}(2, \mathbb{R})$ in the first j factors of $\mathfrak{sl}(2, \mathbb{R})^d$. Then, the intersection

$$A_{\mathbb{R}} \cap \bigcap_j W_0^j \cap F^0$$

is contained in the $\mathfrak{sl}(2, \mathbb{R})^d$ -invariants.

Proof. We proceed by induction on d . By 3.3, the intersection is contained in the invariants of the first factor $\mathfrak{sl}(2, \mathbb{R})$. This space A' is a sub Hodge structure of A acted on by $\mathfrak{sl}(2, \mathbb{R})^{d-1}$, and 3.4 follows from the induction assumption applied to A' . $\square$

4 Proof of 2.16

4.1

In this section we prove 2.7, and hence 2.1, reasoning by contradiction as explained in 2.19. We fix a sequence $(z(n), u(n))$ with the following properties: $0 \leq x(n)_i \leq 1$, $\inf_i y(n)_i \rightarrow \infty$, $u(n) \in V_{\mathbb{Z}}$, the Hodge norm $\|u(n)\|$ of $u(n)$ at $z(n)$ is bounded and, for some fixed $\alpha > 0$, $u(n) \sim_{z(n)} \Phi^0(z(n))$.

We have to show that for a suitable subsequence,

$$u(n) \text{ is constant, and} \tag{28}$$

$$\text{its constant value is in } W_0 \text{ as well as in } F^0 \text{ for some limiting Hodge filtration } F. \tag{29}$$

Taking a subsequence, we may and shall assume that for suitable $\theta_1, \dots, \theta_d$ in $\mathbb{R}^r$, one has

$$y(n) = \tau_1(n)\theta_1 + \dots + \tau_d(n)\theta_d + b(n) \tag{30}$$

with $t_j(n) := \frac{\tau_j(n)}{\tau_{j+1}(n)} \rightarrow \infty$, $t_d(n) = \tau_d(n) \rightarrow \infty$ and $b(n)$ bounded. Then, in addition to (28) and (29), we will show that, with $v := u(n)$ and $T_j := \theta_j N = \sum_i \theta_{ij} N_i$,

$$T_j v = 0. \tag{31}$$

The proof is by induction on $d \geq 1$.

4.2

If θ_j is a linear combination of preceding θ_i (this includes the case $j = 1$, $\theta_1 = 0$), (30) can be replaced by a similar expansion with θ_j omitted. This lowers d and the claims (28), (29), (30) are unaffected: by induction, we may and shall assume that the θ_j are linearly independent.

We assumed that $\inf_i y(n)_i$ tends to ∞ . This means that for each i , one of $\theta_{1i}, \dots, \theta_{di}$ is nonzero and that the first to be nonzero is positive. The θ 's are not uniquely determined by $y(n)$ —only the flag is.

Adding θ_1 to a large enough multiple of θ_2 , then to θ_3 , ..., a large enough multiple of θ_2 , ..., we may and shall assume that $\theta_1 \geq \theta_2 \geq \dots \geq \theta_d$.

For simplicity of notation, we will reorder the coordinates z_i so that the i for which the i th coordinate of θ_j is not zero form an initial segment $1 \leq i \leq a(j)$. We let W^j be the monodromy weight filtration $W(C(j))$, for the cone

$$C(j) = \left\{ \sum_{i=1}^{a(j)} \lambda_i N_i : \lambda_i > 0 \right\}.$$

Let $\Phi' := \Phi_{un, [1, a(1)]}$ be the nilpotent orbit in the z_i , $i \leq a(1)$, approximating Φ . By 2.3, and the fact that $\inf_{i \leq a(1)} (y(n)_i)$ is comparable to $\sup(y(n)_i)$, we still have $u(n)$ bounded in the Φ' -Hodge norm at $z(n)$, and $u(n) \sim_{z(n)} \Phi'^0(z(n))$. The monodromies, as well as the limiting Hodge filtrations being the same for Φ and Φ' , we may replace Φ by Φ' : we may and shall assume that Φ is a nilpotent orbit in $z_1, \dots, z_{a(1)}$. It follows that (W^1, Φ) is a mixed Hodge structure.

For each w , $\text{Gr}_w^{W^1}$ is a variation of Hodge structures of weight w . It is independent of $z_1, \dots, z_{a(1)}$: if $a(1) = r$, it is a constant Hodge structure. If $a(1) < r$, it corresponds to a period map Φ_1 defined on $\mathcal{H}^{[a(1)+1, r]}$. Let $z(n)_1$ be the projection of $z(n)$ on $\mathcal{H}^{[a(1)+1, r]}$. The projection of θ_1 being zero, (30) projects to

$$y(n)_1 = \tau_2(n)\bar{\theta}_2 + \dots + \tau_d(n)\bar{\theta}_d + \bar{b}(n) \quad (32)$$

with $\bar{\cdot}$ denoting projection. This is an expansion like (30), but with $d-1$ θ 's. This is one of two mechanisms by which induction will proceed.

We polarize $\text{Gr}_w^{W^1}(V)$ using any rational element N in $C(1)$: it polarizes the mixed Hodge structure, inducing a polarization of the graded.

Our first task is to prove:

Proposition 4.1. *For n large enough, $u(n)$ is in W_0^1 .*

We say that $u \in V$ is in the position $\ell = (\ell_1, \dots, \ell_d)$ relative to the filtrations W^j if, for A a $\mathbb{Z}^d$ -grading splitting the W^j , ℓ is the largest multi-index, in the lexicographic order, for which the ℓ -component of u does not vanish. By (25), this does not depend on the choice of A .

If u is in position $(\ell_1, \dots, \ell_d)$, then it is in $W_{\ell_1}^1$, with a nonzero image u_1 in $\text{Gr}_{\ell_1}^{W^1}(V)$, and u_1 is in the position $(\ell_2, \dots, \ell_d)$ relative to the filtrations induced by $W^2[\ell_1], \dots, W^d[\ell_1]$, where $W[m]_w = W_{m+w}$.

Those induced filtrations are also the monodromy weight filtrations for the action of any $N \in C(j)$ on $\text{Gr}^{W^1}(V)$: this expresses that W^j is the relative monodromy weight filtration of W^1 and any $N \in C(j)$ ([1, (3.3)]).

Taking a subsequence, we may and shall assume that, for a suitable $(\ell_1, \dots, \ell_d)$, all $u(n)$ are in the position $(\ell_1, \dots, \ell_d)$.

Lemma 4.2. *For each j , one has $\ell_1 + \dots + \ell_j \geq 0$.*

Proof. We will prove by induction on d the following more general statement: one takes Φ to be a variation of weight $w \geq 0$; z, u, θ, τ are as before (except that the θ are not assumed linearly independent); the $u(n)$ are in position $(\ell_1, \dots, \ell_d)$; one assumes $u(n) \sim_z \Phi^0$ and

$$\|u(n)\| \leq c\tau_1(n)^{-w/2}.$$

One claims

$$w + \ell_1 + \dots + \ell_j \geq 0 \text{ for each } j.$$

If θ_j is a linear combination of previous θ , we have $W^j = W^{j-1}$ (resp. trivial if $j = 1, = 0$) and $\ell_j = 0$. Reasoning as in 4.2, we may, using the induction hypothesis, assume that the θ are linearly independent.

As in 4.2, we may and shall assume that Φ is a nilpotent orbit in the variables $z_1, \dots, z_{a(1)}$. Let Φ_1 be the period mapping on $\mathcal{H}^{[a(1)+1, r]}$ corresponding to $\text{Gr}_w^{W^1}(V)$; let $u(n)_1$ (weight $w + \ell_1$) be the image of $u(n)$ in this graded.

Lemma 4.3. *Assumptions and notations being as above,*

$$(i) \quad w + \ell_1 \geq 0$$

$$(ii) \quad \|u(n)_1\| \leq c\tau_1(n)^{-(w+\ell_1)/2} \text{ (Hodge norm)}$$

$$(iii) \quad u(n)_1 \sim_{z_1} \Phi_1^0(z(n)_1).$$

Proof. Let us transform $u(n) \sim_z \Phi^0(z(n))$ by $e(\tau(n))$ (notations of 3.6). In the Hodge norm for $e(\tau(n))\Phi(z(n))$, we continue to have

$$e(\tau(n))u(n) \sim_z e(\tau(n))\Phi^0(z(n)).$$

The subspace on the right tends to $F_\#^0$. Any limiting value of the ray spanned by the real vector $e(\tau(n))u(n)$ is hence in

$$W_{w+\ell_1}^1 \cap F_\#^0 \cap \bar{F}_\#^0.$$

As $(W^1[w], F_\#)$ is a mixed Hodge structure, this intersection can be non zero only for $w + \ell_1 \geq 0$. This proves (i).

Comparing the asymptotics of the Hodge norm for V and for Gr^{W^1} , one finds (cf. 3.9) that for any $v \in W_\ell^1$, with image v_1 in the graded, one has at $z(n)$

$$\|v_1\|_k^2 \leq c\tau_1(n)^{-\ell_1} \|v\|_k^2,$$

proving (ii).

For (iii), we will consider angles in a fixed metric. As explained in 2.17(ii), we still have $u(n) \sim_z \Phi^0$, and it suffices to prove $u(n)_1 \sim_{z_1} \Phi_1^0$, in this new sense.

Fix z_0 with big enough imaginary part so that $(W^1, \Phi_{un}(z_0))$ is a mixed Hodge structure. As $\exp(\sum z(n)_i N_i)$ is bounded by some $\sup(y(n)_i)^k$, we have

$$\exp(\sum (z(n) - z_0) N_i) u(n) \sim_z \exp(\sum (z(n) - z_0) N_i) \Phi^0(z(n)).$$

The filtration $\exp(\sum (z(n) - z_0) N_i) \Phi(z(n))$ tends to $\Phi_{un}(z_0)$. Together with W^1 , both filtrations define a mixed Hodge structure. It follows that some complex endomorphism γ_n tending to 1, respects W^1 , transforms the latter into the former, and satisfies

$$v(n) := \gamma_n^{-1} \exp(\sum (z(n) - z_0) N_i) u(n) \sim_z \Phi_{un}^0(z_0).$$

As $v(n) \in W_{\ell_1}^1$, one also has

$$v(n) \sim_z W_{\ell_1}^1 \cap \Phi_{un}^0(z_0). \quad (33)$$

The projection $v(n)_1$ of $v(n)$ in $\text{Gr}_{\ell_1}^{W^1}$ is the transform by $\gamma_n^{-1} \exp(\sum (z(n) - z_0) N_i)$ of a nonzero element of $\text{Gr}_{\ell_1}^{W^1}(V_\mathbb{Z})$. Its size is hence at least like $c \sup(y(n)_i)^{-k}$, while that of $v(n)$ is at most like $c \sup(y(n)_i)^k$. It follows that the exponential closeness (33) continues to hold mod $W_{\ell_1-1}^1$:

$$v(n)_1 \sim_{z_1} \Phi_{un}^0(z_0) \text{ in } \text{Gr}_{\ell_1}^{W^1}(V).$$

Applying $\exp(\sum (z(n) - z_0) N_i) \gamma_n$, we get 4.5(iii). $\square$

Proof of 4.2: If $a(1) = r$, one has $W^j = W^1$ for all j , $\ell_j = 0$ for $j \geq 2$ and 4.2 follows from 4.3(i). If $a(1) < r$, we apply the induction hypothesis to Φ_1 , of weight $w + \ell_1 \geq 0$ (cf. 4.3(i)), to $u(n)_1$ and to $z(n)_1$ which has an expansion like (30), with $d - 1$ θ 's. By 4.3 (ii), (iii), the required estimates hold.

Proof of 4.1: Fix a rational decomposition $V_\mathbb{Z} = \bigoplus A_\ell$ as in (3.7.1). The projection of $V_\mathbb{Z}$ in A_ℓ is a lattice. By (26), it follows that if $u(n)$ has a nonzero projection in A_a , then

$$\|v(n)\|^2 \gg \tau_1(n)^{a_1} \tau_2(n)^{a_2} \cdots \tau_d(n)^{a_d}.$$

Take $a = (\ell_1, \dots, \ell_d)$. By definition, the projection of $v(n)$ is nonzero. One has

$$\tau_1(n)^{\ell_1} \dots \tau_d(n)^{\ell_d} = \frac{\tau_1(n)^{\ell_1}}{\tau_2(n)^{\ell_1}} \cdot \frac{(\tau_1(n)\tau_2(n))^{\ell_1+\ell_2}}{\tau_3(n)^{\ell_1+\ell_2}} \dots \tau_d(n)^{\ell_1+\dots+\ell_d}.$$

As $\|v(n)\|$ is bounded, we conclude that

$$\ell_1 = \ell_2 = \dots = \ell_d = 0.$$

In particular, $\ell_1 = 0$, proving 4.1. $\square$

4.3

We now apply the induction hypothesis to $\text{Gr}_0^{W^1}(V)$, corresponding to a period mapping Φ_1 , and to the projection $u(n)_1$ of $u(n)$ in $\text{Gr}_0^{W^1}(V)$.

If $a(1) = r$, the Hodge structure $\text{Gr}_0^{W^1}(V)$ is constant. By 4.3(ii), or directly by 3.9(ii), $u(n)_1$ is bounded. Being in a lattice, it can take only a finite number of values. We may and shall assume it is constant and define $u^1 := u(n)_1$. As $u^1 \sim_{z_1} \Phi^0$, one has $u^1 \in F^0$.

If $a(1) < r$, the induction hypothesis applies by (32) and 4.3. Again, we find that $u(n)_1$ may and shall be assumed to have a constant value u^1 . By (31), u^1 is killed by T_j ($j \geq 2$) and in particular sits in the W_0^j ($j \geq 2$). It is also in F^0 for some limiting Hodge filtration F (all this in $\text{Gr}_0^{W^1}(V)$).

T_1 maps W_0^1 to W_{-2}^1 , inducing a morphism of Hodge structures

$$\text{Gr}_0^{W^1}(V) \rightarrow \text{Gr}_{-2}^{W^1}(V)(-1). \quad (34)$$

We next prove

Proposition 4.4. u^1 is in the kernel of the morphism (34).

Proof. If we transform the assumption

$$u(n) \sim_z \Phi^0(z(n))$$

by $e(\tau(n))$ (notations of 3.6.1), we obtain

$$e(\tau(n))u(n) \sim_z e(\tau(n))\Phi^0(z(n))$$

where angles are now taken in the Hodge metric for $e(\tau(n))\Phi(z(n))$. The Hodge norm of $e(\tau(n))u(n)$ in this metric –equal to that of $u(n)$ in the Hodge metric for $\Phi(z(n))$ – is bounded by assumption. The $e(\tau(n))\Phi(z(n))$ tend to the Hodge filtration $F_\#$ (cf. 3.6). The Hodge metric for $e(\tau(n))\Phi(z(n))$ tends to that for $F_\#$. It follows that $e(\tau(n))u(n)$ remains bounded and, taking a subsequence, we may and shall assume that it has a limit u_0 . We have

$$u_0 \in F_\#^0.$$

Consider the decomposition $V = \bigoplus A_\ell$: $e(\tau)$ acts on A_ℓ as multiplication by $\tau^{\ell/2}$. By 3.3, u_0 is in the sum of the A_ℓ with $\ell_1 = 0$ and is in the kernel of T_1 .

The components $u(n)(\ell)$ of $u(n)$, for $\ell_1 = 0$, depend only on u^1 : they are independent of n . That u^1 is killed by the T_j ($j \geq 2$) implies it is in the W_0^j and that, for $\ell_1 = 0$, $u(n)(\ell)$ can be nonzero only for $\ell_2 + \dots + \ell_j \leq 0$.

The V_ℓ with $\ell_1 = 2$ project to a decomposition of $\text{Gr}_{-2}^{W^1}(V)$. By (22), 4.4 and the fact that u^1 is in W_0^j , $T_1 u^1$ has a nonzero component in V_ℓ ($\ell_1 = 2$) only when $(\ell_2, \dots, \ell_d) = (0, \dots, 0)$, $\ell_1 + \dots + \ell_j \leq 0$.

The $e(\tau(n))\Phi(z(n))$ belong to a compact family of Hodge filtrations F , for which (W^1, F) is mixed Hodge. By 3.2, $e(\tau(n))u(n)$ is a sum $\chi + \varepsilon$, χ in F , $\|\varepsilon\| \leq \exp(-\alpha \sup_i y(n)_i)$. By compactity, or 3.2, χ can be taken in W_0^1 . Apply T_1 ; by (22),

$$e(\tau(n))T_1u(n) = \tau_1(n)^{-1}T_1(e(\tau(n))u(n)) = \tau_1(n)^{-1}(T_1\chi + T_1\varepsilon)$$

and we find that T_1u^1 is sum of an element in $\Phi^{-1}(z(n))$ plus an exponentially small term. Passing to the limit, we obtain

$$T_1u^1 \in F_{\sharp}^{-1},$$

where $F_{\sharp}^{-1}$ is the filtration in $\text{Gr}_{-2}^{W^1}(V)$ induced by $F_{\sharp}^{-1}$. If we go to $\text{Gr}_{-2}^{W^1}(V)(-2)$, to have a Hodge structure of weight zero, this reads

$$T_1u^1 \in F_{\sharp}^1(-2)^0.$$

The action of the individual factors of $\text{SL}(2, \mathbb{R})^d$ of index $\neq 1$, and the filtration $F_{\sharp}(-2)$ on $\text{Gr}_{-2}^{W^1}(V)(-2)$, are of the type considered in 3.4. By 3.4, T_1u^1 is in $V^{-2,0,\dots,0}$, hence it is zero—as the $(-2, 0, \dots, 0)$ component has been shown to vanish. This proves 4.4. $\square$

Proposition 4.5. *$T_1u(n)$ is exponentially small: of Hodge norm $\ll \exp(-\beta \sup_i (y(n)_i))$ for suitable β .*

Proof. By 4.4, $T_1u(n)$ is in W_{-3}^1 . It is also the sum of χ in $\Phi^{-1}(z(n))$ and of an exponentially small ε . The same holds after applying $e(\tau(n))$. We gain that the $e(\tau(n))\Phi(z(n))$ belong to a compact family of filtrations F , for which

$$W_{-3}^1 \cap F^{-1} \cap \bar{F}^{-1} = 0,$$

so that $e(\tau(n))T_1u(n)$ must be exponentially small. Hence so is $T_1u(n)$. $\square$

4.4

We now complete the proof of 2.7. With angles measured using a fixed metric, we have

$$u(n) \sim_z \Phi^0(z(n)).$$

Applying $\exp(-\sum_i \tau_1(n)T_i)$, of polynomial size in $\sup(y(n)_i)$:

$$\exp(-\sum_i \tau_1(n)T_i)u(n) \sim_z \Phi^0(z(n) - i\tau_1(n)\theta_1)$$

and, by 4.5,

$$u(n) \sim_z \Phi^0(z(n) - i\tau_1(n)\theta_1).$$

If $d = 1$, we choose the expansion (30) (subtracting a constant to $\tau_1(n)$) so that $y_j(n) - \tau_1(n)\theta_{1j} \geq A > 0$. As $u(n)$ is close to Φ^0 at $z(n)$ and at $z(n) - i\tau_1(n)\theta_1$, its Hodge norm at both places is close to $Q(u(n), u(n))^{1/2}$ and hence is bounded. Moreover, $z(n) - i\tau_1(n)\theta_1$ remains bounded and the corresponding Hodge filtrations remain in a compact set. Being bounded and integral, $u(n)$ can take only finitely many values. Taking a subsequence for which $z(n) - i\tau_1(n)\theta_1$ tends to a limit, we find that u is in the corresponding Φ^0 . It is in W_0 by 4.1. This proves (29), while (31) results from 4.4.

Assume now $d > 1$. The imaginary part of $z(n) - i\tau_1(n)\theta_1$ is

$$\tau_2(n)\theta_2 + \dots + \tau_d(n)\theta_d + b(n),$$

an expansion as in (30), but with only $d-1$ θ 's. Applying the induction assumption, we see that, on a subsequence, $u(n)$ satisfies (28) and (29) and is killed by the T_j ($j > 1$). Being constant, it is also killed by T_1 (cf. 4.5). This finishes the proof. $\square$

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The Exceptional Series of Lie Groups

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Translator's note. This is a direct English working translation from the project TeX transcription. Mathematical notation has been kept close to the source; the prose has been normalized into contemporary English.

Abstract. Numerology of exceptional Lie groups and a conjectural explanation.

Let G_0 be the split adjoint group of one of the following types:

$$A_1, A_2, G_2, D_4, F_4, E_6, E_7, E_8.$$

Fix a pinning of G_0 . Let Γ denote the group of automorphisms of G_0 . For Γ the group of automorphisms of the Dynkin diagram, identified with the group of automorphisms of G_0 preserving the pinning, this is a semidirect product $\Gamma \ltimes G_0$. The group G is an algebraic group over $\mathbb{Q}$.

It would be enough for our purposes to consider the complex Lie group $G(\mathbb{C})$, or even a compact form of $G(\mathbb{C})$. The possibility, for our purpose, of including D_4 in the list was discovered by Cohen and de Man.

Let T be the maximal torus of G supplied by the pinning, and let $\langle \cdot, \cdot \rangle$ be the canonical bilinear form on $\text{Lie}(T)^\vee$: the inverse form to the restriction to $\text{Lie}(T)$ of the Killing form $\text{Tr}(\text{ad } x \text{ ad } y)$. Let α be the highest root and set $k = \langle \alpha, \alpha \rangle$. The values of k are

$$1/2, \quad 1/3, \quad 1/4, \quad 1/6, \quad 1/9, \quad 1/12, \quad 1/18, \quad 1/30.$$

Finally let h be the Coxeter number, equal either to k or to $-(1/6) - k$. Put $a := 1/h$. To this list we also add the trivial group, with $a = 5/6$ or $a = -1$.

Let $\mathfrak{g}$ be the Lie algebra of G , let U be its enveloping algebra, and let $C \in U$ be the Casimir element corresponding to the Killing form. If V is a representation of G , we will call the action of C the Casimir endomorphism of V . If V is irreducible, this is multiplication by a scalar, called the Casimir of V . Recall that the Casimir of the adjoint representation is 1.

Our observation is that, for (G, a) as above, G has virtual representations X_i ($0 \leq i \leq 4$), Y_i ($0 \leq i \leq 3$), Y_i^* ($0 \leq i \leq 3$), A , C , and C^* with the following properties (A)–(G).

(A) Each of these representations is either irreducible, zero, or the negative of an irreducible representation. One has $X_0 = Y_0 = Y_0^* = 1$, the trivial representation, and $X_1 = Y_1 = Y_1^* = \mathfrak{g}$, the adjoint representation.

For each representation R of the symmetric group S_n , there is a functor

$$V \longmapsto [R](V) := \text{Hom}_{S_n}(R, V^{\otimes n}).$$

If V is a representation of a group, the class of $[R](V)$ in the Grothendieck group of representations is a λ -polynomial, that is, a polynomial in the $\wedge^i V$, in the class of V . If R is irreducible and corresponds to a partition p of n , we write $[p]_V$ for the class $[R](V)$. For example,

$$[(n)]_V = \text{Sym}^n V, \quad [(1, \dots, 1)]_V = \wedge^n V, \quad [(2, 1)]_V = V \otimes \wedge^2 V - \wedge^3 V,$$

$$[(3)]_V = V^3 - 2V \cdot \wedge^2 V + \wedge^3 V.$$

(B) In the λ -ring of representations of G , one has

$$\begin{aligned} \text{(B1)} \quad & [(2)]_{\mathfrak{g}} = \text{Sym}^2 \mathfrak{g} = 1 + Y_2 + Y_2^*, \\ \text{(B2)} \quad & [(1, 1)]_{\mathfrak{g}} = \wedge^2 \mathfrak{g} = \mathfrak{g} + X_2, \\ \text{(B3)} \quad & [(3)]_{\mathfrak{g}} = \text{Sym}^3 \mathfrak{g} = \mathfrak{g} + X_2 + A + Y_3 + Y_3^*, \\ \text{(B4)} \quad & [(2, 1)]_{\mathfrak{g}} = 2\mathfrak{g} + X_2 + Y_2 + Y_2^* + A + C + C^*, \\ \text{(B5)} \quad & [(1, 1, 1)]_{\mathfrak{g}} = \wedge^3 \mathfrak{g} = 1 + X_2 + Y_2 + Y_2^* + X_3, \\ \text{(B6)} \quad & Y_2 \otimes \mathfrak{g} = \mathfrak{g} + X_2 + Y_2 + A + Y_3 + C, \\ \text{(B7)} \quad & Y_2^* \otimes \mathfrak{g} = \mathfrak{g} + X_2 + Y_2^* + A + Y_3^* + C^*. \end{aligned}$$

These formulae are permuted by the involution $*$, provided that A and the X_i —in particular $\mathfrak{g}$ —are declared fixed by $*$.

(C) As a consequence of (B), in the λ -ring of representations of G one has

$$\begin{aligned} \text{(C1)} \quad & X_2 = \wedge^2 \mathfrak{g} - \mathfrak{g}, \\ \text{(C2)} \quad & X_3 = \wedge^3 \mathfrak{g} - \mathfrak{g} \cdot \wedge^2 \mathfrak{g} + \text{Sym}^2 \mathfrak{g} + \mathfrak{g} - 1. \end{aligned}$$

Moreover,

$$X_4 = \wedge^4 \mathfrak{g} - \mathfrak{g} \cdot \wedge^3 \mathfrak{g} + \text{Sym}^2 \mathfrak{g} \cdot \wedge^2 \mathfrak{g} - \wedge^2 \mathfrak{g} - (\mathfrak{g} + A) \cdot \mathfrak{g} + \mathfrak{g}.$$

(D) The Casimir has the following values on these representations:

n	X_n	Y_n	Y_n^*	A	C	C^*
n	$n + (n^2 - n)a$	$n + (n^2 - n)a^*$	$8/3$	$3 + 3a$	$3 + 3a^*$	

When the representation in question is zero, this assertion is to be interpreted as empty.

(E) The dimensions of these representations are given by the following formulae, where $\lambda := -6a$:

$$\begin{aligned} \dim X_1 &= -2(\lambda + 5)(\lambda - 6)/\lambda(\lambda - 1), \\ \dim X_2 &= 5(\lambda + 3)(\lambda + 5)(\lambda - 4)(\lambda - 6)/\lambda^2(\lambda - 1)^2, \\ \dim X_3 &= -10(\lambda + 2)(\lambda + 4)(\lambda + 5)(\lambda - 3)(\lambda - 5)(\lambda - 6)/\lambda^3(\lambda - 1)^3, \\ \dim X_4 &= 5(\lambda + 3)(\lambda + 4)(\lambda + 5)(\lambda + 3)(2\lambda - 4)(\lambda - 5)(\lambda - 6)(2\lambda - 5)/\lambda^4(\lambda - 1)^4, \\ \dim Y_2 &= -90(\lambda + 5)(\lambda - 4)/\lambda^2(\lambda - 1)(2\lambda - 1), \\ \dim Y_3 &= -10(\lambda + 5)(5\lambda - 6)(\lambda - 4)(\lambda - 5)(\lambda - 6)/\lambda^3(\lambda - 1)^2(2\lambda - 1)(3\lambda - 1), \\ \dim A &= -27(\lambda + 3)(\lambda + 4)(\lambda + 5)(\lambda - 4)(\lambda - 5)(\lambda - 6)/\lambda^2(\lambda - 1)^2(3\lambda - 1)(3\lambda - 2), \\ \dim C &= 640(\lambda + 3)(\lambda + 5)(\lambda - 3)(\lambda - 5)/\lambda^3(\lambda - 1)(2\lambda - 1)(3\lambda - 2). \end{aligned}$$

For Y_n^* and C^* , the dimension is given by the same formula as for Y_n and C , but with λ replaced by $\lambda^* := 1 - \lambda = -6a^*$. Note that the formulae for the X_i and for A are invariant under $\lambda \mapsto \lambda^*$.

(F) For each of the equalities (B1)–(B7), (C1), and (C2), the formulae (E) express the dimension of the left-hand side, respectively of the right-hand side, as the value at λ of a rational function R_1 , respectively R_2 . One has $R_1 = R_2$ as rational functions.

(G) For each of the equalities (B1)–(B7), (C1), and (C2), the formulae (D), (E), and (1), (2) express the trace of the Casimir on the left-hand side, respectively on the right-hand side, as the value at λ of a rational function R_1 , respectively R_2 . One has $R_1 = R_2$.

If R is a representation of S_n and V is a representation of a semisimple group, the trace of the Casimir endomorphism of $[R](V)$ is obtained from the trace of the Casimir endomorphism of V by the following formula. If χ is the character of R , $n(\sigma)$ is the number of cycles of a permutation $\sigma \in S_n$, and $m(\sigma)$ is the sum of the squares of the cycle lengths of σ , then

$$\mathrm{Tr}(C, [R](V)) = \frac{1}{n!} \sum_{\sigma \in S_n} \chi(\sigma) m(\sigma) (\dim V)^{n(\sigma)-2} \mathrm{Tr}(C, V). \quad (1)$$

For two representations V and W ,

$$\mathrm{Tr}(C, V \otimes W) = \mathrm{Tr}(C, V) \dim(W) + \dim(V) \mathrm{Tr}(C, W). \quad (2)$$

I do not know a satisfactory proof of the preceding assertions. The assertions (B), (C), and (D) are obtained by the brute force of the LiE program. That the dimensions and Casimirs of $\mathfrak{g}$, Y_2 , and Y_2^* have the form (D), (E) for the appropriate a follows in Vogel [2], §6.9, from the fact that the only invariant elements of $\mathrm{Sym}^4(\mathfrak{g})$ are multiples of the square of the invariant element of $\mathrm{Sym}^2(\mathfrak{g})$. The dimension of the X_i follows from that of $\mathfrak{g}$ by (C). If $n = k$, then Y_n is the representation of dominant weight $n\alpha$, with α the highest root, and (D) for Y_n can be deduced from this. The assertion (D) for X_n can be deduced from (C): on the right-hand side, the trace of the Casimir endomorphism is computed by (1).

Assume (D) and the dimension formulae for Y_2 , Y_2^* , and the X_i . If one expresses equality of dimensions and of Casimir traces for both sides of (B3), (B4), (B6), and (B7), one obtains a system of linear equations in $\dim Y_3$, $\dim Y_3^*$, $\dim A$, $\dim C$, and $\dim C^*$, whose right-hand coefficients are rational functions of λ . This gives an expression for these dimensions as rational functions of λ . We used this method to find a polynomial in λ which is a multiple of the denominator, and to estimate the degree of the numerator; the numerator was then obtained by interpolation from the known values corresponding to the pairs (G, a) . That the resulting rational functions factor as in (E) is a miracle. The identities $R_1 = R_2$ of (F), (G) follow simply from the fact that, for a suitable polynomial P in λ , $R_1 P$ and $R_2 P$ are two polynomials in λ which are equal at a number of values of λ strictly greater than their degree.

The conjecture

I conjecture that there exists a semisimple abelian $\mathbb{Q}(t)$ -linear category $\mathcal{C}_t$, equipped with a tensor product $\otimes$ and associativity, commutativity, and unit constraints in the sense of Saavedra [1], Chapter 1, having the following properties.

- (i) The groups $\mathrm{Hom}(X, Y)$ are finite-dimensional over $\mathbb{Q}(t)$.
- (ii) Every object has a dual: $\mathcal{C}_t$ is rigid. Recall that this makes it possible to define, for every object X , its dimension $\dim(X) \in \mathrm{End}(1) = \mathbb{Q}(t)$ as the composite $1 \rightarrow X^\vee \otimes X \rightarrow 1$ of the evaluation and coevaluation morphisms.
- (iii) The tensor category $\mathcal{C}_t$ is of adjoint type in the following sense. There is an object $\mathfrak{g}$, equipped with a Lie bracket $[\cdot, \cdot] : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$, acting functorially on each object X by an action $\rho_X : \mathfrak{g} \otimes X \rightarrow X$. The action is compatible with tensor products: $\rho_{X \otimes Y} = \rho_X \otimes 1 + 1 \otimes \rho_Y$. For $X = \mathfrak{g}$, ρ_X is $[\cdot, \cdot]$. The action ρ_X transposes to a coaction ${}^t\rho_X : X \rightarrow \mathfrak{g}^\vee \otimes X$. For every X , $X^\natural := \mathrm{Ker}({}^t\rho_X)$ is the largest subobject of X which is a sum of copies of $\mathbf{1}$. Finally, every object is a subquotient of a sum of objects $\mathfrak{g}^{\otimes n}$.
- (iv) $\mathfrak{g}$ is a simple object, and the Killing form is nondegenerate.
- (v) There are simple objects X_i ($0 \leq i \leq 4$), Y_i ($0 \leq i \leq 3$), Y_i^* ($0 \leq i \leq 3$), A , C , and C^* satisfying (A)–(E), with $a = t$.

- (vi) Every simple object S is absolutely simple, i.e. $\text{End}(S) = \mathbb{Q}(t)$, and is isomorphic to its dual; more precisely, it admits a nondegenerate symmetric bilinear form $S \otimes S \rightarrow \mathbf{1}$. The dimension $\dim(S) \in \mathbb{Q}(t)$ of a simple object factors into linear factors.
- (vii) The category $\mathcal{C}_t$ has a $\otimes$ -autoequivalence $*$, semilinear with respect to $t \mapsto -(1/6) - t$, sending A and the X_i to themselves, Y_i to Y_i^* , and C to C^* ; the square of $*$ is isomorphic to the identity.
- (viii) In a sense to be made precise, the category of representations of G is a specialization of $\mathcal{C}_t$ at $t = a$.

A model

Here is a model for this conjecture. Let $\mathcal{B}_n^-[t]$ be the following rigid $\mathbb{Z}[t]$ -linear tensor category.

Objects: finite sets.

$\text{Hom}(X, Y)$: let $\mathcal{C}$ be the set of isomorphism classes of compact one-dimensional manifolds with boundary, whose boundary is $X \amalg Y$; these are bordisms from X to Y . The $\mathbb{Z}[t]$ -module $\text{Hom}(X, Y)$ is the quotient of the free $\mathbb{Z}[t]$ -module $\mathbb{Z}[t]^{(\mathcal{C})}$ with basis $[c]$, $c \in \mathcal{C}$, by the relations

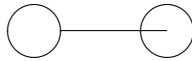
$$[c \cup \text{a circle}] = t[c].$$

Composition: composition of bordisms.

$\otimes$: disjoint union.

For $\lambda \in \mathbb{Q}$, let $\mathcal{B}_\lambda$ be the rigid $\mathbb{Q}$ -linear tensor category obtained from $\mathcal{B}_n^-[t]$ by scalar extension $\mathbb{Z}[t] \rightarrow \mathbb{Q}$, $t \mapsto \lambda$. A morphism $f : X \rightarrow Y$ of $\mathcal{B}_\lambda$ will be called negligible if $\text{Tr}(gf) = 0$ for every $g : Y \rightarrow X$. The negligible morphisms form an ideal I_λ , and if f is negligible then so is $f \otimes g$. Thus the quotient category $\mathcal{B}_\lambda/I_\lambda$ is still rigid tensor. Let $(\mathcal{B}_\lambda/I_\lambda)^{\text{kar}}$ be its Karoubian envelope, obtained by formally adjoining the direct factors corresponding to idempotent endomorphisms.

If λ is an integer $n \geq 0$, this Karoubian envelope is just the category $\text{Rep}(O(n))$ of representations of the orthogonal group $O(n)$. More precisely, let V be a vector space over $\mathbb{Q}$ equipped with a nondegenerate symmetric bilinear form B . Then there is a tensor functor, unique up to unique isomorphism, from $\mathcal{B}_n^-$ to $\text{Rep}(O(V))$, sending the object $\{1\}$ to V and sending the bordism



from $\{1, 2\}$ to $\emptyset$ to $B : V \otimes V \rightarrow \mathbf{1}$. This tensor functor induces an equivalence of $(\mathcal{B}_n/I_n)^{\text{kar}}$ with $\text{Rep}(O(V))$. This is a translation of Weyl's invariant theory for $O(V)$ in $V^{\otimes N}$; see [3], Chapter V, §5.

If λ is an even integer $-2p \leq 0$, the Karoubian envelope $(\mathcal{B}_\lambda/I_\lambda)^{\text{kar}}$ is a variant of the category $\text{Rep}(\text{Sp}(2p))$ of representations of the symplectic group $\text{Sp}(2p)$: the central element $-1 \in \text{Sp}(2p)$ defines a $\mathbb{Z}/(2)$ -grading of each representation V of $\text{Sp}(2p)$, and the commutativity constraint is defined by the Koszul sign rule. Equivalently, in the category of super-representations of $OSp(0|2p)$, one considers the subcategory of subquotients of sums of $V^{\otimes N}$, with V the obvious super-representation.

Similarly, if one considers finite oriented sets S , that is, sets equipped with $\varepsilon : S \rightarrow \{\pm 1\}$, and oriented bordisms, one obtains a rigid $\mathbb{Z}[t]$ -linear tensor category $\mathcal{A}_n^+[t]$. For $\lambda \in \mathbb{Q}$, let $\mathcal{A}_\lambda$ be the rigid $\mathbb{Q}$ -linear category obtained from $\mathcal{A}_n^+[t]$ by the scalar extension $\mathbb{Z}[t] \rightarrow \mathbb{Q}$, $t \mapsto \lambda$. Define the ideal I_λ of negligible morphisms in $\mathcal{A}_\lambda$ as before. If λ is an integer $n \geq 0$, $(\mathcal{A}_\lambda/I_\lambda)^{\text{kar}}$ is just $\text{Rep}(\text{GL}(n))$. If λ is an integer $-n \leq 0$, one obtains the variant of $\text{Rep}(\text{GL}(n))$ defined by $-1 \in \text{GL}(n)$: in the category of super-representations of $\text{GL}(0|n)$, one takes the subcategory of subquotients of sums of $V^{\otimes N} \otimes V^{\vee \otimes M}$, with V the evident super-representation.

In both cases, after extending scalars from $\mathbb{Z}[t]$ to $\mathbb{Q}(t)$, the Karoubian envelope of the category $\mathcal{A}_t$ or $\mathcal{B}_t$ obtained in this way is a semisimple rigid tensor category. Its simple objects S are absolutely simple, and their dimensions $\dim(S) \in \mathbb{Q}(t)$ are polynomials in t that are products of linear factors.

The formulae

Here are the formulae.

Let V_λ^- be the object “one point” of $\mathcal{B}_\lambda^-$, and let V be its image in $\mathcal{B}_\lambda^{-\text{kar}}$, the “evident representation”. To study the subcategory of $\mathcal{B}_\lambda^{-\text{kar}}$ formed by the subquotients of sums of $V^{\otimes i}$, $i \leq N$, one method is to use polynomial continuation from the category $\text{Rep}(O(n))$ for n large. The irreducible objects of $\mathcal{B}_\lambda^{-\text{kar}}$ then correspond to decreasing sequences of integers $\ell_i \geq 0$, with $\ell_i = 0$ for i large enough. The corresponding representation of $O(n)$, for n large, has dominant weight $\sum_i (\ell_i - \ell_{i+1}) \omega_i$ in the notation of the Bourbaki tables. Weyl’s dimension formula gives, by continuation, the dimension of this irreducible object. Let $h((\ell_i))$, or simply h , be the integer such that the representation of the symmetric group defined by the partition (ℓ_i) of $\sum \ell_i$ has dimension $(\sum \ell_i)!/h$. If $\ell_i = 0$ for $i > a$, then

$$h = \prod_{i=1}^a (\ell_i + a - i)! \prod_{1 \leq i < j \leq a} ((\ell_i - i) - (\ell_j - j)).$$

Let $a(n)$, respectively $b(n)$, be the number of pairs i, j , $1 \leq i \leq j$, such that $n = (\ell_i - i) + (\ell_j - j)$, respectively $n = -i - j$. With this notation,

$$\dim V((\ell_i)) = (1/h) \prod (t + n)^{a(n) - b(n)}.$$

Formally, this is

$$(1/h) \frac{\prod (t + \ell_i - i + \ell_j - j)}{\prod (t - i - j)}.$$

It is a polynomial of degree $\sum \ell_i$.

The object $\mathfrak{g} := \wedge^2 V$ plays the role of the adjoint representation: it has a Lie bracket, and acts functorially and compatibly with tensor products on every object. The bilinear form $\text{Tr}(\rho(x)\rho(y), V)$ on $\mathfrak{g}$ is nondegenerate. On $V((\ell_i))$, the corresponding Casimir is

$$\sum_i \ell_i (\ell_i - 2i) + t \sum_i \ell_i,$$

as one checks by continuation from the case of $O(n)$ for n large.

Let V_λ^+ be the object “one positively oriented point” ($\varepsilon = 1$) of $\mathcal{A}_\lambda^+$, and let V be its image in $\mathcal{A}_\lambda^{+\text{kar}}$, the “evident representation”. To study the subcategory of $\mathcal{A}_\lambda^{+\text{kar}}$ formed by the subquotients of sums of $V^{\otimes i} \otimes V^{\vee \otimes j}$, $i, j \leq N$, one method is to use polynomial continuation from the category $\text{Rep}(\text{GL}(n))$ for n large. The irreducible objects of $\mathcal{A}_\lambda^+$ then correspond to pairs of decreasing sequences of positive integers $((\ell_i), (m_i))$, with ℓ_i and m_i zero for i sufficiently large. If the character group of the maximal torus of diagonal matrices in $\text{GL}(n)$ is identified with $\mathbb{Z}^n$ in the usual way, then the corresponding representation of $\text{GL}(n)$, for n large, has dominant weight

$$(\ell_1, \ell_2, \dots, 0, \dots, 0, -m_2, -m_1).$$

The object $\mathfrak{g} := V \otimes V^\vee$ plays the role of the adjoint representation, and the bilinear form $\text{Tr}(\rho(x)\rho(y), V)$ is nondegenerate.

Let $c(n)$, respectively $d(n)$, be the number of pairs i, j , $1 \leq i, j$, such that $n = \ell_i - i + m_j - j + 1$, respectively $n = -i - j + 1$. By continuation of Weyl’s formula,

$$\dim V((\ell_i), (m_i)) = (1/h((\ell_i)) \cdot h((m_i))) \prod (t + n)^{c(n) - d(n)}.$$

Formally, this is

$$(1/h((\ell_i)) \cdot h((m_i))) \frac{\prod (t + \ell_i - i + m_j - j + 1)}{\prod (t - i - j + 1)}.$$

It is a polynomial in t of degree $\sum \ell_i + \sum m_i$. The Casimir relative to $\text{Tr}(\rho(x)\rho(y), V)$ is

$$\sum_i \ell_i(\ell_i - 2i + 1) + \sum_i m_i(m_i - 2i + 1) + t\left(\sum_i \ell_i + \sum_i m_i\right).$$

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This work was inspired by a conversation with P. Vogel, in which he told me that there are three simple Lie groups: linear, orthogonal, and exceptional, depending on parameters. He exploits this point of view in [2], of which numbers 6.8 and 6.9 were especially useful to me.

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The Exceptional Series of Lie Groups II

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Translator's note. This is a direct English working translation from the project TeX transcription. Mathematical notation and numbering have been kept close to the source.

Abstract. Numerology of exceptional Lie groups and their representations.

As in [1] and [2], we consider in this Note the groups G of the exceptional series

$$e, A_1, A_2, G_2, D_4, F_4, E_6, E_7, E_8.$$

Between the trivial group and A_1 we also insert the supergroup $\mathrm{SOSp}(1, 2)$. More than the group itself, what interests us each time is its category of representations. For $\mathrm{SOSp}(1, 2)$, we consider the category of super-representations for which the parity is given by $-1 \in \mathrm{Sp}(2) \subset \mathrm{SOSp}(1, 2)$. The neutral component G^0 of each group is the adjoint group, and in each case the category of representations under consideration is tensor-generated by the adjoint representation.

The dual Coxeter number $h^\vee$ of G^0 is, according to type,

$$6/5, 3/2, 2, 3, 4, 6, 9, 12, 18, 30.$$

For the trivial group, $h^\vee = 6/5$ by fiat. We regard G as depending on the parameter $h^\vee$. Sometimes it is more convenient to use $\mu := 6/h^\vee$, which we now prefer to the parameter λ used in [1] and [2]:

$$\mu = 5, 4, 3, 2, 3/2, 1, 2/3, 1/2, 1/3, 1/5.$$

For G^0 of rank r , consider the following dominant weights $\theta_1, \dots, \theta_r$. For E_8 we stop at θ_7 . Notation: $(a_1, \dots, a_r)$ denotes $\sum a_i \theta_i$, the fundamental weights ω_i being taken in the order of the Bourbaki tables. For G^0 of rank at least 1, that is, $G \neq e$, θ_1 is the highest root, corresponding to the adjoint representation. This defines the θ_i for G of rank at most 1, namely $G^0 = e$, $\mathrm{SOSp}(1, 2)$, or A_1 . After that, we take

$$A_2: (1, 1), (0, 3);$$

$$G_2: (0, 1), (3, 0);$$

$$D_4: (0, 1, 0, 0), (1, 0, 1, 1), (2, 0, 0, 0), (2, 0, 2, 0);$$

$$F_4: (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 2), (0, 0, 2, 0);$$

$$E_6: (0, 1, 0, 0, 0, 0), (0, 0, 0, 1, 0, 0), (1, 0, 0, 0, 0, 1), (0, 0, 1, 0, 1, 0), (0, 0, 0, 0, 1, 1), (1, 0, 0, 0, 2, 0);$$

$$E_7: \text{ the fundamental weights } \omega_1, \omega_3, \omega_6, \omega_4, \text{ and } (0, 1, 0, 0, 0, 0, 1), (0, 1, 0, 0, 1, 0, 0), (0, 0, 0, 0, 1, 0, 1);$$

$$E_8: \text{ the fundamental weights } \omega_8, \omega_7, \omega_1, \omega_6, \omega_2, \omega_5, \omega_3.$$

For A_2 , D_4 , and E_6 , it would be immaterial to replace these θ_i by their images under a common automorphism of the Dynkin diagram.

When G is connected, we write $\mathcal{R}(p_1, \dots, p_r)$ for the irreducible representation of G of dominant weight $\sum p_i \theta_i$. When G is disconnected, this weight must be decorated, as in [1], in order to define $\mathcal{R}(p_1, \dots, p_r)$. The required decoration is as follows. For A_1 : the sign $(-1)^{p_1}$ if $p_2 = 0$, and 0 otherwise. For D_4 : $+$, $0+$, or $0-$. For E_8 : $+$ or $0-$.

Let V_i be the representation $\mathcal{R}(p_1, \dots, p_r)$ for $p_i = \delta_i^j$. The representation V_1 , defined for G of rank at least 1, is the adjoint representation. The representations V_i ($1 \leq i \leq 7$) are the representations $\mathcal{J}$, $\mathcal{X}_2$, $\mathcal{Y}_2^*$, $\mathcal{X}_3$, $\mathcal{O}^*$, $\mathcal{X}_4$, $\mathcal{F}^*$ of [1]. The representation $\mathcal{R}(p_1, \dots, p_r)$ is characterized by the properties that it has dominant weight $\sum p_i \theta_i$ and is a constituent of the tensor product of the $V_i^{\otimes p_i}$.

For $s \leq r$, we will also write $\mathcal{R}(p_1, \dots, p_s)$ for $\mathcal{R}(p_1, \dots, p_s, 0, \dots, 0)$.

Our purpose is to describe, for fixed $p_1, \dots, p_7$, similarities among the representations $\mathcal{R}(p_1, \dots, p_s)$ of the groups in the exceptional series of rank at least s . The analogue in the series A is this: given a decreasing sequence of integers $\ell_1 \geq \ell_2 \geq \dots \geq \ell_k \geq 0$, take the corresponding representation of $\mathrm{GL}(n)$ for each $n \geq k$.

Let G be a simple split pinned group, let $\mathfrak{g}$ be its Lie algebra, and let $\mathfrak{h}$ be the Lie algebra of the pinned split torus T . Identify the group of weights $X := \mathrm{Hom}(T, \mathbb{G}_m)$ with a subgroup of the dual $\mathfrak{h}^\vee$. The canonical bilinear form $(\ , \)$ is the bilinear form on $\mathfrak{h}^\vee$ inverse to the Killing form $\mathrm{Tr}(\mathrm{ad} x \cdot \mathrm{ad} y)$ on $\mathfrak{h}$. For a classical supergroup, for instance $\mathrm{SOSp}(1, 2)$, use the supertrace. If α_0 is the highest root, one definition of the dual Coxeter number $h^\vee$ is that

$$(\alpha_0, \alpha_0) = 1/h^\vee.$$

To each dominant weight λ we will attach a finite system of rational numbers, each taken with a multiplicity $m(q) \in \mathbb{Z}$. In other words, we attach a measure $L_G(\lambda)$ on $\mathbb{Q}$, a $\mathbb{Z}$ -linear combination $\sum m(q) \delta[q]$ of Dirac measures. Let $[R^+]$ be the measure on X given by the sum of the Dirac measures $\delta[\alpha]$ for α a positive root. Let $M(\lambda)$ be the measure on $\mathbb{Q}$ which is the image of $[R^+]$ under $(2\lambda, \cdot) : X \rightarrow \mathbb{Q}$. The factor “12” is explained by the integrality assertion in Theorem 1. If all roots have the same length, or if α is a long root, then $(\lambda, \alpha) = \lambda(H_\alpha)/2h^\vee$. If G has roots of two different lengths and α is short, then

$$(\lambda, \alpha) = ((\alpha, \alpha)/(\alpha_0, \alpha_0)) \lambda(H_\alpha)/2h^\vee.$$

The factor $(\alpha, \alpha)/(\alpha_0, \alpha_0)$ is $1/4$, $1/3$, $1/2$ for $\mathrm{SOSp}(1, 2)$, G_2 , and F_4 , respectively. Let ρ be the half-sum $\frac{1}{2} \int \alpha \cdot [R^+]$ of the positive roots. For a classical supergroup, for example $\mathrm{SOSp}(1, 2)$, these definitions should use instead the sum over positive roots $\sum m(\alpha) \delta[\alpha]$, with $m(\alpha) = 1$ if the corresponding root subspace is even and $m(\alpha) = -1$ if it is odd. We define

$$L_G(\lambda) := M(\lambda + \rho) - M(\rho). \quad (1)$$

When no ambiguity can result, we simply write $L(\lambda)$ for $L_G(\lambda)$. If $L(\lambda) = \sum m(q) \delta[q]$, Weyl’s dimension formula says that the representation $V(\lambda)$ of dominant weight λ has dimension

$$\dim V(\lambda) = \prod_q m(q)^{-1}. \quad (2)$$

Let $F_{2N}^{\mathfrak{g}}$ be the function $\mathrm{Tr}((\mathrm{ad} x)^{2N})$ on $\mathfrak{g}$. Restrict it to $\mathfrak{h}$, and identify $\mathfrak{h}$ with $\mathfrak{h}^\vee$ by the canonical bilinear form. We obtain a function $F_{2N}^{\mathfrak{g}}$ on $\mathfrak{h}^\vee$. By construction,

$$F_{2N}^{\mathfrak{g}}(\lambda + \rho) - F_{2N}^{\mathfrak{g}}(\rho) = 2 \cdot 12^{-N} \sum m(q) q^{2N} L_G(\lambda). \quad (3)$$

The factor 2 comes from the fact that the definition of $L_G(\lambda)$ starts from R^+ and not from the set $R = R^+ \cup (-R^+)$ of all roots. Applying Theorem 2, p. 267, of Duflo [3] to the weights λ and

0, one can make explicit an element of the center of the enveloping algebra which acts on $V(\lambda)$ by $F_{2N}^{\mathfrak{g}}(\lambda + \rho) - F_{2N}^{\mathfrak{g}}(\rho)$. For $N = 1$, it is the Casimir.

For $p_1, \dots, p_7 \in \mathbb{N}$, we will construct a formal linear combination $L(p_1, \dots, p_7)$ of elements of the space $\mathcal{L}$ of linear forms $x \mapsto Ax + B$ with integral coefficients. In other words, it is a $\mathbb{Z}$ -linear combination $\sum m(A, B)\delta[Ax + B]$ of Dirac measures on $\mathbb{Q}$. This combination will specialize to the measure $L_G(\sum p_i \theta_i)$ attached to G .

For $N = 1$, that is, for the scalar by which the Casimir acts on $\mathcal{R}(p_1, \dots, p_7)$, this is even a linear function:

Theorem 1. *With the preceding notation, for arbitrary $p_1, \dots, p_7$, the sum $\sum m(A, B)(Ax + B)^2$ is linear in x ; equivalently,*

$$\sum m(A, B)A^2 = 0. \quad (4)$$

For $r \leq 7$, we write $L(p_1, \dots, p_r)$ for $L(p_1, \dots, p_r, 0, \dots, 0)$ with $7 - r$ final zeroes. Fix $r \leq 7$ and a group G in the exceptional series of rank at least r . To G there corresponds a rational number μ .

Theorem 2. *If $p_1, \dots, p_r \geq 0$, the zero linear form is not in the support of $L(p_1, \dots, p_r)$, and the total mass of the linear forms $Ax + B$ such that $A\mu + B = 0$ is zero. The scalar*

$$\sum_{A\mu+B=0} (A\mu + B)m(A, B)$$

is an integer. It is the length of the restriction to G^0 of $\mathcal{R}(p_1, \dots, p_r)$. According to (2), $\dim \mathcal{R}(p_1, \dots, p_r)$ is therefore the value at μ of the rational function $\prod (Ax + B)^{-m(A, B)}$.

More precisely, if $r \leq \text{rank}(G)$ and $A\mu + B = 0$, the multiplicity $m(A, B)$ of $Ax + B$ in $L(p_1, \dots, p_r)$ is zero except in the following cases. If $G = A_2$ and $p_2 > 0$: 1 for $-2x + 4$ and -1 for $-2x + 2$. If $G = D_4, E_8$ and either $p_3 > 0, p_4 = 0$, or $p_3 = 0, p_4 > 0$: 1 for $-3x + 3$ and -1 for $-x + 1$. If $G = F_4, E_8$ and $p_3, p_4 > 0$: 1 for $-3x + 3$ and $-2x + 2$, and -2 for $-x + 1$. If $G = E_6$ and p_5 or p_6 is > 0 : 1 for $-4x + 2$ and -1 for $-2x + 1$.

The $\mathcal{R}(p_1, \dots, p_7)$ for

$$p_1 + 2p_2 + 2p_3 + 3p_4 + 3p_5 + 3p_6 + 4p_7 \leq 4$$

are the representations $1, \mathcal{J}, \mathcal{A}, \mathcal{C}, \mathcal{O}^*, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{F}^*, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{X}_i$ ($i \leq 4$), $\mathcal{Y}_i$ ($i \leq 4$), and $\mathcal{Y}_i^*$ of [1], which may be read off from E_7 or E_8 . Thus Theorem 3 recovers part of the dimension formulae of [1].

Let $\mathcal{V}(p_1, \dots, p_7)$ be the convolution

$$\mathcal{V}(p_1, \dots, p_7) := L(p_1, \dots, p_7) * (\delta[x] - \delta[x - 1]). \quad (5)$$

If $L(p_1, \dots, p_7) = \sum m(A, B)\delta[Ax + B]$ and $\mathcal{V}(p_1, \dots, p_7) = \sum n(A, B)\delta[Ax + B]$, then the finitely supported functions $m(A, B)$ and $n(A, B)$ determine one another by

$$n(A, B) = m(A, B) - m(A + 1, B), \quad (6)$$

$$m(A, B) = \sum_{A' \geq A} n(A', B). \quad (7)$$

In (10), the sum is over A' ; the factor $(A' \geq A)$ is 1 if $A' \geq A$ and 0 otherwise.

We will take (10) as the definition of $L(p_1, \dots, p_7)$, with $\mathcal{V}(p_1, \dots, p_7)$ defined by specialization from a $\mathbb{Z}$ -linear combination $\mathbf{V}$ of Dirac measures on the space of maps from $\mathbb{Z}^7$ to $\mathbb{Q}$ of the form

$$(B; n; a_1, \dots, a_7) \mapsto \left(2 \sum a_i p_i + n\right)x + B. \quad (8)$$

The measure $\mathbf{V}$ has a simple description in terms of the root system of type E_8 .

Write $(B; n; a_1, \dots, a_7)$ for the function (11), write $\delta[B; n; a_1, \dots, a_7]$ for the Dirac measure supported at this function, and write $\{\varepsilon_1, \dots, \varepsilon_8\}$ for the root $2\delta - \alpha_3$ of E_8 . With this notation, $\mathbf{V}$ is defined as follows.

(A) Each positive root $\{c, e, g, h, f, d, b, a\}$ fixed by s_4 contributes to $\mathbf{V}$

$$\delta[B + h + n - 1; a, b, c, d, e, f, g] - \delta[B + h; n - 1; a, b, c, d, e, f, g]$$

with $5h + n = a + b + c + d + e + f + g + h$.

(B) Each pair of positive roots $\{c, e, g, h', f, d, b, a\}$, $\{c, e, g, h'', f, d, b, a\}$ interchanged by s_4 , with $h'' < h'$ —then $h'' = h' - 1$ —contributes to $\mathbf{V}$

$$\delta[B + h' + n' - 1; a, b, c, d, e, f, g] - \delta[B + h''; n'' - 1; a, b, c, d, e, f, g]$$

with $5h' + n' = a + b + c + d + e + f + g + h'$ and $5h'' + n'' = a + b + c + d + e + f + g + h''$.

(C) $\mathbf{V}$ is the sum of these contributions and of the $\mathbb{Z}$ -linear combination of the $\delta[B; n; 0, \dots, 0]$ such that, for every B and n , the total mass of the $(B; n; a_i, \dots)$ is zero.

Definition 3. (i) $\mathcal{V}(p_1, \dots, p_7)$ is the image of $\mathbf{V}$ under evaluation at $(p_1, \dots, p_7)$:

$$(B; n; a_1, \dots, a_7) \mapsto \left(2 \sum a_i p_i + n\right) x + B \in \mathbb{Q}.$$

(ii) It follows from (C) above that, for each B_0 , the mass in $\mathcal{V}(p_1, \dots, p_7)$ of the $Ax + B$ with $B = B_0$ is zero. Define $L(p_1, \dots, p_7)$ by (8), i.e. by (10).

It is immediate from the definition of $\mathbf{V}$ that

Proposition 4. (i) *On the support of $\mathbf{V}$, one has $B \in \mathbb{Z}$ and $n \geq 0$.*

(ii) *The projections of $\mathbf{V}$ by*

$$(a) \ (B; n; a_1, \dots, a_7) \mapsto (B; n) \in \mathbb{Z}^2,$$

$$(b) \ (B; n; a_1, \dots, a_7) \mapsto (a_1, \dots, a_7) \in \mathbb{Z}^7,$$

are zero.

The vanishing (ii a) is equivalent to $\mathcal{V}(0, \dots, 0) = 0$, i.e. to $L(0, \dots, 0) = 0$. This is compatible with the fact that $\mathcal{R}(0, \dots, 0)$ is, for every G , the trivial representation of dominant weight 0, and that $L_G(0) = 0$.

Taking account of (1) and of the fact that the total mass of $\mathbf{V}$ is zero, the vanishing (ii b) is equivalent to the following statement: for every N , as a function of $p_1, \dots, p_7$,

$$\sum \left(2 \sum a_i p_i + B\right)^{2N} L(p_1, \dots, p_7)$$

has degree at most $2N$. This is compatible, through (4), with the fact that for every simple Lie group G , $F_{2N}^g(\lambda)$ is a function of degree at most $2N$ in the weight λ .

Put $\Lambda = 2 \sum a_i \theta_i + \rho$. Theorem 2 can be reformulated as

$$\int_{\Lambda} (\Lambda + x - 1)(\Lambda + 1) \cdot \mathbf{V} = 0, \tag{9}$$

identically in the indeterminates p_i .

Some indications on the proof of Theorem 1. — Let G be a group of the exceptional series and put $r = \inf(\text{rank}(G), 7)$. Except for E_8 and the simple root α_4 , no positive root is

orthogonal to all the θ_i ($1 \leq i \leq r$). Put $[R^G] := [R^+]$ if $G \neq E_8$, and $[R^G] = [R^+] - \delta[\alpha_4]$ if $G = E_8$. Define the moving part of $L_G(\sum p_i \theta_i)$ by

$$L_G^{\text{mob}}\left(\sum p_i \theta_i\right) = \left\langle M\left(\sum p_i \theta_i + \rho\right), [R^G] \right\rangle. \quad (10)$$

The definition (1) rewrites as $L_G(\sum p_i \theta_i) = L_G^{\text{mob}}(\sum p_i \theta_i) - L_G^{\text{mob}}(0)$.

Theorem 1 for G uses of $\mathbf{V}$ only its image $\mathbf{V}_G$ under

$$(B; n; a_1, \dots, a_r) \mapsto \left(2 \sum a_i p_i + n + B; a_1, \dots, a_r\right). \quad (11)$$

Decompose $\mathbf{V}_G$ as $\mathbf{V}_G^+ - \mathbf{V}_G^-$, the moving part $\mathbf{V}_G^+$, respectively the fixed part $\mathbf{V}_G^-$, having support where $(a_1, \dots, a_r) \neq 0$, respectively $= 0$. Theorem 1 follows from its analogue for the moving parts of L and $\mathbf{V}$: by Proposition 1 (ii a), the fixed part is recovered by putting $x = 0$. For $(a_1, \dots, a_r) \neq 0$, let $R^{(a_1, \dots, a_r)}$ be the set of positive roots α such that $(\alpha, \theta_i) = a_i$ for $1 \leq i \leq r$. The form $(\ , \)$ is as in (5)–(6). For $G \neq E_8$, each $R^{(a_1, \dots, a_r)}$ is a singleton or empty, and their union is R^+ . For $G = E_8$, each $R^{(a_1, \dots, a_r)}$ is an orbit of $\{s_3, s_4\}$ or is empty. Their union is $R^+ - \{\alpha_4\}$. Let $\mathbf{V}_G^{(a_1, \dots, a_r)}$ be the measure on $\mathbb{Q}$ which is the inverse image of $\mathbf{V}_G$ by $q \mapsto (Bh^\vee/6 + n; a_1, \dots, a_r)$: the mass of q is the mass of the $(B; n; a_1, \dots, a_r, \dots)$ with $Bh^\vee/6 + n = q$. Let $[R^{(a_1, \dots, a_r)}]$ be the sum of the $\delta[\alpha]$ for $\alpha \in R^{(a_1, \dots, a_r)}$. Theorem 1, for the moving parts of L and $\mathbf{V}$, decomposes as the sum of the equalities

$$(p_i \cdot [R^{(a_1, \dots, a_r)}]) * (\delta[0] - \delta[-1]) = \mathbf{V}_G^{(a_1, \dots, a_r)}. \quad (12)$$

Our proof of (15) is by brute force. Below are a few properties of the root systems involved which may help explain what is happening.

When all roots have the same length, $(\lambda, \alpha) = \lambda(H_\alpha)$ and (ρ, α) is the sum of the coefficients of α expressed as a linear combination of simple roots. For E_8 , $(\rho, \alpha) = 5$, and (15) follows immediately from the definition of $\mathbf{V}$.

Let R be the root system of type A_1 , A_2 , D_4 , E_6 , or E_7 , and let r be its rank. In the root system of type E_8 , the fundamental weights $\theta_1, \dots, \theta_r$ correspond to simple roots $\alpha_{S(1)}, \dots, \alpha_{S(r)}$. Let S_1 be the complementary set of $8 - r$ simple roots, and let W' be the Weyl group generated by the corresponding reflections. Let $\langle \theta_1, \dots, \theta_r \rangle$ be the vector space spanned by $\theta_1, \dots, \theta_r$. Other descriptions: it is the orthogonal of $\langle \theta_i \mid i \in S_1 \rangle$, or the fixed subspace of W' . One checks case by case that the roots of E_8 lying in $\langle \theta_1, \dots, \theta_r \rangle$ form a root system of type R , and that, if one takes as positive roots of this root system those that are positive for E_8 , the fundamental weights of E_8 are the dominant weights $\theta_1, \dots, \theta_r$ of R .

For R of one of the indicated types, (15) follows from the following assertions, which we verified for D_4 , F_4 , and E_7 :

(a) Let T be an orbit of W' in the set of roots of E_8 . If T is not reduced to a fixed point and, for $t \in T$, (t, θ_i) —constant on T —is > 0 for some θ_i ($1 \leq i \leq r$), then the contribution of T to $\mathbf{V}$ has zero image under (14).

(b) Let ρ_R and $h_R^\vee$ denote the sum of the fundamental weights and the dual Coxeter number of R . If α lies in $\langle \theta_1, \dots, \theta_r \rangle$ and is a positive root, necessarily fixed by s_4 , then, with the notation of part (A) of the definition of $\mathbf{V}$, one has

$$(h^\vee/6)(\rho_R, \alpha) + n = (\rho, \alpha).$$

Examples. — Verify (a) for E_7 . The orbit T is here a pair of roots as in part (B) of the definition of $\mathbf{V}$. The point is to verify that $3h' + n' \geq 3h'' + n'' - 1$. Indeed, $h'' = h' - 1$ and $4h' + n' = 4h'' + n''$, i.e. $n'' = n' + 4$.

Theorem 1 gives the dependence on G of $F_{2N}^g(\lambda + \rho) - F_{2N}^g(\rho)$ for certain λ .

Proposition 5. For $1 \leq N \leq 8$, $F_{2N}(\rho)$ is, as a function of G , a function of $\mu = 6/h^\vee$ of the form $P_{2N}(\mu)/\mu(\mu+1)$, for a polynomial $P_{2N}(X)$ of degree $2N$ satisfying $P_{2N}(X) = P_{2N}(-1-X)$.

It follows from Duflo [3] that $F_{2N}(\rho)$ is the value for G , in the sense of [5], of a linear combination of trivalent graphs with oriented vertices. For $N \leq 3$, arguments imitating [5], 6.8, 6.9, make it possible to deduce the proposition from this presentation of $F_{2N}(\rho)$. Beyond that, we know how to verify it only by brute force. Since we have only ten groups G at our disposal, the proposition remains true, but is empty, for $N \geq 9$.

Correction to [2]. — The formula given for $\dim(\mathcal{X}_4)$ is erroneous. See [1] for the correct formula.

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Local behavior of Hodge structures at infinity

P. Deligne

Abstract

This article is an expanded version of a letter to D. Morrison, inspired by his article [3]. I give a description of “very degenerating” variations of Hodge structures, which is well adapted to the variations arising in the mirror fairy tale. I also try to ascertain how “motivic” this description is.

0. Introduction

In [3], D. Morrison describes some of the results of [1] in a Hodge theoretic language. One ingredient is the variation of Hodge structures defined by a family of Calabi–Yau manifolds near a point (in a completed parameter space) of “maximal degeneracy”. We interpret “maximal degeneracy” as leading to the appearance of variations of mixed Hodge structures of a special type (Hodge-Tate), and give a description of such mixed variations in terms of invertible holomorphic functions. The logarithmic derivatives of those functions are what appears in Morrison’s paper.

I thank E. Cattani for a very helpful letter directing me to parts of [2] I had missed, and the referee for suggesting many improvements.

1 1. Notations

For V a polarized variation of Hodge structures of weight w on a complex variety S , we use the following notations.

$V_{\mathbb{Z}}$: underlying local system of free abelian groups. For $R = \mathbb{Q}, \mathbb{R}, \mathbb{C}$, $\mathcal{O}$ (sheaf of holomorphic functions), $V_R := V_{\mathbb{Z}} \otimes_{\mathbb{Z}} R$.

$\mathbb{Z}(k)$: $(2\pi i)^k \mathbb{Z}$, viewed as a Hodge structure of type $(-k, -k)$. Also: the corresponding constant variation.

$\psi : V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} \rightarrow \mathbb{Z}(-w)$: the polarization form, symmetric for w even and alternating for w odd.

At each point $s \in S$, the Hodge structure of the fiber V_s of V at s is given by a decomposition

$$V_{s, \mathbb{C}} = \bigoplus_{p+q=w} V_s^{p,q}$$

with $V_s^{q,p}$ the complex conjugate of $V_s^{p,q}$. The Weil operator C : multiplication by i^{p-q} on $V_s^{p,q}$, is real and the form

$$(2\pi i)^w \psi(Cv, \bar{v})$$

is hermitian symmetric and positive definite.

The holomorphic vector bundle with integrable connection having $V_{\mathbb{C}}$ as sheaf of horizontal sections is identified with its sheaf $V_{\mathcal{O}}$ of holomorphic sections. We denote

∇ : its connection.

F : the Hodge filtration. At each point, $F^p = \bigoplus_{a \geq p} V_s^{a,b}$. It is a holomorphic filtration and $\nabla F^p \subset \Omega^1 \otimes F^{p-1}$ (Griffiths transversality).

Typical example

Let $(X_s)_{s \in S}$ be a family of nonsingular projective varieties parametrized by S , i.e. a projective and smooth map $f : X \rightarrow S$ (with $X_s = f^{-1}(s)$). Take

$$V_{\mathbb{Z}} := R^w f_* \mathbb{Z} \mod \text{torsion}.$$

It is the local system of the $H^w(X_s)/\text{torsion}$ and $V_s^{p,q} = H^{p,q}(X_s)$.

For X_s purely of dimension n , and $w = n$, on the primitive part of $H^n(X_s)$ (the kernel of the cup-product with the first Chern class of the given ample line bundle), a polarization form is given by

$$\frac{1}{(2\pi i)^n} (-1)^{n(n-1)/2} (2\pi i)^w \int_{X_s} v \cup w.$$

The construction of a polarization form on the whole of the cohomology is by reduction to that case, applied to linear sections of X_s .

2. Asymptotics

Let D be the open unit disc, and $D^* := D - \{0\}$. We take $S = D^{*n}$ ($n \geq 1$) and recall known results on the behavior of $V_{\mathbb{R}}$ for $s \rightarrow 0$. To the extent possible, our references will be to the survey paper [2].

2.1 Monodromy

Let T_j be the monodromy of $V_{\mathbb{Z}}$ around $s_j = 0$. The T_j commute and are known to be quasi-unipotent. For simplicity, we assume they are even unipotent.

We define

$$N_j := -\log(T_j)$$

(note the minus sign). Sign convention, explained for $n = 1$: T is the effect of pushing an element of $(V_{\mathbb{Z}})_{s_0}$ horizontally along the path $\exp(2\pi i u)s_0$ ($0 \leq u \leq 1$).

2.2 Canonical extension

For any complex local system $\mathcal{H}_{\mathbb{C}}$ on D^{*n} , with unipotent monodromy, we continue to write $\mathcal{H}_{\mathcal{O}}$ for the canonical extension of the vector bundle $\mathcal{H}_{\mathcal{O}}$ to D^n . It is characterized by the property that, in a local basis near 0, the matrix of 1-forms defining the connection has logarithmic poles along the $s_j = 0$, with nilpotent residues.

Consider on $\mathcal{H}_{\mathcal{O}}$ the new connection

$$\nabla^0 = \nabla - \frac{1}{2\pi i} \sum_j N_j \frac{ds_j}{s_j}. \quad (2.2.1)$$

The horizontal sections for ∇^0 are the sections of the form $\exp(\sum \log s_j \cdot N_j / 2\pi i) h$ for h ∇ -horizontal. The connection ∇^0 has no monodromy and turns $\mathcal{H}_{\mathcal{O}}$ into a constant bundle. We will say *constant* to mean horizontal for ∇^0 . The canonical extension of $\mathcal{H}_{\mathcal{O}}$ is obtained by extending it as a constant bundle.

The construction of ∇^0 depends on the choice of coordinates near 0, while the canonical extension does not. In fact, the constant vector bundle attached to $\mathcal{H}_{\mathcal{O}}$ should be viewed as living on the tangent space T_0 of D^n at 0. Local coordinates s_j (with s_i/s_j invertible) define (a) an isomorphism φ between a neighborhood of 0 in D^n and a neighborhood of 0 in T_0 , (b) a constantification of $\mathcal{H}_{\mathcal{O}}$.

If we use φ to transplant the local system $\mathcal{H}_{\mathbb{C}}$ and the constantification of $\mathcal{H}_{\mathcal{O}}$ to T_0 , the result is independent of the local coordinates used.

If a filtration F of the canonical extension $\mathcal{H}_\mathcal{O}$ is given, the corresponding nilpotent orbit is the constant filtration of $\mathcal{H}_\mathcal{O}$ which coincides with F at 0. Again, to be coordinate free, it should live on T_0 .

2.3 Nilpotent orbit

For a polarized variation V on D^{*n} , the nilpotent orbit theorem states that

- (a) The Hodge filtration F extends to the canonical extension $V_\mathcal{O}$ as a filtration (still denoted F) by locally direct factors ([2] 2.1 (i)),
- (b) The corresponding nilpotent orbit is again, in a neighborhood of 0, a variation of Hodge structures of weight w polarized by ψ ([2] 2.1 (ii)).

Conversely, let $(V_\mathbb{Z}, F^{\text{nilp}})$ be a polarized variation of Hodge structures on D^{*n} which is a nilpotent orbit: the filtration F^{nilp} is assumed constant (for ∇^0 , see (2.2)). Let F be a new filtration of $V_\mathcal{O}$ on D^n , agreeing with F^{nilp} at $s = 0$, and obeying transversality:

$$\nabla F^p \subset \Omega^1 \otimes F^{p-1}.$$

Assume further that $\psi(F^p, F^{w-p+1}) = 0$ for all p . Then $(V_\mathbb{Z}, F)$ is, near 0, a variation of Hodge structures on D^{*n} , polarized by ψ . See [2], 2.8.

2.4 Weight filtration

There exists on $V_\mathbb{Q}$ a unique increasing filtration W , the weight filtration, such that $N_j W_k \subset W_{k-2}$ and that for any positive linear combination $N = \sum \lambda_j N_j$ ($\lambda_j > 0$), N^k induces an isomorphism from $\text{Gr}_{w+k}^W(V_\mathbb{R})$ to $\text{Gr}_{w-k}^W(V_\mathbb{R})$. Further, $(V_\mathbb{Z}, W, F^{\text{nilp}})$ is a variation of mixed Hodge structures ([2] 2.3).

The construction of W is compatible with passage to the dual, as well as to tensor products. It follows that if ψ is a polarization form, then W_a and W_{-a-1} are mutually orthogonal, and ψ induces a perfect pairing $\text{Gr}_a^W(V_\mathbb{Z})$ between $\text{Gr}_{-a}^W(V)$ and $\text{Gr}_{a-w}^W(V)$. Further, the induced pairings, and

$$\psi(\cdot, N^k \cdot) : \text{Gr}_{w+k}^W(V) \otimes \text{Gr}_{w+k}^W(V) \rightarrow \mathbb{Q}$$

have positivity properties reminiscent of those of the cohomology of a nonsingular projective variety of dimension k , with the cup-product pairing to H^{2k} , the role of N being played by the cup-product with the first Chern class of an ample line bundle.

In the terminology of [2] 1.16, (V, W, F^{nilp}) is polarized by ψ and N . The reason for this analogy is not understood.

3 Warning

In general, W and the original Hodge filtration F do not define a mixed Hodge structure in a neighborhood of 0. Consider, for instance, a general pencil $f : X \rightarrow \mathbb{P}^1$ of hypersurfaces of degree d in $\mathbb{P}^M$. The fiber $X_{(\lambda, \mu)}$ at the point with projective coordinates (λ, μ) is the hypersurface with equation

$$\lambda F + \mu G = 0$$

(F, G general forms of degree d).

Let S be the set of critical values of f . On $\mathbb{P}^1 - S$, we consider the variation

$$V_\mathbb{Z} := R^w f_* \mathbb{Z}$$

and restrict it to a punctured disc D^* centered at a critical value (λ_0, μ_0) .

We assume that

- (a) M is even, hence X_s is of odd dimension $w = M - 1$.

The cohomology $H^w(X_s, \mathbb{Z})$ is torsion-free, primitive (i.e. annihilated by cup product with the hyperplane section class) and the polarization form is given by the cup product.

One knows that the global monodromy, image of $\pi_1(\mathbb{P}^1 - S)$, is Zariski dense in the corresponding symplectic group. Provided that $H^w(X_s, \mathbb{Z}) \neq 0$, i.e. $d \neq 1, 2$, N is nilpotent of rank one. The weight filtration is

$$0 \subset W_{w-1} = \text{im}(N) \subset W_w = \ker(N) = V_{\mathbb{Q}}.$$

If $M \geq 4$ and d is large enough, so that $V_s \neq F^{(w-1)/2}$.

If (W, F) were a mixed Hodge structure on D^* , Gr_{w-1}^W , being one-dimensional, would have to be of type $((w-1)/2, (w-1)/2)$:

$$\text{im}(N) \subset F^{(w-1)/2}.$$

By analytic continuation, any image of $\text{im}(N)$ by the global monodromy group would still be in $F^{(w-1)/2}$. Those images span V , by Zariski density of the global monodromy group in the symplectic group, contradicting (b).

4 4

We now consider a case where (W, F) is a mixed Hodge structure. Fix a base point $s_0 \in D^{*n}$, let G be the group of automorphisms of $V_{\mathbb{Q}}$ respecting the form ψ , and let G_0 be the connected component of the identity of G , viewed as a linear algebraic group: it is a special orthogonal group for w even, a symplectic group for w odd.

The filtrations W , F and F^{nilp} correspond to isotropic flags:

$$(P_i)_i := F^{w-i+1}, \quad (Q_i)_i := W_{w+i-1}$$

and similarly for F^{nilp} . The subgroups of G_0 stabilizing them are parabolic subgroups.

Proposition 1 (4.1). *If the parabolic subgroups $P(F^{\text{nilp}})$ and $P(W)$ defined by F^{nilp} and W satisfy*

$$\text{Lie } P(F^{\text{nilp}}) + \text{Lie } P(W) = \text{Lie } G_0, \quad (4.1.1)$$

then, near 0, (W, F) is a mixed Hodge structure, with the same Hodge numbers as (W, F^{nilp}) .

Remarque 2. As the $\exp(2\pi i s_j N_j) \in G_0$ respect W and permute transitively the F^{nilp} taken at various points, the validity of (4.1.1) does not depend on the point s_0 at which it is considered.

Proof. The map $u \mapsto s_0 \exp(2\pi i u)$ maps a product of upper half-planes to D^{*n} . The pullback of the local system $V_{\mathbb{Z}}$ is a trivial local system, and we can consider the Hodge filtration F at $s_0 \exp(2\pi i u)$ as a filtration $\varphi(u)$ of the fixed vector space $V_{\mathbb{C}}$.

If one wants to view $V_{\mathcal{O}}$ as a constant vector bundle as in 2.2, $\varphi(u)$ is to be replaced by $\exp(-uN)\varphi(u)$.

In particular, if we write F^{nilp} for F^{nilp} at s_0 ,

$$F^{\text{nilp}} = \lim \exp(-uN)\varphi(u).$$

Here the limit is taken as the infimum of the $\Im(u_j) \rightarrow \infty$, i.e. $s_0 \exp(2\pi i u) \rightarrow 0$.

From (4.1.1), it then follows that for $\inf \Im(u_i)$ large, the pair of filtrations $(W, \exp(-uN)\varphi(u))$ is conjugate to (W, F^{nilp}) by $g \in G(\mathbb{C})$ close to 1. In particular,

$$\dim(W_a \cap F^{\text{nilp}b}) = \dim(W_a \cap \exp(-uN)\varphi(u)^b).$$

As $\exp(-uN)\varphi(u)$ is close to F^{nilp} , this equality of dimensions implies that the filtration induced by $\exp(-uN)\varphi(u)$ on $\text{Gr}_a^W(V_{\mathbb{Q}} \otimes \mathbb{C})$ tends to that induced by F^{nilp} . As N_j respects W and acts

trivially on $\mathrm{Gr}_*^W(V_{\mathbb{Q}} \otimes \mathbb{C})$, the filtrations $\varphi(u)$ and $\exp(-uN)\varphi(u)$ induce the same filtration on $\mathrm{Gr}_a^W(V_{\mathbb{Q}} \otimes \mathbb{C})$.

For $\inf \Im(u_i)$ large enough, this induced filtration defines a Hodge structure of weight a , being close to the Hodge filtration $\mathrm{Gr}_a^W(F^{\mathrm{nilp}})$.

It follows that (W, F) is mixed Hodge, with the same Hodge numbers as (W, F^{nilp}) . $\square$

5 5. Variant

Fix some Hodge structures H_{α} , and some horizontal tensors t_{α} , integral of type $(0, 0)$, in some tensor powers

$$\bigotimes_a V_{\mathbb{Z}}^{\otimes n(\alpha)} \otimes \bigotimes_b V_{\mathbb{Z}}^{*\otimes m(\alpha)} \otimes H_{\alpha}.$$

One can then repeat 4.1 and its proof with G replaced by the subgroup fixing the t_{α} .

6 6

We now assume that the mixed Hodge structure $(V_{\mathbb{Z}}, W, F^{\mathrm{nilp}})$ is Hodge-Tate, i.e. such that

$$\mathrm{Gr}_k^W \text{ is of type } (-k/2, -k/2) \text{ for } k \text{ even, and } 0 \text{ for } k \text{ odd.} \quad (6.1)$$

This should be viewed as an assumption of maximal degeneracy. It means that $F_p^{\mathrm{nilp}} \oplus W_{2p-2} \rightarrow V_{\mathcal{O}}$ is an isomorphism for all p . If we set

$$V_{(p)} := F_p^{\mathrm{nilp}} \cap W_{2p} = F_p^{\mathrm{nilp}} \cap W_{2p+1},$$

$V_{\mathcal{O}}$ is then the direct sum of the $V_{(p)}$, with

$$F_p^{\mathrm{nilp}} = \bigoplus_{a \geq p} V_{(a)}, \quad W_{2p} = W_{2p+1} = \bigoplus_{b \leq p} V_{(b)}.$$

The assumption (4.1.1) holds and 4.1 is in fact easier to prove in this case. In terms of the nilpotent orbit trivialization of $V_{\mathcal{O}}$, the filtrations W and F^{nilp} are constant. As W and F^{nilp} are opposed, and F and F^{nilp} coincide at 0, W and F remain opposed in a neighborhood of 0:

$$V_{\mathcal{O}} = \bigoplus V_{(a)} \quad \text{with} \quad V_{(a)} := F^p \cap W_{2p} = W_{2p+1} \quad (6.2)$$

implying that $(V_{\mathbb{Z}}, W, F)$ is mixed Hodge and Hodge-Tate. We will assume D^n has been shrunk so that $(V_{\mathbb{Z}}, W, F)$ is mixed Hodge on the whole of D^{*n} .

Our aim is to classify, i.e. to give another description, of variations such as the above.

7 7

In the category of mixed Hodge structures, $\mathrm{Hom}(\mathbb{Z}(0), \mathbb{Z}(1)) = 0$ (notations: see 1): an extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ has no nontrivial automorphism as extension. The extensions are classified by $\mathbb{C}^*$:

$$\mathrm{Ext}^1(\mathbb{Z}(0), \mathbb{Z}(1)) = \mathbb{C}^*, \quad (7.1)$$

with $q \in \mathbb{C}^*$ corresponding to the extension $0 \rightarrow \mathbb{Z}(1) \rightarrow H_q \rightarrow \mathbb{Z}(0) \rightarrow 0$.

$$\begin{aligned} H_{\mathbb{C}} &\cong \mathbb{C}^2, \text{ basis } e_0, e_1; \\ W_{-2} &= \mathbb{C}e_1, \quad F^0 = \mathbb{C}e_0; \\ H_{\mathbb{Z}} &= 2\pi i \mathbb{Z} \cdot e_1 + \mathbb{Z} \cdot (e_0 + \log q \cdot e_1) \subset H_{\mathbb{C}}; \\ \alpha(2\pi i) &= 2\pi i e_1, \quad \beta(e_0) = 1. \end{aligned} \quad (7.2)$$

Note that the choice of the determination of log does not matter.

Equivalent description: H_q has a basis f_0, f_1 with $\alpha(2\pi i) = f_0$, $\beta(f_1) = 1$, $F^0 \subset H_{\mathbb{C}}$ the kernel of $\alpha : H_{\mathbb{C}} \rightarrow \mathbb{C}$.

Variations of mixed Hodge structures H over S that are extensions of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ are similarly classified by invertible holomorphic functions. The extension H defined by q has $H_{\mathcal{O}} = \mathcal{O} \cdot e_0 \oplus \mathcal{O} \cdot e_1$, with basis e_0 (spanning F^0) and e_1 (spanning W_{-2}) and integral lattice and structure maps α, β as in (7.2). The connection is given by

$$\nabla = d - \frac{1}{2\pi i} \frac{dq}{q} \beta. \quad (7.3)$$

We now take $S = D^{*n}$. Let n_j be the order of q along $s_j = 0$:

$$\frac{1}{2\pi i} \frac{(\log)q}{s_j} = i\partial_j \log q + n_j \cdot 2\pi i.$$

The constant sections of $H_{\mathcal{O}}$, in the sense of 2.2, are spanned by e_1 and $e_0 + (\log(q/\prod s_j^{n_j})) \cdot e_1$.

It follows that F extends to the canonical extension of $H_{\mathcal{O}}$, remaining a supplement to W_{-2} , if and only if q belongs to the group $\mathcal{O}_{\text{mer}}^*(D^n)$ of invertible holomorphic functions on D^{*n} , meromorphic along the $s_j = 0$. The filtration F is constant with respect to the trivialization of $H_{\mathcal{O}}$ given in 2.2 if and only if q is a monomial $a \cdot \prod s_j^{n_j}$.

The nilpotent orbit attached to H (cf. 1.3) is again an extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$. Its invariant is the monomial q_0 such that q/q_0 is holomorphic invertible on D^n with value 1 at 0.

The definitions given do not require the choice of which i is i and which is $-i$ in $\mathbb{C}$. If one is willing to make that choice, one could rather say that the extensions of $\mathbb{Z}$ (type $(0,0)$) by $\mathbb{Z}$ (type $(-1,-1)$) are parametrized by $\mathbb{C}^*$. Formulas (7.2)(7.3) should be changed as follows:

$$\begin{aligned} H_{\mathbb{Z}} &= \mathbb{Z}e_1 + \mathbb{Z}(e_0 + \frac{\log q}{2\pi i}) \subset H_{\mathbb{C}}; \\ \alpha(1) &= e_1, \quad \beta(e_0) = 1; \end{aligned} \quad (7.2')$$

replace $\frac{1}{2\pi i} \frac{dq}{q}$ by $\frac{1}{2\pi i} \frac{dq}{q}$ in (7.3).

More generally, for A and B free abelian groups, viewed as purely of type (p,p) and $(p-1, p-1)$, one has for the corresponding constant local system on S

$$\text{Ext}_{\text{mixed Hodge}}^1(A, B) = \text{Hom}(A, B) \otimes \mathcal{O}^*(S). \quad (7.4)$$

The underlying vector bundle with connection $(H_{\mathcal{O}}, \nabla)$ of the extension with class ξ is

$$\begin{aligned} H_{\mathcal{O}} &= A \otimes \mathcal{O} \oplus B \otimes \mathcal{O}, \\ \nabla &= d - \frac{1}{2\pi i} d \log \xi \cdot \beta. \end{aligned} \quad (7.5)$$

For $S = D^{*n}$, the filtration F extends to the canonical extension, remaining opposite to W at 0, if and only if

$$\xi \in \text{Hom}(A, B) \otimes \mathcal{O}_{\text{mer}}^*(D^n).$$

The group $\mathcal{O}_{\text{mer}}^*(D^n)$ contains the group of monomials $\{a \cdot \prod s_j^{n_j}\}$, and retracts to it by a unique homomorphism nilp such that $f/\text{nilp}(f)$ is holomorphic at zero, where it takes the value 1.

Let F^{nilp} be the constant filtration (2.2) agreeing with F at 0. The nilpotent orbit $(H_{\mathbb{Z}}, F^{\text{nilp}})$ is again an extension of A by B . Its invariant is $\text{nilp}(\xi)$, in $\text{Hom}(A, B) \otimes \langle \text{monomials} \rangle$.

8 8

With §§6 as the motivating example, we now consider on D^{*n} variations of mixed Hodge structures $(V_{\mathbb{Z}}, W, F)$ of Hodge-Tate type (6.1) such that (8.1), (8.2) below hold.

8.1

$\mathrm{Gr}_{2k}^W(V_{\mathbb{Z}})$ is a constant local system.

In (8.1), W denotes the filtration of $V_{\mathbb{Z}}$ induced by the filtration W of $V_{\mathbb{Q}}$. We will identify the constant local system $\mathrm{Gr}_{2k}^W(V_{\mathbb{Z}})$ with its constant value: a free abelian group. The conditions (8.1) imply that the monodromy is unipotent.

8.2

F extends to the canonical extension of $V_{\mathcal{O}}$, and, at 0, the filtrations F and W are opposed.

By (8.1)(8.2), the quotient $W_{2k}(V_{\mathbb{Z}})/W_{2k-4}(V_{\mathbb{Z}})$ is an extension of $\mathrm{Gr}_{2k}^W(V_{\mathbb{Z}})$ (type (k, k)) by $\mathrm{Gr}_{2k-2}^W(V_{\mathbb{Z}})$ (type $(k-1, k-1)$) of the type considered in §7. It has an extension class

$$B_k \in \mathrm{Hom}(\mathrm{Gr}_{2k}^W(V_{\mathbb{Z}}), \mathrm{Gr}_{2k-2}^W(V_{\mathbb{Z}})) \otimes \mathcal{O}_{\mathrm{mer}}^*(D^n).$$

Let E be the sum of those classes:

$$E \in \mathrm{End}(\mathrm{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathcal{O}_{\mathrm{mer}}^*(D^n). \quad (8.3)$$

As in (6.1), the weight and Hodge filtration of $V_{\mathcal{O}}$ define a direct sum decomposition $V_{\mathcal{O}} = \bigoplus V^{(a)}$. One has

$$V^{(p)} \xrightarrow{\sim} \mathrm{Gr}_{2p}^W(V_{\mathbb{Z}}) \otimes \mathcal{O},$$

giving

$$V_{\mathcal{O}} \cong \mathrm{Gr}^W(V_{\mathbb{Z}}) \otimes \mathcal{O}. \quad (8.4)$$

Lemme 3 (9). *The isomorphism (8.4) transforms the connection ∇ of $V_{\mathcal{O}}$ into the connection*

$$\nabla = d - \frac{1}{2\pi i} d \log E.$$

Proof. The connection ∇ of $V_{\mathcal{O}}$ respects W , induces the trivial connection on $\mathrm{Gr}^W(V_{\mathbb{Z}}) \otimes \mathcal{O}$, and maps F^p to F^{p-1} . It is hence given by a 1-form with values in endomorphisms of degree -2 of $\mathrm{Gr}^W(V_{\mathbb{Z}}) \otimes \mathcal{O}$. To compute it, it suffices to compute the induced connections on the quotients W_{2k}/W_{2k-4} , and we apply (7.5). $\square$

9 10

If we replace F by F^{nilp} , the constant filtration (2.2) agreeing with F at 0, one obtains the nilpotent orbit $(V_{\mathbb{Z}}, W, F^{\mathrm{nilp}})$ attached to $(V_{\mathbb{Z}}, W, F)$. By §§7, the corresponding invariant $E((V_{\mathbb{Z}}, W, F^{\mathrm{nilp}}))$ is deduced from $E = E((V_{\mathbb{Z}}, W, F))$ by applying

$$\mathrm{nilp} : \mathcal{O}_{\mathrm{mer}}^*(D^n) \rightarrow \{\text{monomials}\}. \quad (10.1)$$

The group of monomials is an extension of $\mathbb{Z}^n$ (the exponents) by $\mathbb{C}^*$. If one projects further to $\mathbb{Z}^n$, the information retained is the $\mathrm{Gr}^W(1 - T_j) = \mathrm{Gr}^W(N_j)$: by reduction to the case of extensions of $\mathbb{Z}$ by $\mathbb{Z}(1)$, one checks that, applying ν_j , the order along $s_j = 0$, one has

$$\mathrm{Gr}^W(1 - T_j) = \mathrm{Gr}^W(N_j) = -\nu_j(E) \in \mathrm{End}(\mathrm{Gr}^W(V_{\mathbb{Z}}))_{-2}. \quad (10.2)$$

The integrability of ∇ can be written

$$d \log E \wedge d \log E = 0 \quad \text{in} \quad \mathrm{End}(\mathrm{Gr}^W) \otimes \Lambda^2 \otimes \mathbb{Q}. \quad (10.3)$$

For $(V_{\mathbb{Z}}, W, F^{\mathrm{nilp}})$, i.e. for the dominant part E_0 , it corresponds to the commutativity of the $\mathrm{Gr}^W(N_i)$.

Let us consider pairs $((V_{\mathbb{Z}}, W, F^{\text{nilp}}), E)$ of a mixed Tate nilpotent orbit $(V_{\mathbb{Z}}, W, F^{\text{nilp}})$, whose invariant will be denoted E_{nilp} , and of

$$E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathcal{O}_{\text{mer}}^*(D^n)$$

such that

$$d \log E \wedge d \log E = 0 \quad \text{and} \quad E_{\text{nilp}} = \text{nilp}(E).$$

We turn those pairs into a category by defining a morphism f from $((V_{\mathbb{Z}}, W, F^{\text{nilp}}), E)$ to $((V'_{\mathbb{Z}}, W', F^{\text{nilp}'}) , E')$ to be $f : V_{\mathbb{Z}} \rightarrow V'_{\mathbb{Z}}$, compatible with W and F^{nilp} , and such that $\text{Gr}^W(f) \circ E = E' \circ \text{Gr}^W(f)$.

Théorème 4 (11). *To $(V_{\mathbb{Z}}, W, F)$ as in §§8, let us attach*

(a) *the corresponding nilpotent orbit $(V_{\mathbb{Z}}, W, F^{\text{nilp}})$,*

(b) *$E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathcal{O}_{\text{mer}}^*(D^n)$.*

This construction is an equivalence from the category of variations $(V_{\mathbb{Z}}, W, F)$ as in §§8 to the category of pairs as above.

Remarque 5. The functor of the theorem is compatible with tensor products, with the E of $V \otimes V'$ defined to be $E \otimes \text{Id}_{V'} + \text{Id}_V \otimes E'$. It is also compatible with duals (with E going to the opposite of its transpose) and, combining the two, with $\underline{\text{Hom}}$.

Proof. A morphism of V to V' is a section of $\text{Hom}(V, V')_{\mathbb{Z}}$ in W_0 and F^0 . Thus, to show that the function is fully faithful, it suffices to show that a global section v of $V_{\mathbb{Z}}$ is in W_0 and F^0 if it is in W_0 , in F_0^{nilp} and if its image $v^{(0)}$ in $\text{Gr}_0^W(V_{\mathbb{Z}})$ is killed by E : $E v^{(0)}$ trivial in $\text{Gr}_{-2}^W(V_{\mathbb{Z}}) \otimes \mathcal{O}^*$.

Being global and horizontal, v is constant, in the sense of (2.2). Let $v^{(0)}$ be its component in $V^{(0)}$ (6.2). Via the isomorphism of $V_{\mathcal{O}}^{(0)}$ with $\text{Gr}_0^W(V_{\mathbb{Z}}) \otimes \mathcal{O}$ (8.4), it corresponds to $\bar{v}$ in $\text{Gr}_0^W(V_{\mathbb{Z}})$. By Lemma 9, as $\xi \cdot \bar{v}$ is trivial, $v^{(0)}$ is horizontal. The two constant sections v and $v^{(0)}$ coincide at 0, as v is in F^{nilp} . It follows that they are equal, and $v = v^{(0)} \in F^0$.

It remains to show that any $((V_{\mathbb{Z}}^{\text{nilp}}, W^{\text{nilp}}, F^{\text{nilp}}), E)$ is the image by the functor of some $(V_{\mathbb{Z}}, W, F)$. The sought after $V_{\mathcal{O}}$, with its weight and Hodge filtration, and connection, is

$$V_{\mathcal{O}} = \text{Gr}^W(V_{\mathbb{Z}}^{\text{nilp}}) \otimes \mathcal{O}.$$

Its canonical extension is $\text{Gr}^W(V_{\mathbb{Z}}^{\text{nilp}}) \otimes \mathcal{O}$ on D^n , and the constantification (2.2) gives

$$\phi : \text{Gr}^W(V_{\mathbb{Z}}^{\text{nilp}}) \otimes \mathcal{O} \rightarrow V_{\mathcal{O}}$$

with $\text{Gr}^W(V_{\mathbb{Z}}^{\text{nilp}})$ mapping to constant sections. The map ϕ is horizontal if, at the source, one uses

$$\nabla = d - \frac{1}{2\pi i} d \log E_{\text{nilp}}.$$

The vector bundle with connection $\text{Gr}^W(V_{\mathbb{Z}}^{\text{nilp}}) \otimes \mathcal{O}, \nabla_0$ is the one underlying $(V_{\mathbb{Q}}^{\text{nilp}}, W^{\text{nilp}}, F^{\text{nilp}})$.

The sought after $(V_{\mathbb{Z}}, W, F)$ is given by

$$V_{\mathbb{Z}} = \phi(V_{\mathbb{Z}}^{\text{nilp}}) \subset V_{\mathcal{O}}.$$

Applying the functor, it gives back $(V_{\mathbb{Q}}^{\text{nilp}}, W^{\text{nilp}}, F^{\text{nilp}})$. To check that it gives back E , it suffices to check it on the quotients W_{2k}/W_{2k-4} . This case in turn reduces to that of extensions of $\mathbb{Z}$ by $\mathbb{Z}(1)$, which is left to the reader. $\square$

10 12

We now apply Theorem 11 in the situation of §§6. To a polarized variation V on D^{*n} , with (V, W, F^{nilp}) of Hodge-Tate type, we attach the corresponding polarized nilpotent orbit (V, F^{nilp}) and $E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathcal{O}_{\text{mer}}^*(D^n)$.

The fact that ψ is a morphism $V \otimes V \rightarrow \mathbb{Z}(-w)$, and hence induces a morphism $(V_{\mathbb{Z}}, W, F) \otimes (V_{\mathbb{Z}}, W, F) \rightarrow \mathbb{Z}(-w)$, translates into

$$\psi(Ex, y) + \psi(x, Ey) = 0 \quad \text{in} \quad \mathcal{O}_{\text{mer}}^*(D^n) \quad (12.1)$$

for $x \in \text{Gr}_k^W(V_{\mathbb{Z}})$, $y \in \text{Gr}_{k'}^W(V_{\mathbb{Z}})$ and $k + k' = 2$.

Conversely, given

- (a) a polarized nilpotent orbit (V, F^{nilp}) with (V, W, F^{nilp}) of Hodge-Tate type,
- (b) $E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathcal{O}_{\text{mer}}^*(D^n)$ with $d \log E \wedge d \log E = 0$, E compatible with E_0 attached to (V, W, F^{nilp}) : $E - E_0$ holomorphically trivial at 0, and E obeying (12.1),

we obtain by 11 a mixed variation (V, W, F) . By (12.1), $\psi : (V_{\mathbb{Z}}, W, F) \otimes (V_{\mathbb{Z}}, W, F) \rightarrow \mathbb{Z}(-w)$ is a morphism. If we forget W , $(V_{\mathbb{Z}}, F)$ admits (V, F^{nilp}) as nilpotent orbit. By 2.3, $(V_{\mathbb{Z}}, F)$ is a polarized variation near 0.

11 13. Example

Let E be a family of elliptic curves parametrized by D^* , and consider the variation $H_1(E_s)$ ($s \in D^*$). Assume the monodromy is unipotent and nontrivial. In that case, one can uniquely write

$$E_s = \mathbb{C}^*/q^{\mathbb{Z}}.$$

with $q(s)$ holomorphic invertible on D^* and tending to 0 with s . We are in the situation of (4.1) and the mixed variation $(H_1(E_s), W, F)$ is the extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ with invariant $q(s)$.

In this case, the mixed Hodge structures $(H_1(E_s), W, F)$ are subquotients of the homology of algebraic varieties: they are “motivic”. Indeed, the extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ with invariant q ($q \neq 1$) is the homology (H_1) of $\mathbb{G}_m(\mathbb{C}) - \{1, q\}$ with the points 0 and ∞ identified. To check this, apply (P. Deligne, Théorie de Hodge III, Publ. Math. IHES 44 (1974), pp. 5–78, construction 10.3.8).

More generally, for a family of polarized abelian varieties A on D^{*n} , and their H_1 , 4.1 applies and the resulting mixed Hodge structures are motivic.

I do not expect this to hold in general. Let V be a direct factor of $H^w(X_s)$, for X a family of projective nonsingular varieties over D^{*n} . The nilpotent orbit (V, W, F^{nilp}) should be motivic but, even if (V, W, F^{nilp}) is Hodge-Tate, the $(V_{\mathbb{Z}}, W, F)$ should not always be. If they were, it would conflict with another conjecture: for an iterated extension E of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)^n$ by $\mathbb{Z}(2)$, defining extension classes

$$q_{1i} \in \text{Ext}^1(\mathbb{Z}(0), \mathbb{Z}(1)^n) = \mathbb{C}^{*n}$$

$$q_{2j} \in \text{Ext}^1(\mathbb{Z}(1)^n, \mathbb{Z}(2)) = \mathbb{C}^{*n},$$

if E is motivic, then $\prod_{i,j} \{q_{1i}, q_{2j}\} \in K_2(\mathbb{Q})$ is trivial. As a consequence,

$$q_{1i} \frac{d}{dq_{1i}} \wedge q_{2j} \frac{d}{dq_{2j}} = 0$$

in $\wedge^2 \mathbb{Q}^*$ and, in the case $n = 1$, q_1 and q_2 are algebraically dependent over $\mathbb{Q}$.

Let us consider the family of quintic threefolds

$$\sum_{i=1}^5 X_i^5 - \psi \prod_{i=1}^5 X_i = 0.$$

Let G be the multiplicative group of 5-uples of fifth roots of 1 with $\prod \zeta_i = 1$. It acts on each quintic in the family: $(X_i) \mapsto (\zeta_i X_i)$, with the quotient Γ of G by the diagonal (all ζ_i equal) acting faithfully. Let V be the part of H^3 fixed by G . By the computations of [1], V is of rank 4 and the monodromy group is Zariski dense in Sp_4 : it contains a unipotent transformation u with $u - 1$ of rank one (monodromy around $\lambda = \zeta$, for $\zeta^5 = 1$), and one with $u - 1$ of rank 3, i.e. one Jordan block (monodromy around $\psi = \infty$). As the Zariski closure of the monodromy is known to be semi-simple, and contained in Sp_4 , this leaves no choice.

Near $\psi = \infty$, the nilpotent orbit is of Hodge-Tate type, so that E gives rise to an iterated extension of $\mathbb{Z}(-3)$ by $\mathbb{Z}(-2)$ by $\mathbb{Z}(-1)$ by $\mathbb{Z}(0)$. Let q_1 (resp. q_2) be the class of the resulting extension of $\mathbb{Z}(-3)$ by $\mathbb{Z}(-2)$ (resp. of $\mathbb{Z}(-2)$ by $\mathbb{Z}(-1)$). If, for every value of λ near ∞ , q_1 and q_2 are algebraically dependent over $\mathbb{Q}$, a Baire category argument and analytic continuation show that $F(q_1, q_2) = 0$ identically for some polynomial F .

For all but finitely many points (q_1^0, q_2^0) on the algebraic curve $F(q_1, q_2) = 0$, in a neighborhood of (q_1^0, q_2^0) , the equation $F(q_1, q_2) = 0$ amounts to $q_2/q_1^0 = \varphi(q_1/q_1^0)$ for some function φ depending on (q_1^0, q_2^0) . We write $\varphi_\infty[q_1^0, q_2^0]$ for that function. It maps 1 to 1, and its ∞ -jet at 1, $\varphi_{\infty\infty}[q_1^0, q_2^0]$, depends algebraically on (q_1^0, q_2^0) , i.e. for each k , the k -jet $\varphi_{\infty k}[q_1^0, q_2^0]$ does so.

The ∞ -jet $\varphi_{\infty\infty}[q_1^0, q_2^0]$ carries the following information. Fix λ_0 near ∞ , giving rise to (q_1^0, q_2^0) . Fix on V a new integral basis e_i ($0 \leq i \leq 3$), with e_i in $F^{-i} \cap W_{-2i}$ and projecting in $\mathrm{Gr}_{-2i}^W(V) \cong \mathbb{Z}(-i)$ to the integral generator. In this new basis $V_{\mathbb{Z}}$ is the split mixed Hodge structure $\bigoplus_0^3 \mathbb{Z}(-i)$. The ∞ -jet $\varphi_{\infty\infty}[q_1^0, q_2^0]$ tells how, in this basis, the Hodge filtration changes with λ , up to a reparametrization $\lambda \mapsto g(\lambda)$.

Let $\mathbb{P}(V)$ be the projective space of rays in V . The symplectic form ψ gives on $\mathbb{P}(V)$ a “null-system” of lines, corresponding to the Lagrangian subspaces of V . The variable Hodge filtration F gives a curve Γ in $\mathbb{P}(V)$: the trajectory of F^3 . At a general point of Γ , the tangent line is a null line, corresponding to F^2 . The weight filtration gives an isotropic flag. The previous discussion gives: for each $\gamma \in \Gamma$, let us consider the relative position of the ∞ order jet R_γ of Γ at γ , and of ψ . Here, “relative position” is an element of the quotient by $\mathrm{Sp}(4)$ of the space of pairs (∞ -order jet of curve, isotropic flag). For variable γ , this relative position is controlled by a $\varphi_{\infty\infty}[q_1^0, q_2^0]$, hence stays on a 1-dimensional algebraic subspace. If we analytically continue and come back to γ , we see that (R_γ, ψ') is again in this subspace, for a Zariski dense set of ψ' , hence for all ψ' . As the space of all isotropic flags is 4-dimensional, there is a 3-dimensional space of ψ' such that all (R_γ, ψ') are in the same relative position.

The ∞ -jet R_γ is hence stable by a subgroup H of $\mathrm{Sp}(4)$, fixing γ , and of dimension ≥ 3 . The curve Γ spans $\mathbb{P}(V)$, hence cannot be pointwise fixed. It must be an orbit of H , hence is an algebraic curve. Contradiction: no algebraic curve has a group of automorphisms of dimension ≥ 3 , all fixing a point γ .

12 14. Example

The following occurs in the study of mirror symmetry.

Let V be a polarized variation of Hodge structure on D^{*n} , of the following type:

(14.1) weight -3 (like an H^3), Hodge numbers $h^{p,q}$ ($p = 0, -1, -2, -3$): $1, n, n, 1$.

(14.2) polarization form ψ of discriminant 1.

(14.3) unipotent monodromy, and associated nilpotent orbit of Hodge-Tate type.

Rank of the $\text{Gr}_i^W(V_{\mathbb{Z}})$ ($i = 0, -2, -4, -6$): $1, n, n, 1$. The form ψ is a perfect duality between Gr_0^W and Gr_{-6}^W .

(14.4) For 1 a basis of $\text{Gr}_0^W(V_{\mathbb{Z}})$, the $N_i(1)$ form a basis of $\text{Gr}_{-2}^W(V_{\mathbb{Z}})$.

Fix a basis $1 \in \text{Gr}_0^W(V_{\mathbb{Z}})$. By definition of the weight filtration and (14.4), $\text{Gr}^W(V_{\mathbb{Q}})$, as a module over the polynomial algebra $\mathbb{Q}[N_1, \dots, N_n]$, is generated by 1. As a quotient of $\mathbb{Q}[N_1, \dots, N_n]$, it inherits an algebra structure, with unit 1 and with $N_i =$ multiplication by $N_i(1)$. The integral graded module $\text{Gr}^W(V_{\mathbb{Z}})$ is a subalgebra: as $\text{Gr}_i^W(V_{\mathbb{Z}}) = 0$ for $i \neq 0, -2, -4, -6$, it suffices to check that it is stable by multiplication by $v \in \text{Gr}_{-2}^W(V_{\mathbb{Z}})$, $i = 0$ or -2 which is clear: $\text{Gr}_0^W(V_{\mathbb{Z}}) = \mathbb{Z} \cdot 1$ and, if L is the free module spanned by $N_1, \dots, N_n$,

$$L \xrightarrow{\sim} \text{Gr}_{-2}^W(V_{\mathbb{Z}}) : \ell \mapsto \ell \cdot 1.$$

The form induced by ψ can be recovered from the linear form t on $\text{Gr}_{-6}^W(V_{\mathbb{Z}})$:

$t(v) := \psi(1, v)$. Indeed, for $\ell \in \text{Gr}_{-2}^W(V_{\mathbb{Z}}), \lambda \in \text{Gr}_{-4}^W(V_{\mathbb{Z}})$, one has

$$\psi(\ell \cdot 1, \lambda) = -\psi(1, \ell \cdot \lambda) = t(\ell \cdot \lambda).$$

The algebra structure, and t , can be recovered from the symmetric trilinear form

$$\Phi(x, y, z) := t(x \cdot y \cdot z)$$

on $L = \text{Gr}_{-2}^{\mathbb{Q}}(V_{\mathbb{Z}})$: the algebra is $\mathbb{Z} \oplus L \oplus L^{\vee} \oplus \mathbb{Z}$, the first $\mathbb{Z}$ spanned by 1, the second by $1^{\vee}$ with $t(1^{\vee}) = 1$. The product of $\lambda \in L^{\vee}$ with $\lambda' \in L^{\vee}$ is $\lambda(\lambda') \cdot 1^{\vee}$, while that of $x, y \in L$ is the linear form $z \mapsto \Phi(x, y, z)$.

We now consider the extension classes

$$E_0 \in \text{Ext}^1(\text{Gr}_0^W(V_{\mathbb{Z}}), \text{Gr}_{-2}^W(V_{\mathbb{Z}})) = L \otimes \mathcal{O}_{\text{mer}}^*(D^n) = \text{Hom}(L^{\vee}, \mathcal{O}_{\text{mer}}^*(D^n))$$

$$E_1 \in \text{Ext}^1(\text{Gr}_{-2}^W(V_{\mathbb{Z}}), \text{Gr}_{-4}^W(V_{\mathbb{Z}})) = L^{\vee} \otimes L^{\vee} \otimes \mathcal{O}_{\text{mer}}^*(D^n) = \text{Hom}(L \otimes L, \mathcal{O}_{\text{mer}}^*(D^n))$$

$$E_2 \in \text{Ext}^1(\text{Gr}_{-4}^W(V_{\mathbb{Z}}), \text{Gr}_{-6}^W(V_{\mathbb{Z}})) = L \otimes \mathcal{O}_{\text{mer}}^*(D^n) = \text{Hom}(L^{\vee}, \mathcal{O}_{\text{mer}}^*(D^n)).$$

The valuation of the E_k along $s_j = 0$ are given by N_j (10.2). For E_0 , recalling that $L = \mathbb{Z}^n$, we find

$$E_0 : (q^1, \dots, q^n)$$

and the q^i form a system of local coordinates near 0. For $\lambda = (a_1, \dots, a_n) \in L^{\vee} \cong \mathbb{Z}^n$, we write $q^{\lambda} = \prod q_i^{a_i}$. We have then

$$E_0 : \lambda \longmapsto q^{\lambda},$$

$$E_1 : \ell \otimes m \longmapsto q^{\Phi(\ell, m)} \cdot U_{\ell, m}, \quad \text{with } U_{\ell, m} \text{ invertible on } D^n,$$

$$E_2 : \lambda \longmapsto q^{\lambda};$$

compatibility with ψ imposes $E_2(\lambda) = E_1(\lambda)$, $E_1(\ell, m) = E_1(m, \ell)$.

Let us expand the $U_{\ell, m}$ as an infinite product, extended over the $\lambda \in L^{\vee}$, $\lambda = (a_1, \dots, a_n)$, with $a_i \geq 0$, $\sum a_i \neq 0$:

$$U_{\ell, m} = C_{\ell, m} \prod_{\lambda} (1 - q^{\lambda})^{c(\ell, m, \lambda)}$$

with $C_{\ell, m}$ a constant, bi-multiplicative in ℓ, m and $c(\ell, m, \lambda)$ complex, bi-additive in ℓ, m . The integrability condition 10.3 reduces to

$$c(\ell_1, m, \ell_2, \lambda) / c(\ell_2, m, \ell_1, \lambda) = c(\ell_1, \ell_2, m, \lambda) / c(\ell_2, \ell_1, m, \lambda).$$

Thus we obtain an expansion

$$E_1(\ell, m) = C_{\ell, m} \cdot q^{\Phi(\ell, m)} \prod_{\lambda} (1 - q^{\lambda})^{\lambda(\ell)\lambda(m)\zeta(\lambda)}$$

In the mirror story, $\mathbb{Z}$, L , $L^\vee$, $\mathbb{Z}$ become the H^0 , H^2 , H^4 , H^6 of a Calabi–Yau threefold, the basis of L some basis of H^2 contained in the ample cone, and λ runs over the cohomology class of rational curves (in H_4), with $\zeta(\lambda)$, the number of rational curves in a given class, suitably defined.

For the components of E_1 in the given basis of $L = \mathbb{Z}^n$, and the dual basis of $L^\vee$, this gives

$$\partial_i \partial_j \partial_k \log E_1(e_i, e_j) = \Phi(e_i, e_j, e_k) + \sum_{\lambda} \zeta(\lambda) \lambda(e_i) \lambda(e_j) \lambda(e_k) \frac{q^\lambda}{1 - q^\lambda}$$

having as logarithmic derivative

$$\Phi(e_i, e_j, e_k) + \sum_{\lambda \neq 0} \zeta(\lambda) \lambda(e_i) \lambda(e_j) \lambda(e_k) q^\lambda.$$

13 15. Integrality

For the degenerating elliptic curve 13 over D^* (coordinate z), the formal power series $\sum a_n z^n$, Taylor expansion of q as a function of z , has a purely algebraic sense: if the coefficients c_α , of an equation for E are expanded as formal power series in z : $c_\alpha = \sum c_{\alpha n} z^n$, then the a_n are polynomials in the $c_{\alpha n}$.

Further, the story makes sense in any characteristic, and is compatible with reduction modulo p . I expect similar integrality to hold in general.

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Some Main Ideas in the Work of A. Grothendieck

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Abstract. This article attempts to explain four fundamental mathematical concepts created by Grothendieck: schemes, topoi, the six operations, and motives.

In *Recoltes et semailles* (part three), Grothendieck writes:

“Take for example the task of proving a theorem that remains hypothetical, to which, for some people, mathematical work would seem to reduce. I see two extreme approaches to going about it. One is that of the hammer and chisel, when the problem posed is seen as a large nut, hard and smooth, whose interior one must reach, the nourishing flesh protected by the shell. The principle is simple: one puts the edge of the chisel against the shell and strikes hard. If necessary, one repeats this at several different places, until the shell breaks—and one is pleased. [...]

“I could illustrate the second approach, still keeping the image of the nut that must be opened. The first parable that came to my mind just now is that one plunges the nut into a softening liquid, simply water, why not; from time to time one rubs it so that it penetrates better, and for the rest one lets time do its work. The shell softens over weeks and months—when the time is ripe, pressure from the hand suffices, and the shell opens like that of an avocado ripened to the point. Or else one leaves the nut to ripen under the sun and the rain, and perhaps also under the frosts of winter. When the time is ripe, it is a delicate shoot emerging from the substantial flesh that will have pierced the shell, as if in play—or better said, the shell will have opened by itself, to let it pass. [...]

“The reader who is at all familiar with some of my work will have no difficulty recognizing which of these two modes of approach is ‘mine.’” [1985, p. 552–553]

A little further on [ibid., p. 554, note (***)], Grothendieck puts forward four examples: Riemann–Roch, the prime-to-the-characteristic structure of π_1 for curves, the rationality of L -functions for schemes of finite type over a finite field, and the semistable reduction theorem for abelian varieties.

I remember my astonishment in 1965–66 after Grothendieck’s lecture [SGA 5] proving the base-change theorem for $Rf_!$: devissages, more devissages, nothing seems to happen, and yet at the end of the lecture a clearly nontrivial theorem is there.

Many of Grothendieck’s ideas have become so familiar to us, and are so perfectly suited to their object, that we forget that they were far from obvious at birth; we even forget their author. My purpose in this article is to describe four of these ideas: schemes, topoi, the six operations, and motives.

1 Schemes

The invention of schemes is the first of Grothendieck’s ideas that one thinks of, perhaps because it was the one accepted most quickly. Serre’s lecture in Stockholm in 1962 begins: “I would like

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to present here some of the recent developments in algebraic geometry. I should specify that I take this last term in the sense it has acquired over the last few years: that of the theory of schemes.” This acceptance was facilitated by the rapid publication, thanks to Dieudonné’s collaboration, of EGA.

The boldness of Grothendieck’s definition is to accept that every commutative ring with unit A defines an affine scheme $\text{Spec}(A)$; that is, not to try to restrict oneself to a category of “good” rings: integral, reduced, noetherian, and so on. This has a price. The points of $\text{Spec}(A)$, namely the prime ideals of A , do not have a manageable geometric meaning, and the structure sheaf $\mathcal{O}$ is not a sheaf of functions. When one has to construct a scheme, one generally does not begin by constructing its set of points.

Perhaps more important is the decision to build a relative theory, witnessed by the omnipresent diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

in Grothendieck’s lectures. The classical case of a variety defined over a field k becomes the special case $S = \text{Spec}(k)$. In a relative theory, with an S -scheme X , i.e. a scheme X over S , or equivalently with a morphism of schemes $f : X \rightarrow S$, one systematically considers the scheme X' deduced from X by a base change $u : S' \rightarrow S$, that is, the fiber product $X' := S' \times_S X$ and its projection f' to S' :

$$\begin{array}{ccc} X' & \longrightarrow & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{u} & S \end{array}$$

In the category of schemes, fiber products always exist: if allowing every commutative ring to define an affine scheme gives bizarre schemes a right to exist, it also provides a category of schemes with good properties.

A property of X over S will be called *geometric* if it has good invariance properties under base change. Classical analogue: for a variety X defined over a field k , the set of points of X over an algebraically closed extension $\bar{k}$ of k (for example Weil’s universal domain) is considered “geometric,” while the set of k -points is “arithmetic.”

If X is a scheme over S , and $u : S' \rightarrow S$ is a morphism of schemes, an S' -point p of X is a morphism of S -schemes:

$$\begin{array}{ccc} & & X \\ & \nearrow p & \downarrow \\ S' & \longrightarrow & S \end{array}$$

We write $X(S')$ for the set of S' -points of X . It identifies with the set of sections of $X' \rightarrow S'$.

Example 1. Let X be the affine scheme over a field k defined by equations $P_\alpha(X_1, \dots, X_n) = 0$:

$$X = \text{Spec}(k[X_1, \dots, X_n]/(P_\alpha)).$$

Let k' be a k -algebra. The set $X(k') := X(\text{Spec}(k'))$ is the set of solutions in k'^n of the equations $P_\alpha = 0$. For k' an algebraically closed extension $\bar{k}$ of k , this is the set $X(\bar{k})$ that Weil views as underlying X .

Example 2. The scheme GL_N over $\text{Spec}(\mathbb{Z})$ is such that, for every commutative ring A —automatically a $\mathbb{Z}$ -algebra, so $\text{Spec}(A)$ is a scheme over $\text{Spec}(\mathbb{Z})$ — $\text{GL}_N(\text{Spec}(A))$ “is” $\text{GL}_N(A)$.

In these examples, the geometric intuition one has of X , a scheme over S , is well reflected in $X(S')$ —better than in the underlying set of X . In the case of Example 1.2, this underlying

set is not a group, whereas every $\mathrm{GL}_N(S')$ is one. More precisely, it is useful to attach to the scheme X over S the contravariant functor from S -schemes to sets

$$S' \longmapsto \text{the set } X(S') \text{ of } S'\text{-points of } X.$$

In the category of S -schemes, this is simply the representable functor

$$h_X : S' \longmapsto \mathrm{Hom}(S', X)$$

attached to X . By Yoneda's lemma, the functor $X \mapsto h_X$ is fully faithful. Rather than thinking of an S -scheme X as a ringed space, endowed with $X \rightarrow S$ and satisfying suitable properties, it is often convenient to think of it as a functor

$$S\text{-points} : (\mathrm{Schemes}/S)^0 \longrightarrow (\mathrm{Sets}),$$

with the virtue of being representable. When one wants to define a “moduli space,” the first step is to define the corresponding functor. Typically, defining this functor requires a relative theory.

Example 3. Let X be projective over a field k . Question: what does “the moduli space of closed subschemes of X ” mean? For $u : S' \rightarrow \mathrm{Spec}(k)$, let X' over S' be deduced from X by base change. Let $\mathcal{H}(S')$ be the set of closed subschemes Z' of X' , flat over S' —flat and of finite presentation if one does not want to assume S' noetherian. Answer: it is a scheme $\mathrm{Hilb}(X)$ representing the functor $S' \mapsto \mathcal{H}(S')$.

To be viable, this point of view requires methods for verifying when a functor is representable. Most of the lectures of FGA are devoted to this problem. A definitive solution, extending those works, was obtained by M. Artin [1969], at least if one agrees to replace the category of schemes by the more natural category of algebraic spaces. More natural: in the same sense in which the étale topology is more natural than the Zariski topology.

2 Topoi

Let $u : S' \rightarrow S$ be a morphism of schemes, the “base-change morphism.” Descent theory [Grothendieck FGA, Séminaire Bourbaki 190, 1959–60] considers problems of the following types. *Descent of properties*: let X be a scheme over S , and let X' over S' be deduced from X by base change. Suppose that X'/S' has a property P . Can one conclude that X/S has P ? *Descent of morphisms*: let X, Y be over S , let X', Y' be deduced by base change, and let $g' : X' \rightarrow Y'$ be a morphism of S' -schemes. When does g' come by base change from a morphism $g : X \rightarrow Y$? *Descent of objects*: let X' be over S' . What descent datum on S' is required to construct X over S from which X' is obtained by base change?

If S' is the disjoint union of the opens $(U_i)_{i \in I}$ of a covering of S , base change to S' is essentially restriction to each U_i , and the preceding problems are problems of localization on S and of gluing. Gluing typically requires considering the pairwise intersections $U_i \cap U_j$, and in terms of $u : S' \rightarrow S$, the disjoint union of the $U_i \cap U_j$ is simply the fiber product $S' \times_S S'$.

The theory of topoi makes it possible to transpose topological intuition into descent theory. For $u : S' \rightarrow S$, a base-change morphism of a kind considered in descent theory, for instance faithfully flat and quasi-compact (fqc), base change from S to S' becomes a localization. A descent datum is the analogue of a gluing datum.

An antecedent of descent theory is Galois descent, corresponding to S the spectrum of a field k and S' the spectrum of a Galois extension. Here $S' \times_S S'$ is a sum of copies of S' indexed by $\mathrm{Gal}(k'/k)$. The proofs, and in particular Galois theory, are nevertheless simpler in the more general framework of descent theory. In Cartier's phrase: Grothendieck proves Galois theory, and Galois descent, by Galois descent.

The tool provided by the theory of topoi made possible the construction of étale cohomology of schemes, and this is its best-known success. A sheaf on X for the Zariski topology is a contravariant functor from the category of Zariski opens of X to sets, satisfying a gluing condition for $(U_i)_{i \in I}$, a Zariski open cover of U . In the étale topology, the Zariski site considered above is replaced by the étale site: the category of Zariski opens is replaced by the category of maps $f : U \rightarrow X$ étale over X , and coverings are replaced by covering families $(U_i)_{i \in I}$, i.e. a surjective morphism of schemes over X from the disjoint union of the U_i to U . An antecedent was Serre's introduction of the notion of an isotrivial principal homogeneous space, namely one locally trivial for a topology close to the étale topology. In his Kansas [1955] and Tohoku [1957] articles, Grothendieck had shown that, once a category of sheaves is given, a notion of cohomology groups follows from it. Thus the étale topology provides étale cohomology. That the resulting H^1 groups are reasonable is not surprising. The miracle is that the higher H^i are reasonable as well.

For Grothendieck, the importance of topos theory goes far beyond the sole case of the étale topology. The title given to SGA 4 bears witness to this: "Theory of topoi and étale cohomology of schemes."

Other applications:

(A) *Crystalline cohomology.* Crystalline cohomology is the cohomology of the crystalline topos, and this definition makes its functoriality clear. The crystalline topos is nevertheless delicate to use, and it is often necessary to pass to an interpretation in terms of de Rham complexes.

(B) *Rigid analytic spaces.* Tate's rigid analytic sheaves are coherent sheaves on a suitable ringed topos, and their cohomology is the corresponding cohomology.

(C) *Foliations.* A variety X endowed with a foliation $\mathcal{F}$ defines a "quotient" $X/\mathcal{F}$, which is a topos locally isomorphic to the topos of sheaves on a variety. This viewpoint seems to have been eclipsed by Connes's, which instead associates to $\mathcal{F}$ a noncommutative C^* -algebra.

I would also like to mention Grothendieck's use of "large" sites, or topoi, especially to interpret classifying spaces.

3 The six operations

The formalism of the six operations presupposes the formalism of derived categories. For the history of the latter, I refer to Illusie's text [1990]. The basic idea is this: for all sorts of cohomology groups, their definition supplies not only these groups but also a complex K whose cohomology groups they are. Typically, this complex is not uniquely determined, but it is determined up to quasi-isomorphism: for two variants K' and K'' of K , there is a K''' and morphisms $K' \leftarrow K''' \rightarrow K''$ inducing isomorphisms on cohomology. In the derived category, K becomes unique up to unique isomorphism.

The six operations are functors between derived categories. They are the derived tensor product $\overset{L}{\otimes}$, the internal Hom functor RHom , which gives rise to the local $\mathcal{E}xt^i$, and, for a morphism of schemes $f : X \rightarrow Y$, two direct-image functors Rf_* and $Rf_!$, and two inverse-image functors Lf^* and $Rf^!$.

The formalism of these operations was first isolated in the cohomology of coherent sheaves [Grothendieck 1963]. Here, for direct images, one should consider only proper morphisms, for which $Rf_* = Rf_!$. The formalism gives a relative version of Serre duality, in the form of an adjunction between the functors $Rf_!$ and $Rf^!$. It also gives a fixed-point formula more general than the contemporaneous "Woods Hole fixed point formula" (Summer Institute on Algebraic Geometry, Whitney estate, Woods Hole, Massachusetts, 1964).

Miraculously, the same formalism applies in étale cohomology, with very different proofs. It again provides a duality formalism, of Poincaré type, and Lefschetz-type fixed-point formulae.

4 Motives

Let X be an algebraic variety over an algebraically closed field k . For each prime number ℓ prime to the characteristic, the étale topology provides ℓ -adic cohomology groups $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$. If k is a subfield of $\mathbb{C}$, one has comparison isomorphisms

$$H^i(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} H_{\text{ét}}^i(X, \mathbb{Z}_\ell).$$

For k of characteristic > 0 , there is no functorial integral cohomology giving rise to such isomorphisms. Nevertheless, the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$, as ℓ varies, have a “family resemblance.” For $i = 1$, and X projective and smooth, $\text{Pic}^0(X)$ plays the role of the nonexistent integral theory: it recovers the $H_{\text{ét}}^1(X, \mathbb{Z}_\ell)$ and is an object “over $\mathbb{Z}$,” in the sense that the groups of homomorphisms between abelian schemes are finitely generated.

The theory of motives is first of all an attempt to find a substitute for the nonexistent integral cohomology, explaining the family resemblance among the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$, especially for X projective and smooth. One recognizes the Master’s touch in the idea that the problem is not to define what a motive is: the problem is to define the category of motives and to isolate the structures it carries. These structures should allow one to prove the Weil conjecture on the model of Serre [1960]. See Grothendieck [1969].

It was on this occasion that Grothendieck invented the notion of a Tannakian category, developed in Saavedra’s thesis [1972]. The point was to formalize the tensor product of motives, corresponding by the Künneth formula to the product of varieties.

Even though the theory of motives has not achieved its original goal, its influence has been great. I refer to the Seattle conference on motives [Jannsen et al., 1994] for a panorama of its applications.

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Central Extensions of Reductive Groups by K_2

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1 Introduction

If a group G is equal to its commutator subgroup (G, G) , it has a universal central extension

$$E \rightarrow G \rightarrow 1. \quad (1)$$

(C. Moore (1968) no. 1). Suppose that G is the group $G(k)$, for G a split simple simply-connected algebraic group over a field k , and let us forget about some small groups over small finite fields. Then R. Steinberg (1962) describes the central extension E by generators and relations, while

H. Matsumoto (1969) describes the kernel by generators and relations. Except for G of type C_n ($n > 1$), the kernel is $k^* \otimes k^* / \langle u \otimes v \mid u + v = 1 \rangle$, the group we now call $K_2(k)$, following J. Milnor (1971). For G of any type, including C_n , the following remains true: the adjoint group $G_{\text{ad}}(k)$ acts on $G(k)$, hence on the corresponding central extension (1), and the group of coinvariants is $K_2(k)_{G_{\text{ad}}(k)}$. The universal central extension hence gives rise to a central extension

$$K_2(k) \rightarrow \tilde{G}(k) \rightarrow G(k). \quad (2)$$

The central extension (2) is canonically an extension by $K_2(k)$: the action on $G(k)$ of the group of automorphisms of G (and in particular of $G_{\text{ad}}(k)$) lifts to an action on $\tilde{G}(k)$, trivial on $K_2(k)$. We will call (2) Matsumoto's central extension. Notation: the universal symbol $\{, \} : k^* \times k^* \rightarrow K_2(k)$ maps u, v to the image of $u \otimes v$ in $K_2(k)$.

Suppose now that k is a global field, i.e. a number field or the field of rational functions of a curve over a finite field. Let m be the group of roots of unity in k . Matsumoto's central extensions, for k and its completions k_v , then give rise to a topological central extension of the adelic group $G(\mathbb{A})$ by m , and to a splitting of this central extension over $G(k)$:

$$\begin{array}{ccc} & G(k) & \\ & \downarrow & \nwarrow \\ m & \longrightarrow \tilde{G}(\mathbb{A}) & \longrightarrow G(\mathbb{A}) \end{array} \quad (3)$$

See C. Moore (1968).

In this paper, we consider reductive groups G over k , where reductive is meant to imply connected. We classify central extensions of $G(k)$ by $K_2(k)$ which, roughly speaking, are functorial in k . More precisely, we define objects called central extensions of G by K_2 , which give rise to a central extension of $G(k)$ by $K_2(k)$ and, for k a global field, to a diagram (3). We classify the central extensions of G by K_2 , see below, and we determine their functoriality in G and k . We hope that for k a global field this will prove useful in the study of “metaplectic” automorphic forms, that is, the harmonic analysis of functions on $G(\mathbb{A})/G(k)$.

Notation. — For X a scheme, the $K_n(X)$ are Quillen's higher K -theory groups of the exact category of vector bundles on X . For U open in X , $U \mapsto K_n(U)$ is a presheaf on X . We denote by $\mathcal{K}_n$ the associated sheaf.

An algebraic group G over a field k defines a sheaf $S \mapsto G(S) = \text{Hom}_{\text{Spec}(k)}(S, G)$ on the big Zariski site $\text{Spec}(k)_{\text{Zar}}$ (see 1.4) of $\text{Spec}(k)$. A central extension of G by K_2 is a central extension, on $\text{Spec}(k)_{\text{Zar}}$, of the sheaf of groups G by the sheaf of abelian groups $\mathcal{K}_2$. By SGA7 (tI, exp. VII, no. 1), it can be viewed as a K_2 -torsor P on G , provided with a multiplicative structure: an isomorphism $m : \text{pr}_1^* P + \text{pr}_2^* P \rightarrow m^* P$ of K_2 -torsors on $G \times G$, for $m : G \times G \rightarrow G$ the group law, obeying an associativity condition on $G \times G \times G$.

The following alternate description, to be explained in §8, played for us a useful heuristic role. It is valid when k is infinite and G , as an algebraic variety over k , is connected and unirational. The face maps of the simplicial classifying space BG , that is, the simplicial scheme G^{Δ^n}/G , induce homomorphisms of fields of rational functions

$$k \rightrightarrows k(G) \rightrightarrows k(G \times G) \rightrightarrows k(G \times G \times G) \cdots$$

Applying the functor K_2 and taking the usual alternating sum of face maps, one obtains

$$K_2 k \rightarrow K_2 k(G) \rightarrow K_2 k(G \times G) \rightarrow K_2 k(G \times G \times G) \rightarrow \cdots$$

This complex incarnates in degree 2 the category of central extensions of G by K_2 . This means that: (a) a 2-cocycle $c \in K_2 k(G \times G)$ defines a central extension $E(c)$ of G by K_2 and any isomorphism class of central extensions can be obtained in this way; (b) if c_0 and c_1 are two

2-cocycles, a 1-cochain b such that $c_1 - c_0 = db$ defines an isomorphism $E(b) : E(c_0) \rightarrow E(c_1)$, and this construction is a bijection

$$\{b \mid c_1 - c_0 = db\} \xrightarrow{\sim} \text{Isom}(E(c_0), E(c_1));$$

(c) obvious compatibilities hold. The idea to go from the cocycle $c \in K_2k(G \times G)$ to the corresponding extension of $G(k)$ by $K_2(k)$ is as follows. One can write c as a finite sum $\sum n_i \{f_i, g_i\}$ with f_i, g_i in $k(G \times G)^*$. The f_i and g_i are all regular on some dense open subset U of $G \times G$ and

$$c(x, y) = \sum n_i \{f_i(x, y), g_i(x, y)\} \quad (4)$$

is defined for $(x, y) \in U$. Because c is a cocycle, $c(x, y)$ obeys the cocycle condition $c(y, z) - c(xy, z) + c(x, yz) - c(x, y) = 0$ for (x, y, z) in some dense open subset of $G \times G \times G$.

That such a generic cocycle suffices to define a group extension is similar to Weil's theorem that a birational group law determines an algebraic group (SGA3, t2, XVIII 3.7). We were also inspired by Mackey's theorem that for separable locally compact topological groups, borelian 2-cocycles define topological extensions (G. W. Mackey (1957)).

Example. — For G split simple simply-connected, Matsumoto's central extension carries a set theoretic section for which the cocycle describing the extension is expressed in terms of universal symbols. On some open subset U of $G \times G$, this cocycle is given by a formula (4). As this formula continues to define a cocycle after extension of scalars from k to $k(G \times G \times G)$, the element $\sum n_i \{f_i, g_i\} \in K_2k(G \times G)$ satisfies the cocycle condition in $K_2k(G \times G \times G)$: Matsumoto's central extension of $G(k)$ by $K_2(k)$ comes from an extension of G by K_2 .

We now sketch the classification of the central extensions of a reductive group G by K_2 . If G is simply-connected, the central extensions of G by K_2 have no nontrivial automorphisms. In contrast, it can happen that $G(k)$ is not its own commutator subgroup and that the group of automorphisms $\text{Hom}(G(k), K_2(k))$ of a central extension of $G(k)$ by $K_2(k)$ is not trivial. If G is in addition simple (over k), the group of isomorphism classes of central extensions of G by K_2 is $\mathbb{Z}$. Corresponding to $1 \in \mathbb{Z}$, we have a canonical central extension of G by K_2 . It induces a canonical central extension of $G(k)$ by $K_2(k)$, well-defined up to unique isomorphism. Being canonical, it is acted upon by the group of automorphisms of G , and in particular by $G_{\text{ad}}(k)$. For G split, the central extension of $G(k)$ by $K_2(k)$ obtained is the one constructed by Matsumoto.

Fix a maximal torus T in G , and a Galois extension k' of k over which T splits. Let Y be the dual of the character group of T over k' . It is acted upon by $\text{Gal}(k'/k)$ and by the Weyl group W . Rather than to say that central extensions of G by K_2 are classified by integers, it is better to say that they are classified by the integer-valued quadratic forms Q on Y invariant by W and by $\text{Gal}(k'/k)$. The classification then remains valid for G semi-simple simply-connected, and is functorial in G and k . For G simple, invariant quadratic forms correspond to integers by $Q \mapsto Q(a^\vee)$, for $a^\vee$ the (short) coroot corresponding to a long root a .

Let G be a reductive group over k . The group of automorphisms of any central extension of G by K_2 is $\text{Hom}(G, \mathcal{K}_2)$. The Hom group is in the category of sheaves on the big Zariski site of k . It is the subgroup $H^0(G, \mathcal{K}_2)_{\text{prim}}$ of primitive elements of $H^0(G, \mathcal{K}_2)$. It is also the subgroup of elements b of $K_2k(G)$ for which $\text{pr}_2^*b - m^*b + \text{pr}_1^*b = 0$ in $K_2k(G \times G)$. Example: if G is the quotient of a reductive group G_1 by a central μ_n , Kummer theory attaches to the μ_n -torsor G_1 over G a section f of $\mathcal{O}^*/\mathcal{O}^{*n}$ on G and for z a root of unity of k of order dividing n , $b := \{z, f\}$ belongs to $H^0(G, \mathcal{K}_2)_{\text{prim}}$. Other example: $\text{Hom}(\mathbb{G}_m, \mathcal{K}_2) = k^*$, with $a \in k^*$ giving rise to $x \mapsto \{a, x\}$.

Because of those automorphisms, the classification problem we consider, for G reductive, is not that of determining the set of isomorphism classes of central extensions by K_2 . It is that of determining the category of central extensions by K_2 . This turns out to be an easier problem. It means: to define an equivalence from the category of central extensions of G by K_2 to a more down-to-earth category $\mathcal{C}$.

Suppose first that G is a torus T . Fix a Galois extension k' of k over which T splits and let Y be the dual of the character group of T over k' . The category $\mathcal{C}$ is here the category of pairs (Q, E) , where Q is a $\text{Gal}(k'/k)$ -invariant integer-valued quadratic form on Y and E a $\text{Gal}(k'/k)$ -equivariant central extension of Y by k'^* , for which (5) below holds. Let B be the bilinear form associated to Q : $B(y_1, y_2) := Q(y_1 + y_2) - Q(y_1) - Q(y_2)$. The condition is that the commutator of liftings $\tilde{y}_1, \tilde{y}_2$ to E of $y_1, y_2 \in Y$ should be given by

$$(\tilde{y}_1, \tilde{y}_2) = (-1)^{B(y_1, y_2)} \in k'^*. \quad (5)$$

In $\mathcal{C}$, there can be a morphism from (Q', E') to (Q'', E'') only if $Q' = Q''$. If $Q' = Q''$, morphisms are isomorphisms of equivariant central extensions from E' to E'' .

Let $T^\vee$ be the torus over k , split over k' , whose character group is Y . It follows from the equivalence of the category of central extensions of T by K_2 with $\mathcal{C}$ that

$$T^\vee(k) \xrightarrow{\sim} \text{Hom}(T, \mathcal{K}_2).$$

Suppose that T is a maximal torus in a semi-simple simply-connected group G . A Weyl group and $\text{Gal}(k'/k)$ -invariant integer-valued quadratic form Q on Y defines a central extension E of G by K_2 . Restricting E to T , we obtain a central extension E_T of T by K_2 , classified by a pair (Q, E) . One has $Q = Q_0$. In 11.7 we will describe E in terms of the root spaces decomposition of the Lie algebra of G over k' , relative to the action of T .

The classification for general reductive groups is a mixture of the torus and simply-connected cases. Let G be a reductive group over k , with maximal torus T split over k' . Our main result, a particular case of Theorem 7.2, is:

Theorem 1.1. *The category of central extensions of G by K_2 is naturally equivalent to the category of triples (Q, E, j) as follows: Q is a Weyl and Galois invariant integer-valued quadratic form on Y ; E is a Galois equivariant central extension of Y by k'^* , obeying (5). Let $f : G_{\text{sc}} \rightarrow G$ be the simply connected covering of the derived group of G , $T_{\text{sc}} := f^{-1}(T)$ and $Y_{\text{sc}} \subset Y$ be the dual character group of T_{sc} . The form Q induces Q_{sc} on Y_{sc} , from which we get a Galois-equivariant central extension E_{sc} of Y_{sc} by k'^* , and j is a Galois equivariant morphism from E to E_{sc} making the diagram*

$$\begin{array}{ccccc} k'^* & \longrightarrow & E_{\text{sc}} & \longrightarrow & Y_{\text{sc}} \\ \parallel & & \uparrow j & & \updownarrow \\ k'^* & \longrightarrow & E & \longrightarrow & Y \end{array}$$

commute.

Let H be the group of multiplicative type over k , split over k' , with character group Y/Y_{sc} . It follows from the equivalence of the category of central extensions of G by K_2 with $\mathcal{C}$ that

$$H(k) \xrightarrow{\sim} \text{Hom}(G, \mathcal{K}_2).$$

For X connected and smooth over k , pointed by $x \in X(k)$, K_2 -torsors P over X , given with a trivialization of x^*P , obey Galois descent. This results from the Quillen resolution of K_2 and from B. Kahn (1993) (corollary 1 to theorem 3.1).

Galois descent reduces the classification problem for central extensions by K_2 to the split case. It matters here that the classification problem considered is that of describing a category, not just a set of isomorphism classes. The split case is handled using the Bruhat decomposition and known results on cohomology with coefficients in K_2 for affine spaces or multiplicative group bundles (Quillen (1973), Sherman (1979)).

In 1978, the second named author constructed a canonical central extension of $G(k)$ by $K_2(k)$, for G absolutely simple and simply-connected, under the assumption that G is rational as an algebraic variety over k . This construction was exposed in a seminar at the IHES in

1977–1978. It used the results of Tate on the structure of $K_2k(T)$, and its consequence that $H^1(\text{Gal}(k'/k), K_2k'(T)/K_2k') = 0$. In retrospect, one sees that this was a substitute for Galois descent.

Let n be an integer invertible in k . For $\bar{k}$ a separable closure of k , let us write simply $H^2(k, \mathbb{Z}/n(2))$ for $H^2(\text{Gal}(\bar{k}/k), \mu_n(\bar{k})^{\otimes 2})$. By Tate (1976) one has a map

$$K_2(k)/nK_2(k) \rightarrow H^2(k, \mathbb{Z}/n(2)).$$

A central extension by $K_2(k)$ hence provides one by $H^2(k, \mathbb{Z}/n(2))$. For G simply-connected, Deligne (1996) attaches a central extension of $G(k)$ by $H^2(k, \mathbb{Z}/n(2))$ to each étale cohomology class in $H^4(BG \bmod Be, \mathbb{Z}/n(2))$. We will show that the central extension of $G(k)$ by $H^2(k, \mathbb{Z}/n(2))$ deduced from the canonical central extension by $K_2(k)$ is canonically isomorphic to that of Deligne (1996), applied to a natural generator of $H^4(BG \bmod Be, \mathbb{Z}/n(2))$.

In this introduction, we have emphasized the case of a ground field k . In the text, we consider a more general base S as well. This is useful to handle actions of an algebraic group H on an extension of G by K_2 . Cf. 1.7. For the ease of exposition, we often present the ground field case first, and explain later the changes required to work over S . Our reliance on the Quillen resolution of K_2 usually forces us to assume that S is regular of finite type over a field. Using Bloch (1986), one could also handle the case where S is smooth over the spectrum of a discrete valuation ring.

We now review the successive sections of the paper. In §1, we define central extensions of G by K_2 , we review from SGA7 their description as multiplicative torsors and we explain how to compute at the cocycle level. In §2, we prove the Galois descent theorem for pointed K_2 -torsors. We also consider a relative version, for a base S which is regular of finite type over a field, with Galois descent generalized to descent for étale surjective maps.

In §§3, 4, 5 and 6, we consider split groups. We successively consider tori, semi-simple simply-connected groups, semi-simple groups and reductive groups. Our results partially duplicate those of §§3 and 4 of Esnault et al. (1998).

In §7, we put together §§2 and 6 to classify the central extensions of not necessarily split reductive groups by K_2 . In §8, we explain the formalism of generic cocycles. In §9, we compare the central extensions constructed with those of Deligne (1996). In §10, we explain how, over global fields, to construct diagram (3). The central extensions of Y by k^* alluded to above are made explicit in §11. A number of examples are collected in §12.

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1.1 Notation and terminology

1.1.1

We will systematically use the following notation:

- T : an algebraic torus over a base scheme;
- X : the character group $\text{Hom}(T, \mathbb{G}_m)$ of T ;
- Y : the dual $\text{Hom}(\mathbb{G}_m, T)$ of X .

If T is a split torus over a field k : T isomorphic to $\mathbb{G}_m^n$ for some n , one can think of X as simply being the free abelian group $\text{Hom}(T, \mathbb{G}_m)$. For T a split torus group scheme over a base S , X is a locally constant sheaf of free abelian groups on S . For T not necessarily split, one should use the étale topology: for T over S , X is a locally constant sheaf of free abelian groups on the étale site of S . For S the spectrum of a field k , and k' a Galois extension of k on which

T splits, one also writes X for the free abelian group $\text{Hom}(T_{k'}, \mathbb{G}_m)$ with the natural action of $\text{Gal}(k'/k)$, which carries the same information.

If a torus is denoted by T , with some decoration, for instance the subscript sc , its character and dual character groups will be X and Y , with the same decoration.

1.1.2

When considering a reductive group G , we will reserve the notation T for a maximal torus of G , and we denote by

$N(T)$: the normalizer of T ;

W : the Weyl group $N(T)/T$.

Over a field, and for T split, one can think of W as simply being a finite group. Over S , or for T not necessarily split, it is a locally constant sheaf of groups on the étale site of S .

1.1.3

When considering a quadratic form Q , we will write B for the associated bilinear form

$$B(x, y) := Q(x + y) - Q(x) - Q(y).$$

The construction $Q \mapsto B$ is bijective, from integer-valued quadratic forms on a free abelian group Y to even symmetric bilinear forms on Y .

1.1.4

Let

$$A \rightarrow E \rightarrow G$$

be a central extension of a group G by an abelian group A . For g in G , we will denote by $\text{int}[g]$, or int_g , the inner automorphism of E defined by any lifting $\tilde{g}$ of g to E :

$$\text{int}[g] : x \mapsto \tilde{g}x\tilde{g}^{-1}. \quad (6)$$

For g and d in G , we will denote by $(g, d)^\sim$ the commutator in E of liftings $\tilde{g}$ of g and of $\tilde{d}$ of d :

$$(g, d)^\sim := \tilde{g}\tilde{d}\tilde{g}^{-1}\tilde{d}^{-1} = \text{int}[\tilde{d}](\tilde{g}) \cdot \tilde{g}^{-1}. \quad (7)$$

Both (6) and (7) are independent of the choices of liftings.

If a group D acts on the central extension E , inducing the trivial action on A , for g in G and d in D , we will write

$$(d, g)^\sim := d(\tilde{g}) \cdot \tilde{g}^{-1} \quad (8)$$

for any lifting of g . For the action (6) of G on E , we recover (7).

The same constructions make sense for a sheaf of central extensions, if liftings are replaced by local liftings.

1.1.5

The group law in K_n will be written additively, and the product map will be denoted by a dot. For f in the multiplicative group, $\{f\}$ is its image in K_1 ; one defines $\{f, g\} := \{f\} \cdot \{g\}$ in K_2 .

1.1.6

For K a cochain complex and n an integer, $t_{\leq n}K$ (resp. $t_{>n}K$) is the subcomplex (resp. quotient complex) of K defined by

$$(t_{\leq n}K)^i = \begin{cases} 0 & i > n \\ \text{cycles in } K_n & i = n \\ K^i & i < n \end{cases} \quad \text{resp.} \quad (t_{>n}K)^i = \begin{cases} K^i & i > n \\ K^n / \text{coboundaries} & i = n \\ 0 & i < n. \end{cases}$$

For $n \leq m$, $t_{[n,m]}K := t_{>n}t_{\leq m}K$ has as cohomology groups H^i those of K for $n \leq i \leq m$, 0 otherwise.

1.1.7

A strict simplicial object of a category $\mathcal{C}$ is a contravariant functor with values in $\mathcal{C}$ from the category of the finite ordered sets Δ_n ($n \geq 0$) and injective increasing maps. Similarly for strict cosimplicial: face maps are given, but not degeneracies.

2 Central extensions, torsors and cocycles

2.1

If A is a sheaf of groups over some space, or site, S , an A -torsor (also called “principal homogeneous space”) on S is a sheaf P on which A acts on the right and which locally on S is isomorphic to the following standard model: A , with the action of A on itself by right translations. If P_i is an A_i -torsor ($i = 1, 2$), $P_1 \times P_2$ is a $A_1 \times A_2$ -torsor. A morphism $f : A \rightarrow B$ induces a functor f_* (also called “pushing by f ”) from A -torsors to B -torsors; one has a map of sheaves $f : P \rightarrow f_*(P)$ obeying $f(pa) = f(p)f(a)$ and this characterizes $f_*(P)$ up to unique isomorphism. Special case: for A commutative, and for the morphism $+$: $A \times A \rightarrow A$, one obtains the bifunctor “addition of A -torsors”.

2.2

We recall from SGA7 (tI, exp. VII, no. 1) Grothendieck’s description of central extensions in terms of torsors. We first consider the set-theoretic case. Let $A \rightarrow E \xrightarrow{p} G$ be a central extension of a group G by an abelian group A . The set E , with its projection p to G , and the action $e \mapsto ea$ of A on E , is then an A -torsor on G (viewed as a discrete topological space). The product map $E \times E \rightarrow E$ sits in a commutative diagram

$$\begin{array}{ccc} E \times E & \longrightarrow & E \\ \downarrow & & \downarrow \\ G \times G & \xrightarrow{m} & G \end{array}$$

where m is the product map of G . The top line can be reinterpreted as corresponding to a morphism of sets mapping to $G \times G$:

$$m_0 : \text{pr}_1^*E \times_{G \times G} \text{pr}_2^*E \rightarrow m^*E.$$

This map is such that $m_0(e_1a_1, e_2a_2) = m_0(e_1, e_2)a_1a_2$, hence defines an isomorphism of A -torsors over $G \times G$

$$m : \text{pr}_1^*E + \text{pr}_2^*E \rightarrow m^*E. \tag{9}$$

If $m_{123} = mm_{12} = mm_{23} : G \times G \times G \rightarrow G$ is the triple product map, the associativity of the multiplication of E is expressed by the commutativity of the following diagram of maps deduced from m between A -torsors on $G \times G \times G$:

$$\begin{array}{ccc}
\mathrm{pr}_1^*E + \mathrm{pr}_2^*E + \mathrm{pr}_3^*E & \xrightarrow{\mathrm{pr}_{12}^*m} & \mathrm{pr}_{12}^*m^*E + \mathrm{pr}_3^*E \\
\mathrm{pr}_{23}^*m \downarrow & & \downarrow m \\
\mathrm{pr}_1^*E + \mathrm{pr}_{23}^*m^*E & \xrightarrow{m} & m_{123}^*E.
\end{array} \tag{10}$$

From the central extension E we have deduced a multiplicative A -torsor on G , that is, an A -torsor E provided with a morphism (9) of A -torsors on $G \times G$ for which the diagram (10) is commutative. This construction is an equivalence from the category of central extensions of G by A to the category of multiplicative A -torsors on G .

2.3

As explained in SGA7 loc. cit., the natural framework for this construction is to work in any category of sheaves $\mathcal{T}$, not just in the category of sets (= sheaves over the space with one point). Here, “sheaf” means sheaf over some site S , not just the case of sheaves on a topological space: $\mathcal{T}$ is any topos (SGA4 IV).

For X in $\mathcal{T}$, one can “think of X as being a space”, i.e. consider the topos $\mathcal{T}/X$ of pairs (F, f) with F in $\mathcal{T}$ and f a map from F to X . In the case of sheaves on a topological space S , this amounts to looking at a sheaf X on S as being a topological space mapped to S by a local homeomorphism. One has a morphism of topos $\mathcal{T}/X \rightarrow \mathcal{T}$.

Let G be a sheaf of groups and A be a sheaf of abelian groups. For $n = 1, 2, 3$, let us “think of G^n as being a space”, and let us keep writing A for the pull-back of A to G^n . The same construction as in 1.2 then defines an equivalence from the category of central extensions of G by A to the category of multiplicative A -torsors on G , that is A -torsors on G provided with a multiplicative structure (9) on G^2 obeying (10) on G^3 .

2.4

A big Zariski site S_{Zar} of a scheme S is a full subcategory $\mathcal{C}$ of the category of schemes over S , with the Zariski topology. If X is in $\mathcal{C}$, any open subscheme of X should be in $\mathcal{C}$ too, and $\mathcal{C}$ should be large enough to contain the schemes over S we are interested in. For us, the precise choice of S_{Zar} will be immaterial. To fix ideas, we take for S_{Zar} the category of schemes of finite type over S .

For A a sheaf on S_{Zar} and for X in S_{Zar} , define the sheaf A_X on the topological space X by $A_X(U) := A(U)$. This construction identifies sheaves on S_{Zar} with systems (A_X) of sheaves on the topological spaces X , for X in S_{Zar} , contravariant in X , and with $A_U = (A_X \text{ restricted to } U)$ for U open in X : a morphism $f : X \rightarrow Y$ in S_{Zar} defines $j(f) : f^*A_Y \rightarrow A_X$, with a compatibility for composition of morphisms, and $j(f)$ is an isomorphism if f is an open embedding.

The Zariski sheaves $\mathcal{K}_n$ form such a system. We denote by K_n the corresponding sheaf on S_{Zar} . For $n = 0$, it is the constant sheaf $\mathbb{Z}$. For $n = 1$, it is the sheaf $\mathcal{O}^*$.

We now unravel the description 1.3 of central extensions of sheaves in the case of sheaves over S_{Zar} , when G is representable.

For Y in S_{Zar} , the representable functor $h_Y : T \mapsto Y(T) := \mathrm{Hom}_S(T, Y)$ is a sheaf.

To “think of h_Y as being a space”, as in 1.3, simply means to consider the topos of sheaves over Y . If A is a sheaf of abelian groups on S_{Zar} , its “pull-back to h_Y , thought of as a space”, is simply the sheaf on Y whose sections on $T \rightarrow Y$ is $A(T)$.

A further simplification is that the restriction from the big to the small Zariski site induces an equivalence of categories from A -torsors on S_{Zar} to A_S -torsors on S .

Let G be a group scheme of finite type over S . What matters is that the fiber powers G^n of G over S are in S_{Zar} . By 1.3, central extensions of h_G by A can be described as multiplicative A -torsors. In terms of the system of Zariski sheaves A_X , the description can be rephrased as follows. Notation: for $f : X \rightarrow Y$ in S_{Zar} and P a A_Y -torsor on Y , we write f^*P for the A_X -torsor on X obtained by pushing by $f^* : f^*A_Y \rightarrow A_X$ the f^*A_Y -torsor on X inverse image of P . With this notation, a multiplicative A -torsor is an A_G -torsor on G , provided with a morphism (9) of $A_{G \times G}$ -torsors on $G \times G$ for which the associativity diagram (10) is a commutative diagram of $A_{G \times G \times G}$ -torsors on $G \times G \times G$.

A section over T of the corresponding central extension E of G by A is a pair $(g, e) : g \in G(T) : T \rightarrow G$ and $e \in \Gamma(T, g^*E)$. Attaching to g the isomorphism class of the A_T -torsor e^*E on T , one obtains an exact sequence

$$1 \rightarrow H^0(T, A_T) \rightarrow E(T) \rightarrow G(T) \rightarrow H^1(T, A_T). \quad (11)$$

By abuse of language, we will often say “central extension of G by A ” for “central extension of h_G by A ”.

Remark 2.1. Suppose that S is the spectrum of a field k and write $A(k)$ for $A(S)$. If E is the central extension of G by A corresponding to a multiplicative A -torsor E , (11) for $T = S$ reduces to

$$1 \rightarrow A(k) \rightarrow E(k) \rightarrow G(k) \rightarrow 1,$$

making $E(k) := E(S)$ a central extension of $G(k)$ by $A(k)$.

Our strategy to construct central extensions of $G(k)$ by $K_2(k)$ will be to apply this construction to central extensions of G by $\mathcal{K}_2$.

2.5 Variant

Let X be a scheme in S_{Zar} , given with a section e . A pointed A -torsor on (X, e) is a A_X -torsor P on X , provided with a trivialization, on S , of the A_S -torsor e^*P .

If G is the trivial group scheme over S : G reduced to its neutral section, i.e. $G \xrightarrow{\sim} S$, a multiplicative structure on a G -torsor E is an isomorphism $E + E \xrightarrow{\sim} E$. It amounts to the data of a trivialization of E .

For a group scheme G , with neutral section e , it follows by restriction to e that a multiplicative torsor on G is automatically pointed. Multiplicative torsors could be redefined as pointed torsors on G , provided with a morphism (9) of pointed torsors on $G \times_S G$, obeying an associativity (10) on $G \times_S G \times_S G$.

2.6

Let E be a multiplicative A -torsor on a group scheme of finite type G over S . An automorphism s of G transforms E into a new multiplicative A -torsor $s_*(E)$ on G . If s is inner, i.e. is $\text{int}[g]$ for some g in $G(S)$, then E and $s_*(E)$ are isomorphic. This is clear if E is interpreted as a central extension E : the construction (6) (1) lifts $\text{int}[g]$ to an automorphism of E trivial on A .

Let G and H be group schemes of finite type over S , and let r be an action of H on G .

Construction 2.2. We will construct an equivalence of categories from the category of multiplicative A -torsors E on the semi-direct product $H \ltimes G$ to the category of triples (a) (b) (c) as follows: (a) a multiplicative A -torsor E on G ; (b) a multiplicative A -torsor F on H ; (c) an action of H on (G, E) lifting the action of H on G .

Here are equivalent definitions (1) (1') (2) (3) of what an action (c) is.

(1) For any T of finite type over S , let (G_T, E_T) be deduced from (G, E) by base change to T . The group $H(T)$ acts on G_T . Data (c) is a lifting of this action r to an action $\tilde{r}$ of $H(T)$ on (G_T, E_T) , functorial in T .

If H is smooth over S , we have a variant

(1') Same as (1), except that T is restricted to be smooth over S .

The definition (2) is (1), in a universal case:

(2) Let $h_0 \in H(H) = \text{Hom}(H, H)$ be the identity map of H . The group $H(H)$ acts on G_H . Data (c) is a lifting of the automorphism $r(h_0)$ of G_H to an automorphism $\tilde{r}(h_0)$ of (G_H, E_H) . If $m : H \times H \rightarrow H$ is the multiplication map, it is required that $m^*\tilde{r}(h_0) = \text{pr}_1^*\tilde{r}(h_0) \circ \text{pr}_2^*\tilde{r}(h_0)$ over $H \times H$.

One goes from (1) or (1') to (2) by taking $\tilde{r}(h_0) = \tilde{r}(h_0)$. For any $h \in H(T)$, h is the pull-back of $h_0 \in H(H)$ by $h : T \rightarrow H$ and one has $\tilde{r}(h) = h^*(\tilde{r}(h_0))$: the lifted action is determined by $\tilde{r}(h_0)$, and the condition on $H \times H$ in (2) is what is needed for $h \mapsto h^*(\tilde{r}(h_0))$ to be an action.

We wrote (2) to make clear the equivalence of (1) and (1') when H is smooth over S . It is in order to construct actions in the sense of (1) or (1') that we will systematically be considering multiplicative torsors not only on algebraic groups over a field, but also for group schemes.

(3) The multiplicative torsor E corresponds by 1.4 to a central extension E of the sheaf of groups h_G on S_{Zar} by the sheaf of abelian groups A . The sheaf of groups h_H acts on h_G . An action (c) is an action of h_H on E , lifting the action of h_H on h_G and trivial on A .

To construct 1.7, we will use the interpretation 1.4 of multiplicative torsors as central extensions, and the interpretation (3) of data (c). Its equivalence with interpretation (1) is easily checked. To simplify notations, we write $G, H, H \times G$ for $h_G, h_H, h_{H \times G}$.

Let E be a central extension of $H \times G$ by A . Let E_0 (resp. F) be the inverse image of G (resp. H) in E_0 . It is a central extension of G (resp. H) by A . The central extension $F \subset E_0$ acts on $E_0 \subset E$ by conjugation. The action is trivial on $A \subset F \subset E_0$ hence factors through an action of H on E , which lifts the action of H on G . This gives data (a) (b) (c), from which E is recovered as the quotient of the semi-direct product of $F \times E$, a central extension of $H \times G$ by $A \times A$, by the antidiagonal copy of A in $A \times A$.

2.7

For any central extension E of G by A , if g and h in G commute, the lifted commutator (7) is in A : on the subscheme $C \subset G \times_S G$ of commuting pairs of elements, the lifted commutator is a morphism of sheaves from C to A , i.e. is a section

$$\text{comm} \in H^0(C, A). \quad (12)$$

If G is commutative, then $C = G \times G$ and the section (12) of A over $G \times G$ is bimultiplicative.

More generally, if an algebraic group H acts on G , and if this action is lifted to an action on E , trivial on A , we have a lifted commutator (8) from $H \times G$ to E . If $C \subset H \times G$ is the subscheme of pairs (h, g) for which h fixes g , this lifted commutator defines

$$\text{comm} \in H^0(C, A), \quad (13)$$

and if H acts trivially on G , the section (13) of A over $H \times G$ is bimultiplicative.

2.8

To go back and forth between cohomological computations and categorical interpretations, we will rely on the following construction of SGA4 XVIII 1.4. Let K be a complex and n an integer. Let $A \rightarrow B$ be the truncation $t_{[n-1, n]}K$ of K : one has $A := K^{n-1}/\text{coboundaries}$ and $B := n\text{-cocycles}$. We define a category $\mathcal{C}_n(K)$ as the category of n -cocycles, a morphism from b_0 to b_1 being $a \in A$ such that $b_1 - b_0 = da$. Composition of morphisms is given by addition. Addition of n -cocycles defines an addition law for objects, turning $\mathcal{C}_n(K)$ into a strictly commutative Picard category. The group of isomorphism classes of objects of $\mathcal{C}_n(K)$ is $H^n(K)$, and the automorphism group of any object is $H^{n-1}(K)$. A morphism of complexes

$f : K \rightarrow L$ induces a functor $f : \mathcal{C}_n(K) \rightarrow \mathcal{C}_n(L)$, and if the morphism f induces isomorphisms on H^n and H^{n-1} , the functor f is an equivalence.

We will say that K *incarnates* a category $\mathcal{C}$, in degree n , if we are given an equivalence (usually: an equivalence of commutative Picard categories) from $\mathcal{C}_n(K)$ to $\mathcal{C}$. When no n is mentioned, $n = 1$ is assumed.

The heuristics for using this definition is as follows. Let $\mathcal{C}$ be a strictly commutative Picard category. Let $h^1(\mathcal{C})$ be the group of its isomorphism classes of objects and $h^0(\mathcal{C})$ be the group of automorphisms of any object. Suppose that $h^1(\mathcal{C})$ and $h^0(\mathcal{C})$ can be interpreted as cohomology groups H^n and H^{n-1} . Then, if K is any complex which naturally computes the cohomology in question, the complex K should incarnate $\mathcal{C}$ in degree n . The construction of the equivalence of $\mathcal{C}_n(K)$ with $\mathcal{C}$ will mimic the construction of the isomorphism of $H^n(K)$ with $h^1(\mathcal{C})$.

Examples. — (i) Let F be a sheaf of abelian groups on a space S and let F^* be a resolution of F , such that the complex $GF(S, F^*)$ computes the cohomology of F in degrees 0 and 1: the canonical maps $H^i GF(S, F) \rightarrow H^i(S, F)$ are isomorphisms for $i = 0$ and 1. Then, $GF(S, F^*)$ incarnates the category of F -torsors on S . The equivalence $\mathcal{C}_{GF}(S, F^*) \rightarrow (F\text{-torsors})$ is as follows: a 1-cocycle c gives rise to the F -torsor $P(c)$ of local sections s of F^0 with $ds = c$, and if $c_1 - c_0 = df$, f defines the isomorphism $s \mapsto s + f : P(c_0) \rightarrow P(c_1)$.

(ii) Suppose that a group G acts on (S, F) and that the resolution F^* is G -equivariant. The group G then acts on (S, F^*) and we can for each i form the standard complex $C^*(G, GF(S, F_i))$ computing $H^*(S, F_i)$. These complexes organize into a double complex, whose component in bidegree (p, q) is the group $C^q(G, GF(S, F_p))$ of q -cochains with values in $GF(S, F_p)$; it is the group of maps $G^q \rightarrow GF(S, F_p)$. The associated simple complex incarnates the category of equivariant F -torsors.

Standard operations on complexes can be given categorical meaning.

Examples. — (iii) Let $f : K \rightarrow L$ be a morphism of complexes. We view $f : K \rightarrow L$ as a double complex, with K (resp. L) in second degree 0 (resp. 1) and form the associated simple complex $s(K \rightarrow L)$. This shifted mapping cone construction gives rise to a long exact sequence of cohomology

$$\cdots \rightarrow H^i(s(K \rightarrow L)) \rightarrow H^i(K) \rightarrow H^i(L) \rightarrow \cdots$$

Suppose that $K^i = L^i = 0$ for $i < 0$. Then, $s(K \rightarrow L)$ incarnates the category of objects of $\mathcal{C}(K)$ given with a trivialization (= isomorphism with the object 0) of its image in $\mathcal{C}(L)$.

(iv) We now view $f : K \rightarrow L$ as a double complex with K in second degree -1 and L in second degree 0. The long exact sequence of cohomology is

$$\cdots \rightarrow H^i(K) \rightarrow H^i(L) \rightarrow H^i(s(K \rightarrow L)) \rightarrow \cdots$$

Suppose that $K^i = L^i = 0$ for $i < 0$, and that $H^i(K)$ injects into $H^i(L)$ for $i = 0$ and 2. By the long exact sequence, $H^1(L)$ maps onto $H^1(s(K \rightarrow L))$: the category $\mathcal{C}(s(K \rightarrow L))$ is equivalent to its subcategory with objects the 1-cocycles of L . This category has as objects the objects of $\mathcal{C}(L)$, a morphism from P to Q being an isomorphism class of pairs (A, a) with A an object of $\mathcal{C}(K)$ and a a morphism from $P + f(A)$ to Q .

(v) Let $K \xrightarrow{f} L \xrightarrow{g} M$ be morphisms of complexes with zero composite. They give rise to functors $\mathcal{C}(K) \xrightarrow{f} \mathcal{C}(L) \xrightarrow{g} \mathcal{C}(M)$ and to an isomorphism of the composite functor with the constant functor with value the object 0. We view $K \rightarrow L \rightarrow M$ as a double complex, with K (resp. L, M) in second degree 0 (resp. 1, 2). We assume that the cohomology of K, L and M is zero in degree $i < 0$. The category $\mathcal{C}(s(K \rightarrow L \rightarrow M))$ does not change (up to a natural equivalence) if K, L, M are replaced by their quasi-isomorphic truncation $t_{>0}$ and, using this remark, one checks that it is the category of objects P of $\mathcal{C}(K)$, given with a trivialization $0 \rightarrow f(P)$ in $\mathcal{C}(L)$ whose image $0 \rightarrow gf(P)$ in $\mathcal{C}(M)$ is the identity automorphism of the object zero of $\mathcal{C}(L)$.

In our applications, we will start from a cosimplicial system of complexes $K(p)$. We won't need degeneracies and so could start from a strict cosimplicial system (0.N.7). We assume that $H^i K(p) = 0$ for $i < 0$.

For each n one can consider $K(n) \rightarrow K(n+1) \rightarrow K(n+2)$, taking as morphisms of complexes the usual $(-1)^i \partial_i$. The $\mathcal{C}(K(p))$ form a strict cosimplicial system of Picard categories. For $n = 0$ (resp. $n = 1$), $\mathcal{C}(s(K(n) \rightarrow K(n+1) \rightarrow K(n+2)))$ has the following interpretation.

Case $n = 0$: Objects P of $\mathcal{C}(K(0))$, given with an isomorphism $f : \partial_0(P) \rightarrow \partial_1(P)$ in $\mathcal{C}(K(1))$ obeying a cocycle condition in $\mathcal{C}(K(2))$: for ∂_i ($a = 0, 1, 2$) the three face maps from Δ_0 to Δ_2 , commutativity of the diagram

$$\begin{array}{ccc} \partial_2(P) & \xrightarrow{\partial_1(f)} & \partial_0(P) \\ & \nwarrow \partial_0(f) \quad \nearrow \partial_2(f) & \\ & \partial_1(P) & \end{array}$$

Case $n = 1$: Objects P of $\mathcal{C}(K(1))$ given with a multiplicativity isomorphism $\partial_0(P) + \partial_2(P) \rightarrow \partial_1(P)$ in $\mathcal{C}(K(2))$ obeying an associativity condition in $\mathcal{C}(K(3))$.

(vi) The case $n = 1$ above will be met in the following context, explaining the terminology. Let K be a sheaf on the big Zariski site S_{Zar} , and G be a group scheme of finite type over S . In the inhomogeneous notation, the simplicial classifying space $BG = (G^{\Delta^n}/G)$ of G is

$$\cdots G^3 \rightrightarrows G^2 \rightrightarrows G \rightrightarrows e,$$

the three maps $\partial_0, \partial_1, \partial_2$ from G^2 to G being pr_2, m and pr_1 . Suppose that on each G^n we are given a resolution K^* of K_{G^n} , contravariant in the face maps, and which computes $H^i(G^n, K_{G^n})$ for $i = 0$ or 1 . By the example (i), $GT(G^n, K^*)$ incarnates the category of K_{G^n} -torsors on G^n . Applying example (v) to

$$GT(G, K^*) \rightarrow GT(G^2, K^*) \rightarrow GT(G^3, K^*),$$

one obtains a complex which incarnates the category of multiplicative torsors on G . This complex can be replaced by the subcomplex of simplicial degree > 1 of $s(GT(G^*, K^*))$: the complex

$$s(GT(G^n, K^*) \text{ for } n > 1) \tag{14}$$

incarnates multiplicative torsors in degree 2.

Let us write e for the trivial group scheme G^0 . For each n , $GT(e, K^*)$ maps to $GT(G^n, K^*)$, defining a cosimplicial map from $GT((Be)^*, K^*)$ to $GT((BG)^*, K^*)$. The composite

$$s(GT(G^n, K^*) \text{ for } n > 1) \rightarrow sGT((BG)^*, K^*) \rightarrow s(sGT((Be)^*, K^*) \rightarrow sGT((BG)^*, K^*))$$

is a quasi-isomorphism. It follows that

$$s(s(GT((Be)^*, K^*) \rightarrow s(GT((BG)^*, K^*)))) \tag{15}$$

which computes the relative cohomology of BG mod Be with values in K , also incarnates the category of multiplicative K -torsors in degree 2.

(vii) With notation as in example (vi), if $H^i(G^n, K_{G^n}) = 0$ for $i > 0$, we can, on each G^n , take for K^* the complex reduced to K_{G^n} in degree 0. We find that the category of multiplicative torsors on G is incarnated by the complex

$$GT(G^n, K) \text{ for } n > 1.$$

This is the familiar description of central extensions by 2-cocycles. One could as well use the complex

$$H^0(G^n, K), \quad d = \sum (-1)^i \partial_i,$$

where H^0 is the direct summand $GT(G^n, K)/GT(e, K)$ of $GT(G^n, K)$.

2.9

The constructions 1.9 can be sheaffied. Let K be a complex of sheaves on S and n be an integer. As in 1.9, consider the truncation $t_{[n-1, n]}(K) : A \rightarrow B$ of K . Define $\mathcal{C}_n(K)$ to be the stack on S having as objects over U the A -torsors P over U provided with a trivialization of the B -torsor $d(P)$. It is a strictly commutative Picard stack. For S a point, i.e. for a complex of abelian groups, one recovers a category incarnated by K in degree n : to the cocycle $b \in B$ attach the trivial A -torsor A with the trivialization b of the trivial B -torsor $d(A)$. Indeed, over a point any A -torsor is isomorphic to the trivial one.

The analogue of 1.9, Example (i), is that for $f : X \rightarrow S$, F a sheaf on X , and $R^i f_* F^*$ a resolution of F such that $f_* F^*$ computes $f_* F$ in degree 0 and 1, $f_* F^*$ incarnates the stack on S having as objects over U the F -torsors on $f^{-1}(U)$.

3 Descent for K_2 -torsors

The main result of this section is Theorem 2.2, which will allow us to reduce the construction of multiplicative K_2 -torsors on connected reductive groups to the case of split groups. From 2.6 on, we prove a relative variant of 2.2. This variant will not be needed for our main results, or rather for their restriction to the case of reductive groups over fields.

Our main tool is the following theorem of Colliot-Thélène and Suslin, see B. Kahn (1993) (corollary 1 to theorem 3.1):

Theorem 3.1. *Let X be a smooth geometrically irreducible variety over k . Assume that X has a k -rational point e . If k' is a finite Galois extension of k , with Galois group G , and $X' := X \otimes_k k'$, one has, denoting by $k(X)$ and $k(X')$ the fields of rational functions on X and X'*

$$K_2(k(X))/K_2(k) \xrightarrow{\sim} [K_2(k(X'))/K_2(k')]^G \quad (16)$$

and

$$H^1(G, K_2(k(X'))/K_2(k')) = 0. \quad (17)$$

In characteristic zero, (16) is Colliot-Thélène (1983) Th. 1. In any characteristic, it is in Suslin (1987), 3.6 and 5.8. The vanishing (17) can be deduced from Merkurjev-Suslin (1983) 14.1. The point of view of B. Kahn (1993) is that the Lichtenbaum complex $\Gamma(2)$ has good étale localization properties and that theorem 2.1 is a corollary of these. See also B. Kahn (1996). Under the assumptions of 2.1, one also knows that $K_2(k)$ injects as a direct summand in $K_2 k(X)$.

The group G acts semi-linearly on $(X', e') = (X, e) \times_{\text{Spec}(k)} \text{Spec}(k')$. A (Galois) descent data on a pointed K_2 -torsor P' on (X', e') is an action of G on P' , covering the action of G on (X', e') .

Theorem 3.2. *Under the assumptions of 3.1, the pull-back functor is an equivalence from*

- (a) *the category of pointed K_2 -torsors on (X, e) , to*
- (b) *the category of pointed K_2 -torsors on (X', e') , provided with a descent data.*

Proof. The category of K_n -torsors on X is incarnated by the global sections of the Quillen resolution G_n^* of K_n by flabby sheaves on X . Let $X^{(i)}$ be the set of points of codimension i of X . One has

$$G_n^i = \bigoplus_{x \in X^{(i)}} K_{n-i}(k(x)).$$

For $n = 2$, the complex of global sections is

$$K_2 k(X) \rightarrow \bigoplus_{x \in X^{(1)}} k(x)^* \rightarrow \bigoplus_{x \in X^{(2)}} \mathbb{Z}.$$

We want to use the resolution G_n^* of K_n to construct a complex incarnating the category of pointed K -torsors. The fact that the resolution G_n^* , while contravariant for smooth maps, is not contravariant for closed embeddings, creates a difficulty. We go around it as follows. The point e , viewed as a section of $f : X \rightarrow \text{Spec}(k)$, defines an equivalence of the category of K_n -torsors on X with the product of the categories of pointed torsors on (X, e) and the category of $K_n(k)$ -torsors. This allows us to view the category of pointed torsors as a quotient category, rather than a subcategory, of the category of K -torsors: it is naturally equivalent to the category of relative torsors having as objects the K_n -torsors on X , a map from P to Q being a class mod $K_n(k)$ of maps of torsors from P to Q . The inverse of the forgetting functor from pointed torsors to relative torsors is the functor $P \mapsto P - f^{**}e^*P$.

By 1.9, Example (iv), the category of relative K_2 -torsors on X is incarnated by the complex

$$K_2k(X)/K_2(k) \rightarrow \bigoplus_{x \in X^{(1)}} k(x)^* \rightarrow \bigoplus_{x \in X^{(2)}} \mathbb{Z}. \quad (18)$$

By a variant of 1.9, Example (ii), the category of relative K_2 -torsors on X' , provided with a descent data, is incarnated by the simple complex associated to the double complex

$$C^*(G, K_2k(X')/K_2(k')) \rightarrow \bigoplus_{x' \in X'^{(1)}} C^*(G, k(x')^*) \rightarrow \bigoplus_{x' \in X'^{(2)}} C^*(G, \mathbb{Z}). \quad (19)$$

These incarnations reduce 3.2 to

Lemma 3.3. *The natural map of complexes (18) $\rightarrow$ s(19) induces an isomorphism on H^0 and H^1 .*

Proof of 3.3. More precisely, we will check that the cone on the natural map of complexes (18) $\rightarrow$ s(19) has vanishing H^{-1} , H^0 and H^1 . This cone is the associated simple complex of the double complex (19), coaugmented by (18) in second (= group cohomology) degree -1 . It vanishes outside of $0 \leq p \leq 2$, $q \geq -1$.

The $p = 0$ column is $C^*(G, K_2k(X')/K_2(k'))$, coaugmented by $K_2k(X)/K_2(k)$ in degree -1 . By 3.1, its cohomology vanishes in degree -1 , 0 and 1 .

The $p = 1$ column is the sum over $x \in X^{(1)}$ of the $C^*(G, (k(x) \otimes_k k')^*)$, coaugmented by $k(x)^*$ in degree -1 . As $k(x)^* \xrightarrow{\sim} ((k(x) \otimes_k k')^*)^G$, its cohomology vanishes in degree -1 and 0 .

The $p = 2$ column is the sum over $x \in X^{(2)}$ of the $C^*(G, \mathbb{Z}^{I(x)})$, where $k(x) \otimes k'$ is a product of fields indexed by $I(x)$, coaugmented by $\mathbb{Z}$ in degree -1 . As the diagonal map from $\mathbb{Z}$ to $\mathbb{Z}^{I(x)}$ is injective, its cohomology vanishes in degree -1 .

The spectral sequence for the filtration by p gives the required vanishing. $\square$

If we apply 3.2 to the group of automorphisms of an object (resp. to the set of isomorphism classes of objects), we obtain

Corollary 3.4. *Under the assumptions of 3.1,*

$$(i) \ H^0(X, \mathcal{K}_2)/K_2(k) \xrightarrow{\sim} (H^0(X', \mathcal{K}_2)/K_2(k'))^G;$$

(ii) *one has an exact sequence*

$$0 \rightarrow H^1(G, H^0(X', \mathcal{K}_2)/K_2(k')) \rightarrow H^1(X, \mathcal{K}_2) \rightarrow H^1(X', \mathcal{K}_2)^G \rightarrow H^2(G, H^0(X', \mathcal{K}_2)/K_2(k')).$$

Remark 3.5. (i) If further $K_2(k) \xrightarrow{\sim} H^0(X', \mathcal{K}_2)$, then $K_2(k) \xrightarrow{\sim} H^0(X, \mathcal{K}_2)$ and $H^1(X, \mathcal{K}_2) \xrightarrow{\sim} H^1(X', \mathcal{K}_2)^G$. This results from 3.4.

(ii) Suppose that X is open and dense in a smooth geometrically irreducible variety X_1 over k and that X_1 has a k -rational point. Then 3.2 remains valid, with the same proof, if “pointed K_2 -torsor” is replaced by “relative K_2 -torsor”. $\square$

3.1

Let $f : X \rightarrow S$ be a smooth morphism, with a section e . We assume that

$$S \text{ is regular and of finite type over some field } k; \quad (20)$$

$$\text{For each generic point } \eta \text{ of } S, \text{ the generic fiber } X_\eta/\eta \text{ is geometrically irreducible.} \quad (21)$$

Assumption (20) is to have Quillen resolutions. Assumption (21) is to have 3.1 for X_η/η .

3.2

Let $u : S_0 \rightarrow S$ be an étale surjective map. We define S_n to be the $(n+1)$ st fiber power $(S_0/S)^{\Delta n}$ of S_0 over S . Let (X_0, e_0) be the pull-back of (X, e) over S_0 ; $X_n := X \times_S S_n$ is also the fiber power $(X_0/X)^{\Delta(n+1)}$ of X_0 over X . The X_n ($n \geq 0$) form a simplicial scheme, augmented to $X_{-1} := X$.

A descent data for a K_2 -torsor P on X_0 is an isomorphism between its two inverse images on X_1 , obeying a compatibility condition on X_2 .

Theorem 3.6. *Under the assumptions of 2.6, the pull-back functor is an equivalence from*

- (a) *the category of pointed K_2 -torsors on (X, e) , to*
- (b) *the category of pointed K_2 -torsors on (X_0, e_0) , provided with a descent data.*

Proof. If S is the spectrum of a field k , 3.6 is little more than a rephrasing of 3.2. The general case follows similarly using the Quillen resolution. $\square$

4 Split tori

Our aim in this section is to compute the category of multiplicative K_2 -torsors on a split torus. We also treat the case of split unipotent groups.

4.1

Let S be regular and of finite type over a field k . If $p : \mathbb{A}_S^1 \rightarrow S$ is the affine line over S , Sherman (1979) proves that

$$p_* K_j = K_j \quad \text{and} \quad R^i p_* K_j = 0 \text{ for } i > 0 \quad (\text{for } p : \mathbb{A}_S^1 \rightarrow S). \quad (22)$$

If we remove the 0-section, we obtain the multiplicative group $p : \mathbb{G}_{mS} \rightarrow S$ over S . Sherman (1979) proves that

$$p_* K_j = K_j \oplus K_{j-1} \quad \text{and} \quad R^i p_* K_j = 0 \text{ for } i > 0 \quad (\text{for } p : \mathbb{G}_{mS} \rightarrow S). \quad (23)$$

Proposition 4.1. *If S is regular of finite type over a field, and if the group scheme G over S is an extension of G' smooth over S by a split unipotent group scheme U , the pull back by $p : G \rightarrow G' = G/U$ is an equivalence from the category of central extensions of G' by $\mathcal{K}_2$ to the category of central extensions of G by $\mathcal{K}_2$.*

For G the trivial group scheme e , 4.1 tells that a central extension of a split unipotent group scheme by $\mathcal{K}_2$ is trivial, and admits a unique trivialization.

4.2

Let $p : T \rightarrow S$ be a split torus over S and X, Y be as in (0.N.1). If we choose an isomorphism of Y with $\mathbb{Z}^n$, we obtain $T \xrightarrow{\sim} \mathbb{G}_{mS}^n$. Using induction on n we deduce from (23) that $R^i p_* K_j = 0$ for $i > 0$ and we obtain a description of $p_* K_j$.

Lemma 4.2. *The sheaf of graded rings $p_* K_*$ on S is generated by K_* and by X in degree 1, with the only relations that*

- (i) $X \rightarrow p_* K_1$ is additive;
- (ii) for x in X , $x \cdot x = x \cdot \{-1\}$.

4.3

If we filter $p_* K_*$ by “how many factors in X ”, we obtain an increasing filtration V with

$$\mathrm{Gr}_m^V p_* K_* = \Lambda^m X \otimes K_{*-m}. \quad (24)$$

The bottom piece $V_0(p_* K_*) = K_*$ is a direct factor: the pullback by the 0-section e is a retraction to the embedding of K_* in $p_* K_*$. The kernel of e^* is the reduced direct image $\tilde{p}_* K_*$.

Construction 4.3. Under the assumptions of 3.1 and 3.3, the sheaf $\tilde{p}_* K_2$ is the sheaf of pairs (A, q) , with A an alternating form on Y and q a map of sheaves $Y \rightarrow \mathcal{O}_S^*$ such that

$$q(y+z)/q(y)q(z) = (-1)^{A(y,z)}. \quad (25)$$

In 4.3 and below, “form” means integer-valued form.

Corollary 4.4. (i) *Under the assumptions of 3.1 and 3.3, the group of multiplicative sections of K_2 over T is $H^0(S, X \otimes \mathcal{O}_S^*)$.*

- (ii) *For split tori T_1, T_2 over S , a section of K_2 over $T_1 \times_S T_2$ is bimultiplicative as soon as it has a trivial restriction to $T_1 \times e$ and $e \times T_2$. The group of bimultiplicative sections is $H^0(S, X_1 \otimes X_2)$.*

4.4

If S is the spectrum of a field, it results from 1.9 Example (vii) and from the vanishing of $H^i(T, K_n)$ for $i > 0$ that the category of central extension of T by K_n is incarnated in degree 2 by the complex $H^0(T^p, K_n)$. The cohomology of the complex $H^0(T^*, K_m)$ is computed in Esnault et al. (1998) 4.6:

$$H^n(H^0(T^*, K_m)) = \mathrm{Sym}^n(X) \otimes K_{m-n}(k). \quad (26)$$

For $m = 2$, one deduces from (26) that the complex $H^0(T^p, K_2)$ has cohomology only in degree 1 and 2, with

$$\begin{aligned} H^1 H^0(T^*, K_2) &= X \otimes k^*, \\ H^2 H^0(T^*, K_2) &= \mathrm{Sym}^2(X). \end{aligned} \quad (27)$$

4.5

Let E be a multiplicative K_2 -torsor on T . By 3.9, for some bilinear form C on Y , E is isomorphic to the trivial torsor, with the multiplicative structure given by the cocycle image of C by

$$x_1 \otimes x_2 \mapsto \{\mathrm{pr}_1^* x_1, \mathrm{pr}_2^* x_2\}. \quad (28)$$

The group of isomorphism classes of multiplicative K_2 -torsors is $\text{Sym}^2(X)$, and the class of E is given by the quadratic form $C(y, y)$ on Y .

Let us extend the scalars from k to $k[t, t^{-1}]$. By 1.4, we deduce from E a central extension, on $\text{Spec } k[t, t^{-1}]$, of the sheaf of local sections of T by $\mathcal{K}_2$. As $H^1(\text{Spec } k[t, t^{-1}], \mathcal{K}_2) = 0$ ((23)), taking global sections, we obtain a central extension

$$H^0(\text{Spec}(k[t, t^{-1}]), \mathcal{K}_2) \rightarrow E \rightarrow T(k[t, t^{-1}]). \quad (29)$$

We take the pull back of this central extension by

$$Y \rightarrow T(k[t, t^{-1}]) = k[t, t^{-1}]^* \otimes Y : y \mapsto t \otimes y, \quad (30)$$

and the push out by

$$H^0(\text{Spec}(k[t, t^{-1}]), \mathcal{K}_2) = K_2(k) \oplus k^* \rightarrow k^*, \quad (31)$$

obtaining a central extension

$$k^* \rightarrow E \rightarrow Y. \quad (32)$$

Proposition 4.5. *The construction 3.10 is an equivalence of categories from the category of multiplicative K_2 -torsors on T to the category of pairs (Q, E) , where Q is a quadratic form on Y and E is a central extension of Y by k^* for which the commutator map (0.N.4) (2): $Y \times Y \rightarrow k^*$, is given by*

$$(y, z) = (-1)^{B(y, z)}. \quad (33)$$

In (33), B is the bilinear form associated to Q (0.N.3).

Remark 4.6. In 3.10, instead of extending scalars from k to the ring $k[t, t^{-1}]$, we could have extended scalars to the field $k((t))$, obtaining from E instead of (29) a central extension

$$K_2 k((t)) \rightarrow E_1 \rightarrow T(k((t))).$$

We still have a morphism

$$Y \rightarrow T(k((t))) = k((t))^* \otimes Y : y \mapsto t \otimes y \quad (34)$$

and the tame symbol defines

$$K_2 k((t)) \rightarrow k^*. \quad (35)$$

Pushing by (35) and pulling back by (34), we obtain a central extension of Y by k^* , canonically isomorphic to the central extension (32).

Proposition 4.7. *Let E be a central extension of T by K_2 , that is a multiplicative K_2 -torsor on T (1.4). Let Q be the corresponding quadratic form on Y , and let B be the associated bilinear form. The corresponding commutator comm of (1.8.1), in $H^0(T \times T, K_2)$, is bimultiplicative. It is the image of B , by the identification 4.4 (ii).*

Corollary 4.8. *If $B = \sum b_{ij} x_i \otimes x_j$, the commutator for the central extension*

$$K_2(k) \rightarrow E(k) \rightarrow T(k)$$

is given by $(t_1, t_2) \mapsto \sum b_{ij} \{x_i(t_1), x_j(t_2)\}$.

4.6

We now return to the case of a split torus T over S regular of finite type over a field. Let p be the projection of T^n to S . Ambiguity on n will be avoided by writing $p_*(T^n, K_2)$ for the direct image of K_2 from T^n to S . Let $\tilde{p}_*(T^n, K_2)$ be the reduced direct image $p_*(T^n, K_2)/K_2$. Applying the variant 1.10 of 1.9 Example (vii), one deduces from the vanishing (23) that the complex of sheaves $\tilde{p}_*(T^n, K_2)$ on S incarnates in degree 2 the stack $U \mapsto$ category of multiplicative K_2 -torsors on the split torus $T = p^{-1}(U)$ over U .

Theorem 4.9. *For T a split torus over S regular of finite type over a field k , the construction 3.15 induces an equivalence of commutative Picard categories from*

- (i) *the category of multiplicative K_2 -torsors on T , to*
- (ii) *the category of pairs (Q, E) : Q a quadratic form on the sheaf Y and E a central extension of Y by the sheaf $\mathcal{O}_S^*$, for which, as in 4.5,*

$$(y, z) = (-1)^{B(y, z)}.$$

The commutator (1.8.1) defined by a multiplicative K_2 -torsor is given as in 4.7.

5 Split semi-simple simply-connected groups

For G split semi-simple simply-connected over a field k , with split maximal torus T and Weyl group W , we will prove in this section results which, via the formalism of 1.9, amounts to $H^i(G, \mathcal{K}_2) = 0$ for $i \neq 1$, $H^i(BG, \mathcal{K}_2) = 0$ for $i \neq 2$, quadratic forms on Y being dual to $H^1(G, \mathcal{K}_2)$ and the dual of $H^2(BG, \mathcal{K}_2)$ being the group of W -invariant quadratic forms on Y . Our proofs are similar to those in Esnault et al. (1996) (3.2, 4.7, 4.8), but closer to the cocycle level and at places more elementary.

5.1

Let G be a split reductive group over a field k . We fix a split maximal torus T of G . We will use the notation X, Y, W of (0.N.1) and (0.N.2), as well as the following:

Φ : the set of roots (a subset of X). For α a root, we denote by

$\alpha^\vee$: the corresponding coroot; $\alpha^\vee$ is in Y and $\langle \alpha^\vee, \alpha \rangle = 2$;

U_α : the corresponding radical subgroup; it is isomorphic to $\mathbb{G}_a$ and T acts on it by α ;

s_α : the corresponding reflection in W ;

S_α : the corresponding $\mathrm{SL}(2)$ or $\mathrm{PGL}(2)$ subgroup of G , generated by $\alpha^\vee(\mathbb{G}_m)$, U_α and $U_{-\alpha}$.

The walls are the hyperplanes $\langle x, \alpha^\vee \rangle = 0$ of $X \otimes \mathbb{R}$. For $w \in W$, we will write $\dot{w}$ for a lifting of w in the normalizer $N(T)$ of T .

We fix a Borel subgroup B containing T . We denote by

Φ^+ : the corresponding set of positive roots: a root α is in Φ^+ if and only if U_α is in the unipotent radical U of B ;

I : the corresponding system of simple roots. A simple root $i \in I$ will often be written α_i ;

$\ell(w)$: the length of $w \in W$, relative to the system of generators s_i ($i \in I$);

Φ^- : the complement of Φ^+ in Φ . It is also $-\Phi^+$;

C_0 : the fundamental chamber in $X \otimes \mathbb{R}$, defined by $\langle x, \alpha^\vee \rangle > 0$ for α positive.

For $w \in W$, let $U_{\Phi^+ \cap w\Phi^-}$ be the group generated by the U_α with $\alpha \in \Phi^+ \cap w\Phi^-$. As a scheme, it is isomorphic to $\mathbb{A}^n$ for some n .

The product map

$$U_{\Phi^+ \cap w\Phi^-} \times wT \times U \rightarrow BwB \tag{36}$$

is an isomorphism. The map

$$p : BwB \rightarrow wT : p = \text{pr}_2 \circ (36)^{-1} \quad (37)$$

is the unique retraction of BwB to wT such that $p(uwtv) = wt$ for u and v in U , t in T and w any lifting of w in $N(T)$. By 3.1,

$$H^i(BwB, K_j) = 0 \quad \text{for } i > 0 \quad (38)$$

and $p : BwB \rightarrow wT$ induces an isomorphism

$$H^0(wT, K_j) \xrightarrow{\sim} H^0(BwB, K_j). \quad (39)$$

The length $\ell(w)$ of w is $|\Phi^+ \cap w\Phi^-|$: it is the number of walls separating the fundamental chamber C_0 from its transform $w(C_0)$.

5.2

Lemma 5.1. (i) *With the notation above, the valuation of $\dot{x}_w$ along Bw_1B is given by*

$$v(\dot{x}_w) = \langle x, w''\alpha_i^\vee \rangle. \quad (40)$$

(ii) *If $v(\dot{x}_w) = 0$, let $\dot{x}_w|_{Bw_1B}$ be the restriction to Bw_1B of the extension of $\dot{x}_w$ to $BwB \cup Bw_1B$. Then, $\dot{x}_w|_{Bw_1B}$ is a constant multiple of $\dot{x}_{w_1}$.*

(iii) *If further $w = w's_iw''$ with s_i in S_i and $w_1 = w'w''$, then*

$$\dot{x}_w|_{Bw_1B} = \dot{x}_{w_1}. \quad (41)$$

Writing $w = w' \cdot (w_1s_iw')$, one can rewrite (40) as follows: $w_1 = ws_i$ for s_i a reflection in W , and if α is the corresponding positive root, then

$$v(\dot{x}_w) = \langle x, \alpha^\vee \rangle. \quad (42)$$

5.3

The Bruhat decomposition induces a filtration F on any sheaf $\mathcal{F}$ on G , $F^p\mathcal{F}$ being the sheaf of sections with support in the union of the BwB of codimension $> p$.

5.4

Proposition 5.2. *If G is simply-connected the complex $(4.4.1)_{n=0}$ is acyclic, and $(4.4.1)_{n=1}$ has cohomology only in degree 1. In the model $(4.4.3)_{n=2}$, the morphism $C \mapsto C(y, y)$ identifies the H^1 of $(4.4.1)$ with the group of W -invariant quadratic forms on Y .*

As a consequence,

$$H^0(G, \mathcal{K}_2) = K_2(k), \quad \text{hence } H^0(G, \mathcal{K}_2) = 0; \quad (43)$$

$$H^1(G, \mathcal{K}_2) = \{W\text{-invariant quadratic forms on } Y\}; \quad (44)$$

$$H^i(G, \mathcal{K}_2) = 0 \quad \text{for } i > 2. \quad (45)$$

Theorem 5.3. *For G split and simply-connected, pointed K_2 -torsors on G have no nontrivial automorphism. They admit a unique multiplicative structure. They are classified by W -invariant quadratic forms on Y .*

Proof. The first statement amounts to (43). The last statement expresses (44). $\square$

5.5

For S regular of finite type over a field and G a split reductive group over S , with split maximal torus T and Borel subgroup B containing T , the arguments used in this section can be repeated, when sheafified over S .

If a (resp. a_w) is the projection of G (resp. BwB) to S , the complex $a_*G_j^*$ is quasi-isomorphic to the subcomplex

$$\bigoplus_{w \in W(p)} a_{w*}K_{j-p}[-p] \quad (46)$$

which, for $j = 2$, can be filtered with associated graded Gr_n the following complexes of sheaves on S

$$\begin{aligned} n = 0 : & \quad K_2; \\ n = 1 : & \quad X \otimes \mathcal{O}_S^* \rightarrow \bigoplus_{w \in W(1)} \mathcal{O}_S^*; \\ n = 2 : & \quad \Lambda^2 X \rightarrow \bigoplus_{w \in W(1)} X \rightarrow \bigoplus_{w \in W(2)} \mathbb{Z}. \end{aligned} \quad (47)$$

For G simply-connected the proof of 5.2 gives

$$\tilde{a}_*K_2 = K_2, \quad \text{hence } \tilde{a}_*K_2 = 0 \text{ for the reduced direct image;} \quad (48)$$

$$R^1 a_*K_2 = \text{the locally constant sheaf of } W\text{-invariant forms on } Y; \quad (49)$$

$$R^i a_*K_2 = 0 \quad \text{for } i > 2. \quad (50)$$

As in 5.3, it follows that pointed K_2 -torsors on G have no nontrivial automorphisms, have a unique multiplicative structure, and are classified by (49).

Lemma 5.4. *The quadratic forms on Y and Y' attached to a multiplicative K_2 -torsor on G correspond to each other by the isomorphism of Y with Y' .*

Compatibility 5.5 (4.9). Assume that G is simply-connected. Let E be a multiplicative K_2 -torsor on G , and E_T be its restriction to T . The quadratic form 3.11 describing the isomorphism class of the multiplicative K_2 -torsor E_T on T is the same as the quadratic form 4.7.

5.6

Let G_{ad} be the adjoint group of G and T_{ad} be the maximal split torus of G_{ad} image of T . The group G_{ad} acts on G (adjoint action). Multiplicative K_2 -torsors on G have no nontrivial automorphism and their classification is discrete. It follows that if E is a multiplicative K_2 -torsor on G , the action of G_{ad} on G extends uniquely to an action of G_{ad} on (G, E) .

5.7

We pass to the language 1.4 of central extensions by $\mathcal{K}_2$. As T_{ad} fixes T , the commutator section (1.8.2) is a bimultiplicative section

$$\text{comm} \in H^0(T_{\text{ad}} \times T, K_2). \quad (51)$$

By 4.4 (ii), it corresponds to an element of $X_{\text{ad}} \otimes X$, or equivalently to a bilinear pairing between Y_{ad} and Y .

The pullback of (51) to $T \times T$ is induced by the commutator section (1.8.1). It can be computed using 5.5 and 4.8. This fixes (51) uniquely:

Proposition 5.6. *Suppose that E corresponds by 5.3 to the W -invariant quadratic form Q on Y . Then, the associated bilinear form B extends to a pairing B_1 between Y_{ad} and Y and, with the identifications of 4.4 (ii), the bimultiplicative section (51) of K_2 corresponds to B_1 .*

5.8

Fix E as in 4.10 to 4.12. Taking k -points, we deduce from E a central extension

$$K_2(k) \rightarrow E(k) \rightarrow G(k) \quad (52)$$

as in (1.4). The action 4.10 of G_{ad} on E induces an action of $G_{\text{ad}}(k)$ on this central extension. By restriction to $T_{\text{ad}}(k)$ and $T(k)$, the corresponding “commutator” (0.N.4) (3): $G_{\text{ad}}(k) \times G(k) \rightarrow E(k)$ induces a bimultiplicative map

$$T_{\text{ad}}(k) \times T(k) \rightarrow K_2(k). \quad (53)$$

Corollary 5.7. *The pairing (53):*

$$(Y_{\text{ad}} \otimes k^*) \times (Y \otimes k^*) \rightarrow K_2(k)$$

is given by $(y_1 \otimes a, y \otimes b) \mapsto B_1(y_1, y)\{a, b\}$.

Proposition 5.8. *Suppose that G is simple and that Q takes the value 1 on short coroots. Then the central extension (52) is Matsumoto’s central extension (0.2).*

Proof. The group $G(k)$ is its own commutator subgroup (except for small finite fields k , for which $K_2(k)$ is trivial anyway). It follows that a central extension E of $G(k)$ by an abelian group A has no nontrivial automorphism (trivial on $G(k)$ and A). Suppose that the action of $G_{\text{ad}}(k)$ on $G(k)$ extends to E , the action being trivial on A . As explained in Deligne (1996) 3.8, the central extension E is then determined up to unique isomorphism by the “commutator” pairing $T_{\text{ad}}(k) \times T(k) \rightarrow A$. $\square$

Remark 5.9. We keep the notations of 4.1, with G simply-connected. The choice of a Borel subgroup B is not needed. By 5.3, a W -invariant quadratic form Q on Y defines a canonical multiplicative K_2 -torsor E on G . By restriction to T , we obtain a multiplicative K_2 -torsor E_T on T well defined up to unique isomorphism, and whose isomorphism class is given by Q . It defines a central extension

$$k^* \rightarrow E \rightarrow Y$$

with commutator map $(-1)^{B(y,z)}$. In 11.7, we will describe this central extension in terms of Q and of the root space decomposition of the Lie algebra of G .

6 Split semi-simple groups

We keep the notation of 4.1.

6.1

Let $(f_w)_{w \in W(1)}$ be a 1-cochain in the complex (4.3.2) for $j = 2$. Each f_w is an invertible function on BwB . Write f_i for $f_{w_0 s_i}$ and, as in the proof of 4.9, let f_i be the character of T such that $f_i(gt) = f_i(g)f_i(t)$. Whether (f_w) is a cocycle depends only on its image in the quotient (4.4.1) $_{n=2}$ of the complex (4.3.2). As in 4.4, we identify this image with a bilinear form $C \in X_{\text{sc}} \otimes X$ pairing Y_{sc} and Y . The form C is defined by $C(\alpha_i^\vee, y) = \langle f_i, y \rangle$. By 4.5, (f_w) is a cocycle if and only if the quadratic form $C(y, y)$ on Y_{sc} is W -invariant.

Until the end of this section, we assume that G is semi-simple. This means that Y_{sc} is of finite index in Y : $Y_{\text{sc}} \otimes \mathbb{Q} \xrightarrow{\sim} Y \otimes \mathbb{Q}$. Dually, $X \otimes \mathbb{Q} \xrightarrow{\sim} X_{\text{sc}} \otimes \mathbb{Q}$ and X_{sc} is of finite index in X .

Lemma 6.1. *Suppose that (f_w) is a cocycle and that E is the corresponding K_2 -torsor. The following conditions are then equivalent:*

- (i) the inverse image of E by $\text{pr}_1 : G \times T \rightarrow G$ is isomorphic to its inverse image by the multiplication map m ;
- (ii) the inverse image of E by $\text{pr}_2 : T \times G \rightarrow G$ is isomorphic to its inverse image by the multiplication map m ;
- (iii) the bilinear pairing C of Y_{sc} with Y extends to a bilinear pairing of Y with itself.

If the K_2 -torsor E on G admits a multiplicative structure, the torsors m^*E and $\text{pr}_1^*(E) + \text{pr}_2^*(E)$ on $G \times T$ are isomorphic. As, by 3.3, the restriction of E to T is a trivial torsor, 5.2 (i) then holds. The main result of this section is a converse to be proven in 5.7.

Proof (i) $\Leftrightarrow$ (iii). The right action of T respects the Bruhat stratification. The difference $\text{pr}_1^*E - m^*E$ is given by the difference of the inverse image cocycles. We have $f_i(gt)/f_i(g) = f_i(t)$ and this difference is the cocycle $\text{pr}_2^*(f_i)$ on $(Bw_0s_iB) \times T$. $\square$

Lemma 6.2. *If G is a product $G_1 \times G_2$, with $T = T_1 \times T_2$, the conditions of (i) (ii) (iii) 5.2 are equivalent to*

- (iv) the inverse image of E by $\text{pr}_2 : T_1 \times G \times T_2 \rightarrow G$ is isomorphic to its inverse image by the product map.

Lemma 6.3. *For $G = G_1 \times G_2$ as in 6.2, the functor of restriction to G_1 and G_2 is an equivalence from*

- (i) the category of pointed K_2 -torsors on G , for which the equivalent conditions of 5.3 hold,
- to
- (ii) the category of pairs (E_1, E_2) of pointed K_2 -torsors on G_1 and G_2 , for which the equivalent conditions of 5.2 hold.

Proposition 6.4. *If G is split semi-simple, a pointed K_2 -torsor E on G has at most one multiplicative structure. It has one if and only if the equivalent conditions of 5.2 hold.*

Proof. We have already shown, in the comment after 5.2, that if E has a multiplicative structure, 5.2 (i) holds.

Conversely, suppose that 5.2 (i) (ii) hold. Using that $t(gh)t^{-1} = (tg)(ht^{-1})$, one sees that 5.3(iv) holds for $m^*(E)$ on $G \times G$. Applying 6.3 one finds that there is a unique isomorphism

$$m : \text{pr}_1^*E + \text{pr}_2^*E \rightarrow m^*E$$

which extends the obvious isomorphisms on $G \times \{e\}$ and $\{e\} \times G$. It is the only possible multiplicative structure and it is a multiplicative structure, as the two isomorphisms (1.1.2) deduced from m on $G \times G \times G$ have the same restriction to $G \times e \times e$, $e \times G \times e$ and $e \times e \times G$ and, by (5.4), are characterized by those three restrictions. $\square$

6.2

The arguments leading to 6.4 extend to the case of G split semi-simple over a base S regular of finite type over a field.

Proposition 6.5. *If G is split semi-simple over S regular of finite type over a field, a pointed K_2 -torsor on G has at most one multiplicative structure. It has one if and only if the locally equivalent conditions of 5.2 hold locally on S .*

7 Split reductive groups

7.1

Let S be regular of finite type over a field k and let G be a split reductive group scheme over S . We fix a split maximal torus T . Let G_{der} be the commutator subgroup of G , G_{sc} be its simply-connected covering and G_{ad} be the adjoint group:

$$G_{\text{sc}} \rightarrow G_{\text{der}} \rightarrow G \rightarrow G_{\text{ad}}.$$

The inverse images of T in G_{der} and G_{sc} , and its image in G_{ad} , are split maximal tori in the respective reductive groups. Notation: $T_{\text{der}}, T_{\text{sc}}, T_{\text{ad}}$. With the notation (0.N.1), (0.N.2), the morphisms $T_{\text{sc}} \rightarrow T_{\text{der}} \rightarrow T \rightarrow T_{\text{ad}}$ induce W -equivariant morphisms $Y_{\text{sc}} \rightarrow Y_{\text{der}} \rightarrow Y \rightarrow Y_{\text{ad}}$ and $X_{\text{sc}} \leftarrow X_{\text{der}} \leftarrow X \leftarrow X_{\text{ad}}$. The group Y_{sc} (resp. X_{sc}) is the subgroup of Y (resp. X) generated by the coroots (resp. roots). The group Y_{der} is $Y_{\text{sc}} \otimes \mathbb{Q} \cap Y$, and Y_{sc} and Y_{der} are subgroups of finite index in Y_{ad} .

If E is a multiplicative K_2 -torsor on G , its restriction to T is a multiplicative torsor E_T on T . By the equivalence of categories 4.9, E_T gives (Q, E) , with Q a quadratic form on Y and E a central extension, on S , of Y by $\mathcal{O}_S^*$, obeying (33). Locally on S , any $w \in W$ can be lifted to G and, by 1.6, the automorphism w of T preserves the isomorphism class of E_T . It follows by transport of structure that the quadratic form Q is W -invariant.

The pullback E_{sc} of E to G_{sc} is a multiplicative torsor on G_{sc} . By 4.9, the corresponding quadratic form 4.7 on Y_{sc} is the restriction Q_{sc} of Q to Y_{sc} . The multiplicative torsor E_{sc} is determined up to unique isomorphism by Q_{sc} . Its restriction to T_{sc} gives rise to a central extension 3.15 of sheaves on S

$$\mathcal{O}_S^* \rightarrow E_{\text{sc}} \rightarrow Y_{\text{sc}} \quad (54)$$

depending only on Q_{sc} . As E restricted to T_{sc} is the pullback to T_{sc} of E_T , we have a commutative diagram

$$\begin{array}{ccccc} \mathcal{O}_S^* & \longrightarrow & E_{\text{sc}} & \longrightarrow & Y_{\text{sc}} \\ \parallel & & \downarrow j & & \downarrow \\ \mathcal{O}_S^* & \longrightarrow & E & \longrightarrow & Y \end{array} \quad (55)$$

Theorem 7.1. *Let G be a split reductive group scheme over S regular of finite type over a field. The construction 6.1 is an equivalence of categories from*

- (i) *the category of multiplicative K_2 -torsors E on G , to*
- (ii) *the category of triples (Q, E, j) , where*
 - (a) *Q is a W -invariant quadratic form on Y . By 4.16, the restriction of Q to Y_{sc} defines a central extension E_{sc} of Y_{sc} by $\mathcal{O}_S^*$;*
 - (b) *E is a central extension of Y by $\mathcal{O}_S^*$, obeying (3.11.1);*
 - (c) *$j : E_{\text{sc}} \rightarrow E$ makes the diagram (6.12) commute.*

The morphisms in the category of triples (Q, E, j) are as follows. There can be a morphism from (Q', E', j') to (Q'', E'', j'') only if $Q' = Q''$. If $Q' = Q''$, a morphism is an isomorphism f of central extensions

$$\begin{array}{ccccc} \mathcal{O}_S^* & \longrightarrow & E' & \longrightarrow & Y \\ \parallel & & \downarrow f & & \parallel \\ \mathcal{O}_S^* & \longrightarrow & E'' & \longrightarrow & Y \end{array}$$

compatible with j' and j'' : commutativity of the diagram

$$\begin{array}{ccc} E_{\text{sc}} & \xrightarrow{j'} & E' \\ & \searrow j'' & \downarrow f \\ & & E'' \end{array}$$

Proof of 6.2, when G is semi-simple. The group G is assumed to be semi-simple and for the ease of exposition, we begin by assuming that S is the spectrum of a field k .

We first show that any W -invariant $\mathbb{Z}$ -valued quadratic form Q on Y comes from some multiplicative K_2 -torsor on G . There exists a possibly nonsymmetric $\mathbb{Z}$ -valued bilinear form C on Y , such that $C(y, y) = Q(y)$. We have $C \in X \otimes X \subset X_{\text{sc}} \otimes X$ and there exists a 1-cochain (f_w) as in 5.1 giving rise to C . By 5.1, based on 4.5, this cochain is a cocycle. Let E be the corresponding K_2 -torsor, and choose a trivialization of E at $e \in G$. By 6.4, the pointed torsor E has a unique multiplicative structure. Applying 4.9 to the restriction of the multiplicative torsor E to G_{sc} , we obtain that the quadratic form 6.1 attached to E is Q .

As any W -invariant quadratic form on Y comes from some multiplicative K_2 -torsor, it suffices to prove 6.2 for the full subcategories of the categories described in (i) (ii) given by the condition $Q = 0$. Those categories can be incarnated as follows. For (i), if E is defined by a cocycle as in 5.1, giving rise to C in $X \otimes X$ by 5.5, the condition $Q = 0$ means that C is in the image of $\Lambda^2 X$. The category is incarnated by $(4.4.1)_{n=1}$, i.e. by the complex (4.4.2):

$$X \otimes k^* \rightarrow X_{\text{sc}} \otimes k^*. \quad (56)$$

For (ii), we have to consider the category of pairs (E, j) , where E is a commutative central extension of Y by k^* , and j is splitting of E over Y_{sc} . The extension E is commutative because $Q = 0$. This category is incarnated by (56) as well: an element $f \in X_{\text{sc}} \otimes k^* = \text{Hom}(Y_{\text{sc}}, k^*)$ defines the following object $[f]$: the trivial extension $k^* \times Y$ of Y by k^* , with the splitting f on Y_{sc} . An element g of $X \otimes k^* = \text{Hom}(Y, k^*)$ defines an isomorphism from $[f]$ to $[fg]$.

We leave it to the reader to check that the functor 6.2 is incarnated by the identity automorphism of (56).

Over a base S , one similarly finds that any Q can be locally obtained, and local incarnations of the subcategory of 6.2(i) (ii) where $Q = 0$ are given by

$$X \otimes \mathcal{O}_S^* \rightarrow X_{\text{sc}} \otimes \mathcal{O}_S^*. \quad (57)$$

□

Proof of 6.2 (general case). The quotient Y/Y_{der} is torsion free. It follows that Y_{der} is a direct factor in Y . We choose a decomposition $Y = Y_0 \oplus Y_{\text{der}}$. It gives us a product decomposition $T = T_0 \times T_{\text{der}}$, and G is the semi-direct product

$$G = T_0 \ltimes G_{\text{der}}. \quad (58)$$

We will apply 1.7 to the semi-direct product decomposition (58) of G . For this, even if we cared only for the case when S is the spectrum of a field, we need the semi-simple case of 6.2 over a more general base, to describe actions of T_0 on a central extension of G_{der} by $\mathcal{K}_2$.

The inner action of G on itself factors through G_{ad} . The action of T_0 on G_{der} is hence the composite of $q : T_0 \rightarrow T_{\text{ad}}$ and of the action of T_{ad} on G_{der} . We continue to write q for the induced map

$$q : Y_0 \rightarrow Y_{\text{ad}}. \quad (59)$$

Lemma 7.2. *Let E_{der} be a multiplicative K_2 -torsor on G_{der} , and let Q_{der} be the corresponding quadratic form on Y_{der} .*

- (i) If the action of T_0 on G_{der} can be lifted to E_{der} , the lifting is unique;
- (ii) A lifting exists if and only if the quadratic form Q_{der} can be extended to a W -invariant quadratic form on Y .

Lemma 7.3. *Let Q_{der} be a W -invariant quadratic form on Y_{der} . By 5.6, the restriction to Y_{sc} of the associated bilinear form extends to a pairing B of Y_{ad} with Y_{sc} .*

- (i) *The form Q_{der} extends to a W -invariant quadratic form on Y if and only if $B(q(y_0), y_{\text{sc}})$ extends to a pairing between Y_0 and Y_{der} .*
- (ii) *If Q_{der} extends to a W -invariant quadratic form on Y , the W -invariant extensions Q correspond one to one to quadratic forms on Y_0 , by $Q \mapsto Q|_{Y_0}$.*

The rest of the proof proceeds by using the dictionary 1.4 between multiplicative torsors and central extensions, and translating conditions (a) (b) (c) of 1.7 using 6.2 (semi-simple case) and 7.2. This concludes the proof of 7.1. $\square$

8 Reductive groups

8.1

Let S be regular of finite type over a field k and G be a reductive group scheme over S . Locally for the étale topology of S , G is split (SGA3 XXII 2.3): there exists a surjective étale map $u : S_0 \rightarrow S$ such that G over S_0 deduced from G by base change is a split reductive group over S_0 . By 3.6, the change of base from S to S_0 is an equivalence of categories from pointed K_2 -torsors on (G, e) to pointed K_2 -torsors on (G_0, e) provided with a descent data. Applying this to the G^m , and using that this equivalence of categories is compatible with pull-backs, one sees that giving a multiplicative K_2 -torsor on G amounts to giving one on G_0 , and a descent data from S_0 to S .

Suppose T is a maximal torus of G . As in 6.1, one deduces from G semi-simple group schemes G_{sc} , G_{der} and G_{ad} over S , with maximal tori T_{sc} , T_{der} and T_{ad} . With the notations (0.N.1), (0.N.2), let Q be a W -invariant quadratic form on Y , and Q_{sc} be the induced quadratic form on Y_{sc} . Locally on S for the étale topology, G and T , hence G_{sc} and T_{sc} , are split, and by 5.3, amplified by 4.8, Q_{sc} defines a multiplicative K_2 -torsor E_{sc} on G_{sc} . Locally on S for the étale topology, more precisely after any étale change of base $S' \rightarrow S$ such that T splits on S' , the multiplicative torsor E_{sc} , restricted to T_{sc} , defines by 4.9 a central extension of Zariski sheaves on S'

$$1 \rightarrow \mathcal{O}^* \rightarrow E_{\text{sc}} \rightarrow Y_{\text{sc}} \rightarrow 1, \quad (60)$$

obeying (3.11.1). It is functorial in S' , hence defines locally on S for the étale topology a central extension (60) of sheaves for the étale topology. To define a central extension of sheaves, it suffices to define it locally: we obtain a central extension (60) of étale sheaves on S obeying (3.11.1).

We can now consider triples (Q, E, j) as in 6.2, with all sheaves for the étale topology. For triples (Q, E, j) , étale descent is trivially valid: to give a triple on S amounts to giving one on S' , and a descent data. By descent, one hence deduces from 6.2:

Theorem 8.1. *By étale descent, the equivalence of categories 6.2 induces an equivalence of categories from*

- (i) *the category of multiplicative K_2 -torsors on G , to*
- (ii) *the category of triples (Q, E, j) as in 6.2, with all sheaves taken in the étale topology.*

8.2 Special cases

- (i) If G is semi-simple simply-connected, the data of (Q, E, j) reduces to that of Q : multiplicative K_2 -torsors on G have no nontrivial automorphisms, and are classified by W -invariant quadratic forms Q on Y .
- (ii) For G reductive, the group $\mathrm{Hom}(G, \mathcal{K}_2)$ of automorphisms of a central extension E of G by $\mathcal{K}_2$ is the group of automorphisms for the corresponding (E, j) . For variable S , they form the étale sheaf $\mathrm{Hom}(Y/Y_{\mathrm{sc}}, \mathcal{O}^*)$.
- (iii) Suppose $S = \mathrm{Spec}(k)$. A triple (Q, E, j) determines a multiplicative K_2 -torsor. This torsor in turn defines a central extension $E(k)$ of $G(k)$ by $K_2(k)$. Automorphisms of (Q, E, j) induce automorphisms of the central extension $E(k)$.

Proposition 8.2. *Fix S as in 7.1.*

- (i) *For $p : T \rightarrow S$ a torus over S , $p_* K_2$ is the sheaf of pairs (A, q) as in 4.3.*
- (ii) *The group of multiplicative sections of $\mathcal{K}_2$ over T is $H^0(S_{\mathrm{\acute{e}t}}, X \otimes \mathcal{O}^*)$.*
- (iii) *For tori T_1 and T_2 over S , with sheaves of character groups X_1 and X_2 , the group of bimultiplicative sections of $\mathcal{K}_2$ over $T_1 \times_S T_2$ is $H^0(S_{\mathrm{\acute{e}t}}, X_1 \otimes X_2)$.*

8.3 Special case

Suppose $S = \mathrm{Spec}(k)$ and let T_1 and T_2 be tori over S . For $\bar{k}$ a separable closure of k , the character groups of T_1 and T_2 are $\mathrm{Gal}(\bar{k}/k)$ -modules X_1 and X_2 . Bimultiplicative sections of K_2 are in bijection with the group of $\mathrm{Gal}(\bar{k}/k)$ -invariants in $X_1 \otimes X_2$. Such an invariant element B hence gives rise to a bimultiplicative map

$$(\ , \)_B : T_1(k) \times T_2(k) \rightarrow K_2(k). \quad (61)$$

This, completed by 7.6 below, answers the question 7.11 of Deligne (1996).

8.4

Fix G over S as in 7.1, and a multiplicative K_2 -torsor on G . We view it as an extension E of G by $\mathcal{K}_2$, viewed as sheaves on a big Zariski sites. If $C \subset G \times_S G$ is the subscheme of commuting elements, the commutator map $G \times G \rightarrow E$ induces a morphism from C to $\mathcal{K}_2$, i.e. an element

$$\mathrm{comm} \in H^0(C, \mathcal{K}_2). \quad (62)$$

For T a maximal torus in G , the restriction of comm to $T \times_S T$ is a bimultiplicative section of $\mathcal{K}_2$. To determine which, one may first localize for the étale topology. Applying 4.7, one sees that it corresponds by 8.2 to the bilinear form B associated to the quadratic form Q attached to the multiplicative torsor we started with.

In particular, for $S = \mathrm{Spec}(k)$, we have a central extension of $G(k)$ by $K_2(k)$ and on $T(k)$ the commutator map is (61).

9 Generic cocycles

9.1

Let k be a field and K be a sheaf on the big Zariski site of $\mathrm{Spec}(k)$. For X a reduced and irreducible scheme of finite type over k , with field of rational functions $k(X)$, we define $K(k(X))$

to be the fiber of K at the generic point of X . It is the inductive limit over U of $K(U)$, for U a nonempty open subscheme of X . For $K = \mathcal{K}_2$, we get $K_2(k(X))$.

Let G be a connected algebraic group over k . Each component G^p of the simplicial classifying space $BG = (G^{\Delta^p}/G)$ is geometrically irreducible and the face maps $\partial_i : G^p \rightarrow G^{p-1}$ map the generic point of G^p to the generic point of G^{p-1} . The generic points of the components G^p of BG hence form a strict (0.N.7) simplicial system of spectrum of fields $k(G^p)$.

Applying the functor K to the one obtains a strict cosimplicial abelian group. The complex of generic cochains is the corresponding complex

$$K(k(G^p)), \quad d = \sum (-1)^i \partial_i^*.$$

Construction 9.1. Assume that the field k is infinite, and that the connected algebraic group G over k is unirational. Then the category of multiplicative K -torsors on G is incarnated in degree 2 by the complex of generic cochains.

The unirationality assumption holds if G is reductive (SGA3 XIV 6.7).

In cohomological degree one, the construction boils down to the following statement: the map “restriction to the generic point of G ” is an isomorphism from the group of multiplicative sections of K over G to the group of elements of $K(k(G))$ for which $s(g_1 g_2) = s(g_1) + s(g_2)$ in $K(k(G \times G))$.

Proposition 9.2. *Under the assumptions of 9.1, the morphism of restriction to the generic points*

$$s(G\Gamma(G^p, K^q), \text{ for } p > 1) \rightarrow s(K^q(k(G^p)), \text{ for } p > 1) \quad (63)$$

is a quasi-isomorphism.

Corollary 9.3. *Under the assumptions of 9.1, the complexes*

$$K(k(G^p)) \quad (p > 1) \quad (64)$$

as well as

$$\text{coker}(K(k) \rightarrow K(k(G^p))) \quad (65)$$

are quasi-isomorphic to the complex (1.9.2) computing the cohomology of $H^(BG \bmod Be, K)$.*

Proof. The first quasi-isomorphism follows from 9.1 applied to G and to the trivial group.

For each p , the map $K(k) \rightarrow K(k(G^p))$ is injective: indeed, $K(k(G^p))$ is the inductive limit of the $K(U)$, for $U \subset G^p$ open and nonempty, and each map $K(k) \rightarrow K(U)$ is injective, because by unirationality of G , U has a rational point. It follows that the complex (65) is quasi-isomorphic to (64). $\square$

9.2

Under the assumptions of 9.1, it follows that the category of central extensions of $G(k)$ by A is incarnated in degree 2 by the complex $C_{\text{gen}}^*(G(k), A)$ of generic cochains. In particular, a generic 2-cochain c defines a central extension $E(c)$, unique up to unique isomorphisms. More precisely, c is the image of a cocycle $\tilde{c}$ in $C_1^*(G(k), A)$, defining $E(c)$ and if $\tilde{c}', \tilde{c}''$ are two such liftings of c , there exists a unique 1-cochain d , vanishing on some $U(k)$, with U open and Zariski dense in G , with coboundary $\tilde{c}'' - \tilde{c}'$. It defines an isomorphism from $E(c')$ to $E(c'')$.

10 Comparison with Galois cohomology

10.1

Let k be a field and n an integer prime to the characteristic. We write $\mathbb{Z}/n(1)$ for the group of roots of unity in a separable closure $\bar{k}$ of k , $\mathbb{Z}/n(d)$ for its d^{th} tensor power and $H^p(k, \mathbb{Z}/n(d))$ for a Galois cohomology group $H^p(\text{Gal}(\bar{k}/k), \mathbb{Z}/n(d))$. Let cl be the Kummer coboundary

$$\text{cl} : k^* \rightarrow H^1(k, \mathbb{Z}/n(1)).$$

By Tate (1970), if $u+v=1$, one has $\text{cl}(u) \cup \text{cl}(v) = 0$ in $H^2(k, \mathbb{Z}/n(2))$. As $K_2(k) = k^* \otimes k^* / \langle u \otimes v \mid u+v=1 \rangle$, the morphism $u, v \mapsto \text{cl}(u) \cup \text{cl}(v)$ induces a morphism

$$K_2(k) \rightarrow H^2(k, \mathbb{Z}/n(2)). \quad (66)$$

More generally, if n is invertible on a scheme S , $\mathbb{Z}/n(1)$ is the étale sheaf of n^{th} roots of 1 and $\mathbb{Z}/n(d)$ its d^{th} -tensor power.

10.2

Let G be a simply-connected absolutely simple algebraic group over a field k . Let T be a maximal torus, Y be its cocharacter group (an étale sheaf over $\text{Spec}(k)$, i.e. a $\text{Gal}(\bar{k}/k)$ -module) and Q be the minimal Weyl group invariant positive quadratic form on Y : one has $Q(\alpha^\vee) = 1$ for α a long root. Applying 7.3(i), we obtain a multiplicative K_2 -torsor E on G , hence a central extension of $G(k)$ by $K_2(k)$. Pushing it by (66), one obtains a central extension

$$H^2(k, \mathbb{Z}/n(2)) \rightarrow E(k) \rightarrow G(k). \quad (67)$$

Construction 10.1. We construct an isomorphism of the central extension (67) with the opposite of the canonical central extension of Deligne (1996) 3.6.

For simplicity, we assume k infinite. If k is finite, both central extensions are by the trivial group.

Lemma 10.2. *Both central extensions are deduced, as in 1.4, from a multiplicative H^2 -torsor on G .*

Lemma 10.3. *In the spectral sequence (9.4.1),*

- (i) $E_2^{0,q} = 0$
- (ii) *the morphism $H^4 \rightarrow E_2^{1,3}$ vanishes*
- (iii) *the resulting morphism $H^4 \rightarrow E_2^{2,2} = H^2(BG \bmod Be, H^2)$ carries c_Q to the class of the multiplicative H^2 -torsor on G we constructed.*

Lemma 10.4. (i) *In the spectral sequence (9.4.1), the morphism $H^4 \rightarrow E_2^{2,2} = H^2(BG \bmod Be, H^2)$ (defined thanks to 9.5 (i) (ii)) is an isomorphism.*

- (ii) *One has $E_2^{1,2} = H^1(BG \bmod Be, H^2) = 0$: multiplicative H^2 -torsors on G have no non-trivial automorphisms.*

Proof of 10.1. We will construct the isomorphism 9.3 as the image of an isomorphism between the multiplicative H^2 -torsors of 10.2. By 10.4 (ii), such an isomorphism, if it exists, is unique: we are left with showing that the two multiplicative H^2 -torsors of 10.2 are isomorphic. The group $H^2(BG \bmod Be, H^2)$ of isomorphism classes coincides with $H_{\text{ét}}^4(BG \bmod Be, \mathbb{Z}/n(2))$ (10.4 (i)). This group is $\mathbb{Z}/n$, independently of k . To prove the isomorphism of the two multiplicative H^2 -torsors of 10.2, it therefore suffices to do so after a field extension k'/k : we may and shall assume that G is split.

The two H^2 -torsors to be compared have the same functoriality in G . As in loc. cit. 3.7, we can hence reduce the general case to that of $G = \text{SL}(2)$, and reduce the case of $\text{SL}(2)$ to that of $\text{SL}(3)$. $\square$

10.3

Let T_1 and T_2 be two tori over a field k . We let Y_1 and Y_2 be their groups of cocharacters and view them as Galois-modules. To a bilinear form $B : Y_1 \otimes Y_2 \rightarrow \mathbb{Z}$, supposed to be Galois invariant, we have attached a bimultiplicative map (61)

$$T_1(k) \times T_2(k) \rightarrow K_2(k). \quad (68)$$

Fix an integer n prime to the characteristic of k . Composing (68) with the map (66): $K_2(k) \rightarrow H^2(k, \mathbb{Z}/n(2))$, we obtain

$$T_1(k) \times T_2(k) \rightarrow H^2(k, \mathbb{Z}/n(2)). \quad (69)$$

The short exact sequences of sheaves for the étale topology,

$$0 \rightarrow Y_i \otimes \mathbb{Z}/n(1) \rightarrow T_i \xrightarrow{n} T_i \rightarrow 1,$$

give rise to coboundaries

$$T_i(k) \rightarrow H^1(k, Y_i \otimes \mathbb{Z}/n(1)).$$

Taking a cup product, and applying B , we obtain

$$T_1(k) \times T_2(k) \rightarrow H^2(k, Y_1 \otimes Y_2 \otimes \mathbb{Z}/n(2)) \rightarrow H^2(k, \mathbb{Z}/n(2)). \quad (70)$$

Proposition 10.5. *The maps (69) and (70) are equal.*

11 Local and global fields

11.1

Let F be a local field: F a finite extension of $\mathbb{R}$, $\mathbb{Q}_p$ or $\mathbb{F}_p((T))$ for some p . The group $K_2(F)$ is the target of the universal symbol $\{, \}$, where a symbol with values in an abelian group A is a bimultiplicative map $F^* \times F^* \rightarrow A$, with $(x, y) = 0$ for $x + y = 1$. If one considers only locally constant symbols, there is also a universal one. Its target $K_2^{\text{cont}}(F)$ is trivial if $F \cong \mathbb{C}$, and is the group μ_F of roots of unity of F otherwise, with $\{, \} = \text{Hilbert symbol}$. See Moore (1968), Theorem 3.1.

Lemma 11.1. *Let X be a scheme of finite type over F and let s be a section of $\mathcal{K}_2$ over X . Evaluating s at each $x \in X(F)$, one obtains a map $s_1 : X(F) \rightarrow K_2(F) \rightarrow K_2^{\text{cont}}(F)$. The map s_1 is locally constant.*

Construction 11.2. Let E be a multiplicative K_2 -torsor on the algebraic group G over F . The resulting central extension

$$K_2^{\text{cont}}(F) \rightarrow \tilde{G}(F) \rightarrow G(F) \quad (71)$$

deduced from 1.4 by pushing by $K_2(F) \rightarrow K_2^{\text{cont}}(F)$ is a topological central extension of the locally compact group $G(F)$ by the finite (and discrete) group $K_2^{\text{cont}}(F)$.

For G absolutely simple and simply-connected, the central extensions (71) coincide with the central extensions constructed in Deligne (1996). See §9 and loc. cit. 5.4. For E a generator of the group of central extensions of G by $\mathcal{K}_2$, Prasad-Rapinchuk (1996) 8.4 proves that if F is nonarchimedean and G isotropic, the central extension (71) is a universal topological central extension of $G(F)$.

11.2

Let now F be a global field. For each place v of F , let F_v be the completion of F at v . Let μ_F , or simply μ , be the group of roots of unity of F . For each place v , if v is not a complex place, let μ_v be the group of roots of unity in F_v , and let $[\mu_v : \mu]$ be the index of μ in μ_v . For v a complex place, define μ_v to be trivial. The reciprocity law for the Hilbert symbols says that for x, y in F_v^* , $((x, y))_v$ is in the kernel of the map

$$(z_v) \mapsto \prod_v z_v^{[\mu_v : \mu]} \quad (72)$$

from $\bigoplus_v \mu_v$ to μ . By Moore (1968), Theorem 7.4 one has more precisely an exact sequence

$$K_2(F) \rightarrow \bigoplus_v K_2^{\text{cont}}(F_v) \xrightarrow{(72)} \mu \rightarrow 0. \quad (73)$$

From this, we will deduce that for G a linear algebraic group over F and E a multiplicative K_2 -torsor over G , one has a topological central extension of the adelic group $G(\mathbb{A})$ by μ , canonically split over $G(F)$:

$$\begin{array}{ccccc} & & G(F) & & \\ & & \downarrow & \nwarrow & \\ \mu & \longrightarrow & \tilde{G}(\mathbb{A}) & \longrightarrow & G(\mathbb{A}) \end{array} \quad (74)$$

12 Associated central extensions of dual characters groups

If G is a split simply connected group over a field k , with split maximal torus T , a Weyl group invariant integer-valued quadratic form Q on the dual character group Y of T determines a central extension of Y by k^* . This central extension is an ingredient in the classification 6.2 of central extensions of split reductive groups by K_2 . Our aim in this section is to describe it in terms of the root spaces decomposition of $\text{Lie}(G)$. The classification 6.2 depends on the choice of a split maximal torus in the split reductive group. We will explain also how the classifications for different maximal torus are related.

To simplify the exposition, we will work in this section over a field k .

12.1

We first consider the case when G is isomorphic to $\text{SL}(2)$. Choose a split maximal torus T . Let U^+ , U^- be the root subgroups normalized by T . We write α^+ and α^- for the corresponding roots. In the constructions which follow, U^+ and U^- play symmetric roles: the constructions are invariant by the group of automorphisms of (G, T) , i.e. by the normalizer of T in the adjoint group G_{ad} .

Define $U^{\pm*} := U^{\pm} - \{e\}$ and $N_1 := N(T) - T$. We will use the trijection of Tits (1966) 1.1 between U^{+*} , U^{-*} and N_1 . Trijection means: transitive system of isomorphisms. For the standard $\text{SL}(2)$ and its torus of diagonal matrices, the triples in trijection are the

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}. \quad (75)$$

The elements e^+ , e^- and n are in trijection if and only if $n = e^+ e^- e^+$. One then has

$$n = e^+ e^- e^+ = e^- e^+ e^-. \quad (76)$$

If e^+ , e^- and n are in trijection, so are their inverses and the conjugation by n exchanges e^+ and e^- .

Let E be the extension of G by K_2 defined by a quadratic form Q on Y . By 4.1, it is trivial on the unipotent subgroup U^+ (resp. U^-). We note $u \mapsto \tilde{u}$ the unique trivializing section. We define a section of E over N_1 by $n \mapsto \tilde{n} = \tilde{e}^- \tilde{e}^+$, for n decomposed as in (76).

Fix n in N_1 . The right translation by n induces an isomorphism of schemes from T to N_1 , and $h \mapsto s[n](h) := (\tilde{h}n)\tilde{n}^{-1}$ is a section $s[n]$ of E over T , trivial at e . One has

$$\tilde{n}^{-1} = (\tilde{n}^{-1})^\sim = (\alpha^{\vee+}(-1) \cdot n)^\sim = s[n](\alpha^{\vee+}(-1)). \quad (77)$$

Steinberg's cocycle $c \in H^0(T \times T, K_2)$ is the 2-cocycle defined by

$$s[n](h_1) \cdot s[n](h_2) = s[n](h_1 h_2) \cdot c(h_1, h_2). \quad (78)$$

Let us identify T with $\mathbb{G}_m$ and Y with $\mathbb{Z}$, using $\alpha^{\vee+}$. The cocycle $c(h_1, h_2)$ is trivial for $h_1 = e$ or $h_2 = e$. By 4.4 (ii), it is of the form $c(h_1, h_2) = q \cdot \{h_1, h_2\}$ for some integer q . Applying 4.7 and 4.9, one obtains that the quadratic form Q is qn^2 :

$$c(\alpha^{\vee+}(a), \alpha^{\vee+}(b)) = Q(\alpha^{\vee+})\{a, b\}. \quad (79)$$

The adjoint group G_{ad} acts on G , with T respecting T . One has $Y_{\text{ad}} = \frac{1}{2}Y$. The group U^+ is naturally a vector space, with

$$\alpha^{\vee+}(a)[e^+] = ae^+. \quad (80)$$

The action of G_{ad} on G extends uniquely to E (4.10). If $\alpha^{\vee+}(a)$ transforms n into n' , it will transform the section $s[n]$ of E over T into $s[n']$. By 5.6,

$$s[n'](t) = s[n](t) \cdot \{Q(\alpha^{\vee+})\alpha^{\vee+}(a), t\}. \quad (81)$$

The central extension E of Y by k^* attached to Q is obtained as follows. One extends the scalars from k to $k((t_*))$, one obtains from E a central extension of $T(k((t_*)))$ by $K_2k((t_*))$, one pulls it back by $Y \rightarrow T(k((t_*))) : y \mapsto y(t_*)$ and pushes it out by the residue map $\text{Res} : K_2k((t_*)) \rightarrow k^*$. We normalized the residue map to be the tame symbol

$$\text{Res}\{f, g\} = (-1)^{v(f)v(g)} \cdot (g^{v(f)} f^{-v(g)})(0). \quad (82)$$

For e^+ , e^- and n in Tits trijection, we define $[e^+]$ to be the image in E of $s[n](\alpha^{\vee+}(t_*))$. The element $[e^+]$ (resp. $[e^-]$) of E is a lifting to E of the generator $\alpha^{\vee+}$ (resp. $\alpha^{\vee-}$) of Y . By (80), (81) and (82), one has

$$[ae^+] = a^{-Q(\alpha^{\vee+})}[e^+] \quad (83)$$

in E . By (78), (79) and (82), one has

$$[e^+][e^-] = (-1)^{Q(\alpha^{\vee+})}. \quad (84)$$

12.2

Suppose that G is a split semi-simple simply-connected algebraic group, with split maximal torus T . We will use the notation of 4.1. For each root α , we denote by T_α the maximal torus $\alpha^\vee(\mathbb{G}_m)$ of the $\text{SL}(2)$ -subgroup S_α , and define $N_1^\alpha := N(T_\alpha) - T_\alpha$ in S_α . Let E be the central extension of G by K_2 defined by a quadratic form Q on Y , and E be the corresponding central extension of Y by k^* . Let us apply 11.1 to S_α .

Proposition 12.1. *With the notations of 11.2, for n_α in N_1^α , for t in T , and for $\tilde{t}$ a lifting of t in E , one has*

$$\text{int}[n_\alpha](\tilde{t}) = \tilde{t} \cdot s[n_\alpha](\alpha^\vee(t)^{-1}). \quad (85)$$

Lemma 12.2. *With the notation of 11.2 and 4.1, Y admits the following presentation. Generators: the coroots. Relations:*

$$s_\alpha(\beta^\vee) = \beta^\vee - \langle \alpha, \beta^\vee \rangle \alpha^\vee. \quad (86)$$

Proposition 12.3. *Under the assumption of 11.2, the central extension E of Y by k^* defined by Q (4.16) is characterized up to unique isomorphism as being a central extension of Y by k^* , provided for each root α with a map $[\]$ from U_α^* to the inverse image of $\alpha^\vee$, obeying (3.11.1), (11.2.1) and (11.6.1).*

The unicity claim results from 12.2.

This description extends by Galois descent to the non split case.

Proposition 12.4. *With the notations of 11.8, for n_α in $N_1^\alpha \subset G_{\text{sc}}$ and for t in T , lifted to $\tilde{t}$ in E , one has*

$$\text{int}[q(n_\alpha)](\tilde{t}) = \tilde{t} \cdot q(s[n_\alpha](\alpha^\vee(t)^{-1})). \quad (87)$$

Proposition 12.5. *With the notation of 11.10, let B be the bilinear form associated to Q .*

(i) *For y in Y , the linear form $B(y, \)$ on Y can be identified with a character of T . For t in T , and $\tilde{y}$ in E lifting y , the inner automorphism $\text{int}[t]$ acts on E by*

$$\text{int}[t](\tilde{y}) = \tilde{y} \cdot B(y, t)^{-1}. \quad (88)$$

(ii) *For e_α , $e_{-\alpha}$ and n_α in Tits trijection,*

$$\text{int}[n_\alpha](\tilde{y}) = \tilde{y} \cdot j([e_\alpha])^{-\alpha(y)} \cdot (-1)^{\epsilon(-\alpha(y))Q(\alpha^\vee)}. \quad (89)$$

13 Examples

13.1 $\text{SL}(2)$

Let G be a semi-simple simply-connected algebraic group over a field k , and let E be a multiplicative K_2 -torsor on G . Arguing as in the proof of 4.9, one finds the following description for the central extension $E(k)$ of $G(k)$ by $K_2(k)$: it is the group of automorphisms of the pair (space G , K_2 -torsor E on G), acting on G as a left translation.

Take $G = \text{SL}(2)$. Let U be the unipotent subgroup of matrices $\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$. The quotient $\text{SL}(2)/U$ is identified with the punctured affine plane $\mathbb{A}^2 - \{0\}$ by $g \mapsto g \cdot (1, 0)$. The projection $p : \text{SL}(2) \rightarrow \text{SL}(2)/U = \mathbb{A}^2 - \{0\}$ is a U -torsor. The pull-back functor p^* is hence an equivalence of categories from K_2 -torsors on $\mathbb{A}^2 - \{0\}$ to K_2 -torsors on $\text{SL}(2)$.

Proposition 13.1. *Central extensions E of $\text{SL}(2)$ by K_2 correspond one to one to isomorphism classes of K_2 -torsors on $\mathbb{A}^2 - \{0\}$. For the central extension E corresponding to P , $E(k)$ is the group of g in $\text{SL}(2, k)$ given with a lifting of their action on $\mathbb{A}^2 - \{0\}$ to an action on $(\mathbb{A}^2 - \{0\}, P)$.*

The punctured affine plane $\mathbb{A}^2 - \{0\}$ is covered by the two open sets $x \neq 0$ and $y \neq 0$. The group $H^1(\mathbb{A}^2 - \{0\}, \mathcal{K}_2)$ of isomorphism classes of K_2 -torsors on $\mathbb{A}^2 - \{0\}$ is $\mathbb{Z}$, generated by the Čech cocycle $\{x, y\}$ on the intersection $x \neq 0, y \neq 0$.

13.2

Let G be a reductive group over k , with maximal torus T , and $r : G \rightarrow \text{SL}(V)$ be a unimodular representation of G . Let E_0 be the central extension of $\text{SL}(V)$ by K_2 corresponding to the quadratic form Q_0 taking the value 1 on each coroot. By pullback to G , it gives a central extension E of G by K_2 . To compute the quadratic form Q on Y defined by E , we may first extend the scalars to a splitting field for T : we may and shall assume that T is split.

In characteristic zero, one can identify Y with a subgroup of the Lie algebra t of T , by attaching to $y : \mathbb{G}_m \rightarrow T$ the element $dy(1)$ of t . The quadratic form Q_0 is then induced by the quadratic form $\frac{1}{2} \text{Tr}(x^2)$ on the Lie algebra $\mathfrak{sl}(V)$, and the quadratic form Q is hence induced by the quadratic form $\frac{1}{2} \text{Tr}(r_*(x)^2)$ on the Lie algebra of G .

In any characteristic, similar arguments show that

$$Q(y) = \frac{1}{2} \sum m_i(y)^2,$$

where the sum runs over the weights m_i of V , counted with their multiplicity.

Suppose G split simple and simply connected, fix a simple root system, let $\alpha^\vee$ be the largest root and let $\alpha^\vee = \sum n_i \alpha_i^\vee$ be the decomposition of the coroot $\alpha^\vee$ into simple coroots. When V is the adjoint representation, $Q(\alpha^\vee)$ is twice the dual Coxeter number $h^\vee = 1 + \sum n_i$ of G . One can case by case check that for any V , each n_i divides $Q(\alpha^\vee)$ and that for V nontrivial of the smallest possible dimension, $Q(\alpha^\vee)$ is the l.c.m. of the n_i .

13.3 $\text{PGL}(n)$

Let T_1 be the image in $\text{PGL}(n)$ of the torus T of diagonal matrices in $\text{GL}(n)$. The cocharacter group Y of T_1 is the quotient of $Y = \mathbb{Z}^n$ by the diagonal $\mathbb{Z}$. The Weyl group invariant integer-valued quadratic forms Q on Y are the Weyl group invariant integer-valued quadratic forms on $\mathbb{Z}^n$, which vanish on the diagonal $\mathbb{Z}$. For n even, the minimal positive Q is

$$Q_1(y) := \sum_i y_i^2 - \frac{1}{n} \left(\sum_i y_i \right)^2.$$

For n odd, it is $\frac{1}{2}Q_1$. One has $Q_1(\alpha^\vee) = 2n$. The dual Coxeter number is n , and Q_1 is obtained as in 12.3, for the adjoint representation.

13.4 $\text{GL}(n)$

Let T be the split maximal torus of diagonal matrices in $\text{GL}(n)$. On $Y = \mathbb{Z}^n$, take the quadratic form

$$Q(y) = \frac{1}{2} \left(\sum_i y_i^2 - \left(\sum_i y_i \right)^2 \right).$$

The construction of the §7 of Deligne (1996) agrees with those of 11. Applying loc. cit. 7.10, one finds that for G an inner form of $\text{GL}(n)$, to give a central extension of G by K_2 corresponding to the form Q amounts to giving a representation V of G , which is a form of the defining representation of $\text{GL}(n)$. Such a representation exists only if G is isomorphic to $\text{GL}(n)$.

13.5

Over $\mathbb{R}$, any torus is a product of tori T of the following three types: $\mathbb{G}_m$, U_1 and $\text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$, with groups of points $\mathbb{R}^*$, the circle group and $\mathbb{C}^*$. The corresponding Y are $\mathbb{Z}$ with the trivial action of $\text{Gal}(\mathbb{C}/\mathbb{R})$, $\mathbb{Z}$ with the sign action: $F_\infty(n) = -n$ and $\mathbb{Z} \oplus \mathbb{Z}$, with F_∞ interchanging $e_1 = (1, 0)$ and $e_2 = (0, 1)$. For E a central extension of T by $\mathcal{K}_2$, we will determine the resulting topological central extensions of the groups of real points by $\mu_{\mathbb{R}} = \{\pm 1\}$ (10.3).

Proposition 13.2. *The morphism (12.6.1) is $(q, n) \mapsto q + n$.*

13.6

Let K be a local field, by which we mean: the field of fractions of a complete discrete valuation ring. Let V be the valuation ring, and k be the residue field. Let E be a central extension by K_2 of an algebraic group G over K . Pushing the central extension $E(K)$ by the residue map $K_2(K) \rightarrow k^*$, we obtain a central extension of $G(K)$ by k^* , which we denote by $\tilde{G}(k)$:

$$k^* \rightarrow \tilde{G}(k) \rightarrow G(K). \quad (90)$$

Proposition 13.3. *One has $\tilde{G}(k) \xrightarrow{\sim} \tilde{G}_0(k')^{\text{Gal}(K'/K)}$.*

13.7

Suppose that $K = k((t))$, and that G is absolutely simple and simply-connected, and splits over some unramified extension $k((t'))$ of K . In this situation, Garland (1980) constructed a central extension of $G(K)$ by k^* . It agrees with the central extension 12.8.1, for E defined by the quadratic form Q taking the value 1 on small coroots. Indeed, both obey 12.9, and for G split, Garland's extension is derived from Matsumoto's central extension by pushing by $K_2(K) \rightarrow k^*$.

Questions 13.4. (i) Suppose that G is reductive, and that E is given as in 8.1, for T a maximally split maximal torus of G . Suppose that G_V is given as in Bruhat-Tits (1984) 4.6. It would be interesting to compute the central extension $\tilde{G}_s$ in that case, especially for $G_V(V)$ a maximal bounded subgroup of $G(K)$, given by a vertex of the building of G .

(ii) Suppose that $V = k[[t]]$ and that G is a split reductive group. How to relate the point of view where central extensions of $G(K)$ by k^* are viewed as infinite dimensional groups of k , corresponding to affine Cartan matrices, with our point of view of central extensions by K_2 , classified by 6.2?

(iii) For $V = k[[t]]$, not all natural central extensions by k^* are captured by 12.8. For example, if G is the multiplicative group, $G(K) = K^*$ acts projectively on the semi-infinite exterior algebra of $k((t))$, relative to $k[[t]]$, giving rise to a central extension E of K^* by k^* . See E. Arbarello, C. De Concini and V. Kac, "The infinite wedge representation and the reciprocity law for algebraic curves", in: Theta functions, Proc. Symp. Pure Math, 49¹ AMS (1987), p. 171–190. The square of this commutator is the commutator for the central extension (90) defined by the tame symbol used as a cocycle. We expect that "natural" central extensions of $G(K)$ by k^* are attached to data as follows: a Weyl group and Galois group invariant integer-valued symmetric bilinear form B on Y , even on Y_{sc} ; a central extension E of Y by $\mathbb{G}_m$, for which the commutator is $(y_1, y_2) = (-1)^{B(y_1, y_2) + \epsilon(y_1)\epsilon(y_2)}$ for $\epsilon(y) \equiv B(y, y) \pmod{2}$; $j : E_{\text{sc}} \rightarrow E$ as in 6.2.

13.8

Let X be a projective and smooth curve over a field k . Let K be its field of rational functions. For x a closed point of X , let V_x be the completion of the local ring of X at x , and K_x be its field of fractions, the completion of K at x .

For G a linear algebraic group over K , we write $G(\mathbb{A})$ for the restricted product of the $G(K_x)$. It is also the group of $\mathbb{A}$ -points of G , for $\mathbb{A} = \prod'_x K_x$.

Let E be a central extension of G by K_2 . For some Zariski dense open subset U of X , G extends as an affine group scheme of finite type G_U over U , and E as a multiplicative K_2 -torsor over G_U . By extension of scalars to K_x , E defines central extensions (90) of $G(K_x)$ by $k(x)^*$, split over $G(V_x)$ for x in U . For $U_1 \subset U$ Zariski dense, we get commutative diagrams with

inductive limit

$$\begin{array}{ccccccc}
1 & \longrightarrow & K_2(K) & \longrightarrow & \tilde{G}(K) & \longrightarrow & G(K) \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & \bigoplus_x k(x)^* & \longrightarrow & \tilde{G}(\mathbb{A}) & \longrightarrow & G(\mathbb{A}) \longrightarrow 1.
\end{array} \tag{91}$$

By the reciprocity law for tame symbols, the first vertical arrow has its image in the kernel of the product of the norm maps $N : \bigoplus_x k(x)^* \rightarrow k^*$. Pushing by N , we hence get from (91) a central extension of $G(\mathbb{A})$ by k^* , canonically split over $G(K)$:

$$\begin{array}{ccccc}
& & G(K) & & \\
& & \downarrow & \swarrow & \\
k^* & \longrightarrow & \tilde{G}(\mathbb{A}) & \longrightarrow & G(\mathbb{A})
\end{array} \tag{92}$$

the geometric analogue of 10.4.3. For G simply-connected, it is also an analogue, in algebraic geometry, of the analytic constructions of Segal (1988) and Brylinski-McLaughlin (1994).

As in 12.13(iii), we don't obtain all natural diagrams (92) by this method.

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A counterexample to a 1961 “theorem” in homological algebra

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Abstract

In 1961, Jan-Erik Roos published a “theorem”, which says that in an $[AB4^*]$ abelian category, $\lim^1$ vanishes on Mittag-Leffler sequences. See Propositions 1 and 5 in [4]. This is a “theorem” that many people since have known and used. In this article, we outline a counterexample. We construct some strange abelian categories, which are perhaps of some independent interest.

These abelian categories come up naturally in the study of triangulated categories. A much fuller discussion may be found in [3]. Here we provide a brief, self contained, non-technical account. The idea is to make the counterexample easy to read for all the people who have used the result in their work.

In the appendix, Deligne gives another way to look at the counterexample.

0. Introduction

Abelian categories are old, venerable objects in mathematics, playing an important rôle. We are very accustomed to working with examples, such as categories of modules over a ring R , or more generally categories of sheaves of modules. Much of our intuition comes from these examples.

Here we will see an amusing construction of new and very different abelian categories. Let me explain the phenomenon we will observe.

Suppose $\mathcal{A}$ is an abelian category, satisfying $[AB3]$. That is, all small coproducts (and hence all small direct limits) exist in $\mathcal{A}$. Suppose further that the category $\mathcal{A}$ has enough injectives. Let us be given a sequence of monomorphisms in $\mathcal{A}$

$$a_0 \longrightarrow a_1 \longrightarrow a_2 \longrightarrow \cdots$$

Because the category $\mathcal{A}$ has enough injectives, there is an injective object i and an embedding $a_0 \hookrightarrow i$. Because i is injective and the map $a_0 \rightarrow a_1$ is a monomorphism, the map $a_0 \rightarrow i$ factors as $a_0 \rightarrow a_1 \rightarrow i$. Because $a_1 \rightarrow a_2$ is a monomorphism, the map $a_1 \rightarrow i$ factors as $a_1 \rightarrow a_2 \rightarrow i$.

Proceeding by induction, we obtain for every n a map $a_n \rightarrow i$, which combine to give a map

$$\operatorname{colim}\{a_n\} \longrightarrow i.$$

In other words, the monomorphism $a_0 \rightarrow i$ factors as

$$a_0 \longrightarrow \operatorname{colim}\{a_n\} \longrightarrow i;$$

it follows that the map $a_0 \rightarrow \operatorname{colim}\{a_n\}$ is a monomorphism.

Thus, in an abelian category with enough injectives, given any sequence of monomorphisms

$$a_0 \longrightarrow a_1 \longrightarrow a_2 \longrightarrow \cdots$$

then the map $a_0 \rightarrow \operatorname{colim}\{a_n\}$ is a monomorphism.

Of course, even in the absence of injectives, this still often happens. For example, if $\mathcal{A}$ satisfies [AB5] (that is, filtered direct limits are exact), then the sequence of monomorphisms

$$\begin{array}{ccccccc} a_0 & \longrightarrow & a_1 & \longrightarrow & a_2 & \longrightarrow & \cdots \\ \downarrow & & \downarrow & & \downarrow & & \\ a_0 & \longrightarrow & a_1 & \longrightarrow & a_2 & \longrightarrow & \cdots \longrightarrow \operatorname{colim}\{a_n\} \end{array}$$

has a monomorphism for its direct limit; hence $a_0 \rightarrow \operatorname{colim}\{a_n\}$ is a monomorphism.

In this article, we will construct new and unusual abelian categories. In particular, we will construct an abelian category $\mathcal{A}$ satisfying [AB4] and [AB4*] (that is, coproducts and products exist and are exact), but in $\mathcal{A}$ we will construct a sequence of monomorphisms

$$a_0 \longrightarrow a_1 \longrightarrow a_2 \longrightarrow \cdots$$

so that $\operatorname{colim}\{a_n\} = 0$.

To realise just how counterintuitive this example is, the reader should check Proposition 5 in [4], or Lemma 1.15 on page 213 of [2]. What we have here amounts to a counterexample to the Propositions stated there. This point is discussed, in infinitely more detail, in Sect. A.5 of [3].

The abelian categories we produce and study are genuinely strange. They are not categories of sheaves on any site; they are not Grothendieck abelian categories, and neither are their duals. In the book [3], they are studied for the information they provide, in the study of triangulated categories. But since they are of some independent interest, and might well arise elsewhere, the author thought this brief note might be worthwhile. It contains a very brief sketch of some of the properties that make these unusual abelian categories interesting.

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2 The construction

We begin with two definitions.

Definition 2.1. Let S be an essentially small additive category. The category $\operatorname{Cat}(S^{\operatorname{op}}, \operatorname{Ab})$ is defined to be the category of all additive functors

$$F : S^{\operatorname{op}} \longrightarrow \operatorname{Ab}.$$

So far, we have done nothing unusual. The category $\operatorname{Cat}(S^{\operatorname{op}}, \operatorname{Ab})$ is an old friend, which can be expressed as a category of sheaves on a suitable site. In fact, it is very nearly a category of modules over a ring.

Definition 2.2. Let α be an infinite cardinal. Let S be an essentially small additive category, closed under the formation of coproducts of $\leq \alpha$ of its objects.

The category $\operatorname{Ex}(S^{\operatorname{op}}, \operatorname{Ab}) \subset \operatorname{Cat}(S^{\operatorname{op}}, \operatorname{Ab})$ is defined to be the full subcategory of all additive functors F , which take coproducts [of $\leq \alpha$ objects in S] to products of abelian groups.

Let us remind the reader what Definition 2.2 means. Let Λ be a set of cardinality $\leq \alpha$. Suppose we are given a family of objects in S , of the form $\{s_\lambda, \lambda \in \Lambda\}$. By the hypothesis on S , the coproduct exists in S ; there is a coproduct $\coprod_{\lambda \in \Lambda} s_\lambda$. For each $\lambda \in \Lambda$, the contravariant functor F gives a map

$$F\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow F(s_\lambda).$$

The universal property of the product assembles these to a map

$$F\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow \prod_{\lambda \in \Lambda} F(s_\lambda).$$

The hypothesis on F is that all such maps are isomorphisms.

Lemma 2.3. *Let α be an infinite cardinal. Suppose S is an essentially small additive category, containing coproducts for any collection of $\leq \alpha$ of its objects. The category $\text{Ex}(S^{\text{op}}, \text{Ab})$ is an abelian subcategory of $\text{Cat}(S^{\text{op}}, \text{Ab})$. That is, $\text{Ex}(S^{\text{op}}, \text{Ab})$ is an abelian category, and the inclusion*

$$\text{Ex}(S^{\text{op}}, \text{Ab}) \subset \text{Cat}(S^{\text{op}}, \text{Ab})$$

is an exact functor.

Proof. Suppose $F \rightarrow F'$ is a morphism in $\text{Ex}(S^{\text{op}}, \text{Ab})$. That is, F and F' are functors $S^{\text{op}} \rightarrow \text{Ab}$ taking coproducts of fewer than α objects to products, and $F \rightarrow F'$ is a natural transformation. We need to show that the kernel and cokernel of the natural transformation, which are clearly objects of the big category $\text{Cat}(S^{\text{op}}, \text{Ab})$, actually lie in the subcategory $\text{Ex}(S^{\text{op}}, \text{Ab})$.

Complete the map $F \rightarrow F'$ to an exact sequence in $\text{Cat}(S^{\text{op}}, \text{Ab})$

$$0 \longrightarrow K \longrightarrow F \longrightarrow F' \longrightarrow Q \longrightarrow 0.$$

Let $\{s_\lambda, \lambda \in \Lambda\}$ be a set of $\leq \alpha$ objects in S . Because F and F' lie in $\text{Ex}(S^{\text{op}}, \text{Ab})$, the natural maps

$$F\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow \prod_{\lambda \in \Lambda} F(s_\lambda), \quad F'\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow \prod_{\lambda \in \Lambda} F'(s_\lambda)$$

are both isomorphisms. We deduce that in the commutative square

$$\begin{array}{ccc} F\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) & \longrightarrow & F'\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \\ \downarrow \cong & & \downarrow \cong \\ \prod_{\lambda \in \Lambda} F(s_\lambda) & \longrightarrow & \prod_{\lambda \in \Lambda} F'(s_\lambda) \end{array}$$

the vertical maps are both isomorphisms. But Ab satisfies [AB4*]. Hence the product of the exact sequences

$$0 \longrightarrow K(s_\lambda) \longrightarrow F(s_\lambda) \longrightarrow F'(s_\lambda) \longrightarrow Q(s_\lambda) \longrightarrow 0$$

over $\lambda \in \Lambda$ is an exact sequence.

In the comparison map

$$\begin{array}{ccccccc} K\left(\coprod s_\lambda\right) & \longrightarrow & F\left(\coprod s_\lambda\right) & \longrightarrow & F'\left(\coprod s_\lambda\right) & \longrightarrow & Q\left(\coprod s_\lambda\right) \\ \downarrow & & \downarrow \cong & & \downarrow \cong & & \downarrow \\ \prod K(s_\lambda) & \longrightarrow & \prod F(s_\lambda) & \longrightarrow & \prod F'(s_\lambda) & \longrightarrow & \prod Q(s_\lambda) \end{array}$$

both the top and bottom rows are exact. It easily follows that the natural maps

$$K\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow \prod_{\lambda \in \Lambda} K(s_\lambda), \quad Q\left(\coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow \prod_{\lambda \in \Lambda} Q(s_\lambda)$$

are both isomorphisms. $\square$

This completes the construction. Out of any essentially small additive category S , closed under coproducts of $\leq \alpha$ objects, we have produced an abelian category $\text{Ex}(S^{\text{op}}, \text{Ab})$.

3 Properties of the construction

It is now incumbent on us to study the properties of this construction. We begin with the easy and the well-known.

Lemma 3.1. *Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Then the abelian category $\text{Ex}(S^{\text{op}}, \text{Ab})$ satisfies $[AB4^*]$; it contains arbitrary small products, and products are exact.*

Proof. Since products of functors taking coproducts to products also take coproducts to products, the product, in $\text{Cat}(S^{\text{op}}, \text{Ab})$, of a family of objects in the smaller $\text{Ex}(S^{\text{op}}, \text{Ab}) \subset \text{Cat}(S^{\text{op}}, \text{Ab})$ lies in $\text{Ex}(S^{\text{op}}, \text{Ab})$. It follows that not only do products exist in $\text{Ex}(S^{\text{op}}, \text{Ab})$; the inclusion into $\text{Cat}(S^{\text{op}}, \text{Ab})$ preserves them. Since products are exact in Ab , they are exact in $\text{Cat}(S^{\text{op}}, \text{Ab})$, and hence also in $\text{Ex}(S^{\text{op}}, \text{Ab})$. $\square$

Notation 3.2. Next we will want to study the existence and exactness of colimits in $\text{Ex}(S^{\text{op}}, \text{Ab})$. The colimits that obviously exist are the α -filtered colimits. Recall that a category I is called α -filtered if every subcategory $I' \subset I$ of cardinality $\leq \alpha$ can be embedded in a subcategory $I'' \subset I' \subset I$, where I'' has a terminal object. The colimit of any functor $I \rightarrow \mathcal{A}$, where I is α -filtered, is called an α -filtered colimit. It is very classical that α -filtered colimits, taken in $\text{Cat}(S^{\text{op}}, \text{Ab})$, of functors

$$I \longrightarrow \text{Ex}(S^{\text{op}}, \text{Ab})$$

actually lie in $\text{Ex}(S^{\text{op}}, \text{Ab})$. The colimits therefore exist in $\text{Ex}(S^{\text{op}}, \text{Ab})$, and agree with the colimits in $\text{Cat}(S^{\text{op}}, \text{Ab})$.

In this section, we will consider colimits both in $\text{Ex}(S^{\text{op}}, \text{Ab})$ and in $\text{Cat}(S^{\text{op}}, \text{Ab})$. This being the case, we need to adopt some notational conventions to stop us from getting confused. In this section, when we write $\varinjlim F_\mu$, then we assume that the F_μ form an α -filtered system. If all the F_μ 's lie in $\text{Ex}(S^{\text{op}}, \text{Ab})$, then it does not matter whether the colimit is being computed in $\text{Ex}(S^{\text{op}}, \text{Ab})$ or in $\text{Cat}(S^{\text{op}}, \text{Ab})$. The only colimits we will consider in this section which are not α -filtered are coproducts. Coproducts in $\text{Cat}(S^{\text{op}}, \text{Ab})$ (resp. in $\text{Ex}(S^{\text{op}}, \text{Ab})$) will be denoted

$$\coprod F_\mu \quad \text{resp.} \quad \coprod^{\text{Ex}} F_\mu.$$

Note that we do not yet know that the coproduct on the right exists.

Remark 3.3. Since S is an additive category, the representable functors $S(-, s)$ are additive. And representable functors always take coproducts to products. Therefore, all the functors $S(-, s)$ are objects in $\text{Ex}(S^{\text{op}}, \text{Ab})$.

The Yoneda map, which is usually written as a functor $S \rightarrow \text{Cat}(S^{\text{op}}, \text{Ab})$, can be factored

$$S \longrightarrow \text{Ex}(S^{\text{op}}, \text{Ab}) \subset \text{Cat}(S^{\text{op}}, \text{Ab}).$$

It is classical that $S(-, s)$ is a projective object in the large category $\text{Cat}(S^{\text{op}}, \text{Ab})$. It must therefore also be projective in the exact subcategory $\text{Ex}(S^{\text{op}}, \text{Ab})$. It is also a well-known consequence of Yoneda's lemma that every object in $\text{Cat}(S^{\text{op}}, \text{Ab})$ is the quotient of a direct sum of representables $S(-, s)$. We will next prove

Lemma 3.4. *Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Then the Yoneda map, $S \rightarrow \text{Ex}(S^{\text{op}}, \text{Ab})$, preserves coproducts of $\leq \alpha$ objects.*

Proof. Suppose $\{s_\lambda, \lambda \in \Lambda\}$ is a family of objects in S . Suppose that F is an object in $\text{Ex}(S^{\text{op}}, \text{Ab})$, and suppose that, for each $\lambda \in \Lambda$, we are given a morphism

$$S(-, s_\lambda) \longrightarrow F.$$

By Yoneda's lemma, each of the given maps $S(-, s_\lambda) \rightarrow F$ corresponds, uniquely, to an element $r_\lambda \in F(s_\lambda)$. This gives us an element,

$$r \in \prod_{\lambda \in \Lambda} F(s_\lambda) = F\left(\coprod_{\lambda \in \Lambda} s_\lambda\right).$$

The last equality, $\prod F(s_\lambda) = F(\coprod s_\lambda)$, is because $F \in \text{Ex}(S^{\text{op}}, \text{Ab})$.

Now applying Yoneda's lemma again, the above corresponds to a unique map

$$S\left(-, \coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow F.$$

In other words, any collection of $\leq \alpha$ maps in $\text{Ex}(S^{\text{op}}, \text{Ab})$,

$$S(-, s_\lambda) \longrightarrow F$$

factor through a unique map

$$S\left(-, \coprod_{\lambda \in \Lambda} s_\lambda\right) \longrightarrow F.$$

This precisely says that

$$S\left(-, \coprod_{\lambda \in \Lambda} s_\lambda\right) = \prod_{\lambda \in \Lambda}^{\text{Ex}} S(-, s_\lambda). \quad \square$$

Proposition 3.5. *Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Then the inclusion $\text{Ex}(S^{\text{op}}, \text{Ab}) \subset \text{Cat}(S^{\text{op}}, \text{Ab})$ has a left adjoint.*

In the interest of keeping this paper self contained, we give the entire proof. It is a modification of the argument in [1].

We want to produce a functor

$$L : \text{Cat}(S^{\text{op}}, \text{Ab}) \longrightarrow \text{Ex}(S^{\text{op}}, \text{Ab}),$$

left adjoint to the inclusion. That is, for every $F \in \text{Cat}(S^{\text{op}}, \text{Ab})$, we wish to produce an object $LF \in \text{Ex}(S^{\text{op}}, \text{Ab})$, so that for any $G \in \text{Ex}(S^{\text{op}}, \text{Ab})$,

$$\text{Cat}(S^{\text{op}}, \text{Ab})(F, G) = \text{Ex}(S^{\text{op}}, \text{Ab})(LF, G).$$

Let us begin by treating the special case where

$$F(-) = \coprod_{i \in I} S(-, s_i)$$

is the direct sum of representables (see Remark 3.3). In this special case, we define

$$LF(-) = \text{colim}_{I' \subset I, \#I' \leq \alpha} S\left(-, \coprod_{i \in I'} s_i\right).$$

The notation means that we take the colimit over all $I' \subset I$, where the cardinality of I' is $\leq \alpha$, and hence $\coprod_{i \in I'} s_i$ makes sense. Notice that the right hand side is an α -filtered colimit of objects in $\text{Ex}(S^{\text{op}}, \text{Ab})$, and hence lies in $\text{Ex}(S^{\text{op}}, \text{Ab})$. By Lemma 3.4, $S(-, \coprod_{i \in I'} s_i)$ is the coproduct in $\text{Ex}(S^{\text{op}}, \text{Ab})$ of the objects $S(-, s_i)$. Given any object $G \in \text{Ex}(S^{\text{op}}, \text{Ab})$, to give a map

$$S\left(-, \coprod_{i \in I'} s_i\right) \longrightarrow G(-)$$

is just to give, for every $i \in I'$, maps $S(-, s_i) \rightarrow G$. Putting this together, we have that for any $G \in \text{Ex}(S^{\text{op}}, \text{Ab})$

$$\begin{aligned} \text{Hom}(LF, G) &= \lim_{I' \subset I, \#I' \leq \alpha} \text{Hom}\left(S\left(-, \coprod_{i \in I'} s_i\right), G\right) \\ &= \lim_{I' \subset I, \#I' \leq \alpha} \prod_{i \in I'} \text{Hom}\{S(-, s_i), G\} \\ &= \prod_{i \in I} \text{Hom}\{S(-, s_i), G\} \\ &= \text{Hom}(F, G). \end{aligned}$$

That is, LF satisfies the required universal property.

But now note that, by Remark 3.3, every object $F \in \text{Cat}(S^{\text{op}}, \text{Ab})$ has a projective presentation

$$F'' \longrightarrow F' \longrightarrow F \longrightarrow 0,$$

with

$$F''(-) = \coprod_{i \in I} S(-, s_i), \quad F'(-) = \coprod_{j \in J} S(-, s_j).$$

Define LF to be the quotient in the exact sequence

$$LF'' \longrightarrow LF' \longrightarrow LF \longrightarrow 0,$$

with LF'' and LF' as above. For any object $G \in \text{Ex}(S^{\text{op}}, \text{Ab})$ we have a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(LF, G) & \longrightarrow & \text{Hom}(LF', G) & & \\ & & \downarrow \cong & & \downarrow \cong & & \\ 0 & \longrightarrow & \text{Hom}(F, G) & \longrightarrow & \text{Hom}(F', G) & & \\ & & \longrightarrow & \text{Hom}(LF'', G) & & & \\ & & & \downarrow \cong & & & \\ & & & \longrightarrow & \text{Hom}(F'', G) & & \end{array}$$

from which it immediately follows that $\text{Hom}(LF, G)$ is naturally isomorphic to $\text{Hom}(F, G)$. $\square$

Corollary 3.6. *Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Then the category $\text{Ex}(S^{\text{op}}, \text{Ab})$ satisfies [AB3]. It contains arbitrary small coproducts of its objects.*

Proof. Let L be the left adjoint to the inclusion $\text{Ex}(S^{\text{op}}, \text{Ab}) \subset \text{Cat}(S^{\text{op}}, \text{Ab})$. The existence of L is guaranteed by Proposition 3.5. Let $\{F_\mu, \mu \in M\}$ be a family of objects in $\text{Ex}(S^{\text{op}}, \text{Ab})$. Because coproducts exist in $\text{Cat}(S^{\text{op}}, \text{Ab})$, there is a coproduct in $\text{Cat}(S^{\text{op}}, \text{Ab})$. We denote it

$$\coprod_{\mu \in M} F_\mu.$$

The functor L is a left adjoint, and hence preserves coproducts. It follows that

$$L\left(\coprod_{\mu \in M} F_\mu\right)$$

is the coproduct in $\text{Ex}(S^{\text{op}}, \text{Ab})$, of $LF_\mu = F_\mu$. $\square$

Now make the following definition

Definition 3.7. Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Let $s \rightarrow s' \rightarrow s''$ be two morphisms in S , whose composite is zero. The sequence is called *exact* if it induces an exact sequence in the abelian category $\text{Ex}(S^{\text{op}}, \text{Ab})$. That is, if the sequence of functors

$$S(-, s) \longrightarrow S(-, s') \longrightarrow S(-, s'')$$

is exact.

Remark 3.8. Suppose the category $\text{Ex}(S^{\text{op}}, \text{Ab})$ satisfies [AB4]; that is, coproducts are exact. Then it would follow in particular that if we take a family of $\leq \alpha$ exact sequences in S , of the form

$$s_\lambda \longrightarrow s'_\lambda \longrightarrow s''_\lambda,$$

then the sequence

$$\coprod_{\lambda \in \Lambda} s_\lambda \longrightarrow \coprod_{\lambda \in \Lambda} s'_\lambda \longrightarrow \coprod_{\lambda \in \Lambda} s''_\lambda$$

is also exact in S . After all, by Lemma 3.4, coproducts in S agree with those in $\text{Ex}(S^{\text{op}}, \text{Ab})$.

There is therefore a necessary condition for the category $\text{Ex}(S^{\text{op}}, \text{Ab})$ to satisfy [AB4]. If all coproducts are to be exact in $\text{Ex}(S^{\text{op}}, \text{Ab})$, then at the very least the coproducts of $\leq \alpha$ objects must be exact in S . The next Proposition asserts that, under reasonable hypotheses, this necessary condition is also sufficient.

Proposition 3.9. *Let α be an infinite cardinal. Let S be an essentially small additive category, closed under coproducts of $\leq \alpha$ of its objects. Suppose the following two conditions hold*

2.9.1 Any morphism $s' \rightarrow s''$ in S may be completed to an exact sequence $s \rightarrow s' \rightarrow s''$.

2.9.2 The coproduct of any collection of $\leq \alpha$ exact sequences in S is exact in S .

Then the category $\text{Ex}(S^{\text{op}}, \text{Ab})$ satisfies [AB4].

Proof. Consider the following full subcategory $\mathcal{A}(S) \subset \text{Ex}(S^{\text{op}}, \text{Ab})$. An object $F \in \mathcal{A}(S)$ is any functor $F : S^{\text{op}} \rightarrow \text{Ab}$ admitting a presentation

$$S(-, s) \longrightarrow S(-, t) \longrightarrow F(-) \longrightarrow 0.$$

It is well-known that under Hypothesis 2.9.1 above, this is an abelian subcategory of $\text{Ex}(S^{\text{op}}, \text{Ab})$; the proof is basically the same as the proof showing that coherent sheaves on a noetherian scheme form an abelian category. The objects $F \in \mathcal{A}(S)$ will be referred to as “coherent functors”.

Next observe that any object of $\text{Ex}(S^{\text{op}}, \text{Ab})$ is an α -filtered colimit of coherent functors. Let F be an object of $\text{Ex}(S^{\text{op}}, \text{Ab})$. It admits a projective presentation in $\text{Cat}(S^{\text{op}}, \text{Ab})$

$$\coprod_{i \in I} S(-, s_i) \longrightarrow \coprod_{j \in J} S(-, s_j) \longrightarrow F(-) \longrightarrow 0.$$

But then F is the α -filtered colimit of all quotients

$$\coprod_{i \in I'} S(-, s_i) \longrightarrow \coprod_{j \in J'} S(-, s_j) \longrightarrow F'(-) \longrightarrow 0,$$

where the cardinalities of $I' \subset I$ and of $J' \subset J$ are bounded by α . Now the functor L has a right adjoint, and preserves colimits. Hence $LF = F$ is the α -filtered colimit of LF' , and LF' has a presentation

$$S\left(-, \coprod_{i \in I'} s_i\right) \longrightarrow S\left(-, \coprod_{j \in J'} s_j\right) \longrightarrow LF'(-) \longrightarrow 0.$$

A similar argument shows that any short exact sequence

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

in $\text{Ex}(S^{\text{op}}, \text{Ab})$ is an α -filtered colimit of short exact sequences in $\mathcal{A}(S)$; we leave the details to the reader. Hence to prove that coproducts of short exact sequences in $\text{Ex}(S^{\text{op}}, \text{Ab})$ are exact, it suffices to assume that all the sequences for which we take the coproduct lie in $\mathcal{A}(S) \subset \text{Ex}(S^{\text{op}}, \text{Ab})$. Furthermore, a coproduct over a set Λ is the α -filtered colimit of coproducts over subsets $\Lambda' \subset \Lambda$, of cardinality $\leq \alpha$. We may therefore assume the coproduct is over an index set Λ of cardinality $\leq \alpha$.

Assume therefore that Λ is a set of cardinality $\leq \alpha$, and for every $\lambda \in \Lambda$ we have a short exact sequence in $\mathcal{A}(S)$

$$0 \longrightarrow F_\lambda \longrightarrow G_\lambda \longrightarrow H_\lambda \longrightarrow 0.$$

Because F_λ and H_λ lie in $\mathcal{A}(S)$, each admits a presentation

$$S(-, f_\lambda^1) \longrightarrow S(-, f_\lambda^0) \longrightarrow F_\lambda \longrightarrow 0,$$

$$S(-, h_\lambda^1) \longrightarrow S(-, h_\lambda^0) \longrightarrow H_\lambda \longrightarrow 0.$$

By 2.9.1, these presentations may be continued to a projective resolution

$$\cdots \longrightarrow S(-, f_\lambda^2) \longrightarrow S(-, f_\lambda^1) \longrightarrow S(-, f_\lambda^0) \longrightarrow F_\lambda \longrightarrow 0,$$

$$\cdots \longrightarrow S(-, h_\lambda^2) \longrightarrow S(-, h_\lambda^1) \longrightarrow S(-, h_\lambda^0) \longrightarrow H_\lambda \longrightarrow 0.$$

By standard homological algebra, these may be combined to give a projective resolution of the short exact sequence

$$0 \longrightarrow F_\lambda \longrightarrow G_\lambda \longrightarrow H_\lambda \longrightarrow 0.$$

We want to show that the coproduct of these $\leq \alpha$ short exact sequences is exact. It suffices to show that the coproduct in $\text{Ex}(S^{\text{op}}, \text{Ab})$ of their resolutions is a resolution of a short exact sequence. But this is immediate from 2.9.2.

[This trick, of reducing statements about $\text{Ex}(S^{\text{op}}, \text{Ab})$ to α -filtered direct limits of statements about representables, is very much in the spirit of Gabriel and Ulmer [1]. See, for example, the proof of Satz 5.9 on page 60 of [1]. A completely different proof of [AB4], somewhat more explicit, may be found in [3], Lemma 6.3.2.] $\square$

4 A counterexample

In this section, we look at a special case of a category S . We let our cardinal α be $\aleph_0$, the first infinite cardinal. Now we define the category S .

Definition 4.1. Let S be the category whose objects are complete, non-archimedean, normed abelian groups of cardinality $\leq 2^{\aleph_0}$, and whose morphisms are the contractions.

Remark 4.2. Definition 4.1 is quite a mouthful, so let us paraphrase it. An object of S is an abelian group A of cardinality $\leq 2^{\aleph_0}$, having a norm map. That is

3.2.1 For every $a \in A$, there is a number $\|a\| \in \mathbb{R}$. These numbers satisfy the inequality $\|a\| \geq 0$, with equality if and only if $a = 0$.

3.2.2 The norm is non-archimedean. It satisfies the inequality

$$\|a - b\| \leq \max(\|a\|, \|b\|).$$

3.2.3 The group A is complete with respect to the metric induced by the norm.

The morphisms in the category S are the contractions. They are homomorphisms of abelian groups $f : A \rightarrow B$ satisfying

$$\|f(a)\| \leq \|a\|.$$

Lemma 4.3. *The category S contains coproducts of $\leq \aleph_0$ of its objects.*

Proof. Suppose we are given $\leq \aleph_0$ objects of S , that is countably many objects $\{A_0, A_1, A_2, \dots\}$. The A_i are all abelian groups of cardinality $\leq 2^{\aleph_0}$. Therefore the set theoretic product group

$$\prod_{i=0}^{\infty} A_i$$

has cardinality

$$\leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \times \aleph_0} = 2^{\aleph_0}.$$

Define a norm map on $\prod_{i=0}^{\infty} A_i$ by the formula

$$\left\| \prod_{i=0}^{\infty} a_i \right\| = \sup_{i=0}^{\infty} \|a_i\|.$$

This norm takes its value in $\mathbb{R} \cup \{\infty\}$. The coproduct of the objects A_i in the category S is the subset of all elements of the set theoretic product, which are sequences whose norm tends to zero. That is,

$$\prod_{i=0}^{\infty} A_i \subset \prod_{i=0}^{\infty} A_i,$$

and the condition for a sequence $\{a_0, a_1, a_2, \dots\} \in \prod_{i=0}^{\infty} A_i$ to lie in the smaller $\prod_{i=0}^{\infty} A_i$ is that

$$\lim_{i \rightarrow \infty} \|a_i\| = 0.$$

We need to establish that this satisfies the universal property of the coproduct.

Suppose for each $0 \leq i < \infty$ we have, in the category S , a map $f_i : A_i \rightarrow B$. That is, we have a contraction. Define

$$f : \prod_{i=0}^{\infty} A_i \longrightarrow B$$

by the formula

$$f(a_0, a_1, a_2, \dots) = \sum_{i=0}^{\infty} f_i(a_i).$$

This sum converges since as $i \rightarrow \infty$, we have first $\|a_i\| \rightarrow 0$, but as $\|f_i(a_i)\| \leq \|a_i\|$, we deduce $\|f_i(a_i)\| \rightarrow 0$. Since the norm is non-archimedean,

$$\left\| \sum_{i=m}^n f_i(a_i) \right\| \leq \sup_{i=m}^n \|f_i(a_i)\| \rightarrow 0$$

as $m, n \rightarrow \infty$. The partial sums form a Cauchy sequence, which converges in the complete metric space B .

The uniqueness of f is obvious. □

Lemma 4.4. *The category S is an additive category.*

Proof. Given two morphisms $f, g : A \rightarrow B$ in S , we form $f - g$ by the formula

$$\{f - g\}(a) = f(a) - g(a).$$

Since f and g are contractions, $\|f(a)\| \leq \|a\|$ and $\|g(a)\| \leq \|a\|$. This makes

$$\|\{f - g\}(a)\| = \|f(a) - g(a)\| \leq \max(\|f(a)\|, \|g(a)\|) \leq \|a\|.$$

Hence $f - g$ is a contraction, that is a morphism in S .

This gives the Hom-sets $S(A, B)$ the natural structure of abelian groups.

Now observe that by Lemma 4.3 the category S contains countable coproducts of its objects, hence certainly finite coproducts. The reader can easily check that finite coproducts, as given in the proof of Lemma 4.3, also satisfy the universal property of finite products. Hence the category S is additive. $\square$

Lemma 4.5. *The category S contains kernels for all its morphisms.*

Proof. Let $f : A \rightarrow B$ be a morphism in S . The set theoretic kernel of f , given the subspace norm in A , is a closed subgroup of A and hence complete. It is the categorical kernel. $\square$

Lemma 4.6. *Suppose we are given countably many morphisms in S*

$$\{f_i : A_i \rightarrow B_i \mid 0 \leq i < \infty\}.$$

The kernel of the coproduct map

$$\coprod_{i=0}^{\infty} f_i : \coprod_{i=0}^{\infty} A_i \longrightarrow \coprod_{i=0}^{\infty} B_i$$

is the coproduct of the kernels.

Proof. Both the kernel of the coproduct map and the coproduct of the kernels consist of sequences $\{a_0, a_1, a_2, \dots\}$, with $a_i \in A_i$, so that $\|a_i\| \rightarrow 0$ and $f_i(a_i) = 0$. $\square$

Proposition 4.7. *Let α be the cardinal $\aleph_0$. Then S is an additive category closed under coproducts of $\leq \alpha$ of its objects, any map $s' \rightarrow s''$ may be completed to an exact sequence $s \rightarrow s' \rightarrow s''$, and coproducts of $\leq \alpha$ exact sequences in S are exact.*

Proof. The fact that S is additive is Lemma 4.4. The fact that it is closed under countable coproducts is Lemma 4.3. That any map $s' \rightarrow s''$ may be completed to a short exact sequence $s \rightarrow s' \rightarrow s''$ follows from Lemma 4.5; more precisely, s may be chosen to be the kernel of $s' \rightarrow s''$. The only fact that we still have not completely proved is that coproducts of countably many exact sequences in S are exact.

Let $f : s' \rightarrow s''$ be a morphism in S , and let k be its kernel. As we said in the previous paragraph, the sequence

$$0 \longrightarrow k \longrightarrow s' \longrightarrow s''$$

is exact in S . After all, by the universal property of the kernel, the sequence

$$0 \longrightarrow S(-, k) \longrightarrow S(-, s') \longrightarrow S(-, s'')$$

is exact in $\text{Ex}(S^{\text{op}}, \text{Ab})$.

Now suppose that $s \rightarrow s' \rightarrow s''$ is an exact sequence in S . Then

$$S(-, s) \longrightarrow S(-, s') \longrightarrow S(-, s'')$$

is an exact sequence of functors. But if k is the kernel of $s' \rightarrow s''$ as above, then we have a map $k \rightarrow s'$ whose composite $k \rightarrow s' \rightarrow s''$ vanishes. By the exactness of

$$S(k, s) \longrightarrow S(k, s') \longrightarrow S(k, s'')$$

we deduce that it factors as

$$k \xrightarrow{f} s \longrightarrow s'.$$

Also, since k is the kernel of $s' \rightarrow s''$ and the composite $s \rightarrow s' \rightarrow s''$ vanishes, the map $s \rightarrow s'$ must factor uniquely as

$$s \xrightarrow{g} k \longrightarrow s'.$$

Because the composite

$$k \xrightarrow{f} s \xrightarrow{g} k \longrightarrow s'$$

is the inclusion $k \hookrightarrow s'$, it follows that $gf : k \rightarrow k$ is the identity. Thus the sequence

$$s \longrightarrow s' \longrightarrow s''$$

is a direct sum of the sequence

$$k \longrightarrow s' \longrightarrow s''$$

and the sequence

$$k' \longrightarrow 0 \longrightarrow 0.$$

Any exact sequence $s \rightarrow s' \rightarrow s''$ in S can be decomposed as a direct sum of a kernel, and a trivial exact sequence.

Now we need to show that a countable coproduct of exact sequences is exact. The above argument shows that it suffices to show that a countable coproduct of kernels is a kernel, and we showed that in Lemma 4.6. $\square$

Corollary 4.8. *It follows that $\text{Ex}(S^{\text{op}}, \text{Ab})$ is an abelian category (Lemma 2.3) satisfying $[AB_4^*]$ (Lemma 3.1) and $[AB_4]$ (Proposition 3.9). $\square$*

Construction 4.9. Consider now the sequence of objects and morphisms in S

$$\begin{array}{ccccccc} \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \cdots \\ \downarrow & & \downarrow p & & \downarrow p^2 & & \\ \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \cdots \end{array}$$

where $\mathbb{Z}_p$ is the p -adic numbers with the usual norm, and the connecting maps are multiplication by p . The Yoneda functor

$$S \longrightarrow \text{Ex}(S^{\text{op}}, \text{Ab})$$

takes this to a sequence in $\text{Ex}(S^{\text{op}}, \text{Ab})$. We remind the reader: the Yoneda functor takes an object $s \in S$ to the representable functor $S(-, s)$. In the rest of this section, we will freely confuse the sequence in S with its image in $\text{Ex}(S^{\text{op}}, \text{Ab})$.

Lemma 4.10. *The sequence*

$$\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \cdots$$

is a sequence of monomorphisms in $\text{Ex}(S^{\text{op}}, \text{Ab})$.

Proof. The kernel of $p : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ is trivial, and hence the map

$$S(-, \mathbb{Z}_p) \xrightarrow{p} S(-, \mathbb{Z}_p)$$

is injective. $\square$

Lemma 4.11. *The sequence in $\text{Ex}(S^{\text{op}}, \text{Ab})$*

$$\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \dots$$

has a vanishing colimit (and also a vanishing colim^1).

Proof. The colimit and colim^1 are, respectively, the cokernel and kernel of the map

$$\prod_{i=0}^{\infty} S(-, \mathbb{Z}_p) \xrightarrow{1-p\{\text{shift}\}} \prod_{i=0}^{\infty} S(-, \mathbb{Z}_p).$$

By Lemma 3.4, the natural map gives an isomorphism

$$\prod_{i=0}^{\infty} S(-, \mathbb{Z}_p) \longrightarrow S\left(-, \prod_{i=0}^{\infty} \mathbb{Z}_p\right);$$

in the commutative square below the vertical maps are isomorphisms

$$\begin{array}{ccc} \prod_{i=0}^{\infty} S(-, \mathbb{Z}_p) & \xrightarrow{1-p\{\text{shift}\}} & \prod_{i=0}^{\infty} S(-, \mathbb{Z}_p) \\ \downarrow \cong & & \downarrow \cong \\ S\left(-, \prod_{i=0}^{\infty} \mathbb{Z}_p\right) & \xrightarrow{1-p\{\text{shift}\}} & S\left(-, \prod_{i=0}^{\infty} \mathbb{Z}_p\right) \end{array}$$

It therefore suffices to show that the map

$$\prod_{i=0}^{\infty} \mathbb{Z}_p \xrightarrow{1-p\{\text{shift}\}} \prod_{i=0}^{\infty} \mathbb{Z}_p$$

is an isomorphism. But its inverse is given by

$$\{1 - p\{\text{shift}\}\}^{-1} = 1 + p\{\text{shift}\} + p^2\{\text{shift}\}^2 + \dots$$

and the right hand side clearly converges. □

Consider next the map of sequences in S

$$\begin{array}{ccccccc} A & = & \mathbb{Z}_p & \xrightarrow{1} & \mathbb{Z}_p & \xrightarrow{1} & \mathbb{Z}_p & \xrightarrow{1} & \dots \\ & & \downarrow & & \downarrow p & & \downarrow p^2 & & \\ & & \mathbb{Z}_p & & \mathbb{Z}_p & & \mathbb{Z}_p & & \\ B & = & \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \mathbb{Z}_p & \xrightarrow{p} & \dots \end{array}$$

If we apply the Yoneda functor $S \rightarrow \text{Ex}(S^{\text{op}}, \text{Ab})$ to this map of sequences, we get a monomorphism of sequences in the abelian category $\text{Ex}(S^{\text{op}}, \text{Ab})$. We can form the quotient, deducing a short exact sequence of sequences in $\text{Ex}(S^{\text{op}}, \text{Ab})$

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0,$$

with B the sequence of Lemma 4.11. We now prove

Proposition 4.12. *In the category $\text{Ex}(S^{\text{op}}, \text{Ab})$, there exists a sequence C of monomorphisms*

$$C_0 \longrightarrow C_1 \longrightarrow C_2 \longrightarrow \dots$$

with a non-zero colim^1 .

Proof. Let C be the sequence of monomorphisms in the short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

above, where B is the sequence of Lemma 4.11. Applying the derived functor of the colimit, we have an exact sequence

$$\operatorname{colim}^1 C \longrightarrow \operatorname{colim} A \longrightarrow \operatorname{colim} B.$$

By Lemma 4.11, $\operatorname{colim} B = 0$. But A is a constant sequence

$$S(-, \mathbb{Z}_p) \xrightarrow{1} S(-, \mathbb{Z}_p) \xrightarrow{1} S(-, \mathbb{Z}_p) \xrightarrow{1} \cdots$$

and hence $\operatorname{colim} A = S(-, \mathbb{Z}_p) \neq 0$. The exact sequence implies that $\operatorname{colim}^1 C \neq 0$. $\square$

Remark 4.13. The sequence B , of Lemma 4.11, is a counterexample to Proposition 5 in [4]. It is a sequence of monomorphisms, but the map $B_0 \rightarrow \operatorname{colim} B$ is not mono. In the absence of Proposition 5, Proposition 1 of [4] does not imply that colim^1 vanishes for sequences of monomorphisms. Proposition 4.12 gives an explicit example where this fails. We have an abelian category satisfying [AB4], and in it a sequence C of monomorphisms with a non-vanishing colim^1 .

5 Appendix, by P. Deligne*: a special case

4.1

We fix a ring with unit R . Modules are left R -modules. By a projective system we will mean a projective system indexed by the ordered set $\mathbb{N}$ of integers. If $M = (M_n)_{n \in \mathbb{N}}$ is a projective system of modules, with transition maps $\varphi : M_{n+1} \rightarrow M_n$, the derived projective limits $\lim^0 M = \lim M$ and $\lim^1 M$ of M are the kernel and cokernel of the morphism

$$1 - \varphi : \prod M_n \rightarrow \prod M_n : (x_n) \mapsto (x_n - \varphi(x_{n+1})), \quad (4.1.1)$$

while the $\lim^i M$ vanish for $i \geq 2$. We define $\mathcal{A}$ to be the category of projective systems of modules for which the derived projective limit vanishes, that is, for which (4.1.1) is invertible.

Example 5.1. If almost all M_n are zero, then M is in $\mathcal{A}$. Indeed, the filtration of M by the $\bigoplus_{n < N} M_n$ is then a finite filtration, it is stable by $1 - \varphi$, and $1 - \varphi$ induces the identity on the associated graded.

Example 5.2. A module V with a decreasing filtration F indexed by $\mathbb{N}$ defines the projective system of the F^n . This construction is an equivalence, noted $V \mapsto F(V)$, from the category of filtered modules (V, F) with F exhaustive ($F^0 = V$) to the category of projective systems of modules with injective transition maps. If F is an exhaustive filtration of V , then $F(V)$ is in $\mathcal{A}$ if and only if the filtration F is separated and complete, that is, $V \xrightarrow{\sim} \lim V/F^n$. Indeed, we have a short exact sequence of projective systems

$$0 \rightarrow F^n \rightarrow V \rightarrow V/F^n \rightarrow 0,$$

the constant projective system V has surjective transition maps, hence a vanishing $\lim^1$, and one applies the long exact sequence of $\lim^i$:

$$0 \rightarrow \lim F^n \rightarrow V \rightarrow \lim V/F^n \rightarrow \lim^1 F^n \rightarrow 0.$$

Fix M in $\mathcal{A}$. For $m \geq n$, we denote $\varphi_{n,m} : M_m \rightarrow M_n$ the iterated transition map. For (x_n) in M_n , the defining property of $(s_n) := (1 - \varphi)^{-1}((x_n))$ is that

$$s_n = \varphi(s_{n+1}) + x_n \quad (4.1.4)$$

If almost all x_n are zero, the sums $\sum_{m \geq n} \varphi_{n,m}(x_m)$ reduce to finite sums. They obey (4.1.4), hence

$$s_n = \sum_{m \geq n} \varphi_{n,m}(x_m) \quad (\text{when almost all } x_n \text{ are zero}). \quad (4.1.5)$$

In general, we define linear maps “ $\sum_{m \geq n}$ ” $\varphi_{n,m}(x_m) : \prod M_i \rightarrow M_n$, by

$$“\sum_{m \geq n}” \varphi_{n,m}(x_m) := (1 - \varphi)^{-1}((x_n)_{n \in \mathbb{N}})_n. \quad (4.1.6)$$

As the notation suggests, “ $\sum_{m \geq n}$ ” $\varphi_{n,m}(x_m)$ depend only on the x_m for $m \geq n$. This follows by linearity from the fact that if $x_i = 0$ for $i \geq n$, then s_n , given by (4.1.5), vanishes.

Proposition 5.3. *The category $\mathcal{A}$ is an abelian category in which small products and small coproducts exist and are exact, that is, AB_4 and AB_4^* hold.*

Proof. Let $f : M \rightarrow N$ be morphism in $\mathcal{A}$. Define K and C to be the projective systems of kernels and cokernels of the components $f_n : M_n \rightarrow N_n$ of f . They are the kernel and cokernel of f in the abelian category $\mathcal{P}$ of all projective systems of modules. By exactness of products in the category of modules, the products $\prod K_n$ and $\prod C_n$ are the kernel and cokernel of $(f_n) : \prod M_n \rightarrow \prod N_n$. The endomorphism (4.1.1) is functorial. As it is an automorphism for M and N , it is one for K and C , which hence are in $\mathcal{A}$. The formation of (4.1.1) is compatible with products. If $(M_a)_{a \in A}$ is a family of objects of $\mathcal{A}$, the projective system of products $\prod_a M_{a,n}$ is hence again in $\mathcal{A}$: the category $\mathcal{A}$ is a full subcategory of the abelian category $\mathcal{P}$ stable by kernels, cokernels and products. It inherits from $\mathcal{P}$ being an abelian category in which small products exist and are exact. A sequence in $\mathcal{A}$ is exact if and only if it is exact in $\mathcal{P}$.

Let I be the set of the functions $f : A \rightarrow \mathbb{N} \cup \{\infty\}$ such that for any n in $\mathbb{N}$, the set of a in A with $f(a) \leq n$ is finite. It is suggestive to write this condition $f(a) \rightarrow \infty$ for $a \rightarrow \infty$. For each n , let $I(n)$ be the subset of I consisting of those f which are $\geq n$. When taking colimits indexed by $I(n)$, $I(n)$ will be ordered by $\geq$. As, if f and g are in $I(n)$, so is their infimum, such colimits are filtering.

Lemma 5.4. *Let $(M_a)_{a \in A}$ be a family of objects of $\mathcal{A}$, and define $M_{\infty,a} := 0$. The projective system S of the*

$$S_n := \text{colim}_{f \in I(n)} \prod_a M_{a,f(a)}$$

is then in $\mathcal{A}$ and is the coproduct, in $\mathcal{A}$, of the M_a .

In the colimit, the transition map $\prod_a M_{a,f(a)} \rightarrow \prod_a M_{a,g(a)}$, defined for $g \leq f$, is the product of the $\varphi_{g(a),f(a)}$ (resp. 0) for $f(a) \neq \infty$ (resp. $f(a) = \infty$). The map $S_{n+1} \rightarrow S_n$ is induced by the inclusion $I(n+1) \subset I(n)$.

The components i_a of the natural morphism $i_a : M_a \rightarrow S$ are as follows: if $f \in I(n)$ is such that $f(a) = n$, $i_{a,n}$ is the composite

$$M_a \rightarrow \prod_b M_{b,f(b)} \rightarrow S_n.$$

As filtering colimits and products are exact in a category of modules, Lemma 5.4 implies the existence and exactness of small coproducts in $\mathcal{A}$, and its proof will complete that of 5.3.

Example 5.5. Suppose that each M_a has injective transition maps, hence is deduced as in 5.2 from a module V_a with an exhaustive, separated and complete filtration F_a . We define $F^\infty(V_a) := 0$. The module S_n is then the submodule of $\prod_a F^n(V_a)$ consisting of the (x_a) such that for some f in $I(n)$ each x_a is in $F^{f(a)}(V_a)$. It is the completed direct sum $\varprojlim_m \bigoplus_a F^n(V_a)/F^m(V_a)$, and the projective system S is deduced as in 5.2 from the completed direct sum of the V_a .

Proof of Lemma 5.4. We first prove that S is in $\mathcal{A}$. One has

$$S_n = \operatorname{colim}_{f \in I(n)} \prod_a M_{a,f(a)} = \operatorname{colim}_{(f_n) \in \prod I(n)} \prod_a M_{a,f_n(a)}.$$

We map I to $\prod I(n)$ by $f \mapsto (\sup(f, n))_{n \in \mathbb{N}}$. This map is cofinal (for the order $\geq$). Indeed, for (f_n) in $\prod I(n)$, the infimum f of the f_n is in I , and $f \geq \sup(f_n, n)$. The colimit over the product of the $I(n)$ can hence be replaced by a colimit over I :

$$S_n = \operatorname{colim}_{f \in I} \prod_a M_{a, \sup(f(a), n)} = \operatorname{colim}_{f \in I} \prod_a M_{a, \sup(f(a), n)} :$$

the morphism (4.1.1) for S is the colimit over f in I of the product of the morphisms (4.1.1) for the following projective systems:

$$\begin{aligned} & \text{for } f(a) \neq \infty : M_{a, \sup(f(a), n)} \\ & \text{for } f(a) = \infty : 0. \end{aligned} \tag{4.3.2}$$

For $f(a) \neq \infty$, the projective system (??) coincides with M_a for $n \geq f(a)$. As M_a is in $\mathcal{A}$, it follows from 5.1 that (??) is in $\mathcal{A}$ too, and its map (4.1.1) is invertible. By passage to the colimit in f , (4.1.1) is invertible for S , and S is in $\mathcal{A}$.

It remains to check that for X in $\mathcal{A}$, the map

$$u \longmapsto (\nu \circ i_a) : \operatorname{Hom}(S, X) \rightarrow \prod \operatorname{Hom}(M_a, X) \tag{5.1}$$

is bijective. We first prove injectivity, i.e., that $\nu : S \rightarrow X$ is uniquely determined by the $\nu_a := \nu \circ i_a : M_a \rightarrow X$.

Fix f in $I(n)$ and x_a in $M_{a,f(a)}$ with image $\bar{x}$ in S_n . We have then

$$\bar{x} = \text{“} \sum \text{”} \sum_{f(a)=m} i_{a,m}(x_a), \tag{5.2}$$

where the inner sum is a finite sum in S_m , and the outer “sum” is given by (4.1.6). Indeed, for $m \geq n$, let $x_{\geq m}$ in $\prod_a M_{a, \sup(f(a), m)}$ have as components x_a , for $f(a) \geq m$, and 0 otherwise. Let $\bar{x}_{\geq m}$ be its image in S_m . We have

$$\bar{x}_{\geq m} = \varphi(\bar{x}_{\geq m+1}) + \sum_{f(a)=m} i_a(x_a)$$

and, as $\bar{x}_{\geq n} = \bar{x}$, the claim follows. The “sums” (4.1.6) being functorial, we get

$$\nu(\bar{x}) = \text{“} \sum \text{”} \sum_{f(a)=m} \nu_{a,m}(x_a), \tag{5.3}$$

computing ν in terms of the ν_a .

Given a family of morphisms $\nu_a : M_a \rightarrow X$, (5.3) defines a morphism $\nu : S \rightarrow X$ which induces the ν_a . To prove the surjectivity of (5.1), one has to check that this definition of ν is legitimate, i.e., that for $f \geq g$ in $I(n)$, the diagram

$$\begin{array}{ccc} \prod_a M_{a,f(a)} & \longrightarrow & \prod_a M_{a,g(a)} \\ & \searrow & \swarrow \\ & X_n & \end{array}$$

is commutative. Fix (x_a) in $\prod_a M_{a,f(a)}$. For $p \geq q$, let y_{pq} in X_p be the sum of the $\nu_{a,p}(x_a)$ for $f(a) = p$ and $g(a) = q$. For each fixed q , the set of a for which $g(a) = q$ is finite, hence only finitely many y_{pq} are not zero. We have to check that

$$\text{“} \sum \text{”} \sum_{q \leq n} y_{p,q} = \text{“} \sum \text{”} \sum_{q \geq n} \varphi_{q,p}(y_{p,q}).$$

The difference is “ $\sum_{p \geq n}$ ” applied to the sum in X_n

$$\begin{aligned} & \sum_{q \leq n} (y_{p,q} \text{ in } X_p) - (\varphi_{q,p}(y_{p,q}) \text{ in } X_q) \\ &= (1 - \varphi) \left(\sum_{p \geq i > q} (\varphi_{i,p}(y_{p,q}) \text{ in } X_i) \right) \end{aligned}$$

This sum is $(1 - \varphi)$ applied to an element of $\prod_{p > n} X_p$, and its “ $\sum_{p \geq n}$ ” hence vanishes. $\square$

4.4

For a an integer, define the filtered module $R(a)$ to be R , purely in filtration a . By 5.1 or 5.2, the corresponding projective system T_a is in $\mathcal{A}$. It is given by $T_{a,n} = R$ for $n \leq a$, 0 otherwise. The morphism $R(a) \rightarrow R(a+1)$ which is the identity on the underlying modules induces a monomorphism of projective systems $T_a \rightarrow T_{a+1}$:

$$\begin{array}{ccccccccccc} T_a & : & \cdots & \rightarrow & 0 & \rightarrow \cdots \rightarrow & 0 & \rightarrow & R & \rightarrow \cdots \rightarrow & R \\ & & & & \downarrow & & \downarrow & & \downarrow & & \\ T_{a+1} & : & \cdots & \rightarrow & 0 & \rightarrow \cdots \rightarrow & 0 & \rightarrow & R & \rightarrow \cdots \rightarrow & R \end{array}$$

Proposition 5.6. *In the category $\mathcal{A}$, $\text{colim } T_a = 0$.*

Proof. One has to show that for any X in $\mathcal{A}$, $\varprojlim_a \text{Hom}(T_a, X) = 0$. Indeed,

$$\text{Hom}(T_a, X) = X_a, \quad (4.4.2)$$

and $\varprojlim_a X_a = 0$. $\square$

Remark 5.7. As in 3.12, one can deduce from 5.6 the existence of inductive sequences with injective transition maps in $\mathcal{A}$, for which the left derived colim^1 does not vanish. Example: the inductive system of the coker($T_0 \rightarrow T_a$). In the opposite category, this becomes a projective system with surjective transition maps for which lim^1 does not vanish.

4.5

We now assume that R is a field. In 1.2, we take $\alpha = \aleph_0$. Let S be the category of filtered modules (V, F) as in 5.2, with F exhaustive, separated and complete, and in addition such that each $\text{Gr}_F^n(V)$ is of countable dimension. Countable coproducts exist in S : the coproduct in S of a countable family of objects (V_i, F_i) of S is the completed direct sum

$$\widehat{\bigoplus_i V_i} = \varprojlim_n \bigoplus_i V_i / F_i^n.$$

We found the category $\mathcal{A}$ by unravelling the definition of $\text{Ex}(S^{\text{op}}, \text{Ab})$. With the notation $R(n)$ of 4.4, if T is an additive contravariant functor from S to Ab , each $T(R(n))$ has a natural structure of R -vector space. Indeed, in S , $R(n)$ has a right R -module structure. We define $M(T)$ to be the projective system of the vector spaces $T(R(n))$.

Proposition 5.8. *With the notation above, the functor $T \mapsto M(T)$ induces an equivalence of categories from $\text{Ex}(S^{\text{op}}, \text{Ab})$ to $\mathcal{A}$.*

Proof. We first prove that when T is in $\text{Ex}(S^{\text{op}}, \text{Ab})$, $M(T)$ is in $\mathcal{A}$. Indeed,

$$M_n(T) = \prod_{\alpha} T(R(n)) = T\left(\prod_{\alpha} R(n)\right),$$

the maps $\coprod_{\alpha} R(n) \rightarrow \coprod_{\alpha} R[t]$, $x \mapsto xt^n$, identify the ordinary direct sum of the $R(n)$ with $R[t]$, provided with the t -adic filtration, and the completed direct sum $\widehat{\bigoplus_n} R(n)$ is $R[[t]]$, with the t -adic filtration. The endomorphism (4.1.1) for $M(T)$ is the image by T of the endomorphism $1 - t$ of $R[[t]]$. As the latter is invertible, $M(T)$ is in $\mathcal{A}$.

Let S_0 be the full subcategory of S consisting of the finite sums of $R(n)$ (the (V, F) in S with V finite dimensional). Evaluation at the $R(n)$ is an equivalence from the category $\text{Hom}_{\text{add}}(S_0^{\text{op}}, \text{Ab})$ of contravariant additive functors from S_0 to Ab , to the category $\mathcal{P}$ of projective systems. The functor $T \mapsto M(T)$, from $\text{Ex}(S^{\text{op}}, \text{Ab})$ to $\mathcal{A}$, becomes in this way the functor $T \mapsto T_0$ of restriction to S_0 .

The construction 5.2 turns S into a full subcategory of $\mathcal{A}$, and, by 5.5, countable direct sums in S are direct sums in $\mathcal{A}$. If to each A in $\mathcal{A}$ we attach the restriction to S of the representable functor h_A , we hence obtain

$$\mathcal{A} \rightarrow \text{Ex}(S^{\text{op}}, \text{Ab}).$$

By (4.4.2), the composite $\mathcal{A} \rightarrow \text{Ex}(S^{\text{op}}, \text{Ab}) \rightarrow \mathcal{A}$ is isomorphic to the identity. It remains to check that the composite $\text{Ex}(S^{\text{op}}, \text{Ab}) \rightarrow \mathcal{A} \rightarrow \text{Ex}(S^{\text{op}}, \text{Ab})$ is isomorphic to the identity as well. This composite sends a functor T to the functor

$$V \mapsto \text{Hom}(h_V^0, T_0)$$

(Hom in $\text{Hom}_{\text{add}}(S_0, \text{Ab})$). By Yoneda's lemma, the functor T can be identified with $V \mapsto \text{Hom}(h_V, T)$. Restriction to S_0 defines

$$T(V) = \text{Hom}(h_V, T) \rightarrow \text{Hom}(h_V^0, T_0), \quad (4.5.2)$$

and it remains to check that for V in S and T in $\text{Ex}(S^{\text{op}}, \text{Ab})$, (4.5.2) is an isomorphism. In other words, a system of $t_{X,e} \in T(X)$, functorial in X , $e : X_0 \rightarrow V$, X in S and $e : X \rightarrow V$, should come from a unique $t \in T(V)$.

For each n , let us lift in $F^n(V)$ a basis $(e_{n,\alpha})_{\alpha \in A_n}$ of F^n/F^{n+1} . This gives us morphisms $(e_{n,\alpha}) : R(n)_{\alpha} \rightarrow V$, where $R(n)_{\alpha}$ is a copy of R . The resulting morphism

$$\coprod_{n,\alpha} R(n)_{\alpha} \rightarrow V$$

is the countable coproduct in S , indexed by $A = \coprod_n A_n$, of 1-dimensional objects. By assumption, $T(V) \xrightarrow{\sim} \prod_{n,\alpha} T(R(n)_{\alpha})$. The injectivity of this map proves that of (4.5.2).

If $(t_{X,e})$ is in $\text{Hom}(h_V^0, T_0)$, the $t_{R(n)_{\alpha}, e(n,\alpha)}$ come from a unique $t \in T(V)$. To check surjectivity of (4.5.2), it remains to see that if the $t_{R(n)_{\alpha}, e(n,\alpha)}$ are zero, so are all $t_{X,e}$, and it suffices to check this when X is a $R(k)$. The map $e : R(k) \rightarrow V$ is a sum $f_n + f'_n$, where f_n factors through a finite sum of $R(n)_{\alpha}$. For $n \geq k$, the morphism f'_n factors through $e'_n : R(n) \rightarrow V$.

The difference of $e'_n : R(n) \rightarrow V$ and of $R(n) \xrightarrow{\varphi} R(n+1) \xrightarrow{e'_{n+1}} V$ factors through a finite sum of $R(n)_{\alpha}$, hence $t_{R(n), e'_n - e'_{n+1} \circ \varphi} = 0$ and $\varphi(t_{R(n+1), e'_{n+1}}) = t_{R(n), e'_n}$. As $\lim_n R(n) = 0$, it follows that the $t_{R(k), e'_n}$ and in particular $t_{R(k), e}$ are zero. $\square$

$\square$

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On the exceptional series, and its descendants

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Abstract

Many of the striking similarities which occur for the adjoint representation of groups in the exceptional series (cf. [1–3]) also occur for certain representations of specific reductive subgroups. The tensor algebras on these representations are easier to describe (cf. [4,5,7]), and may offer clues to the original situation.

The subgroups which occur form a Magic Triangle, which extends Freudenthal’s Magic Square of Lie algebras. We describe these groups from the perspective of dual pairs, and their representations from the action of the dual pair on an exceptional Lie algebra.

La série exceptionnelle, et sa descendance

Résumé

Les articles [1–3] exhibent des ressemblances entre les propriétés des représentations adjointes des groupes de la série exceptionnelle. Nous obtenons des ressemblances analogues pour certaines représentations préférées de séries de sous-groupes. L’algèbre tensorielle de ces représentations est plus accessible (cf. [4,5,7]). Ceci pourrait aider à comprendre ce qui se passe.

Les sous-groupes en question forment un «triangle magique» qui prolonge le carré magique d’algèbres de Lie de Freudenthal. Nous décrivons ces sous-groupes en termes de paires duales, et leur représentations préférées en termes de leur action sur l’algèbre de Lie du groupe de la série exceptionnelle ambiant.

We consider the series of algebraic groups

$$e \subset A_1 \subset A_2 \subset G_2 \subset D_4 \subset F_4 \subset E_6 \subset E_7 \subset E_8, \quad (1)$$

where an indexed capital letter stands for the corresponding simple simply connected complex group. Each group is a subgroup of the next, the inclusion morphism being unique up to conjugacy. The inclusions of the trivial group e in A_1 in A_2 and of E_6 in E_7 in E_8 are given by inclusions of Dynkin diagram. The long roots of G_2 (resp. F_4) form a root system of type A_2 (resp. D_4), providing $A_2 \subset G_2$ and $D_4 \subset F_4$. Finally, G_2 (resp. F_4) is the subgroup of D_4 (resp. E_6) fixed by the group S_3 (resp. S_2) of automorphisms respecting a pinning (épinglage).

Table 1. – The centralizer of H in K .

	A_1	A_2	G_2	D_4	F_4	E_6	E_7	E_8
A_1	$\leftarrow$						$E_7 \supset \mu_2$	$\mathbb{G}_m$
A_2		$\leftarrow$				$E_6 \supset \mu_3$		A_1
G_2			$\leftarrow$		$F_4 \supset 1$		μ_2^2	$\mathbb{G}_m^2$
D_4				$\leftarrow$	$D_4 \supset \mu_2^2$	A_1/μ_2	A_2	C_3
F_4					$\leftarrow$	$G_2 \supset 1$	μ_3	μ_2^2
E_6						$\leftarrow$	$A_2 \supset \mu_3$	$\mathbb{G}_m$
E_7							$\leftarrow$	$A_1 \supset \mu_2$
E_8								$\leftarrow$
ν :	1/3	1/2	2	3	2/3	1	3/2	5

The centralizers in E_8 of the subgroups (1) form (a conjugate of) the same series of subgroups of E_8 , taken in the reverse order, giving rise to the dual pairs (e, E_8) , (A_1, E_7) , (A_2, E_6) , (G_2, F_4) and (D_4, D_4) in E_8 .

For each pair $H \subset K$ of distinct subgroups in (1), Table 1 gives the centralizer of H in K . Rows are indexed by H , columns by K . If H' and K' are the centralizers of H and K in E_8 , the centralizer $H' \cap K$ of H in K is also the centralizer of K' in H' . This explains the symmetry of the table. If we go over to Lie algebras, the lower right 4×4 square is Freudenthal's "magic square" of Lie algebras.

For each pair $H \subset K$, H injects into the automorphism group $\text{Aut } \mathfrak{k}$ of the Lie algebra $\mathfrak{k}$ of K . Our magic triangle is the system of the centralizers G of H in $\text{Aut } \mathfrak{k}$. As H is connected, the centralizer of H in the adjoint group K^{ad} is the image of the centralizer of H in K : to obtain G from the corresponding entry of Table 1, one divides by the center of K and takes an extension with, as quotient group, the group of outer automorphisms of K . This modifying isogeny is indicated under each column. In all but one case, the required extension is a semi-direct product, with the quotient group acting as a group of automorphisms respecting a pinning. The exception is the nonsplit extension $(\mathbb{G}_m/\mu_3).2$ at $A_1 \subset A_2$.

As H injects into $\text{Aut}(\mathfrak{k})$, the center $Z(H)$ of H is a central subgroup of G . It is shown at the right of Table 1. In most cases, G and H form a dual pair in $\text{Aut}(\mathfrak{k})$, that is, they are each other's centralizer, hence have the same center. The exceptions are $E_6 \subset E_7$ and $F_4 \subset E_6$.

The last row of the magic triangle, corresponding to $H = e$, is the exceptional series of [2]: the groups of automorphisms of the Lie algebras $\mathfrak{a}_1$, $\mathfrak{a}_2$, $\mathfrak{g}_2$, $\mathfrak{d}_4$, $\mathfrak{f}_4$, $\mathfrak{e}_6$, $\mathfrak{e}_7$, $\mathfrak{e}_8$. Under each column, we also give the value of a parameter ν . For the exceptional series, it is $h^\vee/6$, for $h^\vee$ the dual Coxeter number: the inverse of the parameter μ used in [3].

For each pair $H \subset K$, giving rise to G , the representation of $G \times H$ on $\mathfrak{k} = \text{Lie}(K)$ has the following structure:

$$\mathfrak{k} \cong \mathfrak{g} \otimes 1 + 1 \otimes \mathfrak{h} + \bigoplus_{\alpha} V_{\alpha} \otimes W_{\alpha},$$

with W_{α} running over the irreducible representations of H occurring in $\mathfrak{k}/(\mathfrak{g} + \mathfrak{h})$. If $H = A_1, A_2, D_4, E_6$ or E_7 , the W_{α} are the minuscule representations of H . They can be indexed by their central character, which is an arbitrary nontrivial character of the center, and are conjugate by the outer automorphisms of H . For $H = G_2$ or F_4 , there is only one W : the representation whose nonzero weights are the short roots of H .

The V_{α} are always irreducible. In this way, for each row of the magic triangle, we obtain a system of irreducible representations of G . When H is simply laced, it is indexed by the nontrivial characters of $Z(H)$. When $H = G_2$ or F_4 , it is reduced to a single representation. For $H = e$, it is empty. We call those representations the *preferred representations*.

Table 2. – The preferred representations V of G .

	A_1	A_2	G_2	D_4	F_4	E_6	E_7	E_8
$\dim V$	1	2.1	2	2	3	3	1	
# pref. rep.	(1)	(2)	(1)	(3)	(1)	(2)	(1)	
tensor types	$\wedge$	$\wedge, V' \otimes V''$	$\wedge, S^2$	$S^2, V' \otimes V'' \otimes V'''$	S_2, S_3	$S^3, V' \otimes V''$	$\wedge, S^4$	
	4	8	14	26				
	2.1	5	8	14	20	32	56	
	1	3.1	6	9	15	27		

The preferred representations of G are conjugate by outer automorphisms and, when there are two of them, are in duality. In particular, the preferred representations V of G are all of the same dimension, given in Table 2. When the restriction $V|_{G_0}$ of V to the centralizer G_0 of H in K^{ad} is reducible, the dimension is given in the form $a.b$, a being the length of $V|_{G_0}$ and b the dimension of its irreducible constituents.

At $G_2 \subset D_4$ (resp. $F_4 \subset E_6$), we get the irreducible representation of S_3 of dimension 2 (resp. the sign character of S_2). In all other cases, there is a subgroup A of index a of $\text{Out}(K)$ such that V is induced from a representation V_1 of $A \rtimes G_0$ with A acting trivially on the dominant weight subspace of V_1 . As could be inferred from the knowledge of central characters, at $A_1 \subset F_4$ (resp. $G_2 \subset E_7$), the dominant weight of the representation of C_3 is $[0, 0, 1]$ (resp. $[0, 1, 0]$). For the rows 1 to 7, the group G is the subgroup of $\text{GL}(V_\alpha)$ fixing tensors whose types depend only on the row. The number of preferred representations, and those types, are given in Table 2 as well; an entry such as S^3 stands for $S^3(V_\alpha)$, all α .

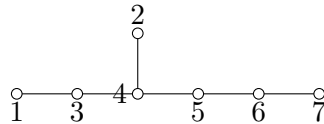
One of us (B.G.) first encountered the Magic Triangle (for Lie algebras) in the Harvard Ph.D. thesis of K. Rumelhart. The approach to it here was suggested by the work of G. Savin on dual pairs.

In each row, the dimension of a preferred representation is a linear function of ν : for the row of centralizers to $E_7, \dots, A_1$, it is

$$E_7 : \nu - 3, \quad E_6 : \nu - 2, \quad F_4 : 2\nu - 3, \quad D_4 : 2\nu - 2, \quad G_2 : 6\nu - 4, \quad A_2 : 6\nu - 3, \quad A_1 : 12\nu - 4. \quad (2)$$

If, in the definition of the magic triangle, one allows $H = K$, prefixing each row by a trivial group e and taking for its preferred representation(s) the zero representation(s), the dimension formulas (2) remain valid.

Other representations correspond to each other across the groups in a row. They give rise to uniform decomposition formulas, and their dimensions are the values at $\nu(G)$ of a rational function, which is the quotient of products of rational linear factors. The case of the last row, the exceptional series, is considered in [1–3]; the adjoint representations correspond to each other, and they have the dimension $2(5\nu - 1)(6\nu + 1)/(\nu + 1)$. This formula, due to Vogel, was the beginning of the story on the exceptional series. The case of the seventh row, corresponding to $H = A_1$, is strikingly similar. The vertices of the Dynkin diagram E_7 are labelled



and we write $\alpha(i)$, ϖ_i , $s(i)$ for the simple roots, fundamental weights, and generators of the Weyl group so numbered. Let V_r be the $(\mathbb{Q}-)$ linear span of the ϖ_i ($i \leq r$); it is the orthogonal complement of the $\alpha(i)$, $i > r$. For $r = 1, 3, 5, 6$ the trace on V_r of the E_7 -root system is empty, resp. a root system A_1^3 , A_5 , D_6 . We take as its positive roots those which are positive for E_7 . For $r = 1, 3$, the orthogonal projection to V_r of the E_7 -root system is a root system A_1 , resp. C_3 . We take as positive roots the nonzero projections of positive roots. For G in row seven and of

rank r , this construction allows us to view the ϖ_i ($i \leq r$) as dominant weights of the connected component G_0 of G .

To an irreducible representation $V(\lambda)$ of E_7 , whose dominant weight λ is linear combination of the ϖ_i ($i \leq r$), will correspond the following representation $V(G, \lambda)$ of G_0 : take the irreducible representation $V(G_0, \lambda)$ of the “same” dominant weight λ , and the sum of its distinct conjugates by the component group G/G_0 . This representation is the restriction to G_0 of an irreducible representation of G . We expect there is a best choice for this representation of G , giving rise to nicer decomposition formulas, but we have not pinned it down. Corresponding representations have the same central character χ , and are orthogonal (resp. symplectic) when χ is the trivial (resp. nontrivial) character of μ_2 .

For each G , $V(\varpi_1)$ is the preferred representation. For $\text{rank}(G) \geq 2$, $V(\varpi_2)$ is the adjoint representation, and $S^2 V(\varpi_1) = V(2\varpi_1) + V(\varpi_2)$. For $\text{rank}(G) \geq 3$, $\wedge^2 V(\varpi_1) = V(\varpi_3) + 1$. For $\text{rank}(G) \geq 3$ and $p \geq 1$, one has

$$V(\varpi_1) \otimes V(p\varpi_1) = \bigoplus_{\mu} V(p\varpi_1 + \mu)$$

with μ running over the weights $\varpi_1, -\varpi_1, \varpi_2 - \varpi_1$ and $\varpi_3 - \varpi_1$. Experiments using the program LiE [6] suggest that more generally

$$V(p\varpi_1) \otimes V(q\varpi_1) = \bigoplus V(a\varpi_1 + b\varpi_2 + c\varpi_3),$$

the sum being over the triples (a, b, c) with a of the same parity as $p + q$ and $|p - q| \leq a \leq p + q - 2(b + c)$, and that for $\text{rank}(G) \geq 4$, a similar formula holds for $V(p\varpi_1) \otimes V(q\varpi_2)$.

We will exhibit rational functions $D(\lambda; X)$ depending on a dominant weight λ of E_7 which is a linear combination of the ϖ_i ($i \leq 6$), with the property that if the coefficient of ϖ_i in λ vanishes whenever $i > \text{rank}(G)$, then

$$\dim V(G, \lambda) = D(\lambda, \nu(G))$$

for $\nu(G)$ the parameter of G , given in Table 1.

To describe $D(\lambda; X)$, we will use the language of multisets (sets with integer multiplicities, positive or negative). For E a set, a multiset $\mathcal{E}$ in E is simply a function $m_{\mathcal{E}} : E \rightarrow \mathbb{Z}$, its multiplicity function. We will only use multisets whose multiplicity function has finite support. For e in E , the characteristic function of $\{e\}$ is noted $\delta[e]$. Multisets can be added, and $\mathcal{E} = \sum m_{\mathcal{E}}(e)\delta[e]$.

Let $(,)$ be the Weyl group invariant symmetric bilinear form on the E_7 root system for which roots have square length 2. It induces a bilinear form $(,)$ on V_r , and hence on the root systems $A_1, A_1^3, C_3, A_5, D_6$. For those of rank $\geq r$, define

$$\begin{aligned} \mathcal{W}(G_0, \lambda) = & (\text{multiset of the } (\lambda + \rho, \alpha), \text{ for } \alpha \text{ a positive root}) \\ & - (\text{multiset of the } (\rho, \alpha), \text{ for } \alpha \text{ a positive root}). \end{aligned}$$

In this definition, ρ is the sum of fundamental weights of G_0 and the multiplicity $m(x)$ of a rational number x in $\mathcal{W}(G_0, \lambda)$ is the number of positive roots such that $(\lambda + \rho, \alpha) = x$, minus the number of positive roots such that $(\rho, \alpha) = x$. Weyl’s dimension formula says

$$\dim V(G_0, \lambda) = \prod_{x \in \mathcal{W}(G_0, \lambda)} x^{m(x)}.$$

It is not only the integers $\dim V(G, \lambda)$ which we will interpolate from group to group, but also the multisets $\mathcal{W}(G_0, \lambda)$. Define $\mathcal{W}'(G_0, \lambda)$ by its multiplicity function

$$m'(x) := m(x) - m(x + 1). \tag{3}$$

One has $m(x) = \sum_{n \geq 0} m'(x + n)$. We will exhibit a multiset $\mathcal{W}'$ of affine linear forms in the coordinates p_i ($i \leq 6$) of $\lambda = \sum p_i \varpi_i$, and an indeterminate X , such that each $\mathcal{W}'(G_0, \lambda)$ is deduced from $\mathcal{W}'$ by specialization at λ and $\nu(G)$. Further, for a given dominant weight $\lambda = \sum p_i \varpi_i$, if we specialize $\mathcal{W}'$ at λ to obtain a multiset $\mathcal{W}'_\lambda$ of linear forms in X , $\mathcal{W}'_\lambda$ is derived by (3) from a finite multiset $\mathcal{W}_\lambda$, and

$$D(\lambda, X) = \prod_{w \in \mathcal{W}_\lambda} w.$$

As in [3], the discrepancy between $\dim V(G_0, \lambda)$ and $\dim V(G, \lambda)$ is accounted for by true values of factors 0/0 (see example below).

Let $\mathcal{W}'_{\text{mv}}$ be the sum of the following contributions (“mv” is for “moving”: linear forms actually depending on λ). A positive root $\alpha = \sum a_i \alpha(i)$ of E_7 fixed by $s(7)$ contributes

$$\delta \left[\sum a_i p_i - \delta_{a_7} X + n - 1 + \sum a_i p_i \right] + \delta \left[\delta_{a_7} X + n + \sum a_i p_i \right]$$

with $5a_7 + n = \sum_{i \leq 7} a_i$. A pair of positive roots exchanged by $s(7)$, $\alpha = \sum a_i \alpha(i)$ and $\alpha' = \alpha - \alpha(7) = \sum a'_i \alpha(i)$, contributes

$$\delta \left[\delta_{a_7} X + n + \sum a_i p_i \right] - \delta \left[\delta_{a'_7} X + n' - 1 + \sum a'_i p_i \right]$$

with $5a_7 + n = \sum_{i \leq 7} a_i$ and $5a'_7 + n' = \sum_{i \leq 7} a'_i$. One has $a'_7 = a_7 - 1$, $a'_i = a_i$ for $i \leq 6$ and $n' - 1 = n + 3$.

Finally one obtains $\mathcal{W}'$ by adding to $\mathcal{W}'_{\text{mv}}$ a sum of $\delta[BX + n]$, such that $\mathcal{W}'$ specialized at $\lambda = 0$ is the empty multiset of linear forms $BX + n$. It follows that

$$D(\lambda, X) = \prod_{\mathcal{W}'_{\text{mv}}} (BX + n + 1) \cdots \left(BX + n + \sum a_i p_i \right),$$

the product being over $\mathcal{W}'_{\text{mv}}$. Example: for $\lambda = p\varpi_1$,

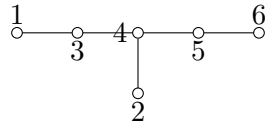
$$\dim V = \frac{(4X - 3 + 1) \cdots (4X - 3 + p) \cdot (3X - 2 + 1) \cdots (3X - 2 + p) \cdot (2X + p - 1)}{(X - 1 + 1) \cdots (X - 1 + p) \cdot (1) \cdots (p) \cdot (2X - 1)}$$

evaluated at $X = \nu(G)$. For $p = 1$, we obtain the dimension of the preferred representation:

$$\dim V(G, p\varpi_1) = \frac{(4X - 2)(3X - 1)(2X)}{X \cdot 1 \cdot (2X - 1)}$$

evaluated at $X = \nu(G)$. For $G = (\mathbb{G}_m/\mu_3).2$, one has $\nu := \nu(G) = 1/2$; the numbers $4\nu - 2$ and $2\nu - 1$, $3\nu - 1$ and ν , as well as 2ν and 1 cancel to give $\mathcal{W}(\mathbb{G}_m, \varpi_1) = \emptyset$, and $(4X - 2)/(2X - 1)$ ($= 0/0$ at $X = 1/2$) contributes the factor 2 in the dimension 2.1 of V .

We expect a similar story for row 6, if the vertices of E_6 are labelled



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THE HODGE CONJECTURE

PIERRE DELIGNE

1 Statement

We recall that a pseudo complex structure on a C^∞ -manifold X of dimension $2N$ is a $\mathbf{C}$ -module structure on the tangent bundle TX . Such a module structure induces an action of the group $\mathbf{C}^*$ on TX , with $\lambda \in \mathbf{C}^*$ acting by multiplication by λ . By transport of structures, the group $\mathbf{C}^*$ acts also on each exterior power $\wedge^n TX$, as well as on the complexified dual $\Omega^n := \text{Hom}(\wedge^n TX, \mathbf{C})$. For $p+q=n$, a (p, q) -form is a section of Ω^n on which $\lambda \in \mathbf{C}^*$ acts by multiplication by $\lambda^{-p} \bar{\lambda}^{-q}$. From now on, we assume X complex analytic. A (p, q) -form is then a form which, in local holomorphic coordinates, can be written as

$$\sum a_{i_1, \dots, i_p, j_1, \dots, j_q} dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q},$$

and the decomposition $\Omega^n = \oplus \Omega^{p,q}$ induces a decomposition $d = d' + d''$ of the exterior differential, with d' (resp. d'') of degree $(1, 0)$ (resp. $(0, 1)$).

If X is compact and admits a Kähler metric, for instance if X is a projective non-singular algebraic variety, this action of $\mathbf{C}^*$ on forms induces an action on cohomology. More precisely, $H^n(X, \mathbf{C})$ is the space of closed n -forms modulo exact forms, and if we define $H^{p,q}$ to be the space of closed (p, q) -forms modulo the $d' d''$ of $(p-1, q-1)$ -forms, the natural map

$$\bigoplus_{p+q=n} H^{p,q} \longrightarrow H^n(X, \mathbf{C}) \quad (1)$$

is an isomorphism. If we choose a Kähler structure on X , one can give the following interpretation to the decomposition (1) of $H^n(X, \mathbf{C})$: the action of $\mathbf{C}^*$ on forms commutes with the Laplace operator, hence induces an action of $\mathbf{C}^*$ on the space $\mathcal{H}^n$ of harmonic n -forms. We have $\mathcal{H}^n \xrightarrow{\sim} H^n(X, \mathbf{C})$ and $H^{p,q}$ identifies with the space of harmonic (p, q) -forms.

When X moves in a holomorphic family, the Hodge filtration $F^p := \bigoplus_{a \geq p} H^{a, n-a}$ of $H^n(X, \mathbf{C})$ is better behaved than the Hodge decomposition. Locally on the parameter space T , $H^n(X_t, \mathbf{C})$ is independent of $t \in T$ and the Hodge filtration can be viewed as a variable filtration $F(t)$ on a fixed vector space. It varies holomorphically with t , and obeys Griffiths transversality: at first order around $t_0 \in T$, $F^p(t)$ remains in $F^{p-1}(t_0)$.

So far, we have computed cohomology using C^∞ forms. We could as well have used forms with generalized functions coefficients, that is, currents. The resulting groups $H^n(X, \mathbf{C})$ and $H^{p,q}$ are the same. If Z is a closed analytic subspace of X , of complex codimension p , Z is an integral cycle and, by Poincaré duality, defines a class $\text{cl}(Z)$ in $H^{2p}(X, \mathbf{Z})$. The integration current on Z is a closed (p, p) -form with generalized function coefficients, representing the image of $\text{cl}(Z)$ in $H^{2p}(X, \mathbf{C})$.

The class $\text{cl}(Z)$ in $H^{2p}(X, \mathbf{Z})$ is hence of type (p, p) , in the sense that its image in $H^{2p}(X, \mathbf{C})$ is. Rational (p, p) -classes are called Hodge classes. They form the group

$$H^{2p}(X, \mathbf{Q}) \cap H^{p,p}(X) = H^{2p}(X, \mathbf{Q}) \cap F^p \subset H^{2p}(X, \mathbf{C}).$$

In [6], Hodge posed the

Conjecture 1 (Hodge Conjecture). *On a projective non-singular algebraic variety over $\mathbf{C}$, any Hodge class is a rational linear combination of classes $\text{cl}(Z)$ of algebraic cycles.*

2 Remarks

- (i) By Chow's theorem, on a complex projective variety, algebraic cycles are the same as closed analytic subspaces.
- (ii) On a projective non-singular variety X over $\mathbf{C}$, the group of integral linear combinations of classes $\text{cl}(Z)$ of algebraic cycles coincides with the group of integral linear combinations of products of Chern classes of algebraic (equivalently by GAGA: analytic) vector bundles. To express $\text{cl}(Z)$ in terms of Chern classes, one resolves the structural sheaf $\mathcal{O}_Z$ by a finite complex of vector bundles. That Chern classes are algebraic cycles holds, basically, because vector bundles have plenty of meromorphic sections.
- (iii) A particular case of (ii) is that the integral linear combinations of classes of divisors (= codimension 1 cycles) are simply the first Chern classes of line bundles. If $Z^+ - Z^-$ is the divisor of a meromorphic section of $\mathcal{L}$, $c_1(\mathcal{L}) = \text{cl}(Z^+) - \text{cl}(Z^-)$. This is the starting point of the proof given by Kodaira and Spencer [7] of the Hodge conjecture for H^2 : a class $c \in H^2(X, \mathbf{Z})$ of type $(1, 1)$ has image 0 in the quotient $H^{0,2} = H^2(X, \mathcal{O})$ of $H^2(X, \mathbf{C})$, and the long exact sequence of cohomology defined by the exponential exact sequence

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}^* \longrightarrow 0$$

shows that c is the first Chern class of a line bundle.

- (iv) The relation between algebraic cycles and algebraic vector bundles is also the basis of the Atiyah and Hirzebruch theorem [2] that the Hodge conjecture cannot hold integrally. In the Atiyah–Hirzebruch spectral sequence from cohomology to topological K -theory,

$$E_2^{pq} = H^p(X, K^q(\text{Pt})) \implies K^{p+q}(X);$$

the resulting filtration of $K^n(X)$ is by the

$$F^p K^n(X) = \text{Ker}(K^n(X) \rightarrow K^n((p-1)\text{-skeleton, in any triangulation})).$$

Equivalently, a class c is in F^p if for some topological subspace Y of codimension p , c is the image of a class $\tilde{c}$ with support in Y . If Z is an algebraic cycle of codimension p , a resolution of $\mathcal{O}_Z$ defines a K -theory class with support in Z : $c_Z \in K^0(X, X - Z)$. Its image in $F^p K^0(X)$ agrees with the class of Z in $H^{2p}(X, \mathbf{Z})$. The latter hence is in the kernel of the successive differentials d_r of the spectral sequence.

No counterexample is known to the statement that integral (p, p) classes killed by all d_r are integral linear combinations of classes $\text{cl}(Z)$. One has no idea of which classes should be effective, that is, of the form $\text{cl}(Z)$, rather than a difference of such.

On a Stein manifold X , any topological complex vector bundle can be given a holomorphic structure and, at least for X of the homotopy type of a finite CW complex, it follows that any class in $H^{2p}(X, \mathbf{Z})$ in the kernel of all d_r is a $\mathbf{Z}$ -linear combination of classes of analytic cycles.

- (v) The assumption in the Hodge conjecture that X be algebraic cannot be weakened to X being merely Kähler. See Zucker's appendix to [11] for counterexamples where X is a complex torus.
- (vi) When Hodge formulated his conjecture, he had not realized it could hold only rationally (i.e. after tensoring with $\mathbf{Q}$). He also proposed a further conjecture, characterizing the subspace of $H^n(X, \mathbf{Z})$ spanned by the images of cohomology classes with support in a suitable closed analytic subspace of complex codimension k . Grothendieck observed that this further conjecture is trivially false, and gave a corrected version of it in [5].

3 The Intermediate Jacobian

The cohomology class of an algebraic cycle Z of codimension p has a natural lift to a group $J^p(X)$, extension of the group of classes of type (p, p) in $H^{2p}(X, \mathbf{Z})$ by the intermediate jacobian

$$J^p(X)_0 := H^{2p-1}(X, \mathbf{Z}) \backslash H^{2p-1}(X, \mathbf{C}) / F^p.$$

This expresses that the class can be given an integral description (in singular cohomology), as well as an analytic one, as a closed (p, p) current, giving a hypercohomology class in $\mathbb{H}^{2p}$ of the subcomplex $F^p \Omega_{\text{hol}}^\bullet := (0 \rightarrow \cdots \rightarrow 0 \rightarrow \Omega_{\text{hol}}^p \rightarrow \cdots)$ of the holomorphic de Rham complex, with an understanding at the cocycle level of why the two agree in $H^{2p}(X, \mathbf{C})$. ‘Understanding’ means a cochain in a complex computing $H^*(X, \mathbf{C})$, whose coboundary is the difference between cocycles coming from the integral, resp. analytic, constructions. Indeed, $J^p(X)$ is the hypercohomology $\mathbb{H}^{2p}$ of the homotopy kernel of the difference map $\mathbf{Z} \oplus F^p \Omega_{\text{hol}}^\bullet \rightarrow \Omega^\bullet$.

In general, using that all algebraic cycles on X fit in a denumerable number of algebraic families, one checks that the subgroup $A^p(X)$ of $J^p(X)$ generated by algebraic cycles is the extension of a denumerable group by its connected component $A_p^0(X)$, and that for some sub-Hodge structure H_{alg} of type $\{(p-1, p), (p, p-1)\}$ of $H^{2p-1}(X)$, $A_p^0(X)$ is $H_{\text{alg}, \mathbf{Z}} \backslash H_{\text{alg}, \mathbf{C}} / F^p$. ‘Sub-Hodge structure’ means the subgroup of the integral lattice whose complexification is the sum of its intersections with the $H^{a,b}$. The Hodge conjecture (applied to the product of X and a suitable abelian variety) predicts that H_{alg} is the largest sub-Hodge structure of $H^{2p-1}(X)$ of type $\{(p-1, p), (p, p-1)\}$.

No conjecture is available to predict what subgroup of $J^p(X)$ the group $A^p(X)$ is. Cases are known where $A^p(X)/A_p^0(X)$ is of infinite rank. See, for instance, the paper [9] and the references it contains. This has made generally inapplicable the methods introduced by Griffiths (see, for instance, Zucker [11]) to prove the Hodge conjecture by induction on the dimension of X , using a Lefschetz pencil of hyperplane sections of X . Indeed, the method requires not just the Hodge conjecture for the hyperplane sections H , but that all of $J^p(H)$ comes from algebraic cycles.

4 Detecting Hodge Classes

Let $(X_s)_{s \in S}$ be an algebraic family of projective non-singular algebraic varieties: the fibers of a projective and smooth map $f : X \rightarrow S$. We assume it is defined over the algebraic closure $\overline{\mathbf{Q}}$ of $\mathbf{Q}$ in $\mathbf{C}$. No algorithm is known to decide whether a given integral cohomology class of a typical fiber X_0 is somewhere on S of type (p, p) . The Hodge conjecture implies that the locus where this happens is a denumerable union of algebraic subvarieties of S (known: see [4]), and is defined over $\overline{\mathbf{Q}}$ (unknown).

The Hodge conjecture is not known even in the following nice examples.

Example 2. For X of complex dimension N , the diagonal Δ of $X \times X$ is an algebraic cycle of codimension N . The Hodge decomposition being compatible with Künneth, the Künneth components $\text{cl}(\Delta)_{a,b} \in H^a(X) \otimes H^b(X) \subset H^{2N}(X \times X)$ ($a+b=2N$) of $\text{cl}(\Delta)$ are Hodge classes.

Example 3. If $\eta \in H^2(X, \mathbf{Z})$ is the cohomology class of a hyperplane section of X , the iterated cup product $\eta^p : H^{N-p}(X, \mathbf{C}) \rightarrow H^{N+p}(X, \mathbf{C})$ is an isomorphism (hard Lefschetz theorem, proved by Hodge. See [10, IV.6]). Let $z \in H^{N-p}(X, \mathbf{C}) \otimes H^{N-p}(X, \mathbf{C}) \subset H^{2N-2p}(X \times X)$ be the class such that the inverse isomorphism $(\eta^p)^{-1}$ is $c \mapsto \text{pr}_{1!}(z \cup \text{pr}_2^* c)$. The class z is Hodge.

5 Motives

Algebraic varieties admit a panoply of cohomology theories, related over $\mathbf{C}$ by comparison isomorphisms. Resulting structures on $H^*(X, \mathbf{Z})$ should be viewed as analogous to the Hodge

structure. Examples: If X is defined over a subfield K of $\mathbf{C}$, with algebraic closure $\bar{K}$ in $\mathbf{C}$, $\text{Gal}(\bar{K}/K)$ acts on $H^*(X, \mathbf{Z}) \otimes \mathbf{Z}_\ell$ and $H^*(X, \mathbf{C}) = H^*(X, \mathbf{Z}) \otimes \mathbf{C}$ has a natural K -structure $H_{\text{DR}}^*(X \text{ over } K)$, compatible with the Hodge filtration. Those cohomology theories give rise to conjectures parallel to the Hodge conjecture, determining the linear span of classes of algebraic cycles. Example: the Tate conjecture [8]. Those conjectures are open even for H^2 .

Grothendieck's theory of motives aims at understanding the parallelism between those cohomology theories. Progress is blocked by a lack of methods to construct interesting algebraic cycles. If the cycles of Examples 1 and 2 of §4 were algebraic, Grothendieck's motives over $\mathbf{C}$ would form a semi-simple abelian category with a tensor product, and be the category of representations of some pro-reductive group-scheme. If the algebraicity of those cycles is assumed, the full Hodge conjecture is equivalent to a natural functor from the category of motives to the category of Hodge structures being fully faithful.

6 Substitutes and Weakened Forms

In despair, efforts have been made to find substitutes for the Hodge conjecture. On abelian varieties, Hodge classes at least share many properties of cohomology classes of algebraic cycles: they are “absolutely Hodge” [3], even “motivated” [1]. This suffices for some applications — for instance, the proof of algebraic relations among periods and quasi periods of abelian varieties predicted by the Hodge conjecture [3], but does not allow for reduction modulo p . The following corollaries of the Hodge conjecture would be particularly interesting. Let A be an abelian variety over the algebraic closure $\mathbf{F}$ of a finite field $\mathbf{F}_q$. Lift A in two different ways to characteristic 0, to complex abelian varieties A_1 and A_2 defined over $\bar{\mathbf{Q}}$. Pick Hodge classes z_1 and z_2 on A_1 and A_2 , of complementary dimension. Interpreting z_1 and z_2 as ℓ -adic cohomology classes, one can define the intersection number κ of the reduction of z_1 and z_2 over $\mathbf{F}$. Is κ a rational number? If z_1 and z_2 were $\text{cl}(Z_1)$ and $\text{cl}(Z_2)$, Z_1 and Z_2 could be chosen to be defined over $\bar{\mathbf{Q}}$ and κ would be the intersection number of the reductions of Z_1 and Z_2 . Same question for the intersection number of the reduction of z_1 over $\mathbf{F}$ with the class of an algebraic cycle on A .

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Voevodsky's Lectures on Motivic Cohomology 2000/2001

Pierre Deligne

Voevodsky's Lectures on Motivic Cohomology

Abstract

These lectures cover several important topics of motivic homotopy theory which have not been covered elsewhere. These topics include the definition of equivariant motivic homotopy categories, the definition and basic properties of solid and ind-solid sheaves and the proof of the basic properties of the operations of twisted powers and group quotients relative to the A -equivalences between ind-solid sheaves.

1 Introduction

The lectures which provided the source for these notes covered several different topics which are related to each other but which do not in any reasonable sense form a coherent whole. As a result, this text is a collection of four parts which refer to each other but otherwise are independent. In the first part we introduce the motivic homotopy category and connect it with the motivic cohomology theory discussed in [7]. The exposition is a little unusual because we wanted to avoid any references to model structures and still prove the main theorem 2. We were able to do it modulo 6 where we had to refer to the next part. The second part is about the motivic homotopy category of $\mathcal{S}$ -schemes where $\mathcal{S}$ is a finite flat group scheme with respect to an equivariant analog of the Nisnevich topology. Our main result is a description of the class of A -equivalences (formerly called A -weak equivalences) given in Theorem 4 (also in Theorem 5). For the trivial group we get a new description of the A -equivalences in the non equivariant setting. In the third part we define a class of sheaves on $\mathcal{S}$ -schemes which we call solid sheaves. It contains all representable sheaves and quotients of representable sheaves by subsheaves corresponding to open subschemes. In particular the Thom P. Deligne e-mail: deligne@math.ias.edu N.A. Baas et al. (eds.), Algebraic Topology, Abel Symposia 4,

355 356 P. Deligne spaces of vector bundles are solid sheaves. The key property of solid sheaves can be expressed by saying that any right exact functor which takes open embeddings to monomorphisms is left exact on solid sheaves. A more precise statement is

Theorem 1 (6). *In the fourth part we study two functors. One is the extension to pointed sheaves of the functor from $\mathcal{S}$ -schemes to schemes which takes X to X_{solid} . The other one is extension to pointed sheaves of the functor which takes X to $X_{\text{solid}}^{\mathcal{S}}$ where $\mathcal{S}$ is a finite flat $\mathcal{S}$ -scheme. We show that both functors take solid sheaves to solid sheaves and preserve local and A -equivalences between termwise (ind-)solid sheaves. The material of all the parts of these notes but the first one was originally developed with one particular goal in mind – to extend non-additive functors, such as the symmetric product, from schemes to the motivic homotopy category. More precisely, we were interested in functors given by*

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$<]$ where $\mathcal{S}$ is a finite flat group scheme, $\mathcal{S}$ is a finite flat $\mathcal{S}$ -scheme and any $\mathcal{S}$ -scheme of finite type. The equivariant motivic homotopy category was introduced to represent $<$ as a composition $]]]$ and solid sheaves as a natural class of sheaves on which the derived functor $L<$ coincides with $<$. In the present form these notes are the result of an interactive process which involved

all listeners of the lectures. A very special role was played by Pierre Deligne. The text as it is now was completely written by him. He also cleared up a lot of messy parts and simplified the arguments in several important places. Princeton, NJ, 2001. Vladimir Voevodsky

2 Motivic Cohomology and Motivic Homotopy Category

We will recall first some of last year results (see [7]).

2.1 Last Year

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1.1 We work over a field which sometimes will have to be assumed to be perfect. The schemes over we consider will usually be assumed separated and smooth of finite type over k . We note their category. Three Grothendieck topologies on $\mathcal{C}$ will be useful: Zariski, Nisnevich and étale. For each of these topologies a

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Voevodsky's Lectures on Motivic Cohomology 2000/2001 (!

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/ / 1.2 The definition of the motivic cohomology groups of smooth over k has the sheaf on amounts to the data for smooth over k of a sheaf $\mathcal{F}$ on the small site $(\mathcal{C}_{\text{ét}}, \tau_{\text{ét}})$ (resp. $(\mathcal{C}_{\text{Zar}}, \tau_{\text{Zar}})$) of the open subsets of X (resp. of étale), with $\mathcal{F}$ functorial in $U \rightarrow X$: a map $U \rightarrow X$ induces $\mathcal{F}(U) \rightarrow \mathcal{F}(X)$.

following form:

(a) One defines for each Z a complex of presheaves of abelian groups Z on $\mathcal{C}$. It is in fact a complex of sheaves for the étale topology, hence a fortiori for the Nisnevich and Zariski topology. For any abelian group A the same applies to $Z \otimes A$. (b) The motivic cohomology groups of X with coefficients in A are the hypercohomology groups of the $Z \otimes A$, in the Nisnevich topology:

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%

%

% % $H^i(X, A)$. %

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For Z we will write simply $H^i(X, A)$. Motivic cohomology has the following properties:

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1. The complex Z is zero for $i < 0$. For any i it lives in cohomological degree i . As a complex of Nisnevich sheaves it is quasi-isomorphic to Z for $i \geq 0$ and to G for $i < 0$. J 2. for any i .

3. For any i in one has

$H^i(X, \mathbb{Z}) = 0$ for $i < 0$, $H^0(X, \mathbb{Z}) = \mathbb{Z}$, where $\mathbb{Z}(n)$ is the n -th higher Chow group of cycles of codimension n . 4. In the étale topology, for ℓ prime to the characteristic of k , the complex $Z \otimes \mathbb{Z}(\ell)$

is quasi-isomorphic to D , giving for the étale analog of the formula $H^i(X, \mathbb{Z}(\ell)) = H^i(X, \mathbb{Z}) \otimes \mathbb{Z}(\ell)$. 1.3 The category $(=AB)$ is the category with objects separated schemes smooth of finite type over k , for which a morphism $C \rightarrow D$ is a cycle $C \in Z^0(D)$ on each of whose irreducible components C is finite over k and projects onto a connected component of D . A morphism $C \rightarrow D$ can be thought of as a finitely valued map from C to D . For C , with residue field ℓ , it defines a zero cycle C on D , and the assumption made on C implies that the degree of this 0-cycle is locally constant on D . A morphism of schemes defines a morphism in $(=AB)$: the graph of f . This graph construction defines a faithful functor

from $(!)$ to the additive category $(=AB!)$. A presheaf with transfers is a contravariant additive functor from the category $(=AB!)$ to the category of abelian groups. The embedding of $(!)$ in

358 P. Deligne $(=AB!)$ allows us to view a presheaf with transfers as a presheaf on $(!)$ endowed with an extra structure. A sheaf with transfers (for a given topology on $(!)$, usually the Nisnevich topology) is a presheaf with transfers which, as a presheaf on $(!)$, is a sheaf. The Nisnevich and the étale topologies have the virtue that if $/$ is a presheaf with transfers, the associated sheaf $\gamma/$ carries a structure of a sheaf with transfers. This structure is uniquely determined by $/ \rightarrow \gamma/$ being a morphism of presheaves with transfers. For any sheaf with transfers

, one has
 $\gamma/ \rightarrow \text{Hom}/$
 Hom

(Hom of presheaves with transfers). All of this fails for the Zariski topology. The complexes Z (or γ) start life as complexes of sheaves with transfers.

% 1.4 A presheaf $/$ on $(!)$ is called homotopy invariant if $/ \rightarrow A$. As the point of A defines a section of the projection of A to $*$, for any presheaf of abelian groups $/$, $/$ is naturally a direct factor of $/ \rightarrow A$; it follows that the condition “homotopy invariant” is stable by kernels, cokernels and extensions of presheaves. The following construction is a derived version of the left adjoint to the inclusion

$A(AA, \dots)$. (a) For a finite set, let Δ be the affine space freely spanned (in the sense of barycentric calculus) by $*$. Over $\mathbb{C}$, or $\mathbb{R}$, Δ contains the standard topological simplex spanned by $*$. The schemes Δ^n form a cosimplicial scheme. (b) For $/$ a presheaf, $\Delta/$ (the “singular complex of $/$ ”) is the simplicial presheaf $\Delta/ \rightarrow 0$. Arguments imitated from topology show that for $/$ a presheaf of abelian groups, the cohomology presheaves of the complex $\Delta/$, obtained from $\Delta/$ by taking alternating sum of the face maps, are homotopy invariant. If $/$ has transfers so do the $\Delta/$ and hence the $\Delta/$. A basic theorem proved last year is:

Theorem 2 (1). *Let $/$ be a homotopy invariant presheaf with transfers over a perfect field with the associated Nisnevich sheaf $\gamma/$. Then the presheaves with transfers $\gamma/$ are homotopy invariant as well.*

The particular case of this theorem for $\gamma/$ claims the homotopy invariance of the sheaf with transfers $\gamma/$. Last year, the equivalence of a number of definitions of Z was proven. Equivalence means: a construction of an isomorphism in a suitable derived category, implying an isomorphism for the corresponding motivic cohomology groups. For our present purpose the most convenient definition is as follows.

$\gamma/$

Let Z be the sheaf with transfers represented by $(\text{on the category } (=AB!))$. We set $\gamma/$ for $J Z$ $\rightarrow A \rightarrow Z \rightarrow A$ for and $Z = *$.

2.2 Motivic Homotopy Category

A

The motivic homotopy category (pointed A-homotopy category of $\Delta/$), A for a finite dimensional noetherian scheme, will be the category deduced from a category of simplicial sheaves by two successive localizations. One starts with the category of schemes smooth over $*$, with the Nisnevich topology, and the category of pointed simplicial sheaves on $*$. For any site S (for instance $*$), there is a notion of local equivalence of (pointed) S -simplicial sheaves. It proceeds as follows:

(
(

(a) A sheaf defines a simplicial sheaf maps the identities. The functor

$\mathcal{H}om$
 $\mathcal{H}om$
 $($
 with all and all simplicial has a left adjoint :
 $\mathcal{H}om$
 The sheaf can be described as the equalizer of sheaf associated to the presheaf
 $\mathcal{H}om$
 $\mathcal{H}om$
 $\mathcal{H}om$, as well as the
 The same holds in the pointed context. We will often write simply for . is a simplicial sheaf, and u a section of over , one also disposes of (b) If sheaves u over : the sheaves associated to the presheaves

$\mathcal{H}om$
 $\mathcal{H}om$
 $\mathcal{H}om$
 $\mathcal{H}om$ / $\mathcal{H}om$ u /
 (c) A morphism is a local equivalence, if it induces an isomorphism on as well as, for any local section u of , an isomorphism on all . This applies also to pointed simplicial sheaves: one just forgets the marked point.

$\mathcal{H}om$
 $\mathcal{H}om$ (($\mathcal{H}om$
 One defines as the category derived from the category of pointed simplicial sheaves on . by formally inverting local equivalences. Until made more concrete, this definition could lead to set-theoretic difficulties, which we leave the reader to solve in its preferred way. 1

In the Appendix we have assembled the properties of “localization” to be used in this talk and in the next.

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For a pointed sheaf on , Proposition 7 applies to tion by local equivalences: one has

and to the localiza-

$\mathcal{H}om / \mathcal{H}om / \mathcal{H}om$ / (0.1)

Definition 3 (1). An object of $\mathcal{H}om$ (is called $\mathcal{H}om$ -local if for any simplicial sheaf , one has $\mathcal{H}om^*$
 $\mathcal{H}om^*$
 $\mathcal{H}om^*$ $\mathcal{H}om$ $\mathcal{H}om$

At the right hand side, $\mathcal{H}om$ means that in the product, $\mathcal{H}om$ is contracted to a point, the new base point.

Proposition 4 (1). For a pointed sheaf on if and only if is homotopy invariant.

(, the simplicial sheaf

is $\mathcal{H}om$ -local

$\mathcal{H}om$, so that by (0.1) “ $\mathcal{H}om$ -local” means that

Proof. We have $\mathcal{H}om$ for any pointed sheaf , one has

A $\mathcal{H}om$ A morphism can be viewed as the data, for each . , of . , functorial in and marked point going to marked point. A morphism can similarly be described as data for . of . A . A Homotopy invariance hence implies $\mathcal{H}om$ -locality. The converse is checked by taking for the disjoint sum of a representable sheaf and the base point.

□

Definition 5 (2). (1) A morphism in $\mathcal{H}om$ (is an $\mathcal{H}om$ -equivalence if for any $\mathcal{H}om$ -local , one has in $\mathcal{H}om$ ($\mathcal{H}om$

Hom

Hom (2) The category $\mathcal{A}\mathcal{A}$ (is deduced from $\mathcal{A}$ (by formally inverting Hom
 $\mathcal{A}$ -equivalences.

Remark 6 (1). If a morphism in $\mathcal{A}$ (becomes an isomorphism in $\mathcal{A}\mathcal{A}$ (it is an $\mathcal{A}$ -equivalence. Indeed, if in $\mathcal{A}$ (is $\mathcal{A}$ -local, an application of Proposition 7 shows that for any $\text{Hom}^* \text{Hom}^* \mathcal{A}$
 $\mathcal{A}$ ($\mathcal{A}\mathcal{A}$ (which
is bijective. If in has an image in
is an isomorphism, it follows that for any $\mathcal{A}$ -local , one has Hom^* Such an
 Hom^*
is an $\mathcal{A}$ -equivalence.

Example 7 (1). Arguments similar to those given before show that if is a homotopy invariant pointed sheaf, then for any simplicial pointed sheaf , one has

$/ \text{Hom}^* (/ \text{Hom} / \mathcal{A} (/$ in particular, if is smooth over and if is the disjoint union of and of a base point, Hom^*

$\mathcal{A}$

$\mathcal{A}$

2.3 Derived Categories Vs. Homotopy Categories

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For any topos , which for us will be the category of sheaves on some site , the pointed homotopy category as well as the derived category are

obtained by localization. For the derived category, one starts with the category of complexes of abelian sheaves. The subcategory of complexes living in homological degree is naturally equivalent, by the Dold Puppe construction, to the category of simplicial sheaves of abelian groups. The equivalence is

$\mathcal{A}$

$\mathcal{E}$

) simplicial /

complex

(

!LB> >

We will write for the inverse equivalence. For a point, and an abelian group,
is indeed the Eilenberg–MacLane space . For a complex not assumed to live in homological degree , we define

\$

\$

=

= $G =$ where $G =$ is the subcomplex in = of the form =

=

=

!LB-

= $G =$ is right adjoint to the inclusion functor complexes in homological degree " all complexes so that) form a pair of adjoint functors: simplicial abelian sheaves complexes of sheaves

Theorem 8 (2). Assume that $\text{Spec}!$ with ! perfect. Then, for / a presheaf with transfers, and as above, and , $\text{Hom}^* /$, $H . = /$ Note that

$\mathcal{A}$

/ , /

In this theorem is the simplicial sheaf of abelian groups whose normalized chain complex is in homological degree . To prove the theorem we establish the chain of equalities,

$$\cdot = / \text{Hom}^* = / , \text{Hom}^* = / , \text{Hom}^* / , H$$

A

(1.1)

A

the first equality is proved right before Proposition 4, the second right after Proposition 5 and the last one follows from Lemma 1. be the forgetting functor from abelian sheaves to sheaves of sets. Its Let left adjoint is Z : the sheaf associated to the presheaf

$ABL /$

$/$ abelian group freely generated by $/$

In the pointed context , the adjoint is

$/ Z\mathcal{E} / Z / Z$ We have the same adjunction for (pointed) simplicial objects.

Proposition 9 (2). *On a site with enough points (and presumably always), one has*

$/$

$) /$

Z from pointed simplicial sheaves to complexes of (1) The functor abelian groups transforms local equivalences into quasi-isomorphisms (2) The right adjoint transforms quasi-isomorphisms to local equivalences.

=

$ABL =$

(! with the Nisnevich topology: for

$/, LO >$ is a point, and they

The assumption “enough points” applies to any in and any point of , form a conservative system.

(!

$/$

Proof. Local equivalence (resp. quasi-isomorphism) can be checked point by point, and the two functors considered commute with passage to the fiber at a point. This reduces our proposition to the case when is just a point, i.e. to usual homotopy theory. In that case, (1) boils down to the fact that a weak equivalence induces an isomorphism on reduced homology, a theorem of Whitehead, and (1.1) reduces to the fact: for a complex , , computed using any base point, is . The are easy to compute because has the Kan property.

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=

Applying Proposition 8, we deduce from Proposition 2 the following. □

Proposition 10 (3). *Under the same assumptions as in Proposition 2, for simplicial sheaf and a complex of abelian sheaves, one has*

=

Hom^*

$/ =$

$\text{Hom}^?$

$) Z\mathcal{E} / =$

$/$ a pointed (3.1)

$/$

$/$

$/$

Let be a sheaf and be deduced from by adjunction of a base point. We also write and for the corresponding “constant” simplicial sheaf. One has

$\mathcal{A}(\mathcal{C})$ is the group of $\mathcal{C}$ -valued cocycles of $\mathcal{C}$ in degree zero. For the pointed simplicial sheaf $\mathcal{C}$, the group $\mathcal{A}(\mathcal{C})$ is which now occurs at the right-hand side of (3.1) can be interpreted as hypercohomology of $\mathcal{C}$ viewed as a space, i.e. in the topos of sheaves over $\mathcal{C}$. For $\mathcal{C}$ defined by an object of the site $\mathcal{C}$, this is the same as hypercohomology of the site $\mathcal{C}$. As we do not want to assume $\mathcal{C}$ is bounded below (in cohomological numbering), checking this

requires a little care. For a complex of sheaves over a site $\mathcal{C}$, not necessarily bounded below, H can be defined as the Hom group in the derived category $D(\mathcal{C})$. For $\mathcal{C}$ in a topos and the topos $\mathcal{C}$ viewed as a space, besides the morphism

$$\mathcal{A}(\mathcal{C}) \rightarrow \mathcal{A}(\mathcal{C})$$

of toposes $\mathcal{C} \rightarrow \mathcal{C}$, i.e. the adjoint pair $\mathcal{C} \rightleftarrows \mathcal{C}$, we have for abelian sheaves an adjoint pair $\mathcal{C} \rightleftarrows \mathcal{C}$, with $\mathcal{C}$ and $\mathcal{C}$ both exact. By Proposition 8, $\mathcal{C} \rightleftarrows \mathcal{C}$ induce an adjoint pair for the corresponding derived categories. As $\mathcal{C} \rightarrow \mathcal{C}$, we get (3.2) $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$

hence

$\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$ (3.3) Let us consider the particular case of $\mathcal{C}$ with the Nisnevich topology. For any complex of sheaves, (3.3) gives for smooth over $\mathcal{C}$ $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$. (3.4) Here, $\mathcal{C}$ is the site $\mathcal{C}$ with the Nisnevich topology. It has however the same hypercohomology as the small Nisnevich site $\mathcal{C}$. Indeed, one has a morphism $\mathcal{C} \rightarrow \mathcal{C}$ and the functors $\mathcal{C}$ and $\mathcal{C}$ are exact. One again applies Proposition 8. If we apply (3.4) to a translate (shift) of $\mathcal{C}$, we get (3.5) $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$. Applying (3.5) to $\mathcal{C}$ we get the first equality in (1.1).

Proposition 11 (4). Let $\mathcal{C}$ be a complex of abelian sheaves on $\mathcal{C}$. The following $\mathcal{A}(\mathcal{C})$ conditions are equivalent:

1. $\mathcal{C}$ is $\mathcal{A}$ -local. $\mathcal{C}$, the functor H is homotopy invariant.
2. For $\mathcal{C}$, one has in the derived category $D(\mathcal{C})$ $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$.
3. For any complex in cohomological degree n $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$.

Proof. By Proposition 3, condition (1) can be rewritten: for any pointed simplicial sheaf $\mathcal{C}$

$\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$ The operation $\mathcal{C} \rightarrow \mathcal{C}$ is better written as a smash product $\mathcal{C} \wedge \mathcal{C}$

$\mathcal{C} \wedge \mathcal{C} = \mathcal{C} \wedge \mathcal{C}$. It follows that with $\mathcal{A}$. For pointed sets and $\mathcal{C}$, $\mathcal{A}(\mathcal{C}) = H(\mathcal{C})$

$\mathcal{C} \wedge \mathcal{C}$

$\mathcal{C} \wedge \mathcal{C} = \mathcal{C} \wedge \mathcal{C}$

$\mathcal{C}$

$\mathcal{C} \wedge \mathcal{C} = \mathcal{C} \wedge \mathcal{C}$

(isomorphisms of simplicial sheaves), hence

$\mathcal{C} \wedge \mathcal{C} = \mathcal{C} \wedge \mathcal{C}$ It follows that (1) is the particular case of (3) for of the form $\mathcal{C} \wedge \mathcal{C}$. Similarly, (2) is the particular case of (3) for of the form $\mathcal{C}$, with $\mathcal{C}$ of a simplicial pointed sheaf is its smash product with The suspension the simplicial sphere (the n -simplex modulo its boundary). It follows that

$\mathcal{C}$

$\mathcal{C}$

$\mathcal{C} \wedge \mathcal{C} = \mathcal{C} \wedge \mathcal{C}$

$\mathcal{C}$

(isomorphism of simplicial sheaves), and by Eilenberg–Zilber, the normalized complex $\mathcal{C}$ is homotopic to the tensor product of the normalized complexes of $\mathcal{C}$ and $\mathcal{C}$. The latter is simply $\mathcal{C}$:

$$\begin{aligned} & \rightarrow \mathbb{Z}/9\mathbb{Z} \\ & \rightarrow \mathbb{Z}/9\mathbb{Z} \oplus \mathbb{Z}/9\mathbb{Z} \end{aligned}$$

This is just a high-brown way of telling that the reduced homology of a suspension is just a shift of the reduced homology of the space one started with. Applying this to ΣX , one obtains that (1)(3). Indeed, ΣX is homotopic to $\Sigma \mathbb{Z}/2\mathbb{Z}$. We now prove that (2)(1). For a complex, let $(*)$ be the statement that the conclusion of (3) holds for all n . The assumption (2) is that $(*)$ holds for reduced to $\mathbb{Z}$ in degree n , and we will conclude that it holds for all in cohomological degree by “devissage”:

$$\begin{aligned} & \rightarrow \mathbb{Z}/9\mathbb{Z} \\ & \rightarrow \mathbb{Z}/9\mathbb{Z} \oplus \mathbb{Z}/9\mathbb{Z} \end{aligned}$$

9

, in degree zero, follows from Corollary 10. (a) The case of a sum of $\mathbb{Z}$ and that $(*)$ holds (b) Suppose that n is bounded, is in cohomological degree for all n .

$$\begin{aligned} & \text{The functors} \\ & A_n = A_n(\mathbb{Z}) \\ & (3.6) \end{aligned}$$

are contravariant cohomological functors, hence give rise to convergent spectral sequences

One has a morphism of spectral sequences

for for which is an isomorphism for n , and both vanish for $n > J$ or n large. It

follows that for n , i.e. that satisfies $(*)$. The same argument can be expressed as an induction on the number of n such that

" . If n is the largest (with n), the induction assumption applies to $\mathbb{Z}/9\mathbb{Z}$, even to $\mathbb{Z}/9\mathbb{Z}$, and one concludes by the long exact sequence defined by $\mathbb{Z}/9\mathbb{Z}$

(c) Expressing n as the inductive limit of the n_i and using Proposition 11, one sees that we need not assume that n is bounded. n is a quasi-isomorphism, $\mathbb{Z}/9\mathbb{Z} \rightarrow \mathbb{Z}/9\mathbb{Z}$ is one too (d) If (flatness of $\mathbb{Z}/9\mathbb{Z}$), and $(*)$ holds for n if and only if it holds for n_i . For instance, (e) Any abelian sheaf is a quotient of a direct sum of sheaves $\mathbb{Z}$ the sum over n , of $\mathbb{Z}$ mapping to by n . It follows that admits a resolution by such sheaves. By (a) and (c), n satisfies $(*)$. It follows from (d) that n satisfies $(*)$ and then by (c) that any complex in degree satisfies $(*)$.

$$\begin{aligned} & \mathbb{Z}/9\mathbb{Z} \\ & \mathbb{Z}/9\mathbb{Z} \end{aligned}$$

□

2.3 Application to Presheaves with Transfers

/

Let $\mathcal{F}$ be a presheaf with transfers. A formal argument [7] shows that the presheaves with transfers are homotopy invariant. By the basic result (Theorem 1) recalled in the first lecture, it follows that for any n , one has

$$\begin{aligned} & = \mathbb{Z}/9\mathbb{Z} \\ & = \mathbb{Z}/9\mathbb{Z} \\ & H \\ & A = \mathbb{Z}/9\mathbb{Z} \\ & (4.1) \end{aligned}$$

Indeed, as n and A are of finite cohomological dimension, both sides are abutment of a convergent spectral sequence

$$= \mathbb{Z}/9\mathbb{Z} = \mathbb{Z}/9\mathbb{Z} \text{ and the same for } A. \text{ By Theorem 1 applied to } n = \mathbb{Z}/9\mathbb{Z}, n = \mathbb{Z}/9\mathbb{Z} = \mathbb{Z}/9\mathbb{Z}$$

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is the same for n or for Proposition 4((2)(1)) that

Proposition 12 (5). *For*

, is A -local.

A. Applying (3.5), we conclude from

! perfect, if $/$ is a presheaf with transfers, for all $„ = /$

Combining Proposition 5 with Proposition 7 we get the second equality in (1.1).

2.5 End of the Proof of Theorem 2

=

For any pointed simplicial sheaf $\mathcal{F}$, is a pointed bisimplicial sheaf of which one can take the diagonal $\mathcal{F}$. For any pointed sheaf $\mathcal{G}$, one has a natural map $\mathcal{F} \rightarrow \mathcal{G}$, and for a pointed simplicial sheaf $\mathcal{H}$, those maps for the induce

=

$0 =$

Proposition 13 (6). *The morphism*

γ

$0 =$

γ

$0 =$ is an A -equivalence.

Proof. We deduce Proposition 6 from Proposition 20.

The two maps

A are equalized by A , hence

become equal in the A -homotopy category. If two maps of pointed simplicial sheaves factor as A , they also become equal. By

the adjunction of A and of A , such a factorization can be rewritten as

$/$

$/$

$/$

$/$

$/$

$/$

$/$

$/ = A($

$/ =$ Particular case: the maps $=$, become equal in the homotopy category.

Evaluated on $\mathcal{F}$, these maps are the restriction maps A .

The affine space A is homotopic to a point in the sense that $A \xrightarrow{A} A$ interpolates between the identity map (for $\mathcal{F}$) and the constant map (for $\mathcal{F}$). The map induces

$A \xrightarrow{A} A$ and, composing with $\mathcal{F}$, in A , we obtain that A

A the identity map, and the map induced by A

A , are equal in the A -homotopy category. The map of simplicial sheaves is hence an A -equivalence. It has as inverse in the A -homotopy category the map induced by

=

Spec!

A and one applies Remark 1. We now apply Proposition 20 to the bisimplicial sheaves

$= 0$. For fixed $\mathcal{F}$, this is just

A ,

and to the natural map and Proposition 20 gives Proposition 6.

To prove the last equality in (1.1), it suffices to show that:

$/, /, = /$.

□

Lemma 14 (1). *For any abelian sheaf $\mathcal{F}$ lence from $\mathcal{F}$ to*

$/$,

Proof. For

$\mathcal{A} = \mathcal{A} / \sim$, induces an $\mathcal{A}$ -equivalence
a monoid (with unit), the pointed simplicial set
is given by

' functors from the ordered set $\mathbb{N}$ viewed as a category $\mathcal{C}$ to
viewed as a category with one object

This construction can be sheafified, and can be applied termwise to a simplicial sheaf of
monoids, leading to a pointed bisimplicial sheaf of which one can take the diagonal

$$0 \\ 0 =$$

This construction commutes with the construction $\mathcal{A}$. Indeed, $\mathcal{A}$ is naturally isomorphic to $\mathcal{A}$, the
operation commutes with products, and $\mathcal{A}$ and $\mathcal{A}$

are both diagonals of the trisimplicial pointed sheaf $\mathcal{A}$. For abelian simplicial sheaves, the
operation gives again abelian simplicial

sheaves, hence can be iterated, and $\mathcal{A}$ commutes with Via Dold–Puppe construction, corre-
sponds, up to homotopy, to the shift of complexes:

$$= \\ 0 = \\ 0 = \\ = \\ 0 = \\) \\)$$

This can be viewed as an application of the Eilenberg–Zilber Theorem (see [9, Theorem
8.5.1]): one has

$$) \\ \text{Tot}$$

(Eilenberg–Zilber),

and for each n , the normalization of $\mathcal{A}$ is just $\mathcal{A}$, so that the double complexes $\mathcal{A}$ and for n , otherwise,
have homotopic $\mathcal{A}$. If $\mathcal{A}$ is an abelian simplicial sheaf, applying Proposition 6 to $\mathcal{A}$, we obtain that

$$, \\ 0 = \\ 0 = \\ < \mathcal{A} \\ (6.1)$$

is an $\mathcal{A}$ -equivalence. The functor transforms chain homotopy equivalences into simplicial
equivalences. For any simplicial abelian group (to be or $\mathcal{A}$), we hence have a simplicial homotopy
equivalence

$$0 = \\)) ,$$

Simplicial homotopy equivalences being $\mathcal{A}$ -equivalences, we conclude that (6.1) induces an
 $\mathcal{A}$ -equivalence

$$) , \\) 0 = ,$$

□

2.6 Appendix: Localization

$\mathcal{A}$ be a category and $\mathcal{A}$ be a set of morphisms of $\mathcal{A}$. The localized category $\mathcal{A}[\mathcal{A}^{-1}]$ is deduced from
 $\mathcal{A}$ by “formally inverting all $\mathcal{A}$ ”. With this definition, it is clear that one has a natural functor
 $\text{MA} : \mathcal{A} \rightarrow \mathcal{A}[\mathcal{A}^{-1}]$, bijective on the set of objects, and that for any category $\mathcal{E}$, $\mathcal{A} / \mathcal{A} \rightarrow \mathcal{A}[\mathcal{A}^{-1}] \rightarrow \text{Hom}(\mathcal{A}[\mathcal{A}^{-1}], \mathcal{E})$
 $\text{Hom}(\mathcal{A}[\mathcal{A}^{-1}], \mathcal{E})$ is a bijection from $\text{Hom}(\mathcal{A}, \mathcal{E})$ to the set of functors from $\mathcal{A}$ to $\mathcal{E}$ transforming morphisms
in $\mathcal{A}$ into isomorphisms. Let

If one remembers that the categories form a 2-category, and if one agree with the principle that one should not try to define a category more precisely than up to equiv given alence (unique up to unique isomorphism), the universal property of above is doubly unsatisfactory. The easily checked and useful universal property is

the following: $\mathcal{A}$ is an equivalence from the category to the full subcategory of consisting of the functors which map to isomorphisms.

$$\begin{aligned} & / \\ & / \text{MA } \mathcal{A}(= \mathcal{E} \\ & = \mathcal{A}(= \mathcal{E} / \end{aligned}$$

in $=$ is such that the functor $* \text{L} \& \text{Hom} \mathcal{Q} \text{X} =$ transforms maps in into bijections, then

Proposition 15 (7). *If*

$$\mathcal{A}(\mathcal{Q} \mathcal{A}(\mathcal{Q} \text{UK } \mathcal{V}$$

Proof. By Yoneda construction X , $=$ embeds into the category $=$ of con

travariant functors from $=$ to Sets, while $=$ embeds into $=$, identified by (a) with the full subcategory of $=$ consisting of $/$ transforming into bijections. For in $=$, with image in $=$, and for any $/$ in $=$, one has in $=$

$$\text{X}/$$

$\text{X}\$$ / Indeed, by (a) and Yoneda lemma for $=$ and $=$ both sides are $/$. This means that $\text{X}\$$ is the solution of the universal problem of mapping X into an object of $=$. In particular, for as in (b), i.e. in $=$, $\text{X}\$$ coincides with Hom

$$\text{Hom}$$

X , as claimed by (b).

$$\text{S}$$

$$= < \mathcal{E} <$$

□

Proposition 16 (8). *Let be a pair of adjoint functors between categories and . Let and be sets of morphisms in and . Assume that maps to and and induced that maps to . Then the functors , between by and again form an adjoint pair.*

$$\mathcal{E}$$

$$\mathcal{S}$$

$$< <$$

$$=$$

Proof. The functors and induced by diagrams

$$\text{S}$$

$$=$$

$$\text{O}$$

$$\mathcal{E}$$

$$\text{S}$$

$$/ \text{ and}$$

$$\mathcal{E}$$

$$=$$

$$/$$

are characterized by commutative

$$\mathcal{E}$$

$$\mathcal{S}$$

$$@$$

$$=$$

$$\mathcal{S}$$

@ $E \leq$ = Adjunction can be expressed by the data of F PS and 2 S P - such that the compositions S S S S S are the identity automorphisms of S and respectively (see, e.g. [6]). S , By the universal property of localization, F induces a morphism F indeed, to define such a morphism amounts to defining a morphism loc S loc , S and S loc $locS$. Similarly, 2 induces 2 P - . The morphism S is induced by S , similarly for S S S S , $O = E \leq$ and the proposition follows. □

Proposition 17 (9). *Suppose that:*

$=$ = Then, a coproduct in $=$ is also a coproduct in $=$.

$=$ gives rise to a right calculus of fractions. 1. The localization 2. Coproducts exist in $=$, and is stable by coproducts.

For the definition of “gives rise to a right calculus of fractions” see [10]. It implies that for in $=$, the category of with in is filtering, and that

$$\begin{aligned} &= \\ &\mathcal{E} \mathcal{E} \text{Hom} Q \text{UK} V \text{colim} - - \text{Hom} Q \\ &= \end{aligned}$$

Proof. For in $=$, let For the coproduct of

It is cofinal: for

be the filtering category of morphisms $=$, $\mathcal{E}$, one has a functor “coproduct”:

$\mathcal{E}$

in $=$, one can construct a diagram

$\mathcal{E}$

with

in $=$.

in $=$, and $\mathcal{E}$ dominates $\mathcal{E}$. For any $=$, it follows that $\text{Hom} Q \text{UK} V \text{colim} \text{KI} - \text{Hom} Q$

$\text{colim} \text{KI} - + \text{Hom}$

$\text{colim} \text{KI} - \text{Hom}$

$\text{colim} \text{KI} - \text{Hom} \text{Hom} Q \text{UK} V$ meaning that is also the coproduct of the in $=$. $+$

$+$

$\mathcal{E}$

□

Corollary 18 (10). *Suppose that in the abelian category arbitrary direct sums exist and are exact. Then, arbitrary direct sums exist in the derived category $=$, and the localization functor*

$E \mathcal{E}$

$= \mathcal{E} E \mathcal{E}$ commutes with direct sums.

$= \mathcal{E}$

$E \mathcal{E}$

$\mathcal{E} = \mathcal{E} \mathcal{E}$

$\text{Hom} \text{Hom}$ is exact for abelian groups. Exactness of $=$ in $\mathcal{E}$ ensures that a direct sum as of quasi-isomorphisms is again a quasi-isomorphism, and Proposition 9 applies to $\mathcal{E}$ and the set of quasi-isomorphisms, proving the corollary.

Proof. The functor factors through the category of complexes and maps up to homotopy. Direct sums in $=$ are also direct sums in $=$. Indeed, $\text{Hom} G 2 \cdot \text{Hom} \cdot$

If

$\mathcal{E}$, $'$ is an inductive system of objects of $\mathcal{E}$, the colimit of $\mathcal{E}$ is the cokernel $\cdot \mathcal{E} = \cdot \mathcal{E}$

colim

$\mathcal{E}$

$\mathcal{E} \mathcal{E}$. If taking $-$ is injective: it is the

of the difference of the identity map and of the sum of the the inductive limit of a sequence is an exact functor, the map colimit of the

. %
. %
%

each of which is injective, as its graded for the filtration by the . is the identity inclusion. Under the assumptions of Corollary 10 if a complex is the colimit of an inductive sequence , and if the sequence

. = .
(10.1)

is exact, then for any , the long exact sequence of cohomology reads

Hom

=

Hom

-

Hom

The kernel of is simply the projective limit of the Hom

. One concludes.

M' (

. The cokernel is

%

□

Proposition 19 (11). *Suppose that in countable direct sums exist and are exact. If the complex is the colimit of the , and if the sequence (10.1) is exact, for instance if either:*

%

1. *In inductive limits of sequences are exact.*

2. *In each degree , each , is the inclusion of a direct factor.*

\$

then, one has a short exact sequence

$M'(\text{Hom}$

Hom

$M'(\text{Hom}$

Proof. It remains to check that condition (2) implies the exactness of (10.1). This is

to be seen degree by degree. By assumption, the , have decompositions

compatible with the transition maps . . A corresponding decomposition of (10.1) in direct sum follows, and we are reduced to check exactness of the particular case (10.1) 2 of (10.1) when for " , i.e. when is a fixed from , and is otherwise. Let (10.1) be the sequence (10.1)Z in for Z. It is a split exact sequence of free abelian groups. Because direct sums exist, , for a free abelian group is defined and functorial in . It is a sum of copies of , indexed by a basis of , and is characterized by

%

%

% %

, \$

, \$

%

%8

% %

$\text{Hom}\%$). The sequence (10.1)2 is (10.1) % and, (10.1) being split Hom

%

(functorial in exact, it splits and in particular is exact.

Hom

V

V

of a complex The truncation is the subcomplex which coincides with in homological degree and is in homological degree . For any complex , one has colim

$\$$

$5\$$

V

and this colimit satisfies condition (2) of Proposition 11. It follows that $\square$

Corollary 20 (12). *Under the assumptions of Corollary 10, for any short exact sequence*

$M'(Hom V$

Hom

$M'(Hom V$

3 A -Equivalences of Simplicial Sheaves on

2.4 Sheaves on a Site of

and , one has a

-Schemes

-Schemes

We fix a base scheme , supposed to be separated noetherian and of finite dimension; fiber product K will be written simply as . We also fix a group scheme over , supposed to be finite and flat. We note - the representable sheaf defined by . Let be the category of schemes quasi-projective over , given with an action of . Any in admits an open covering by affine open subschemes which are -stable. This makes it possible to define a reasonable quotient in the category of schemes over (rather than in the larger category of algebraic spaces). For each , is defined as the spectrum of the equalizer

[
[

O and is obtained by gluing the . It is a categorical quotient, i.e. the coequalizer of . The map is finite, open, and the topological space is the coequalizer of the map of topological spaces . One can show that is again quasi-projective. Remark 2 below shows that this fact, while convenient for the exposition, is irrelevant. One defines on [a pretopology [3, II.1.3] by taking as coverings the family of etale maps with the following property: admits a filtration by closed equivariant subschemes $\#$ such that for each $+$, some map has a section over . The Nisnevich topology on [is the topology generated by this pretopology. The category [with the Nisnevich topology is the Nisnevich site [. . O

Remark 21 (1). The corresponding topos is not the classifying topos of [2, IV.2.5]. A morphism can become a Nisnevich covering after forgetting the action of , and not be a Nisnevich covering. Example: $\mathrm{Spec} \ , \mathbb{Z} \ , \ ,$ and permutes two copies of in .

!

Remark 22 (2). Let affine . be the site defined as above, with “quasi-projective’ replaced by “affine’. It is equivalent to . , in the sense that restriction to affine . is an equivalence from the category of sheaves on . to the category of sheaves on affine . .

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L

Remark 23 (3). If π_1 is the trivial group, the definition given above recovers the usual Nisnevich topology. For π_1 , the condition usually considered: “every point of X is the image of a point with the same residue field of some U ”, is indeed equivalent to the condition imposed above. This is checked by noetherian induction: if a generic point of X can be lifted to U , some open neighborhood of x can be lifted to U , and one applies the induction hypothesis to $X \setminus U$.

L

3

3

We write $[\cdot]$ for the category of quasi-projective schemes over k , with the Nisnevich topology.

$\mathcal{U}$ is a covering of X in $[\cdot]$, there is a U in $[\cdot]$ whose pull-back to X is finer than $\mathcal{U}$. as in the definition of the Nisnevich

Proof. Fix a filtration $\#$ topology. We write π for the quotient map. For x in X , $x \in U$ is in some U_i , by equivariance of the π , and one of the maps

π_i has an equivariant section s_i over U_i . Let π_i be the henselization of π at x . The map π_i being finite, the pull-back of π_i to U_i is the coproduct of the π_i for $i \in I$. The map from U_i to U being étale, the section s_i , restricted

to U_i , extends uniquely to a section (automatically equivariant) of π_i over

U_i . Writing U as the limit of étale neighborhoods of x , one finds that U has an étale neighborhood V such that V has an equivariant section over U . The V form the required covering $\mathcal{V}$.

We define the π_1 -local henselian schemes to be the schemes obtained in the following way. For x in X , π_i a point of U_i , and π_i the henselization of π at x , take the fiber product $\pi_i \times_{\pi} U_i$. As π_i is finite over U_i , this fiber product is a finite disjoint union of local henselian schemes, and π_1 -local henselian schemes are simply the π_1 -equivariant finite disjoint unions of local henselian schemes, for which π_1 is local. $\square$

Lemma 24 (2). *If U covering V of*

is π_1 -local henselian, the functor $A(\cdot)$ is a $[\cdot]$, i.e. it defines a morphism of the punctual site $\text{Sets} [\pi_1 : N]$, the corresponding fiber functor is $\pi_1 / A(\cdot)$ ($\pi_1 : N$), the colimit being taken over the étale neighborhoods of x in X . The collection of fiber functors so obtained is conservative.

Proof. The functor $A(\cdot)$ commutes with finite limits. It follows from $\square$

Lemma 2 that it transforms coverings into surjective families of maps, hence is a morphism of sites $L \& [\cdot]$.

Proposition 25 (13). *If point of the site to X . If*

To check that the resulting set of fiber functors is conservative, it suffices to check that a family of étale is a covering if for any π_1 -local henselian,

$A(\cdot)$

is onto. The proof, parallel to that of Lemma 2 is left to the reader.

3.2 The Brown–Gersten Closed Model Structure on Simplicial Sheaves on π_1 -Schemes We recall that a commutative square of simplicial sets (or pointed simplicial sets)

(13.1)

)

is homotopy cartesian (or a homotopy pull-back square) if, when

)

weakly equivalent to it and mapping to by (Kan) fibration: map from to is a weak equivalence.

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Definition 26 (3). A simplicial presheaf on $\mathcal{C}$ is flasque if tractible and if for any (upper) distinguished square:

is replaced by

$\mathcal{C}$, the

$\mathcal{C}/\#$ is con-

$\mathcal{C}$

$(\mathcal{C}, - \text{ etale}, + \text{ open embedding}, \mathcal{C}, \mathcal{C}$ and $\mathcal{C}$), the square $\mathcal{C}/\mathcal{C}$

$\mathcal{C}/\mathcal{C}$

is homotopy cartesian.

$\mathcal{C}/\mathcal{C}/\mathcal{C}/\mathcal{C}$

Theorem 27 (3). *Let*

be a morphism of flasque simplicial presheaves.

If the induced morphism of simplicial sheaves is a local equivalence,

then, for any $\mathcal{C}$ in $\mathcal{C}$, $\mathcal{C}$ is a weak equivalence.

$\mathcal{C}/\mathcal{C}$

$\mathcal{C}/\mathcal{C}/\mathcal{C}/\mathcal{C}$

Proof. For a $\mathcal{C}$ -scheme let $\mathcal{C}$ be the small Nisnevich site of $\mathcal{C}$ and for a presheaf on $\mathcal{C}$ let $\mathcal{C}$ be the restriction of $\mathcal{C}$ to $\mathcal{C}$. Our assumption that

$\mathcal{C} \rightarrow \mathcal{C}$ is a local equivalence implies that $\mathcal{C} \rightarrow \mathcal{C}$ is a local equivalence defines a morphism of sites hence. The map $\mathcal{C} \rightarrow \mathcal{C}$ and $\square$

Lemma 2 implies that the direct image functor commutes with the associated sheaf functor and takes local equivalences to local equivalences. Therefore the morphism $\mathcal{C} \rightarrow \mathcal{C} \rightarrow \mathcal{C}$ is a local equivalence. The presheaves

and are flasque on $\mathcal{C}$ and by [8, Lemma 3.1.18] we conclude that $\mathcal{C} \rightarrow \mathcal{C}$

$\mathcal{C}/\mathcal{C}$

$\mathcal{C}, \mathcal{C}/\mathcal{C}, \mathcal{C}/\mathcal{C}$

$\mathcal{C}/\mathcal{C}$,

$\mathcal{C}$

$\mathcal{C}/\mathcal{C}$,

$\mathcal{C}, \mathcal{C}/\mathcal{C}$

$\mathcal{C}/\mathcal{C}$

$\mathcal{C}, \mathcal{C}/\mathcal{C}, \mathcal{C}/\mathcal{C}$

is a weak equivalence.

In [1], Brown and Gersten define a simplicial closed model structure on the category of pointed simplicial sheaves on a Noetherian topological space of finite dimension. As in Jardine [11], the equivalences are the local equivalences. The homotopy category is hence the same as Joyal's, but the model structure is different: less cofibrations, more fibrations. The arguments of [1] work as well in the Nisnevich topology, for the big as well as for the small Nisnevich site, or for $\mathcal{C}$, once Theorem 3 is available.

We review the basic definitions, working in $\mathcal{C}$ be the sub. $\mathcal{C}$. Let

simplicial set of $\mathcal{C}$, union of all faces but the n -th face. For $\mathcal{C}, \#$. One takes as generating trivial cofibrations the maps of the form :

$\mathcal{C}$

$\mathcal{C} \rightarrow 0$

!

$\mathcal{C}$

$\mathcal{C}/\mathcal{C} \rightarrow \mathcal{C} - 0 - \mathcal{C}$; $\mathcal{C}$ for an open embedding, $0 >$

$\mathcal{C} -$

$\mathcal{C}$

$\mathcal{C} \rightarrow \mathcal{C}$;

0 - ,

One then defines the fibrations to be the morphisms having the right lifting property with respect to generating trivial cofibrations (see, e.g. [5]), the (weak) equivalences to be the local equivalences, the trivial fibrations to be fibrations which are also (weak) equivalences, and the cofibrations to be the morphisms having the left lifting property with respect to trivial fibrations. Following [1] and using Theorem 3, one proves that the trivial fibrations can be equivalently described as morphisms having the right lifting property with respect to the following class of morphisms :

P

$P \# \rightarrow 0 \rightarrow P \%$ for open embedding, $0 \rightarrow$ The maps of the form

$\sim \%$

$\rightarrow 0 \rightarrow$

$0 \rightarrow$

P are called generating cofibrations.

376 P. Deligne For and pointed simplicial sheaves, one defines a pointed simplicial set by A(0 Following [1], one sees that the classes of cofibrations, (weak) equivalences, fibrations, and are a simplicial closed model structure in the sense of [1]. This has the following consequences.

Corollary 28 (14). *If is cofibrant and fibrant, for any pointed simplicial set has in the relevant homotopy categories*

, one

Hom^* In particular, taking ! 0 one gets $\text{Hom}^* \text{Hom}^*$

Corollary 29 (15). *If is a cofibration and C a cofibrant object, then C C is a cofibration.*

Corollary 30 (16). *If is cofibrant and is fibrant, then for any C $\text{Hom}^* C \text{Hom} \text{Hom}^* C$ (16.1)*

In (16.1), Hom is the pointed simplicial sheaf with components the sheaves of homomorphisms from to . We now apply this framework to prove the following criterion for A -locality.

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0

[[

. If, as a simplicial

Proposition 31 (17). *Let be a pointed simplicial sheaf on presheaf, is flasque, then is A -local if and only if, for any in ,*

/

/

/ / A is a weak equivalence. We recall that A -local means that for any category one has the following in the homotopy

$A(*) / A(*) A /$

(17.1)

Lemma 32 (3). *A fibrant pointed simplicial sheaf is flasque.*

/

;

Proof. The right lifting property relative the morphisms % means that for an open embedding, the morphism is a Kan fibration. As is a sheaf, an upper distinguished square

/

/

/

% gives rise to a Cartesian square

/ /

$/\% / \mathcal{A}_s$

$/ /$ is a Kan fibration, this square is also homotopy Cartesian. $/$ is flasque' is replaced by

□

Lemma 33 (4). *Proposition 17 holds of the assumption “ the assumption “ is fibrant’.*

$/$

P

$\$$

says that for any

Proof. “Only if” # for

□

Corollary 14, for any pointed simplicial set , one has

, $>$ is cofibrant. By

$> / \text{Hom}^* > /$ and $> /$ is just $/$. If in (17.1) we take $>$, so that $> \mathcal{A}$ we get $\mathcal{A} \text{Hom}^* / \mathcal{A} \text{Hom}^* /$ That this holds for any means that $/ / \mathcal{A}$ becomes an isomorphism Hom^*

in the homotopy category , hence is a weak equivalence. “If” We apply Corollary 16. As $\mathcal{A}$ is cofibrant and fibrant,

$\text{Hom}^* \mathcal{A} /$

$/ \text{Hom}^* \text{Hom} \mathcal{A} /$

and it suffice to show that

$/$

$\mathcal{A} /$

Hom

is a local equivalence. This Hom is a simplicial sheaf claim follows.

$/ \mathcal{A}$ and the

$/$

$/$

be a fibrant

Proof. We can now finish the proof of Proposition 17. Let

replacement of . As and are flasque, is a weak equivalence for any . That all $\mathcal{A}$ be weak equivalences is hence equivalent

to all $\mathcal{A}$ be weak equivalences, while is $\mathcal{A}$ -local if and only

if is.

$/$

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$/ / / /$

$/ /$

$/$

3.3 -Closed Classes The proof of the main theorem of this section will be postponed.

of morphisms of pointed simplicial sheaves is 0-closed if: 1. (Simplicial) homotopy equivalences are in . 2. If two of , and are in then so is the third. 3. is stable by finite coproducts. 4. If $/$ is a morphism of pointed bisimplicial sheaves, and if all $/$ are in , so is the diagonal $0/0$.

-closed if, in addition, it is stable by arbitrary coprodDefinition 5. The class is 0 ucts and colimits of sequences $/$ with the property that, degree by degree, $/ /$ (resp.) is isomorphic to an embedding $\% \%$ of pointed sheaves. -closure of the union of the

□

Theorem 34 (4). *The class of $\mathcal{A}$ -equivalences is the 0*

Definition 35 (4). A class

classes of:

1. Local equivalences 2. Morphisms $\mathcal{A}$

for in (! -closed. In particular, the class of $\mathcal{A}$ -equivalences is 0

3.4 The Class of A -Equivalences Is -Closed The properties 4(1), 4(2), 4(3) are clear. The last property is proved in Proposition 20.

%

Lemma 36 (5). *Let $\mathcal{S}$ be a pointed simplicial set and $\mathcal{S}$ is fibrant and A -local, then $\mathcal{S}$ is A -local.*

Proof. Because

$\mathcal{S}$ a pointed simplicial sheaf. If

$\mathcal{S}$ is fibrant, for any A , one has in the homotopy category

$2 \text{ Hom}^* \mathcal{S}(A, \mathcal{S}) \xrightarrow{\sim} \text{Hom}^* \mathcal{S}(A, \mathcal{S})$ Applying this to A and using Hom^*

(17.1)

$\mathcal{S}(A, \mathcal{S}) \xrightarrow{\sim} \mathcal{S}(A, \mathcal{S})$ one deduces from the A -locality of that of $\mathcal{S}$.

□

Lemma 37 (6). *Let f be a morphism of pointed simplicial sheaves and $\mathcal{S}$ be a pointed simplicial set. If f is an A -equivalence, then so is $\mathcal{S}(f, \mathcal{S})$.*

one has in the homotopy category $\text{Hom}^ \mathcal{S}(f, \mathcal{S}) \xrightarrow{\sim} \text{Hom}^* \mathcal{S}(f, \mathcal{S})$ Replacing by a fibrant replacement, one may assume $\mathcal{S}$ fibrant. Applying (17.1)*

Proof. One has to check that for any A -local

one is reduced to Lemma 5.

□

Lemma 38 (7). *Let f be a morphism of pointed simplicial sheaves. If f and $\mathcal{S}$ are cofibrant, then f is a A -equivalence if and only if for any fibrant A -local $\mathcal{S}$, the morphism of simplicial sets $\mathcal{S}(f, \mathcal{S})$ is a weak equivalence.*

Proof. “If” Taking

one deduces from the assumptions that

$\text{Hom}^* \mathcal{S}(f, \mathcal{S}) \xrightarrow{\sim} \text{Hom}^* \mathcal{S}(f, \mathcal{S})$ “Only if” The assumptions imply that f and $\mathcal{S}$ are fibrant. For any pointed simplicial set

$\mathcal{S}$ one has $\mathcal{S}(f, \mathcal{S}) \xrightarrow{\sim} \mathcal{S}(f, \mathcal{S})$ and similarly for

and one applies Lemma 6.

□

Proposition 39 (18). *The coproduct of a family of A -equivalences is an A -equivalence.*

Proof. There are commutative diagrams

+

+

where morphisms on the first line are cofibrations, and where the vertical maps are local equivalences, and similarly for $\mathcal{S}$. Replacing (resp. $\mathcal{S}$) by (resp. $\mathcal{S}$) we may and shall assume that f and $\mathcal{S}$ are cofibrant. The coproducts $\mathcal{S}$, are then cofibrant too. One has

and similarly for the $\mathcal{S}$, and one applies Lemma 7, and the fact that a product of a family of weak equivalences of fibrant pointed simplicial sets is a weak equivalence.

□

Proposition 40 (19). *The colimit*

$\text{colim} \mathcal{S}_i$

$\text{colim} \mathcal{S}_i$

of an inductive sequence of A -equivalences is an equivalence.

/

/ is again an A -

Proof. One inductively constructs an inductive sequence of commutative squares

/

in which the vertical maps are local equivalences, the and are cofibrant and the transition maps

, are cofibrations. A colimit of local equivalences being a local equivalence, it is sufficient to prove the proposition for the sequence . We hence may and shall assume that

is a sequence of cofibrations and similarly for the . The

colimits and of those sequences are then cofibrant. is the limit of the sequence of If is fibrant and A -local, weak equivalences

/

/ In the sequences and / the transition maps are fibrations of fibrant objects. It follows that the limit is again a weak equivalence: the of the limit map onto the limit of , with fibers M'(-torsors. It remains to apply

□

Lemma 41 (7). /

Proposition 42 (20). *Let be a morphism of pointed bisimplicial sheaves. If*

all . are A -equivalences, so is

/

0/ 0 /

To prove Proposition 20 we will functorially attach to an inductive sequence of pointed simplicial sheaves , whose colimit maps to by a local equivalence. We will then inductively prove that is an A -equivalence, and apply Proposition 19. We begin with preliminaries to the construction of the .

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Proposition 43 (21). *Let be the category of finite ordered sets and increasing injective maps. For any category with finite coproducts,*

=

the forgetting functor

| 0* =

0* =

| ,

has a left adjoint : “formally adding degenerate simplices”:

coproduct, over all and all increasing surjective maps of

ℰ 0

|

?B 0 =

| is the 0 , of copies

$0 =$
 $* *$ We define the wrapping functor as the composite . For the category of sets or of pointed sets one has the following.

$\|$
 $=$

Lemma 44 (8). *The adjunction map*

$?B$

$\gamma ? B$ is a weak equivalence.

$=$

Proof. We will prove it for the category of sets. The pointed case is similar. The fundamental groupoid of is the category with set of objects , in which all maps are isomorphisms, and universal for the property that:

V defines a morphism $V \rightarrow V \rightarrow V$. $G, \rightarrow G \rightarrow G \rightarrow G$. One has $? B$. To handle and it suffice to show that γ induces an isomorphism of fundamental groupoids. For any and any , , $\&$, is the identity of , . This results from (2) applied to $\& \&$, which gives $\& , \& , \&$, (1) (2) For

As generators of the fundamental groupoid, it hence suffices to take non degenerate . For relations, it then suffices to take those coming from non degenerate : the degenerate give nothing new. If we apply this to , we find as set of generators , and relations indexed by , the same relations as for . The functor commutes with passage to connected components and to passage to a covering. To handle higher , this reduces us to the case where (and hence) is connected and simply connected. In this case it suffices to check that induces an isomorphism in homology. It does because one has a commutative diagram

degeneracies

$V \rightarrow G$

$G \rightarrow ? B$

$?B$

$? B \rightarrow \gamma$

$=$

6

$= ? B \rightarrow \gamma =$ degeneracies

in which the first arrow is an isomorphism, the second the effect of and the composite is a homotopy equivalence.

$\$$

$\&! \&!$

γ on homology,

□

Proposition 45 (22). *For a pointed simplicial sheaf, let be the n -th skeleton of , i.e. simplicial subsheaf of for which*

is the union of the images of the degeneracies . One has push-out squares for

$>0 \mathcal{E}! ? B$

(22.1)

$0 \mathcal{E}! ? B$ Let now $/$ be bisimplicial. Each $/$ is simplicial, and they form a simplicial system of pointed simplicial sheaves. Let us apply $? B$ and $\mathcal{E}!$ to the first variable, i.e. to the simplicial sheaf $/$ for each fixed (. We again have diagrams (22.1) and, taking the diagonal 0, one obtains push-out squares: $/ >0 0\mathcal{E}! ? B/$

(22.2)

$/ 0 0\mathcal{E}! ? B/$ where $/$ now stands for the pointed simplicial sheaf $/$. This way the simplicial sheaf $0 ? B/$, which by Lemma 8 maps to $0/$ by a local equivalence, appears as an inductive limit of (22.2).

Proof of Proposition 20: With the notations of 22 it suffice to show that the

$0 \mathcal{E}! ? B / 0 \mathcal{E}! ? B \mathcal{E}! ? B \$ \$ 0 \mathcal{E}! ? B / \$ \$$
 (with applied in the first variable) are A -equivalences. We prove it by induction on . For , ,
 and is assumed to be an A equivalence. From to , we have a morphism of push out squares

$$\begin{array}{ccc} & & \\ & & \\ & & \\ (22.2) \text{ for} & & \\ (22.2) \text{ for} & & \\ / \text{ is an } A\text{-equivalence, by Lemma 6, so are its smash product with } >0 \text{ and } 0 . \text{ It remain to} \\ \text{apply the As} \end{array}$$

Lemma 46 (9). Suppose given a morphism of push out squares

$$\begin{array}{ccc} 1 & 2 & \\ 3 & & \\ 1 & & \\ 2 & & \\ 3 & & \\ 4 & & \\ 4 & & \end{array}$$

 which is an A -equivalence in positions 1 2 and 3. If in each square the first vertical map is
 injective, then the morphism of squares is an A -equivalence in position 4 as well.

Proof. Replacing the push out squares by push out squares of local equivalent objects, we may
 and shall assume that all objects considered are cofibrant, and that the vertical maps are cofi-
 brant.

If is fibrant applying to each of the squares we get a morphism of cartesian squares, of pointed
 simplicial sets in which the vertical maps are cofibrations:

$$\begin{array}{ccc} 4 & 3 & \\ 4 & 3 & \\ 2 & 1 & \\ 2 & 1 & \end{array}$$

 If is in addition A -local, it is a weak equivalence in positions 1 2 and 3,
 hence also in position 4. By Lemma 7, this proves Lemma 9, finishing the proof of Proposition
 20 as well as of the claim that the class of A -equivalences is -closed.

0

□

3.5 The Class of A -Equivalences as a -Closure In this section we finish the proof of Theorem
 4.

Proposition 47 (23). The homotopy push-out of a diagram

$$\begin{array}{ccc} [& \text{is the push-out} & \\ (23.1) & & \\ A & \text{of} & \\ 5 & & \\ [& & \\ 5 & (23.2) & \\ 0 & \text{where the vertical map } 0 \rightarrow 0 \text{ is induced by } > > 0 & \\ 0 & \text{mapping } 0 \text{ to (resp.) in } 0 . \text{ In the case of simplicial sets, } A \text{ maps to } 0 \text{ with fibers above} & \\ , \text{ above } , \text{ and above } . \text{ The homotopy push-out } A \text{ is the diagonal of the bisimplicial object with} & & \\ \text{columns } 5 \neq 5 \text{ obtained by formally adding degeneracies to} & & \\ 5 & & \\ 0^* 0 & & \\ > & & \end{array}$$

$\begin{array}{c} > \\ [\\ \end{array}$
in $\ast$ (cf. 21) [maps to , maps to]. If a morphism of diagrams (23.1), the induced morphism from A to
 $[$ *is*
 A is hence in
the closure of the three components of diagonal $(4(3), (4))$. A commutative square for the operation of finite coproduct and induces a morphism

Example 48 (1). Let

$\begin{array}{c} A \\) \\) \end{array}$
 be a morphism. The homotopy push-out of
 $=$
 is the cylinder
 (23.3)
 $\cdot M$ of $\cdot$. The morphisms $\cdot M$
 are homotopy equivalences. To check that the composite cyl homotopic to the identity, one observes that cyl is the push-out of
 $\begin{array}{c} \text{is} \\ \text{cyl} \\ 0 > 0 \ 0 \ 0 \ 0 \\ 0 \end{array}$
 (the vertical map induced by mapping to $)$ and that the composite
 is induced by $\cdot$, homotopic to the identity by a homotopy fixing $\cdot$. Similar arguments would show that the homotopy push out cyl of
 $\begin{array}{c} \cdot M \\ \cdot M \\ \text{is homotopic to} \\ 0 \\ \text{cyl} \\ \text{by} \\ = \\ \cdot \end{array}$

Example 49 (2). In any category with finite coproducts, a coprojection is a map isomorphic to the natural map for some α and β . If in a push-out square of pointed simplicial sheaves

$\begin{array}{c} \% \\ \% \\ \% \end{array}$
 the morphism of the squares
 $) \ 5 \ \%$, the square (23.4) is the coproduct
 is a coprojection:
 $=$
 $\%$
 $=$
 (23.4)
 and
 (23.5)
 $\%$

)

is a homotopy equivalence, being the coproduct of the resulting morphism A and the homotopy equivalences of Example 1 resulting from the two squares (23.5). The same conclusion applies if A is a coprojection. A morphism of pointed simplicial sheaves is a termwise coprojection if each A_n is a coprojection of pointed sheaves. Example: for any diagram

(23.1), the morphisms are termwise coprojections. For any morphism A , this applies in particular to A .

.M

Proposition 50 (24). *If in a cocartesian square (23.3) either A is a termwise coprojection, then the resulting morphism from the closure of the empty set of morphisms.*

or A to

is

) is in the 0-

\$

Proof. For each n , we have a cocartesian square of pointed sheaves

[

)

Let us view it as a cocartesian square of pointed simplicial sheaves. By Example 2, it gives rise to a homotopy equivalence A . One concludes by observing that A is the diagonal of this simplicial system of morphisms.

)

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□

Corollary 51 (25). *If in a cocartesian square (23.3):*

)

or A is a termwise coprojection, then: 1. A is in the 0-closure of A . 2. A is in the 0-closure of A .

Proof of (1): The morphism of cocartesian squares

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defines a commutative square

A

A

)

0 0

in which the vertical maps are in the 0-closure of the empty set by Proposition 24, while the first horizontal map is in the 0-closure of A by 8.

Proof of (2): One similarly uses

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)

/

Proposition 52 (26). *A pointed simplicial sheaf is reliably compact if it coincides with its n -skeleton for some n and each A_n is compact in the sense that the functor commutes with filtering colimits. This property is stable by*

for a finite pointed simplicial set (finite number of non degenerate simplices) and implies that is compact.

$\$ A(/$

$\$$

$/$

$/$

$/$

$/$

Construction 27. Let

and $)$ be classes of morphisms such that:

(a) Sources and targets are reliably compact. (b) Each in is a termwise coprojection.

)

from pointed simplicial sheaves to pointed simplicial P - such that:

We will construct a functor sheaves and a morphism (i) For any (ii) If

$/, /$

$/$ is in the 0 -closure of is in $,$ the morphism

is a weak equivalence. is in $,$ the morphism (27.1) is a Kan fibration. (iii) If

)

(27.1)

Let us factorize as cyl . As the second map is a homotopy equivalence, the first is in the -closure of $.$ In the corresponding factorization of (27.1):

0

$/ cyl / /$ the first map is a homotopy equivalence. To obtain (ii), it hence suffices that cyl be a weak equivalence. Replacing each in by the corresponding cyl , this reduces us to the case where

$/ /$

in is a termwise coprojection, and we will construct in this case a functor such that (iii) for in , (27.1) is a trivial fibration. (c) each

The conditions (ii), (iii) are lifting properties: for

in , in squares: $>0 /$

0 / for

in) , in squares: $Y /$

0 / In the first case, the data are morphisms 0 / and >0

agreeing on >0 , i.e. a morphism 0 and we want it to extend to by :

Y for in :

0

0

$\wedge G$

>0

$/$

. Similarly in the second case, with

>0

$\wedge G$

0

$/$

>0 replaced

$/$

(27.2)

for

in) : 0

Y

$$\begin{array}{c} 0 \\ / \\ \backslash G \\ (27.3) \end{array}$$

The left vertical maps are termwise coprojections, and their sources are compact. One now uses the standard trick of defining as the inductive limit of the iterates of functors , where is deduced from by push out, simultaneously for all

$$\begin{array}{c} </ \\ / </ \\ / \\ >0 \\ / \\ in \\ Y \\ / \\ in \\ / \\ 0 \\ \wedge G \\ and \\ 0 \\ G \\ \backslash \end{array}$$

The push out is by sources

$$\begin{array}{c} 0 \\) \end{array}$$

a morphism which is a termwise coprojection. By 26, to check that the resulting is in the -closure of , it suffices to check that the left vertical morphism in (27.2) (resp. (27.3)) is in the -closure of (resp. of the empty set).

$$\begin{array}{c} / \\ / \\ 0 \\ 0 \end{array}$$

For (27.2), this is the map marked 3 in >0 0

$$\begin{array}{c} 1 \\ >0 \\ 3 \\ 0 \\ 2 \end{array}$$

The morphisms 1 and 3 $\mathcal{A}$ 2 are in the the 2 out of 3 property. 0-closure of . So is 2 by 26 and one applies

For (27.3), the diagram is

$$\begin{array}{c} Y \\ Y \\ 1 \\ 0 \\ 2 \\ 3 \\ 0 \end{array}$$

with 1 and 3 $\in$ 2 in the contractible.

0-closure of the empty set. Indeed, Y and 0 are both

Remark 53 (4). Let $\mathcal{P}$ be a property of pointed simplicial schemes stable by coproduct, and suppose that: (a) For (b) For

5%

) %

in $\mathcal{P}$, the $\mathcal{P}$ and have property $\mathcal{P}$. in $\mathcal{P}$, is in degree isomorphic to the natural map for some having property $\mathcal{P}$.

\$

The functor constructed in Construction 27 is then such that for any $\mathcal{P}$, each morphism where has is isomorphic to some

5 property $\mathcal{P}$. In particular, if the have property $\mathcal{P}$, so have the $\mathcal{P}$.

%

%

Proposition 54 (28). (Proof of Theorem 4) We apply Construction 11, on the site taking for and the following classes.

)

: For any in the site, the morphism A

[,

(28.1)

and for any upper distinguished square

(28.2)

% the morphism

) : For any in the site,

A

(28.3)

#

(28.4)

/

)

, that (28.4) is in ensures If a pointed simplicial sheaf is of the form that each is Kan. That (28.3) is in ensures that for each upper distinguished square (28.2), the morphism

A is a weak equivalence. As each is Kan, A is the homotopy fiber product of % over $\mathcal{P}$, and % is homotopy cartesian:

is flasque.

, for each $\mathcal{P}$, A

Further, as (28.1) is in

is a weak equivalence: by Theorem 3, is A -local. Suppose now that is a A -equivalence. In the commutative diagram

/

/

/-closure of the morphisms (28.1) and (28.2), the later the vertical maps are in the 0 being local equivalences. In particular, they are A -equivalences and / is an A -equivalence between A -local objects $\mathcal{P}$, hence is a local equivalence. -closure, proving Theorem 4. It follows that is in the required 0 The functor used introduced in 28 can also be used to prove the following lemma.

/

/

Lemma 55 (10). If $\mathcal{P}$ is a filtering system of A -equivalences, then is again an A -equivalence.

$AM'(\mathcal{P})$

/

Proof. Consider the square:

$$\begin{array}{ccc} & / & \\ \text{AM}'(& / & \\ / \text{AM}'(& / & \end{array}$$

Since the functor commutes with filtering colimits, the bottom arrow is a filtering colimit of local equivalences, hence a local equivalence. The vertical maps are A -equivalences, hence the top map is an A -equivalence. □

2.5 One More Characterization of Equivalences

[

Denote by the full subcategory in the category of pointed sheaves on generated by all coproducts of sheaves of the form - .

[

Theorem 56 (5). *The class of local equivalences (resp. A -equivalences) in is the smallest class which contains morphisms A upper distinguished and has the following properties:*

$$\begin{array}{c} [[\\ ? \end{array}$$

$$0^*$$

for

1. *Simplicial homotopy equivalences (resp. and A -homotopy equivalences) are in .*

$$?$$

, and are in ? then so is the third. / / is a filtering system of termwise coprojections in ? , then /

$\text{AM}'(/$ is again in ? . / is a morphism of bisimplicial objects, and if all / / are 4. If / in ? , so is the diagonal $0/ 0/$. 2. If two of 3. If

The proof is given in 32.

$$/ /$$

, / > , then is

is such that, for each

Lemma 57 (11). *If the morphism is a weak equivalence, and if the and are all of the form in the -closure of the empty set.*

$$0$$

The proof will use the following construction.

$$=$$

$$= =$$

$$,$$

$$>$$

$$\%>$$

$$= /$$

be a set of objects of , such that Construction 29. Let be a category, and let any isomorphism class has a representative in . Let be the functor which to a presheaf of pointed sets on attaches the family of pointed sets $> Q$. It has a left adjoint :

$$=$$

$$,$$

family

$$\%> > Q$$

$$>$$

disjoint sum over the $\mathcal{C}$ and $\mathcal{D}$ of $\mathcal{F}$ is
 If $\mathcal{C}$ is viewed as a category whose only morphisms are identities, the natural functor
 $\mathcal{F} = \mathcal{F} \circ \mathcal{C}$ defines a morphism of sites $\mathcal{C} \rightarrow \mathcal{D}$, both endowed with the trivial topology (any presheaf
 a sheaf), and $\mathcal{F}_*$, $\mathcal{F}^*$ are the corresponding direct and inverse image
 of pointed sheaves. By a general story valid for any pair of adjoint functors, for any pointed
 presheaf

on $\mathcal{C}$, the form a pointed simplicial presheaf augmented to :

$$\begin{array}{c} / \\ / \\ = \\ ' ' / \\ S/ \end{array}$$

7 $S/$ / Further:

$$\begin{array}{c} / \\ ' \% \\ 7 \end{array}$$

, i.e. of the form $\mathcal{F} = \mathcal{F} \circ \mathcal{C}$, is a homotopy equivalence. (a) If $\mathcal{C}$ is of the form $\mathcal{C} = \mathcal{C} \circ \mathcal{D}$ is a homotopy equivalence:
 for each in $\mathcal{C}$, (b) For any $\mathcal{C}$, $\mathcal{F}$ is a homotopy equivalence.

$$\begin{array}{c} / \\ / ' 7 \\ = S/ \\ / \text{ we define } S/ 0 \text{ simplicial system of } S/ \end{array}$$

For a simplicial presheaf

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[

Proposition 58 (30). (Proof of Lemma 11) Let us say that $\mathcal{C}$ is connected if it is not empty
 and is not a disjoint union: $\mathcal{C}$ should act transitively on the set of connected components of $\mathcal{C}$. Let
 $\mathcal{C}_0$ be the full subcategory of connected objects. A sheaf on $\mathcal{C}$ is determined by its restriction to $\mathcal{C}_0$.
 Indeed,

. To apply Construction 29, we will use this remark to identify the category of sheaves on $\mathcal{C}$ to
 a full subcategory of the category of presheaves on $\mathcal{C}$. For any $\mathcal{F}$ as in Construction 29, the functor
 takes values in sheaves, that is in the restriction of sheaves to $\mathcal{C}_0$. Indeed, for connected, $\mathcal{F}$ is the $\mathcal{C}$
 same, whether $\mathcal{C}$ and $\mathcal{C}_0$ are taken in the sheaf or in the presheaf sense.

$$\begin{array}{c} / \\ / / = = \\ = [[= \\ [\\ = \\ , \end{array}$$

Fix $\mathcal{C}$ as in Lemma 11. For each $\mathcal{C}$, the assumption on $\mathcal{C}$ ensures that $S/ \mathcal{C}$ is in the 0-closure
 of the empty set, and similarly for $\mathcal{C}_0$. For each (connected) $\mathcal{C}$, the morphism of pointed simplicial
 sets $\mathcal{C} \rightarrow \mathcal{C}_0$

-closure of the empty set. It follows that $\mathcal{C}$ is a weak equivalence, hence in the 0 $' ' / ' '$: the
 5 over $=$ of the $\mathcal{C} / \mathcal{C}_0$

-closure of the empty set. Iterating one finds the same for $' ' /$ is in the 0
 $' '$, and S/ S is in this 0 -closure too. It remains to apply the two out of three property to
 S/ S

$\mathcal{C} / \mathcal{C}_0$ is a local equivalence and if the $\mathcal{C}$ and $\mathcal{C}_0$ are all of for $[$ upper, then $\mathcal{C}$ is in the 0 -closure of
 the A

Lemma 59 (12). *If the form $>$ distinguished.*

/

Proof. We will use the Construction 27 for the class of morphisms A for upper distinguished, and for the class of morphisms $.$ By Remark 4, if the are of the form $,$ so are the $> .$ In the commutative diagram

[
)

/

/

/ the vertical maps are in the required 0-closure. They are in particular local equivalences and so is $.$ One verifies as in 28 that $/$ and are flasque. By Construction 27(ii), for each $,$ is a weak equivalence, and it

remains to apply Lemma 11 to $\text{Ex}(f).$

/

□

Lemma 60 (13). *If $is an A$ -equivalence and if the the form $,$ then $is in the$ -closure of the $A > distinguished and A$ for $.$*

0

[

Proof. Similar to the proof of Lemma 12.

/ and are all of for [upper

/

S/

/

□

Proposition 61 (31). *Since for any simplicial sheaf the map is a local equivalence Lemmas 12 and 13 imply that for any local (resp. A -) equivalence $,$ the morphism belongs to the -closure of the A for upper distinguished (resp. the A for upper distinguished and A for $).$*

/ [

S

[

0

[

Proposition 62 (32). *Proof of Theorem 5: We consider only the case of A -equivalences.*

Proposition 20 and Lemma 10 imply that A -equivalences contain the class $.$ In view of Lemma 13 it remains to see that is -closed. The only condition to check is that it is closed under coproducts. Let

?

? 0

/ % be a family of morphisms in ? . For a finite subset P in % set $aD D$

2D

/

$P ;$ we have a morphism $aD aF$ and the map is isomorphic to the $a \text{ colim} D 2aD$ It remains to show that $aD aD$ is in ? . This morphism is of the form $P- / /$ where $is in ? .$ Using the fact that ? is closed for diagonals we reduce to the case $> .$ Using the same reasoning as above we further reduce to the case $> .$ Consider the class of such that $P-$ is in ? . This class clearly contains morphisms $A ,$ has the two out of three property and is closed under filtering colimits. It also

contains simplicial homotopy equivalences. It contains morphisms of the form $\mathcal{A} \rightarrow \mathcal{B}$ because such morphisms are For map

$\mathcal{A} \rightarrow \mathcal{B}$
 $\mathcal{A} \rightarrow \mathcal{B}$
 $\mathcal{A} \rightarrow \mathcal{B}$ -homotopy equivalences.

3 Solid Sheaves

3.1 Open Morphisms and Solid Morphisms of Sheaves

We fix and repeated in
as in Sect. 3.1, and will work in

$(\mathcal{A}, \mathcal{B})$
 $\mathcal{A} \rightarrow \mathcal{B}$
 $\mathcal{A} \rightarrow \mathcal{B}$. The story could be

Definition 63 (6). A morphism of sheaves is open if it is relatively representable by open embeddings, i.e. if for any morphism u (that is, u is

u in $(\mathcal{A}, \mathcal{B})$, the fiber product $\mathcal{A} \times_{\mathcal{B}} \mathcal{C}$ mapping to $\mathcal{A}$ is isomorphic $\mathcal{A}$ -stable open subset of $\mathcal{A}$. $\mathcal{A}$ for a In other words: should identify $\mathcal{A} \times_{\mathcal{B}} \mathcal{C}$ with a subsheaf of $\mathcal{A}$ and, for any $\mathcal{C}$, there is open in $\mathcal{A}$ and $\mathcal{A}$ -stable such that the pull-back of $\mathcal{C}$ with respect to u is in $\mathcal{A}$ if and only if u maps to $\mathcal{A}$.

The property “open” is stable under composition. It is also stable by pull-back: if in a cartesian square

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{f} & \mathcal{B} \\ \downarrow u & & \downarrow v \\ \mathcal{C} & \xrightarrow{g} & \mathcal{D} \end{array} \quad (32.1)$$

if u is open, then if f is open and u is an epimorphism, then g is open. Indeed,

Lemma 64 (14). For f - a morphism, the property that f is represented by open in is local on (for the Nisnevich topology).

Proof. Suppose that the cover $\mathcal{A} \rightarrow \mathcal{B}$, and that each $\mathcal{A}_i \rightarrow \mathcal{B}$ is represented by $\mathcal{A}_i \rightarrow \mathcal{B}$. For $\mathcal{A} \rightarrow \mathcal{B}$, $\mathcal{A}/\mathcal{B} \rightarrow \mathcal{B}$ is then represented by $\mathcal{A}$ with $\mathcal{A}$ the inverse image of $\mathcal{B}$. By descent for open embedding, the come from some $\mathcal{A}$, we have locally on $\mathcal{A}$ an isomorphism $\mathcal{A} \rightarrow \mathcal{B}$ and by descent for

isomorphisms of sheaves one has $\mathcal{A} \rightarrow \mathcal{B}$

$\mathcal{A} \rightarrow \mathcal{B}$. Given a square of the form (32.1) with open u and u an epimorphism, if $\mathcal{C}$ is in $\mathcal{A}$, $\mathcal{C}$ can locally be lifted to a section of $\mathcal{A}$. As u is open, it follows that locally on $\mathcal{A}$, $\mathcal{C} \rightarrow \mathcal{B}$ is represented by an open subset. Applying Lemma 14 one concludes that g is open. The same argument shows that if we have cartesian diagrams u is open. This follows from transitivity of pull-backs. Conversely,

$\mathcal{A} \rightarrow \mathcal{B}$
with each
open and
 $\mathcal{A} \rightarrow \mathcal{B}$
 $\mathcal{A} \rightarrow \mathcal{B}$
 $\mathcal{A} \rightarrow \mathcal{B}$
onto, then
 $\mathcal{A} \rightarrow \mathcal{B}$ is open.

□

Proposition 65 (33). The property “open” is stable by push-outs.

Proof. Suppose

$/$

$/$

$+$

is a cocartesian diagram, with open . In particular is a monomorphism, and it follows that is a monomorphism and that the square is cartesian as well. The

is an is onto. The pull-back of by morphism

isomorphism (being a monomorphism) hence open . The pull-back of by is just , open by assumption. It follows that is open .

$/$

We now fix a class following stabilities

$=$ of open embeddings

6 in $[$. We require the

6 are open embeddings and if 6 is in $=$, 6 6 is an open embedding in $=$, and if 6 6 is etale, with 2. If

for the complement of in 6 , then 6 is in $=$. The classes $=$ we will have to consider are the following: 6 with 6 smooth. 1. The open embeddings 6 with 6 smooth such that the action of is free 2. The open embeddings on 6 . Equivalently: 6 is the union of $\text{and the open subset on which the Conditions 34. 1. If so is}$.

$/$

action of is free . 3. When working in

($\text{: all open embeddings. } /$

$= /$

$/$

$/$

□

Definition 66 (7). A morphism is -solid if it is a composite where each is deduced by push-out from some

in $\text{. } > \text{-}$, is -solid . A sheaf is solid if $\#$ In the pointed context, a pointed sheaf is (pointed) -solid if the morphism is -solid .

$/$

,

$/$

$= /$

$/$

$/ =$

$=$

$/ =$

Example 67 (1). For open in $\text{, let } \text{-I> be the sheaf } \text{-}$, with the subsheaf $>$ contracted to a point. If is in , then -I> is solid: it is the push-out of $>$ by $\text{. Thom spaces are of this form: starting from } >$ a vector bundle on $\text{, one contracts, in the total space of this vector bundle, the complement of the zero section to a point.}$

6

,

,

$=$

,

,

The class of solid morphisms is the smallest class closed by compositions and pushouts which contains all $>$ in $\text{. By Proposition 33 solid - for morphisms are open. For a monomorphism of}$

sheaves, define to be the pointed sheaf obtained by contracting to a point: one has a cocartesian square

$$\begin{array}{c} / \\ = \\ / \\ / \\ / \\ , / \end{array}$$

By transitivity of push-out, any cocartesian diagram

$$\begin{array}{c} / \\ / / . \text{ induces an isomorphism } / \end{array}$$

Proposition 68 (35). *A morphism of sheaves $/$ is $=$ -solid if and only if it is a composite $/ /$ of monomorphisms where each $//$ is isomorphic to some $TI > T >$ for 6 in $=$. $/$ is deduced by push-out from*

Proof. If a morphism isomorphic to $TI >$. From this, “only if” results. Conversely, if we have

$$\begin{array}{c} / \\ 6 , / \text{ is} \\ > \\ T \\ T I > \\ (35.1) \\ T I > \\ / \end{array}$$

cocartesian, and if T is deduced by push-out from $TI >$ lifts to $,$ then $> T$. Indeed, the diagrams (35.1) being cartesian as well as cocartesian, we have a cartesian

$$\begin{array}{c} > \\ T \\ / \text{ If} \\ \text{is deduced from} \\ T \text{ by push-out:} \\ > \\ > \\ / / \\ T \\ / 6 \\ T \end{array}$$

then and it follows that . Let us suppose only that we have $v \&$ etale, inducing an isomorphism from $\& v$ to and a lifting of $T \&$. If $\& v , TI >$ to is an isomorphism. This is most easily checked by applying the fiber $T I > T \& I > \&$ functors defined by a $-$ local henselian : a morphism $,$ if it does not map to $,$ lifts uniquely to a morphism to $\&$. The assumptions made hence imply that is a push-out of $> \& , T \&$. Note that by the second stability property of $\& \&$ is in $.$ We will reduce the proof of “if” to that case. We have to show that if a monomorphism is such that $TI >$ with in $,$ then is solid. The cocartesian square

$$\begin{array}{c} 6 \\ / \\ 6 \\ 6 \\ 6 = / \\ / \\ 6 \end{array}$$

$\mathcal{G}$

$\mathcal{G}$

$\mathcal{G} =$

$\mathcal{G}$

$\mathcal{G}$

(35.2)

$\mathcal{G}$

$\mathcal{G}$

$\mathcal{G} \subset \mathcal{I} \mathcal{G}$,

$\mathcal{G}$ induces an epimorphism $\mathcal{I} \mathcal{G} \rightarrow \mathcal{G}$. The natural section of $\mathcal{I} \mathcal{G}$ on $\mathcal{G}$ hence locally be lifted to $\mathcal{G}$ or to $\mathcal{G}$ for some filtration $\#$

of by closed equivariant subschemes, we have etale maps

with a (equivariant) section over $\mathcal{G}$, and a lifting of $\mathcal{X}$ or to $\mathcal{I} \mathcal{G}$ to $\mathcal{G}$. Note $\mathcal{G}$. We may:

$\mathcal{G}$

$\mathcal{G} \subset \mathcal{C} \mathcal{G} \subset \mathcal{G} \subset \mathcal{C} \mathcal{G}$, taking $\mathcal{G}$: here the lifting is to $\mathcal{G}$, 1. Start with $\mathcal{G} \subset \mathcal{G}$. Assume $\mathcal{G} \subset \mathcal{G}$; the succeeding liftings then cannot be to $\mathcal{G}$, $\mathcal{G}$: they must be to $\mathcal{G}$. Shrink $\mathcal{G}$, first so that it maps to $\mathcal{G}$, next so that it induces an isomorphism from $\mathcal{I} \mathcal{G}$ to $\mathcal{C} \mathcal{G}$. As $\mathcal{G}$ is a monomorphism the cocartesian (35.2) is cartesian as well. The composition $\mathcal{G} \rightarrow \mathcal{I} \mathcal{G} \rightarrow \mathcal{I} \mathcal{G}$ gives by pull-back a factorization of $\mathcal{G}$ as $\mathcal{G} /$

$\mathcal{C}$

with each

$\mathcal{G}$

$\mathcal{G}$

cartesian and cocartesian, hence $\mathcal{G} / \mathcal{G} \rightarrow \mathcal{I} \mathcal{G}$. Further, the morphism $\mathcal{I} \mathcal{X}$, hence $\mathcal{X} \rightarrow \mathcal{I} \mathcal{G}$ lifts to $\mathcal{I} \mathcal{G}$ lifts to $\mathcal{G}$. It follows that $\mathcal{G} / \mathcal{G}$ is a push-out of $\mathcal{I} \mathcal{G}$, which is in $\mathcal{G}$, and $\mathcal{I} \mathcal{G}$

$\mathcal{G}$

solidity follows.

$\mathcal{G}$

□

Remark 69 (1). Another formulation of Proposition 35 is: a morphism solid if and only if the pointed sheaf is an iterated extension of in $\mathcal{G}$, in the sense that there are morphisms

$\mathcal{G}$

$\mathcal{G} =$

$\mathcal{G}$, of the form $\mathcal{I} \mathcal{G}$.

Proposition 70 (36). *If $\mathcal{G}$ is open and*

is

=

$\mathcal{I} \mathcal{G}$'s with

$\mathcal{G}$

with each

is

=

= -solid, then is = -solid.

Proof. In the proof we say “solid” instead of “-solid”. Let $(*)$ be the property of a sheaf that any open is solid. If $\mathcal{G}$ is solid, sits at the end of a

$\mathcal{G}$

chain $\#$

with each

push out of some for in . We prove by induction on that $>$ satisfies (*). For , is in . If - is representable and $\#$ is open, it is of the form $>$ open in , hence solid by Condition 34(1). - for It remains to check that if in a cocartesian square

$$\begin{array}{c} \text{,} \\ = \\ \text{,} \\ = \\ / \\ > \\ - \end{array}$$

(36.1)

the sheaf satisfies (*), so does . In (36.1), $>$ - is a monomorphism and the square (36.1) hence cartesian as well as cocartesian. Fix open, and take the pull-back of (36.1) by . It is again a cartesian and cocartesian square and, being open, it is of the form

$$\begin{array}{c} / \\ T \\ X \end{array}$$

(36.2)

$/ /$ where is open in and 6 . The diagram $/ /$

$$\begin{array}{c} / \\ / \\ / \end{array}$$

expresses as an extension of $-I > X$ by

□

Remark 1 using (*) for and the fact that

$X \text{ IT}$

$-I >$

$-I > T$

$/$ and one concludes by is in $=$.

Proposition 71 (37). *The pull-back of a solid morphism by an open morphism solid. In particular, if is open and if is solid, so is .*

$$\begin{array}{c} / \\ / \end{array}$$

$\mathcal{E}$ is

Proof. Since the pull-back of an open morphism is open, it suffices to check the proposition for a push-out of $>$ - for open in :

$>$

-

Pulling back by , we obtain a cocartesian square

$>$

-

$/ /$ with open in and open in . This shows that $/ /$ is solid.

Suppose now that we are given two classes $=$ and $=$ of open embeddings satisfying conditions 34. We define $=$ as the smallest class stable by Condition 34 containing the

$6 \ 6 \ 6 \ 6$ for 6 in $=$ and 6 in $=$.

□

Example 72 (2). If $=$ is a class of all open embeddings and $=$ is the class of those 6 for which acts freely outside , then $= = =$.

Proposition 73 (38). *If the pointed sheaves $/$ and $/$ are respectively $=$ and $=$ -solid, the $/ /$ is $= =$ -solid.*

Proof. By assumption, f is an iterated extension in the sense of Remark 1 of pointed sheaves T $I \rightarrow$ with 6 in $=$. Similarly for f , with 6 in $=$. The smash product f/f is then an iterated extension of the $T I \rightarrow T I \rightarrow T T I \rightarrow T T \rightarrow$ $=$ solid. □

Definition 74 (8). A morphism is called ind-solid relative to $=$ if it is a filtering colimit of $=$ -solid morphisms. Exercise 39. We take $\mathbb{A}^1$ to be the trivial group. A section on $\mathbb{A}^1$ of a push-out taken for instance in the lexicographical order, hence it is

$$\begin{array}{c} > \\ 7 \\ - \\ / \\ 6 \\ ? \\ I / 6 \text{ a } 6 I 7 6 \end{array}$$

can be described as follows. For an open subset U of $\mathbb{A}^1$ and a section s of $\mathcal{O}_U$ on consider on the small Nisnevich site $\mathcal{C}_U$ of the presheaf which sends

$$\begin{array}{c} \text{to the set of morphisms such that } U \times_{\mathbb{A}^1} \rightarrow U \text{ and } s \circ \text{pr}_1 = \text{pr}_2^* s \\ 7 ? I \\ N \\ 6 I / 6 \end{array}$$

1. An open subset of $\mathbb{A}^1$. 2. A section of $\mathcal{O}_U$ on U .

' 7. $a \in I$ on 6 where ι is the closed embedding and $7.$ denotes the associated Nisnevich sheaf. Exercise 40. In the notations of Exercise 39, if f is a sheaf for the etale topology, so is $\mathcal{O}_U$. For any U , the $6 I$ as in (1),(2) above form a sheaf for the etale topology. It hence suffices to prove that for $6 I$ fixed, the datum (3) forms a sheaf for the etale topology. This is checked by using the following criterion to check if a Nisnevich

sheaf is etale. For U , $\mathcal{O}_U$, and for a finite separable extension of $\mathbb{A}^1$, let $\mathcal{O}_U \otimes \mathcal{O}_N$

be deduced by "extension of the residue field" from the henselization $\mathcal{O}_N$ of $\mathcal{O}_U$ at

U . The criterion is that $\mathcal{O}_U \otimes \mathcal{O}_N$ should be an etale sheaf on $\mathcal{C}_U \otimes \mathcal{C}_N$. Exercise 41. It follows from Exercises 39 and 40 that if f is ind solid, and if $\mathcal{O}_U$ is etale, then $\mathcal{O}_U \otimes \mathcal{O}_N$ is etale. In particular, a solid sheaf, as well as a pointed $\mathbb{A}^1$ -sheaf. A section of

$$\begin{array}{c} 6 \\ \text{solid sheaf, are etale sheaves.} \end{array}$$

Remark 75 (2). The same formalism of open and solid morphisms holds in the site of all schemes of finite type over $\mathbb{A}^1$ with the etale topology.

3.2 A Criterion for Preservation of Local Equivalences

We work with pointed sheaves on following result

$$\begin{array}{c} [\cdot \text{ . Our aim in this section is to prove the} \\ a \text{ a } / \end{array}$$

Theorem 76 (6). *Let f be a functor from pointed sheaves to pointed sets. Suppose that f commutes with all colimits, and that for any open embedding $U \rightarrow V$, $f(U) \rightarrow f(V)$ is a monomorphism. Then if f is a $\mathbb{A}^1$ -local equivalence and if f and $\mathcal{O}_U$ are (pointed) ind-solid, then $\mathcal{O}_U \otimes \mathcal{O}_N$ is a weak equivalence.*

$$\begin{array}{c} a \\ a \\ \text{Suppose that} \\ [\\ / a \end{array}$$

$\%$

$[$

is an upper distinguished square. Adding a base point, we obtain . The morphism is then a local equivalence. Let us check that is a $A \rightarrow A$ weak equivalence. As commutes with coproducts, this morphism is deduced from the commutative square

a

a

$a \rightarrow a \rightarrow X$

$a \rightarrow \mathbb{Z} \rightarrow a \rightarrow$

a

$[$

by applying the same construction (23.3). This square is cocartesian because is. The top horizontal line being a monomorphism, it is homotopy cocartesian, and the claim follows. As commutes with colimits, this special case implies that more generally one has

a

Lemma 77 (15). If is in the -closure of the a weak equivalence.

0

A

as above, then a is

$/$, $S/ /$ is a $[S/ /$ is $/$ a local S/ S

Proposition 78 (42). (Proof of Theorem 6) For any pointed sheaf local equivalence. Indeed for any connected in , a weak equivalence by Construction 29. It follows that for equivalence,

$/$

aS

is a commutative square of local equivalences. By Lemmas 15 and 12, is a weak equivalence. It remains to see that is a weak equivalence – and the same for . For this it suffices to see that for a pointed ind-solid sheaf , is a weak equivalence. As and commute with filtering colimits, the ind-solid reduces to solid, and by the inductive definition of solid, it suffices to prove the following lemma.

$aS/$

$aS/ a/ a S$

$a/$

$/$

be an open embedding. If in a cartesian square of pointed $>$ -

$[$

$/ /$ is such that $aS/ a/$ is a weak equivalence, the same holds for .

Lemma 79 (16). Let sheaves

Proof. Consider the cocartesian square

$[$

$S \rightarrow S \rightarrow$

$S/$

$S [$

One can easily see that the top morphism is a monomorphism. It follows that is point by point homotopy cocartesian, and is a local equivalence. The functor of Construction 29 transforms a monomorphism into a coprojection of the form 5 . It follows that each is of the form $> >$ and, by Lemmas 15 and 12, is a weak equivalence. It remains to show that is a weak equivalence.

$\text{” } \%$

S

$\%$
 $aS \ aS \ a$
 aS
 $S \ S$
 $a \ S \ a[$
 $[[$
 $S \ a \ aS / \ a / \ a[\ a[\ aS \ a \ aS \ a$

Let us apply to the morphism of cocartesian squares . By Construction 29 both and are homotopy $> >$ equivalences, and remain so by applying . We assumed to be a and weak equivalence. As are cocartesian with a top morphism which is a monomorphism (by the assumption on , for), it follows that is a weak equivalence. Hence so is . □

4 Two Functors

5.1 The Functor

One has a natural morphism of sites

$2 [.$

$[.$

given by the functor

$2 [[$ with the trivial $-$ -action Indeed, the functor 2 commutes with finite limits and transforms covering families

into covering families. In particular the functor presheaf

2 is continuous: if $/$ is a sheaf on $[.$, the $/$ with the trivial $-$ -action is a sheaf on $[.$. The functor 2 has a left adjoint Q . As 2 is continuous, the functor Q is cocontinuous, and the functor 2 from sheaves on $[.$ to sheaves on $[.$ is $/$ sheaf associated to the presheaf $/$

Proposition 80 (43). *The cocontinuous functor Q is also continuous, that is, if $/$ is a sheaf on $[.$, the presheaf $/$ on $[.$ is a sheaf.*

Proof. By Lemma 2 it is sufficient to test the sheaf property of $/$ for a covering of deduced by pull-back from a Nisnevich covering 6 of . Passage to quotient commutes with flat base change. Taking as base , this gives that $-I: 6 \rightarrow 6$ identifies 6 with the quotient of $-I: 6$ by . Similarly, if $6 \rightarrow -I: 6$, the quotient by of the pull-back to of 6 is 6 again. This reduces the sheaf property of $/$, for the covering of by the $-I: 6$, to the sheaf property of $/$ for the covering 6 of .

$Q /$

Q

$Q \ Q \ Q /$

The functor

gives rise to a pair of adjoint functors between the categories of presheaves on and , with being

. As is continuous , it induces a similar pair of adjoint functors between the categories of sheaves. This pair is

$[$

$[$

$2\#$

associated sheaf $\mathcal{A} \in Q \rightarrow 2 \rightarrow Q$ so that one has a sequence of adjunctions $2\# \rightarrow 2 \rightarrow 2$. If $/$ on $[$ is representable: $/ -$, then $2\# / -I: .$ In particular, $2\#$ transforms the final sheaf K on $[.$, also called “point”, into the final sheaf on $[.$, and $2\# \rightarrow 2$ is a pair of adjoint functors in the category of pointed sheaves as well. It is clear that $2\#$ takes solid sheaves to solid sheaves. We also have the following. □

Proposition 81 (44). *Let $/$ be a pointed sheaf solid with respect to open embeddings $\hookrightarrow$ of smooth schemes such that the action of π_1 on π_0 is free outside $\emptyset$. Then $2\# /$ is solid with respect to open embeddings of smooth schemes.*

Proof. If $\emptyset$ is the open subset of $\emptyset$ where the action of π_1 is free, then $\emptyset \hookrightarrow \emptyset$ and if $\emptyset \hookrightarrow \emptyset$, a push-out of $\emptyset$ is also a push-out of $\emptyset$: we gained that the action is free everywhere. The next step is applying $2\#$, from pointed sheaves on $\mathcal{C}$ to pointed sheaves on $\mathcal{C}$. This functor is a left adjoint, hence respects colimits and in particular push-outs. It transforms π_1 to π_1 , and in particular, for $\emptyset$, the final object into the final object. To check that it respects solidity it is hence sufficient to apply:

smooth over $\emptyset$, then $\emptyset$ is smooth. *Proof.* If $\emptyset$ is finite etale, for instance $\emptyset$, the case which most interests us, this is clear, resulting from being etale. In general one proceeds as follows. The assumption that π_1 acts freely on π_0 implies that π_1 is a π_1 -torsor over $\emptyset$. In particular, π_1 is faithfully flat. As π_1 is flat over $\emptyset$, this forces π_1 to be flat over $\emptyset$. To check smoothness of $\emptyset$ it is hence sufficient to check it geometric fiber by geometric fiber. For $\emptyset$ a geometric point of $\emptyset$, smoothness of $\emptyset$ amounts to regularity. As $\emptyset$ is smooth over $\emptyset$, hence regular, and $\emptyset \rightarrow \emptyset$ is faithfully flat, this is [4] (an application of Serre's cohomological criterion) $\square$

Lemma 82 (17). *If π_1 acts freely on π_0 for regularity).*
2

Proposition 83 (45). *The functor $\#$ respects local (resp. A -) equivalences between termwise ind-solid simplicial sheaves.*

$/ \hookrightarrow /$ be a local equivalence between termwise ind-solid simpli[. To verify that $2\#$ is a local equivalence it is sufficient to in [and the map

$$2\# / \hookrightarrow 2\# / \text{, LO- } 2\# / \hookrightarrow 2\# / \text{, LO-}$$

Proof. Let cial sheaves on check that for any

$2\#$ is a left adjoint, the functor

$$/ \hookrightarrow 2\# / \text{, LO- (45.1)}$$

$\emptyset$ in $\mathcal{C}$, $\emptyset$ commutes with colimits. For an open embedding $\emptyset \hookrightarrow \emptyset$ is again an open embedding and we can apply to (45.1) Theorem 6. $/$ be an A -equivalence. Consider the square Let $/$ is a weak equivalence of simplicial sets. Since

$$@ S / S /$$

$/ \hookrightarrow /$ By the first part of proposition $2\#$ takes the vertical maps to local equivalences. Since $2\#$ commutes with colimits, Lemma 13 implies that $2\# S$ is in the 0 closure of the class which contains $2\# A$ for $\mathcal{C}$ upper distinguished and $2\# A$ for in $\mathcal{C}$. By Theorem 4 it suffice to prove that

morphisms of these two types are A -equivalences. For morphisms of the first type it follows from the first half of the proposition. For the morphism of the second type

$$2$$

$$\text{and } 2\# D = !$$

it follows from the fact that morphisms $\# A$

A are mutually inverse A -homotopy equivalences.

Define $L2\# A$

$$A \text{ (and similarly on } AA \text{) setting } 2\# S / L2\# /$$

$$S /$$

$$2$$

where $\emptyset$ is defined in Construction 29. Proposition 45 shows that $L \#$ is well defined and that for a termwise ind-solid one has $L \# \#$. $\square$

5.2 The Functor

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$2/2/$

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As in Sect. 3.1, we fix $\mathcal{F}$ and $\mathcal{G}$. We also fix $\mathcal{H}$ in which is finite and flat over $\mathcal{F}$. a presheaf on $\mathcal{F}$, we define $\mathcal{H}$ to be the internal hom object $\text{For Hom } \mathcal{H}$. Its value on $\mathcal{F}$ is K . If $\mathcal{F}$ is a sheaf, so is $\mathcal{H}$.

//

$[//?//]$

Example 84 (6). Take $\mathcal{F}$ and $\mathcal{G}$ deduced from the finite group acting on $\mathcal{F}$ by permutations. In that case, if $\mathcal{F}$ is represented by $\mathcal{F}$, with a trivial action of $\mathcal{F}$, then $\mathcal{H}$ is represented by $\mathcal{H}$, on which acts by permutation of the factors.

Remark 85 (1). If $\mathcal{F}$ is representable (resp. $\mathcal{G}$ represented by smooth over $\mathcal{F}$), so is $\mathcal{H}$. More precisely, if $\mathcal{F}$ is represented by $\mathcal{F}$ in $\mathcal{F}$, consider the contravariant functor on $\mathcal{F}$ of morphisms of schemes from $\mathcal{F}$ to $\mathcal{F}$, that is the functor

$\mathcal{A}(\mathcal{F} \rightarrow \mathcal{F}) = \mathcal{K} \times \mathcal{K}$. This functor is representable, represented by some quasi-projective over $\mathcal{F}$ (resp. and smooth). This carries an obvious action $\mathcal{A}$ of $\mathcal{F}$, and $\mathcal{A}$ in $\mathcal{F}$ represents $\mathcal{H}$. Proof: by attaching to a morphism $\mathcal{F} \rightarrow \mathcal{F}$ its graph, one maps the functor considered into the functor of finite subschemes of $\mathcal{F} \times \mathcal{F}$, of the same rank as $\mathcal{F}$, that is the functor

'subschemes of $\mathcal{F} \times \mathcal{F}$ finite and flat over $\mathcal{F}$, $\mathcal{K}$

$\mathcal{K}$

with the same rank as

$\mathcal{F} \times \mathcal{F}$ over $\mathcal{F}$.

The later functor is represented by a quasi-projective scheme Hilb , by the theory of Hilbert schemes. The condition that $\mathcal{K}$ be the graph of a morphism from $\mathcal{F}$ to $\mathcal{F}$ is an open condition. This means: let $\mathcal{K} \times \mathcal{K}$ be a subscheme finite and flat over $\mathcal{F}$. There is open in such that for any base change $\mathcal{F}$, the pull-back $\mathcal{T}$ of $\mathcal{K} \times \mathcal{K}$ is the graph of some $\mathcal{F}$ -morphism from $\mathcal{K}$ to $\mathcal{K}$ if and only if $\mathcal{K}$ maps to $\mathcal{F}$. This gives the existence of the required $\mathcal{H}$, and that it is open in Hilb . If $\mathcal{F}$ is smooth the smoothness of $\mathcal{H}$ follows from the infinitesimal lifting criterion. The quasi-projectivity follows from that of Hilb . On the functors of $\mathcal{F}$ is clear. For $\mathcal{H}$ represented, the action $\mathcal{A}$, one has

$\mathcal{A} : \mathcal{F} \rightarrow \mathcal{F}$

$6 : \mathcal{F} \rightarrow \mathcal{F} \times \mathcal{F}$

$\mathcal{F} \times \mathcal{F} \subset [\text{Hom} \mathcal{A} \mathcal{F} : \mathcal{F} \times \mathcal{F} : \mathcal{F} \times \mathcal{F} : \mathcal{K} \times \mathcal{K} : \text{Hom} \mathcal{A} \mathcal{F} : \mathcal{K} \times \mathcal{K} : \mathcal{F} \times \mathcal{F}] \subset$

?

=

Let $\mathcal{H}$ be a class of open embeddings in " $\mathcal{F}$ -solid".

=

$[\mathcal{F} \times \mathcal{F}]$. We will simply say "solid" for

$\mathcal{F}$ is a solid sheaf on $[\mathcal{F} \times \mathcal{F}]$, so is $\mathcal{H}$. $\mathcal{F}$ is open, i.e. relatively representable by $\mathcal{F}$. If a morphism of sheaves $\mathcal{F} \rightarrow \mathcal{F}$ open (equivariant) embeddings, there is a natural sequence of sheaves intermediate between $\mathcal{F}$ and $\mathcal{H}$. In the case considered in Example 6, and for $\mathcal{F} \times \mathcal{F}$, they are represented by the open equivariant subschemes of $\mathcal{F} \times \mathcal{F}$ consisting of those n -uples for which at least n of the $\mathcal{F}$ are in $\mathcal{H}$. The formal

Theorem 86 (7). If

definition is as follows. A section of $\mathcal{H}$ over $\mathcal{F}$ is a section of $\mathcal{H}$ over $\mathcal{K}$. Let $\mathcal{H}$ be the $\mathcal{F}$ equivariant open subscheme of $\mathcal{K}$ on which $\mathcal{H}$ is in $\mathcal{H}$. The sheaf $\mathcal{H}$ is $\mathcal{H}$ the subsheaf of $\mathcal{H}$ consisting of those $\mathcal{H}$ such that all fibers over $\mathcal{F}$ of $\mathcal{H}$ are of degree at least n . The condition that the fiber at $\mathcal{F}$ of degree n is open in $[\mathcal{F} \times \mathcal{F}]$, and it follows that the inclusion of $\mathcal{H}$ is open. For $\mathcal{F}$, in

$[\mathcal{F} \times \mathcal{F}]$ is simply $\mathcal{H}$. For large n , it is $\mathcal{H}$.

//

$$\begin{array}{c} / \\ < \\ ? \\ < \\ \mathcal{E} / \% \\ ! \\ ! \\ \% \\ \% \\ \mathcal{E} \\ / \\ / \\ ? \\ \mathcal{E} \\ / \\ ! \\ \% \\ < \mathcal{E} \\ \mathcal{E} / \% \mathcal{E} < ! ! / \% \end{array}$$
is deduced by push-out from an open map

Lemma 87 (18). *Suppose that , so that we have a cocartesian square*

$$(45.2) \quad \begin{array}{ccc} \% & / & \\ ! & & \end{array}$$

Then, for each , the cartesian square
fiber product $\%$
 $/ \%$
 $\%$
 $]$

$$(45.3) \quad \begin{array}{ccc} / & \% & \\ & & \end{array}$$

is cocartesian as well.

[. has enough points: as a consequence of Lemma 2, for , the functor $/ AM' (/ -I$: 6 the limit being taken over the Nisnevich neighborhoods of in , is a point (a fiber functor). The class of all such functors is clearly conservative. Such a functor depends only on $-I$: , where is the henselization of at , and can be any equivariant $-$ scheme which is a finite disjoint union of local henselian schemes essentially of finite type over , and for which is local. We call such a scheme $-$ local henselian, and write $/ /$ for the corresponding

Proof. The site each in and

[fiber functor. We will show that (45.3) becomes cocartesian after application of any of the fiber] functors defined above. It suffices to show that for any in , the fiber of (45.3) above is cocartesian in . This fiber is of the form

$$\begin{array}{c} / \\ / \\ \& \\ \& / \% \\ L \\ \& \end{array}$$

and such a square is cocartesian if and only if whenever U or V is empty, the other is reduced to one element. Here, we also know that π is injective. It hence suffice to check that if U is empty, then V is reduced to one element. Fix in U , and view it as a section of π over K . Let K be

U the open equivariant subset where it is in U . The assumption that π be in

means that the degree of the fiber of π at a closed point of U is at least n . By G -equivariance of π , this degree is independent of the chosen u . We have to show that if U is not in U , that is if this degree is exactly n , then U is the

π image of a unique element of U .

&

π !

&

π ? π

π π π

& ? & π . . ! &

The scheme π K is a disjoint union of G -local henselian schemes π K . By assumption, (45.2)? K is cocartesian, hence if π is not in π on π K , then on π K it comes from a unique π in U . Let π K be the union of those π K on which π is in π , and π K be the union of π K on which it is not. That π is in π but not in π , means that π K is of degree n over U . On π K , π is in π . On π K , it comes from a unique π in U . The section π & in π on π K , & on π K of π over π K is a section of π on lifting &. It is the unique such lifting: any other lifting &, viewed as a section of π on π K , can be in π at most on π K , hence must be in π on the whole of π K which has just the required degree over U . This determines & uniquely on π K , where it is in π , as well as on π K , where it is the unique lifting of & to U .

Proof of Theorem 7: The induction which works to prove Theorem 7 is the following. As π is solid, it sits at the end of a sequence

π

#

π / ' /

π

π [

where each π . We prove π is a push-out of some open embedding in π by induction on that for any π , π is solid. For π , π is representable, X

π] hence so is (1). A fortiori, π is solid. For the induction step X X one applies the following lemma to X

X .

π

π

□

Lemma 88 (19). *Let*

π

, /

π

>

-

π] be a cocartesian square with open in U . Suppose that for any C , π is [] solid. Then π is solid.

Proof. As π is open by Proposition 33, the π] are defined. It suffices to prove that for each $+$, the open morphism π] /] is solid.

By Lemma 18, this morphism sits in a cartesian and cocartesian square U V fiber product π /

$UV /]$

$/$

-

(45.4)

$/]$

$] \text{ By assumption, is solid. It follows that }]$

□

Proposition 37 to the open morphism -

$/$

$] \text{ }$

$/ /$

$] \text{ - is solid too (apply }] \text{). As is open, -}$

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so is $]$, and by Proposition 36, of a solid map.

is solid. The map is then solid as a push-out

$\% ?$

$/ / / , / / / / , /$

Example 89 (7). It is not always true that if $]$ is a solid morphism, so is $]$. Take the trivial group and two points (i.e. $]$). Then $]$. For any sheaf $]$, the inclusion of $]$ in $]$ is solid (deduced by push-out from $\#$), and applying $]$, we obtain $]$. Pulling back by the natural map from $]$ to one of the summands (an open map), we see that if $]$ is solid, so is $]$.

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$/$

$/$

Corollary 90 (46). If $/$

In particular, if

$/$

is open and solid, then $]$ is pointed solid, so is $]$.

$] /]$

$/$

$] \text{ }$

is solid.

$] \text{ is open and one applies Theorem 7 and Proposition 36.}$

We now define for pointed sheaves on $[$ a “smash’ variant of the construction $/ /]$. If we assume that the marked point $]$ is open, it is $/] /] /]$ that is $/]$ with $/]$ contracted to the new base point. This definition can be repeated for any pointed sheaf if, for any monomorphism $\% /$, we define $/ \%]$ to be the following subsheaf of $/]$: a section $\mathcal{E}$ of $/] / ?$ is in $/ \%]$ if for any non empty $]$, there exists a commutative diagram ?

Proof. with non empty and $\&$ in $\%$ on $]$.

□

Example 91 (8). Under the assumptions of Example 6, if $]$ is open in $]$ with $]$ is

ment and if $]$ has the natural $>$, then $- - [$, where action of $]$. In particular, if $]$ is the Thom space of a vector bundle over $]$ (that is, $/$ with $/ X$ contracted to the base point), then $]$ is the Thom space of the $]$ -equivariant vector bundle $]$ on $]$.

C

$/$

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$/$

$/$

$, B$

Proposition 92 (47). *If a pointed sheaf $/$ is pointed solid, that is if $, /$ is solid, then so is $/$.*

Proof. As $/$ is solid, the open morphism $/,] /$ is solid too (Proposition 36), and $, /$ is deduced from it by push-out.

The definition of $/$ immediately implies the following: □

Lemma 93 (20). *Let be a $-$ local henselian scheme. Then $* /] /$?*

where

? are $-$ local henselian schemes such that ?

?

$/$ respects local and A -equivalences.

Proof. Let $/$ be a local equivalence. To check that $/]$ is a local equivalence it is enough to show that for any $-$ local henselian $, the map $/] / ? ?]$$

□

Proposition 94 (48). *The functor*

is a weak equivalence of simplicial sets. This follows from Lemma 20. Let be an A -equivalence. By the first part it is sufficient to show that $]]$ is an A -equivalence. We use the characterization of $]]$ commutes with filtering col A -equivalences given in Theorem 5. Since imits and preserves local equivalences it suffices to check that it takes A -homotopy equivalences to A -homotopy equivalences. This is seen using the natural map

$$\begin{array}{c} S \\ S/ \\ S \\ /] A \\ / \\ / \\ / A] \end{array}$$

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Numerical Equivalence, Cohomological Equivalence, and Lefschetz Theory for Abelian Varieties over Finite Fields

P. Deligne (edited by L. Clozel)*

To Jean-Pierre Serre, in testimony of our admiration

Abstract

We determine the group of automorphisms generated, inside the endomorphisms of the cohomology of an abelian variety over a finite field, by the Lefschetz operators and by complex multiplications.

Keywords: Algebraic geometry (abelian varieties, étale cohomology); 14F20, 14K22

1.

Let k be an algebraic closure of the field $\mathbf{F}_p$ with p elements. We denote by ℓ a prime number different from p . We fix, over k , an isomorphism $\mathbf{Q}_\ell \xrightarrow{\sim} \mathbf{Q}_\ell(1)$.

Let A be an abelian variety over k . We write Z , or $Z(A)$, for the group of cycles with rational coefficients on A , graded by codimension:

$$Z = \bigoplus_i Z^i.$$

On Z^i we have the relations of numerical equivalence and of cohomological equivalence, the latter for étale cohomology with coefficients in $\mathbf{Q}_\ell$. Two cohomologically equivalent cycles are numerically equivalent.

Let A_j be the simple factors of A , let E_j be the center of $\text{End}(A_j) \otimes \mathbf{Q}$, and let F be the compositum, inside an algebraic closure of $\mathbf{Q}$, of the Galois closures of the E_j ; thus F is either totally real or a CM field. If ℓ is a prime number, let $D_\ell \subset \text{Gal}(F/\mathbf{Q})$ be a decomposition group at ℓ ; let $c \in \text{Gal}(F/\mathbf{Q})$ be complex conjugation.

Theorem 1. *If $c \in D_\ell$, then numerical equivalence and cohomological equivalence, for ℓ -adic étale cohomology, coincide for A .*

A less precise theorem, in which the condition on ℓ depended, for a simple variety, on the choice of a maximal field of complex multiplications, was proved in [2].

The following remarks are immediate:

(a) For a given A , the condition of Theorem 1 is satisfied, by the Čebotarev theorem, for a set $\mathcal{L}$ of prime numbers ℓ of nonzero density, either analytic or naive.

(b) If the condition of Theorem 1 is satisfied for A , it is also satisfied for the powers A^n of A . There is therefore a set $\mathcal{L}$, of density > 0 , such that the conclusion is true for every power A^n . This had been observed by Milne [5].

Let $H = \bigoplus_{i=0}^{2g} H^i(A, \mathbf{Q}_\ell) = \bigoplus_{i=0}^{2g} H^i$, where $g = \dim(A)$. Let $D = \text{End}(A) \otimes \mathbf{Q}$.

*Edition based on a note of Deligne and on a letter of 12 July 1997.

The group $D^\times$ acts on H , as does the group, isomorphic to $\mathrm{SL}(2, \mathbf{Q})$, supplied by Lefschetz theory. Both preserve algebraic cohomology classes. Let $G \subset \mathrm{GL}(H)$ be the Zariski closure of the group generated by $D^\times$ and $\mathrm{SL}(2)$. The key point in our proof of Theorem 1 is a description of G , or rather of the group $G_{\mathbf{Q}_\ell}$ over $\mathbf{Q}_\ell$ obtained from it by extension of scalars. In §4 we shall show how this description reduces to the following two cases.

First let E be a supersingular elliptic curve and let $A = E^n$. The Zariski closure of the action of $\mathrm{End}(A)^\times$ on $H \otimes \mathbf{Q}_\ell = \Lambda^\bullet(H^1 \otimes \mathbf{Q}_\ell)$ is the group $\mathrm{GL}(H^1 \otimes \mathbf{Q}_\ell)$. Let $V = H^1 \otimes \mathbf{Q}_\ell$, and let V^* be its dual.

Theorem 2. *The group $G_{\mathbf{Q}_\ell}$ is isomorphic to the group $\mathrm{CSpin}(V \oplus V^*)$ of spin similitudes, where $V \oplus V^*$ is endowed with its natural quadratic form.*

It acts on $H \otimes \mathbf{Q}_\ell = \bigoplus_i H^i(A, \mathbf{Q}_\ell)$ as the sum of its two half-spin representations. In particular, the representation on classes of even degree, respectively odd degree, is irreducible.

At the other extreme, let A be an irreducible variety such that $\mathrm{End}(A) \otimes \mathbf{Q} = D$ is a division algebra, of reduced degree n , over an imaginary quadratic field E ; A has dimension n .

Let $c \in \mathrm{Gal}(E/\mathbf{Q})$ be complex conjugation, and choose an embedding j of E into $\mathbf{Q}_\ell$. The E -module structure on $H^1(A, \mathbf{Q}_\ell)$ gives a decomposition $H^1 = H^{10} \oplus H^{100}$, where $\lambda \in E$ acts on H^{10} , respectively on H^{100} , by multiplication by $j(\lambda)$, respectively by $j(c\lambda)$. Let $V = V_1 \oplus V_2$ be the decomposition dual to $H^1(A, \mathbf{Q}_\ell)$. Over $\mathbf{Q}_\ell$, the Zariski closure of $D^\times$ becomes the group $\mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$.

Endow

$$H^\bullet(A, \mathbf{Q}_\ell) = \bigoplus_{p,q} \Lambda^p V_1 \otimes \Lambda^q V_2$$

with the grading given by $k := p - q$:

$$\mathrm{gr}^k = \bigoplus_{p-q=k} \Lambda^p V_1 \otimes \Lambda^q V_2. \quad (1)$$

The part of degree $(-n)$ is therefore $\Lambda^0 V_1 \otimes \Lambda^n V_2$, of dimension 1, and the part of degree $(-n+1)$ is $V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2$, of dimension $2n$.

Theorem 3. (i) *For an isomorphism of $\bigoplus_k \Lambda^p V_1 \otimes \Lambda^q V_2$ with $\Lambda^\bullet(\mathrm{gr}^{-n+1})$ carrying gr^k to $\Lambda^{\frac{n+k}{2}} \mathrm{gr}^{-n+1}$, the group $G_{\mathbf{Q}_\ell}$ identifies with the group generated by $\mathrm{GL}(\mathrm{gr}^{-n+1})$ acting on $\Lambda^\bullet \mathrm{gr}^{-n+1}$ and by the homotheties $x \mapsto \lambda x$.*

(ii) *The gr^k ($-n \leq k \leq n$) are therefore irreducible, pairwise nonisomorphic representations of G . The subspaces gr^k and gr^ℓ are orthogonal for the cup product if $k + \ell \neq 0$, and are in perfect duality if $k + \ell = 0$.*

2.

We now show how to reduce Theorem 1, and in fact a slightly stronger variant; see §5, to the analysis of the group G .

(a) Let Z_0 be the $\mathbf{Q}$ -vector subspace of $H^\bullet(A, \mathbf{Q}_\ell)$ which is the image of Z , and let Z_1 be the $\mathbf{Q}_\ell$ -vector subspace generated by Z_0 . We shall prove that the intersection form is nondegenerate on Z_1 . Its restriction to Z_0 , which is rational, then has zero kernel: it is nondegenerate and induces a surjection from Z_1 to $\mathrm{Hom}(Z_0, \mathbf{Q}_\ell)$. Hence $\dim_{\mathbf{Q}_\ell}(Z_1) \geq \dim_{\mathbf{Q}}(Z_0)$, the elements of a basis of Z_0 are linearly independent over $\mathbf{Q}_\ell$, and $Z_0 \otimes \mathbf{Q}_\ell \xrightarrow{\sim} Z_1$.

(b) We shall write $\mathbf{Q}_\lambda$ for a sufficiently large finite Galois extension of $\mathbf{Q}_\ell$. It is then enough to verify the nondegeneracy of the intersection form on $Z_\lambda := Z_1 \otimes_{\mathbf{Q}_\ell} \mathbf{Q}_\lambda \subset H_\lambda := H^\bullet(A, \mathbf{Q}_\lambda)$.

(c) Let L be the Lefschetz operator on H deduced from a polarization of A , and let $\Lambda : H^i \rightarrow H^{i-2}$ be the associated operator [4]. The exponentials of the nilpotent endomorphisms L and Λ of H generate a subgroup S , isomorphic to $\mathrm{SL}(2, \mathbf{Q})$. Let G be the Zariski closure, in $\mathrm{GL}(H)$,

of the subgroup generated by S and by $(\text{End}(A) \otimes \mathbf{Q})^\times$. It preserves Z_1 , hence also Z_λ ; the fact that Λ preserves algebraic cycles is due to Lieberman; see [4]. Since G contains the group $\mathbf{G}_m$ corresponding to the grading of $H^\bullet(A)$, coming from the action of $(\mathbf{Q} \cdot 1)^\times \subset \text{End}(A) \otimes \mathbf{Q}$, and also the group corresponding to the same grading shifted by $\dim(A)$, coming from the diagonal matrices in S , it contains the homotheties $x \mapsto \lambda x$.

(d) Suppose that $\mathbf{Q}_\lambda$ is a sufficiently large Galois extension of $\mathbf{Q}_\ell$. We shall check that H_λ is a sum of absolutely irreducible, pairwise nonisomorphic representations W of G , and that the intersection form defines nondegenerate pairings on $W \times W'$, where $W \mapsto W'$ is an involution of the factors. Suppose there is a $\sigma \in \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$ exchanging the factors in duality. Since Z_λ is stable under the Galois group, the intersection form is then nondegenerate; this will finish the proof.

Let E be the center of $\text{End}(A) \otimes \mathbf{Q}$: E is a product of CM fields E_i ; by convention a totally real field is allowed here as a CM field. Suppose that E has the following property, equivalent to the hypothesis “ $c \in D_\ell$ ” of Theorem 1:

there exist a Galois extension $\mathbf{Q}_\lambda$ of $\mathbf{Q}_\ell$ and $\sigma \in \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$ such that (2)

for every i and every embedding $\tau : E_i \rightarrow \overline{\mathbf{Q}_\ell}$, one has $\sigma\tau = \tau \circ c$,

where c is complex conjugation on E_i .

We shall then verify below the existence of a σ exchanging W and W' . Under this hypothesis, Theorem 1 follows.

3.

We recall some linear algebra, referring to [3], which is based on [1]. Let V be a vector space over a field K ; here K is one of $\mathbf{Q}_\ell$, $\mathbf{Q}_\lambda$, or $\mathbf{Q}_\ell$. Let V^* be its dual, and endow $W = V \oplus V^*$ with the nondegenerate quadratic form

$$Q(v + \lambda) = \langle \lambda, v \rangle \quad (v \in V, \lambda \in V^*).$$

Let C be the associated Clifford algebra, the quotient of the tensor algebra $T(V \oplus V^*)$ by the ideal generated by the elements $x \cdot x - Q(x)$, for $x \in V \oplus V^*$. It is a $\mathbf{Z}/2\mathbf{Z}$ -graded algebra [3]; let $C = C^+ \oplus C^-$ denote this grading. Let W act on the exterior algebra $\Lambda^\bullet V$ of V by

$$v \cdot x = v \wedge x \quad (v \in V, x \in \Lambda^\bullet V) \quad \lambda \cdot x = \lambda \rfloor x \quad (\lambda \in V^*, x \in \Lambda^\bullet V) \quad (3)$$

where $\lambda \rfloor x = i(\lambda)x$. The relation defining C is satisfied, and (3) therefore extends to a C -module structure

$$C \longrightarrow \text{End}(\Lambda^\bullet V). \quad (4)$$

The morphism (4) is an isomorphism: see [1], Theorem 2; alternatively, if $V = V' \oplus V''$, then C is the tensor product, in the $\mathbf{Z}/2$ -graded sense, of the analogous algebras C' and C'' , reducing the assertion to the easy case where V has rank one.

The image of C^+ under (4) is the algebra of endomorphisms of $\Lambda^\bullet V$ preserving the $\mathbf{Z}/2$ -grading of $\Lambda^\bullet V$.

The group $C^\times$ acts on the algebra C by conjugation. The subgroup H which normalizes $W \subset C$ respects the $\mathbf{Z}/2$ -grading of C , and is therefore made of even or odd elements. Its action on W defines a map $H \rightarrow \text{O}(W)$ whose kernel is the center of $C^\times$. The subgroup $(C^+)^\times \cap H$ of $(C^+)^\times$ which normalizes W is the group of spin similitudes $\text{CSpin}(W)$. It maps to $\text{SO}(W)$:

$$1 \longrightarrow \mathbf{G}_m \longrightarrow \text{CSpin}(W) \longrightarrow \text{SO}(W) \longrightarrow 1.$$

Thus $\text{CSpin}(W)$ is the group of automorphisms of $\Lambda^\bullet V$ preserving the $\mathbf{Z}/2$ -grading and the subspace W of $\text{End}(\Lambda^\bullet V)$. Its Lie algebra is the Lie algebra of even endomorphisms T of

$\Lambda^\bullet V$ such that, with W identified as a subspace of $\text{End}(\Lambda^\bullet V)$, one has $[T, W] \subset W$. The representation $\Lambda^\bullet V$ of $\text{CSpin}(W)$ is the sum of the two half-spin representations.

It is clear that $\text{GL}(V)$, acting on $\Lambda^\bullet V$, is a subgroup of $\text{CSpin}(W)$. It lifts, in CSpin , the Levi subgroup of $\text{SO}(W)$ which is the intersection of the two opposite parabolic subgroups preserving the maximal isotropic subspaces V and V^* of W .

The maximal isotropic subspace $V^* \subset W \subset C$ and the line $K \cdot 1$ in $\Lambda^\bullet V$ determine one another: V^* is the subspace of W annihilating 1, and $K \cdot 1$ is the subspace of $\Lambda^\bullet V$ annihilated by V^* . By the same rule, V corresponds to the line $\Lambda^{\max} V$.

The same story repeats for every maximal isotropic subspace of W . The case of interest to us is the one where $V = V_1 \oplus V_2$, and where W is decomposed as the sum of two maximal isotropic subspaces by

$$W = (V_1 \oplus V_2^*) \oplus (V_1^* \oplus V_2).$$

The annihilator of $V_1^* \oplus V_2$ is the line $\Lambda^{\max} V_2 \subset \Lambda^\bullet V$. Fix a generator Ω of it. The inclusion into W of the complementary isotropic subspace $V_1 \oplus V_2^*$ extends to an algebra morphism

$$\Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow C,$$

and

$$I : \Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow \Lambda^\bullet V, \quad \lambda \longmapsto \lambda \cdot \Omega \quad (5)$$

is an isomorphism. The supercommutator with $x \in W$ stabilizes $\Lambda^\bullet(V_1 \oplus V_2^*) \subset C$ and induces a superderivation. For $x \in V_1^* \oplus V_2$, identified with the dual of $V_1 \oplus V_2^*$, this derivation is the interior product $x \rfloor$. Via the isomorphism I , the action of $v \in V_1 \oplus V_2^* \subset W$ therefore becomes exterior multiplication, and the action of $x \in V_1^* \oplus V_2 \subset W$, which kills Ω , becomes interior multiplication.

The exterior product in $\Lambda^\bullet(V_1 \oplus V_2^*)$, transported by I , defines a new product, denoted $*$, on $\Lambda^\bullet V$. If $\Lambda^\bullet(V_1 \oplus V_2^*)$ is endowed with the grading for which V_1 , respectively V_2^* , has degree 1, respectively -1 , then the product is compatible with the grading. The isomorphism I transports this to the evident grading of $\Lambda^\bullet V$, translated by $\dim(V_2)$ so as to place Ω in degree 0. Thus the product $*$ is compatible with the evident translated grading. The natural grading of $\Lambda^\bullet(V_1 \oplus V_2^*)$, for which V_2^* has degree 1, is also compatible with the product. One then obtains on $\Lambda^\bullet V$ the grading gr^k of (1), shifted by $\dim(V_2)$; the element $1 \in \Lambda^\bullet V_2$, of degree 0 for gr^k , is of degree $\dim(V_2)$.

For $v_1 \in V_1$, $v_2 \in V_2$, let us compute $(v_1 \wedge v_2)^\wedge : \Lambda^\bullet V \rightarrow \Lambda^\bullet V$ using the isomorphism (5). We have the diagram

$$\begin{array}{ccc} \Lambda^\bullet(V_1 \oplus V_2^*) & \longrightarrow & \Lambda^\bullet(V_1 \oplus V_2^*) \\ \downarrow I & & \downarrow I \\ \Lambda^\bullet V & \longrightarrow & \Lambda^\bullet V \end{array}$$

and the lower map is a tensor product of maps. If $a \in \Lambda^\bullet V_1$ and $b \in \Lambda^\bullet V_2$, then

$$(v_1 \wedge v_2)(a \wedge b) = (-1)^{\deg a} (v_1 \wedge a) \wedge (v_2 \wedge b).$$

For the isomorphism $\Lambda^\bullet V_2^* \rightarrow \Lambda^\bullet V_2$ defining I , the map $b \mapsto v_2 \wedge b$ is transported to $b^* \mapsto v_2 \rfloor b^*$. It follows that $(v_1 \wedge v_2)^\wedge$ is transported to the endomorphism $v_1 \wedge v_2 \rfloor$ of $\Lambda^\bullet(V_1 \oplus V_2^*)$, which is immediately checked to be a derivation. This derivation preserves $V_1 \oplus V_2^*$.

4.

Computing the group G from §1, after extension to a sufficiently large extension of $\mathbf{Q}_\ell$, is amusing. Here is the story.

(a) Let E be the center of $D = \text{End}(A) \otimes \mathbf{Q}$, a product of CM fields E_i ; D is central simple over E .

(b) After extending scalars from $\mathbf{Q}$ to a sufficiently large $\mathbf{Q}_\lambda$, the algebra E becomes a sum of copies of $\mathbf{Q}_\lambda$, indexed by the set Σ of $\mathbf{Q}$ -algebra morphisms $E \rightarrow \mathbf{Q}_\lambda$. After extending scalars from $\mathbf{Q}_\ell$ to $\mathbf{Q}_\lambda$, $H^1(A)$ becomes $\bigoplus H_\sigma$ for $\sigma \in \Sigma$, where

$$H_\sigma = \{\alpha \in H^1(A, \mathbf{Q}_\lambda) : [z]\alpha = \sigma(z)\alpha \ (z \in E)\},$$

with $[z]$ denoting the action of E on $H^1(A)$ deduced from its action on A . The algebra $D \otimes \mathbf{Q}_\lambda$ is $\prod_\sigma \text{End}(H_\sigma)$.

The group G , acting on $\Lambda^\bullet(\bigoplus_\sigma H_\sigma)$, is generated by the product of the $\text{GL}(H_\sigma)$ and by a “diagonal” $\text{SL}(2)$. Let $L : H^i(A, \mathbf{Q}_\lambda) \rightarrow H^{i+2}(A, \mathbf{Q}_\lambda)$ be the Lefschetz operator, which we identify with a cohomology class in $H^2(A, \mathbf{Q}_\lambda) = \Lambda^2(\bigoplus_\sigma H_\sigma)$; Lemma 2.1 of [2] implies that

$$L = \sum_v L_v$$

where v runs through the quotient set $\Sigma/\{1, c\}$, complex conjugation acting on Σ through its action on E , and where

$$L_v \in \Lambda^2(H_\sigma \oplus H_{c\sigma})$$

for $v = \{\sigma, c\sigma\}$. The embedding σ may be fixed by c , in which case $L_v \in \Lambda^2 H_\sigma$. Each L_v lies in the closure of the orbit of L under $\prod \text{GL}(H_\sigma)$, and therefore in the Lie algebra of G .

The same arguments give an analogous decomposition of the operator Λ ; the commutation relations are preserved by the components L_v and Λ_v . Consequently, G is also generated by the product of the $\text{GL}(H_\sigma)$ and by one copy of $\text{SL}(2)$, defined over $\mathbf{Q}_\lambda$, for each orbit of c in Σ . Thus G is a product acting on a tensor product.

(c) *The story for $\{\sigma\}$, where $\sigma = c\sigma$.*

This corresponds to a possible $\mathbf{Q}$ -factor of E , hence to a factor of A isogenous to $B = E^n$, where E is a supersingular elliptic curve over k .

Let $V = H_\sigma = H^1(B, \mathbf{Q}_\lambda)$, a vector space of even dimension. Let $W = V \oplus V^*$. The Lefschetz class $L \in \Lambda^2 V$ is an isomorphism from V to V^* . We shall show that the group generated by $\text{GL}(V)$ and by the subgroup $\text{SL}(2)$ associated with L is $\text{CSpin}(W)$ in its spin representation on $H^\bullet(B) = \Lambda^\bullet V$.

One immediately checks that, for its action on $\Lambda^2 V$ by left multiplication, L satisfies

$$[L, W] \subset W.$$

Consequently, L is an element of $\text{Lie}(\text{CSpin}(W))$; it belongs to the Lie algebra of the unipotent radical N of the Levi subgroup defined by $W = V \oplus V^*$ (§3), and is a nonsingular element of it. The conjugates by $\text{GL}(V)$ of the exponential of L therefore generate N . Similarly, Λ and its conjugates generate the Lie algebra of the opposite unipotent subgroup. Hence the group generated by $\text{GL}(V)$ and $\text{SL}(2)$ contains the group Spin ; we have already seen that it contains the homotheties (§2, (c)). Its Zariski closure over $\mathbf{Q}_\ell$ is therefore $\text{CSpin}(V \oplus V^*)$, proving Theorem 2.

(d) *The story for $\{\sigma, c\sigma\}$, where $\sigma \neq c\sigma$.*

In this case $H_v = H_\sigma \oplus H_{c\sigma} = V_1 \oplus V_2$, and $(\text{End}(A) \otimes \mathbf{Q}_\lambda)^\times$ acts on H_v by $\text{GL}(V_1) \times \text{GL}(V_2)$. Write, as in §3,

$$\Lambda^\bullet(V_1 \oplus V_2) \xleftarrow{I^{-1}} \Lambda^\bullet(V_1 \oplus V_2^*).$$

We endow $\Lambda^\bullet(V_1 \oplus V_2)$ with the product $*$, which under I^{-1} becomes the exterior product on the right. We may write ([2, Lemma 2.1])

$$L = \sum e_i \wedge f_i \in \Lambda^2(V_1 \oplus V_2)$$

with $e_i \in V_1$ and $f_i \in V_2$. The isomorphism I^{-1} exchanges the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ with the action on the right-hand side given by

$$\sum e_i \wedge f_i].$$

This is a derivation (§3); indeed, in the bases of V_1 and V_2^* given by (e_i, f_i^*) , it is the sum of the endomorphisms

$$x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$$

of k^2 , hence the endomorphism of k^{2n} thus obtained, extended as a Lie-algebra action to $\Lambda^\bullet(V_1 \oplus V_2^*) = \Lambda^\bullet(k^{2n})$. Therefore the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ leaves the product $*$ invariant in the Lie-algebra sense; that is, it is a derivation.

Moreover the element $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ of $\mathfrak{sl}(2)$ acts on $\Lambda^i V$ by $i - n = \text{gr}^1(\Lambda^i V)$, where gr^1 is the translated natural grading (§3). Since this grading is compatible with the product $*$, the latter is invariant under the Borel subalgebra of $\mathfrak{sl}(2)$. But in a representation of $\mathfrak{sl}(2)$, a vector annihilated by a Borel subalgebra is annihilated by all of $\mathfrak{sl}(2)$. It follows that $*$ is invariant under $\text{SL}(2)$.

Consider the decomposition of $\Lambda^\bullet V$ given by (1):

$$\text{gr}^k \Lambda^\bullet V = \bigoplus_{p+q=k} \Lambda^p V_1 \otimes \Lambda^q V_2.$$

Note that this decomposition is invariant under L and Λ . Indeed, E , via its embedding in $\mathbf{Q}_\lambda$, acts on $\Lambda^\bullet V$. Let $\text{U}(1)$ be the group $\{\lambda \in E : N_{E/F} \lambda = 1\}$, where F is the subalgebra of E formed by the fixed points of c . If $\lambda \in \text{U}(1)$, then λ acts on gr^k by the scalar λ^{p-q} in $\mathbf{Q}_\lambda$; it commutes with L and Λ , and the λ^{p-q} are distinct characters of $\text{U}(1)$. Summarizing:

Lemma 4. *Endow $\Lambda^\bullet V$ with the product $*$ and with the grading (1).*

- (i) *The algebra $\Lambda^\bullet V$ is generated by $\text{gr}^{1-n} = V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2$.*
- (ii) *The product $*$: $\text{gr}^i \times \text{gr}^j \rightarrow \text{gr}^{i+j}$ is invariant under $\text{SL}(2)$, $\text{SL}(V_1)$, and $\text{SL}(V_2)$.*
- (iii) *The grading of $\Lambda^\bullet V$ is preserved by G .*

The image of G in $\text{GL}(\text{gr}^{n-1})$ contains $\text{GL}(V_1)$, $\text{GL}(V_2) \simeq \text{GL}(\Lambda^{n-1} V_2)$, an isomorphism $\Lambda^0 V_1 \otimes \Lambda^n V_2 \xrightarrow{\sim} V_1 \otimes \Lambda^n V_2$ given by L , and its inverse given by Λ . Hence this image is $\text{GL}(\text{gr}^{n-1})$. Property (i) identifies $\Lambda^\bullet V$ with $\Lambda^\bullet(\text{gr}^{n-1})$, compatibly with the action of $\text{SL}(2) \times \text{SL}(V_1) \times \text{SL}(V_2)$. The group generated by $\{zs_1, z^{-1}s_2\}$, where z is a scalar and $s_i \in \text{SL}(V_i)$, inside $\text{GL}(V_1) \times \text{GL}(V_2)$ acts on $\Lambda^\bullet V$ by the action of $\text{GL}(\text{gr}^{1-n})$ described in Theorem 3. The group G is therefore generated by $\text{GL}(\text{gr}^{1-n})$, isomorphic to $\text{GL}(2n)$, and by a one-dimensional torus acting by global homotheties on $\Lambda^\bullet V$; this proves Theorem 3(i). It is clear that, already for the action of $\text{GL}(\text{gr}^{1-n})$, the representations obtained are absolutely irreducible and pairwise nonisomorphic. Finally, if $k + \ell \neq 0$, the spaces gr^k and gr^ℓ are not dual representations, even for the action of $\text{GL}(2n) \subset G$. Since the cup product is nondegenerate, gr^k and gr^ℓ are therefore in perfect duality precisely in the complementary case.

5.

Once the description of G has been obtained, here is the proof of Theorem 1.

Write

$$H_\lambda = H^\bullet(A, \mathbf{Q}_\lambda) = \bigotimes_{v \in \Sigma/c} H_v^\bullet$$

where $H_v^\bullet$ is as in §4. We suppose $\mathbf{Q}_\lambda$ sufficiently large and Galois over $\mathbf{Q}_\ell$.

Under the action of the product, henceforth again denoted G , of the groups described in §4, the space H_λ decomposes as a direct sum indexed by the functions $\sigma \mapsto k(\sigma)$, for $\sigma \in \Sigma$, where

$$k(\sigma) \in \mathbf{Z} \text{ if } \sigma \neq \sigma c; \text{ then } k(\sigma c) = -k(\sigma), \quad (6)$$

$$k(\sigma) \in \mathbf{Z}/2\mathbf{Z} \text{ if } \sigma = \sigma c, \text{ the possible supersingular factor.} \quad (7)$$

In (6), $k(\sigma)$ is given by Theorem 3, extended as in §4 to the case where E is a CM field rather than an imaginary quadratic field. Notice that replacing σ by σc interchanges the spaces V_1 and V_2 of §4, which gives the relation (6).

Let K be the group of functions satisfying (6) and (7). Since the $\mathbf{Q}_\lambda$ -vector space generated by algebraic cycles is stable under G and under the cup product, it corresponds to a submonoid Alg of K . Moreover,

$$\text{Alg} \subset K_0 = \{k : \sum_{\sigma \bmod c} k(\sigma) \text{ is even}\}, \quad (8)$$

$$H_\lambda(k) \text{ is orthogonal to } H_\lambda(\ell) \text{ if } k + \ell \neq 0; \quad (9)$$

$$H_\lambda(k) \text{ and } H_\lambda(-k) \text{ are in duality;} \quad (10)$$

$$\text{Alg is stable under } \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell). \quad (11)$$

Suppose that the Galois group contains an element acting as (-1) on K . Then, on every $\mathbf{Q}_\ell$ -rational subspace of H_λ which is invariant under G , the duality given by the intersection form is nondegenerate. This applies in particular when, with the notation of §1, complex conjugation $c \in \text{Gal}(E/\mathbf{Q})$ belongs to the decomposition group of ℓ ; hence Theorem 1 follows.

A little less suffices: for every stable submonoid to be a subgroup of K , and therefore for the space of algebraic cycles to contain, together with the space associated with k , its dual associated with $-k$, it is enough that $K \otimes \mathbf{Q}$ have no fixed line under the action of $\text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$. In other words, if ℓ is unramified in the factors of E , it is enough that Frob_ℓ acting on Σ satisfy:

$$\text{for every } \sigma \in \Sigma, \text{ there exists } a \in \mathbf{Z} \text{ such that } \text{Frob}_\ell^a(\sigma) = \sigma c.$$

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Extended Euler congruence

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Abstract. The matricial Euler congruence $\mathrm{Tr}(A^{p^n}) \equiv \mathrm{Tr}(A^{p^{n-1}})$ modulo p^n , previously announced in Arnold, Japanese J. Math. 1(1), 1–24, 2006 for $A \in M_N(\mathbb{Z})$, is given a proof based on extending it to the ring of Witt vectors of length n .

Keywords. Euler congruence · Witt vector

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The matricial Euler congruence

$$\mathrm{Tr}(A^{p^n}) \equiv \mathrm{Tr}(A^{p^{n-1}}) \pmod{p^n}, \quad (1)$$

which was announced in [1] for $A \in M_N(\mathbb{Z})$, can be extended as follows. We replace $\mathbb{Z}$ with $\mathbb{Z}/p^n\mathbb{Z}$ and consider $\mathbb{Z}/p^n\mathbb{Z}$ as being $W_n(\mathbb{F}_p)$ (the ring of Witt vectors of length n). As geometers, we dislike considering the case of the field $\mathbb{F}_p$ alone: the main part of the following proof depends on the discovery of an extension to the case of $W_n(R)$, where R is of characteristic p .

Proposition 1. *Let R be an $\mathbb{F}_p$ algebra (commutative with unity). The endomorphism $F: x \mapsto x^p$ of R defines by functoriality an endomorphism F of $W_n(R)$. For A in $M_N(W_n(R))$, the equality*

$$\mathrm{Tr}(A^{p^n}) = F\left(\mathrm{Tr}(A^{p^{n-1}})\right) \quad (2)$$

holds in $W_n(R)$.

Proof. It suffices to consider the universal case, where R is the ring of the polynomials of N^2n variables $X_{ij}^{(k)}$ over $\mathbb{F}_p$ and A_{ij} is the Witt vector whose components are $X_{ij}^{(k)}$. We must prove a polynomial identity in these variables. It suffices to prove it in $W_n(R[1/D])$, where D is the discriminant of the characteristic polynomial of $(X_{ij}^{(0)})$. If k is an algebraically closed field containing $R[1/D]$, it suffices to prove (2) in $W_n(k)$: we may and shall assume that $R = k$ is algebraically closed and that the image of A in $M_N(k)$ is a matrix with distinct eigenvalues.

To simplify, we may also suppose that the image of A in $M_N(k)$ is invertible. In this case, the matrix A is conjugate to a diagonal matrix. It remains to prove that for $a_i \in W_n(k)^*$ ($1 \leq i \leq n$), we have

$$\sum_i a_i^{p^n} = F\left(\sum_i a_i^{p^{n-1}}\right) = \sum_i F(a_i^{p^{n-1}}).$$

As F is an endomorphism, the task is thus reduced to the case $N = 1$.

It remains to show that for $a \in W_n(k)$, the element $a^{p^{n-1}}$ is a multiplicative representative (i.e., a Witt vector of the form $(\lambda, 0, 0, \dots, 0)$). Indeed, F acts on a multiplicative representative by raising it to the p -th power.

Writing $a = \alpha u$, where α is a multiplicative representative and $u \equiv 1$ in k , we must prove that $u^{p^{n-1}} = 1$.

The ring $W(k)$ of the Witt vectors is a complete discrete valuation ring with residue field k and maximal ideal p , and $W_n(k)$ is its quotient by p^n . We have

$$u \equiv 1 \pmod{p},$$

whence $u^{p^a} \equiv 1 \pmod{p^{a+1}}$ follows inductively, as can be seen from the development of $(1 + p^{a-1}x)^p$.

And, finally, (1) follows from (2) because F is the identity of $\mathbb{F}_p$. □

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The Nagata Embedding Theorem

Pierre Deligne

Translator’s note. This is a working English translation of the uploaded LaTeX transcription of Deligne’s *Le théorème de plongement de Nagata*. It is intended as a literal mathematical translation, not a modern recast. Mathematical notation and the structure of the transcription have been kept as close as possible to the source.

Abstract. Except for a few corrections added in 2010, this article reproduces notes sent to Nagata in the 1970s, translating his papers [2] and [3] into the language of schemes. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to avoid the use of valuations. A more detailed exposition and additional results may be found in Conrad [1]. Unless expressly stated otherwise, all schemes considered below are assumed to be Noetherian.

0. Recollections on blow-ups

0.1.

Let $\mathfrak{a}$ be an ideal of the ring A . The scheme obtained from $X = \mathrm{Spec}(A)$ by blowing up $\mathfrak{a}$ is the union of the open sets $D^+(x)$, where x ranges over a system of generators of $\mathfrak{a}$, and $D^+(x)$ is the spectrum of the subring $A[\mathfrak{a}x^{-1}]$ of $A[x^{-1}]$. Moreover,

$$D^+(x) \cap (X - V(\mathfrak{a})) = \mathrm{Spec}(A[x^{-1}]).$$

Let $\mathfrak{c}$ be a second ideal of A , defining a closed subscheme $V(\mathfrak{c})$ of X . The trace on $D^+(x)$ of the proper transform of $V(\mathfrak{c})$ is defined by the ideal of $A[\mathfrak{a}x^{-1}]$ obtained as the trace of the ideal of $A[x^{-1}]$ generated by $\mathfrak{c}$.

Lemma 1. *Let X be a scheme, let $\mathfrak{a}$ and $\mathfrak{b}$ be two sheaves of ideals on X , and let $\tilde{X}$ be the blow-up of X along $\mathfrak{a} + \mathfrak{b}$. Then the open subset of $\tilde{X}$ which is the union of the $D^+(x)$ for $x \in \mathfrak{b}$ contains the proper transform of $V(\mathfrak{a})$.*

Indeed, $\tilde{X}$ is the union of the $D^+(x)$ for $x \in \mathfrak{a}$ or $x \in \mathfrak{b}$, and for $x \in \mathfrak{a}$ the open set $D^+(x)$ is disjoint from the proper transform of $V(\mathfrak{a})$.

Lemma 0.3 of the original text was false. The corrected version that follows dates from March 1996. It led to a change of notation. The original lemma used the following notation, to which the rest of the text refers: instead of F and G , one considered their complements U and V , which, together with $Y' := U \cap p^{-1}(Y)$, formed a commutative diagram

$$\begin{array}{ccc} Y' & \hookrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X. \end{array}$$

Here Y' is closed in $p^{-1}(V)$.

Lemma 2. *Let $p : Z \rightarrow X$ be a morphism, let $F \supset G$ be closed subschemes of X , and let Y be a closed subscheme of Z . Assume that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G) \quad \text{scheme-theoretically.}$$

Blow up X along G to obtain $\tilde{X}$, and let $\tilde{p} : \tilde{Z} \rightarrow \tilde{X}$ be obtained from $p : Z \rightarrow X$ by base change. Then the closure

$$\overline{(Y \cap p^{-1}(G) \text{ in } \tilde{Z})}$$

of $Y \cap p^{-1}(G)$ in $\tilde{Z}$ and

$$\widetilde{\tilde{p}^{-1}(F \cap G \text{ in } \tilde{X})}$$

are disjoint closed subsets of $\tilde{Z}$.

We may assume that Z and X are affine. Let G denote the ideal of G . Cover $\tilde{X}$ by the open subsets $D^+(g)$ for $g \in G$ (a system of generators of G suffices), and cover $\tilde{X} - \tilde{G}$ by the open sets

$$\tilde{U}(g) = D^+(g) \cap (\tilde{X} - \tilde{G}) \subset \tilde{X} - \tilde{G},$$

with

$$\tilde{U}(g) \subset D^+(g).$$

It is enough to see that, for each g , the traces on $\tilde{p}^{-1}D^+(g)$ of

$$\widetilde{\tilde{p}^{-1}(F \cap G \text{ in } \tilde{X})}$$

and of

$$\overline{(Y \cap p^{-1}G \text{ in } \tilde{Z})}$$

are disjoint. They are, respectively, the inverse image of the closure of $F \cap \tilde{U}(g)$ in $D^+(g)$, and the closure of $Y \cap p^{-1}(\tilde{U}(g))$ in $\tilde{p}^{-1}D^+(g)$.

Write $\sim$ for inverse image from $\tilde{X}$ to Z , or from $D^+(g)$ to $\tilde{p}^{-1}D^+(g)$. Since $p^{-1}G \supset Y \cap p^{-1}(F)$, the inverse image $\tilde{g}$ of g on $\tilde{Z}$ can be written

$$\tilde{g} = y + f,$$

where y vanishes on Y and f vanishes on $p^{-1}(F)$. There are functions a_i on $\tilde{Z}$ and elements f_i in the ideal of F such that

$$f = \sum a_i f_i \sim.$$

Since $F \supset G$, the f_i belong to the ideal of G , and the f_i/g are functions on $D^+(g)$. They vanish on $F \cap \tilde{U}(g)$, hence on

$$\overline{(F \cap \tilde{U}(g) \text{ in } D^+(g))},$$

and $\sum a_i (f_i/g) \sim$ vanishes on

$$\tilde{p}^{-1} \overline{(\tilde{U}(g) \text{ in } D^+(g))}.$$

On $\tilde{p}^{-1}D^+(g)$ one has

$$1 = y/g + f/g = y/g + \sum a_i (f_i/g) \sim,$$

and the same sum is therefore equal to 1 on $Y \cap p^{-1}\tilde{U}(g)$, hence on its closure

$$\overline{(Y \cap \tilde{U}(g) \text{ in } \tilde{p}^{-1}D^+(g))}.$$

This proves the required disjointness.

Remark 3. Suppose instead that we are given closed subsets $F \supset G_0$ of X , with G_0 such that $Y \cap p^{-1}(F)$ is set-theoretically contained in $p^{-1}(G_0)$. Then there exists an r such that the r -th infinitesimal neighborhood G of G_0 in F satisfies the hypotheses of the lemma. If J is the ideal defining

$$\overline{(Y \cap p^{-1}(F))}_{\text{red}}$$

in $Y \cap p^{-1}(F)$, it is enough to choose r such that $J^{r+1} = 0$.

Thus the lemma makes it possible, by a blow-up that does not affect the complement V of G_0 , to “separate” the proper transform of F and

$$\overline{(Y \cap p^{-1}(G_0))}$$

in Z .

Lemma 4. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{p} & X \\ \uparrow & & \uparrow \\ \tilde{X}' & \xrightarrow{q} & X' \end{array}$$

with U a dense open subset of both X and X' , with p and q separated of finite type, and with $q^{-1}(X) = \tilde{X}'$.

Let F be a closed subset of X' , and let G be a closed subset of $\tilde{X}'$ containing $p^{-1}(F)$. Then there exists an ideal $\mathfrak{a}$ with

$$V(\mathfrak{a}) \subset \overline{F} - F$$

such that, after base change by the blow-up morphism defined by $\mathfrak{a}$, $\tilde{X} \rightarrow X$, and after replacing the new $\tilde{X}'$ by the closure of U , one has

$$\tilde{q}^{-1}(F) \subset \tilde{G}.$$

In Nagata’s language, to take the closure of F in the Zariski–Riemann space of X (the “bubble space”), it is enough to take the “intersection” of the closures of F in the blow-ups of X whose centers are contained in $\overline{F} - F$.

Apply Lemma 0.3 to

$$\begin{array}{ccc} \tilde{X}' & \longrightarrow & q^*F \\ \downarrow & & \\ X - F & \hookrightarrow & X - (\overline{F} - F) \hookrightarrow X. \end{array}$$

This gives $\mathfrak{a}$. After the base change $\tilde{X} \rightarrow X$, the new closure $\tilde{F}$ is disjoint from the image of the closure of the old $\tilde{X}' - q^*F$ in $\tilde{X}' - G$. In particular, after replacing $\tilde{X}'$ by $\tilde{U}$, this new closure is disjoint from the image of $\tilde{X}' - G$, which is what was required.

Lemma 5. *Let there be given a diagram of separated S -schemes of finite type*

$$\begin{array}{ccccc} & & \tilde{X}_i & & \\ & \nearrow & & \searrow & \\ U & \hookrightarrow & X_i & \longrightarrow & \tilde{X}_i \quad (1 \leq i \leq n) \\ & \searrow & & & \\ & & X & & \end{array}$$

with $\tilde{X}_i$ proper over S and U a dense open subset of X and of the $\tilde{X}_i$. Let X^* be the closure of U in the product of the $\tilde{X}_i$. Finally, let F_i be a closed subset of X_i ; assume that the intersection of the inverse images of the F_i in X^* is empty.

Then, after replacing each X_i by a blow-up with center in $\overline{F_i} - F_i$, one can obtain an analogous situation in which the intersection of the inverse images of the F_i is empty.

Let X^{**} be a blow-up of X^* ,

$$X^{**} \rightarrow \prod_i \tilde{X}_i,$$

such that in X^{**} one has $p_i^{-1}(F_i) = \emptyset$; this is obtained by blowing up $p_i^{-1}(F_i)$. Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccc} p_i^{-1}(\tilde{X}_i) & \hookrightarrow & X^{**} \\ \downarrow & & \downarrow \\ U & \hookrightarrow & X_i \hookrightarrow \tilde{X}_i \end{array}$$

and to $F_i \subset X_i$, $G_i = p_i^{-1}(F_i)$. This provides the desired blow-ups of the X_i . Indeed, in the new situation, in the closure of U in X^{**} one has

$$p_i^{-1}(F_i) \subset G_i = \emptyset.$$

Since $X^{**} \rightarrow X^*$ is surjective, the same remains true in X^* .

1. The theorems

Definition 6. Let X and Y be two S -schemes, with Y separated over S . A quasi-domination $f : X \dashrightarrow Y$ is a pair consisting of a dense open subset U of X and a morphism of S -schemes $f : U \rightarrow Y$ whose graph is closed in $X \times_S Y$.

Let X and Y be two S -schemes, with Y separated over S , let U be a quasi-compact open subset of X , and let $f : U \rightarrow Y$. We say that X is quasi-dominant over Y relative to U and f if there exists a quasi-domination of the scheme-theoretic closure $\overline{U}$ of U , with values in Y , extending f . There is at most one such quasi-domination; its graph is the closure of the graph of f .

A quasi-domination f is called proper, or complete, if the morphism $f : U \rightarrow Y$ is proper.

Theorem 7 ([2, Theorem 3.2]). *Let X and Y be two S -schemes, and let $U \subset V \subset X$ be two dense open subsets of X . Assume that Y is separated of finite type over S . Let $f : U \rightarrow Y$ be a quasi-domination of V into Y . Then there exists an ideal $\mathfrak{a}$ with*

$$V(\mathfrak{a}) \subset X - V$$

such that the blow-up $\tilde{X}$ of X along $V(\mathfrak{a})$ quasi-dominates Y .

By the base change $X \rightarrow S$, one reduces to the case $X = S$. Then Y is a scheme over X , f is a section of Y over U , and by hypothesis $f(U)$ is closed over V .

Case 1. $U = V$, $p : Y \rightarrow X$ quasi-affine, X affine.

By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \xhookrightarrow{\quad} & E \\ & \searrow p & \swarrow \\ & X & \end{array}$$

with E an affine space of finite type over X . If X quasi-dominates E relative to f , then *a fortiori* X quasi-dominates Y . Thus we may assume $Y = E_X$. A preliminary blow-up makes it possible to assume that $X - U$ is a divisor D . Locally on X , the section f therefore has coordinates $(f_i u^{-n})$, where u is a local equation for D , the f_i are regular, and n is chosen large enough once and for all. Let $\mathfrak{a}$ be the ideal generated by u and the f_i . After blowing up $V(\mathfrak{a})$, the section f extends to a section of the projective completion of E_X , and this proves the case.

Case 2. $U = V$, p quasi-affine.

By Case 1, if $(U_i)_{0 \leq i \leq n}$ is an affine open covering of X , then on each U_i there exists an ideal $\mathfrak{a}_i$ which works. This ideal has an extension $\tilde{\mathfrak{a}}_i$ to X , with $V(\tilde{\mathfrak{a}}_i) \subset X - V$, and it is enough to blow up the product of the $\tilde{\mathfrak{a}}_i$.

Case 3. p quasi-affine.

By Lemma 0.3, applied to

$$\begin{array}{ccc} f(U) & \hookrightarrow & Y \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X, \end{array}$$

a preliminary blow-up makes it possible to assume that

$$p(\overline{f(U)}) \cap (V - U) = \emptyset.$$

There is therefore a quasi-compact open subset X' of X with $X' \cap V = U$ and $p(\overline{f(U)}) \subset X'$. By Case 2, there exists an ideal $\mathfrak{a}$ on X' , with $V(\mathfrak{a}) \subset X' - U$, such that the blow-up $\widetilde{X'}$ quasi-dominates Y , equivalently $\overline{f(U)}$. It remains only to extend $\mathfrak{a}$ to an ideal on X , with $V(\mathfrak{a}) \subset X - V$.

General case.

Let (Y_i) be a finite covering of Y by affine open subsets, and set $U_i = f^{-1}(Y_i)$, $X_i = \overline{U_i}$, $V_i = X_i \cap V$.

Case 3 applies to the diagrams

$$\begin{array}{ccccc} U_i & \hookrightarrow & V_i & \hookrightarrow & X_i \\ f_i \downarrow & & & & \downarrow \\ Y_i & \hookrightarrow & Y & & X, \end{array}$$

and gives an ideal $\mathfrak{a}_i$ on X_i . This ideal defines an ideal, still denoted $\mathfrak{a}_i$, on X , again with $V(\mathfrak{a}_i) \subset X - V$. Blow up the product of the $\mathfrak{a}_i$. The resulting blow-up $q : \tilde{X} \rightarrow X$ is such that $q^{-1}(X_i)$ dominates the blow-up $\tilde{X}_i$. If $X'_i = q^{-1}(X_i)$, there exists an open subset U'_i of X'_i , with $U'_i \supset q^{-1}(U_i)$, such that, over X'_i , $f(U'_i)$ is a section of $f : U'_i \rightarrow Y$. We conclude that after the blow-up,

$$f(U) = \bigcup f_i(U'_i).$$

To prove Theorem 1.2, we may replace Y by $\overline{f(U)}$, and we conclude by Case 3 and the following lemma.

Lemma 8. $\overline{f(U)}$ is quasi-affine over X .

Indeed, $\overline{f(U)}$ is quasi-finite over X .

Corollary 9 (A variant of Chow's lemma). *Let $f : X \rightarrow S$ be a separated morphism of finite type. Then for every open subset U which is quasi-projective over S and dense in X , there exists a diagram of S -schemes*

$$X \xleftarrow{q} \tilde{X}' \xrightarrow{j} \tilde{X}$$

where q is a blow-up morphism defined by an ideal satisfying $V(\mathfrak{a}) \subset \tilde{X}' - U$, and where $\tilde{X}$ is projective over S .

In particular, q is projective and surjective, and $q^{-1}(U) \xrightarrow{\sim} U$.

Let U^* be a projective S -scheme containing U as a schematically dense open subset. By Theorem 1.2 applied to the inclusion of U in X , one can choose U^* quasi-dominant over X : one has a diagram

$$\begin{array}{ccc} & U & \\ \swarrow & & \searrow \\ V & \xleftarrow{\phi} & U^* \\ \searrow & & \swarrow \\ & \bar{U} & \longrightarrow X, \end{array} \quad (1)$$

and here, since U^* is proper over S , ϕ is proper.

Lemma 10. *Let there be given a diagram of separated S -schemes of finite type*

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \\ X & \xleftarrow{\phi} & Y \end{array}$$

with U dense in X , dense in Y , and schematically dense in V , and with ϕ a quasi-domination of Y into X . Let $\mathfrak{a}$ be an ideal of X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ of X along $\mathfrak{a}$ quasi-dominates V relative to U , and such that U is schematically dense in $\tilde{X}$. Thus we obtain a diagram

$$\begin{array}{ccc} & \tilde{U} & \\ \swarrow & & \searrow \\ \tilde{V} & \xleftarrow{\beta} & \tilde{W} \\ \searrow \psi & & \swarrow \alpha \\ & \tilde{X}. & \end{array}$$

Then:

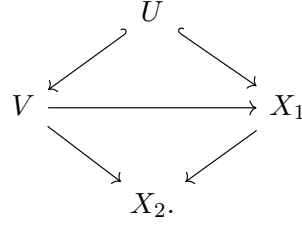
- (a) ψ is the blow-up morphism of V along $\phi^*\mathfrak{a}$;
- (b) the graph of ψ is closed in $\tilde{X} \times_S Y$.

Let $\tilde{V}$ be the blow-up of V along $\phi^*\mathfrak{a}$.

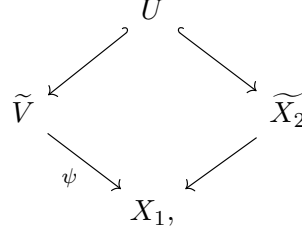
- (1) By the universal property of $\tilde{X}$, there is a unique $\tilde{X}$ -morphism from $\tilde{V}$ to $\tilde{X}$; denote it by α .
- (2) By the universal property of $\tilde{V}$, there is a unique V -morphism from $\tilde{W}$ to $\tilde{V}$; denote it by β .

$$\begin{array}{ccc} \tilde{W} & \xrightarrow{\beta} & \tilde{V} \\ \alpha \downarrow & & \downarrow \psi \\ \tilde{X} & & V \hookrightarrow Y \end{array} \quad \begin{array}{ccc} U & \hookrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ V & \longrightarrow & \tilde{V} \longrightarrow \tilde{X}. \end{array}$$

By Theorem 1.2, after blowing up X_1 without changing U , we reduce to the case where X_1 quasi-dominates X_2 .



Apply Lemma 1.5 to this diagram so as to obtain



where

- (a) ψ is the blow-up morphism of an ideal $\mathfrak{a}$ of V with $V(\mathfrak{a}) \subset V - U$;
- (b) the graph of ψ is closed in $X_1 \times_S \tilde{X}_2$; and
- (c) $\tilde{X}_2$ is a blow-up of X_2 .

Let $\mathfrak{a}'$ be an ideal of X_1 extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset X_1 - U$, and let $q : \tilde{X}_1' \rightarrow X_1$ be the blow-up of X_1 along $\mathfrak{a}'$. Then ψ induces an isomorphism between $\tilde{W}$ and $q^{-1}(V)$, and the graph of this isomorphism is closed in $\tilde{X}_1' \times \tilde{X}_2$. Define $\tilde{X}$ by gluing $\tilde{X}_1'$ and $\tilde{X}_2$ along this isomorphism.

Remark 11. The proof shows that, for the maps $p_i : U_i \rightarrow X_i$ with $U_i \subset X_i$, one may take a blow-up morphism of an ideal supported outside U .

Theorem 12 ([2, Lemma 4.1 and Theorem 4.3]). *Let $f : X \rightarrow S$ be separated of finite type. Then there exists an immersion of X into an S -scheme proper over S .*

The graph of ϕ is, trivially, proper over V ; the same is true for its inverse image on $V \times_S X$, and it follows that the graph of ψ is proper over V , hence that ψ is proper. Thus β is proper, and has schematically dense image, since U is already schematically dense in V . On the other hand, β is a local section of α on $\tilde{W}$ because $\beta|_U$ is one on U ; hence β is an isomorphism.

By hypothesis, the graph of ψ is closed in $V \times_S X$, and is contained in the inverse image of the graph of ϕ , which is closed in $Y \times X$. It follows that $\Gamma(\psi)$ is closed in $Y \times \tilde{X}$.

Apply Lemma 1.5 to (1). There exists an ideal $\mathfrak{a}$ on X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ quasi-dominates $\bar{U} \subset X - V$. Moreover, one can arrange that U is schematically dense in $\tilde{X}$. By Lemma 1.5, and because ϕ is proper, $\tilde{X}$ is therefore the blow-up of V along $\phi^*\mathfrak{a}$. If $\mathfrak{a}'$ is an ideal of U^* extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset U^* - U$, we then take

$$\tilde{X}' = \tilde{X}, \quad \tilde{X} = U^*.$$

Lemma 13 ([2, Lemma 4.2]). *Let there be given a diagram of separated S -schemes of finite type*

$$X_2 \longleftarrow U \longrightarrow X_1$$

with U a dense open subset of the X_i . Then there exists an embedding $j : U \hookrightarrow X$, with X separated of finite type over S and U schematically dense in X , and there exist proper quasi-dominations $p_i : X \dashrightarrow X_i$.

Let U_i be a finite open covering of X by dense quasi-projective open subsets. For each U_i there exists, by Corollary 1.4, a commutative diagram of S -schemes

$$\begin{array}{ccc} & & \tilde{X}_i \\ & \nearrow j_i & \downarrow \\ U_i & \xrightarrow{q_i} & X_i \end{array}$$

with q_i proper, $\tilde{X}_i$ proper over S , and U_i dense in $\tilde{X}_i$. Let $F_i = X_i - U_i$ and $F'_i = q_i^{-1}(F_i)$. Finally, let X^* be the closure of U in the product of the $\tilde{X}_i$, and write $p_i : X^* \rightarrow \tilde{X}_i$.

By Lemma 0.5, after a preliminary blow-up of X_i which does not affect $\tilde{X}'_i$, we may assume that

$$p_i^{-1}(F'_i) = \emptyset.$$

Let M be the separated S -scheme obtained by gluing the $\tilde{X}_i$ and $X_i - F'_i$ along U_i .

$$\begin{array}{ccc} X_i - F'_i & & \\ \downarrow & & \\ \tilde{X}_i & \longrightarrow & M. \end{array}$$

Apply Lemma 1.7 to X and to the M_i , to obtain $X \rightarrow M$ where M properly quasi-dominates each M_i . It remains to prove that M is proper over S .

Indeed, one has maps

$$X^* - p_i^{-1}(F'_i) \rightarrow M_i$$

which glue over $U = \bigcup U_i$. After replacing X^* by a blow-up X^{**} , one obtains maps

$$X^{**} - p_i^{-1}(F'_i) \rightarrow M_i$$

which glue. This gives a map from the scheme-theoretic closure of U in X^{**} with values in M . The scheme M , being the image of this map, is proper over S .

Comments

I hope that, by being still more constructive, one can prove Theorem 1.6 for $f : X \rightarrow S$ separated of finite type with S quasi-compact and quasi-separated. The rules to follow are:

- (a) consider only quasi-compact open subsets and closed subsets defined by ideals of finite type;
- (b) never take a scheme-theoretic closure, since that would take us outside (a); instead, take only a closed subset containing the given part and sufficiently small for the later constructions;
- (c) do not speak of a dense quasi-projective open subset; cf. (b). Remember that, to construct such open subsets, one could start with an affine cover U_i and take $U'_i = U_i - \bigcup_{j < i} U_j$. Translate the arguments in terms of the original U_i , and apply (a)+(b).

On the other hand, I have no idea in the case of algebraic spaces.

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Counting Local Systems with Principal Unipotent Local Monodromy

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Abstract

Let X_1 be a curve of genus g , projective and smooth over $\mathbb{F}_q$. Let $S_1 \subset X_1$ be a reduced divisor consisting of N_1 closed points of X_1 . Let (X, S) be obtained from (X_1, S_1) by extension of scalars to an algebraic closure $\mathbb{F}$ of $\mathbb{F}_q$. Fix a prime ℓ not dividing q . The pullback by the Frobenius endomorphism Fr of X induces a permutation Fr^* of the set of isomorphism classes of rank n irreducible $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X - S$. It maps to itself the subset of those classes for which the local monodromy at each $s \in S$ is unipotent, with a single Jordan block. Let $T(X_1, S_1, n, m)$ be the number of fixed points of Fr^{*m} acting on this subset.

Under the assumption that $N_1 \geq 2$, we show that $T(X_1, S_1, n, m)$ is given by a formula in which the main term is an integral power of $q^{mD/2}$, where $D = (2g - 2)(n^2 - 1) + N(n^2 - n)$.

We use Lafforgue to reduce the computation of $T(X_1, S_1, n, m)$ to counting automorphic representations of $\text{GL}(n)$, and the assumption $N_1 \geq 2$ to move the counting to the multiplicative group of a division algebra, where the trace formula is easier to use.

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Introduction

We keep the notations of the abstract. They are elaborated upon in §0. What are “ $\overline{\mathbb{Q}}_\ell$ -local systems” is explained in 1.1. A better terminology, which we will use from now on, is “ $\overline{\mathbb{Q}}_\ell$ -smooth sheaf”. Actions of Frobenius are explained in 1.2.

The present work is motivated by Drinfeld’s 1981 paper [D]. In it, Drinfeld considers irreducible $\overline{\mathbb{Q}}_\ell$ -smooth sheaves of rank 2 on X , and computes the number of fixed points of Fr^{*m} on the set of their isomorphism classes. He uses the trace formula for $\mathrm{GL}(2)$ and the then not yet wholly available correspondence between irreducible $\overline{\mathbb{Q}}_\ell$ -smooth sheaves of rank 2 over X_1 and everywhere unramified cuspidal automorphic representations of $\mathrm{GL}(2, \mathbb{A})$, where $\mathbb{A}$ is the adèle ring of the function field $\mathbb{F}_1$ of X_1 . He shows in particular that, as a function of m , the number of fixed points is of the form $\sum n_i \gamma_i^m$. Our result is a simple yet higher dimensional analogue of Drinfeld’s deep and beautiful analysis. We hope similar counting formulas of Lefschetz type hold for any number of ramification points and any imposed local monodromy at those points. See 6.29, 6.30. The case of tame local monodromy, given at each point by n characters of the residue field, has been considered by Arinkin (unpublished), under a “generic position” assumption for the characters involved.

We now describe what is done in each section. In §1, we review relations between $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X_1 - S_1$, ℓ -adic representations of $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_1)$, and automorphic representations of $\mathrm{GL}(n)$ and of the multiplicative groups of some division algebras.

Our proof relies on the relation between automorphic representations for $\mathrm{GL}(n)$ and for multiplicative groups of division algebras. More precisely, our results use Statement 1.13. How to extract this statement from the literature is explained in the Appendix. As computations of numbers of various kinds of automorphic representations for multiplicative groups of some division algebras, our results are independent of 1.13. It is the interpretation of our results as computations of numbers of fixed points which makes our results interesting.

In §2, we state a first form (2.3) of our result. This form is the one to which an application of the trace formula (compact quotient case) leads. It does not make clear that, as a function of m , the number $T(X_1, S_1, n, m)$ of fixed points of Fr^{*m} has the form $m \mapsto \sum n_i \gamma_i^m$.

After a discussion of division algebras and Tamagawa numbers in §3, the proof of 2.3 is given in §4 and §5. It applies the trace formula in the compact quotient case of multiplicative groups of division algebras. A categorical approach to the proof of the trace formula in the compact quotient case we need is presented in §4, and the building of $\mathrm{SL}(n)$ is used in the computations of §5.

The reader is invited, at first reading, to jump from §2 to §6. In §6, 2.3 is massaged into a form which cries for a geometric explanation, and makes clear that, as a function of m , $T(X_1, S_1, n, m)$ has the form

$$m \mapsto \sum n_i \gamma_i^m, \tag{1}$$

where the n_i are nonzero integers and the γ_i are distinct q -Weil numbers: for each γ_i there exists an integer w , its weight, such that all complex conjugates of γ_i have absolute value $q^{w/2}$. For $g = 0$ and $\deg(S_1) = 2$, as well as for $g = 0$, $n = 2$ and $\deg(S_1) = 3$, the sum (1) is empty: $T(X_1, S_1, n, m)$ is identically zero.

After excluding these cases, we find the γ_i with the largest complex absolute value. It occurs with multiplicity one and is an integral power of q . We also show that each γ_i is q times an algebraic integer. Next, we consider how $T(X_1, S_1, n, m)$ varies with (X_1, S_1) , and in 6.3.1 we ask for a topological understanding.

We do not understand the meaning of the change of summations from $\sum_a \sum_b$ in the expression (6.5.2) for $T(X_1, S_1, n, 1)$ to $\sum_b \sum_a$ in the expression (6.6.4). It leads to a decomposition of $T(X_1, S_1, n, m)$ as a sum over the divisors b of n , each term of which has the form (1).

In §7, we look at one of the simplest examples: the case where X_1 is of genus 0 and where S_1 consists of four points. An amusing trick allows us in that case to remove the assumption $N_1 \geq 2$. The same trick implies a symmetry property of some automorphic representations, of which we do not know of a proof not using [L].

1 Notations

1.1

We fix a finite field $\mathbb{F}_q$ with q elements and its algebraic closure $\mathbb{F}$. The characteristic is p , and $\mathbb{F}_{q^m}$ is the degree m extension of $\mathbb{F}_q$ in $\mathbb{F}$.

1.2

We fix a projective and smooth curve X_1 over $\mathbb{F}_q$, absolutely irreducible of genus g , and a reduced divisor $S_1 \subset X_1$ consisting of N_1 closed points. As $\mathbb{F}_q$ is a perfect field, it is automatically étale over $\text{Spec}(\mathbb{F}_q)$. The field of rational functions on X_1 will be denoted by $\mathbb{F}_1$. It is a global field of characteristic p , with field of constants $\mathbb{F}_q$. Closed points of X_1 correspond one to one to places of $\mathbb{F}_1$; we identify the two. If s is a closed point, we denote the local ring of X_1 at s by $\mathcal{O}_{(s)}$. It is a discrete valuation ring. We denote its completion by $\mathcal{O}_s$. This is contrary to the usual usage. The residue field at s we denote by $k(s)$. The completion of $\mathbb{F}_1$ at the place s we denote by $\mathbb{F}_{1s}$. They are, respectively, the quotient of the complete valuation ring $\mathcal{O}_s$ by its maximal ideal $\mathfrak{m}_s$, and the fraction field of $\mathcal{O}_s$.

1.3

Suppressing the index 1 indicates an extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}$, and replacing it by m an extension of scalars to $\mathbb{F}_{q^m}$. For instance, N is the number of points of $X = X_1 \otimes_{\mathbb{F}_q} \mathbb{F}$ in the inverse image S of S_1 . We write $\mathbb{A}$ for the ring of $\mathbb{F}_1$ -adèles.

Exception: from §3 on, we do not use the function field $\mathbb{F} = \mathbb{F}_1 \otimes_{\mathbb{F}_q} \mathbb{F}$ of X . To simplify notations we will write simply $\mathbb{F}$ for $\mathbb{F}_1$, and D for a division algebra with center $\mathbb{F}$ (renamed $\mathbb{F}$).

1.4

We fix a prime $\ell \neq p$ and an algebraic closure $\overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}_\ell$.

1.5

The element $\sigma : \mathbb{F} \rightarrow \mathbb{F}$, $x \mapsto x^q$, of $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$ is called the Frobenius substitution. Its inverse, denoted Frob , is the geometric Frobenius.

1.6

For any scheme Y over $\mathbb{F}_q$, let Φ_Y be the endomorphism of Y which is the identity on the underlying set, and for which $\Phi_Y^*(f) = f^q$. For any étale sheaf $\mathcal{F}$ on Y , the pullback $\Phi_Y^* \mathcal{F}$ is canonically isomorphic to $\mathcal{F}$. If one pictures $\mathcal{F}$ as an (algebraic) space over Y , this expresses

the functoriality of Φ , which gives rise to a commutative diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\Phi_{\mathcal{F}}} & \mathcal{F} \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\Phi_Y} & Y \end{array} \quad (1)$$

Special case: Φ_{X_1} is an endomorphism of the $\mathbb{F}_q$ -scheme X_1 . Extending scalars to $\mathbb{F}$, one obtains the Frobenius endomorphism Fr of X .

1.7

The multiplicative group of an algebra A is denoted by A^* .

2 Dictionaries

The reader familiar with $\overline{\mathbb{Q}}_\ell$ -smooth sheaves and actions of Frobenius is invited to jump to 1.4.

2.1 $\overline{\mathbb{Q}}_\ell$ -smooth sheaves

Warning: $\overline{\mathbb{Q}}_\ell$ -smooth sheaves, called “ $\overline{\mathbb{Q}}_\ell$ -local systems” in the abstract, are not defined to be locally constant for the étale topology. Rather, the category of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves is defined by limiting processes from categories of local systems with finite fibers, which are locally constant for the étale topology. One proceeds in three steps:

(a) Let E_λ be a finite extension of $\mathbb{Q}_\ell$ in $\overline{\mathbb{Q}}_\ell$, $\mathcal{O}_\lambda$ its valuation ring, and $\mathfrak{m}_\lambda$ the maximal ideal of $\mathcal{O}_\lambda$. The category of $\mathcal{O}_\lambda$ -smooth sheaves on a scheme Y is the category of projective systems $(\mathcal{F}_k)_{k \geq 1}$ of locally constant sheaves of $\mathcal{O}_\lambda/\mathfrak{m}_\lambda^k$ -modules of finite type, with $\mathcal{F}_k/\mathfrak{m}_\lambda^{k'} \mathcal{F}_k \xrightarrow{\sim} \mathcal{F}_{k'}$ for $k' \leq k$.

(b) [Assuming Y to be normal] The category of E_λ -smooth sheaves on Y is obtained from it by tensoring over $\mathcal{O}_\lambda$ by E_λ : same objects, and $\text{Hom}(F, G) = \text{Hom}_{\mathcal{O}_\lambda}(F, G) \otimes_{\mathcal{O}_\lambda} E_\lambda$. When Y is not assumed to be normal, a more complicated definition is required if one wants categories of E_λ -smooth sheaves to form a stack for the étale topology. We will not need this more general setting.

(c) The set of finite extensions E_λ of $\mathbb{Q}_\ell$ contained in $\overline{\mathbb{Q}}_\ell$, ordered by inclusion, is filtering. The category of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves is the 2-inductive limit, along this set, of the categories of E_λ -smooth sheaves. Those categories do not form an inductive system in the category of categories, only a 2-inductive system in the 2-category of categories, hence the appearance of 2-inductive limits.

Suppose the normal scheme Y is connected. If o is a geometric point of Y , the functor “fiber at o ” is an equivalence of categories from the category of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves to the category $\text{Rep}(\pi_1(Y, o), \overline{\mathbb{Q}}_\ell)$ of continuous linear representations of the profinite fundamental group of Y :

$$\pi_1(Y, o) \rightarrow \text{GL}(V),$$

for V a finite dimensional vector space over $\overline{\mathbb{Q}}_\ell$. In this statement, the topology used on $\text{GL}(n, \overline{\mathbb{Q}}_\ell)$ is any such that the $\text{GL}(n, E_\lambda)$ are closed subgroups, and the induced topology on $\text{GL}(n, E_\lambda)$ is the ℓ -adic topology. A Baire category argument shows that a continuous homomorphism from $\pi_1(Y, o)$ to $\text{GL}(n, \overline{\mathbb{Q}}_\ell)$ factors through a $\text{GL}(n, E_\lambda)$.

2.2 Actions of Frobenius

The Galois group $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on $\mathbb{F}$, hence on the scheme $X - S = (X_1 - S_1) \otimes_{\mathbb{F}_q} \mathbb{F}$, and, by transport of structures, on the set of isomorphism classes of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X - S$.

Transport of structures (Bourbaki Ens Ch. IV) is the principle that any isomorphism $Y_1 \rightarrow Y_2$ extends to objects constructed from Y_1 and Y_2 . When $Y_1 = Y_2$: symmetries extend. The construction has to be canonical: not involving choices.

Example: If $\tau : K_1 \rightarrow K_2$ is an isomorphism between fields, the corresponding isomorphism $[\tau] : \text{Spec}(K_1) \rightarrow \text{Spec}(K_2)$ is such that $[\tau]^*(a) = \tau^{-1}(a)$ for a in K_2 . If τ is an automorphism of K , the action by transport of structures of τ on $\text{Spec}(K)$ is $[\tau] : \text{Spec}(K) \rightarrow \text{Spec}(K)$ such that $[\tau]^*(a) = \tau^{-1}(a)$.

Lemma 2.1. *The action, by transport of structures, of the geometric Frobenius Frob in $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$ (0.5) on the set of isomorphism classes of $\overline{\mathbb{Q}_\ell}$ -smooth sheaves on $X - S$ coincides with the pullback by the Frobenius endomorphism $\text{Fr} : X - S \rightarrow X - S$.*

More precisely, if $\mathcal{F}$ is $\overline{\mathbb{Q}_\ell}$ -smooth sheaf on $X - S$, one has a canonical isomorphism between $\text{Fr}^\mathcal{F}$, and the $\overline{\mathbb{Q}_\ell}$ -smooth sheaf obtained from $\mathcal{F}$ by transport of structures using Frob . This is deduced from a similar result for étale sheaves, which we now explain, following SGA 5 XIV, §1,2.*

Proof. The endomorphism Φ_{X-S} (0.6) of $X - S = (X_1 - S_1) \times_{\text{Spec}(\mathbb{F}_q)} \text{Spec}(\mathbb{F})$ is the composite of the endomorphisms induced by $\Phi_{X_1-S_1}$ and by $\Phi_{\text{Spec}(\mathbb{F})}$. The first is $\text{Fr} : X - S \rightarrow X - S$. By the example in 1.2, $\Phi_{\text{Spec}(\mathbb{F})} = [\text{Frob}]$. By the commutative diagram, the pullback functor Φ_{X-S}^* is isomorphic to the identity. It is the composite of Fr^* , and of $(X_1 - S_1) \times_{\text{Spec}(\mathbb{F}_q)} [\text{Frob}]^*$. The latter is the inverse of the action of Frob by transport of structures, and the claim follows. $\square$

2.3

We now explain how the action 2.1 is expressed in the languages of representations of Galois groups or of fundamental groups.

The function field $\mathbb{F}_1$ of X_1 is a global field of characteristic p , with field of constants $\mathbb{F}_q$. The function field $\mathbb{F}$ of X is $\mathbb{F}_1 \otimes_{\mathbb{F}_q} \mathbb{F}$. Fix a separable closure $\overline{\mathbb{F}}$ of $\mathbb{F}$. The Galois group $\text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1)$ is an extension

$$1 \rightarrow \text{Gal}(\overline{\mathbb{F}}/\mathbb{F}) \rightarrow \text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1) \rightarrow \text{Gal}(\mathbb{F}/\mathbb{F}_q) \rightarrow 1 \quad (2)$$

and

$$\text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1) \xrightarrow{\sim} \text{Gal}(\mathbb{F}/\mathbb{F}_q). \quad (3)$$

We recall the identification 0.2 of closed points and places. If s is a closed point of X , with image in X_1 , the choice of a place $\bar{s}$ of $\overline{\mathbb{F}}$ above s defines a decomposition (= inertia) group $I_s \subset \text{Gal}(\overline{\mathbb{F}}/\mathbb{F})$ as well as a decomposition group $D_{s_1} \subset \text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1)$, and I_s is the inertia subgroup of D_{s_1} . If the closed point s_1 is of degree d over $\mathbb{F}_q$, the residue field $k(s_1)$, naturally embedded in $k(s) \xleftarrow{\sim} \mathbb{F}$, is $\mathbb{F}_{q^d}$. We have a comparison morphism relating the extension above (used to replace $\text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1)$ by $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$) and its local analogue:

$$\begin{array}{ccccccc} 1 & \longrightarrow & I_s & \longrightarrow & D_{s_1} & \longrightarrow & \text{Gal}(\mathbb{F}/\mathbb{F}_{q^d}) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & \text{Gal}(\overline{\mathbb{F}}/\mathbb{F}) & \longrightarrow & \text{Gal}(\overline{\mathbb{F}}/\mathbb{F}_1) & \longrightarrow & \text{Gal}(\mathbb{F}/\mathbb{F}_q) \longrightarrow 1. \end{array} \quad (4)$$

On the right, $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$ is $\hat{\mathbb{Z}}$, generated by Frob , $\text{Gal}(\mathbb{F}/\mathbb{F}_{q^d})$ is $\hat{\mathbb{Z}}$, generated by its d th power, and the vertical map is $b \mapsto db$.

The geometric point $\text{Spec}(\mathbb{F}) \rightarrow X - S$ of $X - S$, as well as its image $\text{Spec}(\mathbb{F}) \rightarrow X_1 - S_1$ in $X - S_1$, will be denoted by o . Let $\mathbb{F}_S$ be the maximal extension of $\mathbb{F}$ in $\overline{\mathbb{F}}$ which is unramified outside of S . The fundamental group $\pi_1(X - S, o)$ (resp. $\pi_1(X_1 - S_1, o)$) is the Galois group $\text{Gal}(\mathbb{F}_S/\mathbb{F})$ (resp. $\text{Gal}(\mathbb{F}_S/\mathbb{F}_1)$), and the homotopy sequence of fundamental groups for the

fibration $X_1 - S_1 \rightarrow \text{Spec}(\mathbb{F}_q)$ is the quotient of the second line above obtained by replacing $\overline{\mathbb{F}}$ by $\mathbb{F}_S$:

$$1 \rightarrow \pi_1(X - S, o) \rightarrow \pi_1(X_1 - S_1, o) \rightarrow \text{Gal}(\mathbb{F}/\mathbb{F}_q) \rightarrow 1. \quad (5)$$

As explained in 1.1, the functor “fiber at o ” is an equivalence of categories from the category of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X - S$ to the category of continuous representations of $\pi_1(X - S, o)$ on finite dimensional $\overline{\mathbb{Q}}_\ell$ -vector spaces. To compute in this language the action by transport of structures of τ in $\text{Gal}(\mathbb{F}/\mathbb{F}_q)$, one should first lift τ to an automorphism of the geometric point o , i.e. of $\mathbb{F}$. The induced automorphism $\tilde{\tau}$ of $\mathbb{F}_S$ is an element of $\pi_1(X_1 - S_1, o)$. It acts on $\pi_1(X_1 - S_1, o)$ by the corresponding inner automorphism. The induced action on $\pi_1(X - S, o)$ gives the action on representations: $\rho \mapsto (g \mapsto \rho(\tilde{\tau}^{-1}g\tilde{\tau}))$.

As was clear a priori, the action obtained on isomorphism classes of representations (i.e. $\overline{\mathbb{Q}}_\ell$ -smooth sheaves) does not depend on the lifting chosen. This will be used for $\tau = \text{Frob}$.

2.4

To state the relation between Frobenius fixed points and automorphic representations, it is convenient to replace Galois groups by Weil groups. Let $W(X_1 - S_1, o)$ be the inverse image in $\pi_1(X_1 - S_1, o)$ of the subgroup $\mathbb{Z}$ of $\text{Gal}(\mathbb{F}/\mathbb{F}_q) = \hat{\mathbb{Z}}$. One gives it the topology for which $\pi_1(X - S, o)$ is an open subgroup.

Locally, with the notations $s, s_1, \bar{s}, I_s, D_{s_1}$ and d of 1.4, let W_{s_1} be the inverse image in D_{s_1} of the subgroup $\mathbb{Z}$ of $\text{Gal}(\mathbb{F}/\mathbb{F}_{q^d}) = \hat{\mathbb{Z}}$, with the topology for which I_s is an open subgroup. From the comparison diagram we get

$$\begin{array}{ccccccc} 1 & \longrightarrow & I_s & \longrightarrow & W_{s_1} & \longrightarrow & \mathbb{Z} \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & \pi_1(X - S, o) & \longrightarrow & W(X_1 - S_1, o) & \longrightarrow & \mathbb{Z} \longrightarrow 1. \end{array} \quad (6)$$

The left vertical map is trivial when $s \notin S$, in which case we get from the diagram a map $\mathbb{Z} \rightarrow W(X_1 - S_1, o)$. The image of 1 is called a Frobenius at s_1 .

From the computation 1.4 of the Frobenius action, we get

Lemma 2.2. *Let $\mathcal{F}$ be a $\overline{\mathbb{Q}}_\ell$ -smooth sheaf on $X - S$. Its isomorphism class is fixed by $\text{Frob} \in \text{Gal}(\mathbb{F}/\mathbb{F}_q)$ (cf. 2.1) if and only if the corresponding representation of $\pi_1(X - S, o)$ can be extended to $W(X_1 - S_1, o)$.*

2.5

In the case of function fields, the notion of cuspidal automorphic representation is purely algebraic. One can use as field of coefficients any algebraically closed field k of characteristic zero, for instance $\overline{\mathbb{Q}}_\ell$. Let $\mathbb{A}$ be the adèle ring of $\mathbb{F}_1$. In the case of $\text{GL}(n)$, the multiplicity one theorem allows us not to make a distinction between cuspidal automorphic representations π of $\text{GL}(n, \mathbb{A})$, viewed as subrepresentations of the space of k -valued locally constant cuspidal functions on $\text{GL}(n, \mathbb{F}_1) \backslash \text{GL}(n, \mathbb{A})$, on which $\text{GL}(n, \mathbb{A})$ acts by right translations, or viewed as isomorphism classes of irreducible representations of $\text{GL}(n, \mathbb{A})$ occurring in that space. As representations, they are restricted tensor products of representations π_s of the local groups $\text{GL}(n, \mathbb{F}_{1s})$, s a place of $\mathbb{F}_1$. We will say that π is unramified at s if π_s admits a nonzero vector fixed by the maximal compact subgroup $\text{GL}(n, \mathcal{O}_s)$ of $\text{GL}(n, \mathbb{F}_{1s})$.

By Lafforgue [L], the isomorphism classes of n -dimensional irreducible $\overline{\mathbb{Q}}_\ell$ -linear continuous representations of $W(X_1 - S_1, o)$ are in a natural bijective correspondence with the $\overline{\mathbb{Q}}_\ell$ -cuspidal automorphic representations of $\text{GL}(n, \mathbb{A})$ which are unramified outside of S_1 (global Langlands correspondence). If ρ corresponds to π , the restriction of ρ to W_{s_1} corresponds to π_{s_1} by the local Langlands correspondence.

A surprising consequence: an algebraic isomorphism $\overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \overline{\mathbb{Q}}_{\ell'}$ induces a bijection between isomorphism classes of irreducible $\overline{\mathbb{Q}}_\ell$ -linear continuous representations of $W(X_1 - S_1, o)$, and the same for $\overline{\mathbb{Q}}_{\ell'}$, in spite of the fact that we are considering continuous representations for the ℓ - or ℓ' -adic topologies. Representations correspond if the characteristic polynomials of Frobeniuses do. Caveat: if one wants to use the so-called unitary correspondence between representations of $W(X_1 - S_1, o)$ and automorphic representations, one needs to choose a square root of q in $\overline{\mathbb{Q}}_\ell$.

2.6

Let $\mathcal{F}$ be a $\overline{\mathbb{Q}}_\ell$ -smooth sheaf on $X - S$. Let $V := \mathcal{F}_o$ be the corresponding representation of $\pi_1(X - S, o)$. Suppose that the isomorphism class of $\mathcal{F}$ is fixed by the Frobenius, and choose an extension of the representation V to a representation V_1 of $W(X_1 - S_1, o)$ (2.2).

In general, irreducibility of V does not imply irreducibility of V_1 (equivalently: of $\mathcal{F}$). The 1981 work of Drinfeld [D], in which $S = \emptyset$, suggests that it is the irreducibility of $\mathcal{F}$ we should be concerned with. We are interested in the case when $S \neq \emptyset$ and when the local monodromy at each $s \in S$ is principal unipotent. This case is simpler.

As in 1.4, let I_s be an inertia group at s . The largest pro- ℓ quotient of I_s is isomorphic to $\mathbb{Z}_\ell$. “Principal unipotent local monodromy at s ” means that the action of I_s factors through this $\mathbb{Z}_\ell$, with an element of I_s with image a in $\mathbb{Z}_\ell$ acting as $\exp(aN)$, where N is nilpotent with one Jordan block.

Lemma 2.3. (i) *If at one $s \in S$ the local monodromy is principal unipotent, then V is irreducible as soon as V_1 is.*

(ii) *If V is irreducible, any extension V'_1 of V to a representation of $W(X_1 - S_1, o)$ is of the form $V_1 \otimes \chi$, for $\chi : \mathbb{Z} \rightarrow \overline{\mathbb{Q}}_\ell^*$ a character of the quotient $\mathbb{Z}$ of $W(X_1 - S_1, o)$, and the $V_1 \otimes \chi$ are all nonisomorphic.*

Proof of (i). Any subrepresentation V' of V , being stable by the $\exp(aN)$ ($a \in \mathbb{Z}_\ell$), and hence by N , will be of the form $\sum N^k V'$. It will hence be the only subrepresentation of V of its dimension. As $\pi_1(X - S, o)$ is an invariant subgroup of $W(X_1 - S_1, o)$, any element of $W(X_1 - S_1, o)$ will map V' to a subrepresentation of V , of the same dimension, hence to itself: V' is a subrepresentation of V_1 . $\square$

Proof of (ii). By assumption, V_1 and V'_1 are identical as representations of $\pi_1(X - S, o)$. By Schur’s lemma,

$$\mathrm{Hom}_{\pi_1(X-S, o)}(V_1, V'_1)$$

is reduced to the line of multiplication by scalars. It is a representation of the quotient $\mathbb{Z}$ of $W(X_1 - S_1, o)$, and is given by a character χ of $\mathbb{Z}$. That

$$V_1 \otimes \mathrm{Hom}_{\pi_1(X-S, o)}(V_1, V'_1) \xrightarrow{\sim} V'_1$$

gives that V'_1 is $V_1 \otimes \chi$. Conversely, if we take $V'_1 = V_1 \otimes \eta$, the character χ we obtain is η , so that twists by distinct characters are nonisomorphic. $\square$

2.7

The divisor of an idèle $a = (a_s)$ is $\sum_s \mathrm{valuation}(a_s) \cdot s$, where the sum is over all closed points of X_1 . The degree of a is the degree $\sum_s \mathrm{valuation}(a_s) \deg(s)$ of its divisor. Equivalently, the degree map, from the idèle class group $\mathbb{F}_1^* \backslash \mathbb{A}^*$ to $\mathbb{Z}$, is characterized by

$$\|a\| = q^{-\deg(a)}. \quad (7)$$

Let $\chi : \mathbb{Z} \rightarrow \overline{\mathbb{Q}}_\ell^*$ be a character of $\mathbb{Z}$, where $\mathbb{Z}$ is viewed as the quotient $\deg \circ \det : \mathrm{GL}(n, \mathbb{A}) \rightarrow \mathbb{Z}$ of $\mathrm{GL}(n, \mathbb{A})$ (resp. as the quotient $\mathbb{Z}$ of $W(X_1 - S_1, o)$). The twist by χ , denoted $\pi \otimes \chi$ (resp.

$V_1 \otimes \chi$), of a representation π of $\mathrm{GL}(n, \mathbb{A})$ (resp. V_1 of $W(X_1 - S_1, o)$) is $\pi \otimes \chi$ (resp. $V_1 \otimes \chi$). Such twists we name $\mathbb{F}_q$ -twists. If $\mathcal{F}$ is a $\overline{\mathbb{Q}}_\ell$ -smooth sheaf on $X - S$, if V_1 is the restriction of the corresponding representation of $\pi_1(X_1 - S_1, o)$ to $W(X_1 - S_1, o)$, and if χ is with values in the units of $\overline{\mathbb{Q}}_\ell^*$, hence extends to $\hat{\mathbb{Z}}$, then $V_1 \otimes \chi$ is obtained from the tensor product of $\mathcal{F}$ with the pullback to $X - S_1$ of a rank one $\overline{\mathbb{Q}}_\ell$ -sheaf on $\mathrm{Spec}(\mathbb{F}_q)$. This is what motivates the terminology.

If π is automorphic, hence is a space of functions on $\mathrm{GL}(n, \mathbb{F}_1) \backslash \mathrm{GL}(n, \mathbb{A})$, then so is $\pi \otimes \chi$: it is the space of functions $f \cdot \chi(\deg \circ \det)$ for f in π . With the notations of 1.7, if an irreducible representation V_1 of $W(X_1 - S_1, o)$ corresponds by the global Langlands correspondence to the cuspidal automorphic representation π of $\mathrm{GL}(n, \mathbb{A})$, then $V_1 \otimes \chi$ corresponds to $\pi\chi$.

By the local Langlands correspondence, the condition that $\mathcal{F}$ has principal unipotent local monodromy at $s \in S$, with image in S_1 , corresponds to the condition on the automorphic representation π that the local component π_{s_1} be of the form

$$\text{Steinberg representation} \otimes \chi(\det), \quad (1.10.2)$$

for χ an unramified character $\mathbb{F}_{1s_1}^* \rightarrow \overline{\mathbb{Q}}_\ell^*$.

Applying 2.3 we conclude:

Scholium 2.4. *Assume that S_1 is not empty.*

(i) *There is a bijective correspondence between*

(A) *isomorphism classes of rank n irreducible $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X - S$, fixed by the Frobenius, and with principal unipotent monodromy at each $s \in S$, and*

(B) *classes modulo $\mathbb{F}_q$ -twisting of cuspidal automorphic $\overline{\mathbb{Q}}_\ell$ -representations π of $\mathrm{GL}(n, \mathbb{A})$, unramified outside of S_1 , such that for each $s \in S_1$ the representation π_s is of the form (1.10.2).*

(ii) *For π as in (B), the $\mathbb{F}_q$ -twists are all distinct.*

2.8

Suppose that D is a rank n^2 division algebra over $\mathbb{F}_1$, unramified outside a subset S_0 of the set of places S_1 , and for which at each $s \in S_0$ the completion D_s is a division algebra over $\mathbb{F}_{1s}$. Such a division algebra exists if and only if $|S_0| \geq 2$.

By abuse of notations, we denote by D^* the algebraic group over $\mathbb{F}_1$ such that for any commutative $\mathbb{F}_1$ -algebra R , the group $D^*(R)$ of R -points of D^* is the multiplicative group $(D \otimes_{\mathbb{F}_1} R)^*$ of $D \otimes_{\mathbb{F}_1} R$. The group of $\mathbb{F}_1$ -points of D^* is simply the multiplicative group of D . The reduced norm defines a homomorphism $\det$ of algebraic groups from D^* to the multiplicative group $\mathbb{G}_m$, and $\mathbb{F}_q$ -twists of automorphic representations of $D^*(\mathbb{A})$ are defined as for $\mathrm{GL}(n)$.

Our results depend on the following statement. How to extract this statement from the literature is explained in the Appendix.

Statement 2.5. *Suppose that $n \geq 2$. There is then a bijective correspondence, compatible with $\mathbb{F}_q$ -twists, between*

(A) *cuspidal automorphic representations π of $\mathrm{GL}(n, \mathbb{A})$, whose local components at each $s \in S_0$ is of the form (1.10.2), and*

(B) *automorphic representations π' of $D^*(\mathbb{A})$, other than one-dimensional, whose local components at each $s \in S_0$ is one-dimensional and of the form $\chi(\det)$ for χ an unramified character of $\mathbb{F}_{1s}^*$.*

The representation π (resp. π') occurs with multiplicity one in the cuspidal spectrum of $\mathrm{GL}(n, \mathbb{A})$ (resp. $D^(\mathbb{A})$). If π corresponds to π' , at each $v \notin S_0$ (so that $D_v^* \simeq \mathrm{GL}(n, \mathbb{F}_{1v})$), π_v is isomorphic to π'_v .*

The number of classes 2.4(A), or (B), is hence also the number of classes modulo $\mathbb{F}_q$ -twisting of automorphic representations π' of $D^*(\mathbb{A})$ other than one-dimensional, with local components at each $s \in S_0$ as in 2.5(B), with local components at each $s \in S_1 - S_0$ of the form (1.10.2), and unramified outside of S_1 . Further, for π' as in (B), the $\mathbb{F}_q$ -twists $\pi' \otimes \chi$ are all distinct.

So far, we have considered $\overline{\mathbb{Q}}_\ell$ -automorphic representations. The theory being algebraic, nothing changes if one considers instead the usual $\mathbb{C}$ -automorphic representations.

3 Statement of the theorem: first form

3.1

Let $T(n)(X, S)$, or simply $T(n)$, be the set of isomorphism classes of rank n irreducible $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $X - S$, with principal unipotent local monodromy at each $s \in S$ (see 1.8).

The geometric Frobenius $\text{Frob} \in \text{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on $T(n)$ by transport of structures. This action coincides with the pullback by the Frobenius endomorphism $\text{Fr} : X - S \rightarrow X - S$ (1.3). Our aim is to compute the number $T(X_1, S_1, n)$ of its fixed points. For each $m \geq 1$, the Frobenius endomorphism for $(X_m - S_m)/\mathbb{F}_{q^m}$ is Fr^m . The number $T(X_1, S_1, n, m)$ of fixed points of the m th iterate of Frobenius is hence given by

$$T(X_1, S_1, n, m) = T(X_m, S_m, n). \quad (2.1.1)$$

Before stating the result, we introduce some notations.

3.2

The zeta function of X_1 is a function of a complex variable usually called s . We will call it z , to avoid confusion with points of X_1 . Rather than this variable, we will systematically use the variable $t = q^{-z}$. The function $Z(X_1, t)$ is defined by

$$Z(X_1, t) = \zeta(X_1, z) \quad \text{when } t = q^{-z}. \quad (8)$$

It depends not only on the scheme X_1 , but also on q such that X_1 is a $\mathbb{F}_q$ -scheme. It has the cohomological description

$$Z(X_1, t) = \prod_i \det(1 - \text{Frob} \cdot t, H^i(X))^{(-1)^{i+1}} = \frac{\det(1 - \text{Frob} \cdot t, H^1(X))}{(1-t)(1-qt)} \quad (9)$$

where Frob stands for the geometric Frobenius $\text{Frob} \in \text{Gal}(\mathbb{F}/\mathbb{F}_q)$, acting on the ℓ -adic cohomology groups $H^*(X)$ by transport of structures (1.2). Put

$$f(t) := \det(1 - \text{Frob} \cdot t, H^1(X)).$$

It is a polynomial of degree $2g$ with integral coefficients. We will write α for a quantity running over the inverse roots of $f(t)$, counted with their multiplicities. One has

$$f(t) = \prod_{\alpha} (1 - \alpha t). \quad (10)$$

When $X_1/\mathbb{F}_q$ is replaced by $X_m/\mathbb{F}_{q^m}$, the corresponding polynomial f_m is

$$f_m(t) = \prod_{\alpha} (1 - \alpha^m t). \quad (11)$$

In the above, the relation between the variable t and z is $t = q^{-mz}$. The value of f_m at $t = 1$ is the number h_m of $\mathbb{F}_{q^m}$ -points of the jacobian $\text{Pic}^0(X_1)$ of X_1 :

$$f_m(1) = \prod_{\alpha} (1 - \alpha^m) = h_m := |\text{Pic}^0(X_1)(\mathbb{F}_{q^m})|. \quad (12)$$

We define

$$(n/S_1) := \text{the largest divisor of } n \text{ which is prime to all } \deg(s), \text{ for } s \text{ in } S_1. \quad (13)$$

$$T_1 := f(1) \cdot q^{n-1} \cdot \prod_{j=1}^{n-1} (1 - q^j)^{-2} \cdot f(q^j) \cdot \prod_{s \in S_1} (1 - q^{j \deg s})^{-1}. \quad (2.2.7)$$

$$T_m := T_1 \text{ for } (X_m, S_m) \text{ over } \mathbb{F}_{q^m}. \quad (2.2.8)$$

$$c_m := \#\{x \in \mathbb{F}_{q^m}^* \mid x \text{ generates } \mathbb{F}_{q^m} \text{ over } \mathbb{F}_q\}. \quad (2.2.9)$$

If m is prime to the degree of a closed point s_1 of X_1 , then $\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} k(s_1)$ is a field and

$$[\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} k(s_1) : \mathbb{F}_{q^m}] = [k(s_1) : \mathbb{F}_q].$$

In geometric terms: there is a unique point s_m of X_m above s_1 , and its degree, as a closed point of $X_m/\mathbb{F}_{q^m}$, coincides with $\deg(s_1)$. It follows that, when $m|(n/S_1)$, the étale divisor S_m maps bijectively to S_1 and if $s_m \in S_m$ has image $s_1 \in S_1$, the degree $[k(s_m) : \mathbb{F}_{q^m}]$ of the closed point s_m of X_m is equal to the degree $[k(s_1) : \mathbb{F}_q]$ of the closed point s_1 of X_1 : $N_m = N_1$, and T_m is given by the formula (2.2.7) with f replaced by f_m and q by q^m .

Theorem 3.1. *Suppose that $n \geq 2$ and $N_1 \geq 2$. One then has*

$$(-1)^{N_1(n-1)} T(X_1, S_1, n) = \sum_{m|(n/S_1)} c_m \cdot \frac{T_m}{m \cdot n/m}. \quad (2.3.1)$$

In section 4 (resp. 5), we will prove (2.3.1) under the assumption that n is odd or N_1 even (resp. that $N_1 \geq 3$ and that n or N_1 is odd).

4 Division algebras and Tamagawa numbers

We will no more use the field $\mathbb{F}_1 \otimes_{\mathbb{F}_q} \mathbb{F}$ previously noted $\mathbb{F}$. To lighten the notations we will simply write $\mathbb{F}$ for $\mathbb{F}_1$. We fix a finite set S_1 of places of $\mathbb{F}$.

4.1

The central simple algebras D over $\mathbb{F}$ of dimension n^2 are classified by their local invariants $\text{inv}(D_v) \in (\mathbb{Q}/\mathbb{Z})_n$, where D_v is the completion $D \otimes_{\mathbb{F}} \mathbb{F}_v$ of D at the place v of $\mathbb{F}$. The only constraints on the local invariants are that almost all are zero and that their sum is zero. The completion D_v is a matrix algebra (resp. a division algebra) if and only if $\text{inv}(D_v) = 0$ (resp. $\text{inv}(D_v)$ is of exact order n).

Proposition 4.1. *Assume that $|S_1| \geq 2$. Then, except in the case where n is even and $|S_1|$ is odd, there exists a D as in 3.1, such that D_v is a matrix algebra for $v \notin S_1$, and a division algebra for $v \in S_1$.*

Proof. For D as in 3.1, define $a_v = n \text{inv}(D_v) \in \mathbb{Z}/n$. The problem is to find a family of a_v , for $v \in S_1$, of exact order n and sum zero. If $|S_1|$ is even, take half of the a_v to be 1, and the other half to be -1 . If $|S_1| \geq 3$ is odd and n is also odd, take two a'_v 's to be $(n-1)/2$, one to be 1, and the others in $(1, -1)$ pairs. $\square$

4.2

If γ in D^* is of finite order, it generates over $\mathbb{F}_q \subset \mathbb{F}$ a finite extension, isomorphic to $\mathbb{F}_{q^m}$ for some m : γ is in the image of a morphism of $\mathbb{F}_q$ -algebras: $\mathbb{F}_{q^m} \rightarrow D$. Such morphisms correspond one-to-one to morphisms of $\mathbb{F}$ -algebras: $\mathbb{F}_m = \mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} \mathbb{F} \rightarrow D$. By a special case of the Skolem-Noether theorem, a proof of which is recalled at the end of the proof of 4.3, any two such morphisms are conjugate by some d in the multiplicative group D^* of D . With the notation (n/S_1) of (2.2.6), one has:

Proposition 4.2. *Let D be as in 4.1. There exists a morphism of $\mathbb{F}$ -algebras: $\mathbb{F}_m \rightarrow D$ if and only if m divides (n/S_1) .*

Before giving the proof of 4.2, we observe that 3.3 and 4.2 imply that there exists a morphism of $\mathbb{F}$ -algebras $\varphi : \mathbb{F}_{(n/S_1)} \rightarrow D$, inducing $\varphi_0 : \mathbb{F}_{q^{(n/S_1)}} \rightarrow D$, that any conjugacy class of elements of finite order of D^* meets $\varphi_0(\mathbb{F}_{q^{(n/S_1)}}^*)$, and that two elements of $\mathbb{F}_{q^{(n/S_1)}}^*$ have images in the same conjugacy class if and only if they are $\text{Gal}(\mathbb{F}_{q^{(n/S_1)}}/\mathbb{F}_q)$ -conjugate.

We will deduce 4.2 from the following well-known lemma:

Lemma 4.3. *Let $\mathbb{F}$ be a field, D a division algebra with center $\mathbb{F}$ of dimension n^2 over $\mathbb{F}$, and $\mathbb{F}_m$ a field extension of degree m of $\mathbb{F}$. The central simple algebra $D_m := \mathbb{F}_m \otimes_{\mathbb{F}} D$ over $\mathbb{F}_m$ is isomorphic to a matrix algebra $M_k(C)$ over a division algebra C with center $\mathbb{F}_m$. One has $k|m$ and the following conditions are equivalent:*

- (i) $\mathbb{F}_m$ can be embedded, as an $\mathbb{F}$ -algebra, in D ;
- (ii) $k = m$.

The proof will repeat that of the Skolem-Noether theorem.

Proof. Conjugacy classes of $\mathbb{F}$ -algebra embeddings $\mathbb{F}_m \rightarrow D$ correspond one-to-one to isomorphism classes of $(\mathbb{F}_m, D)$ -bimodules, of dimension one over D . By “bimodule” we mean a bimodule for which the two induced $\mathbb{F}$ -module structures are equal; this is the same as a right D^* -module. Indeed, if M is a $(\mathbb{F}_m, D)$ -bimodule, of dimension one over D , each $e \neq 0$ in M is a basis of M over D , the map $x \mapsto e^{-1}xe : \mathbb{F}_m \rightarrow D$ such that $xe = e(e^{-1}xe)$ is an embedding, and the $x \mapsto e^{-1}xe$ for $e \neq 0$ in M form a conjugacy class of embeddings $\mathbb{F}_m \rightarrow D$. If φ belong to that conjugacy class, corresponding to $e \in M$, M is isomorphic to D , with D acting by right multiplication and $x \in \mathbb{F}_m$ acting by left multiplication by $\varphi(x)$. The isomorphism is given by $1 \mapsto e$. The same construction shows that any embedding φ is obtained from some (M, e) : take $M = D$, $e = 1$ and actions of $\mathbb{F}_m$ and D as above.

If C is of dimension c^2 over $\mathbb{F}_m$, one has $n = kc$, and simple right D_m -modules are of dimension kc^2 over $\mathbb{F}_m$, hence mkc^2 over $\mathbb{F}$, and $mkc^2/n^2 = m/k$ over D . It follows that k divides m , and that the dimensions over D of $(\mathbb{F}_m, D)$ -bimodules are the multiples of m/k . The dimension one is possible if and only if $k = m$. If $k = m$, the $(\mathbb{F}_m, D)$ -bimodules of dimension one over D are the simple D_m -modules, they are all isomorphic, and this proves that the embeddings $\mathbb{F}_m \rightarrow D$ are all conjugate (Skolem-Noether theorem). $\square$

Proof of 4.2. In our case, 4.3 shows that $\mathbb{F}_m$ embeds in D if and only if m divides n and the local invariants of D_m are of order dividing n/m . If the place v of $\mathbb{F}$ is above the place s of $\mathbb{F}_1$, by extension of scalars the local invariant gets multiplied by $[\mathbb{F}_{m,v} : \mathbb{F}_s]$. This degree is divisible by m if and only if s is inert in $\mathbb{F}_m$. This is the case if and only if $\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} \mathbb{F}_s$, or equivalently $\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} k(s)$, is a field, i.e. when m is prime to the degree of $k(s)$ over $\mathbb{F}_q$. We need this to be the case at each s in S . $\square$

4.3 Orders

We fix D as in 4.1. By an “order of D ” we will mean an order containing the maximal order of $\mathbb{F}$. Orders cannot be defined, as in the number field case, as suitable subalgebras. Such a

description is available only on each affine chart of the projective curve X_1 . The constant sheaf D on X_1 is a quasi-coherent sheaf of algebras. An order of D (over X_1) is a coherent subsheaf of algebras, whose fiber at the generic point is D . Being contained in the constant sheaf D , an order $\mathcal{O}_D$ is torsion free, hence locally free as a sheaf of $\mathcal{O}$ -modules. It is of rank n^2 , and over some open subset of X_1 is a sheaf of Azumaya algebras.

Similarly, modules over an order $\mathcal{O}_D$ are sheaves of $\mathcal{O}_D$ -modules, quasi-coherent as sheaves of $\mathcal{O}$ -modules. We will denote by $\text{Mod}(\mathcal{O}_D)$ the category of right $\mathcal{O}_D$ -modules which are coherent as sheaves of $\mathcal{O}$ -modules, and $\text{Mod}^*(\mathcal{O}_D)$ the subcategory of those which are invertible, that is, locally free of rank one over $\mathcal{O}_D$. In parallel to the notations $\mathcal{O}_{(s)}, \mathcal{O}_s, \mathbb{F}_s$ of (0.2), we denote by $\mathcal{O}_{D,(s)}$ the stalk of $\mathcal{O}_D$ at the closed points of X_1 , $\widehat{\mathcal{O}}_{D,s} = \mathcal{O}_{D,(s)} \otimes_{\mathcal{O}_{(s)}} \mathcal{O}_s$ its completion, and $D_s = \widehat{\mathcal{O}}_{D,s} \otimes_{\mathcal{O}_s} \mathbb{F}_s$ the completion of D at s . Any other order $\mathcal{O}'_D$ coincides with $\mathcal{O}_D$ outside of a finite set of closed points, call it T , and is determined by the $\mathcal{O}'_{D,t} \subset D_t$ for $t \in T$. The $\mathcal{O}'_{D,t}$ are arbitrary orders of the D_t : $\mathcal{O}_t$ -subalgebras, generated as $\mathcal{O}_t$ -modules by a basis of D_t over $\mathbb{F}$.

If $v \notin S_1$, D_v is isomorphic to $\text{End}(V)$ for V a vector space of dimension n over $\mathbb{F}_v$. The maximal orders of $\text{End}(V)$ are the $\text{End}_{\mathcal{O}_v}(V_0)$ for V_0 an $\mathcal{O}_v$ -lattice in V : an $\mathcal{O}_v$ -submodule generated over $\mathcal{O}_v$ by a basis of V over $\mathbb{F}_v$. If $\mathcal{O}_D$ is a maximal order of D , $\mathcal{O}_{D,v}$ is Azumaya at v .

If $v \in S_1$, the division algebra D_v admits a valuation and its unique maximal order is its valuation ring. The order $\mathcal{O}_D$ is maximal if and only if, at each closed point v of X_1 , $\mathcal{O}_{D,v}$ is a maximal order in D_v .

4.4

As in 1.12, we denote by D^* the obvious affine algebraic group over $\mathbb{F}$ with group of rational points the multiplicative group of D . For any order $\mathcal{O}_D \subset D$ of D , the adelic group $D^*(\mathbb{A})$ ($\mathbb{A}$ the ring of adèles of $\mathbb{F}$) is the restricted product of the D_v^* , relative to the compact open subgroups $\mathcal{O}_{D,v}^*$. Its center is the group $\mathbb{A}^*$ of idèles of $\mathbb{F}$.

4.5 Tamagawa numbers

We will use that the Tamagawa number $\tau(D^*)$ is 1. As D^* is not semisimple, we use Ono's definition of Tamagawa numbers for reductive groups [O]. We need to explain how Ono's definition extends to the function field case.

Let G be a unimodular connected smooth linear algebraic group over $\mathbb{F}$. A translation invariant differential form of maximal degree on G , or equivalently, $\omega \in \bigwedge^{\max} \text{Lie}(G)^\vee$, defines at each place v of $\mathbb{F}$ a measure ω_v on $G(\mathbb{F}_v)$. One would like to define the Tamagawa measure on $G(\mathbb{A})$ to be the product measure $\omega = \prod_v \omega_v$, independent of $\omega \neq 0$ by the product formula, times $q^{(1-g)\dim G}$.

When G is semisimple, this makes sense: the product is absolutely convergent. When G is reductive and does not admit the multiplicative group $\mathbb{G}_m$ as a quotient, the product is conditionally convergent in the following sense: if $K = \prod_v K_v$ is a compact open subgroup of $G(\mathbb{A})$, the product of the $\omega_{K,v} := \omega_v(K_v)^{-1} \omega_v$ converges if one first groups together the factors for which v has a given degree: the product over $j \in \mathbb{Z}$ of

$$\prod_{\deg(v)=j} \omega_{K,v}$$

is absolutely convergent.

When G admits $\mathbb{G}_m$ as a quotient, the same product vanishes, essentially because $\prod_v (1 - q^{-\deg v})$ does. As it is the only case we will need, we will explain what is to be done under the additional assumption that G is given with $d : G \rightarrow \mathbb{G}_m$, an epimorphism whose kernel is the derived group of G . This applies to $\text{GL}(n)$ and to D^* , with $d = \det$.

For $t < 1/q$, the rational function $Z(X_1, t)$ (see 2.2) is the product over all places of the local factors $Z_v(t) = (1 - t^{\deg v})^{-1}$. At the simple poles $t = 1, 1/q$, a regularized value Z^* is defined to be the negative of the residue of $Z(X_1, t) \frac{dt}{t}$.

Under our assumptions Ono's Tamagawa measure on $G(\mathbb{A})$ is

$$\mu := Z^*(X_1, 1/q)^{-1} \cdot q^{(1-g) \dim G} \cdot \prod_v Z_v(1/q) \cdot \omega_v. \quad (3.8.1)$$

Let $G(\mathbb{A})^0$ be the kernel of the homomorphism $\deg \circ d : G(\mathbb{A}) \rightarrow \mathbb{Z}$, and $k\mathbb{Z}$ be its image. If $G(\mathbb{A})^{(i)}$ is the inverse image of i , the $G(\mathbb{F}) \backslash G(\mathbb{A})^{(i)}$ for $i \in k\mathbb{Z}$ have all the same volume. It follows that $\mu(G(\mathbb{F}) \backslash G(\mathbb{A}))$ is infinite. This is a counterpart to the divergence of $Z(X_1, t)$ at $t = 1/q$, which forced us to put $Z^*(X_1, 1/q)$ rather than $Z(X_1, 1/q)$ in (3.8.1). Under our assumptions Ono's Tamagawa number $\tau(G)$ of G is

$$\tau(G) := \mu(G(\mathbb{F}) \backslash G(\mathbb{A})^0) / k. \quad (14)$$

For $G = \mathrm{GL}(n)$ or D^* , $\deg \circ d$ is onto and we have simply $\tau(G) := \mu(G(\mathbb{F}) \backslash G(\mathbb{A})^0)$. Ono's general result is simply that

$$\tau(\mathrm{GL}(n)) = 1 \text{ and } \tau(D^*) = 1. \quad (3.8.3)$$

Remark 4.4. Ono considers the number field case. He uses the complex variable z (which we call z , see 2.2), and the morphism $\log \|d\|_k : G(\mathbb{A}) \rightarrow \mathbb{R}$ to define $\tau(G)$. His proofs work as well in the function field case. One should, however, be aware of two differences between his definition of $\tau(G)$ and the one we explained. They can cancel each other.

(a) Ono uses as regularized value $\zeta^*(X_1, 1)$ the value at $z = 1$ of $(z - 1)\zeta(X_1, z)$, equal to the residue at 1 of $\zeta(X_1, z) dz$. As $dt/t = -\log q dz$, one has

$$Z^*(X_1, 1/q) = \log q \cdot \zeta^*(X_1, 1).$$

In the case we are considering, the measure μ_O used by Ono in [O] is hence $\log q$ times the measure μ defined by (3.8.1).

(b) The right side of the formula for $\tau(G)$ should be viewed as the volume modulo $G(\mathbb{F})$ of the “complex” $G(\mathbb{A}) \rightarrow \mathbb{Z}$, for the measure μ on $G(\mathbb{A})$ and the counting measure on $\mathbb{Z}$. Ono uses the “complex” $\log \|d\|_k : G(\mathbb{A}) \rightarrow \mathbb{R}$, μ_O on $G(\mathbb{A})$ and dx on $\mathbb{R}$. In the number field case, this amounts to using the measure μ_O/dx on $G(\mathbb{A})^0$.

In the function field case, the two constructions are possible. They are related by the commutative diagram

$$\begin{array}{ccc} G(\mathbb{A}) & \xrightarrow{\deg \circ d} & \mathbb{Z} \\ -\log q \downarrow & & \downarrow \\ G(\mathbb{A}) & \xrightarrow{\log \|d\|_k} & \mathbb{R} \end{array}$$

When one uses the measure μ on $G(\mathbb{A})$, the counting measure on $\mathbb{Z}$, and the measure dx on $\mathbb{R}$, the volume modulo $G(\mathbb{F})$ of the second line is equal to the volume modulo $G(\mathbb{F})$ of the first line, divided by $\log q = \int_{\mathbb{R}/\log q \cdot \mathbb{Z}} dx$.

The following is well known.

Proposition 4.5. *For $G = \mathrm{GL}(n)$, the Tamagawa volume of $\mathrm{GL}(n, \mathcal{O}_v)$ is*

$$(Z^*(1) \cdot Z(q) \cdots Z(q^{n-1}))^{-1}.$$

Sketch of Proof. For the additive group $\mathbb{G}_a$, and $\omega = dx$, the volume of $\mathcal{O}_v$ for $|\omega|_v$, and the volume of $\mathcal{O}_v$ for $\|\omega\|_v$, are 1. The intersection $\mathbb{F} \cap \mathcal{O}_v$ is $\mathbb{F}_q$, $\mathbb{F} \backslash \mathbb{A} / \prod_v \mathcal{O}_v$ is $H^1(\mathcal{O})$ and the volume of $\mathbb{F} \backslash \mathbb{A} = \mathbb{G}_a(\mathbb{F}) \backslash \mathbb{G}_a(\mathbb{A})$ for $\|\omega\|$ is hence q^{g-1} . This, and the wished for multiplicativity

in G , are the reasons for the factor $q^{(1-g)\dim G}$ in the definition of the Tamagawa measure: it ensures that $\tau(\mathbb{G}_a) = 1$.

Let us temporarily forget that $\prod_v (1 - q^{-\deg v})$ diverges. The Tamagawa measure of $\mathrm{GL}(n, \mathcal{O}_v)$ would then be the measure of $\mathrm{GL}(n, \mathcal{O}_v)$, viewed as a subspace of $M_n(\mathcal{O}_v)$. The case of the additive group tells us that the Tamagawa volume of $M_n(\mathcal{O}_v)$ is $q^{(1-g)n^2}$. For $\mathrm{GL}(n, \mathcal{O}_v)$, the Tamagawa volume is then

$$q^{(1-g)n^2} \cdot \prod_v \frac{|\mathrm{GL}(n, k(v))|}{|M_n(k(v))|} = q^{(1-g)n^2} \cdot \prod_v ((1 - q_v^{-1}) \cdots (1 - q_v^{-n}))$$

where $q_v = |k(v)|$. The required regularization replaces $Z(q^{-1})$ by $Z^*(q^{-1})$. One then uses the functional equation

$$Z(t) = (qt^2)^{g-1} Z(1/qt).$$

As $t \mapsto 1/qt$ maps dt/t to $-dt/t$, the same holds up to sign for the regularized values at $t = 1, 1/q$. We get

$$\begin{aligned} Z^*(q^{-1}) Z(q^{-2}) \cdots Z(q^{-n}) &= (q^{-1} q^{-(2n-1)})^{g-1} Z^*(1) Z(q) \cdots Z(q^{n-1}) \\ &= q^{(1-g)n^2} Z^*(1) Z(q) \cdots Z(q^{n-1}) \end{aligned}$$

and the proposition. $\square$

One has $Z(t) = f(t)/(1-t)(1-qt)$ and $Z^*(1)$ is the value at 1 of $f(t)/(1-qt)$. One can hence rewrite the proposition as

$$\mu \left(\prod_v \mathrm{GL}(n, \mathcal{O}_v) \right) = \frac{1}{f(1)} \cdot \frac{1}{q^{n-1}} \cdot \prod_{j=1}^{n-1} (1 - q^j)^{-2} \cdot f(q^j). \quad (3.10.1)$$

The analogous question for D^* is reduced to the above by a standard argument:

Proposition 4.6. *Let $\mathcal{O}_D$ be a maximal order of D as in 4.1. Define $q_v = |k(v)| = q^{\deg(v)}$. Then, for $G = D^*$, the Tamagawa volume of $\prod_v \mathcal{O}_{D,v}^*$ is*

$$\mu \left(\prod_v \mathcal{O}_{D,v}^* \right) = \prod_{v \in S_1} \delta_v \cdot \mu \left(\prod_v \mathrm{GL}(n, \mathcal{O}_v) \right) \quad (3.11.1)$$

with

$$\delta_v = ((q_v - 1) \cdots (q_v^{n-1} - 1))^{-1}. \quad (3.11.2)$$

Plugging (3.10.1) in (3.11.1) and using that $(n-1)|S_1|$ is even, one can rewrite (3.11.1) as

$$\mu \left(\prod_v \mathcal{O}_{D,v}^* \right) = \left[f(1) \cdot q^{n-1} \cdot \prod_{j=1}^{n-1} (1 - q^j)^{-2} \cdot f(q^j) \cdot \prod_{s \in S_1} (1 - q^{j \deg(s)})^{-1} \right]^{-1}. \quad (3.11.3)$$

For $S = S_1$, (3.11.3) is the inverse of T_1 (2.2.7).

Sketch of Proof. The algebra D over $\mathbb{F}$ is obtained from the $n \times n$ matrix algebra M_n by twisting by a $\mathrm{PGL}(n)$ -torsor. The algebraic group D^* as well as $\mathrm{Lie}(D^*)^\vee$ are similarly obtained from $\mathrm{GL}(n)$ and $\mathrm{Lie}(\mathrm{GL}(n))^\vee$. As $\mathrm{PGL}(n)$ acts trivially on $\mathrm{Lie} \mathrm{GL}(n)^\vee$, $\bigwedge^{\max} \mathrm{Lie}(D^*)^\vee$ is canonically isomorphic to $\bigwedge^{\max} \mathrm{Lie}(\mathrm{GL}(n))^\vee$. This isomorphism is compatible with extensions of scalars. It induces a correspondence between the Haar measures on $\mathrm{GL}(n, \mathbb{F}_v)$ and D_v^* . Taking products, it induces also a correspondence between the Haar measures of $\mathrm{GL}(n, \mathbb{A})$ and $D^*(\mathbb{A})$. The

Tamagawa measures μ correspond to each other. If the μ_v are corresponding Haar measures for the local groups, this gives

$$\mu \left(\prod_v \mathcal{O}_{D,v}^* \right) / \mu \left(\prod_v \mathrm{GL}(n, \mathcal{O}_v) \right) = \prod_v \mu_v(\mathcal{O}_{D,v}^*) / \mu_v(\mathrm{GL}(n, \mathcal{O}_v)). \quad (3.11.4)$$

It suffices to extend the product on the right only over the set S of places where D is not isomorphic to $M_n(\mathbb{F}_v)$.

For $v \in S$, $\mathcal{O}_{D,v}$ is an Iwahori subgroup of D_v^* . Let T_v be the subalgebra of $M_n(\mathcal{O}_v)$ consisting of the matrices whose reduction in $M_n(k(v))$ is upper triangular. The multiplicative group $I_v := T_v^*$ is an Iwahori subgroup of $\mathrm{GL}(n, \mathbb{F}_v)$. As multiplicative groups of $\mathcal{O}_v$ -algebras which as $\mathcal{O}_v$ -modules are free of finite type, $\mathcal{O}_{D,v}$ and I_v are the groups of $\mathcal{O}_v$ -points of group schemes smooth over $\mathrm{Spec}(\mathcal{O}_v)$.

Suppose that G is a reductive group over $\mathbb{F}_v$, $\mathbb{F}_{v'}$ an unramified extension of $\mathbb{F}_v$ and $\mathcal{O}_{v'}$ its valuation ring. By Bruhat-Tits, an Iwahori subgroup of G , that is of $G(\mathbb{F}_v)$, is in a natural way the group of $\mathcal{O}_v$ -points of a smooth group scheme $\mathcal{I}$ over $\mathrm{Spec}(\mathcal{O}_v)$, with generic fiber G . Further, $\mathcal{I}(\mathcal{O}_{v'})$ is an Iwahori subgroup of $G' := G \otimes_{\mathbb{F}_v} \mathbb{F}_{v'}$ and $\mathcal{I} \otimes_{\mathcal{O}_v} \mathcal{O}_{v'}$ is the corresponding smooth group scheme over $\mathrm{Spec}(\mathcal{O}_{v'})$. We use here that, as the residue fields are finite, we are in the residually split case. This is in fact how Iwahori subgroups are constructed: one finds an Iwahori subgroup $\mathcal{I}'$ of G' fixed by $\mathrm{Gal}(\mathbb{F}_{v'}/\mathbb{F}_v)$, and one constructs $\mathcal{I}$ from $\mathcal{I}'$ by étale descent.

In our case, choose $\mathbb{F}_{v'}$ such that D_v and M_n become isomorphic after extension of scalars to $\mathbb{F}_{v'}$. For $\mathrm{GL}(n)$, an Iwahori subgroup $\mathcal{I}'$ determines the corresponding order $T_{v'}$ of M_n , of which it is the multiplicative group. Let $\mathcal{I}'$ be an Iwahori subgroup of $(D_v \otimes_{\mathbb{F}_v} \mathbb{F}_{v'})^* \simeq \mathrm{GL}(n, \mathbb{F}_{v'})$ and $T_{v'}$ be the order of $D_v \otimes_{\mathbb{F}_v} \mathbb{F}_{v'}$ of which $\mathcal{I}'$ is the multiplicative group. If $\mathcal{I}'$ is stable by $\mathrm{Gal}(\mathbb{F}_{v'}/\mathbb{F}_v)$, $T_{v'}$ descends to an order T_v of D_v . As $\mathcal{I}'$ is the Iwahori subgroup of D_v^* , T_v must be $\mathcal{O}_{D,v}$.

We conclude that $\mathcal{O}_{D,v} \otimes_{\mathbb{F}_v} \mathbb{F}_{v'} \subset D_v \otimes_{\mathbb{F}_v} \mathbb{F}_{v'} \simeq M_n(\mathbb{F}_{v'})$ is conjugate to $T_{v'}$ (we will also check this directly in 3.12), and that the Haar measures μ_v on D_v^* and $\mathrm{GL}(n, \mathbb{F}_v)$ defined by generators of $\bigwedge^{\max} \mathrm{Lie}(\mathcal{O}_{D,v})^*$ and $\bigwedge^{\max} \mathrm{Lie}(T_v)^*$ correspond to each other.

On $\mathcal{O}_{D,v}^*$, μ_v is induced by the Haar measure on D_v^* for which $\mathcal{O}_{D,v}$ has volume one. On $\mathrm{GL}(n, \mathcal{O}_v)$, μ_v is determined by the Haar measure on $M_n(\mathbb{F}_v)$ for which T_v has volume one. Using that $\mathcal{O}_{D,v}$ is a valuation ring with residue field $\mathbb{F}_{q_v^n}$, we get

$$\mu_v(\mathcal{O}_{D,v}^*) = 1 - q_v^{-n}.$$

The μ_v -volume of $\mathrm{GL}(n, \mathcal{O}_v)$ is

$$\begin{aligned} \frac{|\text{image of } \mathrm{GL}(n, \mathcal{O}_v) \text{ in } M_n(k(v))|}{|\text{image of } T_v \text{ in } M_n(k(v))|} &= q_v^{n^2} \cdot (1 - q_v^{-1}) \cdots (1 - q_v^{-n}) / q_v^{n(n+1)/2} \\ &= (q_v - 1) \cdots (q_v^{n-1} - 1) \cdot (1 - q_v^{-n}). \end{aligned}$$

This gives

$$\mu_v(\mathcal{O}_{D,v}^*) / \mu_v(\mathrm{GL}(n, \mathcal{O}_v)) = ((q_v - 1) \cdots (q_v^{n-1} - 1))^{-1}$$

and (3.11.1) follows from (3.11.4). $\square$

4.6

Let k be the residue field of $\mathbb{F}_v$, and choose an isomorphism of $\mathbb{F}_v$ with $k((t))$. Let k' be a degree n extension of k , and $\mathbb{F}_{v'} := k'((t))$. For some generator τ of $\mathrm{Gal}(\mathbb{F}_{v'}/\mathbb{F}_v) \simeq \mathrm{Gal}(k'/k) \simeq \mathbb{Z}/n$, the division algebra D_v admits the following model as a crossed product: one adds to $\mathbb{F}_v$ an element π such that $\pi^n = t$ and $\pi f = \tau(f)\pi$ for f in $\mathbb{F}_{v'}$. The elements of D_v can be written

as Laurent series $\sum_n a_n \pi^n$, with a_n in k' , and the product is such that $\pi a = \tau(a)\pi$ for a in $\mathbb{F}_{v'}$: D_v is a twisted Laurent formal power series field $k'((\pi))$. It admits the valuation

$$v : \sum a_n \pi^n \mapsto \inf\{n \mid a_n \neq 0\}.$$

The valuation ring is the twisted formal power series ring $k'[[\pi]]$.

The tensor product $\mathbb{F}_{v'} \otimes_{\mathbb{F}_v} \mathbb{F}_{v'}$ is a product of copies of $\mathbb{F}_{v'}$, indexed by $\text{Gal}(\mathbb{F}_{v'}/\mathbb{F}_v)$. The coordinates of the isomorphism are the

$$\text{pr}_\alpha : \mathbb{F}_{v'} \otimes_{\mathbb{F}_v} \mathbb{F}_{v'} \rightarrow \mathbb{F}_{v'} : x \otimes y \mapsto \alpha(x)y.$$

They are permuted by τ , acting on the first factor $\mathbb{F}_{v'}$: $\text{pr}_\alpha(\tau(x) \otimes y) = \text{pr}_{\alpha\tau}(x \otimes y)$. The algebra $D_{v'} := D_v \otimes_{\mathbb{F}_v} \mathbb{F}_{v'}$ is obtained by adding to this product of copies of $\mathbb{F}_{v'}$ an element π such that $\pi^n = t$ and $\pi(x_\alpha) = (x_{\alpha\tau})\pi$.

Let us identify $\text{Gal}(k'/k)$ with $\mathbb{Z}/n$ by using τ as a generator. One then defines an isomorphism of $D_{v'}$ with $M_n(\mathbb{F}_{v'})$ by mapping $\mathbb{F}_{v'} \otimes_{\mathbb{F}_v} \mathbb{F}_{v'} = \mathbb{F}_{v'}^{\mathbb{Z}/n}$ to the diagonal matrices, and π to the matrix 1 just above the diagonal, t on the lower left corner, and 0 elsewhere. With T_v as in 3.11, by this isomorphism, $\mathcal{O}_{D,v} \otimes k'[[t]]$ maps to $T_v \otimes k'[[t]]$.

5 Proof of 2.3: Masses of categories

In this section, we prove 2.3 under the assumption that there exists a division algebra D with center $\mathbb{F}$ of dimension n^2 over $\mathbb{F}$, such that D_v is a division algebra for $v \in S_1$, and a matrix algebra over $\mathbb{F}_v$ for $v \notin S_1$. As explained in 4.1, this amounts to assuming that n is odd, or that N_1 (assumed to be ≥ 2) is even. In both cases, $N_1(n-1)$ is even and the sign $(-1)^{N_1(n-1)}$ in (2.3.1) is $+1$.

5.1

As we recalled in 2.4 and 2.5, the number $T(X_1, S_1, n)$ we want to compute is also the number of classes modulo $\mathbb{F}_q$ -twists of automorphic representations π of $D^*(\mathbb{A})$ such that

- (i) π is unramified outside of S_1 ;
- (ii) for v in S_1 , the local component π_v is of the form $\chi(\det)$ for χ an unramified character of $\mathbb{F}_v^*$;
- (iii) π is not one-dimensional.

To count these representations up to $\mathbb{F}_q$ -twists, we will proceed in two steps. Let a be an idèle of $\mathbb{F}$ of positive degree: $\deg(a) > 0$. Identifying $\mathbb{A}^*$ with the center of $D^*(\mathbb{A})$, one has $\deg \det(a) = \deg(a^n) = n \deg(a)$. It follows that for χ a character of the quotient $\mathbb{Z}$ of $D^*(\mathbb{A})$, $\omega_{\pi\chi}(a) = \omega_\pi(a)\chi(n \deg a)$ and any π has an $\mathbb{F}_q$ -twist π' such that $\omega_{\pi'}(a) = 1$. This π' is not unique: one remains free to twist it by a character of $\mathbb{Z}/n \deg(a)$. We conclude that $T(X_1, S_1, n)$ is the number of automorphic representations of $D^*(\mathbb{A})$ obeying (i), (ii), (iii) and

- (iv) $\omega_\pi(a) = 1$,

taken modulo $\mathbb{F}_q$ -twists by a character of $\mathbb{Z}/n \deg a$. These twists being distinct, we have

Lemma 5.1. *The number $T(X_1, S_1, n)$ is $1/n \deg(a)$ times the number of automorphic representations of $D^*(\mathbb{A})$ obeying (i), (ii), (iii), (iv) of 4.1.*

In the rest of this section, we take a to be of degree one.

5.2

Let us fix a maximal order $\mathcal{O}_D \subset D$ of D (3.6). The space of locally constant functions on $D^* \backslash D^*(\mathbb{A})/a^{\mathbb{Z}}$ is the direct sum of the automorphic representations π for which $\omega_\pi(a) = 1$. This direct sum decomposition, being $D^*(\mathbb{A})$ -equivariant, is compatible with taking the invariants by $\mathcal{O}_{D,v}^*$. The π for which (i), (ii) hold are those for which the subspace of vectors fixed by $\mathcal{O}_{D,v}^*$ is nontrivial. This subspace is then of dimension one. It follows that the number of automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii) holds is the dimension of the space of functions on $D^* \backslash D^*(\mathbb{A})/\prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$:

Lemma 5.2. *The number of automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii), (iv) hold is the number of elements of the finite set*

$$D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}. \quad (4.4.1)$$

Lemma 5.3. *There are nh_1 one-dimensional automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii), (iv) hold.*

Proof. The map $\det$ maps $D^*(\mathbb{A})$ (resp. D^* , resp. $\mathcal{O}_{D,v}^*$) onto $\mathbb{A}^*$ (resp. $\mathbb{F}^*$, resp. $\mathcal{O}_v^*$). The representations considered may hence be identified with the unramified characters χ of the idèle class group such that $\chi(a^n) = 1$. As $\mathbb{F}^* \backslash \mathbb{A}^* / \prod_v \mathcal{O}_v^* \cdot a^{n\mathbb{Z}}$ is an extension of $\mathbb{Z}/n$ by $\text{Pic}^0(X_1)(\mathbb{F}_q)$, the lemma follows. $\square$

From the above lemmas, we get

Lemma 5.4. *We have*

$$T(X_1, S_1, n) + h_1 = \frac{1}{n} |D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}|.$$

5.3

Let φ_0 be the Haar measure with mass one of $\prod_v \mathcal{O}_{D,v}^*$, extended by zero to a measure on $D^*(\mathbb{A})$. Convolution with φ_0 is an idempotent projection from locally constant functions on $D^* \backslash D^*(\mathbb{A})/a^{\mathbb{Z}}$ to functions on $D^* \backslash D^*(\mathbb{A})/\prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$. Its trace is $|D^* \backslash D^*(\mathbb{A})/\prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}|$.

The trace formula (compact quotient case) expresses it as a sum over the conjugacy classes of γ in D^* contained in a $D^*(\mathbb{A})$ -conjugate of $\prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$.

Lemma 5.5. *Any γ in D^* contained in a $D^*(\mathbb{A})$ -conjugate of $\prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$ is of finite order.*

Proof. As $\deg \det(\gamma) = 0$, γ will be in a $D^*(\mathbb{A})$ -conjugate of $\prod_v \mathcal{O}_{D,v}^*$. For any d in $D^*(\mathbb{A})$, the intersection $D^* \cap d \prod_v \mathcal{O}_{D,v}^* d^{-1}$ is discrete and compact, hence finite. $\square$

With essentially no change in content, the trace formula computation can be expressed using masses of suitable categories (cf. 4.18). The direct use of the trace formula has the advantages of being more directly applicable in the number field case, and of clearly separating local and global questions. The use of masses is more geometric. As the equivalence between the two approaches seemed to us interesting, we will use masses. We first review their formalism.

5.4 Masses of categories

The mass of a category $\mathcal{C}$ is the sum, over the isomorphism classes of objects,

$$\text{mass}(\mathcal{C}) := \sum_X 1/|\text{Aut}(X)|, \quad (4.10.1)$$

of $\mathcal{C}$.

Let $\mathcal{C} \xrightarrow{\sim}$ be the subcategory with the same objects, and for which the morphisms are the isomorphisms in $\mathcal{C}$. The category $\mathcal{C} \xrightarrow{\sim}$ is a groupoid. It has the same mass as $\mathcal{C}$. A category and its opposite have the same mass. Two equivalent categories have the same mass. If $\mathcal{C}$ is a finite sum (resp. product) of categories $\mathcal{C}_i$, one has

$$\text{mass}(\coprod_i \mathcal{C}_i) = \sum_i \text{mass}(\mathcal{C}_i) \quad (4.10.2)$$

$$\text{mass}(\prod_i \mathcal{C}_i) = \prod_i \text{mass}(\mathcal{C}_i). \quad (4.10.3)$$

More generally, suppose each object X of $\mathcal{C}$ is given a weight $w(X)$, and that isomorphic objects have the same weight. The weighted mass of $(\mathcal{C}, w)$ is the sum, over isomorphism classes of objects,

$$\text{mass}(\mathcal{C}, w) = \sum_X w(X) / |\text{Aut}(X)|. \quad (4.10.4)$$

In our applications, the groups of automorphisms $\text{Aut}(X)$ will all be finite.

Suppose $T : \mathcal{C} \rightarrow \mathcal{D}$ is a functor. The fiber $\mathcal{C}_Y$ of $\mathcal{C}/\mathcal{D}$ at the object Y of $\mathcal{D}$ is the category of objects X of $\mathcal{C}$, given with an isomorphism $\alpha : T(X) \xrightarrow{\sim} Y$. A morphism from (X, α) to (X', α') is $u : X \rightarrow X'$ such that $\alpha' \circ T(u) = \alpha$. Maybe this should be called “homotopy fiber”. The naive definition of fiber: $(T^{-1}(Y), T^{-1}(\text{Id}_Y))$ is no good, as it is not compatible with equivalences. If, however, $\mathcal{C}$ is fibered over $\mathcal{D}$, the two definitions give equivalent categories. If Y and Y' are isomorphic, the fibers $\mathcal{C}_Y$ and $\mathcal{C}_{Y'}$ are equivalent, hence have the same mass.

Lemma 5.6. *Let $T : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. The mass of $\mathcal{C}$ is the weighted mass of $\mathcal{D}$, weighted by the mass of the fibers. Special case: if all fibers have the same mass, one has*

$$\text{mass}(\mathcal{C}) = \text{mass}(\mathcal{D}) \cdot \text{mass}(\text{any fiber}). \quad (4.11.1)$$

Proof. One reduces to the case where all morphisms in $\mathcal{C}$ and $\mathcal{D}$ are isomorphisms. Using (4.10.2) and its analogue for weighted masses, one may assume that $\mathcal{C}$ and $\mathcal{D}$ have only one isomorphism class of objects. Replacing $\mathcal{C}, \mathcal{D}$ by equivalent categories, one may assume that $\mathcal{C}$ (resp. $\mathcal{D}$) has only one object X (resp. Y). The functor T is then given by a morphism of groups T from $G = \text{Aut}(X)$ to $H = \text{Aut}(Y)$.

An object of the fiber $\mathcal{C}_Y$ is the data of $h \in H$: $T(X) = Y \xrightarrow{h} Y$. A morphism from (X, h) to (X, h') is g in G such that $h' \circ T(g) = h$. This identifies the set of isomorphism classes in $\mathcal{C}_Y$ with $H/T(G)$, and the group of automorphisms of any object with $\text{Ker}(T)$. Then (4.11.1) reduces to

$$1/|G| = 1/|H| \cdot (|H/T(G)| / |\text{Ker}(T)|). \quad \square$$

The same proof shows that if $\mathcal{C}$ is given with a weight, its weighted mass is the weighted mass of $\mathcal{D}$, for the weighted masses of the fibers.

5.5

Suppose a group Γ acts on a set E . The category $[\Gamma \backslash E]$ is defined as follows: the set of objects is E , a morphism from x to y is γ in Γ such that $y = \gamma x$, and the composition of morphisms is the product in Γ . Variant: suppose given a (left) action of Γ_1 , and a right action of Γ_2 , that is an action of the opposite group Γ_2^0 . If the actions commute, $\Gamma_1 \times \Gamma_2^0$ acts and one defines $[\Gamma_1 \backslash E / \Gamma_2] = [\Gamma_1 \times \Gamma_2^0 \backslash E]$. If Γ_2 acts freely, the natural functor $[\Gamma_1 \backslash E / \Gamma_2] \rightarrow [\Gamma_1 \backslash (E / \Gamma_2)]$ is an equivalence. More generally, if the normal subgroup Γ_0 of Γ acts freely on E , $[\Gamma \backslash E] \rightarrow [(\Gamma / \Gamma_0) \backslash (\Gamma_0 \backslash E)]$ is an equivalence.

The mass of $[\Gamma \backslash E]$ is the sum, over representatives of the orbits of Γ ,

$$\text{mass}([\Gamma \backslash E]) = \sum_{\text{orbits}} 1/|\text{Stabilizer}(e)|. \quad (4.12.1)$$

Lemma 5.7. *If Γ' is a subgroup of finite index of Γ , then*

$$\text{mass}([\Gamma' \backslash E]) = |\Gamma/\Gamma'| \cdot \text{mass}([\Gamma \backslash E]). \quad (4.13.1)$$

Proof. We apply (4.11.1) to the natural functor $[\Gamma' \backslash E] \rightarrow [\Gamma \backslash E]$. An object of the fiber at e is (e', γ) with $\gamma e' = e$. It is determined by γ , has no nontrivial automorphism, and (e', γ) is isomorphic to $(\delta^{-1}e, \delta)$ if and only if γ and δ have the same image in $\Gamma \backslash \Gamma'$. All fibers have hence mass $|\Gamma/\Gamma'|$. $\square$

Example 5.8. If E and Γ are finite,

$$\text{mass}([\Gamma \backslash E]) = |E/\Gamma|. \quad (4.14.1)$$

Indeed, take Γ' in the lemma to be trivial.

Example 5.9. Let a finite group Γ act on itself by conjugation. For this action, the stabilizer of γ in Γ is the centralizer $Z(\gamma)$ of γ . Applying (4.12.1) and (4.14.1), we get that the sum over representatives of conjugacy classes

$$\sum 1/|Z(\gamma)| = 1 \quad (4.15.1)$$

(sum over conjugacy classes).

The same holds for twisted conjugacy and twisted centralizer: if Γ_1 is an extension of $\mathbb{Z}$ by the finite group Γ , one applies (4.14.1) to the conjugation action of Γ on the inverse image of 1.

For $\mathcal{C}$ a category, let $\mathcal{C}^*$ be the category of objects X of $\mathcal{C}$, given with an automorphism α . A morphism from (X, α) to (X', α') is a morphism $f : X \rightarrow X'$ such that $\alpha' \circ f = f \circ \alpha$.

Proposition 5.10. *If the groups of automorphisms of objects of $\mathcal{C}$ are finite, the mass of the category $\mathcal{C}^*$ is the number of isomorphism classes of $\mathcal{C}$.*

Proof. As $(\mathcal{C}^*) \xrightarrow{\sim} (\mathcal{C} \xrightarrow{\sim})^*$, we may and shall assume that $\mathcal{C} = \mathcal{C} \xrightarrow{\sim}$. We apply 4.11 to the forgetful functor $(X, \alpha) \mapsto X$ from $\mathcal{C}^*$ to $\mathcal{C}$. The fiber at X is equivalent to the discrete category with set of objects $\text{Aut}(X)$. The mass of $\mathcal{C}^*$ is the weighted mass of $\mathcal{C}$, for the weight $|\text{Aut } X|$: the sum over isomorphism classes in $\mathcal{C}$ of $|\text{Aut}(X)|/|\text{Aut}(X)| = 1$. $\square$

If we take for $\mathcal{C}$ a category with one object whose group of automorphisms is Γ , and compute the mass of $\mathcal{C}^*$ by (4.10.1), we recover (4.15.1).

Example 5.11. For E and Γ as in 4.12, the objects of $[\Gamma \backslash E]^*$ are the pairs (e, γ) with $e \in E^\gamma$. The objects isomorphic to (e, γ) are the $(\delta e, \delta \gamma \delta^{-1})$. This shows that the conjugacy class of γ depends only on the isomorphism class of (e, γ) , and that if we fix a set R of representatives of the conjugacy classes, any object of $[\Gamma \backslash E]^*$ is isomorphic to a (e, γ) with γ in R . A morphism from (e, γ) to (e', γ) is δ in Γ such that $\delta e = e'$ and $\gamma \delta = \delta \gamma$.

The category $[\Gamma \backslash E]^*$ is hence equivalent to the disjoint sum, over representatives γ of the conjugacy classes, of the categories $[Z(\gamma) \backslash E^\gamma]$. We get

$$\text{mass}([\Gamma \backslash E]^*) = \sum_{\gamma} \text{mass}([Z(\gamma) \backslash E^\gamma]), \quad (4.17.1)$$

the sum being over a set of representatives of conjugacy classes of Γ . It suffices to consider those for which E^γ is not empty.

Example 5.12. Let G be a totally disconnected locally compact group, K an open compact subgroup, Γ a cocompact discrete subgroup and consider the action of Γ on $E = G/K$. The formula (4.17.1) expresses the number of double cosets of Γ, K in G as a sum over conjugacy classes in Γ . For γ in Γ , E^γ is the set of $g^{-1}K$ such that $\gamma g^{-1}K = g^{-1}K$, that is $\gamma \in g^{-1}Kg$. Let $Z_\Gamma(\gamma)$ (resp. $Z_G(\gamma)$) be the centralizer of γ in Γ (resp. G). Let us check that the term $\text{mass}([Z_\Gamma(\gamma) \backslash E^\gamma])$ in (4.17.1) is an orbital integral. Recall that orbital integrals associate a function on the set of conjugacy classes to a density. The density one takes here is the Haar measure of K giving it volume 1, extended by 0, that is $1_K dg$, for 1_K the characteristic function of K and dg the Haar measure of G giving K the volume 1:

$$\text{mass}([Z_\Gamma(\gamma) \backslash E^\gamma]) = \int_{Z_G(\gamma)/Z_\Gamma(\gamma)} 1_K(g\gamma g^{-1}) dg. \quad (4.18.1)$$

Indeed, the double coset $Z_\Gamma(\gamma)g^{-1}K$ (resp. $KgZ_\Gamma(\gamma)$) contributes to the mass (resp. integral) if and only if $g\gamma g^{-1} \in K$, in which case the contribution is $1/|gZ_\Gamma(\gamma)g^{-1} \cap K|$.

The formula (4.17.1) hence gives

$$|\Gamma \backslash G/K| = \sum_{\gamma} \int_{Z_G(\gamma)/Z_\Gamma(\gamma)} 1_K(g\gamma g^{-1}) dg \quad (4.18.2)$$

(sum over conjugacy classes in Γ). The trace formula (compact quotient case), after telling the same, introduces a Haar measure dz on the centralizer $Z_G(\gamma)$ of γ in G and rewrites each term as

$$\int_{Z_G(\gamma)/Z_\Gamma(\gamma)} dz \cdot \int_{Z_G/Z_G(\gamma)} 1_K(g\gamma g^{-1}) dg/dz.$$

5.6

The number of double cosets $D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$ appearing in 4.6 will now be computed using these methods. By 4.9, only the conjugacy classes of elements of finite order of D^* need to be considered. We recall that (n/S_1) is the largest divisor of n prime to the degrees $\deg(v)$ for v in S_1 . It will be convenient to choose an $\mathbb{F}_q$ -algebra embedding of $\mathbb{F}_{q^{(n/S_1)}}$ in D (4.2), and to choose the maximal order $\mathcal{O}_D \subset D$ to contain this $\mathbb{F}_{q^{(n/S_1)}}$. This is possible: as $\mathbb{F}_{q^{(n/S_1)}}$ is finite, it is contained in a maximal compact subgroup of $D^*(\mathbb{A})$, and these are the $\prod_v \mathcal{O}_{D,v}^*$ for $\mathcal{O}_D$ a maximal order.

Fix γ in $\mathbb{F}_{q^{(n/S_1)}}^*$. Let m be the degree $[\mathbb{F}_q(\gamma) : \mathbb{F}_q]$: the element γ is in the image of $\mathbb{F}_{q^m}$ and generates it over $\mathbb{F}_q$. The contribution of γ to the number of double cosets is

$$\text{mass}([Z(\gamma) \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}})^\gamma]), \quad (4.19.1)$$

where $Z(\gamma)$ is the centralizer of γ in D^* and $(\dots)^\gamma$ is the set of fixed points of γ . The centralizer $Z(\gamma)$ is the multiplicative group of D_γ , the subalgebra of D consisting of the elements which commute with γ or, what amounts to the same, with each element of $\mathbb{F}_q(\gamma) \subset \mathbb{F}_{q^{(n/S_1)}} \subset D$. As we used in the proof of 4.9, $(D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}})^\gamma = (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma / a^{\mathbb{Z}}$, and $(D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma$ consists of the cosets of d in $D^*(\mathbb{A})$ such that γ , or, what amounts to the same, $\mathbb{F}_{q^m}^*$, is contained in $d \prod_v \mathcal{O}_{D,v}^* d^{-1}$. Such a coset is fixed, not only by γ , but by all elements of $\mathbb{F}_{q^m}^*$. It follows that (4.19.1) depends only on m . As the orbits of $\text{Gal}(\mathbb{F}_{q^{(n/S_1)}}/\mathbb{F}_q)$ acting on $\mathbb{F}_{q^{(n/S_1)}}^*$ map bijectively onto the conjugacy classes of elements of finite order of D^* (4.2), the number of double cosets is

$$|D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}| = \sum_{m|(n/S_1)} c_m \cdot (\text{mass (4.19.1) for } \mathbb{F}_q(\gamma) = \mathbb{F}_{q^m}). \quad (4.19.2)$$

The category $\mathcal{C} = [D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*]$, the disjoint summand $[D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma]$ of $\mathcal{C}^*$, and their analogues when one divides also by $a^\mathbb{Z}$, have a concrete interpretation, which we now explain (4.20–4.23). Our proof of 2.3 will not rely on this interpretation.

Construction 5.13. The category $[D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*]$ is naturally equivalent to the category $\text{Mod}_{\sim}^*(\mathcal{O}_D)$.

Construction. To $x = (x_v)$ in $D^*(\mathbb{A})$ one attaches the coherent sub- $\mathcal{O}_D$ -module of D whose completion at v is $x_v \mathcal{O}_{D,v} \subset D_v$. Call it $x\mathcal{O}_D$. It depends only on the coset $x \cdot \prod_v \mathcal{O}_{D,v}^*$. For any invertible right $\mathcal{O}_D$ -module E , a trivialization of E at the generic point defines an isomorphism of it with some $x\mathcal{O}_D$. An $\mathcal{O}_D$ -module isomorphism from $x\mathcal{O}_D$ to $y\mathcal{O}_D$ has the form $f \mapsto df$, where d in D^* is such that $dx\mathcal{O}_{D,v} = y\mathcal{O}_{D,v}$. This defines the announced equivalence to $\text{Mod}_{\sim}^*(\mathcal{O}_D)$. $\square$

Define $\text{Mod}(\mathbb{F}_{q^m}, \mathcal{O}_D)$ to be the category of $(\mathbb{F}_{q^m}, \mathcal{O}_D)$ -bimodules, where the two implied $\mathbb{F}_q$ -module structures are assumed to agree. A decoration $*$ means that we consider only those which are invertible as $\mathcal{O}_D$ -modules.

Construction 5.14. The category $[D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma]$ is naturally equivalent to the category $\text{Mod}_{\sim}^*(\mathbb{F}_{q^m}, \mathcal{O}_D)$.

Construction. The coset $x \cdot \prod_v \mathcal{O}_{D,v}^*$ is fixed by γ if and only if $x\mathcal{O}_D \subset D$ is a left $\mathbb{F}_{q^m}$ -module. If the cosets of x and y are fixed by γ , an $\mathcal{O}_D$ -module isomorphism, defined as in 4.20 by d in D^* , respects the $\mathbb{F}_{q^m}$ -module structure if and only if d is in D_γ^* . This defines the functor, and proves it is fully faithful. To see that it is essentially surjective, one uses that an $\mathbb{F}_{q^m}$ -module structure on $x\mathcal{O}_D$ is given by an embedding $\mathbb{F}_{q^m} \rightarrow D$, and that two such embeddings are D^* -conjugate. $\square$

Variants 5.15. (i) Define A to be the divisor of the idèle a . The $\mathcal{O}_D$ -module $xa\mathcal{O}_D$ is naturally isomorphic to $x\mathcal{O}_D(-A) = x\mathcal{O}_D \otimes_{\mathcal{O}} \mathcal{O}(-A)$, and the $x\mathcal{O}_D(kA)$ for $k \in \mathbb{Z}$ are nonisomorphic: their degrees are distinct. It follows that $[D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^\mathbb{Z}]$ is naturally equivalent to the category $\text{Mod}_{\sim}^*(\mathcal{O}_D)/a$ of invertible right $\mathcal{O}_D$ -modules E , taken up to $E \mapsto E(kA)$. In this category, E' and E'' are isomorphic if and only if for some k, k' , the modules $E'(kA)$ and $E''(k'A)$ are isomorphic. In this case, (k, k') is unique up to $(k, k') \mapsto (k + c, k' + c)$, and $\text{Hom}_{/a}(E', E'')$ is any of the $\text{Isom}(E'((k + c)A), E''((k' + c)A))$, between which one has a transitive system of bijections.

(ii) Similarly, $[Z(\gamma) \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^* \cdot a^\mathbb{Z})^\gamma]$ is naturally equivalent to the category $\text{Mod}_{\sim}^*(\mathbb{F}_{q^m}, \mathcal{O}_D)/a$ of bimodules E in $\text{Mod}_{\sim}^*(\mathbb{F}_{q^m}, \mathcal{O}_D)$, taken up to $E \mapsto E(kA)$.

Remark 5.16. The decomposition $[D^* \backslash D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*]^* = \coprod_\gamma [D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma]$ becomes – via 4.20 and 4.21 – the fact that the algebra of endomorphisms of an invertible left $\mathcal{O}_D$ -module E is a finite field, and that the category of the (E, α) , α an automorphism of E which generates an extension of degree m of $\mathbb{F}_q$ and has a specified minimal polynomial over $\mathbb{F}_q$, is equivalent to the category $\text{Mod}_{\sim}^*(\mathbb{F}_{q^m}, \mathcal{O}_D)$ of $(\mathbb{F}_{q^m}, \mathcal{O}_D)$ -bimodules.

5.7

The tensor product $\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ is the product of copies of $\mathbb{F}_{q^m}$, indexed by $\text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$. The projections are the $x \otimes y \mapsto \tau(x)y : \mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m} \rightarrow \mathbb{F}_{q^m}$. An $\mathbb{F}_{q^m} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ -module structure on M hence gives a decomposition of M in submodules M_τ , which we will view as a grading, with group of degrees $\text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q) \simeq \mathbb{Z}/m$.

We can apply this to D , as well as to $\mathcal{O}_D$, on which $\mathbb{F}_{q^m}$ acts by left and right multiplications. On the τ -component, $\lambda d = d\tau(\lambda)$, for λ in $\mathbb{F}_{q^m}$. The degree zero component of D (resp. $\mathcal{O}_D$) is the commutant D_γ (resp. $\mathcal{O}_\gamma^D$) of $\mathbb{F}_{q^m}$. The embedding of $\mathbb{F}_{q^m}$ in D (resp. $\mathcal{O}_D$) extends to

an embedding of $\mathbb{F}_m = \mathbb{F} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ (resp. $\mathcal{O}_m = \mathcal{O} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$), and D_γ (resp. $\mathcal{O}_\gamma^D$) is also the commutant of $\mathbb{F}_m$ (resp. $\mathcal{O}_m$). The division algebra D_γ is of degree $(n/m)^2$ over its center $\mathbb{F}_m$.

So far, we have viewed $\mathcal{O}_m$ and $\mathcal{O}_\gamma^D$ as sheaves over X_1 . The sheaf $\mathcal{O}_m$ is the direct image, from X_m to X_1 , of the structural sheaf of X_m . As X_m is finite over X_1 , the direct image functor is an equivalence from coherent sheaves on X_m to coherent sheaves of $\mathcal{O}_m$ -modules on X_1 . We will tacitly use this equivalence to view $\mathcal{O}_\gamma^D$ as an order, over X_m , of the central simple algebra D_γ over $\mathbb{F}_m$.

To compute the category (4.19.1), we will first consider the fixed point set of γ , acting on $D^*(\mathbb{A})/\prod_v \mathcal{O}_{D,v}^*$. It is the restricted product of the fixed points of γ acting on the $D_v^*/\mathcal{O}_{D,v}^*$.

Proposition 5.17. *If $v \notin S_1$, $\mathcal{O}_{\gamma,D,v}^*/\mathcal{O}_{\gamma,D,v}^*$ is a maximal compact subgroup in $D_{\gamma,v}^*$ and*

$$D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^* \xrightarrow{\sim} (D_v^*/\mathcal{O}_{D,v}^*)^\gamma.$$

Proof. Let V be a vector space of dimension n over $\mathbb{F}_v$. As $v \notin S_1$, there exists an isomorphism $D_v \xrightarrow{\sim} \text{End}(V)$. We choose one, and identify D_v with $\text{End}(V)$. The maximal order $\mathcal{O}_{D,v} \subset D_v$ is $\text{End}(E_0)$, for E_0 a lattice in V : a free $\mathcal{O}_v$ -module such that $\mathbb{F}_v \otimes_{\mathcal{O}_v} E_0 \xrightarrow{\sim} V$. The completion $\mathbb{F}_{m,v} = \mathbb{F}_v \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ is a product of local fields $\mathbb{F}(i)$: the product of the completions of $\mathbb{F}_m$ at the points of X_m above v . The completion $\mathcal{O}_{m,v} = \mathcal{O}_v \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ is the product of their valuation rings $\mathcal{O}(i)$.

By assumption, $\mathbb{F}_{m,v}$ embeds in $\text{End}(V)$, turning V into an $\mathbb{F}_{m,v}$ -module, i.e. into a product of vector spaces $V(i)$ over the $\mathbb{F}(i)$, while $\mathcal{O}_{m,v}$ embeds into $\text{End}(E_0)$, turning E_0 into a product of lattices $E_0(i)$ in $V(i)$.

The map $d \mapsto dE_0$ induces a bijection from $D_v^*/\mathcal{O}_{D,v}^*$ to the set of lattices in V . Indeed, any lattice E_1 , being a free $\mathcal{O}_v$ -module, is the image of E_0 by some element of $\text{GL}(V)$. The fixed points by γ are the lattices which are $\mathcal{O}_{m,v}$ -modules, that is product of lattices $E(i)$ in $V(i)$. They are the images of E_0 by some d in $\prod_i \text{GL}_{\mathbb{F}(i)}(V(i)) = D_{\gamma,v}^*$, with d unique up to $\prod_i \text{GL}_{\mathcal{O}(i)}(E_0(i)) = \mathcal{O}_{\gamma,D,v}^*$. The claim follows. $\square$

5.8

For v in S_1 , D_v is a division algebra with center $\mathbb{F}_v$ and $\mathcal{O}_{D,v}$ is its valuation ring. The map $\det : D_v^* \rightarrow \mathbb{F}_v^*$ is onto, and induces a bijection from $D_v^*/\mathcal{O}_{D,v}^*$, the group of the valuation of D_v , to the group $\mathbb{Z}$ of the valuation of $\mathbb{F}_v$. The action of d in D_v^* is by “adding the valuation of d at v ”. Special case: γ , being of finite order, acts trivially.

Recall that m is prime to $\deg(v)$, and that the field $(\mathbb{F}_m)_v = \mathbb{F}_v \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ is contained in D_v , with commutant D_γ . The reduced norms for D_γ , with center $\mathbb{F}_{m,v}$, and for D_v , with center $\mathbb{F}_v$, are related by the commutative diagram

$$\begin{array}{ccc} D_{\gamma,v}^* & \longrightarrow & \mathbb{F}_{m,v}^* \\ \downarrow & & \downarrow \text{Norm} \\ D_v^* & \longrightarrow & \mathbb{F}_v^* \end{array} \quad (15)$$

Identifying the groups of the valuations of $D_{\gamma,v}$ and D_v with $\mathbb{Z}$, a quotient of the diagram is

$$\begin{array}{ccc} D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^* & \xrightarrow{\sim} & \mathbb{Z} \\ \downarrow & & \downarrow m \\ D_v^*/\mathcal{O}_{D,v}^* & \xrightarrow{\sim} & \mathbb{Z}. \end{array}$$

We conclude that

$$(D_v^*/\mathcal{O}_{D,v}^*)^\gamma = D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^* = (D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*). \quad (4.26.2)$$

5.9

From (4.26.2) we get a decomposition of the category

$$[D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma]$$

as the disjoint sum of m^{N_1} subcategories, indexed by $(\mathbb{Z}/m)^{S_1}$. By 4.25, the subcategory with index 0 is

$$[D_\gamma^* \backslash D_\gamma^*(\mathbb{A}) / \prod_v \mathcal{O}_{\gamma,D,v}^*].$$

The others are equivalent to it: if π_v is of valuation one in D_v ($v \in S_1$), the equivalences are given by right multiplication by the $\prod_v \pi_v^{a_v}$, $0 \leq a_v < m$.

Similarly,

$$[D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_v \mathcal{O}_{D,v}^*)^\gamma / a^\mathbb{Z}]$$

is the disjoint sum of m^{N_1} subcategories equivalent to

$$[D_\gamma^* \backslash D_\gamma^*(\mathbb{A}) / \prod_v \mathcal{O}_{\gamma,D,v}^* \cdot a^\mathbb{Z}]. \quad (4.27.1)$$

In (4.27.1), D_γ is a central simple division algebra of dimension $(n/m)^2$ over $\mathbb{F}_m$. The adelization $D_\gamma(\mathbb{A}) = D_\gamma \otimes_{\mathbb{F}_m} \mathbb{A}$ ($\mathbb{A}$ adèles of $\mathbb{F}$) can be viewed also as the adelization of D_γ , viewed as an $\mathbb{F}_m$ -algebra, that is $D_\gamma \otimes_{\mathbb{F}_m} \mathbb{A}_m$ for $\mathbb{A}_m$ the adèles of $\mathbb{F}_m$, while $\prod_v \mathcal{O}_{\gamma,D,v}^*$ can be viewed as the product of the completions of a maximal order of D_γ over X_m . As an adèle of $\mathbb{F}_m$, a is still of degree one, and the corresponding divisor on $X_m/\mathbb{F}_{q^m}$ is of degree one. One has $S_m \xrightarrow{\sim} S_1$, D_γ is a division algebra at each v in S_m , and a matrix algebra elsewhere. The same assumptions as those we started with, but over X_m , and a dimension $(n/m)^2$ for D_γ .

5.10

We now work over X_m . The mass of the category (4.27.1) is the volume of the double coset $D_\gamma^* \backslash D_\gamma^*(\mathbb{A}) / a^\mathbb{Z}$ for the Haar measure on $D_\gamma^*(\mathbb{A})$ which gives $\prod_v \mathcal{O}_{\gamma,D,v}^*$ the volume one. The product is over the places of X_m , and by 4.25 and 4.26 $\mathcal{O}_{\gamma,D,v}$ is a maximal order in D_γ . Let $D_\gamma^*(\mathbb{A})^0$ be the subgroup of d in $D_\gamma^*(\mathbb{A})$ on which the degree of $\det(d)$ is zero, and $D_\gamma^*(\mathbb{A})^{(i)}$ the coset on which it is i . The $D_\gamma^* \backslash D_\gamma^*(\mathbb{A})^{(i)}$ have all the same volume, and their disjoint sum, for $0 \leq i < n/m$, maps bijectively onto $D_\gamma^* \backslash D_\gamma^*(\mathbb{A}) / a^\mathbb{Z}$. If μ is the Tamagawa measure on $D_\gamma^*(\mathbb{A})$, the Tamagawa number $\tau(D_\gamma^*) = \mu(D_\gamma^* \backslash D_\gamma^*(\mathbb{A})^0)$ is 1 and we conclude that

$$\text{mass}(4.27.1) = \frac{n}{m} \cdot \mu \left(\prod_v \mathcal{O}_{\gamma,D,v}^* \right)^{-1}. \quad (4.28.1)$$

Lemma 4.6 and (4.19.2) give

Lemma 5.18. *We have*

$$T(X_1, S_1, n) + h_1 = \sum_{m|(n/S_1)} c_m \cdot \frac{1}{m} \cdot m^{N_1} \cdot \mu \left(\prod_v \mathcal{O}_{\gamma,D,v}^* \right)^{-1}$$

where γ is such that $\mathbb{F}_q(\gamma) = \mathbb{F}_{q^m}$, μ is the Tamagawa measure for D_γ^* , viewed as a reductive group over $\mathbb{F}_m$, and the product is over the places v of $\mathbb{F}_m$.

Proof of 2.3 when $N_1(n-1)$ is even and $N_1 \geq 2$. As $\mathcal{O}_\gamma^D$ is a maximal order of D_γ over X_m , the Tamagawa volumes in 4.29 have been computed in (3.11.1). Applying (3.11.3) to the $(X_m, S_m, D_\gamma, \mathcal{O}_\gamma^D)$, one obtains 2.3. $\square$

6 Proof of 2.3: Using the building

In this section, we prove 2.3 under the assumption that for some w in S_1 , there exists a division algebra D with center $\mathbb{F}$ of dimension n^2 over $\mathbb{F}$ such that D_v is a division algebra for $v \in S_w := S_1 - \{w\}$, and a matrix algebra over $\mathbb{F}_v$ for $v \notin S_w$. As explained in 4.1, this amounts to assuming that $N_1 \geq 3$, and that n or N_1 is odd; the place w can then be chosen freely in S_1 . The integers $N_1(n-1)$ and $n-1$ have the same parity, and the sign $(-1)^{N_1(n-1)}$ in (2.3.1) is $(-1)^{n-1}$.

The constructions made in section 4 apply to D and S_w . With the notation of 2.2, one has that (n/S_1) divides (n/S_w) which divides n . As in 4.19, we choose an $\mathbb{F}_q$ -algebra embedding of $\mathbb{F}_{q^{(n/S_w)}}$ in D , and a maximal order $\mathcal{O}_D$ of D containing this $\mathbb{F}_{q^{(n/S_w)}}$.

6.1

As recalled in 2.4 and 2.5, the number $T(X_1, S_1, n)$ we want to compute is equal to the number of classes modulo $\mathbb{F}_q$ -twists of automorphic representations π of $D^*(\mathbb{A})$ such that

- (i) π is unramified outside of S_1 ;
- (ii) for v in S_w , the local component π_v is of the form $\chi(\det)$ for χ an unramified character of $\mathbb{F}_v^*$;
- (iii) the local component of π at w is of the form Steinberg representation $\otimes \chi(\det)$ for χ an unramified character of $\mathbb{F}_w^*$.

Let a be an idèle of degree > 0 . Let (iv) be the condition

- (iv) The central character ω_π of π is trivial on a .

The same proof as in 4.2 gives

Lemma 6.1. *The number $T(X_1, S_1, n)$ is $1/n \deg(a)$ times the number of automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii), (iii), (iv) hold.*

6.2

Let (iii)' be the condition

- (iii)' the local component of π at w is of the form $\chi(\det)$, for χ an unramified character of $\mathbb{F}_w^*$.

The algebraic subgroup $SD^* := \text{Ker}(\det : D^* \rightarrow \mathbb{G}_m)$ of D^* , being a form of $\text{SL}(n)$, is simply connected. By the strong approximation theorem applied to SD^* , an automorphic representation π of $D^*(\mathbb{A})$ for which (iii)' holds factors through $\det : D^*(\mathbb{A}) \rightarrow \mathbb{A}^*$. Indeed, for any automorphic function f in π , and any d in $D^*(\mathbb{A})$, f is constant on $SD^*(\mathbb{F}) \cdot d \cdot SD^*(\mathbb{F}_w) = SD^*(\mathbb{F}) \cdot SD^*(\mathbb{F}_w) \cdot d$, which is dense in $SD^*(\mathbb{A})d$. As in 4.5, if π satisfies (i), (ii), (iii)', (iv), it is of the form $\chi(\det)$, for χ an unramified character of the idèle class group of $\mathbb{F}$, trivial on a^n . There are $n \deg(a) h_1$ such characters.

6.3

The space of locally constant functions on $D^* \backslash D^*(\mathbb{A}) / a^{\mathbb{Z}}$ is the direct sum of the automorphic representations π of $D^*(\mathbb{A})$ for which (iv) holds. Those for which (i), (ii) hold are those whose local component π_v have, for $v \neq w$, a nontrivial subspace of vectors fixed by $\mathcal{O}_{D,v}^*$, and this subspace is then of dimension 1. The space L of locally constant functions on $D^* \backslash D^*(\mathbb{A}) / \prod_{v \neq w} \mathcal{O}_{D,v}^* \cdot a^{\mathbb{Z}}$ is hence isomorphic, as a representation of D_w^* , to the direct sum of the components π_w of the automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii), (iv) hold. We get:

Lemma 6.2. *The number of automorphic representations of $D^*(\mathbb{A})$ for which (i), (ii), (iii), (iv) hold is the sum, over the unramified characters χ of $\mathbb{F}_w^*$ for which $\chi^n(a) = 1$,*

$$T(X_1, S_1, n) = \frac{1}{n \deg(a)} \sum_{\chi} [L : \text{Steinberg} \otimes \chi(\det)]. \quad (5.5.1)$$

6.4

Let E be a nonarchimedean local field, $\text{val} : E^* \rightarrow \mathbb{Z}$ its valuation, $\mathcal{O}$ its valuation ring and t a uniformizing parameter. The case we need is $E = \mathbb{F}_w$. Define $\text{GL}(n, E)^0$ by the short exact sequence

$$1 \rightarrow \text{GL}(n, E)^0 \rightarrow \text{GL}(n, E) \xrightarrow{\text{val det}} \mathbb{Z} \rightarrow 1. \quad (5.6.1)$$

Define $\text{GL}(n, E)^{(i)}$ to be the coset of $\text{GL}(n, E)^0$ on which $\text{val}(\det)$ is i .

Dividing by the center E^* of $\text{GL}(n, E)$, its intersection $\mathcal{O}^*$ with $\text{GL}(n, E)^0$ and its image $n\mathbb{Z}$ in $\mathbb{Z}$, we obtain from (5.6.1) an exact sequence

$$1 \rightarrow \text{PGL}(n, E)^0 \rightarrow \text{PGL}(n, E) \xrightarrow{\text{val det}} \mathbb{Z}/n \rightarrow 1. \quad (5.6.2)$$

The group $\text{GL}(n, E)$ acts by conjugation on $\text{SL}(n, E)$, and hence on its building. This action factors through $\text{PGL}(n, E)$.

The vertices of the building $\mathcal{B}$ of $\text{SL}(n, E)$ are the lattices ($= \mathcal{O}$ -submodules, free of rank n) Λ in E^n , taken up to dilations $\Lambda \mapsto t^i \Lambda$. If $e_1, \dots, e_n$ is a basis of Λ , the valuation of $e_1 \wedge \dots \wedge e_n \in \bigwedge^n E^n \simeq E$ depends only on Λ . Its class in $\mathbb{Z}/n$ is the same for Λ and $t^i \Lambda$. We call it the type of the corresponding vertex. The group $\text{SL}(n, E)$ acts transitively on the vertices of each type. The group $\text{GL}(n, E)$ permutes the types: g in $\text{GL}(n, E)$ maps the vertex $\{t^i \Lambda\}$ to the vertex $\{t^i g \Lambda\}$, of type that of $\{t^i \Lambda\}$ plus $\text{val}(\det g)$.

The types of vertices (of the building) are the vertices of the affine Dynkin diagram of $\text{SL}(n, E)$. The subgroup $\text{PGL}(n, E)^0$ of $\text{PGL}(n, E)$ is the subgroup acting trivially on this Dynkin diagram. By [BT, I, 1.2.13–1.2.17], applied to the natural homomorphism $\varphi : \text{SL}(n, E) \rightarrow \text{PGL}(n, E)^0$ or to $\text{SL}(n, E) \rightarrow \text{GL}(n, E)^0$, the building of $\text{SL}(n, E)$ is also the building of a Tits system of $\text{PGL}(n, E)^0$, as well as of $\text{GL}(n, E)^0$.

A chamber of the building is spanned by vertices represented by lattices Λ_i ($0 \leq i \leq n-1$) forming a cyclic chain of distinct lattices

$$\Lambda_0 \subset \Lambda_1 \subset \dots \subset \Lambda_{n-1} \subset t^{-1} \Lambda_0.$$

If $(e_i)_{1 \leq i \leq n}$ is the standard basis of E^n , the fundamental chamber C is the chamber obtained when Λ_i is spanned by $t^{-1}e_1, \dots, t^{-1}e_i, e_{i+1}, \dots, e_n$.

The type of a face ($=$ simplex) of $\mathcal{B}$ is the set of types of its vertices. By the properties of Tits systems, any face of $\mathcal{B}$ is $\text{GL}(n, E)^0$ -conjugate to the unique face of C of the same type, and the stabilizer in $\text{GL}(n, E)^0$ of a face (a parahoric subgroup) fixes it. We will orient each face by the ordering of its vertices induced by the ordering $0 < 1 < \dots < n-1$ of the types of vertices. If σ is the face of type $S \subset \mathbb{Z}/n$ of C and $K_\sigma \subset \text{GL}(n, E)^0$ its stabilizer, the set of (oriented) faces of $\mathcal{B}$ of type S is the orbit $\text{GL}(n, E)^0/K_\sigma$.

For any i , one still has a bijection

$$\text{GL}(n, E)^{(i)}/K_\sigma \xrightarrow{\sim} \text{set of faces of } \mathcal{B} \text{ of type } S+i. \quad (5.6.3)$$

The component $C_d(\mathcal{B})$ of the chain complex $C_*(\mathcal{B})$ of oriented chains, computing the homology of $\mathcal{B}$, is

$$C_d(\mathcal{B}) = \bigoplus_{\dim \sigma = d} \mathbb{C}(\text{GL}(n, E)^0/K_\sigma). \quad (5.6.4)$$

The group $\text{GL}(n, E)^0$ acts on this complex. As $\mathcal{B}$ is contractible, it is a resolution of the trivial representation 1. We will not use the fact that $\text{GL}(n, E)$ acts too. Its action introduces signs, because $\text{GL}(n, E)$ permutes types and does not respect the orientations of faces which we used.

6.5

By representation we will always mean C^∞ representation: the stabilizers of vectors are open. The cohomology used below is the analogue, in that setting, of continuous cohomology, as in Casselman [C]. To apply 5.5, we need to detect, in a unitary representation of $\mathrm{GL}(n, E)$, the occurrences of the representations $\mathrm{Steinberg} \otimes \chi(\det)$, for χ a unitary unramified character of E^* . The following lemma is a corollary of [C]. (We write unitary for unitarizable.)

Lemma 6.3. *Put $H_*(\pi)$ for $H_*(\mathrm{GL}(n, E)^0, \pi)$. The irreducible unitary representations π of $\mathrm{GL}(n, E)$ with $H_*(\pi) \neq 0$ are the representations $\mathrm{Steinberg} \otimes \chi(\det)$ and $\chi(\det)$, for χ a unitary unramified character of E^* . The nonzero H_i are a one-dimensional H_{n-1} in the Steinberg case, a one-dimensional H_0 in the other.*

Proof. If the cohomology is not zero, the center $\mathcal{O}^*$ of $\mathrm{GL}(n, E)^0$ must act trivially. Twisting by a $\chi(\det)$, which is trivial on $\mathrm{GL}(n, E)^0$, has no effect on the cohomology. Replacing π by a twist, we are reduced to considering only representations with trivial central character, i.e. of $\mathrm{PGL}(n, E)$. As $\mathcal{O}^*$ is compact,

$$H_*(\mathrm{PGL}(n, E)^0, \pi) \xrightarrow{\sim} H_*(\mathrm{GL}(n, E)^0, \pi).$$

On the left, we have the cohomology of π , restricted to $\mathrm{PGL}(n, E)^0$. As $\mathrm{PGL}(n, E)^0$ is of finite index in $\mathrm{PGL}(n, E)$, it is the same as the cohomology of $\mathrm{PGL}(n, E)$, with coefficient in the induced (= coinduced) of this restriction to $\mathrm{PGL}(n, E)$. This is the direct sum of the twists $\pi \otimes \chi(\mathrm{val}(\det))$, for χ a character of $\mathbb{Z}/n$. By Casselman, $H_* \neq 0$ is nonzero if and only if one of these twists is Steinberg or trivial, and such a twist contributes respectively to a one-dimensional H_{n-1} or H_0 . As the twists are nonisomorphic, the lemma follows. $\square$

6.6

For L a representation of $\mathrm{GL}(n, E)$, to compute

$$H_*(\mathrm{GL}(n, E)^0, L) = \mathrm{Ext}_{\mathrm{GL}(n, E)^0}^*(1, L)$$

one can use the resolution (5.6.4) of the trivial representation 1 of $\mathrm{GL}(n, E)^0$. It gives us a complex $C_*(L)$ computing the cohomology, with components the

$$C_d(L) = \bigoplus_{\dim \sigma = d} L^{K_\sigma} \quad (5.9.1)$$

(sum over faces σ of the fundamental chamber C).

In particular, if the L^{K_σ} are finite dimensional, we have

$$\sum (-1)^i \dim H_i(\mathrm{GL}(n, E)^0, L) = \sum_{\sigma} (-1)^{\dim \sigma} \dim L^{K_\sigma}.$$

Moreover, if L is the direct sum of irreducible unitary representations of $\mathrm{GL}(n, E)$, one has

$$\sum_{\sigma} (-1)^{\dim \sigma} \dim L^{K_\sigma} = \sum_{\chi} [L : \chi(\det)] + (-1)^{n-1} \sum_{\chi} [L : \mathrm{Steinberg} \otimes \chi] \quad (5.9.2)$$

with χ running over the unitary unramified characters of E^* .

6.7

We now take $E = \mathbb{F}_w$, choose an isomorphism of D_w with the matrix algebra $M_n(\mathbb{F}_w)$, and identify D_w^* with $\mathrm{GL}(n, \mathbb{F}_w)$. For L the representation 5.4 of D_w^* can be identified with the space of functions on the finite set $D^* \backslash D^*(\mathbb{A}) / \prod_{v \neq w} \mathcal{O}_{D,v}^* \cdot K_\sigma \cdot a^\mathbb{Z}$.

By 5.3 and 5.4, the sum of the one-dimensional subrepresentations of L which are isomorphic to a $\chi(\det)$ with χ unramified, is of dimension $n \deg(a) h_1$. By (5.5.1) and (5.9.2), one has

Lemma 6.4. *With the notations of 5.10, we have*

$$h_1 + (-1)^{n-1} T(X_1, S_1, n) = \frac{1}{n \deg(a)} \sum_{\sigma} (-1)^{\dim \sigma} |D^* \backslash D^*(\mathbb{A}) / \prod_{v \neq w} \mathcal{O}_{D,v}^* \cdot K_\sigma \cdot a^\mathbb{Z}|, \quad (5.11.1)$$

the sum being over the faces of the fundamental chamber of the building of $\mathrm{SL}(n, \mathbb{F}_w)$.

The trace formula (compact quotient case) expresses the right side of (5.11.1) in terms of alternating sums ($\sum (-1)^{\dim \sigma} \dots$) of orbital integrals (cf. 4.18). Similar sums appear in Kottwitz [K], and we might have quoted the computations of [K], except that they are written for the characteristic zero case, and mainly for semisimple simply connected groups. We prefer to use the methods of [K], restricted to the case we need, couched in the language of masses.

6.8

We use the arguments of 4.16–4.18 to express the numbers of double cosets appearing in (5.11.1) as sums of masses of categories. Each sum is over conjugacy classes in D^* . As in 4.19, only the conjugacy classes of elements of finite order need to be considered. They have representatives in our chosen $\mathbb{F}_{q(n/S_w)} \subset D$, and this representative is unique up to the $\mathrm{Gal}(\mathbb{F}_{q(n/S_w)}/\mathbb{F}_q)$ action. Fix γ in $\mathbb{F}_{q(n/S_w)}^*$. Define $m := [\mathbb{F}_q(\gamma) : \mathbb{F}_q]$. As in 4.19 and 4.24, the commutant D_γ of γ in D is also the commutant of $\mathbb{F}_{q^m} \subset \mathbb{F}_{q(n/S_w)}$, as well as of $\mathbb{F}_m$. The contribution of the conjugation class of γ to the right side of (5.11.1) is

$$\frac{1}{n \deg(a)} \sum_{\sigma} (-1)^{\dim \sigma} \mathrm{mass}([D_\gamma^* \backslash (D^*(\mathbb{A}) / \prod_{v \neq w} \mathcal{O}_{D,v}^* \cdot K_\sigma \cdot a^\mathbb{Z})^\gamma]). \quad (5.12.1)$$

It depends only on m , not on the chosen γ . Indeed, K_σ is the multiplicative group of an order in $\mathcal{O}_{D,w}$, and one argues as in 4.19. The right side of (5.11.1) becomes

$$\sum_{m|(n/S_w)} c_m \cdot (5.12.1) \text{ for } \gamma \text{ such that } \mathbb{F}_q(\gamma) = \mathbb{F}_{q^m}. \quad (5.12.2)$$

It will be convenient to choose $a = (a_v)$ such that $a_v = 1$ for $v \neq w$, and that a_w is a uniformizing parameter of $\mathbb{F}_w$. The degree of a is $\deg(w)$. By abuse of notations, a will also be denoted a . The fixed point sets occurring in (5.12.1) are the restricted product over v of

- for $v \in S_w$: $(D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*)$, as in (4.26.2);
- for $v \notin S_1$: $D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*$, as in 4.25;
- for $v = w$: $(D_w^* / K_\sigma \cdot a^\mathbb{Z})^\gamma$.

The corresponding categories decompose into the disjoint sum of $m^{|S_w|}$ subcategories, each equivalent to the one obtained by replacing $D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*$ by $D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*$. The proof is as in 4.27. Denoting restricted direct product by $\prod'$, we conclude:

Lemma 6.5. *The contribution (5.12.1) of the conjugacy class of γ to the right side of (5.11.1) is*

$$\frac{1}{n \deg(a)} m^{N_1-1} \sum_{\sigma} (-1)^{\dim \sigma} \mathrm{mass}[D_\gamma^* \backslash \prod'_{v \neq w} (D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*) \times (D_w^* / K_\sigma \cdot a^\mathbb{Z})^\gamma], \quad (5.13.1)$$

where the sum is over the faces σ of the fundamental chamber of $\mathrm{SL}(n, \mathbb{F}_w)$.

6.9

Let $D_\gamma^*(\mathbb{A})^0$ (resp. $D_{\gamma,v}^*(\mathbb{A})^0$) be the kernel of $\deg \det : D_\gamma^*(\mathbb{A}) \rightarrow \mathbb{Z}$ (resp. $\text{val det} : D_{\gamma,v}^* \rightarrow \mathbb{Z}$), and $D_\gamma^*(\mathbb{A})^0$ (resp. $D_{\gamma,v}^*(\mathbb{A})^0$) be its intersection with $D_\gamma^*(\mathbb{A})$ (resp. $D_{\gamma,v}^*$). As in 4.27 and 4.28, D_γ is a division algebra with center $\mathbb{F}_m$, and $D_\gamma^*(\mathbb{A})$ can be identified with its adelic multiplicative group, over $\mathbb{F}_m$. The diagram

$$\begin{array}{ccccc} D_\gamma^*(\mathbb{A}) & \xrightarrow{\deg \det} & \text{idèles of } \mathbb{F}_m & \xrightarrow{\deg} & \mathbb{Z} \\ \downarrow & & & & \downarrow \cdot m \\ D^*(\mathbb{A}) & \xrightarrow{\deg \det} & \text{idèles of } \mathbb{F} & \xrightarrow{\deg} & \mathbb{Z} \end{array}$$

(with Norm in the middle) is commutative. The notation $D_\gamma^*(\mathbb{A})^0$ above hence agrees with that (“over $\mathbb{F}_m$ ”) of 4.28.

If δ in $D_{\gamma,v}^*$ fixes x in $D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*$, its image in $D_{\gamma,v}$ is in the conjugate $x\mathcal{O}_{\gamma,D,v}x^{-1}$ of $\mathcal{O}_{\gamma,D,v}$. Hence it is in $D_{\gamma,v}^*(\mathbb{A})^0$. By the product formula, $\deg(\det(\delta)) = 0$. It follows that if δ fixes x in $D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*$, then its image in $D_{\gamma,v}$ is in $D_{\gamma,v}^*(\mathbb{A})^0$. As D_γ is a division algebra, D_γ^* is cocompact in $D_\gamma^*(\mathbb{A})^0$: it acts with finitely many orbits in $\prod'_{v \neq w} D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*$, and for each x in $\prod'_{v \neq w} D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*$, the stabilizer Γ_x of x in D_γ^* is cocompact in $D_{\gamma,w}^*(\mathbb{A})^0$. The group Γ_x admits torsion free subgroups of finite index (Serre [S, Th. 4(b)]). For any place $v \neq w$, the kernel of the reduction mod v , from Γ_x to $(x\mathcal{O}_{\gamma,D,v}x^{-1} \otimes \mathcal{O}_v/\mathfrak{m}_v)^*$, is such a subgroup.

6.10

Recall that we have chosen an isomorphism of D_w with $M_n(\mathbb{F}_w)$, that $D_w^*(i)$ is the coset of $D_w^*(\mathbb{A})^0 \subset D_w^*$ on which the valuation of $\det$ is i , and that for each i the $D_w^*(i)/K_\sigma = \text{GL}(n, \mathbb{F}_w)^{(i)}/K_\sigma$ are the facets of one copy $\mathcal{B}(i)$ of the building of $\text{SL}(n, \mathbb{F}_w)$ (5.6.4). Because γ is in $\text{GL}(n, \mathbb{F}_w)^0$, the fixed locus $\mathcal{B}(i)^\gamma$ of γ on $\mathcal{B}(i)$ is a union of facets. Because γ is of finite order, this fixed locus is not empty, hence is contractible. As in [K, p. 635], all the conditions of [S, 3.3] are satisfied by the action of $D_{\gamma,w}^*(\mathbb{A})^0$ on $\mathcal{B}(i)^\gamma$.

If Γ'_x is a torsion free subgroup of finite index of Γ_x , its action on $\mathcal{B}(i)$ is free, its cohomology is that of $\Gamma'_x \backslash \mathcal{B}(i)$, and its Euler-Poincaré characteristic is

$$\chi(\Gamma'_x) = \sum_{\sigma} (-1)^{\dim \sigma} |\Gamma'_x \backslash (D_w^*(i)/K_\sigma)^\gamma|.$$

Dividing both sides by $[\Gamma_x : \Gamma'_x]$, one obtains that the virtual Euler-Poincaré characteristic of Γ_x is

$$\chi(\Gamma_x) = \sum_{\sigma} (-1)^{\dim \sigma} \text{mass}[\Gamma_x \backslash (D_w^*(i)/K_\sigma)^\gamma]. \quad (5.15.1)$$

Let $\mu_{w,EP}$ be the invariant measure on $D_{\gamma,w}^*$ whose restriction to $D_{\gamma,w}^*(\mathbb{A})^0$ is the Euler-Poincaré measure of $D_{\gamma,w}^*(\mathbb{A})^0$. By the definition of $\mu_{w,EP}$,

$$\sum_{\sigma} (-1)^{\dim \sigma} \text{mass}[\Gamma_x \backslash (D_w^*(i)/K_\sigma)^\gamma] = \mu_{w,EP}(\Gamma_x \backslash D_{\gamma,w}^*(\mathbb{A})^0).$$

As $(D_w^*/K_\sigma \cdot a^{\mathbb{Z}})^\gamma$ is the disjoint sum of the $(D_w^*/K_\sigma)^\gamma$ for $0 \leq i < n$, we also have

$$\sum_{\sigma} (-1)^{\dim \sigma} \text{mass}[\Gamma_x \backslash (D_w^*/K_\sigma \cdot a^{\mathbb{Z}})^\gamma] = n \mu_{w,EP}(\Gamma_x \backslash D_{\gamma,w}^*(\mathbb{A})^0).$$

Hence

$$\sum_{\sigma} (-1)^{\dim \sigma} \text{mass}([\Gamma_x \backslash \prod'_{v \neq w} (D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*) \times (D_w^*/K_\sigma \cdot a^{\mathbb{Z}})^\gamma]) = n \sum_x \mu_{w,EP}(\Gamma_x \backslash D_{\gamma,w}^*(\mathbb{A})^0), \quad (5.15.2)$$

the sum being over representatives of the orbits of D_γ^* acting on $\prod'_{v \neq w} D_{\gamma,v}^*/\mathcal{O}_{\gamma,D,v}^*$.

6.11

Let c be the greatest common divisor of m and $\deg(w)$ and suppose that $c > 1$. There are c places of X_m above w , and $(\mathbb{F}_m)_w = \mathbb{F}_w \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ is the product of the completions of $\mathbb{F}_m$ at these places. The completion $(\mathbb{F}_m)_w$ is hence the product of $c > 1$ fields, and $D_{\gamma,w}$ is a product of matrix algebras over these fields. The multiplicative group $D_{\gamma,w}^*$ admits a quotient $\mathbb{Z}^c$, and $D_{\gamma,w}^{*0}$ a quotient $\mathbb{Z}^{c-1}$. As $c > 1$, its Euler-Poincaré measure is 0. If $m = [\mathbb{F}_q(\gamma) : \mathbb{F}_q]$ is not prime to $\deg(w)$, (5.15.2) implies that

$$\sum_{\sigma} (-1)^{\dim \sigma} \text{mass}([D_{\gamma}^* \setminus \prod_{v \neq w}^{\prime} (D_{\gamma,v}^* / \mathcal{O}_{\gamma,D,v}^*) \times (D_w^* / K_{\sigma} \cdot a^{\mathbb{Z}})^{\gamma}]) = 0$$

and γ does not contribute to the right side of (5.11.1).

6.12

Suppose now that m is prime to $\deg(w)$, that is that $m \mid (n/S_1)$. In this case, $(\mathbb{F}_m)_w$ is the local field of X_m at the unique place above w , and $D_{\gamma,w}$ is a matrix algebra $M_{n/m}(\mathbb{F}_{m,w})$. Define μ_{EP} to be the measure on $D_{\gamma}^*(\mathbb{A})$ which is the product of the Haar measures on the $D_{\gamma,v}^*$ giving volume one to $\mathcal{O}_{\gamma,D,v}^*$, for $v \neq w$, and of $\mu_{w,EP}$ on $D_{\gamma,w}^*$. As $D_{\gamma,w}^{*0}$ is of index $m \cdot n$ in $D_{\gamma}^* / a^{\mathbb{Z}}$, (5.15.2) is equal to

$$m \sum_x \mu_{w,EP}(\Gamma_x \setminus D_{\gamma,w}^* / a^{\mathbb{Z}}) = m \mu_{EP}(D_{\gamma}^* \setminus D_{\gamma}^*(\mathbb{A}) / a^{\mathbb{Z}}); \quad (5.16.1)$$

the sum is over representatives of orbits, as in (5.15.2). As D_{γ} is central simple of dimension $(n/m)^2$ over $\mathbb{F}_m$, and $\deg(a)$ is the same, whether a is viewed as an idèle of $\mathbb{F}/\mathbb{F}_q$, or of $\mathbb{F}_m/\mathbb{F}_{q^m}$, when we plug (5.15.2) and (5.16.1) in 5.13, we get that (5.13.1) equals

$$\frac{1}{n \deg(a)} m^{N_1} \mu_{EP}(D_{\gamma}^* \setminus D_{\gamma}^*(\mathbb{A}) / a^{\mathbb{Z}}) = \frac{1}{n \deg(a)} m^{N_1} \frac{n}{m \cdot \deg(a)} \mu_{EP}(D_{\gamma}^* \setminus D_{\gamma}^*(\mathbb{A})^0). \quad (5.16.2)$$

By (5.12.2), (5.11.1) becomes

$$h_1 + (-1)^{n-1} T(X_1, S_1, n) = \sum_{m \mid (n/S_1)} c_m \cdot \frac{1}{m} \cdot m^{N_1} \cdot \frac{1}{\mu(\prod_v \mathcal{O}_{\gamma,D,v}^*)} \cdot \frac{\mu(\prod_v \mathcal{O}_{\gamma,D,v}^*)}{\mu_{w,EP}}. \quad (5.16.3)$$

If μ is the Tamagawa measure on D_{γ}^* , and μ_w the measure on $D_{\gamma,w}^*$ giving $\mathcal{O}_{\gamma,D,w}$ the volume one, the Tamagawa number $\tau(D_{\gamma}^*) = \mu(D_{\gamma}^* \setminus D_{\gamma}^*(\mathbb{A})^0)$ is 1 and the term of index m in the sum (5.16.3) can be rewritten as

$$c_m \cdot \frac{m^{N_1}}{m} \cdot \frac{1}{\mu(\prod_v \mathcal{O}_{\gamma,D,v}^*)} \cdot \frac{\mu_w}{\mu_{w,EP}}. \quad (5.16.4)$$

The factor in curly brackets appeared in 4.29. As there, it is $T_{n/m}$ for (X_m, S_m) . The subgroups $\text{SL}(n/m, \mathbb{F}_{m,w})$ of $\text{GL}(n/m, \mathbb{F}_{m,w})^0$, and $\text{SL}(n/m, \mathcal{O}_{m,w})$ of $\text{GL}(n/m, \mathcal{O}_{m,w})$ are both of the same finite index $q_w - 1$. It follows that the restriction of $\mu_{w,EP}$ to $\text{SL}(n/m, \mathbb{F}_{m,w})$ is $(q_w - 1)$ times its Euler-Poincaré measure, and that the ratio of measures in (5.16.4) is the Euler-Poincaré volume of the subgroup $\text{SL}(n/m, \mathcal{O}_{m,w})$ of $\text{SL}(n/m, \mathbb{F}_{m,w})$. By [S, Th. 7], it is

$$\prod_{i=1}^{n/m-1} (1 - q_w^{mi})^{-1}$$

and 2.3 follows.

7 Lefschetz type form of the theorem

We keep the notation of section 2.

7.1

Formula (2.3.1), applied to (X_m, S_m) , gives the number $T(X_1, S_1, n, m) = T(X_m, S_m, n)$ of fixed points of Fr^{*m} acting on the set $T(n)$ of 2.1. In the form given, it is not helpful to understand how this number varies with m . One of the difficulties is that when (X_1, S_1) over $\mathbb{F}_q$ is replaced by (X_m, S_m) over $\mathbb{F}_{q^m}$, the divisor (n/S_1) of n can change. The cardinality N_1 of S_1 changes too (it becomes N_m), as well as the degrees of the elements of S_1 . Our first aim in this section is to give the right side of (2.3.1) a more convenient form. In this rewriting, until 6.16 we do not assume that $n \geq 2$. We do assume that $n \geq 1$ and that $N_1 \geq 2$.

7.2

Let B' be the sum of the multisets (a) and (b_s) for s in S_1 described below.

(a) The multiset of the eigenvalues of Frob acting on $H^1(X)$, counted with their multiplicity. The polynomial $f(t) := \det(1 - \text{Frob} \cdot t, H^1(X))$ has integral coefficients. It is the product, over the multiset (a), of the $(1 - \alpha t)$. The complex absolute value of each α is $q^{1/2}$.

(b_s) The set of $\deg(s)$ th roots of unity.

The multiset (a) and the sets (b_s) are viewed as multisets in a fixed algebraically closed extension $\overline{\mathbb{Q}}$. It does not matter which.

The number 1 appears once in each (b_s) . It hence appears N_1 times in B' . We define

$$B := B' - \text{twice } \{1\}. \quad (6.2.1)$$

The sum of the sets (b_s) is the multiset of the eigenvalues of the Frob acting on $\mathbb{Q}_l^S$. Indeed, S is the disjoint union of the fibers of the projection $S \rightarrow S_1$. The fiber at $s \in S_1$ has $\deg(s)$ elements, permuted cyclically by the Frobenius. The exact sequence

$$0 \rightarrow \mathbb{Q}_l \rightarrow \mathbb{Q}_l^S \rightarrow H_c^1(X - S) \rightarrow H^1(X) \rightarrow 0 \quad (6.2.2)$$

shows that B is the multiset of eigenvalues of Frob acting on $H_c^1(X - S)$, minus $\{1\}$. It has $2g + N - 2$ elements.

Fix a divisor m of n . We will write $\prod_{\zeta^m=1}$ (resp. $\prod'_{\zeta^m=1}$) or simply $\prod$ and $\prod'$, for the product over the m th roots of unity (resp. m th roots of unity other than 1). We define

$$T'_m := \prod'_{\zeta^m=1} \prod_{\beta \in B} (1 - \zeta\beta)^{-1} (1 - \zeta\beta q^j)^{-1}. \quad (6.2.3)$$

Proposition 7.1. (i) *If the divisor m of n does not divide (n/S_1) , then $T'_m = 0$.*

(ii) *If m divides (n/S_1) , one has*

$$\frac{m^{N_1-2}}{n/m} T_m = \frac{f(1)}{q^n - 1} \cdot T'_m. \quad (6.3.1)$$

Proof of (i). Suppose $m \nmid (n/S_1)$. For some s in S_1 , m is not prime to $\deg(s)$: there is a $\deg(s)$ th-root of unity $\beta \neq 1$ which is an m th root of unity, and the m th-root of unity $\zeta := \beta^{-1}$ contributes a factor $1 - \zeta\beta = 0$ to T'_m . $\square$

Proof of (ii). Suppose that m divides (n/S_1) . As observed at the end of 2.2, $T_{n/m}$ is then the product of

$$f_m(1) = \prod_{\alpha} (1 - \alpha^m) = \prod_{\zeta^m=1} \prod_{\alpha} (1 - \zeta\alpha) = f(1) \cdot \prod'_{\zeta^m=1} \prod_{\alpha} (1 - \zeta\alpha)$$

$$= \frac{(q^m)^{n/m} - 1}{q^n - 1} \cdot f(1)$$

and for each j , $1 \leq j < n/m$,

$$\begin{aligned} (1 - q^{mj})^{-2} f_m(q^{mj}) \cdot \prod_{s \in S_1} (1 - q^{mj \deg(s)}) &= \prod_{\beta \in B} (1 - \beta^m q^{mj}) \\ &= \prod_{\zeta^m=1}^{\prime} (1 - \zeta \beta q^j). \end{aligned}$$

The right side of (6.3.1) is the product of these same factors, and of

$$\prod_{\zeta^m=1}^{\prime} \prod_{\beta} (1 - \zeta \beta) \text{ for } \beta \text{ in the multiset sum of the sets } (b_s), \text{ minus twice } \{1\}. \quad (6.3.2)$$

For each s in S_1 , $\zeta \mapsto \zeta^{\deg(s)}$ is a permutation of the m th roots of unity other than 1. This allows the rewriting of (6.3.2) as

$$\prod_{\zeta^m=1}^{\prime} (1 - \zeta)^{-2} \cdot \prod_{s \in S_1} \prod_{\eta^{\deg(s)}=1}^{\prime} (1 - \zeta \eta) = \prod_{\zeta^m=1}^{\prime} (1 - \zeta)^{N_1-2}.$$

One has

$$\prod_{\zeta^m=1}^{\prime} (1 - \zeta) = \lim_{t \rightarrow 1} \frac{t^m - 1}{t - 1} = m,$$

and (6.3.2) is equal to the factor m^{N_1-2} on the left of (6.3.1). $\square$

Corollary 7.2. *The right side of (2.3.1) is equal to*

$$\frac{f(1)}{q^n - 1} \sum_{m|n} c_m T'_m. \quad (6.4.1)$$

We recall that the Möbius function $\mu(a)$ is $(-1)^d$ when a is the product of d distinct primes, and is 0 otherwise.

Lemma 7.3. *One has*

$$c_m = \sum_{a|m} \mu(a) (q^{m/a} - 1). \quad (6.5.1)$$

Proof. Any element of $\mathbb{F}_{q^m}^*$ is, for some divisor a of m , a generator of $\mathbb{F}_{q^a}$ over $\mathbb{F}_q$: for each m ,

$$q^m - 1 = \sum_{a|m} c_a.$$

The formula (6.5.1) follows by Möbius inversion. $\square$

Substituting (6.5.1) in the last factor of (6.4.1), we get

$$\sum_{m|n} c_m T'_m = \sum_{a, b \geq 1, ab|n} \mu(a) (q^b - 1) T'_{ab}, \quad (6.5.2)$$

(the sum on the right is over all integers $a, b \geq 1$ with $ab|n$), which, plugged in (6.4.1), gives

Proposition 7.4. *The right side of (2.3.1) is equal to*

$$\frac{f(1)}{q^n - 1} \sum_{ab|n} \mu(a)(q^b - 1)T'_{ab}. \quad (6.6.1)$$

Let us, in the second factor of (6.6.1), “consider q and the β in B as indeterminates”. We introduce an indeterminate Q and $2g + N - 2$ indeterminates X_i , and define for $m|n$ (resp. $b|n$) polynomials with integral coefficients, symmetric in the X_i

$$T'_m(Q; (X_i)) := \prod_{\zeta^m=1}^I \prod_i (1 - \zeta X_i) \prod_{1 \leq j < n/m} (1 - \zeta X_i Q^j), \quad (6.6.2)$$

$$S_{b|n}(Q; (X_i)) := \sum_{a:ab|n} \mu(a)T'_{ab}(Q; (X_i)) \quad (\text{sum over } a). \quad (6.6.3)$$

The second factor of (6.6.1) is obtained by evaluating Q at q , and the X_i at the β in B , in the sum over a, b given by

$$\sum_{ab|n} \mu(a) \frac{Q^b - 1}{Q^n - 1} T'_{ab}(Q; (X_i)) = \sum_{b|n} \frac{Q^b - 1}{Q^n - 1} S_{b|n}(Q; (X_i)). \quad (6.6.4)$$

Proposition 7.5. *For each divisor b of n , the rational function*

$$\frac{Q^b - 1}{Q^n - 1} S_{b|n}(Q; (X_i))$$

lies in $\mathbb{Z}[Q, (X_i)]$.

Proof for $b = 1$. We have to show that the polynomial with integral coefficients $S_{1|n}(Q; (X_i))$ is divisible by the polynomial $(Q^n - 1)/(Q - 1)$. This is equivalent to the vanishing of the polynomial in the X_i

$$S_{1|n}(u; (X_i)) = \sum_{a|n} \mu(a) \prod_{\zeta^a=1}^I \prod_i (1 - \zeta X_i) \prod_{0 \leq j < n/a} (1 - \zeta X_i u^j)$$

whenever u is an n th root of unity other than 1. It suffices to prove this vanishing for the product of $S_{1|n}(u; (X_i))$ with $\prod_i (1 - X_i)$. This product is

$$\sum_{a|n} \mu(a) \prod_i \prod_{\zeta^a=1} \prod_{0 \leq j < n/a} (1 - \zeta X_i u^j). \quad (6.7.1)$$

We will show that the terms in the sum over a cancel two by two. More precisely, let r be a prime which divides the order m of u . We will show that the coefficients of $\mu(a)$, for $a = a_0$ prime to r , and for $a = a_0 r$, are equal (recall that $\mu(a)$ is zero unless a is a product of distinct primes). This means the equality, for these two values of a , of the multisets

$$\text{the } \zeta u^j \text{ for } \zeta^a = 1 \text{ and } 0 \leq j < n/a. \quad (6.7.2)$$

The multiset (6.7.2) is the inverse image, by $z \mapsto z^a$, of the multiset of the u^{aj} ($0 \leq j < n/a$). The root of unity u^a has order $m/(m, a)$. As the order of u^a divides n/a , the multiset of the u^{aj} is a multiple of the set of roots of unity of order dividing $m/(m, a)$. Its inverse image by $z \mapsto z^a$ is the same multiple of the set of roots of unity of order dividing $am/(a, m)$, the lowest common multiple of a and m . As $r|m$, this l.c.m. is the same for a_0 and for $a_0 r$. The multiplicity will be the same as well, as both multisets have the same number n of elements. $\square$

Proof of 6.7 (general case). For each divisor b of n , we have

$$\prod_{\zeta^{ab}=1} (1 - \zeta X_i Q^j) = \prod_{\zeta^a=1} (1 - \zeta X_i^b Q^{bj})$$

and

$$\prod'_{\zeta^{ab}=1} (1 - \zeta X_i) = \frac{1 - X_i^{ab}}{1 - X_i^b} = \prod'_{\zeta^a=1} (1 - \zeta X_i^b).$$

It follows that

$$\frac{Q^b - 1}{Q^n - 1} S_{b|n}(Q; (X_i)) = \frac{(Q^b - 1)}{(Q^b)^{n/b} - 1} \cdot \frac{\prod_i (1 - X_i^b)}{\prod_i (1 - X_i)} \cdot S_{1|n/b}(Q^b; (X_i^b)).$$

This identity reduces 6.7 to the case $b = 1$, n being replaced by n/b . $\square$

Corollary 7.6. *When $(X_1, S_1)/\mathbb{F}_q$ is replaced by $(X_m, S_m)/\mathbb{F}_{q^m}$, the second factor of (6.6.1), as a function of m , has the form*

$$\sum_j n_j \gamma_j^m. \quad (6.8.1)$$

In (6.8.1), the n_j are integers, and each γ_j is the product of a root of unity and of a monomial in q and the eigenvalues of Frob acting on $H^1(X)$.

The expression (6.8.1) is not unique. It becomes unique if one imposes that the γ_j be distinct and $n_j \neq 0$. We then call γ_j the eigenvalues occurring with nonzero multiplicity; we call n_j the multiplicity of γ_j . The motivation for this terminology is 6.25.

Proof. The polynomial (6.6.4) depends only on n and the number of X_i . One obtains a description (6.8.1) of the right side of (6.6.1), for (X_m, S_m) , by decomposing it as a linear combination of monomials. Indeed, when passing from $(X_1, S_1)/\mathbb{F}_q$ to $(X_m, S_m)/\mathbb{F}_{q^m}$, q is replaced by q^m , and the β in B by their m th powers: B is the multiset of eigenvalues of Frob acting on $H_c^1(X - S)$, minus $\{1\}$. By definition, the β in B are roots of unity or eigenvalues of Frobenius acting on $H^1(X)$. $\square$

Corollary 7.7. *The same property holds for (6.6.1), the right side of (2.3.1).*

Proof. The product of two functions of m of the form (6.8.1) is again of this form, and the first factor $f(1)$ of (6.6.1) is $\prod_\alpha (1 - \alpha)$ for α the eigenvalues of Frob acting on $H^1(X)$. Again, when one goes from $X_1/\mathbb{F}_q$ to $X_m/\mathbb{F}_{q^m}$, Frob is replaced by Frob^m . $\square$

7.3

We define polynomials R_k in the variables X_i by

$$\begin{aligned} Q_k R_k(X_i) &:= \frac{Q^b - 1}{Q^n - 1} S_{b|n}(Q; (X_i)) \\ &= \sum_{ab|n} \mu(a) \frac{Q^b - 1}{Q^n - 1} T'_{ab}(Q; (X_i)). \end{aligned} \quad (6.10.1)$$

Each R_k is a symmetric polynomial with integer coefficients.

Example 7.8. If $n = 2$, one has

$$R_0 = 1, \quad (6.11.1)$$

$$R_k = (-1)^{k+1} \sigma_j, \quad (k \geq 1), \quad (6.11.2)$$

$$\sum_{j>k}$$

where the σ_j are the elementary symmetric polynomials.

Proof. When n is prime, the sum (6.10.1) over (a, b) has only the three terms for $(a, b) = (1, 1), (n, 1)$ or $(1, n)$. When $ab = 1$, the factor in $T'_{a'b}(Q; (X_i))$ is one. When $ab = n$, the product over j in $T'_{a'b}$ is one. For n prime, (6.10.1) hence reduces to

$$\frac{Q-1}{Q^n-1} \prod_i \prod_{1 \leq j < n} (1 - X_i Q^j) \cdot \prod_{\zeta^n=1}' \prod_i (1 - \zeta X_i) + \prod_{\zeta^n=1}' \prod_i (1 - \zeta X_i).$$

For $n = 2$, this reduces to

$$\frac{1}{Q+1} \prod_i (1 - X_i Q)(1 + X_i) + \prod_i (1 + X_i)$$

which we write as

$$\begin{aligned} & \frac{1}{1 - (-Q)^j} \sigma_j + \sigma_j \\ &= \sum_j \sigma_j ((-Q)^j + 1) = \sum_{k \geq 0} (-1)^{k+1} Q^k \sum_{j > k} \sigma_j \\ &= 1 + \sum_{k \geq 1} (-1)^{k+1} Q^k \sum_{j > k} \sigma_j. \end{aligned} \quad \square$$

Example 7.9. If $n = 1$, or if B is empty (which occurs only for $g = 0$ and $N = 2$), (6.10.1) reduces to 1.

Proof. If $n = 1$, only the term $(a, b) = (1, 1)$ occurs in (6.10.1), and all its factors are 1.

If B is empty, the polynomials T'_m are 1, and (6.10.1) reduces to

$$\sum_{ab|n} \mu(a) \frac{Q^b - 1}{Q^n - 1} = \sum_{b|n} \frac{Q^b - 1}{Q^n - 1} \sum_{a|(n/b)} \mu(a).$$

The sum over a vanishes, except when $n = b$ and the claim follows. $\square$

Proposition 7.10. $R_0 = 1$.

Proof. The polynomial R_0 is obtained by taking $Q = 0$ in (6.10.1). At $Q = 0$, the quotient $(Q^b - 1)/(Q^n - 1)$ is one, and (6.10.1) reduces to

$$\sum_{ab|n} \mu(a) T'_{ab}(Q; (X_i)).$$

Except for $ab = 1$, the sum over a vanishes, and at $Q = 0$, $T'_1(Q; (X_i))$ is one. $\square$

Proposition 7.11. (i) In the following two cases, (6.10.1) is reduced to 1: $|B| = 0$ (that is $g = 0$, $N = 2$); $|B| = 1$ (that is $g = 0$, $N = 3$) and $n = 2$.

(ii) Let d be the largest of the integers k such that $R_k \neq 0$. Excluding the cases (i), one has

$$d = |B| \cdot \frac{n(n-1)}{2} + 1 - n, \quad (6.14.1)$$

$$R_d = (-1)^{N(n-1)} \prod_i X_i^{n-1}. \quad (6.14.2)$$

(iii) Each R_k is a linear combination of monomials dividing $\prod_i X_i^{n-1}$.

Proof. The cases (i) are covered by 6.11 and 6.12. To compute d and R_d , we will expand (6.10.1) around $Q = \infty$. This means embedding the ring of polynomials in Q and the X_i in the ring of Laurent formal power series in Q^{-1} , with coefficients polynomials in the X_i , that is in $\mathbb{Z}[(X_i)][[Q^{-1}]]$, and computing there, using that

$$\frac{Q^b - 1}{Q^n - 1} = Q^{b-n} \frac{1 - Q^{-b}}{1 - Q^{-n}} = Q^{b-n} (1 - Q^{-b}) \sum_{r \geq 0} Q^{-nr}. \quad (6.14.3)$$

This expansion shows that to prove (iii), it suffices to prove that for each divisor m of n , when one expands the product (6.6.2) defining $T'_m(Q; (X_i))$, the monomials occurring are of the form

$$(\text{power of } Q) \cdot \text{monomial dividing } \prod_i X_i^{n-1}.$$

The largest power of X_i occurring is indeed

$$(n/m - 1) + m(n/m - 1) = n - 1.$$

Expanding (6.6.2), one sees that the highest power with which Q occurs in $T'_m(Q; (X_i))$ is

$$|B| \cdot \sum_{1 \leq j < n/m} j = |B| \cdot \frac{(n/m)(n/m - 1)}{2} = |B| \cdot \frac{n(n - m)}{2m^2}.$$

In $\mu(a) \frac{Q^b - 1}{Q^n - 1} T'_{ab}(Q; (X_i))$, it is the same plus $b - n$, which at a given m is maximum for $b = m$. This maximum is

$$|B| \cdot \frac{n(n - m)}{2m} + m - n = |B| \cdot \frac{n^2}{2m} - |B| \cdot \frac{n}{2} + m - n. \quad (6.14.4)$$

Define $A := |B| \cdot n^2/2$. For $x \geq 0$, the function $A/x + x$ decreases from $x = 0$ to its minimum at $x = \sqrt{A} = |B|/2 \cdot n$. If $|B| \geq 2$, the divisor m of n at which $A/m + m$ takes its largest value is hence $m = 1$, while for $|B| = 1$ it is 1 or n . For $|B| = 1$, the values of $A/m + m$ for $m = 1$ and n are

$$\frac{n^2}{2} + 1 \quad \text{and} \quad \frac{3n}{2}.$$

When $|B| = 1$, we assumed that $n \geq 3$, and the value at $m = 1$ is again the largest.

We conclude that in (6.10.1) only one term contributes the largest power of Q : the term $(a, b) = (1, 1)$, and 6.14 is now easily checked. $\square$

The product of the β in B is the determinant of Frob acting on $H_c^1(X - S)$. By (6.2.2), this determinant is the product of the following two factors: $\varepsilon(S)$, the determinant of Frob acting on $\mathbb{Q}_l^S$, that is, the signature of the permutation Frob of S , and the determinant of Frob acting on $H^1(X)$, equal to q^g :

$$\prod_{\beta \in B} \beta = \varepsilon(S) q^g. \quad (6.14.5)$$

Corollary 7.12. *If we exclude the cases 6.14(i), in the decomposition (6.8.1) of the second factor of (6.6.1) for $(X_m, S_m)/\mathbb{F}_{q^m}$, the eigenvalue $\varepsilon(S)^{n-1} q^{D'/2}$, with*

$$D' = (2g - 2)(n^2 - 1) + N(n^2 - n),$$

occurs with multiplicity $(-1)^{N(n-1)}$. The other eigenvalues occurring with nonzero multiplicity have strictly smaller complex absolute values.

Proof. One applies the description of the eigenvalues given in the proof of 6.8, applied to (6.10.1). The β in B have complex absolute value 1 or $q^{1/2}$, and by 6.14 the top eigenvalue is

$$\prod_i \beta_i^{n-1} q^d,$$

occurring with multiplicity $(-1)^{N(n-1)}$. By (6.14.5), it equals

$$\varepsilon(S)^{n-1} q^{g(n-1)+d},$$

and

$$\begin{aligned} g(n-1) + d &= [(g-1)(n^2-1) + (n-1)] + [(2g-2+N)\frac{n(n-1)}{2} - (n-1)] \\ &= (g-1)((n-1) + n(n-1)) + N\frac{n(n-1)}{2} \\ &= \frac{1}{2}[(2g-2)(n^2-1) + N(n^2-n)]. \end{aligned} \quad \square$$

From now on, we again assume that $n \geq 2$.

7.4

Over $\mathbb{C}$, let us consider a compact Riemann surface of genus g , a set S of N points of Σ , the inclusion j of $\Sigma - S$ in Σ , and, on $\Sigma - S$, irreducible complex local systems of rank n , with trivial determinant and – at each $s \in S$ – with local monodromy which is unipotent with one Jordan block (principal unipotent). Let $\mathcal{M}$ be the moduli space of these local systems. In the cases 6.14(i) it is empty. We exclude these cases.

The deformation theory of local systems V as above is controlled by $H^*(\Sigma, j_* \text{End}^0(V))$, where $\text{End}^0(V)$ is the local system of trace zero endomorphisms of V . The local system $\text{End}^0(V)$ is self dual, for the pairing $\text{Tr}(uv)$. From this pairing we get a perfect pairing between H^0 and the H^2 of $j_* \text{End}^0(V)$, and a symplectic form on the H^1 . By Schur's lemma, the H^0 vanishes. By duality, so does H^2 : the deformation theory is unobstructed, and $\mathcal{M}$ is smooth. Its tangent space at V is $H^1(\Sigma, j_* \text{End}^0(V))$, and the autoduality of this H^1 turns $\mathcal{M}$ into a complex symplectic manifold, of dimension

$$\dim \mathcal{M} = \dim H^1(\Sigma, j_* \text{End}^0(V)) = -\chi(j_* \text{End}^0(V)).$$

The local system $\text{End}^0(V)$ on $\Sigma - S$ is of rank $n^2 - 1$, and the fiber of $j_* \text{End}^0(V)$ at each $s \in S$ is of rank $n - 1$ (it is the centralizer of the local monodromy in a nearby fiber of $\text{End}(V)^0$). By additivity of Euler-Poincaré, this gives

$$\begin{aligned} \chi(j_* \text{End}^0 V) &= \chi(\Sigma - S) \cdot (n^2 - 1) + N(n - 1) \\ &= \chi(\Sigma)(n^2 - 1) - N(n^2 - n) = (2 - 2g)(n^2 - 1) - N(n^2 - n), \end{aligned}$$

showing that the complex dimension of $\mathcal{M}$ is D' . We do not understand why q at the power half the dimension of $\mathcal{M}$ appears in 6.15.

7.5

Let us write $R_k(B)$ for the value of the polynomial R_k at a point (x_i) where the x_i run over the multiset B . By (2.3.1), (6.6.1) and the definition 6.10 of the R_k , one has

$$T(X_1, S_1, n) = (-1)^{N_1(n-1)} [-f(1) + f(1) \sum_{k \geq 0} q^k R_k(B)]. \quad (6.17.1)$$

By 6.13, the $k = 0$ term in the sum cancels $-f(1)$. If a permutation σ of N letters has N_1 cycles, its signature is $(-1)^{N-N_1}$. Plugging this in (6.17.1), we get

Theorem 7.13. (second form of theorem 2.3) *One has*

$$T(X_1, S_1, n) = (-1)^{N(n-1)} \varepsilon(S)^{n-1} f(1) \sum_{k \geq 1} q^k R_k(B). \quad (6.18.1)$$

In the exceptional cases 6.14(i), the right side vanishes. Otherwise, the sum over k ranges from 1 to d (6.14.1), and $d \geq 1$.

When we go from $(X_1, S_1)/\mathbb{F}_q$ to $(X_m, S_m)/\mathbb{F}_{q^m}$, the sign $\varepsilon(S) = \det(\text{Frob}, \mathbb{Q}_l^S)$ is replaced by its m th power. As a function of m , all factors $\varepsilon(S)$, $f(1) = \prod_{\alpha} (1 - \alpha)$, $q^k R_k(B)$ of (6.18.1), and hence $T(X_m, S_m, n)$ have the form (6.8.1): a sum $\sum n_j \gamma_j^m$, reminiscent of a Lefschetz trace formula. If we exclude the cases 6.14(i), the top eigenvalue for $m \mapsto T(X_m, S_m, n)$ is the product of that for $m \mapsto f_m(1)$, that is q^g , of $\varepsilon(S)^{n-1}$, and of the top eigenvalue computed in 6.15, that is $\varepsilon(S)^{n-1} q^{D'/2}$. Its multiplicity is $(-1)^{N(n-1)}$ times the multiplicity $(-1)^{N(n-1)}$ of 6.15: it is one.

Corollary 7.14. (i) *As a function of m , the number $T(X_1, S_1, n, m) = T(X_m, S_m, n)$ of fixed points of Fr^{*m} has the form explained in (6.8.1):*

$$\sum_i n_i \gamma_i^m. \quad (6.19.1)$$

The γ_i are products of a strictly positive power of q , of a root of unity, and of eigenvalues of Frob acting on $H^1(X)$.

(ii) *If $g = 0$, $N = 2$ or if $g = 0$, $N = 3$, $n = 2$, $T(X_1, S_1, n, m) = 0$. Otherwise, the eigenvalue $q^{D/2}$, with $D = D' + 2g = (2g - 2)(n^2 - 1) + N(n^2 - n) + 2g$, occurs in (6.19.1) with multiplicity one, and the other eigenvalues occurring with nonzero multiplicity have strictly smaller complex absolute values.*

7.6

In parallel to 6.16, and excluding the cases 6.14(i), D is the dimension of the space of irreducible complex local systems of dimension n and principal unipotent local monodromy at each point of S .

A similar phenomenon occurs for $n = 2$ in the case of no ramification, studied by Drinfeld [D], and in cases of fixed and “generic” tame local ramifications, studied by Arinkin.

Corollary 7.15. *The number $T(X_1, S_1, n)$ is divisible by q .*

7.7

Somewhat abusively, we will write “virtual object of $\mathcal{A}$ ” to mean “element of the Grothendieck group of the abelian category $\mathcal{A}$ ”. The polynomial R_k , being symmetric and with integral coefficients, is the character of a virtual polynomial representation of $\text{GL}(2g + N - 2)$, evaluated at the diagonal matrix with diagonal entries the X_i . We write R_k also for this virtual representation. It is a difference of two representations, say R_k^+ and R_k^- .

Example 7.16. The elementary symmetric polynomial σ_j is the character of $\bigwedge^j V$, for V the defining representation of the linear group. For $n = 2$ and $k \geq 1$, 6.11 tells that the virtual representation R_k is

$$R_k = (-1)^{k+1} \sum_{j \geq k+1} \bigwedge^j V \quad (\text{for } n = 2, k \geq 1). \quad (6.23.1)$$

7.8

Representations of the linear group $\mathrm{GL}(M)$ can be identified (equivalence of categories) with functors from the category of vector spaces of dimension M , with isomorphisms as morphisms, to the category of vector spaces and linear maps. For instance, to the representation $\bigwedge^j V$, for V the defining representation of $\mathrm{GL}(M)$, corresponds the functor $\bigwedge^j$ (the j -th exterior power). The equivalence is (functor T) $\mapsto$ (representation $T(V)$). Such a functor T can be applied to smooth ℓ -adic sheaves of rank M .

7.9

Let H_{c-}^1 be the quotient of $H_c^1(X - S)$ by a line fixed by Frob. The assumption $N_1 \geq 2$ ensures there is one. The eigenvalues of Frob acting on H_{c-}^1 are the β in B . It follows that $R_k(B)$ of 6.17 is

$$R_k(B) = \mathrm{Tr}(\mathrm{Frob}, R_k(H_{c-}^1)) := \mathrm{Tr}(\mathrm{Frob}, R_k^+(H_{c-}^1)) - \mathrm{Tr}(\mathrm{Frob}, R_k^-(H_{c-}^1)). \quad (6.25.1)$$

The other pieces in (6.18.1) have a similar interpretation: q is the eigenvalue of Frob acting on $\mathbb{Q}_l(-1) = H^1(\mathbb{G}_m)$, $\varepsilon_l(S)$ is the eigenvalue of Frob acting on the top exterior power of $\mathbb{Q}_l^S$, and

$$f(1) = \sum_i (-1)^i \mathrm{Tr}(\mathrm{Frob}, \bigwedge^i H^1(X)).$$

Let us write $W(-k)$ for $W \otimes \mathbb{Q}_l(-1)^{\otimes k}$, and $\varepsilon_l(S)$ for the top exterior power of $\mathbb{Q}_l^S$. We get

$$T(X_1, S_1, n) = \mathrm{Tr} \left(\mathrm{Frob}, (-1)^{N(n-1)} \varepsilon_l(S)^{\otimes(n-1)} \otimes \sum_i (-1)^i \bigwedge^i H^1(X) \otimes \sum_{k=1}^d R_k(H_{c-}^1)(-k) \right). \quad (6.25.2)$$

Example 7.17. If $n = 2$, (6.23.1) gives

$$R_k(B) = (-1)^{k+1} \mathrm{Tr} \left(\mathrm{Frob}, \sum_{j \geq k+1} \bigwedge^j H_{c-}^1 \right) \quad (k \geq 1).$$

Define $H_c := H_c^1(X - S)$. As $\bigwedge^j H_c$ is an extension of $\bigwedge^j H_{c-}^1$ by $\bigwedge^{j-1} H_{c-}^1$, one also has

$$R_k(B) = (-1)^{k+1} \mathrm{Tr} \left(\mathrm{Frob}, \bigwedge^{k+2j} H_c \right) \quad (\text{for } n = 2, k \geq 1) \quad (6.26.1)$$

$$\sum_{j \geq 1}$$

from which it follows that, under our standing assumption $N_1 \geq 2$,

$$T(X_1, S_1, 2) = \mathrm{Tr} \left(\mathrm{Frob}, (-1)^N \varepsilon_l(S) \otimes \sum_i (-1)^i \bigwedge^i H^1(X) \otimes \sum_{k \geq 1} (-1)^{k+1} \bigwedge^{k+2j} H_c(-k) \right). \quad (6.26.2)$$

7.10

Let $\mathcal{M}_{g,[N]}$ be the moduli stack, over $\text{Spec}(\mathbb{Z})$, of curves of genus g given with a set S of N distinct points. More precisely, a morphism $Y \rightarrow \mathcal{M}_{g,[N]}$, or equivalently an object of $\mathcal{M}_{g,[N]}$ over the scheme Y , is a proper and smooth morphism $a : X \rightarrow Y$ whose geometric fibers are irreducible curves of genus g , given with a relative divisor $S \subset X$, finite étale of degree N over Y .

The moduli stack $\mathcal{M}_{g,N}$ of curves of genus g given with an ordered set of N distinct points is a Galois covering of $\mathcal{M}_{g,[N]}$, with Galois group the symmetric group $\mathfrak{S}_N$. More precisely, it is an $\mathfrak{S}_N$ -torsor over $\mathcal{M}_{g,[N]}$.

Fix a decomposition $N = N' + N''$, with $N', N'' \geq 1$. Corresponding to the subgroup $\mathfrak{S}_{N'} \times \mathfrak{S}_{N''}$ of $\mathfrak{S}_N$, we have between $\mathcal{M}_{g,N}$ and $\mathcal{M}_{g,[N]}$ the moduli stack $\mathcal{M}_{g,[N',N']}$ of curves of genus g given with disjoint sets of N' and N'' distinct points. An object of $\mathcal{M}_{g,[N',N']}$ over Y is $a : X \rightarrow Y$ as above, given with disjoint relative divisors S', S'' , finite étale of degrees N', N'' over Y . We put $S := S' \cup S''$.

For each such stack $\mathcal{M}$, we denote by $\mathcal{M}[1/\ell]$ the open substack where ℓ is invertible.

A $\overline{\mathbb{Q}}_\ell$ -smooth sheaf on a stack $\mathcal{M}$ is the data, for each $Y \rightarrow \mathcal{M}$, of a $\overline{\mathbb{Q}}_\ell$ -smooth sheaf whose formation is compatible with pullbacks by maps $Y' \rightarrow Y$. Here are examples of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $\mathcal{M}_{g,[N]}[1/\ell]$, defined by giving their value on $a : (X, S) \rightarrow Y$, object of the stack over Y . By pullback, they give similar smooth sheaves on $\mathcal{M}_{g,[N',N']}[1/\ell]$.

- H : $R^1(a : X \rightarrow Y)_* \overline{\mathbb{Q}}_\ell$. It is of rank $2g$.
- S : $a(S \rightarrow N \cdot Y)_* \overline{\mathbb{Q}}_\ell$. It is of rank N .
- $\varepsilon(S) := \bigwedge^N S$, of rank one.
- H_c : $R^1(a : X - S \rightarrow Y)_! \overline{\mathbb{Q}}_\ell$. It is of rank $2g + N - 1$ and sits in an exact sequence whose geometric fibers are given by (6.2.2):

$$0 \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow S \rightarrow H_c \rightarrow H \rightarrow 0. \quad (6.27.1)$$

On $\mathcal{M}_{g,[N',N']}[1/\ell]$, H_c contains a copy of the constant sheaf $\overline{\mathbb{Q}}_\ell$. Indeed, S decomposes into the sum of $S' : (a : S' \rightarrow Y)_* \overline{\mathbb{Q}}_\ell$ and $S'' : (a : S'' \rightarrow Y)_* \overline{\mathbb{Q}}_\ell$, the map $\overline{\mathbb{Q}}_\ell \rightarrow a_* a^* \overline{\mathbb{Q}}_\ell$ embeds $\overline{\mathbb{Q}}_\ell$ into both S' and S'' , and (6.27.1) embeds $\overline{\mathbb{Q}}_\ell \simeq \overline{\mathbb{Q}}_\ell \oplus \overline{\mathbb{Q}}_\ell / (\text{diagonal } \overline{\mathbb{Q}}_\ell)$ into H_c . On $\mathcal{M}_{g,[N',N']}[1/\ell]$, we define $H_c / \overline{\mathbb{Q}}_\ell$ to be the quotient of H_c by this copy of $\overline{\mathbb{Q}}_\ell$.

If $\mathcal{W}$ is a $\overline{\mathbb{Q}}_\ell$ -smooth sheaf, or more generally a $\overline{\mathbb{Q}}_\ell$ -sheaf, on a stack $\mathcal{M}$, if $x : \text{Spec}(\mathbb{F}_q) \rightarrow \mathcal{M}$ is an $\mathbb{F}_q$ -point of $\mathcal{M}$, and $\bar{x} : \text{Spec}(\mathbb{F}) \rightarrow \text{Spec}(\mathbb{F}_q) \rightarrow \mathcal{M}$ a corresponding geometric point, the geometric Frobenius $\text{Frob} \in \text{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on the fiber of $\mathcal{W}$ at $\bar{x}$. One defines

$$\text{Tr}(\text{Frob}_x, \mathcal{W}) := \text{Tr}(\text{Frob}, \mathcal{W}_{\bar{x}}).$$

This definition extends by additivity to the case of virtual smooth sheaves, that is of elements of the Grothendieck group of the category of $\overline{\mathbb{Q}}_\ell$ -smooth sheaves.

The formula (6.25.2) can now be translated as follows.

Proposition 7.18. *Let x be an $\mathbb{F}_q$ -point of $\mathcal{M}_{g,[N',N']}[1/\ell]$, that is (X_1, S'_1, S''_1) over $\mathbb{F}_q$, and let $S_1 := S'_1 \cup S''_1$. Then $T(X_1, S_1, n)$ is the trace of Frob_x on the virtual smooth sheaf*

$$(-1)^{N(n-1)} \varepsilon_l(S)^{\otimes(n-1)} \otimes \sum_i (-1)^i \bigwedge^i H \otimes \sum_{k=1}^d R_k(H_c / \overline{\mathbb{Q}}_\ell)(-k). \quad (6.28.1)$$

Conjecture 7.19. *Let us drop our standing assumption $N_1 \geq 2$. We conjecture that for any $g \geq 0$, $N \geq 0$ and $n \geq 2$, there exists a virtual $\overline{\mathbb{Q}}_\ell$ -smooth sheaf $\mathcal{W}(n)$ on $\mathcal{M}_{g,[N]}[1/\ell]$ such that for x an $\mathbb{F}_q$ -point of $\mathcal{M}_{g,[N]}[1/\ell]$, that is (X_1, S_1) over $\mathbb{F}_q$, one has*

$$T(X_1, S_1, n) = \text{Tr}(\text{Frob}_x, \mathcal{W}(n)).$$

This implies a dependence on m of the form (6.8.1) for $T(X_m, S_m, n)$. An optimistic version of the conjecture would be that for some virtual polynomial representations $R_k(n)$ of $\mathrm{GL}(2g+N-1)$, $\mathcal{W}(n)$ is of the form

$$\mathcal{W}(n) = \varepsilon_l(S)^{\otimes(n-1)} \otimes \sum_i (-1)^i \bigwedge^i H \otimes \sum_k R_k(n)(H_c)(-k).$$

If $N > 0$, one might also hope that the sum over k is over positive k 's.

For $n = 2$ and $N = 0$, the conjecture (in its optimistic form) is a corollary of what Drinfeld proves in [D].

For $n = 2$ and $N_1 \geq 1$, one of us (Y.F.) has checked, using the trace formula in its full glory, that (6.26.2) continues to hold for $N_1 = 1$. From this the optimistic version of the conjecture (for $n = 2$ and $N_1 \geq 1$) readily follows.

For $N \geq 2$, and $N = N' + N''$ with $N', N'' \geq 1$, the conjectural $\mathcal{W}(n)$ on $\mathcal{M}_{g,[N]}$ would have (6.28.1) as inverse image on $\mathcal{M}_{g,[N',N'']}$. For $n = 2$, this made (6.26.2) a plausible guess. One should, however, beware that there are virtual $\overline{\mathbb{Q}}_\ell$ -smooth sheaves on $\mathcal{M}_{g,[N]}$ all of whose inverse images to the $\mathcal{M}_{g,[N',N'']}$ ($N = N' + N''$, $N', N'' \geq 1$) are zero. Examples are $\sum_i (-1)^i \bigwedge^i H_c$ and $\sum_i (-1)^i \bigwedge^i (S/\overline{\mathbb{Q}}_\ell)$. The latter boils down to the following fact. Given a finite group H , with representation ring $R(H)$, and a family of subgroups H_j , if $\bigcup H_j$ does not meet all conjugacy classes in H , the restriction map

$$R(H) \rightarrow \prod_j R(H_j)$$

is not injective. In the case of $\mathfrak{S}_N$, the family of subgroups $\mathfrak{S}_{N'} \times \mathfrak{S}_{N''}$ ($N = N' + N''$, $N', N'' \geq 1$) misses the conjugacy class of cycles of length N . If V is the representation $\mathbb{C}^N / (\text{diagonal } \mathbb{C})$ of $\mathfrak{S}_N$, its restriction to $\mathfrak{S}_{N'} \times \mathfrak{S}_{N''}$ contains a copy of the trivial representation 1, and as

$$\bigwedge^j (W + 1) = \bigwedge^j W \oplus \bigwedge^{j-1} W$$

(with $\bigwedge^{-1} W := 0$), the restriction of $\sum_j (-1)^j \bigwedge^j V$ to each $\mathfrak{S}_{N'} \times \mathfrak{S}_{N''}$ vanishes. The character of $\sum_j (-1)^j \bigwedge^j V$ vanishes outside of the conjugacy class of cycles of length N , where its value is N .

7.11

The conjecture should not be specific to the case of principal unipotent local monodromy. It should hold, with other virtual $\overline{\mathbb{Q}}_\ell$ -smooth sheaves $\mathcal{W}$, when one imposes the local monodromy at each $s \in S$. When the local monodromy imposed is tame (and makes sense in characteristic not dividing P), the relevant moduli stack $\mathcal{M}$ will be over $\mathbb{Z}[1/\ell P]$. It is the moduli stack of curves of genus g given with a set of N points decorated by the imposed monodromy. When the local monodromy imposed is not tame, one might have to stay in a specific characteristic, and consider points decorated by a local parameter given up to some order.

Remark 7.20. The virtual $\overline{\mathbb{Q}}_\ell$ -smooth sheaf (6.28.1) is the formal difference of local systems built out of the local systems of cohomology of X , S' and S'' . We used ℓ -adic cohomology, but (6.28.1) would make sense for any of the standard cohomology theories. It is “motivic”. Over $\mathbb{C}$, the same construction gives a virtual variation of Hodge structures on $\mathcal{M}_{g,[N',N'']}(\mathbb{C})$.

Over $\mathbb{C}$, to each curve X of genus g given with a set $S = S' \cup S''$ of $N = N' + N''$ points, one can attach the moduli space $\mathcal{T}_{X,S}$ of rank n irreducible complex local systems on $X - S$, with principal unipotent local monodromy at each $s \in S$.

These spaces are the fibers of a morphism $a : \mathcal{T} \rightarrow \mathcal{M}_{g,[N',N'']}$. Can one relate the virtual variation (6.28.1) and this family of spaces? From the complex analytic point of view, this family

is locally constant, because it can be interpreted in terms of representations of the fundamental group of $X - S$, itself locally constant. This should be related to the fact that $T(X_1, S_1, n)$ is controlled by a virtual $\overline{\mathbb{Q}}_\ell$ -smooth sheaf, rather than a virtual $\overline{\mathbb{Q}}_\ell$ -sheaf.

The analogue of 6.19(ii) is that the top weight part of the variation (6.28.1) is a $\overline{\mathbb{Q}}_\ell(-D/2)$, of Hodge type $(D/2, D/2)$, where D is the (complex) dimension of the $\mathcal{T}_{X,S}$.

8 Example: $g = 0$, $N = 4$, $n = 2$

The case of rank 2 local systems over the projective line minus an étale divisor of degree four is among the simplest nontrivial cases. It has been investigated numerically by Kontsevich (cf. [Ko, 0.1]). It might be a useful testing ground to tentative answers to question 7.20.

In this section, we do not assume $N_1 \geq 2$: X_1 is a projective line over $\mathbb{F}_q$ and the degree four reduced divisor S_1 is allowed to consist of one closed point of degree four.

Proposition 8.1. *With the above notations, one has*

$$T(X_1, S_1, 2) = q. \quad (7.1.1)$$

As $g = 0$, one has $f(1) = 1$. When $N_1 \geq 2$,

$$D = D' = (2g - 2)(n^2 - 1) + N(n^2 - n) = 2$$

and the dominant term q computed in 6.19(ii) is the only term by 6.19(i). This leaves out the case where S_1 consists of a single point of degree 4. We will explain in 7.8 how this case can be reduced to the case $N_1 \geq 2$.

8.1

Let E be a set with four elements. We denote by V_E the Vierergruppe of E , that is the subgroup of the symmetric group of E consisting of the identity and of the three $(2, 2)$ permutations of E . The action of V_E on E is simply transitive: when viewed as a right action, it turns E to a V_E -torsor. When we need to emphasize this, we write $t(E)$ for E . The abelian group V_E being killed by 2, $t(E)$ is its own opposite: the torsor sum $t(E) + t(E)$ is the trivial V_E -torsor V_E . As V_E acts on E , the twist $E_{t(E)}$ of E by $t(E)$ is defined. It is just $t(E) + t(E) = V_E$. It has a canonical point 0.

8.2

Let P be a projective line over a field k . If (s_1, s_2, s_3, s_4) and (s'_1, s'_2, s'_3, s'_4) are two quadruples of distinct k -points of P with the same cross ratio, there is a unique automorphism of P over k mapping the first quadruple to the second. Special case: the cross ratio being invariant by a permutation of a quadruple of points which belongs to the Vierergruppe, if $S \subset P(k)$ consists of four points, the action of V_S on S extends (uniquely) to an action on P . A twist $P_{t(S)}$ of P by the torsor $t(S)$ is hence defined. It contains the twist $S_{t(S)} = V_S$ of S by $t(S)$. Concrete description: one takes four copies $P[s]$ of P indexed by S ; one has a transitive system of isomorphisms between the $P[s]$: as isomorphism from $P[s]$ to $P[t]$, one takes the action of the unique element of V_S mapping s to t ; the twist $P_{t(S)}$ is the “common value” (projective limit) of the $P[s]$. In other words,

$$P_{t(S)} = (P \times S)/V_S$$

(diagonal action). The canonical point $0 \in V_S = S_{t(S)} \subset P_{t(S)}$ is the image of the diagonal S of $S \times S \subset P \times S$.

By étale descent, this construction continues to make sense for P a projective and smooth curve of genus 0 over k , and for S a divisor of degree 4, étale over k . The group V_S is now a group scheme étale over k . It acts on S , on P , on $P_{t(S)} = (P \times S)/V_S$ and $0 \in V_S = S_{t(S)} \subset P_{t(S)}$ is again the image of the diagonal of $S \times S$.

8.3

We now suppose that $k = \mathbb{C}$. Let $T((P, S)/\mathbb{C})$ denote the set of isomorphism classes of irreducible rank 2 complex local systems on $P - S$, with principal unipotent local monodromy at each $s \in S$ (6.16). The group V_S acts on this set by transport of structures.

Proposition 8.2. *The action of V_S on $T((P, S)/\mathbb{C})$ is trivial.*

Fix an involution σ in V_S , and label S by $\mathbb{Z}/4$, in such a way that σ is $s_i \mapsto s_{i+2}$.

Proposition 7.5 results from the more general

Proposition 8.3. *Let V be an irreducible rank 2 complex local system on $P - S$. Assume that the local monodromy transformations are in $\mathrm{SL}(2)$, and that for each $s \in S$ the local monodromies around s and $\sigma(s)$ are conjugate. Then, V is isomorphic to σ^*V .*

Proof. All involutions in $\mathrm{Aut}(P)$ are conjugate: we may and shall choose a coordinate z such that the automorphism σ of P is $z \mapsto -z$. The claim is invariant under deformation of S . We may assume that S consists of the points $\pm 1, \pm i$, labelled by $\mathbb{Z}/4$ as in the following picture.

Take the fixed point $z = 0$ of σ as base point. The fundamental group $\pi_1(P - S, 0)$ is generated by the loops γ_i ($i \in \mathbb{Z}/4$) pictured in the figure, with $\gamma_1\gamma_2\gamma_3\gamma_4 = 1$ as the only relation, and σ maps γ_i to γ_{i+2} .

To give a local system V on $P - S$ amounts to giving its fiber V_0 at 0, and the action of $\pi_1(P - S, 0)$ on it. Let A_i be the image of γ_i . For our V , if we choose an isomorphism of V_0 with $\mathbb{C}^2$, the representation of π_1 is given by four A_i in $\mathrm{SL}(2, \mathbb{C})$, obeying

$$A_1 A_2 A_3 A_4 = 1. \quad (7.6.2)$$

Our assumptions are that the representation is irreducible and that A_i is conjugate to A_{i+2} . Our claim is that the quadruples (A_1, A_2, A_3, A_4) and (A_3, A_4, A_1, A_2) are conjugate, in other words that the corresponding representations of π_1 are isomorphic. For this, it suffices (by irreducibility) to check that they have the same character: that for any word $\prod A_{i(k)}^{\varepsilon(k)}$ in the $A_i^{\pm 1}$,

$$\mathrm{Tr} \prod A_{i(k)}^{\varepsilon(k)} = \mathrm{Tr} \prod A_{i(k)+2}^{\varepsilon(k)}. \quad (7.6.3)$$

We are in rank $n = 2$. By Procesi [P, Theorem 3.4(a) p. 316], applied to the A_i and their inverses, it suffices to check (7.6.3) for words of length $\leq 2n - 1 = 3$. Words can be viewed as circular words. By the identities $A_i^{-1} = \mathrm{Tr}(A_i) \cdot A_i - 1$ and $A_i^2 = \mathrm{Tr}(A_i)A_i - 1$, it suffices to consider words in the A_i , for which consecutive A_i have distinct indices. For circular words of length 3, this means all indices distinct. We now check (7.6.3) for these words. We will use that in $\mathrm{SL}(2)$, $\mathrm{Tr}(A) = \mathrm{Tr}(A^{-1})$.

We assumed $\mathrm{Tr}(A_i) = \mathrm{Tr}(A_{i+2})$. That $\mathrm{Tr}(A_i A_{i+2}) = \mathrm{Tr}(A_{i+2} A_i)$ is clear. By (7.6.2), one has

$$\mathrm{Tr}(A_3 A_4) = \mathrm{Tr}((A_1 A_2)^{-1}) = \mathrm{Tr}(A_1 A_2)$$

and similarly for $A_i A_{i+1}$. The case of $A_1 A_2 A_3$ is reduced to that of A_4 by

$$\mathrm{Tr}(A_1 A_2 A_3) = \mathrm{Tr}(A_4^{-1}) = \mathrm{Tr}(A_4).$$

The case of $\mathrm{Tr}(A_1 A_3 A_2)$ follows, thanks to the identity

$$\mathrm{Tr}(ABC) + \mathrm{Tr}(ACB) = \mathrm{Tr}(A) \mathrm{Tr}(BC) + \mathrm{Tr}(B) \mathrm{Tr}(CA) + \mathrm{Tr}(C) \mathrm{Tr}(AB) + \mathrm{Tr}(A) \mathrm{Tr}(B) \mathrm{Tr}(C) = 0, \quad (7.6.4)$$

which follows from the vanishing of the antisymmetrization operator $a = \sum_{\tau} \varepsilon(\tau) \tau$ of $(\mathbb{C}^2)^{\otimes 3}$: one expands

$$\mathrm{Tr}(A \otimes B \otimes C \circ a) = 0.$$

Similarly for $(1, 2, 3)$ replaced by $(i, i + 1, i + 2)$. This concludes the required check. $\square$

Suppose now that k is any algebraically closed field, and consider $\overline{\mathbb{Q}_\ell}$ -smooth sheaves in the étale sense of 1.1.

Corollary 8.4. *For $(P, S)/k$ as above, 7.6 remains valid provided the local monodromy is tame. As a consequence, 7.5 remains valid.*

Tameness is needed already to make sense of “conjugate local monodromy”.

Proof for $k = \mathbb{C}$. The proof of 7.6 is algebraic, hence holds also for local systems of $\overline{\mathbb{Q}_\ell}$ -vector spaces on $P(\mathbb{C}) - S$. It remains to observe that the functor

$$(\overline{\mathbb{Q}_\ell}\text{-smooth sheaves on } P - S) \rightarrow (\text{local systems of } \overline{\mathbb{Q}_\ell}\text{-vector spaces on } P(\mathbb{C}) - S)$$

is fully faithful. Indeed, both categories are 2-inductive limits of similar categories, with $\overline{\mathbb{Q}_\ell}$ replaced by a finite extension E_λ of $\mathbb{Q}_\ell$ in $\overline{\mathbb{Q}_\ell}$. This reduces us to the E_λ case. Let $\mathcal{O}_\lambda$ be the valuation ring of E_λ and fix a basepoint 0. Local systems on $P(\mathbb{C}) - S$ are representations of $\pi_1(P(\mathbb{C}) - S, 0)$ on an E_λ -vector space of finite dimension. The E_λ -smooth sheaves on $P - S$ are those representations V which extend to continuous representations of the profinite completion of π_1 , that is for which V contains a lattice V_0 (a free $\mathcal{O}_\lambda$ -submodule of V which generates V over E_λ) stable by the action. Morphisms are the same. $\square$

Proof in characteristic zero. This case follows from the case $k = \mathbb{C}$, by invariance of the algebraic π_1 upon extension of scalars from one algebraically closed field of characteristic zero to another one. $\square$

Proof in characteristic p . Grothendieck proved that the tame π_1 is a quotient of the characteristic 0 group π_1 . We will show that his proof reduces the characteristic p case to the characteristic zero case. Let $W(k)$ be the ring of Witt vectors over k . Let $\bar{K}$ be an algebraic closure of the field of fractions K of $W(k)$. Consider a lifting (P_W, S_W) of (P, S) to $W(k)$. The action of V_S lifts. A $\overline{\mathbb{Q}_\ell}$ -local system on $P - S$ which is tamely ramified along S lifts uniquely to $P_W - S_W$, and pulls back to $(P_W, S_W) \otimes_W \bar{K}$. By Grothendieck, the resulting functor from tame local systems on $P - S$ to local systems on $P_{\bar{K}} - S_{\bar{K}}$ is fully faithful, reducing us to the characteristic zero case. $\square$

8.4 End of Proof of (7.1.1)

Define (X'_1, S'_1) to be the twist of (X_1, S_1) by $t(S_1)$. Over $\mathbb{F}$, we have a natural system of four isomorphisms between (X, S) and (X', S') , exchanged by the Vierergruppe. By 7.5, they all induce the same bijection from $T(2)(X, S)$ to $T(2)(X', S')$. By transport of structure, this bijection is compatible with the action of Frob. It follows that

$$T(X_1, S_1, 2) = T(X'_1, S'_1, 2).$$

The divisor S'_1 contains a rational point. For (X'_1, S'_1) , one hence has $N'_1 \geq 2$, and (7.1.1) for (X_1, S_1) results from (7.1.1) for (X'_1, S'_1) .

Translating the generalization 7.7 of 7.5 using the global Langlands correspondence [L], one obtains the following result, of which we do not know of a proof not using [L].

Proposition 8.5. *Let S_1 be an étale divisor of degree four of $\mathbb{P}^1/\mathbb{F}_q$, and σ an involutive automorphism of $(\mathbb{P}^1, S_1)$ which acts on S_1 by an element of V_{S_1} . Suppose that the automorphic representation π of $\mathrm{GL}(2, \mathbb{A})$ is cuspidal, unramified outside of S_1 , and that its local component at each $s \in S_1$ is of the form*

$$\text{Steinberg} \otimes \chi(\det)$$

with χ unramified.

Under these assumptions, $\sigma(\pi)$ is an $\mathbb{F}_q$ -twist of π : there exists a sign $\varepsilon = \pm 1$ such that $\sigma(\pi)$ is the twist of π by the character $a \mapsto \varepsilon^{\deg(a)}$ of the idèle class group. As a consequence, for any

closed point $x \notin S_1$, the Hecke eigenvalue of π at x is $\varepsilon^{\deg(a)}$ times the Hecke eigenvalue of π at $\sigma(x)$.

The pair $(\mathbb{P}^1, S_1)$ admits an involution σ as in 7.9 as soon as S_1 does, that is, except in the case where S_1 contains a rational point and a point of degree 3.

Appendix. Transfer of special automorphic representations

Yuval Z. Flicker

The purpose of this appendix is to extract Statement 1.13 from the literature.

The correspondence, relating discrete spectrum automorphic representations π of any inner form G' of $G = \mathrm{GL}(n)$ (multiplicative group of a simple algebra of dimension n^2) with discrete spectrum representations of G , is known unconditionally when the base field is a number field F , by Arthur's work on the trace formula. The case where π and π' have a cuspidal ([BZ]) component at a place v where G'_v is $\mathrm{GL}(n, F_v)$ had been proven in [FK] (after previous work of [BDKV] and [F1] on π, π' with two such components), using the simple trace formula of [FK]. The latter method applies also when the base field F is a function field, but does not cover the case which we need, which concerns automorphic representations π' of $D^*(\mathbb{A})$, where D is a central division algebra over F , such that no component π'_v of π' corresponds to a cuspidal representation π_v of $G_v = \mathrm{GL}(n, F_v)$ by the local correspondence.

To establish the correspondence in the case stated in 1.13 we shall use the trace formula for $\mathrm{GL}(n)$ over a function field F as developed by Lafforgue [Laf], where the formula is proven for any inner form of $\mathrm{GL}(n)$. The case we need is where at two places $v = v_1, v_2$ (denoted 0 and ∞ in [Laf]) the test function $f = \otimes_v f_v$ (denoted h in [Laf]) has a discrete component f_v . A test (compactly supported locally constant) function f_v is called discrete if for every proper standard parabolic subgroup $P = MN$ of $G = \mathrm{GL}(n)$, with unipotent radical N and standard Levi subgroup M , it satisfies the identity

$$\int_{K_v} \int_{N_v} f_v(k_v^{-1} n_v k_v) dn_v dk_v = 0$$

for every $m \in M_v$. A discrete pseudo coefficient of the Steinberg representation is constructed in [Lau, Theorem (5.1.3), p. 133], after previous work by Kottwitz. Replacing f_v by $g \mapsto \int_{K_v} f_v(k_v^{-1} g k_v) dk_v$, $K_v = \mathrm{GL}(n, \mathcal{O}_v)$, we may assume f_v satisfies $f_v(k_v^{-1} g k_v) = f_v(g)$ ($k_v \in K_v, g \in G_v$). For f with a discrete component f_v the truncation ([Laf, pp. 225-227]) is trivial.

Theorem 10, p. 241, V.2.d, of [Laf] implies that for f with two discrete components the geometric side of the trace formula reduces to

$$\int_{Z_G(F) \backslash G(\mathbb{A}) / a^{\mathbb{Z}}} \sum_{\gamma} f(g^{-1} \gamma g) dg = \sum_{\gamma} \int_{Z_G(\gamma)(F) \backslash G(\mathbb{A}) / a^{\mathbb{Z}}} f(g^{-1} \gamma g) dg$$

where the sum on the left ranges over the elements γ in $G(F)$ whose characteristic polynomial is a power of an irreducible polynomial, while the sum on the right ranges over a set of representatives for the conjugacy classes of such γ in $G(F)$, and $Z_G(\gamma)$ denotes the centralizer of γ in G . To simplify things we may choose f with a component f_{v_3} which vanishes on the singular set (the set of γ with at least two equal eigenvalues). As explained in the last paragraph below, this does not reduce the applicability of our techniques. Then the γ in the sum are elliptic regular (regular: distinct eigenvalues): $F[\gamma]$ is a separable field extension of F of degree n .

This sum is equal to the analogous sum for the inner form G' , recorded in (2), p. 191, of [FK], for matching test functions $f = \otimes_v f_v$ on $G(\mathbb{A})$ and $f' = \otimes_v f'_v$ on $G'(\mathbb{A})$. In particular $f_v = f'_v$ at the v where $G_v \simeq G'_v$, and f_v, f'_v have matching orbital integrals (at all regular elements γ' in G'_v and the γ in G_v with the same characteristic polynomials; the orbital integral

of f'_v at the regular elements which do not come from G_v in this sense are zero, for all v). Note that the usage of Theorem 10, p. 241, V.2.d, of [Laf] is made for convenience. The method of “ n -admissible spherical functions” of [FK, p. 192], could be used too.

For a test function f with a cuspidal component (thus at some place v , for every proper parabolic subgroup $P_v = M_v N_v$ of G_v we have $\int_{N_v} f_v(xny) dn = 0$ for all $x, y \in G_v$), the convolution operator $r(f)$ splits through the projection to the cuspidal spectrum. The spectral side of the trace formula becomes the sum $\sum_{\pi} m(\pi) \operatorname{tr} \pi(f)$, where π ranges over the equivalence classes of the irreducible representations in the cuspidal spectrum of $G(\mathbb{A})$, and $m(\pi)$ denotes the multiplicity of π in the cuspidal spectrum ($m(\pi)$ is known to be 1 for $G = \operatorname{GL}(n)$). For a general test function f , which may not have a cuspidal component, one needs to use the spectral decomposition of the space of automorphic forms and compute the spectral side of the trace formula. This is done in [Laf, Theorem 12, p. 309, VI.2.f], in the case we need, namely $\operatorname{GL}(n)$ over a function field F .

In the number field case Arthur has shown (“a splitting property”) that for a test function $f = \otimes f_v$ with two discrete components the spectral side reduces to a discrete sum $\sum_{\pi} m(\pi) \operatorname{tr} \pi(f)$, where π ranges over the equivalence classes of the irreducible representations π in the discrete spectrum of $G(\mathbb{A})$, and $m(\pi)$ is the multiplicity of π in the discrete spectrum. In the function field case this has not yet been done, so we proceed differently (and in fact deduce this result).

Fix a new place u ($\neq v_1, v_2$) of F . Let the component f_u be spherical (K_u -biinvariant). Denote by $f_u^{\vee}$ the Satake transform of f_u . Then the spectral side, described by Theorem 12, p. 309, VI.2.f, of [Laf], has the form

$$\sum_i c_i f_u^{\vee}(t_i) + \sum_j \int_{\tilde{T}_j} c_j(t) f_u^{\vee}(t) d\mu_j(t)$$

where the c_i are complex numbers, the t_i lie in the compact Hausdorff space $\tilde{T}$ defined in the first lines of [FK, proof of Theorem 2, p. 197], the $\tilde{T}_j$ are compact submanifolds of $\tilde{T}$ (all irreducible components of a $\tilde{T}_j$ have the same dimension j ($1 \leq j < n$)), the $c_j(t)$ are complex valued functions on $\tilde{T}_j$ which are measurable with respect to a bounded measure $d\mu_j$ on $\tilde{T}_j$ which has the property that $\operatorname{vol}(\tilde{T}_j(\varepsilon)(t))/\varepsilon$ is bounded uniformly in ε (see [FK, first paragraph in the proof of the Proposition, p. 198]), and

$$\sum_i |c_i| + \sup_{t \in \tilde{T}_j} |c_j(t)| + \sum_j \int_{\tilde{T}_j} |c_j(t)| d\mu_j(t)$$

is finite.

Now the trace formula asserts that the spectral side equals the geometric side of the trace formula. As we saw above, the geometric side is a sum of orbital integrals (for f with discrete f_{v_1}, f_{v_2}). For matching test functions f and f' the geometric sides of the trace formula for f on $G(\mathbb{A}) = \operatorname{GL}(n, \mathbb{A})$ and for f' on $G'(\mathbb{A}) = D^*(\mathbb{A})$, are equal. Hence the spectral sides are equal. As is well known (see, e.g., [FK, Proposition, p. 191]), the spectral side in the anisotropic case (of $G' = D^*$) is discrete: has the form $\sum_i c'_i f_u^{\vee}(t'_i)$. We combine this last sum with the sum $\sum_i c_i f_u^{\vee}(t_i)$, for new c 's. The Proposition on p. 198 of [FK], prepared precisely for a situation as the present one, implies that all (new) c_i are zero (namely $c'_i = c_i$ and $t'_i = t_i$).

We conclude that for a test function $f = \otimes f_v$ with two discrete components, the spectral side, as described in Theorem 12, p. 309, VI.2.f, of [Laf], reduces to a discrete sum. Then for matching test functions $f = \otimes f_v$ on $G(\mathbb{A}) = \operatorname{GL}(n, \mathbb{A})$ and $f' = \otimes f'_v$ on $G'(\mathbb{A}) = D^*(\mathbb{A})$ we have the identity

$$\sum_{\pi} m(\pi) \operatorname{tr} \pi(f) = \sum_{\pi'} m(\pi') \operatorname{tr} \pi'(f'),$$

where the sum on the left (resp. right) ranges over the equivalence classes of the irreducible representations π (resp. π') in the discrete spectrum of $G(\mathbb{A})$ (resp. $G'(\mathbb{A})$), and $m(\pi)$ (resp. $m(\pi')$) signifies the multiplicity of π (resp. π') in the discrete spectrum.

A standard argument of “generalized linear independence of characters” implies that on fixing a representation $\pi_0^S = \otimes_{v \notin S} \pi_{0v}$ of $G(\mathbb{A}^S) = G'(\mathbb{A}^S)$, where S is the set of places of F such that $G'_v \simeq G_v$ for all $v \notin S$, the sums over π and π' can be taken to range over the subsets of π with $\pi_v = \pi_{0v}$ and π' with $\pi'_v = \pi_{0v}$ for all $v \notin S$. Rigidity theorem (“strong multiplicity one theorem”) implies that the sum over π reduces to at most one term (with $m(\pi) = 1$, by multiplicity one theorem for $\mathrm{GL}(n)$).

Since in our case D is a division algebra, a result of Godement-Jacquet can be used as in [F2] to show that the sum over π' is finite. As explained in [F2], using matching functions f_v on G_v and f'_v on G'_v for $v \in S$ which are supported only on the regular set (in particular we may use f_{v_3} as above), linear independence of characters implies character relations between π_v and π'_v for all $v \in S$. The character determines π_v and π'_v uniquely since it is locally integrable (a function field analogue of this result of Harish-Chandra in the characteristic 0 case was proven by Lemaire [Le]). This implies the correspondence which we need: if π is a cuspidal representation of $G(\mathbb{A})$ whose components at $v \in S$ are Steinberg twisted by an unramified character, there is precisely one (thus $m(\pi) = 1$) cuspidal representation π' of $G'(\mathbb{A})$ with $\pi'_v \simeq \pi_v$ for all $v \notin S$. Then π'_v corresponds to π_v by the local correspondence, thus π'_v is one dimensional unramified character, for all $v \in S$. Conversely, given cuspidal π' with $\dim \pi' > 1$ there exists a unique corresponding π .

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On complete reducibility in characteristic p

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in memoriam Vikram Mehta

Abstract

Let G be a reductive group over a field k which is algebraically closed of characteristic $p \neq 0$. We prove a structure theorem for a class of subgroup schemes of G when p is bounded below by the Coxeter number of G . As applications, we derive semi-simplicity results generalizing earlier results of Serre proven in 1998 and also obtain an analogue of Luna's étale slice theorem for suitable bounds on p .

Keywords. Group schemes, saturation, reductive, étale slice, Dynkin height, Coxeter number

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Titre. Complète réductibilité en caractéristique p

Résumé. Soit G un groupe réductif sur un corps k qui est algébriquement clos de caractéristique $p \neq 0$. Nous prouvons un théorème de structure pour une classe de schémas en sous-groupes de G lorsque p est minoré par le nombre de Coxeter de G . En guise d'applications, nous en déduisons des résultats de semi-simplicité qui généralisent des résultats antérieurs de Serre établis en 1998 et nous obtenons également un analogue du théorème de Luna (étales slice) pour des bornes convenables sur p .

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1 Introduction

We work over an algebraically closed field k of characteristic $p > 0$.

In [22], Serre showed that if semi-simple representations V_i of a group Γ are such that $\sum(\dim V_i - 1) < p$, then their tensor product is semi-simple. In [23], he more generally considers the case where Γ is a subgroup of $G(k)$, for G a reductive group, and where Γ is G -cr, meaning that whenever Γ is contained in a parabolic subgroup P , it is already contained in a Levi subgroup of P . For $G = \mathrm{GL}(V)$, this is equivalent to the semi-simplicity of the representations V_i of Γ . For a representation V of G , one can then ask under what conditions does V become semi-simple, when considered as a representation of Γ . In [23, Theorem 6, page 25], Serre shows this is the case when the Dynkin height $\mathrm{ht}_G(V)$ is less than p . For $G = \mathrm{GL}(V)$ and $V = \bigotimes V_i$, one has $\mathrm{ht}_G(V) = \sum(\dim V_i - 1)$.

In [10], the results of [22] were generalized to the case when the V_i are semi-simple representations of a group scheme $\mathcal{G}$. In this paper, we consider the case when $\mathcal{G}$ is a subgroup scheme of a reductive group G and generalize [23] (see 4.8) and [10] (see 2.3). As in [10], we first have to prove a structure theorem (2.3) on doubly saturated (see 2.2) subgroup schemes $\mathcal{G}$ of reductive groups G . The proof makes crucial use of a result of Zhiwei Yun on root systems. The appendix contains the result.

In Section 5, we consider a reductive group acting on an affine variety X and a point x of X whose orbit $G \cdot x$ is closed in X . We prove a schematic analogue of [2, Propositions 7.4, 7.6] under some conditions on the characteristic of k . More precisely, if X embeds in a G -module V of low height, then we obtain, as a consequence of 2.3(2), an analogue of Luna's étale slice theorem (5.1, 5.6). In [2] the language of schemes was not used and as a consequence the orbit $G \cdot x$ had to be assumed "separable". An orbit $G \cdot x$ is separable if and only if the stabilizer G_x is reduced.

2 Saturation and Infinitesimal saturation

2.1

Let G be a reductive algebraic group over k . Our terminology is that of [26]: reductive implies smooth and connected. By an algebraic group we will mean an affine group scheme of finite type over k . Fix a maximal torus T and a Borel subgroup containing T which determines a root system R and a set of positive roots R^+ . Let $\langle \cdot, \cdot \rangle$ be the natural pairing between the characters and co-characters and for each root α , let $\alpha^\vee$ be the corresponding coroot.

If the root system R associated to G is irreducible, the Coxeter number h_G of R and of G , admits the following equivalent descriptions:

1. It is the order of the Coxeter elements of the Weyl group W . This shows that R and the dual root system $R^\vee$ have the same Coxeter number.
2. Let $\tilde{\alpha}$ be the highest root and $\tilde{\alpha} = \sum n_i \alpha_i$ its expression as a linear combination of the simple roots. One has:

$$h_G = 1 + \sum n_i. \tag{1}$$

3. Applying this to the dual root system, one gets $h_G = \langle \rho, \tilde{\alpha}^\vee \rangle + 1$ where ρ is half the sum of positive roots and $\tilde{\alpha}^\vee$ is the highest coroot. Indeed, ρ is also the sum of the fundamental weights ϖ_i and $\langle \varpi_i, \alpha_j^\vee \rangle = \delta_{ij}$.

For a general reductive group G , define the Coxeter number h_G to be the largest among the ones for the irreducible components of R .

It follows from (2) above that if G is a reductive group, U the unipotent radical of a Borel subgroup, and $\mathfrak{u} := \mathrm{Lie}(U)$, the descending central series of $\mathfrak{u}$, defined by $Z^1 \mathfrak{u} = \mathfrak{u}$ and $Z^i \mathfrak{u} =$

$[\mathfrak{u}, Z^{i-1}\mathfrak{u}]$, satisfies

$$Z^{h_G}\mathfrak{u} = 0. \quad (2)$$

For the group U , similarly $Z^{h_G}U = (1)$.

The Lie algebra $\mathfrak{g}$ of G is a p -Lie algebra. Let $X \mapsto X^{[p]}$ denote the p -power operation on $\mathfrak{g}$. If $\xi \in \mathfrak{g}$ is nilpotent and if $p \geq h_G$, then $\xi^{[p]} = 0$. To check this, we may assume that ξ is in the Lie algebra $\mathfrak{u}$ of the unipotent radical $U \subset B$ of a Borel subgroup. For each positive root α , let X_α be a basis for the root subspace $\mathfrak{g}_\alpha$. Express ξ as $\sum_{\alpha \in R^+} a_\alpha X_\alpha$. Each X_α is the infinitesimal generator of an additive group. It follows that $X_\alpha^{[p]} = 0$ for each α . Observe that

$$\xi^{[p]} = \sum_{\alpha \in R^+} a_\alpha^p X_\alpha^{[p]} \pmod{Z^p \mathfrak{u}} \quad (3)$$

and $Z^p \mathfrak{u} = 0$, as $p \geq h_G$ (see [16, page 10]).

2.2

Let $\mathfrak{g}_{\text{nilp}}$ (resp. G^u) be the reduced subscheme of $\text{Lie}(G)$ (resp. G) with points the nilpotent (resp. unipotent) elements. Let U be the unipotent radical of a Borel subgroup. For $p \geq h_G$, the Campbell-Hausdorff group law makes sense in characteristic p and turns $\mathfrak{u} := \text{Lie}(U)$ into an algebraic group over k . This is so since $Z^p \mathfrak{u} = 0$. Further, there is a unique isomorphism

$$\exp : (\text{Lie}(U), \cdot) \xrightarrow{\sim} U \quad (4)$$

equivariant for the action of B and whose differential at the origin is the identity. If in addition the simply connected covering of the derived group of G is an étale covering (which is the case for $p > h_G$, and could fail when $p = h_G$ due to the presence of $\text{SL}(p)$ as factors in the covering), then there is a unique G -equivariant isomorphism ([23, Theorem 3, page 21] see Section 6 for details):

$$\exp : \mathfrak{g}_{\text{nilp}} \xrightarrow{\sim} G^u \quad (5)$$

which induces (4) on each unipotent radical of a Borel subgroup. Let $\log : G^u \rightarrow \mathfrak{g}_{\text{nilp}}$ denote its inverse.

For u a unipotent element of $G(k)$, one defines the “ t -power map” $t \mapsto u^t$, from $\mathbb{G}_a$ to G , by

$$t \mapsto \exp(t \log u) : \mathbb{G}_a \rightarrow G. \quad (6)$$

For $G = \text{GL}(V)$ such a map $t \mapsto u^t$ is more generally defined for any u in G such that $u^{[p]} = 1$. It is given by the truncated binomial expression [22, 4.1.1, page 524]:

$$u^t := \sum_{i < p} \binom{t}{i} (u - 1)^i. \quad (7)$$

Similarly, X in $\text{End}(V)$ such that $X^{[p]} = 0$ defines a morphism $t \mapsto \exp(tX)$ from $\mathbb{G}_a$ to $\text{GL}(V)$ given by the truncated exponential series:

$$t \mapsto I + tX + \frac{(tX)^2}{2!} + \cdots + \frac{(tX)^{p-1}}{(p-1)!}. \quad (8)$$

Until the end of §3, we make the following assumptions on the reductive group G .

Assumption 2.1. Let $\tilde{G}$ be the simply connected covering of the derived group G' . We assume that $p \geq h_G$ and that the map $\tilde{G} \rightarrow G'$ is étale.

In particular, by 2.1 and 2.2, the exponential map (5) is defined, every unipotent element in $G(k)$ is of order p , and every nilpotent in $\text{Lie}(G)$ is a p -nilpotent. One can then define the notions of saturation and infinitesimal saturation of subgroup schemes $\mathcal{G} \subset G$ as follows (see Remark 2.17 for the case when $G = \text{PGL}(p)$).

Definition 2.2 ([22, §4], [10, Definition 1.5]).

1. A subgroup scheme $\mathcal{G} \subset G$ is called *saturated* if for every u in $\mathcal{G}(k)$ which is unipotent, the homomorphism $t \mapsto u^t$ (6) from $\mathbb{G}_a$ to G factors through $\mathcal{G}$.
2. A subgroup scheme $\mathcal{G} \subset G$ is called *infinitesimally saturated* if for every nilpotent X in $\text{Lie}(G)$, the morphism $t \mapsto \exp(tX)$ (5) from $\mathbb{G}_a$ to G factors through $\mathcal{G}$.
3. $\mathcal{G}$ is *doubly saturated* if it is saturated and infinitesimally saturated.

An element of $\text{Lie}(\mathcal{G})$ is nilpotent if and only if it is nilpotent as an element of $\text{Lie}(G)$. The reference to the exponential map (5) in (2) therefore makes sense. One way to see that the notions of nilpotence for elements of $\text{Lie}(\mathcal{G})$ and $\text{Lie}(G)$ coincide is to observe that the inclusion of $\text{Lie}(\mathcal{G})$ in $\text{Lie}(G)$ is a morphism of p -Lie algebras and that X is nilpotent if and only if it is killed by an iterated p -power map, i.e. $X^{[p^\ell]} = 0$.

Let $\mathcal{G}^0$ be the identity component of $\mathcal{G}$ and $\mathcal{G}_{\text{red}}^0$ the reduced subscheme of $\mathcal{G}^0$.

Theorem 2.3. *Let $\mathcal{G} \subset G$ be a k -subgroup scheme which is infinitesimally saturated. Assume that if $p = h_G$, $\mathcal{G}_{\text{red}}^0$ is reductive. Then*

1. *The group $\mathcal{G}_{\text{red}}^0$ and its unipotent radical $R_u(\mathcal{G}_{\text{red}}^0)$ are normal subgroup schemes of $\mathcal{G}$ and the quotient group scheme $\mathcal{G}_{\text{red}}^0/R_u(\mathcal{G}_{\text{red}}^0)$ is of multiplicative type.*
2. *If $\mathcal{G}_{\text{red}}^0$ is reductive, there exists a central, connected subgroup scheme of multiplicative type $M \subset \mathcal{G}$ such that the morphism $M \times \mathcal{G}_{\text{red}}^0 \rightarrow \mathcal{G}$ realizes $\mathcal{G}$ as a quotient of $M \times \mathcal{G}_{\text{red}}^0$.*

We note that since k is assumed to be algebraically closed, a group scheme of multiplicative type is simply a diagonalisable group scheme.

The proof of part (2) of 2.3 will occupy most section 2, until 2.14. Part (1) will be proven in section 3.

By [10, Lemma 2.3], the conclusions of 2.3 hold for $\mathcal{G}$ if and only if they hold for the identity component $\mathcal{G}^\circ$.

Until the end of section 3, we will assume that $\mathcal{G}$ is connected.

Lemma 2.4. *If $\mathcal{G}$ is an infinitesimally saturated subgroup scheme of G , every nilpotent element of $\text{Lie}(\mathcal{G})$ is in $\text{Lie}(\mathcal{G}_{\text{red}})$.*

Proof. As $\mathbb{G}_a$ is reduced, the morphism $t \mapsto \exp(t\xi)$ maps $\mathbb{G}_a$ to $\mathcal{G} \subset \mathcal{G}_{\text{red}}$. Identifying $\text{Lie}(\mathbb{G}_a)$ with a copy of the ground field, the image of 1 in $\text{Lie}(\mathcal{G}_{\text{red}})$ is ξ . $\square$

Part (2) of 2.3 is a corollary of 2.4 and of the following theorem, which does not refer to $\mathcal{G}$ anymore.

Theorem 2.5. *Let G be a connected algebraic group such that*

- (a) *G_{red} is reductive,*
- (b) *any nilpotent element of $\text{Lie}(G)$ is in $\text{Lie}(G_{\text{red}})$.*

Then, the conclusion of 2.3(2) holds. As a consequence, G_{red} is a normal subgroup scheme of G and G/G_{red} is of multiplicative type.

Let T be a maximal torus of G_{red} , and let H be the centralizer of T in G . One has $H \cap G_{\text{red}} = T$. It follows from (b) that any nilpotent element of $\text{Lie}(H)$ is in $\text{Lie}(T)$, hence vanishes. By the following lemma, H is of multiplicative type, and in particular commutative.

Lemma 2.6. *Let H be a connected algebraic group over k . If all the elements of $\text{Lie}(H)$ are semi-simple, then H is of multiplicative type.*

Proof. (See also [11, IV, §3, Lemma 3.7].) $\text{Lie}(H)$ is commutative: Fix x in $\text{Lie}(H)$, and let us show that it is central in $\text{Lie}(H)$. As adx is semi-simple, it suffices to show that if y is in an eigenspace of adx , i.e. $[x, y] = \lambda y$, then x and y commute, i.e. $\lambda = 0$. Let W be the vector subspace of $\text{Lie}(H)$ generated by the $y^{[p^\ell]}$ ($\ell \geq 0$). The $y^{[p^\ell]}$ commute. The map $z \mapsto z^{[p]}$ therefore induces a p -linear map from W to itself, injective by assumption. This implies that W has a basis e_i ($1 \leq i \leq N$) consisting of elements such that $e_i^{[p]} = e_i$, and $(\sum a_i e_i)^{[p]} = \sum a_i^p e_i$. To see this, we view W as an algebraic group. The morphism $W \rightarrow W$ given by $x \mapsto x^{[p]} - x$ is étale. Let K be its kernel. In any basis it is defined by $d = \dim(W)$ equations of degree p , namely: $(x_i^{[p]} - x_i = 0; i = 1, 2, \dots, d$, with (the system of homogeneous equations $(x_i^{[p]}) = 0$ has an étale set of solutions and no solution at ∞ by assumptions). By Bezout, the group of solutions is hence a $(\mathbb{Z}/p)^d$. Let e_i be a basis for it. If the e_i were not linearly independent in W , there would be a linear dependence $\sum a_i e_i = 0$, with the $a_i \in k$ non-zero, involving a minimal number of e_i . We also have $\sum a_i^p e_i = 0$. Hence, by the minimality of the e_i , for some λ and $a_i = \lambda a_i^p$. Rescaling by $\lambda^{p-1} = 1$, we get a contradiction. $\square$

Lemma 2.7. *For $b = (b_i)$ in $k^\mathbb{N}$, define $b^{[p]} := (b_i^p)$. Then, any b in $k^\mathbb{N}$ is a linear combination of the $b^{[p^\ell]}$ for $\ell \geq 1$.*

Proof. The $b^{[p^a]}$ ($a \geq 0$) are linearly dependent. A linear dependence relation can be written

$$\sum_{j \geq m} c_j b^{[p^j]} = 0 \quad (9)$$

with $c_m \neq 0$. Extracting p^m -roots, we get

$$\sum_{j \geq 0} d_j (b^{[p^j]}) = 0 \quad (10)$$

where $d_j = c_{m+j}^{1/p^m}$. In particular, $d_0 \neq 0$, proving the lemma. $\square$

From the lemma above, y is a linear combination of the $y^{[p^\ell]}$ ($\ell > 0$).

End of proof of commutativity: The bracket $[y^{[p^\ell]}, x]$ vanishes for ($\ell > 0$). Indeed, it is $(\text{ady})^{[p^\ell]}(x)$ which vanishes because $[y, [y, x]] = [y, \lambda y] = 0$. It follows that $[y, x]$ vanishes too.

The p -Lie algebra $\text{Lie}(H)$ is commutative and hence its p -power operation $\text{Lie}(H) \rightarrow \text{Lie}(H)$ is injective. It therefore has a basis $e_i = e_i^{[p]}$. The dual of its restricted enveloping algebra is the affine bigebra of the kernel K of the Frobenius morphism $F : H \rightarrow H^{(p)}$, where $H^{(p)}$ is obtained from H by extension of scalars $k \rightarrow k$, $k \mapsto k^p$. It follows that K is a product of α_p 's.

The same holds for $H^{(p)}$, which is obtained from H using an automorphism of k . The same holds for each $H^{(p^\ell)}$ as well.

For any n , the kernel K_n of the iterated Frobenius map $F^n : H \rightarrow H^{(p^n)}$ is an iterated extension of subgroups of the kernel of the Frobenius of the $H^{(p^i)}$, $i < n$. It is hence of multiplicative type, being an iterated extension of connected groups of multiplicative type [?, SGA3, XVII, 7.1.1]

The K_n form an increasing sequence. By the proof of [10, Proposition 1.1], there exists a subgroup of multiplicative type M of H containing all K_n . As H is connected and as M contains all infinitesimal neighbourhoods of the identity element, one has $M = H$.

Whenever a group M of multiplicative type acts on a group K , its action on $\text{Lie}(K)$ defines a weight decomposition:

$$\text{Lie}(K) = \bigoplus_{\beta \in X(M)} \text{Lie}(K)_\beta. \quad (11)$$

If $v \in \text{Lie}(K)_\beta$, then $v^{[p]}$ is in $\text{Lie}(K)_{p\beta}$. Indeed, after any extension of scalars R/k , if m is in $M(R)$,

$$m(v^{[p]}) = (m(v))^{[p]} = (\beta(m)v)^{[p]} = \beta(m)^p v^{[p]}. \quad (12)$$

Lemma 2.8. *Let M be a multiplicative group acting on a group K and let (11) be the weight space decomposition for the corresponding action on $\mathrm{Lie}(K)$. If $\beta \in X(M)$ is not torsion, $\mathrm{Lie}(K)_\beta$ consists of nilpotent elements.*

Proof. Indeed, if β is not torsion, the $p^\ell \beta$ are all distinct and must vanish for $\ell \gg 0$. It follows that the elements of $\mathrm{Lie}(K)_\beta$ are nilpotent. $\square$

Let us apply this to the action of T on G by inner automorphisms.

Lemma 2.9. *If β in $X(T)$ is not zero, the weight spaces $\mathrm{Lie}(G)^\beta$ equals $\mathrm{Lie}(G_{\mathrm{red}})^\beta$.*

Proof. Indeed, $\mathrm{Lie}(G)^\beta$ consists of nilpotent elements. By our assumption 2.5(b), it is contained in $\mathrm{Lie}(G_{\mathrm{red}})$. $\square$

Let B a Borel subgroup of G_{red} containing T , and let U be its unipotent radical.

Lemma 2.10. *Under the assumption of 2.5, H normalizes U .*

Proof. Let C in $X(T) \otimes \mathbb{R}$ be the cone generated by the positive roots relative to B and define

$$C^* := C \cap [0, 1/h]. \quad (13)$$

As in 2.9, we let T act on G by conjugation (t acts by $g \mapsto tgt^{-1}$). This action induces actions on $\mathfrak{g} := \mathrm{Lie}(G)$, the affine algebra A of G , its augmentation ideal $\mathfrak{m}$ (defining the unit element), and the dual $\mathfrak{g}^\vee = \mathfrak{m}/\mathfrak{m}^2$ of $\mathfrak{g}$. Similarly T acts on U , its Lie algebra $\mathfrak{u}$, its affine algebra A_U and its augmentation ideal $\mathfrak{m}_U$. For the action on affine algebras, $t \in T$ transforms $f(g)$ into $f(t^{-1}gt)$. From these actions, we get $X(T)$ -gradings. By 2.5(b)

$$\mathfrak{u} = \bigoplus_{\beta \in C^*} \mathfrak{g}_\beta. \quad (14)$$

It follows that $\mathfrak{u}^\vee = \mathfrak{m}_U/\mathfrak{m}_U^2$ is the sum of the $(\mathfrak{g}^\vee)_\beta$, for β in the negative $-C^*$ of C^* . For $n > 0$, the weights by which T acts on $\mathfrak{m}_U^n/\mathfrak{m}_U^{n+1}$ are in nC^* . As U is connected, the intersection of the $\mathfrak{m}_U^n$ is 0. The ideal I of A generated by the graded components $\mathfrak{m}_U^\beta$ for $\beta \notin -nC^*$, defines a closed subscheme of G . For any n , the graded component $\mathfrak{m}_U^\beta$ is contained in I , the graded component A_0 of A (the T -invariants) is reduced to the constants. As T is connected it follows that U_1 is connected. As the image of $\mathfrak{m}_U^\beta$ in $\mathfrak{m}/\mathfrak{m}^2$ is $(\mathfrak{m}/\mathfrak{m}^2)^\beta$, the image of I in $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{g}^\vee$ is the orthogonal of $\mathfrak{u} \subset \mathfrak{g}$ and the tangent space at the origin of U_1 is $\mathfrak{u}$.

Claim: The subscheme U_1 of G is a subgroup scheme, i.e. the coproduct $\Delta : A \rightarrow A \otimes A$; $f(g) \mapsto f(gh)$ maps I to $I \otimes A + A \otimes I$. Indeed, Δ respects the gradings; for k the constants, one has $A = k \oplus \mathfrak{m}$, and Δ maps $\mathfrak{m}$ to $(k \otimes \mathfrak{m}) \oplus (\mathfrak{m} \otimes k) \oplus (\mathfrak{m} \otimes \mathfrak{m})$, and hence $\mathfrak{m}_\beta$ maps to

$$(k \otimes \mathfrak{m}_\beta) \oplus (\mathfrak{m}_\beta \otimes k) \oplus \bigoplus_{\beta = \beta' + \beta''} \mathfrak{m}_{\beta'} \otimes \mathfrak{m}_{\beta''}. \quad (15)$$

As $-C^*$ is stable by addition, if $\beta = \beta' + \beta''$ and that β is not in $-C^*$, one of β' or β'' is not in $-C^*$, and the corresponding $\mathfrak{m}_{\beta'}$ or $\mathfrak{m}_{\beta''}$ is contained in I . The claim follows.

To summarize, U_1 is connected, and the inclusion $U \subset U_1$ induces an isomorphism $\mathrm{Lie}(U) \xrightarrow{\sim} \mathrm{Lie}(U_1)$. As U is smooth, this implies that $U_1 = U$.

Since H centralizes T , the ideal I is stable by H , meaning that H normalizes U . $\square$

Corollary 2.11. *Under the assumption of 2.5, H normalizes G_{red} .*

Proof. Let B^- be the Borel subgroup of G_{red} containing T and opposite to B , and let U^- be its unipotent radical. As U^- , T and U are normalized by H , the big cell $U^- \cdot T \cdot U \subset G_{\mathrm{red}}$ is stable by the conjugation action of H , and so is its schematic closure G_{red} . $\square$

In what follows, we identify schemes with the corresponding fppf sheaves. A quotient such as G/H represents the quotient of the sheaf of groups G by the subsheaf H : G is a H -torsor over G/H .

Lemma 2.12. *The morphism of schemes*

$$G_{\text{red}}/T \rightarrow G/H \quad (16)$$

is an isomorphism.

Proof. As T is the intersection of G_{red} and H , the morphism (16), as a morphism of fppf sheaves, is injective. Testing on $\text{Spec}(k)$ and $\text{Spec}(k[\epsilon]/(\epsilon^2))$, one sees that it is bijective on points and injective on tangent space at each point. It is hence radicial and unramified. Therefore, on some open set of G_{red}/T , this morphism is an immersion. The G_{red} -homogeneity then shows that it is a closed embedding. As G_{red}/T is smooth and (16) is bijective on k -points, to prove that the closed embedding (16) is an isomorphism, it suffices to prove that at each point it induces an isomorphism of tangent spaces. By homogeneity, it suffices to check this at the origin. The tangent space at the origin of G_{red}/T is $\text{Lie}(G_{\text{red}})/\text{Lie}(T)$.

For the tangent space of G/H we proceed as follows. Let J be the ideal defining H in G and $\bar{\mathfrak{m}}$ be the ideal defining the origin in G/H . As G is a H -torsor on G/H , the pull-back of $\bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2$ to G is J/J^2 , and $\bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2$ is the fiber of J/J^2 at the origin e . By [10, 2.15] applied to T acting on G by conjugation, we have an isomorphism $(J/J^2)_e \xrightarrow{\sim} \bigoplus_{\beta \neq 0} (\mathfrak{m}/\mathfrak{m}^2)^\beta = (\text{Lie}(G_{\text{red}})/\text{Lie}(T))^\vee$. Taking duals, we see that (16) is an isomorphism near the origin, hence everywhere, proving 2.12. $\square$

Corollary 2.13. *Under the assumptions of 2.5, G_{red} is normal in G , and there exists in G a central connected subgroup scheme of multiplicative type M such that the morphism $M \times G_{\text{red}} \rightarrow G$ realizes G as a quotient of $M \times G_{\text{red}}$.*

Proof. By 2.12, the product map $G_{\text{red}} \times H \rightarrow G$ is onto, as a morphism of sheaves. As both G_{red} and H normalize G_{red} , so does G . $\mathcal{G} \subset G$.

To complete the proof of 2.13 (and thereby of 2.5), we follow [10, §2.25]. Let M be the subgroup of H which centralizes G_{red} (as always, in the scheme theoretic sense). By [11, Corollary 2.4(a), page 476], M is also multiplicative. Since G_{red} is reductive, the group scheme $\text{Aut}_T(G_{\text{red}})$ of automorphisms which preserve T is precisely T_{ad} , the image of T in the adjoint group. Hence the conjugation action of H on G_{red} gives the exact sequence:

$$1 \rightarrow M \rightarrow H \rightarrow T_{\text{ad}} \rightarrow 1 \quad (17)$$

surjects onto T_{ad} and T and M generate H . Since M is generated by $M \cap G_{\text{red}}$ and M^0 , and since H and G_{red} generate G we see that M^0 and G_{red} generate G . Moreover, M is central. Thus

$$M^0 \times G_{\text{red}} \rightarrow G \quad (18)$$

is an epimorphism. This concludes the proof of 2.5 and in particular, 2.3(2). $\square$

Lemma 2.14. *Suppose that H is a maximal connected subgroup scheme of multiplicative type of an algebraic group G . Let $Z^0(H)$ be the identity component of the centralizer of H in G and define $U_G := Z^0(H)/H$. Then the sequence:*

$$0 \rightarrow \text{Lie}(H) \rightarrow \text{Lie}(Z^0(H)) \rightarrow \text{Lie}(U_G) \rightarrow 0 \quad (19)$$

associated to the central extension

$$1 \rightarrow H \rightarrow Z^0(H) \rightarrow U_G \rightarrow 1 \quad (20)$$

is exact.

Proof. Left exactness of (19) is clear. The maximality of H implies that U_G is unipotent (see [10, §2.5, page 590]).

Embed H in $\mathbb{G}_m^r$ as a subgroup scheme. The quotient $H'' = \mathbb{G}_m^r/H$ is a torus, being a quotient of one. The central extension (20), by a pushforward, gives a central extension (see SGA3, Exposé XVII, Lemma 6.2.4)

$$1 \rightarrow \mathbb{G}_m^r \rightarrow E \rightarrow U_G \rightarrow 1 \quad (21)$$

and a diagram of groups:

$$\begin{array}{ccccccc} & & 1 & & 1 & & 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & H & \longrightarrow & Z^0(H) & \longrightarrow & U_G \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \parallel \\ 1 & \longrightarrow & \mathbb{G}_m^r & \longrightarrow & E & \longrightarrow & U_G \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & H'' & \xlongequal{\quad} & H'' & & 1 \\ & & \downarrow & & \downarrow & & \\ & & 1 & & 1 & & \end{array} \quad (22)$$

Since H'' is multiplicative every nilpotent in $\text{Lie}(E)$ maps to 0 in $\text{Lie}(H'')$ and hence comes from a nilpotent in $\text{Lie}(Z^0(H))$.

Since $\mathbb{G}_m^r$ is smooth, by [26, SGA3, Exposé VII, Proposition 8.2] the sequence (21) gives an exact sequence

$$0 \rightarrow \text{Lie}(\mathbb{G}_m^r) \rightarrow \text{Lie}(E) \rightarrow \text{Lie}(U_G) \rightarrow 0. \quad (23)$$

Since U_G is unipotent, any element z in $\text{Lie}(U_G)$ is nilpotent. Let z_0 in $\text{Lie}(E)$ be a lift of z . The Jordan decomposition makes sense for any p -Lie algebra over a perfect field k and uses only the p -power map (see for example [29, Corollary 4.5.9, page 135]). Thus, by using the Jordan decomposition of the lift z_0 in $\text{Lie}(E)$ and noting that the semi-simple part gets mapped to zero in $\text{Lie}(U_G)$, we can assume that z_0 can also be chosen to be nilpotent.

Since every nilpotent in $\text{Lie}(E)$ comes from a nilpotent in $\text{Lie}(Z^0(H))$, we conclude that z gets lifted to a nilpotent in $\text{Lie}(Z^0(H))$. This implies that (19) is also right exact. $\square$

Lemma 2.15. *Suppose that $\mathcal{G}$ is a subgroup scheme of G which is infinitesimally saturated in G . Then the subgroup scheme $Z_{\mathcal{G}}^0(H)$ is infinitesimally saturated in G ; in particular, every non-zero nilpotent in $\text{Lie}(Z_{\mathcal{G}}^0(H))$ lies in $\text{Lie}(Z_{\mathcal{G}}^0(H)_{\text{red}})$.*

Proof. If ξ in $\text{Lie}(Z_{\mathcal{G}}^0(H))$ is nilpotent, the map $\varphi : t \mapsto \exp(t\xi) : \mathbb{G}_a \rightarrow G$ factors through $\mathcal{G}$. Since H is central in $Z_{\mathcal{G}}^0(H)$, the action of H by inner automorphisms fixes ξ . Since the map “exp” is compatible with conjugation the entire curve φ is fixed by H . Therefore, φ factors through $Z_G(H)$ and hence through $Z_{\mathcal{G}}^0(H)_{\text{red}}$, since $\mathbb{G}_a$ is reduced and connected, and hence $\xi \in \text{Lie}(Z_{\mathcal{G}}^0(H)_{\text{red}})$. This completes the proof of the lemma. $\square$

We have the following extension of [10, Lemma 2.7, Corollary 2.11].

Lemma 2.16. *Let $\mathcal{G}$ be as in 2.15. Let $H \subset \mathcal{G}$ be a maximal connected subgroup scheme of multiplicative type. Then the central extension (20) splits and $U_{\mathcal{G}}$ is a smooth unipotent group.*

Proof. We claim that $U_{\mathcal{G}}$ is smooth. By Lemma 2.14, every element z in $\text{Lie}(U_{\mathcal{G}})$ comes from a nilpotent ξ in $\text{Lie}(Z_{\mathcal{G}}^0(H))$. By 2.15, ξ is in $\text{Lie}(Z_{\mathcal{G}}^0(H)_{\text{red}})$. Its image z is hence in $\text{Lie}(U_{\mathcal{G}_{\text{red}}})$. Thus, $\text{Lie}(U_{\mathcal{G}}) = \text{Lie}(U_{\mathcal{G}_{\text{red}}})$ proving the claim.

Now we have a central extension (20) with the added feature that $U_{\mathcal{G}}$ is smooth. By [26, SGA3, Exposé XVII, Theorem 6.1.1] it follows that (20) splits (uniquely) and

$$Z_{\mathcal{G}}^0(H) = H \times U_{\mathcal{G}}. \quad (24)$$

□

Remark 2.17. Recall that in 2.2, for the existence of an exponential map (5) we assumed that the covering morphism $\tilde{G} \rightarrow G'$ is étale, which therefore became a part of the standing assumption (2.1). The case which gets excluded is when the simply connected cover of the derived group has factors of $\mathrm{SL}(p)$. For instance, when G' is simple with $p = h_G$, the only case excluded is $G' = \mathrm{PGL}(p)$.

Let G be a reductive group for which $p = h_G$. The tables of Coxeter numbers of simple groups show that except for type A , where $h_{\mathrm{SL}(n)} = n$, Coxeter numbers are even greater than 2. It follows that G is a product of $\mathrm{SL}(p)$'s and other simple factors with p larger than their Coxeter numbers.

Even when the morphism $\tilde{G} \rightarrow G'$ is not étale, we can still define the notions of saturation (resp infinitesimal saturation) of subgroup schemes $\mathcal{G} \subset G$ as follows.

Say $\mathcal{G} \subset G$ is saturated (resp infinitesimally saturated) if the inverse image of $\mathcal{G}$ in $\tilde{G}$ is saturated (resp infinitesimally saturated). With this definition, Theorem 2.3 remains true for $p \geq h_G$.

This notion of saturation (resp infinitesimal saturation) can also be seen in terms of suitably defined t -power maps and exponential maps. We restrict ourselves to the case when $G' = \mathrm{PGL}(p)$.

At the level of Lie algebras, the induced morphism:

$$\mathfrak{sl}(p) \rightarrow \mathfrak{pgl}(p) \quad (25)$$

is a radicial map on the locus of nilpotent elements. If $A \in \mathfrak{gl}(p)$ is a matrix representing an element of $\mathfrak{pgl}(p)$, it is nilpotent if all but the constant coefficients of the characteristic polynomial vanishes, i.e. $\mathrm{Tr}(\wedge^i(A)) = 0$ for $i < p$ and the characteristic polynomial reduces to $T^p - \det(A)$. This condition is stable under $A \mapsto A + \lambda \cdot I$ as $(T - \lambda)^p - \det(A + \lambda \cdot I) = T^p - (\det(A) + \lambda^p) = T^p - \det(A)^{1/p} \cdot I$.

We get the unique lift $\tilde{A} \in \mathfrak{sl}(p)$ by taking $\tilde{A} := A - \det(A)^{1/p} \cdot I$. Now one has $\mathrm{Tr}(\wedge^i(\tilde{A})) = 0$ for $1 \leq i \leq p$ and hence we can define the exponential morphism $\mathbb{G}_a \rightarrow \mathrm{PGL}(p)$ as:

$$t \mapsto \exp(tA) := \exp(t\tilde{A}). \quad (26)$$

Likewise, in the case of unipotents in the group $\mathrm{PGL}(p)$, the restriction of this map to the locus of unipotent elements:

$$\mathrm{SL}(p)^u \rightarrow \mathrm{PGL}(p)^u \quad (27)$$

is radicial. Thus, any unipotent $u \in \mathrm{PGL}(p)$ has a unique unipotent lift $\tilde{u} \in \mathrm{SL}(p)$ and one can define the t -power map $\mathbb{G}_a \rightarrow \mathrm{PGL}(p)$ as:

$$u^t := \mathrm{image}(\tilde{u}^t). \quad (28)$$

With the notions of t -power map and exponential morphisms in place, we can define the notions of saturation (resp infinitesimal saturation) of subgroup schemes $\mathcal{G} \subset G'$ exactly as in Definition 2.2 using (28) (resp (26)) and these coincide with the definitions made above.

We note however that these “punctual” maps, i.e. defined for each nilpotent A (resp each unipotent u), are not induced by a morphism from $\mathfrak{pgl}(p)_{\mathrm{nilp}}$ to $\mathrm{PGL}(p)^u$ (resp. $\mathbb{G}_a \times \mathrm{PGL}(p)^u \rightarrow \mathrm{PGL}(p)^u$: $(t, u) \mapsto u^t$ is not a morphism).

3 Completion of proof of the structure theorem 2.3

We begin by stating a general result on root systems whose proof is given in the appendix.

Proposition 3.1. *Let R be an irreducible root system with Coxeter number h and let X be the lattice spanned by R . Let $\varphi : X \rightarrow \mathbb{R}/\mathbb{Z}$ be a homomorphism. Then there exists a basis B for R such that if $\alpha \in R$ satisfies $\varphi(\alpha) \in [0, 1/h) \bmod \mathbb{Z}$, then α is positive with respect to B .*

3.1

Assumption 2.1 on the reductive group G continues to be in force. Let $\mathcal{G}$ be a subgroup scheme of G , and H a maximal connected subgroup scheme of multiplicative type of $\mathcal{G}$. For the existence of H see [10, Proposition 2.1].

Let $X(H)$ be the group of characters of H . The action of H on $\mathrm{Lie}(G)$ by conjugation gives an $X(H)$ -gradation $\mathrm{Lie}(G) = \mathrm{Lie}(Z_G^0(H)) \oplus \bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)^\alpha$. Recall that if M is a connected group scheme of multiplicative type over k , the group of characters $X(M)$ is finitely generated and does not have torsion prime to p . As k is algebraically closed, this means that M is a product of subgroups isomorphic to $\mathbb{G}_m$ or μ_{p^j} with $j > 0$.

Corollary 3.2 (cf. [10, Lemma 2.12]). *Let M be a connected subgroup scheme of multiplicative type of G and suppose that $p > h_G$. The action of M on $\mathrm{Lie}(G)$ by conjugation gives an $X(M)$ -gradation $\mathrm{Lie}(G) = \bigoplus_{\alpha \in X(M)} \mathrm{Lie}(G)^\alpha$. If $\alpha \neq 0$, then for $\xi \in \mathrm{Lie}(G)^\alpha$ one has $\xi^{[p]} = 0$. In particular, if $M = H$ and $\xi \in \mathrm{Lie}(G)^\alpha$ we have $\xi^{[p]} = 0$.*

Proof. If α is of infinite order, it suffices to apply 2.8. If α is of finite order, there is a subgroup N of M isomorphic to μ_{p^j} to which α restricts non-trivially and it suffices to prove 3.2 for N and the restriction of α to N . Let us choose an isomorphism of $X(N)$ with $X(\mu_{p^j}) = \mathbb{Z}/p^j$ such that the image of α in $\mathbb{Z}/p^j$ is of the form p^b with $0 \leq b < j$.

In a smooth algebraic group, the maximal connected subgroups of multiplicative type are (maximal) tori. Indeed, if T is such a maximal subgroup, its centralizer $Z_G(T)$ is smooth, being the fixed locus of a linearly reductive group acting on a smooth variety (see for example [11, Theorem 2.8, Chapter II, §5]). Define $U := Z_G(T)^\circ/T$. The group scheme U is smooth, as a quotient of $Z_G(T)^\circ$ which is smooth, and is unipotent by maximality of T . By [26, SGA3, Exposé XVII, Theorem 6.1.1], we have a splitting $Z_G(T)^\circ \simeq T \times U$, and hence T is smooth.

In our case N is hence contained in a maximal torus T of G . Let

$$\varphi : X(T) \rightarrow \mathbb{R}/\mathbb{Z} \tag{29}$$

be the composite of $X(T) \rightarrow X(N) = \mathbb{Z}/p^j$ and of the inclusion $x \mapsto x/p^j$ of $\mathbb{Z}/p^j$ in $\mathbb{R}/\mathbb{Z}$.

The N -weight space $\mathrm{Lie}(G)^\alpha$ is the sum of T -weight spaces $\mathrm{Lie}(G)^\beta$ for β a root such that $\varphi(\beta) = 1/p^{j-b}$. As $0 < 1/p^{j-b} \leq 1/p < 1/h$ we may apply 3.1, to conclude that $\mathrm{Lie}(G)^\alpha$ is contained in the Lie algebra of the unipotent radical of a Borel subgroup of G . In particular, it consists of nilpotent elements, proving 3.2. $\square$

3.2

The proof of 2.3 now follows [10, pages 594–599] verbatim with a sole alteration; recall that in [10] the group G was the linear group $\mathrm{GL}(V)$ and the condition on the characteristic was $p > \dim(V)$. For an arbitrary connected reductive G , this condition now gets replaced by $p > h_G$, which makes 3.2 applicable.

Remark 3.3. If $G_i = \mathrm{GL}(V_i)$ one has $h_{G_i} = \dim(V_i)$. In the case $G = \prod \mathrm{GL}(V_i)$ with for each i , $p > \dim(V_i)$, the case $p > h_G$ of Theorem 2.3 gives us [10, Theorem 1.7].

Example 3.4 ((Brian Conrad)). Here is an example in any characteristic $p > 0$, of a connected group of multiplicative type M acting on a reductive group G , and of a non-trivial character α of M , such that the weight space $\text{Lie}(G)^\alpha$ contains elements which are not nilpotent. We take $G = \text{SL}_p$ (so that $h_G = p$) and $M = \mu_{p^2}$, and for α the character $\zeta \mapsto \zeta^{[p]}$.

We embed M into the maximal torus of diagonal matrices of SL_p by

$$\zeta \mapsto \text{diag}(\zeta^0, \zeta^{-p}, \zeta^{-2p}, \dots, \zeta^{-(p-1)p}).$$

The restriction to M of each simple root and of the lowest root is the character $\alpha : \zeta \mapsto \zeta^{[p]}$, and $\text{Lie}(G)^\alpha$ is the sum of the corresponding root spaces. In the standard visualization of SL_p this weight space inside $\text{Lie}(G) = \mathfrak{sl}_p$ is the span of the super-diagonal entries and the lower-left entry.

A sum of non-zero elements in those root lines contributing to $\text{Lie}(G)^\alpha$ is a $p \times p$ matrix $X \in \mathfrak{sl}_p$ which satisfies

$$X(e_1) = t_p e_p, \quad X(e_2) = t_1 e_1, \quad \dots, \quad X(e_p) = t_{p-1} e_{p-1}. \quad (30)$$

Iterating p times gives $X^p = \text{diag}(t, \dots, t)$, with $t := \prod t_j \neq 0$. Hence, $X \in \text{Lie}(G)^\alpha$ is not nilpotent.

Example 3.5. A variant of the above example leads to an example of an infinitesimally saturated group scheme $\mathcal{G} \subset \text{SL}(V)$ with $\dim(V) = p$ and such that V is an irreducible representation and $\mathcal{G}_{\text{red}}$ is a unipotent group. This in particular implies that $\mathcal{G}_{\text{red}}$ is not normal in $\mathcal{G}$.

Let V be the affine algebra of μ_p , that is

$$V = \mathcal{O}(\mu_p) := \frac{k[u]}{(u^{[p]} - 1)}. \quad (31)$$

The vector space V admits the basis $\{u^i \mid i \in \mathbb{Z}/p\}$.

The multiplicative group $\mathcal{O}(\mu_p)^*$ acts by multiplication on V . For $f \in \mathcal{O}(\mu_p)^*$, $f^{[p]}$ is constant. Define N_f to be the constant value of $f^{[p]}$. It is in fact the norm of f . The action of $\mathcal{O}(\mu_p)^*$ on V induces an action of the subgroup $N := \{f \in \mathcal{O}(\mu_p)^* \mid N_f = 1\}$.

On V we have also the action of $\mathbb{Z}/p$ by translations and this action normalizes the group $\mathcal{O}(\mu_p)^*$ and its subgroup N . Consider the group scheme:

$$\mathcal{G} := (\mathbb{Z}/p) \ltimes N. \quad (32)$$

We make a few observations on $\mathcal{G}$:

1. The group $\mathcal{G}$ is infinitesimally saturated.
2. In $\mathcal{G}$ consider the first factor $\mathbb{Z}/p$ and u_i , $i \in \mathbb{Z}/p$. These subgroups generate a subgroup H , which is a central extension of $\mathbb{Z}/p$ by $\mathbb{Z}/p$ by $\mathbb{Z}/p$.
3. The representation V is irreducible as an H -module. Indeed, a $\mathbb{Z}/p$ -submodule is generated by some of the u_i , and if non-zero and $\mathbb{Z}/p$ -stable, contains all of them. A fortiori, V is an irreducible $\mathcal{G}$ -module.
4. The reduced group $\mathcal{G}_{\text{red}}$ can be identified with the unipotent group $\{f \in \mathcal{O}(\mu_p)^* \mid f(1) = 1\}$.
5. The subgroup $\mathcal{G}_{\text{red}}$ is not normal as the point 1 of $\mathbb{Z}/p$ is not invariant by translations.

4 Semi-simplicity statements

Let G be a reductive group. Let C be an algebraic group and $\rho : C \rightarrow G$ be a morphism.

Definition 4.1 ([23, page 20]). One says that ρ is *cr* if, whenever ρ factorsthrough a parabolic P of G , it factors through a Levi subgroup of P .

When $G = \mathrm{GL}(V)$, ρ is *cr* if and only if the representation V of C is completely reducible (or equivalently, semi-simple) and hence the terminology.

The property of ρ being *cr* depends only on the subgroup scheme of G which is the (schematic) image of C . It in fact only depends on the image of C in the adjoint group G_{ad} .

Indeed, the parabolic subgroups of G are the inverse images of the parabolic subgroups of G_{ad} , and similarly for the Levi subgroups. A subgroup scheme $\mathcal{G}$ of G will be called *cr* if its inclusion in G is so.

For an irreducible root system R , let $\tilde{\alpha} = \sum n_i \alpha_i$ be the highest root and its expression as a linear combination of the simple roots. The characteristic p of k is called *good* for R if p is larger than each n_i . For a general root system R , p is good if it is so for each irreducible component of R .

Proposition 4.2. *Suppose C is an extension*

$$1 \rightarrow B \rightarrow C \rightarrow A \rightarrow 1 \quad (33)$$

*with A° of multiplicative type and $A = A^\circ$ a finite group of order prime to p , and suppose that p is good. Let $\rho : C \rightarrow G$ be a morphism. If the restriction of ρ to B is *cr*, then ρ is *cr*.*

We don't know whether the proposition holds without the assumption that p is good.

Proof. Let P be a parabolic subgroup, U its unipotent radical, and $\mathfrak{u}$ the Lie algebra of U . The parabolic P is said to be *restricted* if the nilpotence class of U is less than p . If P is a maximal parabolic corresponding to a simple root α , the nilpotence class of U is the coefficient of α in the highest root. It follows that p is good if and only if all the maximal parabolic subgroups are restricted.

By [21, Proposition 5.3] (credited by the author to Serre, see ??), if P is restricted, one obtains by specialization from characteristic 0, a P -equivariant isomorphism:

$$\exp : (\mathfrak{u}, \cdot) \xrightarrow{\sim} U \quad (34)$$

from $\mathfrak{u}$ endowed with the Campbell-Hausdorff group law, to U .

We will first show that whenever ρ factors through a restricted parabolic subgroup P as above, if the restriction of ρ to B factors through some Levi subgroup of P , then ρ itself factors through some Levi subgroup of P .

The group $U(k)$ acts on the right on the set $\mathcal{L}(k)$ of Levi subgroups of P by

$$u \in U(k) \text{ acts by } L \mapsto \nu^{-1} L \nu.$$

This action turns $\mathcal{L}(k)$ into a $U(k)$ -torsor. This expresses the fact that two Levi subgroups are conjugate by a unique element of $U(k)$. The group $P(k)$ acts on $\mathcal{L}(k)$ and on $U(k)$ by conjugation. This turns $\mathcal{L}(k)$ into an equivariant $U(k)$ -torsor.

We will need a scheme-theoretic version of the above. Fix a Levi subgroup L_o . Let $\mathcal{L}$ be the trivial U -torsor (i.e. U with the right action of U by right translations). We have the family of Levi subgroups $L_u := u^{-1} L_o u$ parametrized by $\mathcal{L} = U$. We let P act on U by conjugation, and on $\mathcal{L}$ as follows: $p = v\ell$ in $P = U \rtimes L$ acts on $\mathcal{L} = U$ by $u \mapsto v^{-1} \ell u \ell^{-1} v$. This turns $\mathcal{L}$ into an equivariant U -torsor. When we pass to k -points, and attach to u in $\mathcal{L}$ the Levi subgroup $u L_o u^{-1}$, we recover the previously described situation.

The morphism $\rho : C \rightarrow P$ turns $\mathcal{L}$ into an equivariant U -torsor. A point x of $\mathcal{L}$ corresponding to a Levi subgroup L_x is fixed by C (scheme-theoretically) if and only if ρ factors through L_x . This expresses the fact that a Levi subgroup is its own normalizer in P .

Let U_B be the subgroup of U fixed by B , for the conjugation action. If B has a fixed point in $\mathcal{L}$, the fixed locus $\mathcal{L}^B$ is a U_B -torsor. As B is a normal subgroup of C , C acts on $\mathcal{L}^B$ and U_B , and the action factors through A .

The isomorphism $\exp : (\mathfrak{u}_B, \cdot) \xrightarrow{\sim} U_B$ is compatible with the action of B by conjugation. Hence it induces an isomorphism from $(\mathfrak{u}_B)$ to U_B . Let $Z^i(\mathfrak{u}_B)$ be the central series of $\mathfrak{u}_B$ and define $Z^i(U_B) := \exp(Z^i(\mathfrak{u}_B))$. The isomorphism $\exp$ induces an isomorphism between the vector group $\mathrm{Gr}_i(\mathfrak{u}_B)$ and $Z^i(U_B)/Z^{i+1}(U_B)$, compatible with the action of A . On $\mathrm{Gr}_i(\mathfrak{u}_B)$, this action is linear.

The assumption on A amounts to saying that A is linearly reductive, that is, all its representations are semi-simple. Equivalently, if k is the trivial representation, any extension

$$0 \rightarrow V \rightarrow E \rightarrow k \rightarrow 0 \quad (35)$$

splits. Passing from E to $b^{-1}(1)$, such extensions correspond to A -equivariant V -torsors, and the extension splits if and only if A has a scheme-theoretic fixed point on the corresponding torsor.

Define $U_{B,i}$ to be $U_B/Z^i(U_B)$, and $\mathcal{L}_{B,i}$ to be the $U_{B,i}$ -torsor obtained from $\mathcal{L}$ by pushing by $U_B \rightarrow U_{B,i}$. We prove by induction on i that A has a fixed point on $\mathcal{L}_{B,i}$.

As $U_B \rightarrow U_{B,1}$ is trivial, the case $i = 1$ is trivial. If x is a fixed point of A in $\mathcal{L}_{B,i}$, the inverse image of x in $\mathcal{L}_{B,i+1}$ is an equivariant A -torsor on $\mathrm{Gr}_i(\mathfrak{u}_B) \simeq Z^i(U_B)/Z^{i+1}(U_B)$. By linear reductivity, A has a fixed point on the inverse image.

As the central descending series of $\mathfrak{u}$, and hence of $\mathfrak{u}_B$ terminates, this proves 4.2 for restricted parabolic subgroups.

We now prove 4.2 by induction on (the dimension) of G . Suppose that ρ factors through a proper parabolic subgroup P . As p is good, there exists a restricted proper parabolic Q containing P , and P is the inverse image by the projection $Q \rightarrow Q/R_u(Q)$ of a parabolic P' of $Q/R_u(Q)$. Let L be a Levi subgroup of Q through which ρ factors and let P_L be the parabolic subgroup of L obtained as the inverse image of P' by the isomorphism $L \xrightarrow{\sim} Q/R_u(Q)$. Levi subgroups of P_L are Levi subgroups of P , and it remains to apply the induction hypothesis to L , for which p is good too. $\square$

Remark 4.3. For several results related to 4.2 but in the setting of reduced subgroups, see [3, Theorem 3.10] and [4, Theorem 1.1 and Corollary 3.7].

Fix in the reductive group G , a maximal torus T , and a system of simple roots corresponding to a Borel subgroup B containing T . Let U be the unipotent radical of B .

Definition 4.4 ((cf. [12], [22], [14])). The *Dynkin height* $\mathrm{ht}_G(V)$ of a representation V of G is the largest among $\langle \lambda, \alpha^\vee \rangle$, for λ a weight for the action of T on V , $\alpha > 0$.

This notion and this terminology go back to Dynkin [12, pages 331–332] where it is called “height” as in [14], while in Serre ([22] and [23]), it is simply “ $n(V)$ ”. If V is an irreducible representation, with dominant weight λ_+ and smallest weight λ_- , it is the sum of the coefficients of $\lambda_+ - \lambda_-$, expressed as linear combination of the simple roots.

It follows that the product in $\mathrm{End}(V)$ of the action of $\mathrm{ht}_G(V) + 1$ elements of $\mathrm{Lie}(U)$ vanishes and for ξ nilpotent in $\mathrm{Lie}(G)$, one has $\xi^{\mathrm{ht}_G(V)+1} = 0$ in $\mathrm{End}(V)$.

4.1

The representation $\rho : G \rightarrow \mathrm{GL}(V)$ is said to be of *low height* if $p > \mathrm{ht}_G(V)$. By [23, Theorem 6, page 25], representations of low height are semi-simple. One can show that if G admits a

representation V of low height which is almost faithful, meaning that its kernel is of multiplicative type, then G satisfies the assumption 2.1. That $p \geq h_G$ results from the more precise statement that $\text{ht}_G(V) \geq h_G - 1$ ([24, (5.2.4), page 213]). For the property that $\tilde{G} \rightarrow G'$ is étale, one uses the fact that the non-trivial irreducible representation of $\text{PGL}(p)$ of the smallest height is the adjoint representation which is of height $2p - 2$.

We now assume that V is of low height, and the assumption 2.1 holds for G . It follows that any nilpotent ξ in $\text{Lie}(G)$ satisfies $\xi^{[p]} = 0$, and further, the exponential map (5) is defined. The image $d\rho(\xi)$ of ξ in $\text{Lie}(\text{GL}(V)) = \text{End}(V)$ also has a vanishing p th-power, hence $\exp(d\rho(\xi) \cdot t)$ is defined.

4.2

Compatibility: If ξ in $\text{Lie}(G)$ is nilpotent

$$\rho(\exp(t\xi)) = \exp(t d\rho(\xi)). \quad (36)$$

As a consequence, if $u^{[p]} = 1$ in G , one has

$$\rho(u^t) = \rho(u)^t. \quad (37)$$

The following theorem is a schematic analogue of [8], for p large enough.

Theorem 4.5. *Suppose that the reductive group G admits a low height almost faithful representation $\rho : G \rightarrow \text{GL}(V)$, and that $p > h_G$. Then, for any non-trivial unipotent subgroup U of G , there exists a proper parabolic subgroup P of G containing the normalizer $N_G(U)$ of U , and whose unipotent radical contains U .*

The condition $p > h_G$ implies that G satisfies the assumption 2.1. If G is simple simply connected, it implies the existence of an almost faithful low height representation except for the G of type F_4 , E_6 , E_7 or E_8 , in which case the lowest height of a non-trivial representation and the Coxeter number are respectively $16 > 12$, $16 > 12$, $27 > 18$ and $58 > 30$. For these groups, we do not know whether the conclusion of the theorem is valid assuming only $p > h_G$.

Proof. Let V^U be the invariants of U acting on V . It is not zero, because U is unipotent. It is not V , because the representation V is almost faithful, hence faithful on U . It does not have a U -stable supplement V' in V , because U would have invariants in V' .

Let H be the subgroup scheme of G which stabilizes V^U . It contains the normalizer $N_G(U)$ of U . It is a doubly saturated subgroup scheme of G . Indeed, if h in $H(k)$ is of order p , by (37), $\rho(h^t) = \rho(h)^t = \sum_{i < p} \binom{t}{i} (\rho(h) - 1)^i$, which stabilizes V^U , and similarly if ξ in $\text{Lie}(H)$ is nilpotent, the $\exp(t\xi)$ are in H .

As $p > h_G$, Theorem 2.3 ensures that H_{red}^0 is a normal subgroup scheme of H and that the quotient H/H_{red}^0 is an extension of a finite group of order prime to p by a group of multiplicative type. It follows that $U \subset H_{\text{red}}^0$.

Lemma 4.6. H_{red}^0 is not reductive.

Proof. If it were, V would be a representation of low height of H_{red}^0 ([23, Corollary 1, page 25]), hence a semi-simple representation of H_{red}^0 , and V^U would have in V a H_{red}^0 -stable supplement. As V^U does not admit a supplement stable under $U \subset H_{\text{red}}^0$, this is absurd. $\square$

Proof of 4.5 continued: If S is a doubly saturated subgroup scheme of G , we will call $R_u(S_{\text{red}}^0)$ the unipotent radical of S and denote it simply by $R_u(S)$. By 2.3, it is a normal subgroup of S and $S/R_u(S^0)$ does not contain any normal unipotent subgroup. This justifies the terminology.

Define $U_1 := R_u(H)$. By 4.6, it is a non-trivial unipotent subgroup of G , and we can iterate the construction. We define for $i \geq 1$

$$\begin{aligned} H_i &:= \text{stabilizer of } V^{U_i} \\ U_{i+1} &:= R_u(H_i). \end{aligned}$$

One has $U_i \subset H_{i,\text{red}}^0$ and

$$U_i \subset N_G(U_i) \subset H_i \subset N_G(U_{i+1}) \subset H_{i+1} \subset \cdots \quad (38)$$

The H_i form an increasing sequence of smooth connected subgroups of G . It stabilizes, hence so do the sequences of the $U_{i,\text{red}}$ and of the H_i . If $H_i = H_{i+1}$, one has $H_i = N_G(U_{i+1}) = H_{i+1}$ and $U_{i+1} = R_u(H_i) = R_u(H_{i+1}) = R_u(N(U_{i+1})_{\text{red}}^\circ)$.

By [8, Proposition 2.3, page 99] (or for example [13, Section 30.3, Proposition on page 186]), this implies that $N(U_{i+1}) = (H_{i+1})_{\text{red}}$ is a proper parabolic subgroup of G . Call it Q . A parabolic subgroup of G is its own normalizer scheme (cf. [26, SGA3, XII, 7.9], [9, page 469]). As $(H_{i+1})_{\text{red}} = Q$ is normal in H_{i+1} , it follows that $H_{i+1} = Q$ and that $N_G(U) \subset Q$.

A Levi subgroup L of Q is a reductive subgroup of G , hence satisfies the assumptions of 4.5. It is isomorphic to $Q/R_u(Q)$. If U is not contained in $R_u(Q)$, we can repeat the argument for the image $\bar{U}$ of U in $Q/R_u(Q)$, which is isomorphic to L . One obtains a proper parabolic subgroup of $Q/R_u(Q)$ which contains the normalizer of $\bar{U}$. Its inverse image in Q is a parabolic subgroup, properly contained in Q and containing the normalizer of U . Iterating, one eventually finds a parabolic P containing $N_G(U)$ and such that $U \subset R_u(P)$. $\square$

Remark 4.7. Let $\tilde{G}$ be the simply connected central extension of the derived group G' , and let $\tilde{G}_i$ be its simple factors: $\tilde{G} = \prod \tilde{G}_i$. A representation of low height V of $\tilde{G}$ is almost faithful if and only if its restriction to each $\tilde{G}_i$ is not trivial. It suffices to check this for each $\tilde{G}_i$ separately. Thus we may assume $\tilde{G}$ is simply connected. The existence of a non-trivial V of low height implies that $p > 2$ for G of type B_n, C_n ($n \geq 2$) or F_4 and $p > 3$ for G of type G_2 . Let $\bar{G}$ be the image of $\tilde{G}$ in $\text{GL}(V)$.

If V is non-trivial, $u : \tilde{G} \rightarrow \bar{G}$ is an isogeny. We want to show that it is a central isogeny. If it is not, the structure of isogenies ([26, SGA3, XXII, 4.2.13]) shows that $\ker(u)$ contains the kernel of the Frobenius. The weights of V are the p th-powers and $\text{ht}_G(V)$ is a multiple of p , contradicting the low height assumption.

4.3

If G is a reductive group for which assumption 2.1 holds, and $\mathcal{G}$ is a subgroup scheme of G , the double saturation $\mathcal{G}^*$ of $\mathcal{G}$ is the smallest doubly saturated subgroup scheme of G containing $\mathcal{G}$. It is the intersection of the doubly saturated subgroup schemes containing $\mathcal{G}$, and is obtained from $\mathcal{G}$ by iterating the construction of taking the group generated by $\mathcal{G}$, the additive groups $\exp(t\xi)$ for ξ nilpotent in $\text{Lie}(\mathcal{G})$ and u^t for u of order p in $\mathcal{G}(k)$.

Corollary 4.8. *Let V be a low height almost faithful representation of a reductive group G . Assume that $p > h_G$. Let $\mathcal{G}$ be a subgroup scheme of G , and let $\mathcal{G}^*$ be its double saturation. Then the following conditions are equivalent:*

- (i) V is a semi-simple representation of $\mathcal{G}$
- (ii) V is a semi-simple representation of $\mathcal{G}^*$
- (iii) $\mathcal{G}$ is cr in G
- (iv) $\mathcal{G}^*$ is cr in G

(v) the unipotent radical of $(\mathcal{G}^*)_{\text{red}}^0$ is trivial.

Proof. (i) $\Leftrightarrow$ (ii): If W is a subspace of V , the stabilizer in G of W is doubly saturated, as we saw in the beginning of the proof of 4.5. If $\mathcal{G}$ stabilizes W , it follows that $\mathcal{G}^*$ also stabilizes W : the lattice of sub-representations of V is the same for $\mathcal{G}$ and $\mathcal{G}^*$, hence the claim.

(iii) $\Leftrightarrow$ (iv): Similarly, the parabolic subgroups of G and their Levi subgroups are doubly saturated, hence contain $\mathcal{G}$ if and only if they contain $\mathcal{G}^*$.

not (v) $\Rightarrow$ not (ii): Let U be the unipotent radical of $(\mathcal{G}^*)_{\text{red}}^0$. If it is non-trivial, $V^U \neq V$, because V is faithful on U and does not have a U -stable supplement. As U is normal in $\mathcal{G}^*$ (2.3), V^U is a sub-representation for the action of $\mathcal{G}^*$ on V . This contradicts the semi-simplicity of V .

(v) $\Rightarrow$ (ii): The representation V of the reductive group $(\mathcal{G}^*)_{\text{red}}^0$ is of low height, hence semi-simple. By 2.3, $(\mathcal{G}^*)_{\text{red}}^0$ is a normal subgroup of $\mathcal{G}^*$ and the quotient A is linearly reductive. If W is a sub- $\mathcal{G}^*$ -representation of V , A acts on the affine space of $(\mathcal{G}^*)_{\text{red}}^0$ -invariant retractions $V \rightarrow W$. It has a fixed point, whose kernel is a supplement to W .

not (v) $\Rightarrow$ not (iv): Let U be the unipotent radical of $(\mathcal{G}^*)_{\text{red}}^0$. If it is not trivial, there exists a parabolic P containing its normalizer, hence $\mathcal{G}^*$, and the unipotent radical of P contains U (4.5). Thus, no Levi subgroup of P can contain $\mathcal{G}^*$.

(v) $\Rightarrow$ (iv): By [23, Theorem 7, page 26], $(\mathcal{G}^*)_{\text{red}}^0$ is cr in G , and one applies 4.2. $\square$

Corollary 4.9. *Let $\rho : G \rightarrow \text{GL}(V)$ be an almost faithful low height representation, and let v in V be an element such that the G -orbit of v in V is closed. Then there exists a connected multiplicative central subgroup scheme $M \subset G_v$ and a surjective homomorphism $M \times G_{v,\text{red}}^0 \rightarrow G_v^0$.*

Proof. The orbit being closed in V and hence affine, the reduced stabilizer $G_{v,\text{red}}$ is reductive ([6]). Since $p \geq h_G$ and since stabilizers are doubly saturated and 2.3(2) holds for $\mathcal{G} = G_v$, we get the required result. $\square$

We now observe that the results of [10, Section 6] can be obtained as a consequence of 4.8. Note that by the remarks in [10, page 607] it suffices to prove the semi-simplicity results in the case when k is algebraically closed.

Theorem 4.10. *Let G be an algebraic group. Let $(V_i)_{i \in I}$ be a finite family of semi-simple G -modules and let m_i be integers ≥ 0 . If*

$$\sum m_i(\dim V_i - m_i) < p \tag{39}$$

the G -module $\bigotimes_j \bigwedge^{m_j} V_j$ is semi-simple.

Proof. Let $G = \prod \text{GL}(V_j)$ and $V = \bigotimes \bigwedge^{m_j} V_j$. Then, (39) is simply the inequality $p > \text{ht}_G(V)$. Replacing G by its image in G we may and shall assume that G is a subgroup scheme of G . Since V_i are semi-simple G -modules, it follows that G is G -cr, for $G = \prod \text{GL}(V_j)$. By [10, §6.2], we may also assume that $p > \dim(V_j) = h_{\text{GL}(V_j)}$. Hence by working with the image of G (and G) in $\text{GL}(V)$ and applying 4.8, we conclude that V is semi-simple as a G -module. $\square$

Remark 4.11. If (V_i, q) is a non-degenerate quadratic space with $\dim V_i = 2d_i$ on which G acts by similitudes, then by passing to a subgroup of index at most 2 and mapping to the group of similitudes rather than $\text{GL}(V)$, one can replace the term $m_i(\dim V_i - m_i)$ by $m_i(\dim V_i - m_i - 1)$, when $m_i \leq 2d_i$.

Complete reducibility in the classical case

By 4.1, a subgroup scheme $\mathcal{G} \subset G$ is called cr if for every parabolic subgroup $P \supset G$ containing G , there exists an opposite parabolic subgroup P' such that $G \subset P \cap P'$. Suppose that $\text{char}(k) \neq 2$ and let G be $\text{SO}(V)$ (or $\text{Sp}(V)$ in any characteristic), relative to a non-degenerate symmetric or alternating bilinear form B on V . In this situation, the notion of cr can be interpreted as follows: $\mathcal{G}$ is cr in G if and only if for every $\mathcal{G}$ -submodule $W \subset V$ which is totally isotropic, there exists a totally isotropic $\mathcal{G}$ -submodule W' of the same dimension, such that the restriction of B to $W + W'$ is non-degenerate (cf. [24, Example 3.3.3, page 206]).

Lemma 4.12. *Let the subgroup scheme $\mathcal{G}$ of G be cr. Then the $\mathcal{G}$ -module V is semi-simple and conversely.*

Proof. Let $W \subset V$ be a $\mathcal{G}$ -submodule. Then we need to produce a $\mathcal{G}$ -complement.

Consider $W_1 := W \cap W^\perp$. If $W_1 = (0)$, then $W \oplus W^\perp = V$ and we are done. So let $W_1 \neq (0)$. Then W_1 is a $\mathcal{G}$ -submodule which is totally isotropic and hence by the cr property, we have a totally isotropic $\mathcal{G}$ -submodule W_{10} of the same dimension as W_1 , such that the form B is non-degenerate on $W_1 + W_{10}$.

In particular, $W_1 \cap W_{10} = (0)$. Since $W_1 \subset W^\perp$, we see that $W \subset W_1^\perp$.

Let $w \in W_2 := W \cap W_{10}^\perp \cap W_1^\perp \cap W_{10}$ and suppose $w \neq 0$. Since $w \in W_{10}$, there exists $w' \in W_1$ such that $B(w, w') \neq 0$. On the other hand, since $w \in W_2 \subset W_1^\perp$, $B(w, v) = 0$ for all $v \in W$ and in particular $B(w, w') \neq 0$ which contradicts the assumption that $w \neq 0$. Hence it follows that $W \cap W_{10} = (0)$.

Thus, $W \oplus W_{10} \subset V$ is a $\mathcal{G}$ -submodule. We proceed similarly and get $X_1 = X \cap X^\perp$ such that $X \oplus X_{10} \subset V$. If $X_1 = (0)$, then $V = X \oplus X^\perp$, so get a $\mathcal{G}$ -decomposition of V as $W \oplus W_{10} \oplus \cdots$, that is a $\mathcal{G}$ -complement of W .

Conversely, let V be semi-simple as a $\mathcal{G}$ -module. Let $W \subset V$ be a totally isotropic $\mathcal{G}$ -submodule of $\dim(W) = d$. Note that $d < \dim(V)/2$. We therefore have a $\mathcal{G}$ -submodule $Z \subset V$ such $V = W \oplus Z$. The non-degenerate form B gives a $\mathcal{G}$ -equivariant isomorphism $\varphi : V \xrightarrow{\sim} V^* = W^* \oplus Z^*$ and since W is totally isotropic $\varphi(W) \cap W^* = (0)$. Hence $\varphi(W) \subset Z$.

Again, since W is totally isotropic, the restriction of B to Z is non-degenerate and hence we get an isomorphism $\psi : Z^* \xrightarrow{\sim} Z$. Define $W^0 := \psi(\varphi(W))$.

Then it is easily seen that W^0 is of dimension d and also totally isotropic $\mathcal{G}$ -invariant submodule of V . Finally, B is non-degenerate on $W \oplus W^0$. Hence $\mathcal{G}$ is cr in G . $\square$

5 Étale slices in positive characteristics

Let G act on an affine variety X and let x be a point of X whose orbit $G \cdot x$ is closed in X . We prove a schematic analogue of Luna's étale slice theorem (5.1, 5.6) for suitable bounds on the characteristic p . In order to make this precise, one needs to extend the notion of a slice in [2, Definition 7.1] to a scheme theoretic setting (see 5.6) and in the process one has to work with schematic stabilizers of closed orbits. The key issue then becomes the semi-simplicity of the tangent space to x in X as a representation of the stabilizer of G at x . Here the structure theorem 2.3 become crucial and the bounds on p are forced as a consequence. In [2] the scheme-theoretic aspects are eschewed forcing them to make the assumption that the orbit $G \cdot x$ is "separable", i.e. the stabilizer G_x is reduced. The bounds on the characteristic therefore do not show up in [2].

We begin this section with the following (linear) analogue of the Luna étale slice theorem in positive characteristics.

Theorem 5.1. *Let V be a G -module with low height i.e. such that $p > \text{ht}_G(V)$. Let v in V be an element such that the orbit $G \cdot v$ is a closed orbit in V . Then there exists a G_v -invariant*

linear subspace S of V giving rise to a commutative diagram:

$$\begin{array}{ccc} G \times^{G_v} S & \xrightarrow{\varphi} & V \\ f \downarrow & & \downarrow q \\ (G \times^{G_v} S) // G & \xrightarrow{\ell} & V // G \end{array} \quad (40)$$

and G -equivariant open subsets $U \subset (G \times^{G_v} S)$ containing the closed orbit $G \cdot v$ and an open subset U_0 of V containing v , for which (40) induces a cartesian diagram

$$\begin{array}{ccc} U & \xrightarrow{\varphi} & U_0 \\ f|_U \downarrow & & \downarrow q|_{U_0} \\ U // G & \xrightarrow{\ell|_U} & U_0 // G \end{array} \quad (41)$$

such that the morphism $\ell|_U$ is étale.

Remark 5.2. The above theorem was stated in the note [18] whose proof contained serious gaps (as was pointed to the authors by Serre in a private correspondence).

We note that G/G_v is constructed in [11, III, Proposition 3.5.2]. It represents the quotient in the category of fppf sheaves. Furthermore if $\rho_v : G \rightarrow V$; $g \mapsto g \cdot v$, the image $\text{im}(\rho_v)$, as a locally closed sub-scheme of V with its reduced scheme structure, can be identified with the scheme G/G_v . We call this locally closed sub-scheme, the orbit $G \cdot v$ and have the identification $G/G_v \simeq G \cdot v$ (see also [11, Proposition and Definition 1.6, III, §3, page 325]).

Proposition 5.3. *Let V be an arbitrary G -module. Let $v \in V$ and suppose that there exists a G_v -submodule S of the tangent space $T_v(V)$, such that $T_v(V)$ splits as $T_v(G \cdot v) \oplus S$. Let G act on $G \times S$ by $h : (g, s) \mapsto (hg, s)$. Then the G -morphism $\Phi : G \times S \rightarrow V$ given by $(g, s) \mapsto g \cdot v + g \cdot s$ descends to a G -morphism*

$$\varphi : G \times^{G_v} S \rightarrow V \quad (42)$$

which is étale at $(e, 0)$, e being the identity of G .

Proof. To ensure that Φ descends to a morphism φ , we need to check for every commutative k -algebra A , that Φ is constant on all $G_v(A)$ -orbits. For simplicity of exposition we will suppress the A .

Let γ in G_v act on $G \times S$ by $\gamma : (g, s) \mapsto (g \cdot \gamma, \gamma^{-1} \cdot s)$. Observe that $\Phi \circ \gamma : (g, s) \mapsto \Phi(g \cdot \gamma, \gamma^{-1} \cdot s) = g \cdot \gamma \cdot v + g \cdot \gamma \cdot \gamma^{-1} \cdot s = g \cdot v + g \cdot s$ (since γ fixes v). Therefore it is constant on the G_v -orbits. Since the action of G_v on $G \times S$ is scheme-theoretically free, Φ descends to a morphism $\varphi : G \times^{G_v} S \rightarrow V$.

Clearly the actions of G and G_v on $G \times S$ commute and hence the descended morphism is also a G -morphism.

Observe that the quotient morphism $G \rightarrow G/G_v$ is a torsor for the group scheme G_v , locally trivial under the fppf topology. Since the action of G_v on S is linear, we see that the associated fibre space $G \times^{G_v} S \rightarrow G/G_v$ is a locally free sheaf of rank $= \dim(S)$. In particular, $G \times^{G_v} S$ is a smooth k -scheme of finite type. Observe further that under the morphism φ , the zero section of the vector bundle $\pi : G \times^{G_v} S \rightarrow G/G_v$ canonically maps onto the orbit $G \cdot v \subset V$, while the fibre $\pi^{-1}(e \cdot G_v)$ of the identity coset $e \cdot G_v : G/G_v$ maps isomorphically to the affine subspace $S + v \subset V$.

Since $T_v(V) = T_v(G \cdot v) \oplus S$ by assumption, it follows that the differential $d\varphi_z$, at $z = (e, 0)$, is an isomorphism.

We now apply [10, Lemme 2.9], to conclude that the morphism $\varphi : G \times^{G_v} S \rightarrow V$ is étale at $z = (e, 0)$. $\square$

Proposition 5.4. *Let V be a G -module such that $p > \text{ht}_G(V)$, and let v in V be an element such that the G -orbit of v in V is closed. Then there exists a G_v -submodule $S \subset V$ such that $V = T_v(G \cdot v) \oplus S$ as a G_v -module. In particular, the consequences of 5.3 hold.*

Proof. One knows that G_v^0 is the kernel of the natural map $G_v \rightarrow \pi_0(G_v)$, to the group of connected components, and hence is a normal subgroup of finite index. Further, we note that $G_v = G_v^0 = G_{v,\text{red}} \cdot |G_v/G_v^0|$. Note also that since G_v is a saturated subgroup of G , by [23, page 23, Property 3] the index $|G_v/G_{v,\text{red}}^0|$ is prime to p .

Since $G \cdot v$ is a closed orbit by 4.9 (and 2.3) we have an exact sequence:

$$1 \rightarrow G_{v,\text{red}}^0 \rightarrow G_v^0 \rightarrow Q \rightarrow 1 \quad (43)$$

where Q is a multiplicative group scheme.

Now V is semi-simple as a $G_{v,\text{red}}^0$ -module, $G_{v,\text{red}}^0$ being a saturated subgroup of G and also as a Q -module since it is multiplicative (and hence linearly reductive). Thus by [10, Lemme 4.2] V is semi-simple as a G_v^0 -module and therefore as a G_v -module as well.

In particular, we have a G_v -supplement S for the G_v -invariant subspace $T_v(G \cdot v) \subset T_v(V) = V$, i.e. we have a G_v -decomposition $S \oplus T_v(G \cdot v)$ for V . $\square$

We recall the ‘‘Fundamental lemma of Luna’’ which holds in positive characteristics as well and which is essential to complete the proof of Theorem 5.1.

Lemma 5.5 ([20, page 152]). *Let $\bar{s} : X \rightarrow Y$ be a G -morphism of affine G -schemes. Let $F \subset X$ be a closed orbit such that:*

1. $\bar{s}$ is étale at some point of F
2. $\bar{s}(F)$ is closed in Y
3. $\bar{s}$ is injective on F
4. X is normal along F .

Then there are affine G -invariant open subsets $U \subset X$ and $U_0 \subset Y$ with $F \subset U$ such that $\bar{s} : U \rightarrow U_0$ is étale and $(\bar{s}, p_U) : U \rightarrow U_0 \times_{U_0//G} U//G$ is an isomorphism (where $p_U : U \rightarrow U//G$ is the quotient morphism).

Proof of 5.1. As a first step, we need to show that the categorical quotient $G \times^{G_v} S//G$ exists as a scheme. As we have seen in 5.3, action of G on $G \times S$ (via $g : (a, s) = (ga, s)$) and the (twisted) action of G_v on $G \times S$ commute and hence

$$k[G \times^{G_v} S]^G = (k[G \times S]^{G_v})^G = (k[G \times S]^G)^{G_v} = k[S]^{G_v}. \quad (44)$$

Thus it is enough to show that $k[S]^{G_v}$ is finitely generated and we would have

$$G \times^{G_v} S//G \simeq S//G_v. \quad (45)$$

As we have observed in the proof of 5.4, $G_v^0 \subset G_v$ is a normal subgroup of finite index. Thus, the finite generation of $k[S]^{G_v}$ is reduced to checking finite generation of $k[S]^{G_v^0}$. We may therefore replace G_v by G_v^0 in the proof. Further, since $G \cdot v$ is a closed orbit, $G_{v,\text{red}}$ is reductive. Therefore, by 4.9, we have a surjection $M \times G_{v,\text{red}} \rightarrow G_v^0$. Since M in G_v is central $k[S]^{G_v^0} = k[S]^{M \times G_{v,\text{red}}}$.

Again, since $G_{v,\text{red}}$ is reductive, the ring of invariants $k[S]^{G_{v,\text{red}}}$ is a finitely generated k -algebra. Since M is a group of multiplicative type over an algebraically closed field, it is a product of $\mathbb{G}_m$ ’s and a finite group scheme. Therefore, by [20, Theorem 1.1] and [19, Page 113], the ring of invariants $k[S]^{M \times G_{v,\text{red}}}$ is also finitely generated. This proves the finite generation of $k[S]^{G_v}$.

The commutativity of the diagram (40) now follows by the property of categorical quotients by the group G .

Now we check the conditions of Lemma 5.5. We consider the given closed G -orbit $F = G \times^{G_v} \{v\}$. Then we need to check that $\varphi(F)$ is a closed orbit in V . In fact, $\varphi(F)$ is precisely the closed orbit $G \cdot \varphi(v) = G \cdot v \subset V$. Thus we have verified conditions (2) and (3) of Lemma 5.5. Condition (1) is precisely the content of 5.4 and (4) holds since $G \times^{G_v} S$ has been seen to be smooth.

The isomorphism $(\bar{s}, p_U)$ shows that the diagram (41) is cartesian. This completes the proof of 5.1.

Corollary 5.6. *Let X be an affine G -scheme embeddable as a closed G -subscheme in low height G -module. Let $x \in X$ be such that the orbit $G \cdot x \subset X$ is closed. Then there exists a locally closed G_x -invariant subscheme (a “slice”) X_1 , of X with $x \in X_1$ such that the conclusions of 5.1 hold for the G -morphism $G \times^{G_x} X_1 \rightarrow X$.*

Proof. This follows from 5.1 (which corresponds to [2, Proposition 7.4]) exactly as in [2, Proposition 7.6]. $\square$

The following consequence of the slice theorem has many applications so we state it here without proof.

Corollary 5.7 ((see [2, Proposition 8.5, page 312])). *Let F be an affine G -subvariety of $\mathbb{P}(V)$, with V as a G -module with low height, and suppose that F contains a unique closed orbit F_{cl} . Then there exists a G -retract $F \rightarrow F_{cl}$.*

Let V be a finite dimensional G -module and let

$$\mathrm{ht}_G(\hat{\mathfrak{g}}(V)) := \max_i \mathrm{ht}_G(\wedge^i(V)) \quad (46)$$

We then have the following application to the theory of semistable principal bundles in positive characteristics.

Theorem 5.8. *Let E be a stable G -bundle with G semi-simple and $\rho : G \rightarrow \mathrm{SL}(V)$ be a representation such that $p > \mathrm{ht}_G(\hat{\mathfrak{g}}(V))$. Then the associated bundle $E(V)$ is polystable of degree 0.*

Proof. The proof follows [1, Theorem 9.11] verbatim where the notion of “separable index” is replaced by the Dynkin height, the key ingredient being Corollary 5.7. $\square$

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6 Added in proof

In Section 2.2, after making suitable assumptions on G , we refer to Serre in (5) for the construction of the exponential map $\exp : \mathfrak{g}_{\mathrm{nilp}} \xrightarrow{\sim} G^u$. The proof that Serre gives in ([23, Theorem 3, page 21]) is only a sketch. At his suggestion, we complete it. We work in principle over an algebraically closed field but phrase the argument so that it remains valid over more general bases.

Let T be a maximal torus, B a Borel subgroup containing T and U its unipotent radical. We assume $p \geq h$. By this assumption on the characteristic, the Campbell-Hausdorff group law on $\mathrm{Lie}(U)$ is well-defined. The first step is the construction of a B -equivariant isomorphism

$$\exp : (\mathrm{Lie}(U), \cdot) \xrightarrow{\sim} U \quad (47)$$

For this step we give complete proofs. The second step is to glue these isomorphisms into

$$\exp : \mathfrak{g}_{\text{nilp}} \xrightarrow{\sim} G^u. \quad (48)$$

This requires the additional assumption that the simply connected covering $\tilde{G}$ of the derived group G' is étale over G' . Here we will mainly give references.

6.1

Let R^+ be the set of positive roots: the set of roots of T acting on $\text{Lie}(U)$. For α in R^+ , let U_α be the corresponding root subgroup, and let x_α be an isomorphism from U_α to $\mathbb{G}_a$.

Fix a total order on R^+ . The corresponding product map from $\prod_\alpha U_\alpha$ to U is an isomorphism of schemes: it is a morphism between smooth connected schemes which is étale at the origin, bijective by [5, 13-05], and one applies Zariski's main theorem as in [26, SGA3, XXII, 4.1] (or see [7, IV, 14.5]).

This isomorphism identifies U with the affine space with coordinates (x_α) ($\alpha \in R^+$). In these coordinates, the action of T on U by conjugation is given by

$$t \in T \text{ acts by } (x_\alpha) \mapsto (\alpha(t)x_\alpha)$$

This action respects the group law. It follows that the α -coordinate of $x \cdot y$ is a linear combination of the $x_\beta^i y_\gamma^j$ for $i\beta + j\gamma = \alpha$.

We grade the polynomial algebra $\mathcal{O}(U)$ by defining degree of x_α to be the height of α (if the expression of α as a linear combination of simple roots is $\sum n_i \alpha_i$, the height of α is $\sum n_i$). The corresponding increasing filtration $\text{Fil}_\bullet$ of $\mathcal{O}(U)$ is invariant by T -conjugation as well as by the action on the right and left by U -translations. It is hence invariant by B -conjugation.

The assumption $p \geq h$ is equivalent to $\mathcal{O}(U)$ being generated by Fil_{p-1} . The definition of the exponential (47) is a special case of the following construction.

6.2

Let U be an algebraic group and $\text{Fil}_\bullet$ be an increasing filtration of its affine algebra. Assume that $\mathcal{O}(U)$ is a polynomial algebra $k[(x_\alpha)_\alpha]$ and that for some integers $d(\alpha) > 0$, $\text{Fil}_\bullet$ is the increasing filtration deduced from the grading for which x_α is homogeneous of degree $d(\alpha)$. Assume that the α -coordinate of $x \cdot y$ is a polynomial of degree $\leq d(\alpha)$ in the (weighted) coordinate of x and y . This is equivalent to the compatibility:

$$\Delta(\text{Fil}_i) \subset \sum_{j+k=i} \text{Fil}_j \otimes \text{Fil}_k \quad (49)$$

of the filtration $\text{Fil}_\bullet$ with the coproduct.

We will view $\text{Lie}(U)$ as the Lie algebra of the left invariant vector fields on U . Let D_α be the element of $\text{Lie}(U)$ which is x_α at e . If $[\epsilon]$ is the $k[\epsilon]/((\epsilon)^2)$ valued point of U with coordinate $x_\alpha = \epsilon$, $x_\beta = 0$ for $\beta \neq \alpha$, $D_\alpha f$ is the coefficient of ϵ in $f(x \cdot [\epsilon])$. From this it follows that $D_\alpha(x_\beta)$ is of degree $\leq d(\beta) - d(\alpha)$, and as a consequence

$$D_\alpha \text{Fil}_i \subset \text{Fil}_{i-d(\alpha)} \quad (50)$$

i.e. D_α is of (descending) filtration $d(\alpha)$.

We now assume that $\mathcal{O}(U)$ is generated by Fil_{p-1} , i.e. that all $d(\alpha)$ are less than p . Then any Lie polynomial of degree $\geq p$ in the derivations $D_\alpha \in \text{Lie}(U)$ vanishes on the x_α and hence vanishes identically, implying that the Campbell-Hausdorff group law on $\text{Lie}(U)$ is defined.

Definition 6.1. The exponential map $\exp$ from $\text{Lie}(U)$ to U is the map having as coordinates the truncated exponential

$$\exp(D)_\alpha := \sum_{n < p} \frac{D^n}{n!}(x_\alpha) \text{ at } e \quad (51)$$

Lemma 6.2. *For f in Fil_{p-1} , one has*

$$f(\exp(D)) = \sum_{n < p} \frac{D^n}{n!}(f) \text{ at } e \quad (52)$$

Proof. This holds by the definition of $\exp$ when f is a coordinate function x_α , and it suffices to check that, for g in Fil_a and h in Fil_b with $a + b < p$, one has the following equality:

$$\frac{D^n}{n!}(fg) = \sum \frac{D^n}{n!}(f) \frac{D^n}{n!}(g) \quad (53)$$

the sums being truncated at $n < p$. Both sides can be identified with sums $\frac{D^n}{n!}(f) \frac{D^m}{m!}(g)$, where on the left we have $n + m < p$ and on the right $n < p$ and $m < p$. If $n + m \geq p$, then either $n > a$ or $m > b$, in which case $D^n(f) = 0$ (resp. $D^m(g) = 0$). The terms missing on the left side therefore vanish. $\square$

Lemma 6.3. *The morphism $\exp$ is a group homomorphism*

$$\exp : (\text{Lie}(U), \cdot) \rightarrow U. \quad (54)$$

Proof. If f is in Fil_{p-1} , so are its left-translates. As the derivations D_α in $\text{Lie}(U)$ are left invariant, (52) applied to the left-translate of f gives:

$$f(x \cdot \exp(D)) = \sum_{n < p} \frac{D^n}{n!}(f) \text{ at } x \quad (55)$$

hence

$$f(\exp(D') \cdot \exp(D)) = \sum_{n, m < p} \frac{D'^m}{m!} \frac{D^n}{n!}(f) \text{ at } e \quad (56)$$

In the completed free associative algebra $\mathbb{Q}\langle\langle X, Y \rangle\rangle$, if $H_d(X, Y)$ is the Lie polynomial which is the degree d component of the Campbell-Hausdorff group law, one has

$$\exp\left(\sum_d H_d(X, Y)\right) = \exp(X) \cdot \exp(Y). \quad (57)$$

Modulo terms of degree $\geq p$, one still has

$$\sum_{n \leq p} \frac{H_n(X, Y)}{n!} = \sum_{n < p} \frac{X^n}{n!} \cdot \sum_{n < p} \frac{Y^n}{n!} \quad (58)$$

and this continues to hold in $\mathbb{Z}[\frac{1}{(p-1)!}]\langle\langle X, Y \rangle\rangle$. Substituting D' and D for X and Y , we get

$$f(\exp(D') \cdot \exp(D)) = f(\exp(D' \cdot D)) \quad (59)$$

for f in Fil_{p-1} . Hence $\exp(D') \cdot \exp(D) = \exp(D' \cdot D)$. For f in Fil_{p-1} , (55) reads

$$\frac{D^n}{n!}f = \sum_{0 < n < p} \frac{D^n}{n!}(f) \text{ at } \exp(D) \quad (60)$$

In $\mathbb{Q}[[X]]$, one has $\log \exp(X) = X$. Modulo monomials of degree $\geq p$, one still has

$$\sum_{0 < n < p} \frac{(-1)^n}{n} \left(\sum_{0 < m < p} \frac{X^m}{m!} \right)^n = X \quad (61)$$

and this continues to hold in $\mathbb{Z}[\frac{1}{(p-1)!}][[X]]$.

Substituting D for X , in (61), we obtain from (60) that for f in Fil_{p-1} , one has

$$Df = \sum_{0 < n < p} (-1)^n \frac{(\log \Delta)^n}{n} f \quad (62)$$

This gives a formula for a left inverse $\log \Delta$ of $\exp : \text{Lie}(U) \rightarrow U$; viewed as a derivation,

$$\log \Delta = \sum_{n < p} (-1)^n \frac{\Delta^n}{n} \quad (63)$$

when applied to f in Fil_{p-1} . This shows that the exponential is injective and hence, being an étale homomorphism of connected groups, it is an isomorphism.

In the case of the unipotent radical U of a Borel subgroup, the given construction of the exponential map, depending only on the filtration $\text{Fil}_\bullet$ of $\mathcal{O}(U)$, is invariant by B -conjugation. $\square$

Remark 6.4.

- (i) Let $I \subset R^+$ be the set of simple roots, and for $J \subset I$, let P_J be the corresponding parabolics subgroup and U_J its unipotent radical. Let $R_J^+ \subset R^+$ be the set of $\alpha \in R^+$ which are linear combinations of simple roots in J . For $\alpha = \sum n_i \alpha_i$ in $R^+ \setminus R_J^+$, define $d_J(\alpha) = \sum_{i \notin J} n_i$. Then if all $d_J(\alpha)$ are $< p$, the construction we have given of the exponential map gives a similar map

$$\exp : \text{Lie}(U_J) \xrightarrow{\sim} U_J. \quad (64)$$

It suffices to replace d by d_J in the earlier proofs.

- (ii) More generally, let $\Gamma \subset R^+$ be a closed set of positive roots, and let U_Γ be the corresponding subgroup of U . For any fixed total order on Γ , the corresponding product map

$$\prod_{\alpha \in \Gamma} U_\alpha \xrightarrow{\sim} U_\Gamma \quad (65)$$

is an isomorphism of schemes. For α in Γ , define $d_\Gamma(\alpha)$ to be the largest number of elements of Γ of which α is the sum (with repetitions allowed).

In [21], Seitz states that when all $d_\Gamma(\alpha)$ are $< p$, an exponential map from $\text{Lie}(U_\Gamma)$ to U_Γ is defined. He writes (before [21, Proposition 5.1, page 483]) that the argument in Section 4 of [22] yields this result. The reference is mistaken and should be replaced by Section 2.1 of [25]. The argument given in the preceding paragraphs, with d replaced by d_Γ , also yields the result.

6.3

Let $\mathcal{B}$ be the variety of Borel subgroups of G . We have fiber spaces $X_{\mathcal{B}}$ and $Y_{\mathcal{B}}$ over $\mathcal{B}$ where the fibers at a Borel subgroup B in $\mathcal{B}$ are respectively the unipotent radical U and its Lie algebra.

The given construction of the exponential map works over any base on which the primes $\ell < h$ are invertible. Over $\mathcal{B}$, we therefore obtain an isomorphism $\exp : Y_{\mathcal{B}} \xrightarrow{\sim} X_{\mathcal{B}}$. The natural map from $X_{\mathcal{B}}$ to G (resp. $Y_{\mathcal{B}}$ to $\text{Lie}(G)$) is proper, being the restriction to a closed subset of $\mathcal{B} \times G$ (resp. $\mathcal{B} \times \text{Lie}(G)$) of the second projection.

The reduced subscheme $\mathfrak{g}_{\text{nilp}}$ (resp. G^u) of $\text{Lie}(G)$ (resp. G) with points the nilpotent (resp. unipotent) elements, is the schematic image of $Y_{\mathcal{B}}$ in $\text{Lie}(G)$ (resp. of $X_{\mathcal{B}}$ in G). We want to complete the following G -equivariant diagram in the solid arrows by an isomorphism at the dotted arrow:

$$\begin{array}{ccc} Y_{\mathcal{B}} & \xrightarrow[\sim]{\exp} & X_{\mathcal{B}} \\ \downarrow & & \downarrow \\ \mathfrak{g}_{\text{nilp}} & \cdots \cdots \cdots & G^u \end{array} \quad (66)$$

Replacing G by its derived group G' does not change the diagram, nor does replacing a semi-simple G by its simply connected covering $\tilde{G}$, provided $\tilde{G}$ is étale over G . Under our assumptions this reduces us to the case where G is semi-simple and simply connected. Here are references for the existence of the dotted arrow isomorphism in (66).

- (A) By [27, 6.11, 8.1] G^u is normal with a smooth dense orbit whose complement is of codimension ≥ 2 . This holds in any characteristic for any simply connected G . Normality results from G^u being the complete intersection in G defined by the equations $\chi(g) = \chi(e)$ for χ the characters of the fundamental representations, and being smooth outside codimension 2.
- (B) The same holds for $\mathfrak{g}_{\text{nilp}}$ when G is simply connected and p good, because $\mathfrak{g}_{\text{nilp}}$ is then equivariantly isomorphic to G^u . A nice proof is in [2, Corollary 9.3.4]. The isomorphism uses a representation (V, ρ) such that the pairing $\text{Tr}(\rho(X)\rho(Y))$ on $\text{Lie}(G)$ is non-degenerate. It is then defined as induced by the map $g \mapsto \rho(g) - 1$ from G to $\text{End}(V)$, composed with the orthogonal projection of $\text{End}(V)$ (endowed with the symmetric bilinear form $\text{Tr}(XY)$) to $\text{Lie}(G) \subset \text{End } V$. This does not work for $\text{SL}(n)$ for which one uses $X \mapsto 1 + X$.
- (C) Over the dense orbits of $\mathfrak{g}_{\text{nilp}}$ and G^u , (66) is a system of isomorphisms, and the composite isomorphism extends across codimension 2 by normality.

A Coxeter number and root systems, by Zhiwei Yun

Proposition 1. Let R be an irreducible root system associated to a simple group G with Coxeter number h and let X be the lattice spanned by R . Let $\varphi : X \rightarrow \mathbb{R}/\mathbb{Z}$ be a homomorphism. Then there exists a basis B for R such that if $\alpha \in R$ satisfies $\varphi(\alpha) \in [0, 1/h) \bmod \mathbb{Z}$, then α is positive with respect to B .

Proof. Pick any basis $B_0 = \{\alpha_1, \dots, \alpha_r\}$ of R . We lift φ to a linear map $y : X \rightarrow \mathbb{R}$ which is viewed as a point in $V = \text{Hom}(X, \mathbb{R})$.

Let W be the Weyl group of R . Then there is an action of the extended Weyl group $\text{Hom}(X, \mathbb{Z}) \rtimes W$ on V and the validity of the statement for the linear map y is invariant for this action. Therefore we may assume that y lies in the fundamental alcove attached to B_0 , i.e.

$$\alpha_j(y) \geq 0, \quad 1 \leq j \leq r \quad (67)$$

$$\tilde{\alpha}(y) \leq 1 \quad (68)$$

where $\tilde{\alpha}$ is the highest root with respect to B_0 . Then for all $\alpha \in R$, we have

$$|\alpha(y)| \leq 1. \quad (69)$$

We claim that there exists a basis B of R such that for any $\beta \in R$ with

$$|\beta(y)| \in [1/2, 1/2 + 1/h) \setminus [0, 1/h) \bmod \mathbb{Z}, \quad (70)$$

β is positive with respect to B .

Write $\tilde{\alpha} = \sum_{i=0}^r n_i \alpha_i$ and let $\alpha_0 = -\tilde{\alpha}$ and $n_0 = 1$ as usual. Then we have the equality $\sum_{i=0}^r n_i \alpha_i = 0$, viewed as affine functions on V and we have

$$h = \sum_{i=0}^r n_i. \quad (71)$$

Therefore, for some $0 \leq i_0 \leq r$, we have $\alpha_{i_0}(y) \geq 1/h$.

If $i_0 = 0$, then $\tilde{\alpha}(y) \leq 1 - 1/h$ and in this case, we can take $B = B_0$. In fact, for any negative root β with respect to B_0 , we have $0 \geq \beta(y) \geq -1 + 1/h$, so that β cannot satisfy the condition (70).

If $i_0 > 0$, let v_{i_0} be the vertex of the fundamental alcove opposite to the root hyperplane α_{i_0} . So $\alpha_j(v_{i_0}) = 0$ if $j \neq i_0$ and $\alpha_{i_0}(v_{i_0}) = 1 = n_{i_0}$. We take a basis B of R such that the vector $y - v_{i_0}$ is dominant, i.e. for any $\beta \in R$ positive for B , we have the relation: $\beta(y) \geq \beta(v_{i_0})$.

Such a basis B of R satisfies the requirements of the claim. In fact, let $\beta \in R$ be negative with respect to B .

Write $\beta = \sum_{j=1}^r m_j \alpha_j$ for some m_j . Then, $\beta(v_{i_0}) = m_{i_0}/n_{i_0}$. On the other hand, $\beta(y) = \sum_{j=1}^r m_j \alpha_j(y)$. Since β is negative for B , we have

$$\beta(y) \leq \beta(v_{i_0}) = m_{i_0}/n_{i_0} \quad (72)$$

and we have two cases:

- If $\beta > 0$, with respect to B_0 , then $\beta(y) \geq 0$ and $0 \leq m_j \leq n_j$ for all j . If $m_{i_0} = 0$, then (72) implies that $\beta(y) = 0$ and so β fails to satisfy (70). If $m_{i_0} \geq 1$, then $\beta(y) \geq \alpha_{i_0}(y) \geq 1/h$, again not satisfying (70).
- If $\beta < 0$, with respect to B_0 , then $-1 \leq \beta(y) \leq 0$ and $-n_j \leq m_j \leq 0$ for all j . If $m_{i_0} = n_{i_0}$, then $\beta(v_{i_0}) = -1$ which forces $\beta(y) = -1$ by (72). If $m_{i_0} \leq n_{i_0} - 1$, then comparing $\beta(y)$ with $\tilde{\alpha}(y)$, it misses at least one copy of $\alpha_{i_0}(y)$, therefore $\beta(y) \leq \tilde{\alpha}(y) - \alpha_{i_0}(y) \leq 1 - 1/h$ and yet again, $\beta(y)$ does not satisfy (70).

This proves the claim and the proposition. □

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Multizetas, after Francis Brown

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0. Introduction

Let $k \geq 0$ and let $s = (s_1, \dots, s_k)$ be a sequence of k integers ≥ 1 . The multizeta number $\zeta(s)$ is the iterated sum (the name is explained in 1.1)

$$\zeta(s) := \sum_{n_1 > \dots > n_k > 0} \frac{1}{n_1^{s_1} \dots n_k^{s_k}} \quad (0.1)$$

(the sum being over strictly decreasing sequences of k integers > 0). For $k = 0$, this definition gives $\zeta(s) = 1$. For $k \geq 1$, the sum (0.1) converges if and only if $s_1 \geq 2$. We shall consider only convergent $\zeta(s)$, i.e. those for which (0.1) converges.

The product of two multizeta numbers is a linear combination, with integer coefficients, of multizeta numbers. See 3.6. It is therefore equivalent to determine the polynomial relations with rational coefficients among the multizeta numbers, or to determine the $\mathbb{Q}$ -linear relations among them.

For $k = 1$, the convergent $\zeta(s)$ are the values at the integers $n \geq 2$ of Riemann's zeta function. Euler showed that for n even, $\zeta(n)$ is a rational multiple of π^n ([3]). Computing $\zeta(3)$ to 10 significant figures, he also checked that $\zeta(3)$ was not the product of π^3 by a rational number of small denominator. Following a correspondence with Goldbach, Euler studied in [4], for $k = 2$, the variant of the sums (0.1) in which one sums over $n_1 \geq n_2$:

$$\zeta_{\text{Euler}}(s_1, s_2) = \zeta(s_1, s_2) + \zeta(s_1 + s_2).$$

He proves a series of $\mathbb{Q}$ -linear relations among multizeta numbers, among them $\zeta(2, 1) = \zeta(3)$. One of his motivations was the hope of obtaining information on the values of ζ at the odd integers ≥ 3 (see the end of his letter to Goldbach of January 5, 1743).

The problem of determining all $\mathbb{Q}$ -linear relations among multizeta numbers has two aspects:

- (A) What relations should one expect? Prove them.
- (B) Prove that there are no others.

The available evidence, both numerical and theoretical, suggests that every $\mathbb{Q}$ -linear relation among the $\zeta(s)$ is a sum of isobaric relations, i.e. relations among $\zeta(s)$ of the same weight $\sum s_i$. If one restricts oneself to considering only isobaric relations, one knows nothing about problem (B). We shall speak only of (A).

F. Brown has defined a class of $\mathbb{Q}$ -linear relations among the $\zeta(s)$, the motivic relations. More precisely, he defines a graded $\mathbb{Q}$ -algebra M , homogeneous elements $\zeta^m(s)$ of M , of degree the weight $\sum s_i$ of s , and a homomorphism $\text{real} : M \rightarrow \mathbb{R}$ such that $\text{real}(\zeta^m(s)) = \zeta(s)$. As a vector space, the algebra M is generated by the $\zeta^m(s)$. The motivic relations among the $\zeta(s)$ are those that come from relations among the $\zeta^m(s)$.

A variant of Grothendieck's period conjecture implies that every $\mathbb{Q}$ -linear relation among the $\zeta(s)$ is motivic. This prediction has been verified for many of the known relations. Warning: although some proofs of relations among the $\zeta(s)$ are easy to lift to proofs of the same relations among the $\zeta^m(s)$, this is far from always being the case.

Theorem 0.2 (F. Brown [1]). The $\zeta^m(s_1, \dots, s_k)$ for which each s_i is equal to 2 or 3 form a basis of M over $\mathbb{Q}$.

Each $\zeta(s)$ can therefore be expressed uniquely, in a motivic way, as a linear combination with rational coefficients of $\zeta(s_1, \dots, s_k)$ with $k \geq 0$ and each $s_i \in \{2, 3\}$. Unfortunately, Brown's proof does not provide an algorithm, or at least not a usable algorithm, for finding what this linear combination is. From Theorem 0.2, F. Brown deduces that the category of mixed Tate motives over $\mathbb{Z}$ is generated by the fundamental group of $\mathbb{P}^1 - \{0, 1, \infty\}$. See 7.18.

Some of our notations differ from those of [1]. The correspondence between the notations is involutive: $\zeta(s_1, \dots, s_k)$ becomes $\zeta(s_k, \dots, s_1)$, the composition of paths $\alpha\beta$ becomes $\beta\alpha$, a monomial in the e_t (respectively a sequence of 0 and 1) is to be replaced by the same one read from right to left, and e_1 is to be replaced by $-e_1$.

1. Integral formula

1.1

Let $\omega_1, \dots, \omega_N$ be 1-forms on the interval $[0, 1]$. The iterated integral $\text{It} \int_0^1 \omega_1, \dots, \omega_N$ is the integral, over the simplex

$$\sigma_N := \{(t_1, \dots, t_N) \mid 1 \geq t_1 \geq \dots \geq t_N \geq 0\}, \quad (1.1.1)$$

contained in the cube $[0, 1]^N$, of

$$\text{pr}_1^* \omega_1 \wedge \text{pr}_2^* \omega_2 \wedge \dots \wedge \text{pr}_N^* \omega_N.$$

Let S be the operation that attaches to a 1-form ω the function

$$S[\omega](t) := \int_0^t \omega.$$

The iterated integral $\text{It} \int_0^1 \omega_1, \dots, \omega_N$ can be constructed as follows: apply S to ω_N , multiply the function obtained by ω_{N-1} , apply S , multiply by ω_{N-2} , and so on; at the end, evaluate at 1:

$$\text{It} \int_0^1 \omega_1, \dots, \omega_N = S[\omega_1.S[\omega_2.\dots S[\omega_N]\dots]](1). \quad (1.1.2)$$

It is this construction that explains the name “iterated integral”. A sum such as (0.1) admits an analogous description in terms of the operator P which, to a function $f(n)$ ($n \geq 1$), attaches the function

$$P[f](n) := \sum_{m=1}^{n-1} f(m);$$

this explains the name “iterated sum” given to (0.1).

Let us compute in this way the iterated integral

$$\text{It} \int_0^1 \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_1-1) \text{ times}}, \frac{dt}{1-t}, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_2-1) \text{ times}}, \frac{dt}{1-t}, \dots, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_k-1) \text{ times}}, \frac{dt}{1-t}. \quad (1.1.3)$$

We have

$$\frac{dt}{1-t} = \sum_{n \geq 0} t^n dt.$$

Applying S term by term, one obtains $\sum_{n \geq 1} t^n/n$. Multiplying by dt/t and applying S , one obtains $\sum_{n \geq 1} t^n/n^2$. Iterating $(s_k - 1)$ times, one obtains $\sum_{n \geq 1} t^n/n^{s_k}$. Multiplying by $dt/(1 -$

$t) = \sum_{m \geq 0} t^m dt$ and applying S , one obtains

$$S \left(\sum_{m \geq 0, n \geq 1} \frac{t^{m+n}}{n^{s_k}} dt \right) = \sum_{m, n \geq 1} \frac{t^{m+n}}{(m+n)n^{s_k}} = \sum_{m > n > 0} \frac{t^m}{mn^{s_k}}.$$

Continuing in this way, one finally obtains:

Proposition 1.2. The iterated integral (1.1.3) is equal to $\zeta(s_1, \dots, s_k)$.

1.3

Whereas the notion of an infinite sum is foreign to algebraic geometry, the study of integrals of algebraic quantities is one of its sources. It is thanks to Proposition 1.2 that algebraic geometry, more precisely the theory of mixed Tate motives, is useful in the study of multizeta numbers.

2. Pro-unipotent fundamental group and cohomology

2.1

We shall need a variant Z' of the descending central series Z of a group Γ . For each $i \geq 1$, Γ/Z_i is a nilpotent group. The set T_i of its torsion elements is therefore a subgroup. We define Z'_i as the inverse image of T_i in Γ . Below, we shall always assume Γ to be finitely generated. This hypothesis implies that the Z'_i/Z'_{i+1} are finitely generated free abelian groups.

First suppose Γ nilpotent and torsion-free, i.e. that there is an A such that $Z'_A = 0$. For each i ($1 \leq i < A$), choose a basis of Z'_i/Z'_{i+1} and lift it to Z'_i . Let $\gamma_1, \dots, \gamma_M$ be the elements of Γ obtained, arranged in an arbitrary order, and let $w(j)$ be the integer i such that γ_j lifts an element of a basis of Z'_i/Z'_{i+1} . One checks by induction on A that the map

$$\mathbb{Z}^M \longrightarrow \Gamma : n \longmapsto \gamma_1^{n_1} \cdots \gamma_M^{n_M}$$

is bijective. In such a “system of coordinates”, the group law of Γ is given by polynomials with rational coefficients and integer values on the integers. More precisely, if the coordinate of index j is viewed as having weight $w(j)$, the j -th coordinate of $m \circ n$ is a polynomial of weight $\leq w(j)$ in the m_k and n_l .

The same polynomials define a unipotent algebraic group over $\mathbb{Q}$, the unipotent envelope Γ^{un} of Γ . The group $\Gamma^{\text{un}}(\mathbb{Q})$ of its rational points is $\mathbb{Q}^N$, endowed with the group law given by the same formulae as that of Γ . The group $\Gamma^{\text{un}}(\mathbb{Q})$ is the Malcev completion $\Gamma \otimes \mathbb{Q}$ of Γ .

Let $\mathbb{Q}[\Gamma]$ be the group algebra of Γ , and let I be its augmentation ideal. Quillen gave an elegant description of the affine algebra $O(\Gamma^{\text{un}})$ of Γ^{un} : it is the inductive limit of the duals of the $\mathbb{Q}[\Gamma]/I^N$. More precisely, the inclusion of Γ in $\mathbb{Q}[\Gamma]$ identifies the dual of $\mathbb{Q}[\Gamma]$ with the space of functions $\Gamma \rightarrow \mathbb{Q}$, and under this identification the dual of $\mathbb{Q}[\Gamma]/I^{N+1}$ becomes the space of polynomials in the n_j of weight $\leq N$.

For an arbitrary finitely generated Γ , the preceding applies to the Γ/Z'_i , and one defines Γ^{un} as the projective limit group scheme of the $(\Gamma/Z'_i)^{\text{un}}$. The affine algebra of Γ^{un} is still the inductive limit of the duals of the $\mathbb{Q}[\Gamma]/I^N$.

2.2

Let E be a reasonable topological space, for example a complex algebraic variety. Assume E connected. A path γ from a to b , $[0, 1] \rightarrow E$, gives

$$\gamma_N : [0, 1]^N \rightarrow E^N$$

and, by restriction to the simplex σ_N of (1.1.1), a singular chain of E^N which is a cycle modulo the union of $b = x_1, x_1 = x_2, \dots, x_N = a$. The homology class of this cycle depends only on the homotopy class of γ .

Make $a = b$. Elaborating the preceding construction (cf. [2] §3), one obtains an isomorphism of $\mathbb{Q}[\pi_1(E, a)]/I^{N+1}$ with a relative homology group of E^N and, for its dual $(\mathbb{Q}[\pi_1(E, a)]/I^{N+1})^\vee$, with a relative cohomology group of E^N . From this, by passage to the inductive limit, one obtains a cohomological description of the affine algebra of $\pi_1(E, a)^{\text{un}}$.

No longer suppose that $a = b$. Let $\pi_1(E; b, a)$ be the set of homotopy classes of paths from a to b . The group $\pi_1(E, a)$ acts on the right, by composition of paths, on $\pi_1(E; b, a)$. This action makes $\pi_1(E; b, a)$ a principal homogeneous space, we prefer to say torsor, under $\pi_1(E, a)$. We write $\pi_1(E; b, a)^{\text{un}}$ for the $\pi_1(E, a)^{\text{un}}$ -torsor deduced from it by pushing forward by $\pi_1(E, a) \rightarrow \pi_1(E, a)^{\text{un}}$. Let $\mathbb{Q}[\pi_1(E; b, a)]$ be the space of formal linear combinations of elements of $\pi_1(E; b, a)$. It is a right $\mathbb{Q}[\pi_1(E, a)]$ -module, free of rank one. Set

$$\mathbb{Q}[\pi_1(E; b, a)]/I^N := \mathbb{Q}[\pi_1(E; b, a)]/\mathbb{Q}[\pi_1(E; b, a)]I^N.$$

The preceding construction again provides a cohomological description of the $(\mathbb{Q}[\pi_1(E; b, a)]/I^N)^\vee$, and of the affine algebra of $\pi_1(E; b, a)^{\text{un}}$ which is its inductive limit.

2.3

Let X be an algebraic variety defined over a subfield K of $\mathbb{C}$. The Betti cohomology $H_B^*(X)$ of X is the rational cohomology of the space $X(\mathbb{C})$ of complex points of X , endowed with its classical topology. It is endowed with structures reflecting the algebraic structure of X , in particular with a mixed Hodge structure. One also has the algebraic de Rham cohomology $H_{\text{dR}}^*(X)$. If X is nonsingular, $H_{\text{dR}}^*(X)$ is the hypercohomology of the de Rham complex of sheaves of algebraic differential forms. It is a K -vector space, one has a comparison isomorphism

$$\text{comp}_{B, \text{dR}} : H_{\text{dR}}^i(X) \otimes_K \mathbb{C} \xrightarrow{\sim} H_B^i(X) \otimes_{\mathbb{Q}} \mathbb{C}, \quad (2.3.1)$$

and the filtrations W and F of the mixed Hodge structure of $H_B^i(X)$ come from filtrations of $H_{\text{dR}}^i(X)$. The coefficients of the matrix of the isomorphism (2.3.1), in a basis of $H_{\text{dR}}^i(X)$ and one of $H_B^i(X)$, are called periods.

Thanks to 2.2, all this applies to the affine algebras of the group schemes $\pi_1(X, a)^{\text{un}}$ or of their homogeneous spaces $\pi_1(X; b, a)^{\text{un}}$. We shall write $\pi_1(X, a)_B$ for $\pi_1(X, a)^{\text{un}}$, $\pi_1(X, a)_{\text{dR}}$ for its de Rham variant, and similarly for the $\pi_1(X; b, a)$. We shall put B/dR in the subscript when the statement is valid equally well in Betti and in de Rham.

2.4. Example

Take $K = \mathbb{Q}$ and let X be the multiplicative group $\mathbb{G}_m$, whose group of complex points is $\mathbb{C}^*$. Here the fundamental group, which is commutative, is independent of the base point and is identified with the homology group H_1 . In de Rham, the dual H_{dR}^1 of H_1^{dR} is $\mathbb{Q} dz/z$. In Betti, $H_1(\mathbb{C}^*, \mathbb{Z})$ is $\mathbb{Z}$, generated by a positive turn around 0. If one does not want to speak of a “positive turn”, i.e. to choose which of i and $-i$ is i , it is better to say that $H_1 = \pi_1 = 2\pi i\mathbb{Z}$, with $\ell \in 2\pi i\mathbb{Z}$ corresponding to the loop $\exp(\ell t)$ ($0 \leq t \leq 1$), on which the integral of dz/z is ℓ .

The $H_1^B(\mathbb{G}_m) = H_1(\mathbb{C}^*, \mathbb{Q})$ and $H_1^{\text{dR}}(\mathbb{G}_m)$ are the Betti and de Rham realizations of the Tate motive $\mathbb{Q}(1)$ (see §5). We have $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$ (1 being dual to $dz/z \in H_{\text{dR}}^1(\mathbb{G}_m)$), and $\mathbb{Q}(1)_B$, included in $\mathbb{Q}(1)_{\text{dR}} \otimes \mathbb{C}$ by $\text{comp}_{\text{dR}, B}$, is $2\pi i\mathbb{Q}$; $\mathbb{Q}(1)$ is purely of Hodge type $(-1, -1)$.

The algebraic group $\pi_1(\mathbb{G}_m, 1)_{\text{dR}}$ is the additive group $\mathbb{G}_a$, whose group of rational points is $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$, and whose affine algebra is $\mathbb{Q}[T]$ (more precisely: $\text{Sym}^*(H_{\text{dR}}^1(\mathbb{G}_m))$). The algebraic group $\pi_1(\mathbb{G}_m, 1)_B$ is the additive group whose group of rational points is $2\pi i\mathbb{Q}$. Affine algebra: $\text{Sym}^*(H_B^1(\mathbb{G}_m))$. It is the algebra of functions f on $2\pi i\mathbb{Z}$, with rational values, and such that,

if one identifies $2\pi i\mathbb{Z}$ with $\mathbb{Z}$ by $n \mapsto 2\pi in$, the following equivalent conditions are verified for N large enough: (a) f is a polynomial of degree $< N$; (b) denoting by Δ the operator $g \mapsto (g(n+1) - g(n))$, one has $\Delta^N f = 0$; (c) the linear form on the group algebra $\mathbb{Q}[\mathbb{Z}]$ defined by f is zero on I^N .

By abuse of language, we shall sometimes write $\mathbb{Q}(1)_{B/\mathrm{dR}}$ for these algebraic groups.

3. The case of $\mathbb{P}^1 - \{0, 1, \infty\}$

3.1

If X is the complement in $\mathbb{P}^1$ of a finite set T of rational points, the variety X is defined over $\mathbb{Q}$ and the mixed Hodge structures of 2.2, 2.3 are Hodge-Tate: Gr_n^W is zero for n odd, and for $n = 2m$ is purely of Hodge type (m, m) . For each of the groups M considered in 2.2, let M_{dR} be its de Rham variant (2.3). After renumbering, the weight and Hodge filtrations W and F of M_{dR} are opposite. They define the grading of M_{dR} for which

$$\begin{aligned} W_{-2n}(M_{\mathrm{dR}}) &= \bigoplus_{m \geq n} M_{\mathrm{dR}}^m, \\ F^{-p}(M_{\mathrm{dR}}) &= \bigoplus_{m \leq p} M_{\mathrm{dR}}^m. \end{aligned} \tag{3.1.1}$$

The convention (3.1.1) ensures that $\mathbb{Q}(1)_{\mathrm{dR}}$ is of degree one. It is sometimes convenient to grade by the opposite of the degree, the codegree, and to set $(M_{\mathrm{dR}})_m := (M_{\mathrm{dR}})^{-m}$.

Special case: the affine algebra of $\pi_1(X; b, a)_{\mathrm{dR}}$ is graded. It is in degrees ≤ 0 and is reduced to $\mathbb{Q}$ in degree 0. The augmentation morphism $O(\pi_1(X; b, a)_{\mathrm{dR}}) \rightarrow \mathbb{Q}$ is a point ${}_b 1_a$ of $\pi_1(X; b, a)_{\mathrm{dR}}$. In de Rham, one thus has a canonical path ${}_b 1_a$ from one base point to another: for a certain pro-unipotent group scheme Π , each $\pi_1(X; b, a)_{\mathrm{dR}}$ is identified with a copy ${}_b \Pi_a$ of Π , and composition of paths is the group law.

The group Π has the following description. Let L be the Lie algebra over $\mathbb{Q}$ generated by elements e_t ($t \in T$) subject only to the relation $\sum e_t = 0$. It is endowed with the grading for which each e_t is homogeneous of degree 1. The descending central series Z is the filtration by “degrees $\geq i$ ”. Let $L^\wedge$ be the completion of L for this filtration, and let $\exp(L/Z_i)$ be the unipotent algebraic group with Lie algebra L/Z_i . We have

$$\Pi = \exp(L^\wedge) := \varprojlim \exp(L/Z_i). \tag{3.1.2}$$

Another description: the enveloping algebra A of L is the associative algebra generated by elements e_t ($t \in T$) subject only to the relation $\sum e_t = 0$. It is graded, each e_t being homogeneous of degree one, and the N -th power of the augmentation ideal I is the part of A of degree $\geq N$. The completion $A^\wedge = \varprojlim A/I^N$ of A is endowed with a coproduct

$$\Delta : A^\wedge \rightarrow A^\wedge \hat{\otimes} A^\wedge : e_t \mapsto e_t \otimes 1 + 1 \otimes e_t,$$

$L^\wedge$ is the Lie algebra of primitive elements, and for every $\mathbb{Q}$ -algebra R , if one sets

$$A^\wedge \hat{\otimes} R := \varprojlim_N (A/I^N) \otimes_{\mathbb{Q}} R,$$

the group $\Pi(R)$ of R -points of Π is the group of group-like elements of $(A^\wedge \hat{\otimes} R)^*$:

$$\Pi(R) = \{g \in (A^\wedge \hat{\otimes} R)^* \mid \Delta g = g \otimes g\}. \tag{3.1.3}$$

The affine algebra $O(\Pi)$ of Π (with the discrete topology) and the completed enveloping algebra $A^\wedge$ (with its projective-limit topology) are dual to one another. The topological dual of $A^\wedge$ is

identified with the graded dual $\bigoplus (A_n)^\vee$ of A , and this gives the grading (in degrees ≤ 0) of $O(\Pi)$.

We shall write ω for the logarithmic 1-form on X with values in L :

$$\omega := \sum_{t \neq \infty} e_t \frac{dz}{z-t}. \quad (3.1.4)$$

At each $t \in T$, it has residue e_t .

Suppose that $\infty \in T$ and put $T' := T - \{\infty\}$. Then $A = \mathbb{Q}\langle (e_t)_{t \in T'} \rangle$ and

$$A^\wedge = \mathbb{Q}\langle\langle (e_t)_{t \in T'} \rangle\rangle$$

(associative formal series in the e_t). For every word $w = t_1 \cdots t_k$ in the alphabet T' , let $c(w)$ be the linear form on $A^\wedge$ “coefficient of the monomial $e_{t_1} \cdots e_{t_k}$ ”. We again write $c(w)$ for the restriction of $c(w)$ to $\Pi \subset A^{\wedge*}$, and for a formal linear combination of words we put $c(\sum \lambda_w w) = \sum \lambda_w c(w)$. The $c(w)$ form a vector-space basis of $O(\Pi)$. The product in $O(\Pi)$ corresponds to the shuffle product of words:

$$c(w')c(w'') = c(w' \amalg w''), \quad (3.1.5)$$

where, for $w' = t_1 \cdots t_k$ and $w'' = t_{k+1} \cdots t_{k+\ell}$, $w' \amalg w''$ is the sum of the $t_{\sigma^{-1}(1)} \cdots t_{\sigma^{-1}(k+\ell)}$ for σ running through the set $S_{k,\ell}$ of permutations of $\{1, \dots, k+\ell\}$ increasing on $\{1, \dots, k\}$ and on $\{k+1, \dots, k+\ell\}$.

3.2

A path γ from a to b defines a rational point of $\pi_1(X; b, a)_B = \pi_1(X(\mathbb{C}); b, a)^{\text{un}}$. Let $\text{comp}_{\text{dR}, B}(\gamma)$ be the image of this rational point by the comparison isomorphism

$$\text{comp}_{\text{dR}, B} : \pi_1(X; b, a)_B(\mathbb{C}) \xrightarrow{\sim} \pi_1(X; b, a)_{\text{dR}}(\mathbb{C}) = \Pi(\mathbb{C}).$$

This image is obtained by “integration” along γ , in the noncommutative group Π , of the form ω : it is the value at 1 of the solution $g : [0, 1] \rightarrow \Pi(\mathbb{C})$, equal to 1 at 0, of

$$dg(t) \cdot g(t)^{-1} = \omega \left(\frac{d\gamma}{dt} \right) dt. \quad (3.2.1)$$

3.3

The same formalism applies when one allows the base points a, b to be tangential base points (a nonzero tangent vector at a “rational point at infinity” $t \in T$ of X). Let Θ_t be the tangent space to $\mathbb{P}^1$ at t , $\Theta_t^* := \Theta_t - \{0\}$, and $a \in \Theta_t^*$ a tangential base point at t . Everything happens as if one had a morphism from Θ_t^* to X : one has morphisms

$$\pi_1(\Theta_t^*, a)_{B/\text{dR}} = \mathbb{Q}(1)_{B/\text{dR}} \longrightarrow \pi_1(X, a)_{B/\text{dR}} \quad (3.3.1)$$

compatible with the comparison isomorphisms. On the Betti side, this morphism is induced by the germs of maps φ from $(\Theta_t, 0)$ to $(\mathbb{P}^1, t)$ with differential the identity. The space of these germs is contractible, and each φ induces a map from a small punctured disk around 0 (homotopic to Θ_t^*) to X . In de Rham, and after passage to Lie algebras, (3.3.1) is the morphism from $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$ to $L^\wedge$ which sends 1 to e_t .

The $\mathbb{Q}$ -form $\pi_1(X; b, a)_B$ of $\Pi(\mathbb{C})$ is characterized by the property that the images of paths from a to b are rational points.

Warning: $\pi_1(X; b, a)_{\text{dR}}$ is independent of a and b , and $\pi_1(X; b, a)_B$ is locally constant in a and b , but the comparison isomorphism between their complexifications is not locally constant. The independence of $\pi_1(X; b, a)_{\text{dR}}$ from a and b comes from the fact that one has the canonical de Rham path ${}_b 1_a$ between any two base points. The image of this path by $\text{comp}_{B, \text{dR}}$ is not locally constant.

3.4

The case that interests us is that in which $X = \mathbb{P}^1 - \{0, 1, \infty\}$ and in which, as base points, one takes the tangential base points 1 at 0 and -1 at 1. They will be denoted 0 and 1. Here

$$\omega = e_0 \frac{dz}{z} + e_1 \frac{dz}{z-1}$$

and for every $\mathbb{Q}$ -algebra R ,

$$\Pi(R) = \{\text{group-like elements of } R\langle\langle e_0, e_1 \rangle\rangle^*\}$$

(3.1.3). In Betti, one has the “straight path” dch from 0 to 1. We shall again write dch for its image by $\text{comp}_{\text{dR}, B}$ in $\Pi(\mathbb{C})$. This image is the following limit for $\epsilon, \eta \rightarrow 0$ through values > 0 :

$$\text{comp}_{\text{dR}, B}(dch) = \lim \exp(-\log \eta e_1) \text{comp}_{\text{dR}, B}([\epsilon, 1 - \eta]) \exp(\log \epsilon e_0). \quad (3.4.1)$$

The third (respectively first) factor is to be viewed as computing, in the punctured tangent space at 0 (respectively at 1), $\text{comp}_{\text{dR}, B}([1, \epsilon])$ (respectively $\text{comp}_{\text{dR}, B}([-\eta, -1])$), the form ω being replaced by $(du/u)e_0$ (respectively $(du/u)e_1$).

In general one can express the solution of (3.2.1) as a sum of iterated integrals. Applying 1.2, one obtains that, for convergent $\zeta(s_1, \dots, s_k)$, one has:

Proposition 3.5. The coefficient of $e_0^{s_1-1} e_1 \dots e_0^{s_k-1} e_1$ in $dch \in \Pi(\mathbb{C}) \subset \mathbb{C}\langle\langle e_0, e_1 \rangle\rangle^*$ is $(-1)^k \zeta(s_1, \dots, s_k)$.

This proposition determines the coefficients $c(w)$ of monomials of dch only for w a word in the alphabet $\{0, 1\}$ which does not begin with 1 and does not end with 0. The coefficients $c(0)$ of e_0 and $c(1)$ of e_1 in dch are zero. This, together with the fact that dch is group-like, determines the missing coefficients. More precisely, the vanishing of $c(0)$ and $c(1)$ means that dch belongs to the derived group Π' of Π , whose graded Lie algebra is the part of degree ≥ 2 of that of Π . On Π' , writing $\{p\}$ for a power in the monoid of words,

$$c(0^{\{p\}}) = c(1^{\{p\}}) = 0 \quad \text{for } p > 0, \quad (3.5.1)$$

and, applying (3.1.5) to $w10^{\{p-1\}}$ and 0 (respectively to 1 and $1^{\{p-1\}}0w$), one checks by induction on $p \geq 0$ that

$$c(w10^{\{p\}}) = (-1)^p c((w \amalg 0^{\{p\}})1), \quad (3.5.2)$$

respectively

$$c(1^{\{p\}}0w) = (-1)^p c(0(1^{\{p\}} \amalg w))$$

on Π' .

3.6

The identity (3.1.5), evaluated at dch , expresses the product of two multizeta numbers as a linear combination with integer coefficients of multizeta numbers.

4. Motivic multizetas

4.1

Continue to put $X := \mathbb{P}^1 - \{0, 1, \infty\}$ and to take as base points the tangential base points 0 or 1 (see 3.4). In 5.6 we shall give a precise meaning to the expression “motivic structure on the affine algebras of the $\pi_1(X; b, a)_{B/\text{dR}}$ ”. We shall also say “motivic structure on the $\pi_1(X; b, a)_{B/\text{dR}}$ ”. Heuristically, these are structures coming from geometry, and hence having a meaning both for Betti and for de Rham, and respected by the comparison isomorphisms. Examples: the product

giving the algebra structure, the coproducts coming from composition of paths, the weight filtration, the isomorphism $\pi_1(X; b, a)_{B/\mathrm{dR}} \rightarrow \pi_1(X; \sigma(b), \sigma(a))_{B/\mathrm{dR}}$ induced by the involution $\sigma : x \mapsto 1 - x$ of $\mathbb{P}^1$. Likewise, the morphisms (3.3.1) are a motivic structure on $\mathbb{Q}(1)_{B/\mathrm{dR}}$ and $\pi_1(X, a)_{B/\mathrm{dR}}$. By contrast, neither the Hodge filtration of the affine algebra of $\pi_1(X; b, a)_{\mathrm{dR}}$, nor therefore its grading, nor the independence of $\pi_1(X; b, a)_{\mathrm{dR}}$ from a and b , is motivic. They are available only for de Rham. The structures in question will be structures of multilinear algebra. Let H_B (respectively H_{dR}) be the group scheme over $\mathbb{Q}$ of automorphisms of $\pi_1(X; 1, 0)_B$ (respectively $\pi_1(X; 1, 0)_{\mathrm{dR}}$) respecting all its motivic structures. The weight filtration W of the affine algebra is a filtration by finite-dimensional subspaces, and H_B (respectively H_{dR}) is a projective limit of algebraic subgroups of the $\mathrm{GL}(W_i)$.

4.2

Let $P_{\mathrm{dR}, B}$ be the scheme of isomorphisms from $\pi_1(X; 1, 0)_B$ to $\pi_1(X; 1, 0)_{\mathrm{dR}}$ respecting all motivic structures. The group H_B acts on the right on $P_{\mathrm{dR}, B}$. The comparison isomorphism $\mathrm{comp}_{\mathrm{dR}, B}$ is a complex point of $P_{\mathrm{dR}, B}$. The scheme $P_{\mathrm{dR}, B}$ is therefore nonempty. It is an H_B -torsor (terminology of 2.2).

If Z_B is a subscheme of $\pi_1(X; 1, 0)_B$ stable under H_B , twisting by the H_B -torsor $P_{\mathrm{dR}, B}$ transforms $\pi_1(X; 1, 0)_B$ into $\pi_1(X; 1, 0)_{\mathrm{dR}} = {}_1\Pi_0$, and Z_B into a subscheme Z_{dR} of ${}_1\Pi_0$. Over $\mathbb{C}$, $Z_{\mathrm{dR}}(\mathbb{C})$ is the image of $Z_B(\mathbb{C})$ by $\mathrm{comp}_{\mathrm{dR}, B}$.

I like to think of the H_B -orbit of dch as the space of points ch of $\pi_1(X; 1, 0)_B$ having all the properties of dch expressible in motivic terms. For example: the morphism $\pi_1(X; 1, 0)_B \rightarrow \pi_1(X; 0, 1)_B$ induced by the involution $x \mapsto 1 - x$ of $\mathbb{P}^1$ must transform ch into its inverse.

Let $M_B \subset \pi_1(X; 1, 0)_B$ be the closure of the H_B -orbit of dch . The twist of M_B by $P_{\mathrm{dR}, B}$ is a closed subscheme M_{dR} of ${}_1\Pi_0$. We again write dch for the image of dch by $\mathrm{comp}_{\mathrm{dR}, B}$. It is a complex point of M_{dR} , and $M_{\mathrm{dR}}(\mathbb{C}) \subset {}_1\Pi_0(\mathbb{C})$ is the closure of its $H_{\mathrm{dR}}(\mathbb{C})$ -orbit.

Definition 4.3. Let $s = (s_1, \dots, s_k)$. If $k \neq 0$, we assume $s_1 \geq 2$. Let w be the word $0^{\{s_1-1\}}1 \dots 0^{\{s_k-1\}}1$ in the alphabet $\{0, 1\}$, and let $c(w) \in O(\Pi)$ be the function “coefficient of the monomial $e_0^{s_1-1}e_1 \dots e_0^{s_k-1}e_1$ ” (3.1). We denote by $\zeta^m(s)$ the restriction to M_{dR} of $(-1)^k c(w)$.

The $\zeta^m(s)$ are the motivic multizetas (F. Brown [1] 2.3). The algebra denoted M in the introduction is $O(M_{\mathrm{dR}})$. Before [1], only the restriction of the $\zeta^m(s)$ to the orbit of 1 was considered. This gave rise to painful contortions. By 3.5, the value of $\zeta^m(s)$ at the complex point dch of M_{dR} is $\zeta(s)$. Every polynomial identity with rational coefficients in the $\zeta^m(s)$ therefore gives, by evaluation at dch , a polynomial relation among the $\zeta(s)$. Grothendieck’s yoga of periods predicts that every polynomial relation among the $\zeta(s)$ comes from geometry; more precisely, that it is obtained as above.

For every word w in 0 and 1, similarly write $c^m(w)$ for the restriction of $c(w)$ to M_{dR} . Relying on the remark following 3.5, one can show that the $c^m(w)$ are linear combinations with integer coefficients of the $\zeta^m(s)$.

By construction, H_{dR} acts on M_{dR} . Its Lie algebra therefore acts by derivations on the affine algebra of M_{dR} , a quotient of that of Π . These derivations, which have a meaning only once one has lifted from the $\zeta(s)$ to the $\zeta^m(s)$, will play a key role in F. Brown’s arguments.

5. Mixed Tate motives

5.1

We are here in one of the cases in which the philosophy of motives is not only a precious guide, but allows proofs. Over every field K , Voevodsky has defined a motivic triangulated category V_K , endowed with a tensor product, and containing Tate objects $\mathbb{Z}(n)$ ($n \in \mathbb{Z}$). If K is a

number field, after tensoring with $\mathbb{Q}$ one can construct a subcategory $MT(K)$ of $V_K \otimes \mathbb{Q}$, the category of mixed Tate motives over K , and this category is Tannakian. This rests on our knowledge of the K -theory groups of a number field, due to the transcendental determination of $H_i(BGL(n, K), \mathbb{C})$ for $n \gg i$.

All this is for me a black box. It is difficult to open, and this is why the identities promised by 0.2 are not explicit.

5.2

Let us review some properties of the category $MT(K)$. It is a Tannakian category over $\mathbb{Q}$. It contains a rank-one object $\mathbb{Q}(1)$, the Tate motive. Since this motive has rank one, $\mathbb{Q}(n) := \mathbb{Q}(1)^{\otimes n}$ is defined for $n \in \mathbb{Z}$. The objects of $MT(K)$ are endowed with a finite increasing functorial filtration, compatible with the tensor product, the weight filtration W . It is indexed by even integers. The functor $M \mapsto \text{Gr}^W(M)$ is exact, and $\text{Gr}_{-2n}^W(M)$ is a sum of copies of $\mathbb{Q}(n)$.

Suppose, for simplicity, that $K = \mathbb{Q}$. Then one has Betti and de Rham fiber functors

$$\omega_B, \omega_{\text{dR}} : MT(K) \longrightarrow \mathbb{Q}\text{-vector spaces}$$

and a comparison isomorphism

$$\text{comp}_{\text{dR}, B} : \omega_B \otimes \mathbb{C} \longrightarrow \omega_{\text{dR}} \otimes \mathbb{C}.$$

Note that, because $K = \mathbb{Q}$, the fiber functor ω_{dR} coincides with the fiber functor ω of [2]. For the motive $\mathbb{Q}(1)$, $\omega_B(\mathbb{Q}(1))$, $\omega_{\text{dR}}(\mathbb{Q}(1))$, and $\text{comp}_{\text{dR}, B}$ are as in 2.4.

The functor ω_{dR} is endowed with a Hodge filtration F , opposite, up to a renumbering, to the weight filtration of $\omega_{\text{dR}}(M)$ by the $\omega_{\text{dR}}(W_{2n}M)$. This makes it into a graded functor, as in 3.1.

5.3

If T is a Tannakian category, we shall call a T -algebra an Ind-object A of T , endowed with a product $A \otimes A \rightarrow A$. One defines the category of affine T -schemes as the opposite of that of commutative T -algebras with unit, and an affine T -group scheme as a group in this category. For T a category of motives, one replaces the prefix T by the adjective motivic.

5.4

The $\pi_1(X; b, a)_B$, $\pi_1(X; b, a)_{\text{dR}}$ of 3.1, 3.3, and the comparison isomorphism between their complexifications are motivic: they are images of a motivic group scheme $\pi_1(X; b, a)_{\text{mot}}$ ([2] §4). The morphisms “composition of paths” and (3.3.1) are motivic as well.

If $X = \mathbb{P}^1 - \{0, 1, \infty\}$ and a, b are chosen among the tangential base points 0, 1 of 3.4, the points 0, 1, ∞ remain distinct after reduction modulo p , and the tangent vectors chosen at 0 and 1 remain nonzero. One can deduce that the affine algebras $O(\pi_1(X; b, a)_{\text{mot}})$ are Ind-objects of a Tannakian subcategory $MT(\mathbb{Z})$ of $MT(\mathbb{Q})$, that of mixed Tate motives with good reduction over $\text{Spec}(\mathbb{Z})$ ([2] 1.7, 1.4, 1.8). In $MT(\mathbb{Z})$,

$$\text{Ext}^1(\mathbb{Q}(0), \mathbb{Q}(n)) \text{ is } \begin{cases} \text{one-dimensional over } \mathbb{Q}, & n \text{ odd and } n \geq 3, \\ 0, & \text{otherwise,} \end{cases} \quad (5.4.1)$$

$$\text{Ext}^2(\mathbb{Q}(0), \mathbb{Q}(n)) = 0. \quad (5.4.2)$$

5.5

Let T be a Tannakian category over a field K , and let ω be a fiber functor over an extension K' of K . The family of the $\omega(Y)$, for Y in T , endowed with the $\omega(f) : \otimes \omega(Y_i) \rightarrow \otimes \omega(Z_j)$ for $f : \otimes Y_i \rightarrow \otimes Z_j$ a morphism of T , is a family of K' -vector spaces endowed with a structure of multilinear algebra. The subgroup scheme of $\prod \mathrm{GL}(\omega(Y))$ (product over $\mathrm{Ob}(T)$) that respects this structure is the group scheme $\mathrm{Aut}(\omega)$ of automorphisms of the fiber functor ω . It is denoted $\pi(T, \omega)$ and is called the fundamental group of T at ω . One will see in examples that it is of the “same nature” as the $\omega(Y)$, for example that its affine algebra is graded if the functor ω is graded. The explanation of this fact is that there exists a T -group scheme $\pi(T)$, acting functorially on each object of T , such that for every fiber functor ω , $\pi(T, \omega)$ is $\omega(\pi(T))$.

Suppose $K' = K$. In this case, ω induces an equivalence of T with the category of finite-dimensional linear representations of $\pi(T, \omega)$. If T is a category of motives, the fundamental groups are also called motivic Galois groups.

5.6

Let $\langle(Y_i)\rangle$ be the Tannakian subcategory of T generated (by subquotients, sums, duals, and tensor products) by a family (Y_i) of objects of T . A T -structure on the family of the $\omega(Y_i)$ is the image by ω of a morphism of $\langle(Y_i)\rangle$. More precisely, it is the datum of the image by ω of subquotients of sums of tensor products of the Y_i and $Y_i^\vee$, and of morphisms between such subquotients. If T is a category of motives, one will rather say motivic structure. The subgroup of $\prod \mathrm{GL}(\omega(Y_i))$ which respects all T -structures is therefore the fundamental group of $\langle(Y_i)\rangle$ at ω . Every algebraic subgroup of a group $\mathrm{GL}(V)$ is the subgroup respecting a few subspaces and elements in tensor spaces on V . The equivalence 5.5 therefore ensures that this fundamental group of $\langle(Y_i)\rangle$ at ω is the quotient of $\pi(T, \omega)$ which acts faithfully on the family of the $\omega(Y_i)$. Same terminology if one starts from a family of ind-objects of T : one replaces it by the family of subobjects in T .

Take $T := MT(\mathbb{Z})$. With the notation B/dR of 2.3, set

$$G_{B/\mathrm{dR}} := \pi(MT(\mathbb{Z}), \omega_{B/\mathrm{dR}}). \quad (5.6.1)$$

The group scheme $H_{B/\mathrm{dR}}$ of automorphisms of $\pi_1(X; 1, 0)_{B/\mathrm{dR}}$ “respecting all motivic structures” considered in 4.1 is the quotient of $G_{B/\mathrm{dR}}$ acting faithfully on

$$\pi_1(X; 1, 0)_{B/\mathrm{dR}} = \omega_{B/\mathrm{dR}}(\pi_1(X; 1, 0)_{\mathrm{mot}}).$$

By the equivalence 5.5, a closed subscheme of $\pi_1(X; 1, 0)_B$ stable under H_B is the Betti realization of a closed subscheme of $\pi_1(X; 1, 0)_{\mathrm{mot}}$. Example: there is a unique motivic subscheme M_{mot} of $\pi_1(X; 1, 0)_{\mathrm{mot}}$ whose Betti realization is the orbit closure M_B ; the subscheme M_{dR} (4.2) of $\pi_1(X; 1, 0)_{\mathrm{dR}}$ is its de Rham realization.

5.7

The action of $G_{B/\mathrm{dR}}$ on $\mathbb{Q}(1)_{B/\mathrm{dR}}$ defines a morphism t from $G_{B/\mathrm{dR}}$ to the multiplicative group $\mathbb{G}_m$. For every M in $MT(\mathbb{Z})$, $G_{B/\mathrm{dR}}$ respects the weight filtration of $\omega_{B/\mathrm{dR}}M$, and since $\mathrm{Gr}_{-2n}^W(M)$ is a sum of copies of $\mathbb{Q}(n)$, the action of g in $G_{B/\mathrm{dR}}$ on $\mathrm{Gr}_{-2n}^W(M)$ is multiplication by $t(g)^n$. The kernel $U_{B/\mathrm{dR}}$ of t therefore acts trivially on the $\mathrm{Gr}_{-2n}^W(M)$: it is a pro-unipotent group scheme.

Let $\tau(\lambda)$ be multiplication by λ^n on $\omega_{\mathrm{dR}}^n(M)$. The $\tau(\lambda)$ define a section $\tau : \mathbb{G}_m \rightarrow G_{\mathrm{dR}}$ of t , which makes G_{dR} a semidirect product $\mathbb{G}_m \ltimes U_{\mathrm{dR}}$. The action of $\mathbb{G}_m$ on U_{dR} (by inner automorphisms in G_{dR}) will again be denoted τ . It permits one to define the graded Lie algebra $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$. The vanishing (5.4.2) implies that this graded Lie algebra is free, and

(5.4.1) that it is freely generated by one generator in each odd degree ≥ 3 . The functor ω_{dR} induces an equivalence of categories from $MT(\mathbb{Z})$ with the category of finite-dimensional graded vector spaces, endowed with an action, compatible with the gradings and therefore nilpotent, of $\text{Lie}^{\text{gr}}(U_{\text{dR}})$.

5.8

Let P be the scheme of isomorphisms of fiber functors from ω_{dR} to ω_B . The comparison isomorphism $\text{comp}_{B,\text{dR}}$ is a complex point of P . The scheme P is therefore nonempty, and is a G_{dR} -torsor. Since G_{dR} is an extension of $\mathbb{G}_m$ by U_{dR} , a Galois cohomology argument shows that every G_{dR} -torsor has a rational point: there exists an isomorphism of fiber functors $p : \omega_{\text{dR}} \rightarrow \omega_B$. Write

$$p = \text{comp}_{B,\text{dR}} \cdot a \quad (5.8.1)$$

with $a \in G_{\text{dR}}(\mathbb{C})$. It follows from (5.8.1) that for every M in $MT(\mathbb{Z})$, if one identifies $\omega_{\text{dR}}(M) \otimes \mathbb{C}$ with $\omega_B(M) \otimes \mathbb{C}$ by $\text{comp}_{B,\text{dR}}$, $\omega_B(M) \subset \omega_B(M) \otimes \mathbb{C} = \omega_{\text{dR}}(M) \otimes \mathbb{C}$ verifies

$$\omega_B(M) = a(\omega_{\text{dR}}(M)). \quad (5.8.2)$$

Since $\omega_B(\mathbb{Q}(1)) = 2\pi i \mathbb{Q}$, one has $t(a) \in 2\pi i \mathbb{Q}^*$, and $a = u\tau(\lambda)$ with $u \in U_{\text{dR}}(\mathbb{C})$ and $\lambda \in 2\pi i \mathbb{Q}^*$. A compatibility with complex conjugation makes it possible to show that one can choose ([2] 2.12)

$$a \in U_{\text{dR}}(\mathbb{R}) \cdot \tau(2\pi i). \quad (5.8.3)$$

6. The space $\tilde{H}(2, 3)$ of functions on $\pi_1(X; 1, 0)_{\text{dR}}$

From now on, $X := \mathbb{P}^1 - \{0, 1, \infty\}$, and the notation of 3.4 is in force. By abuse of language, write $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ (respectively $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$) for the group scheme over $\mathbb{Q}$ such that, for every $\mathbb{Q}$ -algebra R , the set of its R -points is $R\langle\langle e_0, e_1 \rangle\rangle$ (respectively $R\langle\langle e_0, e_1 \rangle\rangle^*$). Recall that $\pi_1(X; b, a)_{\text{dR}}$ is a copy ${}_b\Pi_a$ of the subscheme Π of group-like elements (3.1.3) of $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$. We shall regard it as contained in a copy $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle_{b,a}$ of $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$. A point x of $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ will sometimes be denoted ${}_bx_a$ when regarded as belonging to this copy.

6.1

As a $\mathbb{Q}$ -vector space, the affine algebra $O(\Pi)$ of Π has for basis the “coefficients of monomials” $c(w)$ of 3.1. It is graded. If the word w in the alphabet $\{0, 1\}$ has length d , the element $c(w)$ of $O(\Pi)$ is of degree $-d$, of codegree d . When we write B' , with decorations attached, for a set of words, B , with the same decorations attached, will be the corresponding set of $c(w)$.

Definition 6.2. (i) $B'(2, 3)$ is the set of words that are concatenations of the words 01 and 001;

(ii) $\tilde{H}(2, 3)$ is the vector subspace of $O(\Pi)$ with basis $B(2, 3)$.

The restrictions to M_{dR} of the $c(w)$ in $B(2, 3)$ are those of the $(-1)^k \zeta^m(s_1, \dots, s_k)$ where each s_i is 2 or 3.

For every graded vector space Z , with finite-dimensional homogeneous components, the Poincaré series of Z is

$$P(Z, t) = \sum \dim(Z_d) t^d. \quad (6.2.1)$$

If $\tilde{H}(2, 3)$ is graded by the codegree, one has

$$P(\tilde{H}(2, 3), t) = (1 - t^2 - t^3)^{-1}. \quad (6.2.2)$$

This is verified by expanding $(1 - t^2 - t^3)^{-1} = \sum (t^2 + t^3)^n$.

6.3

The words $w \in B'(2, 3)$ are characterized by:

- a. they do not begin with 1 and do not end with 0;
- b. they do not contain two consecutive 1's;
- c. they do not contain three consecutive 0's.

For a function $f \in O(\Pi)$ to be a linear combination of $c(w)$ where w satisfies a, b, or c is equivalent respectively to:

- a'. for γ in Π , $f(\exp(ue_1)\gamma \exp(te_0))$ is independent of t and u ;
- b'. for γ, δ in Π , $f(\gamma \exp(te_1)\delta)$ is of degree ≤ 1 in t ;
- c'. for γ, δ in Π , $f(\gamma \exp(te_0)\delta)$ is of degree ≤ 2 in t .

To verify this, note that the linear combinations of elements of $\Pi(\mathbb{Q})$ are dense in $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$, and that if f is interpreted as a continuous linear form on $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$, condition a' (respectively b', respectively c') is equivalent to the same condition, where one allows γ, δ to be arbitrary in $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$. The conditions a', b', c' are motivic, because they are expressible in terms of composition of paths and of (3.3.1). Here one must interpret γ (respectively γ, δ) as being in ${}_1\Pi_0$ (respectively in ${}_1\Pi_1$ and ${}_1\Pi_0$, respectively in ${}_1\Pi_0$ and ${}_0\Pi_0$).

To be convinced of the motivic character of a', b', c', one can observe that these conditions admit a "Betti" formulation. One obtains it by regarding f as a function on $\pi_1(X; 1, 0)$, and replacing $\exp(te_0)$ (respectively $\exp(te_1)$) by σ_0^t (respectively σ_1^t) with $t \in \mathbb{Z}$ and σ_0 (respectively σ_1) a positive turn around 0 in the punctured tangent space to $\mathbb{P}^1$ at 0 (respectively at 1) (cf. 3.3).

We leave to the reader the verification of the following lemma:

Lemma 6.4. Let $f = \sum \lambda_w c(w)$ in $O(\Pi)$. In order that, for all $\gamma_0, \dots, \gamma_r$ in Π ,

$$f(\gamma_0 \exp(te_0)\gamma_1 \exp(te_0) \cdots \exp(te_0)\gamma_r) \quad (6.4.1)$$

be of degree $\leq d$ in t , it is necessary and sufficient that for every w such that $\lambda_w \neq 0$, the total length of at most r disjoint strings of consecutive 0's in w be $\leq d$.

6.5

Let $B'_\ell(2, 3)$ (respectively $B'_{\leq \ell}(2, 3)$) be the set of words w in $B'(2, 3)$ having exactly (respectively at most) ℓ factors 001. With the convention of 6.1, let N_ℓ be the subspace of $\tilde{H}(2, 3)$ with basis $B_{\leq \ell}(2, 3)$. It follows from 6.4 that for f in $\tilde{H}(2, 3)$ to be in N_ℓ , it is necessary and sufficient that for $r = \ell + 1$, and every choice of the γ_i , (6.4.1) be of degree $\leq 2\ell + 1$ in t . This property is motivic. The action of G_{dR} on $O({}_1\Pi_0)$ therefore respects $\tilde{H}(2, 3)$ and its increasing filtration N .

Proposition 6.6. The action of U_{dR} on $\text{Gr}^N \tilde{H}(2, 3)$ is trivial.

Proof. The graded Lie algebra of U_{dR} is generated by elements ν_d of odd degrees $d \geq 3$, and, if $O({}_1\Pi_0)$ is graded by the degree (6.1), the action is compatible with the gradings. It suffices to show that each ν_d acts trivially. Let w be a word of length n , a concatenation of words 01 and of ℓ words 001. We have $n \equiv \ell \pmod{2}$. The image of $c(w)$ by ν_d is a linear combination of $c(w')$ with w' of length $n - d$ and a concatenation of words 01 and of $\ell' \leq \ell$ words 001. We have $n - d \equiv \ell' \pmod{2}$, and, d being odd, ℓ and ℓ' do not have the same parity. That $\ell' < \ell$ proves 6.6.

6.7

It follows from 6.6 that the action of $\mathrm{Lie}^{\mathrm{gr}} U_{\mathrm{dR}}$ on $N_\ell/N_{\ell-2}$ factors through the abelianization $\mathrm{Lie}^{\mathrm{gr}} U_{\mathrm{dR}}^{\mathrm{ab}}$. The action of ν_d on $N_\ell/N_{\ell-2}$ therefore does not depend, up to a factor, on the choice of the generators ν_d . It is given by a map

$$\mathrm{Gr}^N(\nu_d) : \mathrm{Gr}_\ell^N(\tilde{H}(2, 3)) \longrightarrow \mathrm{Gr}_{\ell-1}^N(\tilde{H}(2, 3)). \quad (6.7.1)$$

The vector space $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))$ has for basis the image of $B_\ell(2, 3)$ (6.5). It is graded by the codegree. Let $B'_\ell(2, 3)_n$ be the part of $B'_\ell(2, 3)$ consisting of words of length n . Write $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))_n$ for the direct factor of codegree n of $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))$. It has for basis the image of $B_\ell(2, 3)_n$. It is nonzero only for $n \equiv \ell \pmod{2}$ and $n - 3\ell \geq 0$.

The morphisms (6.7.1) induce

$$\sum_d \mathrm{Gr}^N(\nu_d) : \mathrm{Gr}_\ell^N(\tilde{H}(2, 3))_n \longrightarrow \bigoplus_d \mathrm{Gr}_{\ell-1}^N(\tilde{H}(2, 3))_{n-d}, \quad (6.7.2)$$

where it suffices to sum over odd d satisfying $3 \leq d \leq 3 + n - 3\ell$. The left-hand side of (6.7.2) has a basis indexed by $B'_\ell(2, 3)_n$. The right-hand side has a basis indexed by the union of the $B'_{\ell-1}(2, 3)_{n-d}$. If $\ell \geq 1$, one obtains a bijection between these sets of words by associating to the word $w \in B'_\ell(2, 3)_n$ the word $\bar{w}$ deduced from it by removing from w its final $(001)(01) \cdots (01)$.

F. Brown computes the morphisms (6.7.2) and proves:

Theorem 6.8. For $\ell \geq 1$, the morphisms (6.7.2) are isomorphisms.

The strategy that he uses to prove 6.8 will be explained in §8.

Corollary 6.9. The restriction morphism to M_{dR}

$$\mathrm{restr} : \tilde{H}(2, 3) \longrightarrow O(M_{\mathrm{dR}}) \quad (6.9.1)$$

is injective.

Applying (6.2.2), one deduces from 6.9:

Corollary 6.10. The Poincaré series of $O(M_{\mathrm{dR}})$ satisfies

$$P(O(M_{\mathrm{dR}}), t) \geq (1 - t^2 - t^3)^{-1} \quad (6.10.1)$$

(coefficientwise inequality), with equality if and only if (6.9.1) is bijective.

Assume 6.8, and prove 6.9.

Lemma 6.11. The morphism (6.9.1) is injective on N_0 .

Proof. The morphism induced by (6.9.1) from N_0 to $O(M_{\mathrm{dR}})$ is compatible with the gradings by the codegree. It therefore suffices to verify its injectivity separately in each codegree. The $c((01)^n)$ form a basis of N_0 , and $c((01)^n)$ is of codegree $2n$. Since these codegrees are distinct, it suffices to verify that for each n the restriction of $c((01)^n)$ to M_{dR} is nonzero. The value of $c((01)^n)$ at $dch \in M_{\mathrm{dR}}(\mathbb{C})$ is $(-1)^n \zeta(2^{\{n\}})$, where $\zeta(2^{\{n\}}) = \zeta(2, \dots, 2)$ with n copies of “2”. One concludes by observing that every multizeta number is > 0 , hence nonzero.

Remark 6.12. Expanding the identities due to Euler

$$\prod_{n>0} (1 - z^2/n^2) = \sin(\pi z)/\pi z = \sum (-1)^n (\pi z)^{2n}/(2n+1)!,$$

one sees that $\zeta(2^{\{n\}}) = \pi^{2n}/(2n+1)!$. That $\zeta(2^{\{n\}})$ is a rational multiple of π^{2n} already followed from the fact that the motivic subspace N_0 of $O(1\Pi_0)$ is fixed under U_{dR} , hence is the de Rham realization of a sum of Tate motives. This forces $\zeta(2^{\{n\}})$ to be a period of $\mathbb{Q}(-2n)$.

6.13. Proof of 6.8 $\Rightarrow$ 6.9

Filter the image $\text{restr}(\tilde{H}(2,3))$ of $\tilde{H}(2,3)$ in $O(M_{\text{dR}})$ by the images of the N_ℓ . It suffices to show the injectivity of

$$\tilde{H}(2,3) \longrightarrow \text{restr}(\tilde{H}(2,3)) \quad (6.13.1)$$

after passage to the associated gradeds. We prove by induction on $\ell \geq 0$ the injectivity on $\text{Gr}_\ell^N(\tilde{H}(2,3))$. Lemma 6.11 proves this injectivity for $\ell = 0$. If $\ell > 0$, consider the morphism

$$\sum \nu_d : \text{Gr}_\ell^N \tilde{H}(2,3) \longrightarrow \bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2,3),$$

where the target is a sum of copies of $\bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2,3)$ indexed by the odd integers $d \geq 3$. Theorem 6.8 ensures that this morphism is bijective. The morphism (6.9.1) commutes with the derivations ν_d . The diagram

$$\begin{array}{ccc} \text{Gr}_\ell^N \tilde{H}(2,3) & \xrightarrow{\text{Gr}_\ell^N(6.13.1)} & \text{Gr}_\ell^N \text{restr}(\tilde{H}(2,3)) \\ \sum \nu_d \downarrow \sim & & \downarrow \sum \nu_d \\ \bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2,3) & \xrightarrow{\text{Gr}_{\ell-1}^N(6.13.1)} & \bigoplus \text{Gr}_{\ell-1}^N \text{restr}(\tilde{H}(2,3)) \end{array}$$

is therefore commutative. The induction hypothesis ensures that $\text{Gr}_{\ell-1}^N(6.13.1)$ is injective. Since the first vertical arrow is injective, $\text{Gr}_\ell^N(6.13.1)$ is injective too.

7. Structure of the orbit closure M_{dR}

7.1

Consider the groupoid of the ${}_b\Pi_a$ for $a, b \in \{0, 1\}$, endowed with $e_0 \in \text{Lie}({}_0\Pi_0)$ and $e_1 \in \text{Lie}({}_1\Pi_1)$. Let V_{dR} be the group scheme of automorphisms of this structure. The action of U_{dR} on the groupoid of the ${}_b\Pi_a$ factors through a morphism

$$I : U_{\text{dR}} \longrightarrow V_{\text{dR}}. \quad (7.1.1)$$

We write IU_{dR} for the image of U_{dR} in V_{dR} .

The group of automorphisms that respect $e_0 \in \text{Lie}({}_0\Pi_0)$ and $e_1 \in \text{Lie}({}_1\Pi_1)$ only up to a factor, the same for e_0 and e_1 , is a semidirect product $\mathbb{G}_m \ltimes V_{\text{dR}}$. We write $\circ$ for its group law, and τ for the inclusion of $\mathbb{G}_m$. The action of $\mathbb{G}_m$ on the ${}_b\Pi_a$ is that of the subgroup $\mathbb{G}_m$ of G_{dR} . It corresponds to the gradings.

By [2] 5.9,

$$v \longmapsto v({}_110) : V_{\text{dR}} \longrightarrow {}_1\Pi_0 \quad (7.1.2)$$

is an isomorphism of schemes. This isomorphism identifies V_{dR} with Π , endowed with a new group law, denoted $\circ$. The action τ of $\mathbb{G}_m$ which defines the grading of $O({}_1\Pi_0)$ fixes ${}_110$. It follows that (7.1.2) is compatible with the gradings. Since it sends identity element to identity element, it defines an isomorphism of graded vector spaces

$$\text{Lie}^{\text{gr}} V_{\text{dR}} \xrightarrow{\sim} \text{Lie}^{\text{gr}} \Pi. \quad (7.1.3)$$

The graded Lie algebra $\text{Lie}^{\text{gr}} V_{\text{dR}}$ is thus identified with $\text{Lie}^{\text{gr}} \Pi$, endowed with a new bracket, the Ihara bracket $\{\cdot, \cdot\}$.

7.2

Write $x \mapsto \gamma_{b,a}\langle x \rangle$ for the action on ${}_b\Pi_a$ of γ in $(\Pi, \circ)$. It is computed as follows. The action on ${}_a\Pi_a$ respects the group structure. For $a = 0$, it is induced by the automorphism

$$\gamma_{00} : e_0 \mapsto e_0, \quad e_1 \mapsto \gamma^{-1}e_1\gamma \quad (7.2.1)$$

of the completed enveloping algebra $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ of $\text{Lie } \Pi$. To verify this, note that $e_0 \in \text{Lie}({}_0\Pi_0)$ and $e_1 \in \text{Lie}({}_1\Pi_1)$ are fixed by definition of V_{dR} . Writing ${}_0(e_1)_0 = ({}_110)^{-1}{}_1(e_1)_1({}_110)$, one obtains as promised

$${}_0(e_1)_0 \mapsto \gamma^{-1}{}_1(e_1)_1\gamma.$$

The automorphism γ_{10} of ${}_1\Pi_0$ is induced by the γ_{00} -semilinear automorphism

$$\gamma_{10} : x \mapsto \gamma \gamma_{00}\langle x \rangle \quad (7.2.2)$$

of the right $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ -module $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$. Verify this as above, starting from ${}_1x_0 = {}_110{}_0x_0$. Since $\gamma \circ x$ sends ${}_110$ to $\gamma_{10}\langle {}_1x_0 \rangle$, this is also

$$\gamma_{10} : x \mapsto \gamma \circ x, \quad (7.2.3)$$

and therefore

$$x \circ y = x x_{00}\langle y \rangle. \quad (7.2.4)$$

7.3

Via $\text{comp}_{\text{dR}, B}$, regard $\pi_1(X; b, a)_B$ as a $\mathbb{Q}$ -structure on the complexification of $\pi_1(X; b, a)_{\text{dR}} = {}_b\Pi_a$, the Betti $\mathbb{Q}$ -structure, denoted $({}_b\Pi_a)_B$. For this new $\mathbb{Q}$ -structure, $2\pi i e_0$ in $\text{Lie}({}_0\Pi_0)_{\mathbb{C}}$ and $2\pi i e_1$ in $\text{Lie}({}_1\Pi_1)_{\mathbb{C}}$ are rational: their exponentials are the σ_0 and σ_1 of 6.3. One deduces:

Lemma 7.4. Let ch be a rational point of $({}_1\Pi_0)_B$. Regard its image by $\text{comp}_{\text{dR}, B}$, still denoted ch , as a complex point of $(\Pi, \circ)$. Then the complex point $ch \circ \tau(2\pi i)$ of $\mathbb{G}_m \ltimes V_{\text{dR}} = \mathbb{G}_m \ltimes (\Pi, \circ)$ transforms the $\mathbb{Q}$ -structure of ${}_b\Pi_a$ into its Betti $\mathbb{Q}$ -structure.

7.5

Let $a = u\tau(2\pi i) \in U_{\text{dR}}(\mathbb{R})\tau(2\pi i) \subset G_{\text{dR}}(\mathbb{C})$ be as in (5.8.3). If ch is as in 7.4, both $ch\tau(2\pi i)$ and $I(u)\tau(2\pi i)$ transform the $\mathbb{Q}$ -structure of the ${}_b\Pi_a$ into their Betti $\mathbb{Q}$ -structure. If $\gamma \in V_{\text{dR}}(\mathbb{C}) = \Pi(\mathbb{C})$ is defined by

$$ch \circ \tau(2\pi i) = I(u) \circ \tau(2\pi i) \circ \gamma, \quad (7.5.1)$$

then γ respects the $\mathbb{Q}$ -structure, and hence is in $V_{\text{dR}}(\mathbb{Q}) = \Pi(\mathbb{Q})$. Writing again τ for the action of $\mathbb{G}_m$ on Π , one can rewrite (7.5.1) in the form

$$ch = I(u) \circ \tau(2\pi i)(\gamma).$$

Apply this to the straight path dch :

Lemma 7.6. For u as above, there exists γ in $\Pi(\mathbb{Q})$ such that

$$dch = I(u) \circ \tau(2\pi i)(\gamma). \quad (7.6.1)$$

Every such γ verifies

$$\tau(-1)(\gamma) = \gamma. \quad (7.6.2)$$

Proof. Equation (7.6.2) follows from the fact that u and dch , and hence also $\tau(2\pi i)(\gamma)$, are real.

7.7

Apply (7.2.3) to (7.6.1). One obtains that the orbit of dch under $U_{\text{dR}}(\mathbb{C})$ is

$$IU_{\text{dR}}(\mathbb{C}) \circ \tau(2\pi i)(\gamma) \subset \Pi_{\text{dR}}(\mathbb{C}).$$

To obtain the orbit under $G_{\text{dR}} = \mathbb{G}_m \ltimes U_{\text{dR}}$, it suffices to replace $\tau(2\pi i)(\gamma)$ by its orbit $\tau(\mathbb{C}^*)(\gamma)$ under $\mathbb{G}_m$:

$$G_{\text{dR}}(\mathbb{C})(dch) = IU_{\text{dR}}(\mathbb{C}) \circ \tau(\mathbb{C}^*)(\gamma). \quad (7.7.1)$$

Indeed, IU_{dR} is already stable under $\mathbb{G}_m$. As we already knew, the orbit (7.7.1) is, in the de Rham sense, defined over $\mathbb{Q}$. It is the set of complex points of

$$IU_{\text{dR}} \circ \tau(\mathbb{G}_m)(\gamma) \subset \Pi. \quad (7.7.2)$$

In (7.7.2), the product is that of $(\Pi, \circ) = V_{\text{dR}}$, and the result is regarded as a subscheme of ${}_1\Pi_0$.

In $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$, write $\gamma = \sum c_w w$. By (7.6.2), the sum runs only over monomials w of even degree. If

$$\rho(\lambda) := \sum \lambda^{\deg(w)/2} c_w w,$$

then $\rho(\lambda)$ is defined for $\lambda \in \mathbb{G}_a$ and verifies

$$\rho(\lambda^2) = \tau(\lambda)(\gamma). \quad (7.7.3)$$

Proposition 7.8. The morphism of schemes

$$IU_{\text{dR}} \times \mathbb{G}_a \longrightarrow \Pi : (u, \lambda) \longmapsto u \circ \rho(\lambda) \quad (7.8.1)$$

induces an isomorphism from $IU_{\text{dR}} \times \mathbb{G}_a$ to $M_{\text{dR}} \subset {}_1\Pi_0$ defined in 4.2.

Proof. If u is in $IU_{\text{dR}} \subset V_{\text{dR}} = \Pi \subset \mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$, it can be written $u = 1 + \sum e_w w$, where the sum runs only over monomials of degree ≥ 3 . This is a consequence of the fact that $\text{Lie}^{\text{gr}} U_{\text{dR}}$ is zero in degree < 3 . The coefficient of $e_0 e_1$ in $u \circ \rho(\lambda)$ therefore coincides with that in $\rho(\lambda)$. The coefficient of $e_0 e_1$ in dch is $-\pi^2/6 = (2\pi i)^2/24$. By (7.7.3), that in γ is therefore $1/24$, and that in $\rho(\lambda)$ is $\lambda/24$. Starting from $x = u \circ \rho(\lambda)$, one recovers λ as $24c(e_0 e_1)(x)$. That x be of the form $u \circ \rho(\lambda)$ is equivalent to saying that u defined by $x = u \circ \rho(24c(e_0 e_1)(x))$ is in IU_{dR} . This is a closed condition, and (7.8.1) is an isomorphism onto a closed subscheme of Π , in which the orbit of dch is dense. This proves 7.8.

7.9

We shall write D for the image of $\mathbb{G}_a$ by ρ . Thus $\rho : \mathbb{G}_a \xrightarrow{\sim} D$, and 7.8 says that the group law $\circ$ induces an isomorphism

$$\circ : IU_{\text{dR}} \times D \xrightarrow{\sim} M_{\text{dR}}. \quad (7.9.1)$$

This isomorphism is compatible with the gradings of affine algebras, if $O(D) = \mathbb{Q}[\lambda]$ is graded by giving λ degree -2 (codegree 2).

7.10

By 6.6 applied to N_0 , the $(-1)^n c((01)^n)$, and therefore their restrictions $\zeta^m(2^{\{n\}})$ to M_{dR} , are U_{dR} -invariant. Transported from M_{dR} to $IU_{\text{dR}} \times D$ by (7.9.1), the action of $u \in U_{\text{dR}}$ is $(a, \lambda) \mapsto (I(u)a, \lambda)$. Viewed as functions on $IU_{\text{dR}} \times D$, the $\zeta^m(2^{\{n\}})$ are therefore inverse images of functions on D . Up to a factor, there is only one such function in each even degree. To determine which one is $\zeta^m(2^{\{n\}})$, it suffices to test on dch . Set $\pi_m^2 := 6\zeta^m(2)$, a definition suggested by the identity $\zeta(2) = \pi^2/6$, and $w_m^{2n} := (\pi_m^2)^n$. By 6.12, one has:

Lemma 7.11. (i) For each n ,

$$\zeta^m(2^{\{n\}}) = \pi_m^{2n} / (2n+1)!.$$

(ii) IU_{dR} is the subscheme of M_{dR} defined by $\pi_m^2 = 0$.

Proposition 7.12. For each integer $n \geq 1$:

- (i) $\zeta^m(2n)$ is a rational multiple of π_m^{2n} (which multiple is seen by evaluating on dch);
- (ii) as a function on $IU_{\text{dR}} \times D$, $\zeta^m(2n+1)$ is the inverse image of a function on IU_{dR} , and this function is a nonzero homomorphism from IU_{dR} to $\mathbb{G}_a$;
- (iii) on abelianized groups, the $\zeta^m(2n+1)$ for $n \geq 1$ induce an isomorphism

$$U_{\text{dR}}^{\text{ab}} \xrightarrow{\sim} IU_{\text{dR}}^{\text{ab}} \xrightarrow{\sim} (\text{product of copies of } \mathbb{G}_a).$$

Proof (sketch). The depth filtration of $\text{Lie } \Pi$ is the filtration by the subalgebras “degree in $e_1 \geq i$ ”. Let $D_i\Pi$ be the corresponding subgroups. If a subgroup Q of Π , viewed as a subgroup of ${}_0\Pi_0$, is stable under V_{dR} , it follows from (7.2.4) that it is also a subgroup of $(\Pi, \circ)$. This applies to the $D_i\Pi$.

The subgroup $D_1\Pi$ is defined by the equation $c(0) = 0$. It contains the derived group Π' , which is defined by $c(0) = c(1) = 0$, hence contains both IU_{dR} and dch . The quotient $D_1\Pi/D_2\Pi$ is also the quotient $D_1\Pi/D_2\Pi$ for the group law $\circ$. It is abelian, and the $c(0^{\{n-1\}}1)$ define an isomorphism of $D_1\Pi/D_2\Pi$ with a product of copies of $\mathbb{G}_a$. Because $IU_{\text{dR}} \subset (\Pi, \circ)$ has a Lie algebra generated in odd degrees, if n is even, $c(0^{\{n-1\}}1)$ is zero on IU_{dR} ; $\zeta^m(n)$, transported by (7.9.1) into a function on $IU_{\text{dR}} \times D$, is the inverse image of a function on D , and (i) is proved by the arguments of 7.10. By (7.6.2), the $c(0^{\{n-1\}}1)$ are zero on D for n odd. That $\zeta(n) \neq 0$ ensures that, for n odd and $n \geq 3$, they are not zero on IU_{dR} . This proves (ii). Assertion (iii) follows from (ii) and from the fact that the graded Lie algebra of U_{dR} has one generator in each odd degree ≥ 3 .

Corollary 7.13. For all integers $a, b \geq 0$, there exist rational numbers $\alpha_{a,b,r}$ ($1 \leq r \leq a+b+1$), unique, such that

$$\zeta^m(2^{\{b\}}32^{\{a\}}) = \sum_{r=1}^{a+b+1} \alpha_{a,b,r} \zeta^m(2r+1) \zeta^m(2^{\{a+b+1-r\}}). \quad (7.13.1)$$

Proof. Regard, by (7.9.1), $\zeta^m(2^{\{b\}}32^{\{a\}})$ as a function on $IU_{\text{dR}} \times D$. Expand it according to powers of a coordinate on D :

$$\zeta^m(2^{\{b\}}32^{\{a\}}) = \sum u_{2r+1} \pi_m^{2(a+b+1-r)},$$

with u_{2r+1} of codegree $2r+1$ on IU_{dR} . By 6.6 and 7.9, a left translate of u_{2r+1} differs from u_{2r+1} only by addition of a constant. It is therefore a homomorphism from IU_{dR} to the additive group. Being of weight $2r+1$, it is by 7.12 a rational multiple of $\zeta^m(2r+1)$, and 7.13 follows from 7.11.

Deus ex Machina 7.14 (D. Zagier [6]). Set

$$A_{a,b}^r = \binom{2r}{2a+2}, \quad B_{a,b}^r = (1-2^{-2r}) \binom{2r}{2b+1}.$$

Then

$$\zeta(2^{\{b\}}32^{\{a\}}) = 2 \sum (-1)^r (A_{ab}^r - B_{ab}^r) \zeta(2r+1) \zeta(2^{\{a+b+1-r\}}). \quad (7.14.1)$$

This theorem suggests that the $\alpha_{a,b,r}$ of (7.13.1) are the $2(-1)^r (A_{a,b}^r - B_{a,b}^r)$. How to prove this starting from 7.14 will be explained in §8.

7.15

The affine algebra $O(U_{\text{dR}})$ of U_{dR} is in duality, in the graded sense, with the enveloping algebra of $\text{Lie}^{\text{gr}}(U_{\text{dR}})$. This enveloping algebra is a free associative algebra in generators of degrees the odd integers ≥ 3 . The Poincare series of $O(U_{\text{dR}})$, graded by the codegree, is therefore

$$P(O(U_{\text{dR}}), t) = \sum_{a \geq 0} \left(\sum_{n \geq 1} t^{2n+1} \right)^a = \left(1 - \sum_{n \geq 1} t^{2n+1} \right)^{-1} = (1 - (t^3/(1-t^2)))^{-1} = \frac{1-t^2}{1-t^2-t^3}.$$

If the affine algebra of the additive group $\mathbb{G}_a$, with coordinate λ , is graded by giving λ codegree 2, one has $P(O(\mathbb{G}_a), t) = (1-t^2)^{-1}$ and

$$P(O(U_{\text{dR}} \times \mathbb{G}_a), t) = (1-t^2-t^3)^{-1}.$$

The affine algebra $O(IU_{\text{dR}})$ is a graded subalgebra of $O(U_{\text{dR}})$. By 7.9, one therefore has:

Proposition 7.16. The Poincare series of $O(M_{\text{dR}})$, graded by the codegree, satisfies

$$P(O(M_{\text{dR}}), t) \leq (1-t^2-t^3)^{-1} \quad (7.16.1)$$

(coefficientwise inequality), with equality if and only if the morphism $I : U_{\text{dR}} \rightarrow IU_{\text{dR}}$ is an isomorphism, i.e. if U_{dR} acts faithfully on $\pi_1(X; 1, 0)_{\text{dR}}$.

Comparing with 6.10, one obtains:

Theorem 7.17. (i) U_{dR} acts faithfully on ${}_1\Pi_0$.

(ii) The morphism of $\mathbb{Q}$ -vector spaces $\tilde{H}(2, 3) \rightarrow O(M_{\text{dR}})$ is an isomorphism.

The affine algebra of $\pi_1(X; 1, 0)_{\text{mot}}$ contains a $\mathbb{Q}(-1)$ as a subquotient. It follows therefore from (i) that the action of G_{dR} on the affine algebra of $\pi_1(X; 1, 0)_{\text{dR}}$ is faithful. The theory of Tannakian categories gives, as a consequence of (i):

Corollary 7.18. The affine algebra $O(\pi_1(X; 1, 0)_{\text{mot}})$, which is an Ind-object of $MT(\mathbb{Z})$, generates the Tannakian category $MT(\mathbb{Z})$.

More precisely, every mixed Tate motive with good reduction over $\text{Spec}(\mathbb{Z})$ is deduced from subobjects in $MT(\mathbb{Z})$ of $O(\pi_1(X; 1, 0)_{\text{mot}})$ by the operations of tensor products, direct sums, duals, and subquotients.

Assertion (ii) of 7.17 is equivalent to saying that the $\zeta^m(s)$ with $s_i \in \{2, 3\}$ form a basis of $O(M_{\text{dR}})$.

8. Sketches of proofs

8.1

The group scheme $V_{\text{dR}} = (\Pi, \circ)$ acts on ${}_1\Pi_0$ (7.1), and therefore on its affine algebra $O({}_1\Pi_0)$. This action is the action of $(\Pi, \circ)$ on $O(\Pi)$ by left translations (7.2.3): x transforms f into $y \mapsto f(x^{-1} \circ y)$. These actions are given by (7.2.1), (7.2.2). Differentiating these formulae, one obtains the action of $\text{Lie}(V_{\text{dR}}) = (\text{Lie } \Pi, \{, \})$ on $O({}_1\Pi_0)$, and dually the coaction of the Lie coalgebra. The latter is m/m^2 , for m the ideal of functions vanishing at the identity element 1, and the coaction is the projection onto $m/m^2 \otimes O(\Pi)$ of

$$f \longmapsto f(x^{-1} \circ y) - f(y)$$

in $m \otimes O(\Pi) \subset O(\Pi \times \Pi)$.

Goncharov gave a convenient formula for this coaction (see [5] Theorem 1.2). For w a word in the alphabet $\{0, 1\}$, consider the subwords cId of $1w0$, with I nonempty: one considers all decompositions $w = w'Iw''$ of w , with I nonempty, and c, d are the letters adjacent to I in $1w0$. For each decomposition, put $w \setminus I := w'w''$. Formula for the coaction:

$$c(w) \mapsto - \sum p_{cd}(I) \otimes c(w \setminus I) \quad (8.1.1)$$

where $p_{cd}(I)$ is the projection in m/m^2 of the following element of $m \subset O(\Pi)$:

$$p_{cd}(I) = \text{projection of } \begin{cases} 0, & c = d, \\ c(I), & c = 1, d = 0, \\ \text{the antipode } c(I)^* \text{ of } c(I), & c = 0, d = 1, \end{cases} \quad (8.1.2)$$

where, if $I = a_1 \cdots a_k$, the antipode is $(-1)^k c(a_k \cdots a_1)$.

8.2

The action of U_{dR} on $O(\Pi_0)$ factors through that of $IU_{\text{dR}} \subset (\Pi, \circ)$, and to obtain the coaction of the Lie coalgebra it remains to compose (8.1.1) with

$$\text{coLie}(\Pi, \circ) \longrightarrow \text{coLie}(IU_{\text{dR}}) \longrightarrow \text{coLie}(U_{\text{dR}}).$$

The $c(I)$ and $c(I)^*$ of (8.1.2) matter no longer except by their restriction to M_{dR} , taken modulo π_m^2 and the square of the maximal ideal at 1.

We shall choose the generators ν_{2r+1} of the Lie algebra $\text{Lie}^{\text{gr}}(U_{\text{dR}})$ so that, writing again ν_{2r+1} for the image of ν_{2r+1} in the tangent space of $IU_{\text{dR}} \subset \Pi$ at 1, one has

$$\langle \nu_{2r+1}, d\zeta^m(2r+1) \rangle = 1$$

(possible by 7.12(ii)). This determines the image of ν_{2r+1} in $\text{Lie}^{\text{gr}}(U_{\text{dR}}^{\text{ab}})$. If $r = a + b + 1$ and $\alpha_{a,b}$ is the coefficient of $\zeta^m(2r+1)$ in (7.13.1), then

$$\langle \nu_{2r+1}, d\zeta^m(2^{\{b\}}32^{\{a\}}) \rangle = \alpha_{a,b}.$$

Proposition 8.3. Formula (7.14.1) remains true with ζ replaced by ζ^m .

Proof. Proceed by induction on $a+b$. If $a+b=0$, the identity (7.14.1)^m to be proved reduces to $\zeta^m(3) = \zeta^m(3)$. Suppose (7.14.1)^m true for $a+b < N$. This ensures that, for $a+b < N$ and $r = a+b+1$, one has, with the notation of 7.14,

$$\zeta^m(2^{\{b\}}32^{\{a\}}) = 2(-1)^r (A_{ab}^r - B_{ab}^r) \zeta^m(2r+1) \quad \text{on } IU_{\text{dR}}. \quad (8.3.1)$$

For $a+b = N$, if the $\alpha_{a,b,r}$ are defined as in 7.13, one has, for the action of ν_{2r+1} ,

$$\nu_{2r+1} \zeta^m(2^{\{b\}}32^{\{a\}}) = -\alpha_{a,b,r} \zeta^m(2^{\{a+b+1-r\}}) : \quad (8.3.2)$$

apply 7.12. For $r \leq N$, compute the left-hand side by (8.1.1), (8.1.2). One must sum over the subwords I of $w = (01)^b(001)(01)^a$, of length $2r+1$, and surrounded by 0, 1 or 1, 0 in $1w0$. The $c(w \setminus I)$, restricted to M_{dR} , are $\pm \zeta^m(2^{\{a+b+1-r\}})$. The $c(I)$ or $c(I)^*$ that appear in (8.1.2), restricted to M_{dR} , are $\pm \zeta^m(2^{\{b'\}}32^{\{a'\}})$ to which (8.3.1) applies by the induction hypothesis, or a $c((01)^m 0)$ which, on Π' , is expressed by (3.5.2) as a sum of $\zeta(2^{\{b'\}}32^{\{a'\}})$ to which (8.3.1) applies. This permits the computation of the $\alpha_{a,b,r}$ for $r \leq N$, and one finds that they have the expected value. This leaves unknown in (7.13.1) only the coefficient $\alpha_{a,b,a+b+1}$. It is determined by evaluating at dch and applying (7.14.1).

8.4

Armed with (8.3.1), one can now compute the morphisms (6.7.1) by (8.1.1), (8.1.2). To prove 6.8, the idea is that, in the matrix obtained for (6.7.2), the dominant terms, in the 2-adic sense, come from the 2^{-2r} which appears in B_{ab}^r (7.14). If, in the target and source of (6.7.2), one orders the basis vectors in suitable orders which correspond by the bijection defined in 6.7, and if in the target one multiplies the basis vectors by suitable powers of 2, the matrix of (6.7.2) is a square matrix with integer coefficients whose reduction modulo 2 is triangular with 1's on the diagonal. This ensures that it is invertible.

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Counting ℓ -adic Sheaves

Pierre Deligne
English working translation

To Gerard Laumon, on the occasion of his sixtieth birthday.

1 Introduction

1.1 1.1

Let X_0 be a smooth, projective, absolutely connected curve of genus g over a finite field $\mathbf{F}_q$ of characteristic p , and let X be obtained from it by extension of scalars to an algebraic closure $\mathbf{F}$ of $\mathbf{F}_q$:

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbf{F}) & \longrightarrow & \mathrm{Spec}(\mathbf{F}_q). \end{array} \quad (1.1.1)$$

The endomorphism of X_0 which is the identity on the underlying topological space and sends $f \mapsto f^q$ on the structure sheaf is an endomorphism of the $\mathbf{F}_q$ -scheme. We write Frob for the endomorphism of the $\mathbf{F}$ -scheme X obtained from it by extension of scalars. This is the Frobenius endomorphism of X . Fix a prime number $\ell \neq p$, an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$, and let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X . Inverse image by Frob induces a permutation of E . We denote it by V .

1.2 1.2

In the article [Dr], which remains as mysterious to me as it was thirty-one years ago, Drinfeld computes the number of fixed points of $V : E \rightarrow E$. A lisse $\overline{\mathbf{Q}}_\ell$ -sheaf L_0 of rank one on $\mathrm{Spec}(\mathbf{F}_q)$ is determined up to isomorphism by the unit $\lambda \in \overline{\mathbf{Q}}_\ell^*$ by which the geometric Frobenius $\mathrm{Fr} \in \mathrm{Gal}(\mathbf{F}/\mathbf{F}_q)$ acts on the fibre of L_0 at the geometric point $\mathrm{Spec}(\mathbf{F})$. The $\mathbf{F}_q$ -twist of F_0 on X_0 by L_0 is the tensor product with the inverse image of L_0 on X_0 . By abuse of language, we shall also say “ $\mathbf{F}_q$ -twist by λ .”

Drinfeld uses the fact that the isomorphism class of a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F on X is fixed by Frob^* if and only if F is the inverse image of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 , and that, if F is irreducible, F_0 is unique up to $\mathbf{F}_q$ -twist. The problem solved in [Dr] is thus to count lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 , taken up to $\mathbf{F}_q$ -twist, considering only those which are irreducible and remain so after inverse image on X .

In 1981 one almost had the correspondence between irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 and everywhere unramified cuspidal automorphic representations for $\mathrm{GL}(2, k(X_0))$. Thanks to this, Drinfeld reduced the problem to an application of the trace formula for $\mathrm{GL}(2)$. This reduction shows that the desired number is independent of ℓ . The formula obtained by Drinfeld has the following miraculous properties (A) and (B).

For $n \geq 1$, let N_n be the number of fixed points of the n -th iterate $V^n : E \rightarrow E$ of V . Computing N_n is the above problem with $X_0/\mathbf{F}_q$ replaced by $(X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n})/\mathbf{F}_{q^n}$.

(A) The function $n \mapsto N_n$ has the form

$$N_n = \sum a_i \beta_i^n, \quad (1.2.1)$$

for integers a_i and suitable Weil numbers β_i . The β_i are monomials in q and in the eigenvalues of the endomorphism Frob^* of $H^1(X, \overline{\mathbf{Q}}_\ell)$. Once the genus g is fixed, which monomials β_i occur, and with what multiplicities a_i , is independent of the curve under consideration.

The trace formula expresses N_n as the sum of several terms. Taken separately, these terms do not all have the form (1.2.1).

Let P be a compact Riemann surface of genus g . Consider complex local systems of vector spaces on P . Let M be the moduli space of those which are irreducible of rank 2. The space M and the set E are empty if $g = 0$ or 1 . Otherwise M is a connected complex symplectic variety. Its complex dimension is therefore an even integer $2N$.

(B) In (1.2.1), the dominant term is q^N : one of the β_i is q^N , its multiplicity a_i is 1, and the other β_i satisfy $|\beta_i| < q^N$.

1.3 1.3

Formula (1.2.1) is reminiscent of a Lefschetz trace formula in which, in good cases, the number of fixed points of the iterates T^n ($n \geq 1$) of an endomorphism T of a space S is

$$\text{Tr}(T^{*n}, H_{\mathbb{Q}}^*(S)) := \sum (-1)^i \text{Tr}(T^{*n}, H_{\mathbb{Q}}^i(S)). \quad (1.3.1)$$

The question mark is there to recall that one must consider a cohomology with support conditions. The support conditions to be imposed depend on the behaviour of T at infinity. For example, if for an exhaustion function f on S one has $f(T(x)) > f(x)$ (respectively $f(T(x)) < f(x)$) for $f(x)$ sufficiently large, the cohomology to consider is compactly supported cohomology (respectively ordinary cohomology).

Assume that $H_{\mathbb{Q}}^*$ is finite-dimensional so that (1.3.1) makes sense. The right hand side of (1.3.1) is then of the form (1.2.1): it is $\sum a(\beta)\beta^n$, where β ranges over the eigenvalues of T^* and the integer $a(\beta)$ is the alternating sum of the multiplicities of β as an eigenvalue of the endomorphisms T^* on the $H_{\mathbb{Q}}^i(S)$.

Unfortunately I have no idea how to regard the set E of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X as a “space” having a cohomology $H_{\mathbb{Q}}^*(E)$ for which one could hope that the number of fixed points of V^n is given by a formula of type (1.3.1).

1.4 1.4

Even though one does not know how to think geometrically of E , if $f \in E$ is the isomorphism class of F , one knows what the formal completion $\widehat{E}_f$ of E at f should be. A finite-dimensional local $\overline{\mathbf{Q}}_\ell$ -algebra Λ is augmented over $\overline{\mathbf{Q}}_\ell$. A deformation of F over $\text{Spec}(\Lambda)$ is a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F_Λ on X , endowed with a Λ -module structure and an isomorphism

$$F_\Lambda \otimes_\Lambda \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} F,$$

such that at one point x (and therefore at every point x of X), $(F_\Lambda)_x$ is a free Λ -module.

The functor on $\text{Spec}(\Lambda)$ of isomorphism classes of deformations of F over $\text{Spec}(\Lambda)$ is pro-representable. Let R_F be the complete local $\overline{\mathbf{Q}}_\ell$ -algebra with residue field $\overline{\mathbf{Q}}_\ell$ such that $\text{Spf}(R_F)$ pro-represents it. There are no obstructions to deformations, and R_F is therefore isomorphic to a formal power series algebra $\overline{\mathbf{Q}}_\ell[[t_1, \dots, t_m]]$.

Because F is irreducible, its only automorphisms are multiplications by $\lambda \in \overline{\mathbf{Q}}_\ell^*$. They extend to every deformation. If F and F' are two representatives of the isomorphism class f , two isomorphisms from F to F' therefore induce the same isomorphism between R_F and $R_{F'}$. Thus $\widehat{E}_f := \text{Spf}(R_F)$ depends, up to unique isomorphism, only on f .

The Zariski tangent space of $\widehat{E}_f$ at f is $H^1(X, \text{End}(F))$. The symmetric bilinear form $\text{Tr}(fg)$ on $\text{End}(F)$ is a self-duality. It induces on $H^1(X, \text{End}(F))$ a symplectic form with values

in $H^2(X, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell(-1)$. The same construction over $\mathbf{C}$ gives the symplectic structure of M . The formal completion $\widehat{E}_f$ is therefore smooth of even dimension, the same as that of M .

The inverse-image functor by $\text{Frob} : X \rightarrow X$ transforms deformations of F into deformations of $\text{Frob}^* F$. It therefore induces a morphism

$$\widehat{V}_f : \widehat{E}_f \longrightarrow \widehat{E}_{V(f)}.$$

The argument showing that, because Frob induces an equivalence of étale sites, V is bijective, also shows that the $\widehat{V}_f$ are isomorphisms. Another proof is as follows. Let ω be the symplectic form defined above on the tangent space at f . If $[t] \in H^1(X, \text{End}(F))$ is the class of the tangent vector t at F , the class of $d\widehat{V}_f(t)$ in $H^1(X, \text{End}(\text{Frob}^* F))$ is $\text{Frob}^*([t])$. For two tangent vectors t_1, t_2 , $\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2))$ is thus the inverse image, by Frob^* , of $\omega(t_1, t_2) \in \overline{\mathbf{Q}}_\ell(-1)$, identified with $H^2(X, \overline{\mathbf{Q}}_\ell)$. Since $\text{Frob} : X \rightarrow X$ has degree q , the endomorphism Frob^* of $H^2(X, \overline{\mathbf{Q}}_\ell)$ is multiplication by q , and

$$\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2)) = q \omega(t_1, t_2),$$

which is also written $(d\widehat{V}_f)^* \omega = q\omega$. The differential $d\widehat{V}_f$ is therefore an isomorphism.

The map $d\widehat{V}_f$ between Zariski tangent spaces is identified with

$$\text{Frob}^* : H^1(X, \text{End}(F)) \longrightarrow H^1(X, \text{End}(\text{Frob}^* F)).$$

If f is a fixed point of V , then F is the inverse image of F_0 on X_0 . One knows that F_0 may be chosen pure, and $\text{End}(F_0)$ is therefore pure of weight 0. The eigenvalues of $d\widehat{V}_f$ on the Zariski tangent space of $\widehat{E}_f$ at f are therefore Weil numbers of weight 1. In particular no eigenvalue of $d\widehat{V}_f$ at f is equal to 1. This is what makes it reasonable that a Lefschetz-type formula (1.3.1) might count the fixed points of V^n , all taken with multiplicity one.

1.5 1.5

If Y_0 is a scheme of finite type over $\mathbf{F}_q$, the Frobenius endomorphism Frob of $Y := Y_0 \otimes_{\mathbf{F}_q} \mathbf{F}$ induces a permutation Frob of $Y(\mathbf{F})$. The preceding discussion shows that the nature of (E, V) is different from that of (Y, Frob) :

- (i) E is “over $\overline{\mathbf{Q}}_\ell$,” hence an object of characteristic 0.
- (ii) The $\widehat{V}_f$ are isomorphisms. One would like to say that V has “degree one.”
- (iii) The fixed points of V are isolated because the eigenvalues of dV at a fixed point are Weil numbers of weight 1. The fixed points of Frob are isolated because $d\text{Frob} = 0$.
- (iv) If E has dimension $2N$, in the sense that the $\widehat{E}_f$ do, the order of magnitude of the number of fixed points of V^n is $(q^n)^N$, rather than $(q^n)^{2N}$.

1.6 1.6

The moduli space M is a complex algebraic variety. Contrary to what a hasty analogy might suggest, E is not naturally the space of $\overline{\mathbf{Q}}_\ell$ -points of an algebraic variety over $\overline{\mathbf{Q}}_\ell$. Let $x \in X(\mathbf{F})$. The set E is identified with the set of isomorphism classes of continuous irreducible representations V of $\pi_1(X, x)$, of dimension 2 over $\overline{\mathbf{Q}}_\ell$. As coordinates on E , one would like to take the $\text{Tr}(\gamma, V)$ for $\gamma \in \pi_1(X, x)$. In the complex case this construction gives M its “Betti” structure as a complex algebraic variety. Here, since $\pi_1(X, x)$ is a profinite group, every representation V admits an invariant lattice, and the $\text{Tr}(\gamma, V)$ take their values in the valuation ring $\mathbf{Z}_\ell$ of the valued field $\overline{\mathbf{Q}}_\ell$.

The point of view “ E as a space of representations of $\pi_1(X, x)$ ” also does not help one understand that the number of fixed points of V is independent of ℓ . Another mystery: why,

once the genus g is fixed, does the number of fixed points of V admit a uniform description in terms of $H^1(X, \overline{\mathbf{Q}}_\ell)$, of its symplectic structure with values in $\overline{\mathbf{Q}}_\ell(-1)$, and of its endomorphism Frob^* , whereas $\pi_1(X, x)$ is a quotient of the profinite completion of the topological π_1 of P , which quotient depends on X ?

1.7 1.7

Let E_r be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r on X . Thanks to Lafforgue [L], the number of fixed points of the bijection $\text{Frob}^* : E_r \rightarrow E_r$ has an automorphic interpretation. For $S_0 \subset X_0$ a finite set of closed points of X , and $S = S_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, one may also consider irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves on $X - S$, with prescribed ramification at each $s \in S$. In all cases that have been computed, the miracles (A) and (B) discovered by Drinfeld persist, mutatis mutandis. The aim of the talk is to give a panorama of what is known.

2 What is to be counted?

2.1 2.1

Fix an integer $r \geq 1$, and let $X_0, S_0/\mathbf{F}_q$ and $X, S/\mathbf{F}$ be as in 1.1 and 1.7. For $s \in S$, let $X_{(s)}$ be the henselization of X at s and $X_{(s)}^* := X_{(s)} - \{s\}$. Suppose given, for each $s \in S$, a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf of rank r , denoted $R(s)$, on $X_{(s)}^*$. Only its isomorphism class matters. The morphism $\text{Frob} : X \rightarrow X$ induces a morphism, again denoted Frob , from $X_{(s)}^*$ to $X_{(\text{Frob}(s))}^*$. We assume that, for every $s \in S$, $R(s)$ is isomorphic to the inverse image of $R(\text{Frob}(s))$ by Frob :

$$R(s) \simeq \text{Frob}^*(R(\text{Frob}(s))). \quad (2.1.1)$$

Let $R := (R(s))_{s \in S}$ be the family of the $R(s)$, and write $E(R)$ for the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves F of rank r on $X - S$ such that, for every $s \in S$, the inverse image $F|_{X_{(s)}^*}$ of F on $X_{(s)}^*$ is isomorphic to $R(s)$. Hypothesis (2.1.1) ensures that inverse image by Frob induces a permutation of $E(R)$. We denote this permutation by V .

Our choice of the letter V , the first letter of “Verschiebung,” is due to an analogy with the case of certain abelian varieties, explained in 2.2 below. In that analogy q is classically a prime number, rather than a power of a prime number.

2.2 2.2

For any scheme Y over $\mathbf{F}_q$, let Frob_Y be the endomorphism of the $\mathbf{F}_q$ -scheme Y which is the identity on the underlying topological space and which on the structure sheaf is given by $\text{Frob}_Y^*(f) = f^q$. This is the Frobenius endomorphism of Y .

Take for Y the abelian variety $\text{Pic}^0(X_0)$ over $\mathbf{F}_q$. We obtain

$$\text{Frob}_{\text{Pic}^0(X_0)} : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(X_0). \quad (2.2.1)$$

The Jacobian $\text{Pic}^0(X_0)$ represents the following functor on $\mathbf{F}_q$ -schemes: the sheaf (for the Zariski topology, or for the étale topology; it is the same) associated to the presheaf attaching to S the group $\text{Pic}^0(S \times_{\mathbf{F}_q} X_0)$ of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$, of degree zero on each fibre of the projection to S . If $f : Y_0 \rightarrow X_0$ is a morphism, inverse image of invertible sheaves by f defines a morphism $f^* : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(Y_0)$.

Take for f the endomorphism Frob_{X_0} of X_0 . The endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ deserves to be called V , for it satisfies

$$\text{Frob}_{X_0}^* \circ \text{Frob}_{\text{Pic}^0(X_0)} = \text{Frob}_{\text{Pic}^0(X_0)} \circ \text{Frob}_{X_0}^* = q. \quad (2.2.2)$$

Proof of (2.2.2). Interpret (2.2.2) as two equalities between endomorphisms of the functor represented by $\text{Pic}^0(X_0)$. For any contravariant functor G on the category of $\mathbf{F}_q$ -schemes, the Frobenius endomorphism Frob_G of G is defined by

$$\text{Frob}_G(S) : G(S) \rightarrow G(S), \quad \text{namely } G(\text{Frob}_S).$$

If G is represented by Y , Frob_G is induced by Frob_Y .

Take for G the functor $S \mapsto \text{Pic}(S \times_{\mathbf{F}_q} X_0)$ of isomorphism classes of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$. The endomorphism $\text{Frob}_G(S)$ of $\text{Pic}(S \times_{\mathbf{F}_q} X_0)$ is inverse image of isomorphism classes of invertible sheaves by $\text{Frob}_S \times \text{id}_{X_0} : S \times_{\mathbf{F}_q} X_0 \rightarrow S \times_{\mathbf{F}_q} X_0$. If one composes this inverse image with the inverse image by Frob_{X_0} , one obtains inverse image by $\text{Frob}_{S \times_{\mathbf{F}_q} X_0}$, which is nothing other than raising to the q -th tensor power. Restricting to invertible sheaves of degree zero on the fibres of the projection to S , and passing from the presheaf obtained to the associated sheaf, gives (2.2.2).

If now one extends scalars from $\mathbf{F}_q$ to $\mathbf{F}$, Frob_{X_0} becomes the endomorphism Frob of X , $\text{Pic}^0(X_0)$ becomes $\text{Pic}^0(X)$, and the endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ becomes the endomorphism of $\text{Pic}^0(X)$ “inverse image by Frob .”

2.3 2.3–2.6

The questions we ask are the following.

- (i) Compute, for $n \geq 1$, the number $N_n(R)$ of fixed points of the n -th iterate V^n of the permutation V of $E(R)$. Replacing (X_0, S_0) by $(X_0, S_0) \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$ replaces V by V^n .
- (ii) Determine whether the miracles 1.2 (A) and (B) persist.
- (iii) If so, why?

Let F be a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf on $X - S$. As in 1.2, one has:

Lemma 2.4. (i) For the isomorphism class of F to be fixed under Frob^* , it is necessary and sufficient that F be isomorphic to the inverse image on X of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 . Moreover, if F is irreducible,

- (ii) such an F_0 is unique up to $\mathbf{F}_q$ -twist, and
- (iii) the $\mathbf{F}_q$ -twists of F_0 by $\lambda \in \overline{\mathbf{Z}}_\ell^*$ are pairwise non-isomorphic.

According to 2.4, the number of fixed points of the permutation V of $E(R)$ is the number of classes, modulo $\mathbf{F}_q$ -twist, of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 of rank r on $X_0 - S_0$, such that

- (a) the inverse image F of F_0 on $X - S$ is irreducible;
- (b) for each $s \in S$, $F|_{X_{(s)}^*}$ is isomorphic to $R(s)$.

Let $D_0 = \sum n_i x_i$ be a divisor of degree one on $X_0 - S_0$. Regard F_0 as a Galois representation (see 2.7), let F_{x_i} be a Frobenius at x_i , and put

$$\langle D_0, F_0 \rangle := \prod \det(F_{x_i}, F_0)^{n_i}.$$

If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , then

$$\langle D_0, F'_0 \rangle = \lambda^r \langle D_0, F_0 \rangle.$$

By 2.4 (ii) and (iii), it is therefore equivalent either to count the F_0 satisfying (a) and (b) up to $\mathbf{F}_q$ -twist, or to count, up to isomorphism, only those which also satisfy $\langle D_0, F \rangle = 1$, and then divide by r . For an automorphic counterpart, see the remark following 2.6.

Put $K := k(X_0)$. For each closed point x of X_0 , let $k(x)$ be the residue field of x , $\deg(x) := [k(x) : \mathbf{F}_q]$ its degree, K_x the completion of K at x , and O_x the valuation ring of K_x . Let $\mathbf{A}$ be the adèle ring of K . The sum $v := \sum \deg(x)v_x : \mathbf{A}^* \rightarrow \mathbf{Z}$ is trivial on K^* . One has $\|a\| = q^{-v(a)}$.

By Lafforgue, F_0 corresponds to a cuspidal automorphic representation π of $\mathrm{GL}(r, \mathbf{A})$. If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , the corresponding representation π' is the $\mathbf{F}_q$ -twist of π by λ , defined as follows. As a representation, it is the tensor product of π by the one-dimensional representation $g \mapsto \lambda^{v \det g}$. As a space of functions on $\mathrm{GL}(r, \mathbf{A})$, it is the space of products of $f \in \pi$ with the character $g \mapsto \lambda^{v \det g}$.

Translated into automorphic language, problem 2.3 (i) is to count, up to $\mathbf{F}_q$ -twist, the cuspidal automorphic representations π of $\mathrm{GL}(r, \mathbf{A})$, unramified outside S_0 , and such that

- (a') for every $n \geq 1$, π remains cuspidal after base change from K to $K \otimes \mathbf{F}_{q^n}$. This condition expresses the irreducibility 2.4 (a) of F . It suffices to check it for n dividing r .
- (b') for each $s_0 \in S_0$ and $s \in S$ above s_0 , the local component π_{s_0} of π satisfies a condition dictated by $R(s)$ and by the local Langlands correspondence.

Lemma 2.6. If the cuspidal automorphic representation π satisfies (a'), then its $\mathbf{F}_q$ -twists are pairwise distinct. This is the translation of 2.4 (iii). If d is an idele satisfying $v(d) = 1$, it is therefore equivalent to count up to $\mathbf{F}_q$ -twist the π satisfying (a') and (b'), or to count only those whose central character ω_π satisfies $\omega_\pi(d) = 1$, and then divide by r .

In finite characteristic the notion of automorphic representation is purely algebraic. This makes it possible to take the automorphic representations π to have coefficients in $\overline{\mathbf{Q}}_\ell$, rather than in $\mathbf{C}$. If one does this, π_{s_0} is an irreducible admissible $\overline{\mathbf{Q}}_\ell$ -representation of $\mathrm{GL}(r, K_{s_0})$, and the local Langlands correspondence associates to it an isomorphism class of continuous F -semisimple $\overline{\mathbf{Q}}_\ell$ -representations of dimension r of the local Weil group $W(\overline{K}_{s_0}/K_{s_0})$. Condition (b') is that its restriction to the inertia subgroup be given by $R(s)$.

Almost all known results on question 2.3 (i) are obtained through the dictionary 2.5.

2.4 2.7–2.16

In 2.1 we used the language of $\overline{\mathbf{Q}}_\ell$ -sheaves. This saved us from having to choose base points. An equivalent language is that of Galois representations. Here is the translation. Let $\overline{k(X)}$ be an algebraic closure of the field of rational functions $k(X)$ of X , and let $k(X)^{\mathrm{nr}}$ be the largest subextension unramified outside S . The Galois group $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$ is the fundamental group of X at the geometric point $\bar{\eta} := \mathrm{Spec}(k(X))$ of X : the fibre functor at $\bar{\eta}$ is an equivalence from lisse $\overline{\mathbf{Q}}_\ell$ -sheaves (respectively irreducible ones of rank r) on $X - S$ to linear $\overline{\mathbf{Q}}_\ell$ -representations of $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$ (respectively irreducible representations of rank r).

The scheme $X_{(s)}^*$ is the spectrum of an extension $k(X_{(s)}^*)$ of $k(X)$. If one chooses an embedding of this extension into $\overline{k(X)}$, the corresponding inertia group $I_s := \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*))$ is likewise the fundamental group of $X_{(s)}^*$ at $\bar{\eta}$. It maps to $\pi_1(X, \bar{\eta})$:

$$\pi_1(X_{(s)}^*, \bar{\eta}) = I_s = \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*)) \rightarrow \mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X)) = \pi_1(X, \bar{\eta}). \quad (2.7.1)$$

The fibre at $\bar{\eta}$ of $R(s)$ is a representation ρ_s of I_s . The condition that $F|_{X_{(s)}^*}$ be isomorphic to $R(s)$ is equivalent to the restriction to I_s being isomorphic to ρ_s . Condition (2.1.1) is equivalent to saying that the ρ_s , for s over s_0 , come from a representation of a decomposition group $D_{s_0} \subset \mathrm{Gal}(k(\overline{X})/k(X_0))$ at s_0 . It implies that the representations ρ_s are quasi-unipotent.

I hope problem 2.3 (i) is more manageable when a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F such that $F|_{X_{(s)}^*} \simeq R(s)$ for every $s \in S$ is automatically irreducible. This is the case if the following condition holds. Consider an integer $0 < r' < r$ and, for each $s \in S$, a lisse sub- $\overline{\mathbf{Q}}_\ell$ -sheaf $R'(s)$ of rank r' of $R(s)$. Put $R''(s) := R(s)/R'(s)$. The condition is:

For all r' and all $R'(s)$, either there exists no lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F' of rank r' on $X - S$ with $F'|_{X_{(s)}^*} \simeq R'(s)$ for all s , or there exists no lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F'' of rank $r - r'$ on $X - S$ with $F''|_{X_{(s)}^*} \simeq R''(s)$ for all s . (2.8.1)

Condition (2.8.1) is satisfied if one of the $R(s)$ is irreducible. For other cases, see 2.10.4, 2.11, and 2.12.

For any field k of characteristic p or 0, containing all roots of unity of order prime to p , set

$$\widehat{\mathbf{Z}}_{p'}(1)(k) := \varprojlim \mu_n(k). \quad (2.9.1)$$

The projective limit is taken over the set of subgroups $\mu_n(k) \subset k^*$ (n prime to p), ordered by inclusion. The transition morphism from $\mu_m(k)$ to $\mu_n(k)$ is $x \mapsto x^{m/n}$. If no ambiguity is created, we omit “ p' ” and the mention of k .

For $k = \mathbf{F}$, and $\mathbf{F}_{q^k}$ the subfield of $\mathbf{F}$ with q^k elements, the subgroups $\mathbf{F}_{q^k}^*$ of $\mathbf{F}^*$ are cofinal among the $\mu_n(\mathbf{F})$. Since $(q^{kd} - 1)/(q^k - 1) = 1 + q^k + \dots + q^{k(d-1)}$, the transition morphism from $\mu_{q^{kd-1}} = \mathbf{F}_{q^{kd}}^*$ to $\mu_{q^{k-1}} = \mathbf{F}_{q^k}^*$ is the norm morphism, and

$$\widehat{\mathbf{Z}}(1) \xrightarrow{\sim} \varprojlim \mathbf{F}_{q^k}^* \quad (2.9.2)$$

(the projective limit being taken with respect to the norm morphisms between the $\mathbf{F}_{q^k}^*$). For k a product $\prod k_\alpha$ of finite extensions of $\mathbf{F}_q$ and $\iota : k \rightarrow k_\alpha \xrightarrow{\sim} \mathbf{F}_{q^k} \hookrightarrow \mathbf{F}$, write pr_ι for the composite

$$\text{pr}_\iota : \widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_{q^k}^* \xleftarrow{\sim} k_\alpha^* \hookrightarrow \bigoplus k_\alpha^* = k^*. \quad (2.9.3)$$

For $k = \mathbf{F}_{q^k}$ and ι the identical inclusion, $\text{Gal}(\mathbf{F}/\mathbf{F}_{q^k})$ acts on $\widehat{\mathbf{Z}}(1)$, and (2.9.3) induces an isomorphism of coinvariants with $\mathbf{F}_{q^k}^*$.

The local fundamental group $\pi_1(X_{(s)}^*, \bar{\eta}) = I_s$ is an extension of $\widehat{\mathbf{Z}}(1)$ by a pro- p -group P_s . The quotient $I_s/P_s \simeq \widehat{\mathbf{Z}}(1)$ is the tame fundamental group of $X_{(s)}^*$. The morphism $\text{Frob} : X_{(s)}^* \rightarrow X_{(\text{Frob}(s))}^*$ induces a morphism between tame fundamental groups. Via the identifications of these groups with $\widehat{\mathbf{Z}}(1)$, this morphism is multiplication by q .

Let $\mathbf{Z}_\ell(1)$ denote the quotient $\varprojlim \mu_{\ell^n}(\mathbf{F})$ of $\widehat{\mathbf{Z}}(1)$. Every quasi-unipotent $\overline{\mathbf{Q}}_\ell$ -representation ρ of $\widehat{\mathbf{Z}}(1)$ is isomorphic to a sum over finite-order characters $\chi : \widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$:

$$\rho \simeq \bigoplus_\chi \chi \otimes \nu_\rho(\chi), \quad (2.9.4)$$

where $\nu_\rho(\chi)$ is a unipotent representation of the quotient $\mathbf{Z}_\ell(1)$ of $\widehat{\mathbf{Z}}(1)$. Let $\lambda_\rho(\chi)$ be the following partition of the integer $\dim \nu_\rho(\chi)$: the decreasing sequence of sizes of the Jordan blocks of the image of a generator of $\mathbf{Z}_\ell(1)$ in the representation $\nu_\rho(\chi)$. The isomorphism class of ρ is determined by the function $\chi \mapsto \lambda_\rho(\chi)$.

Assume that the $R(s)$ are tame, i.e. that the corresponding representations ρ_s of the inertia groups I_s (2.7) factor through a representation of $\widehat{\mathbf{Z}}(1)$. We write $\chi \mapsto \lambda_s(\chi)$ for the function, constructed above, attached to this representation. With this notation, condition (2.1.1) is equivalent to

$$\lambda_{\text{Frob}(s)}(\chi) = \lambda_s(\chi^q). \quad (2.9.5)$$

If L is a $\overline{\mathbf{Q}}_\ell$ -sheaf of rank one on $X - S$, the restriction of L to $X_{(s)}^*$ defines a character l_s of the fundamental group I_s of $X_{(s)}^*$. Since $\ell \neq p$, the restriction of l_s to P_s is of finite order: there is a power P of p such that the l_s^P factor through characters, again denoted l_s^P , of $\widehat{\mathbf{Z}}(1) = I_s/P_s$. One has

$$\prod l_s^P = 1. \quad (2.10.1)$$

The analogue over $\mathbf{C}$ of (2.10.1) is that, for P as in 1.2 and S a finite subset of P , the sum over S of a small positive circle around each $s \in S$ is a boundary in $P - S$.

In rank r , (2.10.1) applied to $\bigwedge^r F$ provides a necessary condition for the existence of F on $X - S$ satisfying $F|_{X_{(s)}^*} \simeq R(s)$, and interesting cases in which the hypothesis of the irreducibility criterion (2.8.1) is satisfied. Let ρ_s be the representation of I_s corresponding to $R(s)$. Assume, as in 2.9, that the representations ρ_s are tame. Let $\mathcal{R}(s)$ be the multiset of characters of $\widehat{\mathbf{Z}}(1)$ forming the semisimplification of ρ_s . With the notation of 2.9, the multiplicity in $\mathcal{R}(s)$ of a character χ is $|\lambda_s(\chi)| = \dim \nu_{\rho_s}(\chi)$. Assume that

$$\prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (2.10.2)$$

By (2.10.1), this is a necessary condition for the existence of an F on $X - S$ such that $F|_{X_{(s)}^*} \simeq R_s$ for all $s \in S$. Consider the condition:

$$\text{For every } 0 < r' < r \text{ and every family } (\mathcal{R}'(s))_{s \in S} \text{ of sub-multisets with } r' \text{ elements} \\ \text{of the multisets } \mathcal{R}(s), \text{ one has } \prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}'(s)} \varepsilon \neq 1. \quad (2.10.3)$$

Lemma 2.11. Condition (2.10.3) implies (2.8.1), and hence the irreducibility of every lisse sheaf F of rank r on $X - S$ satisfying $F|_{X_{(s)}^*} \simeq R_s$ for all $s \in S$.

Remark 2.12. There exist $R(s)$ as above, giving rise to multisets $\mathcal{R}(s)$ satisfying (2.10.2), for which (2.10.3) is false but where condition (2.8.1) is nevertheless satisfied. Here is an example. Take $X = \mathbf{P}^1$, $r = 4$, $|S| = 3$, and semisimple $R(s)$, sums of multisets of characters $\mathcal{R}(s)$ of the following type

$$\{a_1, a_1, a_3, a_4\}, \quad \{b_1, b_1, b_3, b_4\}, \quad \{c_1, c_2, c_3, c_4\}.$$

Assume that the product of all these characters is 1 (2.10.2), and that $a_4 b_4 c_4 = 1$, so that (2.10.3) is false. If the a_i, b_i, c_i are “general,” condition (2.8.1) is satisfied, because there is no F of rank 3 with local monodromy given by

$$\{a_1, a_1, a_3\}, \quad \{b_1, b_1, b_3\}, \quad \{c_1, c_2, c_3\}.$$

The verification is by reduction to the complex case, by lifting (X, S) to characteristic 0. This reduction brings us to the following lemma.

Lemma 2.13. Let $u_1, u_3, v_1, v_3, w_1, w_2, w_3 \in \overline{\mathbf{Q}}_\ell^*$. Assume that $u_1 v_1 w_i \neq 1$ for $i = 1, 2, 3$. Under this hypothesis there do not exist elements U, V, W of $\text{GL}(3, \overline{\mathbf{Q}}_\ell)$, conjugate respectively to the diagonal matrices (u_1, u_1, u_3) , (v_1, v_1, v_3) , (w_1, w_2, w_3) , with $UVW = 1$.

Proof. Let L be the intersection of the kernels of $U - u_1$ and $V - v_1$. These kernels have dimension at least 2, so L is nonzero. On L , U is multiplication by u_1 , V is multiplication by v_1 , and, since $UVW = 1$, W is multiplication by $1/(u_1 v_1)$. Since $1/(u_1 v_1)$ is not an eigenvalue of W , this is impossible.

Let P be a compact Riemann surface of genus g . Suppose an injection of S into P has been chosen; we identify S with a finite set of points of P . If $P_{(s)}$ is a small disk around s , the fundamental group of $P_{(s)}^* := P_{(s)} - \{0\}$ is $\mathbf{Z}$, generated by one positive turn around s . More intrinsically, this local fundamental group is $\mathbf{Z}(1) := 2\pi i \mathbf{Z}$, with $2\pi i$ corresponding to the positive turn around s . For each $n \geq 1$, identify $\mathbf{Z}(1)/n$ with the group of n -th roots of unity in $\mathbf{C}$ by $z \mapsto \exp(z/n)$. Passing to the limit gives an isomorphism of $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$ with the projective limit of the $\mathbf{Z}(1)/n$ for n prime to p .

A complex representation ρ of $\mathbf{Z}(1)$ is isomorphic to a sum over characters $\chi : \mathbf{Z}(1) \rightarrow \mathbf{C}^*$:

$$\rho \simeq \bigoplus_{\chi} \chi \otimes \nu(\chi), \quad (2.14.1)$$

where $\nu(\chi)$ is a unipotent representation of $\mathbf{Z}(1)$. Define the partition $\lambda(\chi)$ of $\dim \nu(\chi)$ as in 2.9. The isomorphism class of ρ is determined by the function $\chi \mapsto \nu(\chi)$. We shall only have

to consider the case where the characters χ occurring in (2.14.1) are of finite order prime to p , and hence may be viewed as complex characters of $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$.

If T is a family $(T_s)_{s \in S}$ of isomorphism classes of complex representations of $\mathbf{Z}(1)$, we write $M_{\mathbf{C}}(T)$ for the moduli space of irreducible complex local systems of rank r on $P - S$, with local monodromy T_s at s . This moduli space has two natural structures of algebraic variety, under the names Betti and de Rham; only the underlying topological space will matter to us.

Choose

an isomorphism between the groups of roots of unity of order prime to p in $\overline{\mathbf{Q}}_{\ell}$ and in $\mathbf{C}$,

(2.14.2)

an isomorphism between the groups of roots of unity of order prime to p in $\mathbf{F}$ and in $\mathbf{C}$.

(2.14.3)

These data are exactly what is needed to attach to an isomorphism class of quasi-unipotent ℓ -adic representations of $\widehat{\mathbf{Z}}(1)$ an isomorphism class of complex representations of $\mathbf{Z}(1)$, as one checks by comparing 2.9 and 2.14.

If the $R(s)$ are tame, giving rise to representations ρ_s of $\widehat{\mathbf{Z}}(1)$, we write R^* for the corresponding family of complex representations $R^*(s)$ of $\mathbf{Z}(1)$, and we say that the moduli space $M_{\mathbf{C}}(R^*)$ corresponds to $E(R)$, or is the complex analogue of $E(R)$.

Conjecture 2.15. Under the hypotheses and with the notation of 2.14:

- (i) As a function of $n \geq 1$, the number of fixed points $N_n(R)$ of V^n acting on $E(R)$ is of the form

$$N_n(R) = \sum a_i \beta_i^n \quad (2.11.1)$$

for integers a_i and suitable numbers β_i .

- (ii) The sum of the a_i is equal to the Euler-Poincaré characteristic of the moduli space $M_{\mathbf{C}}(R^*)$ corresponding to $E(R)$.

- (iii) As in 1.2 (B), the dominant term in (2.15.1) is q to the power $\dim_{\mathbf{C}}(M_{\mathbf{C}}(R^*))/2$.

As will be explained in 6.6, if $g > 0$ then $\chi(M_{\mathbf{C}}(R^*)) = 0$. For a refinement of 2.15 (i) and (ii), see 6.3 and 6.7.

Using the analogy between the exponential and Artin-Schreier sheaves, it is possible to define complex analogues for certain wild representations ρ_s . Rather than moduli spaces of local systems on $P - S$, the complex analogues are moduli spaces of algebraic vector bundles with connection on $P - S$, with prescribed formal completions at each $s \in S$. I conjecture that 2.11 remains valid in this more general setting.

Variant 2.16. Let ρ and ρ' be two quasi-unipotent ℓ -adic representations $\rho, \rho' : \widehat{\mathbf{Z}}(1) \rightarrow \mathrm{GL}(r, \overline{\mathbf{Q}}_{\ell})$ (respectively two complex representations $\rho, \rho' : \mathbf{Z}(1) \rightarrow \mathrm{GL}(r, \mathbf{C})$). We say that the isomorphism class of ρ' is in the closure of the isomorphism class of ρ if ρ' is a limit of conjugates of ρ . With the notation of 2.9 (respectively 2.14), this means that, for each character χ , $\nu_{\rho}(\chi)$ and $\nu_{\rho'}(\chi)$ have the same dimension $r(\chi)$, and that the unipotent conjugacy class of $\mathrm{GL}(r(\chi))$ defined by the partition $\lambda_{\rho'}(\chi)$ of $r(\chi)$ is in the closure of that defined by $\lambda_{\rho}(\chi)$.

Transporting this definition to tame $\overline{\mathbf{Q}}_{\ell}$ -sheaves on $X_{(s)}^*$, one defines $\bar{E}(R)$ as the set of classes, up to semisimplification, of lisse $\overline{\mathbf{Q}}_{\ell}$ -sheaves F of rank r on $X - S$ such that for each $s \in S$, $F|_{X_{(s)}^*}$ lies in the closure of the isomorphism class of $R(s)$. One defines $\overline{M}_{\mathbf{C}}(R^*)$ as the moduli space of local systems of rank r on $P - S$, whose local monodromy at s lies in the closure of $R^*(s)$. These local systems are taken up to semisimplification.

One still has $V : \bar{E}(R) \rightarrow \bar{E}(R)$, and one can repeat Conjecture 2.15 for $\bar{E}(R)$ and for the corresponding $\overline{M}_{\mathbf{C}}(R^*)$. In some cases, the formulas giving the number of fixed points of V become simpler if $E(R)$ is replaced by $\bar{E}(R)$. See 4.5.

The precisions 2.17 to 2.22 that follow, added in July 2013, owe much to the talks I gave at IHES in March and April 2013.

2.5 2.17–2.22

Let us first place ourselves in the everywhere unramified case: S is empty, and E_r is the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r on X . Let $W(\mathbf{F})$ be the ring of Witt vectors of $\mathbf{F}$, let K be its fraction field, and let $\overline{K}$ be an algebraic closure of K . Let X_W be a lifting of X over W , and denote by X_K (respectively $X_{\overline{K}}$) the curve over K (respectively $\overline{K}$) obtained from it. Let M_K (respectively $M_{\overline{K}}$) be the scheme over K (respectively $\overline{K}$) which is the moduli space of geometrically irreducible vector bundles with connection of rank r on X_K (respectively $X_{\overline{K}}$). By extension of scalars from K to $\overline{K}$, M_K becomes $M_{\overline{K}}$.

Let ι be an embedding of $\overline{K}$ into $\mathbf{C}$, and in 2.14 take for P the Riemann surface deduced from $X_{\overline{K}}$ by extension of scalars from $\overline{K}$ to $\mathbf{C}$. For this choice of P , the corresponding space $M_{\mathbf{C}}$ defined in 2.14 is deduced from $M_{\overline{K}}$ by extension of scalars from K to $\mathbf{C}$, and, for every prime number ℓ' , the ℓ' -adic cohomology of $M_{\overline{K}}$ is therefore identified with the cohomology of $M_{\mathbf{C}}$ with coefficients in $\mathbf{Q}_{\ell'}$. The conjectures 2.15 (i) and (ii) (for S empty) are therefore consequences of the following.

Conjecture 2.18. The cohomology of $M_{\overline{K}}$ admits an endomorphism “which deserves to be called V^* ” such that, for each $n \geq 1$, the number N_n of fixed points of V^n acting on E_r is given by

$$N_n = \sum (-1)^i \text{Tr}(V^{*n}, H^i(M_{\overline{K}})). \quad (2.18.1)$$

In (2.18.1), H^i is ℓ' -adic cohomology for any prime number ℓ' . One should also be able to take de Rham cohomology of M_K/K . The clause in quotation marks has no precise sense, but suggests compatibilities. For instance, V^* should be an endomorphism of the cohomology algebra.

There will not generally exist an endomorphism $V : M_{\overline{K}} \rightarrow M_{\overline{K}}$ inducing V^* in cohomology, but I hope that in the analytic space $M_{\overline{K}}^{\text{an}}$ (in the sense of Berkovich) associated to $M_{\overline{K}}$ there is an open $M_{\overline{K}}^0$ such that

- (a) the restriction morphism $H^*(M_{\overline{K}}) = H^*(M_{\overline{K}}^{\text{an}}) \rightarrow H^*(M_{\overline{K}}^0)$ is an isomorphism; and
- (b) a crystalline interpretation defines $V = \text{Frob}^* : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$, inducing V^* in cohomology.

The conjectural morphism $V : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$ should send $M_{\overline{K}}^0$ into a proper open subset of $M_{\overline{K}}^0$, corresponding to vector bundles with connection for which parallel transport has a larger radius of convergence; this is what makes ordinary cohomology, rather than compactly supported cohomology, the natural choice in (2.18.1). A variant of Conjecture 2.18, for the completions 2.16 of E_r and $M_{\overline{K}}$, might be more natural.

Example 2.19. Suppose $r = 1$. In this case, $M_{\overline{K}}$ is the universal additive extension of the Jacobian $\text{Pic}^0(X_{\overline{K}})$, and

$$H^* \text{Pic}^0(X_{\overline{K}}) \xrightarrow{\sim} H^*(M_{\overline{K}}) \quad (2.19.1)$$

is an isomorphism. There is also a natural isomorphism

$$H^*(\text{Pic}^0(X)) \xrightarrow{\sim} H^*(\text{Pic}^0(X_{\overline{K}})). \quad (2.19.2)$$

Over $\mathbf{F}$, one has $V : \text{Pic}^0(X) \rightarrow \text{Pic}^0(X)$, deduced by functoriality from $\text{Frob} : X \rightarrow X$ (cf. 2.2). This endomorphism V of $\text{Pic}^0(X)$ gives

$$V^* : H^*(\text{Pic}^0(X)) \rightarrow H^*(\text{Pic}^0(X)). \quad (2.19.3)$$

Define the endomorphism

$$V^* : H^*(M_{\overline{K}}) \rightarrow H^*(M_{\overline{K}}) \quad (2.19.4)$$

to be transported from (2.19.3) by (2.19.2) and (2.19.1).

Proposition 2.20. If $r = 1$ and V^* is defined as above, then (2.18.1) is true.

Proof for $n = 1$. One reduces to this case by an extension of scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$. The endomorphisms Frob and V of $\text{Pic}^0(X)$ are transposes of one another by the autoduality of the Jacobian $\text{Pic}^0(X)$ of X . Hence

$$\text{Tr}(V^*, H^i(\text{Pic}^0(X))) = \text{Tr}(F^*, H^i(\text{Pic}^0(X)))$$

and

$$\sum (-1)^i \text{Tr}(V^*, H^i(\text{Pic}^0(X))) = |\text{Pic}^0(X_0)(\mathbf{F}_q)|. \quad (2.20.1)$$

Class field theory identifies the set of fixed points of V acting on E_1 with the Pontryagin dual of the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$, with values in the roots of unity of $\overline{\mathbf{Q}}_\ell$ (see 6.1). Thus

$$N_1 = |\text{Pic}^0(X_0)(\mathbf{F}_q)|, \quad (2.20.2)$$

and, by (2.20.1), this verifies (2.18.1). Notice that, again by the autoduality of the Jacobian, the subscheme $\text{Pic}^0(X)^V$ of $\text{Pic}^0(X)$ fixed by V is the Cartier dual of the finite abelian group $\text{Pic}^0(X)^{\text{Frob}=\text{Pic}^0(X_0)(\mathbf{F}_q)}$.

Now pass to the tamely ramified case. Choose over W a lifting (X_W, S_W) of X and of its etale divisor S . Let (X_K, S_K) and $(X_{\overline{K}}, S_{\overline{K}})$ be deduced from (X_W, S_W) by extension of scalars from W to K and to $\overline{K}$. To define in 2.14 a complex analogue $M_{\mathbf{C}}(R^*)$ of $E(R)$, we had to make the choices (2.14.2) and (2.14.3). To define a p -adic analogue we choose

$$\text{an isomorphism between the groups of roots of unity of order prime to } p \text{ in } \overline{\mathbf{Q}}_\ell \text{ and in } K. \quad (2.21.1)$$

There is no need to give an analogue of (2.14.3), because reduction modulo p supplies a canonical isomorphism between the groups of roots of unity of order prime to p in K or $\overline{K}$ and in $\mathbf{F}$.

Let $X_{K(s)}$ be the completion of X_K at (the lifting of) s , and $X_{K(s)}^* := X_{K(s)} - \{s\}$. The datum (2.21.1) gives a natural isomorphism between the finite-order characters $\widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$ and the prime-to- p part of $/\mathbf{Z}$. It is exactly what is needed to attach to $R(s)$ an isomorphism class $R(s)_K^*$ of vector bundles with connection on $X_{K(s)}^*$. Define $M_K(R^*)$ to be the scheme over K which is the moduli space of geometrically irreducible vector bundles with connection of rank r on $X_K - S_K$, with completion at $s \in S$ isomorphic to $R(s)_K$.

Let ι be an embedding of K into $\mathbf{C}$. If in 2.14 one takes P and $S \rightarrow P$ deduced from (X_K, S_K) by extension of scalars from K to $\mathbf{C}$, if (2.14.2) is deduced from (2.21.1) by ι , and if (2.14.3) is induced by ι , then $M_{\mathbf{C}}(R^*)$ defined in 2.14 is deduced from $M_K(R^*)$ by extension of scalars from K to $\mathbf{C}$.

Let $M_{\overline{K}}(R^*)$ be obtained from $M_K(R^*)$ by extension of scalars from K to $\overline{K}$. As in 2.18, I conjecture that the cohomology of $M_{\overline{K}}(R^*)$ admits an endomorphism “which deserves to be called V^* ” such that for $n \geq 1$, the number of fixed points of V acting on $E(R)$ is given by

$$N_n = \sum (-1)^i \text{Tr}(V^*, H^i(M_{\overline{K}}(R^*))), \quad (2.21.2)$$

and this conjecture implies 2.15 (i) and (ii).

Continue to assume that the $R(s)$ are tamely ramified, and assume condition (2.10.2). Write $\det$ for the functor “maximal exterior power” $\bigwedge^r$. It induces a map

$$\det : E(R) \rightarrow E(\det R). \quad (2.22.1)$$

Hypothesis (2.10.2) ensures that $\det R$ satisfies (2.10.1), and that $E(\det R)$ is nonempty, hence a torsor under the group E_1 of isomorphism classes of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank one on X . The action is tensor product.

In 2.16, for $r = 1$ and S empty, we defined a moduli space M_K . Here we denote it by M_{1K} . The functor $\bigwedge^r$ induces

$$\det : M_K(R^*) \rightarrow M_K(\det R^*), \quad (2.22.2)$$

and $M_K(\det R^*)$ is a principal homogeneous space under M_{1K} . Extend scalars from K to $\overline{K}$ and consider the Leray spectral sequence for the morphism $\det$:

$$E_2^{pq} = H^p(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \Rightarrow H^{p+q}(M_{\overline{K}}(R^*), \mathbf{Q}'_\ell).$$

I conjecture the existence of an endomorphism V^* of this spectral sequence, respecting its multiplicative structure. The E_2 -term is given by

$$H^p(M_{\overline{K}}(\det R^*)) \otimes H^0(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \xrightarrow{\sim} E_2^{pq},$$

and the cohomologies of $M_{\overline{K}}(\det R^*)$ and of $M_{1\overline{K}}$ are canonically isomorphic. The conjectured existence of V^* therefore implies 6.7.

3 Arinkin's Method

3.1 3.1–3.3

Let s_0 be a closed point of X_0 . One may identify the closed points s of X both with the $\mathbf{F}$ -points of the $\mathbf{F}$ -scheme X , and with the $\mathbf{F}$ -points of the $\mathbf{F}_q$ -scheme X_0 . In doing so, the s above s_0 are identified with the $\mathbf{F}$ -points of $\text{Spec}(k(s_0))$, i.e. with the embeddings of $k(s_0)$ into $\mathbf{F}$ extending the embedding of $\mathbf{F}_q$ into $\mathbf{F}$. If one chooses such an embedding s , a character $\varepsilon_0 : k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$ gives, by 2.9, a character $\varepsilon := \varepsilon_0 \text{pr}_s : \widehat{\mathbf{Z}}(1) \rightarrow k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$. If $s = \text{Frob}(s')$, the character ε' of $\widehat{\mathbf{Z}}(1)$ defined by ε_0 and s' is ε^q .

At each $s_0 \in S_0$, give a multiset $R(s_0)$ of r characters $k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$. By 3.1 one deduces, for every $s \in S$, a multiset of r characters $\mathcal{R}(s)$ of $\widehat{\mathbf{Z}}(1)$. Let ρ_s be the representation of $\widehat{\mathbf{Z}}(1)$ that is the sum of the one-dimensional representations defined by these characters. The corresponding $\overline{\mathbf{Q}}_\ell$ -sheaves $R(s)$ on the $X_{(s)}^*$ satisfy (2.1.1). Write R for the family $(R(s))_{s \in S}$. We assume that

$$\prod_s \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (3.2.1)$$

By (2.10.1), this is a necessary condition for the existence of a lisse $\overline{\mathbf{Q}}_\ell$ -sheaf F on $X - S$ with $F|_{X_{(s)}^*} \simeq R(s)$. If s_0 has degree d , if ε_0 is a character of $k(s_0)^*$, and if for every s above s_0 , $\varepsilon[s] := \varepsilon_0 \text{pr}_s$ is the corresponding character of $\widehat{\mathbf{Z}}(1)$, then the product of these $\varepsilon[s]$ is the character that 3.1 attaches to $\varepsilon_0^{(q^d-1)/(q-1)} = \varepsilon_0 \text{N}_{k(s_0)/\mathbf{F}_q}$ and to any s over s_0 . It is also the character that 3.1 attaches to $\varepsilon_0|_{\mathbf{F}_q^*}$ and to $\mathbf{F}_q \hookrightarrow \mathbf{F}$. Hypothesis (3.2.1) can therefore be rewritten as

$$\prod_{s_0 \in S_0} \prod_{\varepsilon_0 \in R(s_0)} \varepsilon_0|_{\mathbf{F}_q^*} = 1. \quad (3.2.2)$$

Assume further that the $\mathcal{R}(s)$ satisfy the general-position condition (2.10.3). I learned from Arinkin how, under this hypothesis, to construct an algebraic variety Z over $\mathbf{F}_q$ such that, for every $n \geq 1$, the number $N_n(R)$ of fixed points of the permutation V^n of $E(R)$ is equal to the number of $\mathbf{F}_{q^n}$ -points of Z :

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.3.1)$$

Under the hypotheses of 3.3, Conjecture 2.15 (i) is therefore true. Beware that Z will only be defined after cutting into locally closed pieces.

3.2 3.4–3.6

For each $s_0 \in S_0$, let $\varepsilon[s_0](i)$ ($0 \leq i \leq a(s_0)$) be the characters of $k(s_0)^*$ occurring in $R(s_0)$, arranged in any order. Let $n[s_0](i)$ be the multiplicity of $\varepsilon[s_0](i)$ in the multiset $R(s_0)$, and

let $n[s_0]$ be the family of integers $(n[s_0](i))_{0 \leq i \leq a(s_0)}$. For $s \in S$ above s_0 , write $\varepsilon[s](i)$ for the character $\varepsilon[s_0](i) \circ \text{pr}_s$ of $\widehat{\mathbf{Z}}(1)$ (3.1), and set $n[s] := n[s_0]$. The multiset $\mathcal{R}(s)$ is

$$\mathcal{R}(s) := \{\text{the } \varepsilon[s](i), \text{ with multiplicities } n[s](i)\}. \quad (3.4.1)$$

Let E_0 be a vector bundle of rank r on X_0 . A parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$ on E_0 is the datum, for each $s_0 \in S_0$, of a finite filtration $F(s_0)$ of the $k(s_0)$ -vector space $(E_0)_{s_0}$, such that $\text{Gr}_{F(s_0)}^i((E_0)_{s_0})$ has dimension $n[s_0](i)$. Parabolic structures for E on X are defined similarly. If E_0 on X_0 is endowed with a parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$, its inverse image E on X inherits a parabolic structure α of type $(n(s))_{s \in S}$. We say that (E_0, α_0) is geometrically indecomposable if its inverse image E on X admits no nontrivial decomposition $E = E' \oplus E''$ compatible with the parabolic structure α , i.e. inducing decompositions of the $F^i E_s$. Let $\text{End}(E_0, \alpha_0)$ (respectively $\text{End}(E, \alpha)$) be the algebra of endomorphisms of E_0 (respectively E) respecting the filtrations $F(s_0)$ of the $(E_0)_{s_0}$ (respectively $F(s)$ of the E_s). Each of the following conditions is equivalent to geometric indecomposability:

$$\text{End}(E, \alpha) = \text{End}(E_0, \alpha_0) \otimes_{\mathbf{F}_q} \mathbf{F} \text{ has no idempotent other than 0 and 1.} \quad (3.4.2)$$

$$\text{End}(E_0, \alpha_0) / \text{rad } \text{End}(E_0, \alpha_0) = \mathbf{F}_q. \quad (3.4.3)$$

Fix an integer d . Consider vector bundles E_0 of rank r and degree d on X_0 , endowed with a parabolic structure α of type $(n[s_0])_{s_0 \in S_0}$. Let $Z(\mathbf{F}_q)$ be the set of isomorphism classes of such (E_0, α) that are geometrically indecomposable. Define $Z(\mathbf{F}_{q^n})$ similarly, replacing $X_0/\mathbf{F}_q$ by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}/\mathbf{F}_{q^n}$. I learned from Arinkin the following theorem.

Theorem 3.5. Under the hypotheses of 3.2 and 3.3 and with the notation of 3.4, the number $N_n(R)$ of fixed points of V^n (2.1) is

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.5.1)$$

Replacing X_0 by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$, one reduces to the case $n = 1$. Beware that the proof does not suggest a natural bijection between $E(R)^V$ and $Z(\mathbf{F}_q)$. It gives a family of one-dimensional vector spaces indexed by $E(R)^V$, with direct sum V , another family, with direct sum V' , indexed by $Z(\mathbf{F}_q)$, and an isomorphism of vector spaces $V \simeq V'$.

Sketch of proof of 3.5 $\Rightarrow$ (3.3.1). For every $\mathbf{F}_q$ -scheme T , let $X_T := X_0 \times_{\mathbf{F}_q} T$ and $S_T := S_0 \times_{\mathbf{F}_q} T$. Define $Z(T)$ as the set of isomorphism classes of vector bundles of rank r on X_T , endowed with a parabolic structure along S_T , and such that the conditions of 3.2 and 3.3 are satisfied on each geometric fibre of $X_T \rightarrow T$: degree d , type of the parabolic structure, indecomposability.

Naively one would like to take for the scheme Z of (3.3.1) a scheme representing the functor Z . For several reasons this functor is not representable:

- (a) The objects considered have nontrivial automorphisms, for instance homotheties; thus this functor is not a sheaf, even for the Zariski topology. For the functor Pic , the same problem is bypassed by passing to the associated sheaf. Here the situation is worse, because, for (E_T, α_T) a vector bundle with parabolic structure on X_T , and for (E_t, α_t) the induced bundle on the fibre X_t of $X_T \rightarrow T$ at $t \in T$,
- (b) the dimension of the automorphism group of (E_t, α_t) (equal to $\dim \text{End}(E_t, \alpha_t)$) is not always locally constant in t .
- (c) The set of t for which (E_t, α_t) is geometrically indecomposable is not always locally closed.

Because of (c), one cannot even hope that Z comes from an algebraic stack C (i.e. that $Z(T)$ be the set of isomorphism classes in $C(T)$). Nevertheless:

- (d) The automorphism groups are connected algebraic groups. If $k' \subset k''$ are finite extensions of $\mathbf{F}_q$, this ensures that

$$Z(k') \xrightarrow{\sim} Z(k'')^{\text{Gal}(k''/k')}.$$

- (e) In (b) above, the function $t \mapsto \dim \text{End}(E_t, \alpha_t)$ is constructible; in (c), the set of t such that (E_t, α_t) is geometrically indecomposable is constructible.

Properties (d) and (e) make it possible, by “cutting Z into pieces” and taking the associated (representable) sheaves, to obtain Z as promised by (3.3.1).

3.3 3.7–3.16

Sketch of proof of 3.5 (for $n = 1$). One applies the strategy 2.5. We must show that $|Z(\mathbf{F}_q)|$ is the number of cuspidal automorphic representations π of $\text{GL}(r, \mathbf{A})$, counted up to $\mathbf{F}_q$ -twist, that are unramified outside S_0 and whose local component π_{s_0} at $s_0 \in S_0$ is of the following type. By the local Langlands correspondence, the representation π_{s_0} of $\text{GL}(r, K_{s_0})$ corresponds to an F -semisimple $\overline{\mathbf{Q}}_\ell$ -representation ρ of the Weil group $W(\overline{K}_{s_0}/K_{s_0})$. We first require ρ to be semisimple (in the language of the Weil–Deligne group: $N = 0$; in Arthur’s language: trivial action of the local $\text{SL}(2)$ or $\text{SU}(2)$). The abelianization $W(\overline{K}_{s_0}/K_{s_0})^{\text{ab}}$ is $K_{s_0}^*$. Its inertia group $O_{s_0}^*$ has quotient $k(s_0)^*$. We require the restriction of ρ to inertia to factor through the representation of $k(s_0)^*$ which is the sum of the ε_0 in $R(s_0)$.

Let $P[s_0] \subset \text{GL}(r, O_{s_0})$ be the parahoric subgroup inverse image of the standard parabolic $\overline{P}[s_0]$ of $\text{GL}(r, k(s_0))$ of type $n(s_0)$. The reductive quotient $L[s_0]$ of $\overline{P}[s_0]$ is $\prod \text{GL}(n[s_0](i), k(s_0))$. Let $U[s_0]$ be the kernel of $P[s_0] \rightarrow L[s_0]$, and let $\varepsilon[s_0]$ be the character

$$\varepsilon[s_0] = \prod \varepsilon[s_0](i)(\det g_i) : P[s_0] \rightarrow \overline{P}[s_0] \rightarrow L[s_0] \rightarrow \overline{\mathbf{Q}}_\ell^*$$

of the parahoric $P[s_0]$. By construction, the $\varepsilon[s_0](i)$ are pairwise distinct. We shall admit the following assertion.

Assertion 3.8. The irreducible admissible representations of $\text{GL}(r, K_{s_0})$ of the type defined in 3.7 are those which admit a line stable under P on which P acts through the character $\varepsilon[s_0]$. This line is unique.

Let A' be the set of cuspidal automorphic representations π of $\text{GL}(r, \mathbf{A})$ admitting a line $L(\pi)$ fixed by the $\text{GL}(r, O_x)$ for $x \notin S_0$, and stable by $P[s_0]$ for $s_0 \in S$, with the action of $P[s_0]$ given by $\varepsilon[s_0]$. The line $L(\pi)$ is unique if it exists. By Lafforgue [L] and 3.8, $N_1(R)$ is the number of classes modulo $\mathbf{F}_q$ -twist in A' . Fix an idele d of valuation 1, and let A be the set of π in A' whose central character ω_π is equal to 1 at d . As observed after 2.3.1, one has

$$N_1(R) = |A' \bmod \mathbf{F}_q\text{-twist}| = |A \bmod \mathbf{F}_q\text{-twist by } \zeta \in \mu_r(\overline{\mathbf{Q}}_\ell)| = \frac{1}{r} |A|. \quad (3.9.1)$$

Let $\mathcal{A}$ be the vector space of functions on $\text{GL}(r, \mathbf{A})$ which are

- (a) invariant on the left by $\text{GL}(r, K) \cdot d^{\mathbf{Z}}$;
- (b) invariant on the right under $\text{GL}(r, O_x)$, for $x \notin S_0$;
- (c) transformed by $\varepsilon[s_0]$ for the right action of $P[s_0]$, for $s_0 \in S_0$.

A key point of the proof of 3.5 is the following lemma.

Lemma 3.11. The vector space $\mathcal{A}$ is the direct sum of the lines $L(\pi)$, for $\pi \in A$. It is therefore of dimension $rN_1(R)$.

This lemma is the automorphic counterpart, and a consequence, of the fact that irreducibility of the $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 under consideration follows from the local hypotheses made. One must check that the functions in $\mathcal{A}$ are automatically cuspidal. This done, if one writes the space

of cuspidal functions on $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$ as a direct sum of irreducible representations π of $\mathrm{GL}(r, \mathbf{A})$, the π in A contribute $L(\pi)$ to $\mathcal{A}$, and the others contribute nothing. That the functions in $\mathcal{A}$ are cuspidal follows from the fact that, in the spectral decomposition of functions on $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$, the induced representations giving rise to Eisenstein series admit no nonzero vector satisfying 3.10 (b) and (c).

The space $\mathcal{A}$, the sum of the $L(\pi)$ for $\pi \in A$, is identified with the space of functions f on

$$\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A}) / \prod_{x \notin S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.1)$$

such that, for every $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$. Let v be the map from (3.12.1) to $\mathbf{Z}/r$ induced by $v \det : \mathrm{GL}(r, \mathbf{A}) \rightarrow \mathbf{Z}$. Let $\bar{A}$ be the quotient of A by the $\mathbf{F}_q$ -twist action of $\mu_r(\mathbf{Q}_\ell)$, and for each orbit B , let V_B be the subspace of $\mathcal{A}$ which is the sum of the $L(\pi)$ for $\pi \in B$. By (2.3.1), $\dim(V_B) = r$. For $m \in \mathbf{Z}/r$, write $(3.12.1)_m$ for the part of (3.12.1) where $v = m$, let φ_m be its characteristic function, and put $L(\pi, m) = \{\varphi_m f \mid f \in L(\pi)\}$. The dimension of $L(\pi, m)$ is at most one. For each character η of $\mathbf{Z}/r$, if π' is the $\mathbf{F}_q$ -twist of π by $\eta(1)$, multiplication by η^v transforms $L(\pi)$ into $L(\pi')$. For $\pi \in B$, $L(\pi, m)$ is therefore independent of π ; write $L(B, m) := L(\pi, m)$. The space V_B is stable under the multiplications by the η^v , hence under multiplication by the φ_m : for $\pi \in B$,

$$V_B = \bigoplus_{m \in \mathbf{Z}/r} L(B, m).$$

Each $L(B, m)$ is therefore one-dimensional. The sum $\mathcal{A}_m$ of the $L(B, m)$ is of dimension $|\bar{A}| = N_1(R)$. It is identified with the space of functions f on $(3.12.1)_m$ such that, for each $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$.

Let $\mathrm{GL}(r, \mathbf{A})^{(d)}$ be the set of $g \in \mathrm{GL}(r, \mathbf{A})$ such that $v \det(g) = d$. If $\bar{d}$ is the class of d modulo r , the set

$$\mathrm{GL}(r, K) \backslash \mathrm{GL}(r, \mathbf{A})^{(d)} / \prod_{x \in S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.2)$$

maps bijectively onto $(3.12.1)_{\bar{d}}$. Hence:

Lemma 3.13. $N_1(R)$ is the dimension of the space, again denoted $\mathcal{A}_d$, of functions f on (3.12.2) which, for the right action of the $P[s_0]$ ($s_0 \in S_0$), satisfy $f(gp) = \varepsilon[s_0](p)f(g)$.

The set (3.12.2) is interpreted as the set of isomorphism classes of triples (E_0, α, β) , where E_0 is a vector bundle of rank r and degree d on X_0 , and where α and β are structures of the following types on E_0 :

- (a) α is a parabolic structure of type $(n[s_0])_{s_0 \in S_0}$;
- (b) β is the datum of bases of the vector spaces $\mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0})$.

We identify a basis as in (b) with an isomorphism

$$\beta_{s_0, i} : k(s_0)^{n[s_0](i)} \xrightarrow{\sim} \mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0}).$$

With this interpretation, the space $\mathcal{A}_d$ of 3.13 becomes the space of functions $f(E_0, \alpha, \beta)$, for E_0, α, β as above, such that

- (i) $f(E_0, \alpha, \beta)$ depends only on the isomorphism class of (E_0, α, β) ;
- (ii) for $s_0 \in S_0$ and $g \in L[s_0] = \prod \mathrm{GL}(n_i, k(s_0))$, one has

$$f(E_0, \alpha, \beta g) = \varepsilon[s_0](g)f(E_0, \alpha, \beta).$$

For an automorphism T of (E_0, α) , let $\text{Gr}_F^i(s_0)(T_{s_0})$ be the automorphism of $\text{Gr}_F^i(s_0)((E_0)_{s_0})$ induced by T , and put

$$\chi(T) := \prod_{s_0} \prod_i \varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})). \quad (3.14.1)$$

For any β , hypothesis (i) gives $f(E_0, \alpha, T(\beta)) = f(E_0, \alpha, \beta)$. Hypothesis (ii) gives $f(E_0, \alpha, T(\beta)) = \chi(T)f(E_0, \alpha, \beta)$. If there exists an automorphism T with $\chi(T) \neq 1$, then $f(E_0, \alpha, \beta) = 0$. Conversely, if $\chi(T) = 1$ for every automorphism T of (E_0, α) , then f may be prescribed freely at one (E_0, α, β) , and its value there determines its value at all (E_0, α, β') . Thus the dimension of $\mathcal{A}_d$ is the number of isomorphism classes of (E_0, α) as above such that for every automorphism T , one has $\chi(T) = 1$. Theorem 3.5 follows from the characterization (3.4.3) of geometric indecomposability and the following proposition.

Proposition 3.15. Let E_0 be a vector bundle of rank r on X_0 , endowed with a parabolic structure of type $(n(s_0))_{s_0 \in S_0}$. The following conditions are equivalent:

- (i) The quotient of the $\mathbf{F}_q$ -algebra $\text{End}(E_0, \alpha)$ by its radical is reduced to $\mathbf{F}_q$.
- (ii) For every automorphism T of (E_0, α) , $\chi(T)$, defined by (3.14.1), is equal to 1.

We shall deduce that (i) implies (ii) from (3.2.2), and that the negation of (i) implies the negation of (ii) from the general-position hypothesis (2.10.3).

Proof of (i) $\Rightarrow$ (ii). If λ is the image in $\mathbf{F}_q^*$ of the automorphism $T \in \text{End}(E_0, \alpha)^*$, then

$$\varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})) = \varepsilon(s_0)(i)^{n[s_0](i)}(\lambda),$$

so that (3.2.2) implies $\chi(T) = 1$.

Proof of (ii) $\Rightarrow$ (i). The quotient of the $\mathbf{F}_q$ -algebra $\text{End}(E_0, \alpha)$ by its radical is a product of matrix algebras over finite extensions of $\mathbf{F}_q$. If it is not reduced to $\mathbf{F}_q$, it contains a commutative separable $\mathbf{F}_q$ -algebra of dimension > 1 . This algebra admits a lifting into $\text{End}(E_0, \alpha)$. Suppose then that $\text{End}(E_0, \alpha)$ contains a commutative separable subalgebra k of dimension > 1 . We must show that the character (3.14.1) of $\text{End}(E_0, \alpha)^*$ is nontrivial. We shall prove that its restriction χ to k^* is nontrivial.

Split case. Suppose first that k is a product of copies of $\mathbf{F}_q$, say $k \simeq \mathbf{F}_q^I$, and that the closed points $s_0 \in S_0$ have degree 1, so that $\mathbf{F}_q \xrightarrow{\sim} k(s_0)$. The $\mathbf{F}_q^I$ -module structure of E_0 gives a decomposition

$$E_0 = \bigoplus_{\iota \in I} E_0^\iota \quad (3.15.1)$$

such that $(\lambda_\iota)_{\iota \in I} \in \mathbf{F}_q^I$ acts on E_0^ι by multiplication by λ_ι . The decomposition (3.15.1) is compatible with the parabolic structure. For $\iota \in I$, put

$$n^\iota[s_0](i) = \dim \text{Gr}_{F(s_0)}^i((E_0^\iota)_{s_0}).$$

The sum over i of the $n^\iota[s_0](i)$ is the rank of the bundle E_0^ι . The restriction of χ to the factor with index ι of $(\mathbf{F}_q^I)^* = \bigoplus_{\iota \in I} \mathbf{F}_q^*$ is

$$\lambda \longmapsto \prod_{s_0 \in S_0} \prod_i \varepsilon[s_0](i)^{n^\iota[s_0](i)}(\lambda). \quad (3.15.2)$$

Its composite with the morphism $\widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_q^*$ of 3.1 is the character

$$\prod_{s \in S} \prod_i \varepsilon[s](i)^{n^\iota[s_0](i)}$$

of $\widehat{\mathbf{Z}}(1)$. Hypothesis (2.10.3) ensures that this character is nontrivial.

Reduction of the general case to the split case. It suffices to show that nontriviality of χ is invariant under extension of scalars from $\mathbf{F}_q$ to a finite extension $\mathbf{F}_{q^n}$. Under such an extension of scalars, X_0 becomes a curve X' over $\mathbf{F}_{q^n}$, S_0 becomes a divisor S' of X' , and k becomes $k' := k \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$. For $s' \in S'$ above $s_0 \in S_0$, put $n[s'] := n[s_0]$, and let $\varepsilon[s'](i)$ be the composite of $\varepsilon[s_0](i)$ with the norm $N_{k(s')/k(s_0)}$. The bundle E' on X' inverse image of E_0 inherits a parabolic structure α' of type $(n(s'))_{s' \in S'}$. Repeating for E' on $X'/\mathbf{F}_{q^n}$ the construction (3.14.1), one obtains a character χ' of k'^* . The curve $X/\mathbf{F}$, the integers $n[s](i)$, and the characters $\varepsilon[s](i)$ of $\hat{\mathbf{Z}}(1)$, come equally from $X_0/\mathbf{F}_q$ and from $X'/\mathbf{F}_{q^n}$.

To prove that triviality of χ is equivalent to triviality of χ' , it suffices to prove that

$$\chi' = \chi \circ N_{k'/k}. \quad (3.15.3)$$

Indeed, the norm $N_{k'/k}$ is surjective. Identity (3.15.3) follows from identity (3.16.1) below, applied to each $\mathrm{Gr}_F^i(s_0)(E_{s_0})$, endowed with its $(k, k(s_0))$ -bimodule structure.

Let k_0 be a finite extension of $\mathbf{F}_q$, N_0 a finite-dimensional k_0 -vector space, k a product of finite extensions of $\mathbf{F}_q$, and $\rho : k \rightarrow \mathrm{End}_{k_0}(N_0)$. The (k, k_0) -bimodule N defines a homomorphism

$$[N] : k^* \rightarrow k_0^*, \quad \lambda \mapsto \det(\rho(\lambda)).$$

Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$. One obtains k'_0 , k' , and a (k', k'_0) -bimodule N' . This bimodule defines

$$[N'] : k'^* \rightarrow k'_0^*, \quad \lambda \mapsto \det_{k'_0}(\rho(\lambda)).$$

Lemma 3.16. The diagram

$$\begin{array}{ccc} k'^* & \xrightarrow{[N']} & k'_0^* \\ N_{k'/k} \downarrow & & \downarrow N_{k'_0/k_0} \\ k^* & \xrightarrow{[N]} & k_0^* \end{array} \quad (3.16.1)$$

is commutative.

Proof. Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}$. Diagram (3.16.1) embeds in an analogous diagram (3.16.2), where k and k_0 are replaced by $k_F := k \otimes_{\mathbf{F}_q} \mathbf{F}$, $k_{0F} := k_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, where N_0 is replaced by a (k_F, k_{0F}) -bimodule N , and where the extension $\mathbf{F}_{q^n}$ of $\mathbf{F}_q$ is replaced by a product $\mathbf{F}^J$ of n copies of $\mathbf{F}$:

$$\begin{array}{ccc} (k_F \otimes \mathbf{F}^J)^* & \xrightarrow{[N \otimes \mathbf{F}^J]} & (k_{0F} \otimes \mathbf{F}^J)^* \\ \downarrow & & \downarrow \\ k_F^* & \xrightarrow{[N]} & k_{0F}^*. \end{array} \quad (3.16.2)$$

One has $(k_F \otimes \mathbf{F}^J)^* = (k_F^*)^J$, $(k_{0F} \otimes \mathbf{F}^J)^* \simeq (k_{0F}^*)^J$, and the vertical “norm” maps are the product maps. The first horizontal line is the product of copies, indexed by J , of the second horizontal line. The commutativity of (3.16.2) follows.

4 Trace Formula

4.1 4.1–4.5

In [DF], the counting problem 2.3 (i) is solved when $|S_0| \geq 2$ and the imposed local monodromies are unipotent, with a single Jordan block. In automorphic language this means asking the local components π_{s_0} ($s_0 \in S_0$) to be of the form

$$\text{special representation} \otimes \chi \det g, \quad (4.1.1)$$

for χ an unramified character of $K_{s_0}^*$.

The hypothesis $|S_0| \geq 2$ allows one to pass from automorphic representations for $\mathrm{GL}(r, K)$ to automorphic representations for the multiplicative group of a division algebra D of dimension r^2 over K . It is reassuring to use the trace formula only in a case where the quotient is compact (modulo the centre), but it should be possible instead to use the trace formula for $\mathrm{GL}(n)$, for a suitable test function, with the simplification brought by the fact that one wants to detect automorphic representations which at two places are in the discrete series. After all, this is how one relates $\mathrm{GL}(r, K)$ and D^* . The explanations that follow will be simplified by staying with $\mathrm{GL}(r, K)$, but admitting that the only nonzero terms in the required trace formula are those associated with conjugacy classes of elliptic elements of finite order in $\mathrm{GL}(r, K)$.

Choose an embedding $\mathbf{F}_{q^r} \hookrightarrow M_r(\mathbf{F}_q)$. It defines a morphism $\mathbf{F}_{q^r}^* \hookrightarrow \mathrm{GL}(r, \mathbf{F}_q) \hookrightarrow \mathrm{GL}(r, K)$, through which the set of orbits of $\mathrm{Gal}(\mathbf{F}_{q^r}/\mathbf{F}_q)$ in $\mathbf{F}_{q^r}^*$ maps bijectively to the set of conjugacy classes under consideration.

Let $T(\gamma)$ be the term of the trace formula (giving the number of fixed points of V) thus associated to $\gamma \in \mathbf{F}_{q^r}^*$. It depends only on the subextension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ generated by γ . Put $T(m) := T(\gamma)$. The number c_m of elements of $\mathbf{F}_{q^m}^*$ which generate the extension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ is

$$c_m = \sum_{a|m} \mu(a)(q^{m/a} - 1).$$

The number of fixed points of V is

$$N_1 = \sum_{m|r} \frac{c(m)}{m} T(m) \tag{4.1.2}$$

$$= \sum_{m|r} \sum_{ab=m} \frac{1}{m} \mu(a)(q^b - 1) T(ab) \tag{1}$$

$$= \sum_b \sum_{ab|r} \frac{1}{m} \mu(a)(q^b - 1) T(ab). \tag{2}$$

The number of fixed points of V^n ($n \geq 1$) is obtained similarly, after extending scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$.

Surprise 4.2. When n varies, each $T(m)$ is generally not, as a function of n , of the form (1.2.1). On the other hand, in (4.1.2), each term of the sum over b is of this form.

I have no explanation, neither geometric nor automorphic, of 4.2, only a proof. I likewise have no explanation for the following.

Surprise 4.3. Under the hypotheses of 4.1, the number N_1 of fixed points of V is divisible by q .

If $r = 2$, the trace formula is a little less frightening. Drinfeld [Dr] used it for the case where S_0 is empty. Flicker (unpublished) treated the case where $|S_0| = 1$ and where the local monodromy at each $s \in S$ is unipotent with a single Jordan block, i.e. since $r = 2$, nontrivial. Gathering the information thus obtained, one gets:

Surprise 4.5. Let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy (both nontrivial and trivial). The number of fixed points of $V := \mathrm{Frob}^*$ on E depends only on $X_0/\mathbf{F}_q$ and on $|S|$.

This is the surprise to which 2.16 alludes. If, as in [DF], one requires the local monodromy at each $s \in S$ to be unipotent with a single Jordan block, the number of fixed points depends on X_0 , on $|S|$, and also on the action of Frobenius on S (viewed as a conjugacy class in the symmetric group $S_{|S|}$).

5 Examples

5.1 5.1–5.9

Take $X_0 = \mathbf{P}^1$, $|S| = 4$, $r = 2$, and let E be the set of isomorphism classes of irreducible lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with local monodromy at each $s \in S$ nontrivial unipotent. In this case, the number of fixed points of the iterates of the permutation $V = \text{Frob}^*$ of E is given by

$$N_n = q^n. \quad (5.1.1)$$

If $|S_0| \geq 2$, (5.1.1) is an application of the main result of [DF]. The following curious result ([DF] §7) allows one to deduce the case where S_0 is reduced to one closed point of degree 4.

Proposition 5.2. Each permutation σ of type $(2, 2)$ of S extends to a projectivity of $\mathbf{P}^1$. The action of this projectivity on E is trivial.

The proof is by reduction to an analogous theorem over $\mathbf{C}$.

It is natural to complete the set E of 5.1 to $\bar{E}$, the set of isomorphism classes, up to semisimplification, of lisse $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy at each $s \in S$. Since $X_0 = \mathbf{P}^1$ and $|S| = 4$, this completion $\bar{E}$ differs from E only by the adjunction of one point corresponding to the constant $\overline{\mathbf{Q}}_\ell$ -sheaf $\overline{\mathbf{Q}}_\ell^2$. We again write V for the permutation of $\bar{E}$ induced by Frob^* . The number $\bar{N}_n$ of fixed points of V^n acting on $\bar{E}$ is

$$\bar{N}_n = q^n + 1 \quad (\text{fixed points on } \bar{E}). \quad (5.3.1)$$

The complex analogue of $\bar{E}$ is the moduli space $\bar{M}$ of complex local systems of rank 2 on $\mathbf{P}^1 - S$, for $|S| = 4$, with unipotent local monodromy. These local systems are taken up to semisimplification. If one chooses a base point $o \in \mathbf{P}^1 - S$, the functor $F \mapsto F_o$ is an equivalence from local systems to representations of $\pi_1(\mathbf{P}^1 - S, o)$, and one can take as coordinates on $\bar{M}$ the traces of elements of π_1 . The space $\bar{M}$ is affine, purely of dimension 2, with a unique singular point corresponding to the constant local system $\mathbf{C}^2$. The singularity is of type D_4 . The nonzero Betti numbers of $\bar{M}$ are

$$b_0 = 1, \quad b_2 = 1. \quad (5.4.1)$$

I have not made a complete verification, but the situation seems exactly the same for $X_0 = \mathbf{P}^1$, $|S| = 3$, $r = 3$, with unipotent local monodromy. Again, if one of the local monodromies is not a single Jordan block, the $\overline{\mathbf{Q}}_\ell$ -sheaf is an iterated extension of constant local systems $\overline{\mathbf{Q}}_\ell$. The analogue of 5.3 holds for σ a cyclic permutation of the three points of S . The number of fixed points of V^n is again $q^n + 1$, with the “1” given by reducible local systems with unipotent local monodromy, all having a constant local system as semisimplification. The unique singular point of the complex analogue is of type E_6 .

Take $X_0 = \mathbf{P}^1$, $|S| = 4$, $r = 2$, and assume that, as in 3.4, the imposed local monodromy is, at each $s_0 \in S_0$, given by two distinct characters $\alpha'[s_0]$, $\alpha''[s_0]$ of $k(s_0)^*$. Assume (3.2.2) and the general-position hypothesis (2.10.3). Using (3.5.1), one obtains:

Proposition 5.7. Under the hypotheses of 5.6, one has

$$N_1 = q + 1 + |S_0(\mathbf{F}_q)|. \quad (5.7.1)$$

Arinkin has checked that for $X_0 = \mathbf{P}^1$ and $r = 2$, if a vector bundle E_0 of rank 2 on X_0 with parabolic structure α_0 at S_0 is indecomposable, then $\text{End}(E_0, \alpha)$ is reduced to the scalars. If one defines Z as in 3.6, and passes to the associated sheaf for the Zariski topology, one obtains a functor representable by an algebraic space. In the case $|S| = 4$, if one takes the degree $d = 1$, it is represented by the sum of two copies of X_0 , glued along $X_0 - S_0$. The number of points of this scheme over $\mathbf{F}_q$ is given by (5.7.1).

Pass to the complex analogue. On $\mathbf{P}^1 - S$, with $|S| = 4$, consider the moduli space of complex local systems of rank 2 with prescribed local monodromy at each $s \in S$. At $s \in S$, require the local monodromy to be conjugate to $A_s = \begin{pmatrix} a_s & 0 \\ 0 & b_s \end{pmatrix}$. Assume $\prod_s a_s b_s = 1$, and that if, for every s , c_s is one of a_s or b_s , one never has $\prod c_s = 1$. Put $A := (A_s)_{s \in S}$, and write $M(A)$ for the moduli space, 2.10. It is smooth and purely of dimension 2. One may view $M(A)$, for A_s close to 1, as a deformation of $\bar{M}$ considered in 5.4. In this deformation, the space of vanishing cycles has dimension 4, since the singularity of $\bar{M}$ is of type D_4 . At infinity the geometry does not change, and the nonzero Betti numbers are therefore

$$b_0 = 1, \quad b_2 = 5. \quad (5.8.1)$$

Take $X_0 = \mathbf{P}^1$, $|S| = 3$, $r = 3$, and assume that, as in 3.4, the imposed local monodromy is at each $s_0 \in S_0$ given by three distinct characters of $k(s_0)^*$. Assume (3.2.2) and the general-position hypothesis 2.10.3. Using (3.5.1), one obtains here

$$N_1 = q + 1 + 2|S_0(\mathbf{F}_q)|. \quad (5.9.1)$$

The sheaf associated to Z as in 3.6 is indeed, for $d = 1$, represented by the sum of three copies of X_0 , glued along $X_0 - S_0$. If S_0 consists of three rational points over $\mathbf{F}_q$, (5.9.1) says that $N_1 = (q + 1) + 6$, as suggested by the E_6 singularity of the complex analogue for unipotent local monodromy (cf. 5.5).

6 Rank 1

6.1 6.1–6.7

Let $E(1, \emptyset)$ denote the set $E(R)$ of 2.1 for $r = 1$ and $S_0 = \emptyset$. Tensor product gives it the structure of an abelian group. Class field theory identifies $E(1, \emptyset)^V$ with the group of $\bar{\mathbf{Q}}_\ell^*$ -valued characters of the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$. Indeed, this theory identifies the abelianized fundamental group of X_0 with the profinite completion of $\text{Pic}(X_0)$. The group $\text{Pic}(X_0)$ maps to $\mathbf{Z}$ by the degree map, with kernel the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$ of $\mathbf{F}_q$ -points of the Jacobian $\text{Pic}^0(X_0)$. The isomorphism classes of lisse $\bar{\mathbf{Q}}_\ell^*$ -sheaves F_0 of rank one on X_0 are thus identified with characters χ of $\text{Pic}(X_0)$ with values in $\bar{\mathbf{Z}}_\ell^*$; $\mathbf{F}_q$ -twist by λ corresponds to multiplication by the character λ^n of $\mathbf{Z}$; and, by attaching to F_0 the restriction of χ to $\text{Pic}^0(X_0)(\mathbf{F}_q)$, one obtains a bijection between $\mathbf{F}_q$ -twist classes of lisse F_0 of rank 1 on X_0 and characters of $\text{Pic}^0(X_0)(\mathbf{F}_q)$.

For arbitrary $r \geq 1$ and R as in 2.1, tensor product induces an action of the abelian group $E(1, \emptyset)$ on $E(R)$, and of $E(1, \emptyset)^V$ on $E(R)^V$. If $r = 1$, and R satisfies the compatibility (2.10.1), class field theory shows that $E(R)^V$ is a principal homogeneous space under $E(1, \emptyset)^V$. Thus

$$|E(R)^V| = |E(1, \emptyset)^V| = |\text{Pic}^0(X_0)(\mathbf{F}_q)| \quad (r = 1 \text{ and } (2.10.1)). \quad (6.2.1)$$

If $r > 1$, taking the exterior power $\wedge^r$ gives a map

$$\det : E(R) \rightarrow E(\bigwedge^r R)$$

such that, for the action of $E(1, \emptyset)$, denoted $\otimes$, one has

$$\det(l \otimes F) = l^r \otimes \det(F).$$

Similarly, after passage to the fixed points of $V = \text{Frob}^*$, for $\det : E(R)^V \rightarrow E(\bigwedge^r R)^V$ and the action of $E(1, \emptyset)^V$.

The action of $E(1, \emptyset)^V$ on $E(R)^V$ is not necessarily free, and the fibres of

$$\det : E(R)^V \rightarrow E(\bigwedge^r R)^V$$

do not necessarily all have the same number of elements. Nevertheless, in all cases that have been computed, $|E(1, \emptyset)^V|$ divides $|E(R)^V|$. As a function of n , $|\text{Pic}^0(X_0)(\mathbf{F}_{q^n})|$ has the form (2.15.1):

$$|\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| = \sum (-1)^i \text{Tr}(\text{Frob}^*, H^i(\text{Pic}^0(X), \mathbf{Q}_\ell)) = \sum (-1)^i \text{Tr}(\text{Frob}^*, \bigwedge^i H^1(X, \mathbf{Q}_\ell)).$$

I conjecture the following precise form of Conjecture 2.15 (i).

Conjecture 6.3. As a function of n , the number of fixed points $N_n(R)$ of V^n is of the form

$$N_n(R) = |\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n \quad (6.3.1)$$

for integers c_k and suitable numbers γ_k .

Here is an analogue of 6.3 in algebraic geometry. Let A_0 be an abelian variety over $\mathbf{F}_q$, and let $f_0 : X_0 \rightarrow A_0$ be a proper smooth morphism. Although the fibres of $f : X_0(\mathbf{F}_{q^n}) \rightarrow A_0(\mathbf{F}_{q^n})$ need not have the same number of elements (example: $X_0 = A_0$ and f_0 multiplication by an integer prime to p), $|A_0(\mathbf{F}_{q^n})|$ divides $|X_0(\mathbf{F}_{q^n})|$. Indeed, the $R^i f_! \mathbf{Q}_\ell$ are lisse $\mathbf{Q}_\ell$ -sheaves on A , and the (abelian) fundamental group of A acts on these sheaves. One knows that this action is semisimple. Let $(R^i f_! \mathbf{Q}_\ell)_0$ be the subsheaf of invariants. It comes from a subsheaf $(R^i f_{0!} \mathbf{Q}_\ell)_0$ of $R^i f_{0!} \mathbf{Q}_\ell$ on A_0 , and this last sheaf is the inverse image on A_0 of a sheaf on $\text{Spec}(\mathbf{F}_q)$, denoted $H^i(X_0/A_0)$. One knows that

$$H^i(A, (R^j f_! \mathbf{Q}_\ell)_0) \xrightarrow{\sim} H^i(A, R^j f_! \mathbf{Q}_\ell). \quad (6.4.1)$$

Grothendieck's trace formula for the number of $\mathbf{F}_q$ -points of X_0 can therefore be rewritten as

$$|X_0(\mathbf{F}_{q^n})| = \text{Tr} \left(\text{Frob}^*, \left(\sum (-1)^i H^i(A, \mathbf{Q}_\ell) \right) \otimes \left(\sum (-1)^j H^j(X/A) \right) \right). \quad (6.4.2)$$

If the γ_k are the eigenvalues of Frobenius on the $H^j(X/A)$, and the c_k are the alternating sums of their multiplicities, one deduces from (6.4.2) that

$$|X_0(\mathbf{F}_{q^n})| = |A_0(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n, \quad (6.4.3)$$

a formula analogous to (6.3.1). The promised divisibility of $|X_0(\mathbf{F}_{q^n})|$ by $|A_0(\mathbf{F}_{q^n})|$ follows from the fact that the Frobenius eigenvalues on $H^j(X/A)$, contained in the cohomology of a fibre of $X_0 \rightarrow A_0$, are algebraic integers.

In 6.4 the crucial point is not that f_0 is proper and smooth, but that the $R^i f_! \mathbf{Q}_\ell$ are lisse. One can avoid having to assume that the $R^i f_! \mathbf{Q}_\ell$ are semisimple by considering, rather than the subsheaf of invariants for the action of $\pi_1(A)$, the largest subsheaf on which the action is unipotent. Assume that there is an action of A_0 on X_0 , and an integer r , such that f_0 is equivariant for the action of A_0 on A_0 by $x : y \mapsto rx + y$. Under this hypothesis, the inverse images of the $R^i f_! \mathbf{Q}_\ell$ on A by multiplication by $r : A \rightarrow A$ are constant sheaves. This implies both their lissity and their semisimplicity.

If $g \geq 1$, Conjecture 6.3 implies that in the conjectural expression (2.11.1) for $N_n(R)$, one has $\sum a_i = 0$. Pass to the complex analogues as in 2.14. Let $M(1, \emptyset)$ be the group of isomorphism classes of complex local systems of rank 1 on P . It is isomorphic to $(\mathbf{C}^*)^{2g}$, and, if the compatibility $\prod \det R^*(s) = 1$ is satisfied, $M(\det R^*)$ is a principal homogeneous space under $M(1, \emptyset)$. The group $M(1, \emptyset)$ acts on $M(R^*)$ and on $M(\det R^*)$. Denote the action by $\otimes$. The map induced by $\bigwedge^r$,

$$\det : M(R^*) \rightarrow M(\det R^*),$$

satisfies $\det(l \otimes F) = l^r \otimes \det(F)$. This implies that $\det$ is a fibration, that the $R^i \det_! \mathbf{Z}$ are locally constant, and that the action of $\pi_1 M(\det R^*)$ on the $R^i \det_! \mathbf{Z}$ factors through its quotient $\pi_1 / r\pi_1 \simeq (\mathbf{Z}/r\mathbf{Z})^{2g}$. If $g \geq 1$, one deduces by arguments analogous to those of 6.4 that $\chi(M(R^*)) = 0$, in agreement with 2.15 (ii). Let $(R^i \det_! \mathbf{Q})_0$ be the local subsystem of $R^i \det_! \mathbf{Q}$ of invariants under the action of π_1 , and let $\chi(M(R^*)/M(\det R^*))$ be the alternating sum of the ranks of the $(R^i \det_! \mathbf{Q})_0$.

Conjecture 6.7. With the notation of 6.3 and 6.6, one has

$$\sum c_k = \chi(M(R^*)/M(\det R^*)).$$

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